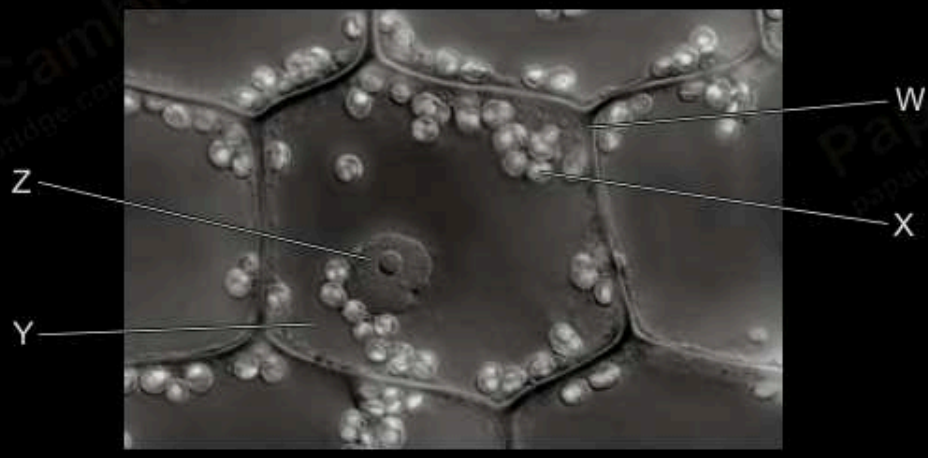
May/June 2026 · Paper 1 · Variant 1 · Question 1

Correct answer: A

1 The photomicrograph shows *Elodea* cells.



Which row identifies the cell wall, a chloroplast, the cytoplasm and the nucleus?

	cell wall	chloroplast	cytoplasm	nucleus
A	W	X	Y	Z
B	W	Z	Y	X
C	Y	X	W	Z
D	Y	W	Z	X

May/June 2026 · Paper 1 · Variant 1 · Question 2

Correct answer: B

2 Human liver cells contain, on average, 1500 mitochondria per cell. The volume of a typical liver cell is 11 305 μm^3 and mitochondria occupy 20% of this volume.

What is the average volume of a single liver cell mitochondrion?

A 0.15 μm^3 **B** 1.5 μm^3 **C** 15.0 μm^3 **D** 150 μm^3

May/June 2026 · Paper 1 · Variant 1 · Question 4

Correct answer: B

4 In a practical lesson, a student makes the following observations about some organisms.

'They are spherical, unicellular organisms which are visible with the light microscope.

Cytoplasm is present and, after staining, a nucleus can be seen.

A vacuole is present.'

What is being described?

A bacterium

B fungus

C plant

D virus

May/June 2026 · Paper 1 · Variant 1 · Question 6

Correct answer: A

6 Which statements about diffusion are correct?

1 Molecules move in a random way.

2 Molecules move down a concentration gradient.

3 Molecules may move through a partially permeable membrane.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

May/June 2026 · Paper 1 · Variant 1 · Question 7

Correct answer: C

7 The roots of two groups of maize plants are placed in solutions containing radioactive nitrate ions. This enables the movement of the nitrate ions to be tracked.

A chemical that prevents respiration is added to the solution in which the experimental plants are placed.

The second group of plants have no chemical added and act as a control.

At regular intervals, the solutions surrounding the roots are tested for radioactive nitrate ions.

The table shows the results.

time / minutes	concentration of radioactive nitrate ions in the solutions surrounding the roots (arbitrary units)	
	control plants	experimental plants
0	7.8	7.8
15	3.6	6.9
30	2.9	6.7
60	2.5	6.5
120	1.3	6.5

What is **not** a valid conclusion from the results?

A Active transport takes place in the control plants because respiration is supplying energy.

B Small amounts of nitrate ions enter the experimental plants by diffusion.

C Small amounts of nitrate ions enter the experimental plants by active transport.

D The chemical that prevents respiration reduces the uptake of nitrate ions by maize plants.

May/June 2026 · Paper 1 · Variant 1 · Question 8

Correct answer: D

8 What is an example of an organ?

A alveolus

B sperm

C neurone

D kidney

May/June 2026 · Paper 1 · Variant 1 · Question 9

Correct answer: C

9 Many nutrients can be identified by simple chemical tests.

Which nutrient causes Benedict's solution to change colour to red?

A fat

B protein

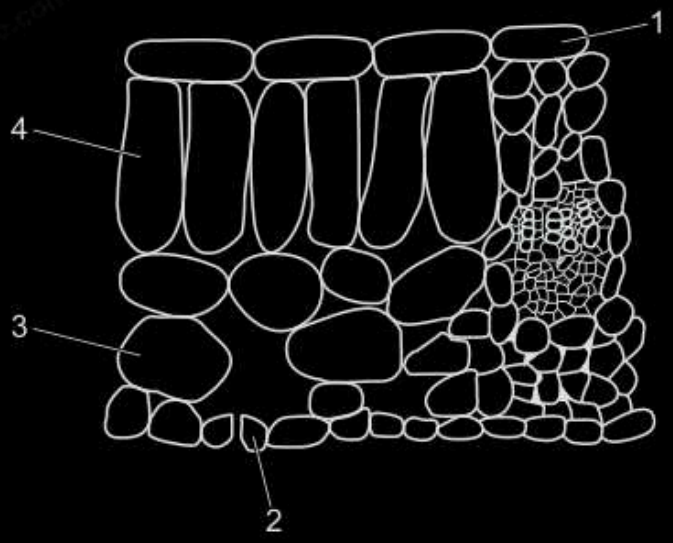
C glucose

D starch

May/June 2026 · Paper 1 · Variant 1 · Question 12

Correct answer: C

12 The diagram shows a section through a leaf.



Which cells in the diagram contain chloroplasts?

A 1, 2, 3 and 4

B 1 and 2 only

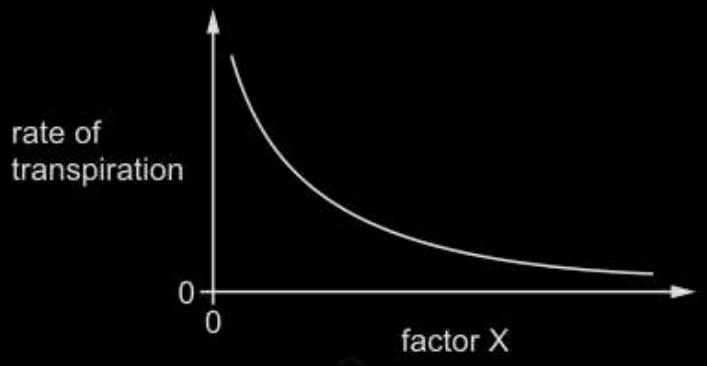
C 2, 3 and 4 only

D 3 and 4 only

May/June 2026 · Paper 1 · Variant 1 · Question 14

Correct answer: B

14 The graph shows how the rate of transpiration in a green plant is affected by a factor, X, when all other factors are kept constant.



What is factor X?

A light intensity

B humidity

C temperature

D wind speed

May/June 2026 · Paper 1 · Variant 1 · Question 15

Correct answer: D

15 Which row does **not** correctly link an enzyme with its substrate and an end-product?

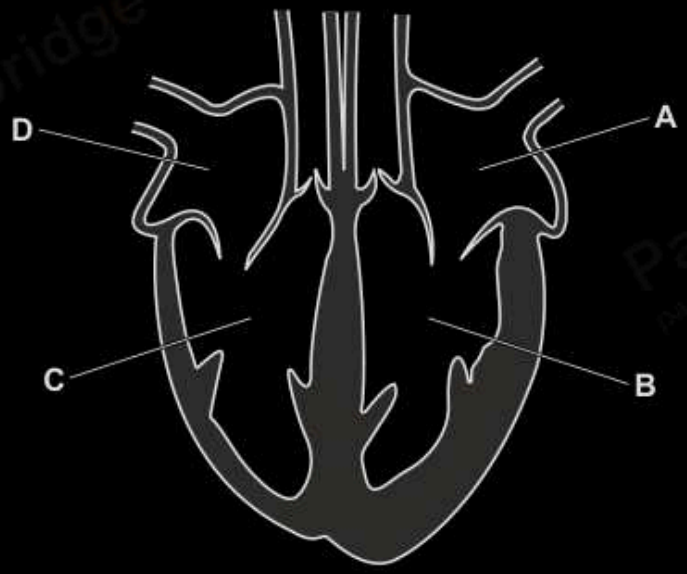
	substrate	enzyme	end-product
A	starch	amylase	maltose
B	lipid	lipase	fatty acids
C	protein	protease	amino acids
D	starch	amylase	glycerol

May/June 2026 · Paper 1 · Variant 1 · Question 18

Correct answer: D

18 The diagram shows the structure of a human heart.

Which part of the heart is the first to receive nutrients absorbed from the small intestine?



May/June 2026 · Paper 1 · Variant 1 · Question 19

Correct answer: C

19 The mean blood pressure is measured in the pulmonary artery, pulmonary vein, aorta and vena cava of a human.

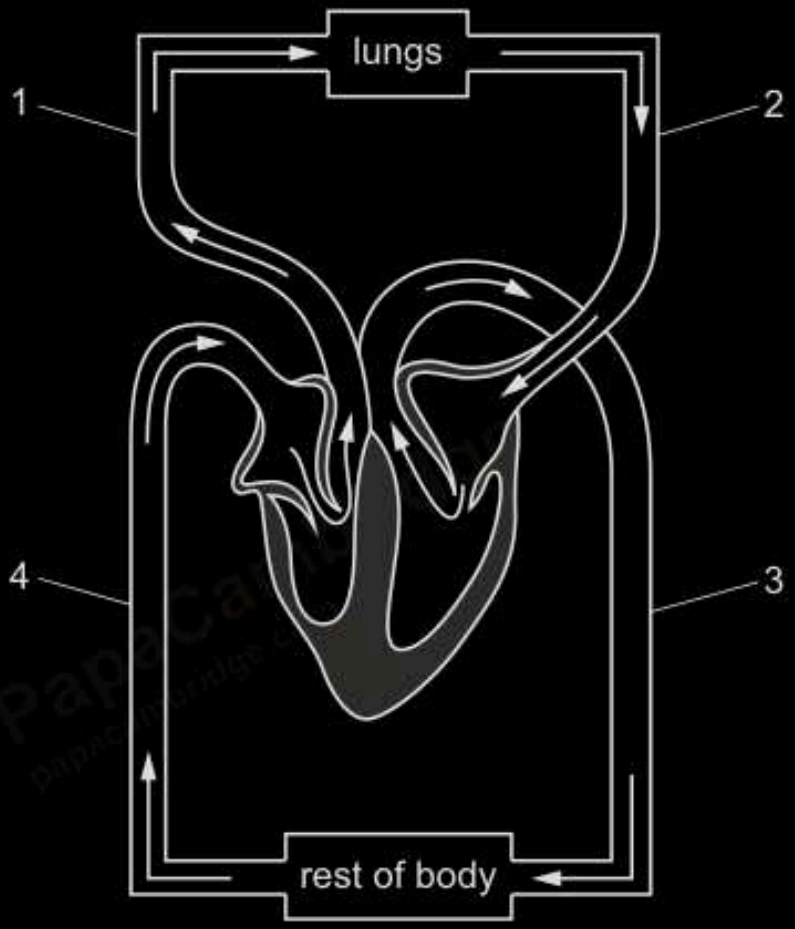
Which blood vessel is most likely to be the pulmonary artery?

	blood pressure mm / Hg
A	3
B	9
C	12
D	120

May/June 2026 · Paper 1 · Variant 1 · Question 20

Correct answer: C

20 The diagram shows part of the human circulatory system.



Which vessels contain oxygenated blood?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

May/June 2026 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 Cholera is caused by a toxin produced by a bacterium.

What are the effects of this toxin?

1 Person becomes dehydrated.

2 Ions are secreted into the small intestine.

3 Water moves into the intestine by osmosis.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

May/June 2026 · Paper 1 · Variant 1 · Question 26

Correct answer: A

26 The pupil reflex action in the eye is a response to changes in light intensity.

The size of the pupil is altered by the action of the radial and circular muscles in the iris.

Which row describes the pupil reflex action?

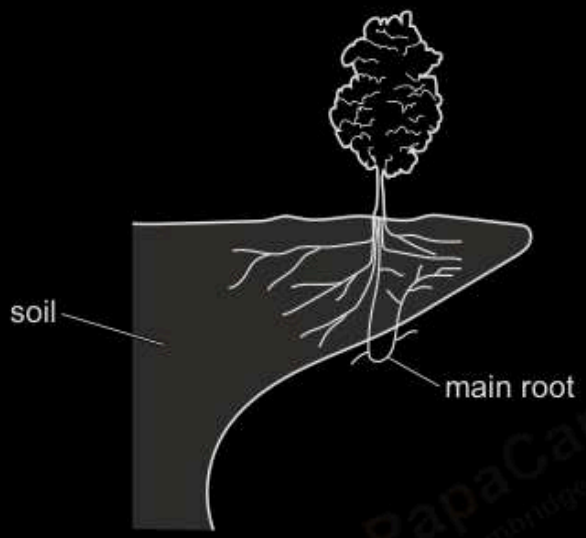
	light intensity	radial muscles	circular muscles	pupil
A	high	relax	contract	gets smaller
B	low	relax	contract	gets bigger
C	high	contract	relax	gets bigger
D	low	contract	relax	gets smaller

May/June 2026 · Paper 1 · Variant 1 · Question 27

Correct answer: D

27 A walker was climbing a hill and noticed a small tree that had grown through an overhanging piece of ground.

The diagram shows how the roots of the tree had grown.



Which statement explains the behaviour of the main root?

A It is preventing the soil from collapsing.

B It must absorb oxygen for the survival of the tree.

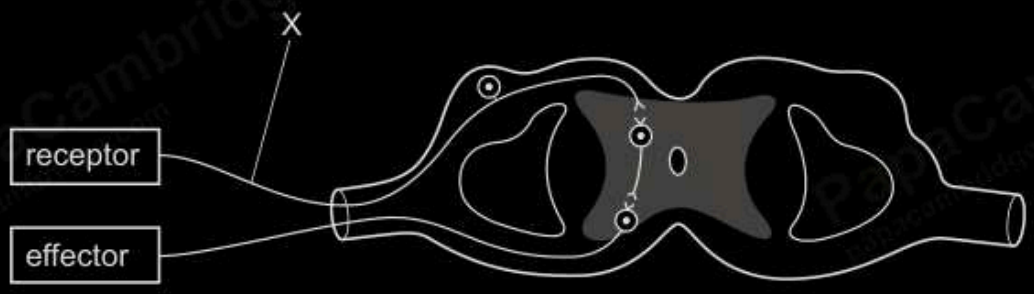
C It receives extra auxin and grows back into the soil.

D It responds negatively to the stimulus of light.

May/June 2026 · Paper 1 · Variant 1 · Question 28

Correct answer: D

28 The diagram shows the parts of the nervous system involved in a reflex action.



Which part of the nervous system is labelled X?

A motor neurone

B synapse

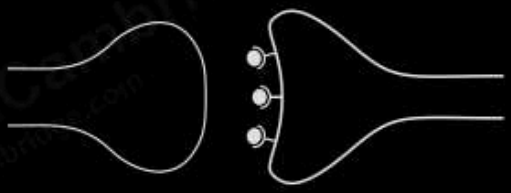
C relay neurone

D sensory neurone

May/June 2026 · Paper 1 · Variant 1 · Question 29

Correct answer: B

29 The diagram shows a synapse.



Which stage of synaptic transmission is shown?

A An impulse stimulates the release of neurotransmitter molecules from vesicles into the synaptic gap.

B The neurotransmitter molecules bind with receptor proteins.

C The neurotransmitter molecules diffuse across the synaptic gap.

D Vesicles containing neurotransmitter move towards the presynaptic neurone membrane.

May/June 2026 · Paper 1 · Variant 1 · Question 30

Correct answer: B

30 The following conversation took place between a doctor (Dr) and a patient (P).

P: My heart seems to be beating faster than usual and I seem to be taking faster and deeper breaths.

Dr: Is this happening just when you are exercising?

P: No. It's happening even when sitting at my desk working.

The doctor concluded that the level of a hormone in the patient's blood might be higher than normal.

Which hormone could this be?

A insulin

B adrenaline

C testosterone

D glucagon

May/June 2026 · Paper 1 · Variant 1 · Question 31

Correct answer: A

31 When the body temperature of a dog is low, muscles attached to body hairs will contract and cause the body hairs to stand upright.

How does this help to maintain the dog's body temperature?

A The upright body hairs trap air and air provides good insulation.

B The upright body hairs trap air and air provides poor insulation.

C The upright body hairs trap water lost through sweating and water provides good insulation.

D The upright body hairs trap water lost through sweating and water provides poor insulation.

May/June 2026 · Paper 1 · Variant 1 · Question 33

Correct answer: B

33 What does auxin stimulate?

A cell division

B cell elongation in shoots

C uptake of mineral ions by active transport

D release of an egg cell

May/June 2026 · Paper 1 · Variant 1 · Question 34

Correct answer: B

34 Some features of cell division are listed.

1 sometimes called reduction division
2 used in asexual reproduction
3 used in the cell division of stem cells
4 results in the production of haploid cells
5 sometimes produces a cancer

Which of these are features of meiosis?

A 1, 4 and 5 **B** 1 and 4 only **C** 2 and 3 **D** 3, 4 and 5

May/June 2026 · Paper 1 · Variant 1 · Question 36

Correct answer: B

36 What is an example of continuous variation?

A ABO blood groups
B body mass
C seed colour in peas
D seed shape in peas

May/June 2026 · Paper 1 · Variant 1 · Question 37

Correct answer: C

37 A strand of DNA has the code CTCAGCA.

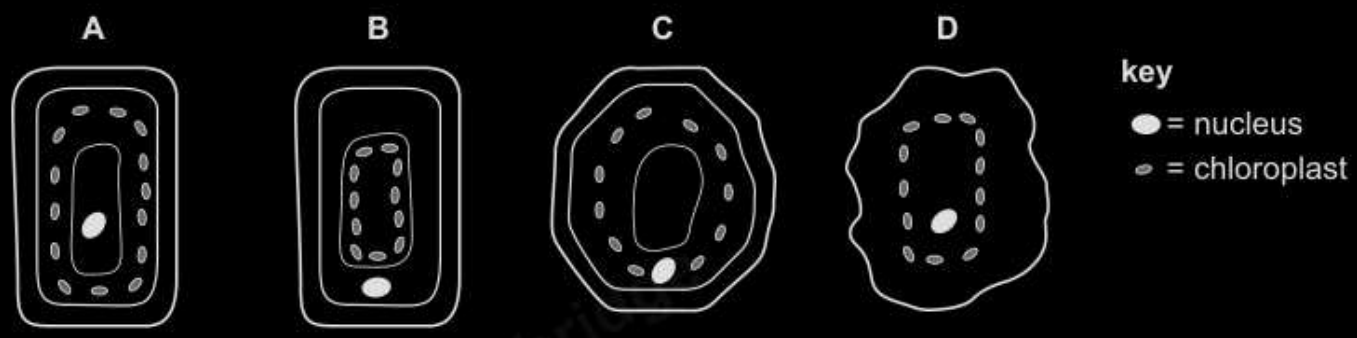
What would be the code for the complementary strand?

A AGACTAC
B CTCAGCA
C GAGTCGT
D TCTGATG

May/June 2026 · Paper 1 · Variant 2 · Question 1

Correct answer: C

1 Which diagram shows the positions of the chloroplasts and the nucleus in a cell?



May/June 2026 · Paper 1 · Variant 2 · Question 2

Correct answer: D

2 What is the formula used to calculate the actual size of an object from an image of that object?

A actual size = magnification × image size

B actual size = magnification + image size

C $\text{actual size} = \frac{\text{magnification}}{\text{image size}}$

D $\text{actual size} = \frac{\text{image size}}{\text{magnification}}$

May/June 2026 · Paper 1 · Variant 2 · Question 3

Correct answer: D

3 Which statement about diffusion is correct?

A Diffusion always involves the movement of water molecules.

B Diffusion depends on the presence of a partially permeable membrane.

C Diffusion only happens to molecules in a solution.

D Diffusion always involves molecules moving down a concentration gradient.

May/June 2026 · Paper 1 · Variant 2 · Question 4

Correct answer: B

4 Which change in conditions will increase the rate of diffusion through a membrane?

A decreasing temperature

B decreasing thickness of membrane (diffusion distance)

C decreasing surface area

D decreasing concentration gradient

May/June 2026 · Paper 1 · Variant 2 · Question 5

Correct answer: D

5 Which statements about osmosis are correct?

1 Glucose molecules can move by osmosis.

2 Water molecules pass through a partially permeable membrane.

3 Molecules move down a water potential gradient.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

Correct answer: A

6 Benedict's solution can be used to find out how much glucose a substance contains.

The table shows the colour of Benedict's solution when heated with substances containing different concentrations of glucose.

colour of Benedict's solution	glucose content / %
blue	0
green	0.5–1.0
yellow	1.1–1.5
orange	1.6–2.0
red	>2.0

Four different substances, **A**, **B**, **C** and **D**, are tested with Benedict's solution.

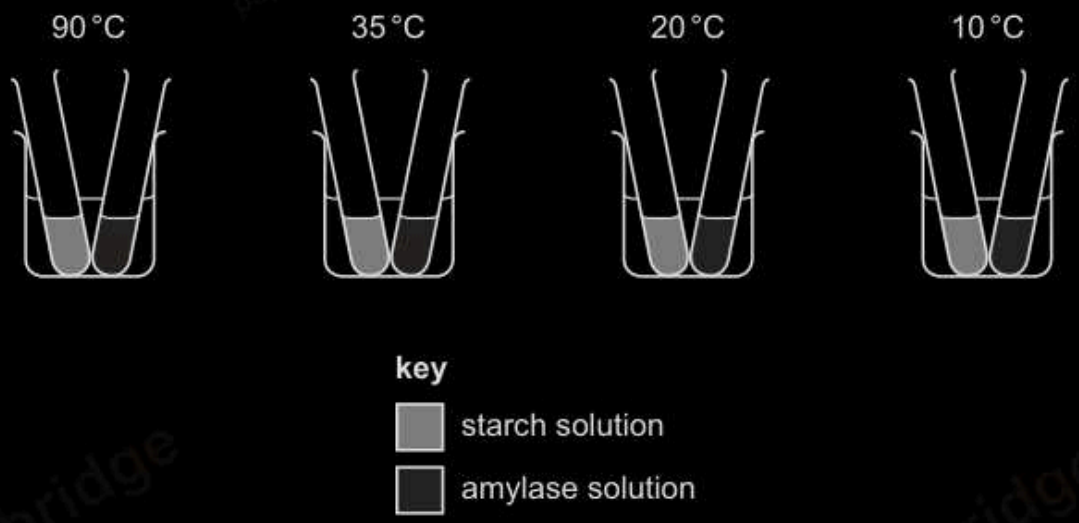
Which substance has a 1.3% glucose content?

	colour of Benedict's solution
A	yellow
B	blue
C	orange
D	green

Correct answer: A

7 The diagram shows an experiment on amylase.

Each beaker contains water maintained at the temperature shown.



Each test-tube of amylase is poured into the test-tube of starch solution in the same beaker, and then samples of each mixture are tested with iodine solution.

Samples of the mixture are then tested with iodine solution again at five minute intervals.

Which results will be obtained?

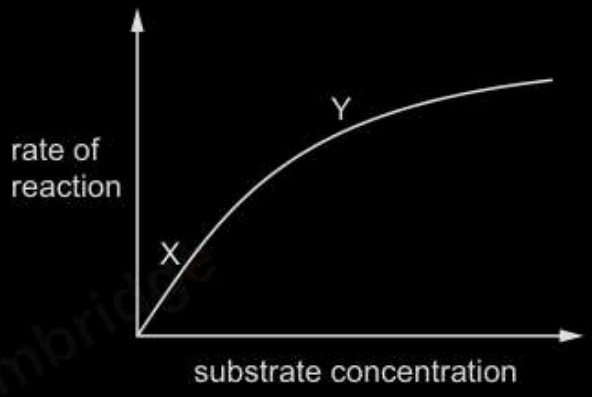
	90 °C	35 °C	20 °C	10 °C
A	blue-black after 30 minutes	goes yellow-brown immediately	goes yellow-brown after five minutes	blue-black after 30 minutes
B	blue-black after 30 minutes	goes yellow-brown after five minutes	goes yellow-brown immediately	blue-black after 30 minutes
C	goes yellow-brown immediately	goes yellow-brown after five minutes	goes yellow-brown after five minutes	blue-black after 30 minutes
D	goes yellow-brown after five minutes	blue-black after 30 minutes	blue-black after 30 minutes	goes yellow-brown after five minutes

May/June 2026 · Paper 1 · Variant 2 · Question 8

Correct answer: B

8 The graph shows the effect of substrate concentration on the rate of an enzyme-controlled reaction.

The results were obtained using a constant enzyme concentration, a temperature of 30 °C and a pH of 7.0.



Which statement explains the shape of the graph from X to Y?

A Denaturation of the enzyme molecules is increasing.

B The enzyme active sites are **not** all occupied.

C The concentration of substrate molecules is decreasing.

D The frequency of effective collisions between enzyme active sites and substrate molecules is decreasing.

May/June 2026 · Paper 1 · Variant 2 · Question 11

Correct answer: D

11 Which parts of the digestive system produce protease enzymes?

A colon and liver

B liver and stomach

C pancreas and colon

D pancreas and stomach

May/June 2026 · Paper 1 · Variant 2 · Question 13

Correct answer: A

13 Which row describes the structures in the lungs?

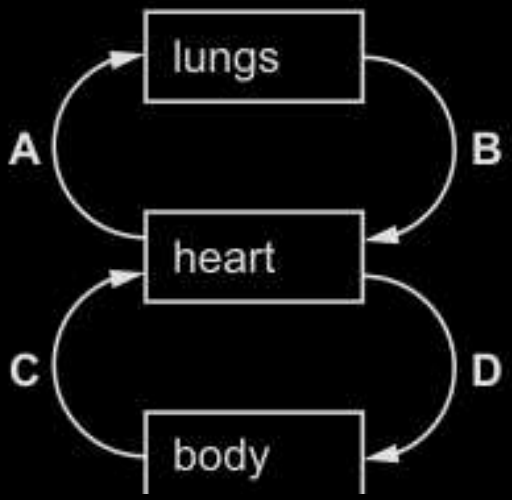
	network of capillaries	total surface area of alveoli	width of walls of alveoli	width of capillary walls
A	large	large	thin	thin
B	large	small	thin	thick
C	small	large	thick	thin
D	small	small	thick	thick

May/June 2026 · Paper 1 · Variant 2 · Question 16

Correct answer: D

16 The diagram shows the double circulatory system in a human.

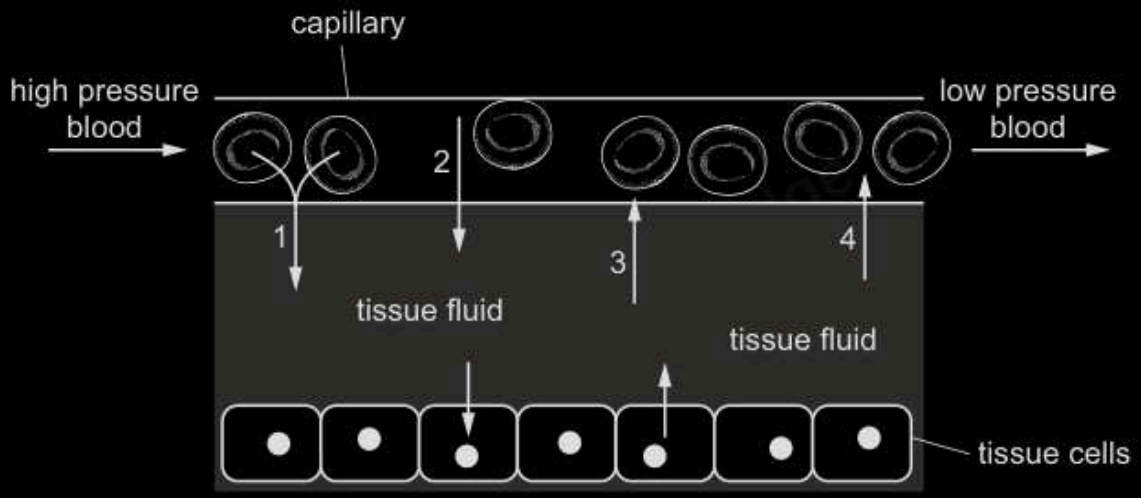
Which arrow shows the blood vessel with the highest pressure?



May/June 2026 · Paper 1 · Variant 2 · Question 17

Correct answer: C

17 The diagram shows the transfer of materials between blood in a capillary and tissue fluid.



Which row shows the correct labels for the arrows in the diagram?

	1	2	3	4
A	carbon dioxide	glucose	oxygen	water
B	glucose	water	oxygen	urea
C	oxygen	glucose	carbon dioxide	water
D	oxygen	water	carbon dioxide	glucose

May/June 2026 · Paper 1 · Variant 2 · Question 18

Correct answer: D

18 When people donate blood they usually give 0.470 dm^3.

If 1 cm^3 of blood contains 5000 million red blood cells, how many red blood cells will be present in one donation?

A 2.35×10^6 **B** 2.35×10^9 **C** 2.35×10^{10} **D** 2.35×10^{12}

May/June 2026 · Paper 1 · Variant 2 · Question 24

Correct answer: C

24 The normal mass of urea excreted in urine in 24 hours by a man is between 12 and 20 g.

One man excretes only 7 g of urea in his urine over a 24-hour period.

What could **not** be the reason for this low mass of urea?

A The man had low levels of protein in his diet.

B The man's kidneys were functioning incorrectly.

C The man had low levels of salts in his diet.

D The man's liver was functioning incorrectly.

May/June 2026 · Paper 1 · Variant 2 · Question 26

Correct answer: C

26 Which statement about plant growth responses is correct?

A A root of a plant that grows downwards into the soil is an example of positive phototropism.

B When a shoot of a plant is growing with light from one side, auxin will be equally distributed throughout the shoot.

C Auxin is a chemical made by plants which stimulates growth in shoots.

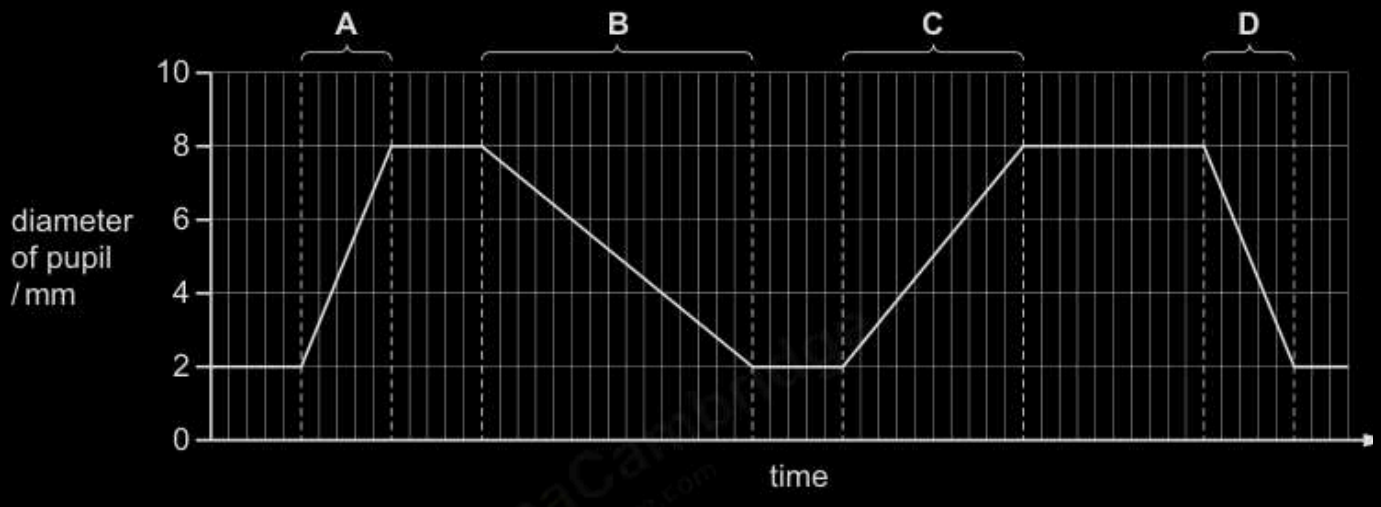
D A root of a plant that grows downwards into the soil is an example of negative gravitropism.

May/June 2026 · Paper 1 · Variant 2 · Question 27

Correct answer: D

27 The diameter of a person's pupil is measured as the light intensity is varied.

During which time period does the light intensity increase fastest?



May/June 2026 · Paper 1 · Variant 2 · Question 28

Correct answer: B

28 What is produced when a cell divides once by mitosis?

A two cells with genetically different chromosomes

B two cells with genetically identical chromosomes

C four cells with genetically different chromosomes

D four cells with genetically identical chromosomes

May/June 2026 · Paper 1 · Variant 2 · Question 29

Correct answer: B

29 The gametes and the embryo are important in plant reproduction.

Which row shows the chromosome content of the nucleus in the female gamete and of a nucleus in the plumule of the plant embryo?

	nucleus in female gamete	nucleus in plumule of plant embryo
A	diploid	diploid
B	haploid	diploid
C	diploid	haploid
D	haploid	haploid

May/June 2026 · Paper 1 · Variant 2 · Question 32

Correct answer: B

32 A section of a DNA molecule contains 400 bases, 80 of which are base A.

Which percentage of the bases in this section of DNA are base C?

A 25 **B** 30 **C** 60 **D** 80

May/June 2026 · Paper 1 · Variant 2 · Question 33

Correct answer: D

33 Which statement about recessive alleles is correct?

A They always affect the phenotype.

B They always cause a harmful effect.

C They can only be inherited by males.

D They can only affect the phenotype if two are present.

May/June 2026 · Paper 1 · Variant 2 · Question 35

Correct answer: C

35 Which statement describes the function of a DNA molecule?

A The copying of a nucleotide in a DNA molecule produces a gene.

B The formation of a nucleotide using a gene manufactures an amino acid.

C The order of the bases in a DNA molecule codes for the sequence of amino acids in a protein.

D The repeating chain of four bases in sequence codes for a chromosome.

May/June 2026 · Paper 1 · Variant 2 · Question 38

Correct answer: C

38 Which enzyme is added to fruit to aid juice production?

A lactase

B lipase

C pectinase

D protease

2 Figure 2.1 shows a tube labelled **P** leading into air sacs in the lungs.

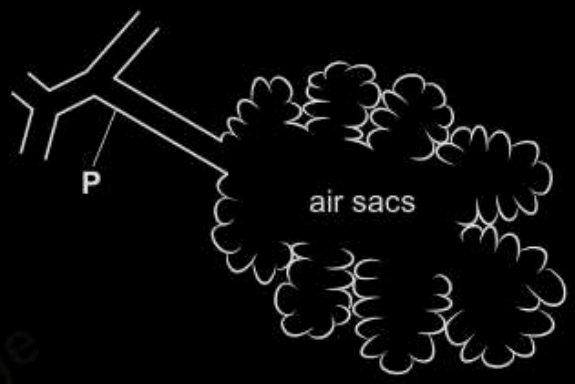


Figure 2.1

(a) Name the type of tube labelled **P** in Figure 2.1.

..... [1]

(b) **(i)** Complete these sentences about how muscles in the chest work to move air into the lungs.

The process of moving air into the lungs occurs because of the contraction of the

.... muscles between the ribs.

These actions cause the ribs to move ..... and

..... .

A sheet of muscle called the ..... also contracts.

[3]

(ii) Explain how the changes described in **2(b)(i)** cause air to move into the lungs.

.....

.....

.....

..... [2]

© Cambridge University Press & Assessment 2026

(c) (i) Figure 2.2 shows the lining of tube **P**.

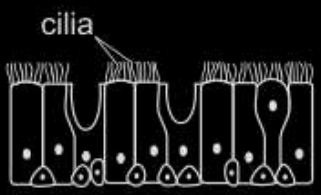


Figure 2.2

Explain how the lining of tube **P** helps to prevent infection in the lungs.

.....

.....

.....

..... [2]

(ii) Describe the effects of smoking tobacco on the lining of tube **P**.

.....

.....

.....

.....

.....

..... [3]

[Total: 11]

PapaCambridge

2(a)	bronchiole ;
2(b)(i)	external intercostal ; up(wards) and out(wards) ; diaphragm ;
2(b)(ii)	any **two** from: increase in the volume of the chest / thorax ; so the pressure in the chest / thorax decreases ; pressure in the chest / thorax is lower than atmospheric pres

© Cambridge University Press & Assessment 2026

	Marks	Guidance
		Two fruits with the mechanisms can be either way round. To award mechanism mark it must be linked to correct fruit.
	1	
	1	**A** idea of fruit attaching to animal
	1	
	1	
	2	**A** to prevent inbreeding

	Marks	Guidance
	1	
	3	**A** either way round
ssure / outside ;	2	**A** lungs for thorax/chest once only

Question	Answer
2(c)(i)	(goblet cells) produce mucus which traps pathogens / microc ciliated cells / cilia waft / move mucus away / up / out (of the l
2(c)(ii)	any **three** from: destruction of cilia / cilia slowed down ; mucus not cleared from P / mucus builds up in P / increased production ; increased infection / bronchitis ; (lung) cancer ; **Max 3**

Question	Answer

3 **(a)** In the spaces provided, name biological molecules that fit these descriptions.

a sugar that is transported in the phloem of plants .....

a carbohydrate that is used to build plant cell walls .....

a vitamin that is needed in the human diet to prevent rickets .....

an excretory molecule that is made in the liver from excess amino acids .....

[4]

(b) Complete these sentences to state the functions of the biological molecules glycogen and glucagon.

Glycogen is .....

.....

Glucagon is .....

.....

[2]

(c) Enzymes are biological molecules with a precise shape.

Explain why the shape of enzyme molecules is important for their function.

.....

.....

.....

.....

.....

..... [3]

[Total: 9]

© Cambridge University Press & Assessment 2026

3(a)	sucrose ; cellulose ; vitamin D ; urea ;
3(b)	*glycogen is* used as a store of glucose / energy ;
	glucagon is (a hormone) used to increase blood glucose / sugar level ;
3(c)	any **three** from: to make it specific to a substrate ; (active site) has complementary <u>shape</u> to substrate ; so that the substrate can fit into active site ; to allow enzyme-substrate complex to form ; **Max 3**

© Cambridge University Press & Assessment 2026

	Marks	Guidance
organisms ; lungs) ;	2	I germs
I mucus	3	I emphysema / heart disease

	Marks	Guidance
	4	
	1	
	1	**A** controls blood sugar level / converts glycogen to glucose **R** decreases blood sugar level
	3	**R** active site has the same shape as the substrate If no other mark is awarded, allow one mark for reference to lock and key hypothesis

Question	Answer	Marks	Guidance

5 Coronary heart disease (CHD) affects many people.

(a) Explain how CHD affects the functioning of heart muscle.

.....

.....

.....

.....

.....

..... [3]

(b) There are a number of risk factors that can affect the chance of developing CHD.

Table 5.1 lists some of these risk factors and gives them points.

The more points a person has, the greater the chance of developing CHD.

Table 5.1

	number of points			
risk factor	**1**	**2**	**3**	**4**
age in years	less than 20	21 to 30	31 to 40	41 to 50
body mass	low	medium	high	very high
number of cigarettes smoked a day	0	1–10	11–20	over 20
blood cholesterol level	low	medium	high	very high
number of parents with CHD	0	1	2	-

(i) State **one** risk factor for CHD that is not included in Table 5.1.

..... [1]

(ii) State the reason why having parents with CHD counts as a risk factor.

.....

..... [1]

© Cambridge University Press & Assessment 2026

(iii) Two people, **A** and **B**, are discussing their chance of developing CHD.

Person **A** uses Table 5.1 and calculates that they have scored 12 points by adding together the points for each of their risk factors.

Person **B** is 32 years old, has high body mass, does not smoke, has a high blood cholesterol level and has one parent with CHD.

Use Table 5.1 to compare the chances of developing CHD for person **B** and person **A**.

.....

.....

.....

..... [2]

[Total: 7]

PapaCambridge

5(a)	blockage / narrowing of the coronary arteries ; less blood / glucose / oxygen to the heart <u>muscle</u> ; therefore, cannot contract correctly / cannot pump blood ;	3	

© Cambridge University Press & Assessment 2026

Question	Answer
5(b)(i)	gender / sedentary lifestyle / stress / diet ;
5(b)(ii)	CHD has a genetic link / can be affected by genes / hered
5(b)(iii)	B scored 12 points ; equal risk ;

Question	Answer

6 **(a)** Figure 6.1 shows a pathogen called CMV.

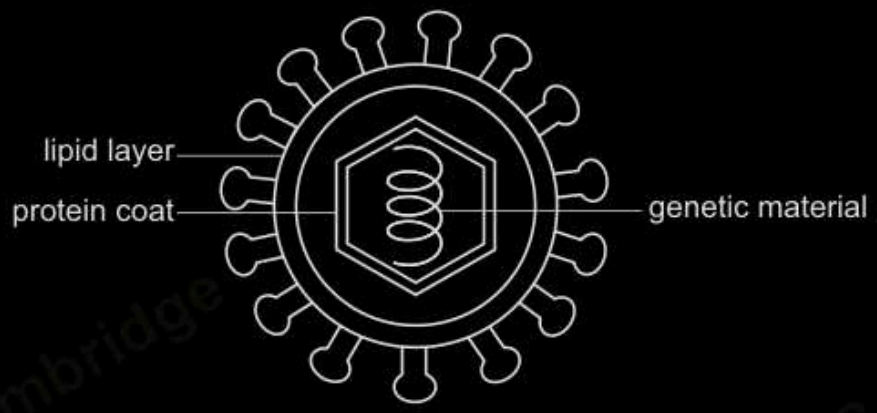


Figure 6.1

(i) What type of pathogen is CMV?

Circle the correct answer in this list.

bacterium **fungus** **protoctist** **virus**

[1]

(ii) CMV can be found in the saliva, urine, and blood of infected people.

Suggest why babies can be born already infected with CMV.

.....

.....

.....

..... [2]

(b) CMV does not cause serious health problems for most people who are infected.

However, in young babies and people with HIV, it can cause more serious symptoms.

Suggest why CMV can cause more serious symptoms in people who have HIV.

.....

.....

.....

..... [2]

© Cambridge University Press & Assessment 2026

(c) After an infection, CMV can remain inside cells in the body for many years.

For this reason, donated blood can only be given to young babies if the donor has not been infected by CMV.

(i) To see if a person can donate blood to a young baby, doctors test the potential donor's blood for a specific antibody.

Suggest why they do this.

.....

.....

.....

..... [2]

(ii) In the United Kingdom, 24 million people can donate blood but only 2% actually do.

60% of all people are infected with CMV.

Calculate how many people in the United Kingdom donate blood that can be given to young babies.

Space for working.

number of people = ... [3]

[Total: 10]

6(a)(i)	virus ;
6(a)(ii)	idea that the mother could have the virus in her blood ; virus could pass across the placenta ;
6(b)	any **two** from: HIV causes reduced lymphocyte numbers / destroys lymp body is unable to produce/does not have enough antibodi weakens the immune system ; **Max 2**
6(c)(i)	any **two** from: after some time CMV may not be in the blood ; CMV may only be inside cells ; the body/lymphocytes will have made antibodies against in the blood) ; **Max 2**

© Cambridge University Press & Assessment 2026

	Marks	Guidance
	1	**A** sex **I** exercise **I** lifestyle unless qualified **A** high blood pressure **I** genetic disposition / pollution
ditary ;	1	**A** similar lifestyles to parents **A** passed from parents to children
	2	incorrect number of points = no marks

	Marks	Guidance
	1	
	2	**R** blood is passed across the placenta
phocytes ; dies ;	2	
CMV (which will be	2	

Question	Answer
6(c)(ii)	192 000 ;;;

Question	Answer

7 **(a)** Different structures in the excretory system transport different substances.

Draw a line from each **structure** to the substance that it **transports**.

structure	transports
renal artery	urine to the bladder
renal vein	urine from the bladder
ureter	deoxygenated blood
urethra	oxygenated blood

[3]

(b) Figure 7.1 shows one kidney nephron that has been stretched out. It also shows changes in the composition of fluid inside this nephron.

Lines **X** and **Y** show how two different environmental conditions affect the composition of the fluid inside this nephron.

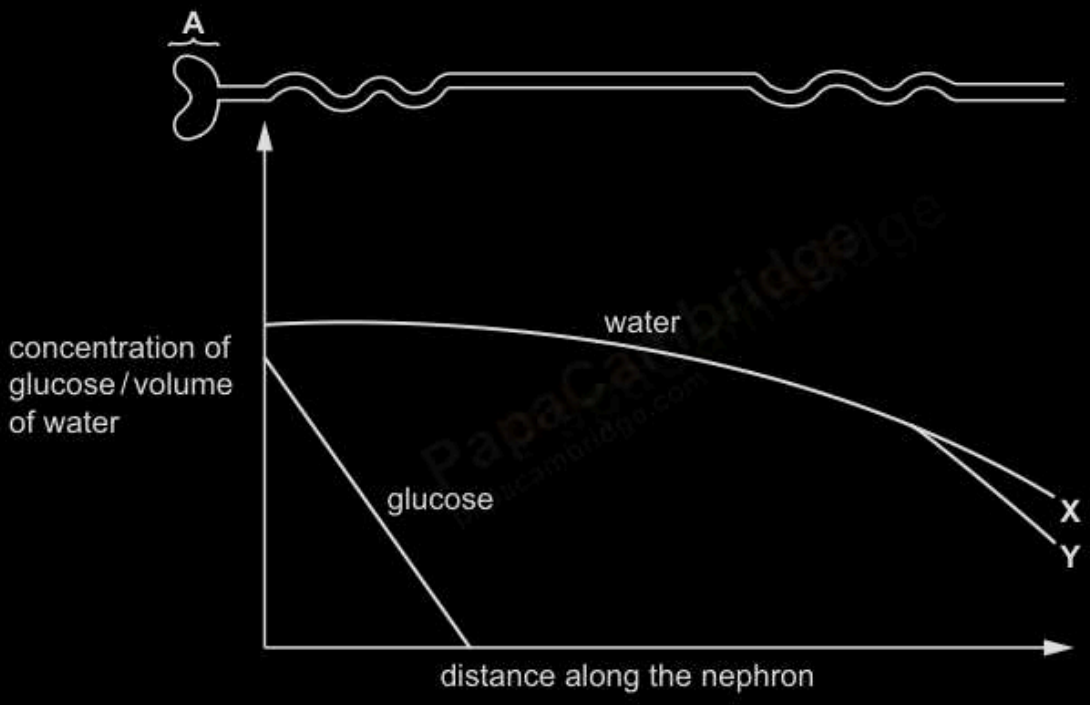


Figure 7.1

(i) Name the part of the nephron labelled **A**.

..... [1]

(ii) Explain the change in the concentration of glucose as it passes down the nephron.

.....

.....

.....

..... [2]

(iii) Line **X** shows the volume of water in the fluid when the person is in an area where the temperature is low.

Line **Y** shows the volume of water in the fluid when the person is in an area where the temperature is high.

Suggest why there is a difference between line **X** and line **Y**.

.....

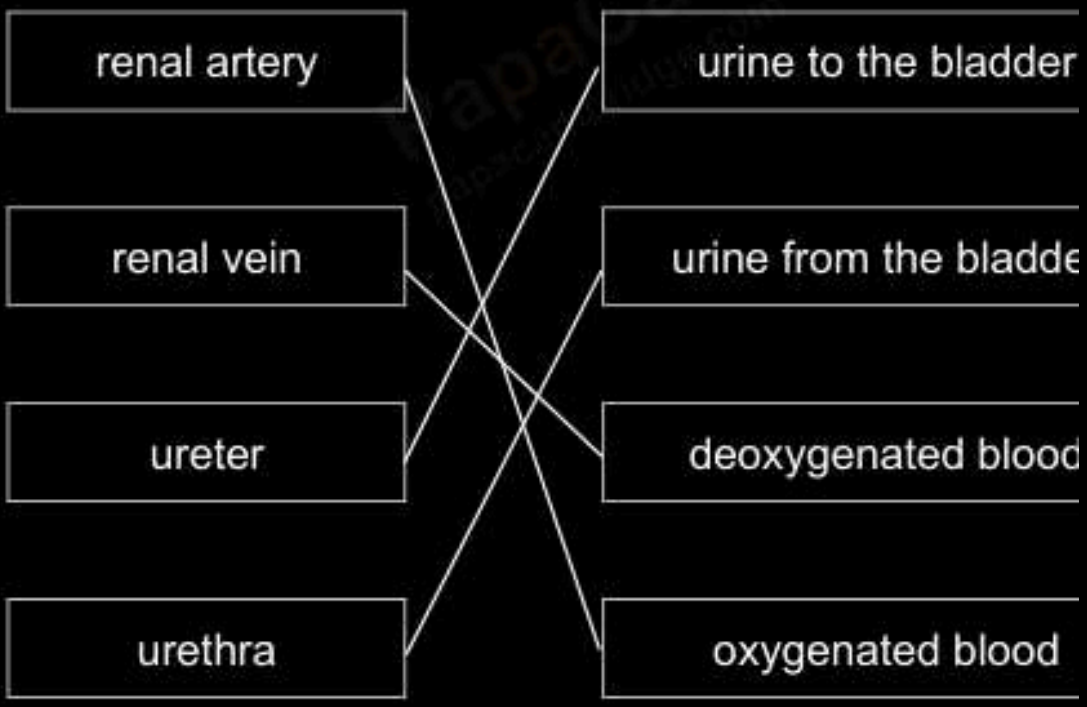
.....

.....

..... [2]

[Total: 8]

PapaCambridge

7(a)	
7(b)(i)	Bowman's capsule ;
7(b)(ii)	glucose is reabsorbed / glucose is absorbed back into the bl selectively / by active transport ;

© Cambridge University Press & Assessment 2026

	Marks	Guidance
	3	**A** 0.192 million / 1.92×10^5 0.192 = 2 marks 0.48 million or 480 000 = 1 mark 9.6 million or 9 600 000 = 1 mark 40% of 480 000 = 2 marks 288 000 = 2 marks (two correct calculations but using 60% rather than 40%)

	Marks	Guidance
r er d	3	all correct = 3 marks three correct = 2 marks two correct = 1 mark
	1	
lood ;	2	**I** taken back into the blood selective reabsorption = 2 marks

Question	Answer
7(b)(iii)	when the temperature is high / line Y, the person will sweat water ; so more water is reabsorbed / absorbed back into the blood

Question	Answer

3 The jackfruit tree grows in hot climates around the world.

Figure 3.1 shows the cross-section of a leaf from the jackfruit tree.

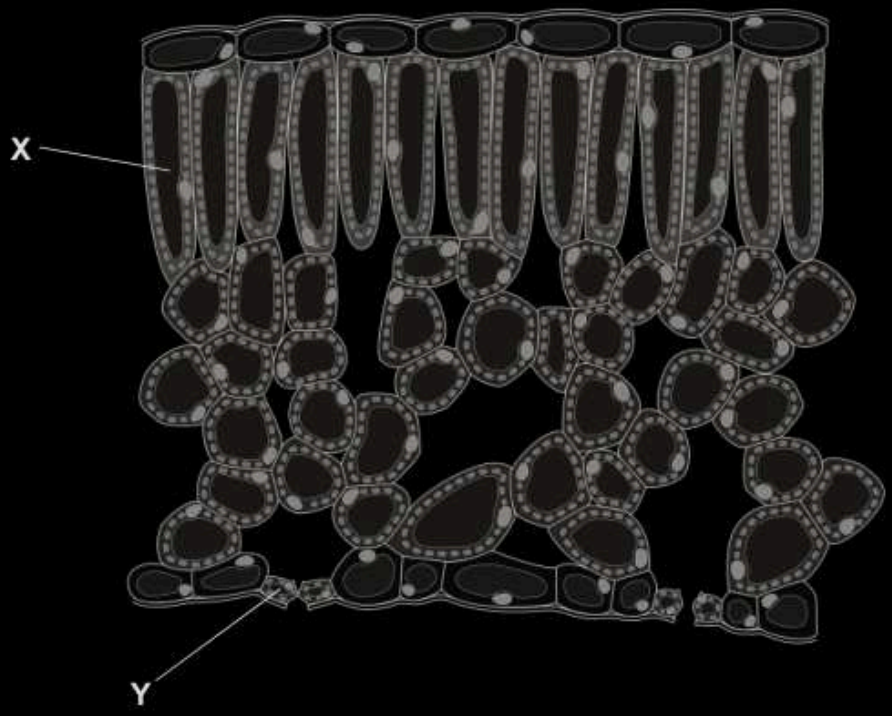


Figure 3.1

(a) From the diagram, identify cells **X** and **Y**

X .....

Y .....

[2]

(b) Like most leaves, the leaves of the jackfruit tree have a large surface area and are thin.

State how each of these features is an adaptation for photosynthesis.

large surface area .....

.....

thin .....

.....

[2]

© Cambridge University Press & Assessment 2026

(c) The jackfruit tree can grow to be 20 metres tall. The tree takes in water through its roots.

Explain how water moves upwards from the roots to the leaves.

.....

.....

.....

.....

.....

..... [3]

(d) **(i)** The fruit from the jackfruit tree can be eaten by humans. The fruit is a source of dietary fibre.

State the importance of fibre in the human diet.

.....

..... [1]

(ii) The fruit from the jackfruit tree is also a source of Vitamin C.

State the name of the disease which is caused by a deficiency of vitamin C in the diet.

..... [1]

[Total: 9]

PapaCambridge

3(a)	X = palisade mesophyll (cell) ;
	Y= guard (cell) ;
3(b)	(large surface area) allows **more / maximum** light to be absorbed / **more / maximum** gas exchange / diffusion of gas
	(thin) for diffusion of gases / carbon dioxide in / oxygen out
3(c)	transpiration pull ; water vapour lost from leaf / stomata ; water (molecules) held together by forces of attraction ; water moves in continuous column ; (transports water) in the xylem (from roots / to leaves) ; **Max 3**

© Cambridge University Press & Assessment 2026

	Marks	Guidance
	3	45 / 900 or 0.05 (mm) = 1 marks 0.05 (mm) × 1000 = 2 marks **OR** 45 × 1000 or 45000 (μm) = 1 mark 45 000 / 900 = 2 marks If no other marks awarded then for 1 mark: Answer × 1000

	Marks	Guidance
	1	**A** palisade (cell)
	1	**I** stomata
ses ;	1	**A** correctly named gases
	1	**A** short distance for gas exchange/carbon dioxide in/oxygen out
	3	**A** transpiration **A** water evaporates / water vapour diffuses out of leaf/stomata **A** cohesion/adhesion of water (molecules) **R** water moves by active transport **R** phloem

Question	Answer
3(d)(i)	softens the faeces / forms the bulk in faeces / shortens the ti spends in body / prevents/helps constipation / helps peristals movements / decreases bowel cancer ;
3(d)(ii)	scurvy ;

Question	Answer

6 Figure 6.1 shows a section through human skin.

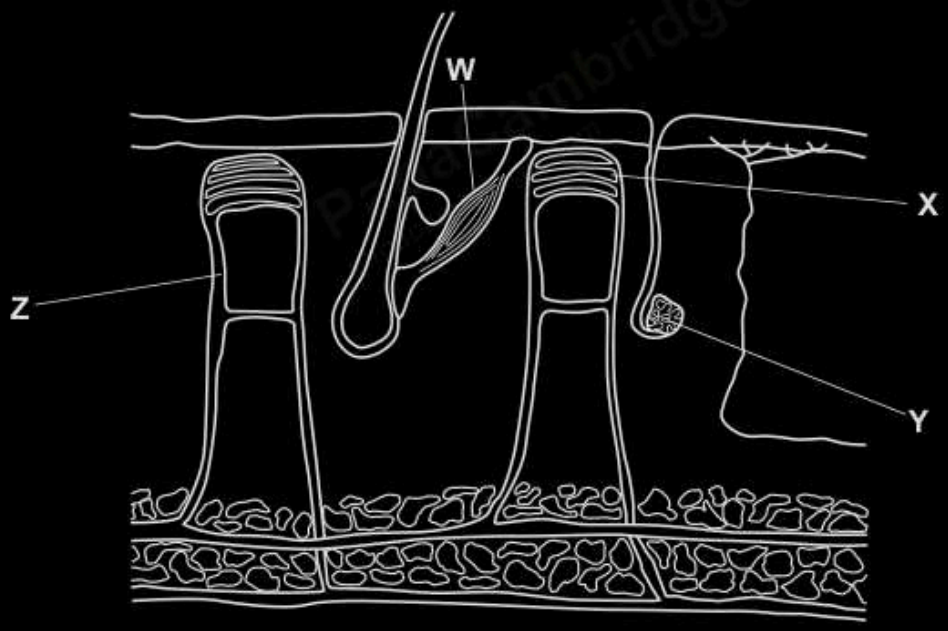


Figure 6.1

(a) Identify the parts labelled **W** and **Y**.

W .....

Y .....

[2]

(b) Parts **X** and **Z** help to maintain a constant internal body temperature. When a person starts exercising, their internal body temperature increases.

(i) Name the part of the brain which monitors body temperature.

..... [1]

(ii) Describe the roles of **X** and **Z** in reducing body temperature back to its normal level after exercise.

.....

.....

.....

.....

.....

..... [3]

© Cambridge University Press & Assessment 2026

(c) A man is wearing shorts and a t-shirt on a sunny day. The weather becomes cloudy and the air temperature decreases. The man's internal body temperature also decreases. The man starts to shiver.

Explain how shivering increases the man's internal body temperature.

.....

.....

.....

..... [2]

[Total: 8]

PapaCambridge

6(a)	W = (hair) erector muscle ;
	Y = sweat gland ;
6(b)(i)	hypothalamus ;
6(b)(ii)	vasodilation ; Z/arterioles dilate ; more blood reaches X / capillaries / surface of the skin ; X / capillaries are close to the surface of the skin ; Heat is lost (from the blood) ; **Max 3**

© Cambridge University Press & Assessment 2026

	Marks	Guidance
	1	
	1	**R** high temperature / heat **A** hotter climate **A** more water / moisture / humidity **I** humid / moist **I** pH
ganic material) ; nic material) ;	2	**I** decay / decompose **A** extracellular digestion = 2 marks **A** obtain nutrients **A** correct named example

	Marks	Guidance
	1	
	1	
	1	**A** thermoregulatory centre
	3	**R** X / capillaries dilate **R** capillaries move closer to the skin surface **I** Z / arterioles close to the surface of the skin **A** increases radiation / conduction from skin **I** keep warm / releases energy unqualified **I** heat loss due to sweating

Question	Answer
6(c)	(shivering is) rapid AW **+** contractions of muscles ;
	releases heat ;

Question	Answer

7 The Devon Rex is a breed of cat. Most breeds of cat have straight hair. Due to a gene mutation the Devon Rex has wavy hair.

(a) Explain what a gene mutation is, and how it has produced a change in the hair of the Devon Rex cat.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) The wavy hair characteristic is caused by a recessive allele.

Define the term allele.

.....

..... [1]

(c) **(i)** A male Devon Rex cat has the genotype **bb**. A female straight-haired cat has the genotype **Bb**. They are bred together.

Complete the genetic diagram to calculate the probability that a kitten will have wavy hair.

probability = ...

[3]

© Cambridge University Press & Assessment 2026

(ii) The male Devon Rex cat and the female straight-haired cat have seven kittens. Only one of the kittens has wavy hair; the others have straight hair.

Explain why the observed ratio of hair type in the kittens is different to the expected ratio.

.....

.....

.....

.....

..... [2]

[Total: 10]

PapaCambridge

<table>
<tr><td>7(a)</td><td>(a mutation) is a change in the base / nucleotide sequence (
(sequence of bases in a gene) determines the sequence of

(DNA / gene) codes for (specific wavy hair) protein / polypept

(different sequences of amino acids) change in the protein s</td></tr>
<tr><td>7(b)</td><td>an alternative form of a / the same gene ;</td></tr>
<tr><td>7(c)(i)</td><td><table>
<tr><td></td><td>b</td><td>b</td></tr>
<tr><td>B</td><td>Bb</td><td>Bb</td></tr>
<tr><td>b</td><td>bb</td><td>bb</td></tr>
</table>
correctly identifying the gametes ;

correct cross / offspring ;

probability = 50% or 0.5 or ½ or 1 in 2 or 1:1 ;</td></tr>
</table>

© Cambridge University Press & Assessment 2026

	Marks	Guidance
	1	**A** sudden / uncontrollable / continuous contraction of muscles
	1	**A** produces heat / thermal energy

	Marks	Guidance
(of DNA / gene) ; amino acids ;)tide ; shape / structure ;	4	**A** changes sequence of amino acids **R** codes for production of amino acids / amino acids code for proteins **A** contains instructions for proteins / controls protein production
	1	**A** variation / form of a / the same gene **R** alternative form of genes
	3	If gametes genotype are wrong = 0 marks If gametes are wrongly placed in punnet square mp2 and 3 can still be awarded **A** 2 in 4

Question	Answer
7(c)(ii)	random fertilisation (of gametes) ; only a small number of offspring / small sample size ; **Max 2**

Question	Answer

9 Figure 9.1 shows a wasp flying towards a girl's eye.

The girl blinks her eyes in response. This is an example of a reflex action.

Figure 9.1

(a) State what is meant by reflex action.

.....

.....

.....

..... [2]

© Cambridge University Press & Assessment 2026

(b) The reflex action occurs as a result of nerve impulses being sent through a reflex arc.

Describe the stages of the reflex arc that cause the girl to blink.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

PapaCambridge

(c) Figure 9.2 shows a synapse which is a junction between two neurones. When the nerve impulse reaches a synapse, it triggers the release of neurotransmitter molecules into the synaptic gap.

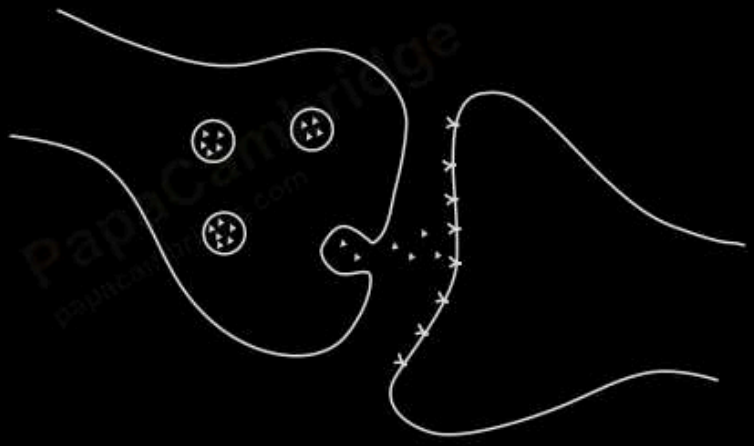


Figure 9.2

Describe how these neurotransmitter molecules enable the impulse to continue.

.....

.....

.....

.....

.....

..... [3]

[Total: 11]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (Cambridge University Press & Assessment) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in our Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge International Education is the name of our awarding body and a part of Cambridge University Press & Assessment, which is a department of the University of Cambridge.

© Cambridge University Press & Assessment 2026

9(a)	rapid ;	1	**A** quick / fast / sudden / immediate
	and automatic response (to a stimulus) ;	1	**A** involuntary / unconscious / instinct action **I** brain
9(b)	1 stimulus / wasp detected **+** by the retina / (photo) receptor / rods / cones **or** (photo)receptor / rods / cones **+** in the retina ; 2 nerve/electrical impulse ; 3 (nerve/electrical impulse) passes through optic / sensory neurone ; 4 (optic / sensory nerve sends impulse) to relay neurone (in CNS) ; 5 (impulse sent from relay neurone / CNS) through motor neurone ; 6 (from motor neurone) to the effector / (eyelid) muscles ; 7 (eyelid muscles) which contract (to close the eye / response) ; **Max 6**	6	**I** eye **I** signal / messages **I** brain / spinal cord / coordinator **R** consciousness involved
9(c)	(neurotransmitter molecules) diffuse across the gap ; (neurotransmitters) bind with receptor (proteins / molecules) ; cause an impulse to be stimulated in the (next) neurone ;	3	

© Cambridge University Press & Assessment 2026

October/November 2025 · Paper 1 · Variant 1 · Question 1

Correct answer: D

1 2 3 4 5 6

When constructing a dichotomous key, which question could be used to separate arthropods 2, 5 and 6 from arthropods 1, 3 and 4?

A Does it have a segmented body?

B Does it have antennae?

C Does it have six legs?

D Does it have wings?

October/November 2025 · Paper 1 · Variant 1 · Question 4

Correct answer: C

4 Contact lenses are placed on the surface of the eye to correct problems with vision.

One type of contact lens is partially permeable and is normally stored in a sterile sodium chloride solution.

The sodium chloride solution has the same concentration as the fluid that covers the surface of the eye.

A person makes a mistake and stores their contact lenses in distilled water rather than sodium chloride solution.

They then place a contact lens on the surface of their eye.

Which pair of statements describe what then happens?

1 Water from the eye fluid enters the contact lens by osmosis.

2 Water leaves the contact lens and enters the eye fluid by osmosis.

3 The contact lens shrinks.

4 The contact lens swells.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

October/November 2025 · Paper 1 · Variant 1 · Question 5

Correct answer: A

5 Which process involves the movement of molecules against a concentration gradient?

A active transport

B diffusion

C osmosis

D transpiration

October/November 2025 · Paper 1 · Variant 1 · Question 6

Correct answer: D

6 Many reactions in cells are dependent on enzyme activity.

Which statement about enzymes is **not** correct?

A They are protein molecules.

B They catalyse biological reactions.

C They remain unchanged at the end of a reaction.

D They have active sites which are the same shape as their substrate molecules.

October/November 2025 · Paper 1 · Variant 1 · Question 14

Correct answer: B

14 Which processes use energy obtained from respiration?

1 contraction of intercostal muscles during breathing

2 nitrate ions entering a root hair cell against the concentration gradient

3 oxygen moving from the lungs into blood capillaries

4 production of antibodies by lymphocytes

5 water entering a bacterial cell by osmosis

A 1, 2 and 3 **B** 1, 2 and 4 **C** 2, 3 and 4 **D** 3, 4 and 5

October/November 2025 · Paper 1 · Variant 1 · Question 16

Correct answer: C

16 The table shows the mean rate of blood flow through some blood vessels.

blood vessel	mean rate of blood flow / mm per s
W	0.4
X	5.0
Y	15.0
Z	1200.0

Which row correctly identifies the blood vessels?

	W	X	Y	Z
A	main artery	small artery	small vein	capillary
B	small vein	small artery	main artery	capillary
C	capillary	small vein	small artery	main artery
D	capillary	small artery	small vein	main artery

October/November 2025 · Paper 1 · Variant 1 · Question 18

Correct answer: B

18 Which statement explains the association between smoking tobacco and emphysema?

A Tobacco smoke can cause mutations in lung cells.

B Tobacco smoke can lead to the breakdown of alveoli.

C Tobacco smoke may reduce the growth of a fetus.

D Tobacco smoke reduces the amount of oxygen carried by red blood cells.

October/November 2025 · Paper 1 · Variant 1 · Question 19

Correct answer: D

19 What prevents pathogens from entering the human body?

A antibodies and lymphocytes

B antigens and memory cells

C red blood cells and lymphocytes

D stomach acid and mucus

October/November 2025 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 The table shows the concentration of five substances in the blood entering the kidney, in fluid entering the nephron and in the urine.

substance	concentration / mg per dm^3		
	blood entering the kidney	fluid entering the nephron	urine
urea	0.4	20.0	20.0
glucose	1.5	1.5	0.0
amino acids	0.8	0.8	0.0
ions	8.0	8.0	16.5
protein	82.0	0.0	0.0

Which substances are totally reabsorbed into the blood from the nephron?

A amino acids and glucose

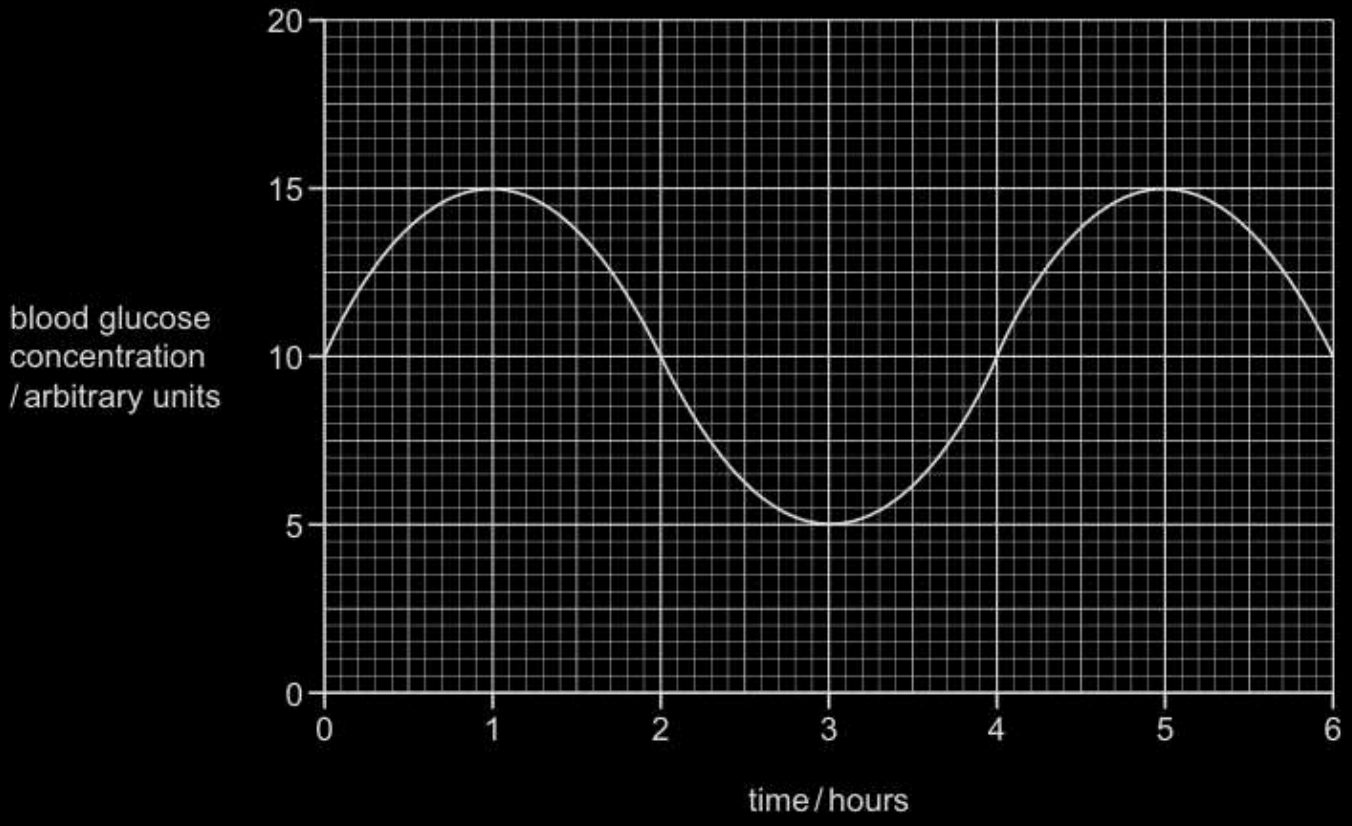
B glucose and ions

C protein and amino acids

D protein and glucose

Correct answer: B

22 The graph shows blood glucose concentration over 6 hours in a day.



What is the set point for blood glucose concentration shown by this graph?

A 5 arbitrary units

B 10 arbitrary units

C 15 arbitrary units

D 20 arbitrary units

October/November 2025 · Paper 1 · Variant 1 · Question 24

Correct answer: C

24 An athlete's resting body temperature is 37.0 °C.

After vigorous exercise for 30 minutes, the athlete's body temperature rises to 38.6 °C.

The athlete rests for 10 minutes.

After resting, the athlete's body temperature is measured again and is 37.2 °C.

Which events take place in the athlete's body during the 10-minute rest to lower their body temperature?

	arterioles near the surface of the skin	heat energy is lost through
A	contract	evaporation of sweat
B	contract	shivering
C	dilate	evaporation of sweat
D	dilate	shivering

October/November 2025 · Paper 1 · Variant 1 · Question 25

Correct answer: D

25 Which endocrine gland produces follicle-stimulating hormone?

A adrenal gland

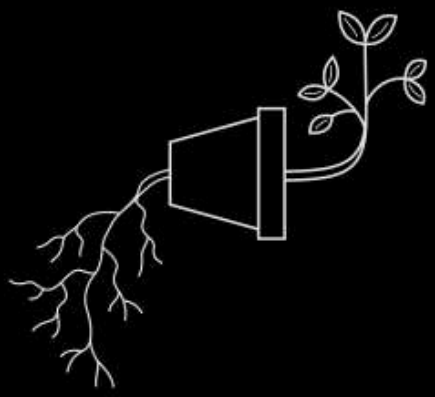
B ovary

C pancreas

D pituitary gland

Correct answer: C

26 The diagram shows a plant grown on its side without light.



Which row explains the responses shown by the roots and the shoot?

	roots	shoot
A	positive gravitropism	positive phototropism
B	negative phototropism	positive phototropism
C	positive gravitropism	negative gravitropism
D	negative gravitropism	positive gravitropism

October/November 2025 · Paper 1 · Variant 1 · Question 27

Correct answer: B

27 The tips of growing shoots of a plant are cut off and placed on small blocks of jelly which contain **no** auxin.

Small pieces of impermeable clear glass are placed in some of the jelly blocks, as shown in diagram X. Some of the shoot tips are divided by impermeable clear glass pieces, as shown in diagram Y.

The tips are left on the small blocks of jelly with light coming from **one** side only.

After 12 hours, the number of arbitrary units of auxin is measured in each of the jelly blocks and the means, shown in the diagrams, are calculated.

diagram X

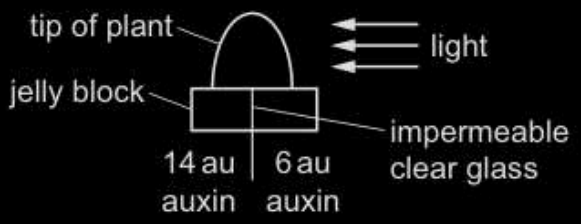
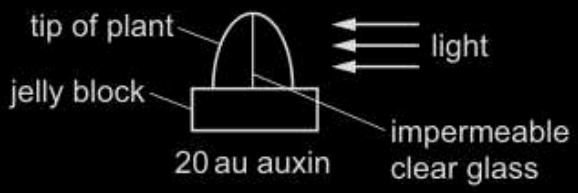


diagram Y



A student wrote down some conclusions from the results.

1 Auxin moves downwards out of the shoot tip.

2 The distribution of auxin in a shoot tip is affected by light intensity.

3 Some auxin is destroyed by light.

4 Auxin moves from areas of more light to areas of less light.

Which of the student's conclusions are correct?

A 1, 2, 3 and 4

B 1, 2 and 4 only

C 1 and 3 only

D 2, 3 and 4 only

October/November 2025 · Paper 1 · Variant 1 · Question 29

Correct answer: A

29 Chromosomes, DNA molecules and genes are pieces of inherited material with different sizes.

Which statement describes the relationship between these pieces of inherited material?

A A chromosome is larger than a gene.

B A gene is larger than a chromosome.

C Multiple chromosomes make up a DNA molecule.

D Multiple DNA molecules make up a gene.

October/November 2025 · Paper 1 · Variant 1 · Question 30

Correct answer: A

30 During the division of a nucleus by meiosis, changes can happen that produce a gamete with an abnormal number of genes or an abnormal number of chromosomes.

Which change can result in a child having Down's syndrome?

A a gamete with one extra chromosome

B a gamete with one less gene

C a gamete with one less chromosome

D a gamete with one extra gene

October/November 2025 · Paper 1 · Variant 1 · Question 31

Correct answer: B

31 Which row shows possible effects of natural selection?

	natural selection can cause mutations to occur	natural selection can lead to the extinction of a species
A	no	no
B	no	yes
C	yes	no
D	yes	yes

October/November 2025 · Paper 1 · Variant 1 · Question 32

Correct answer: C

32 All organisms in the same species show variation.

Some features of different types of variation are listed.

1 an example is body mass

2 has a limited number of possible forms with **no** intermediates

3 has any value from a minimum to a maximum

4 is unaffected by the environment

5 is usually caused by genes only

Which statements refer to discontinuous variation?

A 1, 3 and 5 **B** 1 and 3 only **C** 2, 4 and 5 **D** 2 and 4 only

October/November 2025 · Paper 1 · Variant 1 · Question 35

Correct answer: C

35 An investigation is done to find out how effective different biological washing powders are at digesting protein.

Four equal-sized cubes of cooked egg white (protein) are cut and their masses measured.

The cubes of protein are then placed in test-tubes containing 10 cm^3 of solutions of four different biological washing powders of the same concentration. The cubes of protein are left in the solutions at room temperature.

After 24 hours, the cubes of protein are removed from the test-tubes and their masses remeasured.

biological washing powder	mass of cube of protein / g	
	start of investigation	end of investigation
1	10.2	8.3
2	10.2	7.9
3	10.4	8.4
4	10.3	8.5

What is the percentage decrease in mass of the cube of protein in the **most** effective washing powder given to **one** decimal place?

A 17.5% **B** 19.2% **C** 22.5% **D** 29.1%

October/November 2025 · Paper 1 · Variant 1 · Question 36

Correct answer: C

36 Which enzyme is added to fruit to aid juice production?

A lactase

B lipase

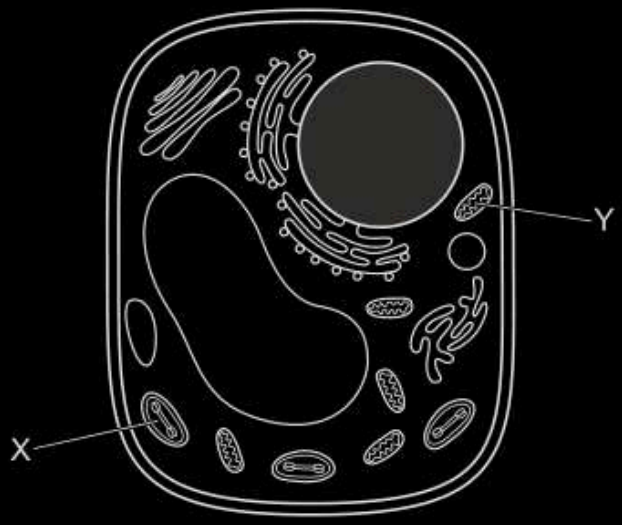
C pectinase

D protease

October/November 2025 · Paper 1 · Variant 2 · Question 1

Correct answer: A

1 Which row identifies structures X and Y on the plant cell diagram?



	structure X	structure Y
A	chloroplast	mitochondrion
B	chloroplast	ribosome
C	nucleus	chloroplast
D	mitochondrion	ribosome

October/November 2025 · Paper 1 · Variant 2 · Question 2

Correct answer: C

2 Viruses can only replicate inside cells.

A single *Herpes simplex* virus replicates rapidly inside a living cell and produces 4000 copies in 12 hours.

What is the maximum number of viruses that could be produced in 36 hours from a single *Herpes simplex* virus?

A 1.2×10^3 **B** 16×10^6 **C** 6.4×10^{10} **D** 2.6×10^{14}

October/November 2025 · Paper 1 · Variant 2 · Question 3

Correct answer: D

3 Which statement about osmosis is correct?

A An animal cell immersed in distilled water will shrink.

B A plant cell immersed in distilled water will become flaccid.

C A plant cell immersed in a strong sodium chloride solution will become turgid.

D A plant cell immersed in a strong sodium chloride solution will become plasmolysed.

October/November 2025 · Paper 1 · Variant 2 · Question 4

Correct answer: C

4 Globulin is a substance that can be detected by the biuret test.

Which type of substance is globulin?

A fat

B oil

C protein

D reducing sugar

October/November 2025 · Paper 1 · Variant 2 · Question 5

Correct answer: C

5 Cellulose and glucose are carbohydrates.

Cellulose is a large molecule made from a long chain of smaller glucose molecules.

One cellulose molecule is 0.01 mm long. One glucose molecule is 0.001 μm long.

How many glucose molecules are found in this cellulose molecule?

A 1×10^2 **B** 1×10^3 **C** 1×10^4 **D** 1×10^5

October/November 2025 · Paper 1 · Variant 2 · Question 6

Correct answer: B

6 Four test-tubes contain starch solution and amylase. They are placed in water-baths at different temperatures and provided with different pHs, as shown in the table. All other conditions are kept the same.

After 30 minutes, iodine solution is added to each test-tube. The contents of three of the test-tubes turns blue-black.

In which test-tube are the contents yellow-brown?

	temperature / °C	pH
A	35	2.5
B	35	6.9
C	75	2.5
D	75	6.9

October/November 2025 · Paper 1 · Variant 2 · Question 8

Correct answer: B

8 Oxygen is produced in leaf cells during photosynthesis and diffuses out of the leaf.

By which pathway does the oxygen diffuse?

A through the upper epidermis and through the cuticle

B through the airspaces and through the stomata

C through the mesophyll and through the phloem

D through the mesophyll and through the xylem

October/November 2025 · Paper 1 · Variant 2 · Question 9

Correct answer: D

9 Why do newly germinated seeds fail to grow into healthy plants if they lack magnesium ions?

A Magnesium ions are a necessary component of all proteins.

B Magnesium ions are needed to convert chlorophyll to starch.

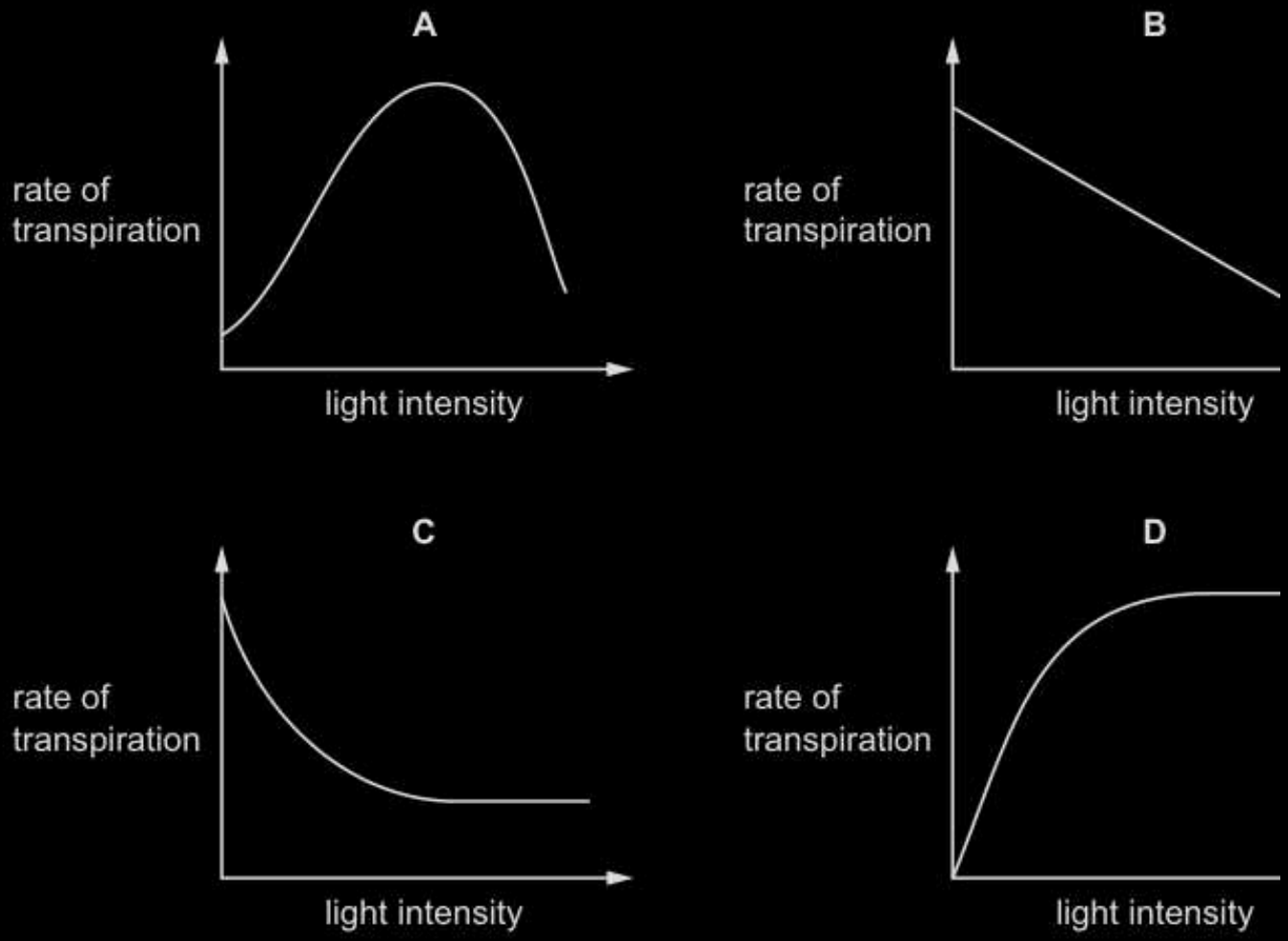
C Magnesium ions are needed to form cell walls.

D Magnesium ions are needed to form chlorophyll molecules.

October/November 2025 · Paper 1 · Variant 2 · Question 10

Correct answer: D

10 Which graph shows the effect of light intensity on the rate of transpiration?



October/November 2025 · Paper 1 · Variant 2 · Question 13

Correct answer: C

13 Villi and alveoli have certain common features.

What is **not** a feature of villi **and** alveoli?

A a rich capillary network

B a single-celled layer of epithelium

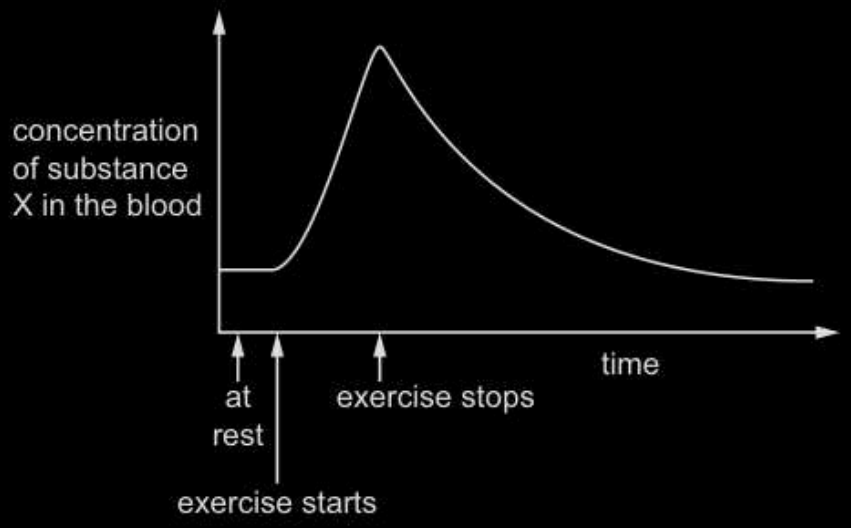
C lacteals connected to the lymphatic system

D a very large total surface area

October/November 2025 · Paper 1 · Variant 2 · Question 14

Correct answer: C

14 The graph shows the concentration of substance X in a person's blood before, during and after exercise.



What is substance X?

A alcohol

B glycogen

C lactic acid

D urea

October/November 2025 · Paper 1 · Variant 2 · Question 17

Correct answer: B

17 What is the function of the hepatic vein?

A to transport blood from the intestine to the liver

B to transport blood from the liver to the vena cava

C to transport blood to the kidney from the aorta

D to transport blood to the liver from the aorta

October/November 2025 · Paper 1 · Variant 2 · Question 18

Correct answer: A

18 A student measures the thickness of the walls of the four chambers of a heart.

The measurements are 15 mm, 9 mm, 3 mm and 3 mm.

Which row shows the correct measurements for the chambers of the heart?

	15 mm	9 mm	3 mm	3 mm
A	left ventricle	right ventricle	right atrium	left atrium
B	left ventricle	left atrium	right ventricle	right atrium
C	right ventricle	left ventricle	right atrium	left atrium
D	right ventricle	left atrium	left ventricle	right atrium

October/November 2025 · Paper 1 · Variant 2 · Question 19

Correct answer: C

19 There are approximately 5 million red blood cells in each mm^3 of blood.

The mean diameter of red blood cells is 0.007 mm.

Which row expresses this information correctly in standard form?

	number of red blood cells / mm^3 of blood	mean diameter of red blood cells / mm
A	5×10^{6}	7×10^{3}
B	5×10^{-6}	7×10^{-3}
C	5×10^{6}	7×10^{-3}
D	5×10^{-6}	7×10^{3}

October/November 2025 · Paper 1 · Variant 2 · Question 23

Correct answer: B

23 Which term can be used to describe the release of oxygen from plants?

A breathing

B excretion

C respiration

D transpiration

October/November 2025 · Paper 1 · Variant 2 · Question 25

Correct answer: D

25 Which event happens when the blood glucose concentration rises?

A Less adrenaline is released.

B Less insulin is released.

C More adrenaline is released.

D More insulin is released.

October/November 2025 · Paper 1 · Variant 2 · Question 26

Correct answer: B

26 What is **not** an example of homeostasis involving a negative feedback mechanism?

A maintaining the core body temperature at around 36.5 °C

B releasing adrenaline from the adrenal glands when frightened

C releasing glucose from liver cells into the blood after missing a meal

D releasing insulin from the pancreas after eating a meal

October/November 2025 · Paper 1 · Variant 2 · Question 27

Correct answer: D

27 What is a correct example of phototropism?

A A stem grows away from gravity.

B A root grows towards light.

C A root grows towards gravity.

D A stem grows towards light.

October/November 2025 · Paper 1 · Variant 2 · Question 29

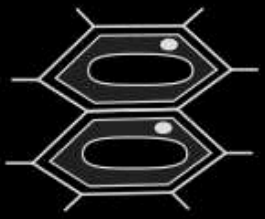
Correct answer: B

29 The diagrams show some plant and animal cells. They are **not** drawn to the same scale.

Which cells contain haploid nuclei?

A

B

C 

D

October/November 2025 · Paper 1 · Variant 2 · Question 30

Correct answer: A

30 Which examples show continuous variation?

1 length of seedlings

2 mass of bananas

3 human skin colour

A 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only

October/November 2025 · Paper 1 · Variant 2 · Question 31

Correct answer: C

31 A man has six fingers on each hand and six toes on each foot. This genetically inherited condition is caused by a dominant allele. The man's genotype is heterozygous.

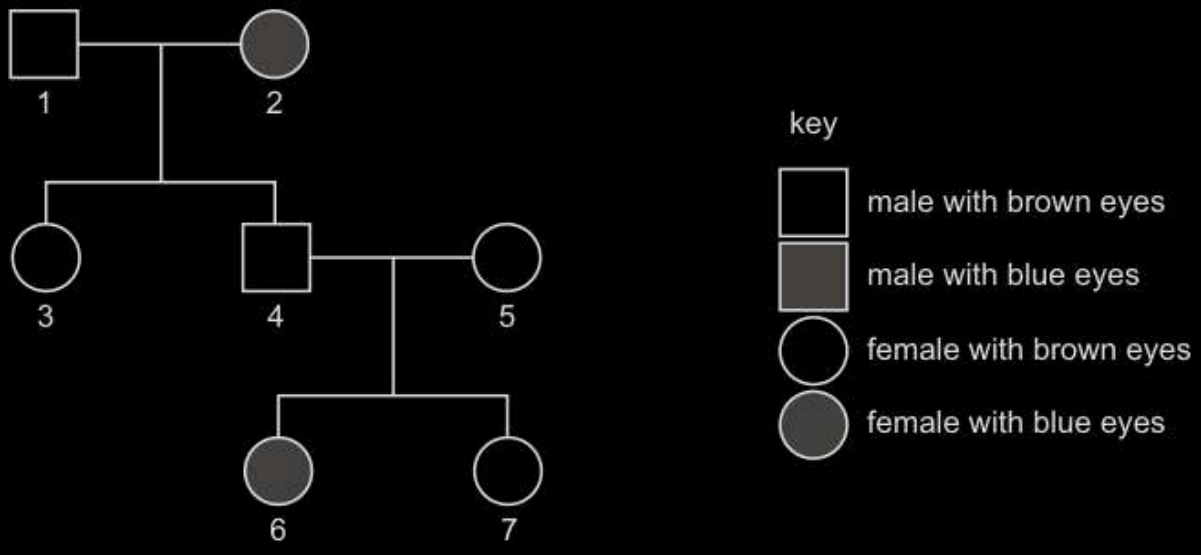
His wife does **not** have the condition.

What is the probability that a child of this couple will be born with the condition?

A 0.00 **B** 0.25 **C** 0.50 **D** 1.00

Correct answer: C

32 The diagram shows a family tree for the inheritance of eye colour. The allele for brown eyes is dominant and the allele for blue eyes is recessive.



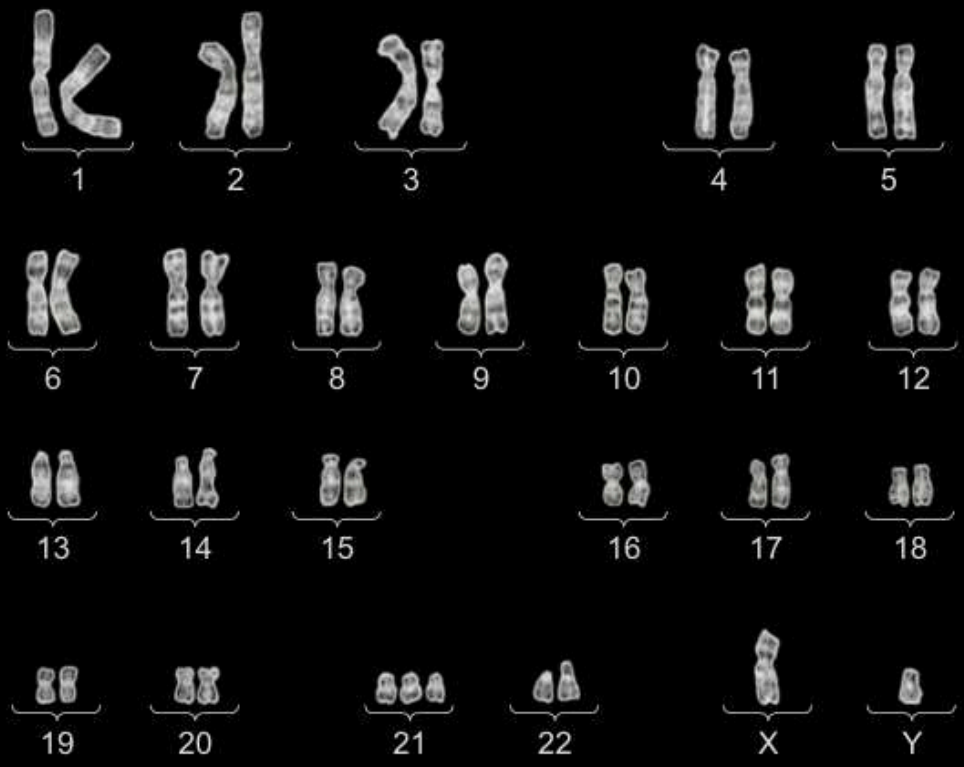
Which people **must** be heterozygous for eye colour?

A 1, 3 and 7 **B** 2 and 6 **C** 3, 4 and 5 **D** 4 and 5 only

October/November 2025 · Paper 1 · Variant 2 · Question 33

Correct answer: D

33 A karyotype is a photograph showing the chromosomes of an individual.



Which description matches an individual who has this karyotype?

A female without Down's syndrome

B male without Down's syndrome

C female with Down's syndrome

D male with Down's syndrome

October/November 2025 · Paper 1 · Variant 2 · Question 34

Correct answer: A

34 Three statements about natural selection are listed.

1 An organism with an advantageous inherited feature does **not** always pass this to its offspring.

2 Many organisms in a population die before they interbreed and produce any offspring.

3 Organisms with more advantageous inherited features are more likely to survive and produce offspring.

Which statements are correct?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

2 Fig. 2.1 shows a section through the human eye.

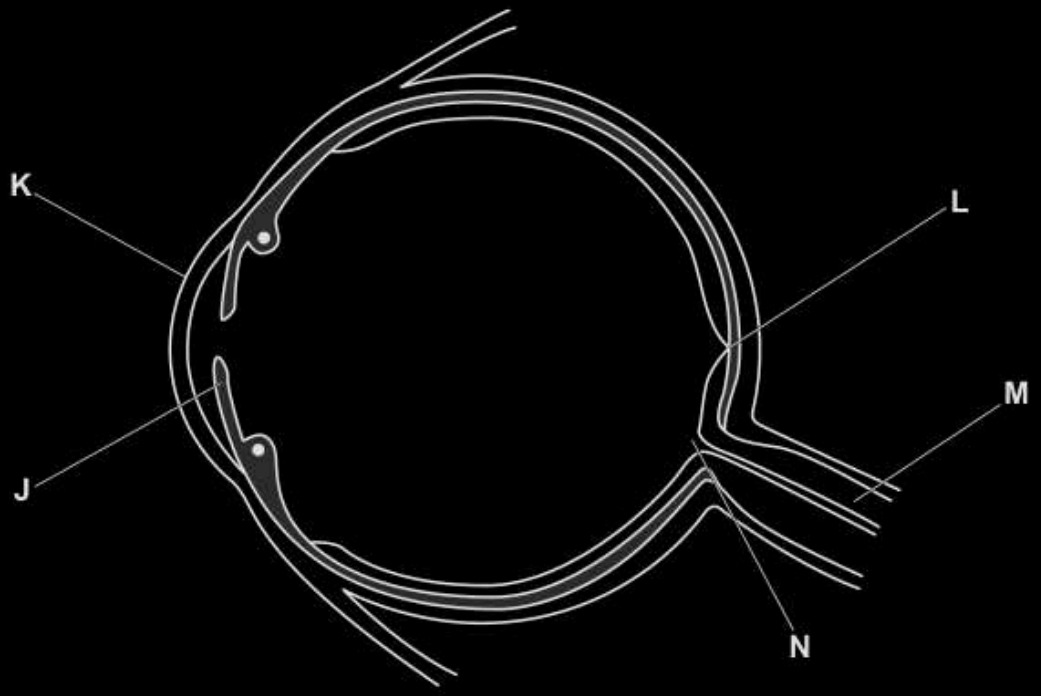


Fig. 2.1

(a) (i) Identify the letter on Fig. 2.1 that has been used to label the part of the eye that:

contains circular and radial muscles .....

has the highest concentration of light receptors. .....

[2]

(ii) On Fig. 2.1, draw **and** label:

- the lens
- the suspensory ligaments.

[2]

(iii) Explain the function of the suspensory ligaments.

.....

.....

.....

.....

.....

.....

..... [3]

(b) Some people have a genetic condition called cornea plana. This condition causes the cornea to be flatter than usual.

(i) Explain why a person with cornea plana **cannot** see some objects clearly.

.....

.....

.....

.....

..... [2]

(ii) One type of cornea plana is caused by a recessive allele.

Two people who do **not** have cornea plana have a child who has the condition.

Explain how this is possible.

.....

.....

.....

.....

..... [2]

[Total: 11]

2(a)(i)	J ; L ;	**2**	
2(a)(ii)	lens labelled and drawn in correct position ; suspensory ligaments labelled and drawn in correct position ;	**2**	
2(a)(iii)	max **three** from: attached to ciliary bodies / muscles ; can be loose / slack or taut / tight ; pulls on the lens / ora ; controls / changes the shape of the lens ; allows <u>accommodation</u> to occur ;	**3**	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance
2(b)(i)	the light will not be bent / refracted enough ; light will not be focussed on the retina ;	2	
2(b)(ii)	parents must be heterozygous / carriers / carry one affected allele ; they both pass the allele on to their child ;	2	

Question	Answer	Marks	Guidance

3 **(a)** Fig. 3.1 shows a summary of protein metabolism in the body.

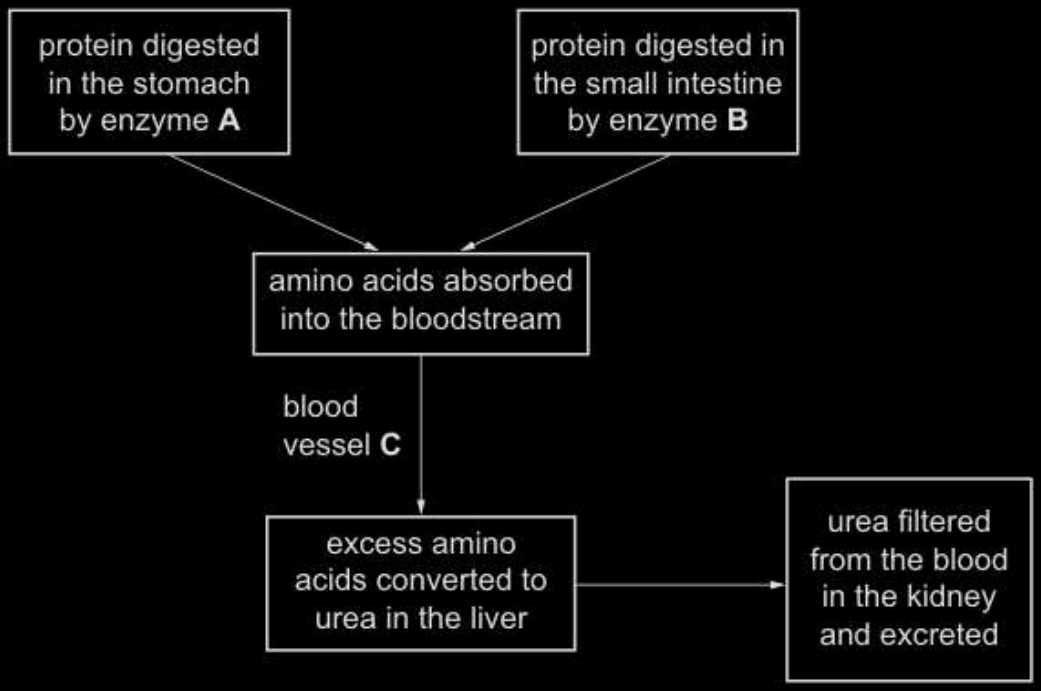


Fig. 3.1

Use Fig. 3.1 to answer these questions.

(i) State the name of enzyme **A** and enzyme **B**.

enzyme **A** .....

enzyme **B** .....

[2]

(ii) State the name of blood vessel **C**.

..... [1]

(iii) Name the capillary structures in the kidney that filter urea from the blood.

..... [1]

(b) Two groups of students, **X** and **Y**, investigate the effect of different diets on the concentration of urea in their urine.

For one day, the two groups eat different foods:

- group **X** eat bread, jam, orange juice, lettuce, fruit
- group **Y** eat nuts, milk, bread, cheese, meat.

Early the next morning, they collect and analyse their urine to measure the concentration of urea.

Table 3.1 shows their results.

Table 3.1

	concentration of urea / g per 100 cm^3 of urine	
	individual students	mean
group **X**	1.5, 1.7, 2.3, 1.6, 2.1, 1.4	1.8
group **Y**	1.8, 1.9, 2.3, 2.5, 1.9, 2.4	

(i) Calculate the mean concentration of urea for the students in group **Y**.

Give your answer to **two** significant figures.

Space for working.

mean concentration of urea = ... g per 100 cm^3 of urine [2]

(ii) Explain the difference in the mean concentration of urea for the two groups of students.

.....

.....

.....

.....

.....

..... [3]

(iii) One of the students said: 'To make a fair comparison, we should have made sure that the students in each group drank the same volume of water'.

Explain why this student is correct.

.....

.....

.....

.....

..... [2]

[Total: 11]

3(a)(i)	pepsin ; trypsin ;	2	
3(a)(ii)	hepatic portal vein ;	1	
3(a)(iii)	glomerulus ;	1	
3(b)(i)	2.1 ;;	2	**A** 12.8 / 6 **OR** 2.133 = 1 mark
3(b)(ii)	group Y ate more protein / ate protein rich foods ; so / in group Y, more amino acids are deaminated to produce urea ; so / in group Y, more urea is filtered / passes into the urine in the kidney ;	3	**A** ora for group X
3(b)(iii)	idea that the volume of water drunk, will affect the volume of water used to form urine ; this / the volume of water drunk, will affect the concentration of urea in the urine ;	2	

Question	Answer	Marks	Guidance

October/November 2025 · Paper 2 · Variant 1 · Question 4

4 Chlorophyll is found in all organisms that photosynthesise.

(a) Complete these sentences about chlorophyll by writing suitable words in the gaps.

A chlorophyll molecule contains carbon, hydrogen, oxygen, nitrogen and an atom of

..... .

Chlorophyll molecules are found in plant cells in structures called

..... .

The type of leaf cells that contain the most chlorophyll are the

..... cells.

Chlorophyll molecules trap sunlight energy and this is used by plants to produce glucose.

Glucose can then be converted to ..... to be transported around the plant.

Glucose can also be converted to ..... to make cell walls.

[5]

(b) Mistletoe is a parasitic green plant that can often be seen growing attached to trees.

Fig. 4.1 shows some mistletoe growing on the branch of a tree.

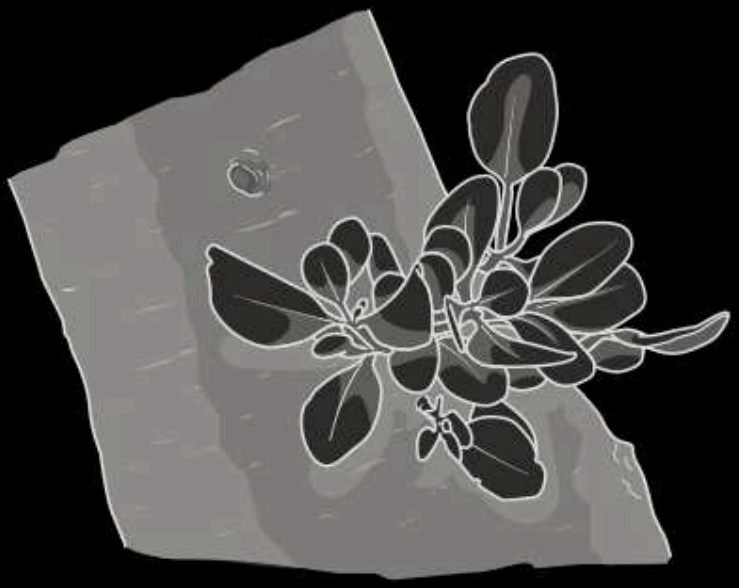


Fig. 4.1

Most plants have about 1 mg of chlorophyll in 100 g of their leaves.

Mistletoe, however, has about 0.4 mg of chlorophyll in 100 g of its leaves.

Mistletoe seeds are stuck to the branches of trees by birds. The seeds germinate to produce special roots that grow into the phloem of the tree branch.

Suggest why mistletoe grows these special roots.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 9]

4(a)	magnesium ; chloroplasts ; palisade (mesophyll) ; sucrose ; cellulose ;	**5**	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance
4(b)	max **four** from: mistletoe / it has less chlorophyll ; so can perform less photosynthesis ; therefore, it makes less sugars ; so the mistletoe obtains sugars / sucrose from the tree ; phloem contains sugars / sucrose ;	**4**	

Question	Answer	Marks	Guidance

5 **(a)** Fig. 5.1 shows a model of the human circulatory system.

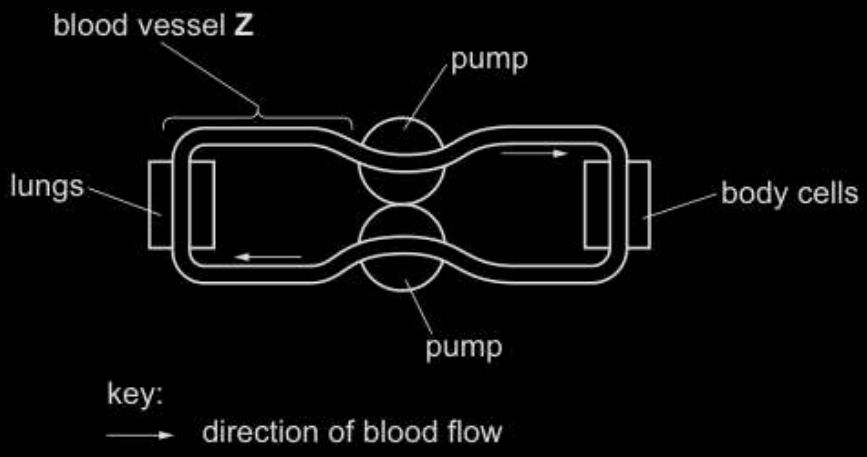


Fig. 5.1

(i) Give the name of the blood vessel represented by **Z** in Fig. 5.1.

.....

[1]

(ii) The system is described as a double circulatory system.

Describe what is meant by a double circulation.

.....

.....

.....

.....

..... [2]

(iii) Explain how the structure of the human heart is adapted to be the pump in a double circulatory system.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Some people have a heart defect called a 'hole in the heart', as shown in Fig. 5.2.

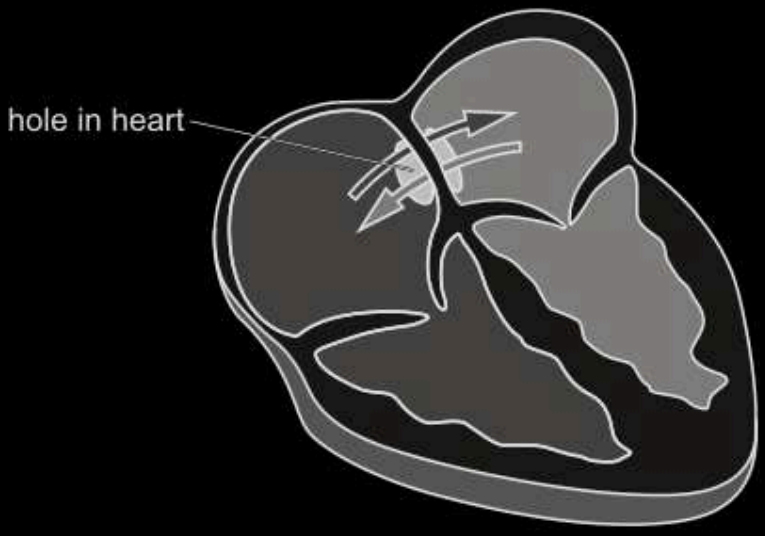


Fig. 5.2

People with a hole in their heart may develop pain in their muscles and quickly become tired when they exercise.

Explain why this happens.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

5(a)(i)	pulmonary vein ;	**1**	
5(a)(ii)	blood travels through pump / heart twice ; for each full circuit / circulation ;	**2**	
5(a)(iii)	max **three** from: it has four chambers / has a septum ; blood on the left and right side needs to be kept separate ; deoxygenated and oxygenated blood kept separate ; larger / thicker wall of left ventricle / ora ; to pump blood around body / ora ;	**3**	
5(b)	max **four** from: blood from both sides mix ; deoxygenated blood mixes with oxygenated blood ; less oxygen reaches the muscles ; muscles respire anaerobically / less aerobic respiration ; (increased) production / build-up of lactic acid ;	**4**	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance

7 HIV and MRSA are two different microorganisms which are pathogens.

(a) (i) Put ticks (✓) in Table 7.1 to show the characteristics of each pathogen.

Table 7.1

characteristic	HIV	MRSA
is a living cell		
contains cytoplasm		
contains genetic material		
has a protein coat		

[2]

(ii) Describe how HIV affects the human immune system.

.....

.....

.....

.....

.....

..... [3]

(iii) For different reasons, HIV and MRSA are **not** destroyed by antibiotics such as penicillin.

Give these reasons.

HIV .....

.....

MRSA .....

.....

[2]

(b) Fig. 7.1 shows the number of people infected by the MRSA pathogen in different countries.

It also shows the total amount of antibiotics used in each country in one year per 1000 people.

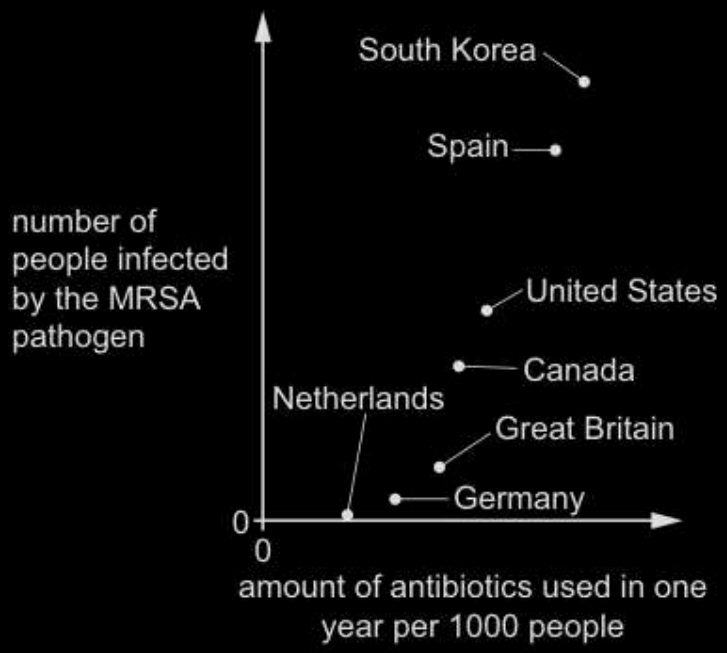


Fig. 7.1

Describe and explain the relationship shown between antibiotic use and the number of people infected by the MRSA pathogen.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

<table>
<tr><td>7(a)(i)</td><td><table>
<tr><th>Characteristic</th><th>HIV</th><th>MRSA</th></tr>
<tr><td>is a living cell</td><td></td><td>✓</td></tr>
<tr><td>contains cytoplasm</td><td></td><td>✓</td></tr>
<tr><td>contains genetic</td><td>✓</td><td>✓</td></tr>
<tr><td>contains a protein coat</td><td>✓</td><td></td></tr>
</table></td><td>2</td><td>mark by column</td></tr>
<tr><td>7(a)(ii)</td><td>weakens the immune system ;
decreased lymphocyte numbers ;
reduced ability to produce antibodies ;</td><td>3</td><td></td></tr>
<tr><td>7(a)(iii)</td><td>HIV is a virus ;
MRSA is resistant (to many antibiotics / penicillin) ;</td><td>2</td><td></td></tr>
<tr><td>7(b)</td><td>max three from:

the more antibiotics used, the more people have the pathogen / MRSA / ora ;
more antibiotics used to try and kill the pathogen / MRSA ;
(however), the antibiotics will not kill resistant microbes / MRSA ;
(use of the antibiotic) increases the number of resistant bacteria ;</td><td>3</td><td></td></tr>
</table>

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance

8 Fig. 8.1 shows a synapse between a sensory neurone and a relay neurone.

The sensory neurone receives information from a pain receptor.

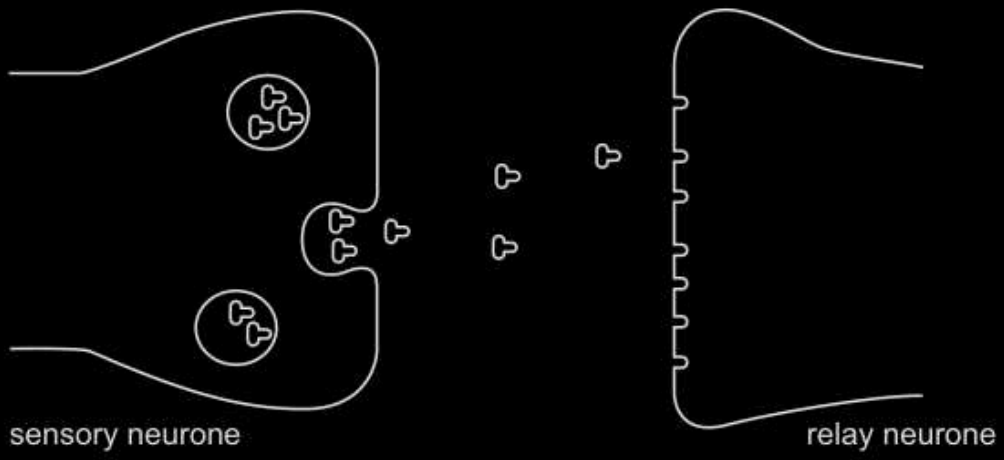


Fig. 8.1

(a) Explain how impulses in the sensory neurone can stimulate the relay neurone.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) Analgesics are drugs that help to stop a person from feeling pain.

Fig. 8.2 shows the shape of some analgesic drug molecules.

Fig. 8.2

Suggest how this analgesic drug can help to stop a person from feeling pain.

Use information from Fig. 8.1 and Fig. 8.2 in your answer.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

8(a)	max **six** from: use of neurotransmitter molecules ; impulse stimulates the release of neurotransmitter molecules ; from vesicles ; into the synaptic gap / cleft ; the neurotransmitter molecules pass across the gap / cleft ; by diffusion ; then bind with receptor (proteins / sites) : cause an impulse to be initiated in the relay / post synaptic neurone ;	**6**	
8(b)	the analgesic / drug has a similar shape to the neurotransmitter ; fits / binds with the receptor (proteins / sites) ; this blocks the receptor (protein / sites) / stops the neurotransmitter binding ; information from the pain receptor cannot reach the brain ;	**4**	

© Cambridge University Press & Assessment 2025

2 A constant supply of water is essential for both animals and plants.

(a) Fig. 2.1 is a diagram showing the average daily water intake and average daily water loss for an adult human.

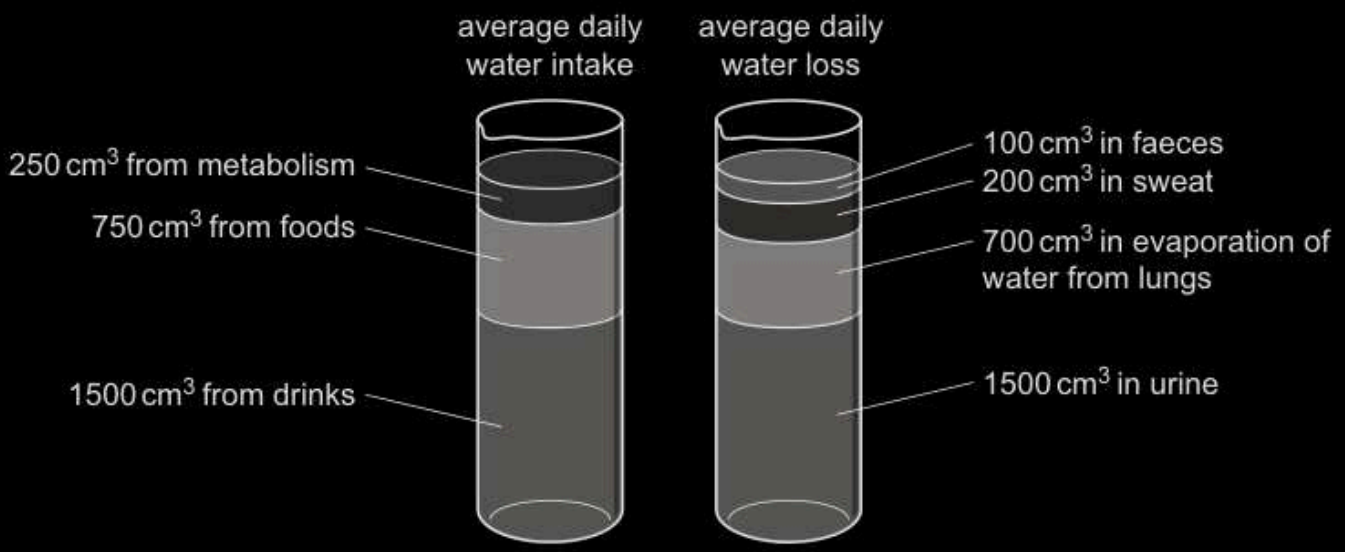


Fig. 2.1

(i) Calculate the percentage of average daily water intake that is lost in sweat from an adult human.

Space for working.

..... % [2]

(ii) The percentage of water lost in sweat is **not** always the same.

Explain why.

.....

.....

.....

.....

.....

.....

..... [3]

(b) A scientist measured the mass of water taken up and lost by a plant over 21 hours. The plant was kept at a constant temperature of 20 °C on a windowsill in a laboratory. Readings were taken every 90 minutes.

At 0 hours it was sunrise and at 15 hours it was sunset.

The results are shown in Fig. 2.2.

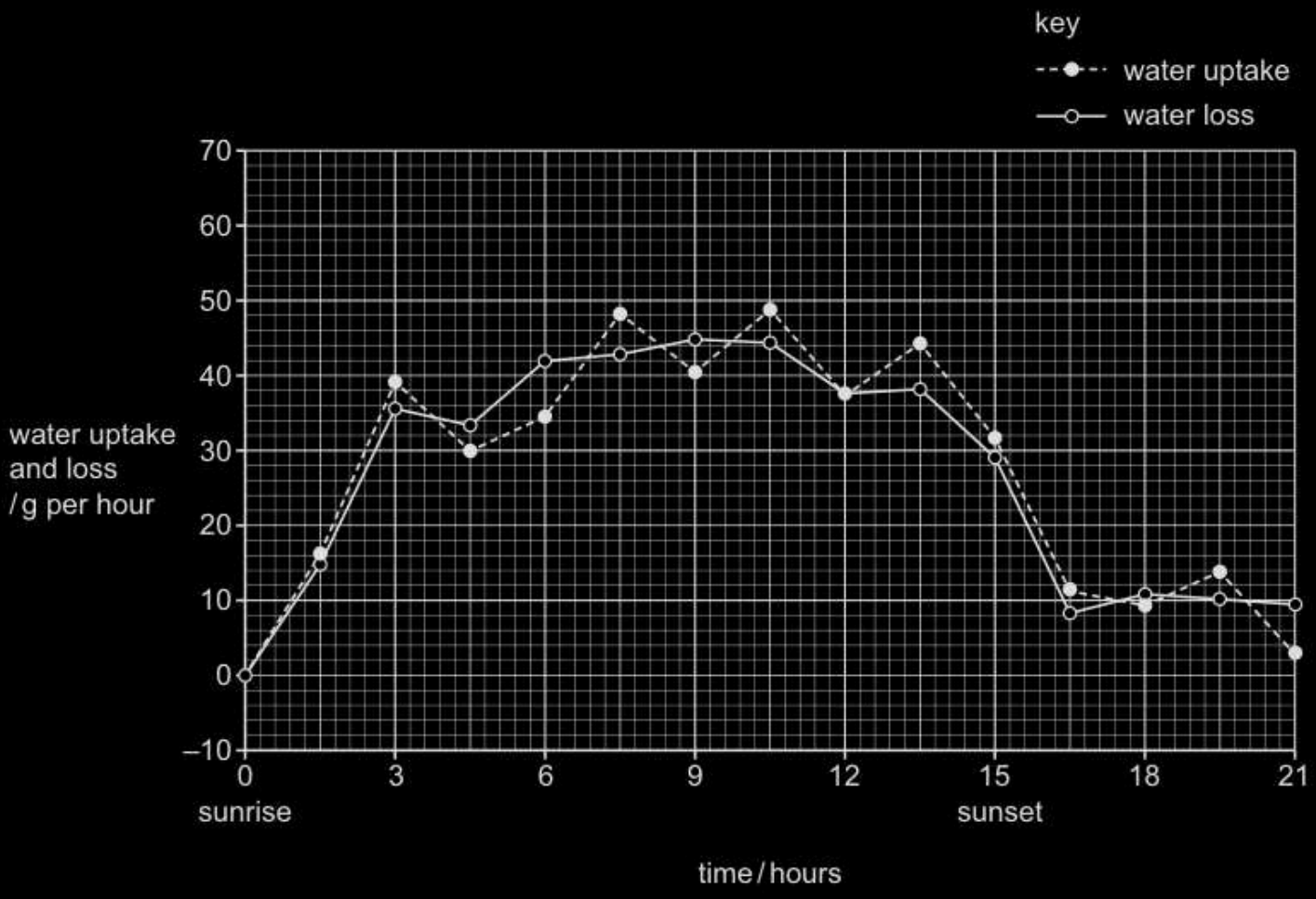


Fig. 2.2

(i) Plants lose water by evaporation from their leaves.

State the name of this process.

..... [1]

(ii) Explain the shape of the curve for **water loss** from the plant over the 21 hours of the experiment.

.....

.....

.....

.....

..... [2]

(iii) Explain why the shape of the curve for water loss is very similar to the shape of the curve for water uptake.

.....

.....

.....

.....

.....

.....

..... [3]

(iv) The scientist repeated the experiment. All conditions were kept the same except the temperature of the laboratory, which was 10 °C lower.

On Fig. 2.2, draw a new curve to show the expected results for **water loss** at the lower temperature. [1]

[Total: 12]

2(a)(i)	8 ;;	**2**	**A** 200 ÷ 2500 **or** 0.08 = 1 mark
2(a)(ii)	max **three** from: 1 homeostasis ; 2 maintaining constant body temperature ; 3 evaporation of sweat to remove heat (energy) ; 4 increased activity / increase in temperature means more sweat / ora ; 5 if volume lost in faeces / lungs / urine changes, then the percentage lost in sweat will change ;	**3**	
2(b)(i)	transpiration ;	**1**	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance
2(b)(ii)	max **two** from: increased water loss **+** during light / day / 0–15 hours / around sunrise **+** stomata open ; decreased water loss **+** during dark / night / 15–21 hours / around sunset **+** stomata close ; greater light intensity **+** faster evaporation / more diffusion / ora ;	**2**	
2(b)(iii)	max **three** from: 1 water evaporates / transpires from leaves ; 2 this reduces water potential of leaves ; 3 water moves up / through xylem vessels ; 4 transpiration pull / transpiration stream ; 5 forces of attraction between water ; 6 prevents loss of turgidity / prevents wilting ;	**3**	
2(b)(iv)	same shape of curve but below the original ;	**1**	

Question	Answer	Marks	Guidance

4 Lipases are enzymes produced by animals, plants and microorganisms. Lipases from different species can have different chemical structures.

(a) All enzymes are biological catalysts.

Explain what is meant by the term catalyst.

.....

.....

..... [2]

(b) Lipase is a digestive enzyme in humans.

(i) Name an organ that secretes lipase into the duodenum.

..... [1]

(ii) Name the substrate and products of lipase.

substrate .....

products .....

[2]

(c) Some scientists collected species of bacteria from three different sources, **A**, **B** and **C**.

The scientists investigated how temperature affected the activity of the lipase produced by the bacteria collected from the three different sources. They wanted to find a source of lipase that could be used in biological washing powders.

Fig. 4.1 shows the results.

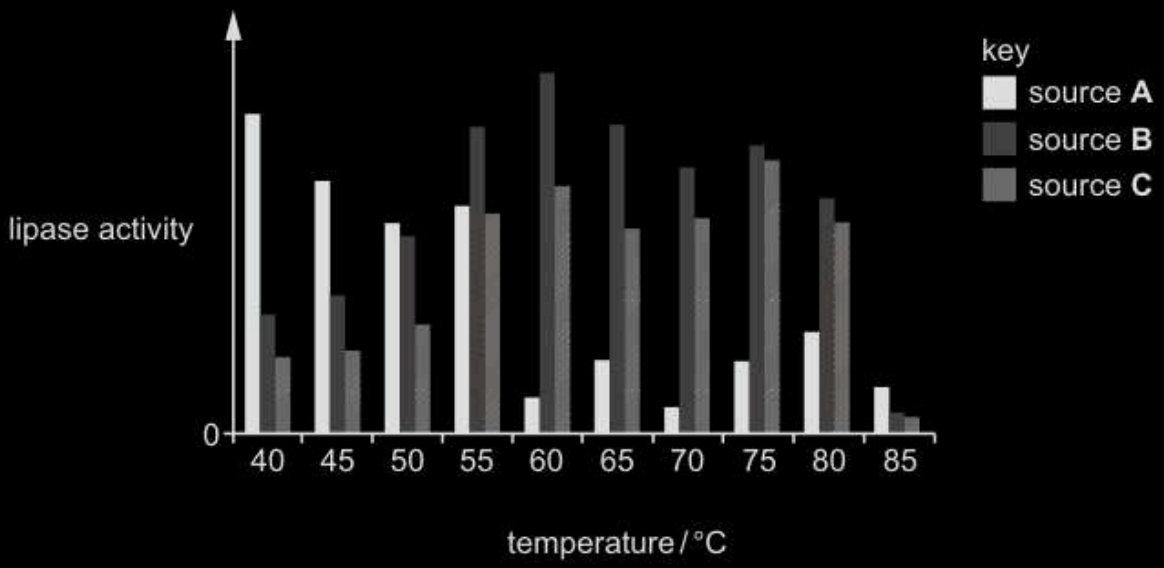


Fig. 4.1

(i) Using Fig. 4.1, discuss the results from this investigation.

.....

.....

.....

.....

.....

.....

..... [3]

(ii) The lipase molecules produced by different species of bacteria have different chemical structures.

Explain:

- what determines the different chemical structures of lipase molecules
- why having different chemical structures can affect their activity at different temperatures.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 11]

4(a)	speeds up (chemical) reactions ; not changed ;	**2**	
4(b)(i)	pancreas ;	**1**	
4(b)(ii)	*substrate* lipid / fat / oil ; *products* fatty acid(s) **+** glycerol ;	**2**	
4(c)(i)	max **three** from: lipases from A, B and C are active between 40 and 85 °C ; all have different optimum temperatures ; A optimum 40 °C / B optimum 60 °C / C optimum 75 °C ; B most suitable **+** highest activity / A most suitable **+** works best at lower temperatures (40 °C) ;	**3**	**A** name all 3 optimums = 2 marks (for mp 2 + 3)
4(c)(ii)	max **two** from: 1 DNA codes for protein / polypeptide ; 2 sequence of bases ; 3 (determines) sequence of amino acids ; 4 enzymes/lipases are proteins / made of amino acids ; 5 sequence of amino acids determines shape / function ; **and** 6 some enzymes / proteins / structures / active sites denature at lower / higher / different temperatures than others ;	**3**	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance

5 A sample of human blood is put in a small tube. The tube is placed in a centrifuge, which spins it around very fast, to separate the components of the blood. The denser components of the blood collect at the bottom of the tube.

Fig. 5.1 shows the composition of this blood sample.

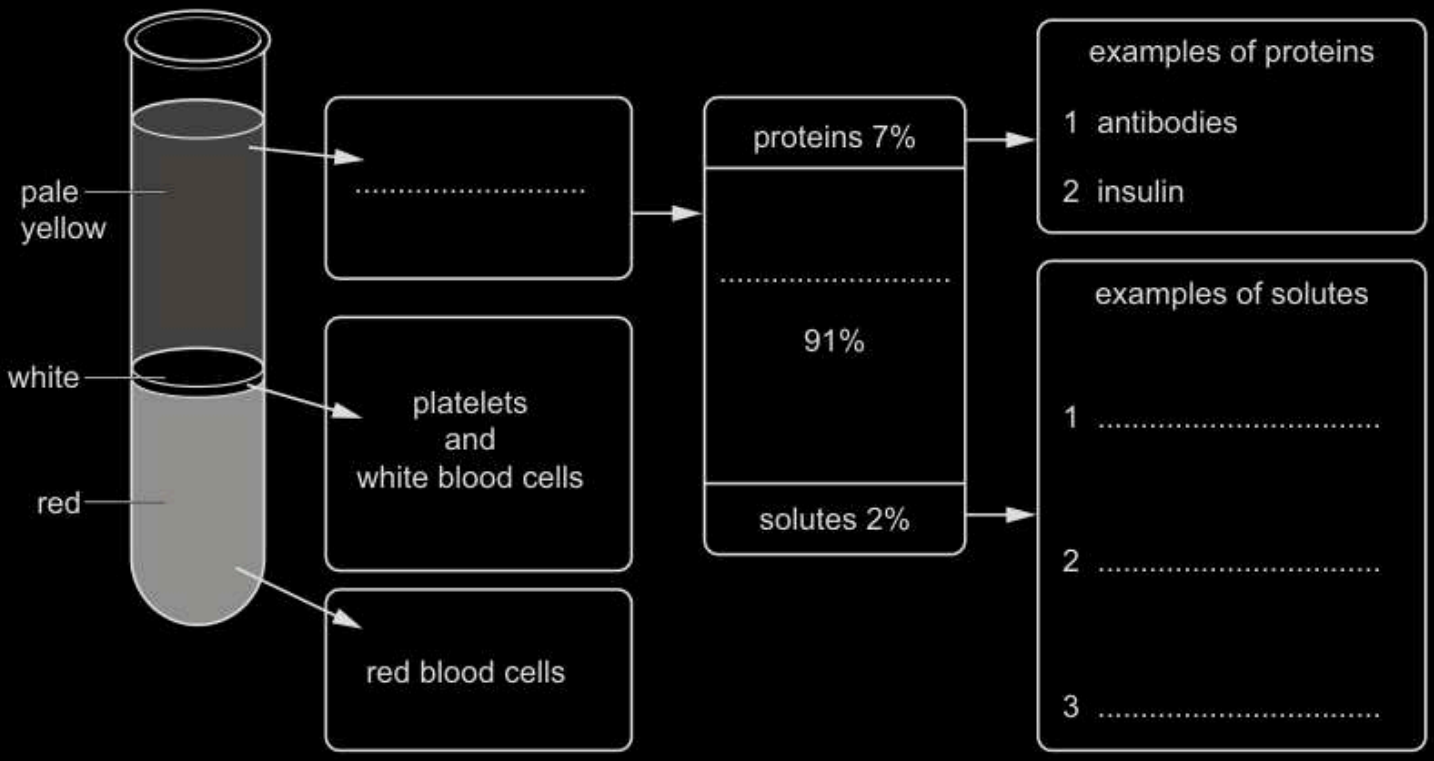


Fig. 5.1

(a) (i) Complete Fig. 5.1 by filling in the names of **five** components shown by the blank spaces. [5]

(ii) Suggest why red blood cells are **more** dense than white blood cells.

.....

.....

.....

.....

..... [2]

(b) The ABO blood groups in humans are determined by three alleles, I^A, I^B and I^O.
There are four blood groups (phenotypes) but six different genotypes for the ABO blood groups.

Complete Table 5.1 to match the blood groups to their possible genotypes.

Table 5.1

blood group (phenotype)	possible genotype(s)
A	
B	
AB	
O	

[4]

(c) A pregnant woman has blood group O.

The fetus in the woman's uterus developed from one of her fertilised eggs and has blood group B.

(i) State the **genotype** of the fetus.

..... [1]

(ii) The woman and the fetus have different blood groups so they have different antigens on their red blood cell membranes.

The woman has antibodies to the antigens on the red blood cell membranes of the fetus.

Explain why the woman's antibodies are **not** usually able to come into contact with the antigens of the fetus.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 15]

<table>
<tr><td>5(a)(i)</td><td>plasma ;
water ;

max three from:

examples of solutes
glucose ;
amino acids ;
urea ;
mineral ions ;
vitamins ;
fatty acids/glycerol ;</td><td>5</td><td>

A named mineral
A named vitamin</td></tr>
<tr><td>5(a)(ii)</td><td>max two from:

1 haemoglobin ;
2 (densely) packed ;
3 contains iron ;
4 less water / solvent in red blood cells / ora ;
5 solutes increase density ;</td><td>2</td><td></td></tr>
<tr><td>5(b)</td><td><table><tr><th>blood group</th><th>genotype</th></tr><tr><td>A</td><td>$I^A I^A + I^A I^O$;</td></tr><tr><td>B</td><td>$I^B I^B + I^B I^O$;</td></tr><tr><td>AB</td><td>$I^A I^B$;</td></tr><tr><td>O</td><td>$I^O I^O$;</td></tr></table></td><td>4</td><td>each correct row in table = 1 mark</td></tr>
<tr><td>5(c)(i)</td><td>fetus = $I^B I^O$;</td><td>1</td><td></td></tr>
</table>

Question	Answer	Marks	Guidance
5(c)(ii)	max **three** from: placenta ; separate blood stream ; antibodies too big ; antibodies cannot pass across placenta / through capillary walls ;	**3**	

Question	Answer	Marks	Guidance

6 Cholera is a disease caused by the pathogen *Vibrio cholerae*.

Outbreaks of cholera often occur after natural disasters such as earthquakes.

(a) (i) *Vibrio cholerae* is a microorganism.

Name the type of microorganism.

..... [1]

(ii) Suggest and explain reasons why outbreaks of cholera are more likely after natural disasters.

.....

.....

.....

.....

.....

.....

..... [3]

(b) An infection with *Vibrio cholerae* can cause severe diarrhoea. This can be treated by giving the patient a solution to drink.

Fig. 6.1 shows the ingredients of a standard solution.

water	1000 cm^3
sodium chloride	3.5 g
potassium chloride	1.5 g
trisodium citrate dihydrate	2.9 g
glucose	20.0 g

Fig. 6.1

(i) Explain how an infection with *Vibrio cholerae* causes diarrhoea.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) Using Fig. 6.1, suggest why giving the patient this solution to drink is an effective treatment for diarrhoea.

.....

.....

.....

.....

..... [2]

[Total: 10]

6(a)(i)	bacterium / bacteria ;	**1**	
6(a)(ii)	max **three** from: 1 transmitted / spread through drinking water ; 2 poor sanitation ; 3 *idea that* water supplies are contaminated ; 4 removal of sewage is disrupted ; 5 difficulties with waste disposal / personal hygiene / food preparation ;	**3**	
6(b)(i)	max **four** from: 1 toxin / choleragen ; 2 stimulates release of chloride ions into small intestine ; 3 water passes into intestine / gut ; 4 from high to low water potential ; 5 (by) osmosis ; 6 makes faeces watery ;	**4**	
6(b)(ii)	replaces water / rehydrates ; chloride / ions replaced ;	**2**	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance

7 The mammalian nervous system is essential for coordination and control.

It is an organ system that contains the eyes.

(a) Explain the differences between organs, tissues and cells, using the eye as an example.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Synapses are found in the mammalian nervous system.

Explain how the structure of a synapse ensures that electrical impulses travel in **one** direction only.

You may use the outline of a synapse in Fig. 7.1 to illustrate your answer.

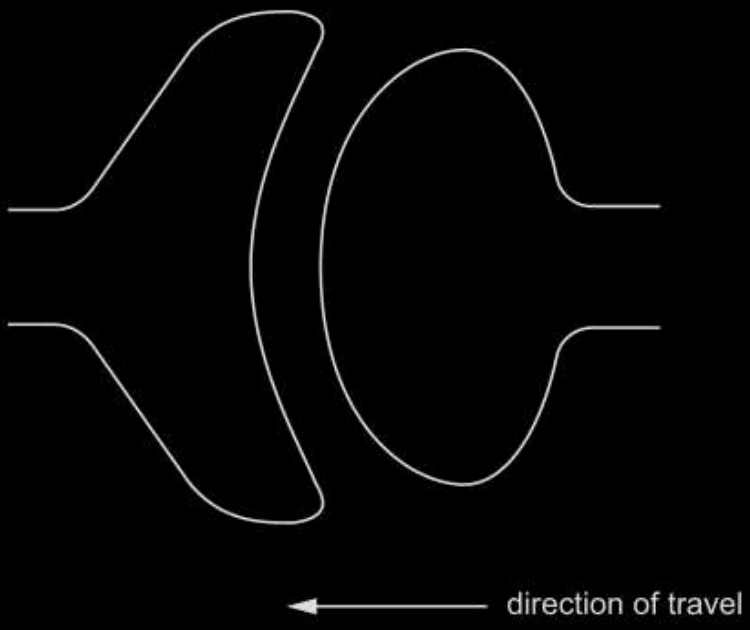


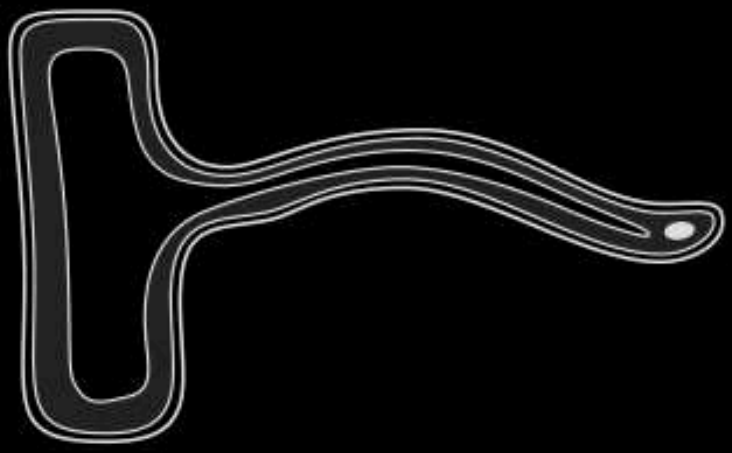
Fig. 7.1

7(a)	max **four** from: 1 cell is the basic unit of life ; 2 cell **+** light receptor / rods / cones ; 3 tissue is a group of cells with a similar structure / function ; 4 tissue **+** retina / cornea / muscle / sclera / iris / conjunctiva / suspensory ligament ; 5 organ is group of tissues with specific function ; 6 organ **+** eye / optic nerve ;	**4**	
7(b)	1 vesicles contain / release neurotransmitter ; 2 vesicles only on one side of (synaptic) gap ; 3 impulse causes neurotransmitter to be released into gap ; 4 (neurotransmitter) diffuses across gap ; 5 (neurotransmitter) binds to receptor / protein ; 6 receptors / proteins in cell membrane only on other side of gap ;	**6**	

© Cambridge University Press & Assessment 2025

Correct answer: D

1 The diagram shows a plant cell.



What is the function of this cell?

A to carry mineral ions along stems

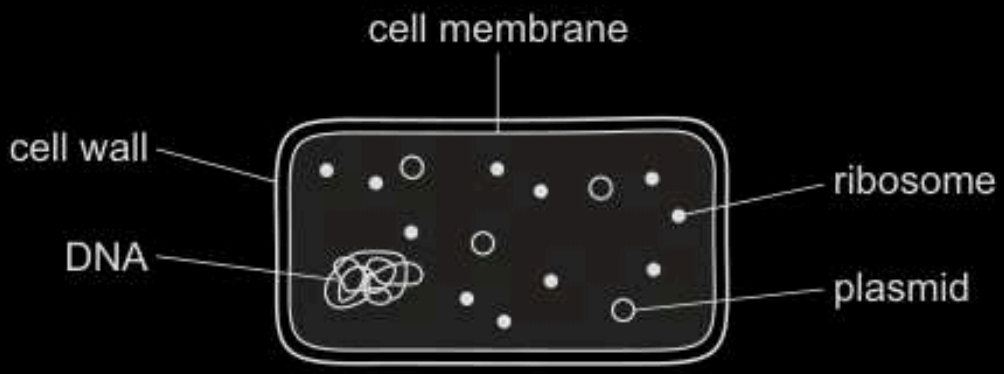
B to photosynthesise

C to take in carbon dioxide

D to take in water

Correct answer: B

2 The diagram shows a structure that was observed using a microscope.



What does the diagram show?

A an animal cell

B a bacterium

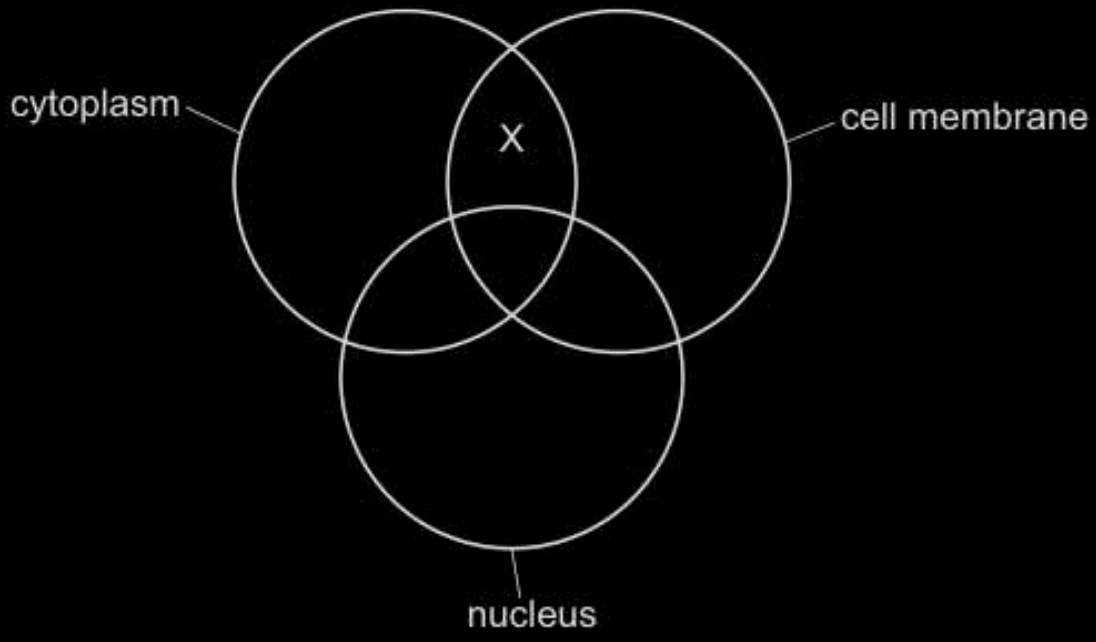
C a plant cell

D a virus

May/June 2025 · Paper 1 · Variant 1 · Question 3

Correct answer: B

3 The diagram shows the structures found in cell X.



Which type of cell is X?

A liver cell

B red blood cell

C root hair cell

D xylem vessel

May/June 2025 · Paper 1 · Variant 1 · Question 4

Correct answer: B

4 What is an example of an organ?

A digestive system

B eye

C blood

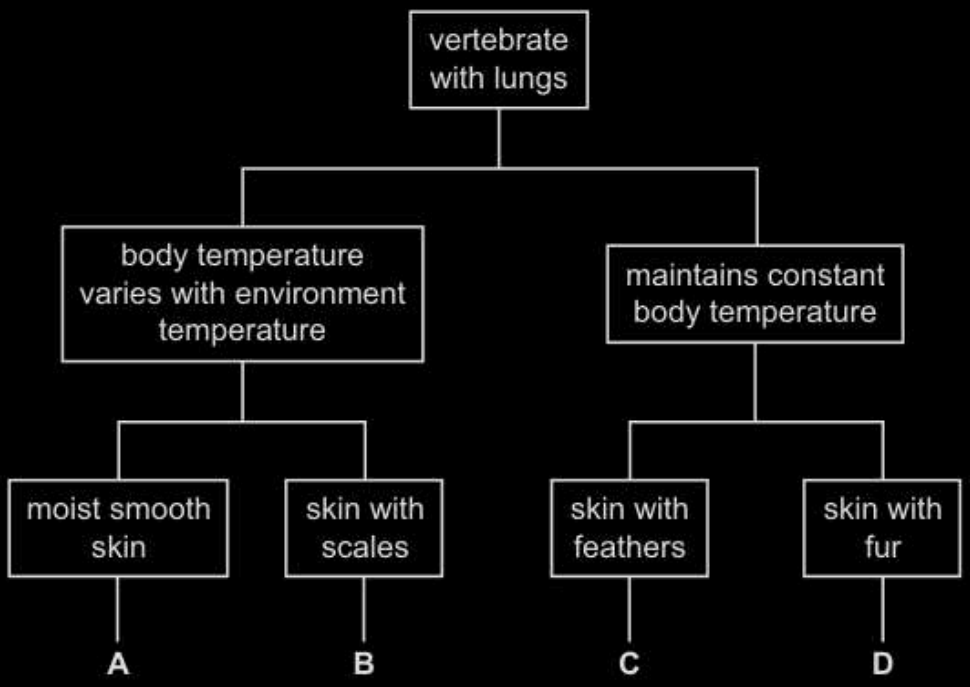
D neurone

May/June 2025 · Paper 1 · Variant 1 · Question 5

Correct answer: B

5 A new type of small vertebrate with lungs is discovered. This is identified as a reptile.

Which vertebrate in the key is a reptile?



May/June 2025 · Paper 1 · Variant 1 · Question 6

Correct answer: A

6 Molecules and ions can be moved into and out of cells by active transport.

Which statement about active transport is **not** correct?

A Active transport takes place only in plants.

B Active transport moves molecules and ions against a concentration gradient.

C Active transport uses energy to move molecules and ions.

D Molecules and ions are moved across partially permeable membranes by active transport.

May/June 2025 · Paper 1 · Variant 1 · Question 8

Correct answer: A

8 The diagram shows an enzyme molecule.

Which diagram shows the substrate molecule for this enzyme?

A

B

C

D

May/June 2025 · Paper 1 · Variant 1 · Question 9

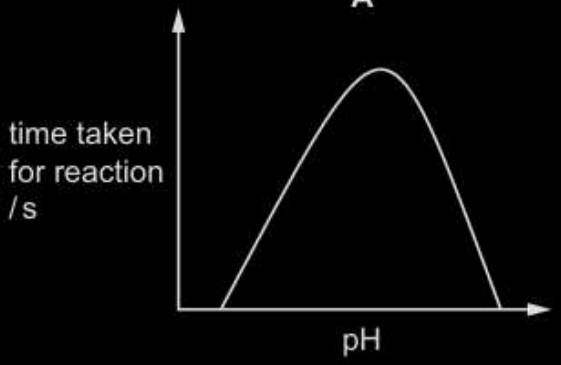
Correct answer: D

9 The enzyme amylase catalyses the breakdown of starch into maltose.

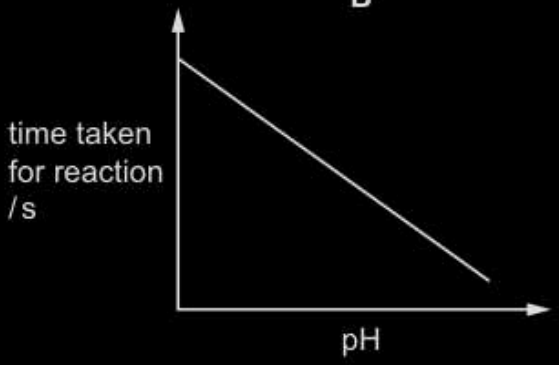
The effect of different pHs on the time it took for this breakdown to occur when all other variables were controlled was investigated.

Which graph shows the results of this investigation?

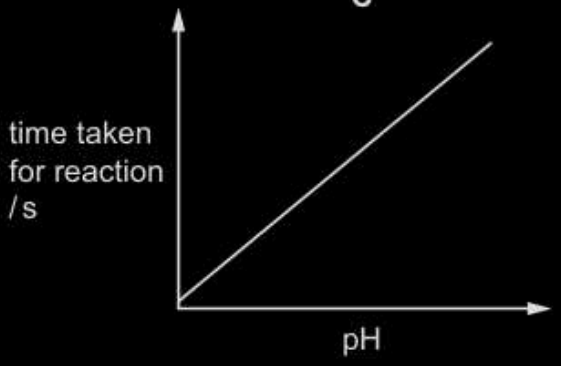
A



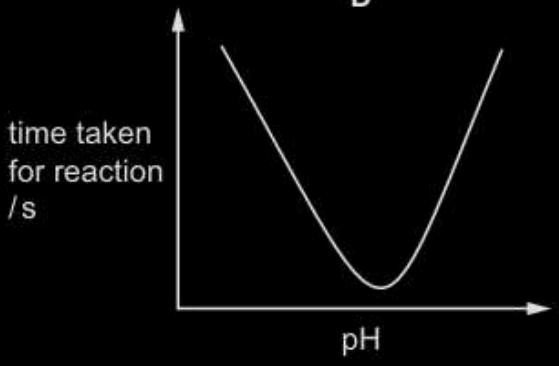
B



C



D



May/June 2025 · Paper 1 · Variant 1 · Question 10

Correct answer: B

10 What are features of enzymes?

1 They act on specific substrates.

2 They are carbohydrates.

3 They are catalysts.

4 They are involved in some, but **not** all, metabolic reactions.

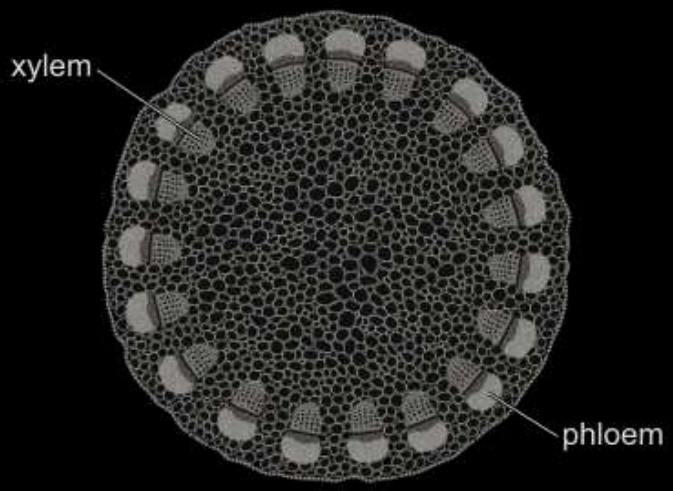
A 1, 2 and 3 **B** 1 and 3 only **C** 1 and 4 **D** 2, 3 and 4

May/June 2025 · Paper 1 · Variant 1 · Question 12

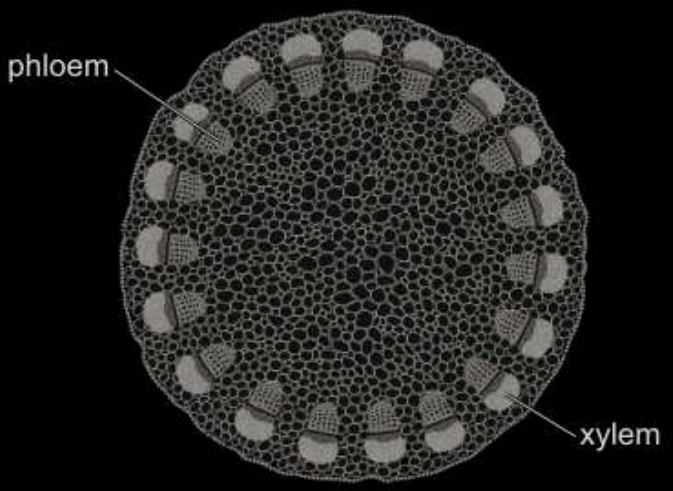
Correct answer: A

12 Which transverse section photomicrograph shows the position of the xylem and phloem in a non-woody dicotyledonous stem?

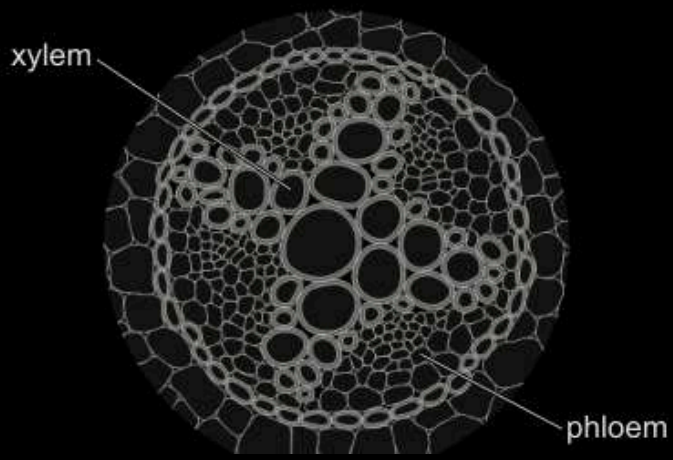
A



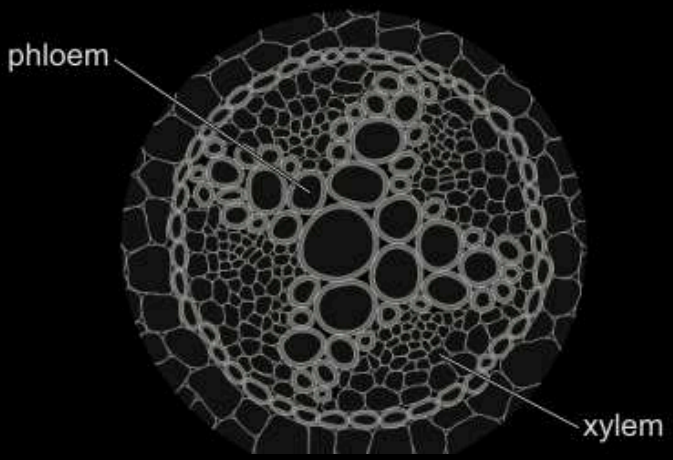
B



C

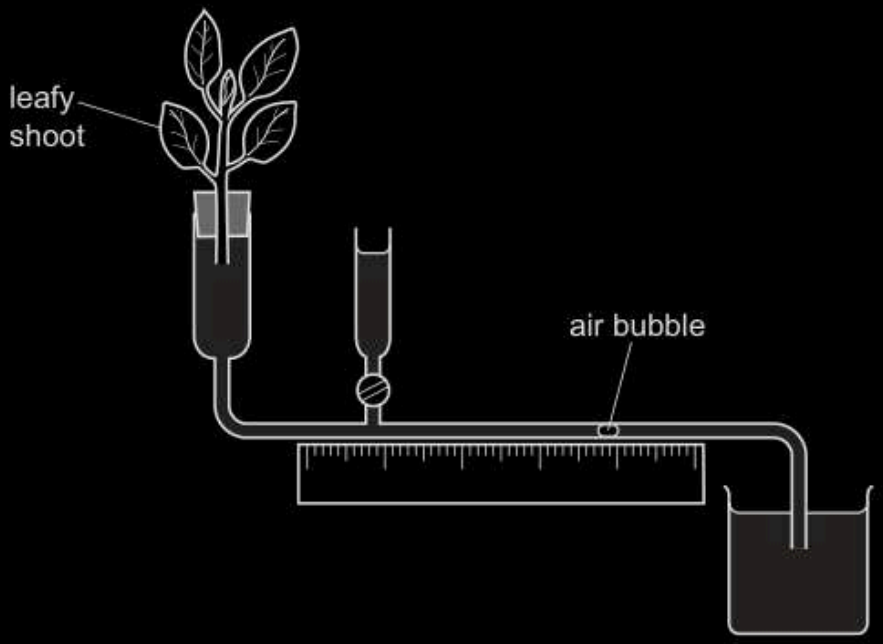


D



Correct answer: D

13 Students set up apparatus to measure the uptake of water by a leafy shoot.

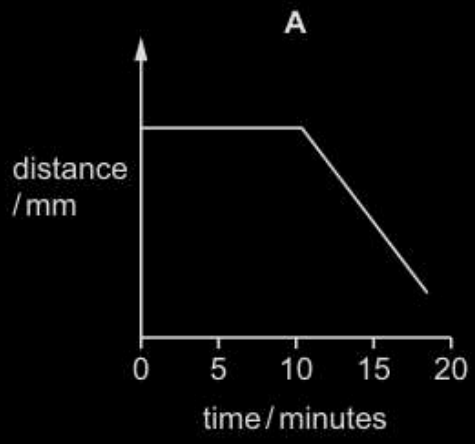


They recorded the position of the air bubble at the start. They observed the apparatus for 10 minutes, and recorded the distance travelled by the air bubble at one-minute intervals.

They then switched on a fan pointed towards the leafy shoot and recorded the distance travelled by the air bubble from the start, each minute, for another 10 minutes.

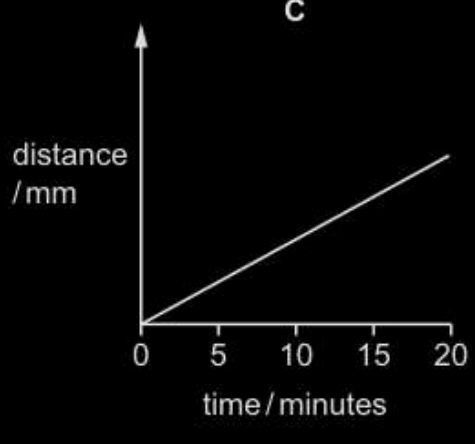
Which graph would they draw from their data?

A

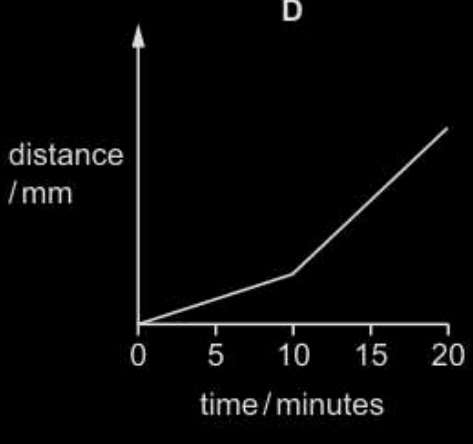


B

C



D

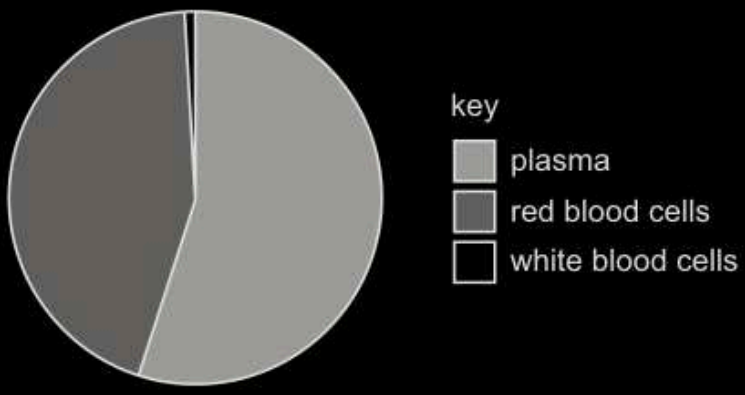


May/June 2025 · Paper 1 · Variant 1 · Question 16

Correct answer: C

16 Blood is made up from a number of different components.

The pie chart shows the percentage volume of each of the components of blood.



What is the percentage volume of blood for the component that has the function of transporting oxygen?

A 1% **B** 30% **C** 44% **D** 55%

May/June 2025 · Paper 1 · Variant 1 · Question 17

Correct answer: C

17 What is the total number of atria and ventricles that the blood passes through during two complete circuits of the human body?

A 2 **B** 4 **C** 8 **D** 16

May/June 2025 · Paper 1 · Variant 1 · Question 18

Correct answer: C

18 The table shows the range of percentages of different types of tissue found in the walls of the aorta of people of different ages.

Collagen tissue is present in the walls of blood vessels to provide strength and support.

type of tissue	percentage (%)	
	young adult	old adult
elastic	38–46	28–40
muscle	14–21	9–37
collagen	13–36	24–51

Which conclusion about the range of percentages is correct?

A The range of collagen tissue found in people of both age groups is 39.

B The range of each type of tissue is smaller in old adults than in young adults.

C The range of muscle tissue in old adults is four times greater than in young adults.

D The range of elastic tissue in people of both age groups is the greatest of all three types of tissue.

May/June 2025 · Paper 1 · Variant 1 · Question 19

Correct answer: B

19 The table shows information about three different blood vessels, P, Q and R.

One of the vessels is an artery, one is a capillary and one is a vein.

blood vessel	P	Q	R
thickness of wall / μm	500	<1	1000
diameter of lumen / mm	5	0.008	4

Using the information from the table, what is the cross-sectional area of the lumen of the vein?

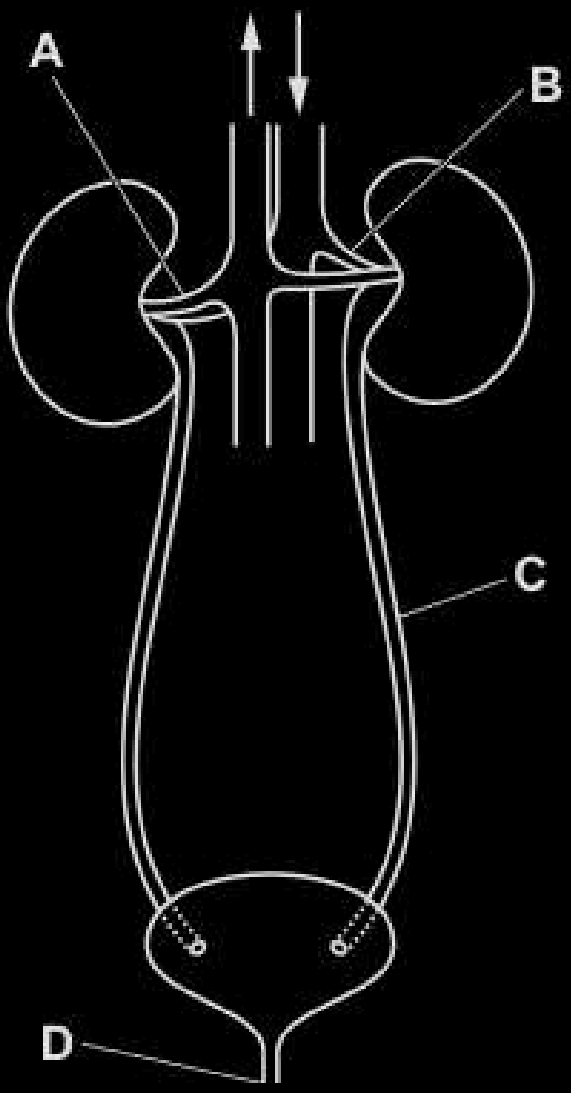
(Assume $\pi = 3.14$)

A $12.6\,mm^2$ **B** $19.6\,mm^2$ **C** $50.2\,mm^2$ **D** $78.5\,mm^2$

May/June 2025 · Paper 1 · Variant 1 · Question 23

Correct answer: C

23 Which structure is the ureter?



May/June 2025 · Paper 1 · Variant 1 · Question 24

Correct answer: A

24 The processes of filtration and reabsorption take place in the kidneys.

The daily amounts of some of the substances entering the kidneys in the blood and of these substances leaving the kidneys in the urine are shown.

substance	amount entering the kidneys in the blood	amount leaving the kidneys in the urine
water	150 dm^3	1.5 dm^3
glucose	150 g	0 g
urea	50 g	30 g

Which percentage of each substance is reabsorbed by the kidneys during a day?

	% of substance reabsorbed by kidneys		
	water	glucose	urea
A	99	100	40
B	148.5	150	20
C	0.99	1.0	0.4
D	1.0	0.0	60

May/June 2025 · Paper 1 · Variant 1 · Question 25

Correct answer: D

25 Where are the receptors for the pupil reflex located?

A brain

B iris

C pupil

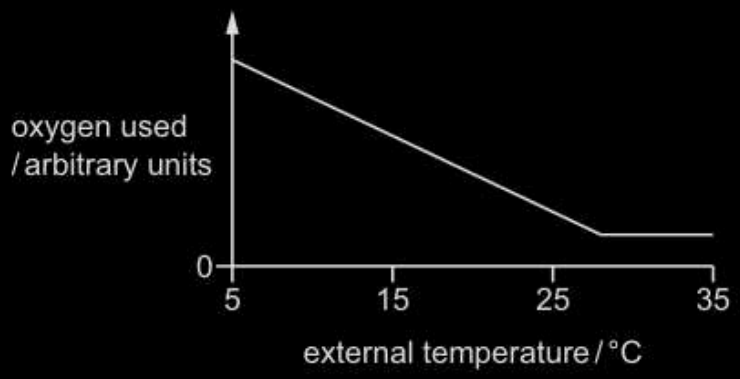
D retina

May/June 2025 · Paper 1 · Variant 1 · Question 26

Correct answer: B

26 The internal body temperature of some animals, including humans, remains constant even when the external temperature changes.

The graph shows how the oxygen used by one of these animals is affected by change in the external temperature.



Which conclusion can be made from this graph?

A The animal stops respiring when the external temperature is greater than 28 °C.

B The animal's respiration rate increases as the external temperature decreases from 28 °C.

C The internal temperature of the animal decreases as the external temperature increases.

D The animal's respiration rate increases as the external temperature increases.

May/June 2025 · Paper 1 · Variant 1 · Question 27

Correct answer: A

27 Which statements about synapses are correct?

1 Synapses are junctions between neurones.

2 Synapses ensure impulses travel in only one direction.

3 Neurotransmitter chemicals diffuse across synaptic gaps.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

May/June 2025 · Paper 1 · Variant 1 · Question 28

Correct answer: B

28 Which process is **not** part of a homeostatic mechanism in mammals?

A change in flow of blood in vessels of the skin

B growth of longer hairs from the skin

C increased rate of sweat production

D contraction of hair erector muscles

May/June 2025 · Paper 1 · Variant 1 · Question 29

Correct answer: B

29 A nurse tests a sample of urine from a patient using Benedict's reagent. The sample tests positive.

What is a possible explanation for this result?

A The patient's liver is **not** deaminating protein efficiently.

B The patient's pancreas is **not** producing sufficient insulin.

C The patient has eaten too much glucose.

D The patient's concentration of plasma proteins is too high.

May/June 2025 · Paper 1 · Variant 1 · Question 30

Correct answer: C

30 Why do plant shoots grow towards light?

A Auxin inhibits cell elongation on one side.

B They grow more rapidly on the lighter side.

C More auxin accumulates on the darker side.

D Auxin stimulates cell division.

May/June 2025 · Paper 1 · Variant 1 · Question 31

Correct answer: B

31 Which statement about nuclear division is correct?

A Meiosis results in the production of cells for the growth and repair of organisms.

B Meiosis results in genetically different daughter cells.

C Mitosis gives rise to cells where the chromosome number is halved.

D Mitosis results in the production of four daughter cells.

May/June 2025 · Paper 1 · Variant 1 · Question 32

Correct answer: B

32 Stem cells are found in human embryos.

Which statements about stem cells are correct?

1 Stem cells are unspecialised cells.

2 Stem cells can divide by mitosis.

3 Stem cells are haploid.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

May/June 2025 · Paper 1 · Variant 1 · Question 33

Correct answer: B

33 In the placenta, which substance passes from the blood of the embryo to the blood of the mother down a concentration gradient?

A amino acids

B carbon dioxide

C glucose

D oxygen

May/June 2025 · Paper 1 · Variant 1 · Question 36

Correct answer: D

36 Which row identifies the factors causing continuous and discontinuous variation?

	continuous variation	discontinuous variation
A	both the environment and the genes	both the environment and the genes
B	the environment only	both the environment and the genes
C	the genes only	the environment only
D	both the environment and the genes	the genes only

May/June 2025 · Paper 1 · Variant 1 · Question 37

Correct answer: D

37 Which row shows what happens in the eye when focusing on a nearby object?

	ciliary muscle	shape of lens
A	relaxes	becomes thinner
B	relaxes	becomes rounder
C	contracts	becomes thinner
D	contracts	becomes rounder

May/June 2025 · Paper 1 · Variant 2 · Question 1

Correct answer: D

1 A student observes a human egg cell through a light microscope.

The cell is 0.12 mm in diameter.

The student draws a diagram of this egg cell.

The diagram has a diameter of 4.2 cm.

What is the magnification of the drawing to the nearest whole number?

A ×5 **B** ×35 **C** ×50 **D** ×350

May/June 2025 · Paper 1 · Variant 2 · Question 2

Correct answer: D

2 Which row correctly identifies a cell, a tissue, an organ and an organ system?

	cell	tissue	organ	organ system
A	chloroplast	mesophyll	liver	digestive
B	sap vacuole	red blood cell	blood	nervous
C	red blood cell	sap vacuole	stomach	liver
D	neurone	muscle	leaf	urinary

May/June 2025 · Paper 1 · Variant 2 · Question 4

Correct answer: C

4 Which combination of factors will result in the fastest rate of diffusion across a membrane?

	surface area of the membrane	temperature
A	small	low
B	small	high
C	large	high
D	large	low

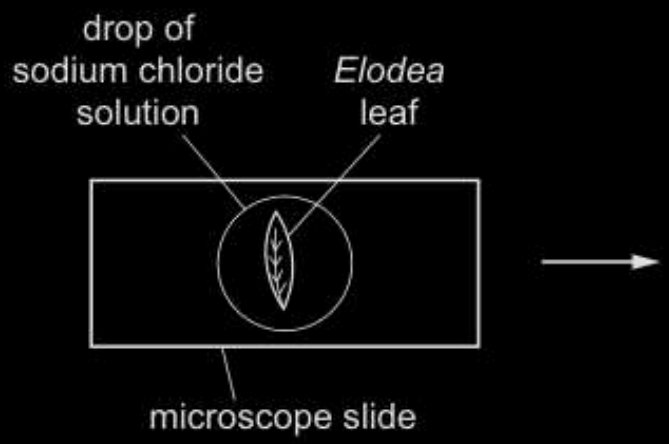
May/June 2025 · Paper 1 · Variant 2 · Question 5

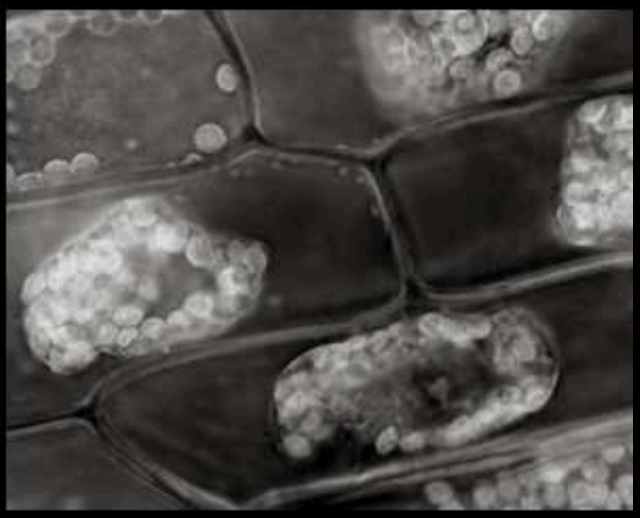
Correct answer: B

5 A leaf is taken from some pondweed (*Elodea*) and placed on a microscope slide.

A drop of sodium chloride solution is added.

The cells of the leaf are then viewed through a light microscope.





view through the microscope

Which statement describes the cells viewed through the microscope?

A The cells are plasmolysed because water has moved in by osmosis.

B The cells are plasmolysed because water has moved out by osmosis.

C The cells are turgid because water has moved in by osmosis.

D The cells are turgid because water has moved out by osmosis.

May/June 2025 · Paper 1 · Variant 2 · Question 6

Correct answer: D

6 What is an example of a biological catalyst?

A bile

B DNA

C insulin

D maltase

May/June 2025 · Paper 1 · Variant 2 · Question 7

Correct answer: B

7 What will be the result if the active site of an enzyme is damaged?

A A different product is produced from the substrate.

B The enzyme **cannot** carry out its specific reaction.

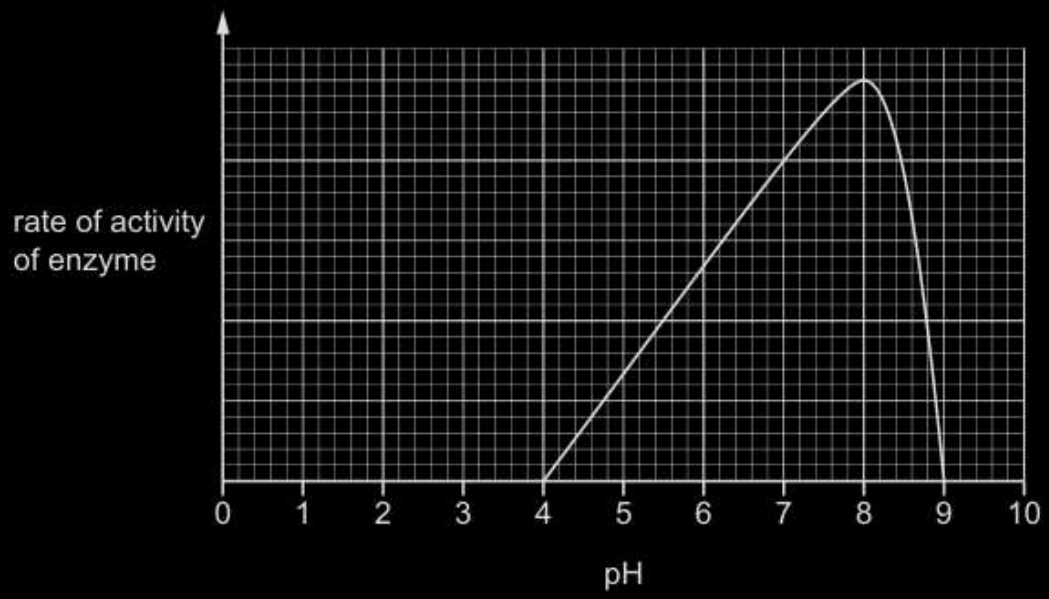
C The enzyme can no longer collide with a substrate molecule.

D Enzyme–substrate complexes will form faster.

May/June 2025 · Paper 1 · Variant 2 · Question 8

Correct answer: D

8 The graph shows the rate of activity of enzyme X at different pH values. All other conditions are kept constant.



Using the data shown in the graph, which statement is correct?

A Enzyme X is denatured in weak acid and alkali conditions.

B The kinetic energy of the substrate increases between pH 6 and 8.

C The kinetic energy of enzyme X is lowest at pH 4.

D The frequency of effective collisions between the enzyme and substrate is greatest at pH 8.

May/June 2025 · Paper 1 · Variant 2 · Question 11

Correct answer: B

11 Glucose produced by photosynthesis can be converted into other carbohydrates or used immediately as glucose.

What is an immediate use of glucose in plants?

A building cell walls

B providing energy in respiration

C storing energy

D transporting energy in the phloem

May/June 2025 · Paper 1 · Variant 2 · Question 12

Correct answer: D

12 Which enzyme catalyses the breakdown of its substrate at an optimum rate in a low pH?

A amylase

B lipase

C maltose

D pepsin

May/June 2025 · Paper 1 · Variant 2 · Question 14

Correct answer: A

14 Which substances are involved in the emulsification and digestion of vegetable oil in the digestive system?

	emulsification of vegetable oil	digestion of vegetable oil
A	bile	lipase
B	hydrochloric acid	lipase
C	bile	protease
D	hydrochloric acid	protease

May/June 2025 · Paper 1 · Variant 2 · Question 15

Correct answer: D

15 What is **not** an example of the assimilation of absorbed food molecules?

A the use of amino acids to produce antibodies

B the use of fatty acids to produce lipids

C the use of glucose to produce glycogen

D the use of proteins to produce amino acids

May/June 2025 · Paper 1 · Variant 2 · Question 18

Correct answer: B

18 Mammals have a double circulatory system.

What is meant by a double circulatory system?

A The blood is always enclosed in vessels or in the heart chambers.

B The blood passes through the heart twice for each complete circuit of the body.

C The circulatory system consists of two types of blood vessel.

D The heart consists of two chambers.

May/June 2025 · Paper 1 · Variant 2 · Question 19

Correct answer: C

19 Pulmonary stenosis is a condition caused by the narrowing of the entrance to the blood vessel leaving the right ventricle.

Where in the body will blood flow be reduced first as a result of pulmonary stenosis?

A reduced blood flow to the brain

B reduced blood flow to the left ventricle

C reduced blood flow to the lungs

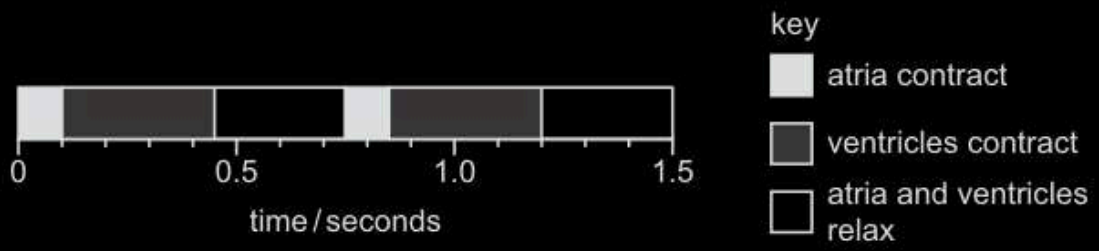
D reduced blood flow to the right atrium

May/June 2025 · Paper 1 · Variant 2 · Question 20

Correct answer: B

20 In one beat of the heart, the muscles in the atria and ventricles contract and relax, pushing blood out of the heart and into the circulatory system.

The diagram shows how long it takes for a heart to beat twice.



What is the heartbeat rate?

A 60 beats per minute

B 80 beats per minute

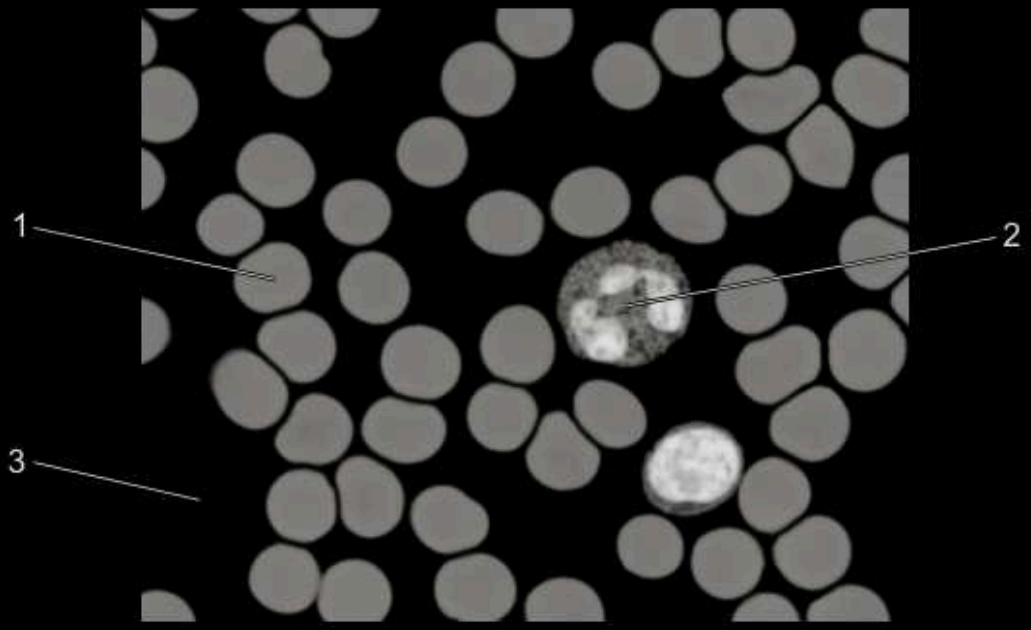
C 100 beats per minute

D 120 beats per minute

May/June 2025 · Paper 1 · Variant 2 · Question 21

Correct answer: A

21 The photomicrograph shows a sample of blood viewed through a light microscope.



Which row shows the functions of the labelled parts of the blood?

	oxygen transport	hormone transport	engulfing pathogens
A	1	3	2
B	2	3	1
C	1	2	3
D	3	2	1

May/June 2025 · Paper 1 · Variant 2 · Question 23

Correct answer: B

23 A disease is caused by a bacterium that infects the small intestine, producing a toxin.

The toxin causes osmotic movement of water from the cells into the small intestine, producing diarrhoea and extreme dehydration.

Which disease is being described?

A AIDS

B cholera

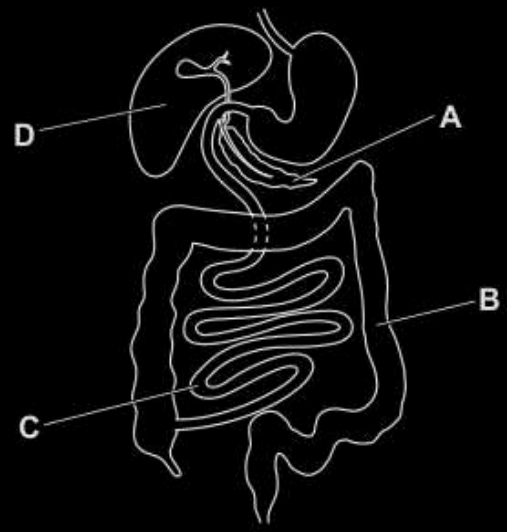
C malaria

D sickle cell anaemia

May/June 2025 · Paper 1 · Variant 2 · Question 25

Correct answer: D

25 Which letter shows where urea is produced?



May/June 2025 · Paper 1 · Variant 2 · Question 26

Correct answer: A

26 Which substance would **not** be present in the urine of a healthy person with a normal diet?

A glucose

B salts

C urea

D water

May/June 2025 · Paper 1 · Variant 2 · Question 27

Correct answer: D

27 Which part of the eye contains the greatest density of light receptors?

A cornea

B optic nerve

C iris

D fovea

May/June 2025 · Paper 1 · Variant 2 · Question 28

Correct answer: B

28 The diagrams show the front of an eye as seen in cross-section from above.

Which diagram shows the eye observing an object in the distance in dim light?

May/June 2025 · Paper 1 · Variant 2 · Question 29

Correct answer: A

29 Different studies have shown that the movement of electrical nerve impulses travelling along neurones slows down with age. Studies **A**, **B**, **C** and **D** have published different results.

The average speed of a nerve impulse in people who are 40 years old is 55 m / s.

Which study showed the **least** reduction in transmission speed between the ages of 40 and 60?

A The speed reduced by an average of 0.14 m / s per year.

B The speed reduced by 8.2 m / s.

C The speed reduced by 10%.

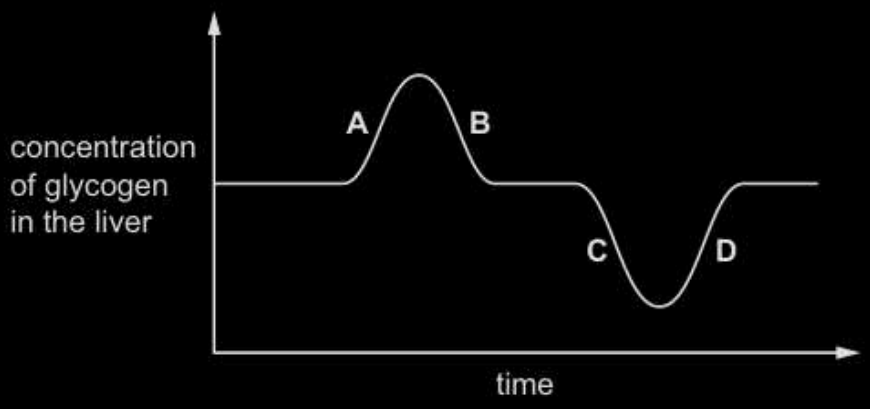
D The speed reduced by 10 m / s.

May/June 2025 · Paper 1 · Variant 2 · Question 30

Correct answer: B

30 The graph shows the concentration of glycogen stored in the liver of a human.

During which period of time is the secretion of adrenaline causing the blood glucose concentration to move towards its set point?



May/June 2025 · Paper 1 · Variant 2 · Question 32

Correct answer: B

32 A muscle cell from a horse contains 64 chromosomes.

How many chromosomes are there in a sperm cell from a horse?

A 23 **B** 32 **C** 64 **D** 128

May/June 2025 · Paper 1 · Variant 2 · Question 34

Correct answer: B

34 Sperm cells are stored in a laboratory before they can be used in artificial insemination.

Substances added to the sperm cells for storage have been shown to damage the acrosomes. This leads to reduction in fertility.

Which statement describes why reduction in fertility may happen?

A The sperm cells are rejected by the egg cells.

B The sperm cells **cannot** penetrate the egg cells.

C The sperm cells **cannot** swim to the egg cells.

D The sperm cells **cannot** detect the egg cells.

May/June 2025 · Paper 1 · Variant 2 · Question 36

Correct answer: B

36 Sickle cell anaemia is caused by a recessive allele.

A couple have two children.

One of the children has sickle cell anaemia.

The parents and the other child do **not** have sickle cell anaemia.

The couple are expecting their third child.

What is the probability that this third child will be male with sickle cell anaemia?

A 0% **B** 12.5% **C** 25% **D** 50%

Correct answer: A

37 Each human gamete has one sex chromosome.

Which statement is correct?

A All egg cells have an X chromosome.

B All egg cells have a Y chromosome.

C All sperm cells have an X chromosome.

D All sperm cells have a Y chromosome.

1 **(a)** Large molecules in cells are made from smaller molecules.

Complete Table 1.1 to show which small molecules make up each large molecule.

Choose the small molecules from this list.

amino acids **fatty acids** **glucose** **glycerol** **nucleotides**

Table 1.1

large molecule	small molecules
protein	
cellulose	
lipid	

[3]

(b) DNA is found in the nucleus of each cell.

Put a tick (✓) next to any statements that correctly describe the structure of DNA.

A DNA molecule has four strands coiled together to form a helix.	
Each strand is made up of a chain of nucleotides.	
Each nucleotide contains a base, a sugar and an amino acid.	
The bases always pair up in the same way, A with T, and C with G.	
Bonds between sugar molecules hold the strands together.	

[2]

(c) Explain how the sequence of bases in a DNA molecule controls how a cell functions.

.....

.....

.....

.....

.....

..... [3]

[Total: 8]

1(a)	proteins = amino acids ; cellulose = glucose ; lipids = fatty acids **+** glycerol ;	**3**	
1(b)	The molecule has four strands coiled together to form a helix. ; Each strand is made up of a chain of nucleotides. ✓ Each nucleotide contains a base, a sugar and an amino acid. ; The bases always pair up in the same way, A with T, and C with G ✓ Bonds between sugar molecules hold the strands together ;	**2**	
1(c)	(DNA base sequence) codes for proteins ; by determining the order of amino acids ; (the order of amino acids) determines the structure/function of the protein ; these proteins may act as enzymes ; **max 3**	**3**	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance

3 Fig. 3.1 shows the human digestive system.

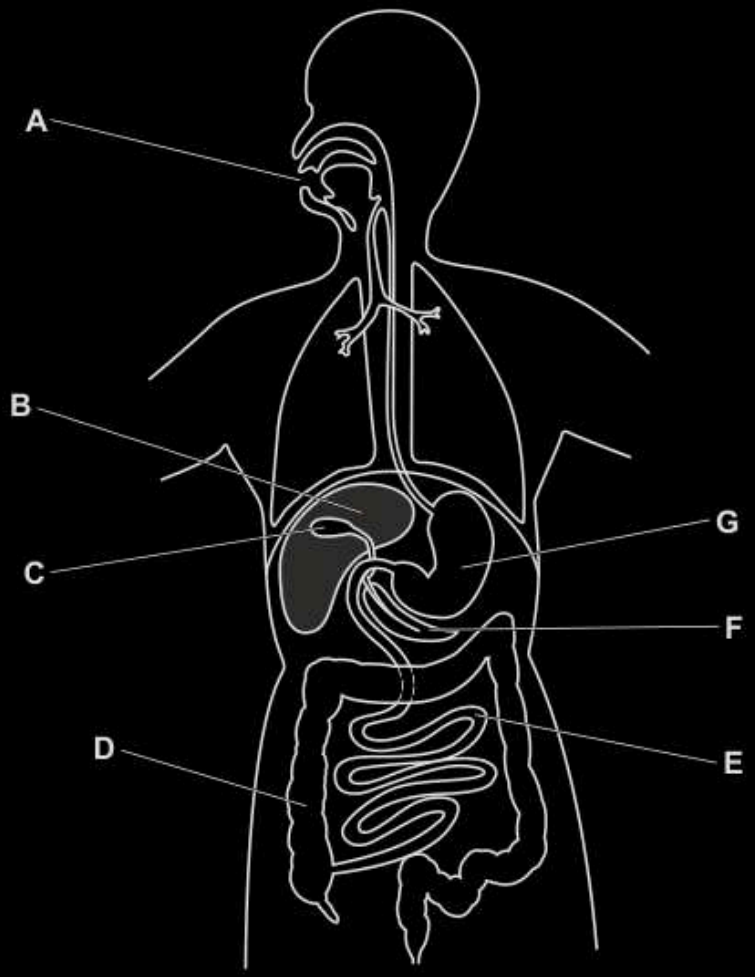


Fig 3.1

(a) (i) In each box write the letter that has been used to label these parts of the digestive system.

The part where starch digestion starts. ☐

The part where protein digestion starts. ☐

[2]

(ii) Describe the roles of organs **B** and **C** in lipid digestion.

.....

.....

.....

.....

..... [3]

(b) Coeliac disease affects several million people in the world. It is linked to a substance called gluten that is found in some foods.

Fig. 3.2 shows a section through organ **E** in a healthy person and in a person with coeliac disease.

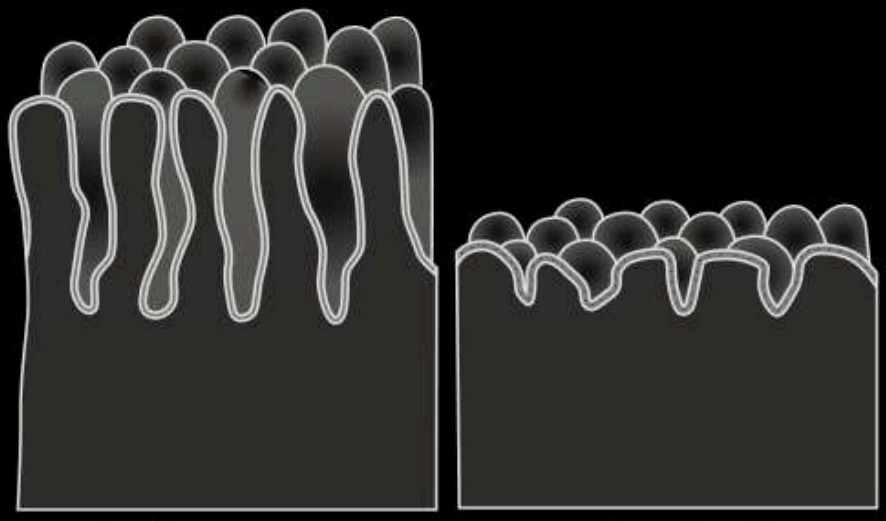


Fig. 3.2

(i) If a child with coeliac disease is **not** treated, their growth may be reduced.
Use Fig. 3.2 to explain why.

.....

.....

.....

.....

..... [3]

(ii) Scientists are investigating a new treatment for coeliac disease. They have produced an enzyme that will digest gluten when it reaches organ **E**. They plan to put the enzyme into a tablet which can be swallowed.

To make this treatment work, the tablet must have a coating that will dissolve in alkaline conditions but **not** in acid conditions.
Explain why.

.....

.....

.....

.....

..... [3]

[Total: 11]

3(a)(i)	the part where starch digestion starts = A ; the part where protein digestion starts = G ;	**2**	
3(a)(ii)	B / liver produces bile ; C / gall bladder stores bile ; bile emulsifies lipids (so speeds up lipid digestion) ;	**3**	
3(b)(i)	villi are smaller / shorter / have lower surface area ; less food materials / nutrients will be absorbed (into the bloodstream) ; less amino acids / proteins are obtained which are needed for growth ;	**3**	
3(b)(ii)	needs to resist the acid in the stomach ; otherwise the enzyme could be damaged / denatured in the stomach ; it will be released in E / the small intestine which is alkaline ;	**3**	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance

4 Transmissible diseases are caused by pathogens.

(a) Draw a line from each transmissible disease to the type of pathogen that causes the disease.

transmissible disease	type of pathogen
AIDS	bacterium
cholera	virus
malaria	protoctist

[2]

(b) (i) Complete these sentences about cholera by writing words in the gaps.

When the cholera pathogen is inside the small intestine it produces a

..... .

This results in the secretion of ..... ions into the small intestine.

The water potential inside the small intestine will ..... , causing water to move into the intestine.

[3]

(ii) In 2017 the World Health Organization estimated that 95 000 people die from cholera each year.

The World Health Organization set up a project to reduce this number of deaths by 95% by 2030.

If the project is successful, calculate the maximum number of cholera deaths in 2030.

Space for working.

maximum number of cholera deaths = ... [2]

(iii) The World Health Organization recommends vaccines to protect people from cholera.

Describe how vaccination can make people immune to a disease.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(iv) Table 4.1 shows details of a vaccine used by the World Health Organization to protect people from cholera.

Table 4.1

doses needed	how taken	storage temperature for vaccine
two doses, one to six weeks apart	vaccine added to a glass of water	2–8 °C

Discuss the practical problems of using this vaccine in different parts of the world.

.....

.....

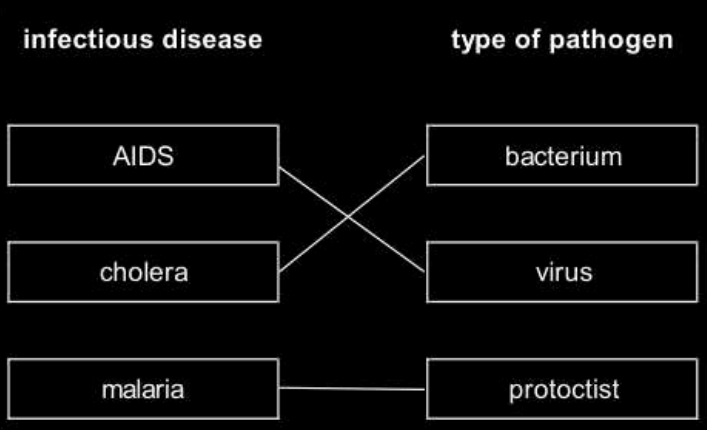
.....

.....

.....

..... [3]

[Total: 14]

4(a)		2	
4(b)(i)	toxin ; chloride ; decrease / reduce ;	3	
4(b)(ii)	4750 ;;	2	
4(b)(iii)	weakened / harmless form of the pathogens are given ; this stimulates production of antibodies by lymphocytes ; memory cells are produced ; idea that if the live pathogen is encountered, then antibodies are rapidly produced ;	4	
4(b)(iv)	may be hard to encourage / find people to give them the second dose ; idea that a safe water supply may not be available ; would need power for refrigeration / access to a refrigerator ;	3	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance

7 The inheritance of ABO blood groups in humans is an example of codominance.
The blood groups are controlled by three different alleles, I^A, I^B, and I^O.

(a) (i) State the blood group of a person who has the genotype $I^A I^O$.

..... [1]

(ii) State the genotype of a person who is blood group O.

..... [1]

(b) Fig. 7.1 is a family tree showing the ABO blood groups of eight different people, numbered (1) to (8).

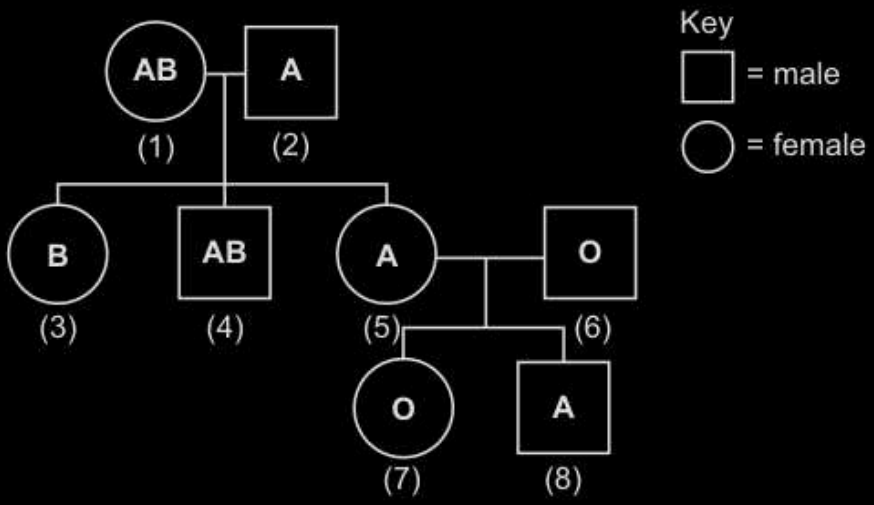


Fig. 7.1

(i) The parents (5) and (6) in Fig. 7.1 are expecting another child.

Complete this Punnett square to show the possible genotypes of the child.

parent 6

parent 5		
I^A		
I^O		

[2]

(ii) What is the probability that this child is male with blood group A?

probability = ... [1]

(iii) How many people in the family tree in Fig. 7.1 are homozygous for the ABO blood group alleles?

..... [1]

[Total: 6]

7(a)(i)	blood group A ;	1	
7(a)(ii)	$I^o I^o$;	1	
7(b)(i)	<table><tr><td></td><td>I^0</td><td>I^0</td></tr><tr><td>I^A</td><td>$I^A I^0$</td><td>$I^A I^0$</td></tr><tr><td>I^0</td><td>$I^0 I^0$</td><td>$I^0 I^0$</td></tr></table>	2	
7(b)(ii)	1 / 4 / 25% / 0.25 / one in four / 1:3 ;	1	
7(b)(iii)	two ;	1	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance

8 Pectinase is an enzyme that is produced by fungi.

In the large-scale industrial production of pectinase, fungi are grown in fermenters.

Fig. 8.1 shows a fermenter.

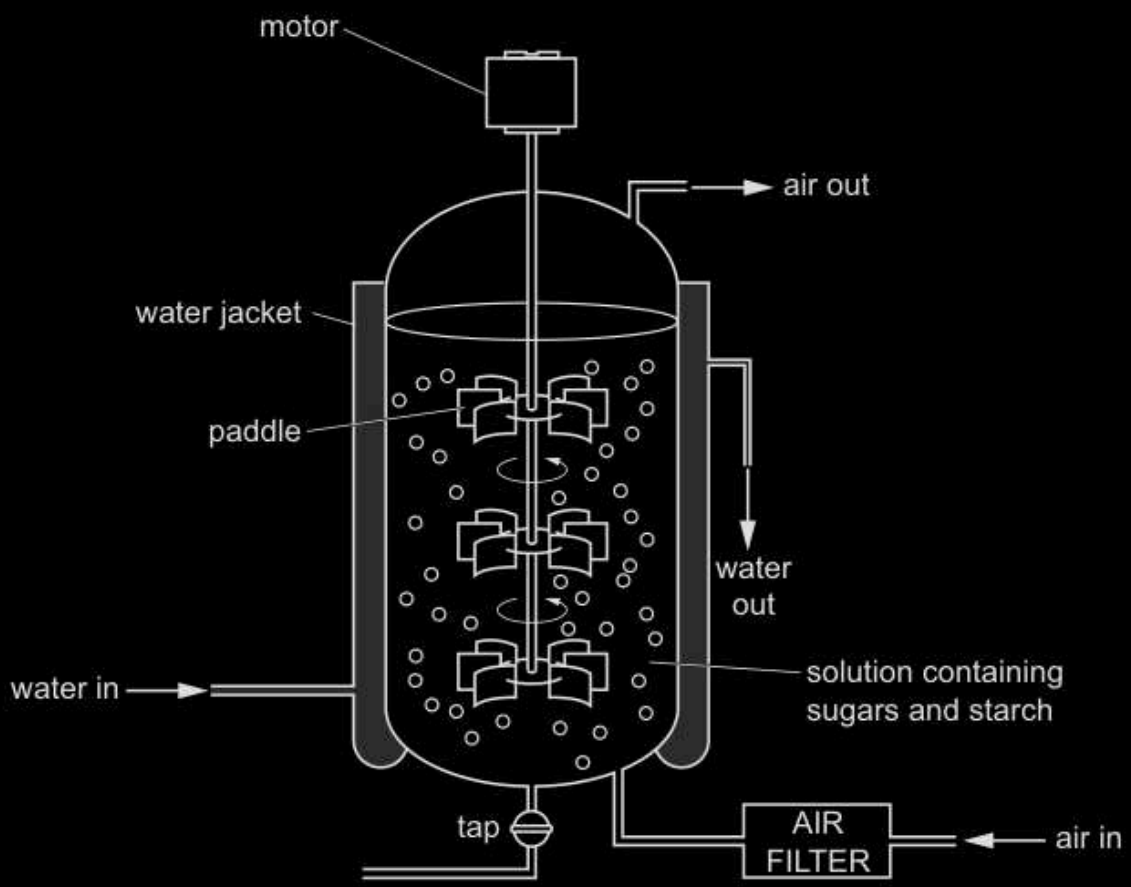


Fig. 8.1

(a) Explain how the features of the fermenter shown in Fig. 8.1 provide suitable conditions for the growth of fungi.

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) In an experiment, scientists added pectinase made by the fungi to a sample of crushed fruit.

They measured changes in:

- the sugar content of the sample of crushed fruit
- the thickness of the sample of crushed fruit.

The scientists' results are shown in Fig. 8.2.

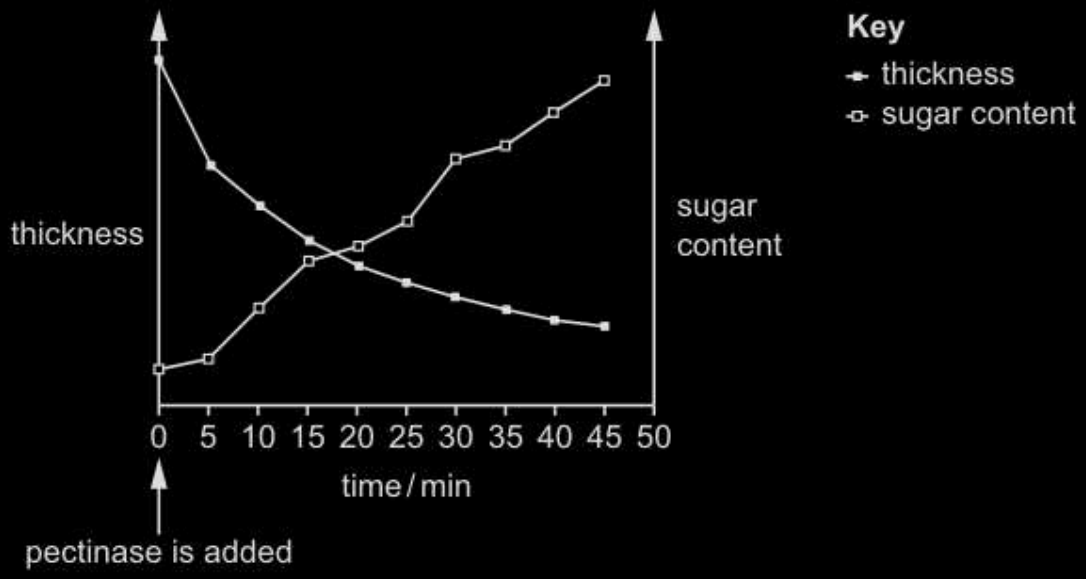


Fig. 8.2

Use your knowledge of the action of pectinase to explain the scientists' results.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

8(a)	*Motor / paddle:* mix the solution ; food supply / heat / waste products / air are evenly distributed ; *Water jacket / water in:* helps to cool the fermenter contents ; prevents high temperatures killing the fungus / denaturing enzyme ; *Air in:* provides oxygen ; allows the fungi to aerobically respire ; *Air filter:* prevents entry of other microorganisms ; prevents contamination of the solution / reduces competition for the fungi ; **max 6**	**6**	
8(b)	pectinase digests / breaks down pectin ; this releases sugars (from the pectin / from the cells) ; making the concentration / content of sugar increase ; pectin makes the solution thicker, so when digested the thickness decreases / sugars will dissolve and so decrease the thickness ;	**4**	

© Cambridge University Press & Assessment 2025

2 Scientists have been thinking about how to feed the human population of the planet in a healthy and sustainable way.

They have produced a model called the planetary health plate. This is shown in Fig. 2.1.

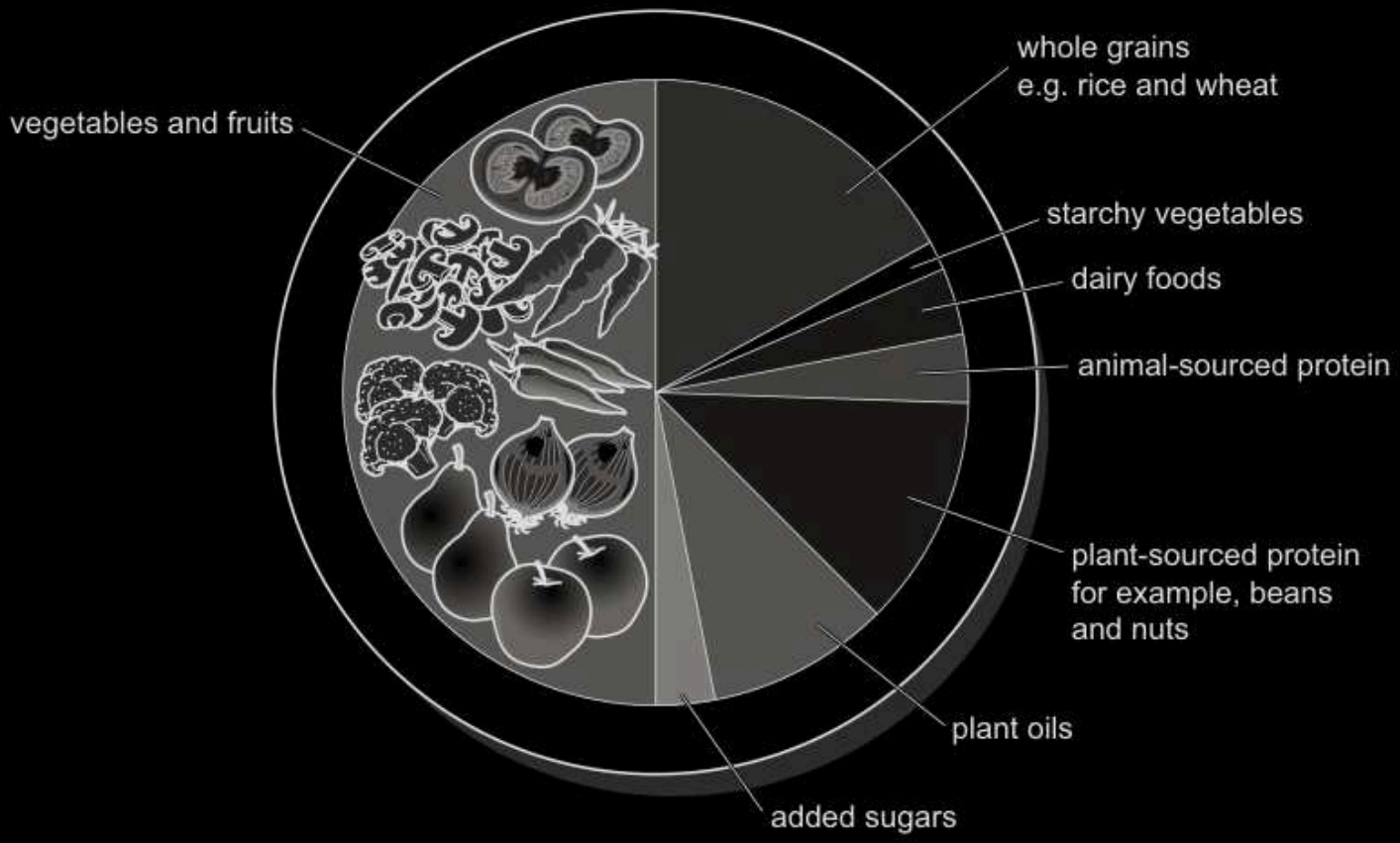


Fig. 2.1

(a) (i) The vegetables and fruits provide some of the protein, carbohydrates and lipids needed in a balanced diet.

Name **three** other components of a balanced diet that can be obtained from vegetables and fruits.

1 .....

2 .....

3 .....

[3]

(ii) State **two** reasons why plant oils and other lipids are important in the human diet.

1 .....

2 .....

[2]

(b) Table 2.1 shows the recommended daily average intake of sources of protein, from animals and plants, in the planetary health plate model.

Table 2.1

food	protein source	recommended average intake / g per day
red meat	animal	14
poultry	animal	29
eggs	animal	13
fish	animal	28
legumes	plant	75
nuts	plant	50

(i) Use Table 2.1 to calculate the percentage that plant protein sources contribute to the recommended average daily protein intake for humans.

Space for working.

..... % [2]

(ii) The global average daily human intake of **red meat** is approximately 300% of the recommended amount.

Calculate how much red meat an average human eats in a year (365 days).

Give your answer in kilograms.

average yearly intake = ... kg [2]

(c) It is more energy efficient for humans to eat plants.

Explain why.

.....

.....

.....

..... [2]

(d) If humans reduce the amount of animal protein they eat, then less agricultural land is required.

Suggest **three** environmental benefits of reducing the amount of land used for agriculture.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

[Total: 14]

2(a)(i)	minerals ; vitamins ; fibre ; water ; **max 3**	**3**	
2(a)(ii)	energy / energy store ; electrical insulation in neurones ; (thermal) insulation ; mechanical protection; components of cell membranes ; **max 2**	**2**	
2(b)(i)	60 / 59.8 / 59.81 ;;	**2**	
2(b)(ii)	15 / 15.3 / 15.33 ;;	**2**	
2(c)	fewer energy transfers involved when eating plants ORA ; energy is lost to environment / energy transfers not 100% efficient ORA ; example of loss ; **max 2**	**2**	
2(d)	less deforestation / more land/habitat available for wildlife ; more biodiversity / less monoculture ; less extinction ; reduced pollution + from insecticides / herbicides / fertilisers ; less erosion / loss of soil ; less flooding ; less methane production + reduced AW global warming / climate change ; **max 3**	**3**	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance

3 *Staphylococcus aureus* is a pathogen which infects humans. It can be destroyed by antibodies and antibiotics.

The cells of this bacterium are spherical and about 1 μm in diameter.

Fig. 3.1 shows a diagram of an *S. aureus* cell.

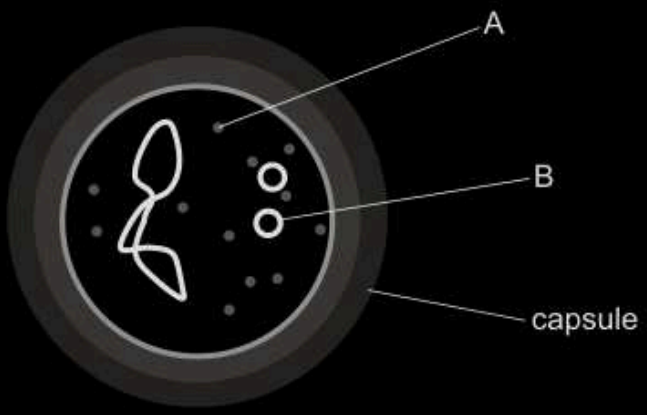


Fig. 3.1

(a) (i) Name parts A and B.

A .....

B .....

[2]

(ii) Part A can also be found in animal and plant cells.

State **two** other components of this bacterial cell that can also be found in animal **and** plant cells.

1 .....

2 .....

[2]

(iii) Antibodies can destroy bacterial cells without destroying human cells.

Explain how this is possible.

.....

.....

.....

.....

..... [3]

(b) There are different types or strains of *S. aureus* cells and some can resist antibiotics better than others.

A disc diffusion test is carried out to investigate the resistance of four strains of *S. aureus* (**K**, **L**, **M** and **N**) to different antibiotics. The four strains are first grown to cover the surface of separate agar plates. Then discs containing six different antibiotics (1–6) are placed on the agar plates.

Fig. 3.2 shows the results of the investigation after several days.

Any clear zones indicate where *S. aureus* has died.

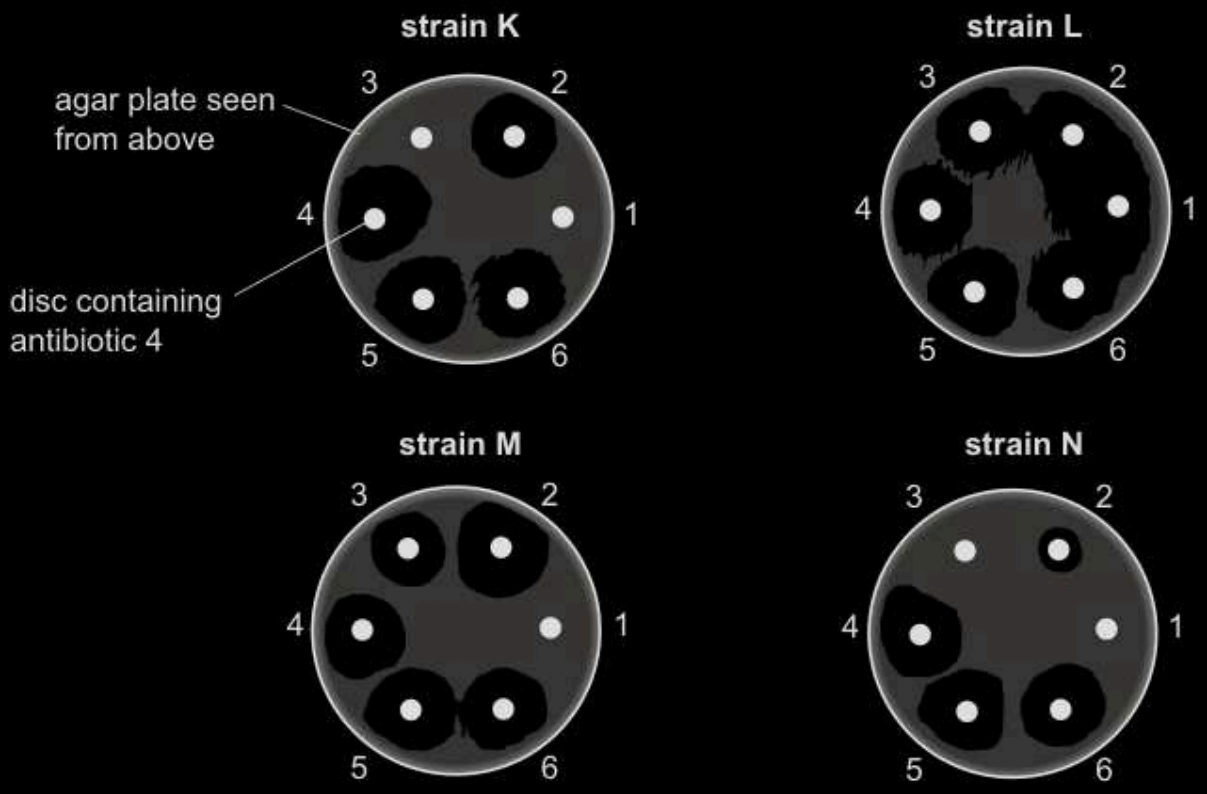


Fig. 3.2

(i) Suggest why this is called a disc diffusion test.

.....

..... [1]

(ii) Describe what these results show about the effectiveness of antibiotics 1 and 5.

antibiotic 1 .....

.....

antibiotic 5 .....

.....

[2]

(c) Over time the percentage of each strain in the *S. aureus* population will change.

(i) Suggest which strain, **K**, **L**, **M** or **N**, in Fig. 3.2 is likely to become the largest percentage of the population.

..... [1]

(ii) Explain why the strain named in **(c)(i)** is likely to become the largest percentage of the population over time.

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 16]

3(a)(i)	A: ribosome ;	1	
	B: plasmid ;	1	
3(a)(ii)	cytoplasm ; cell membrane ; DNA / RNA / genetic material / chromosome ; **max 2**	2	
3(a)(iii)	antigens / proteins ; on surface AW of bacteria ORA ; recognised as foreign ORA ; antibody specific to the antigen ; **max 3**	3	
3(b)(i)	antibiotic, moves / diffuses into the agar / bacteria ;	1	
3(b)(ii)	antibiotic 1: only effective against strain L;	1	
	antibiotic 5: is effective against all four strains ;	1	
3(c)(i)	N	1	
3(c)(ii)	1 variation in population ; 2 N has greatest resistance ; 3 best adapted (to environment) ; 4 they (are more likely to) survive ; 5 they (are more likely to) reproduce ; 6 beneficial alleles passed to next generation / offspring ; 7 natural selection ; **max 5**	5	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance

4 Fig. 4.1 shows a human bronchiole and some of the specialised cells from the inner layer that surrounds the lumen.

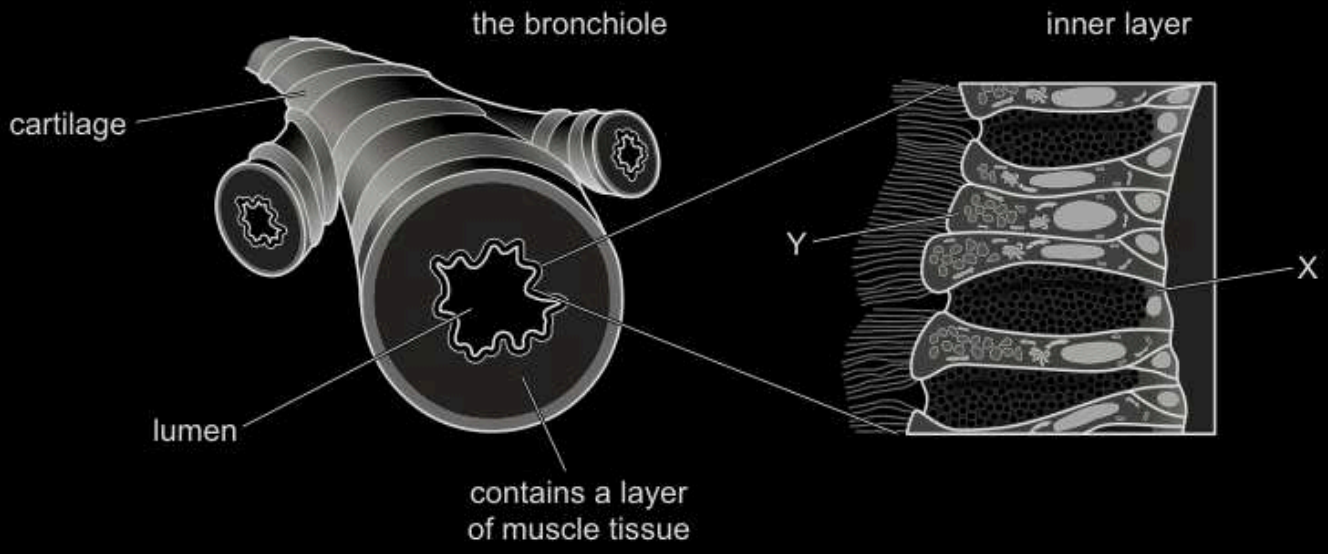


Fig. 4.1

(a) (i) Name the cell labelled X and describe its function.

X .....

function .....

.....

.....

[3]

(ii) Name the cell labelled Y and describe its function.

Y .....

function .....

.....

.....

[2]

(b) The muscle layer in the walls of the bronchi and bronchioles sometimes contracts causing the lumens to narrow. This is called bronchoconstriction. One substance that triggers contraction is nicotine.

(i) Suggest the effects of bronchoconstriction on a person.

.....

.....

.....

..... [3]

(ii) Describe **one** effect of nicotine on the circulatory system.

.....

..... [1]

[Total: 9]

4(a)(i)	(X) goblet ; produces / secretes mucus ; (mucus) traps bacteria / pathogens / dust / particles ;	**3**	
4(a)(ii)	(Y) ciliated ; moves mucus away from lungs AW ;	**2**	
4(b)(i)	1 lower volume of air / oxygen into lungs / alveoli ; 2 reduced oxygen into blood ; 3 decrease in oxygen reaching cells / tissues / muscles ; 4 aerobic respiration limited / less ; 5 (more) anaerobic respiration ; 6 reduced ability to exercise / do activities ; **max 3**	**3**	
4(b)(ii)	increased blood pressure ; artery constriction ; atherosclerosis ; increased clotting tendency ; increased heart rate ; **max 1**	**1**	

Question	Answer	Marks	Guidance

5 *Rhizomucor pusillus* is a fungus found in piles of dead plant vegetation. It has an optimum growth rate between 60 °C and 70 °C.

(a) (i) Identify the name of the kingdom and genus of this organism.

kingdom .....

genus .....

[1]

(ii) Give **two** of the main features of organisms classified in the kingdom named in **(a)(i)**.

1 .....

2 .....

[2]

(iii) Explain how piles of dead plant vegetation can have temperatures much higher than the surrounding environment.

.....

.....

.....

.....

.....

..... [4]

(b) This fungus produces an enzyme called pectinase. Scientists extracted pectinase from the fungus and investigated the effect of pH on its activity at 20 °C.

Fig. 5.1 shows a graph of their results.

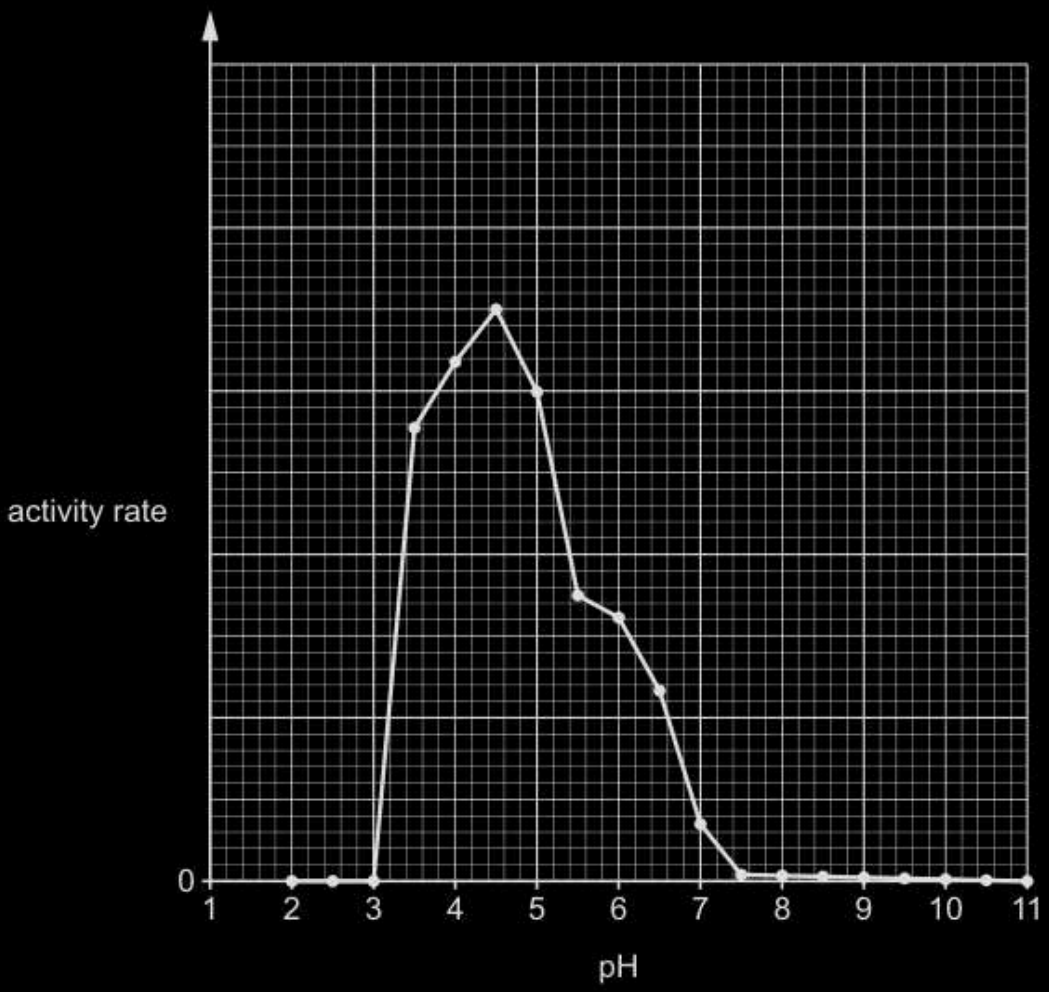


Fig. 5.1

(i) Explain the results for the pH range 2–7.

.....

.....

.....

.....

..... [4]

(ii) The experiment was repeated at a temperature of 55 °C.

On Fig. 5.1, draw a graph to suggest the results expected at 55 °C. [1]

(iii) Name an industrial process that uses pectinase.

..... [1]

[Total: 13]

5(a)(i)	fungi / fungus **+** *Rhizomucor*	**1**	
5(a)(ii)	cell wall made of chitin ; heterotrophic / saprotrophic / saprophytic ; filamentous / hyphae / mycelium ; eukaryotic; **max 2**	**2**	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance
5(a)(iii)	fungi/bacteria / *Rhizomucor pusillus* + decomposition / decomposers ; respiration ; heat / energy released ; vegetation layer insulating AW ;	**4**	
5(b)(i)	(pH) 2 / 3 + denatured ; active site of enzyme deformed / changed shape ; substrate no longer fits the active site ; optimum (pH) at 4.5 ; substrate and active site complementary AW ; ES complexes form ; starts to denature above 4.5 ; **max 4**	**4**	
5(b)(ii)	line with optimum activity at pH 4.5 + drawn above the original ;	**1**	
5(b)(iii)	fruit juice production ;	**1**	

Question	Answer	Marks	Guidance

6 The nucleus of a cell contains DNA molecules that control cell function.

Fig. 6.1 is a diagram showing one of these DNA molecules.

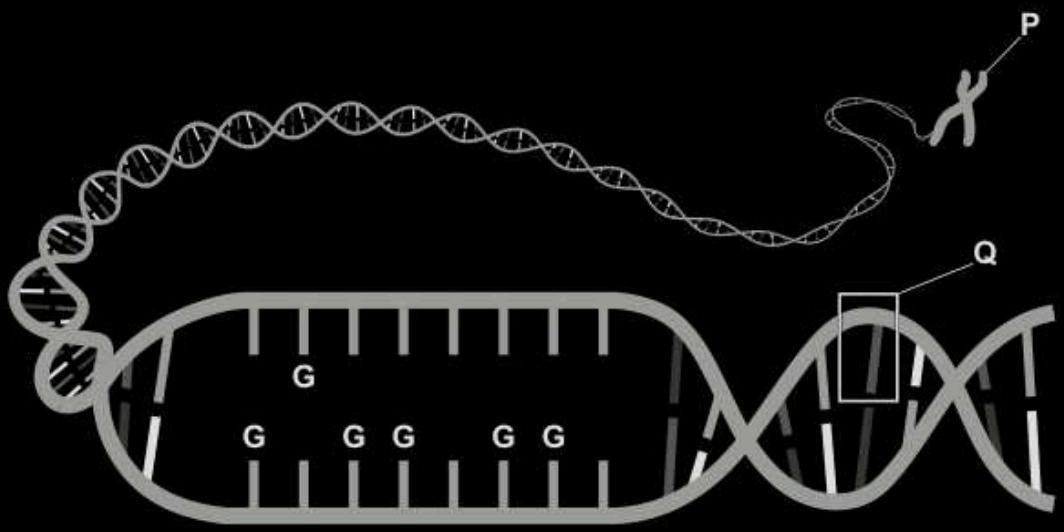


Fig. 6.1

(a) Using information from Fig. 6.1:

(i) state the name of the structure labelled **P**

..... [1]

(ii) state the name of the unit of DNA labelled **Q**

..... [1]

(iii) state the letter of the base that pairs with **G**.

..... [1]

(b) Explain how this DNA molecule controls cell function.

Include an example in your answer.

.....

.....

.....

.....

.....

..... [4]

6(a)(i)	(P) chromosome	1	
6(a)(ii)	(Q) nucleotide	1	
6(a)(iii)	C	1	
6(b)	(DNA) codes for proteins / polypeptides ; order of bases / base sequence ; determines sequence of amino acids; example protein;	4	

© Cambridge University Press & Assessment 2025

Question	Answer	Marks	Guidance

7 Photosynthesis and transpiration are two processes that take place in a plant leaf.

(a) Explain how the structure of a leaf is adapted for photosynthesis.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) Transpiration is both necessary for plants and a problem for plants.

Discuss this statement.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

7(a)	1 large surface area + maximum light absorption ; 2 stomata + gaseous exchange / carbon dioxide in / oxygen out ; 3 thin + diffusion of gases / carbon dioxide in / oxygen out ; 4 air spaces / spongy mesophyll + diffusion of gases / gaseous exchange ; 5 chloroplasts / chlorophyll + absorb light ; 6 chloroplasts / chorophyll + most dense in palisade (mesophyll) ; 7 palisade in upper leaf + receive most light ; 8 xylem (vessels) + bring water (from soil) ; 9 epidermal cells no chloroplasts / transparent + light reaches cells ; **max 6**	**6**	
7(b)	***necessary (max 2):*** 1 provides (leaves with) water for photosynthesis ; 2 transports minerals / ions from roots / to leaves ; 3 provides plant cells with support / turgidity / turgor ; 4 cools the leaf / plant ; ***problem (max 2):*** 5 too much water is lost (through stomata / leaf) ; 6 (water loss) leads to wilting / death ; 7 plants may close stomata to prevent transpiration, which slows down photosynthesis ; **max 4**	**4**	

© Cambridge University Press & Assessment 2025

October/November 2024 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 Which structures are present in plant cells but **not** in animal cells?

A cell membrane, cytoplasm, chloroplasts

B cellulose cell wall, chloroplasts, sap vacuole

C cellulose cell wall, cell membrane, cytoplasm

D cytoplasm, nucleus, chloroplasts

October/November 2024 · Paper 1 · Variant 1 · Question 3

Correct answer: D

3 The bacterium *Vibrio cholerae* that causes the disease cholera acts on the cells of the intestinal wall. It causes chloride ions to move out of the cells into the small intestine.

Which process will then lead to the production of dangerously high amounts of watery diarrhoea?

A active transport

B active uptake

C diffusion

D osmosis

October/November 2024 · Paper 1 · Variant 1 · Question 4

Correct answer: C

4 Which movement of a substance in a plant requires the plant to provide energy?

A absorption of carbon dioxide by a palisade cell

B absorption of oxygen by a mesophyll cell

C nitrate uptake by root hair cells

D transport of water up through the xylem

October/November 2024 · Paper 1 · Variant 1 · Question 5

Correct answer: B

5 Which statements are correct?

1 A molecule of DNA consists of many nucleotides.

2 A molecule of glycogen consists of many glucose molecules.

3 A molecule of lipid consists of glycerol and amino acid molecules.

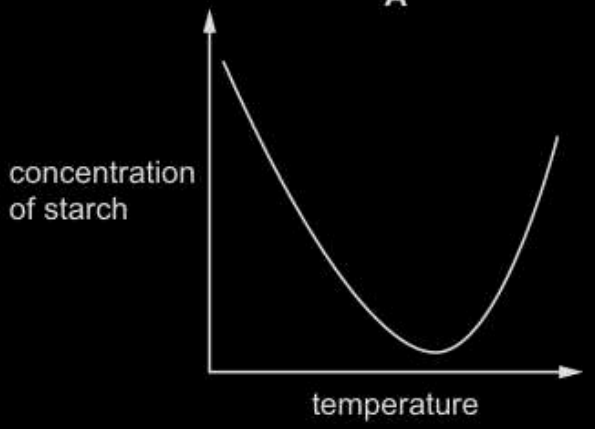
A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

Correct answer: A

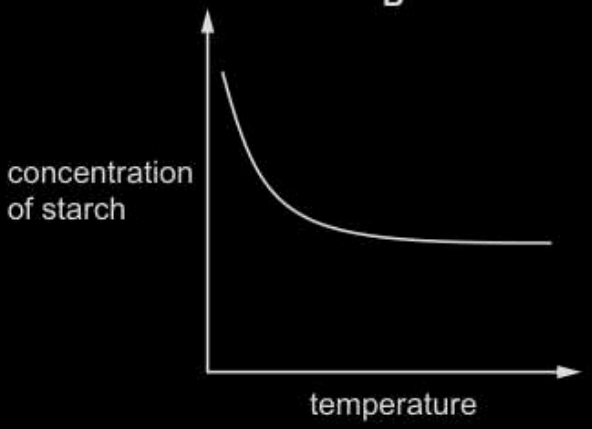
6 Some students investigated the effect of temperature on the digestion of starch to form maltose. They added equal volumes of amylase and starch solution to a number of test-tubes. The test-tubes were left for 30 minutes at a range of temperatures between 10 °C and 60 °C. They then measured the concentration of starch in each test-tube.

Which graph was produced from these results?

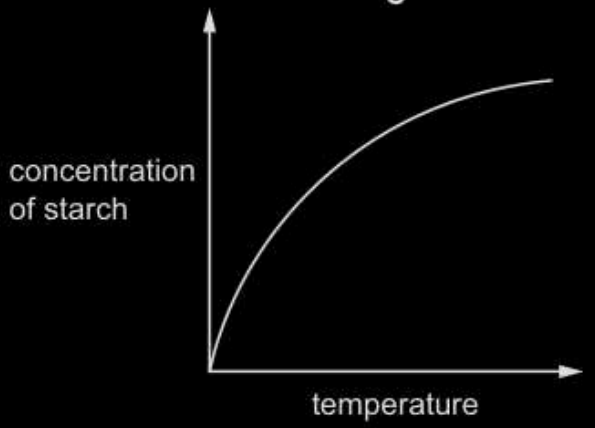
A



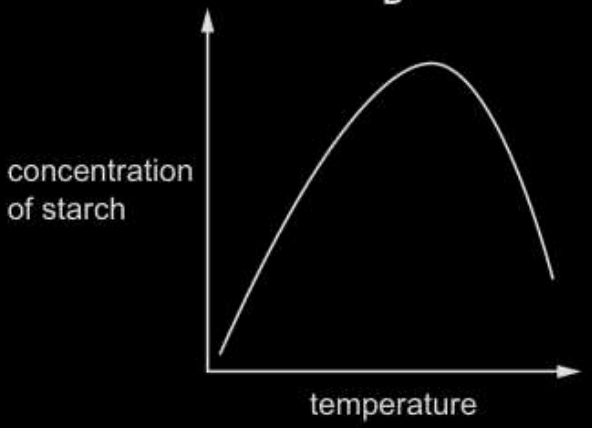
B



C



D

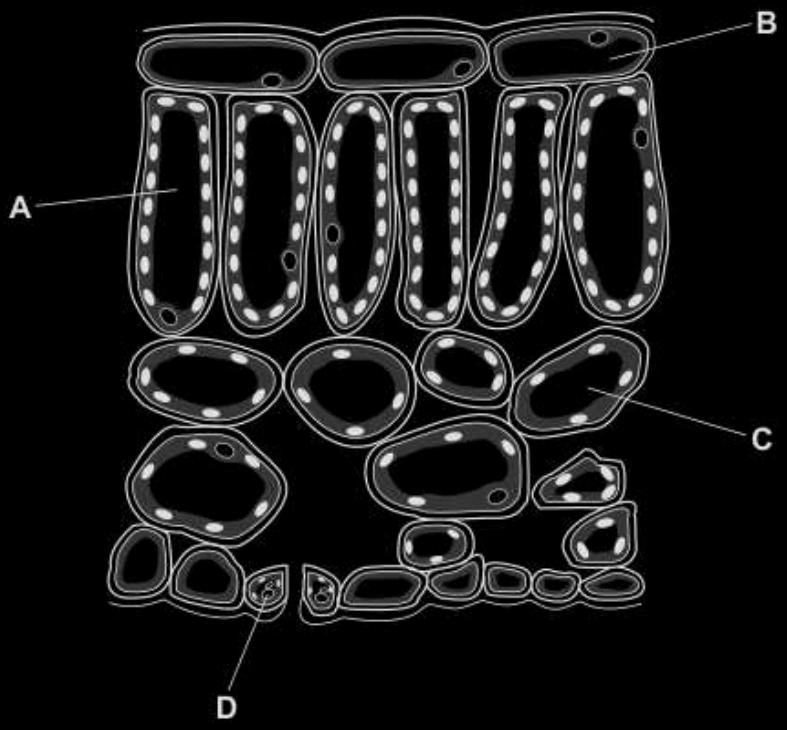


October/November 2024 · Paper 1 · Variant 1 · Question 8

Correct answer: B

8 The diagram shows a section through a dicotyledonous leaf.

Which cell **cannot** photosynthesise?



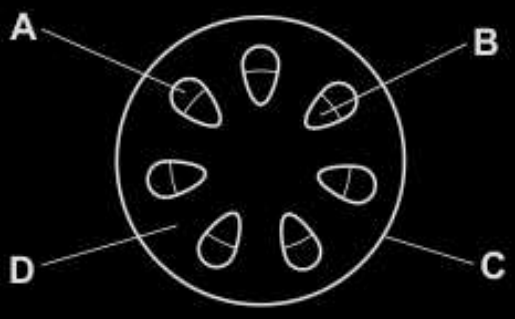
October/November 2024 · Paper 1 · Variant 1 · Question 9

Correct answer: B

9 A leafy stem was cut from a plant and placed in a red dye solution for 18 hours.

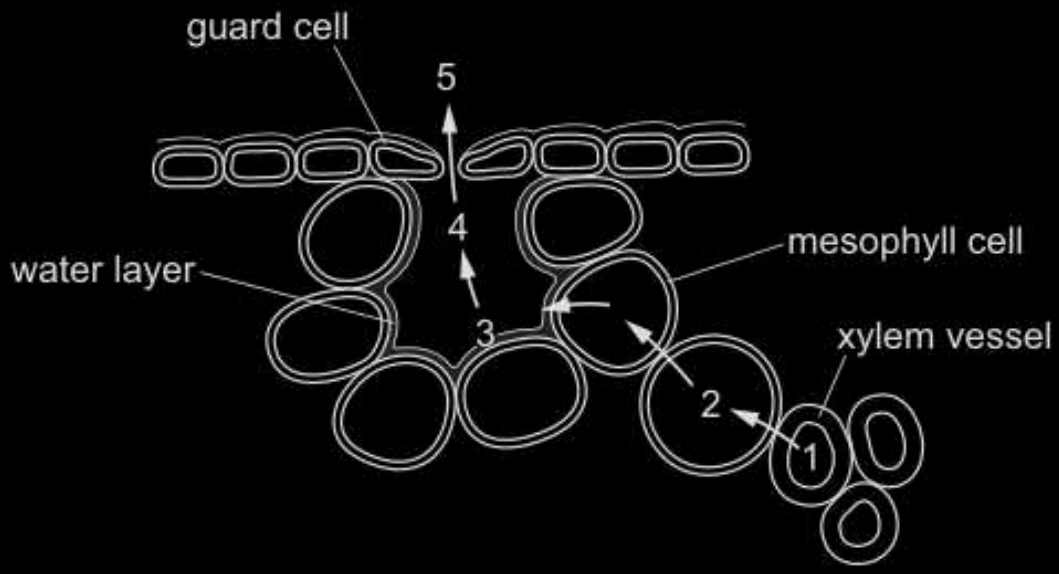
The diagram shows a transverse section of the stem after 18 hours.

Which tissue will be stained red?



Correct answer: D

10 The diagram shows the movement of water through part of a leaf.



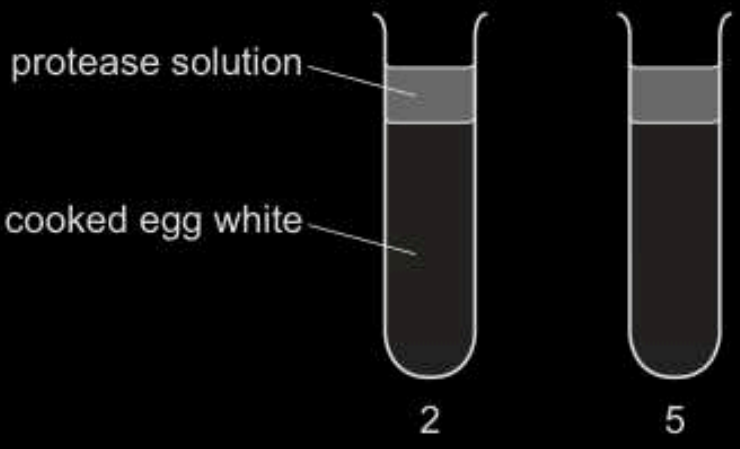
Which processes are involved in the movement of water at these stages?

	1–2	3–4	4–5
A	diffusion	evaporation	osmosis
B	evaporation	diffusion	osmosis
C	osmosis	diffusion	evaporation
D	osmosis	evaporation	diffusion

October/November 2024 · Paper 1 · Variant 1 · Question 12

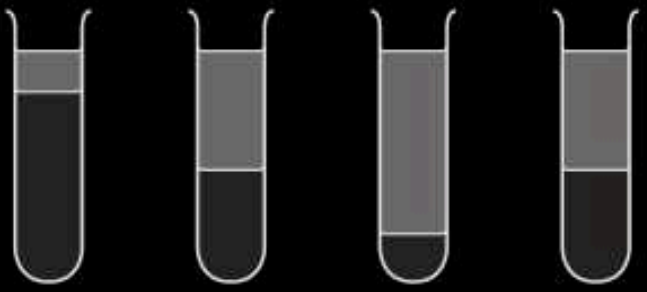
Correct answer: C

12 Five test-tubes containing cooked egg white a
protein. Protease solutions of different pH are a

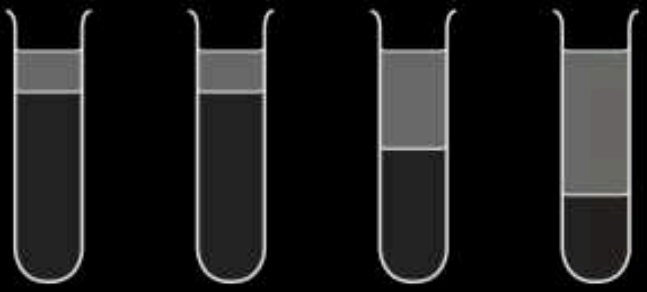


Which diagram shows the results of this experi

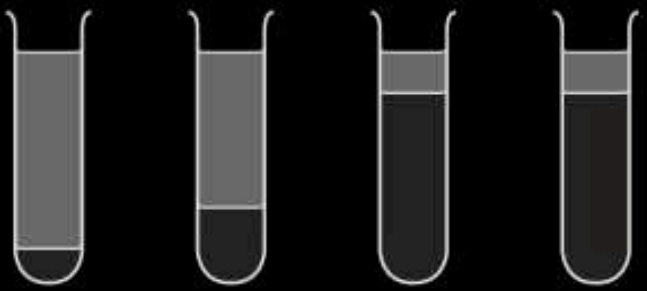
A



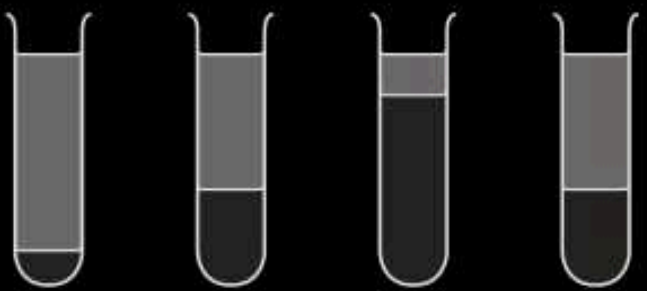
B



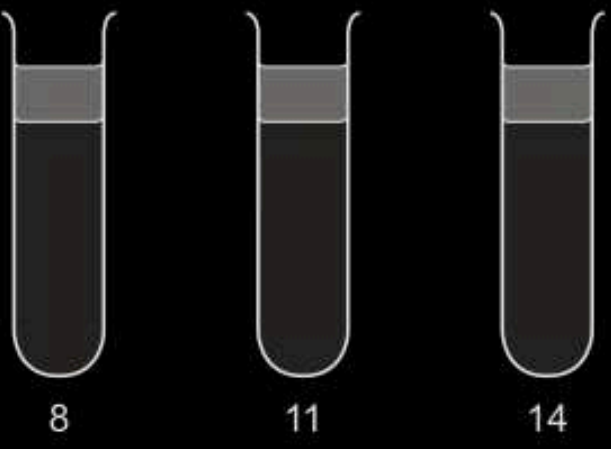
C



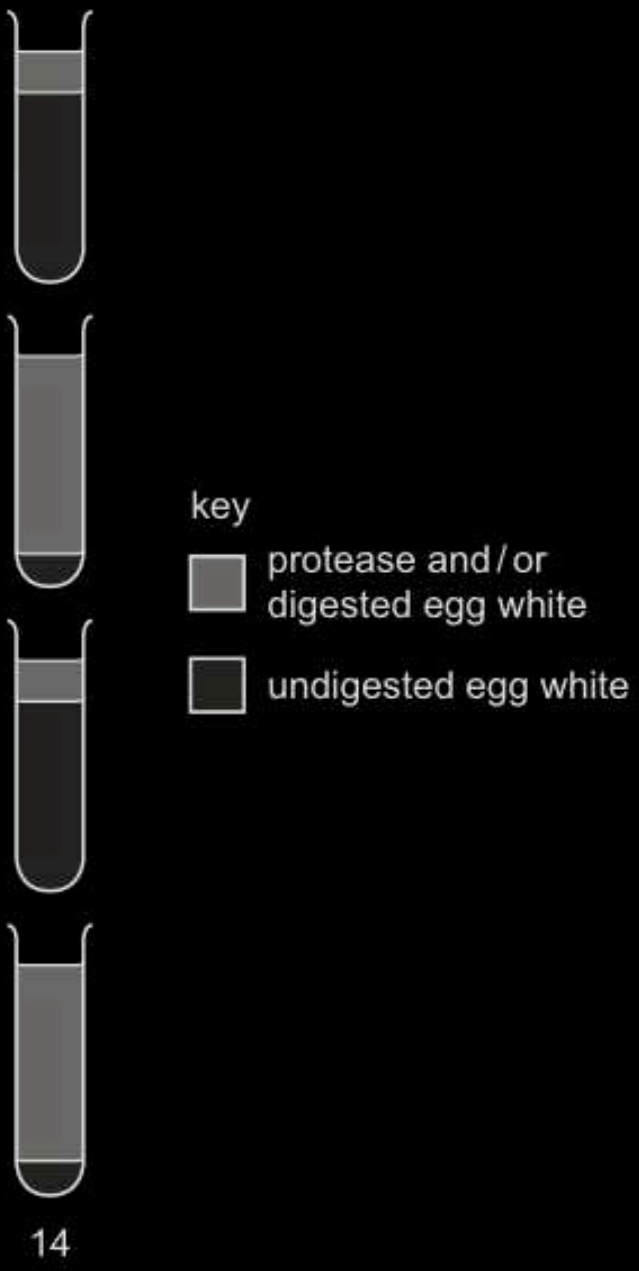
D



are set up as shown. Cooked egg white contains a
added to each tube.



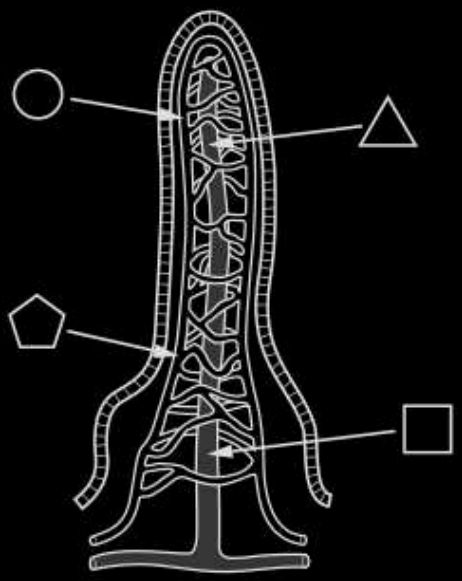
iment for a protease from the stomach?



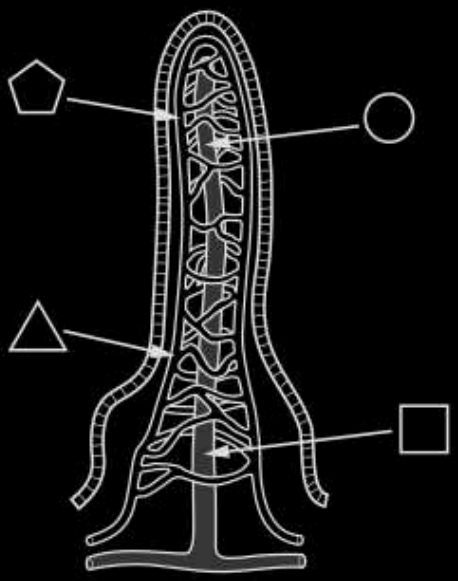
Correct answer: D

13 Which diagram shows the absorption of different food molecules into a villus?

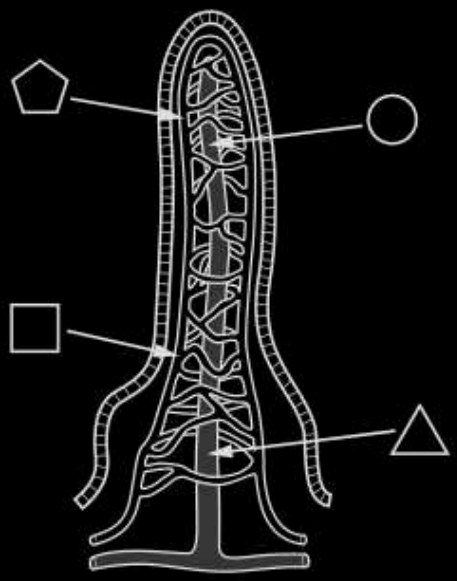
A



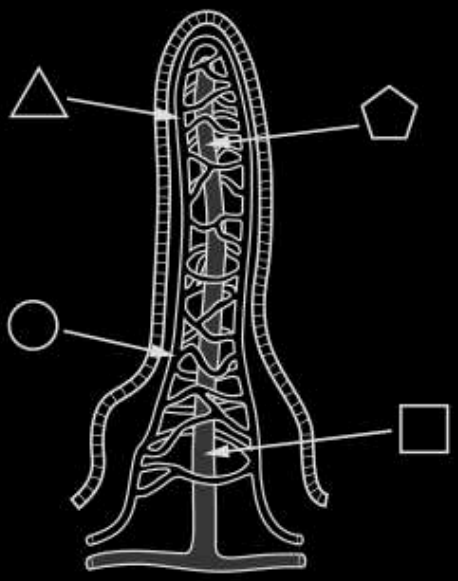
B



C



D



key

- amino acid
- fatty acid
- glucose
- glycerol

October/November 2024 · Paper 1 · Variant 1 · Question 15

Correct answer: D

15 Scientists have concluded that absorption of mineral ions in plants requires energy from respiration.

Which observation best supports this conclusion?

A Carbohydrate is stored in the roots.

B Living roots give off carbon dioxide.

C The root hairs have a large surface area.

D Uptake of mineral ions is reduced in lower oxygen concentrations.

October/November 2024 · Paper 1 · Variant 1 · Question 16

Correct answer: A

16 When people travel in an aircraft and sit still in their seats for long periods of time, their lower legs may swell up.

Which statement explains why this may happen?

A Fewer leg muscle contractions occur to push blood along the veins.

B The rate of the heartbeat is reduced as no exercise is happening.

C There is little or no arterial blood pressure to force the blood upwards.

D Valves in the veins do **not** fill up with blood to prevent its backflow.

October/November 2024 · Paper 1 · Variant 1 · Question 17

Correct answer: A

17 The diagram shows a blood cell as seen under a light microscope.

What is the type of blood cell shown?

A phagocyte

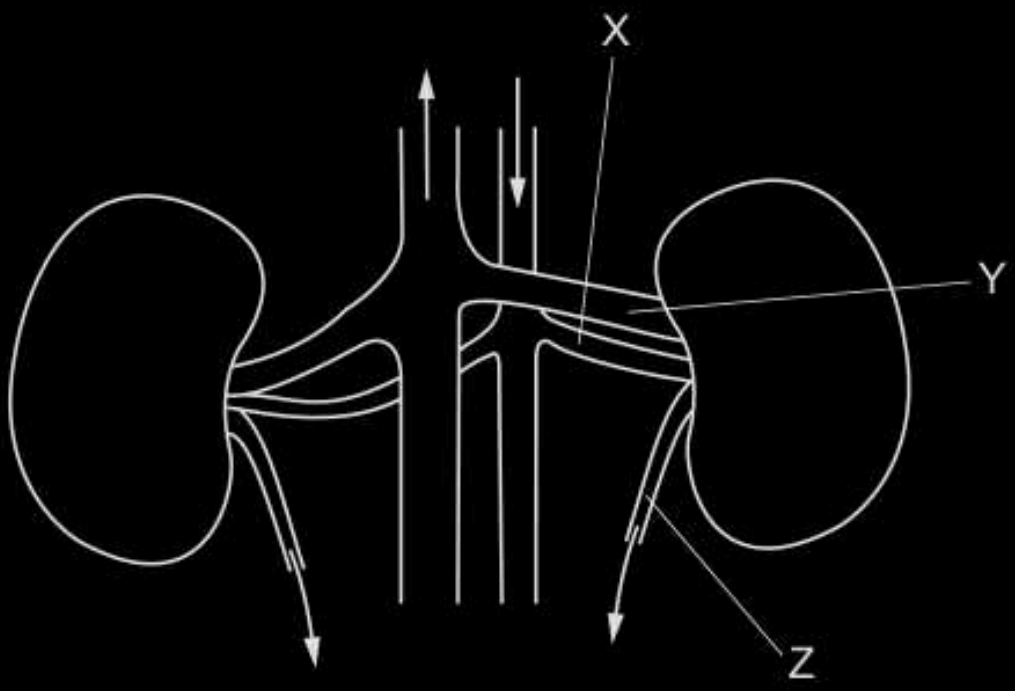
B lymphocyte

C red blood cell

D platelet

Correct answer: A

22 The diagram shows part of the human urinary system.



What are structures X, Y and Z?

	X	Y	Z
A	artery	vein	ureter
B	artery	vein	urethra
C	vein	artery	ureter
D	vein	artery	urethra

October/November 2024 · Paper 1 · Variant 1 · Question 23

Correct answer: D

23 Nerve impulses in neurones can travel:

1 away from the central nervous system

2 towards the central nervous system

3 within the central nervous system.

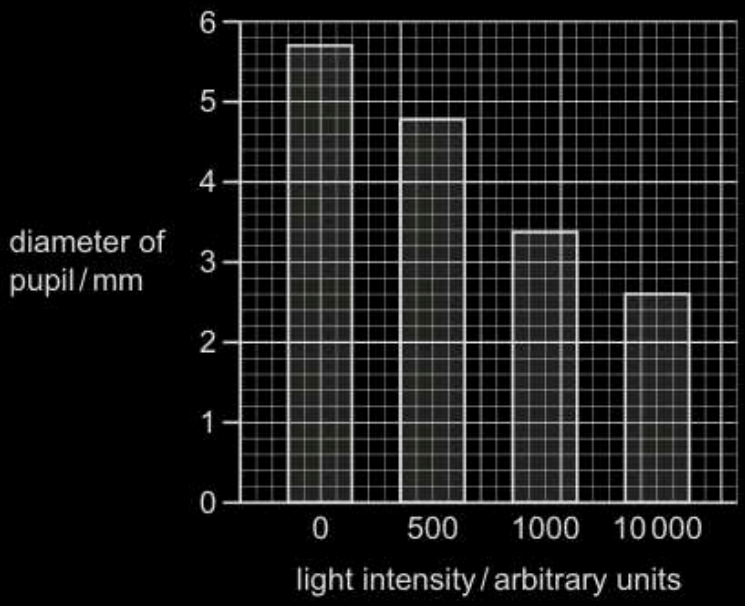
In which direction do impulses in sensory and in relay neurones travel?

	sensory neurones	relay neurones
A	1	2
B	1	3
C	2	1
D	2	3

October/November 2024 · Paper 1 · Variant 1 · Question 24

Correct answer: D

24 The diameter of the pupil of the eye was measured at a range of different light intensities.



What is the percentage decrease in pupil diameter (rounded to 1 decimal place) when a person moves from an area of complete darkness to an area of light intensity 10 000 arbitrary units?

A 45.6% **B** 53.4% **C** 54.3% **D** 54.4%

October/November 2024 · Paper 1 · Variant 1 · Question 25

Correct answer: A

25 When a person is frightened, which substance causes an increase in the blood sugar level?

A adrenaline

B carbon dioxide

C insulin

D lactic acid

October/November 2024 · Paper 1 · Variant 1 · Question 26

Correct answer: A

26 Raynaud's syndrome is a condition that can be triggered by stress. It causes the hands and feet to react as if they were affected by extreme cold.

Which reaction will take place in the hands and feet of a person with Raynaud's syndrome?

A vasoconstriction – blood vessels at the skin surface narrow, reducing blood flow

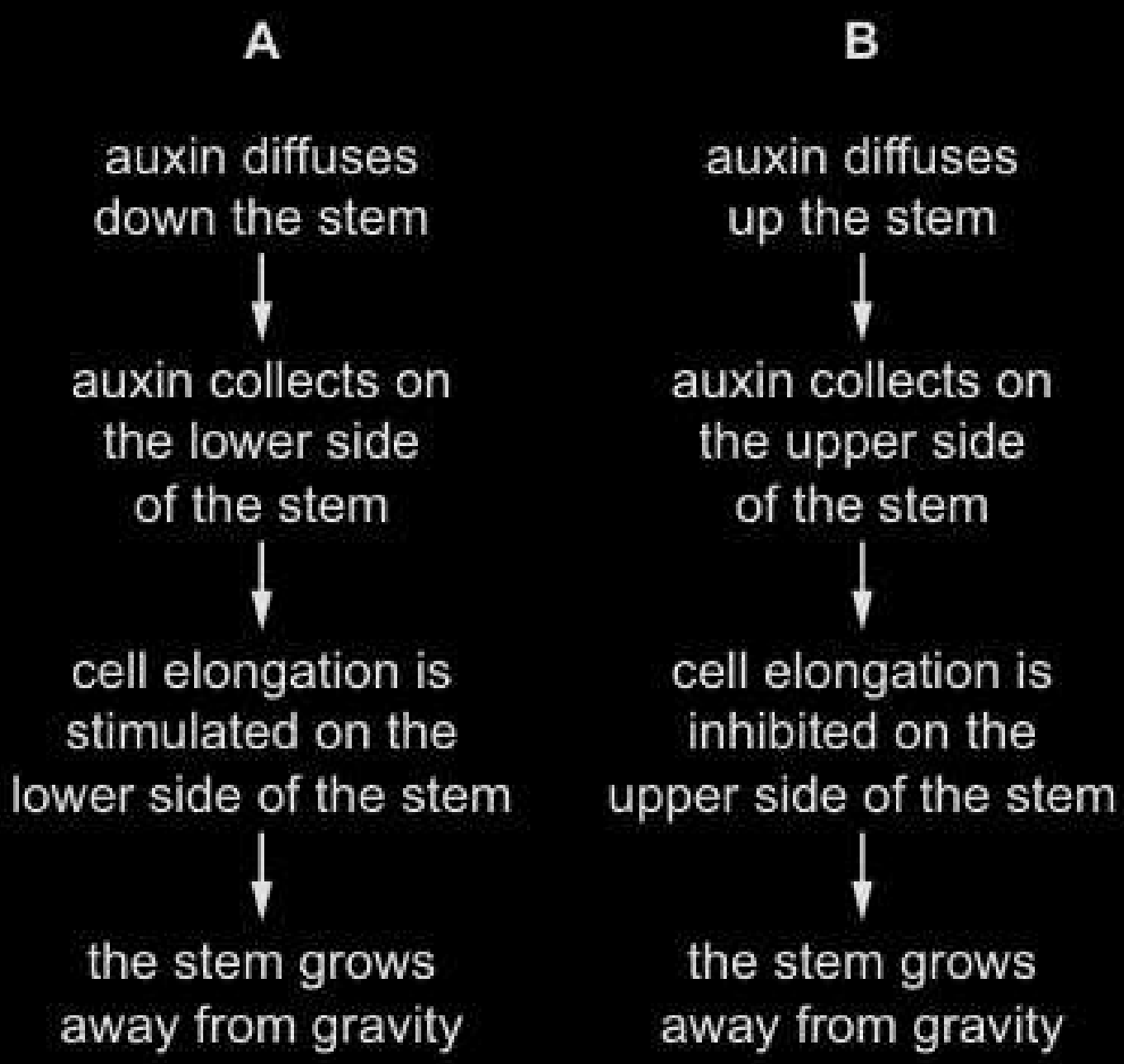
B vasoconstriction – blood vessels at the skin surface widen, increasing blood flow

C vasodilation – blood vessels at the skin surface narrow, reducing blood flow

D vasodilation – blood vessels at the skin surface widen, increasing blood flow

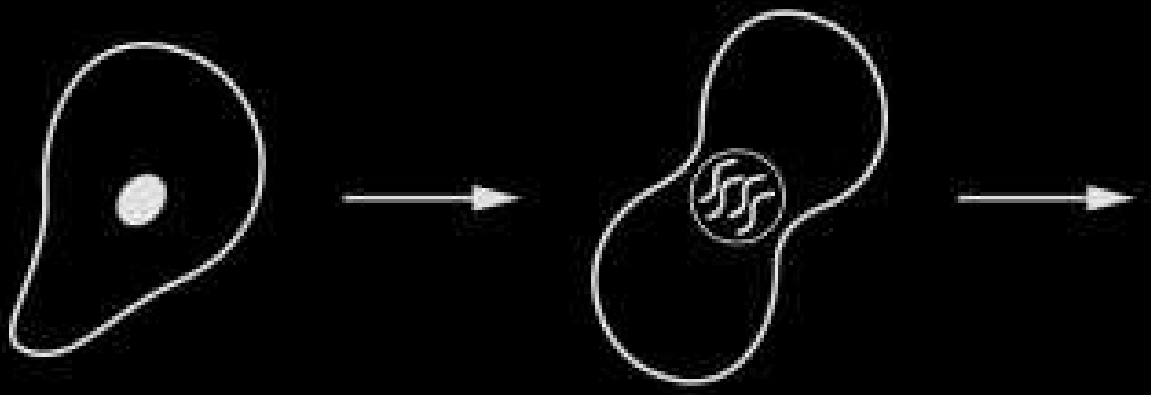
Correct answer: A

27 Which flow diagram describes how auxin affect



Correct answer: C

28 The diagram shows some stages during the as

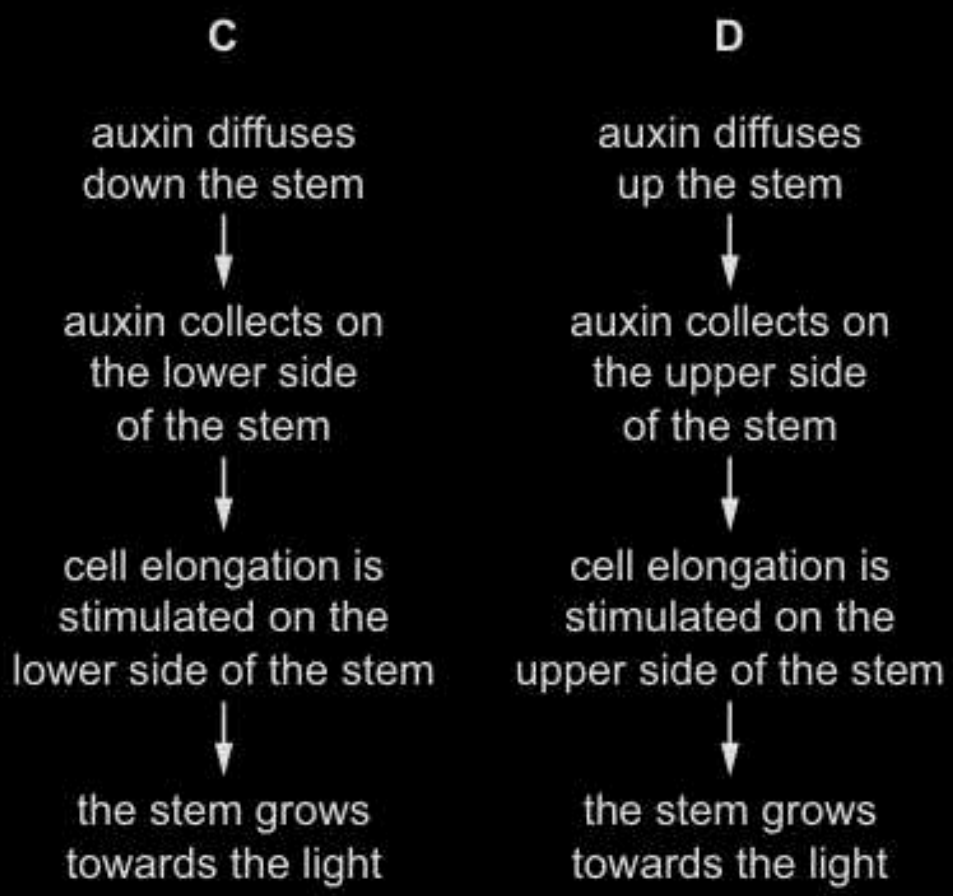


V W

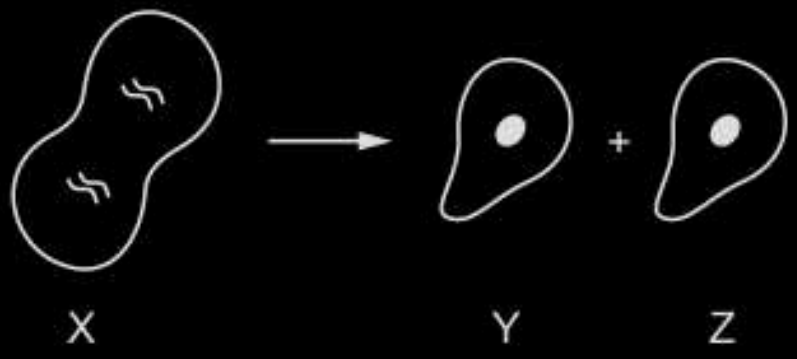
Which row shows the relative amounts of DNA

	V	W	X	Y	
A	1	1	1	2	
B	1	1	2	1	
C	1	2	2	1	
D	2	1	1	2	

:ts growth in negative gravitropism?



sexual reproduction of a single-celled organism.



\ in each of the cells V, W, X, Y and Z?

Z
2
1
1
2

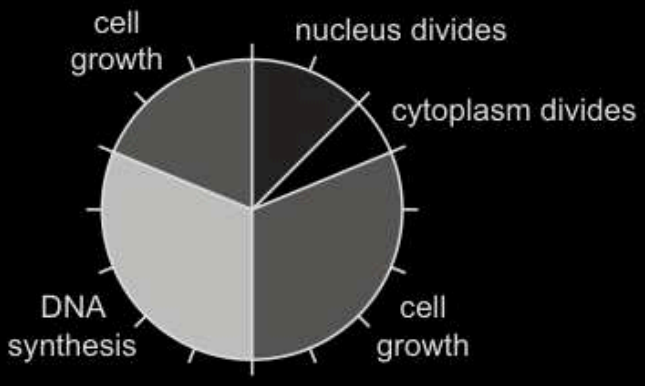
October/November 2024 · Paper 1 · Variant 1 · Question 29

Correct answer: A

29 Asexual reproduction involves the cell cycle in which the cell divides and then growth and DNA synthesis take place before the cell is ready to divide again.

The diagram shows the stages in one complete cell cycle.

This cell cycle takes 1.75 hours to complete.



Using the diagram, which activity in the cycle takes 52.5 minutes in total?

A cell growth

B cytoplasm divides

C nucleus divides

D DNA synthesis

October/November 2024 · Paper 1 · Variant 1 · Question 31

Correct answer: A

31 What is the sequence of organs that a sperm cell must pass through so it can fertilise an egg cell?

A sperm duct → urethra → vagina → cervix → uterus → oviduct

B sperm duct → vagina → urethra → oviduct → cervix → uterus

C uterus → sperm duct → vagina → urethra → cervix → oviduct

D urethra → sperm duct → uterus → vagina → oviduct → cervix

October/November 2024 · Paper 1 · Variant 1 · Question 33

Correct answer: C

33 What is an example of discontinuous variation in humans?

A body weight

B height

C blood group

D skin colour

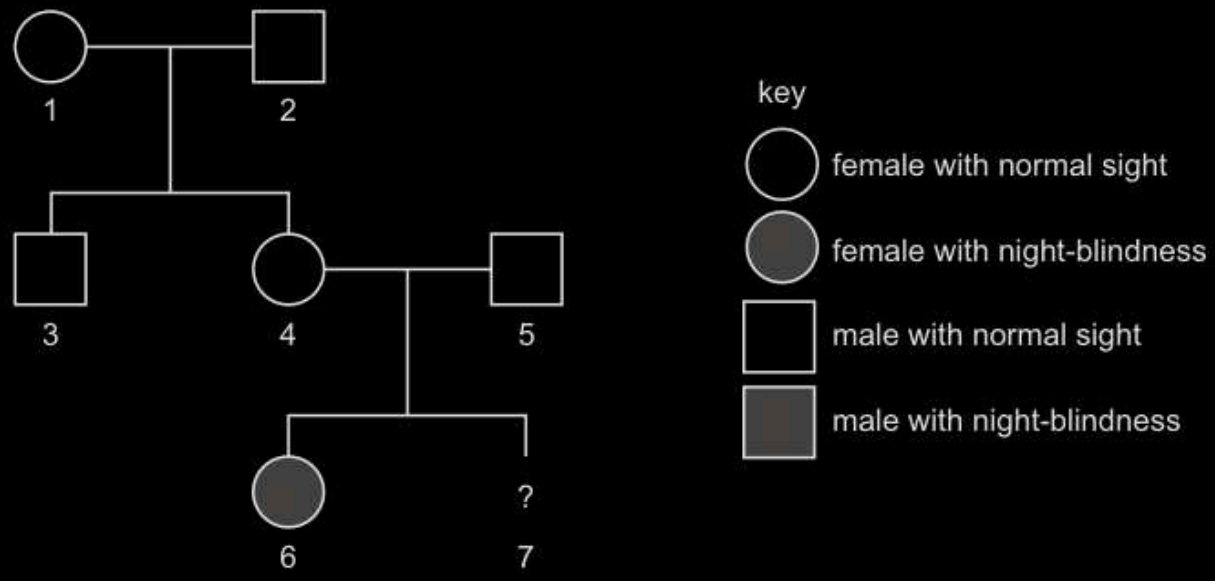
October/November 2024 · Paper 1 · Variant 1 · Question 34

Correct answer: D

34 Night-blindness is an inherited condition in which people have unusually poor vision when light levels are low.

The diagram shows the inheritance of night-blindness in three generations of a family.

Couple 4 and 5 are expecting their second child, individual 7.



What is the probability that individual 7 will be a male and will also show night-blindness as his phenotype?

A 0.500 **B** 0.300 **C** 0.250 **D** 0.125

October/November 2024 · Paper 1 · Variant 1 · Question 35

Correct answer: A

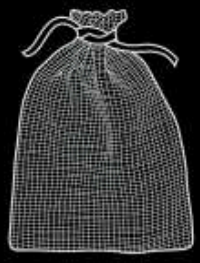
35 What is essential for natural selection to occur?

	competition	variation
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

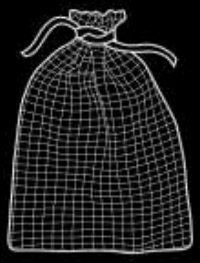
key
✓ = yes
✗ = no

Correct answer: B

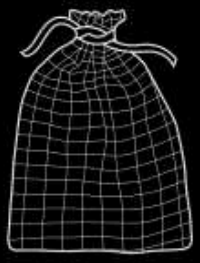
38 Some students set up an experiment to study the decay of leaves in garden soil. They put leaves in bags of different mesh sizes, sealed them and then buried them.



mesh size 0.1 mm



mesh size 1.0 mm



mesh size 5.0 mm

Each month for 5 months the bags were dug up and the total percentage loss in mass from the start of the experiment was calculated.

The results are shown.

	total percentage loss in mass				
mesh size / mm	month 1	month 2	month 3	month 4	month 5
0.1	5	11	16	20	31
1.0	13	23	26	42	48
5.0	21	32	36	54	60

What can the students conclude from these results?

A Decay is dependent on access to oxygen for the decomposers.

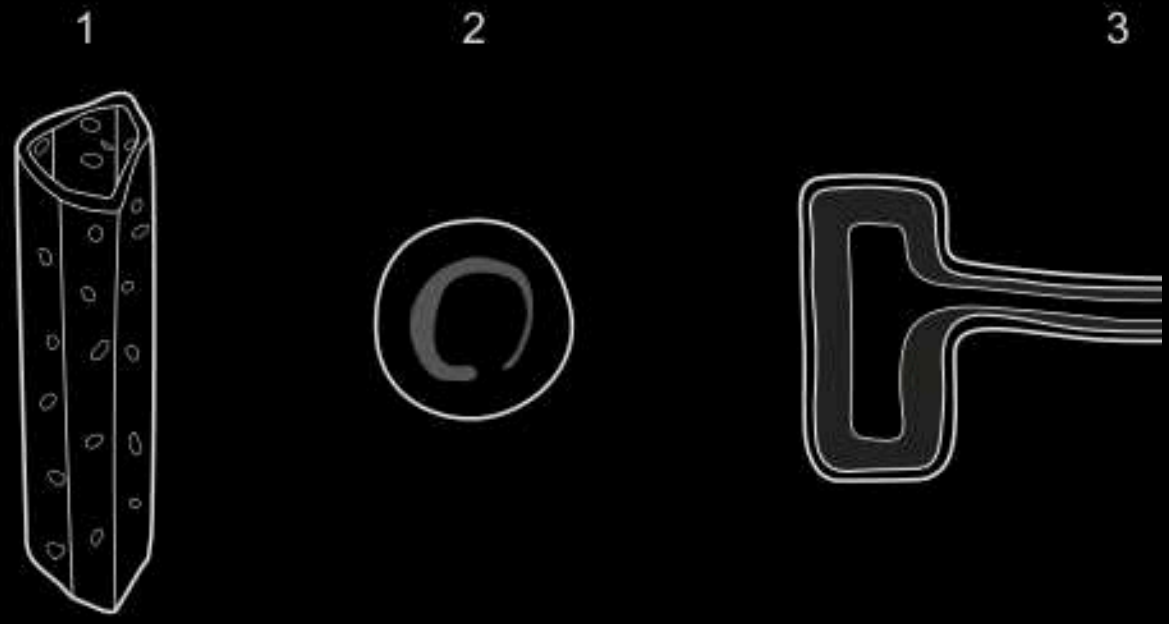
B The larger the mesh size the faster the rate of decay.

C Decay depends on how much water can get to the leaves.

D Nutrients diffuse away from the leaves more easily when mesh size increases.

Correct answer: D

1 The diagram shows three different types of microscopic structure.



What are these structures?

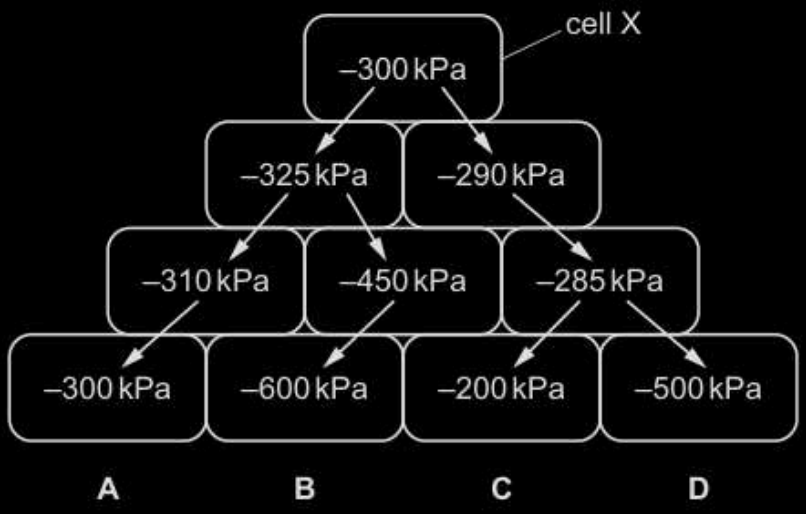
	1	2	3
A	xylem vessel	root hair cell	red blood cell
B	red blood cell	xylem vessel	root hair cell
C	root hair cell	red blood cell	xylem vessel
D	xylem vessel	red blood cell	root hair cell

October/November 2024 · Paper 1 · Variant 2 · Question 3

Correct answer: B

3 The diagram shows the water potentials measured in units of kPa in a group of plant cells. A larger negative number indicates a lower water potential.

Which sequence of arrows shows the net movement of water from cell X?



October/November 2024 · Paper 1 · Variant 2 · Question 4

Correct answer: D

4 Which statements about active transport are correct?

1 Ions move from a region of high concentration to a region of low concentration.

2 Ions move across the cell membrane.

3 Energy released during respiration is used to move ions into or out of a cell.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

October/November 2024 · Paper 1 · Variant 2 · Question 5

Correct answer: A

5 Which biological molecule contains the chemical elements carbon, hydrogen, nitrogen and phosphorus?

A DNA

B cellulose

C lipid

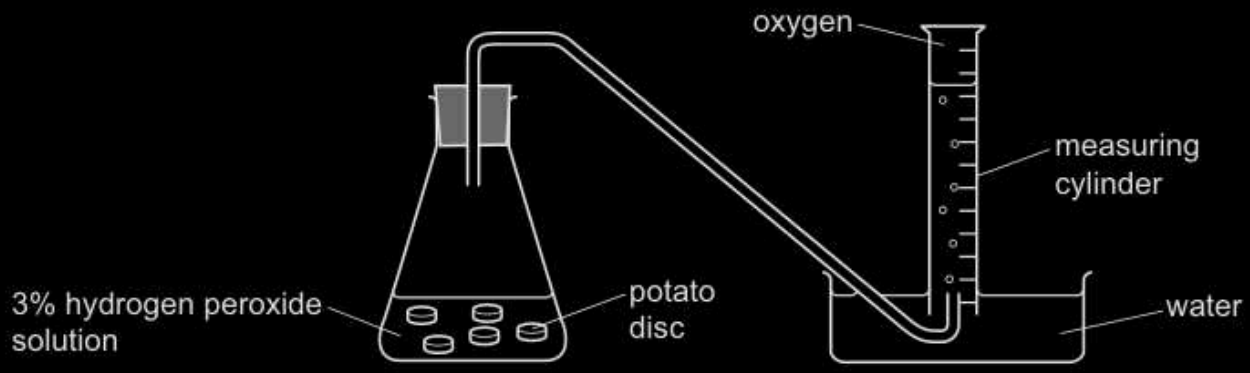
D starch

Correct answer: A

6 Catalase is an enzyme found in potato tissue. It catalyses the breakdown of hydrogen peroxide into water and oxygen.

$$\text{hydrogen peroxide} \xrightarrow{\text{catalase}} \text{water + oxygen}$$

The apparatus shown was used to investigate the activity of catalase.

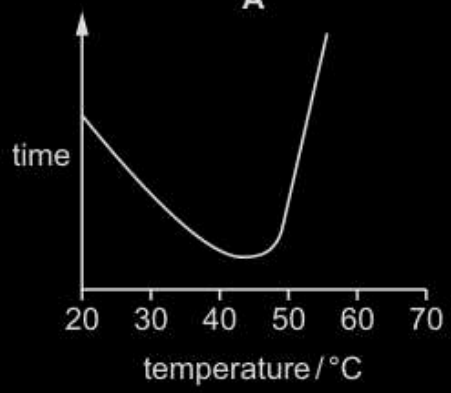


Five identical potato discs were dropped into 15 cm^3 of a 3% hydrogen peroxide solution at a temperature of 20 °C. The time taken for 5 cm^3 of oxygen to be produced was recorded.

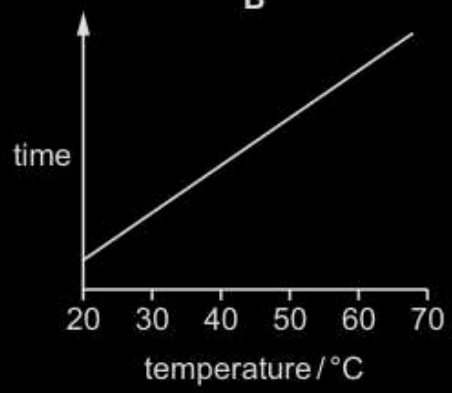
This procedure was repeated at each of the following temperatures: 30 °C, 40 °C, 50 °C, 60 °C and 70 °C.

Which graph shows the results of the investigation?

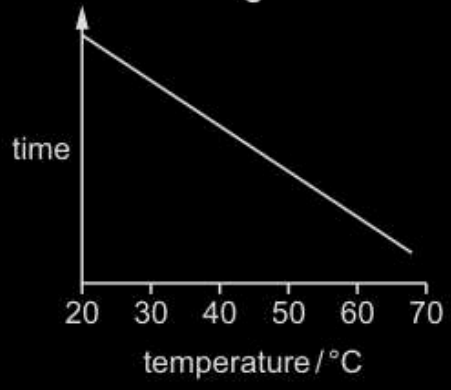
A



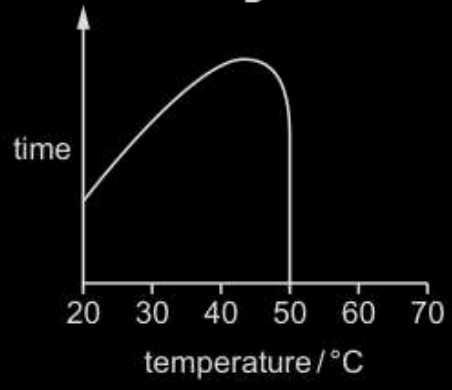
B



C



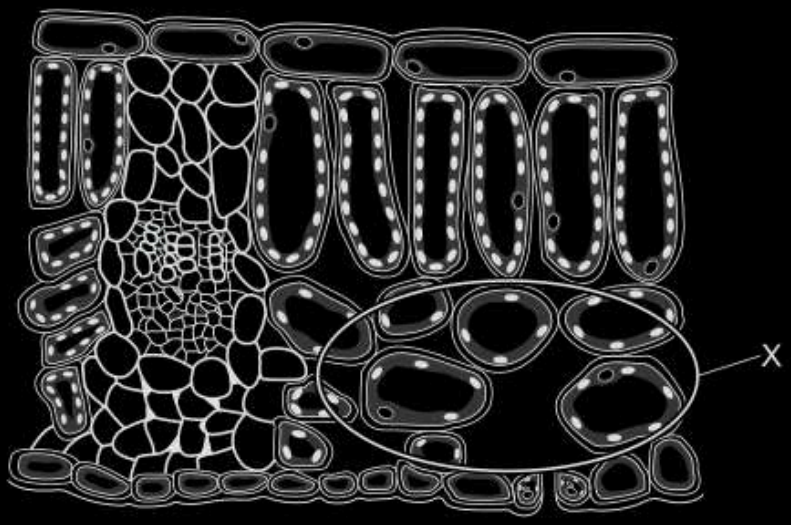
D



October/November 2024 · Paper 1 · Variant 2 · Question 8

Correct answer: B

8 The diagram shows a section through a leaf.



What is the main function of the region labelled X?

A conduction and support

B gaseous exchange

C photosynthesis

D prevention of water loss

October/November 2024 · Paper 1 · Variant 2 · Question 9

Correct answer: D

9 Root hair cells take in water and ions from the soil.

Which row shows how water and ions are taken into root hair cells?

	water	ions
A	active transport	active transport
B	active transport	osmosis
C	osmosis	osmosis
D	osmosis	active transport

Correct answer: C

11 Cubes of boiled egg white are placed in test-tubes containing 5 cm^3 of water. Boiled egg white contains protein. Other substances are added to each test-tube as shown in the table. The test-tubes are left for eight hours and then tested for amino acids.

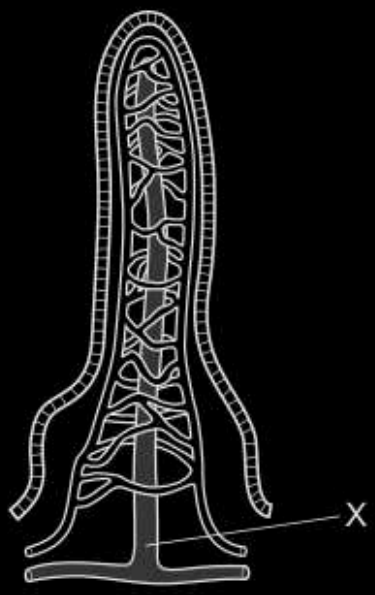
test-tube	solution added	results of test for amino acids
1	pepsin	absent
2	pepsin + alkali	absent
3	none	absent
4	pepsin + acid	large amounts
5	boiled pepsin + acid	traces
6	acid	traces
7	alkali	absent

Which test-tubes show that pepsin is an enzyme?

A 1 and 4 **B** 2 and 7 **C** 4 and 5 **D** 5 and 6

Correct answer: D

12 The diagram shows a section through a villus.



What is the main role of vessel X?

A to carry amino acids to the liver for protein formation

B to deliver deoxygenated blood to the heart

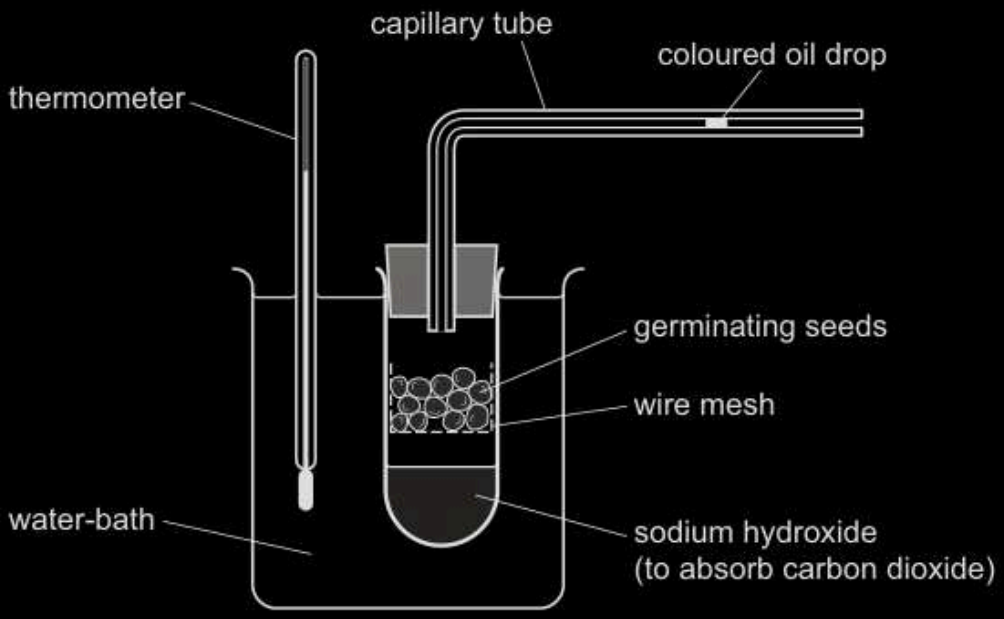
C to supply oxygen to the cells of the villus

D to transfer fatty acids and glycerol to the lymph system

October/November 2024 · Paper 1 · Variant 2 · Question 15

Correct answer: C

15 The diagram shows how some apparatus is set up to investigate respiration in germinating seeds.



The coloured oil drop moves along the capillary tube.

What causes the movement of the coloured oil drop?

A carbon dioxide released

B heat released

C oxygen used

D water used

October/November 2024 · Paper 1 · Variant 2 · Question 16

Correct answer: C

16 A ventricular septal defect is a hole in the septum between the left and right ventricles of the heart.

Which statement describes the effect on blood flow through the heart in a person with a ventricular septal defect?

A Some deoxygenated blood flows into the pulmonary vein.

B Some oxygenated blood flows into the right atrium.

C Some oxygenated blood flows through the pulmonary artery.

D Some deoxygenated blood flows into the left atrium.

October/November 2024 · Paper 1 · Variant 2 · Question 17

Correct answer: B

17 Human blood is composed of plasma, red blood cells, white blood cells and platelets.

Which row states a function of each of these components?

	plasma	red blood cells	white blood cells	platelets
A	clotting	transporting carbon dioxide	transporting oxygen	producing antibodies
B	transporting carbon dioxide	transporting oxygen	producing antibodies	clotting
C	transporting carbon dioxide	transporting oxygen	clotting	producing antibodies
D	transporting oxygen	transporting carbon dioxide	producing antibodies	clotting

October/November 2024 · Paper 1 · Variant 2 · Question 19

Correct answer: B

19 Carbon monoxide is a poisonous gas that combines with haemoglobin to form carboxyhaemoglobin.

Which function of the blood will be affected if a person inhales carbon monoxide?

A the ability of the blood to form clots

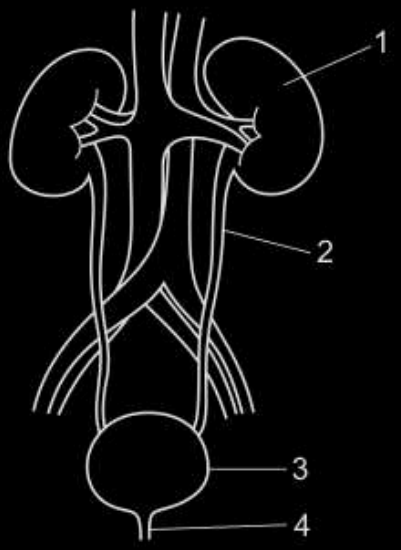
B the carriage of oxygen to the body's organs

C the formation of antibodies by lymphocytes

D the transport of adrenaline in the blood

Correct answer: C

22 The diagram shows the excretory system.



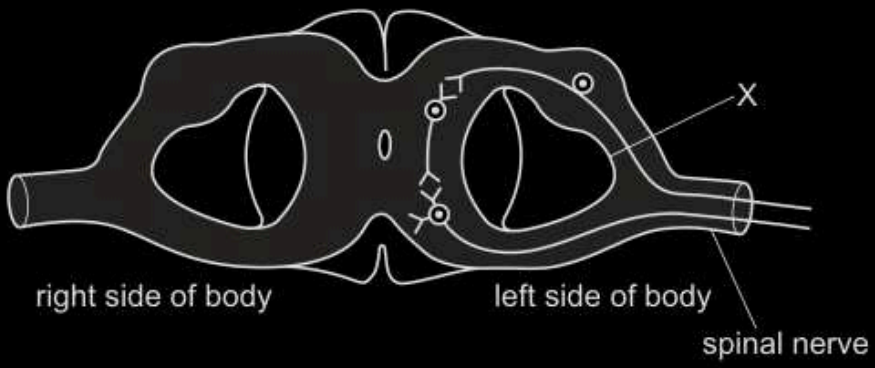
Which row identifies the names of the labelled structures?

	1	2	3	4
A	bladder	ureter	kidney	urethra
B	bladder	urethra	kidney	ureter
C	kidney	ureter	bladder	urethra
D	kidney	urethra	bladder	ureter

Correct answer: B

23 As a result of an accident in a factory, a small piece of metal entered the spinal cord of an engineer and severed the tissue at X.

The diagram shows a section through the spinal cord in the neck at the point of the injury. A nervous pathway is also shown.



Which outcome from the accident is possible?

A loss of all sensation in the head above the spinal cord injury

B loss of some sensation on the left side of the body

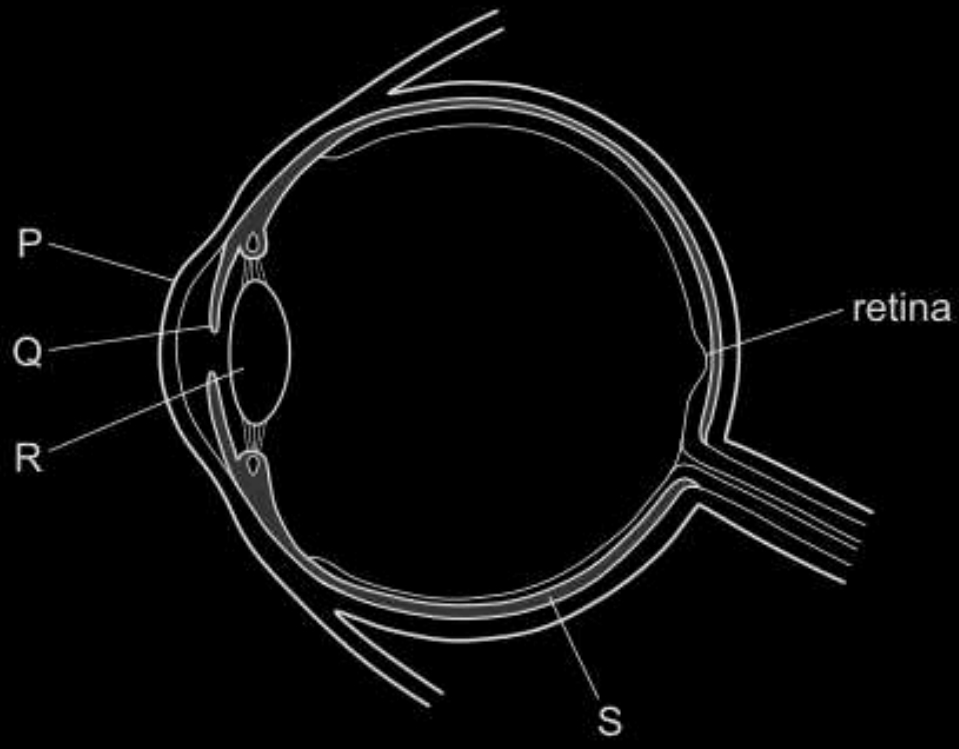
C loss of all sensation on the right side of the body

D some paralysis on the right side of the body

October/November 2024 · Paper 1 · Variant 2 · Question 24

Correct answer: B

24 The diagram shows a section through the eye.



Which two structures focus light rays onto the retina?

A P and Q **B** P and R **C** Q and R **D** Q and S

October/November 2024 · Paper 1 · Variant 2 · Question 26

Correct answer: C

26 When the body temperature is too high, which process causes the body temperature to return to normal?

A contraction of hair erector muscles

B contraction of skeletal muscles, causing shivering

C secretion of sweat onto the surface of the skin

D vasoconstriction of arterioles supplying skin surface capillaries

October/November 2024 · Paper 1 · Variant 2 · Question 28

Correct answer: B

28 Which statements about meiosis are correct?

	meiosis produces genetically identical nuclei	meiosis produces haploid nuclei
A	✗	✗
B	✗	✓
C	✓	✗
D	✓	✓

key

✓ = yes

✗ = no

October/November 2024 · Paper 1 · Variant 2 · Question 29

Correct answer: D

29 What is a feature of asexual reproduction in plants?

A It only needs one type of gamete.

B It requires two parents.

C It uses cell division by meiosis.

D It uses cell division by mitosis.

October/November 2024 · Paper 1 · Variant 2 · Question 33

Correct answer: B

33 DNA is divided into sections called

A chromosomes.

B genes.

C nuclei.

D proteins.

October/November 2024 · Paper 1 · Variant 2 · Question 36

Correct answer: D

36 Single-cell protein can be produced by growing a fungus in a fermenter.

Why is it necessary for the fermenter to have a cooling unit?

A Fungi produce protein faster at low temperatures.

B The air bubbles generate heat as they float upwards.

C The glucose syrup is too warm when it is added to the fermenter.

D The fungus releases heat as it respires.

October/November 2024 · Paper 1 · Variant 2 · Question 39

Correct answer: C

39 Aphids are small insects that feed on plants by sucking fluids from plant cells.

The population of aphids feeding on a bean plant increases rapidly.

Which row shows a combination of factors that explains this rapid growth?

	competition	food supply
A	high	high
B	high	low
C	low	high
D	low	low

1 Fig. 1.1 shows an experiment to measure the water loss from a small plant in 24 hours.

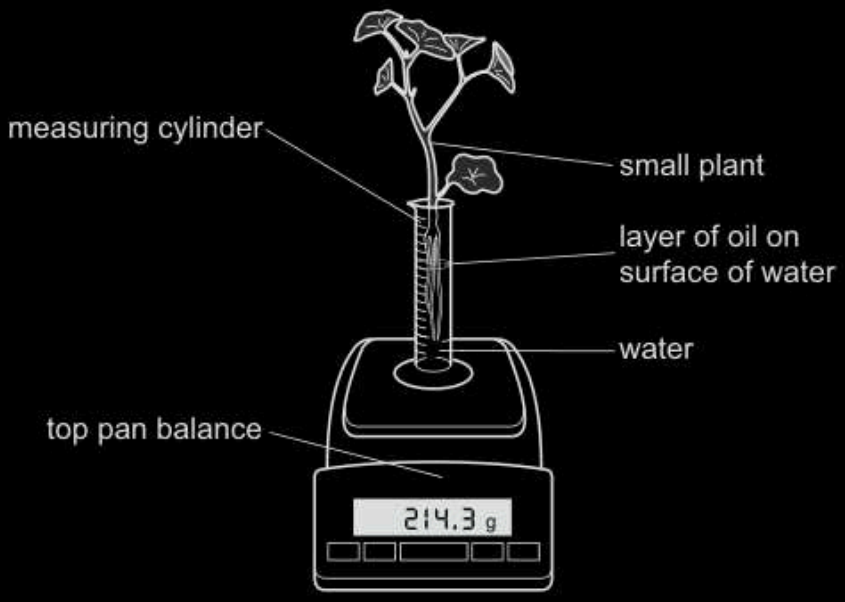


Fig. 1.1

(a) (i) Name the process by which a plant loses water from cells to the atmosphere.

..... [1]

(ii) Suggest the purpose of the layer of oil.

.....

..... [1]

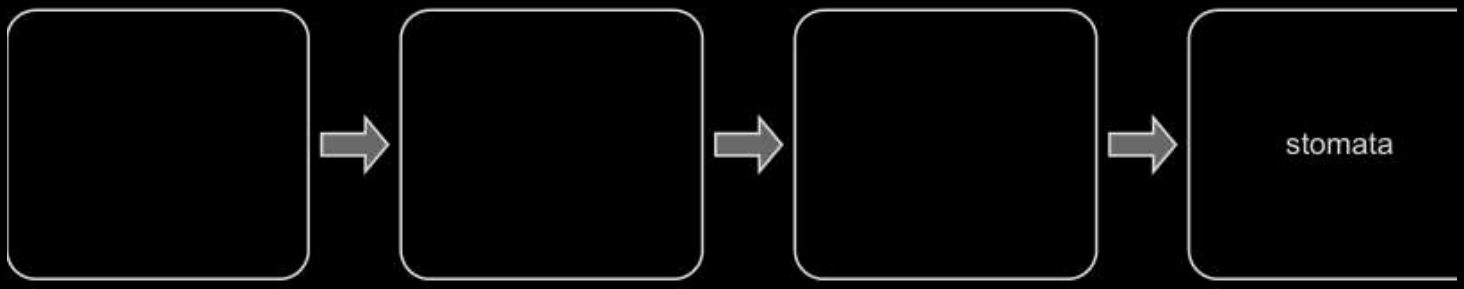
(iii) Describe **one** way in which a student could use this apparatus to obtain a value for the amount of water the plant loses in 24 hours.

.....

.....

..... [2]

(b) Water is lost mainly through the plant's stomata.
Name **three** types of cell that the water passes through before it reaches the stomata in the leaves.
Write the names of the types of cell in the correct order in the boxes.



[3]

(c) The experiment shown in Fig. 1.1 was repeated many times to investigate the effect of humidity on the rate of water loss.
All environmental conditions were kept constant except for humidity which was gradually increased.
Fig. 1.2 shows the axes for a graph.

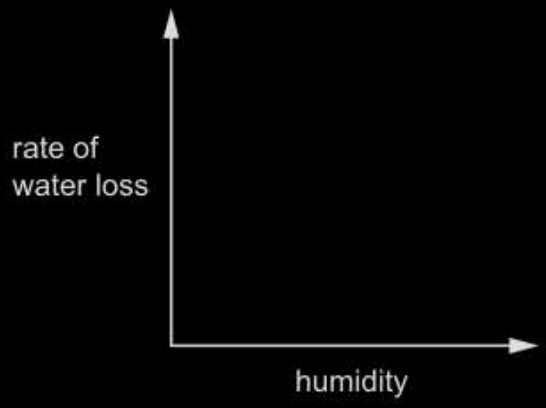


Fig. 1.2

(i) Sketch a graph on these axes of the expected results for this investigation. [1]

(ii) Explain what your graph shows about the effect of humidity on water loss.

.....

.....

.....

.....

..... [3]

[Total: 11]

1(a)(i)	transpiration ;	**1**	
1(a)(ii)	prevent water loss / evaporation of water;	**1**	
1(a)(iii)	measure mass / volume at start **+** at end / after 24 hours ; subtract end value from start value / calculate difference ;	**2**	
1(b)	any **three** in correct sequence from: root hair (cell) ; (root) cortex (cell) ; xylem (vessel element) ; palisade / spongy / mesophyll (cell) ;	**3**	
1(c)(i)	line that is at its highest at a low humidity and at its lowest at a high humidity ;	**1**	
1(c)(ii)	the lower the humidity, the higher the rate of water loss **AW** ; water is lost by diffusion ; when humidity is low, the concentration of water outside will be lower than inside the leaf / atmosphere would not be saturated with water ;	**3**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance

2 Maltase is an enzyme found in the intestines of vertebrates, including humans.

(a) Complete Table 2.1, which provides descriptions of words related to the activity of maltase. The first row has been completed for you.

Table 2.1

word	how it relates to the activity of maltase
maltase	an enzyme that is found in vertebrate intestines
	the substrate of maltase
glucose	
	part of the maltase molecule which fits the substrate
specific	

[4]

(b) Vampire bats live in South America and feed entirely on blood.
They are the only known vertebrates which do **not** have maltase in their intestines.
Suggest an explanation for this.

.....

.....

.....

..... [2]

[Total: 6]

2(a)	maltose ; product / what is produced when maltose is broken down ; active site ; *idea that* maltase only acts on maltose ;	**4**	
2(b)	any **two** from: not much / no maltose in blood / blood contains glucose / blood does not contain starch **AW** ; maltase not released / produced ; do not waste energy producing an unnecessary enzyme ; mutation / evolution ; does not have a functional gene for maltase ;	**2**	

Question	Answer	Marks	Guidance

3 Fig. 3.1 shows the central nervous system and some other parts of a man's body.

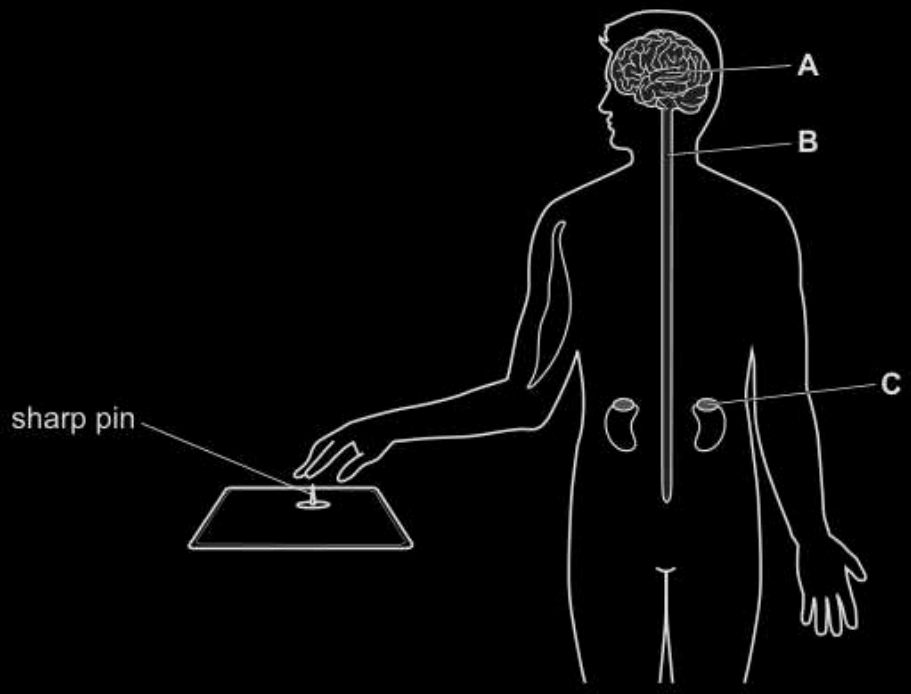


Fig. 3.1

(a) Name the **two** parts of the central nervous system labelled **A** and **B**.

A .....

B .....

[2]

(b) The man accidentally touches his hand on a sharp pin and immediately lifts his lower arm away from the pin in a reflex action.

(i) This reflex action involves a receptor and an effector.
On Fig. 3.1, draw a label line to the location of:

- the receptor and label it **R**
- the effector and label it **E**.

[2]

(ii) Describe the roles of **two** named neurones in this reflex arc.

.....

.....

.....

.....

.....

..... [4]

(c) The part of the body labelled **C** is a gland in the endocrine system.

(i) Name gland **C**.

..... [1]

(ii) Explain the role of gland **C** in triggering an increase in heart rate in response to a sudden shock.

.....

.....

..... [2]

[Total: 11]

3(a)	**A** = brain ; **B** = spinal cord ;	**2**	
3(b)(i)	**R** pointing to the hand ; **E** pointing to muscle ;	**2**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
3(b)(ii)	*sensory neurone* *idea that* a sensory neurone receives impulses / is stimulated by the receptor ; carries impulses to spinal cord / CNS ; *relay neurone* carries impulses from sensory neurone ; to motor neurone ; *motor neurone* carries impulses from relay neurone / spinal cord / CNS ; to the effector / muscle ;	**4**	no neurones named = no marks if more than **two** neurones described, mark best **two** named neurones
3(c)(i)	adrenal (gland) ;	**1**	
3(c)(ii)	any **two** from: produces a hormone / adrenaline ; travels in blood ; prepares the body for action / prepares body for increased activity ;	**2**	

Question	Answer	Marks	Guidance

4 Living organisms excrete carbon dioxide.

(a) **(i)** Describe the meaning of the term excretion.

.....

.....

..... [2]

(ii) Name the process which produces carbon dioxide in living organisms and state how carbon dioxide is excreted by humans.

.....

.....

..... [2]

(b) The volume of carbon dioxide excreted varies over a 24-hour period and the factors that affect this are different for animals and plants.

Suggest and explain reasons why the volume of carbon dioxide excreted varies for both animals and plants.

animals .....

.....

.....

plants .....

.....

.....

[4]

(c) Explain the effect of an increasing concentration of carbon dioxide in the atmosphere.

.....

.....

.....

..... [2]

[Total: 10]

4(a)(i)	any **two** from: removal **+** waste products / **AW** ; removal **+** toxic materials ; removal **+** substances made by the body / metabolism ;	**2**	
4(a)(ii)	respiration ; breathed out / exhaled / removed at the lungs / alveoli ;	**2**	

Question	Answer	Marks	Guidance
4(b)	*animals* activities vary / example given ; so more respiration at some times ; *plants* respiration **+** constant / ongoing ; daylight **+** photosynthesis **AW** ; carbon dioxide used for photosynthesis ;	**4**	max **two** marks for plants
4(c)	any **two** from: acts as a greenhouse gas ; traps thermal / heat energy / global warming ; climate change / specific effect on land or organisms ;	**2**	

Question	Answer	Marks	Guidance

7 **(a)** Explain how long-term active immunity develops in the human body after infection by a pathogen.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

(b) In any group of humans, there is variation in how well each person's immune system responds to infections.

Suggest factors which may contribute to this variation.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

7(a)	any **seven** from: 1 pathogens **+** specific **+** chemicals / proteins ; 2 antigens ; 3 lymphocytes ; 4 production of antibodies ; 5 antibodies complementary to antigens **AW** ; 6 bind ; 7 *idea of* direct destruction of pathogen ; 8 mark / clump together for destruction / engulfing by phagocytes ; 9 memory cells produced / made ; 10 rapid / increased antibody production / action on next infection ;	**7**	
7(b)	any **three** from: numbers of lymphocytes / ability to produce antibodies ; HIV infection ; age ; sex ; genetics ; diet ; previous exposure to pathogen / immunity / vaccination / immunisation ;	**3**	

© Cambridge University Press & Assessment 2024

3 **(a)** Use words from the list to complete the paragraph.

platelets **glucagon** **pancreas** **plasma**
liver **sucrose** **endocrine**
pituitary **insulin** **glycogen**

Glucose is transported in solution by the component of blood called ..... . If the concentration of glucose in the blood increases above normal, the hormone ..... is released into the blood by an ..... gland called the ..... . This hormone promotes the removal of glucose from the blood and its conversion into the chemical ..... .

[5]

(b) Fig. 3.1 is a photomicrograph of a type of blood cell.

×5000

Fig. 3.1

(i) Identify the type of blood cell shown in Fig. 3.1.

.....

[1]

(ii) If the concentration of glucose in the blood increases above normal and is **not** controlled, this can result in changes to some cells of this type. They may:

- lose their specialised shape
- become less flexible
- stick together with many other cells of the same type.

Suggest and explain how these changes will affect the body.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 11]

3(a)	plasma ; insulin ; endocrine ; pancreas ; glycogen ;	**5**	
3(b)(i)	red (blood cell) / erythrocyte ;	**1**	
3(b)(ii)	any **five** from: 1 reduced surface area ; 2 less able to fit through capillaries ; 3 clotting ; 4 less oxygen absorbed / carried / supplied ; 5 (by) haemoglobin ; 6 less aerobic respiration ; 7 less energy ;	**5**	

Question	Answer	Marks	Guidance

4 Fig. 4.1 shows changes in the curved shape of the lens of a person's eye over time.

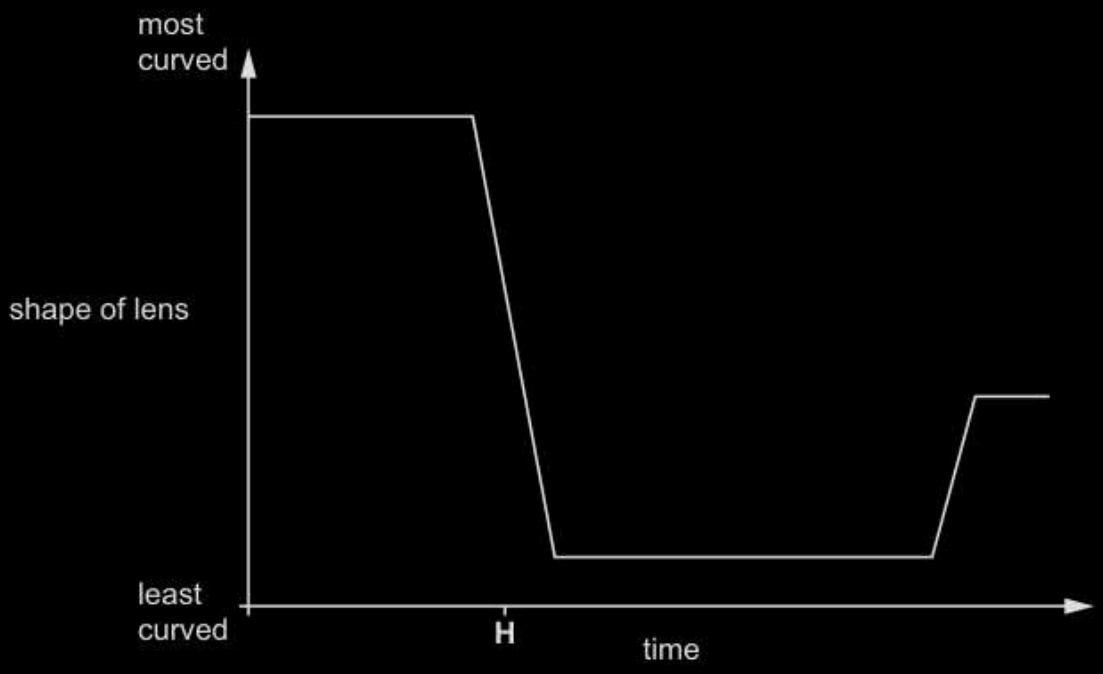


Fig. 4.1

(a) (i) During the period of time shown in Fig. 4.1, the person carried out each of the following actions:

P read the words in a book
Q read the time on a wall clock
R looked out of a window at a bird high in the sky.

Identify in what order the person carried out actions **P**, **Q** and **R**.

Place the letters **P**, **Q** and **R** in the correct order below.

..... → ..... → ..... [1]

(ii) At the time labelled **H** on Fig. 4.1, the shape of the lens is changing. Outline, with reference to **named** structures in the eye, how this change happens.

.....

.....

.....

.....

..... [4]

(b) Fig. 4.2 shows a front view of the eye in dim light.

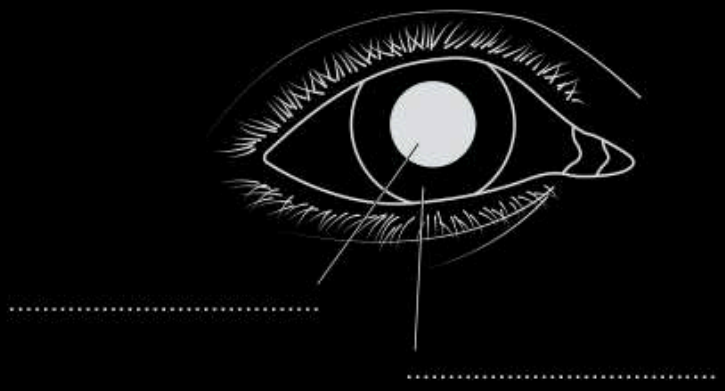


Fig. 4.2

(i) Complete the labels on Fig. 4.2 by naming the two parts of the eye indicated by the label lines. [2]

(ii) Complete Fig. 4.3 to show a view of the same eye as it would appear in bright light. [1]

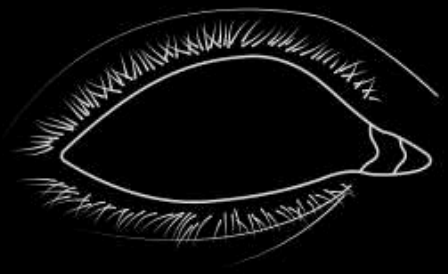


Fig. 4.3

(iii) State the type of nervous response that takes place when the eye changes its appearance from that in Fig. 4.2 to that in Fig. 4.3.

..... [1]

(iv) Suggest the importance to the person of this response taking place in the eye.

.....

.....

..... [2]

[Total: 11]

4(a)(i)	P – R – Q ;	**1**	
4(a)(ii)	ciliary muscles ; (ciliary muscles) relax ; suspensory ligaments ; (suspensory ligaments) tighten / under tension / taut ;	**4**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
4(b)(i)	pupil ; iris ;	**2**	
4(b)(ii)	pupil drawn smaller diameter ;	**1**	
4(b)(iii)	reflex ;	**1**	
4(b)(iv)	any **two** from: protect ; rapidly ; retina / rod cells / cone cells / light receptors **+** from too much light / high light intensity / bright light ;	**2**	

Question	Answer	Marks	Guidance

5 Fish are an important part of the balanced diet of many people, and fishing is a major source of employment in some countries.

Many people therefore believe that fish should be treated as a sustainable resource.

(a) Describe what is meant by the term **sustainable resource**.

.....

.....

.....

..... [2]

(b) The government of a country decided to set fishing quotas for herring fish between 2012 and 2020.

Fig. 5.1 shows the maximum number of tonnes of herring that the quota allowed to be caught each year.

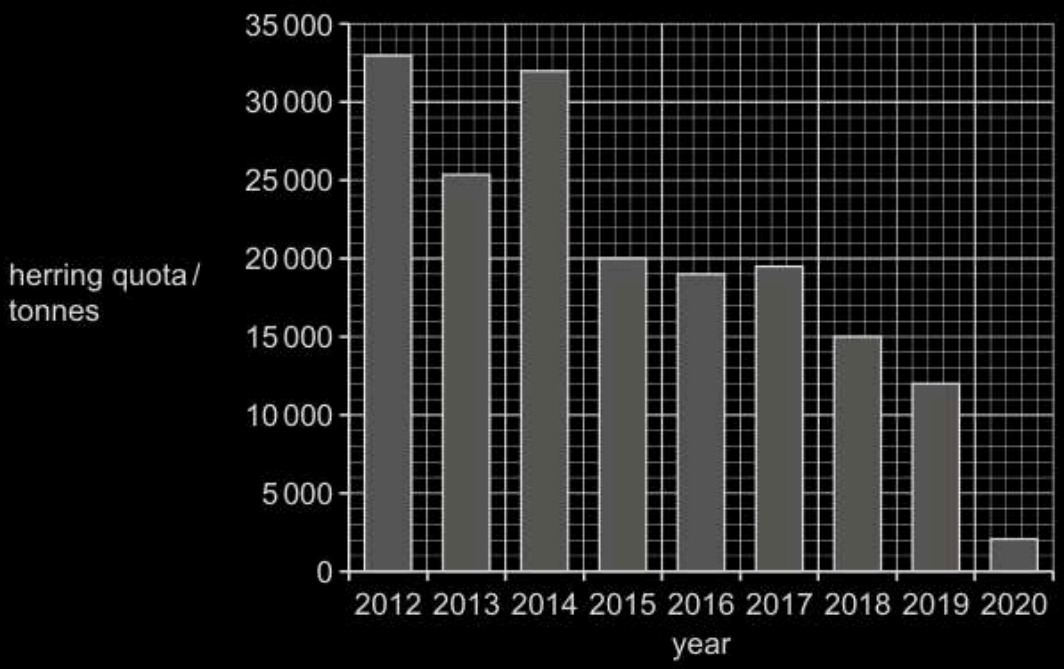


Fig. 5.1

(i) Using the information from Fig. 5.1, describe the trend in quota size and suggest explanations for this trend.

.....

.....

.....

.....

.....

..... [4]

(ii) Setting fishing quotas is one method used when trying to conserve fish stocks.

Explain how **two other** named methods can be used to conserve fish stocks.

name of method

.....

explanation

.....

.....

.....

.....

name of method

.....

explanation

.....

.....

.....

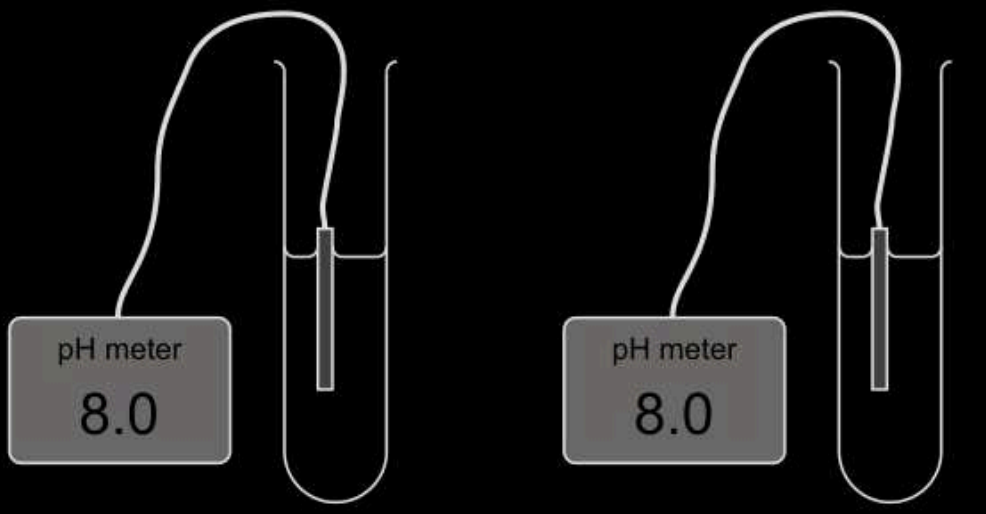
.....

[4]

(c) Fish are a good source of lipids in the form of fish oil.

A student investigated the digestion of fish oil in the laboratory.

The student set up the two experiments shown in Fig. 5.2.



experiment	**1**	**2**
test-tube contents	fish oil + bile solution	fish oil + bile solution + lipase solution

Fig. 5.2

The student recorded the pH of the contents of each test-tube using a digital pH meter at the start of each experiment. Both meters recorded a pH of 8.0.

The student continued to monitor the readings on the pH meters for 30 minutes.

Use your knowledge of how lipids are digested to explain what you would expect to happen in each test-tube.

experiment **1**

.....

.....

.....

.....

experiment **2**

.....

.....

.....

.....

[5]

5(a)	produced as rapidly as it is removed ; does not run out ;	**2**	
5(b)(i)	*description* 1 idea of overall decrease in size of quotas ; *2* comparative data quote ; *explanations* 3 overfishing / too many fish caught ; 4 quota set too high / quota not sustainable ; 5 set lower next year **+** to compensate **AW ;** 6 fish caught before they can breed **AW** ; 7 increase in natural predators / disease / competition / decrease in prey ;	**4**	max **three** marks for explanations

Question	Answer	Marks	Guidance
5(b)(ii)	1.education ; explanation ; 2.closed seasons ; explanation ; 3.protected areas ; explanation ; 4.controlled net types / mesh size ; explanation ; 5.monitoring / enforcement ; explanation ;	**4**	**one** mark for named method and **one** mark for matched explanation to a max of four marks for two named methods, each with a matched explanation
5(c)	(in either experiment **1** or experiment **2** ...) bile **+** emulsification / increase surface area ; (in experiment **1** ...) pH remains at 8.0 ; no digestion / breakdown **+** of lipids / fats / oils ; (in experiment **2** ...) bile provides optimum pH (for lipase) ; lipase **+** digest / breakdown **+** lipids / fats / oils ; (into) fatty acids **+** glycerol ; pH lowers / below 8.0 ; caused by fatty acids ;	**5**	**A** description of emulsification

Question	Answer

7 Compare each of these processes:

(a) diffusion and active transport

.....

.....

.....

.....

.....

..... [4]

(b) transpiration and translocation

.....

.....

.....

.....

.....

..... [4]

(c) phototropism and gravitropism.

.....

.....

.....

.....

.....

..... [4]

[Total: 12]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

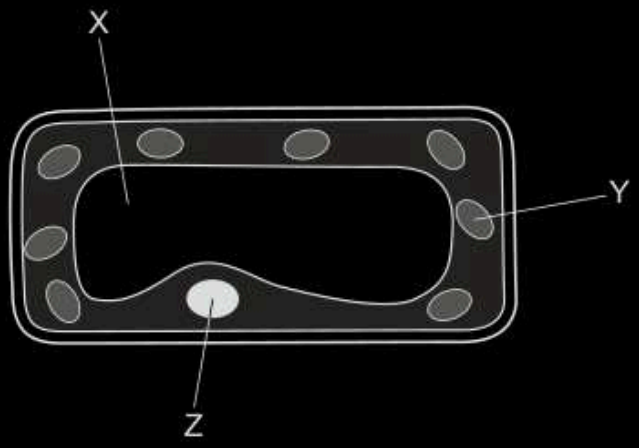
7(a)	any **four** from: *both* movement of **+** molecules / ions / particles; *diffusion* kinetic energy ; high to low concentration / down concentration gradient ; *active transport* low to high concentration / against concentration gradient ; energy required **+** respiration ;	**4**	
7(b)	any **four** from: *transpiration* loss of water **+** vapour ; from leaves ; *translocation* movement of sucrose **+** amino acids ; in phloem ; from source **+** to sink ;	**4**	
7(c)	any **four** from: *both* plant growth (response) ; towards / away (from stimulus) ; *phototropism* light ; *gravitropism* gravity ;	**4**	

© Cambridge University Press & Assessment 2024

May/June 2024 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 The diagram shows some of the structures in a plant cell.



Which labelled parts show a sap vacuole, a nucleus and a chloroplast?

	sap vacuole	nucleus	chloroplast
A	X	Y	Z
B	X	Z	Y
C	Y	Z	X
D	Z	Y	X

May/June 2024 · Paper 1 · Variant 1 · Question 2

Correct answer: B

2 A new organism is discovered. It contains DNA in a cellular structure.

To which group of organisms could it belong and to which group could it **not** belong?

	could belong to	could **not** belong to
A	bacteria	fungi
B	bacteria	viruses
C	fungi	bacteria
D	viruses	bacteria

May/June 2024 · Paper 1 · Variant 1 · Question 3

Correct answer: D

3 When animal cells and plant cells are placed in distilled water, the animal cells burst but the plant cells do not.

Which statement explains this difference?

A Animal cells do not contain starch grains, so there is no osmosis.

B Only plant cells have a permanent vacuole which stores water.

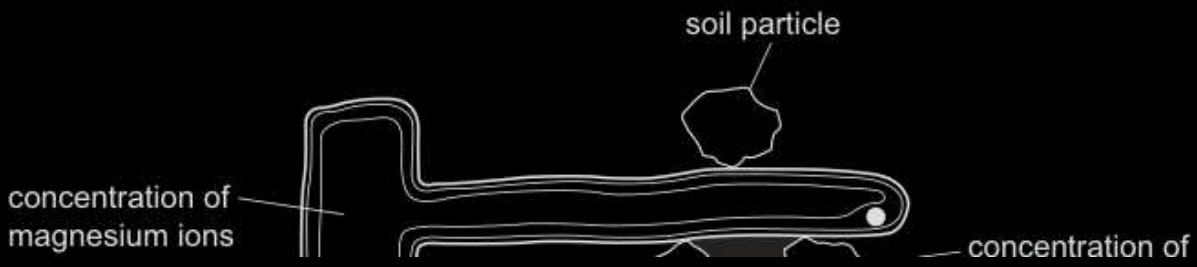
C Plant cells have a stronger cell membrane than animal cells.

D The cell wall of plant cells resists turgor pressure.

May/June 2024 · Paper 1 · Variant 1 · Question 4

Correct answer: A

4 The diagram shows the concentration of magnesium ions in a root hair cell of a healthy plant and in the soil water surrounding the root hair cell. The plant continuously uses up magnesium ions.



Correct answer: B

For the plant to remain healthy, how will the magnesium ions move?

A into the cell by active transport

B into the cell by diffusion

C out of the cell by active transport

D out of the cell by diffusion

5 The key shows shapes representing small food molecules.

Large food molecules are made from smaller food molecules.

Which diagram represents part of a glycogen molecule?

A

B

C

D

May/June 2024 · Paper 1 · Variant 1 · Question 6

Correct answer: D

6 Enzymes are biological catalysts.

Which type of molecule can be an enzyme?

A carbohydrate

B fat

C nucleotide

D protein

May/June 2024 · Paper 1 · Variant 1 · Question 9

Correct answer: D

9 Which pathway is taken by water and mineral ions through a plant?

A root hair cells → mesophyll cells → xylem → cortex cells

B root hair cells → cortex cells → phloem → mesophyll cells

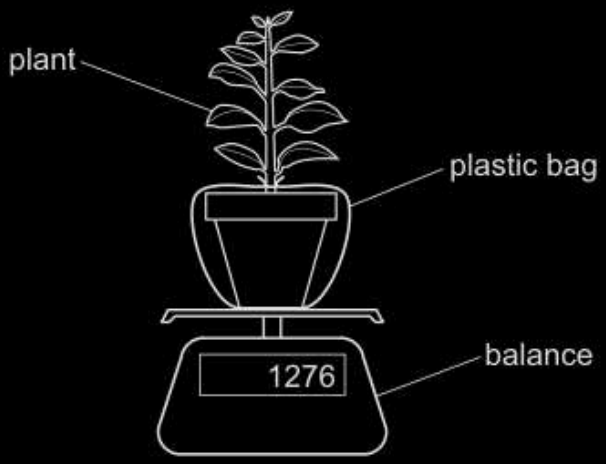
C root hair cells → mesophyll cells → phloem → cortex cells

D root hair cells → cortex cells → xylem → mesophyll cells

Correct answer: D

10 A student investigated the rate of transpiration in a green plant. The plant was growing in soil in a pot.

The student watered the plant and then put a plastic bag around the pot, leaving the leaves of the plant outside the bag. She measured the mass of the plant in the pot at the start and repeated this every two hours for six hours.



The results of the investigation are shown.

time of day	mass of pot and plant / g
09:00	1276
11:00	1270
13:00	1258
15:00	1252

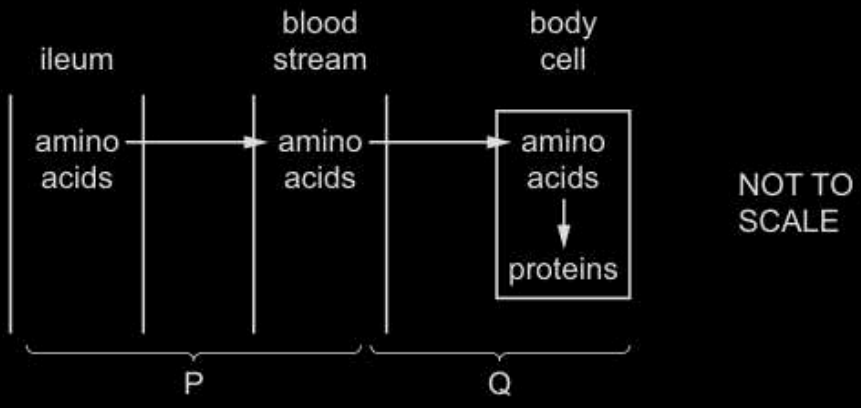
What is the mean rate of transpiration over the six-hour period?

A 43 g / h **B** 12 g / h **C** 8 g / h **D** 4 g / h

May/June 2024 · Paper 1 · Variant 1 · Question 11

Correct answer: A

11 The diagram shows what happens to amino acids in part of the human body.



What describes processes P and Q?

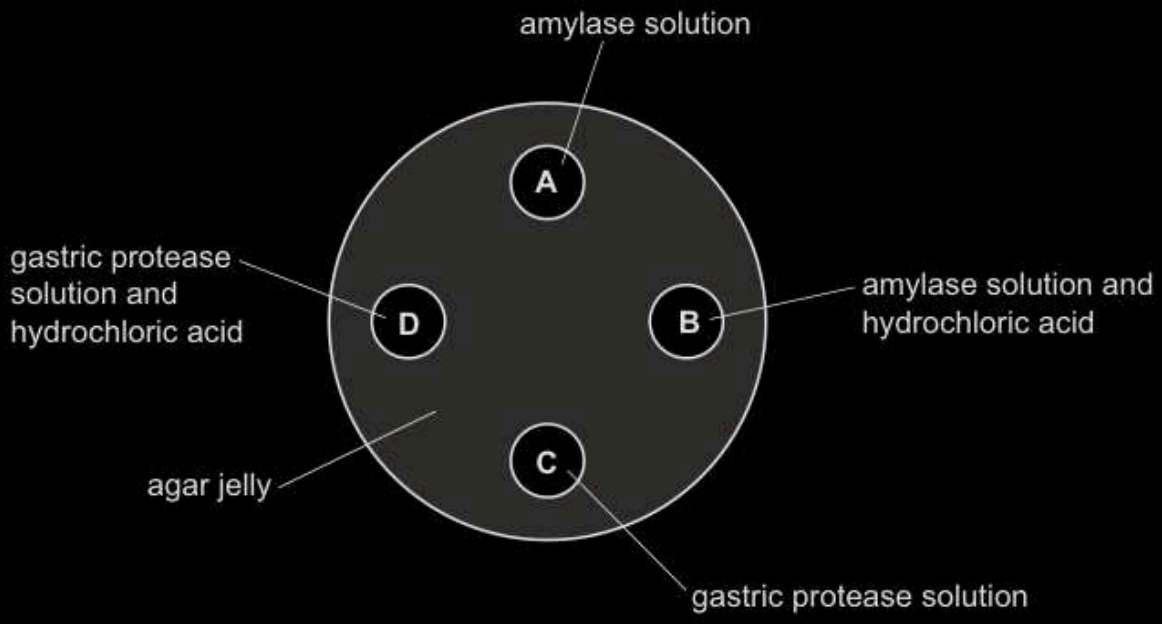
	P	Q
A	absorption	assimilation
B	absorption	digestion
C	digestion	absorption
D	digestion	assimilation

May/June 2024 · Paper 1 · Variant 1 · Question 12

Correct answer: A

12 A dish is filled with agar jelly containing starch. Four holes are cut in the jelly and each hole is filled as shown in the diagram.

After 30 minutes, which hole will be surrounded by the largest area without starch?



May/June 2024 · Paper 1 · Variant 1 · Question 13

Correct answer: B

13 A healthy person eats a meal containing carbohydrates.

Two hours later, which structure will contain a liquid with an increased concentration of glucose?

A colon

B hepatic portal vein

C renal artery

D ureter

May/June 2024 · Paper 1 · Variant 1 · Question 14

Correct answer: B

14 Which cells in the gas exchange system produce mucus?

A ciliated cells

B goblet cells

C lymphocyte cells

D red blood cells

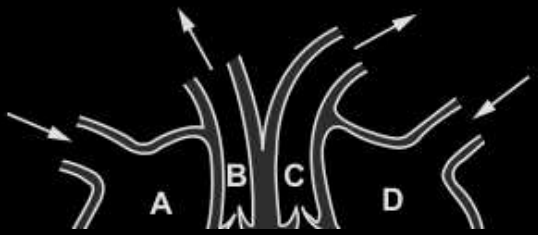
May/June 2024 · Paper 1 · Variant 1 · Question 16

Correct answer: C

16 The heart acts as a pump, pushing blood through the circulatory system.

The arrows on the diagram of the heart indicate the direction in which blood flows.

At which point is the pressure of the blood the greatest?



May/June 2024 · Paper 1 · Variant 1 · Question 17

Correct answer: C

17 A person's cuts and grazes do not heal and their blood does not clot easily.

What may be the cause of these symptoms?

A a higher than normal level of fibrinogen in the blood

B a higher than normal level of red blood cells in the blood

C a lower than normal level of platelets in the blood

D a lower than normal level of iron in the blood

May/June 2024 · Paper 1 · Variant 1 · Question 22

Correct answer: D

22 Which row identifies the contents of urine in a healthy person?

	glucose	urea	water	ions
A	✓	✗	✓	✗
B	✓	✓	✓	✓
C	✗	✗	✓	✓
D	✗	✓	✓	✓

May/June 2024 · Paper 1 · Variant 1 · Question 23

Correct answer: D

23 Which statement describes the roles of components of the nervous system?

A Effectors respond to environmental changes and send impulses to the central nervous system (CNS).

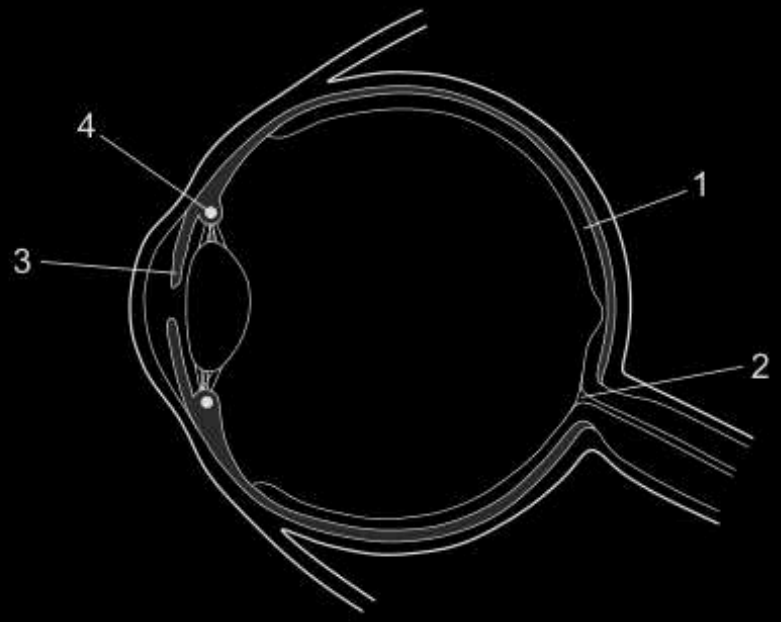
B Parts of the central nervous system (CNS) can only transmit impulses to a single effector at a time.

C Parts of the peripheral nervous system (PNS) carry impulses to effectors and receptors from the central nervous system (CNS).

D Receptors respond to environmental changes and send impulses to the central nervous system (CNS).

Correct answer: A

24 The diagram shows a section through the eye.



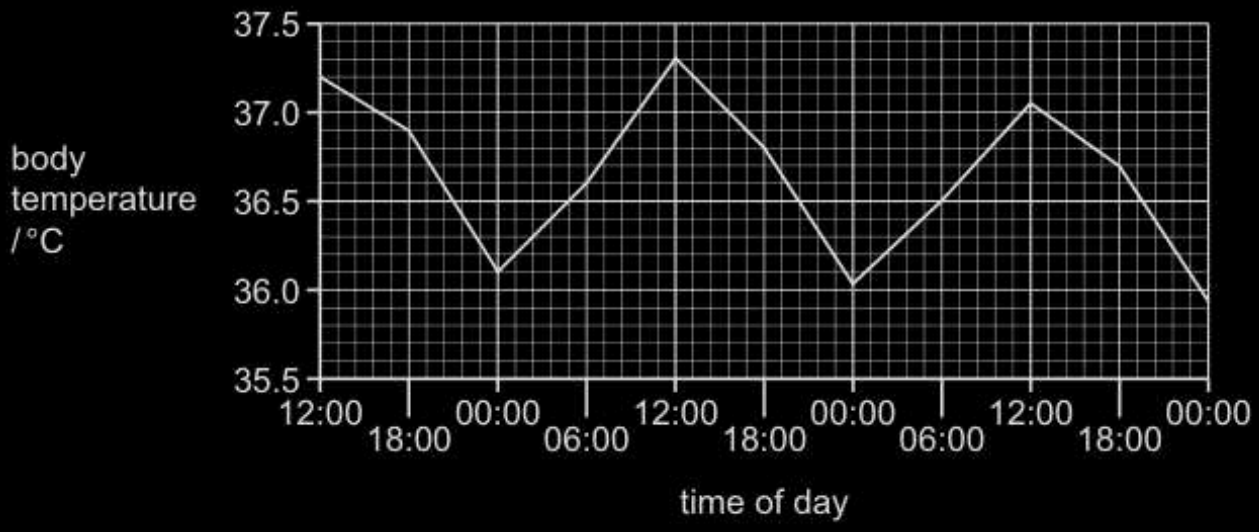
In the pupil reflex, which row gives the sites of the effectors and receptors involved?

	effectors	receptors
A	3	1
B	3	2
C	4	1
D	4	2

May/June 2024 · Paper 1 · Variant 1 · Question 25

Correct answer: D

25 The graph shows changes in the body temperature of a person over a 60-hour period.



Which statement about the graph is correct?

A The temperature of the person is always lower at 18:00 than at 06:00.

B The range shown by the data is 1.0 °C.

C The temperature of the person after 9 hours from the start is 36.8 °C.

D Over this period, the mean temperature of the person at 18:00 is 36.8 °C.

May/June 2024 · Paper 1 · Variant 1 · Question 26

Correct answer: A

26 Which process is stimulated by adrenaline in the cells of the liver?

A breakdown of glycogen, increasing the blood glucose level

B breakdown of excess amino acids, forming urea

C breakdown of proteins, releasing amino acids into the blood

D conversion of excess blood glucose to glycogen

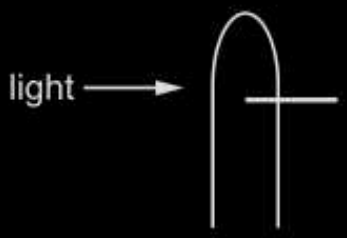
May/June 2024 · Paper 1 · Variant 1 · Question 27

Correct answer: A

27 The diagrams show experiments to investigate the response of plant shoots to light.

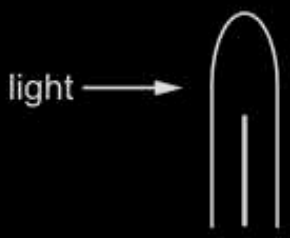
Which shoot will **not** grow towards the light?

A



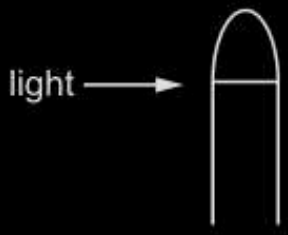
thin glass plate inserted half-way below the tip

B



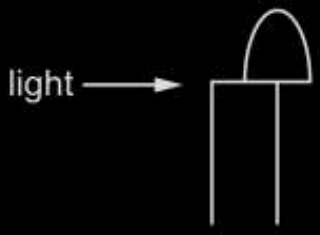
thin glass plate separates the two sides of the shoot lower down

C



the tip of the shoot removed and then replaced

D



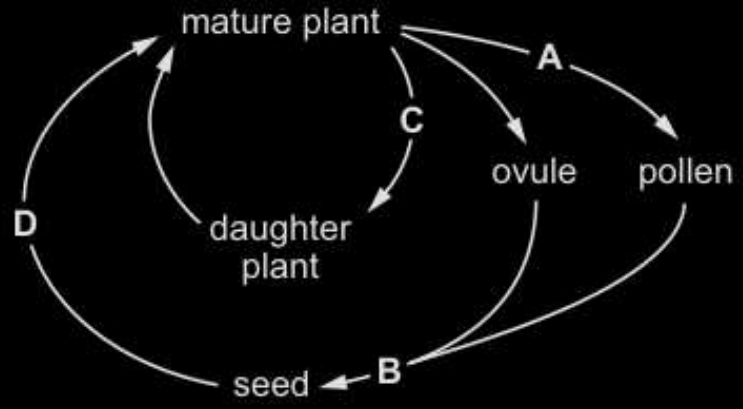
the tip of the shoot removed and replaced to one side

May/June 2024 · Paper 1 · Variant 1 · Question 28

Correct answer: A

28 The diagram shows the life cycle of a species of plant.

During which stage does meiosis (reduction division) occur?



May/June 2024 · Paper 1 · Variant 1 · Question 29

Correct answer: B

29 What is an advantage of sexual reproduction in plants?

A It is faster than asexual reproduction.

B Offspring show genetic variation.

C Only one parent is needed.

D Pollinators are not needed.

May/June 2024 · Paper 1 · Variant 1 · Question 31

Correct answer: C

31 What is the path taken by sperm cells during ejaculation from the male reproductive system?

A sperm duct → testis → urethra

B sperm duct → urethra → testis

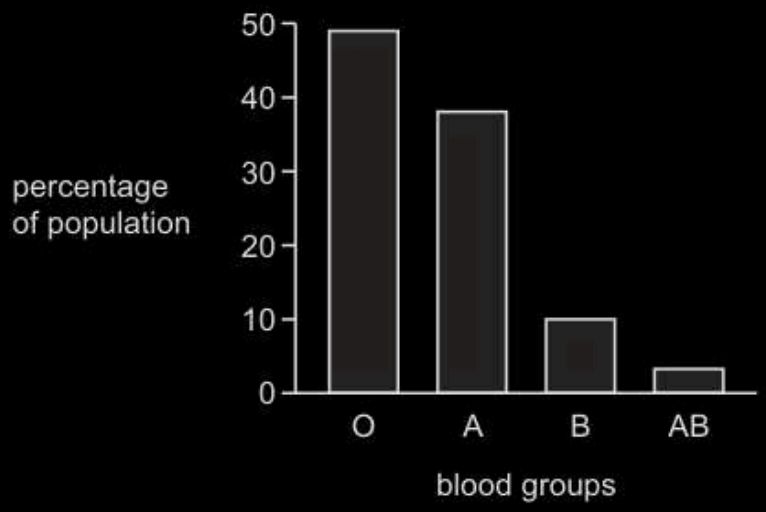
C testis → sperm duct → urethra

D testis → urethra → sperm duct

May/June 2024 · Paper 1 · Variant 1 · Question 33

Correct answer: C

33 The graph shows the percentage of the population with different ABO blood groups in one region.



How does the graph suggest that variation in ABO blood groups is discontinuous?

A There are intermediates.

B The percentage of the population with each blood group is different.

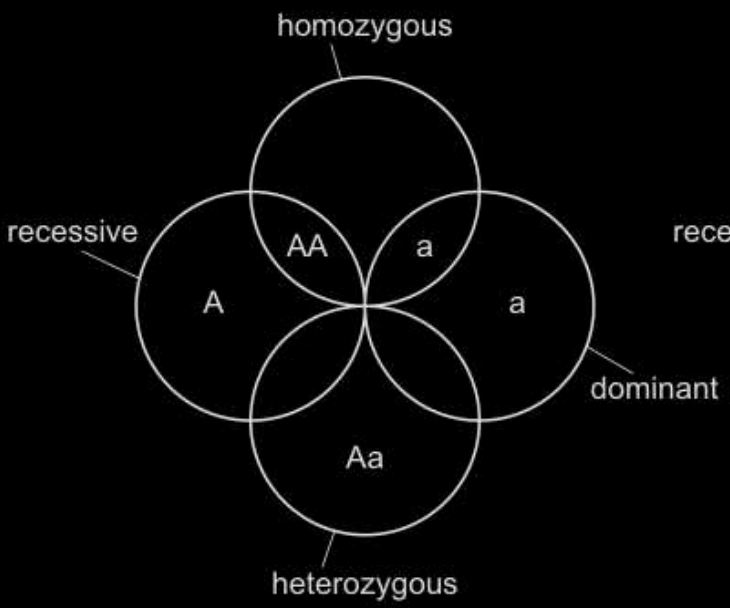
C There is a limited number of phenotypes.

D There is a range of phenotypes between two extremes.

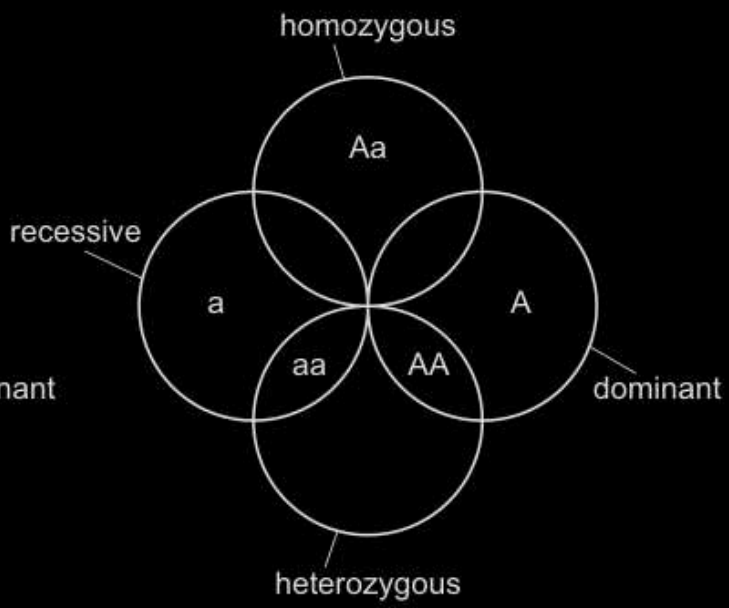
Correct answer: C

34 For a gene with two alleles, A and a, which diagram identifies these alleles and the genotypes they can form?

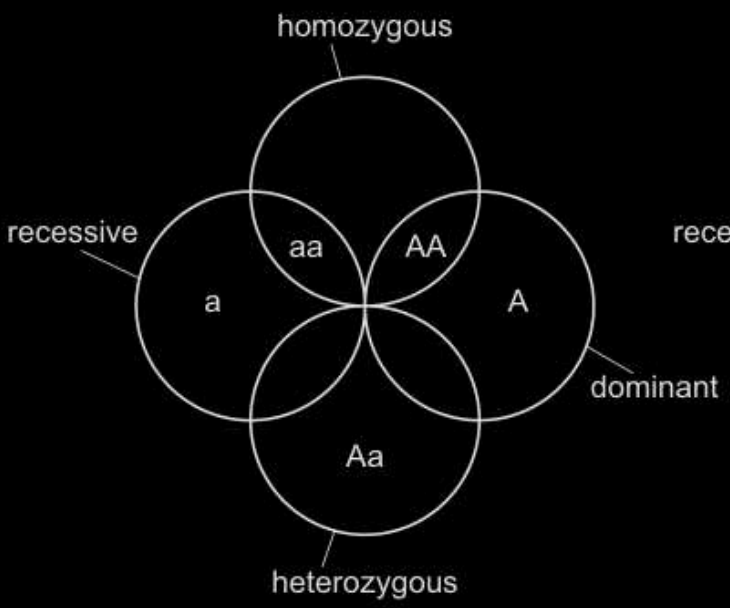
A



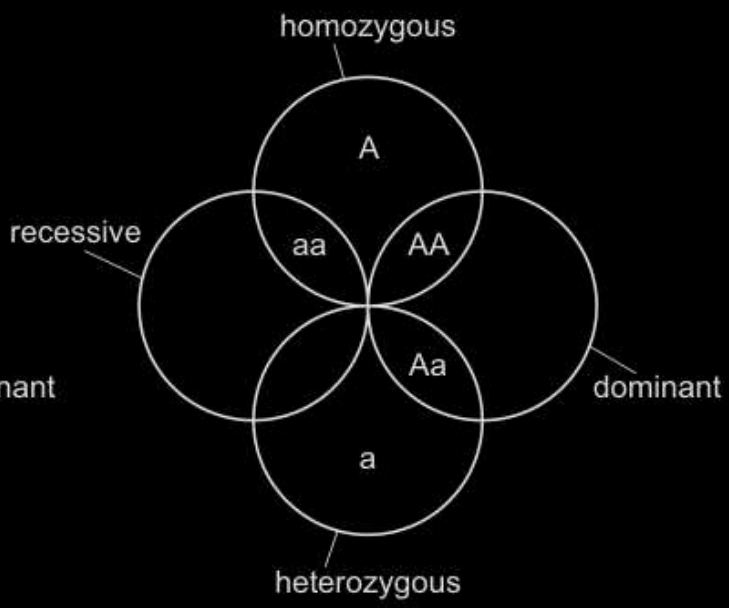
B



C



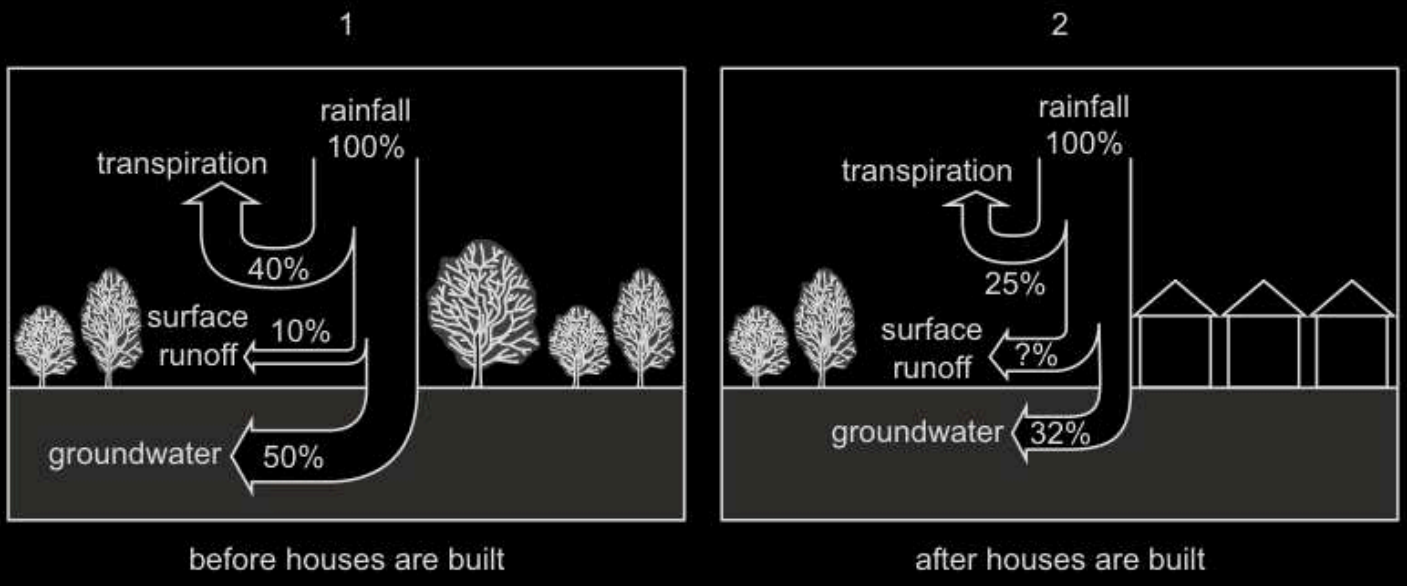
D



May/June 2024 · Paper 1 · Variant 1 · Question 39

Correct answer: A

39 The diagrams show what happens to rain when it falls on an area of forested land before and after houses are built.



before houses are built

after houses are built

What is the difference in the percentage of surface runoff before and after building the houses?

A increases by 33%

B increases by 43%

C decreases to 43%

D decreases to 33%

May/June 2024 · Paper 1 · Variant 2 · Question 1

Correct answer: D

1 Which type of cell has a cellulose cell wall?

A animal

B bacterium

C fungus

D plant

May/June 2024 · Paper 1 · Variant 2 · Question 2

Correct answer: A

2 Some young plants were put into the soil and grew well for a few weeks. They then began to show signs of disease. Samples of the diseased leaves were examined using a microscope.

Which observations of the organism causing the disease show that it could be a fungus?

	long and thread-like structure	chloroplasts **not** present	cell walls present	nuclei surrounded by a membrane
A	✓	✓	✓	✓
B	✓	✓	✓	✗
C	✓	✗	✗	✓
D	✗	✓	✓	✓

key
✓ = yes
✗ = no

May/June 2024 · Paper 1 · Variant 2 · Question 3

Correct answer: D

3 The cell wall of a plant cell is removed using an enzyme.

What would happen if this cell is then placed in distilled water?

A It would take longer for the cell to become turgid.

B Proteins in the cytoplasm would leave through the cell membrane.

C The cell would become smaller as water passes out.

D The cell would burst as water moves into it.

May/June 2024 · Paper 1 · Variant 2 · Question 4

Correct answer: B

4 The table shows the concentrations of four types of particle in two cells.

particle	concentration / arbitrary units	
	cell 1	cell 2
magnesium ions	6	2
nitrate ions	9	4
oxygen molecules	12	9
water molecules	320	520

Which particle can only be moved by active transport between the cells in the direction described?

	particle	direction of movement
A	magnesium ions	from cell 1 to cell 2
B	nitrate ions	from cell 2 to cell 1
C	oxygen molecules	from cell 1 to cell 2
D	water molecules	from cell 2 to cell 1

May/June 2024 · Paper 1 · Variant 2 · Question 5

Correct answer: A

5 Which type of food molecule has the element nitrogen in its structure?

A amino acid

B fatty acid

C glucose

D glycerol

May/June 2024 · Paper 1 · Variant 2 · Question 6

Correct answer: A

6 Which test can be used to determine the presence of glucose?

A Benedict's test

B biuret test

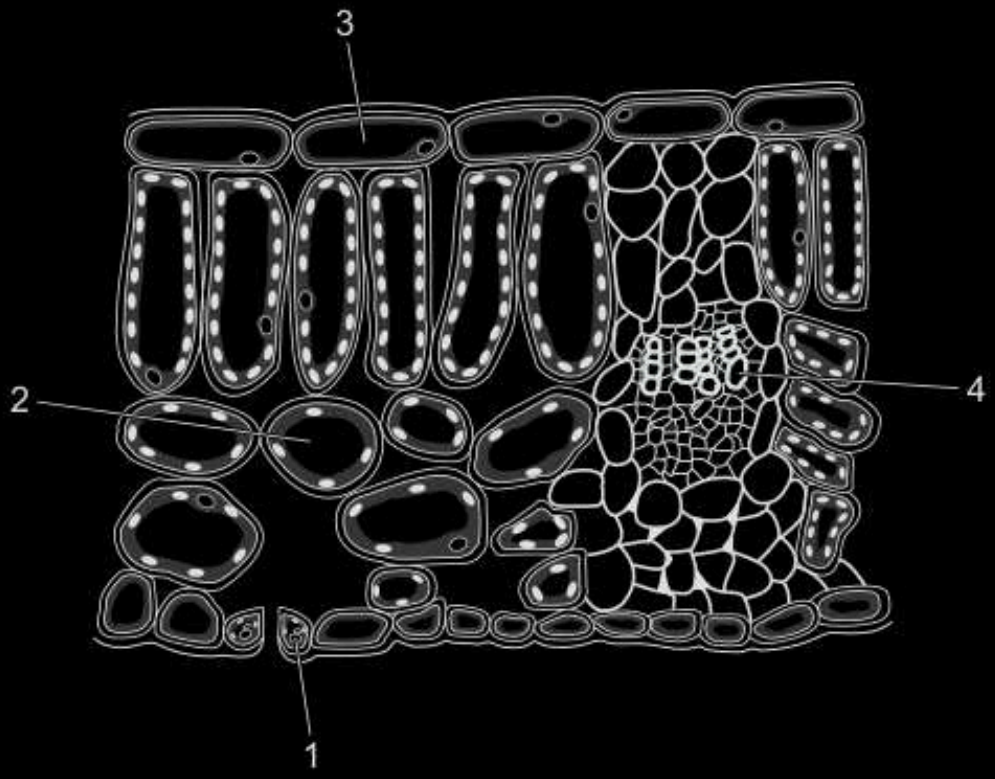
C ethanol emulsion test

D iodine test

May/June 2024 · Paper 1 · Variant 2 · Question 8

Correct answer: A

8 The diagram shows a section of a leaf.



Which numbers indicate cells where both photosynthesis **and** gas exchange occur?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

May/June 2024 · Paper 1 · Variant 2 · Question 9

Correct answer: C

9 Water moves from the soil to the atmosphere through a plant.

The water enters the plant through the root hair cells and moves into the root cortex.

Where does the water move into from the root cortex?

A into the mesophyll cells

B into the phloem cells

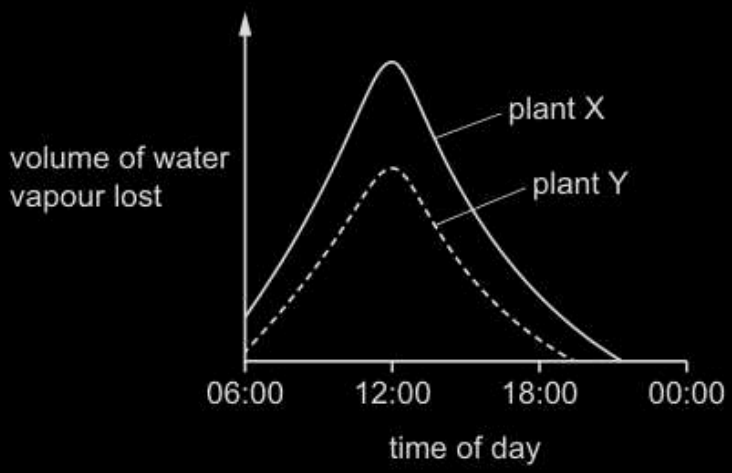
C into the xylem vessels

D through the stomata

May/June 2024 · Paper 1 · Variant 2 · Question 10

Correct answer: C

10 The graph shows the loss of water vapour from two different plants growing in identical conditions.



What is a possible reason for the difference in the volume of water vapour lost from the two plants?

A Plant X has most of its stomata on the lower surfaces of its leaves.

B Plant Y has most of its stomata on the upper surfaces of its leaves.

C The surfaces of the mesophyll cells of plant X have a greater surface area.

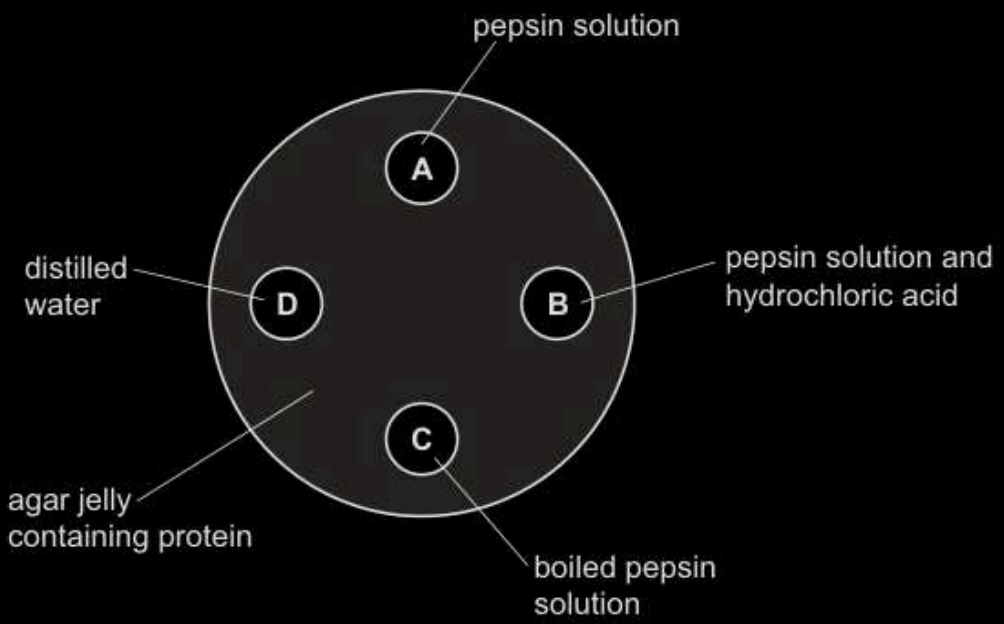
D The surfaces of the mesophyll cells of plant Y have a greater surface area.

May/June 2024 · Paper 1 · Variant 2 · Question 12

Correct answer: B

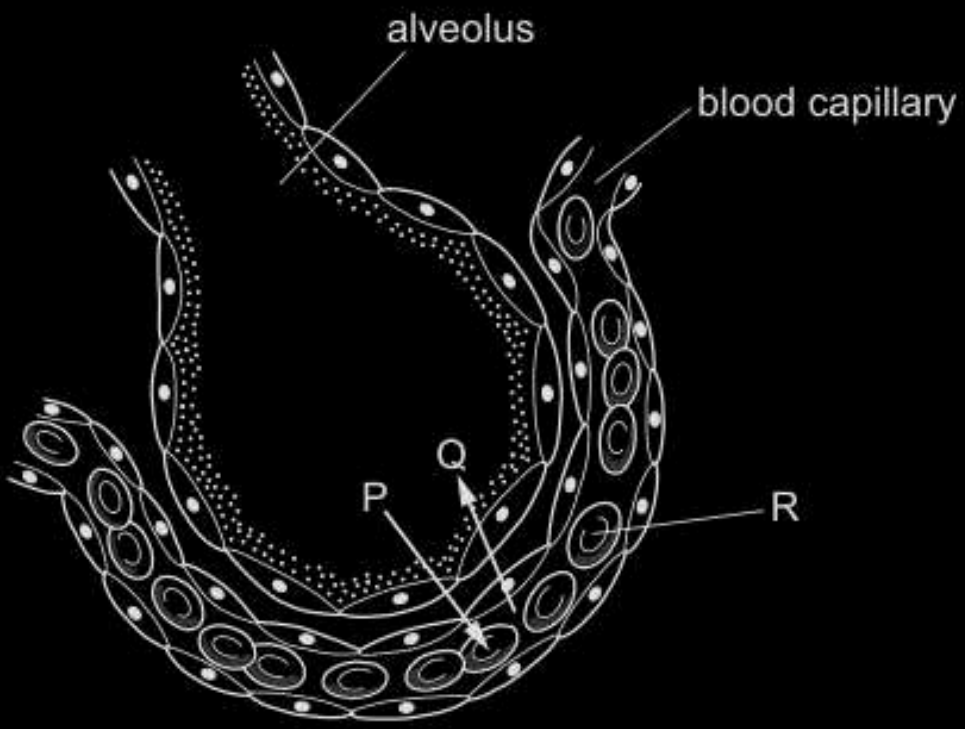
12 A dish is filled with agar jelly containing protein. Four holes are cut in the jelly and each hole is filled as shown in the diagram.

After 30 minutes, which hole will be surrounded by the largest area without protein?



Correct answer: C

14 The diagram shows an alveolus and a blood capillary.



Which row identifies the letters P, Q and R?

	P	Q	R
A	carbon dioxide	oxygen	red blood cell
B	carbon dioxide	oxygen	white blood cell
C	oxygen	carbon dioxide	red blood cell
D	oxygen	carbon dioxide	white blood cell

May/June 2024 · Paper 1 · Variant 2 · Question 16

Correct answer: C

16 A student measured their pulse rate in beats per minute (bpm) three times at rest and three times after running.

The table shows the results.

pulse rate at rest / bpm	pulse rate after running / bpm
62	152
66	157
63	155

What was their mean heart rate after running, to the nearest bpm?

A 64 bpm **B** 154 bpm **C** 155 bpm **D** 157 bpm

May/June 2024 · Paper 1 · Variant 2 · Question 17

Correct answer: A

17 A heart and lung bypass machine is used during heart surgery so that the heart can be stopped to allow an operation to take place.

Blood is diverted into the bypass machine from the blood vessel entering the heart from the body. The machine oxygenates the blood and pumps it back into the blood vessel leaving the heart to return it to the body.

Into which blood vessels are the tubes placed to remove the blood from the body to the bypass machine and to return the blood to the body?

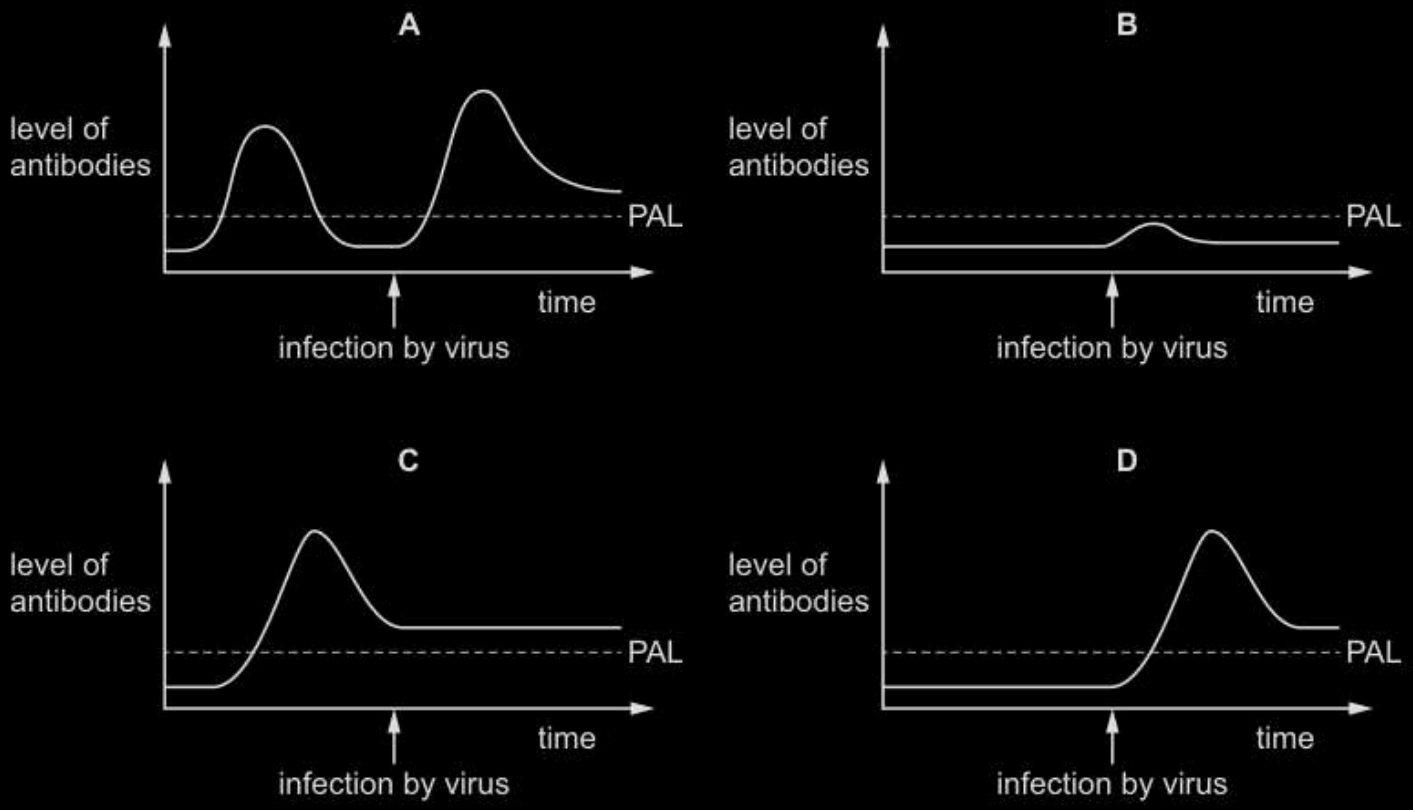
	to remove blood from the body	to return blood to the body
A	vena cava	aorta
B	vena cava	pulmonary artery
C	aorta	pulmonary vein
D	aorta	vena cava

May/June 2024 · Paper 1 · Variant 2 · Question 20

Correct answer: B

20 Four groups of people (**A**, **B**, **C** and **D**) were exposed to infection by a pathogenic virus. Researchers measured the level of antibodies in their blood before and after they were infected. The results are summarised in the graphs. PAL is the Protective Antibody Level – the level required to give protection from the virus.

Which group of people would have suffered the effects of the viral infection?



May/June 2024 · Paper 1 · Variant 2 · Question 21

Correct answer: B

21 What is an example of excretion?

A release of a hormone into the blood

B removal of carbon dioxide from the lungs

C removal of undigested food from the digestive system

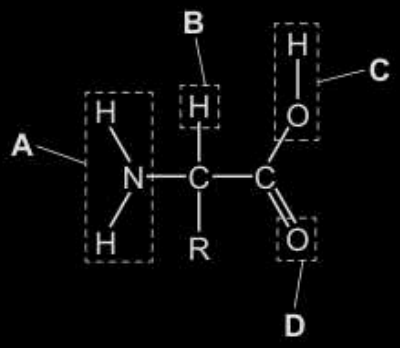
D release of water from the sweat glands

May/June 2024 · Paper 1 · Variant 2 · Question 22

Correct answer: A

22 The diagram shows an amino acid molecule.

Which part of the amino acid molecule is removed and used to make urea?



May/June 2024 · Paper 1 · Variant 2 · Question 23

Correct answer: C

23 What is a function of sensory neurones?

A transmitting impulses from muscles to the spinal cord

B transmitting impulses from receptors to muscles

C transmitting impulses from receptors to the spinal cord

D transmitting impulses from the spinal cord to muscles

Correct answer: D

24 Descriptions of changes that occur in the eye are listed.

1 Ciliary muscles contract.

2 Ciliary muscles relax.

3 Suspensory ligaments become slack.

4 Suspensory ligaments tighten.

5 The lens becomes more spherical.

6 The lens becomes thinner.

7 Light rays are refracted less.

8 Light rays are refracted more.

Which row lists the changes that occur when focusing on distant and near objects?

	focusing on a distant object	focusing on a near object
A	1, 3, 5, 8	2, 4, 6, 7
B	1, 4, 6, 8	2, 3, 5, 7
C	2, 3, 5, 7	1, 4, 6, 8
D	2, 4, 6, 7	1, 3, 5, 8

May/June 2024 · Paper 1 · Variant 2 · Question 25

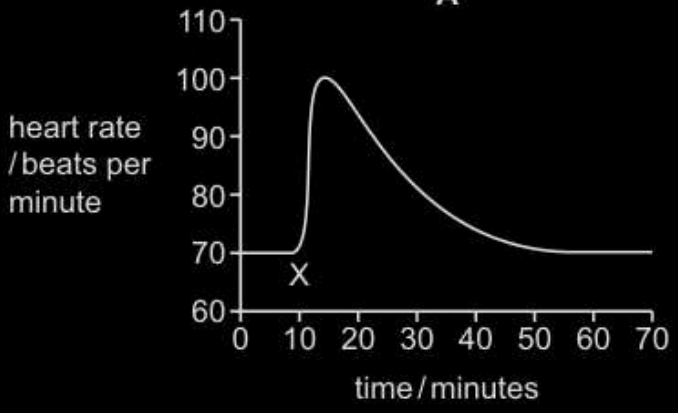
Correct answer: A

25 A student is walking along a road when a friend, who is hiding, jumps out suddenly giving her a shock.

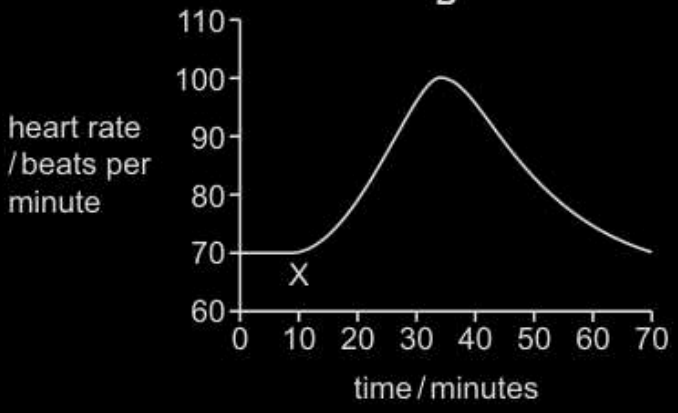
Which graph shows the effect of adrenaline on the heart rate of the student as a result of the shock?

(X on the graph is the point at which the student's friend jumps out suddenly.)

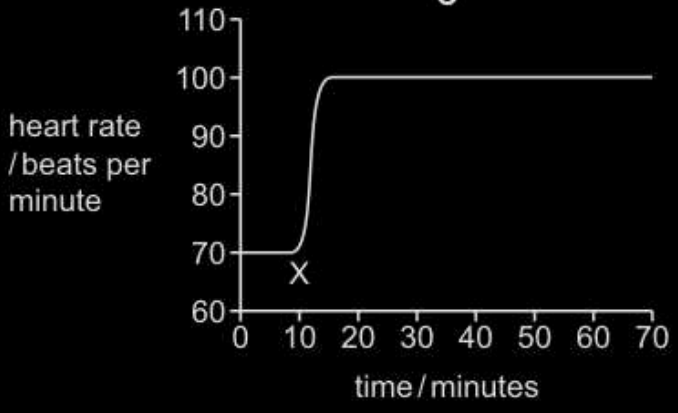
A



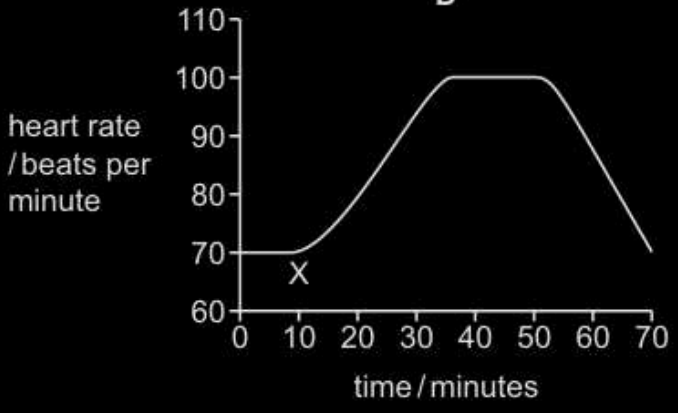
B



C



D



May/June 2024 · Paper 1 · Variant 2 · Question 26

Correct answer: B

26 What is a sign of Type 1 diabetes?

A lack of haemoglobin

B raised blood glucose levels

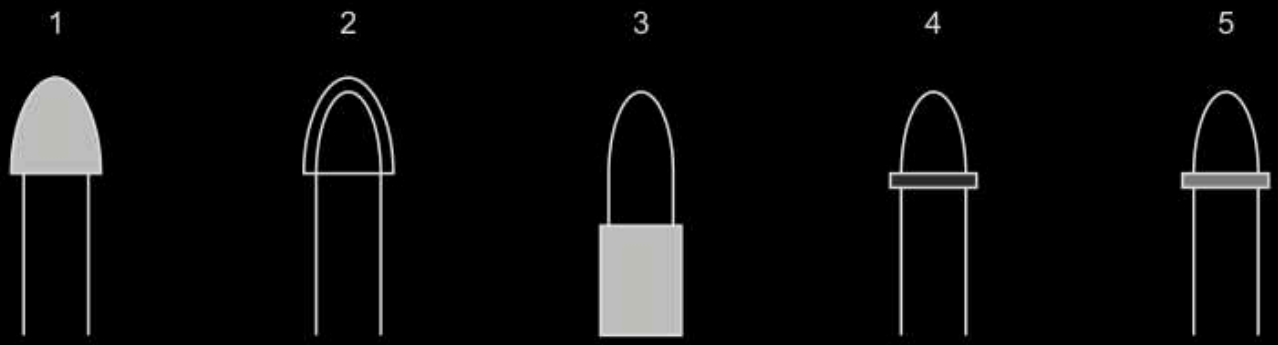
C reduced urine production

D too much insulin

May/June 2024 · Paper 1 · Variant 2 · Question 27

Correct answer: B

27 Five experiments were carried out to investigate the phototropic response of shoots. The diagrams show how the growing shoots of the plants were treated.



1	2	3	4	5
black paper cap covering shoot tip	transparent cap covering shoot tip	black paper collar covering shoot stem	permeable layer inserted between shoot tip and stem	impermeable layer inserted between shoot tip and stem

Which shoot tips showed a positive phototropic response when light was shone on them from one side?

A 1 and 2 **B** 2, 3 and 4 **C** 2, 4 and 5 **D** 2 and 4 only

May/June 2024 · Paper 1 · Variant 2 · Question 28

Correct answer: D

28 A student wrote some statements about chromosomes but made a number of mistakes.

1 There are 46 pairs of chromosomes in a human body cell.

2 In gametes, chromosomes are found in pairs.

3 Males have one X and one Y chromosome in each body cell.

4 Chromosomes contain a long DNA molecule divided into sections called genes.

5 Chromosomes include genes which are divided into sections called DNA molecules.

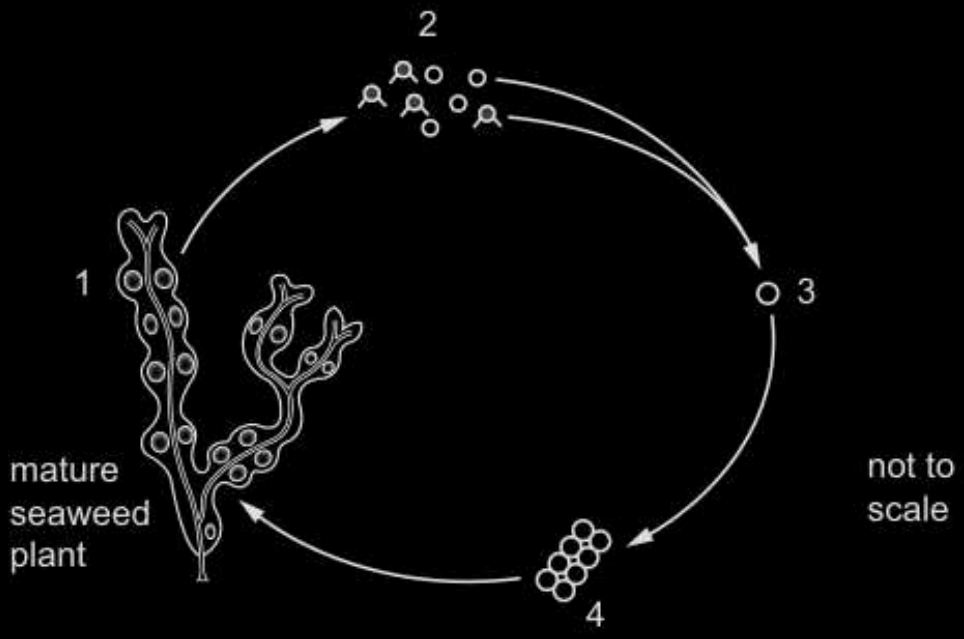
Which two statements are correct?

A 1 and 2 **B** 1 and 3 **C** 2 and 5 **D** 3 and 4

May/June 2024 · Paper 1 · Variant 2 · Question 29

Correct answer: B

29 A seaweed lives in shallow water around the coast. Stages in its life cycle are shown.



At which stages in the life cycle are its cells diploid?

A 1, 2, 3 and 4

B 1, 3 and 4 only

C 2 and 3 only

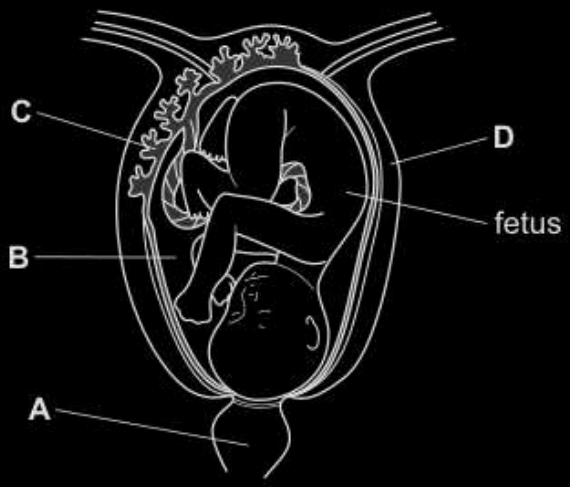
D 2 only

May/June 2024 · Paper 1 · Variant 2 · Question 32

Correct answer: C

32 The diagram shows the female reproductive system during pregnancy.

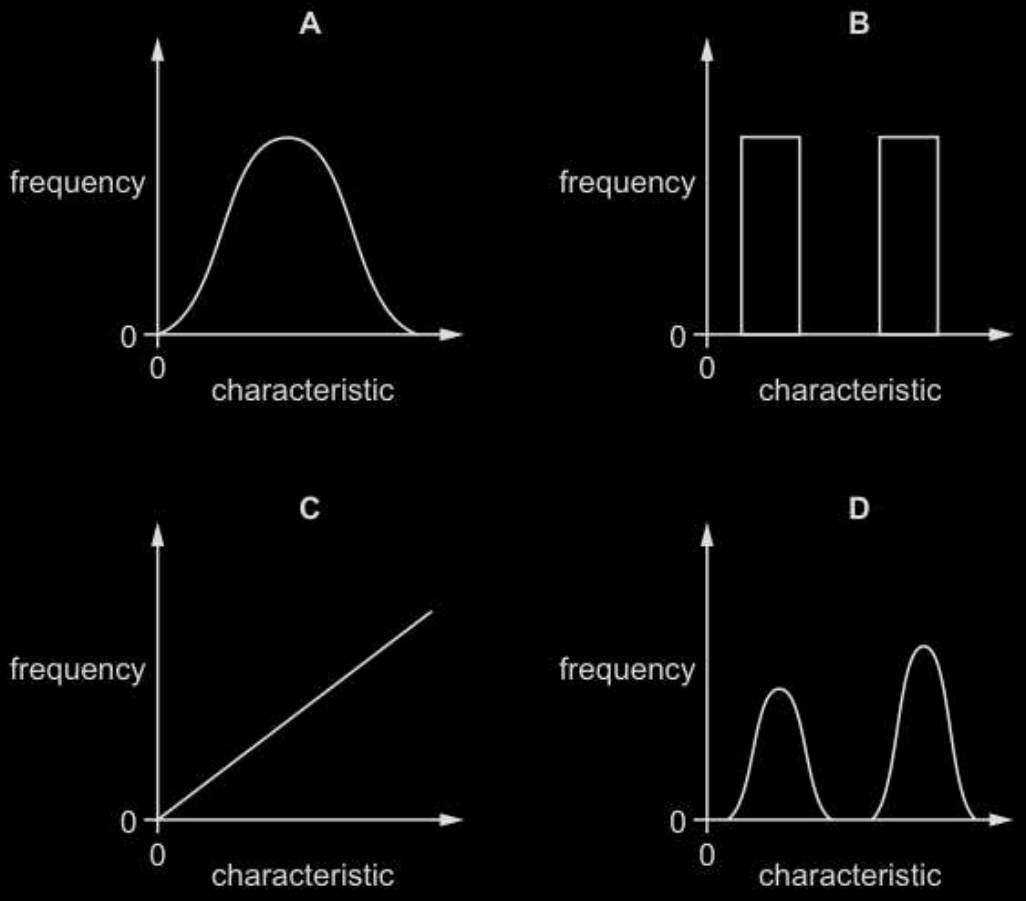
Which labelled part removes the excretory products of the fetus?



Correct answer: C

33 Three of the four graphs shown were constructed from data collected about the variation in particular characteristics in a population.

Which graph was **not** constructed from this kind of data?

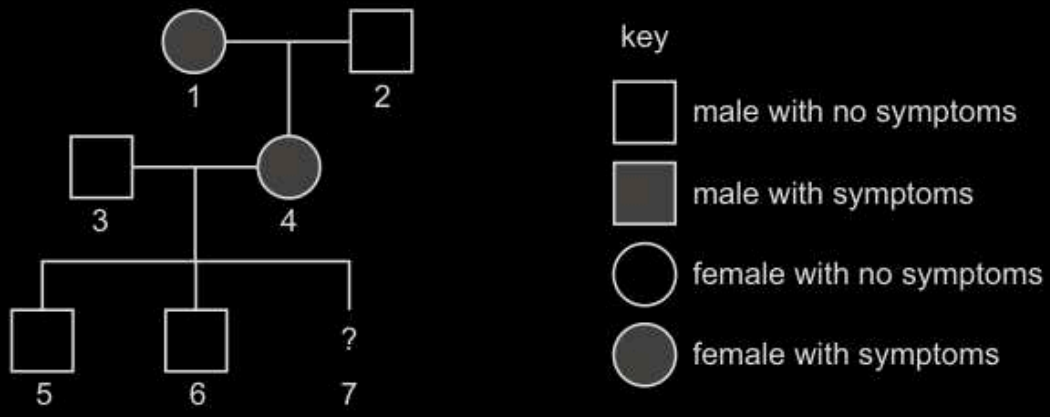


May/June 2024 · Paper 1 · Variant 2 · Question 34

Correct answer: B

34 The diagram shows three generations of a family tree in which an inherited condition that affects the nervous system occurs.

This condition is caused by a dominant allele that normally shows its effect in mature adults.



The woman, 4, shows symptoms of this condition while she is pregnant.

What is the chance that the new baby, 7, will be a girl who will also develop the condition later in life?

A 0.00 **B** 0.25 **C** 0.50 **D** 1.00

May/June 2024 · Paper 1 · Variant 2 · Question 35

Correct answer: C

35 Which process involves reproduction between those members of a species that are best fitted to their environment?

A discontinuous variation

B gene mutation

C natural selection

D survival of the fittest

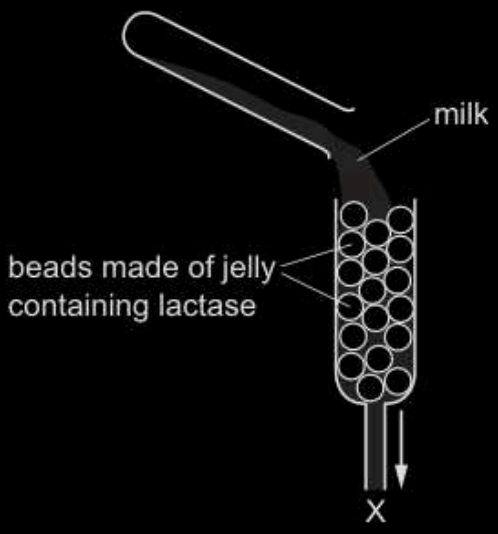
May/June 2024 · Paper 1 · Variant 2 · Question 36

Correct answer: B

36 In an industrial process, milk is poured over beads made of jelly that contain the enzyme lactase.

As the milk passes over the beads, it will come into contact with lactase.

Any digestion of the milk is completed by point X.



What is collected at X?

A milk containing glucose and lactose

B milk containing sugars but no lactose

C milk containing lactose only

D milk containing lactase but no sugars

May/June 2024 · Paper 1 · Variant 2 · Question 38

Correct answer: C

38 Which process is involved in the conversion of ammonia to nitrates?

A denitrification

B excretion

C nitrification

D nitrogen fixation

1 The lungs are part of the human gas exchange system.
Breathing moves air into and out of the lungs.

(a) (i) Fig. 1.1 shows parts of the human gas exchange system and their functions.
On Fig. 1.1, draw a line from each part to its function.
One has already been done for you.

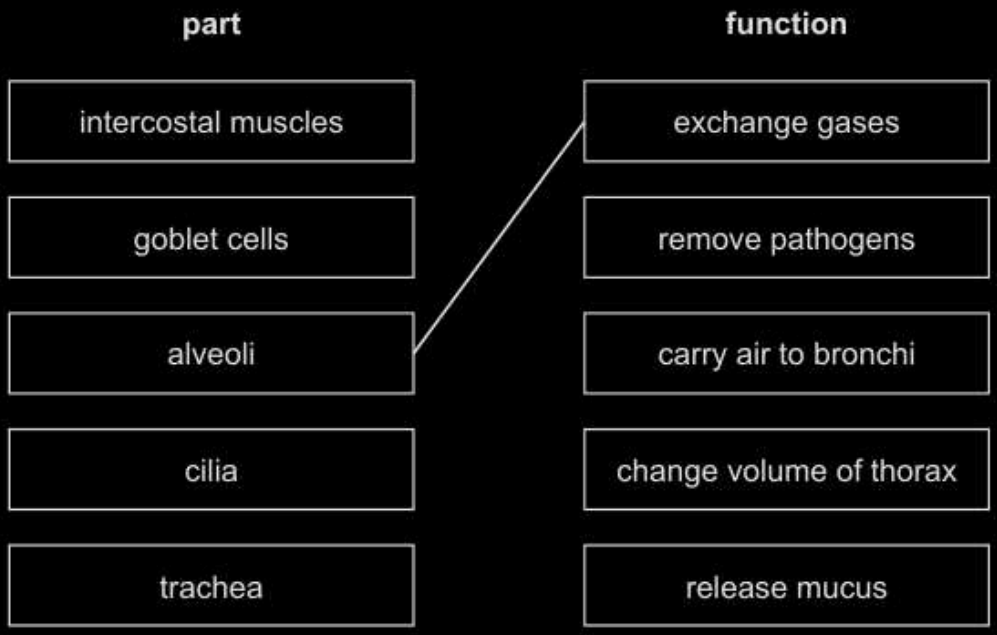


Fig. 1.1

[4]

(ii) State **two** features of alveoli that make them suitable for gas exchange.

1 .....

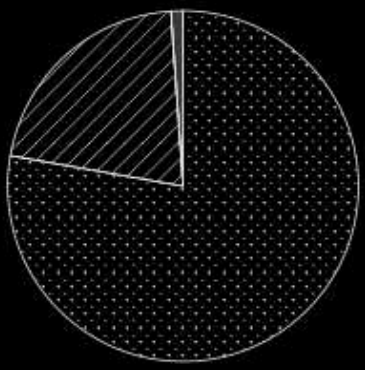
2 .....

[2]

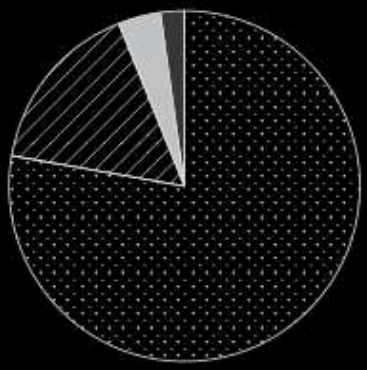
(b) Gas exchange changes the percentages of some gases in the lungs.
The pie charts in Fig. 1.2 show how the percentages of gases in the lungs change during one breath.

Complete the pie chart titles and the key which shows which gases are represented by different types of shading.

..... air



..... air

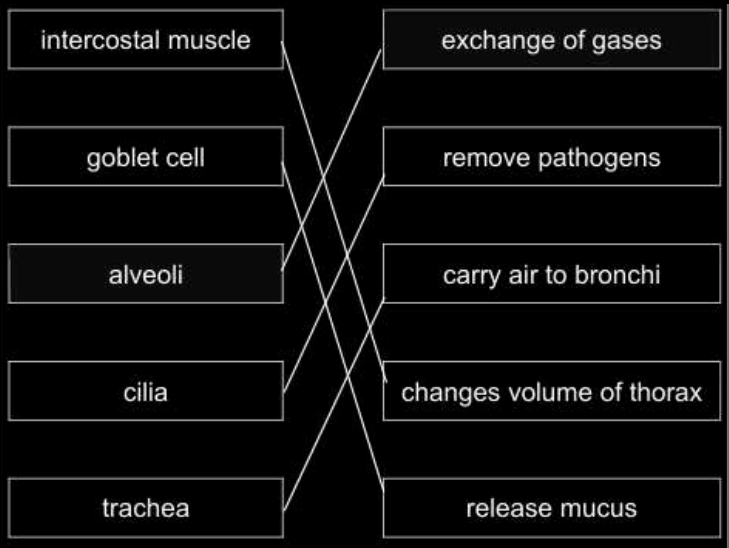


key	
	.....
	.....
	.....
	other gases

Fig. 1.2

[4]

[Total: 10]

1(a)(i)	 four correct = 4 marks three correct = 3 marks two correct = 2 marks one correct = 1 mark	4	
1(a)(ii)	large surface area ; thin/flattened **+** cells / walls ; layer of water **AW** ; close to capillaries ; **Max 2**	**2**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
1(b)	inspired/inhaled air **+** expired/exhaled air in correct order ; nitrogen ; oxygen ; carbon dioxide ;	**4**	

Question	Answer	Marks	Guidance

2 Brown bread, made from the flour of whole wheat grains, contains both starch and fibre.

(a) **(i)** A man eats a piece of brown bread.
In his digestive system, the starch in the bread is chemically digested to maltose and then to glucose.

Describe where and how this digestion takes place as the bread travels through his digestive system.

.....

.....

.....

.....

.....

..... [4]

(ii) Describe what happens to the fibre as it passes through his digestive system.

.....

.....

..... [2]

(b) Glucose passes from the digestive system into the blood by diffusion and active transport.

Give **two** ways in which active transport is different from diffusion.

1 .....

.....

2 .....

.....
[2]

(c) Scientists have found that there are different types of starch which have molecules of different shapes. Some molecules are digested rapidly and others are digested slowly.

The graph in Fig. 2.1 shows the effect on the concentration of glucose in the blood after eating two types of starch.

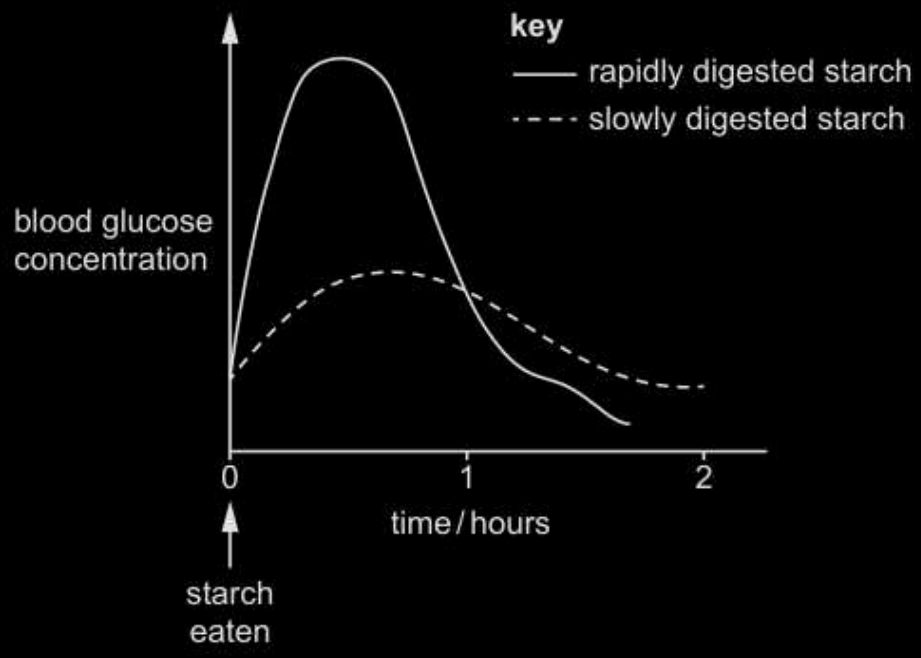


Fig. 2.1

(i) There is already glucose in the blood when the starch is eaten.
Explain how Fig. 2.1 shows this.

.....

..... [1]

(ii) Both graph lines show an initial increase in blood glucose concentration followed by a decrease.
Suggest **two** reasons why the blood glucose concentrations decrease.

1 .....

.....

2 .....

.....
[2]

(iii) Use information from Fig. 2.1 to suggest which type of starch is healthier to have in the diet. Give a reason for your answer.

.....

.....

..... [2]

(d) Scientists want to produce a new wheat variety with an increased fibre content.
Describe **two** methods they could use to do this.

1 .....

.....

.....

2 .....

.....

.....

[4]

[Total: 17]

2(a)(i)	amylase acts on starch / amylase produces maltose ; amylase **+** in the mouth **/** small intestine / duodenum/ileum ; maltase acts on maltose / maltase produces glucose ; maltase **+** in the small intestine / duodenum / ileum ;	**4**	
2(a)(ii)	not digested / not broken down ; is egested / passes out of anus / in faeces ;	**2**	
2(b)	(active transport / it) requires energy / ATP ; (active transport / it) moves molecules against / up a concentration gradient ; (active transport / it) requires carrier proteins ; **Max 2**	**2**	
2(c)(i)	the graph line does not start at the origin / the graph line starts part-way up the y-axis ;	**1**	
2(c)(ii)	glucose being converted to glycogen / insulin released ; glucose being used in respiration / glucose is used for energy ;	**2**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
2(c)(iii)	*slowly-digested starch* less variation in blood glucose concentration **ORA**; pancreas not having to produce so much insulin **ORA** ; less likely to develop diabetes **ORA** ; **Max 2**	**2**	
2(d)	artificial selection / selective breeding ; select / breed **+** plants with greatest fibre content; genetic modification ; transfer / insert / alter **+** gene / allele **+** so plant produces high fibre ;	**4**	

Question	Answer	Marks	Guidance

3 Fig. 3.1 is a photomicrograph showing part of a capillary network of a woman.
The structure labelled **X** is a red blood cell.

Fig. 3.1

(a) (i) The diameter of cell **X** in the photomicrograph is 10 mm.
Calculate the **actual** diameter of cell **X**.
Show your working and express your answer in micrometres (μm).

actual diameter of cell **X** = ... [3]

(ii) **W** and **X** are both red blood cells.
Explain why **W** and **X** appear different from each other in Fig. 3.1.

.....

.....

.....

..... [2]

(iii) **Y** and **Z** are both liquids.
Name the liquids **Y** and **Z** and explain how liquid **Z** forms from liquid **Y**.

Y .....

Z .....

explanation .....

.....

.....

..... [3]

(b) The woman has malaria and symptoms of anaemia.
The malarial pathogens infect red blood cells where they multiply to form many new, genetically identical pathogen cells. These new pathogen cells burst out of the red blood cells.

(i) Describe how malarial pathogens can be transmitted from this woman to another person.

.....

.....

.....

.....

..... [3]

(ii) Name the type of nuclear division that results in new pathogen cells.

..... [1]

(iii) Suggest why infection with the malarial pathogen can cause anaemia.

.....

..... [1]

(iv) State **one** other cause of anaemia.

.....

..... [1]

[Total: 14]

3(a)(i)	10 / 1250 or correct equation ; 0.008 (mm) or 8×10^{-3} ; 8 (μm) ; (3 marks for correct answer)	**3**	
3(a)(ii)	both same shape **AW** ; **W** side view / **X** is a view from above / seen from different angles ;	**2**	
3(a)(iii)	**Y** = plasma **+ Z** = tissue fluid ; plasma / **Y** / blood **+** under pressure / squeezed ; plasma / **Y** passes out of capillaries to form tissue fluid ;	**3**	

Question	Answer	Marks	Guidance
3(b)(i)	any **three** from: mosquito ; vector ; sucks up blood from infected woman / sucks up blood containing pathogen ; injects pathogen **+** into different person ; **Max 3**	**3**	
3(b)(ii)	mitosis ;	**1**	
3(b)(iii)	red blood cells destroyed / reduces the number of red blood cells ;	**1**	
3(b)(iv)	iron deficiency / lack of iron / heavy loss of blood through injury / menstruation ;	**1**	

Question	Answer	Marks	Guidance

4 Some plants can survive extreme dehydration for a long time in a dormant state. They start growing again when water is available. They are adapted to survive in deserts. *Selaginella lepidophylla* is an example of this type of plant.

(a) Fig. 4.1 shows a *S. lepidophylla* plant after several months without water and the same plant three days after rainfall.

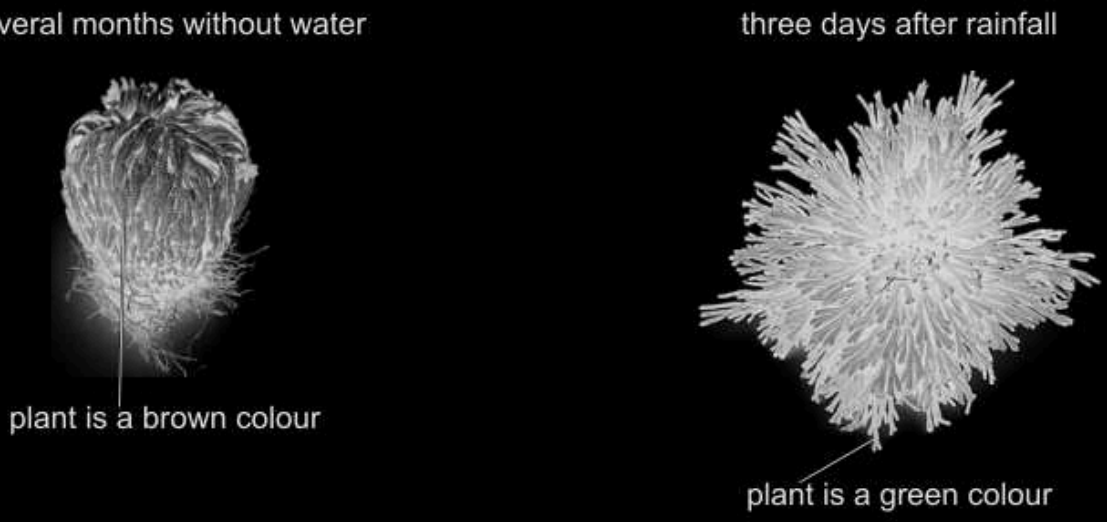


Fig. 4.1

(i) Explain how the changes in the plant, shown in Fig. 4.1, will help it to start growing again after rainfall.

.....

.....

.....

.....

.....

..... [4]

(ii) Fig. 4.2 shows the cell wall and nucleus of a cell from this plant after several months without water.

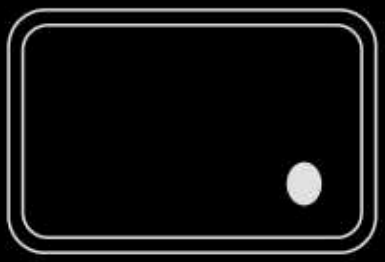


Fig. 4.2

Suggest and explain the appearance of other parts of this cell after several months without water.
You may draw on Fig. 4.2 to help you answer.

.....

.....

.....

.....

.....

..... [4]

(b) *Selaginella lepidophylla* is the scientific name for the plant species, which also has the common name 'resurrection moss'.

(i) Explain what is meant by the term **species**.

.....

.....

..... [2]

(ii) State the name of the scientific system used to name the species and suggest **one** advantage of using the scientific name rather than the common name.

name of system .....

advantage .....

.....
[2]

[Total: 12]

4(a)(i)	Chlorophyll/chloroplasts ; larger surface area (after rainfall) ; photosynthesis ; carbohydrates / glucose / starch produced ; absorption / trap + light (energy) ; **Max 4**	**4**	
4(a)(ii)	loss of water molecules **+** osmosis / diffusion / evaporation ; cytoplasm shrinks ; sap vacuole shrinks ; cell membrane pulls away from cell wall ; lack of turgor pressure ; plasmolysis ; **Max 4** **accept** marks from a labelled diagram	**4**	
4(b)(i)	(a group of organisms that) can reproduce / breed / mate ; produce offspring that are fertile ;	**2**	

Question	Answer	Marks	Guidance
4(b)(ii)	*name:* binomial ; *advantage: relevant advantage given* same name in all parts of the world / international name / common names can differ in different parts of the world / genus name helps identify closely related species ;	**2**	

Question	Answer	Marks	Guidance

5 *Bacillus subtilis* is a species of bacterium that is widely used in biotechnology.
It can be used to manufacture the protein lipase for industrial use.
Fig. 5.1 is a diagram of *B. subtilis*.

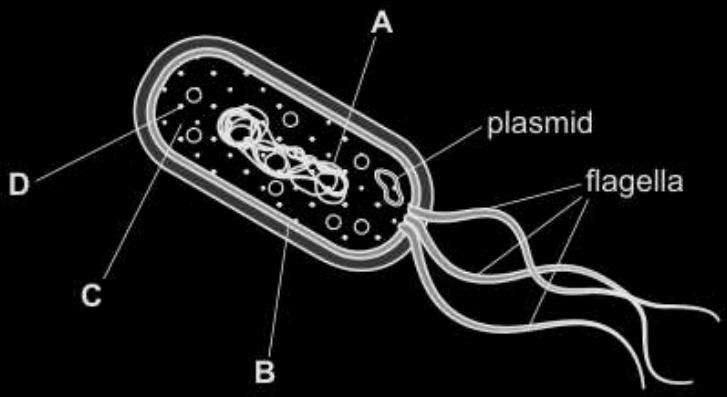


Fig. 5.1

(a) (i) Two of the parts labelled **A**, **B**, **C** and **D** in Fig. 5.1 are involved in the manufacture of proteins such as lipase.
Identify the **two** parts, name them and explain their role in protein manufacture.

letter ..... name .....

role .....

.....

letter ..... name .....

role .....

.....
[4]

(ii) Suggest why *B. subtilis* has flagella.

.....

..... [1]

(iii) Explain why plasmids are useful in biotechnology.

.....

.....

..... [2]

(b) *B. subtilis* can be grown industrially in large containers.
When *B. subtilis* is in ideal conditions it can reproduce rapidly, increasing the population size and the biomass of bacteria in the container.

Graphs **E** and **F** in Fig. 5.2 show how the biomass of *B. subtilis* increases after small populations of bacteria are added to two containers with different environmental conditions.

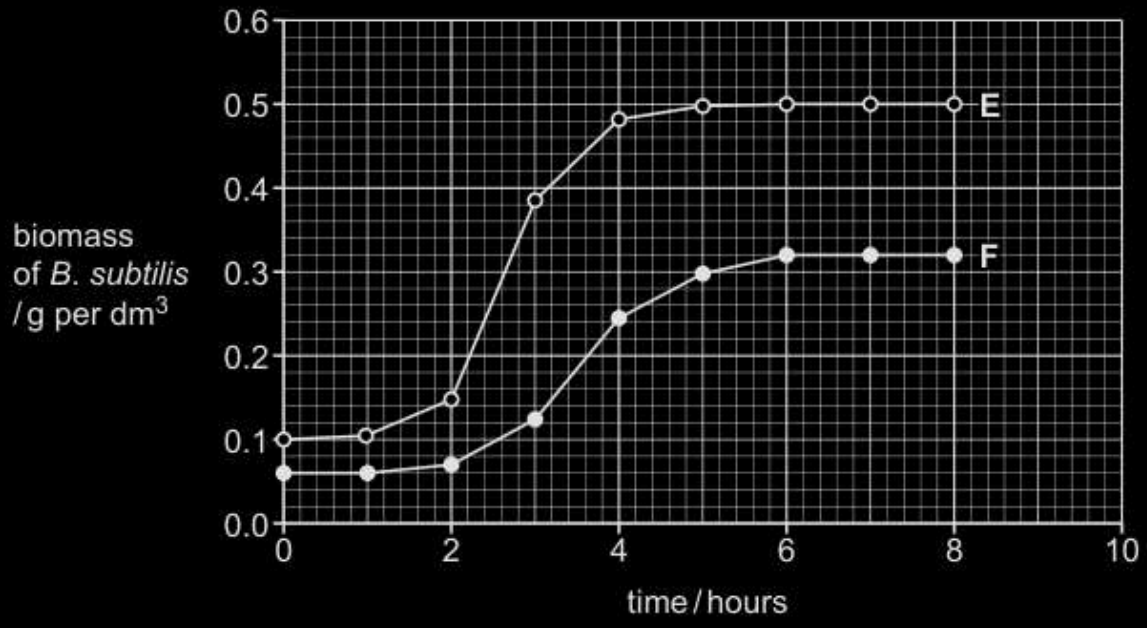


Fig. 5.2

(i) Give the name of the large industrial containers used to grow bacteria.

..... [1]

(ii) In graph **E** the biomass increases as shown in Fig. 5.2.
Indicate using the letter **M** the part of graph **E** where bacteria are reproducing **most** rapidly. [1]

(iii) Calculate the percentage increase in biomass for graph **E** in the first 6 hours.
Space for working.

percentage increase in biomass = ...% [2]

5(a)(i)	**A +** DNA / genetic material ; **D +** ribosome ; Part **A**: contains the gene for the protein / lipase / codes for the production of the protein / lipase / mRNA / codes for sequence of amino acids in protein / lipase ; Part **D**: adds amino acids together to build protein ;	**4**	
5(a)(ii)	motile / can move ;	**1**	
5(a)(iii)	genes / DNA can be inserted into them ; can pass into a (bacterial) cell / vector / can carry a gene into a (bacterial) cell ;	**2**	
5(b)(i)	fermenter ;	**1**	
5(b)(ii)	**M** indicated between 3rd and 4th plots ;	**1**	
5(b)(iii)	0.4 / 0.1 × 100 ; 400 (%) ; (2 marks for correct answer)	**2**	

Question	Answer	Marks	Guidance
5(b)(iv)	**E** has higher final biomass / greater increase in biomass (than **F)**ORA ; **E** reaches max biomass quicker (than **F)** ORA ; **E** starts at a higher biomass (than **F**) ORA ; **E** rate of increase of biomass is faster (than **F**) ORA ; **Max 2** **Accept answers that quote figures**	**2**	
5(b)(v)	pH / oxygen / food supply / nutrients / temperature ;	**1**	
5(c)	lipase digests / hydrolyses / breaks down **+** lipids / fats ; fatty acids and glycerol produced ; bile emulsifies / increases surface area of lipids ; bile neutralises acidic food / provides the optimum pH for digestion ; **Max 3**	**3**	

Question	Answer	Marks	Guidance

6 A beam of bright light shines into a human eye for two seconds.
A similar beam of light shines onto one side of the tip of a plant stem for several days.

(a) Describe the ways in which the human and the plant coordinate their responses to these beams of light.

Your answer should include:

- stimulus detection
- method of responding.

human .....

.....

.....

.....

plant .....

.....

.....

.....

[6]

(b) Describe the role of synapses in a reflex arc.

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

6(a)	*Human:* detected by retina / light receptors; involves neurones / nervous system / electrical impulses; response is muscular contractions (in iris) ; causing constriction / reduction in size of the pupil ; *Plants:* detected by stem tip ; involves hormones / auxin ; more moves / diffuses to the side away from light ; causing more growth / elongation **+** on the shaded side ; positive phototropism / plant stem grows/bends **+** towards the light ; **Max 6**	**6**	
6(b)	transmission of impulse between neurones ; in one direction ; neurotransmitter ; released (from vesicles) into (synaptic) gap ; diffusion **+** across gap ; binds with receptor (proteins) ; **Max 4**	**4**	

© Cambridge University Press & Assessment 2024

1 Fig. 1.1 shows two cells from the leaf of a plant observed using a light microscope.

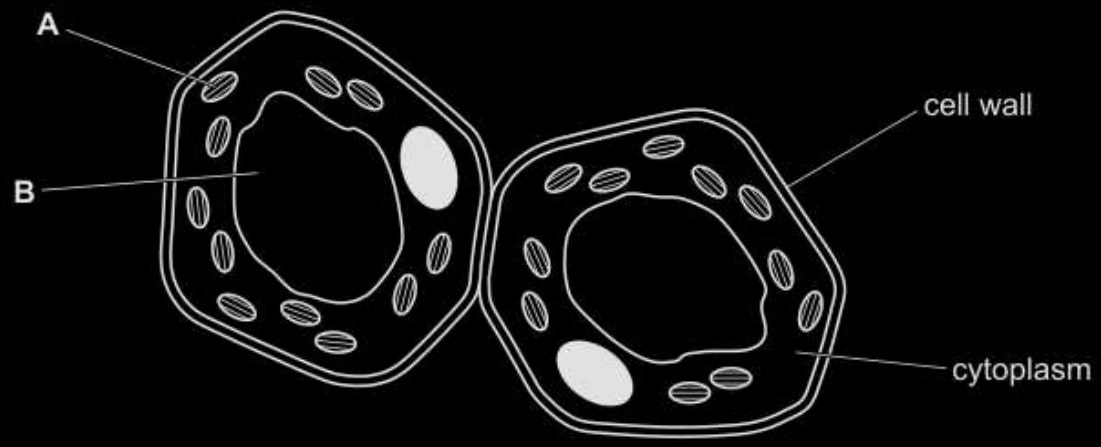


Fig. 1.1

(a) Complete Table 1.1 to identify the structures labelled **A** and **B** and to describe the function of each structure.

Table 1.1

structure	name of structure	description of function
A	.....	....
B	.....	....

[4]

(b) State **one** structure that is **not** visible in Fig. 1.1 but would be visible in the cytoplasm in an electron micrograph of the same two cells.

..... [1]

[Total: 5]

1(a)	**A**) chloroplast ; photosynthesis / light energy to chemical energy / make glucose / make starch ; (**B**) (sap) vacuole ; storage / turgor / support	**4**	
1(b)	ribosomes ; mitochondria ; **max 1** **accept** plant cell structures not on syllabus e.g. rough ER, Golgi apparatus etc.	**1**	

Question	Answer	Marks	Guidance

2 Organisms can be classified into groups using the features they share.

(a) Fish are one of the main groups of vertebrates.

(i) State **three** main features used to classify an organism as a fish.

1 .....

2 .....

3 .....

[3]

(ii) State **two** main groups of vertebrates other than fish.

1 .....

2 .....

[2]

(b) Swordfish are large fish that live in the ocean.

They have a long, bony extension to the skull, called a sword.

Fig. 2.1 is a diagram of a swordfish.

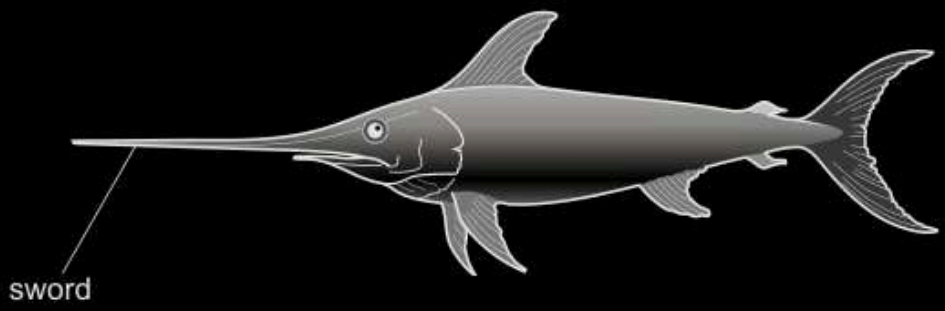


Fig. 2.1

The binomial name of the swordfish is *Xiphias gladius*.

(i) Define the term species.

.....

.....

..... [2]

(ii) State the species name of the organism shown in Fig. 2.1.

..... [1]

(c) Swordfish swim near the surface of the ocean where the water temperature is low.
During evolution, swordfish developed a muscle behind each eye that warms the eyes and the brain to a temperature above that of the ocean.
This feature gives swordfish improved vision.

(i) Name the process that led to the evolution of the muscle behind each eye of the swordfish.

..... [1]

(ii) Suggest reasons why the evolution of this feature has adapted swordfish to survive in the ocean.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 13]

2(a)(i)	gills ; scales / scaly skin ; fins ; **accept** eggs with no shell / surrounded by jelly, cold-blooded, external fertilisation	**3**	
2(a)(ii)	mammals ; birds ; reptiles ; amphibians ; **max 2**	**2**	
2(b)(i)	(group of organisms that) can reproduce / breed / mate ; to produce <u>fertile</u> offspring ;	**2**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
2(b)(ii)	gladius ;	1	
2(c)(i)	natural selection ;	1	
2(c)(ii)	**1** see / hunt **+** prey / food ; **2** so more **+** food / energy ; **3** out-compete other animals ; **4** see **+** predators / named predator ; **5** avoid being eaten / attacked ; **6** see mates ; **7** reproduce / produce offspring / increase population ; **max 4**	4	

Question	Answer	Marks	Guidance

May/June 2024 · Paper 2 · Variant 2 · Question 3

3 (a) Complete Table 3.1, using terms from the list below to match each description to a term.

phenotype **dominant** **gene** **genotype**

heterozygous **homozygous** **allele** **recessive**

Table 3.1

description	term
a form of a gene that codes for one of a pair of contrasting features	.....
a form of a gene that always has an effect when it is present	.....
having two different forms of a gene for a particular feature	.....
having two of the same form of a gene for a particular feature	.....
the combination of alleles that an organism has in its chromosomes	.....

[5]

(b) Fig. 3.1 is a diagram of part of a molecule of DNA taken from a bacterial cell.

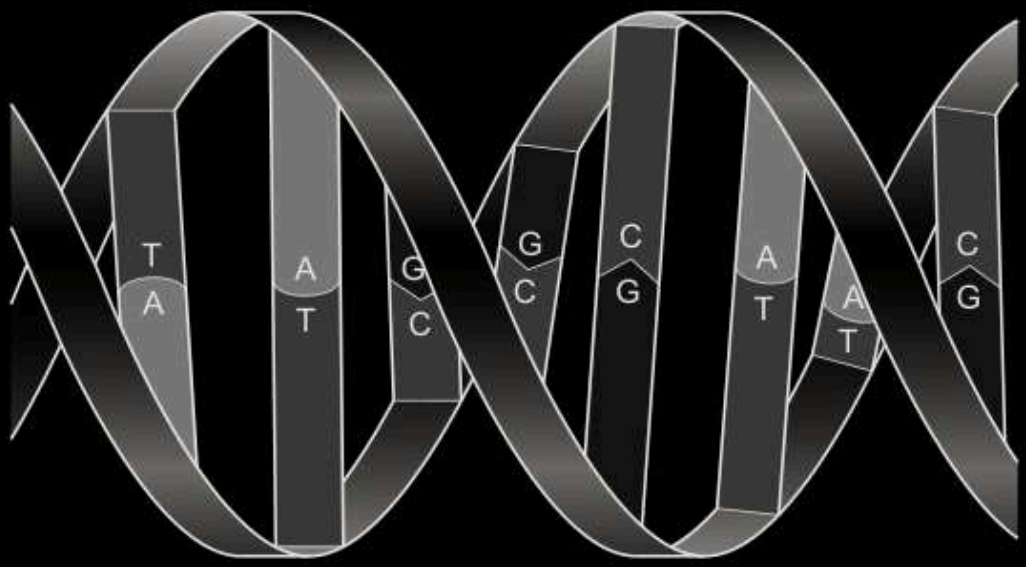


Fig. 3.1

(i) State the term which describes the shape of the DNA molecule.

.....

[1]

(ii) Most of the DNA in bacterial cells is found in one large loop in the cytoplasm.

Name another structure in bacterial cells which contains DNA.

.....

[1]

(iii) State the name of the **type** of molecule that is represented by the letters A, T, G and C in Fig. 3.1.

.....

[1]

(iv) Outline the importance of the sequence of A, T, G and C in DNA.

.....

.....

.....

..... [2]

[Total: 10]

3(a)	allele ; dominant ; heterozygous ; homozygous ; genotype ;	**5**	
3(b)(i)	double helix ;	**1**	
3(b)(ii)	plasmid ;	**1**	
3b(iii)	(nitrogenous / organic) base ;	**1**	
3(b)(iv)	codes for AW **+** proteins / named type of protein ; determines AW **+** sequence of amino acids ; determines AW **+** shape / structure / function of protein ; **max 2**	**2**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance

4 Fig. 4.1 is a simplified diagram of a nephron.

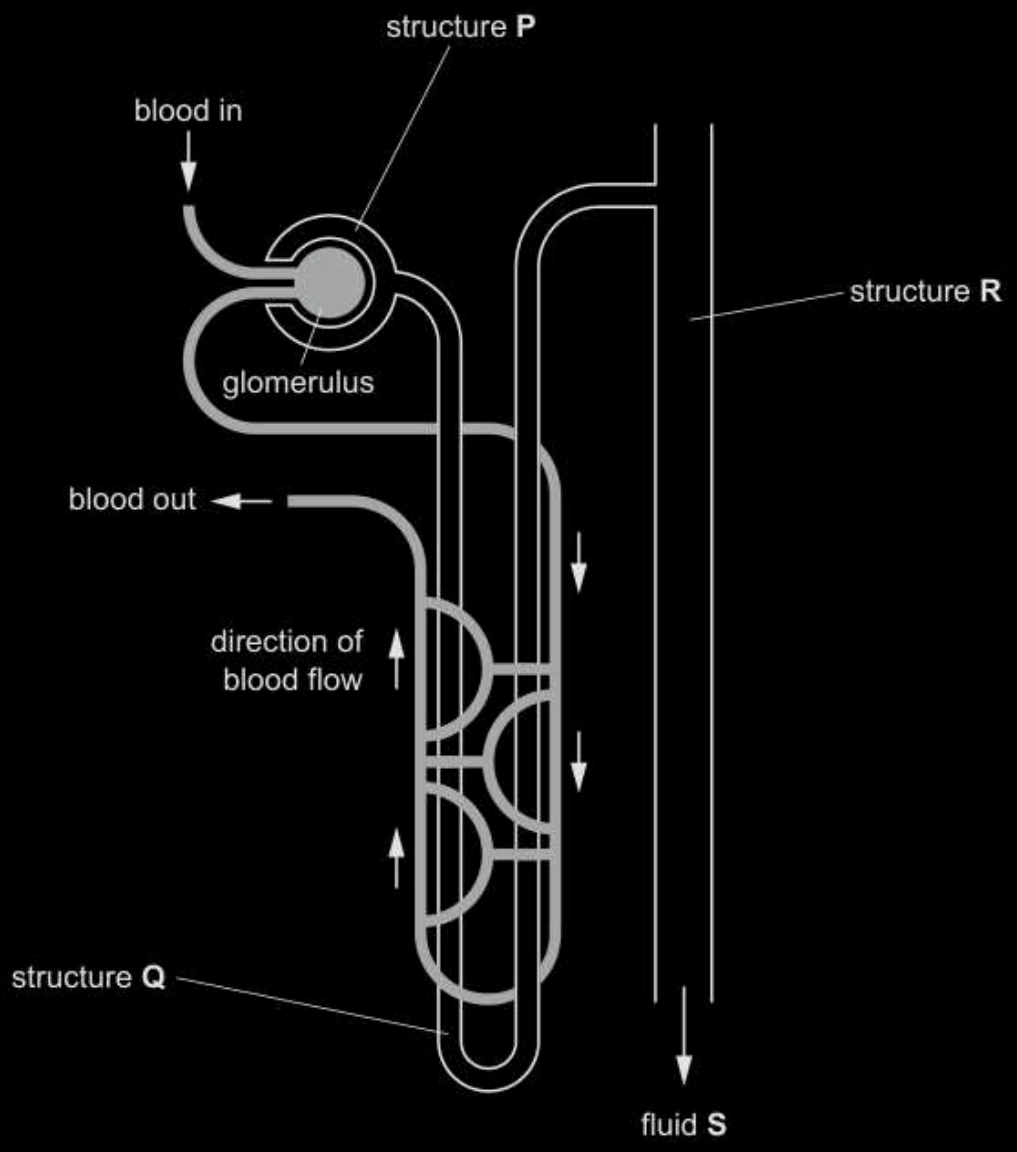


Fig. 4.1

(a) (i) Identify each of the structures labelled **P**, **Q** and **R** in Fig. 4.1.

P .....

Q .....

R .....

[3]

(ii) Name the organ which contains many nephrons.

.....

[1]

(iii) Name the organ that stores fluid **S** before it is released from the body.

.....

[1]

(b) Table 4.1 shows the composition of a sample of fluid **S** from a healthy person.

The person that provided the sample of fluid **S** eats a balanced diet.

Table 4.1

component	concentration / arbitrary units
glucose	0.00
protein	0.00
ions	1.50
urea	2.00

(i) State why it is important that urea is removed from the blood in fluid **S**.

..... [1]

(ii) Use your knowledge of the function of a nephron to explain the concentration of each of the following components of fluid **S**.

glucose

.....

.....

.....

.....

protein

.....

.....

.....

[5]

(iii) Explain what would happen to the concentration of ions in fluid **S** if the person drank a large volume of water several hours before the sample was collected.

.....

.....

.....

..... [3]

[Total: 14]

4(a)(i)	(**P**) Bowman's capsule ; (**Q**) loop of Henle ; (**R**) collecting duct ;	**3**	
4(a)(ii)	kidney ;	**1**	
4(a)(iii)	(urinary) bladder ;	**1**	
4(b)(i)	toxic / toxin ;	**1**	
4(b)(ii)	*glucose* **1** enters Bowman's capsule / enters nephron / removed from blood AW ; **2** by filtration / filtered; **3** back into blood / reabsorbed **+** into blood ; **4** AVP ; *protein* **5** doesn't enter Bowman's capsule / doesn't enter nephron / no filtration / not removed from blood AW ; **6** protein **+** large AW ; **max 5**	**5**	
4(b)(iii)	ion concentration **+** decrease ; more water / higher volume of water / excess water **+** in fluid **S** / urine / excreted; homeostasis / to keep concentration of water in blood constant AW;	**3**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance

5 Fig. 5.1 is a diagram of a pineapple growing on a pineapple plant.

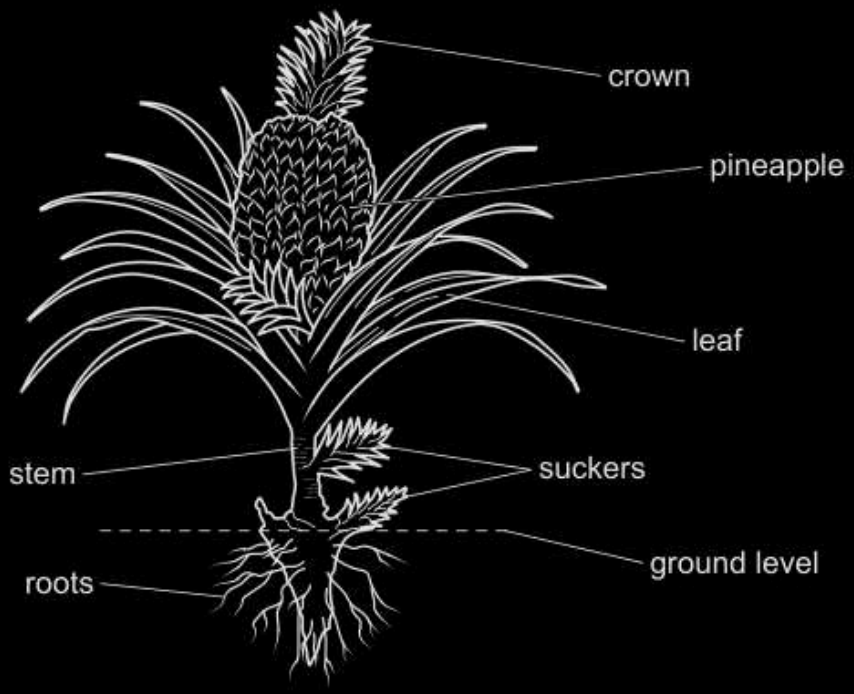


Fig. 5.1

(a) Pineapple plants can be grown on farms by removing the suckers from a parent plant and planting these in the ground to form new plants.

(i) State the type of reproduction used to grow pineapple plants in this way.

.....

[1]

(ii) State **two** advantages and **two** disadvantages of producing pineapple plants in this way.

advantages

1 .....

.....

2 .....

.....

disadvantages

1 .....

.....

2 .....

.....

[4]

(iii) Pineapples are an important crop in Hawaii, USA. They were introduced to Hawaii from South America.

Birds called hummingbirds pollinate pineapple flowers. If pollinated, pineapple fruits develop and they contain hard, inedible seeds. If **not** pollinated, pineapple fruits develop without seeds.

Hummingbirds are **not** found in Hawaii.

Suggest why it is illegal to introduce hummingbirds into Hawaii.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Pineapples contain a mixture of protease enzymes called bromelain.

(i) When bromelain is added to its substrate, a product is formed.

Complete the word equation to show the substrate and product.

..... ⟶ .....

(substrate) (product)

[2]

(ii) The effect of changing pH on the activity of the enzymes found in bromelain was investigated. The results of this investigation are shown in Fig. 5.2.

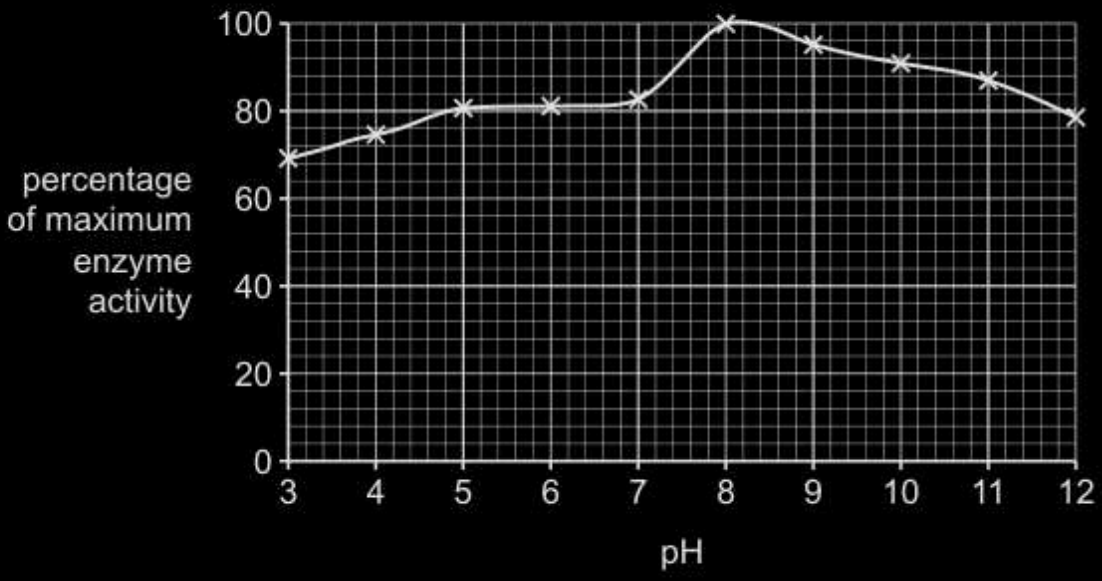


Fig. 5.2

To assist digestion, some people swallow bromelain in the form of a powder with their food.

Use the information in Fig. 5.2 and your knowledge of the human digestive system to discuss the effectiveness of using bromelain in this way.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 15]

5(a)(i)	asexual ;	**1**	
5(a)(ii)	*advantages* rapid AW ; many new plants formed / greater yield ; only one parent plant needed ; plants have same desirable AW **+** feature / allele / named useful feature ; ripen at the same time so easier to harvest ; **max 2** *disadvantages* have same undesirable AW **+** feature / allele / named useful feature ; not adapted if change in environment AW ; ripen at the same time so price goes down / not a constant supply to market ; **max 2**	**4**	
5(a)(iii)	*with hummingbirds* : sexual reproduction / pollination / fertilisation ; seeds formed ; humans / consumers **+** don't like / find inedible AW ; reduced income / business / economy AW ; hummingbird introduces diseases ; hummingbird **+** competes with other species ; (competes) for named resource ; **max 4**	**4**	
5(b)(i)	protein ; ⟶ amino acids ;	**2**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
5(b)(ii)	**1** increased surface area of powder / bromelain ; **2** more enzymes / higher surface area **+** more activity / collisions AW ; **3** optimum / highest activity **+** pH 8 ; **4** high activity across all pH range / pH 3–12 ; **5** bromelain / enzymes **+** not denatured ; **6** stomach **+** acid / H*Cl* / pH 2 ; **7** small intestine / duodenum / ileum **+** alkaline; **8** effective / active **+** in stomach and small intestine ; **max 4**	**4**	

Question	Answer	Marks	Guidance

6 Flowering plants are sometimes grown in soil to provide cut flowers that can then be displayed.

(a) A gardener watered flowering plants with a fertiliser solution containing dissolved chemicals.

Table 6.1 shows the information printed on the label of the container of fertiliser powder used to prepare this fertiliser solution.

Table 6.1

chemical component	mass / g per 100 g fertiliser powder
nitrates	4.0
phosphates	4.0
sulfur	1.0
magnesium	0.5

(i) To prepare the fertiliser solution, the gardener dissolved 20 g of the fertiliser powder in 10 dm^3 of water.

Calculate the total mass of sulfur in the fertiliser solution prepared by the gardener.

Space for working.

..... [2]

(ii) Explain which chemical component of the fertiliser solution is the most important to plants for:

making the leaves of the plants green in colour

.....

.....

the production of enzymes by cells in the leaves of the plants.

.....

.....

[4]

(iii) The chemical components of the fertiliser solution are absorbed by the root hair cells of the plants and are then transported to the leaves.

Complete the words below to name the three main tissues in the pathway taken by the chemical components of the fertiliser solution from the root hair cells to the leaves.

The first letter of each word has been completed for you.

C .....

↓

X .....

↓

M ..... [3]

(b) When flowers from the plants are cut and used for display, the leaves are removed before each stem is placed in water.

A solution containing sucrose is often added to this water.

Sucrose is **not** needed in fertiliser solution used to water plants growing in soil.

Explain the benefit of providing sucrose to cut flowers.

.....

.....

.....

.....

.....

..... [3]

[Total: 12]

6(a)(i)	0.2 ; g ;	**2**	
6(a)(ii)	magnesium ; chlorophyll (makes leaves green); nitrates ; (enzymes made from) amino acids / proteins ;	**4**	
6(a)(iii)	(c)ortex ; (x)ylem ; (m)esophyll ;	**3**	
6(b)	**1** no **+** photosynthesis ; **2** to produce glucose / glucose not produced ; **3** to produce sucrose / sucrose not produced ; **4** sucrose travels in phloem ; **5** for respiration / energy ; **6** flowers / plants **+** last longer / for days **AW**; **max 3**	**3**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance

7 Plastics are produced in very large amounts by industrial processes around the world.

These plastics can enter ecosystems as pollution.

(a) The total mass of plastics produced globally in the year 2020 was 367 million tonnes.

It is estimated that 3% of all plastics produced globally each year enters the ocean as plastic pollution.

Calculate the total mass of plastic pollution that is estimated to have entered the ocean during the year 2020.

Space for working.

..... [2]

(b) **(i)** Some packaging is made from biodegradable carbohydrates.

Microorganisms break down larger carbohydrate molecules into smaller molecules.

Identify this process and name **one** type of microorganism that is involved.

.....

.....

..... [2]

(ii) Most plastics that enter ecosystems as pollution are non-biodegradable.

Describe the harmful effects of non-biodegradable plastics on both aquatic and terrestrial ecosystems.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

[Total: 11]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

7(a)	11(.01) ; million tonnes ; **accept** 11 000 000 / 11 010 000 tonnes / tons ;; converted to standard form = 1.101×10^7 or 1.1×10^7 tonnes **accept** 110.1×10^5 tonnes and variations	**2**	
7(b)(i)	decomposition ; bacteria / fungi ;	**2**	
7(b)(ii)	**1** remains for a very long time / forever AW (in environment) ; **2** breaks up into AW **+** micro-plastics / small particles ; **3** strangles / traps / suffocates / injures **AW +** animals ; **4** animals eat it / chokes on it / blocks digestive systems AW ; **5** enters / affects food chains / webs ; **6** accumulates **+** in top predators / along food chain **AW** ; **7** covers habitat / covers plants / blocks light ; **8** less light for / no **+** photosynthesis ; **9** burning / combustion ; **10** toxins from plastics AW; **max 7**	**7**	

© Cambridge University Press & Assessment 2024

October/November 2023 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 What are characteristics of bacteria?

	contain cytoplasm	have DNA	nuclear membrane present
A	✗	✓	✗
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

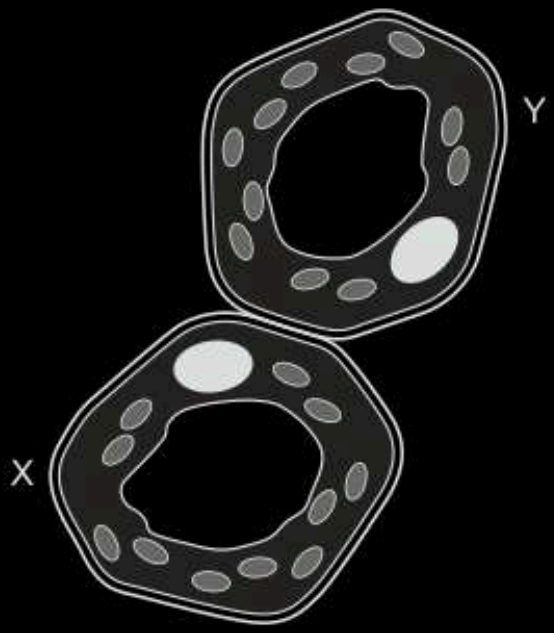
key
✓ = yes
✗ = no

October/November 2023 · Paper 1 · Variant 1 · Question 3

Correct answer: B

3 The diagram shows two plant cells, X and Y.

Cell X has a higher water potential than Y.



In which direction and by which process will water move between these two cells?

	direction	process
A	X to Y	active transport
B	X to Y	osmosis
C	Y to X	active transport
D	Y to X	osmosis

October/November 2023 · Paper 1 · Variant 1 · Question 4

Correct answer: A

4 Which row identifies how ions and molecules move by active transport?

	against the concentration gradient	down the concentration gradient	through a membrane	requiring energy
A	✓	✗	✓	✓
B	✗	✓	✗	✗
C	✗	✓	✓	✓
D	✓	✗	✓	✗

key

✓ = yes

✗ = no

October/November 2023 · Paper 1 · Variant 1 · Question 5

Correct answer: D

5 A student carried out two tests on a liquid. When Benedict's solution was added to half of the liquid and heated, it turned red. When biuret reagent was added to the rest of the liquid, it turned mauve (purple).

Which conclusion is correct?

A The liquid contained solutions of lipids and glucose.

B The liquid contained solutions of lipids and protein.

C The liquid contained solutions of starch and glucose.

D The liquid contained solutions of glucose and protein.

October/November 2023 · Paper 1 · Variant 1 · Question 6

Correct answer: C

6 Enzymes are involved in many reactions in living cells.

Which statement describes the action of enzymes?

A Enzymes are able to start up reactions that would otherwise not occur.

B Enzymes can only act inside living cells or organisms.

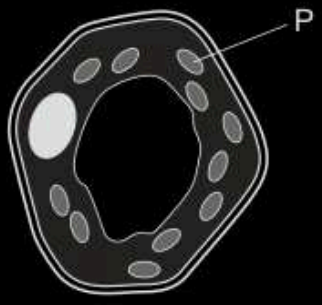
C Enzymes increase the rate of chemical reactions.

D One enzyme can catalyse a number of different reactions.

October/November 2023 · Paper 1 · Variant 1 · Question 8

Correct answer: A

8 The diagram shows a plant cell.



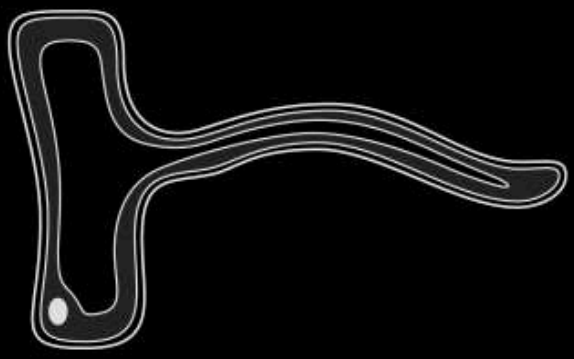
Compared with the rest of the cell, which row describes the concentrations of oxygen and magnesium inside structure P during the daytime?

	oxygen	magnesium
A	high	high
B	high	low
C	low	high
D	low	low

October/November 2023 · Paper 1 · Variant 1 · Question 9

Correct answer: D

9 The diagram shows a root hair cell.



How does the shape of the cell help the plant?

A It allows roots to search for ions.

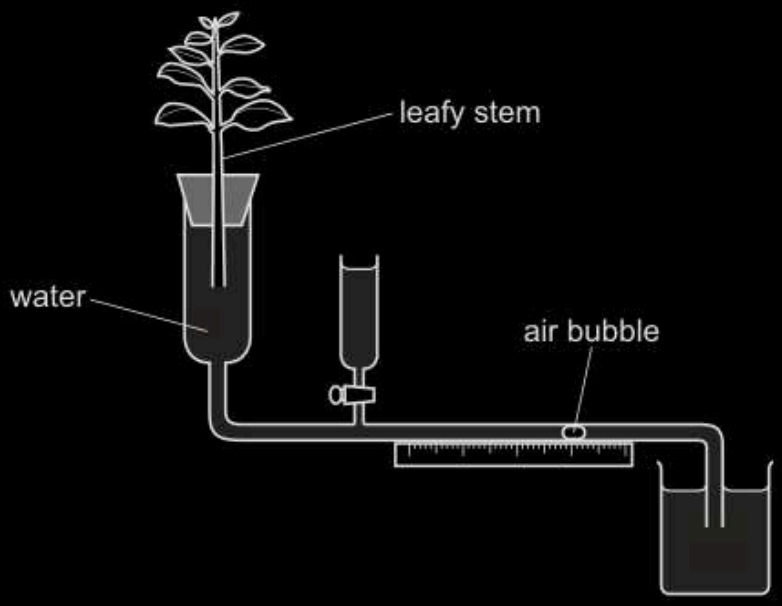
B It allows roots to search for water.

C It increases the surface area for sugar uptake.

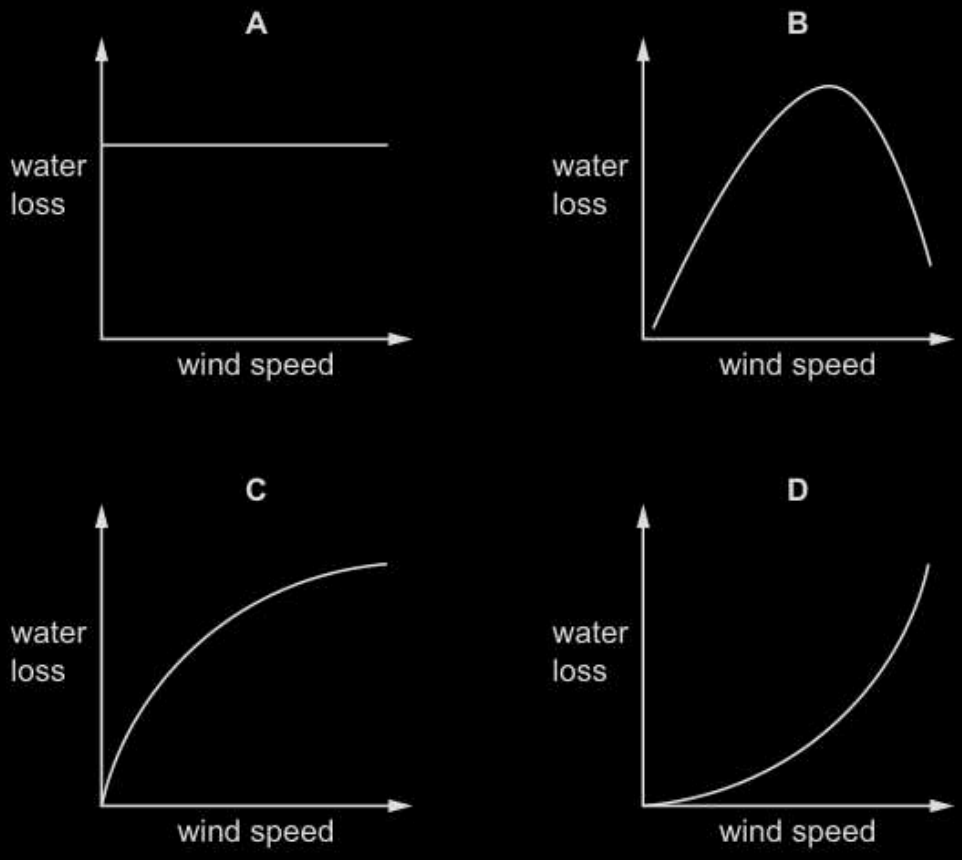
D It increases the surface area for water uptake.

Correct answer: C

10 A student used the apparatus shown to measure the rate of water loss by the leafy stem at increasing wind speeds.

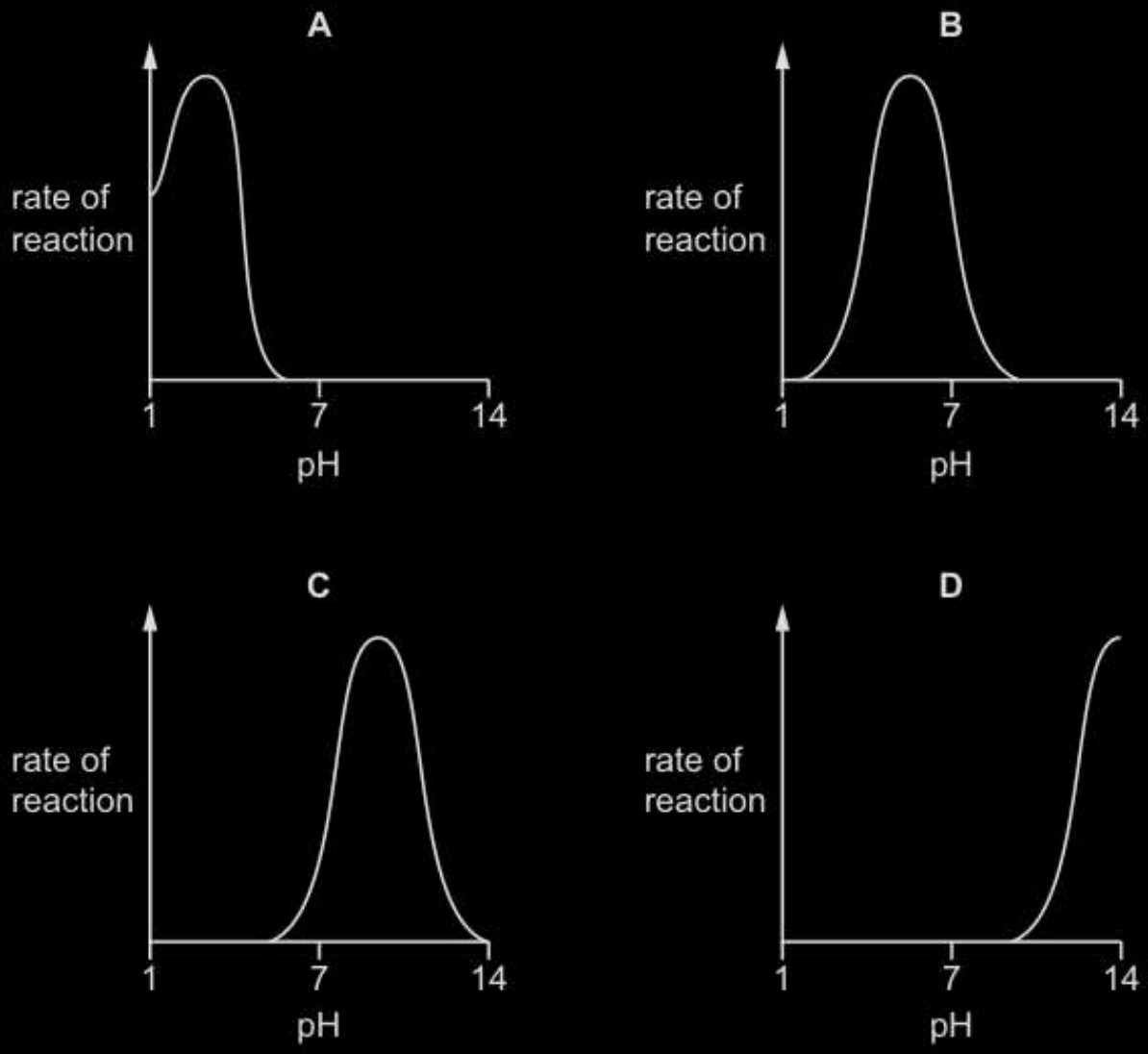


Which graph was produced from their results?



Correct answer: A

13 Which graph best represents the effect of pH on the action of gastric protease?

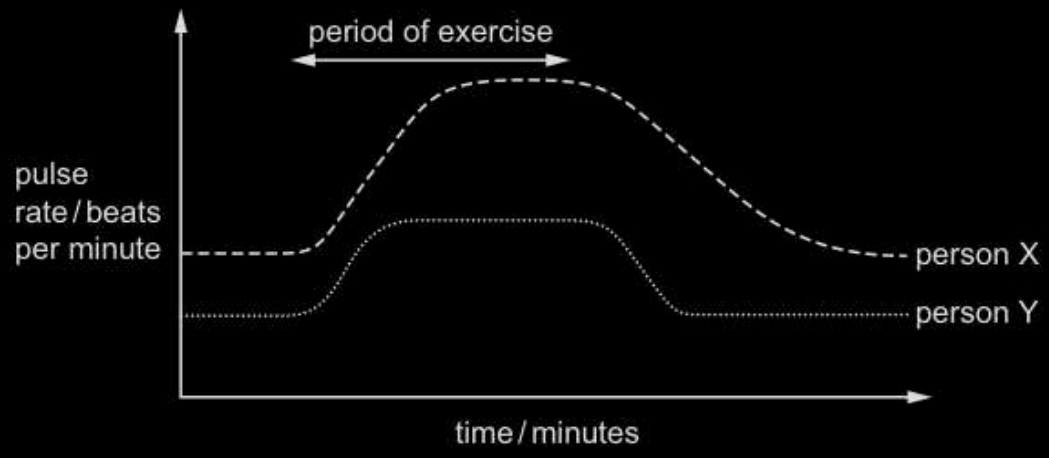


October/November 2023 · Paper 1 · Variant 1 · Question 17

Correct answer: D

17 Two people carried out a short period of exercise on a treadmill. Each of the two people ran on the treadmill for the same amount of time and at the same speed.

The graph shows the pulse rates of these two people before, during and after the period of exercise.



Which interpretation of the graph is correct?

A Person X is fitter, and their heart rate increases more during exercise.

B Person X is fitter, and their heart rate recovers more quickly after exercise.

C Person Y is fitter, and their heart rate increases more during exercise.

D Person Y is fitter, and their heart rate recovers more quickly after exercise.

October/November 2023 · Paper 1 · Variant 1 · Question 18

Correct answer: B

18 Which blood vessels carry deoxygenated blood?

A aorta and vena cava

B pulmonary artery and hepatic vein

C renal vein and pulmonary vein

D vena cava and pulmonary vein

October/November 2023 · Paper 1 · Variant 1 · Question 23

Correct answer: C

23 Some people require a treatment called dialysis to remove waste products and excess salts from their blood.

For these individuals, which part of the body is **not** functioning correctly?

A bladder

B heart

C kidney

D ureter

October/November 2023 · Paper 1 · Variant 1 · Question 24

Correct answer: C

24 Which change occurs in the eye as a person focuses on a vehicle approaching from a long distance?

A ciliary muscles relax

B iris circular muscles contract

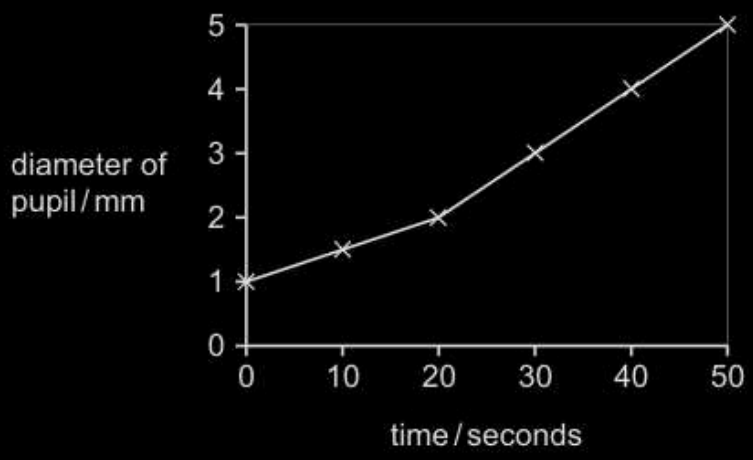
C lens becomes more convex

D pupil becomes larger

Correct answer: C

25 A light was shone into a person's eye for 50 seconds. During this time, the intensity of the light was varied.

The graph shows changes in the person's pupil size.



Which statement explains the change in pupil size?

A The light slowly became brighter.

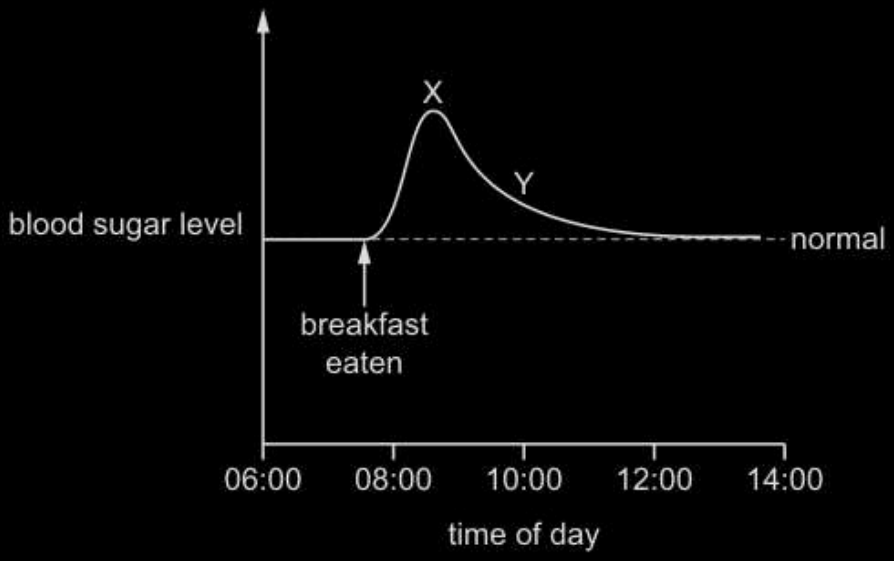
B The light suddenly became brighter.

C The light slowly became dimmer.

D The light suddenly became dimmer.

Correct answer: C

26 The graph shows the blood sugar level for a person over eight hours. They did **not** exercise during this time.



Which statement explains the shape of the graph between X and Y?

A The pancreas produces glucagon which causes excess sugar in the blood to be converted to glycogen.

B The liver produces insulin which causes excess sugar in the blood to be converted to glycogen.

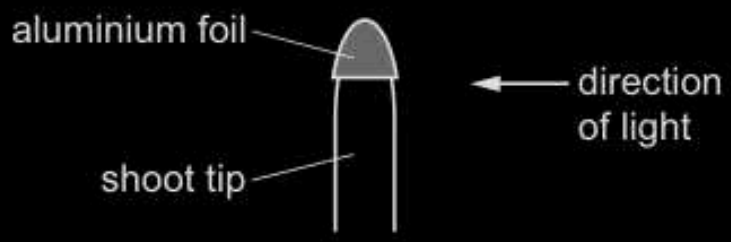
C The pancreas produces insulin which causes excess sugar in the blood to be converted to glycogen.

D The liver produces glycogen which causes excess sugar in the blood to be converted to glucagon.

October/November 2023 · Paper 1 · Variant 1 · Question 28

Correct answer: A

28 An experiment on the growth of a young plant shoot was set up, as shown.



What does the shoot look like after 48 hours?

October/November 2023 · Paper 1 · Variant 1 · Question 29

Correct answer: A

29 A scientist takes four pairs of samples from a wild cherry tree.

In which pair of samples could there be cells with different genotypes?

A two seeds

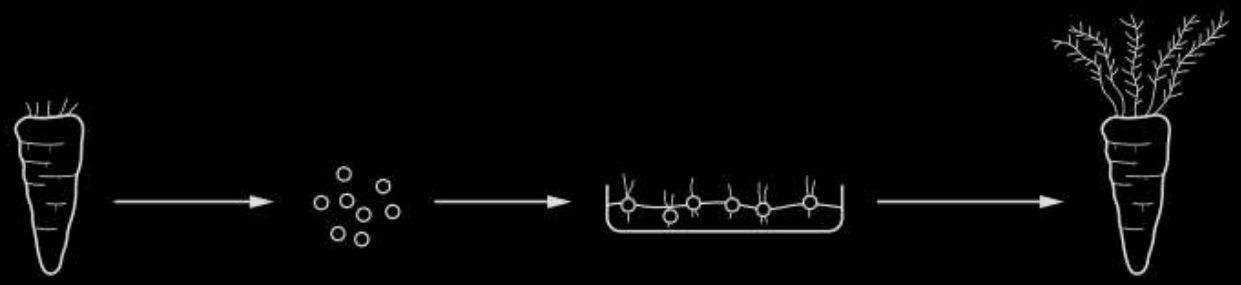
B two leaves

C two petals

D two root cuttings

Correct answer: C

30 The diagram shows how a large number of carrot plants can be produced from small pieces of carrot root.



root of carrot plant

pieces of carrot root are removed

pieces of carrot root are grown into small plants

each small plant grows into a new carrot plant

Which row is correct for this method of plant production?

	asexual reproduction	genetically identical offspring	cell division by mitosis	cell division by meiosis
A	✓	✗	✓	✗
B	✓	✗	✗	✓
C	✓	✓	✓	✗
D	✗	✓	✗	✓

October/November 2023 · Paper 1 · Variant 1 · Question 34

Correct answer: D

34 In a DNA molecule, bonds between pairs of bases hold the strands together.

Which pairs of bases are found in DNA?

1 A-C

2 A-G

3 A-T

4 C-G

5 C-T

6 T-G

A 1, 2, 3, 4, 5 and 6

B 1 and 6 only

C 2 and 5 only

D 3 and 4 only

October/November 2023 · Paper 1 · Variant 1 · Question 35

Correct answer: B

35 The inheritance of the ABO blood groups in humans is controlled by three alleles (I^A, I^B and I^O), only two of which can be present in one individual.

What are the possible blood groups of children born to a homozygous group A woman and a heterozygous group B man?

A AB and B only

B AB and A only

C A, B and AB only

D A, B, AB and O

October/November 2023 · Paper 1 · Variant 1 · Question 38

Correct answer: B

38 Some bacteria and fungi are decomposers.

What is produced when decomposers break down plant and animal protein?

A nitrate ions

B ammonium ions

C nitrogen gas

D nitrite ions

October/November 2023 · Paper 1 · Variant 2 · Question 1

Correct answer: C

1 Which structures have both cytoplasm and cell walls?

A liver cells

B red blood cells

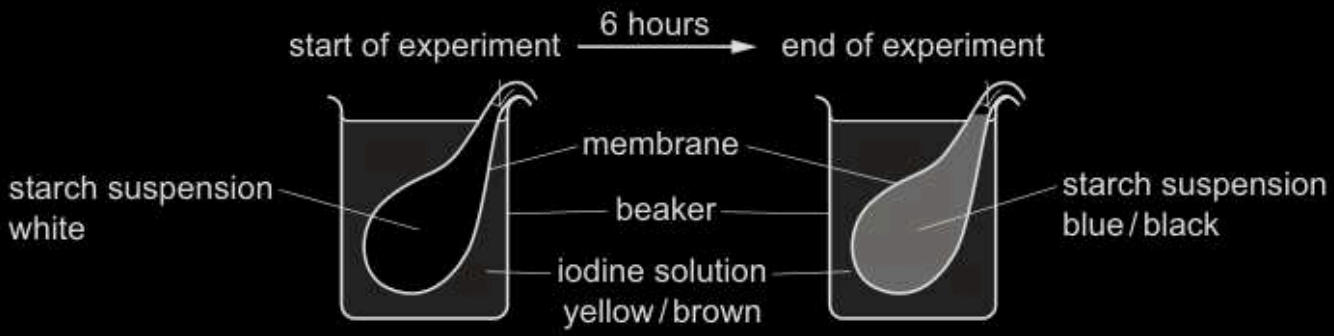
C root hair cells

D xylem vessels

October/November 2023 · Paper 1 · Variant 2 · Question 3

Correct answer: A

3 The diagram shows an experiment.



Why has the starch suspension changed colour?

A Iodine has diffused in through the membrane.

B Iodine has diffused out through the membrane.

C Starch has diffused in through the membrane.

D Starch has diffused out through the membrane.

October/November 2023 · Paper 1 · Variant 2 · Question 4

Correct answer: A

4 The table shows the relative concentrations of two mineral ions in a root cell and in the water in the soil around the plant root.

	concentration of magnesium ions / arbitrary units	concentration of nitrate ions / arbitrary units
water around root	23	19
inside root cell	2475	1306

What is needed for the plant root to increase the concentration of the mineral ions in its cells?

A energy for active transport

B energy for osmosis

C water for active transport

D water for osmosis

October/November 2023 · Paper 1 · Variant 2 · Question 5

Correct answer: B

5 Some health problems can be detected using chemical tests for protein and glucose in the urine.

Which row shows the results for a person with protein and glucose in their urine?

	Benedict's solution	biuret reagent	ethanol	iodine solution
A	blue colour	pale blue colour	milky colour	yellow / brown colour
B	red colour	mauve (purple) colour	colourless	yellow / brown colour
C	blue colour	mauve (purple) colour	milky colour	blue / black colour
D	red colour	pale blue colour	colourless	blue / black colour

October/November 2023 · Paper 1 · Variant 2 · Question 6

Correct answer: C

6 Which statement describes enzyme-catalysed reactions?

A All enzymes work best at the same pH.

B As an enzyme-catalysed reaction progresses, the concentration of reactants increases.

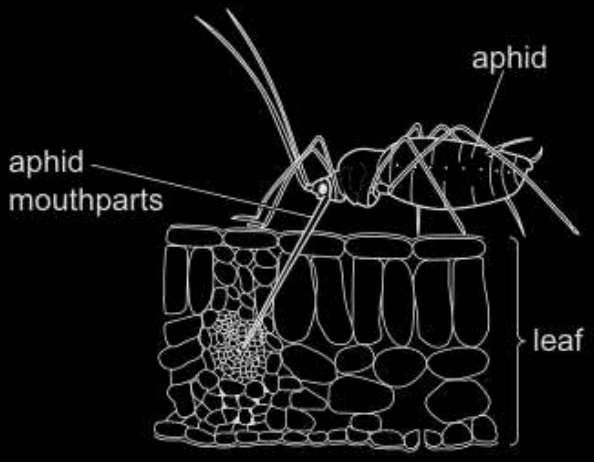
C As the temperature of an enzyme-catalysed reaction increases from 0 °C to 20 °C, the frequency of effective collisions increases.

D Enzymes are denatured as the temperature of an enzyme-catalysed reaction is lowered from 20 °C to 0 °C.

October/November 2023 · Paper 1 · Variant 2 · Question 7

Correct answer: B

7 The diagram shows an aphid feeding on sugar in a leaf.



When the aphid feeds, in which order do its mouthparts pass through the tissues of the leaf?

A palisade cells → phloem → upper epidermis

B upper epidermis → palisade cells → phloem

C palisade cells → upper epidermis → xylem

D upper epidermis → xylem → palisade cells

October/November 2023 · Paper 1 · Variant 2 · Question 8

Correct answer: D

8 For which processes do plants need either nitrate ions or magnesium ions?

	synthesis of cellulose	synthesis of chlorophyll	synthesis of proteins
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✗
D	✗	✓	✓

key

✓ = nitrate ions or magnesium ions needed

✗ = nitrate ions or magnesium ions not needed

October/November 2023 · Paper 1 · Variant 2 · Question 9

Correct answer: A

9 Which row shows the processes involved when nitrate ions and water, from soil, enter a root hair cell?

	nitrate ions	water
A	active transport	osmosis
B	diffusion	osmosis
C	diffusion	active transport
D	osmosis	diffusion

October/November 2023 · Paper 1 · Variant 2 · Question 10

Correct answer: A

10 Which change in conditions will decrease the rate of transpiration?

A a decrease in air currents around the leaf

B a decrease in the humidity of the atmosphere

C an increase in light intensity around the leaf

D an increase in the temperature of the atmosphere

October/November 2023 · Paper 1 · Variant 2 · Question 11

Correct answer: B

11 What are the correct definitions of digestion and assimilation?

1 the intake of food into the body through the mouth

2 the breakdown of large insoluble food molecules into small soluble ones

3 the uptake and use of food molecules in the cells of the body

4 the movement of food molecules through the wall of the intestine into the blood and lymph

	digestion	assimilation
A	1	3
B	2	3
C	1	4
D	2	4

October/November 2023 · Paper 1 · Variant 2 · Question 12

Correct answer: D

12 Which organ produces an acid that kills the bacteria in ingested food?

A liver

B pancreas

C salivary gland

D stomach

October/November 2023 · Paper 1 · Variant 2 · Question 13

Correct answer: C

13 The table shows the rates of absorption of two different sugars, arabinose and glucose, in living and dead intestines. The concentrations of the sugars inside the intestines were the same in each case.

	rate of absorption / arbitrary units	
	arabinose	glucose
living intestine	31	102
dead intestine	31	34

What are the main methods of absorption of arabinose and glucose in living intestine?

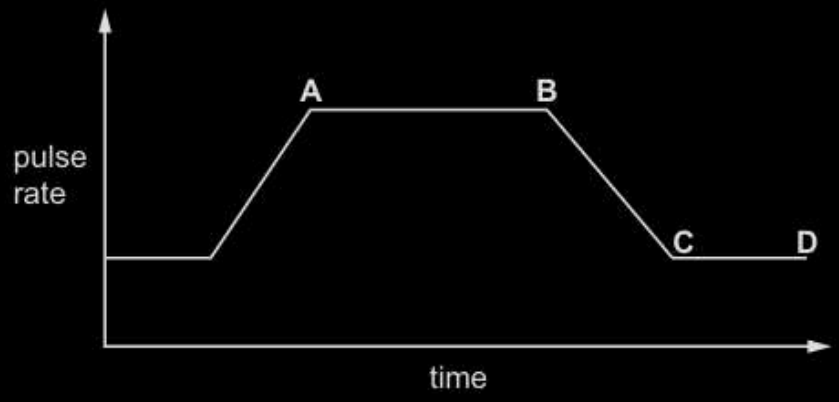
	arabinose	glucose
A	active transport	active transport
B	active transport	diffusion
C	diffusion	active transport
D	diffusion	diffusion

October/November 2023 · Paper 1 · Variant 2 · Question 17

Correct answer: B

17 The graph shows a person's pulse rate over a period of time during which they rest, pedal an exercise bike and then rest again.

At which point did the person stop pedalling?



October/November 2023 · Paper 1 · Variant 2 · Question 18

Correct answer: C

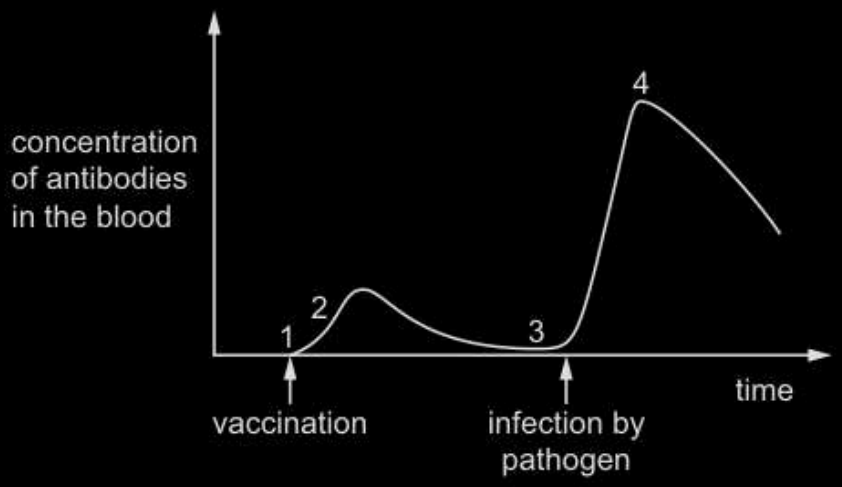
18 Which row shows the names of blood vessels that transport blood to and from the liver and the kidney?

	to the liver	from the liver	to the kidney	from the kidney
A	hepatic portal vein	hepatic vein	renal vein	renal artery
B	hepatic vein	hepatic portal vein	renal vein	renal artery
C	hepatic portal vein	hepatic vein	renal artery	renal vein
D	hepatic artery	hepatic portal vein	renal artery	renal vein

October/November 2023 · Paper 1 · Variant 2 · Question 22

Correct answer: B

22 The graph shows the antibody concentration in a child's blood over a period of time. It shows what happens to the antibody concentration after a child is vaccinated against a particular disease and after the child is infected by the pathogen for that disease.



Based on the graph, which statement is correct?

A Memory cells have disappeared from the child's blood by 3.

B Memory cells were in the child's blood at 2.

C Memory cells were in the child's blood before 1.

D Passive immunity was attained at 4.

October/November 2023 · Paper 1 · Variant 2 · Question 23

Correct answer: D

23 Which label shows the location for the production of urea?

October/November 2023 · Paper 1 · Variant 2 · Question 24

Correct answer: C

24 A student was asked to compare a typical spinal reflex action with a voluntary action.

He has made one mistake in his comparison.

Which row shows the **incorrect** statement?

	reflex action	voluntary action
A	rapid response to a stimulus	response may be slow
B	initiated by the response of a receptor to a stimulus	initiated from the brain by conscious thought
C	the nervous impulse takes the longest pathway	the nervous impulse takes the shortest pathway
D	effectors are muscles or glands	effectors are muscles only

October/November 2023 · Paper 1 · Variant 2 · Question 26

Correct answer: D

26 Descriptions of the body's response to changes in temperature are listed.

1 Arterioles in the skin surface constrict.

2 Arterioles in the skin surface dilate.

3 Hair erector muscles contract.

4 Hair erector muscles relax.

5 Sweat evaporates from the skin.

6 Muscles contract spontaneously.

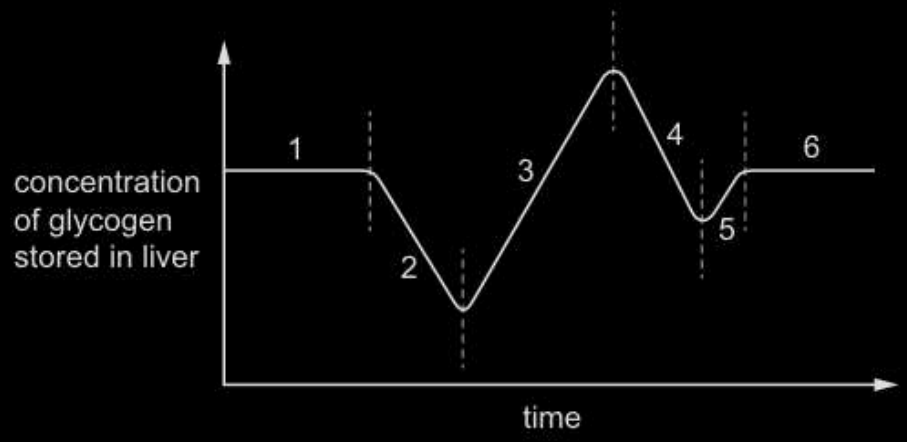
Which descriptions apply to the body's response to hot weather?

A 1, 3 and 6 **B** 1, 4 and 5 **C** 2, 3 and 5 **D** 2, 4 and 5

October/November 2023 · Paper 1 · Variant 2 · Question 27

Correct answer: D

27 The diagram shows how the concentration of glycogen stored in the liver of a human changes over time. The changes help to maintain a constant concentration of glucose in the blood.



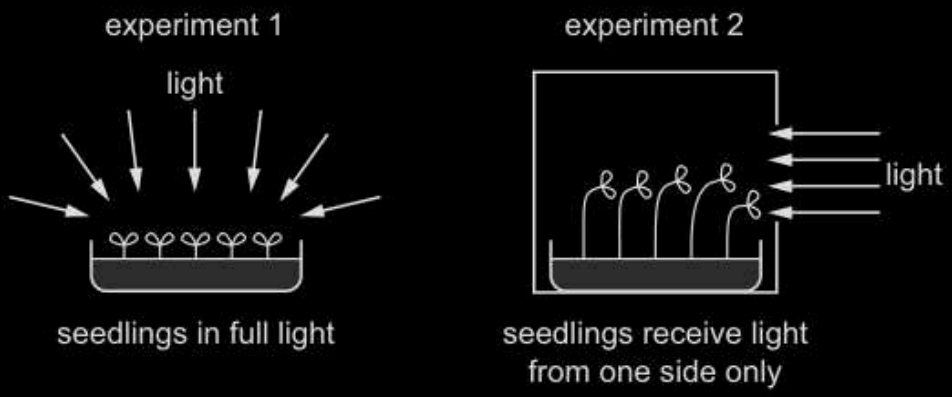
During which periods is the concentration of **glucose** in the blood above average?

A 1 and 6 **B** 2 and 5 **C** 3 and 4 **D** 3 and 5

October/November 2023 · Paper 1 · Variant 2 · Question 28

Correct answer: B

28 The diagram shows some seedlings grown in the light.



Which statement describes the distribution of auxin in the seedlings?

A Auxin is equally distributed in experiments 1 and 2.

B Auxin is equally distributed in experiment 1 only.

C Auxin is equally distributed in experiment 2 only.

D Auxin is **not** distributed equally in experiments 1 and 2.

October/November 2023 · Paper 1 · Variant 2 · Question 29

Correct answer: B

29 A plant has 20 chromosomes in its leaf cells. The plant reproduces both sexually and asexually.

What is the correct number of chromosomes in the gametes and in cells used for asexual reproduction?

	gametes	cells used for asexual reproduction
A	10	10
B	10	20
C	20	10
D	20	20

October/November 2023 · Paper 1 · Variant 2 · Question 30

Correct answer: D

30 What are features of sexual reproduction?

1 fusion of diploid nuclei

2 fusion of haploid nuclei

3 production of clones

4 production of genetically different offspring

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

October/November 2023 · Paper 1 · Variant 2 · Question 32

Correct answer: B

32 The diagram shows the female reproductive system.

Which row correctly labels the diagram?

	1	2	3	4	5	6
A	bladder	urethra	ureter	oviduct	uterus	cervix
B	oviduct	bladder	urethra	uterus	cervix	vagina
C	urethra	bladder	ureter	uterus	cervix	vagina
D	oviduct	bladder	urethra	ureter	uterus	cervix

October/November 2023 · Paper 1 · Variant 2 · Question 34

Correct answer: B

34 Which statements are correct?

1 Each chromosome contains one molecule of DNA.

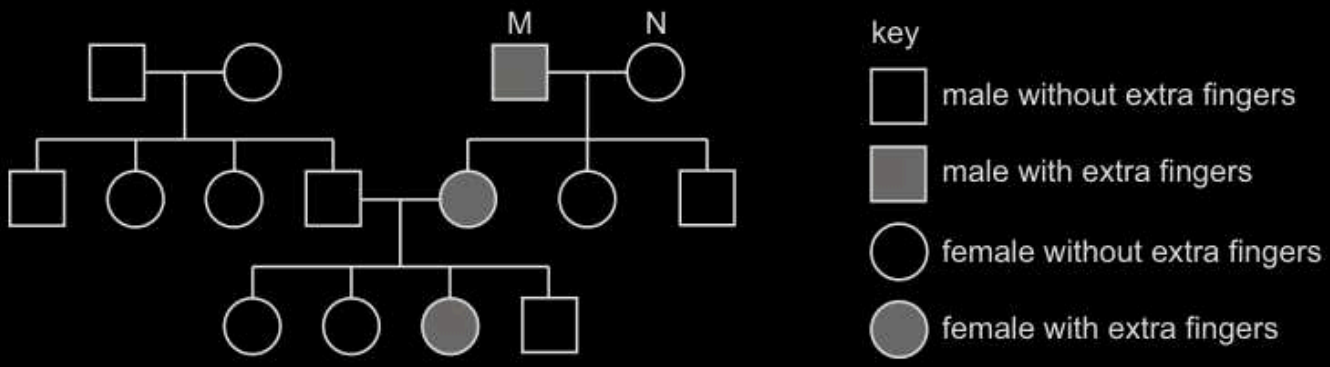
2 A gene is a section of DNA on a chromosome.

3 A gene contains smaller regions called alleles.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

Correct answer: C

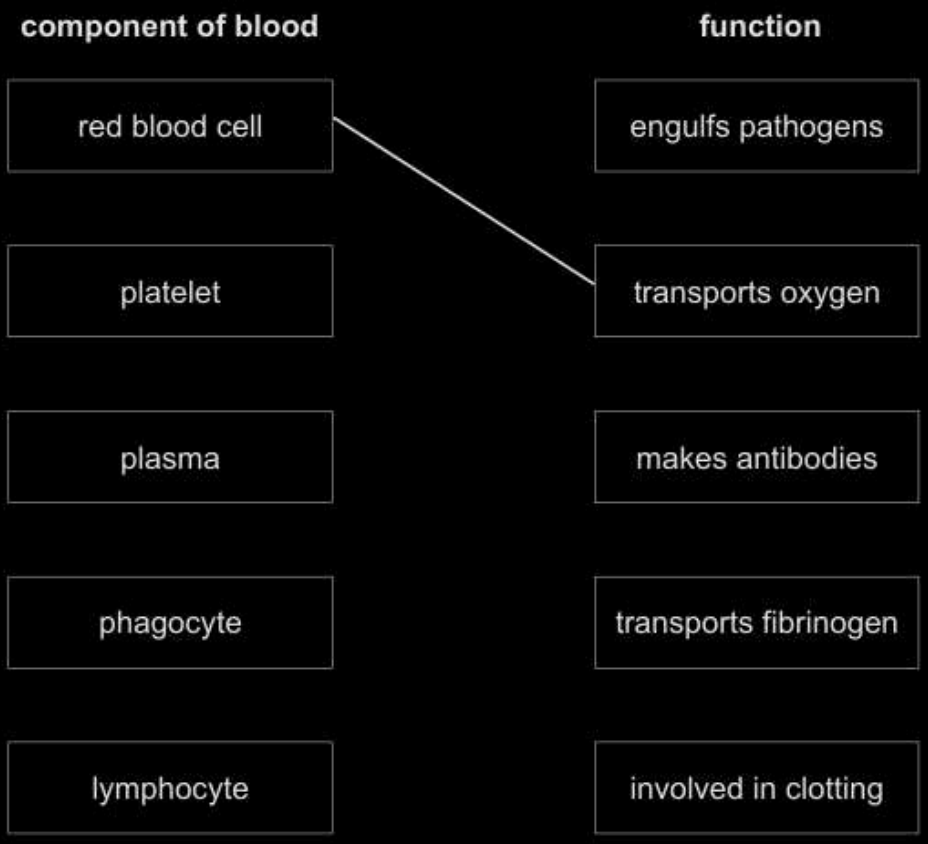
35 The diagram shows the inheritance of the allele, E, for polydactyly (having extra fingers) which is dominant over the allele for no extra fingers, e.



What are the genotypes of persons M and N?

	M	N
A	EE	ee
B	Ee	Ee
C	Ee	ee
D	ee	ee

1 **(a)** Human blood has many different components and functions.
Match each component to its function by drawing a line between them. One has been done for you.



[3]

(b) **(i)** Explain how a red blood cell is specialised for transporting oxygen.

.....

.....

.....

.....

..... [3]

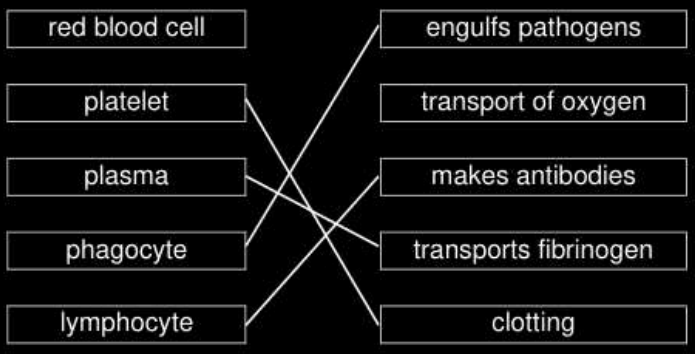
(ii) The number of red blood cells in 1 mm^3 of blood is called a red blood cell count.
A normal red blood cell count for a female is 4.2 – 5.4 million cells in 1 mm^3 of blood.
Suggest **two** factors that could lead to a change in the red blood cell count in a 25 year-old female.

1 .....

2 .....

[2]

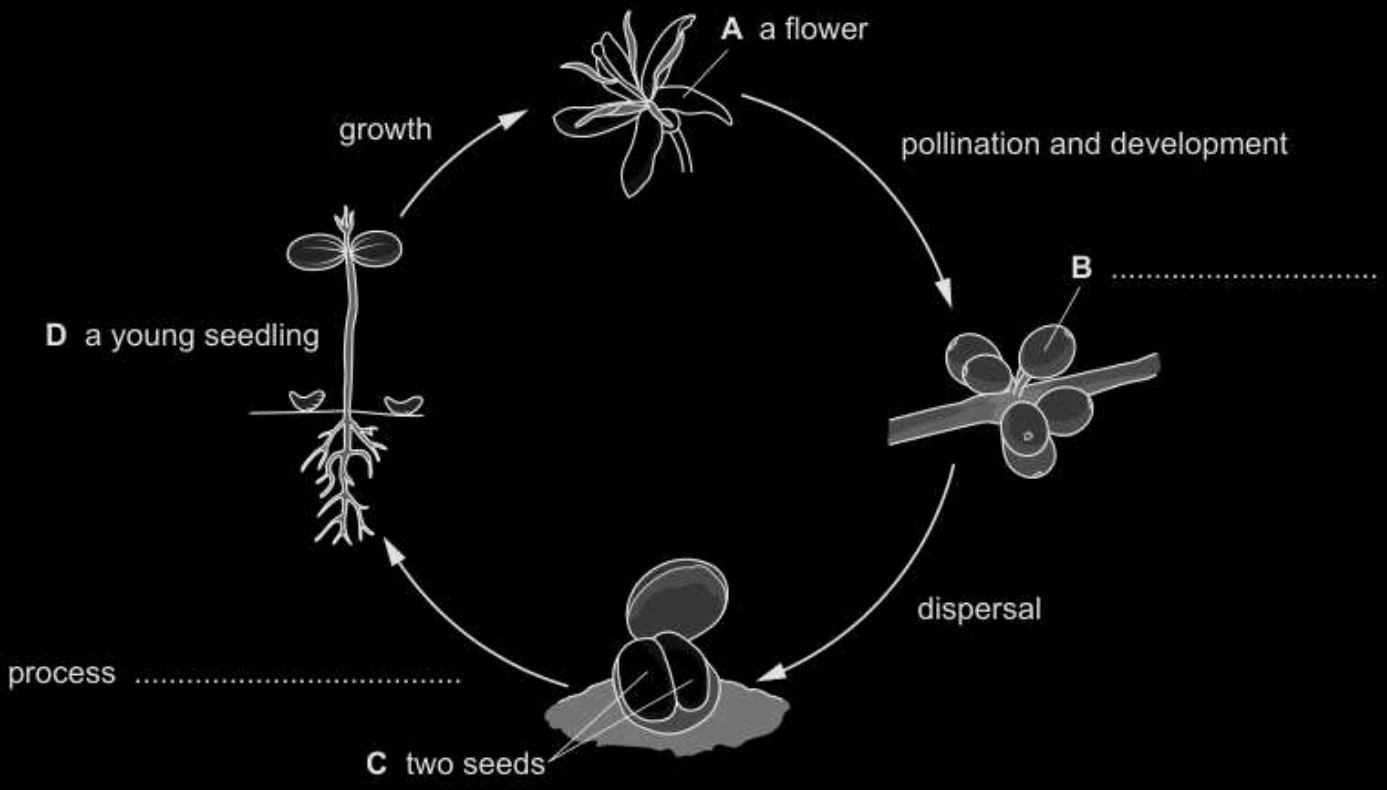
[Total: 8]

1(a)		**3**	all four correct = 3 marks three correct = 2 marks two correct = 1 mark
1(b)(i)	any **three** from: haemoglobin to bind with / carry / transport oxygen ; no nucleus so more room for haemoglobin ; small / flexible / elastic so can squeeze / fit along capillaries ; biconcave shape / large surface area for faster / rapid / more uptake / diffusion (of oxygen) ;	**3**	
1(b)(ii)	any **two** from: iron (in diet) ; anaemia / sickle cell anaemia ; pregnancy ; menstruation; accidental loss ; altitude ;	**2**	

© UCLES 2023

Question	Answer	Marks	Guidance

2 Fig. 2.1 shows the life cycle of the plant *Coffea arabica*.
This plant produces seeds which are used to make coffee.



(not to scale)

Fig. 2.1

(a) (i) Complete Fig. 2.1 by labelling the plant structure **B** and the process taking place between **C** and **D**. [2]

(ii) The flowers produce nectar in nectaries and are pollinated by insects, such as bees.
Suggest where the nectaries are located and how this assists the process of pollination.

.....

.....

..... [2]

(iii) After pollination, pollen tubes grow down the style.
Describe what then happens leading to the formation of **B** and its contents.

.....

.....

.....

.....

..... [3]

(b) *Coffea arabica* plants produce a chemical called caffeine which in high concentrations is toxic to insects.

(i) Suggest why the production of caffeine is useful to the plant.

.....

..... [1]

(ii) Caffeine is found throughout the plant but in some parts the concentration is much higher than in others.
Suggest why it is important to the plant to have different concentrations in different parts.

.....

.....

.....

..... [2]

(c) Coffee is a drink that many humans enjoy.
The caffeine in the coffee causes the body to release adrenaline.

(i) Using caffeine and adrenaline as examples, explain the difference between a drug and a hormone.

.....

.....

.....

.....

..... [3]

(ii) Suggest **two** effects on the body of drinking coffee that contains caffeine.

1 .....

.....

2 .....

.....

[2]

[Total: 15]

2(a)(i)	B = a fruit ; germination ;	**2**	
2(a)(ii)	*position:* at base / inside flower / at the base of the petals / near / at the ovary / anther / stigma ; *how it assists:* idea that the position of the insect allows pollen to be collected from anther / pollen to be deposited on stigma ;	**2**	
2(a)(iii)	any **three** from: male nucleus / nucleus of pollen grain / male gamete **+** fuses with female nucleus / egg (cell) / female gamete ; zygote forms / embryo develops ; ovule becomes a seed ; ovary becomes a fruit ;	**3**	
2(b)(i)	it stops insects from eating the plant **AW** ;	**1**	
2(b)(ii)	some areas / leaves / buds / seeds are more likely to be eaten / attacked; so that pollinators are not killed ;	**2**	
2(c)(i)	caffeine is a drug **+** adrenaline a hormone ; a drug / caffeine is taken in to body **+** affects chemical reactions / body functions / body actions ; a hormone / adrenaline is made in the body / by glands ;	**3**	**A** coffee for caffeine

© UCLES 2023

Question	Answer	Marks	Guidance
2(c)(ii)	any **two** from: increased heart rate ; increased blood pressure ; increased sugar / glucose in bloodstream ; more blood to muscles ; dilates pupils ; dilation of bronchioles ; increased breathing rate ; (mental) alertness **AW** ;	**2**	

Question	Answer	Marks	Guidance

3 The bar chart in Fig. 3.1 shows the percentage of total human body mass for the five most common elements found in the human body.

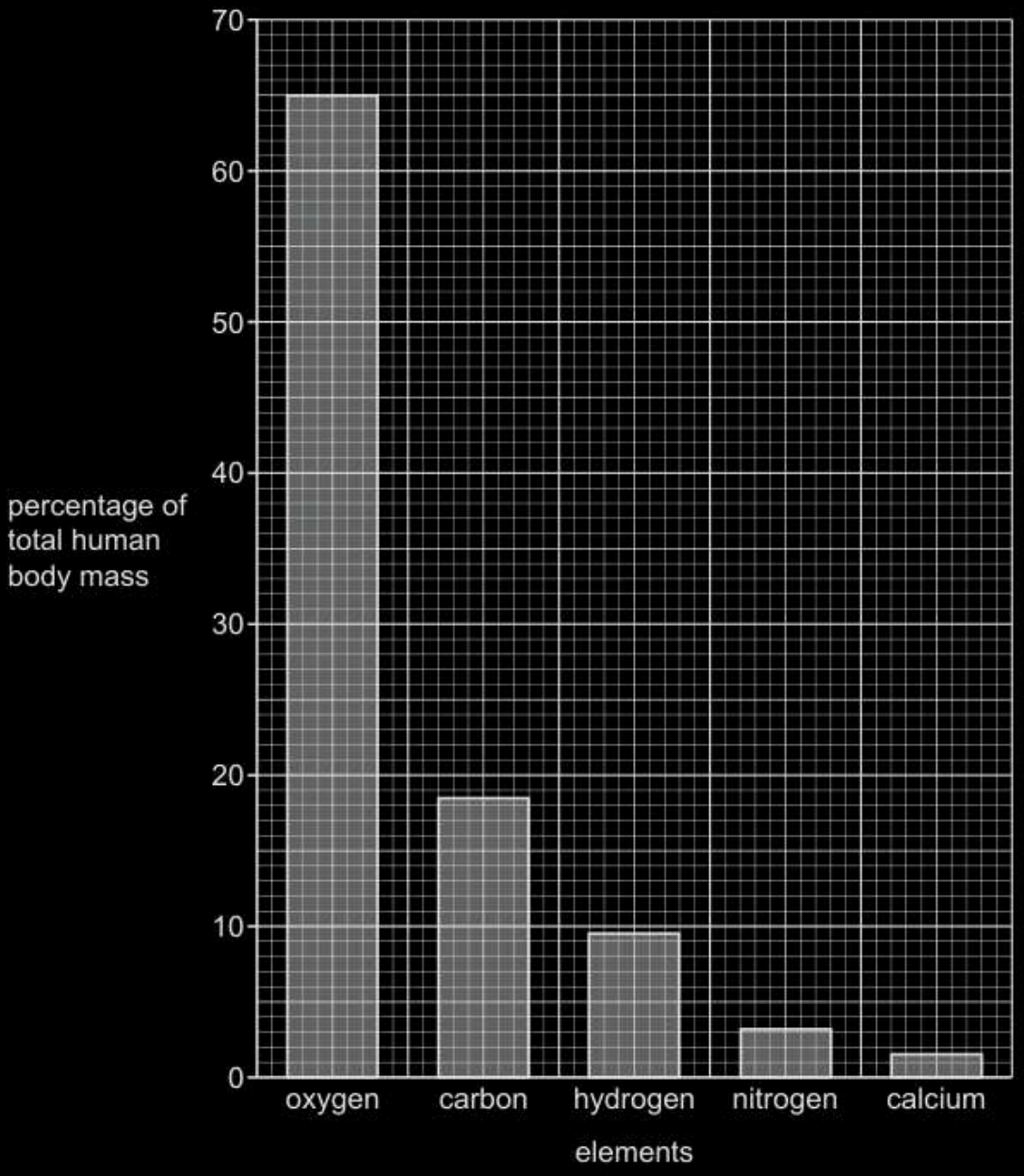


Fig. 3.1

(a) (i) Calculate the percentage of the total human body mass made up from the **three** most common elements.

total = ... % [2]

(ii) Name **two types** of substance found in the human body that contain only carbon, hydrogen and oxygen.

1 .....

2 .....

[2]

(b) Water contains the elements hydrogen and oxygen and makes up a large percentage of an adult's total body mass.

(i) Discuss why water makes up such a high percentage of an adult's total body mass.

.....

.....

.....

.....

..... [3]

(ii) Explain why a cell from human tissue will burst when it is placed in water.

.....

.....

.....

..... [3]

(c) Human tissues contain different percentage masses of the elements shown in Fig. 3.1.
Suggest **one** tissue that contains a relatively high percentage mass of nitrogen and explain your answer.

.....

.....

..... [2]

[Total: 12]

3(a)(i)	93.0 ; ;	**2**	**A** 65, 18.5 & 9.5 for 1 mark
3(a)(ii)	carbohydrates ; fats / lipids ;	**2**	
3(b)(i)	any **three** from: solvent ; named dissolved substance ; cytoplasm is mostly water ; in blood / plasma **+** transport ; in urine **+** excretion ; in sweat **or** evaporation cools body ;	**3**	

Question	Answer	Marks	Guidance
3(b)(ii)	any **three** from: water enters **+** osmosis ; higher water potential in water / lower water potential in tissue / cells ; across selectively / partially / differentially / semi permeable membrane ; cell swells up / expands / cytoplasm volume increases ; bursts due to lack of a cell wall ;	**3**	
3(c)	muscle ; protein ;	**2**	

Question	Answer	Marks	Guidance

4 A sneeze can be triggered by dust irritating receptor cells in the lining of the nose or throat. During a sneeze, air is exhaled from the lungs with some force and the eyelids close.

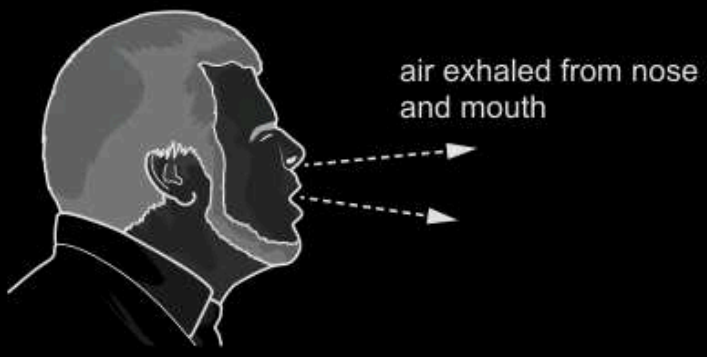


Fig. 4.1

(a) **(i)** Suggest how the closure of the eyelids during a sneeze is coordinated.

.....

.....

.....

.....

.....

..... [4]

(ii) Suggest how sneezing can result in disease transmission.

.....

.....

..... [2]

(b) A sudden increase in light intensity can trigger sneezing in some people. This is called photic sneezing. It is estimated that 18 – 35% of the human population can be triggered to sneeze by an increase in light intensity.
The genetic basis for photic sneezing is not fully understood but it is thought that a dominant allele is involved.

(i) Give the name of this type of variation in a population and explain your answer.

type of variation .....

explanation .....

.....

[2]

(ii) Describe what is meant by the term *dominant allele*.

.....

.....

..... [2]

[Total: 10]

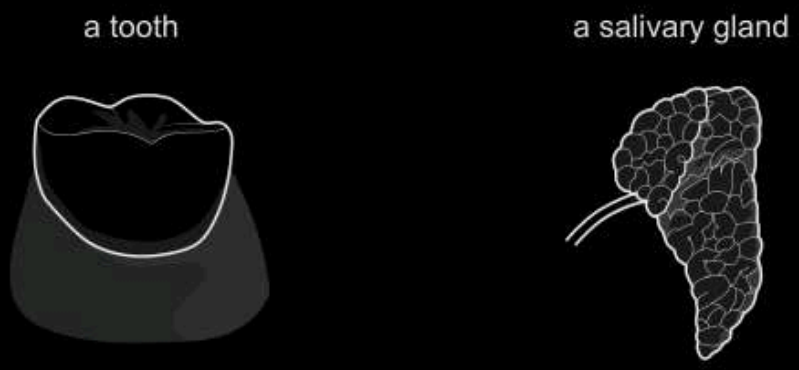
4(a)(i)	any **four** from: impulse in sensory neurone ; passes to CNS / brain ; relay neurone ; motor neurone to effector / muscle ; contraction of muscle closes eyelid ;	**4**	
4(a)(ii)	expelled air contains pathogen / bacteria / viruses ; breathed in by another person / lands on another person / taken in from contaminated surfaces ;	**2**	
4(b)(i)	discontinuous ; only two phenotypes / sneeze or don't sneeze / no intermediates / caused by genes only ;	**2**	

© UCLES 2023

Question	Answer	Marks	Guidance
4(b)(ii)	dominant: always expressed **AW** ; allele: alternative form of a gene / variation of a gene **AW** ;	**2**	

Question	Answer	Marks	Guidance

5 Fig. 5.1 shows two structures found in the human digestive system.



(not to scale)

Fig. 5.1

(a) (i) Identify the type of tooth shown in the diagram.

..... [1]

(ii) Name the outer layer of the tooth.

..... [1]

(iii) Explain how the tooth and the salivary gland contribute to digestion in different ways.

.....

.....

.....

.....

..... [4]

(b) The stomach wall produces different substances that are involved in digestion.

(i) Name **two** of these substances and explain how they are involved in digestion.

.....

.....

.....

.....

.....

..... [4]

(ii) A hormone called gastrin stimulates cells in the stomach wall to release the substances involved in digestion. Some of the gastrin is produced in the duodenum.

State how this gastrin reaches the stomach.

..... [1]

[Total: 11]

5(a)(i)	molar / premolar ;	**1**	
5(a)(ii)	enamel ;	**1**	**Ig** crown
5(a)(iii)	any **two** from: tooth: physical digestion ; grinds / breaks up food into smaller pieces / larger surface area ridged surface / hard-wearing surface ; any **two** from: salivary gland: chemical digestion ; produces amylase / enzyme ; starch broken down to maltose ;	**4**	
5(b)(i)	hydrochloric acid ; optimum / best acidity / correct pH **+** enzyme action ; pepsin / protease ; protein broken down to polypeptides / peptides / amino acids;	**4**	**Ig** references to killing bacteria
5(b)(ii)	in the blood / bloodstream / plasma ;	**1**	

Question	Answer	Marks	Guidance

6 Fig. 6.1 shows a biological molecule.

Fig. 6.1

(a) (i) Name this molecule.

..... [1]

(ii) Describe **three** features of the structure of this molecule.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(b) Animal and bacterial cells contain this molecule.
Describe how the location and shape of the molecule differ in animal and bacterial cells.

location .....

.....

shape .....

.....

[2]

(c) Explain how this molecule controls cell function.

.....

.....

.....

.....

..... [3]

[Total: 9]

6(a)(i)	DNA / deoxyribonucleic acid ;	**1**	
6(a)(ii)	any three from: two / double strands ; double helix ; made of nucleotides ; strands held together by bonds between the bases ; complementary bases / A – T and C – G ; made from elements C, H, O, N, P ;	**3**	**A** sugar phosphate backbone
6(b)	in nucleus in animal cells **+** in cytoplasm in bacterial cells ; linear / straight / ribbon / strand / AW in animal cells + circular in bacterial cells ;	**2**	
6(c)	any **three** from: contains genes ; codes / controls for proteins / polypeptides ; by the order of bases / nucleotides ; example of protein e.g. enzyme / haemoglobin / antibody ;	**3**	

Question	**Answer**	**Marks**	Guidance

1 Cholera is caused by a type of organism that can be passed from one host to another.

(a) State the term used to describe

(i) a disease-causing organism

..... [1]

(ii) the type of disease in which the disease-causing organism is passed from one host to another.

..... [1]

(b) The disease-causing organism that causes cholera is a bacterium.

Complete Table 1.1 to list some of the structures present in a bacterial cell and the function of each structure.

Table 1.1

structure	function
ribosomes	.....
.....	small rings of DNA that carry genes
.....	controls the passage of substances into and out of the cell
cell wall	.....

[4]

(c) Fig. 1.1 shows a photograph of the bacterium that causes cholera.

magnification ×50 000

Fig. 1.1

(i) The length of the bacterium in the photograph is 100 mm. Use this value to calculate the actual length of the bacterium in micrometres (μm).
Space for working.

..... μm
[2]

(ii) Use your knowledge of a type of specialised human cell to suggest the name of structure **Z** and to state its function.

name .....

function .....

.....
[2]

(iii) The binomial name of this type of bacterium is *Vibrio cholerae*.

State the genus of this type of bacterium and **one** advantage of the binomial system of naming species.

genus .....

advantage .....

.....

[2]

(iv) Explain how the presence of bacteria of this type in a human causes diarrhoea, one of the symptoms of cholera.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 16]

1(a)(i)	pathogen ;	1	
1(a)(ii)	transmissible / communicable / infectious ;	1	
1(b)	protein synthesis **AW** ; plasmids ; membrane ; shape / support **AW** / turgidity ;	4	
1(c)(i)	2 ;;	2	1 mark for 100 or 100 000 divided by 50 000
1(c)(ii)	flagellum ; movement / swim / locomotion **AW** ;	2	
1(c)(iii)	*Vibrio* ; international **AW** / universal **AW** ;	2	
1(c)(iv)	**1** bacteria produce / release a toxin ; **2** chloride (ions) / Cl^- ; **3** ions into / increase in + intestine ; **4** reduction in / lower water potential + of intestine **ORA** ; **5** water into intestine / out of cells / water not absorbed from intestine ; **6** osmosis ;	4	

Question	Answer	Marks	Guidance

3 Fig. 3.1 shows diagrams that represent the blood flow into and out of three major organs through blood vessels **P** to **U**.

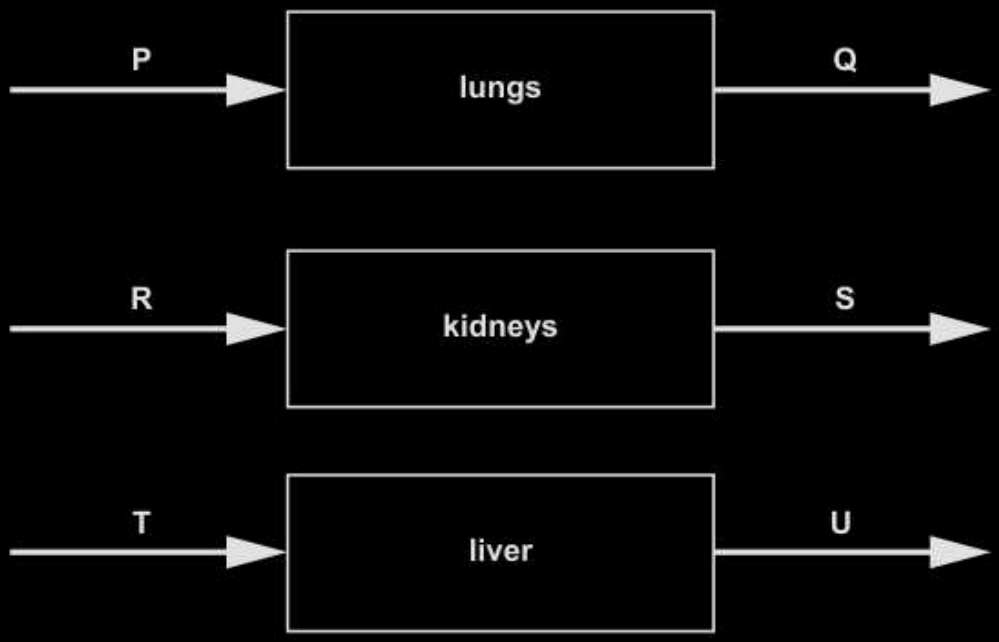


Fig. 3.1

(a) Identify each of the following blood vessels.

Q .....

R .....

U ..... [3]

(b) Compare the structures of blood vessels **Q** and **R**.

.....

.....

.....

.....

.....

..... [3]

(c) Explain each of the following.

(i) Blood in vessel **S** has a lower concentration of urea than blood in vessel **R**.

.....

.....

.....

.....

.....

..... [3]

(ii) Blood in vessel **U** sometimes has a higher concentration of glucose than blood in vessel **T**.

.....

.....

.....

.....

.....

..... [3]

[Total: 12]

3(a)	(**Q**) pulmonary vein ; (**R**) renal artery ; (**U**) hepatic vein ;	**3**	
3(b)	**Q** – (pulmonary) vein / + / **R** – (renal) artery thin + wall / + / thick + wall ; thin muscle / thin elastic / + / thick muscle / thick elastic ; wide + lumen **AW** / + / narrow + lumen **AW** ; valves / + / no valves ;	**3**	/
3(c)(i)	**1** urea + into kidney **OR** kidney removes + urea ; **2** filtration / filtering ; **3** nephron / Bowman's capsule / glomerulus ; **4** urea + excreted / in urine ;	**3**	

Question	Answer	Marks	Guidance
3(c)(ii)	**1** (blood) glucose decreased / sugar low / body needs more sugar ; **2** glycogen ; **3** breakdown / hydrolysis + to glucose ; **4** glucose + passed out of liver / goes into U / hepatic vein / blood ; **5** (stimulated by) glucagon ;		

Question	Answer	Marks	Guidance

4 Fig. 4.1 shows a Siamese fighting fish that lives in fresh water.

Fig. 4.1

Fig. 4.2 shows the effect of changing pH on the solubility of oxygen in water.

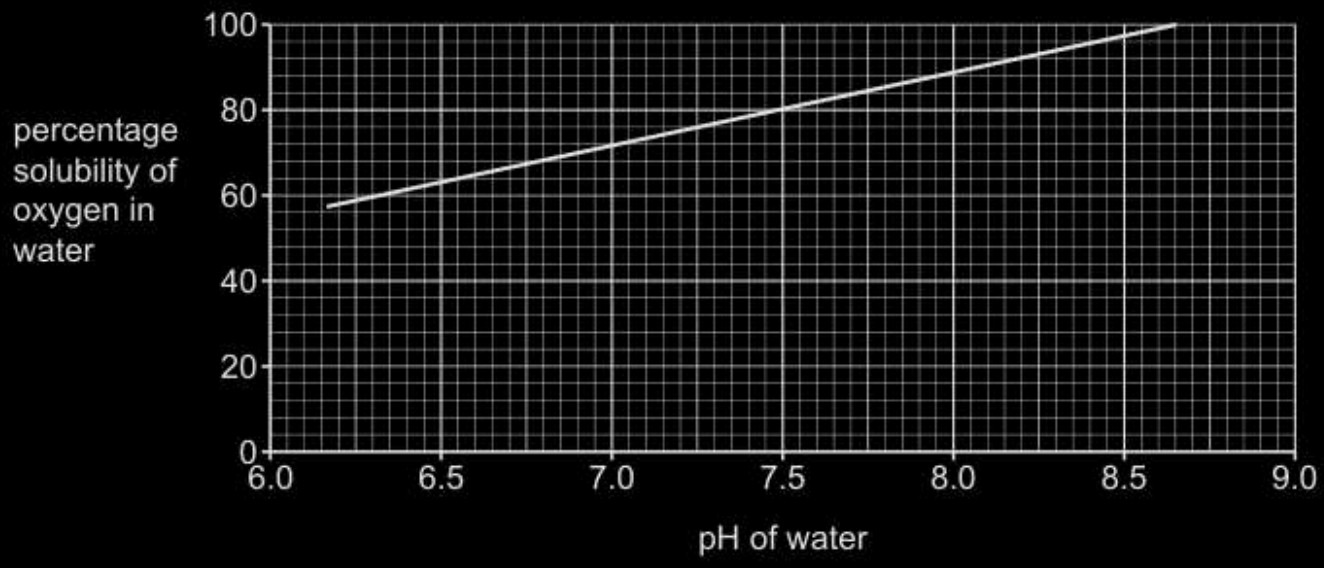


Fig. 4.2

Fig. 4.3 shows the effect of changing temperature on the solubility of oxygen in water.

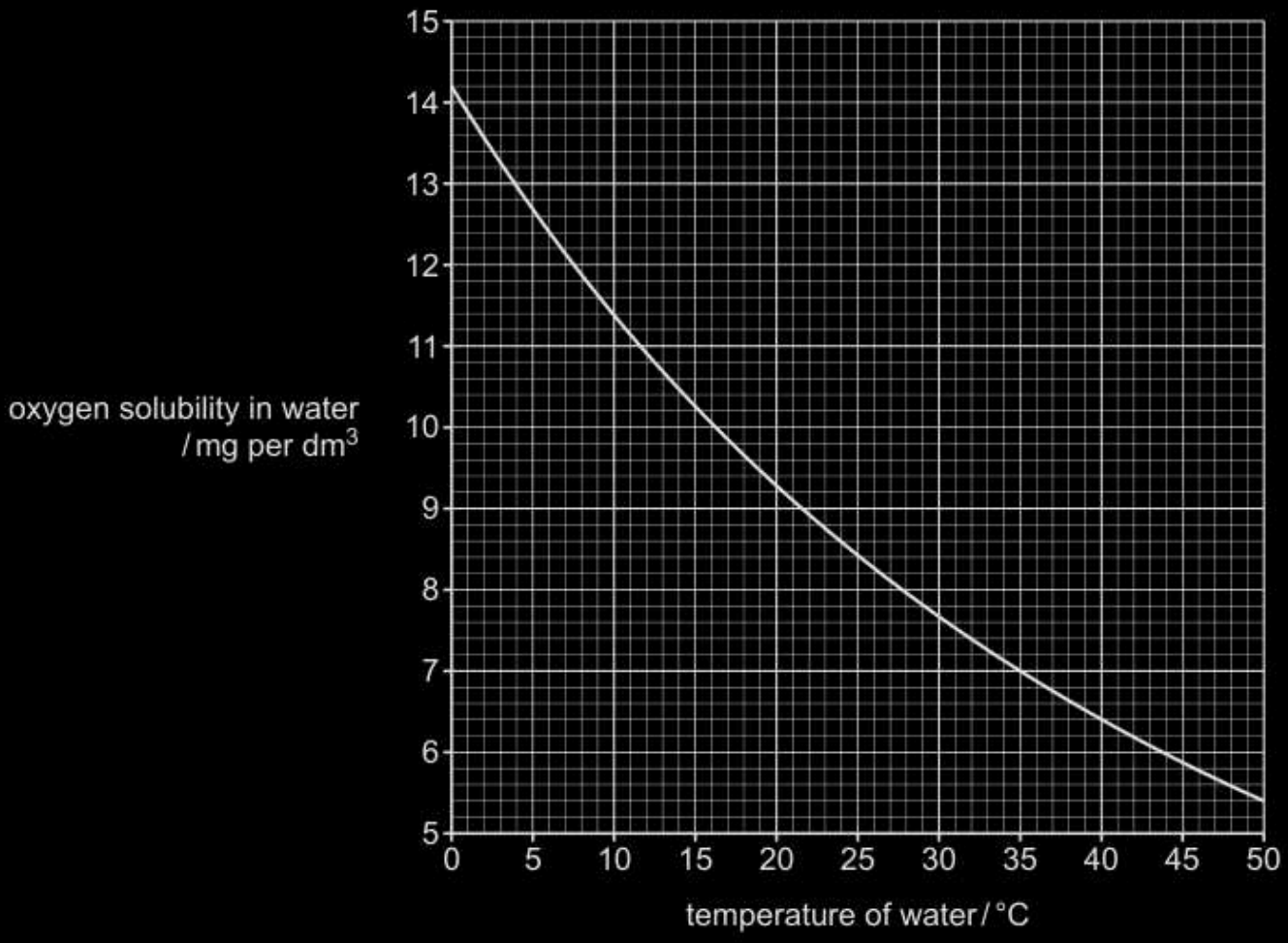


Fig. 4.3

(a) Siamese fighting fish live in shallow water that can range in pH from 6.9 to 8.2 and in temperature from 15 °C to 40 °C.

(i) Use the information in Fig. 4.2 and Fig. 4.3 to explain how the conditions in which these fish live may be challenging.

.....

.....

.....

.....

.....

..... [3]

(ii) All fish have gills that provide a gas exchange surface in water.

Siamese fighting fish also have an organ that has similar features to those of a human lung. These fish swim to the surface of water to take air from the atmosphere into this organ.

Explain the advantage to Siamese fighting fish of also having this organ.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Siamese fighting fish have 42 chromosomes in each diploid cell.

State each of the following.

(i) the named structure in each diploid cell that contains the chromosomes

..... [1]

(ii) the type of nuclear division that produces gametes

..... [1]

(iii) the number of chromosomes in each gamete of the Siamese fighting fish.

..... [1]

(c) The dramatic shape and colour of Siamese fighting fish are produced by selective breeding.

Suggest how the process of selective breeding in Siamese fighting fish can be used to produce these desirable features.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 15]

4(a)(i)	**1** low oxygen availability for fish **AW** ; **2** at low end (6.9) of pH range / at high end (40 ºC) of temperature range ; **3** for (aerobic) respiration ; **4** as pH increases solubility of oxygen increases **ORA** ; **5** as temperature increases solubility of oxygen decreases **ORA** ; **6** evidence of cross-referencing fish range to a graph by data quote ;	**3**	
4(a)(ii)	**1** useful in low / no oxygen ; **2** more + gas exchange / gases exchanged with air ; **3** more + oxygen available for fish ; **4** for (aerobic) respiration ; **5** to release energy ; **6** use of energy ;	**4**	
4(b)(i)	<u>nucleus</u> ;	**1**	
4(b)(ii)	<u>meiosis</u> ;	**1**	
4(b)(iii)	21 ;	**1**	

Question	Answer	Marks	Guidance
4(c)	**1** selection is by human ; **2** choose / select **AW** + desirable feature **AW** ; **3** crossing / breeding / offspring or next generation produced ; **4** (offspring) with best features selected to breed ; **5** repeated many times / many generations ;	5	

Question	Answer	Marks	Guidance

October/November 2023 · Paper 2 · Variant 2 · Question 6

6 Fig. 6.1 is a cross-section of a plant leaf.

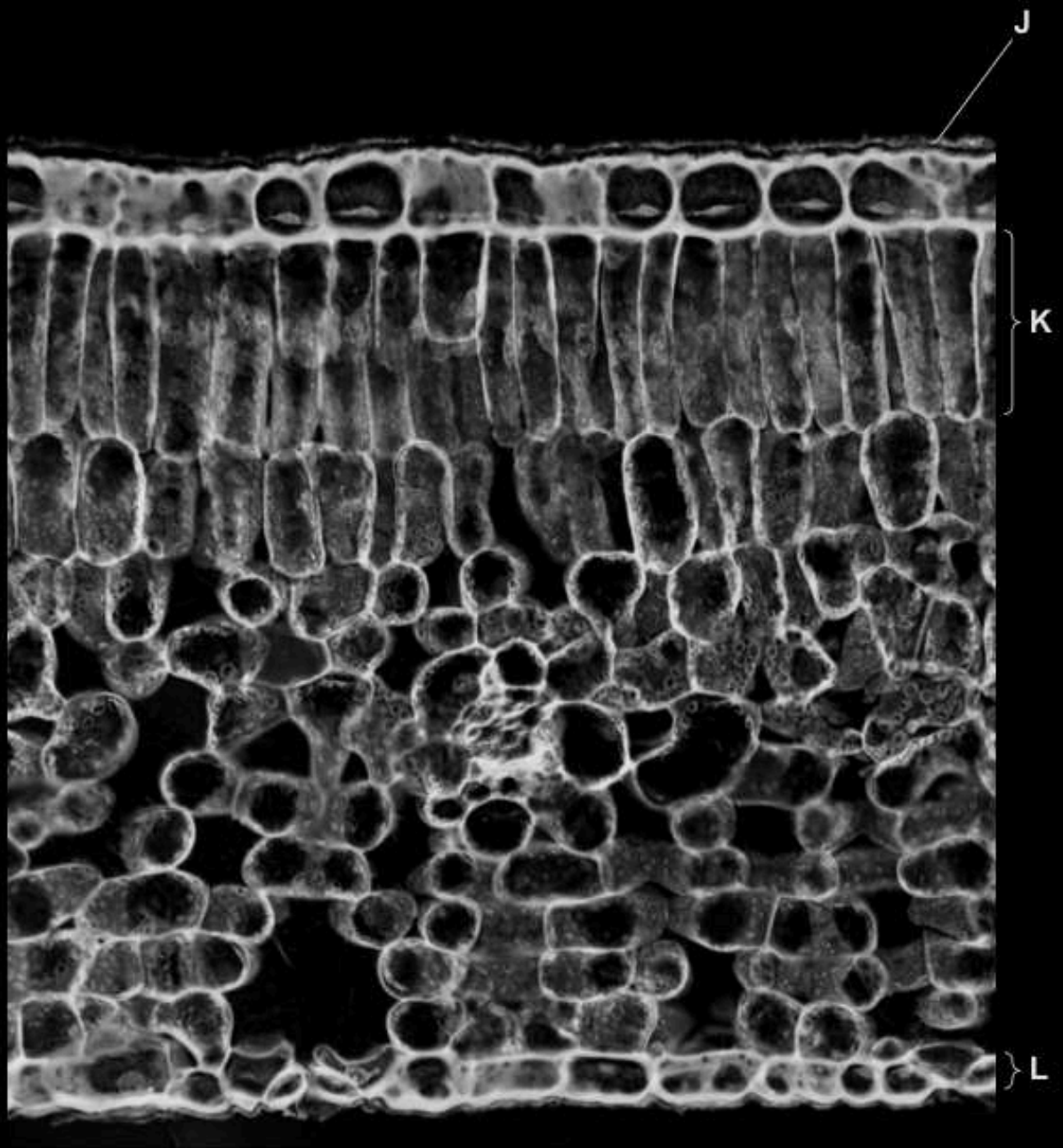


Fig. 6.1

(a) Identify each of **J**, **K** and **L** and explain the importance of each in the process of photosynthesis.

J .....

.....

.....

.....

K .....

.....

.....

.....

L .....

.....

.....

.....

[6]

(b) Name and describe the function of **one** tissue found in the vascular bundle of a leaf.

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

6(a)	(**J**) cuticle ; transparent / allows light through ; (**K**) palisade ; chloroplasts / chlorophyll + absorb light **AW** ; (**L**) lower epidermis ; guard cells / stomata ; carbon dioxide in / oxygen out ;		6
6(b)	xylem ; **OR** phloem ; transpiration; transport + water / mineral ions ; in one direction / upwards **AW** ; support **AW** ;	translocation ; transport + sucrose / amino acids ; up + or down ; source + to sink ;	4

Question	Answer

May/June 2023 · Paper 1 · Variant 1 · Question 1

Correct answer: A

1 Which description applies to a red blood cell?

	cell wall	nucleus
A	absent	absent
B	absent	present
C	present	absent
D	present	present

May/June 2023 · Paper 1 · Variant 1 · Question 3

Correct answer: B

3 Which statement is always correct for both diffusion and osmosis?

A They depend on energy from cells.

B They involve the movement of particles down a concentration gradient.

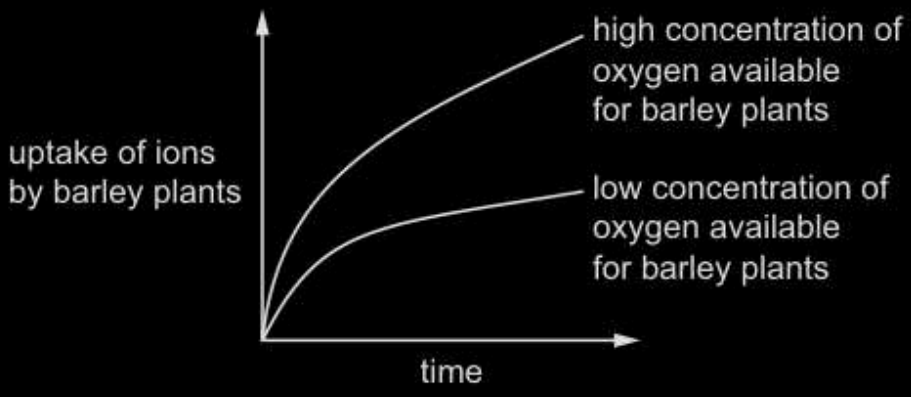
C They involve the movement of water molecules.

D They take place only in liquids.

May/June 2023 · Paper 1 · Variant 1 · Question 4

Correct answer: A

4 The graph shows the effect of oxygen on the uptake of ions by two sets of barley plants.



Except for oxygen availability, both sets of barley plants were grown in identical conditions.

Which process is taking place to explain the difference between the two curves?

A active transport

B diffusion

C osmosis

D photosynthesis

May/June 2023 · Paper 1 · Variant 1 · Question 5

Correct answer: C

5 A student tested a sample of food.

The results are shown.

colour after adding Benedict's solution and heating	colour after adding biuret reagent	colour after adding iodine solution
red	purple	yellow / brown

Which nutrients were in the food?

A protein, starch and reducing sugar

B protein and starch only

C protein and reducing sugar only

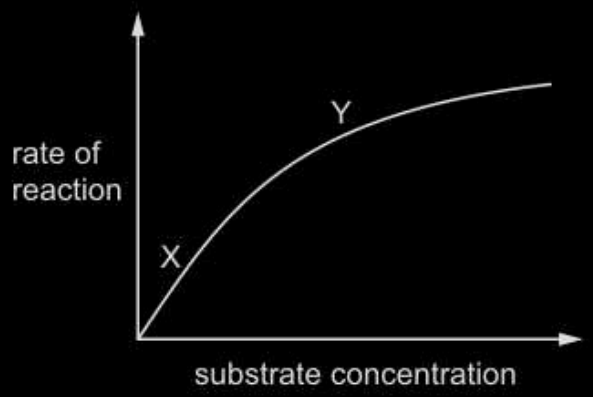
D starch and reducing sugar only

May/June 2023 · Paper 1 · Variant 1 · Question 6

Correct answer: B

6 The graph shows the effect of substrate concentration on the rate of an enzyme-controlled reaction.

The results have been obtained using a fixed concentration of enzyme at a temperature of 30 °C.



Which statement explains the shape of the graph from X to Y?

A Denaturation of the enzyme molecules is increasing.

B The enzyme active sites are not all occupied.

C The concentration of substrate molecules is decreasing.

D The frequency of effective collisions between enzyme active sites and substrate molecules is decreasing.

May/June 2023 · Paper 1 · Variant 1 · Question 9

Correct answer: D

9 Which description of the function of root hairs is correct?

A Root hairs absorb ions by active transport and osmosis.

B Root hairs absorb ions by osmosis only.

C Root hairs absorb water by active transport.

D Root hairs absorb water by osmosis.

May/June 2023 · Paper 1 · Variant 1 · Question 10

Correct answer: B

10 Which conditions result in the highest rate of transpiration from a plant?

	percentage humidity	temperature / °C
A	60	15
B	60	25
C	100	15
D	100	25

May/June 2023 · Paper 1 · Variant 1 · Question 12

Correct answer: A

12 A student states: 'The contraction of circular muscles pushes the contents forward.'

Which process is being described?

A peristalsis

B deamination

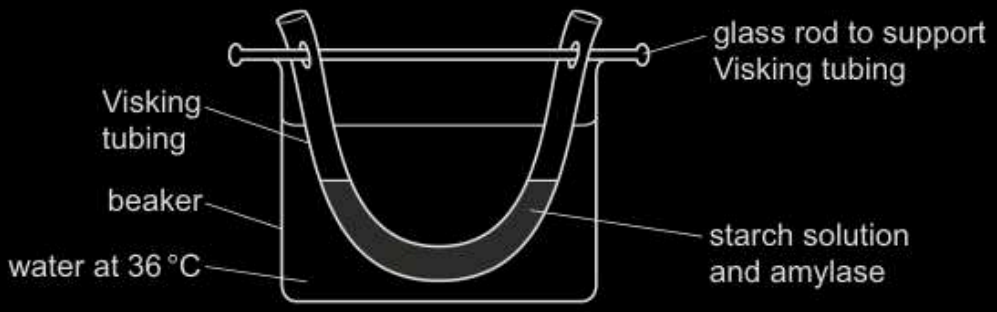
C translocation

D transpiration

May/June 2023 · Paper 1 · Variant 1 · Question 13

Correct answer: D

13 An investigation is carried out on digestion and absorption in the alimentary canal. The diagram shows the apparatus used. The Visking tubing is permeable to small molecules such as glucose but not to large molecules such as starch.



After one hour, samples of **water in the beaker** are tested with Benedict's solution and with iodine solution.

Which colours are obtained?

	colour obtained after heating with Benedict's solution	colour obtained after adding iodine solution
A	blue	blue-black
B	blue	yellow-brown
C	red	blue-black
D	red	yellow-brown

May/June 2023 · Paper 1 · Variant 1 · Question 17

Correct answer: A

17 Blood pressure was measured in the pulmonary artery, pulmonary capillaries and pulmonary vein.

Which blood pressure results could be correct?

	blood pressure in pulmonary artery / kPa	blood pressure in pulmonary capillaries / kPa	blood pressure in pulmonary vein / kPa
A	14	8	5
B	14	5	8
C	5	8	14
D	5	14	8

May/June 2023 · Paper 1 · Variant 1 · Question 18

Correct answer: B

18 Which component of blood engulfs pathogens?

A lymphocytes

B phagocytes

C platelets

D red blood cells

May/June 2023 · Paper 1 · Variant 1 · Question 23

Correct answer: C

23 The blood in the renal vein has a different composition to the blood in the renal artery.

How is the blood in the renal vein different?

A less carbon dioxide, less urea, less water

B less oxygen, more urea, less water

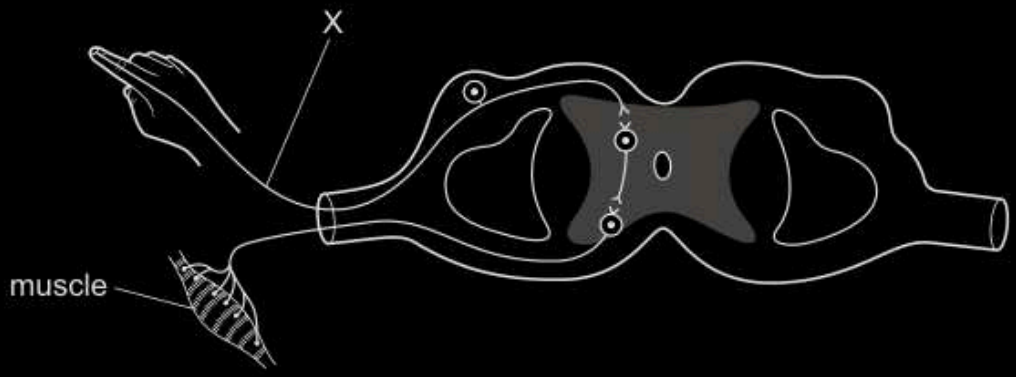
C more carbon dioxide, less urea, less water

D more oxygen, less urea, less water

May/June 2023 · Paper 1 · Variant 1 · Question 24

Correct answer: A

24 The diagram shows a reflex arc.



A person was involved in an accident and their arm was injured. The sensory neurones at X were cut.

Which statement describes what happens if this person touches a hot object with their finger?

A The person is not able to feel their finger touching the hot object but is able to move their finger away from the heat stimulus.

B The person is not able to feel their finger touching the hot object or move their finger away from the heat stimulus.

C The person can feel the heat stimulus but is not able to move their finger away from the heat stimulus.

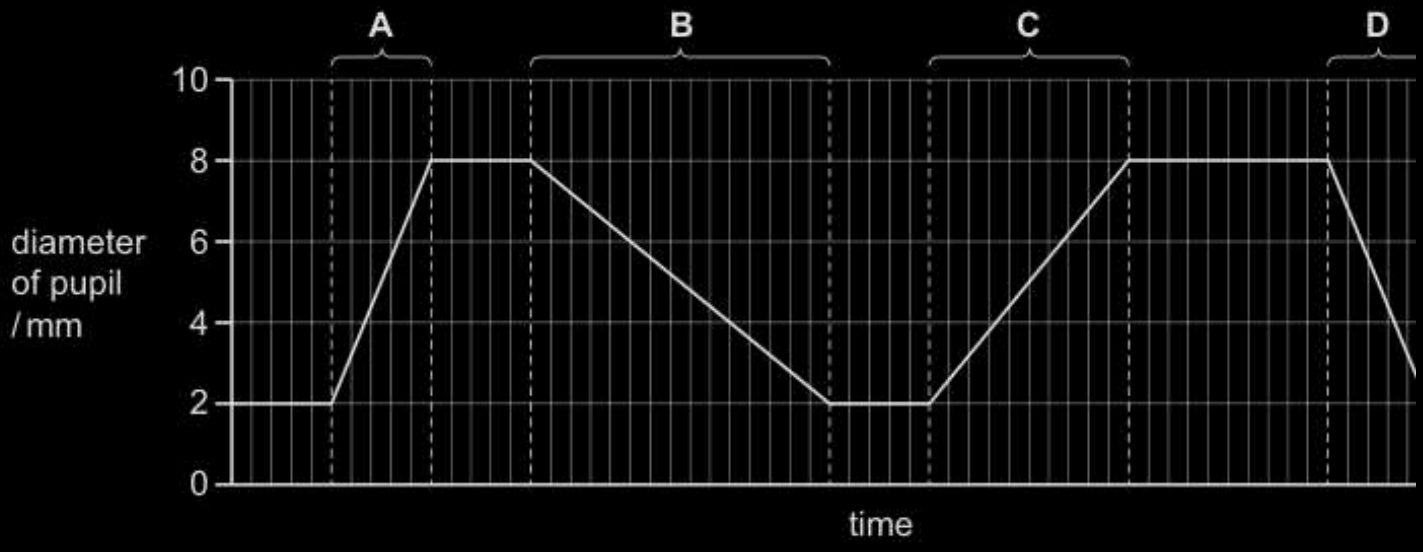
D The receptor cells in the person's finger are not stimulated by the heat.

May/June 2023 · Paper 1 · Variant 1 · Question 25

Correct answer: D

25 The diameter of a person's pupil is measured as the light intensity is varied.

During which time period does the light intensity increase fastest?



May/June 2023 · Paper 1 · Variant 1 · Question 26

Correct answer: D

26 On a warm sunny day, a swimmer climbs out of a pool and sits on the edge to dry.

After a few minutes, the swimmer's skin begins to feel cold.

What has caused this to happen?

A Body temperature drops until it reaches air temperature.

B Muscles and other organs stop producing heat from respiration.

C Sunlight causes the hypothalamus to begin processes that increase heat loss.

D Water on the skin absorbs heat from the body and evaporates.

May/June 2023 · Paper 1 · Variant 1 · Question 27

Correct answer: A

27 A genetic disorder causes too much insulin to be secreted.

What is the effect of the secretion of too much insulin?

A Blood glucose concentration is low.

B Glucose can be detected in the urine.

C Glycogen level in the liver is low.

D Less glucose is converted to glycogen.

May/June 2023 · Paper 1 · Variant 1 · Question 28

Correct answer: A

28 Auxin is a chemical produced by plants.

What is the role of auxin?

A It stimulates cells in the plant shoot to elongate.

B It stimulates meiotic cell division.

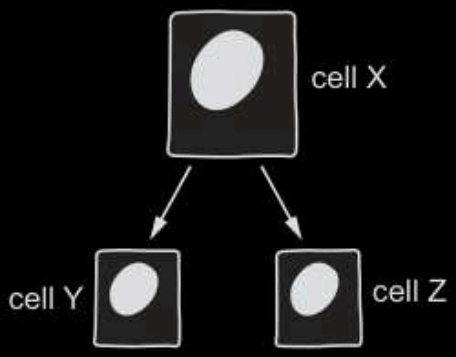
C It stimulates the development of a branching root system.

D It stimulates the process of seed germination.

May/June 2023 · Paper 1 · Variant 1 · Question 29

Correct answer: B

29 The diagram shows cell X which contains 24 chromosomes. It divides by mitosis to produce cells Y and Z.



How many chromosomes does cell Z contain?

A 12 **B** 24 **C** 46 **D** 48

May/June 2023 · Paper 1 · Variant 1 · Question 30

Correct answer: D

30 What is a feature of asexual reproduction in a plant?

A All the offspring will grow to an identical height.

B Any seeds produced will be identical.

C The plant needs to carry out the process of meiosis.

D The offspring will all have identical genotypes.

May/June 2023 · Paper 1 · Variant 1 · Question 34

Correct answer: B

34 Some lizards in a population had longer leg bones than others. They were able to climb better than the lizards with shorter legs, so they were able to escape from their predators on the ground. Over time, lizards with shorter legs died out so that all the lizards in the population now have longer legs.

Which process has caused this change within the lizard population?

A extinction

B natural selection

C conservation

D artificial selection

May/June 2023 · Paper 1 · Variant 1 · Question 35

Correct answer: B

35 The genotype for the height of an organism is written as Tt.

Which conclusion may be drawn?

A The allele for height has at least two different genes.

B There are at least two different alleles of the gene for height.

C There are two different genes for height, each having a single allele.

D There is one allele for height with two different forms.

May/June 2023 · Paper 1 · Variant 2 · Question 1

Correct answer: B

1 How permeable are the cell wall and the cell membrane in a plant cell?

	cell wall	cell membrane
A	fully	fully
B	fully	partially
C	partially	fully
D	partially	partially

May/June 2023 · Paper 1 · Variant 2 · Question 3

Correct answer: B

3 Which statements about diffusion are correct?

	molecules move from a higher to a lower concentration	only occurs within living systems	rate slows down as the temperature increases
A	✓	✓	✗
B	✓	✗	✗
C	✗	✓	✓
D	✗	✗	✓

key

✓ = yes

✗ = no

May/June 2023 · Paper 1 · Variant 2 · Question 4

Correct answer: D

4 Some ways in which molecules may move into and out of cells are listed.

1 Molecules move from a higher concentration to a lower concentration.

2 Molecules move from a lower concentration to a higher concentration.

3 Molecules move from a higher concentration to a lower concentration across a partially permeable membrane.

Which statement is correct?

A 1 describes osmosis and 3 describes active transport.

B 2 describes osmosis and 3 describes diffusion.

C 1 describes active transport and 2 describes osmosis.

D 2 describes active transport and 3 describes osmosis.

May/June 2023 · Paper 1 · Variant 2 · Question 5

Correct answer: A

5 Food tests were carried out on a solution containing biological molecules.

The table shows the results of the tests.

name of test	result of test
biuret test	purple solution
Benedict's test	blue solution

Which conclusion can be drawn from these results?

A The solution contains protein but no glucose.

B The solution contains protein and glucose.

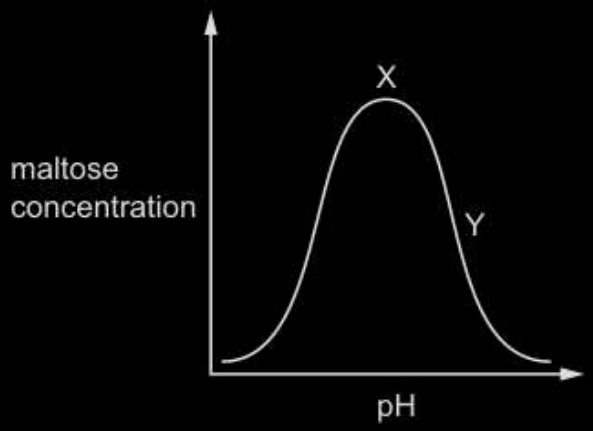
C The solution contains glucose but no protein.

D The solution does not contain glucose or protein.

May/June 2023 · Paper 1 · Variant 2 · Question 6

Correct answer: C

6 The graph shows the concentration of maltose produced when amylase digests starch at different pH values.



What explains the shape of the graph between point X and point Y?

A The enzyme is completely denatured and does not work at all.

B The enzyme molecules have less kinetic energy.

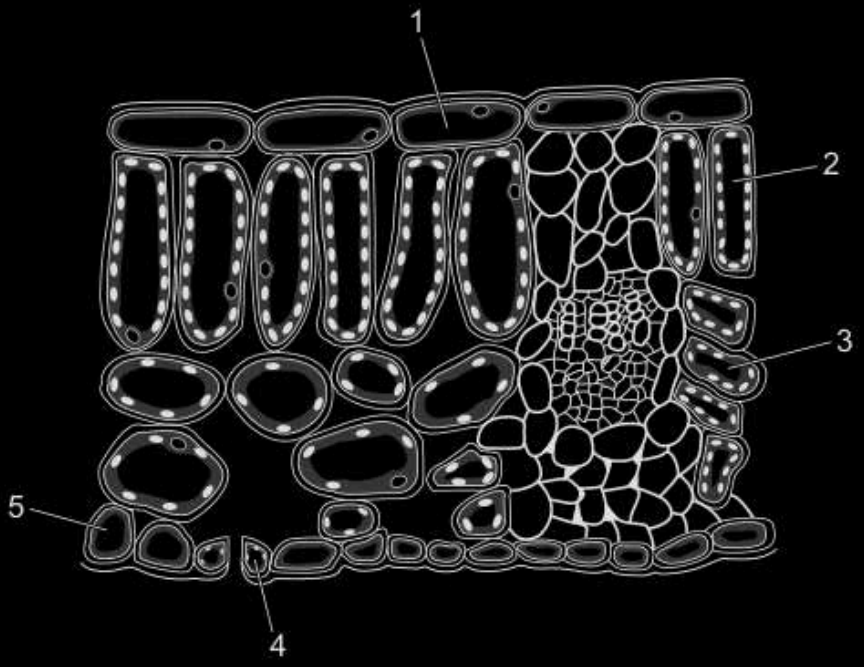
C The shape of the enzyme active site is changing and does not fit the substrate so well.

D The substrate molecules have more kinetic energy.

May/June 2023 · Paper 1 · Variant 2 · Question 7

Correct answer: C

7 The diagram shows a section through part of a leaf from a dicotyledonous plant.



Which cells carry out photosynthesis?

A 1, 2 and 3 **B** 1, 4 and 5 **C** 2, 3 and 4 **D** 3, 4 and 5

May/June 2023 · Paper 1 · Variant 2 · Question 8

Correct answer: D

8 Which row shows why plants need magnesium ions and nitrate ions?

	magnesium ions	nitrate ions
A	to make fatty acids	to make proteins
B	to make amino acids	to make chlorophyll
C	to make chlorophyll	to make cellulose
D	to make chlorophyll	to make amino acids

May/June 2023 · Paper 1 · Variant 2 · Question 9

Correct answer: A

9 Root hairs take in water and ions from the soil.

Which table shows the correct concentrations of water and ions in the root hair?

A

substance	concentration in root hair
water	lower than the soil
ions	higher than the soil

B

substance	concentration in root hair
water	higher than the soil
ions	higher than the soil

C

substance	concentration in root hair
water	higher than the soil
ions	lower than the soil

D

substance	concentration in root hair
water	lower than the soil
ions	lower than the soil

May/June 2023 · Paper 1 · Variant 2 · Question 13

Correct answer: C

13 A person who has coeliac disease has flattened villi in their ileum. If a person with coeliac disease eats the same quantity and types of food as a disease-free person, they will have lower levels of glucose in their hepatic portal vein.

Which row explains this?

	capillaries do not absorb the glucose molecules	the surface area of the ileum is reduced
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = yes

✗ = no

May/June 2023 · Paper 1 · Variant 2 · Question 17

Correct answer: B

17 Which chamber of the heart has the thickest muscular walls?

A left atrium

B left ventricle

C right atrium

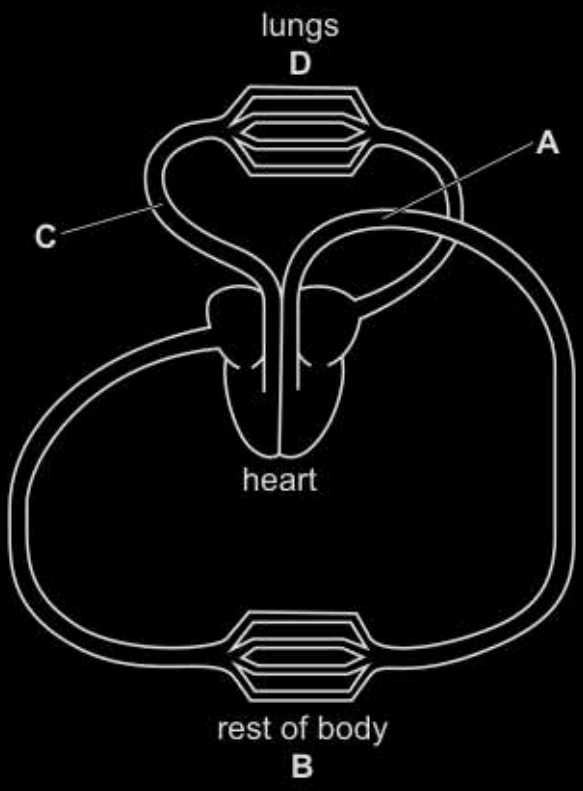
D right ventricle

May/June 2023 · Paper 1 · Variant 2 · Question 18

Correct answer: A

18 The diagram shows the human circulatory system.

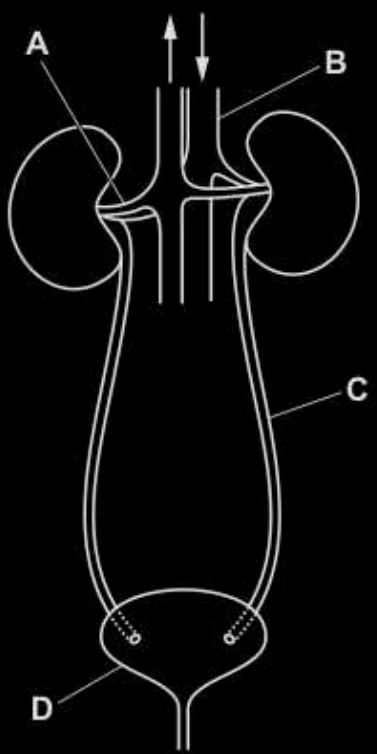
At which point is the pressure of the blood circulating through the blood vessels the highest?



May/June 2023 · Paper 1 · Variant 2 · Question 23

Correct answer: A

23 In which structure is there a fluid with a low concentration of urea?



May/June 2023 · Paper 1 · Variant 2 · Question 24

Correct answer: D

24 A person blinks when an insect flies into their eye.

What explains this reflex action?

A muscles in the eyelid cause the eye to blink → impulses are sent down the motor neurones → relay neurones → sensory neurones → receptor cells in the cornea detect the insect

B muscles in the eyelid cause the eye to blink → impulses are sent down the sensory neurones → relay neurones → motor neurones → receptor cells in the retina detect the insect

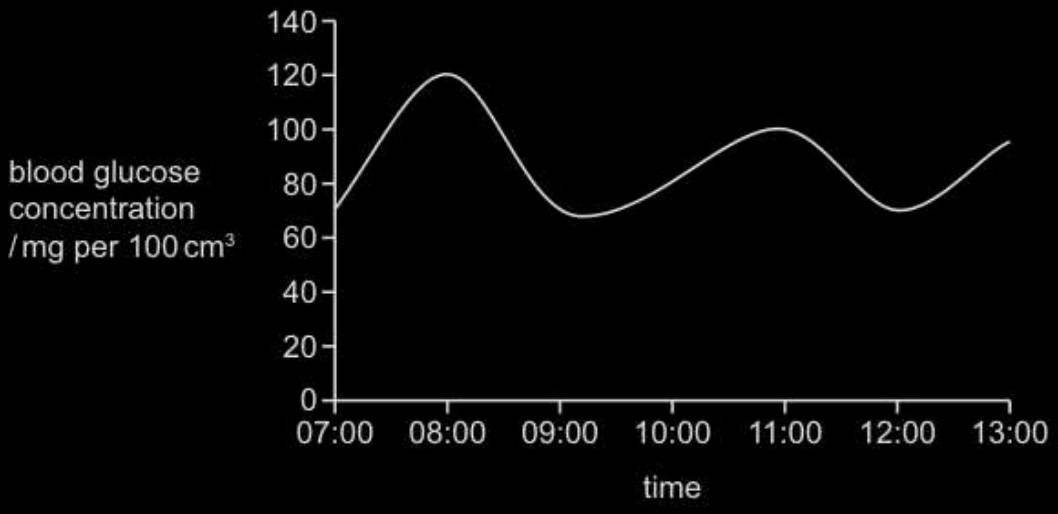
C receptor cells in the retina detect the insect → impulses are sent down the relay neurones → sensory neurones → motor neurones → muscles in the eyelid cause the eye to blink

D receptor cells in the cornea detect the insect → impulses are sent down sensory neurones → relay neurones → motor neurones → muscles in the eyelid cause the eye to blink

May/June 2023 · Paper 1 · Variant 2 · Question 26

Correct answer: B

26 The graph shows the blood glucose concentration for a person in the hours after eating breakfast at 07:00.



Which level of blood glucose concentration is approximately the set point?

A 70 mg / cm^3

B 80 mg / cm^3

C 100 mg / cm^3

D 120 mg / cm^3

May/June 2023 · Paper 1 · Variant 2 · Question 27

Correct answer: C

27 Which hormone causes blood glucose concentration to be lowered?

A adrenaline

B glucagon

C insulin

D testosterone

May/June 2023 · Paper 1 · Variant 2 · Question 28

Correct answer: D

28 Which statement about auxin is correct?

A Auxin is equally distributed in the plant in response to gravity and light.

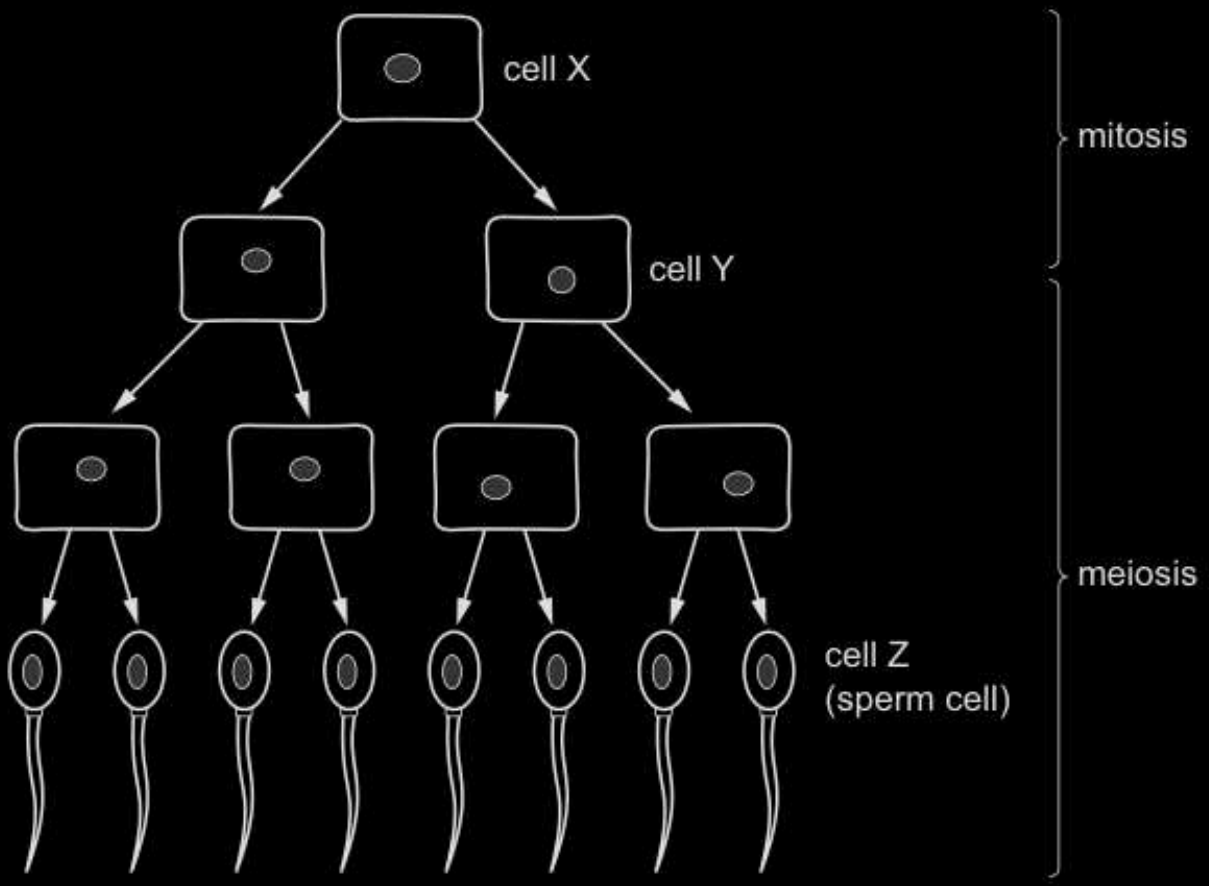
B Auxin inhibits cell elongation.

C Auxin is made in the plant stem.

D Auxin spreads through the plant from the shoot tip.

Correct answer: C

29 The diagram shows some stages in cell division in a fruit fly.



Cell X contains 8 chromosomes.

How many chromosomes are in cell Y and in cell Z?

	cell Y	cell Z
A	4	4
B	4	8
C	8	4
D	8	8

May/June 2023 · Paper 1 · Variant 2 · Question 30

Correct answer: B

30 Which statement applies to organisms that reproduce asexually?

A Offspring always grow to the same height.

B Offspring are genetically identical.

C Organisms only reproduce asexually during extreme conditions.

D Reproduction happens by the fusion of diploid nuclei.

May/June 2023 · Paper 1 · Variant 2 · Question 33

Correct answer: D

33 Which row describes the exchange of substan

	passing from mother to fetus
A	carbon dioxide, glucose and alcohol
B	carbon dioxide, nicotine and glucose
C	oxygen and urea
D	oxygen, glucose and antibodies

May/June 2023 · Paper 1 · Variant 2 · Question 34

Correct answer: B

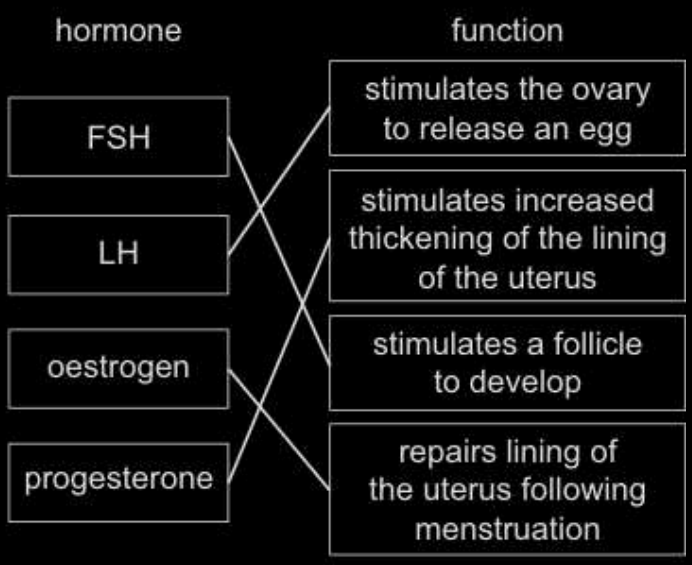
34 In a strand of DNA containing 100 nucleotide p

How many nucleotides in this DNA strand cont

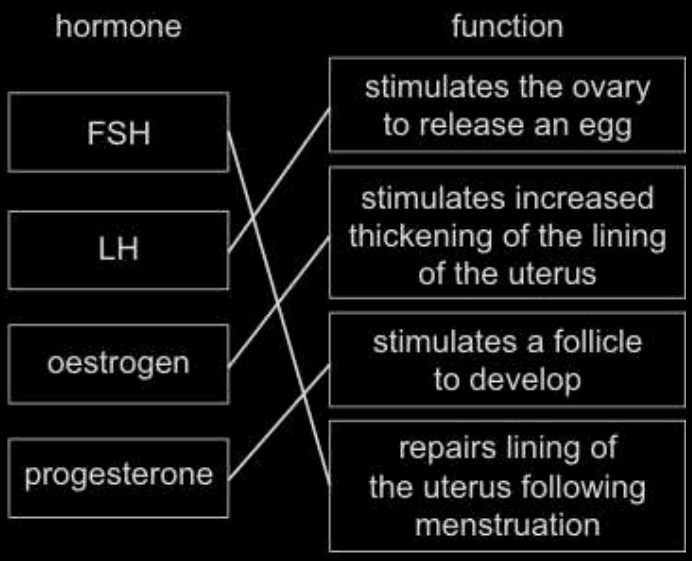
A 23 **B** 46 **C** 5

ıeir functions in preparation for pregnancy?

B



D



ıces at the placenta?

passing from fetus to mother
oxygen and urea
oxygen and urea
carbon dioxide and glucose
carbon dioxide and urea

pairs, 46 nucleotides contain the base A.

ıtain the base T?

54 **D** 154

Correct answer: C

35 Two brothers had different blood groups. One was blood group A and the other was blood group B.

What can be concluded about their parents' blood group genotypes?

A Both of the parents must be heterozygous.

B Both of the parents must be homozygous.

C One parent (at least) must be heterozygous.

D One parent (at least) must be homozygous.

3 Fig. 3.1 shows a young tomato plant growing near a window.
Cells from two positions in the stem are also shown.

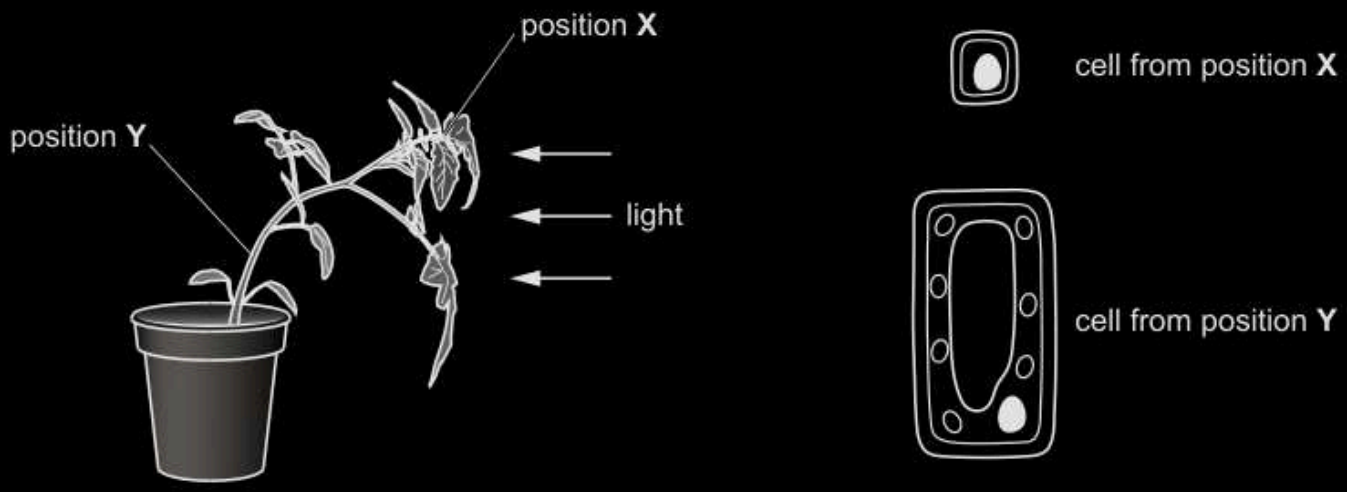


Fig. 3.1

The cells at **X** divide constantly. They then increase in size and some form mature cells like the one shown from position **Y**.

(a) Cells at positions **X** and **Y** differ in size and shape.
Give **two other** differences between these cells.

1 .....

.....

2 .....

.....

[2]

(b) Explain how the movement of water molecules is involved in this increase in cell size.

.....

.....

.....

.....

.....

..... [3]

(c) The tomato plant is responding to the light.
Name this response and explain how auxin and cell elongation are involved.

name of response .....

explanation .....

.....

.....

.....

.....

.....

[5]

(d) The diploid number of chromosomes in a tomato plant cell is 24.
State how many chromosomes a cell from position **Y** will contain and explain your answer.

number .....

explanation .....

.....

.....

[2]

[Total: 12]

3(a)	any **two** from: cell at Y has chloroplasts / cell at X has no chloroplasts ; cell at Y has a (sap) vacuole / cell at X has no vacuole ; cell at Y has more cytoplasm / cell at X has less cytoplasm ; **Max 2**	**2**	

© UCLES 2023

Question	Answer	Marks	Guidance
3(b)	any **three** from: osmosis ; (movement of water molecules) from high to low water potential ; water source, e.g. soil / surrounding cells across a partially / selectively permeable (cell) membrane ; increases volume of sap / cytoplasm ; until equilibrium reached / no net movement / pressure from cell wall prevents further expansion ; **Max 3**	**3**	**A** down a water potential gradient References to concentration must specify solute or water **A** xylem
3(c)	phototropism ;	**1**	
	any **four** from: auxin produced in shoot tip / cells at X ; diffuses down stem / shoot ; more on shaded side / less on side exposed to light ; promotes / stimulates **AW** + cell elongation ; cells on shaded side elongate more than those on light side ; bends/grows towards light ; **Max 4**	**4**	**R** auxin destroyed on light side **Ig** uneven distribution unless qualified **Ig** moves towards the light

© UCLES 2023

Question	Answer	Marks	Guidance
3(d)	24 ;	**1**	
	cells at X dividing by mitosis ;	**1**	**A** references to the production of genetically identical cells / cells at Y are not gametes

Question	Answer	Marks	Guidance

4 Fig. 4.1 shows a label from a snack bar.

Nutrition Facts

One bar: 70 g

	mass in g	percentage of recommended daily value
fat	7	11%
dietary fibre	3	12%
sugars	10	–
protein	30	56%
vitamin C		0%
iron		30%

Fig. 4.1

(a) (i) Explain how the nutrition provided by the snack bar is suitable for a person training for a sporting event.

.....

.....

.....

.....

.....

..... [4]

(ii) Calculate how many grams of dietary fibre are required in the diet to obtain 100% of the recommended daily value.

..... g [2]

(b) Describe what happens to protein in the digestive system.

.....

.....

.....

.....

.....

..... [5]

[Total: 11]

4(a)(i)	any **four** from: protein to build muscles ; iron for haemoglobin / red blood cells ; transport of oxygen ; fats / sugars for energy ; respiration ; enhanced performance of muscles **AW** ; **Max 4**	**4**	
4(a)(ii)	25 (g) ;;	**2**	100 ÷ 12 = 1 mark (suitable working)
4(b)	any **five** from: pepsin / protease ; in stomach ; trypsin ; in duodenum / small intestine ; breaks down / digests to peptides / amino acids ; amino acids absorbed ; **Max 5**	**5**	

© UCLES 2023

Question	Answer	Marks	Guidance

5 Myopia is an eye condition. If a person has myopia, near objects can be seen clearly but not those that are far away.
It can happen when the eyeballs grow slightly too long as shown in Fig. 5.1.

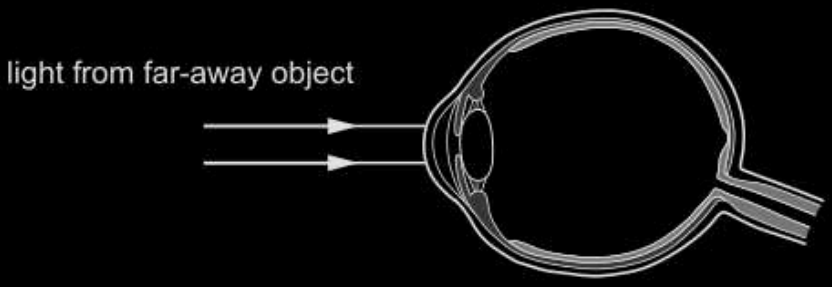


Fig. 5.1

(a) (i) Explain how parts of the eye are involved in producing a clear image of a near object.

.....

.....

.....

.....

.....

..... [4]

(ii) Suggest why individuals with long eyeballs cannot see far-away objects clearly.
You may draw on the diagram of the eye in Fig. 5.1.

.....

.....

..... [1]

(b) A country carried out two national health surveys separated by approximately 30 years.
Table 5.1 shows the information that was obtained about myopia.

Table 5.1

	percentage of people with myopia			
	males		females	
age	1971–72	1999–2004	1971–72	1999–2004
12–17	21.7	30.9	26.4	37.0
18–24	22.5	29.7	32.5	46.4
25–34	20.2	38.9	27.8	49.1

(i) Comment on how the percentage of people with myopia has changed in this population.

.....

.....

.....

.....

.....

..... [3]

(ii) It is thought that myopia is the result of a combination of genetic and environmental factors.
It has been suggested that the changing percentages of people with myopia is a result of changes in human activities.
Suggest a change in human activity and how it could make a person less able to see far-away objects clearly.

.....

.....

.....

..... [2]

(c) Iris colour is genetically determined. The allele for blue colour, **b**, is recessive to the allele for brown colour, **B**.
Explain why it is **not** possible to have a heterozygous blue-eyed person.

.....

.....

.....

.....

..... [3]

[Total: 13]

5(a)(i)	any **four** from: light is refracted / bent by cornea ; ciliary muscles / body contract ; suspensory ligaments become slack ; lens becomes fatter **AW** ; light refracted / bent (more) by lens ; light / image focused / formed on retina; **Max 4**	**4**	
5(a)(ii)	lens bends / refracts light rays too much so they meet before retina / drawn on diagram ;	**1**	Mark written answer first On the diagram light rays must clearly cross before the retina
5(b)(i)	any **three** from: prevalence of myopia has increased ; in both males and females / in all age groups; increased most in 25–34 age group / older people / females ; correct quote of comparative data ; **Max 3**	**3**	
5(b)(ii)	less time spent outdoors / reference to more time spent on activities looking at near objects/computers/mobile phones ;	**1**	
	eyes not trained / not enough practice at looking at far away objects ;	**1**	
5(c)	blue-eyed person has two recessive alleles / would be bb ; blue-eyed person is homozygous / both alleles are the same ; heterozygous means the alleles are different / heterozygous would be Bb; B is dominant / Bb would have brown eyes ; **Max 3**	**3**	

© UCLES 2023

Question	Answer	Marks	Guidance

3 The nervous system coordinates and regulates body functions.

Fig. 3.1 shows two diagrams, each showing parts of the nervous system in a human.

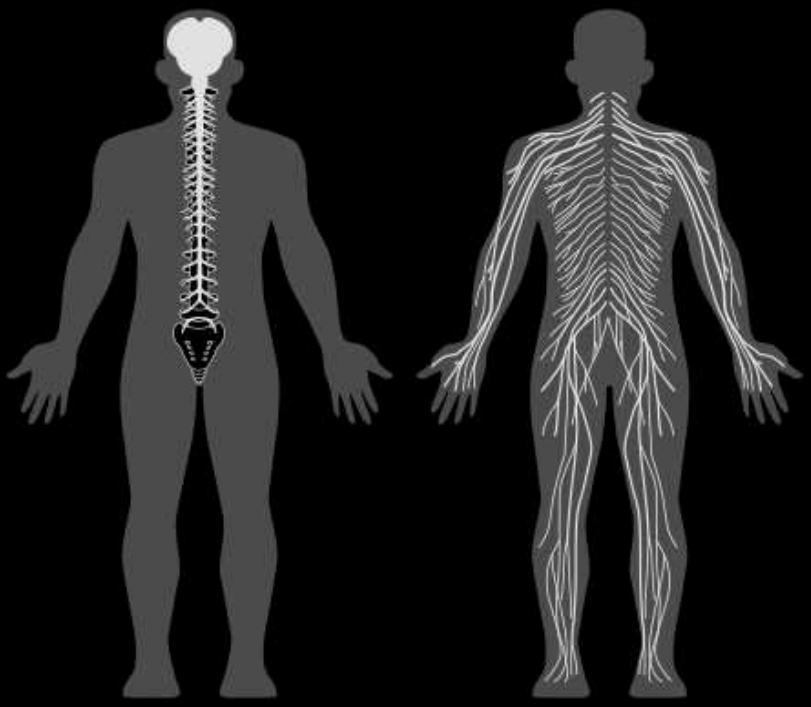


diagram **A** diagram **B**

Fig. 3.1

(a) Describe the parts of the nervous system shown in each diagram.

diagram **A** .....

.....

diagram **B** .....

.....

[4]

(b) Fig. 3.2 shows the junction between two neurones.

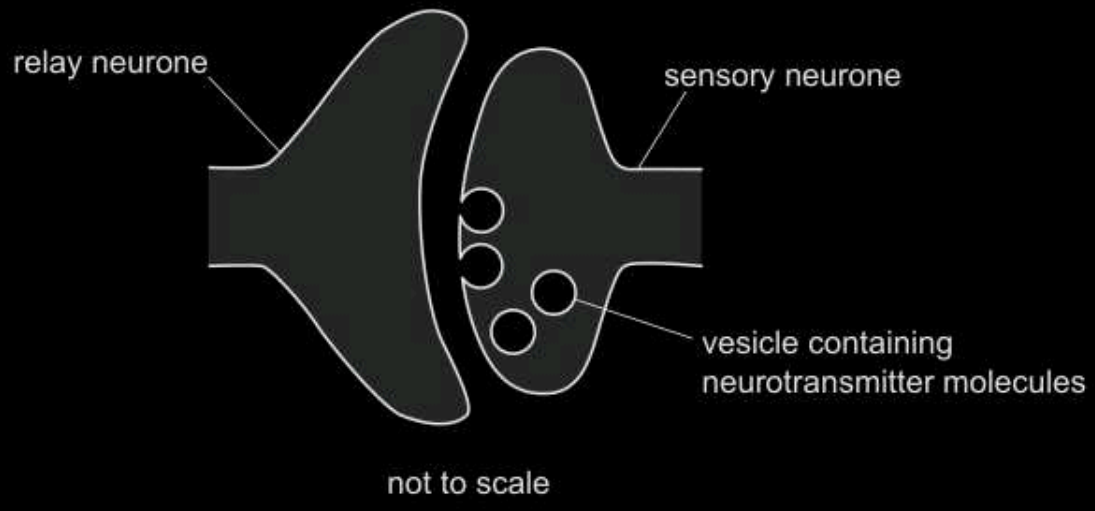


Fig. 3.2

(i) Name the junction shown in Fig. 3.2.

.....

[1]

(ii) Draw an arrow on Fig. 3.2 to show the direction that an impulse will travel. [1]

(iii) The distance between the two neurones shown in Fig. 3.2 is 0.00003 mm.

State this distance in units of μm.

Space for working.

..... μm [1]

(c) Nicotine is a chemical found in cigarette smoke.

A molecule of nicotine has a similar shape to a molecule of neurotransmitter.

Suggest and explain the effect of nicotine at the junctions between neurones.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

3(a)	*(diagram **A**)* central nervous system / CNS ; brain + spinal cord ; *(diagram **B**)* peripheral nervous system / PNS ; nerves / nervous tissue / neurones + outside CNS **AW** ;	**4**	

Question	Answer	Marks	Guidance
3(b)(i)	synapse ;	**1**	**Allow** synaptic cleft / synaptic gap
3(b)(ii)	← ;	**1**	
3(b)(iii)	0.03 ;	**1**	
3(c)	**1** nicotine + binds / blocks **AW** ; **2** receptor (on post-synaptic membrane / relay) ; **3** protein ; **4** impulse + generated / produced / travel / more **AW** ;	**3**	

Question	Answer	Marks	Guidance

5 Fig. 5.1 is a photomicrograph of the centre of a plant root.

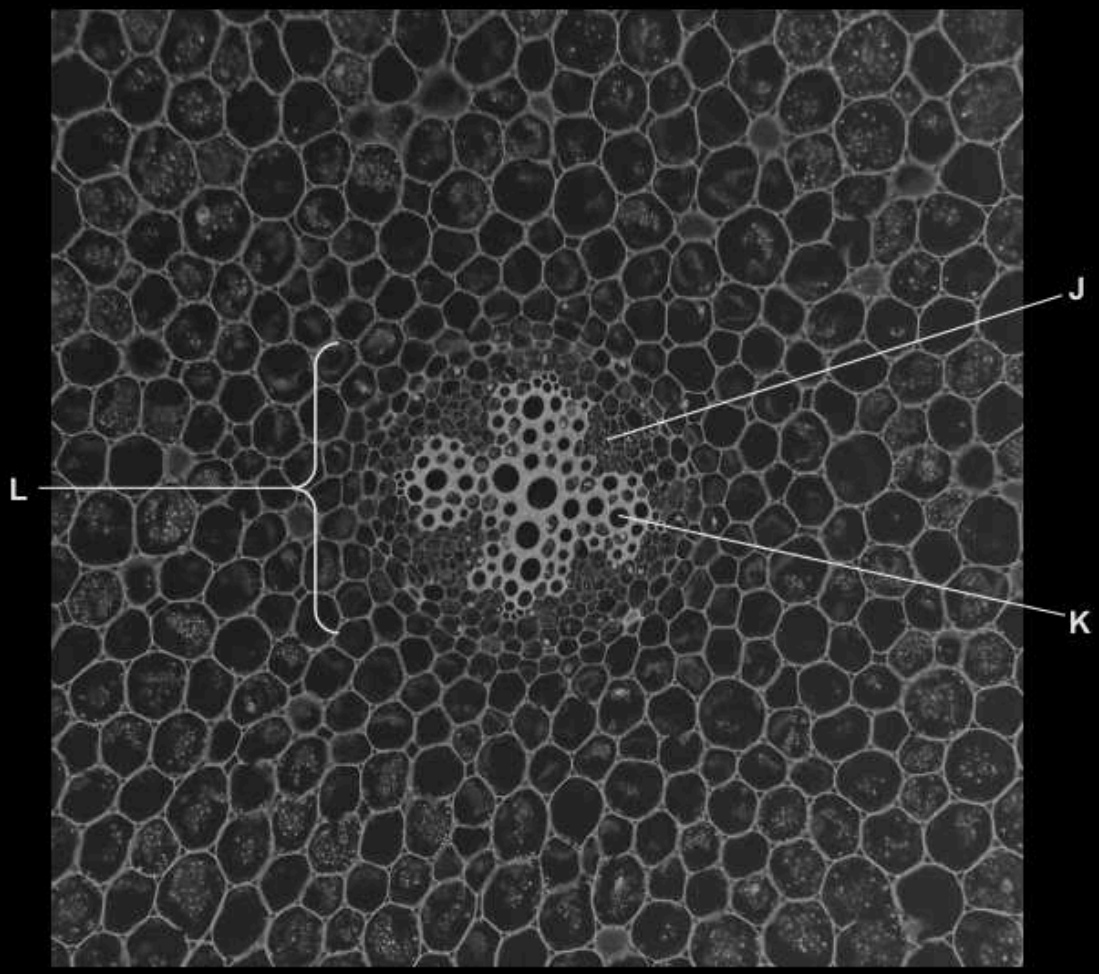


Fig. 5.1

(a) (i) Identify each of the tissues labelled **J** and **K**.

J .....

K .....

[2]

(ii) State the term used to describe the group of tissues labelled **L**.

..... [1]

(iii) State **two** ways in which the structure of a cell in tissue **K** will differ from a cell in tissue **J**.

1 .....

.....

2 .....

.....

[2]

(iv) Outline the function of tissue **J**.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) (i) Name the type of response shown by a plant root when it grows down into soil after the seed has germinated.

..... [2]

(ii) Explain how this response shown by the plant root is necessary for the leaves to grow and to appear green.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 15]

5(a)(i)	*(J)* phloem ; *(K)* xylem ;	2	

Question	Answer	Marks	Guidance
5(a)(ii)	vascular bundle / organ ;	1	
5(a)(iii)	**K** xylem / **ORA J** phloem thick wall ; / thin wall ; lignin / lignified ; / no lignin ; no + cell contents /organelles / named organelles ; / cell + contents / organelles / named organelles ; no + end wall / cross wall / sieve plates ; / sieve plates ; no + companion cell ; / companion cell ;	2	
5(a)(iv)	**1** translocation ; **2** (transport) sucrose ; **3** (transport) amino acids ; **4** from where produced / source + where used / to sink ; **5** bidirectional **AW** flow ;	4	

© UCLES 2023

Question	Answer	Marks	Guidance
5(b)(i)	gravitropism; positive + (gravitropism) ;	2	
5(b)(ii)	**1** absorbs / take up water ; **2** absorb / take up + ions / named ion / minerals ; **3** water + cell elongation **AW** ; **4** photosynthesis ; **5** magnesium + chlorophyll ; **6** nitrates + amino acids / proteins ;	4	

Question	Answer	Marks	Guidance

October/November 2022 · Paper 1 · Variant 1 · Question 1

Correct answer: B

A 1 and 2 **B** 1 and 3 **C** 1 only **D** 2 only

October/November 2022 · Paper 1 · Variant 1 · Question 2

Correct answer: B

2 What are the functions of xylem vessels?

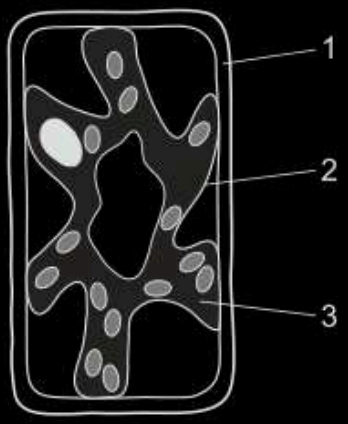
	strengthening roots and stems	supporting leaf shape	transporting sugars
A	✓	✗	✗
B	✓	✓	✗
C	✗	✓	✓
D	✗	✗	✓

key
✓ = yes
✗ = no

October/November 2022 · Paper 1 · Variant 1 · Question 3

Correct answer: D

3 The diagram shows a typical plant cell which has been in a concentrated salt solution for 10 minutes.



Which numbered structures are partially permeable?

A 1 and 2 **B** 1 and 3 **C** 1 only **D** 2 only

October/November 2022 · Paper 1 · Variant 1 · Question 4

Correct answer: C

4 Which row applies to active transport?

	movement occurs from region of higher concentration to region of lower concentration	movement occurs from region of lower concentration to region of higher concentration	an example is ion uptake by root hairs	an example is water uptake by root hairs
A	✓	✗	✓	✗
B	✓	✗	✗	✓
C	✗	✓	✓	✗
D	✗	✓	✗	✓

key

✓ = yes

✗ = no

October/November 2022 · Paper 1 · Variant 1 · Question 5

Correct answer: C

5 Which statements are correct?

1 Enzymes are proteins and function as catalysts.

2 Enzymes are changed by the reaction they catalyse.

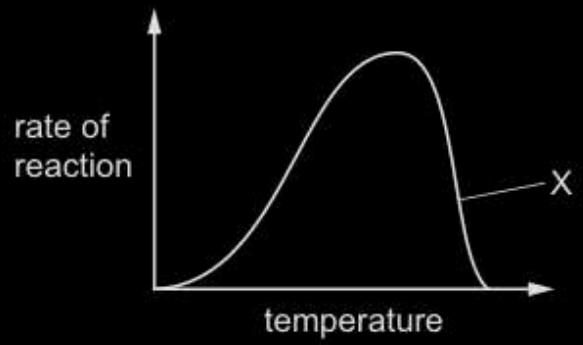
3 The 'lock and key' hypothesis explains the way that enzymes function.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

October/November 2022 · Paper 1 · Variant 1 · Question 6

Correct answer: C

6 The diagram shows the effect of temperature on the rate of an enzyme-controlled reaction.



What is the explanation for the part of the graph labelled X?

A The temperature is dropping.

B The substrate is starting to run out.

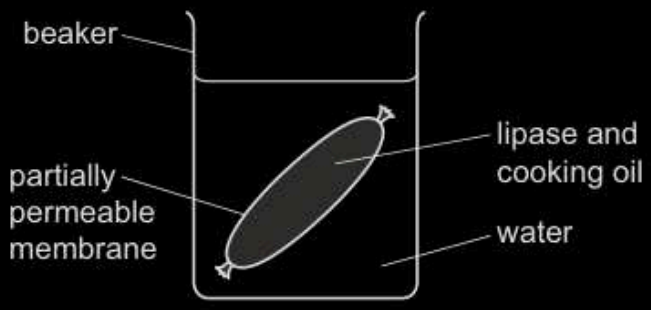
C The enzyme is becoming denatured.

D The enzyme is starting to run out.

October/November 2022 · Paper 1 · Variant 1 · Question 10

Correct answer: B

10 The diagram shows an experiment.



After 30 minutes, tests were carried out on the contents of the membrane bag and on the water in the beaker.

Which row shows the results of the tests?

	biuret test on water in the beaker	biuret test on the contents of the bag	pH of water in the beaker
A	blue	blue	6.0
B	blue	purple	5.5
C	red	purple	7.0
D	purple	blue	8.0

October/November 2022 · Paper 1 · Variant 1 · Question 11

Correct answer: D

11 Which function is **not** carried out by the liver?

A breakdown of alcohol

B conversion of glycogen to glucose

C formation of urea

D secretion of digestive enzymes

October/November 2022 · Paper 1 · Variant 1 · Question 12

Correct answer: A

12 Which statements about root hairs are correct?

1 The cell membrane can allow movement of water and ions into the cell.

2 The cell vacuole extends into the root hair.

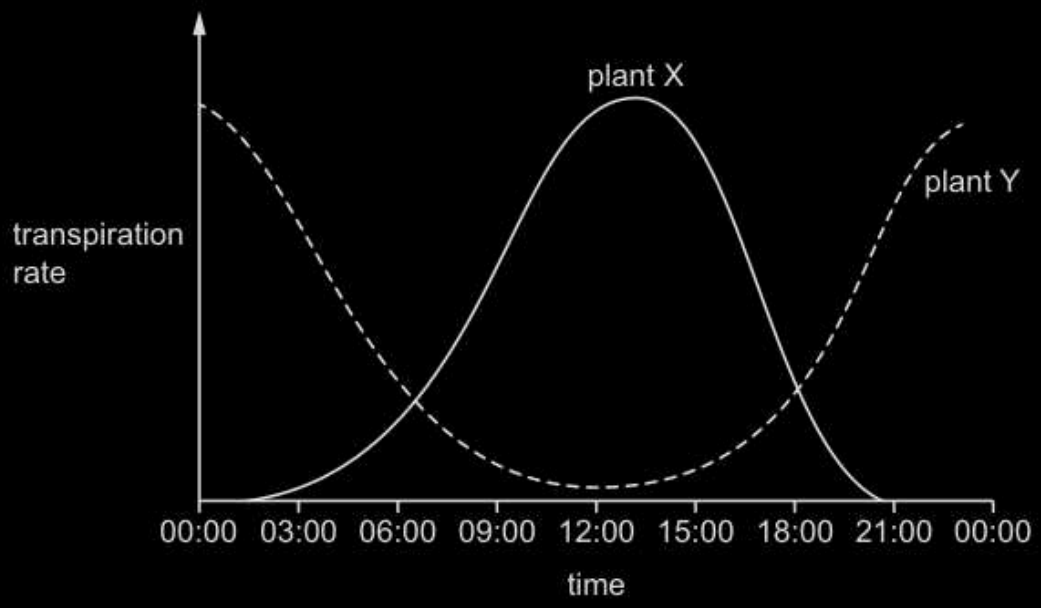
3 They are formed as an extension of the outer layer of root cells.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

October/November 2022 · Paper 1 · Variant 1 · Question 13

Correct answer: B

13 The graph shows the transpiration rates of two plants during one day. Both plants were grown under identical conditions.



Which statement explains what the graph shows?

A Plant X closes its stomata during the brightest part of the day.

B Plant Y closes its stomata during the brightest part of the day.

C Plants X and Y close their stomata during the brightest part of the day.

D Plant Y has no stomata.

October/November 2022 · Paper 1 · Variant 1 · Question 14

Correct answer: A

14 Some people have a rare heart condition in which the lower left chamber of the heart has not developed properly and is much smaller than normal.

The immediate result of this condition is to cause lower than normal blood flow into which blood vessel?

A aorta

B pulmonary artery

C pulmonary vein

D vena cava

October/November 2022 · Paper 1 · Variant 1 · Question 15

Correct answer: D

15 Tissue transplants are normally rejected by the body unless the transplant tissue and the patient are very carefully matched.

Which component of the blood is responsible for the rejection of transplanted tissue?

A plasma

B platelets

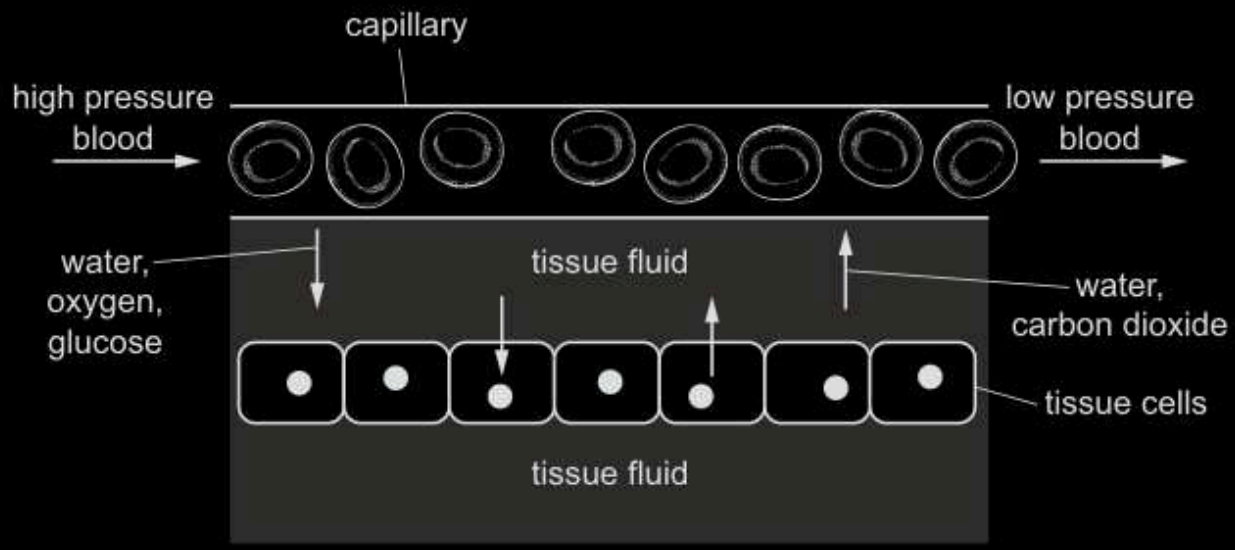
C red blood cells

D white blood cells

October/November 2022 · Paper 1 · Variant 1 · Question 16

Correct answer: C

16 The diagram shows movement of substances between blood in a capillary and tissue fluid.



By which process does water move from tissue fluid to capillary?

A active transport

B water pressure

C osmosis

D assimilation

October/November 2022 · Paper 1 · Variant 1 · Question 20

Correct answer: A

20 How do the concentrations of oxygen and urea in the renal artery compare to their concentrations in the renal vein?

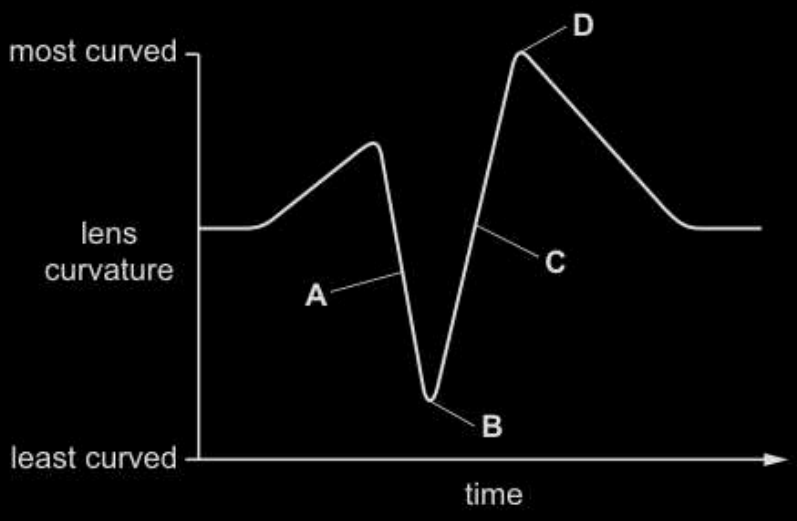
	oxygen in renal artery	urea in renal artery
A	higher	higher
B	higher	lower
C	lower	higher
D	lower	lower

October/November 2022 · Paper 1 · Variant 1 · Question 23

Correct answer: C

23 The graph shows changes in the shape of the lenses of a person's eyes while watching a bird flying.

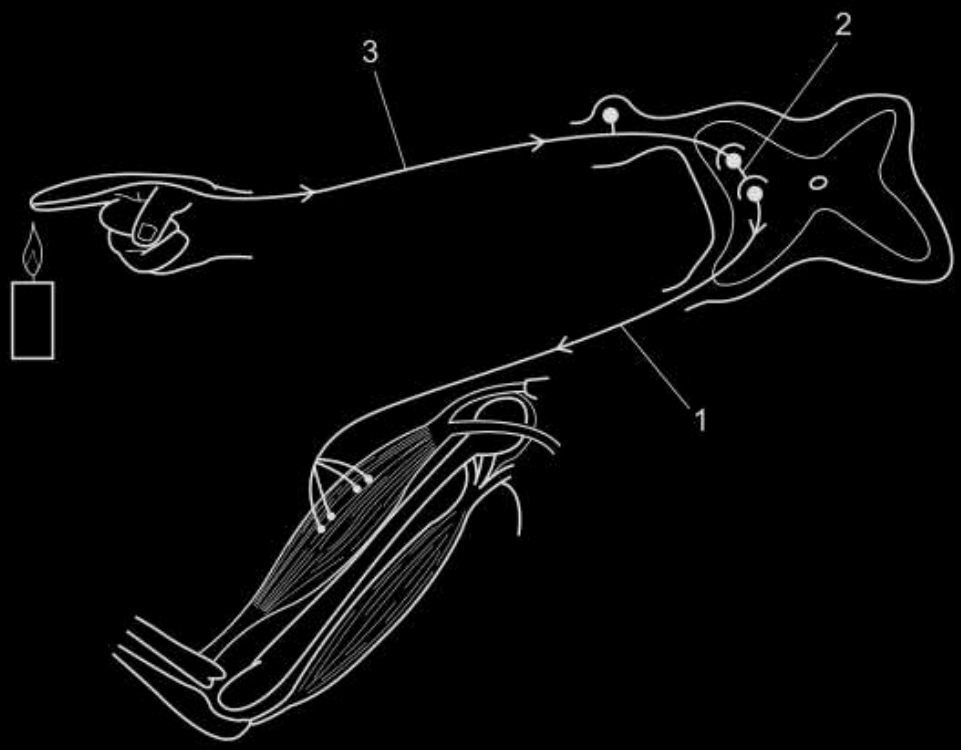
At which point is the bird flying most rapidly towards the person?



October/November 2022 · Paper 1 · Variant 1 · Question 24

Correct answer: A

24 The diagram shows the nerve pathway in a reflex action when a person accidentally touches something hot.



Which row correctly identifies a sensory neurone, a relay neurone and a motor neurone?

	1	2	3
A	motor neurone	relay neurone	sensory neurone
B	sensory neurone	relay neurone	motor neurone
C	relay neurone	motor neurone	sensory neurone
D	motor neurone	sensory neurone	relay neurone

October/November 2022 · Paper 1 · Variant 1 · Question 25

Correct answer: B

25 A patient shows symptoms of unexplained weight loss, severe thirst and frequent need of urination.

A test shows high levels of glucose in the urine.

Which condition is diagnosed?

A cardiovascular disease

B diabetes

C anaemia

D obesity

October/November 2022 · Paper 1 · Variant 1 · Question 28

Correct answer: C

28 The table shows some features of three types of microorganism.

type of microorganism	cell wall	cell membrane	nucleus	cytoplasm
1	✗	✗	✗	✗
2	✓	✓	✗	✓
3	✓	✓	✓	✓

key
✓ = present
✗ = absent

Which row identifies each type of microorganism?

	microorganism 1	microorganism 2	microorganism 3
A	bacterium	virus	fungus
B	fungus	bacterium	virus
C	virus	bacterium	fungus
D	virus	fungus	bacterium

October/November 2022 · Paper 1 · Variant 1 · Question 37

Correct answer: D

37 A variety of snail has an inherited condition that affects the thickness of the shell.

S^tS^t have thick shells.

S^tS^n have thin shells.

S^nS^n do **not** survive.

Two heterozygous snails are mated.

What is the probability that a surviving snail of the next generation is a heterozygote?

A 0.00 **B** 0.25 **C** 0.50 **D** 0.67

October/November 2022 · Paper 1 · Variant 1 · Question 38

Correct answer: B

38 A man of blood group A and his wife of blood group O had two children, both of blood group A. The man concluded that he must be homozygous for the allele I^A, since he thought half his children would be of group O if he were heterozygous.

Why was his conclusion unsound?

A Blood group mutations are common.

B Genetic ratios are unreliable for small numbers.

C His wife might have been heterozygous.

D The expected ratio for a heterozygous father and group O mother is 3 group A : 1 group O.

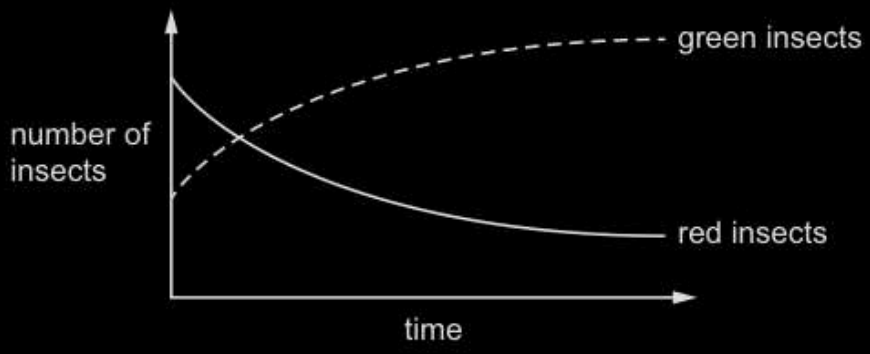
October/November 2022 · Paper 1 · Variant 1 · Question 39

Correct answer: D

39 In a garden there is a species of insect which can be either red or green in colour. The green insects are well camouflaged.

Birds that eat insects start to live in the garden.

The graph shows how the populations of red and green insects then change.



What is this an example of?

A artificial selection

B continuous variation

C incomplete dominance

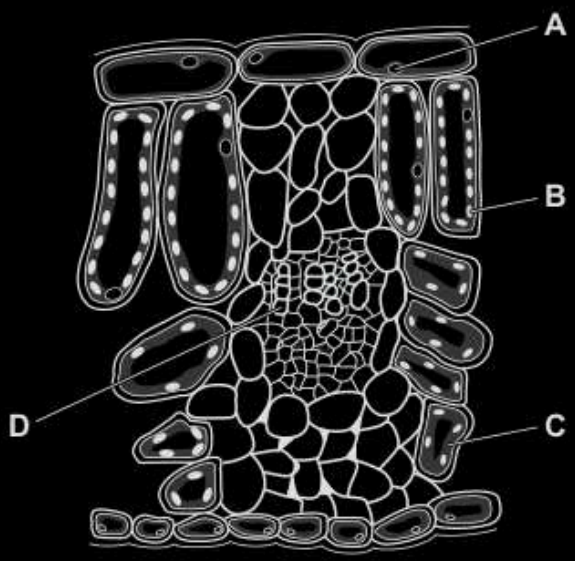
D natural selection

October/November 2022 · Paper 1 · Variant 2 · Question 1

Correct answer: B

1 The diagram shows cells from a plant leaf.

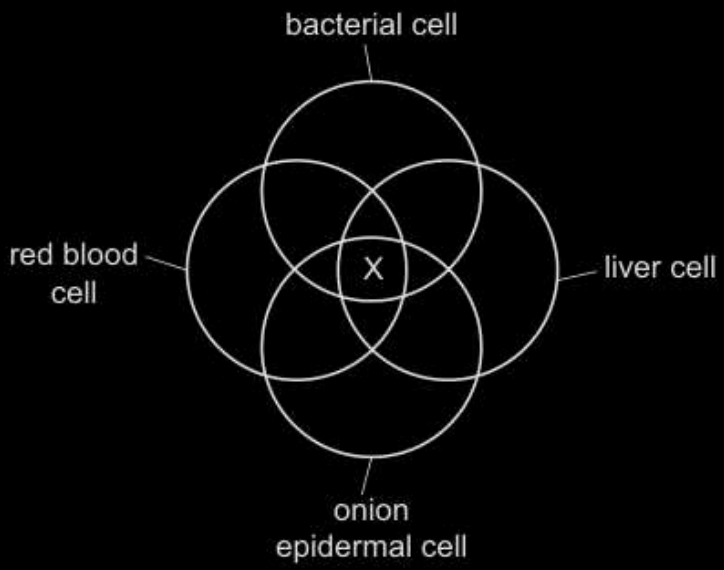
Which structure contains a high concentration of magnesium?



October/November 2022 · Paper 1 · Variant 2 · Question 2

Correct answer: A

2 The diagram represents the cell structures of a bacterial cell, a liver cell, an onion epidermal cell and a red blood cell.



Which structure is represented by X?

A cell membrane

B cellulose cell wall

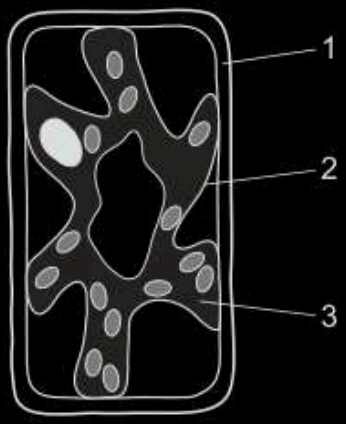
C nucleus

D sap vacuole

October/November 2022 · Paper 1 · Variant 2 · Question 3

Correct answer: D

3 The diagram shows a typical plant cell which has been in a concentrated salt solution for 10 minutes.



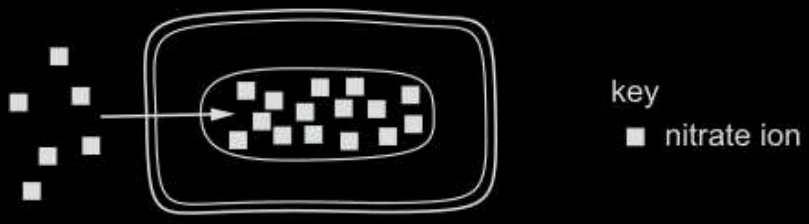
Which numbered structures are partially permeable?

A 1 and 2 **B** 1 and 3 **C** 1 only **D** 2 only

October/November 2022 · Paper 1 · Variant 2 · Question 4

Correct answer: A

4 Which process is moving nitrate ions into the cell shown?



A active transport

B diffusion

C osmosis

D translocation

October/November 2022 · Paper 1 · Variant 2 · Question 5

Correct answer: B

5 Fats are broken down by the enzyme lipase to produce fatty acids and glycerol.

According to the 'lock and key' model of enzyme action, what is the 'lock' and what is the 'key' in this reaction?

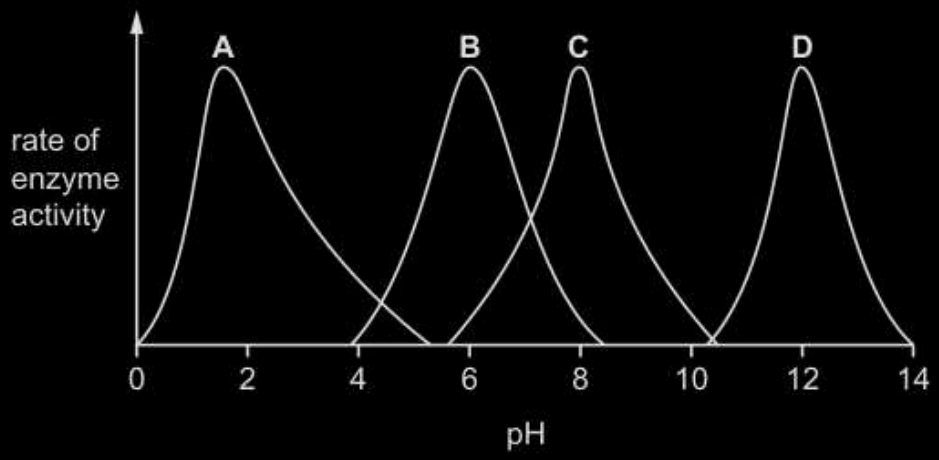
	lock	key
A	fats	lipase
B	lipase	fats
C	fats	fatty acids and glycerol
D	lipase	fatty acids and glycerol

October/November 2022 · Paper 1 · Variant 2 · Question 6

Correct answer: A

6 The graph shows the effect of varying pH values on the activity of four different enzymes, **A**, **B**, **C** and **D**.

Which enzyme is active in the stomach?



Correct answer: D

9 A lack of certain minerals and vitamins can lead to deficiency diseases.

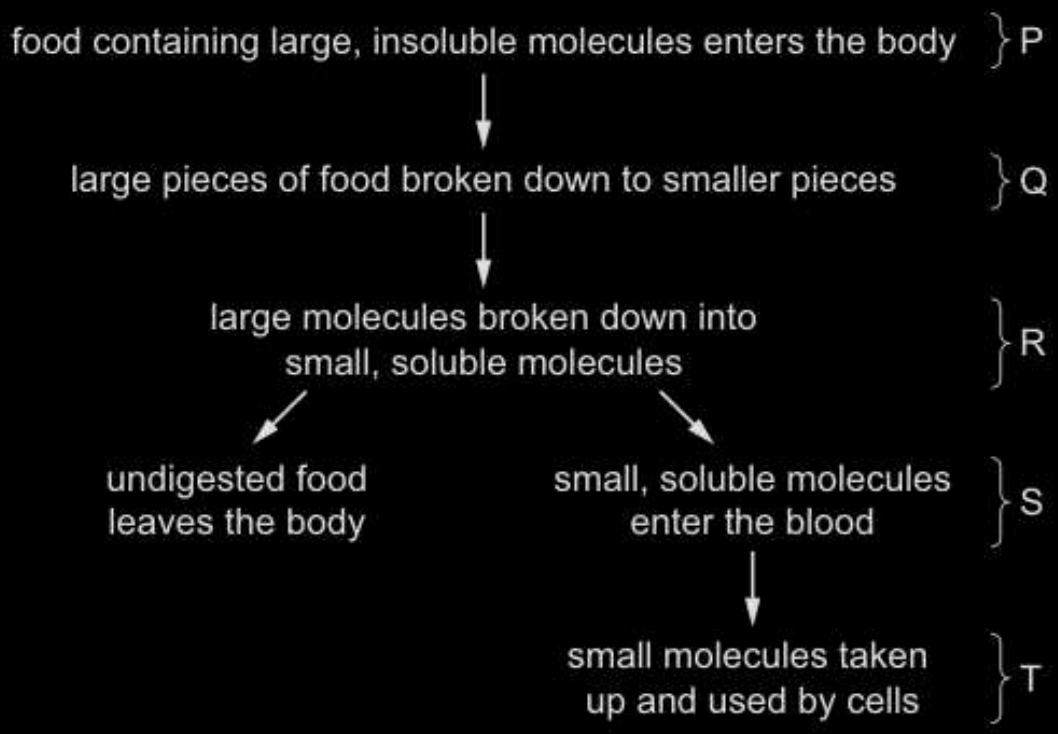
Which row shows the correct symptoms of such diseases?

	deficiencies in diet	symptoms in body
A	calcium and vitamin C	too few red blood cells and deformed bones
B	calcium and vitamin D	brittle bones and bleeding gums
C	iron and vitamin C	brittle bones and bleeding gums
D	iron and vitamin D	too few red blood cells and deformed bones

October/November 2022 · Paper 1 · Variant 2 · Question 10

Correct answer: D

10 The diagram shows how food is processed in the human alimentary canal.



Which row correctly identifies the named stages?

	absorption	assimilation	enzyme digestion
A	P	S	T
B	P	S	Q
C	S	P	Q
D	S	T	R

October/November 2022 · Paper 1 · Variant 2 · Question 11

Correct answer: D

11 Which function is **not** carried out by the liver?

A breakdown of alcohol

B conversion of glycogen to glucose

C formation of urea

D secretion of digestive enzymes

October/November 2022 · Paper 1 · Variant 2 · Question 12

Correct answer: A

12 Which statements about root hairs are correct?

1 The cell membrane can allow movement of water and ions into the cell.

2 The cell vacuole extends into the root hair.

3 They are formed as an extension of the outer layer of root cells.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

October/November 2022 · Paper 1 · Variant 2 · Question 13

Correct answer: B

13 The table shows weather conditions in a field.

Which row shows conditions that would cause a plant to wilt most rapidly?

	air temperature / °C	humidity	time of day	wind speed / km per hour
A	15	raining	00:00	32
B	12	dry	14:00	24
C	6	fog / rain	08:00	3
D	5	dry	22:00	16

October/November 2022 · Paper 1 · Variant 2 · Question 14

Correct answer: A

14 Some people have a rare heart condition in which the lower left chamber of the heart has not developed properly and is much smaller than normal.

The immediate result of this condition is to cause lower than normal blood flow into which blood vessel?

A aorta

B pulmonary artery

C pulmonary vein

D vena cava

October/November 2022 · Paper 1 · Variant 2 · Question 15

Correct answer: C

15 Which route does blood take around the body?

A pulmonary artery → heart → aorta → lungs → pulmonary vein → heart → vena cava

B pulmonary vein → heart → lungs → aorta → pulmonary artery → heart → vena cava

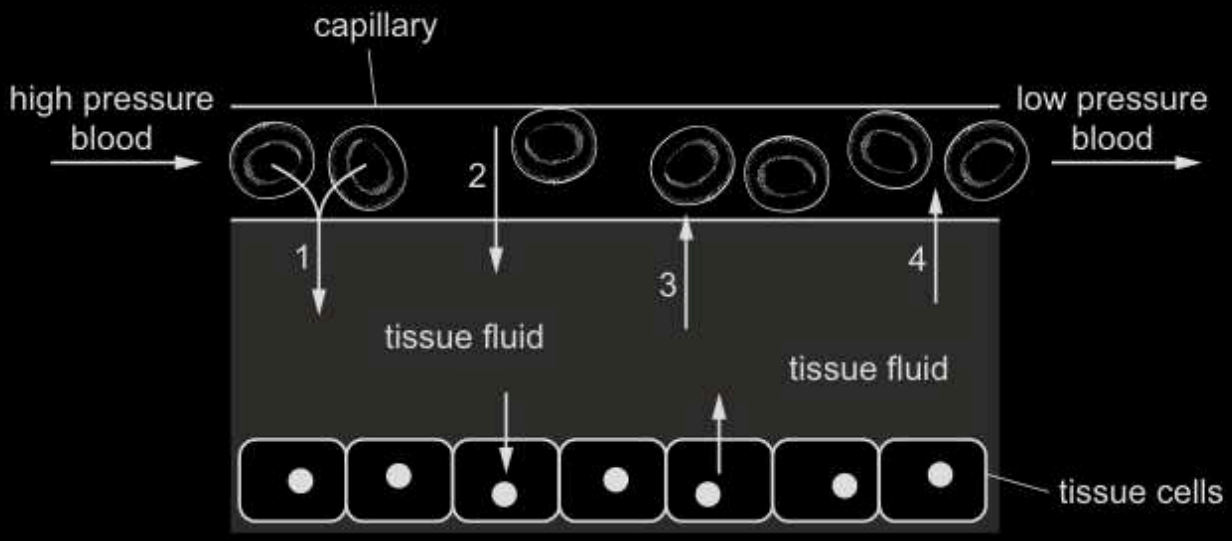
C vena cava → heart → pulmonary artery → lungs → pulmonary vein → heart → aorta

D vena cava → heart → pulmonary vein → lungs → pulmonary artery → heart → aorta

October/November 2022 · Paper 1 · Variant 2 · Question 16

Correct answer: C

16 The diagram shows the transfer of materials between blood in a capillary and tissue fluid.



Which row shows the correct labels for the arrows in the diagram?

	1	2	3	4
A	carbon dioxide	glucose	oxygen	water
B	glucose	water	oxygen	urea
C	oxygen	glucose	carbon dioxide	water
D	oxygen	water	carbon dioxide	glucose

October/November 2022 · Paper 1 · Variant 2 · Question 20

Correct answer: B

20 What are two main functions of the kidneys?

A to make toxins in the blood harmless and to remove excess salt in the blood

B to remove excess salt in the blood and to remove urea from the blood plasma

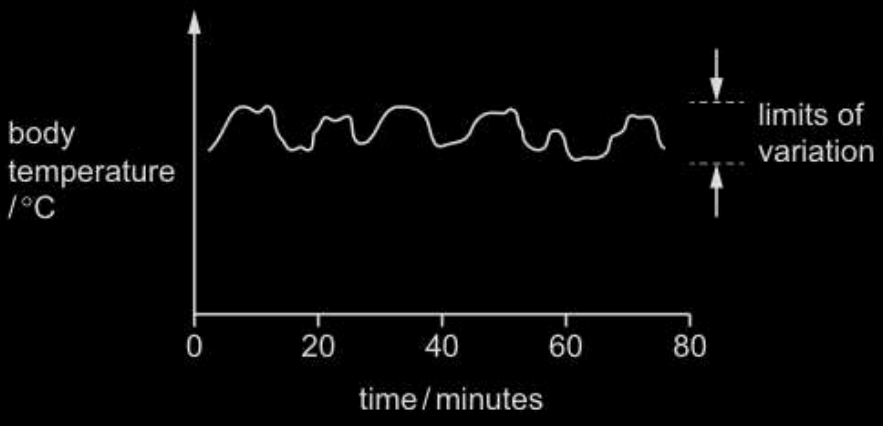
C to remove urea from the blood plasma and to convert glucose to glycogen

D to convert glucose to glycogen and to make toxins in the blood harmless

October/November 2022 · Paper 1 · Variant 2 · Question 22

Correct answer: B

22 The graph shows measurements taken of one aspect of the internal environment of the human body over time.



What controls the internal environment of the human body to keep each aspect within strict limits?

A excretion

B homeostasis

C positive feedback

D respiration

October/November 2022 · Paper 1 · Variant 2 · Question 23

Correct answer: B

23 When a bright light is shone into the eye, the pupil becomes smaller.

Which statement explains this response?

A Ciliary muscles in the eye contract.

B Circular muscles in the iris contract.

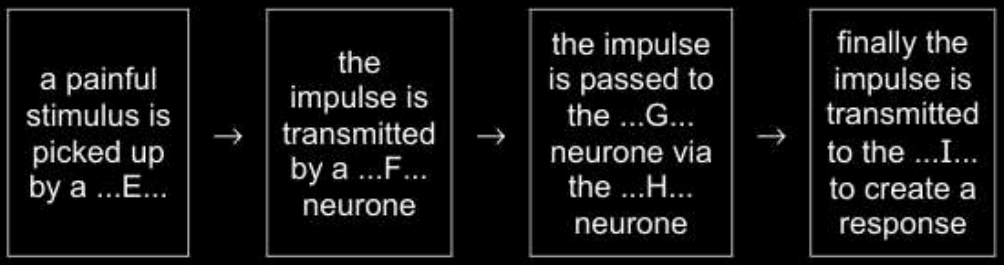
C Sensory cells in the iris detect the bright light.

D Sensory neurones stimulate the ciliary muscles in the iris.

October/November 2022 · Paper 1 · Variant 2 · Question 24

Correct answer: C

24 The flow chart describes a reflex action.



Which row correctly fills the flow chart?

	E	F	G	H	I
A	effector	relay	motor	sensory	brain
B	effector	sensory	relay	motor	receptor
C	receptor	sensory	motor	relay	effector
D	receptor	motor	sensory	relay	brain

October/November 2022 · Paper 1 · Variant 2 · Question 25

Correct answer: B

25 A patient shows symptoms of unexplained weight loss, severe thirst and frequent need of urination.

A test shows high levels of glucose in the urine.

Which condition is diagnosed?

A cardiovascular disease

B diabetes

C anaemia

D obesity

October/November 2022 · Paper 1 · Variant 2 · Question 28

Correct answer: C

28 Bacteria, fungi and viruses are often grouped together as microorganisms.

Which statement about microorganisms is correct?

A All types of microorganism are used to produce antibiotics.

B Bacteria and viruses can only be found inside living cells.

C Decomposition is brought about by bacteria and fungi only.

D All microorganisms contain DNA but not in a nucleus.

October/November 2022 · Paper 1 · Variant 2 · Question 37

Correct answer: A

37 What is the correct order for the sizes of the pieces of genetic material shown in the table, from smallest to largest?

	smallest	⟶	largest
A	gene	molecule of DNA	chromosome
B	molecule of DNA	gene	chromosome
C	chromosome	molecule of DNA	gene
D	gene	chromosome	molecule of DNA

October/November 2022 · Paper 1 · Variant 2 · Question 38

Correct answer: B

38 A man of blood group A and his wife of blood group O had two children, both of blood group A. The man concluded that he must be homozygous for the allele I^A, since he thought half his children would be of group O if he were heterozygous.

Why was his conclusion unsound?

A Blood group mutations are common.

B Genetic ratios are unreliable for small numbers.

C His wife might have been heterozygous.

D The expected ratio for a heterozygous father and group O mother is 3 group A : 1 group O.

Correct answer: D

39 Rising carbon dioxide levels in the atmosphere are causing the Earth's climate to warm up.

In what way might this change of climate affect the process of natural selection?

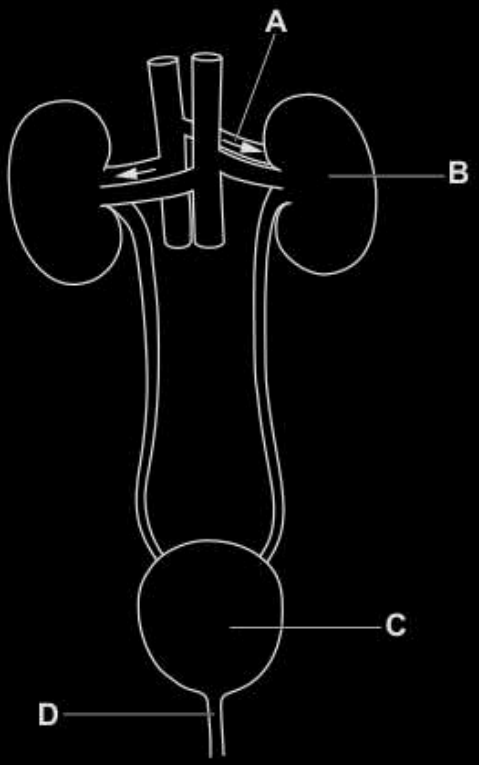
A A small rise in temperature may reduce the rate of photosynthesis in plants.

B Farmers may try to grow different crops in the warmer areas.

C It may allow some animals and plants to colonise new areas.

D Some mutations may prove to be an advantage in warmer conditions.

1 The diagram shows parts of the human body involved in excretion.



(a) Name the liquid that passes out of the human body through tube **D** and the waste product it contains.

liquid .....

waste product .....

[2]

(b) Describe how parts **A**, **B** and **C** are involved in removing the waste product that passes out of tube **D**.

.....

.....

.....

.....

..... [3]

[Total: 5]

1(a)	urine ;	**1**	
	urea ;	**1**	**A** ammonia / excess water / excess minerals
1(b)	renal artery / A **+** brings urea / waste product to B / kidney **AW** ;	**1**	
	kidney / B **+** filters blood to remove urea / waste products ;	**1**	
	bladder / C **+** stores / keeps / collects urine ;	**1**	

Question	Answer	Marks	Guidance

3 Guinea pigs are small mammals. In some countries they are produced for food and in others they are bred as pets.

(a) Guinea pigs can have either straight hair or curly hair. The allele for straight hair, **T**, is dominant to the allele for curly hair, **t**.

(i) Explain the difference between a gene and an allele.

.....

.....

..... [2]

(ii) A breeder has two straight-haired guinea pigs, male and female, and is surprised when one of the offspring has curly hair.

Complete the genetic diagram to show how this is possible.

genotypes of parents ..... × .....

gametes ....

possible offspring genotypes ....

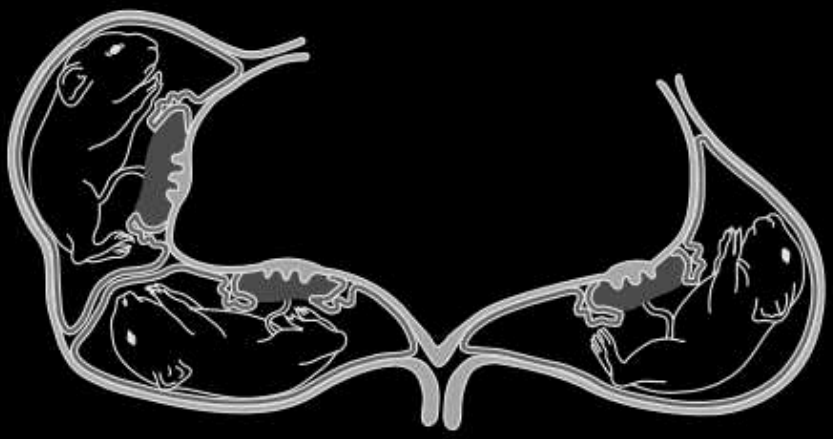
possible offspring phenotypes ....

[4]

(b) Baby guinea pigs develop in the mother's uterus for between 59 and 73 days.

The mother usually gives birth to between two and four offspring.

The diagram shows the uterus and developing offspring.



The offspring of both humans and guinea pigs develop in a uterus.

Using the information provided and your knowledge of human development, discuss the similarities **and** differences in the development of offspring of humans and guinea pigs.

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

3(a)(i)	a gene is a section of DNA ;	**1**	
	an allele is a version of the gene ;	**1**	**A** an alternative form of a gene
3(a)(ii)	genotypes of parents: Tt × Tt ;	**1**	
	gametes: T, t, T, t ;	**1**	
	offspring genotypes: TT, Tt, Tt, tt ;	**1**	
	offspring phenotypes: straight, straight, straight, curly ;	**1**	
3(b)	*max **two** similarities from:* placenta ; umbilical cord ; provides nutrition / oxygen / food ; amniotic sac / amniotic fluid ; receives mechanical protection / offspring / fetus cushioned ;	**2**	
	*max **two** differences from:* shape of uterus / womb ; duration of pregnancy / gestation ; numbers of offspring higher in guinea pig ;	**2**	

© UCLES 2022

Question	Answer	Marks	Guidance

4 Amylases are enzymes that are made in specific cells of animals and plants and also by bacteria. In humans, amylase is secreted into the alimentary canal in saliva and in pancreatic juice.

(a) (i) Explain what is meant by the term enzyme.

.....

..... [2]

(ii) Outline **two** structural features of enzyme molecules.

1 .....

.....

2 .....

.....

[2]

(iii) Name both the substrate and product of amylase.

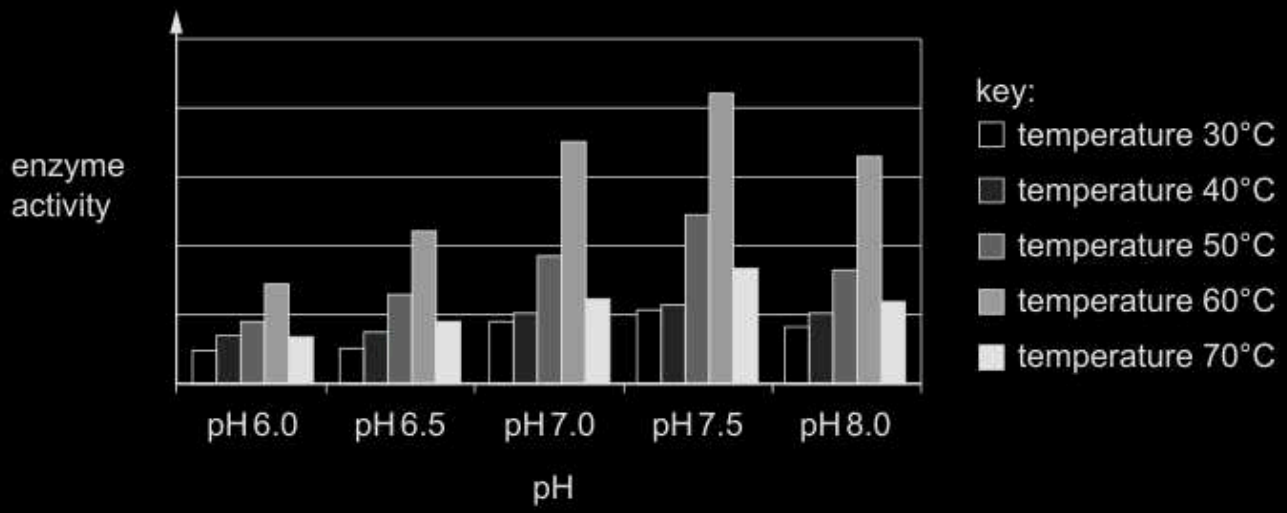
substrate .....

product .....

[2]

(b) Amylase produced by bacteria can be used as part of an industrial process that converts plants such as corn into biofuel. Biofuels can be used instead of diesel or petrol.

The bar chart shows how the activity of amylase is affected by temperature and pH.



(i) Describe what the data in the bar chart show about the activity of the enzyme and explain why this information is useful for the producers of biofuels.

.....

.....

.....

.....

.....

..... [4]

(ii) Biofuels can be burnt in vehicle engines.

Suggest how biofuels can result in pollution and cause environmental change.

.....

.....

.....

..... [2]

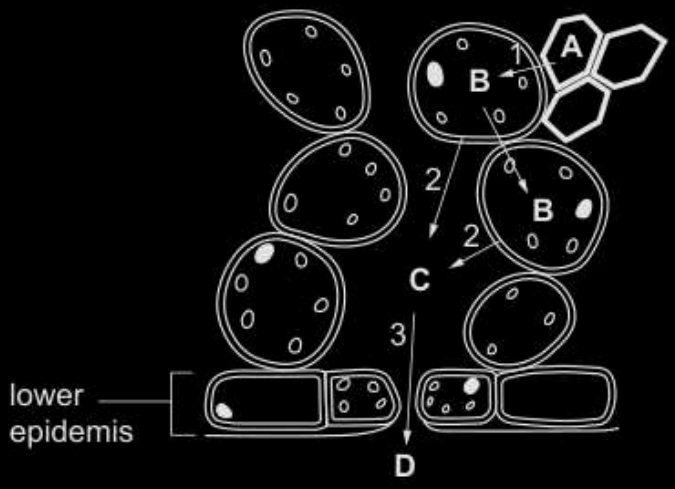
[Total: 12]

4(a)(i)	a (biological) catalyst ; speeds up chemical reactions ; without being changed itself ;	**2**	**A** without being used up / can be reused
4(a)(ii)	made of protein / polypeptides / (chains of) amino acids ; have an active site ; has a specific shape / shape complementary to the substrate ;	**2**	
4(a)(iii)	starch ;	**1**	
	maltose / sugar / glucose ;	**1**	
4(b)(i)	enzyme is active from pH 6 to pH 8 ; optimum pH = 7.5 ; enzyme is active between 30 and 70°C ; optimum temperature = 60°C ; manufacturer can keep optimum / best / correct conditions ; so biofuel produced rapidly / efficiently / cost-effectively / maximum yield ;	**4**	
4(b)(ii)	combustion releases carbon dioxide ; greenhouse gas / contributes to global warming ; climate change ; a change described, e.g. melting glaciers / rising oceans / drought ; an impact of climate change, e.g. loss of habitat / redistribution of species / loss of agricultural land ;	**2**	

© UCLES 2022

Question	Answer	Marks	Guidance

6 The diagram shows some cells from the lower part of a transpiring leaf.



(a) (i) Identify cell type **B**.

B ..... [1]

(ii) Water travels from **A** to **D**. The arrows on the diagram represent the movement of water molecules during transpiration. With reference to these arrows, name and define each process taking place.

arrow 1 .....

.....

.....

arrow 2 .....

.....

.....

arrow 3 .....

.....

.....

[6]

(b) Leaves from different plant species transpire at different rates in the same environmental conditions.
Suggest and explain structural differences in the lower surface of these leaves which might produce differences in the transpiration rate.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

6(a)(i)	spongy mesophyll ;	**1**	
6(a)(ii)	*arrow 1* osmosis ; is the movement (of water molecules) from an area of higher water potential to lower water potential across a partially permeable membrane **AW** ; *arrow 2* evaporation ; is the change from liquid (water) to (water) vapour / gas ; *arrow 3* diffusion ; is the (net random) movement (of molecules) from a high to lower concentration / down a concentration gradient ;	**6**	**A** correct definition if linked to correct process but linked to the incorrect arrow.

© UCLES 2022

Question	Answer	Marks	Guidance
6(b)	the surface area of the leaf ; the greater the surface area of the leaf, the greater the surface for evaporation ; the thickness of the cuticle ; the thicker the cuticle, the greater the barrier for water molecules ; the number / density of stomata ; as density of stomata increase, more places for water to escape from ; size of stomata ; as size of stomata increases, so does area from which water can be lost ; sunken stomata / ridged epidermis / curled up leaves ; sunken stomata / ridged epidermis / curled up leaves increase humidity ; hairs / projections ; hairs / projections increase humidity ;	3	Maximum two marks for structural features only with no explanation.

© UCLES 2022

Question	Answer	Marks	Guidance

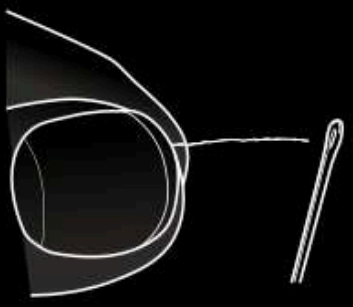
7 Humans have receptors to detect stimuli in their environment and a nervous system to coordinate responses to these stimuli.

(a) Complete the table showing different parts of the nervous system, their location in the human body and some of their functions.
One row has already been completed.

part of the nervous system	location in the body	function
medulla	brain	controls automatic functions such as breathing
	nerve in body	carries nerve impulses from the central nervous system to muscles and glands
	brain	problem solving, language, memory and interpreting visual information
	brain	regulates body temperature
relay neurone		
cerebellum		fine motor control and balance

[5]

(b) A man is threading a needle.



To do this, each of the man's eyes produces a focused image of the needle.

Explain how different parts of the eye function to produce a focused image of a near object.

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

Section C

Answer **either** Question 8 **or** Question 9.

Write your answers in the spaces provided.

7(a)	motor neurone ;	**1**	**A** effector neurone
	cerebrum ;	**1**	**A** cerebral hemispheres
	hypothalamus ;	**1**	
	CNS / brain / spinal cord **+** transfers nerve impulses from sensory to motor neurones ;	**1**	
	brain ;	**1**	
7(b)	light is refracted / bent by cornea ; ciliary muscles / body contract ; suspensory ligaments become slack ; lens becomes fatter **AW** ; light refracted / bent (more) by lens ; light / image focused / formed on retina ; light detected by light receptors in retina / rods / cones ; nerve impulses triggered ;	**5**	

© UCLES 2022

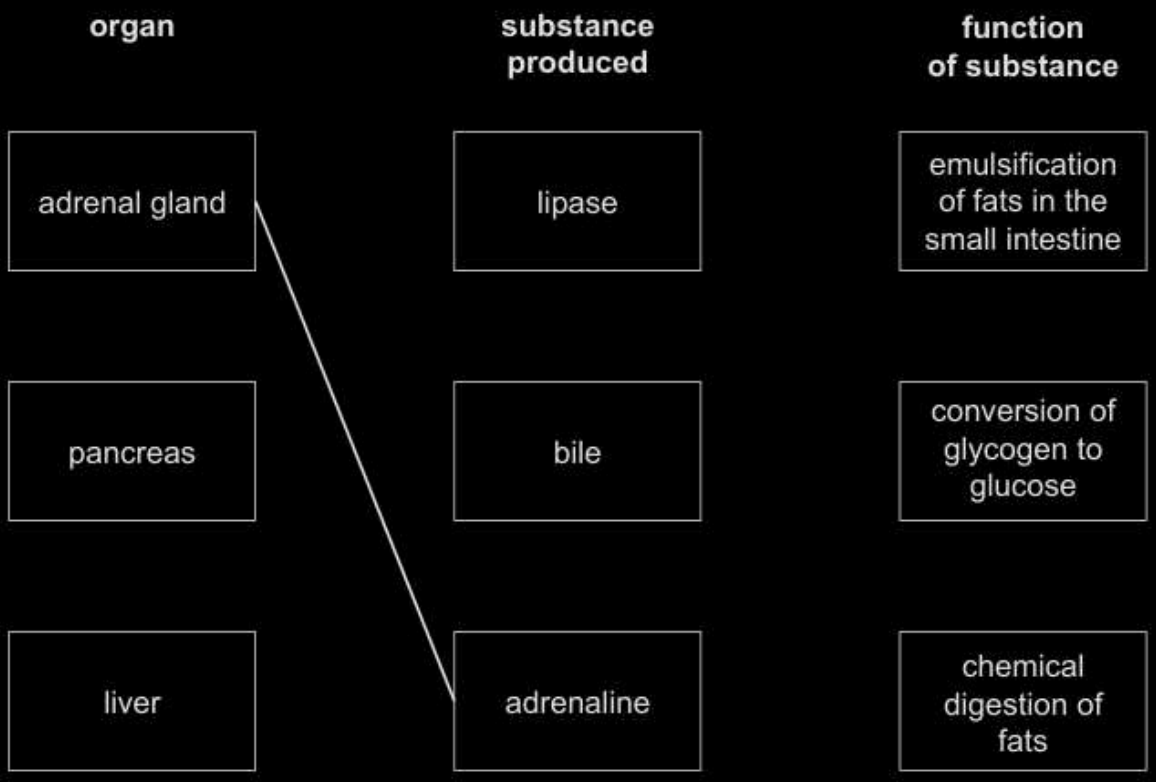
Question	Answer	Marks	Guidance

1 Organs in the human body produce substances which have specific functions.

Draw lines to link each organ with the substance it produces **and** to link each substance with the description of its function.

One line has been drawn for you.

Draw **five more** lines.



[5]

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	'List rule' guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

© UCLES 2022

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

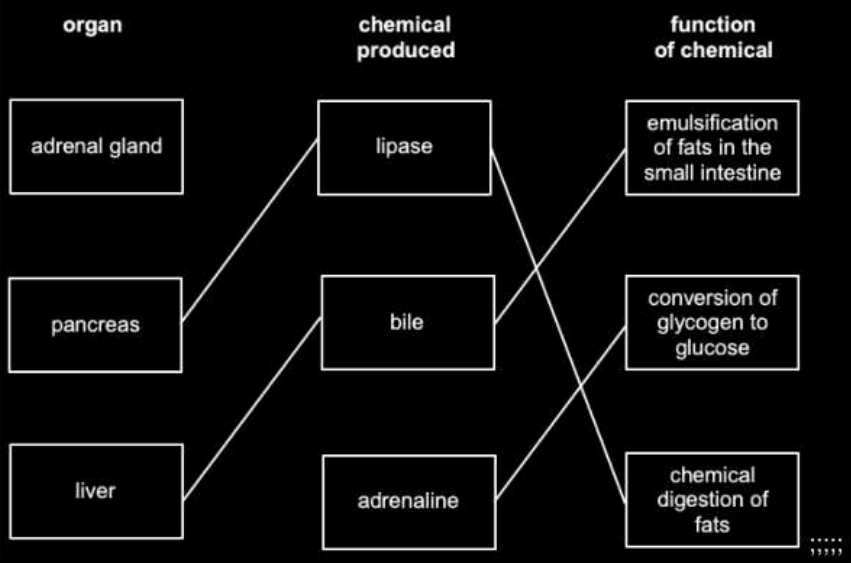
State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

© UCLES 2022

MARK SCHEME ABBREVIATIONS

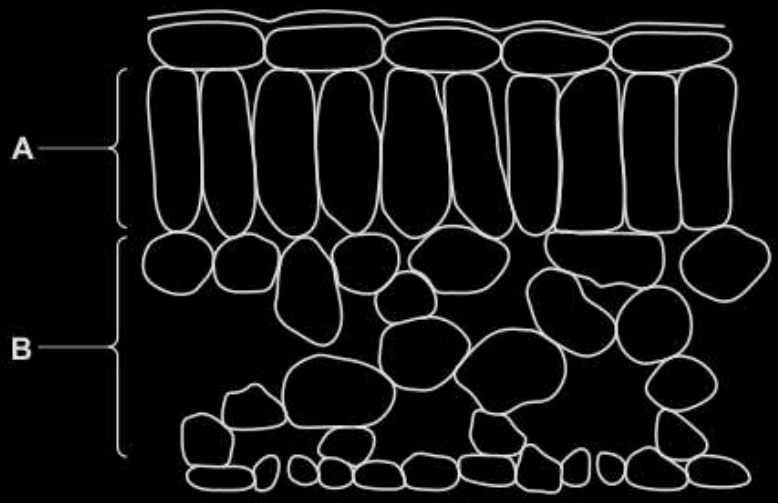
;	separates marking points
/ and **OR**	indicate alternatives within a marking point
()	the contents of brackets are **not required**
Reject	guidance for examiners concerning a point to reject
Allow	guidance for examiners concerning a point to accept
Ignore	guidance for examiners concerning a point to ignore
AW	allow alternative wording to that specified on the mark scheme
ORA	or reverse argument – i.e. the reverse argument may be accepted annotation in RM = 0
underline	the word(s) underlined must be used by candidate for the award of the mark (allow grammatical variants)
ecf	allow error carried forward
max	dictates the maximum number of marks that can be awarded
+	a statement from **both** sides of the **+** is required for that mark

© UCLES 2022

Question	Answer	Marks	Guidance
1	one mark for each correct line  ;;;;; ;;;;;	5	

Question	Answer	Marks	Guidance

2 The diagram shows a cross-section through a leaf when viewed using a light microscope.

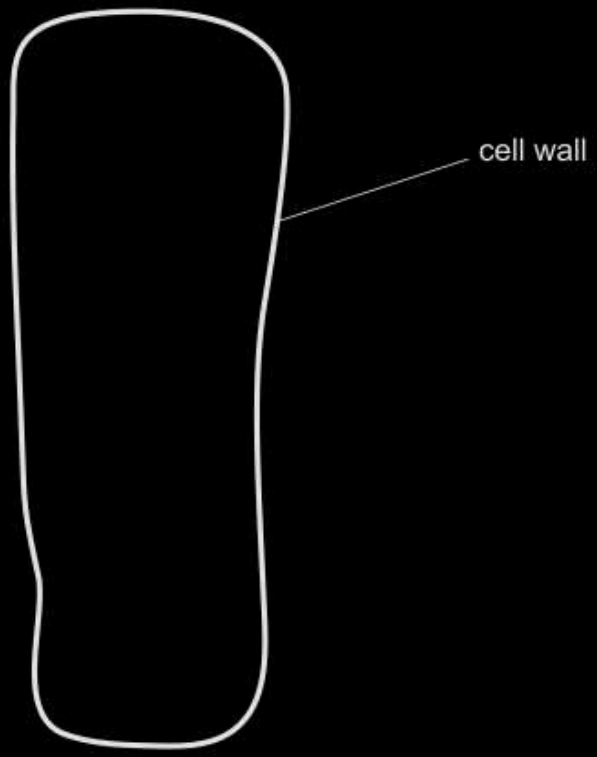


(a) (i) State the term used to describe a group of cells, such as those in part **A** or part **B** of the leaf cross-section.

.....

[1]

(ii) The diagram shows an enlargement of one cell from part **A** of the leaf cross-section.



Complete the diagram of the cell by drawing and labelling to show the position of

- **one** chloroplast
- **three** other types of **named** cell component that will be visible.

[4]

(iii) Use a label line on the diagram of the cross-section through a leaf to name and label **one** cell in the lower epidermis that would also contain chloroplasts. [1]

(b) The cell wall of a plant cell can be removed by treating the cell with a digestive enzyme.

(i) Name the substrate for this enzyme.

..... [1]

(ii) Some plant cells from part **B** of the leaf cross-section were treated with this enzyme and then placed in distilled water on a microscope slide for one hour.

The cells were clearly visible using a light microscope at the start of the hour.

The cells were **not** clearly visible using the same light microscope at the end of the hour.

Explain changes to the structure of the cells that took place between these two observations.

.....

.....

.....

.....

.....

.....

.....

..... [4]

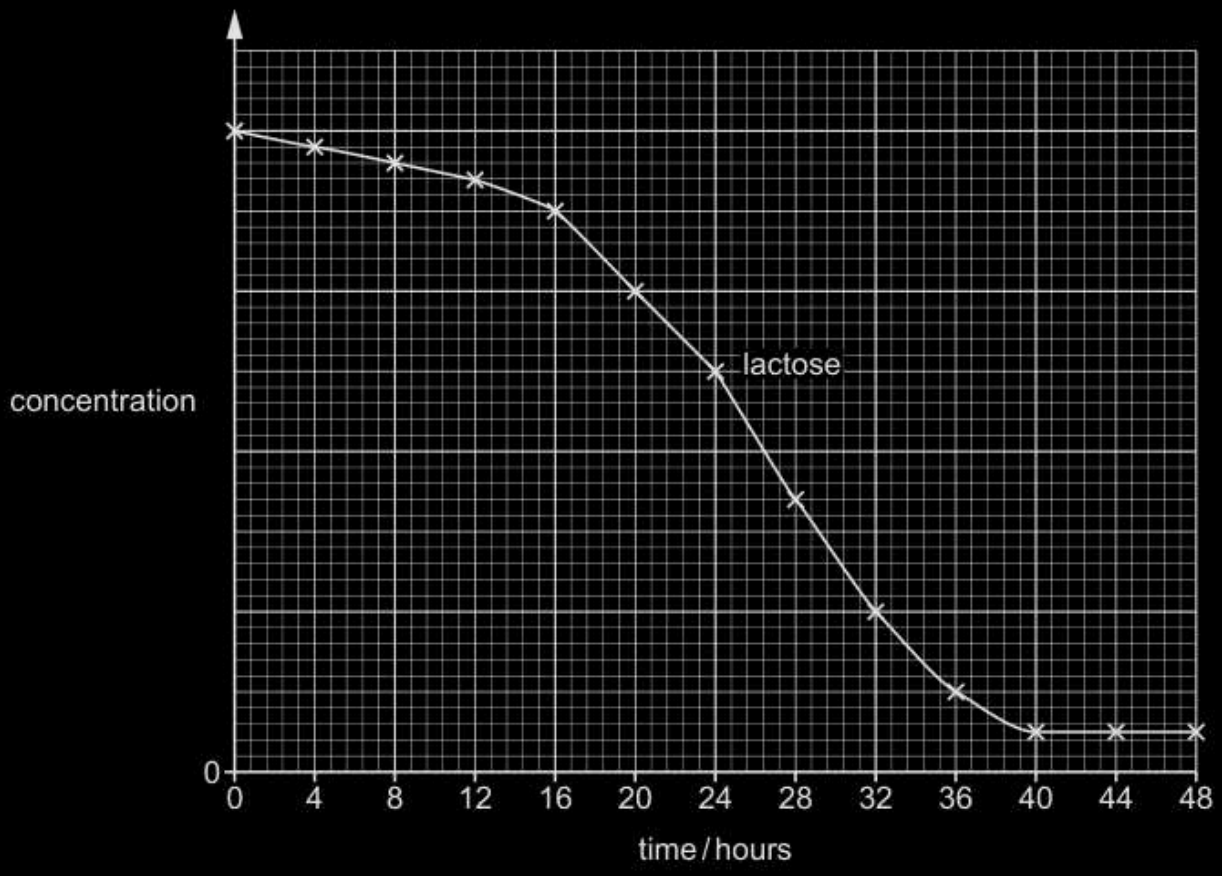
[Total: 11]

2(a)(i)	tissue / mesophyll ;	**1**	**Ignore** spongy / palisade / cells
2(a)(ii)	chloroplast drawn **+** labelled **+** correct position ;	**1**	
	any three drawn + labelled + correct position: nucleus **+** larger than chloroplast if present ; nucleolus / nuclear membrane ; cell membrane ; cytoplasm / protoplasm ; sap / vacuole ; tonoplast ;	**3**	
2(a)(iii)	guard cell **+** label line ends inside / touching guard cell ;	**1**	
2(b)(i)	cellulose ;	**1**	**Reject** cellulase
2(b)(ii)	**1** down water potential gradient **AW** ; **2** water into cell / vacuole ; **3** osmosis / diffusion ; **4** increase **+** size / pressure **AW** ; **5** burst / rupture ; **6** no cell wall to stop rupturing **AW** ;	**4**	

© UCLES 2022

Question	Answer	Marks	Guidance

3 The graph shows how the concentration of lactose sugar changes during the formation of yoghurt from milk over a period of 48 hours.



(a) (i) Name the type of microorganism used in the production of yoghurt.

.....

[1]

(ii) Name the type of cell division that causes the population of this microorganism to increase over the 48-hour period.

.....

[1]

(iii) Name the acid produced by this type of microorganism in the formation of yoghurt.

.....

[1]

(iv) Draw a line on the graph to show how the concentration of this acid will change during the 48-hour period. [2]

(b) Lactose intolerance is a medical condition that results from a genetic change. A person with the condition is unable to produce molecules of the correct enzyme to digest lactose sugar.

(i) Name this type of genetic change and explain how it can result in a person being unable to produce molecules of the correct enzyme.

.....

.....

.....

.....

.....

..... [3]

(ii) The low concentration of lactose sugar in yoghurt makes it a better food than milk for a person with lactose intolerance.

Outline the health benefits to some people with lactose intolerance of continuing to include a dairy product such as yoghurt in the diet.

.....

.....

.....

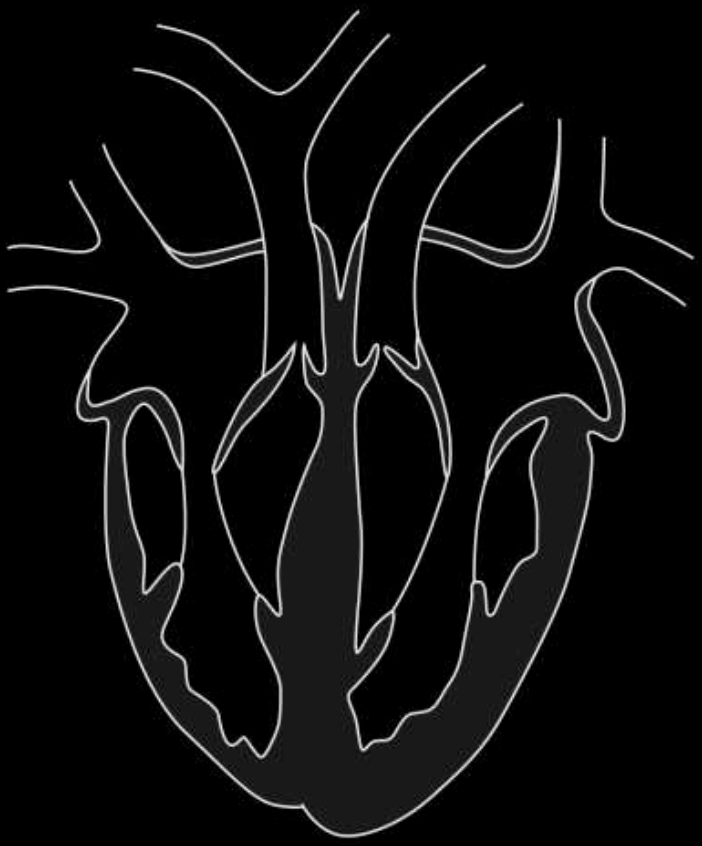
..... [2]

[Total: 10]

3(a)(i)	bacterium / bacteria ;	**1**	
3(a)(ii)	binary fission / mitosis ;	**1**	**Ignore** asexual reproduction
3(a)(iii)	lactic / lactate ;	**1**	
3(a)(iv)	increasing from Y axis and increasing as crosses lactose line ; level off at 40 hours and then constant to 48 hours ;	**2**	
3(b)(i)	**1** mutation; **2** change **+** allele / gene / base sequence / DNA ; **3** genetic material **AW** **+** codes **+** for enzyme / lactase **/** protein ; **4** active site **+** doesn't fit substrate / lactose **AW** ;	**3**	
3(b)(ii)	bones / prevention of rickets ; teeth ; growth / energy ; vitamin D / calcium / protein / fat ;	**2**	

Question	Answer	Marks	Guidance

4 The diagram shows the internal structure of the human heart and associated blood vessels.



(a) Name **two** blood vessels shown in the diagram that carry oxygenated blood.

1 .....

2 .....

[2]

(b) Blood flows through a valve when the left ventricle of the heart contracts. Diagram 1 shows this valve in the open and closed positions.

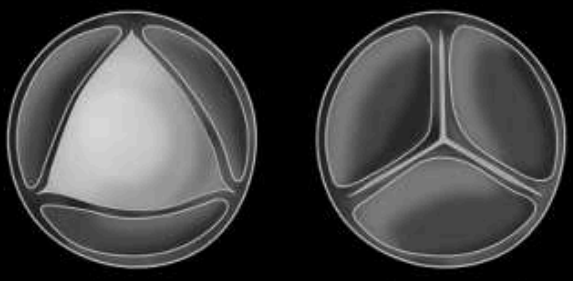
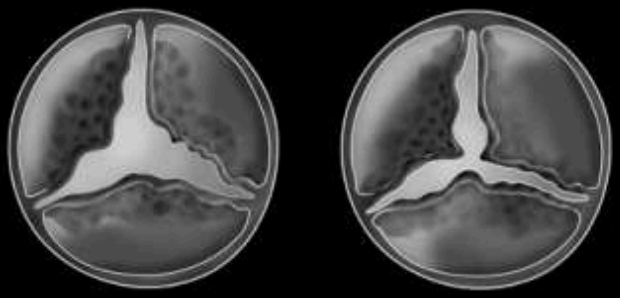


Diagram 1

(i) Label, using the letter **X** on the diagram of the heart, the location of this valve. [1]

A small number of people develop a medical condition that causes changes to this valve. Diagram 2 shows the same heart valve in the open and closed positions in a person with this condition.



valve open valve closed

Diagram 2

(ii) Explain how this medical condition will affect the flow of blood when the left ventricle contracts and relaxes.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(iii) Describe and explain the effect of this condition on the ability of the person to exercise.

.....

.....

.....

.....

.....

..... [3]

(c) It is possible for the valve to be replaced by surgery. The human valve is removed from the patient and can be replaced with a valve containing tissue from another species. This tissue is treated with a chemical to prevent rejection by the human body.

Suggest how the cells of the tissue are modified by the chemical treatment and explain how this will prevent rejection by the human body.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 14]

4(a)	aorta ; pulmonary vein ;	**2**	
4(b)(i)	'**X**' used to indicate valve at base of aorta ;	**1**	
4(b)(ii)	**1** valve + not open / close + fully / properly **AW** ; **2** ventricle contracts + valve opens (not fully) ; **3** contracts + less blood through valve ; **4** ventricle relaxes + valve closes (not fully) ; **5** relaxes + backflow of blood / blood back to ventricle ;	**4**	
4(b)(iii)	**1** less ability to exercise / get tired / (muscle) fatigue **AW** ; **2** less / little **AW** + oxygen / oxygenated blood ; **3** to muscle ; **4** less / little + aerobic respiration ; **5** less / little + energy released ; **6** anaerobic respiration ; **7** lactic acid ;	**3**	
4(c)	**1** antigen / protein ; **2** (antigen / protein) + removed / blocked / coated **OR** cells / antigens made same as human tissue ; **3** no recognition **AW** ; **4** white blood cells / lymphocytes / phagocytes ; **5** reduced + immune response ; **6** less + antibodies / phagocytosis **AW** ;	**4**	

© UCLES 2022

Question	Answer	Marks	Guidance

October/November 2022 · Paper 2 · Variant 2 · Question 8

8 **(a)** Describe and explain the gas exchange that takes place between the leaf of a plant and the air in the atmosphere during a 24-hour period.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

(b) Outline the movement of water through a leaf during the process of transpiration.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

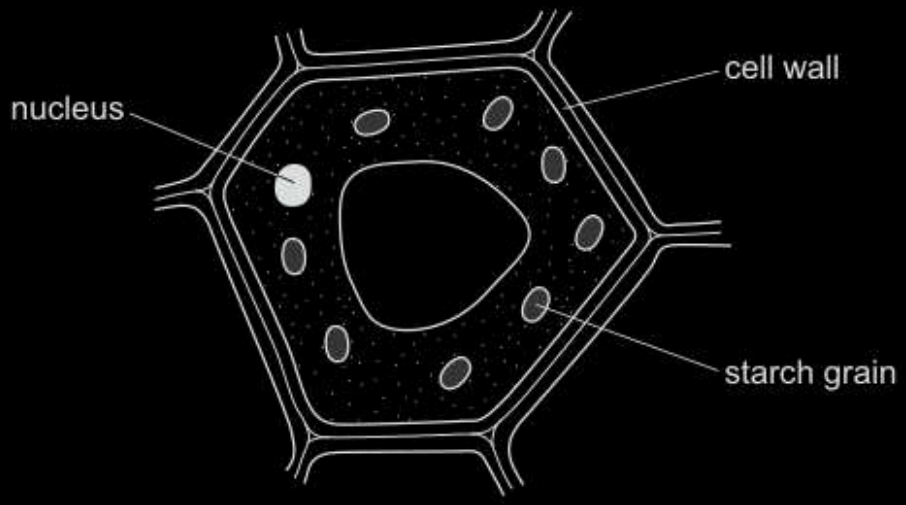
8(a)	*day* **1** light / day **+** photosynthesis ; **2** faster than respiration / respiration uses up oxygen ; **3** day **+** CO_2 used / in **+** O_2 produced / out ; *night* **4** no light / night **+** no photosynthesis ; **5** respiration ; **6** night / evening **+** O_2 used / in **+** CO_2 produced / out ; **7** stomata + open in day / closed at night ; **8** more water vapour lost during day ; **9** compensation point(s) **AW** ;	**5**	
8(b)	**1** xylem ; **2** from xylem into mesophyll ; **3** diffusion / osmosis ; **4** down water potential gradient **AW** ; **5** layer of water **+** mesophyll **+** cell wall / outside cell ; **6** evaporation / water vapour ; **7** air / <u>inter</u>cellular **+** spaces ; **8** exits leaf through stomata / stoma ;	**5**	

Question	Answer	Marks	Guidance

May/June 2022 · Paper 1 · Variant 1 · Question 1

Correct answer: C

1 The diagram shows a plant cell. The cell is stained with iodine solution.



After staining with iodine solution, what are the colours of the cell wall and the starch grain?

	cell wall	starch grain
A	blue-black	blue-black
B	blue-black	orange-brown
C	orange-brown	blue-black
D	orange-brown	orange-brown

May/June 2022 · Paper 1 · Variant 1 · Question 2

Correct answer: C

2 The diagram shows the shape of a red blood cell.

What is an advantage of this shape?

A allows it to carry more haemoglobin

B allows it to move more quickly in arteries

C increases its surface area for diffusion

D prevents it getting trapped in valves in veins

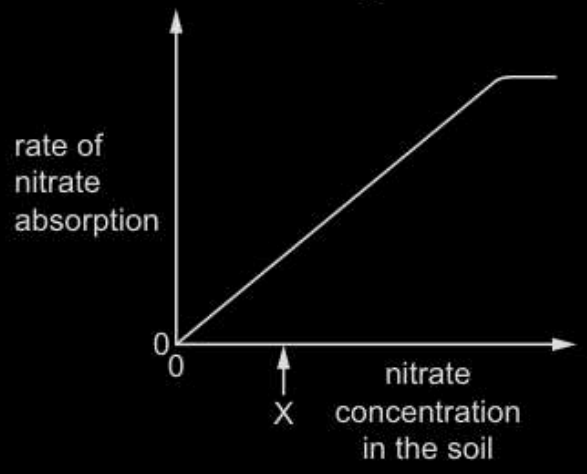
Correct answer: A

4 The rate of nitrate ion absorption by a root hair cell was measured at different soil nitrate concentrations.

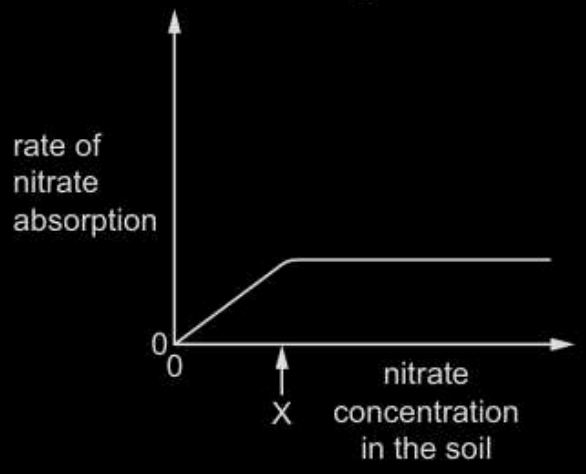
At X, the concentration of nitrate in the soil is the same as in the cell.

Which graph shows how the rate of absorption varies with nitrate concentration in the soil?

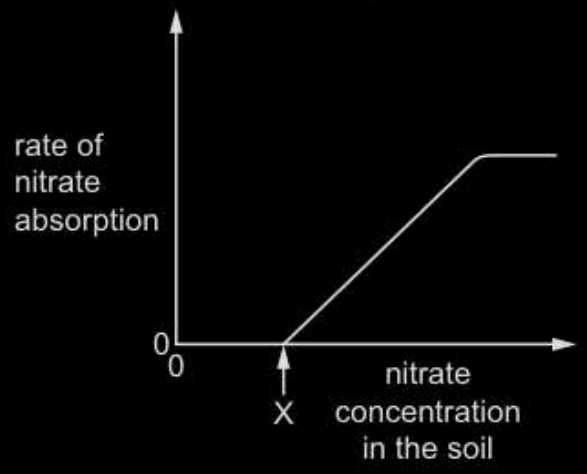
A



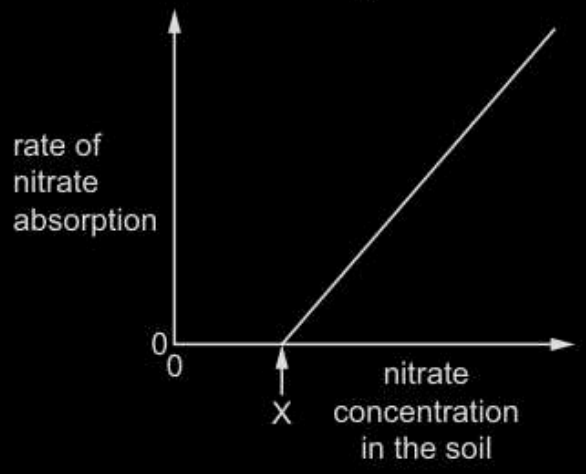
B



C

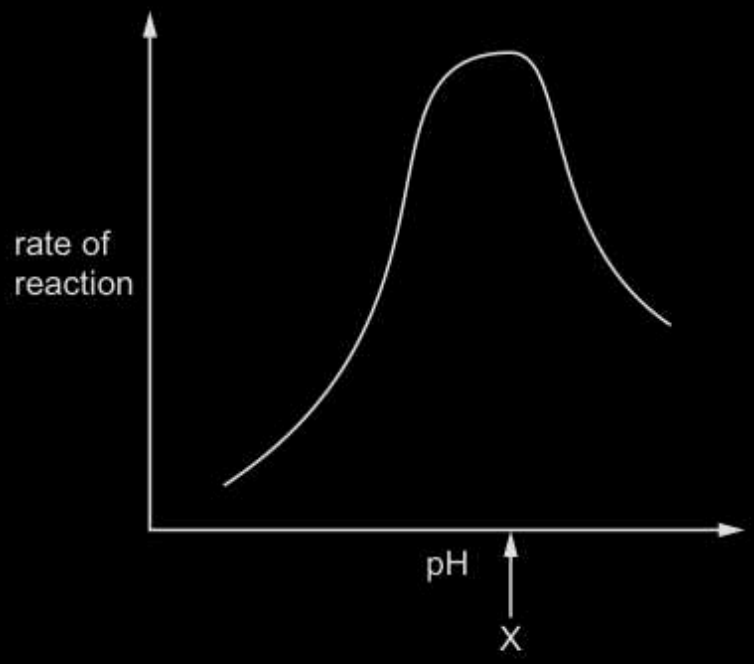


D



Correct answer: D

5 The graph shows how the rate of reaction of a protease from the stomach varies with pH.



What describes the pH at point X?

A the highest pH at which this enzyme is active

B the lowest pH that does not destroy the enzyme

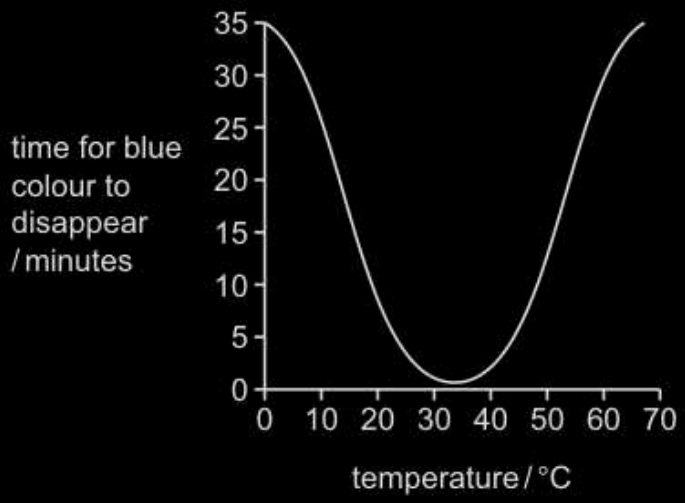
C the pH at which the substrate starts to run out

D the pH that produces the highest rate of enzyme activity

Correct answer: B

6 A dilute starch solution turns blue when very dilute iodine solution is added. This colour disappears as the starch is broken down by amylase.

The graph shows the disappearance of the blue colour at different temperatures.



Which statement explains the shape of the curve?

A Enzyme activity rises as the temperature rises.

B Enzymes have an optimum temperature range for activity.

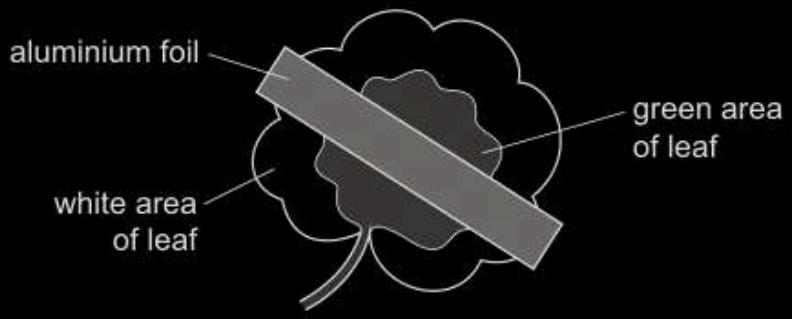
C Enzymes are inactive below 30 °C.

D Enzymes do not work above 30 °C.

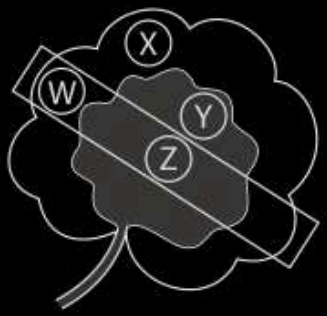
May/June 2022 · Paper 1 · Variant 1 · Question 7

Correct answer: D

7 A plant has leaves with both green and white areas. One of its leaves is partly covered with aluminium foil which blocks light.



The plant is then placed under a lamp for 24 hours. After this time, discs are cut from the areas of the leaf shown, and tested with iodine solution.



Which leaf discs will give a blue-black colour when tested with iodine solution?

A W and X **B** X only **C** Y and Z **D** Y only

May/June 2022 · Paper 1 · Variant 1 · Question 10

Correct answer: A

10 Four food tests were each carried out on four different white powders, **A**, **B**, **C** and **D**.

The results are shown in the table.

Which powder contains a carbohydrate, a fat and a protein?

	results of tests on the powders			
	Benedict's test	biuret test	ethanol emulsion test	iodine solution test
A	blue colour	violet colour	milky	blue-black colour
B	orange colour	blue colour	milky	blue-black colour
C	red colour	violet colour	clear	yellow colour
D	green colour	blue colour	clear	yellow colour

May/June 2022 · Paper 1 · Variant 1 · Question 11

Correct answer: C

11 What is an example of assimilation?

A absorption of glycerol into lacteals

B breakdown of glycogen to glucose in the liver

C building of proteins from amino acids

D release of a hormone from a gland

May/June 2022 · Paper 1 · Variant 1 · Question 13

Correct answer: C

13 What is the main cause of water moving up to the leaves in xylem vessels?

A active transport

B evaporation from the epidermis of the leaf

C evaporation from the walls of the mesophyll cells

D use of water in photosynthesis

May/June 2022 · Paper 1 · Variant 1 · Question 14

Correct answer: B

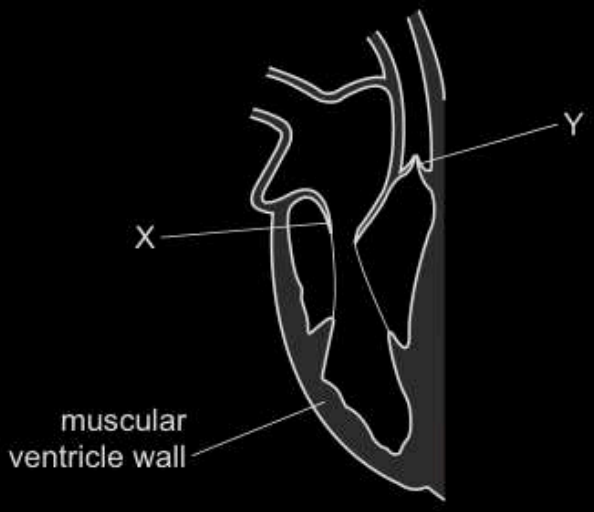
14 Which blood vessels carry blood towards the heart and towards the lungs?

	towards heart	towards lungs
A	aorta	pulmonary artery
B	vena cava	pulmonary artery
C	aorta	pulmonary vein
D	vena cava	pulmonary vein

May/June 2022 · Paper 1 · Variant 1 · Question 15

Correct answer: B

15 The diagram shows the right side of the human heart when the ventricle is relaxed.



Which row describes the positions of valve X and valve Y when the ventricle contracts?

	valve X	valve Y
A	closed	closed
B	closed	open
C	open	closed
D	open	open

May/June 2022 · Paper 1 · Variant 1 · Question 16

Correct answer: B

16 People who have a diet that contains a lot of fat and who smoke have two important risk factors for which disease?

A anaemia

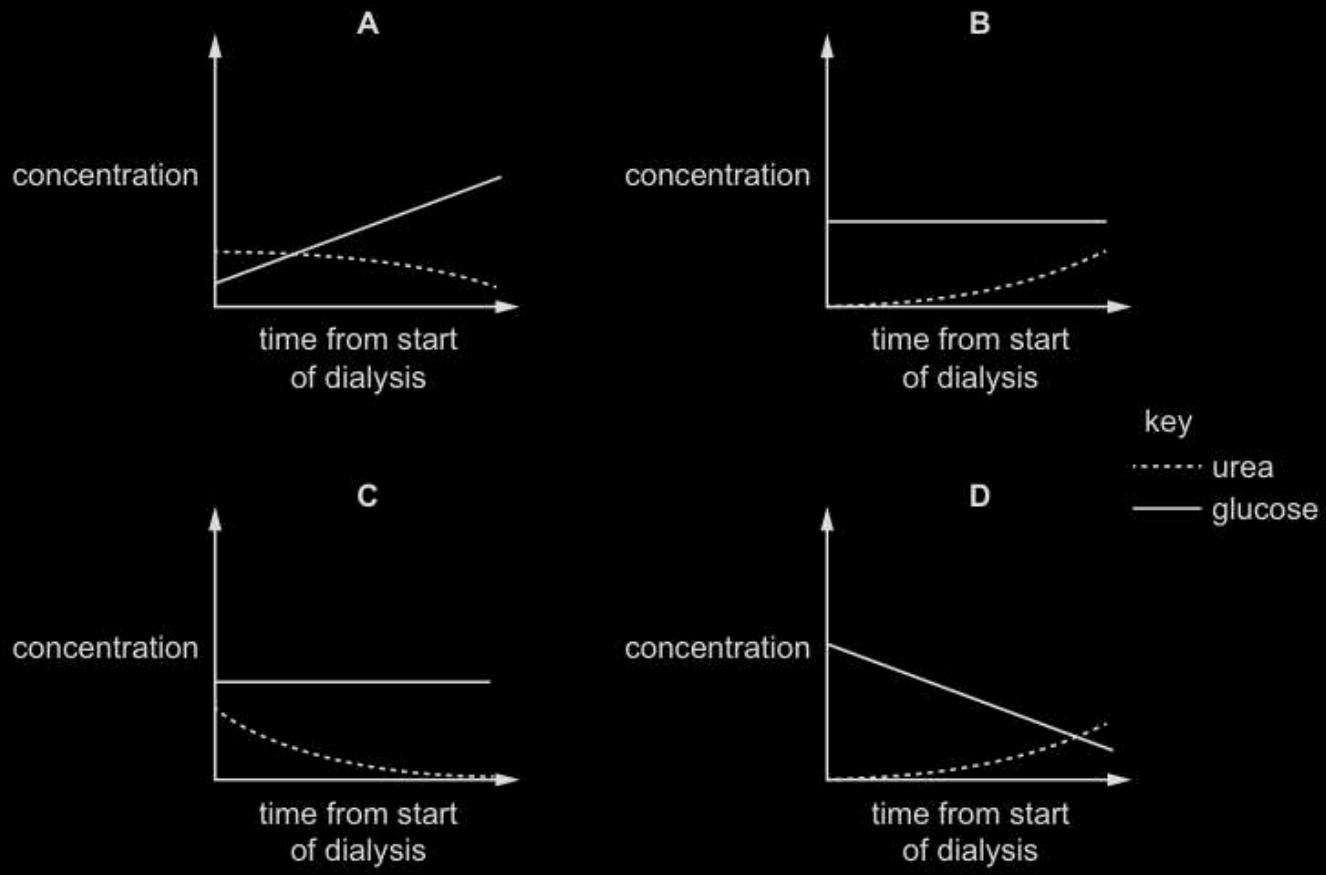
B coronary heart disease

C emphysema

D scurvy

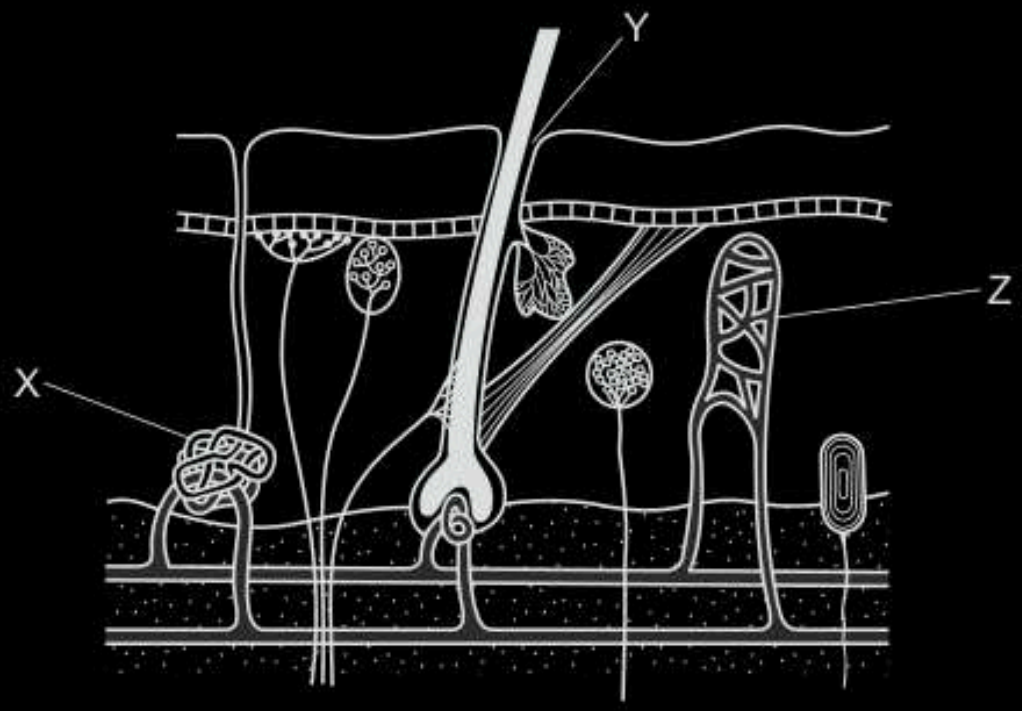
Correct answer: B

20 Which graph shows the change in concentration of molecules in the dialysis fluid during kidney dialysis?



Correct answer: C

21 The diagram shows some structures in human skin.

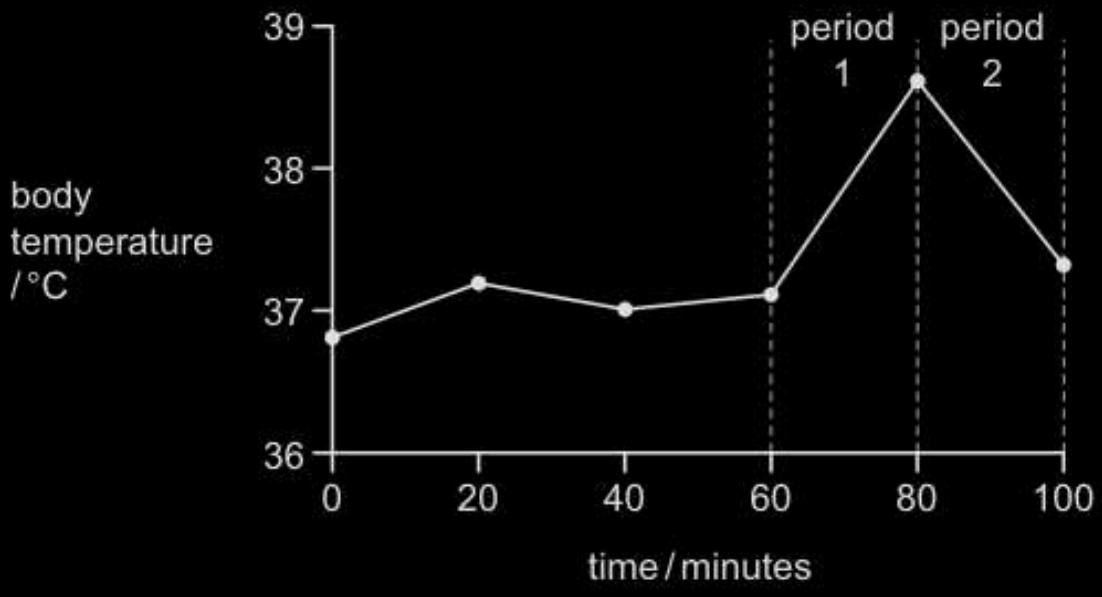


Which row shows the names of structures X, Y and Z?

	X	Y	Z
A	blood vessel	hair follicle	sweat gland
B	hair follicle	blood vessel	sweat gland
C	sweat gland	hair follicle	blood vessel
D	sweat gland	blood vessel	hair follicle

Correct answer: C

22 The graph shows changes in a person's body temperature plotted against time.



What could cause the changes in body temperature in periods 1 and 2?

	period 1	period 2
A	reduced air temperature	increased air temperature
B	reduced air temperature	shivering
C	vigorous exercise	increased sweating
D	vigorous exercise	shivering

May/June 2022 · Paper 1 · Variant 1 · Question 23

Correct answer: D

23 The diagram shows a cross-section of the human brain.

If structure X is damaged, which body functions may be affected?

A control of body temperature

B control of breathing and heartbeat

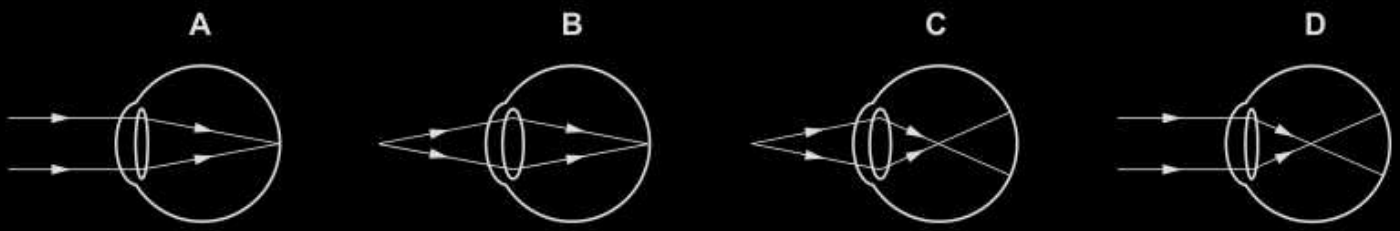
C control of emotion, language, memory and reasoning

D control of movement, posture and balance

May/June 2022 · Paper 1 · Variant 1 · Question 24

Correct answer: A

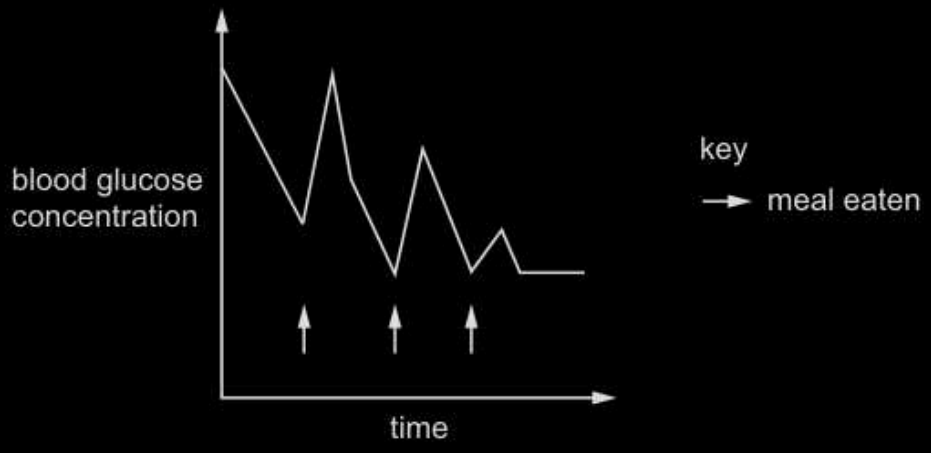
24 Which diagram shows how light from a distant object is focused on the retina to form a clear image?



May/June 2022 · Paper 1 · Variant 1 · Question 25

Correct answer: B

25 The graph shows the changing blood glucose concentration in one person over 24 hours.



Which statement explains the shape of the graph?

A After each meal, adrenaline is released.

B After each meal, insulin is released.

C Before each meal, adrenaline is released.

D Before each meal, insulin is released.

May/June 2022 · Paper 1 · Variant 1 · Question 28

Correct answer: D

28 Which rows show some of the structures found in bacteria, fungi and viruses?

	structure	bacteria	fungi	viruses
1	cell membrane	✓	✓	✗
2	cell wall	✗	✓	✓
3	nucleus	✓	✗	✓
4	protein coat	✗	✗	✓

key
✓ = present
✗ = not present

A 1, 2 and 3 **B** 2, 3 and 4 **C** 3 and 4 only **D** 1 and 4

Correct answer: A

36 The diagram shows a developing fetus in the uterus.

What is the liquid at X called?

A amniotic fluid

B blood

C urine

D water

May/June 2022 · Paper 1 · Variant 1 · Question 37

Correct answer: D

37 Which human feature shows discontinuous variation?

A agility

B body height

C obesity

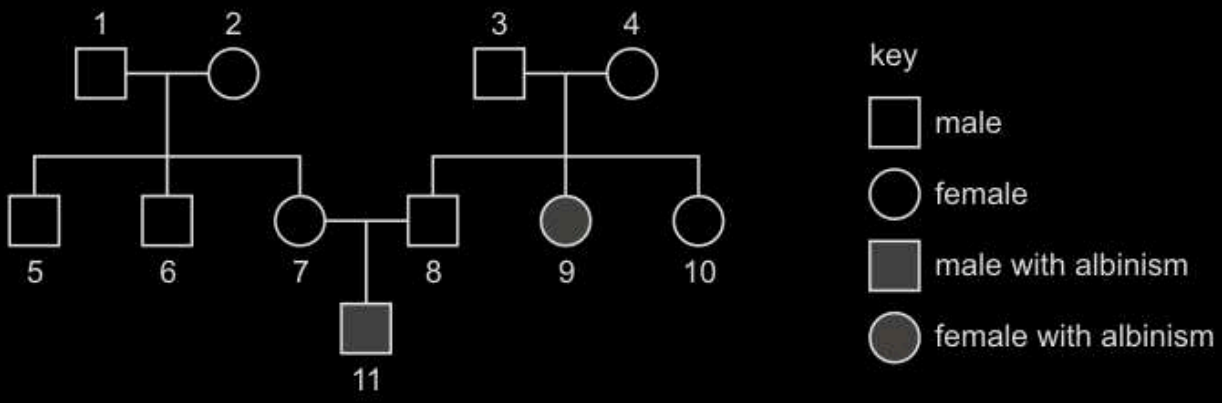
D sickle cell anaemia

May/June 2022 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 Albinism is an inherited condition, caused by a recessive allele.

The diagram shows a family tree for a family affected by this condition.



Individuals 9 and 11 have albinism.

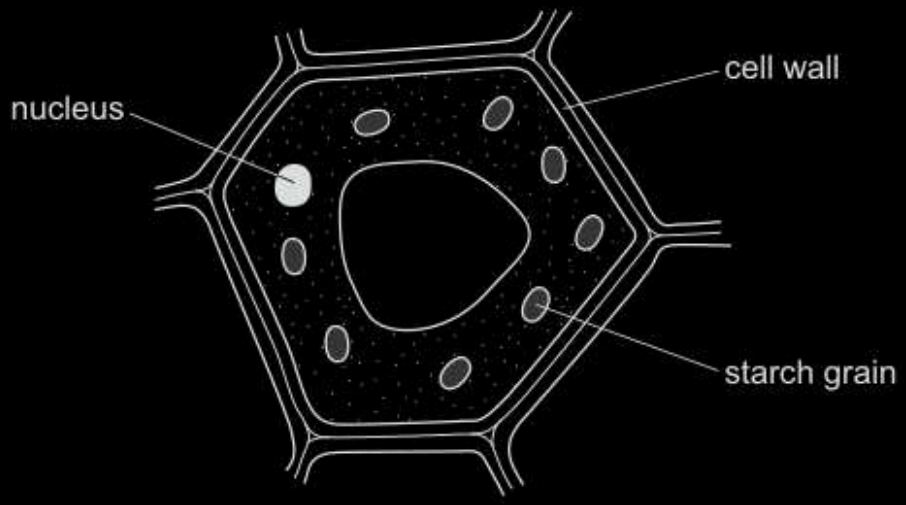
What is the chance that individual 10 is heterozygous?

A 0 in 4 **B** 1 in 4 **C** 2 in 3 **D** 3 in 4

May/June 2022 · Paper 1 · Variant 2 · Question 1

Correct answer: C

1 The diagram shows a plant cell. The cell is stained with iodine solution.



After staining with iodine solution, what are the colours of the cell wall and the starch grain?

	cell wall	starch grain
A	blue-black	blue-black
B	blue-black	orange-brown
C	orange-brown	blue-black
D	orange-brown	orange-brown

May/June 2022 · Paper 1 · Variant 2 · Question 3

Correct answer: B

3 What is diffusion?

A movement of particles by the air

B movement of molecules down a concentration gradient

C movement of molecules in a heated liquid

D movement of particles up a concentration gradient

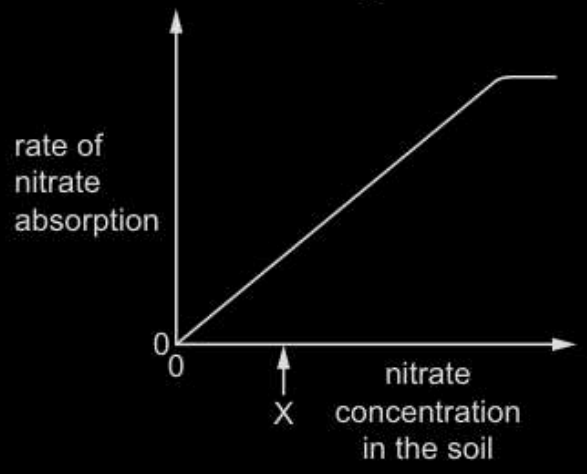
Correct answer: A

4 The rate of nitrate ion absorption by a root hair cell was measured at different soil nitrate concentrations.

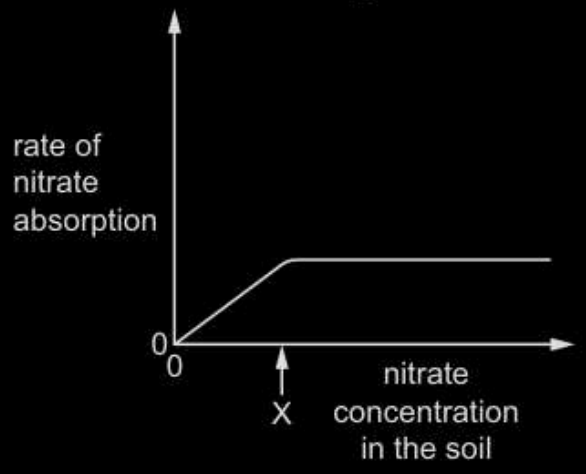
At X, the concentration of nitrate in the soil is the same as in the cell.

Which graph shows how the rate of absorption varies with nitrate concentration in the soil?

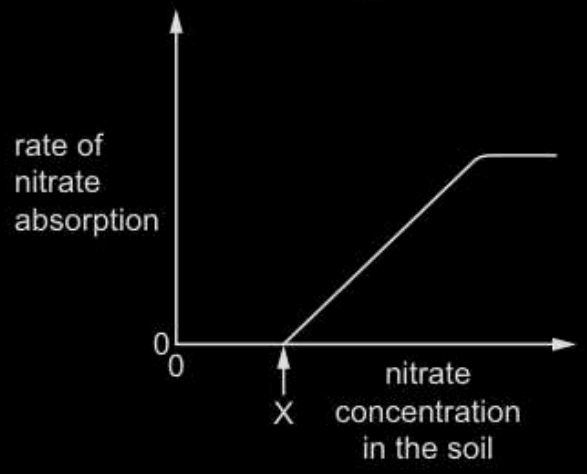
A



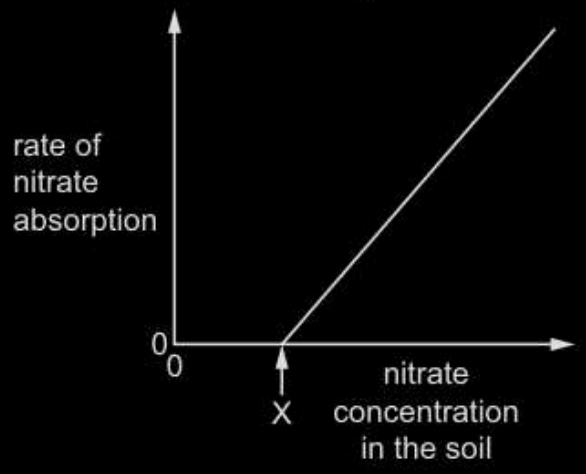
B



C



D



May/June 2022 · Paper 1 · Variant 2 · Question 5

Correct answer: C

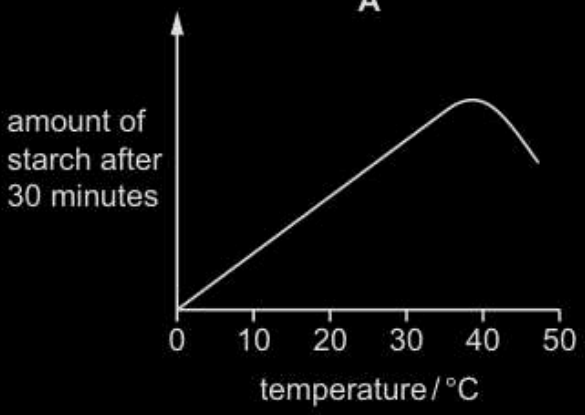
5 Amylase is an enzyme that breaks down starch to maltose.

Students set up an experiment to investigate the effect of different temperatures on the action of amylase on starch solution.

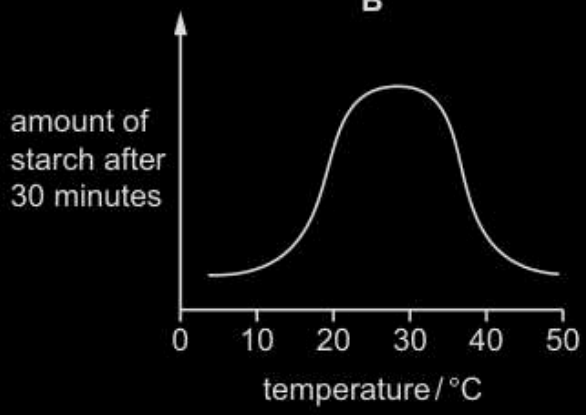
They measured the amount of starch remaining after 30 minutes at different temperatures.

Which graph would you expect the students to draw from their results?

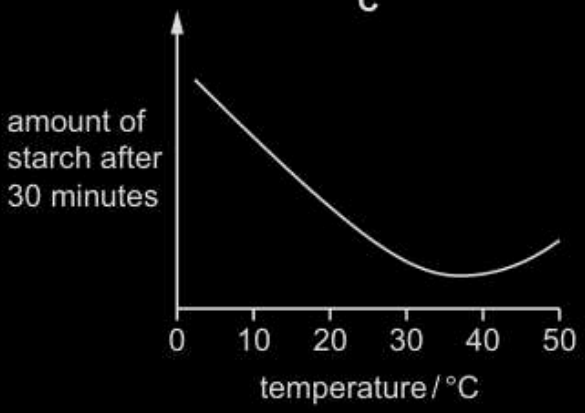
A



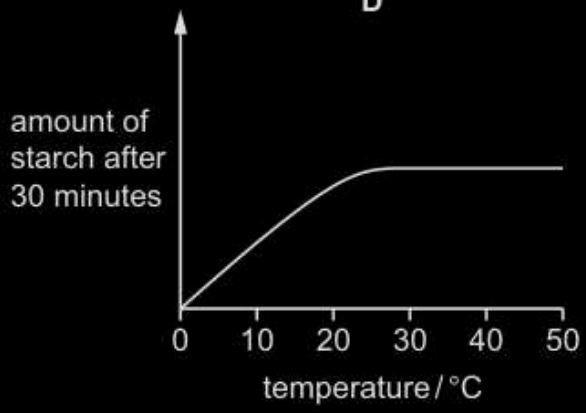
B



C



D



May/June 2022 · Paper 1 · Variant 2 · Question 6

Correct answer: D

6 Which statement identifies the optimum (best) temperature for enzyme activity?

A It is the highest temperature at which any enzyme activity happens.

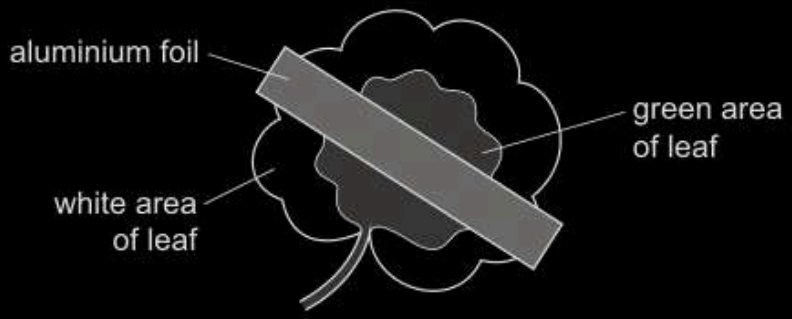
B It is the highest temperature that does not destroy an enzyme.

C It is the lowest temperature that denatures an enzyme.

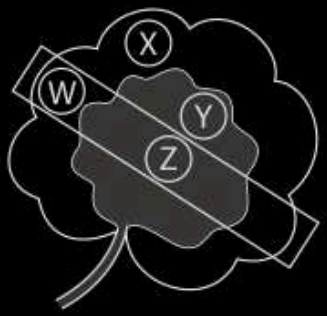
D It is the temperature that produces the highest rate of enzyme activity.

Correct answer: D

7 A plant has leaves with both green and white areas. One of its leaves is partly covered with aluminium foil which blocks light.



The plant is then placed under a lamp for 24 hours. After this time, discs are cut from the areas of the leaf shown, and tested with iodine solution.



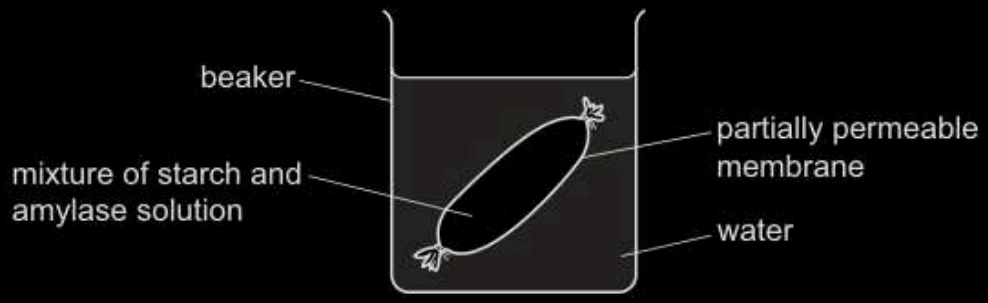
Which leaf discs will give a blue-black colour when tested with iodine solution?

A W and X **B** X only **C** Y and Z **D** Y only

May/June 2022 · Paper 1 · Variant 2 · Question 9

Correct answer: B

9 The experiment shown was set up and left for 30 minutes.



The membrane is permeable to sugar and water but not permeable to starch or protein. After 30 minutes, samples of the water in the beaker were tested with Benedict's reagent, biuret reagent and iodine solution.

Which colours were obtained with these tests?

	Benedict's test	biuret test	iodine test
A	blue	violet	blue-black
B	red	blue	brown
C	red	blue	blue-black
D	red	violet	brown

May/June 2022 · Paper 1 · Variant 2 · Question 10

Correct answer: C

10 Sometimes gall bladders become infected and have to be surgically removed.

How will this affect the functioning of the body?

A reduce the digestion of carbohydrates

B reduce the liver's ability to convert glucose to glycogen

C reduce the amount of glycerol absorbed from the alimentary canal

D reduce the volume of stored urine

May/June 2022 · Paper 1 · Variant 2 · Question 11

Correct answer: C

11 What is an example of assimilation?

A absorption of glycerol into lacteals

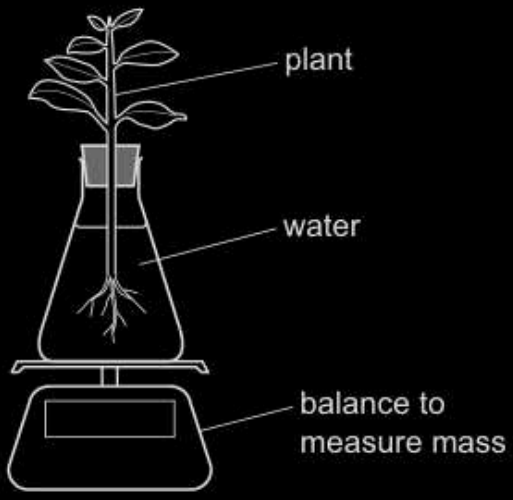
B breakdown of glycogen to glucose in the liver

C building of proteins from amino acids

D release of a hormone from a gland

Correct answer: A

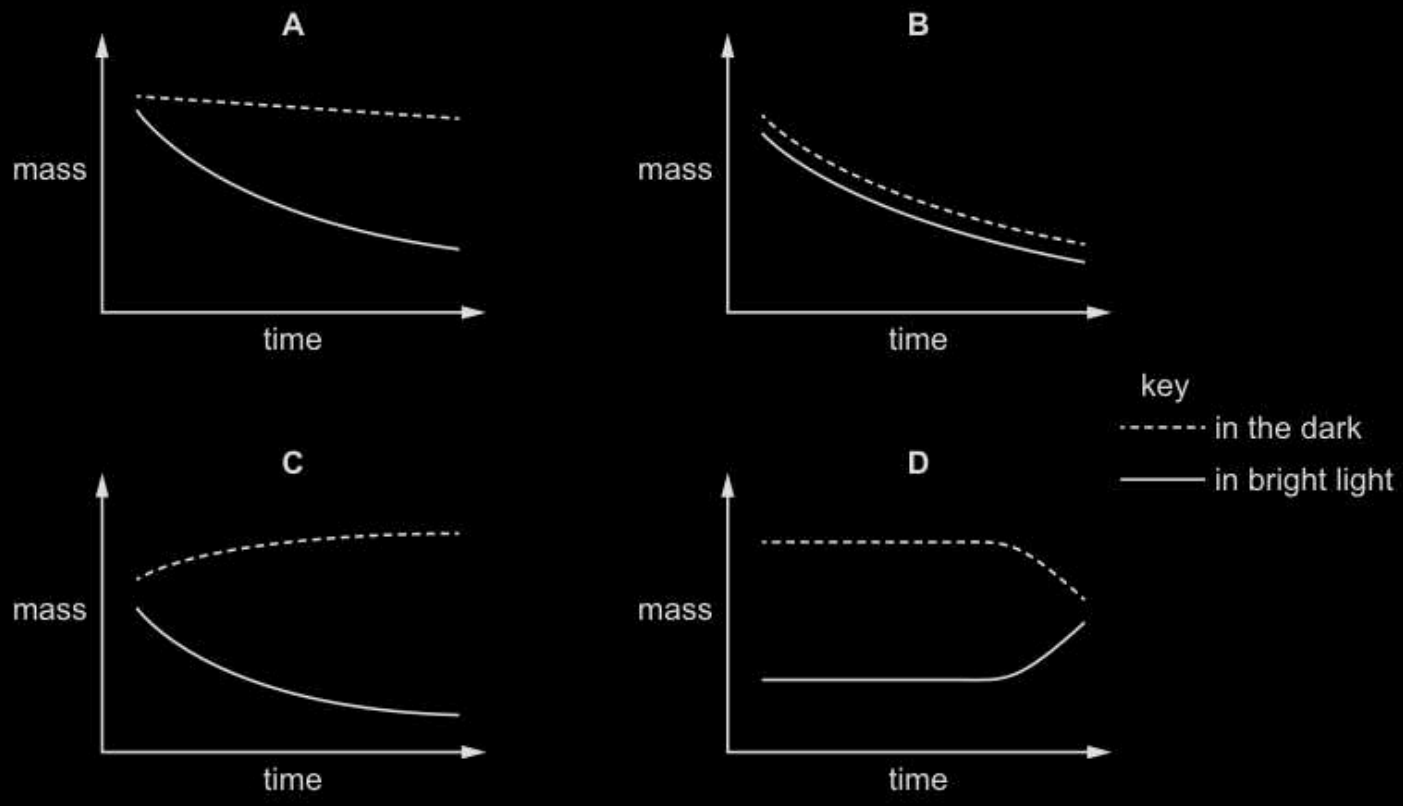
12 A student set up an experiment to investigate the rate of transpiration in a plant.



The student set up two identical sets of apparatus.

One was placed in a room at 20 °C with bright light and one in a room at 20 °C in the dark.

Which graph shows the student's results?



May/June 2022 · Paper 1 · Variant 2 · Question 13

Correct answer: C

13 What is the main cause of water moving up to the leaves in xylem vessels?

A active transport

B evaporation from the epidermis of the leaf

C evaporation from the walls of the mesophyll cells

D use of water in photosynthesis

May/June 2022 · Paper 1 · Variant 2 · Question 14

Correct answer: C

14 Which blood vessels, linked directly to the heart, transport deoxygenated blood?

A aorta and pulmonary artery

B aorta and pulmonary vein

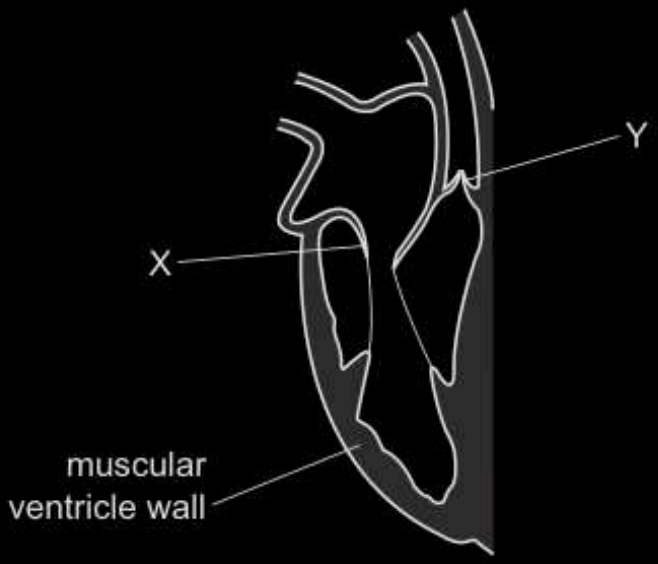
C vena cava and pulmonary artery

D vena cava and pulmonary vein

May/June 2022 · Paper 1 · Variant 2 · Question 15

Correct answer: B

15 The diagram shows the right side of the human heart when the ventricle is relaxed.



Which row describes the positions of valve X and valve Y when the ventricle contracts?

	valve X	valve Y
A	closed	closed
B	closed	open
C	open	closed
D	open	open

May/June 2022 · Paper 1 · Variant 2 · Question 16

Correct answer: B

16 Which component of the blood stimulates the formation of fibres to help seal a wound?

A plasma

B platelets

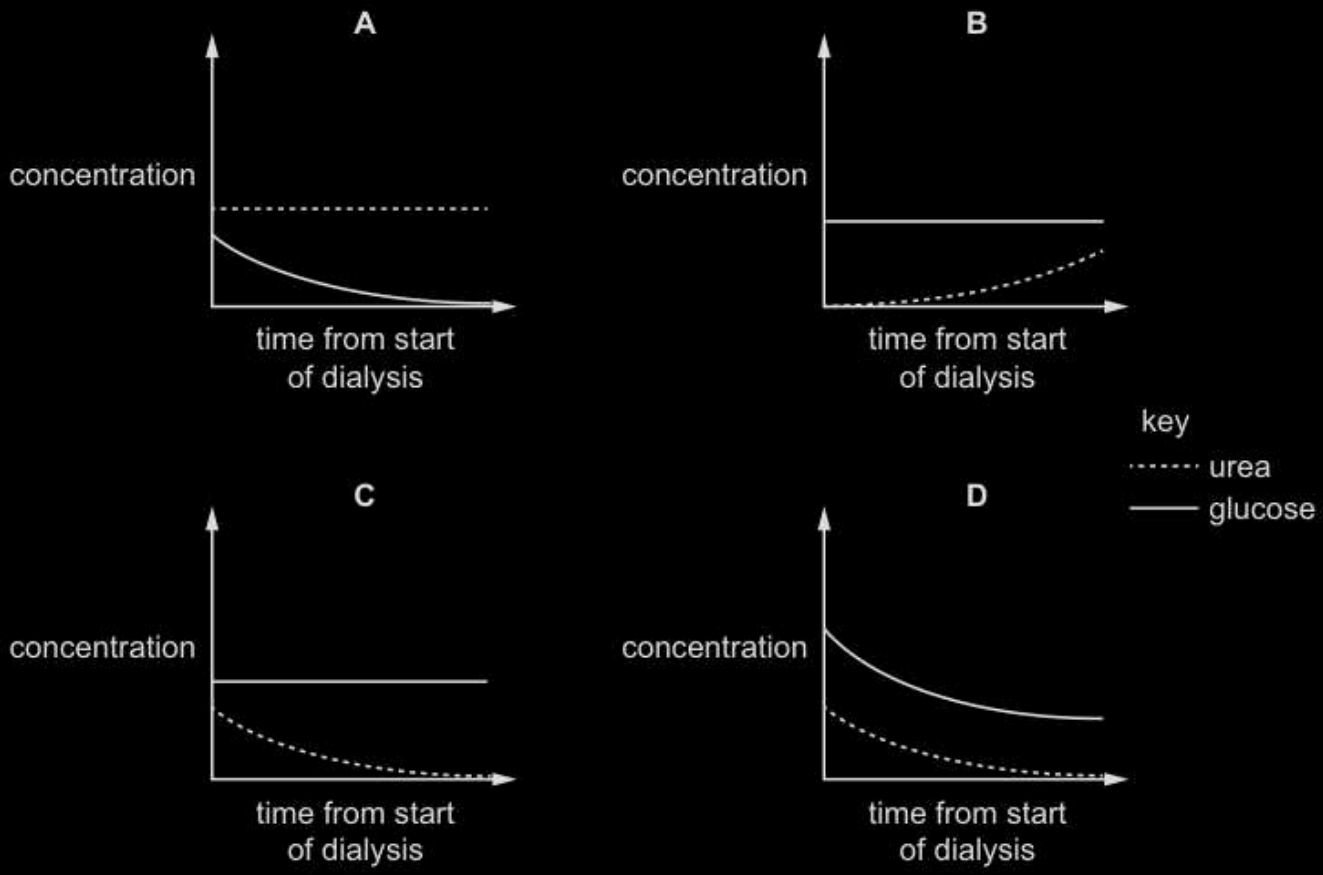
C red blood cells

D white blood cells

May/June 2022 · Paper 1 · Variant 2 · Question 20

Correct answer: C

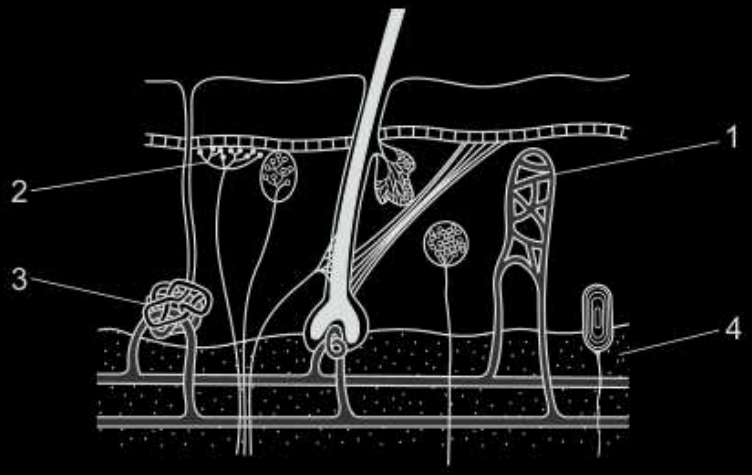
20 Which graph shows the changes in concentration of urea and glucose in the blood during kidney dialysis?



May/June 2022 · Paper 1 · Variant 2 · Question 21

Correct answer: B

21 The diagram shows a cross-section of the human skin.

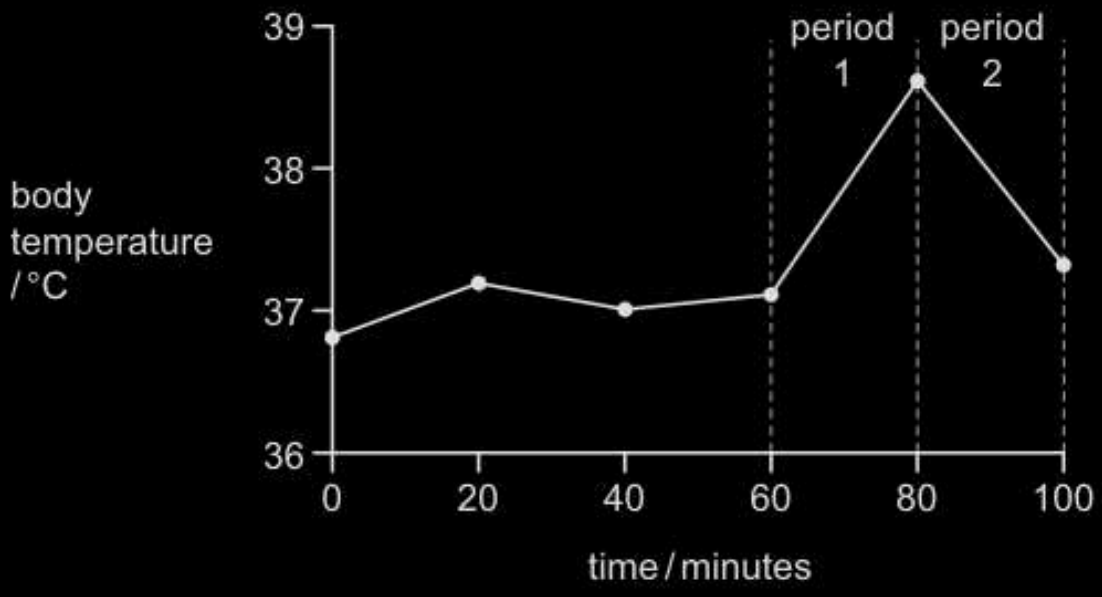


Which structures are involved in maintaining a constant body temperature?

A 1 and 2 only **B** 1, 2, 3 and 4 **C** 2 and 3 only **D** 4 only

Correct answer: C

22 The graph shows changes in a person's body temperature plotted against time.



What could cause the changes in body temperature in periods 1 and 2?

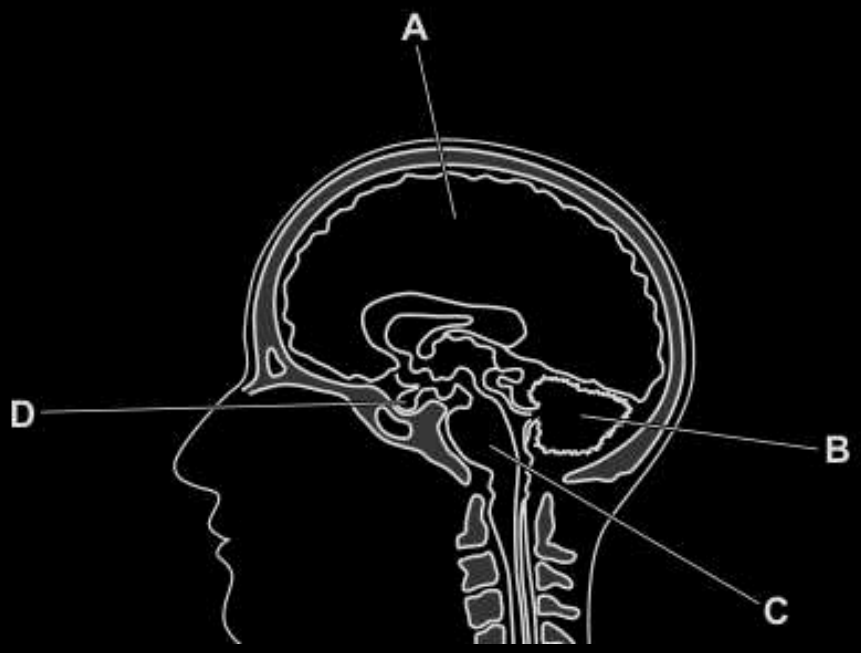
	period 1	period 2
A	reduced air temperature	increased air temperature
B	reduced air temperature	shivering
C	vigorous exercise	increased sweating
D	vigorous exercise	shivering

May/June 2022 · Paper 1 · Variant 2 · Question 23

Correct answer: C

23 The diagram shows a section through the human brain.

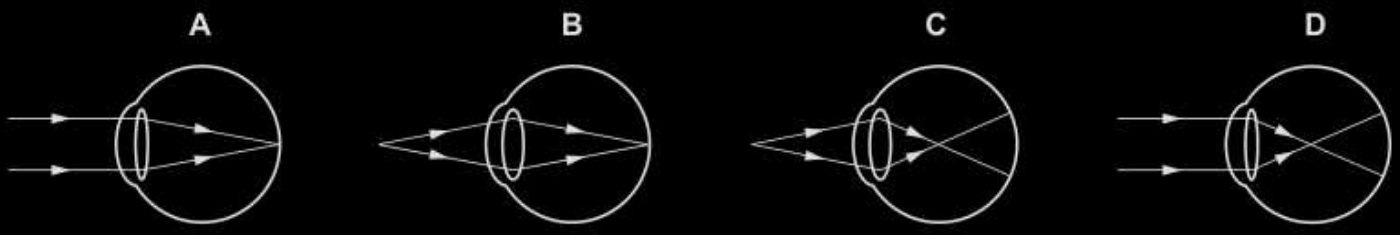
Which part of the brain controls the rate of breathing?



May/June 2022 · Paper 1 · Variant 2 · Question 24

Correct answer: A

24 Which diagram shows how light from a distant object is focused on the retina to form a clear image?



May/June 2022 · Paper 1 · Variant 2 · Question 25

Correct answer: D

25 Which condition causes an increased blood glucose concentration?

A a condition causing sufferers to reduce food intake to extreme levels

B a genetic disorder causing too much insulin to be secreted

C a condition causing a reduction in the number of red blood cells

D a condition causing too much adrenaline to be secreted

May/June 2022 · Paper 1 · Variant 2 · Question 27

Correct answer: C

27 Which short-term effect is caused by excessive consumption of alcohol?

A damage to the tissues of the heart

B faster transmission of nerve impulses

C reduced self-control of behaviour

D shorter reaction times

May/June 2022 · Paper 1 · Variant 2 · Question 28

Correct answer: B

28 Which row shows the characteristics of viruses?

	act as decomposers	can only reproduce inside a host cell	only seen under an electron microscope	genetic material surrounded by a protein coat
A	✓	✗	✓	✓
B	✗	✓	✓	✓
C	✗	✓	✓	✗
D	✓	✗	✗	✓

key

✓ = yes

✗ = no

May/June 2022 · Paper 1 · Variant 2 · Question 31

Correct answer: D

31 An organism such as a mosquito can pass a disease-causing organism from one host to another.

Which term describes the mosquito?

A parasite

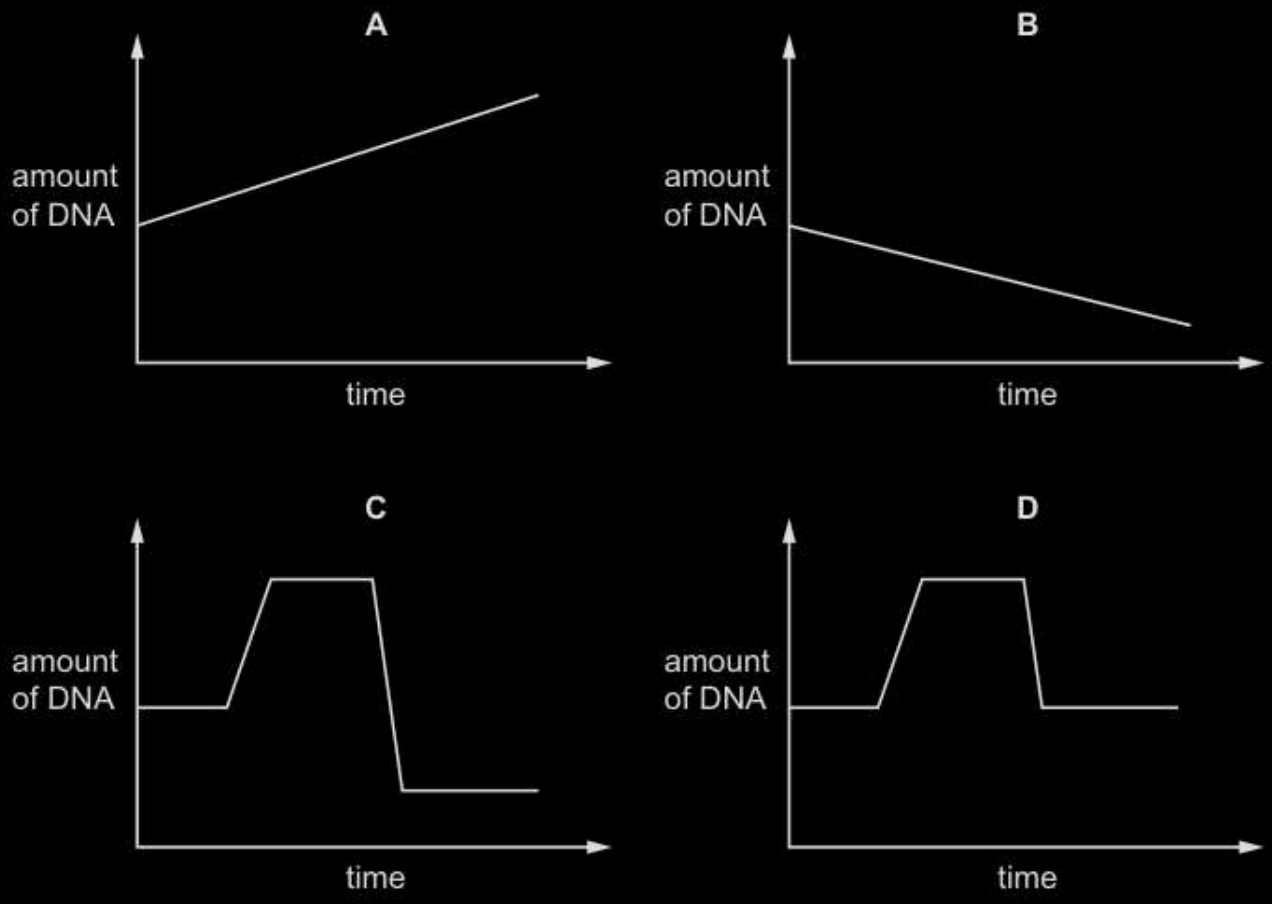
B pathogen

C receptor

D vector

Correct answer: D

33 Which graph shows how the amount of DNA in a cell changes when the cell divides by mitosis?



May/June 2022 · Paper 1 · Variant 2 · Question 36

Correct answer: A

36 The diagram shows a developing fetus in the uterus.

What is the liquid at X called?

A amniotic fluid

B blood

C urine

D water

May/June 2022 · Paper 1 · Variant 2 · Question 37

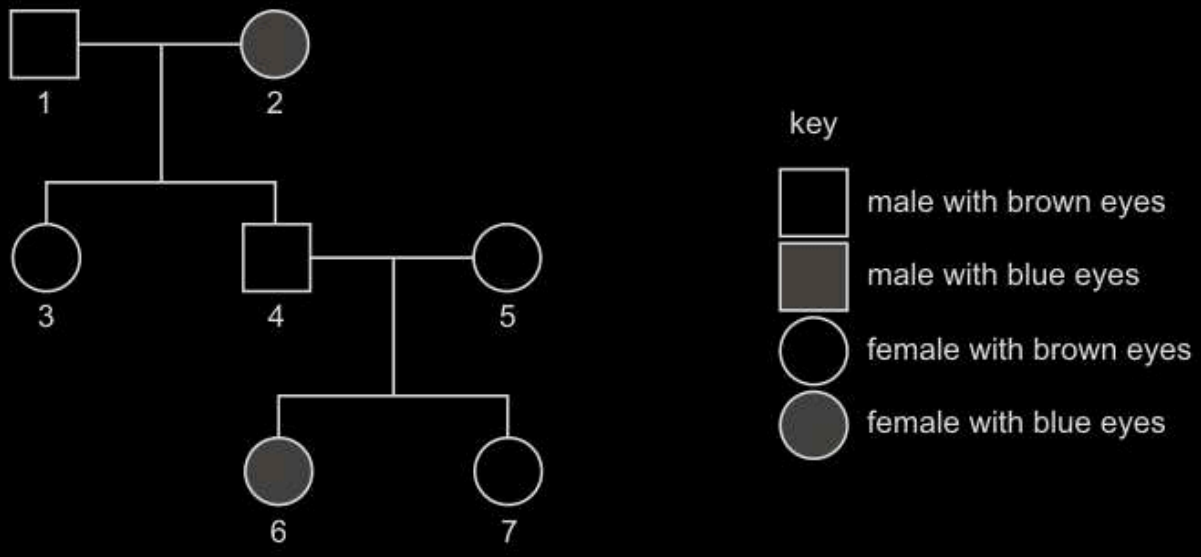
Correct answer: A

37 Which row shows examples of continuous and discontinuous variation in humans?

	continuous variation	discontinuous variation
A	height	blood group
B	blood group	hair colour
C	hair colour	body mass
D	blood group	height

Correct answer: C

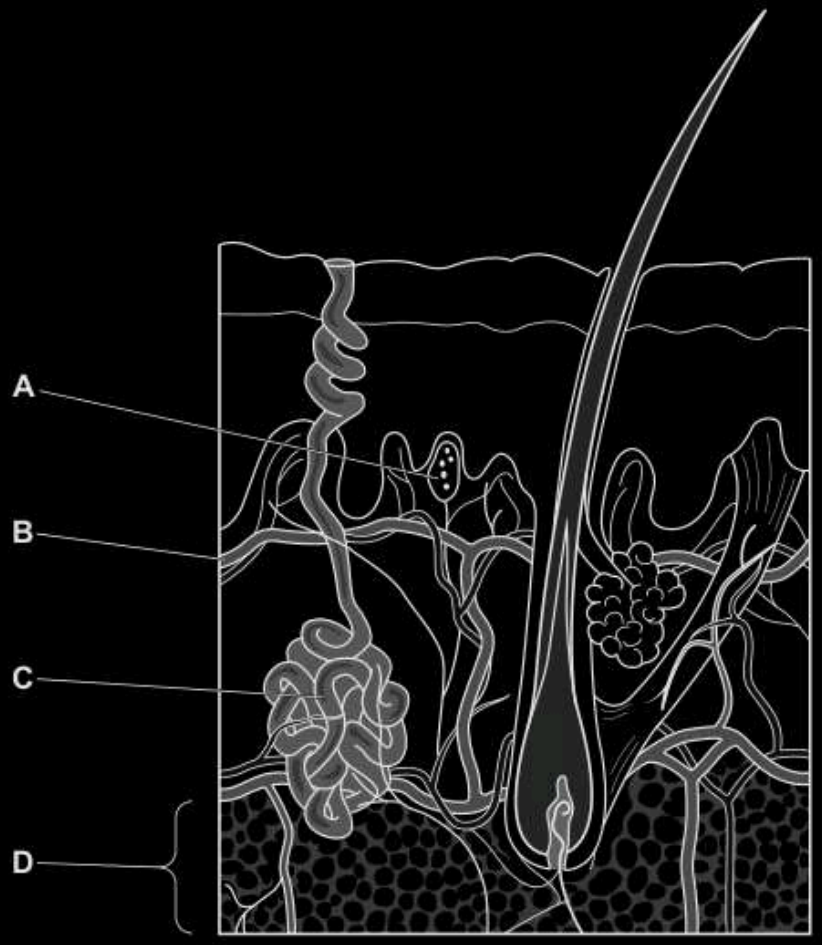
38 The diagram shows a family tree for the inheritance of eye colour. The allele for brown eyes is dominant and the allele for blue eyes is recessive.



Which people **must** be heterozygous for eye colour?

A 1, 3 and 7 **B** 2 and 6 **C** 3, 4 and 5 **D** 4 and 5 only

1 The diagram shows human skin.



(a) (i) Identify the parts labelled **B** and **C**.

B .....

C .....

[2]

(ii) Describe how part **B** is involved in maintaining a constant body temperature when the external temperature drops.

.....

.....

.....

.....

..... [3]

(b) Part **A** is unevenly distributed in human skin.

Name part **A** and suggest reasons for this uneven distribution.

.....

.....

.....

..... [3]

(c) The thickness of layer **D** can change over several months.

Suggest and explain **one** reason for a **decrease** in the thickness.

.....

.....

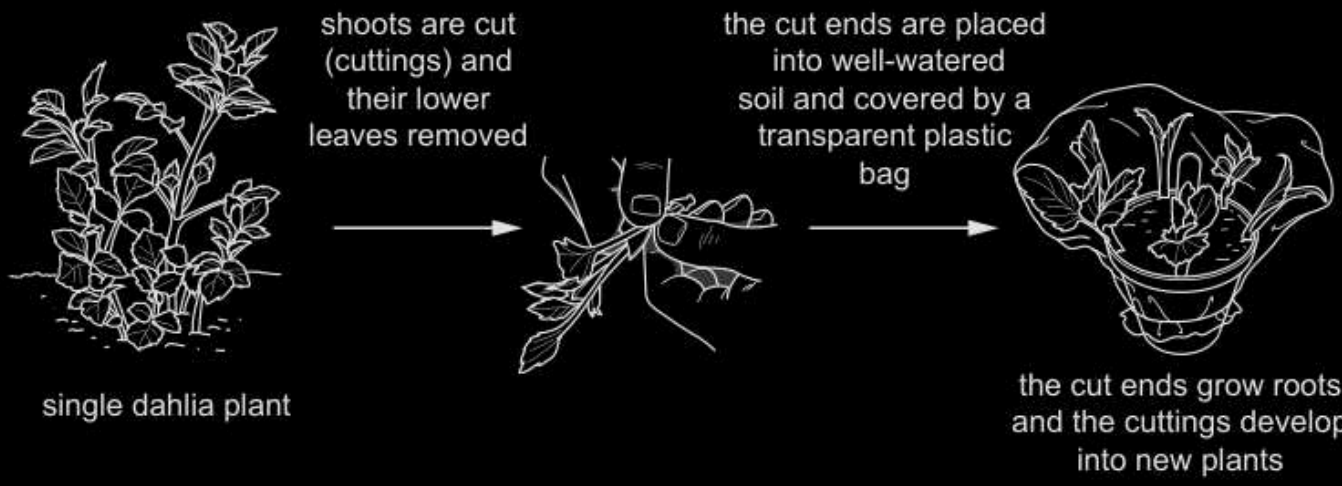
..... [2]

[Total: 10]

1(a)(i)	**B** = blood vessel / capillary / artery / arteriole ;	**1**	
	C = sweat gland ;	**1**	
1(a)(ii)	less blood flows **+** in blood vessel / close to skin surface ;	**1**	
	decrease in radiation / heat loss / **AW** ;	**1**	
	vasoconstriction **OR** blood vessels / arterioles / arteries / part **B +** narrow / constrict ;	**1**	
1(b)	*any **three** from:* **A** is a receptor / nerve ending ; touch / pressure / pain / temperature ; examples of uneven distribution e.g. high density in fingers / lips ; exposed areas **+** need higher sensitivity **AW** ;	**3**	
1(c)	*any **two** from:* fatty / adipose **+** tissue / cell ; less food eaten / less fat in diet / exercise; so fat used by body **+** for respiration / release energy ; body using more energy than it is taking in ;	**2**	

Question	Answer	Marks	Guidance

3 A gardener grows a species of dahlia, a flowering plant.
Each year the gardener chooses his best plants and takes cuttings to make new plants.



(a) (i) Removing the lower leaves and placing the cuttings in a transparent bag helps prevent them wilting until the roots develop.

Suggest and explain how these two steps help to prevent wilting.

removal of lower leaves .....

.....

.....

covering with transparent bag .....

.....

.....

[3]

(ii) Taking cuttings is a commercial application of asexual reproduction.

Define asexual reproduction.

.....

..... [2]

(b) The stems of the dahlia cuttings contain xylem vessels.

Describe **two** functions of xylem vessels in young stems of dahlias.
For each function explain **one** way in which xylem structure is adapted for this function.

function 1 .....

.....

adaptation .....

.....

function 2 .....

.....

adaptation .....

.....

[4]

(c) The gardener's dahlia cuttings grow into full-sized plants with flowers.

The gardener decides that he would like to produce a plant that combines the disease-resistant characteristics of one of these plants with the red petal colour of another.

With reference to named parts of the flowers, describe how he could achieve this.

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 14]

3(a)(i)	*removal of leaves:* reduction **+** surface area / stomata ;	**1**	
	covering with transparent bag: increase humidity ;	**1**	
	to reduce transpiration **OR** evaporation / water loss ;	**1**	
3(a)(ii)	*any **two** from:* production of genetically identical offspring / clones ; mitosis ; from one parent/ **AW**;	**2**	

Question	Answer	Marks	Guidance
3(b)	*function:* xylem vessels transport water and/or minerals ;	**1**	
	adaptation: long / thin / no end cell walls / hollow tubes / no cytoplasm ;	**1**	
	function: (mechanical) support / keeps stem upright ;	**1**	
	adaptation: lignified **OR** woody / strengthened / hard **+** walls ;	**1**	
3(c)	*any **five** from:* **1** artificial selection / selective breeding ; **2** transfer pollen / pollination ; **3** from anther / stamen of one plant **+** e.g. red petal colour ; **4** to stigma / carpel /pistil of another ; **5** fertilisation **OR** gametes / genetic material **+** combined **AW** ; **6** grow seeds ; **7** select the ones with best combination of characteristics ;	**5**	

Question	Answer	Marks	Guidance

4 The human liver contains a high concentration of enzymes called transaminases.
These enzymes are important in the metabolism of amino acids.
Amino acids are absorbed from the alimentary canal and carried to the liver by a blood vessel.

(a) Name the part of the alimentary canal where absorption occurs and the blood vessel that carries the amino acids to the liver.

part of alimentary canal .....

blood vessel .....
[2]

(b) There are different types of transaminase.
Each type has a specific active site.

Explain why this means that each type catalyses a specific reaction.

.....

.....

.....

.....

..... [3]

[Total: 5]

4(a)	ileum / small intestine ;	**1**	
	hepatic portal vein ;	**1**	
4(b)	*any **three** from:* lock and key ; shape **AW +** active site / enzyme / amino acids different shapes ; complementary / fits / matches ; binds **AW** to a substrate / enzyme-substrate complex / ES complex ; wrongly shaped substrates cannot fit **+** active site ;	**3**	

© UCLES 2022

Question	Answer	Marks	Guidance

6 There are many different types of virus that can infect animal cells.

(a) Compare the **structure** of a typical virus with a typical animal cell.

.....

.....

.....

.....

.....

..... [4]

(b) (i) Human immunodeficiency virus (HIV) is one type of virus.

Describe how HIV is transmitted.

.....

.....

.....

..... [3]

The table shows the numbers of people infected with HIV globally and the numbers receiving treatment in different years.

year	millions of people infected with HIV	millions of people receiving treatment for HIV
2000	24.9	0.77
2010	31.7	7.7
2015	35.6	17.0
2018	37.9	23.3

(ii) Use the information in the table to describe changes in the numbers of people infected and treated between the years 2000 and 2018.

.....

.....

.....

.....

..... [3]

[Total: 10]

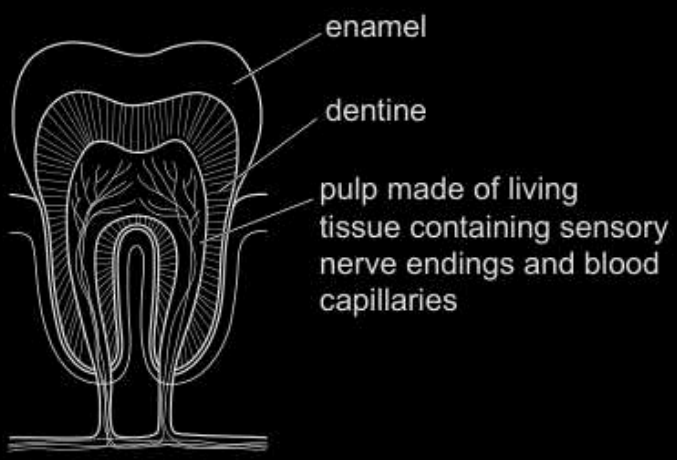
6(a)	*any **four** from:* virus is small<u>er</u> **ORA** ; both contain genetic material ; animal cell always DNA **+** virus DNA or RNA ; animal cell has (cell surface) membrane **ORA** ; virus has a <u>protein</u> coat / capsid **ORA** ; animal cell has a nucleus / nuclear membrane **ORA** ; animal cell contains cytoplasm / organelles / named organelle **ORA** ;	**4**	

© UCLES 2022

Question	Answer	Marks	Guidance
6(b)(i)	*any **three** from:* sexually transmitted ; blood transfer / blood transfusion **AW** ; sharing **AW +** needles / razors ; from mother to baby **+** across placenta / during pregnancy / during birth / breastfeeding ;	3	
6(b)(ii)	*any **three** from:* number infected increasing / more people infected in 2018 than 2000 ; rate of infection decreasing / slowing down ; correct data manipulation **+** data quoted to illustrate either point ; number treated for HIV increasing / more people treated in 2018 than 2000 ; a much bigger proportion of those infected being treated ; correct data manipulation **+** quoted to illustrate the point e.g. in 2000 less than 5% treated in 2000 and in 2018 over 60% treated ;	3	

Question	Answer	Marks	Guidance

7 The diagram shows a section through a molar tooth of a human.



(a) **(i)** State the function of a molar tooth.

.....

..... [1]

(ii) The pulp is a living tissue.

Describe the movement of molecules in the pulp between the blood capillaries and the cytoplasm of the surrounding cells.

.....

.....

.....

.....

..... [4]

(b) Toothache is caused by inflammation of the pulp tissue and is often a sign of dental decay.

Describe the cause of dental decay and how it can be prevented.

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

Section C

Answer **either** Question 8 **or** Question 9.

Write your answers in the spaces provided.

7(a)(i)	crushing /grinding / chewing / increasing surface area of food ;	**1**	
7(a)(ii)	*any **four** from:* blood under pressure ; capillary **+** thin wall /one cell thick ; soluble materials / plasma **+** squeezed / leaks out ; oxygen / glucose **+** from capillary / to cells ; carbon dioxide / urea **+** from cells / to capillary / blood ; diffusion ; across cell membrane / partially permeable **+** membrane ; additional named soluble substance e.g. amino acids **+** direction of movement ;	**4**	

© UCLES 2022

Question	Answer	Marks	Guidance
7(b)	*any **three** from:* bacteria ; bacteria **+** feed on **+** sugar / food ; produce acid ; acid **+** corrodes / destroys **AW +** enamel / dentine ;	**3**	
	*any **two** from:* brushing / cleaning / toothpaste ; flossing / mouthwash ; fluoride ; avoid sugar **AW** ; visit dentist ;	**2**	

Question	Answer	Marks	Guidance

8 The functions of the body are coordinated by nerve impulses and hormones.

(a) Compare these two types of coordination.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) A girl sees a vehicle travelling towards her at speed as she crosses a road. This triggers nerve impulses and the release of adrenaline in her body.

Describe the role of the nerve impulses and adrenaline in helping her to react to this situation.

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

[Total: 10]

8(a)	*any **four** from:* **1** both trigger responses ; **2** both work on specific effectors **AW** ; **3** nerve impulses fast **+** hormones slow ; **4** nerve impulses electrical **+** hormones chemical ; **5** nerve impulses short term effect **+** hormones longer term effect ; **6** nerve impulses travel along nerve cells / neurones **AW OR** hormones travel in blood / plasma ; **7** nerve impulses all the same / hormones are different chemicals ; **8** nerve impulses generated by receptors **AW** / hormones from glands ;	**4**	comparative statements required **accept** faster / slower etc.

© UCLES 2022

Question	Answer	Marks	Guidance
8(b)	*any **six** from:* eye <u>receptors</u> / retina / rods / cones ; impulse **+** sensory neurones ; via relay neurones ; to motor neurones ; to effector / muscle ; adrenaline **+** from adrenal glands ; travels in blood **+** to named specific target organ e.g. heart / lungs ; heart beats faster ; stimulates glycogen to be converted to glucose ; more glucose / oxygen / blood **+** to muscles ; air passageways increase in diameter ; blood diverted **AW +** to muscles ; pupils dilate ;	**6**	

Question	Answer	Marks	Guidance

1 The diagram shows the human brain and the table lists the functions of some parts of the human brain.

part of brain	function
A	memory storage and control of conscious behaviour
B	control of body temperature
C	control of balance and coordination
D	control of unconscious activities such as breathing

(a) Label on the diagram the parts of the brain identified in the table, using label lines and the letters **A** to **D only**. [4]

(b) **(i)** Name **one** structure in the skin that receives impulses from part **B** of the brain to control body temperature.

..... [1]

(ii) Outline how the action of this structure results in the control of body temperature.

.....

.....

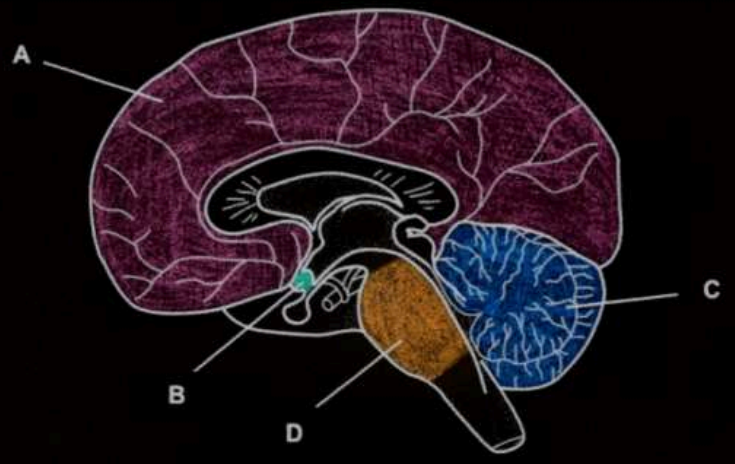
.....

.....

.....

..... [3]

[Total: 8]

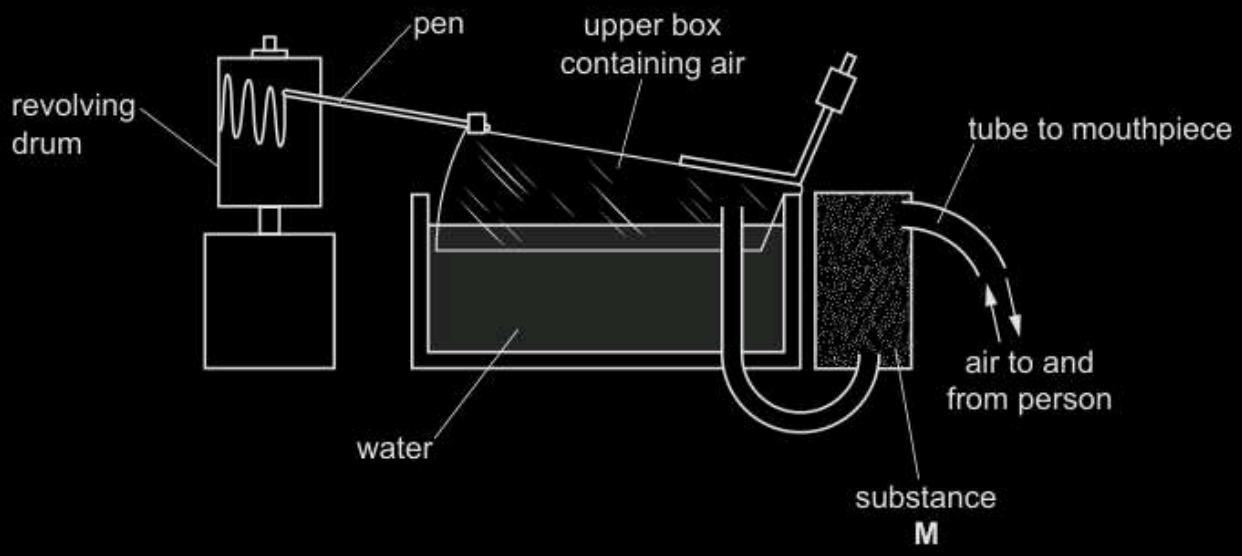
1(a)	one mark for each correct label ;;;;  	4	
1(b)(i)	**1** erector / hair **+** muscle ; **2** <u>sweat gland</u> ; **3** <u>smooth muscle</u> ; **4** blood vessel / artery / arteriole ;	1	**4 Reject** capillary

© UCLES 2022

Question	Answer	Marks	Guidance
1(b)(ii)	(erector muscle) **1** contracts ; **2** raises hair ; **3** traps air / insulation ; **4** reduces heat loss ; (sweat gland) **5** produces / secretes / releases ; **6** sweat / perspiration ; **7** evaporates ; **8** cools ; (smooth muscle / blood vessel / artery / arteriole) **9** contracts ; **10** constricts / narrows / <u>vasoconstriction</u> ; **11** less blood to **+** capillaries / surface of skin ; **12** reduces heat loss ;	3	**Allow** reverse argument for each marking point

Question	Answer	Marks	Guidance

2 The diagram shows apparatus used to study breathing.



(a) (i) Describe what will happen to the upper box containing air as a person takes **one** breath **in** from the apparatus.

..... [1]

(ii) State the percentage of oxygen in atmospheric air.

..... %

Explain how the percentage of **oxygen** in the air in the upper box will change as a person breathes in **and** out through the apparatus several times.

.....

.....

.....

.....

[3]

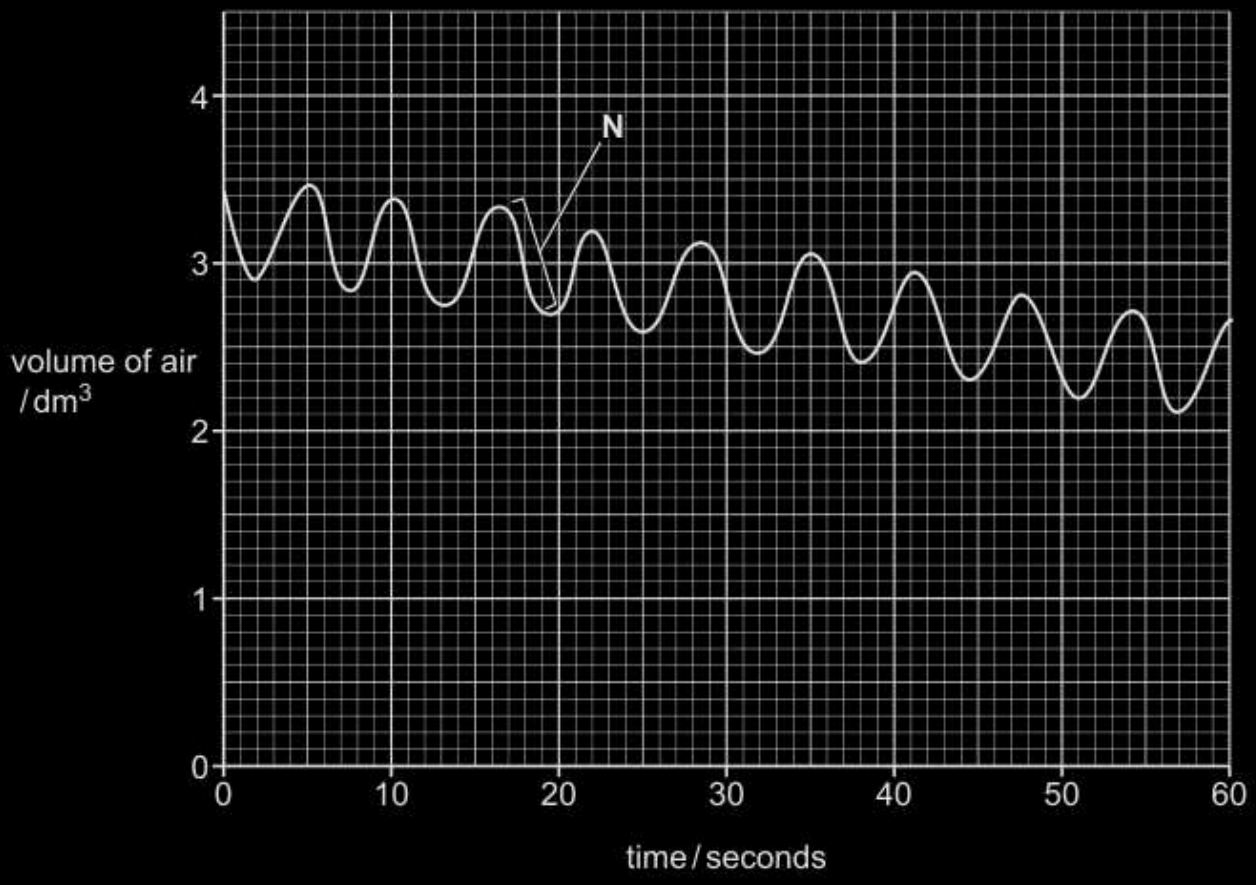
(iii) Substance **M** is present in the apparatus to absorb an excretory product from expired air.

Name this excretory product.

..... [1]

(b) The diagram shows a chart recorded by the pen on the revolving drum.

This chart was recorded for a period of one minute by a person at rest.



(i) Explain how the action of one **named** muscle caused the part of the chart labelled **N**.

.....

.....

.....

.....

.....

..... [3]

(ii) A second chart was recorded for a period of one minute by the same person after vigorous exercise.

State how an increased **rate** of breathing after exercise would be shown on this second chart.

.....

..... [1]

[Total: 9]

2(a)(i)	falls / drops / goes down / lowers / shrinks / collapses **AW** ;	**1**	
2(a)(ii)	**1** any number between 20 and 21 ; (any two from) **2** decreases **AW** ; **3** into blood ; **4** aerobic **+** respiration ;	**3**	
2(a)(iii)	carbon dioxide / CO_2 ;	**1**	**Allow** water / H_2O / water vapour

© UCLES 2022

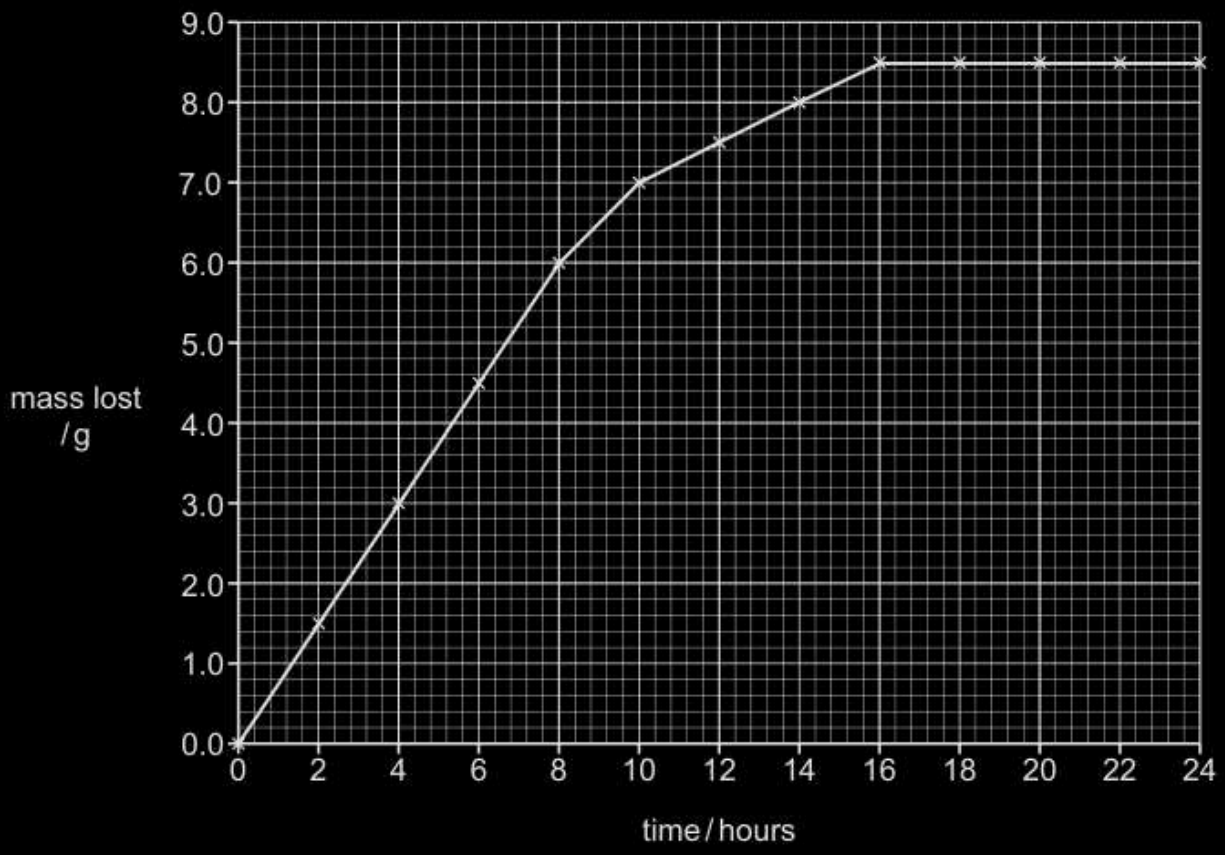
Question	Answer	Marks	Guidance
2(b)(i)	**1** diaphragm / external intercostal muscle **+** contracts ; **2** diaphragm **+** down / flattens **OR** ribs move **+** up / out ; **3** increase volume / decrease pressure **+** thorax / lungs ; **4** air breathed in / air into lungs / inhale ; **5** upper box falls / pen moves down ;	**3**	**2 to 5 Reject** reverse statements
2(b)(ii)	more **+** peaks / troughs / waves **OR** peaks / troughs / waves **+** closer together **OR** shorter wavelength / increased frequency of waves **AW** ;	**1**	

Question	Answer	Marks	Guidance

3 A student carried out two investigations into the action of yeast.

(a) In the first investigation, the student dissolved 20.0 g of glucose in 100 cm^3 of water in a beaker. The student then added 3.5 g of yeast and used a balance to measure the loss in mass from the beaker over the next 24 hours.

The results of this investigation are shown in the graph.



(i) Calculate the rate at which mass was lost during the first 6 hours.

Space for working.

..... g per hour [2]

(ii) Explain why mass was lost during the investigation.

.....

.....

.....

.....

..... [3]

(iii) The student then repeated this first investigation changing **only** the mass of yeast.

Draw a line on the graph on page 6 to show the pattern of results you would expect the student to obtain when using 7.0 g of yeast. [2]

(b) In a second investigation, the student prepared three samples of bread dough.

Each sample of dough had a volume of 50 cm^3 and contained:

- 1 g of yeast
- 25 cm^3 of water
- 40 g of flour.

(i) Starch molecules in the flour are used to provide the yeast with a source of glucose.

Describe how the action of a **named** chemical, produced by yeast cells, makes glucose available from starch.

.....

.....

.....

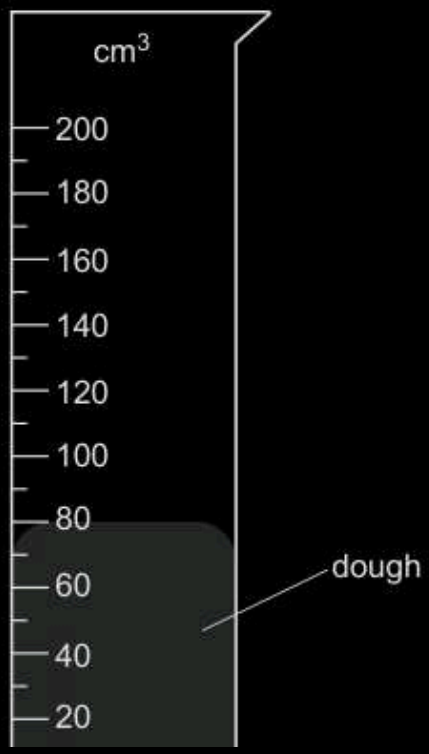
..... [2]

Each sample of dough was placed at the bottom of a 200 cm^3 measuring cylinder.

The three measuring cylinders were then placed in water-baths at different temperatures for 60 minutes. The temperatures chosen were 20 °C, 40 °C and 80 °C.

After this time, the student measured the volume of dough in each measuring cylinder.

The diagram shows the volume of dough in the measuring cylinder from the water-bath at a temperature of 20 °C at the end of the investigation.



(ii) Estimate the volumes of the dough at the end of the investigation in the other two measuring cylinders, from water-baths at temperatures of:

40 °C ..... cm^3

80 °C ..... cm^3

[2]

(iii) Explain the results of this investigation.

.....

.....

.....

.....

.....

..... [3]

[Total: 14]

3(a)(i)	4.5 divided by 6 ; 0.75 ;	**2**	
3(a)(ii)	**1** respiration / fermentation ; **2** carbon dioxide ; **3** released / lost / escapes **+** from beaker / into air **AW** ;	**3**	
3(a)(iii)	**1** increase line to left of that printed **+** starting at (0,0) ; **2** levels off at 8.5 (± half a small square) max mass lost ;	**2**	
3(b)(i)	**1** amylase / maltase ; **2** digests / hydrolyses / breaks down **AW** ;	**2**	
3(b)(ii)	(40 °C) any number between 81 and 200 ; (80 °C) any number between 50 and 79 ;	**2**	

© UCLES 2022

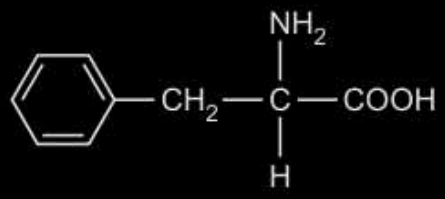
Question	Answer	Marks	Guidance
3(b)(iii)	**1** gas / carbon dioxide **+** produced ; **2** enzyme ; **3** 20 °C **+** less kinetic energy / fewer collisions ; **4** 40 °C **+** more kinetic energy / more collisions / optimum **AW** ; **5** 80 °C **+** denatured / active site changes shape / substrate not fit active site / yeast killed ;	**3**	

Question	Answer	Marks	Guidance

4 There are many different types of amino acid.

One of these is phenylalanine.

The diagram shows the structure of a molecule of phenylalanine.



(a) Phenylalanine is metabolised by an enzyme with the name PAH.

(i) Name the human organ that metabolises amino acids such as phenylalanine.

..... [1]

(ii) Explain how the shape of a PAH enzyme molecule is important to enable phenylalanine to be metabolised.

.....

.....

.....

.....

.....

..... [3]

(b) A rare gene mutation reduces the activity of the PAH enzyme. The mutation results in an increased concentration of phenylalanine in the blood. This condition can cause damage to the brain.

(i) Describe what is meant by the term *gene mutation*.

.....

.....

.....

..... [2]

(ii) Explain how a person with the condition should change their diet to reduce the risk of damage to the brain.

.....

.....

.....

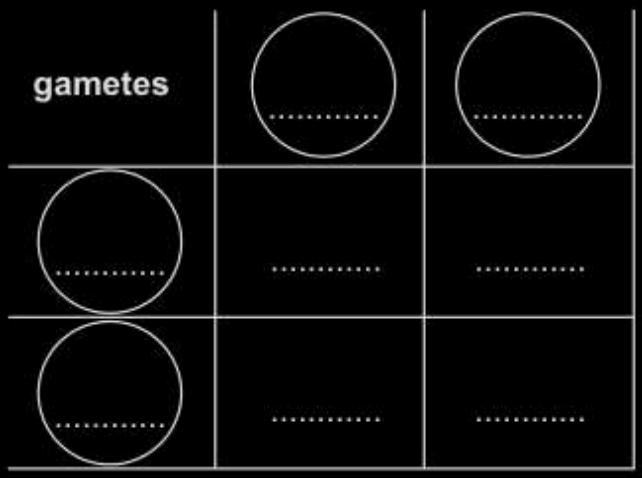
.....

.....

..... [3]

(iii) The gene mutation produces a recessive allele, **h**. A person with a heterozygous genotype is described as a carrier. A carrier will **not** show symptoms of the condition.

Complete the genetic diagram to show the probability of two carriers reproducing to have offspring with the condition.



probability of offspring with the condition .....

[3]

[Total: 12]

4(a)(i)	liver ;	1	
4(a)(ii)	**1** active site ; **2** complementary / specific / lock and key / fits ; **3** binds / bonds to / combines with **AW** ; **4** enzyme-substrate complex / ES complex ;	3	
4(b)(i)	**1** change / alter / modify ; **2** gene structure / DNA structure / base sequence / nucleotide sequence ;	2	
4(b)(ii)	**1** eat less protein / eat food low in phenylalanine ; **2** eat less named protein-containing food ; **3** proteins **+** digested to / broken down to / made of **+** amino acids ;	3	
4(b)(iii)	**1** (gametes) H **+** h **AND** H **+** h ; **2** (offspring) HH **+** Hh **+** Hh **+** hh / correct genotypes for gametes ; **3** (probability) 25% / ¼ / 1 in 4 / 1:3 / 0.25 ;	3	

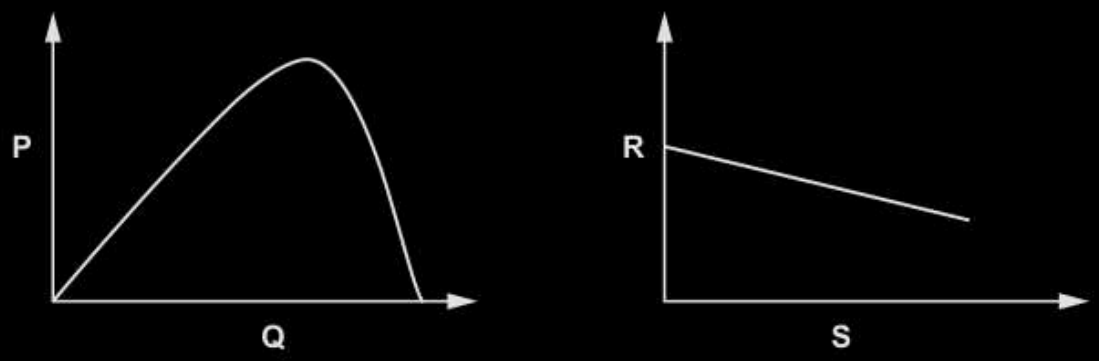
© UCLES 2022

Question	Answer	Marks	Guidance

5 **(a)** Biological processes in plants are affected by environmental factors.

The graphs each show the effect of changing one environmental factor on a biological process.

The labels on the axes for each graph have been replaced with the letters **P**, **Q**, **R** and **S**.



Identify the correct axis label for each of **P**, **Q**, **R** and **S**.

Select from the following possible labels.

temperature **carbon dioxide concentration** **rate of transpiration**

light intensity **rate of photosynthesis** **air humidity**

P .....

Q .....

R .....

S .....

[4]

(b) Describe the process of transpiration in the leaf of a plant.

.....

.....

.....

..... [3]

[Total: 7]

Section B

Answer **both** questions in this section.

Write your answers in the spaces provided.

5(a)	(**P**) rate of photosynthesis ; (**Q**) temperature ; (**R**) rate of transpiration ; (**S**) air humidity ;	**4**	
5(b)	**1** evaporation ; **2** surface of / cell wall / film of water / layer of water **+** mesophyll cells ; **3** water vapour ; **4** air spaces / intercellular spaces ; **5** diffusion / high to low concentration / down concentration gradient ; **6** water **+** through stomata ;	**3**	

Question	Answer	Marks	Guidance

8 **(a)** Compare the pulmonary artery and the vena cava in terms of their structure and function.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) Explain the role of platelets and the importance of this role.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

8(a)	**1** both carry deoxygenated blood ; **2** blood in pulmonary artery from + heart / right ventricle ; **3** blood in vena cava from body ; **4** blood in pulmonary artery to lungs ; **5** blood in vena cava to + heart / right atrium ; (any two from) 3 **6** vena cava + vein ; **7** vena cava + valve / valves ; **8** pulmonary artery + thick wall / muscular wall / elastic wall ; **9** pulmonary artery + narrow lumen / small lumen ; **10** pulmonary artery + high pressure ;	**6**	**7 Allow** reverse for pulmonary artery **8 9 10 Allow** reverse for vena cava
8(b)	**1** blood + clotting ; **2** fibrinogen + to fibrin ; **3** soluble + to insoluble **OR** forms + threads / a mesh ; **4** prevents + blood loss **AW** ; **5** prevents + infection / entry of pathogens **AW** / entry of named type of pathogen ;	**4**	

Question	Answer	Marks	Guidance

9 **(a)** Compare the ureter and the urethra in terms of their functions in adult humans.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Explain how the process of dialysis maintains the correct chemical composition of the blood for a person with kidney disease.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

[Total: 10]

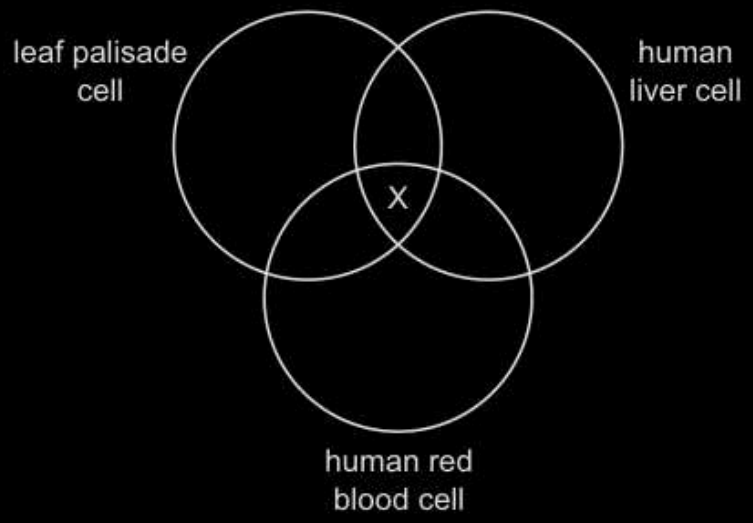
9(a)	1	both carry + urine / urea ;	4	
	2	ureter from kidney ;		
	3	ureter to bladder ;		
	4	urethra from bladder		(any two from)
	5	urethra out of body ;		
	6	male / penis / testis + urethra carries + semen / sperm ;		
9(b)	1	selectively / partially / semi + permeable ;	6	
	2	large surface area ;		
	3	fluid / dialysate ;		
	4	counter-current / continuously changed ;		
	5	urea / toxins / poisons / harmful substances + from blood / to fluid ;		
	6	water / salts / ions + between blood and fluid ;		
	7	diffusion / osmosis ;		
	8	high to low concentration / down concentration gradient ;		
	9	large molecules / proteins + stay in blood / do not enter fluid ;		

© UCLES 2022

October/November 2021 · Paper 1 · Variant 1 · Question 1

Correct answer: C

1 The diagram represents the cell structures of a human liver cell, a leaf palisade cell and a human red blood cell.



Which cell structure is X?

A cell wall

B chloroplast

C cytoplasm

D nucleus

October/November 2021 · Paper 1 · Variant 1 · Question 2

Correct answer: A

2 Which process occurs by diffusion alone?

A movement of carbon dioxide into the palisade cells in a leaf

B uptake of magnesium ions by root hair cells

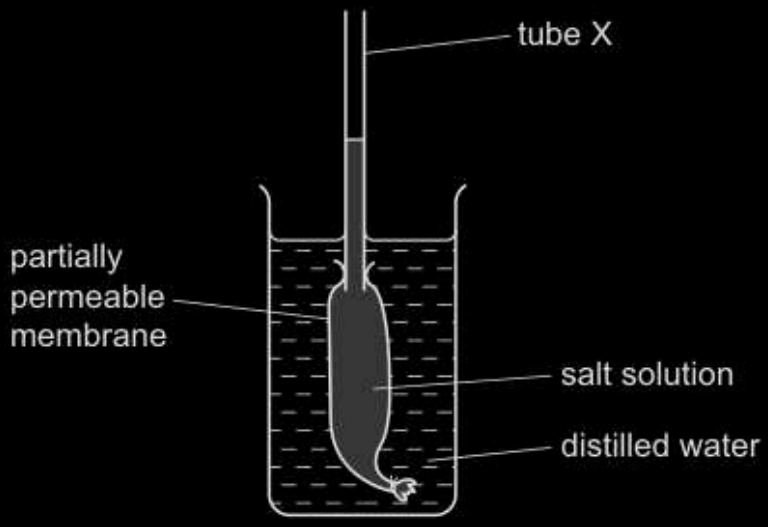
C movement of food from the stomach to the duodenum

D transfer of glucose molecules from the ileum into the blood

October/November 2021 · Paper 1 · Variant 1 · Question 3

Correct answer: C

3 An experiment to investigate osmosis is set up as shown.



What happens to the height of the liquid in tube X and the concentration of the salt solution?

	height of the liquid in tube X	concentration of the salt solution
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

October/November 2021 · Paper 1 · Variant 1 · Question 4

Correct answer: A

4 The diagrams show an enzyme molecule and some substrate molecules.

enzyme molecule

Which diagram shows the substrate molecule for this enzyme?

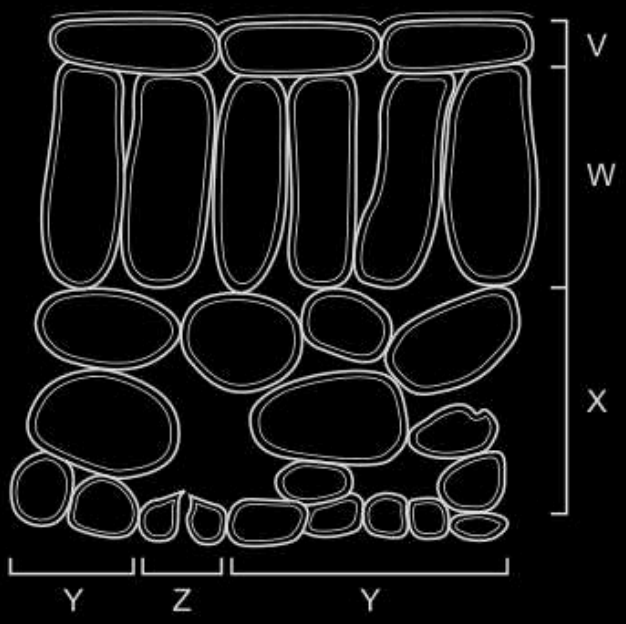
A **B** **C** **D**

October/November 2021 · Paper 1 · Variant 1 · Question 6

Correct answer: D

6 The diagram shows the arrangement of cells inside a green leaf. Different types of cells are indicated by the brackets.

No cell contents are shown.



Which types of cells contain chloroplasts?

A V, W and X **B** V, W and Y **C** W, X and Y **D** W, X and Z

October/November 2021 · Paper 1 · Variant 1 · Question 7

Correct answer: D

7 Why do newly germinated seeds fail to grow into healthy plants if they lack magnesium ions?

A Magnesium ions are a necessary component of all proteins.

B Magnesium ions are needed to convert chlorophyll to starch.

C Magnesium ions are needed to form cell walls.

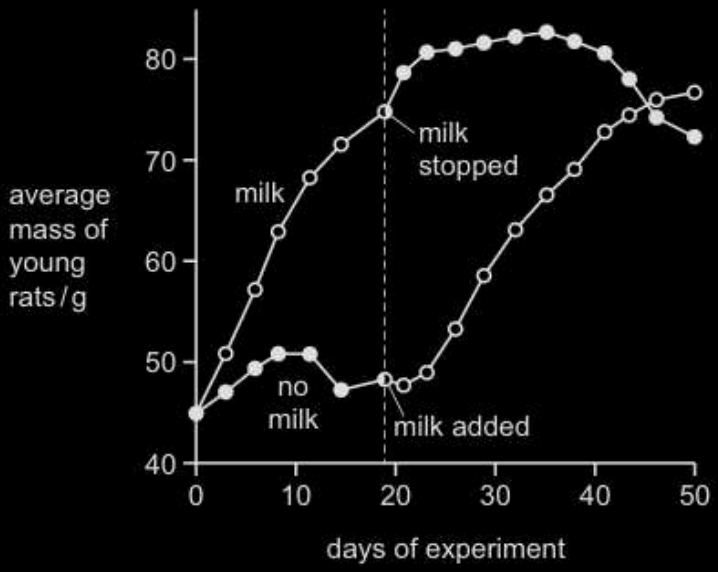
D Magnesium ions are needed to form chlorophyll molecules.

October/November 2021 · Paper 1 · Variant 1 · Question 8

Correct answer: B

8 Milk contains water, carbohydrates, proteins, fats, and some minerals and vitamins.

The graph shows the results of an experiment. One group of young rats was given 3 cm^3 of milk per day for 18 days. On day 18, the milk was stopped for this group but given to a second group of young rats. All other factors were kept constant for the two groups of rats.



What is the explanation for these results?

A Whether or not young rats are given milk, their mass is always the same on day 46.

B Milk contains the nutrients needed for young rats to grow.

C Milk is not important for the growth of young rats.

D Young rats grow fastest after day 18.

October/November 2021 · Paper 1 · Variant 1 · Question 9

Correct answer: C

9 Which statement about chemical digestion in the human alimentary canal is correct?

A Digestion of carbohydrates is completed in the colon.

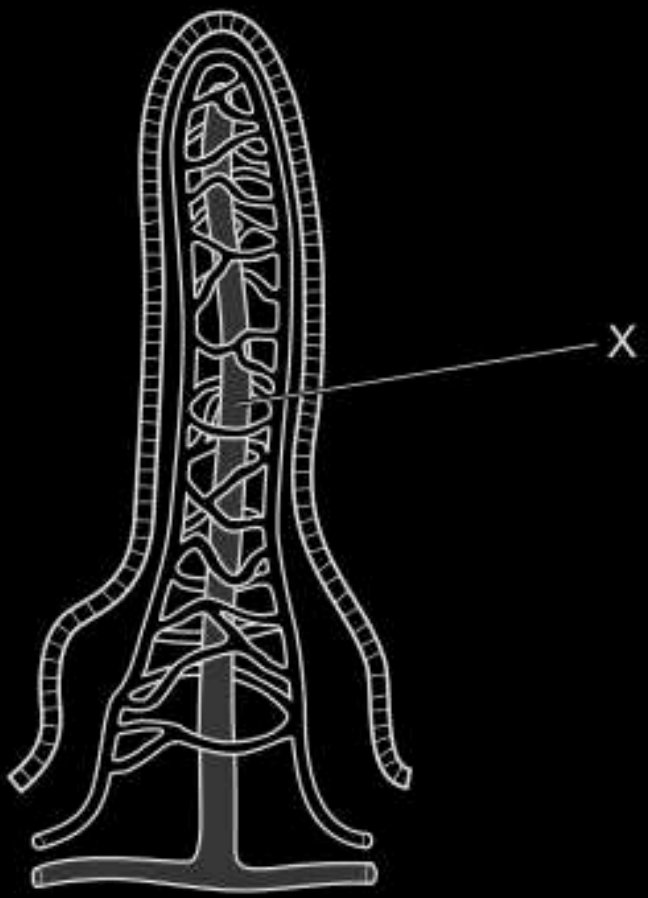
B Enzymes are secreted to break down cellulose in the duodenum.

C Protein digestion is completed in the ileum.

D The stomach secretes enzymes to break down starch.

Correct answer: C

10 The diagram shows a section through a villus.



Which products of digestion are mainly absorbed by vessel X?

A amino acids and glucose

B glucose and fatty acids

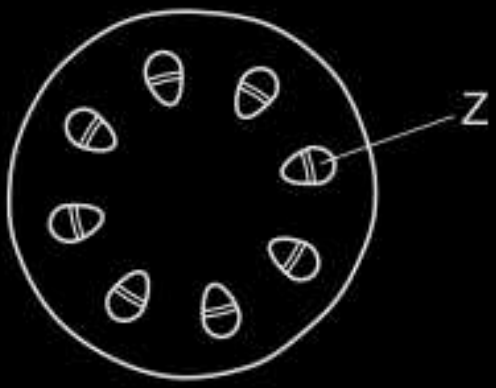
C fatty acids and glycerol

D glycerol and amino acids

October/November 2021 · Paper 1 · Variant 1 · Question 11

Correct answer: C

11 The diagram shows a section through a plant stem.



Which process is responsible for moving substances in region Z?

A diffusion

B osmosis

C translocation

D transpiration

October/November 2021 · Paper 1 · Variant 1 · Question 12

Correct answer: B

12 Which statement describes why a leaf wilts?

A Less water is absorbed by the root hairs when the stomata are closed.

B Less water is absorbed by the root hairs than is lost through the stomata.

C More water is absorbed by the root hairs when the stomata are closed.

D More water is absorbed by the root hairs than is lost through the stomata.

October/November 2021 · Paper 1 · Variant 1 · Question 13

Correct answer: D

13 How do veins differ from arteries?

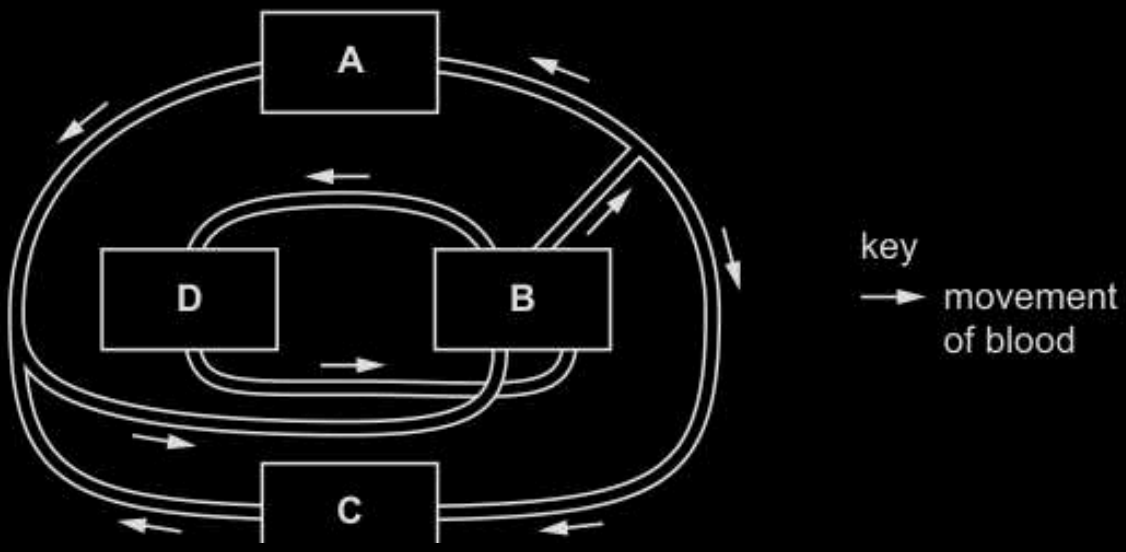
	width of lumen in veins	wall thickness of veins	elastic fibres	muscles in wall
A	narrower	thicker	more	fewer
B	narrower	thinner	fewer	more
C	wider	thicker	more	more
D	wider	thinner	fewer	fewer

October/November 2021 · Paper 1 · Variant 1 · Question 14

Correct answer: B

14 The diagram shows a simplified human circulatory system.

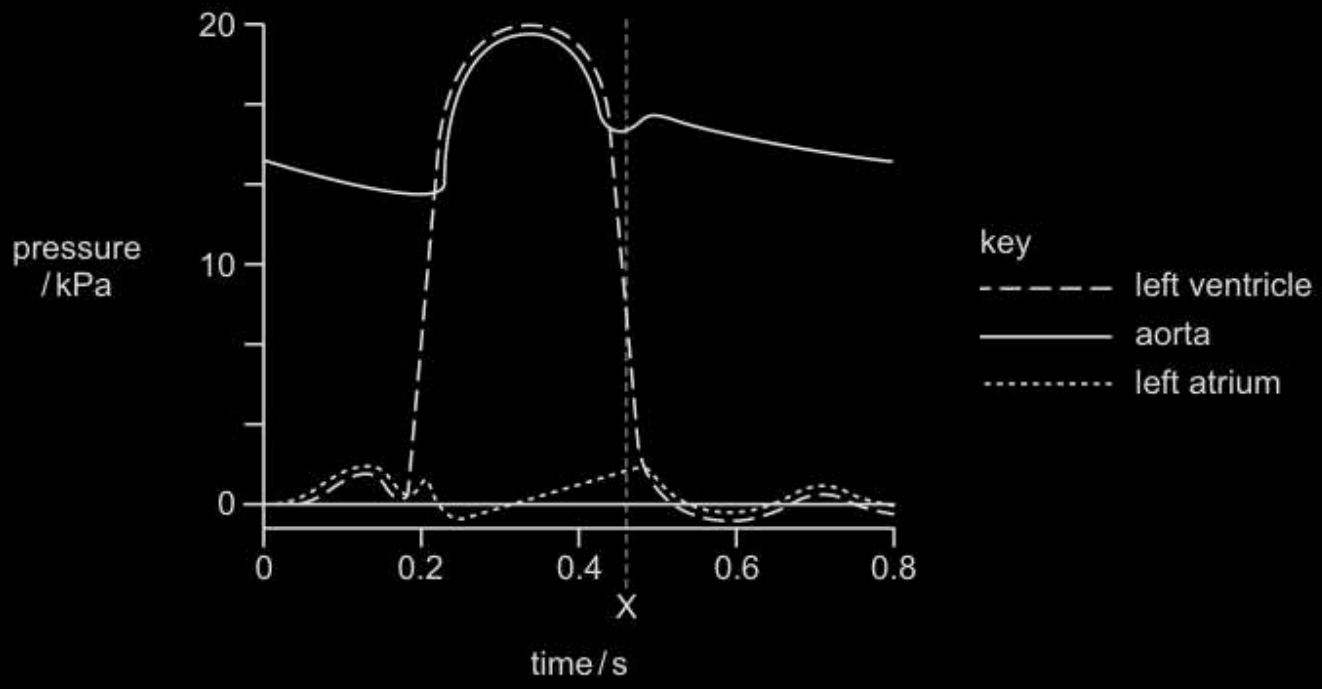
Which structure represents the heart?



October/November 2021 · Paper 1 · Variant 1 · Question 15

Correct answer: A

15 The diagram shows the pressures in the left side of the heart during one heartbeat.



Which valves are open and which are closed at the time marked X?

	bicuspid	semi-lunar
A	closed	closed
B	closed	open
C	open	closed
D	open	open

October/November 2021 · Paper 1 · Variant 1 · Question 16

Correct answer: B

16 Energy is needed for many processes within the body.

Which process does **not** require the use of energy?

A contraction of muscles in the intestines

B movement of oxygen from the alveoli into the blood

C generation of nerve impulses

D protein synthesis in cells

October/November 2021 · Paper 1 · Variant 1 · Question 17

Correct answer: D

17 What are features of the alveoli?

	lined by cilia	lined with moisture	walls are one cell thick
A	✓	✗	✓
B	✗	✓	✗
C	✓	✗	✗
D	✗	✓	✓

key
✓ = yes
✗ = no

October/November 2021 · Paper 1 · Variant 1 · Question 19

Correct answer: C

19 During kidney dialysis, which blood components pass from the blood into the dialysis fluid?

A platelets and proteins

B proteins and salts

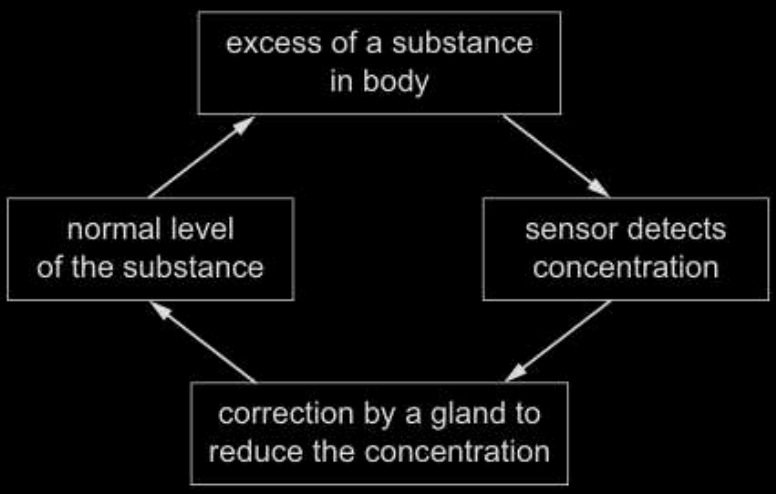
C salts and urea

D urea and platelets

October/November 2021 · Paper 1 · Variant 1 · Question 20

Correct answer: C

20 Which biological process is represented in the diagram?



A antagonistic action

B dialysis

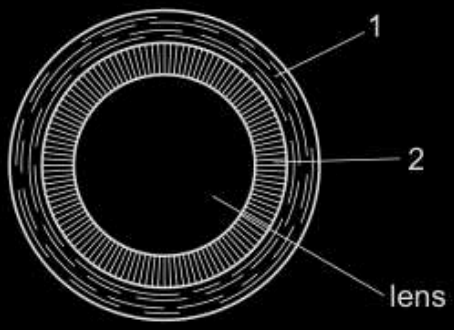
C negative feedback

D reflex action

October/November 2021 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 The diagram shows a front view of the parts of the eye involved in producing a focused image of near and distant objects.



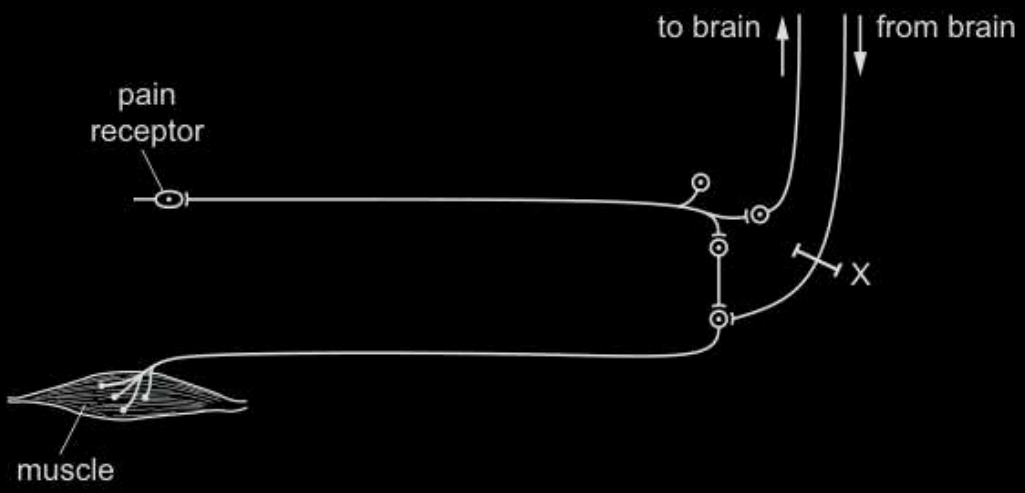
What are structures 1 and 2?

	structure 1	structure 2
A	ciliary muscles	suspensory ligaments
B	iris	pupil
C	pupil	iris
D	suspensory ligaments	ciliary muscles

October/November 2021 · Paper 1 · Variant 1 · Question 22

Correct answer: B

22 The diagram shows some of the nerve pathways associated with a reflex action.



If the pathway at X is damaged, how does this affect the reflex?

A The person will not be aware that the reflex is occurring.

B The reflex cannot be controlled consciously.

C The response will occur without any stimulus.

D There is no response to the stimulus.

October/November 2021 · Paper 1 · Variant 1 · Question 23

Correct answer: D

23 Which statements about hormones are correct?

1 Insulin is made in the pancreas and increases the concentration of glucose in the blood.

2 Adrenaline lowers the amount of glucose in the blood after a meal.

3 Insulin is used to treat people with diabetes mellitus.

A 1, 2 and 3 **B** 1 and 2 only **C** 2 only **D** 3 only

October/November 2021 · Paper 1 · Variant 1 · Question 26

Correct answer: D

26 What is a feature of all bacteria?

A They are parasites.

B They have a nucleus.

C They are made of hyphae.

D They are single-celled organisms.

October/November 2021 · Paper 1 · Variant 1 · Question 27

Correct answer: A

27 Yeast is used in alcohol production.

What is the other product of this process?

A carbon dioxide

B oxygen

C glucose

D water

October/November 2021 · Paper 1 · Variant 1 · Question 33

Correct answer: D

33 Which statement is correct about the offspring produced by asexual reproduction?

A They are genetically identical to both their parents.

B They are only produced by female parent plants.

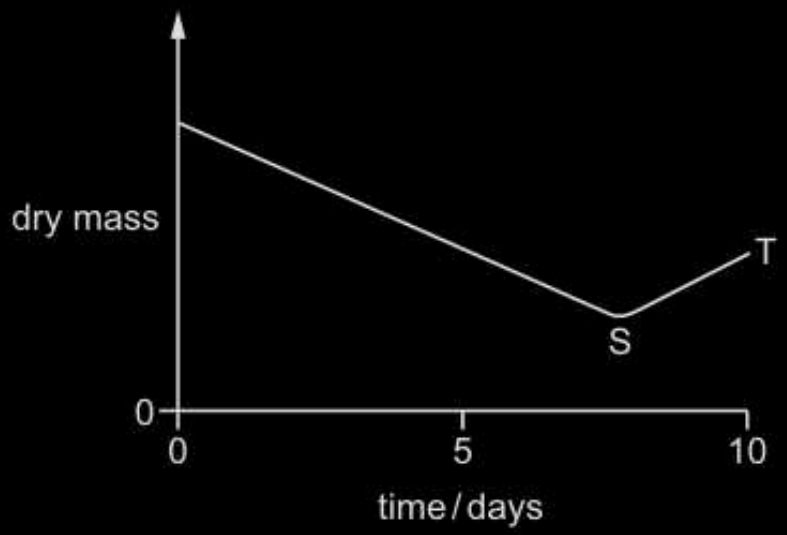
C They are produced by the formation of a zygote.

D They have identical chromosomes to their parent.

October/November 2021 · Paper 1 · Variant 1 · Question 34

Correct answer: B

34 The graph shows changes in the dry mass of a seed as it germinates and grows.



What causes the change shown between points S and T?

A osmosis

B photosynthesis

C respiration

D transpiration

October/November 2021 · Paper 1 · Variant 1 · Question 37

Correct answer: B

37 Some statements about chromosomes are listed.

1 A chromosome includes a long molecule of DNA.

2 Genes are found in chromosomes.

3 In cell division, the chromosome number is kept the same by meiosis.

4 The normal human chromosome number is 46.

Which statements are correct?

A 1, 2, 3 and 4

B 1, 2 and 4 only

C 2 and 3 only

D 3 and 4 only

October/November 2021 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 One gene has two codominant alleles, A^E and A^F, and one recessive allele, A^G.

How many different genotypes and phenotypes are possible?

	genotypes	phenotypes
A	3	3
B	4	6
C	6	4
D	6	6

October/November 2021 · Paper 1 · Variant 1 · Question 39

Correct answer: C

39 What is a result of natural selection?

A dogs that are friendly to humans

B grapes that contain no seeds

C mosquitoes that are resistant to insecticides

D onion crops that have a pleasant taste

October/November 2021 · Paper 1 · Variant 2 · Question 1

Correct answer: B

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

October/November 2021 · Paper 1 · Variant 2 · Question 2

Correct answer: D

2 Which statement about diffusion is correct?

A Diffusion always involves the movement of water molecules.

B Diffusion depends on the presence of a partially permeable membrane.

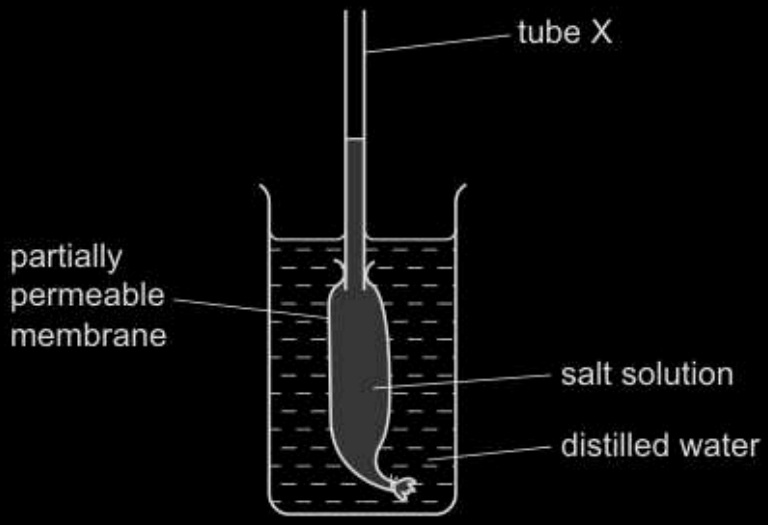
C Diffusion only happens to molecules in a solution.

D Diffusion only involves molecules moving down a concentration gradient.

October/November 2021 · Paper 1 · Variant 2 · Question 3

Correct answer: C

3 An experiment to investigate osmosis is set up as shown.



What happens to the height of the liquid in tube X and the concentration of the salt solution?

	height of the liquid in tube X	concentration of the salt solution
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

October/November 2021 · Paper 1 · Variant 2 · Question 4

Correct answer: B

4 Which statement about enzymes is always correct?

A They change reactions to produce different end products.

B They change the rate of reactions.

C They catalyse reactions inside body cells.

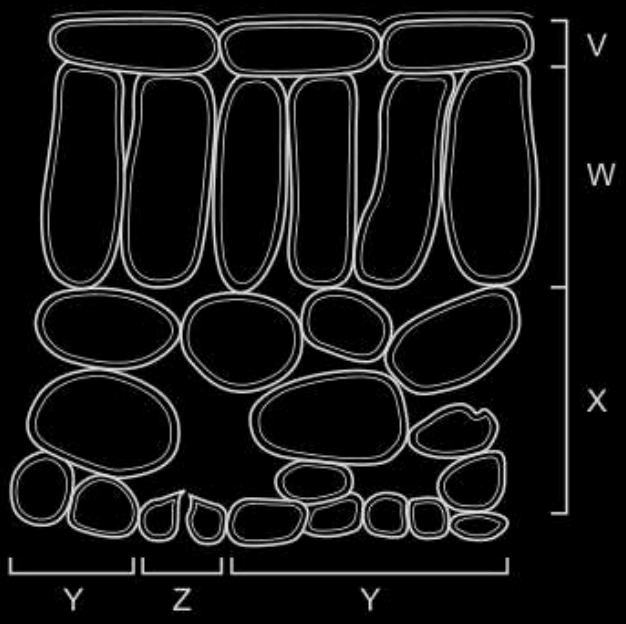
D They catalyse reactions at body temperature.

October/November 2021 · Paper 1 · Variant 2 · Question 6

Correct answer: D

6 The diagram shows the arrangement of cells inside a green leaf. Different types of cells are indicated by the brackets.

No cell contents are shown.



Which types of cells contain chloroplasts?

A V, W and X **B** V, W and Y **C** W, X and Y **D** W, X and Z

October/November 2021 · Paper 1 · Variant 2 · Question 7

Correct answer: C

7 What will happen to a plant that does **not** receive enough magnesium ions from the soil?

A It will have elongated stems and yellow leaves.

B It will have elongated stems and green leaves.

C It will have stunted growth and yellow leaves.

D It will have stunted growth and green leaves.

October/November 2021 · Paper 1 · Variant 2 · Question 8

Correct answer: A

8 The table shows the colours obtained after testing three different foods for carbohydrates, fats and proteins.

test	food X	food Y	food Z
Benedict's	red	blue	blue
iodine	black	brown	brown
biuret	blue	violet	blue
ethanol emulsion	clear	clear	milky white

Which nutrients are present in foods X, Y and Z?

	food X	food Y	food Z
A	carbohydrates	proteins	fats
B	fats	proteins	carbohydrates
C	carbohydrates	fats	proteins
D	proteins	carbohydrates	fats

October/November 2021 · Paper 1 · Variant 2 · Question 9

Correct answer: C

9 Which statement about chemical digestion in the human alimentary canal is correct?

A Digestion of carbohydrates is completed in the colon.

B Enzymes are secreted to break down cellulose in the duodenum.

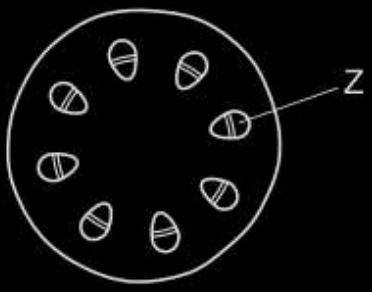
C Protein digestion is completed in the ileum.

D The stomach secretes enzymes to break down starch.

October/November 2021 · Paper 1 · Variant 2 · Question 11

Correct answer: C

11 The diagram shows a section through a plant stem.



Which process is responsible for moving substances in region Z?

A diffusion

B osmosis

C translocation

D transpiration

October/November 2021 · Paper 1 · Variant 2 · Question 12

Correct answer: B

12 What is transpiration?

A evaporation of water from the stomata

B evaporation of water at the surface of mesophyll cells and its loss through the stomata

C the movement of water through the cuticle

D the movement of water in the xylem from the roots to the leaves

October/November 2021 · Paper 1 · Variant 2 · Question 13

Correct answer: D

13 How do veins differ from arteries?

	width of lumen in veins	wall thickness of veins	elastic fibres	muscles in wall
A	narrower	thicker	more	fewer
B	narrower	thinner	fewer	more
C	wider	thicker	more	more
D	wider	thinner	fewer	fewer

October/November 2021 · Paper 1 · Variant 2 · Question 14

Correct answer: D

14 Sickle cell anaemia is an inherited condition which causes abnormal red blood cells.

Which health problem will be caused by this change in the red blood cells?

A Blood will not be able to clot when bleeding occurs.

B Blood cells will not be able to form antibodies when needed.

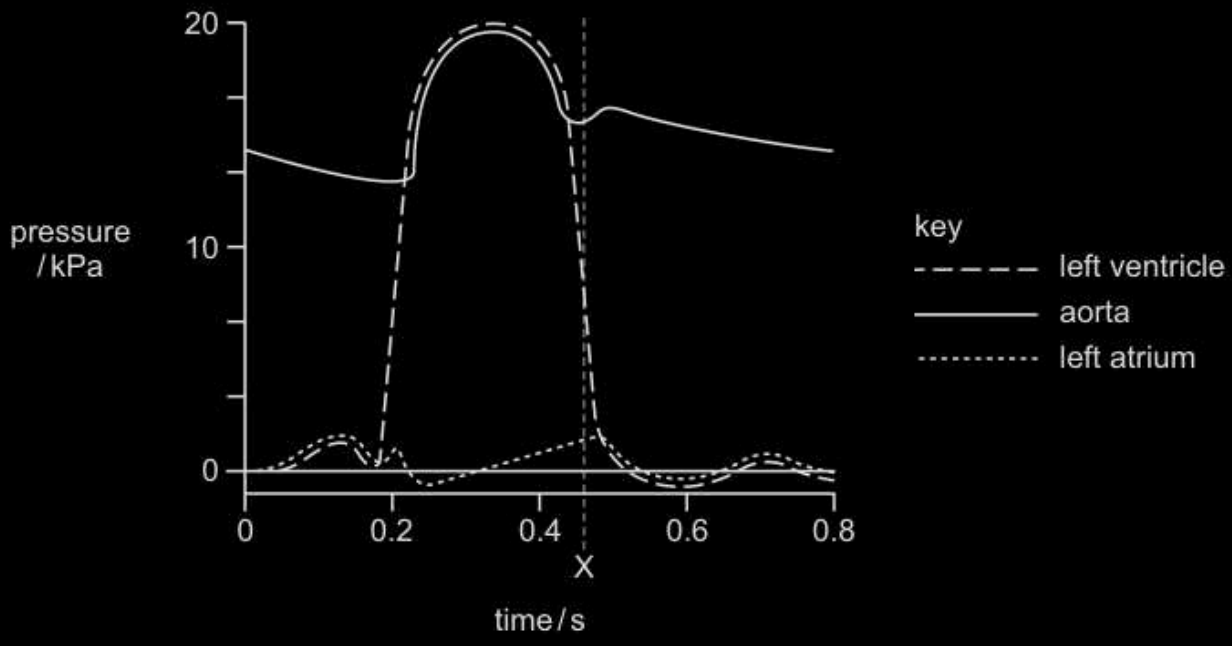
C Glucose levels in the blood plasma will become too low.

D Blood will not be able to carry the normal amount of oxygen.

October/November 2021 · Paper 1 · Variant 2 · Question 15

Correct answer: A

15 The diagram shows the pressures in the left side of the heart during one heartbeat.



Which valves are open and which are closed at the time marked X?

	bicuspid	semi-lunar
A	closed	closed
B	closed	open
C	open	closed
D	open	open

October/November 2021 · Paper 1 · Variant 2 · Question 16

Correct answer: B

16 What are uses of energy from respiration?

	active transport	diffusion	cell division	osmosis
A	✓	✓	✗	✓
B	✓	✗	✓	✗
C	✗	✗	✓	✗
D	✗	✓	✗	✓

key
✓ = yes
✗ = no

October/November 2021 · Paper 1 · Variant 2 · Question 17

Correct answer: D

17 Which statements about gaseous exchange in the alveoli are correct?

1 The alveolus wall is one cell thick.

2 Carbon dioxide can diffuse through the alveolus wall.

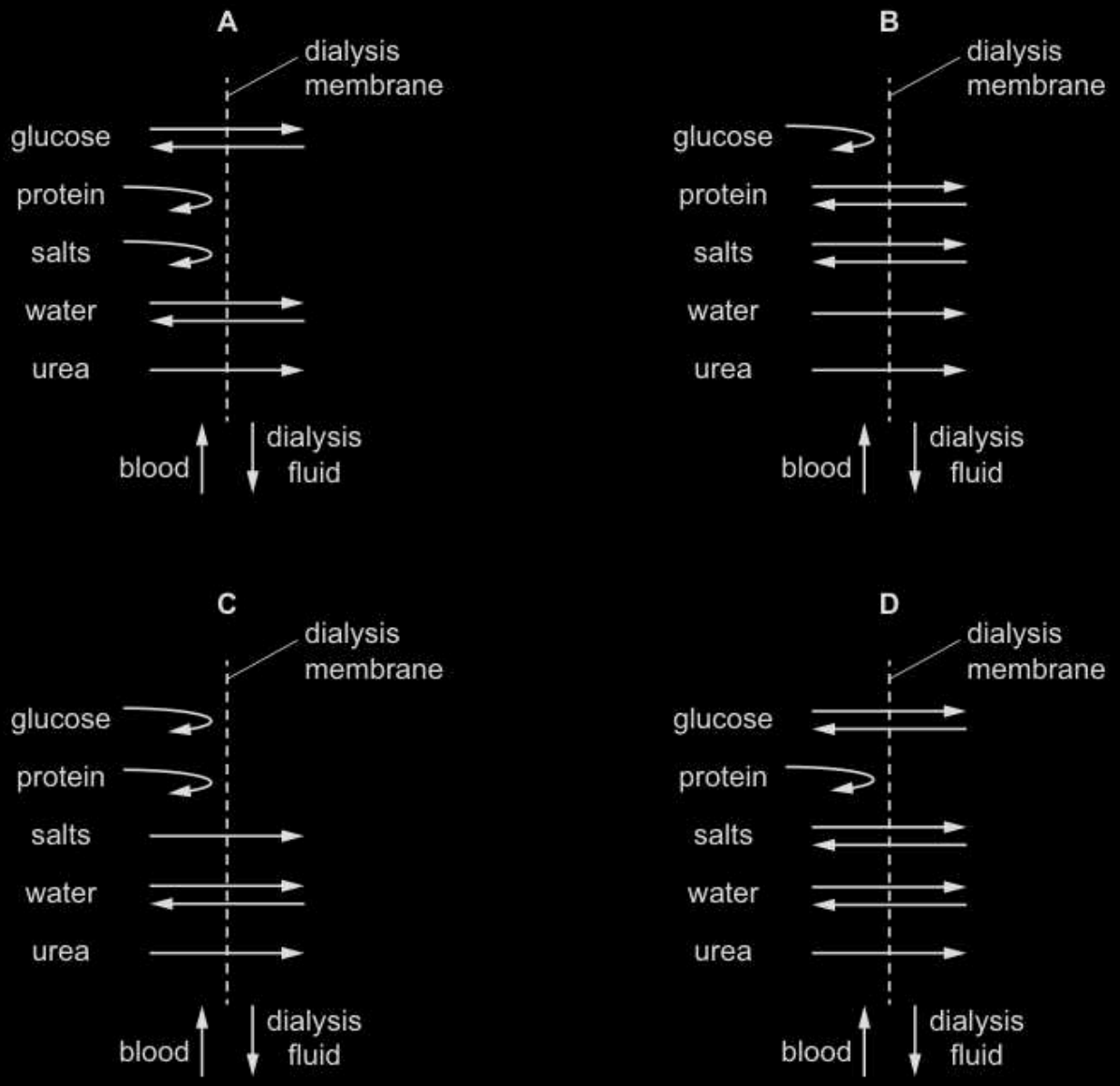
3 Oxygen dissolves in moisture on the surface of the alveolus before diffusing into the blood.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

October/November 2021 · Paper 1 · Variant 2 · Question 19

Correct answer: D

19 Which diagram shows the diffusion of substances between the blood and dialysis fluid during dialysis?



October/November 2021 · Paper 1 · Variant 2 · Question 20

Correct answer: D

20 After exercise, human body temperature often rises above normal.

What will help to return the temperature to normal?

A an increase in the fat layer under the skin

B a hormonal response causing shivering

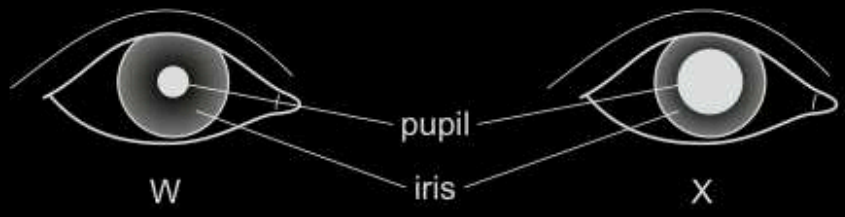
C a positive feedback mechanism

D dilation of blood vessels in the skin

October/November 2021 · Paper 1 · Variant 2 · Question 21

Correct answer: C

21 The diagrams show the front view of the pupil and iris of the eye in different light intensities.



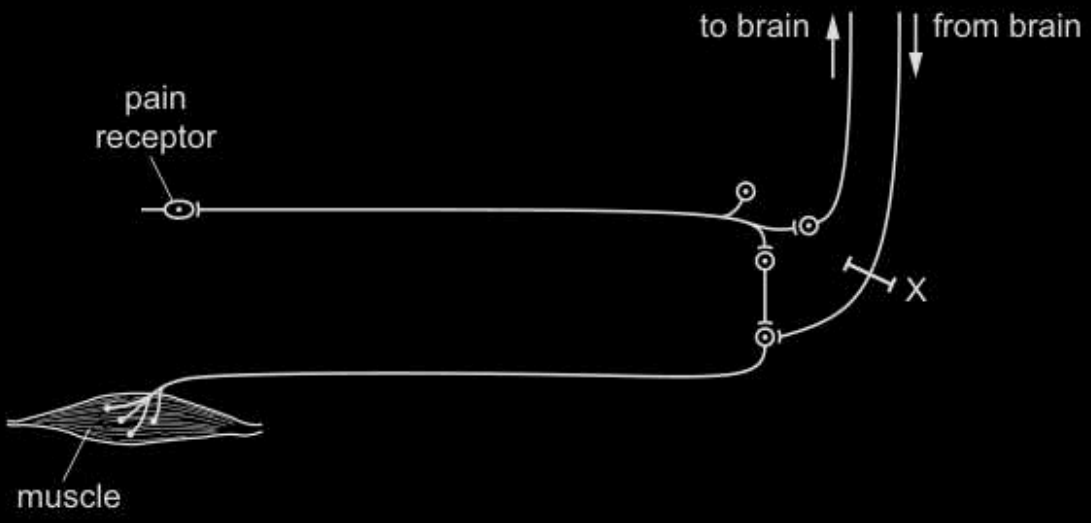
Which row correctly describes diagram X?

	eye is in	circular muscles of the iris are	radial muscles of the iris are
A	bright light	relaxed	contracted
B	bright light	contracted	relaxed
C	dim light	relaxed	contracted
D	dim light	contracted	relaxed

October/November 2021 · Paper 1 · Variant 2 · Question 22

Correct answer: B

22 The diagram shows some of the nerve pathways associated with a reflex action.



If the pathway at X is damaged, how does this affect the reflex?

A The person will not be aware that the reflex is occurring.

B The reflex cannot be controlled consciously.

C The response will occur without any stimulus.

D There is no response to the stimulus.

October/November 2021 · Paper 1 · Variant 2 · Question 23

Correct answer: D

23 What occurs after an injection of insulin?

A More adrenaline is released into the blood.

B More glycogen is released from the liver.

C The blood glucose concentration increases.

D The production of glycogen increases.

October/November 2021 · Paper 1 · Variant 2 · Question 26

Correct answer: D

26 What is a feature of all bacteria?

A They are parasites.

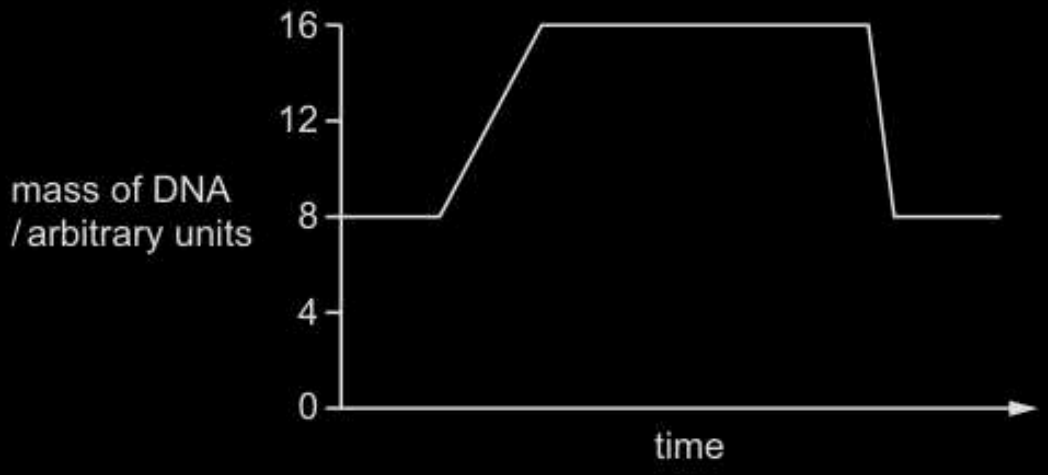
B They have a nucleus.

C They are made of hyphae.

D They are single-celled organisms.

Correct answer: C

33 The diagram shows the mass of DNA in cells which are dividing.



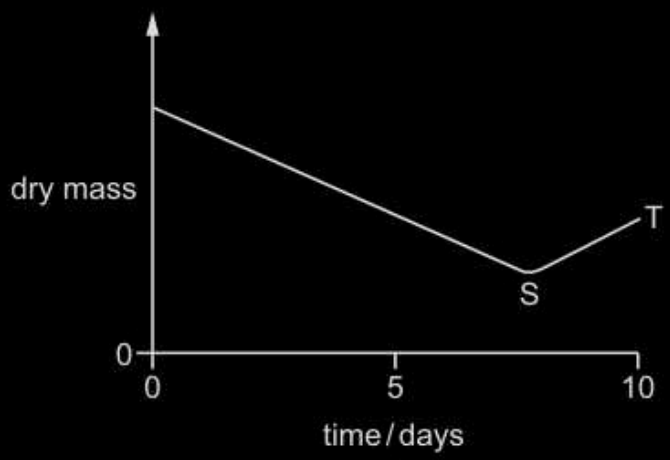
Which row describes this type of cell division?

	type of cell division	type of reproduction using this cell division	this type of cell division gives rise to
A	meiosis	asexual	genetically identical offspring
B	meiosis	sexual	genetically non-identical offspring
C	mitosis	asexual	genetically identical offspring
D	mitosis	sexual	genetically non-identical offspring

October/November 2021 · Paper 1 · Variant 2 · Question 34

Correct answer: B

34 The graph shows changes in the dry mass of a seed as it germinates and grows.



What causes the change shown between points S and T?

A osmosis

B photosynthesis

C respiration

D transpiration

October/November 2021 · Paper 1 · Variant 2 · Question 37

Correct answer: B

37 What is an example of discontinuous variation in humans?

A height

B blood group

C hair colour

D mass

October/November 2021 · Paper 1 · Variant 2 · Question 38

Correct answer: C

38 One gene has two codominant alleles, A^E and A^F, and one recessive allele, A^G.

How many different genotypes and phenotypes are possible?

	genotypes	phenotypes
A	3	3
B	4	6
C	6	4
D	6	6

October/November 2021 · Paper 1 · Variant 2 · Question 39

Correct answer: C

39 What is a result of natural selection?

A dogs that are friendly to humans

B grapes that contain no seeds

C mosquitoes that are resistant to insecticides

D onion crops that have a pleasant taste

October/November 2021 · Paper 1 · Variant 2 · Question 40

Correct answer: A

40 During the division of a nucleus by meiosis, changes can happen that produce a gamete with an abnormal number of genes or chromosomes.

Which change might produce a gamete that can result in a child having Down's syndrome?

A a gamete with one extra chromosome

B a gamete that lacks one gene

C a gamete that lacks one chromosome

D a gamete with one extra gene

3 A man is cooking in a hot kitchen and sweat is being released by sweat glands in his skin.

(a) (i) Explain the term homeostasis with reference to sweating.

.....

.....

.....

..... [2]

(ii) Explain how receptors and blood vessels in the man's skin are also involved in homeostasis when he is in the hot kitchen.

.....

.....

.....

.....

.....

..... [4]

(b) Sweat contains very small quantities of the waste product urea.

State the name of the organ where most urea is made in the body.

..... [1]

(c) The table shows the concentrations of some chemicals found in the sweat, urine and blood plasma of a healthy human.

chemical	concentration of chemical / mmol per dm^3		
	sweat	urine	blood plasma
urea	22	393	6
sodium	66	110	141
chloride	59	103	99

(i) Use the data in the table to compare the chemical composition of urine and blood plasma.

.....

.....

.....

.....

.....

..... [4]

(ii) A patient has kidney disease and is about to start dialysis treatment.

Suggest and explain the concentration of urea in the sweat of the patient compared to that of a person who does not have kidney disease.

.....

.....

.....

..... [3]

[Total: 14]

3(a)(i)	*any **two** from:* (homeostasis is) the maintenance of a constant ; internal environment / temperature ; return to norm / return to set-point / negative feedback ; evaporation of sweat removes heat / thermal energy / cools / decreases temperature ;	**2**	
3(a)(ii)	*any **four** from:* receptors **+** detection of temperature ; impulse to brain / CNS ; vasodilation ; arterioles / blood vessels widen ; increase in blood **+** near skin surface ; greater **+** loss of heat / thermal energy ; by radiation ;	**4**	
3(b)	liver ;	**1**	

© UCLES 2021

Question	Answer	Marks	Guidance
3(c)(i)	*any **four** from:* less urea in blood plasma ; more sodium in blood plasma ; less chloride in blood plasma ; comparative data quote with units ; data manipulation, e.g. 50 × higher urea in urine than blood plasma ;	**4**	
3(c)(ii)	high(er) / over 22 (in patient) ; reduced filtration / reduced removal of urea (from kidney) ; reference to blood plasma ;	**3**	

Question	Answer	Marks	Guidance

6 **(a)** Explain the 'lock and key' hypothesis of enzyme action using a **named** example.

.....

.....

.....

.....

.....

.....

..... [5]

(b) Explain, using examples, why proteins are essential for all living organisms.

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

6(a)	*any **five** from:* enzyme **+** name, e.g. amylase ; substrate **+** name, e.g. amylose/starch ; enzyme **+** has an active site ; specificity of enzyme **AW** ; substrate fits into active site / enzyme-substrate complex formed ; named product formed ; enzyme unchanged at end of reaction ;	**5**	
6(b)	*any **five** from:* growth / new cells ; repair ; energy in absence of carbohydrates/fats ; muscle ; hair / nails / skin ; hormones ; enzymes ; antibodies ; named protein, e.g. haemoglobin, fibrinogen ; cell membrane / carrier protein ; blood clotting ;	**5**	

© UCLES 2021

Question	Answer	Marks	Guidance

7 **(a)** Explain the role of the nucleus and cell division in the inheritance of characteristics from parents to offspring.

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) If people are exposed to ionising radiation, this may affect the inheritance of characteristics by their children.

Suggest how this exposure to radiation may affect inheritance of characteristics.

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

Section C

Answer **either** Question 8 **or** Question 9.

Write your answers in the spaces provided.

7(a)	*any **six** from:* chromosomes ; DNA / genetic material ; genes / alleles ; code **AW +** for proteins ; proteins determine characteristics ; meiosis / meiotic division ; reduction division / produces haploid cells / produces gametes **AW** ; fertilisation / fertilisation described ; combination/mix **+** of genetic material / genes / chromosomes / DNA / characteristics ;	**6**	
7(b)	*any **four** from:* increases the chance / probability of; mutation ; change in chromosome number ; change in gene structure / sequence ; codes for new/different/incorrect protein ; may result in new beneficial variation ; may result in genetic disease / example of genetic condition ;	**4**	

© UCLES 2021

Question	Answer	Marks	Guidance

8 **(a)** Water enters living organisms and is transported to all their cells.

Compare the transport of water in a human and a flowering plant.

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) Water in humans and flowering plants contains dissolved mineral ions.

Suggest reasons why the mineral ion requirements of humans and plants are different.

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

8(a)	*any **six** from:* *similarities* both carried in tubes ; one way movement ; both travel between cells by osmosis ; *differences* <u>plants</u> from soil / root ; in xylem ; detail of xylem vessel, e.g. no end-walls, lignin ; Transpiration ; detail of mechanism, e.g. cohesion / adhesion / continuous water column ; lost through evaporation ; <u>human</u> by ingestion / alimentary canal part; in blood vessels ; detail of blood vessel, e.g. muscular walls elastic walls of artery ; heart pumps ; detail of mechanism, e.g. heart muscles contract, elastic tissue allows pulsing of blood ; lost through sweating / urination **AW** ;	**6**	
8(b)	*any **four** from:* magnesium **+** for chlorophyll ; iron **+** haemoglobin / blood ; calcium **+** bones ; calcium **+** cell walls ; deficiency in plants, e.g. yellow leaves ; deficiency in animals, e.g. rickets, anaemia ; idea that have different cell types / structures / physiology in organism ; produce **+** different chemicals / chemicals in different proportions ;	**4**	

© UCLES 2021

Question	Answer	Marks	Guidance

9 **(a)** Sperm cells and pollen grains are both involved in sexual reproduction.

Compare the functions of a sperm cell and a pollen grain and describe how the structure of each is related to its function.

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) Suggest reasons why sexual reproduction may be better for the survival of a species than asexual reproduction.

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

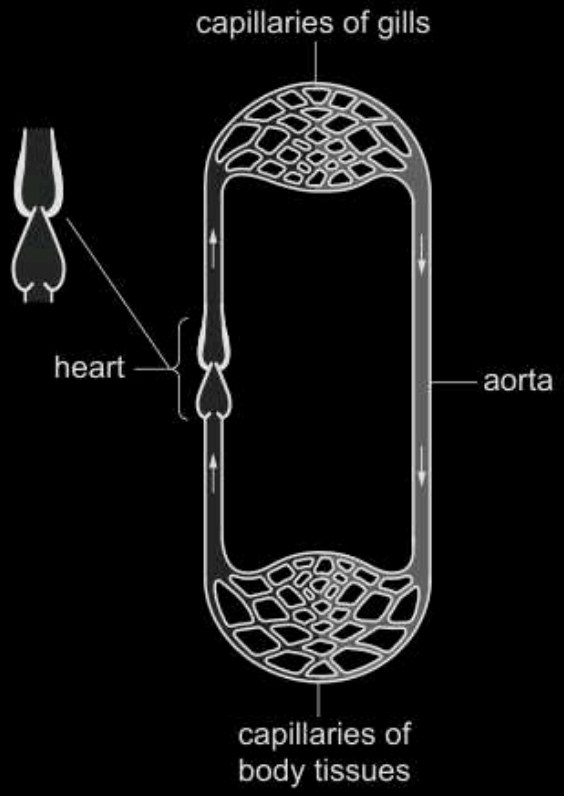
Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

9(a)	*any **six** from:* *similarities* both gametes ; haploid / half chromosome number of body cell / produced by meiosis ; fuses with egg / fertilisation / to produce zygote ; *differences* sperm cell has flagellum / tail **+** move ; enzymes / acrosome **+** to penetrate egg ; wind pollination / carried by wind / insect pollination / carried by insect ; small / light / 'sticky' / sculptured coat ; pollen tube **+** to reach egg / ovule / ovary ;	**6**	
9(b)	*any **four** from:* sexual reproduction **+** genetically different offspring ; sexual reproduction **+** variation ; asexual reproduction **+** genetically identical offspring/clones ; reference to mitosis / meiosis ; condition that affects survival, e.g. disease / temperature / water availability ; some individuals will survive changing environmental condition ; live in wider range of environments ;	**4**	

© UCLES 2021

2 The diagram shows the circulatory system of a fish.

The lungs of a human and the gills of a fish both have a large surface area for gas exchange.



(a) (i) State **two** differences between the structure of the heart of a fish and the structure of the heart of a human.

1 .....

.....

2 .....

.....

[2]

(ii) Draw a (ring) around the correct words to complete the sentence below.

higher than **the same as** **lower than**

The pressure of blood in the aorta of a fish will be ..... the pressure of blood in the aorta of a human. [1]

(b) The photograph shows an Antarctic icefish.

The blood of Antarctic icefish is colourless.

(i) State which component of human blood is **not** present in the blood of an Antarctic icefish.

..... [1]

Antarctic icefish live in the Antarctic Ocean where the water temperature is very cold.

Aerobic respiration is an enzyme-controlled reaction.

More oxygen is able to dissolve in water at a lower temperature.

Antarctic icefish have a larger heart, wider blood vessels and a greater volume of blood than fish of the same size that live in warmer water.

(ii) Suggest how Antarctic icefish with colourless blood are able to survive in the low temperatures of the Antarctic Ocean.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 9]

2(a)(i)	fish **+** 2 chambers **AW** OR human **+** 4 chambers **AW** ; fish **+** 2 valves **AW** OR human **+** 4 valves **AW** ; fish **+** 2 blood vessels **AW** OR human **+** 4 blood vessels **AW** ; fish **+** chambers similar size OR human **+** chambers different sizes ;	**2**	
2(a)(ii)	lower than ;	**1**	**A** any clear reference **only** to lower than
2(b)(i)	red blood cells / erythrocytes / haemoglobin ;	**1**	
2(b)(ii)	less **+** aerobic respiration ; enzyme reactions **+** slower ; less **+** oxygen needed ; more oxygen **+** absorbed **AW** by fish ; more oxygen **+** plasma ; more blood / more oxygen **+** supplied / taken / pumped ; (to) organs / tissues / cells ;	**5**	

Question	Answer	Marks	Guidance

3 The table shows the mass of each component in 250 cm^3 of cow's milk.

component	mass / g
carbohydrate	11
fat	8
protein	8

(a) (i) A 250 cm^3 drink of cow's milk provides 14% of the total mass of protein required each day in the diet of an average adult human.

Calculate the total mass of protein required in the diet of an average adult human each day.

You **must** state your answer to one decimal place and include correct units.

..... [3]

(ii) State **two** important uses of protein in the diet.

1 .....

2 .....

[2]

(b) The table below lists four components of a balanced diet for an average adult human.

For each component listed, the percentage of the daily requirement provided by 250 cm^3 of milk is shown.

component	percentage of daily requirement provided by 250 cm^3 of milk
vitamin C	0
vitamin D	30
calcium	30
iron	0

Discuss how a 250 cm^3 drink of cow's milk contributes to the health of an adult human for each of the components listed in the table.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 9]

3(a)(i)	57 ; (57).1 ; g ;	**3**	
3(a)(ii)	growth / new cells / protoplasm ; repair ; muscle ; hair / nails / skin / keratin ; hormones / named hormone ; enzymes / named enzyme ; antibodies ; haemoglobin / collagen ; membrane component ;	**2**	

© UCLES 2021

Question	Answer	Marks	Guidance
3(b)	no/lacks vitamin C **+** gums / scurvy / immunity **AW /** wound healing ; vitamin D **+** prevents rickets / bones / teeth / absorb calcium ; calcium **+** prevents rickets / bones / teeth / blood clotting ; no/lacks iron **+** haemoglobin / red blood cells / anaemia / lack of energy ;	4	

Question	Answer	Marks	Guidance

5 Some laundry detergents used to wash clothes contain enzymes.

These enzymes break down the molecules that cause stains.

(a) Suggest enzymes that may be components of a laundry detergent that will break down stains made of:

starch .....

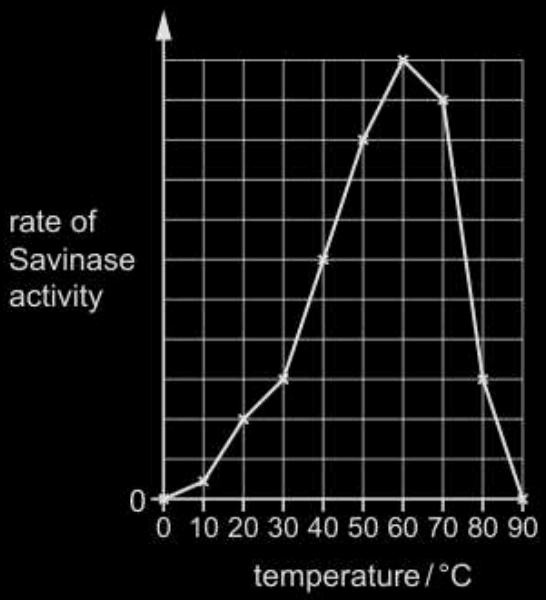
fat. .....

[2]

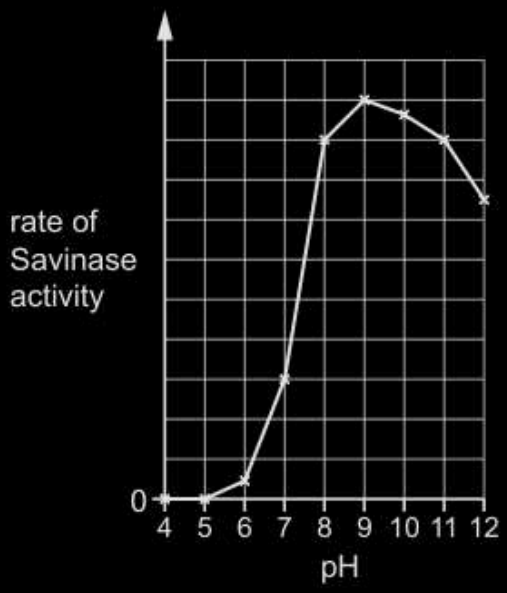
(b) Savinase is a protease enzyme produced by genetically engineered bacteria.

The enzyme is a component of laundry detergents.

The graphs show the results of an investigation into the effects of temperature and pH on the rate of Savinase activity.



effect of temperature



effect of pH

(i) Describe how the effect of temperature on Savinase activity differs from the effect of temperature on a protease that functions in the human stomach.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(ii) Suggest why laundry detergents that contain Savinase also contain chemicals that dissolve to form an alkaline solution.

.....

.....

.....

..... [2]

(iii) After washing clothes using laundry detergents, the waste water is sometimes released into the environment.

This waste water contains inorganic phosphate ions that are also found in fertilisers.

Explain the harmful effect on aquatic life of releasing this waste water into the environment.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 11]

Section B

Answer **both** questions in this section.

Write your answers in the spaces provided.

5(a)	(starch) amylase ; (fat) lipase ;	2	
5(b)(i)	optimum **AW +** 60°C ; (optimum) not 37°C ; correct ref. temperature **+** denaturation for either enzyme ;	3	
5(b)(ii)	savinase / enzyme **+** increased rate / works faster / optimum ; prevents denaturation ; detergent more effective / clothes cleaner ;	2	
5(b)(iii)	pollution ; eutrophication ; **two** correct details of the eutrophication process for **two marks** ;; less oxygen available / kills **+** fish / consumers / animals **AW** ;	4	

Question	Answer	Marks	Guidance

May/June 2021 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 Which structures are present in plant cells but **not** in animal cells?

A cell membrane, cytoplasm, chloroplasts

B cell wall, chloroplasts, sap vacuole

C cell wall, cell membrane, cytoplasm

D cytoplasm, nucleus, chloroplasts

May/June 2021 · Paper 1 · Variant 1 · Question 2

Correct answer: C

2 A piece of plant tissue is transferred from a beaker of water into a concentrated sugar solution.

Which row describes what happens?

	movement of water	volume of tissue cells
A	into the cells	decreases
B	into the cells	increases
C	out of the cells	decreases
D	out of the cells	increases

May/June 2021 · Paper 1 · Variant 1 · Question 3

Correct answer: D

3 Which statements about active transport are correct?

1 Ions move from a region of high concentration to a region of low concentration.

2 Ions move across the cell membrane.

3 Energy released during respiration is used to move ions into or out of a cell.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

May/June 2021 · Paper 1 · Variant 1 · Question 4

Correct answer: B

4 What is a correct description of enzymes?

A biological catalysts made of lipids

B biological catalysts made of proteins

C catalysts that slow down chemical reactions and are changed by the reaction

D catalysts that speed up chemical reactions and are changed by the reaction

May/June 2021 · Paper 1 · Variant 1 · Question 6

Correct answer: D

6 What describes the upper cuticle of a leaf?

A a permeable layer allowing water to enter the leaf

B a single layer of cells containing many chloroplasts

C a single layer of transparent cells allowing light to enter the leaf

D a thin non-cellular layer preventing water loss from the leaf

May/June 2021 · Paper 1 · Variant 1 · Question 7

Correct answer: B

7 Plants growing in soil lacking magnesium ions have poor growth.

What causes their poor growth?

A They cannot form cellulose for new cell walls.

B They cannot form chlorophyll for photosynthesis.

C They cannot produce amino acids to form proteins.

D They cannot take in mineral ions from the soil.

May/June 2021 · Paper 1 · Variant 1 · Question 8

Correct answer: A

8 Which statement about diets is correct?

A A diet providing all the needs of an average man aged 70 can lead to malnutrition in an active boy aged 14.

B Malnutrition can only occur if the body receives less of a nutrient than it requires.

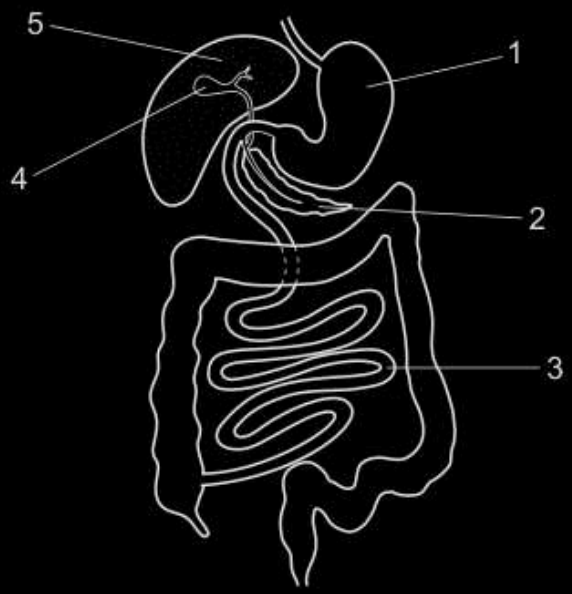
C Protein requirements in the diet increase throughout a person's life.

D Very young children require a greater total energy intake than active adults.

May/June 2021 · Paper 1 · Variant 1 · Question 9

Correct answer: A

9 The diagram shows part of the human alimentary canal and associated organs.



Which row describes the functions of parts shown in the diagram?

	structure	function	structure	function
A	1	digestion of protein	3	absorption of the products of digestion
B	2	emulsifying fats	3	absorption of amino acids and glucose
C	4	production of bile	5	making digestive enzymes
D	4	storing digestive enzymes	2	making digestive enzymes

May/June 2021 · Paper 1 · Variant 1 · Question 10

Correct answer: C

10 What is **not** an example of assimilation?

A synthesis of glycogen from glucose

B synthesis of fats from fatty acids and glycerol

C synthesis of glucose from starch

D synthesis of proteins from amino acids

May/June 2021 · Paper 1 · Variant 1 · Question 11

Correct answer: B

11 Which feature of a root hair cell helps it take up ions from water in the soil?

A The cell has a partially permeable cell wall.

B The cell is able to provide a source of energy.

C The cell vacuole has a high ion concentration.

D The cell vacuole has a low water potential.

May/June 2021 · Paper 1 · Variant 1 · Question 13

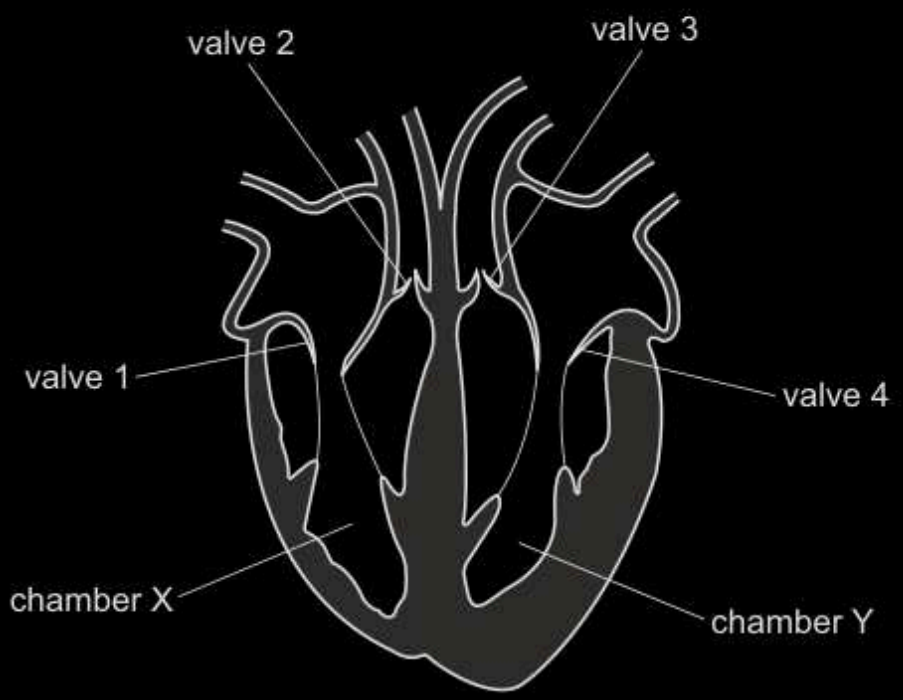
Correct answer: D

13 Which row describes the functions of the blood components?

	plasma	platelets	white blood cells
A	antibody formation	clotting	transport of nutrients
B	clotting	transport of nutrients	antibody formation
C	clotting	antibody formation	transport of nutrients
D	transport of nutrients	clotting	antibody formation

Correct answer: A

14 The diagram shows a section through the human heart.



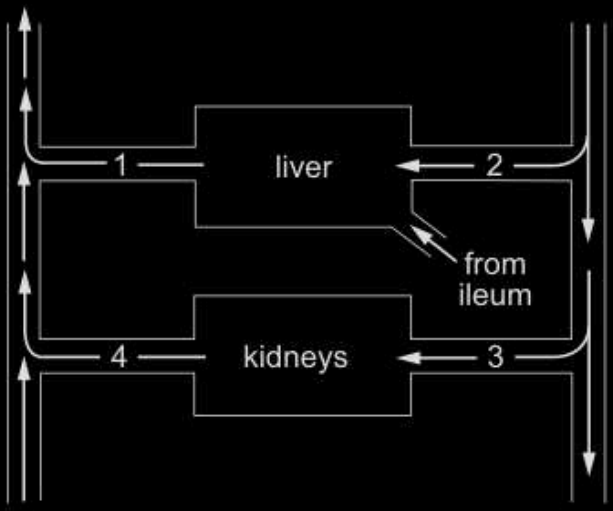
What will be the state of the valves when the walls of chambers X and Y are contracting?

	valves			
	1	2	3	4
A	closed	open	open	closed
B	closed	open	closed	open
C	open	closed	open	closed
D	open	closed	closed	open

May/June 2021 · Paper 1 · Variant 1 · Question 15

Correct answer: B

15 The diagram represents the blood supply to the liver and to the kidneys.



Which vessels contain blood with the highest and lowest concentrations of urea?

	highest	lowest
A	1	2
B	1	4
C	3	2
D	3	4

May/June 2021 · Paper 1 · Variant 1 · Question 17

Correct answer: C

17 In an investigation, scientists treated plant roots with a chemical that stops aerobic respiration. They observed that after the treatment the plant roots absorbed nitrate ions more slowly.

Which statement explains this observation?

A Anaerobic respiration does not release energy.

B The plant cells cannot respire.

C There is less energy for active transport of nitrate ions.

D There is less energy for diffusion of nitrate ions.

May/June 2021 · Paper 1 · Variant 1 · Question 18

Correct answer: C

18 Why does oxygen enter the blood in the lungs?

A Aerobic respiration takes place in the lungs.

B Carbon dioxide diffuses out of the blood into the alveoli.

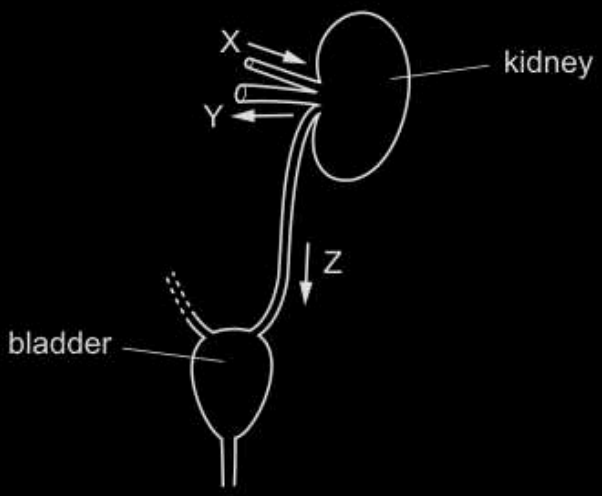
C The air in alveoli contains a higher concentration of oxygen than the blood in the lung capillaries.

D The air in alveoli contains a lower concentration of oxygen than the blood in the lung capillaries.

May/June 2021 · Paper 1 · Variant 1 · Question 19

Correct answer: D

19 The diagram shows structures in a human which form and store urine. Liquids pass through tubes X, Y and Z in the directions shown by the arrows.



Which statement about the volume of liquid passing through Y in one day is correct?

A It is greater than that passing through X.

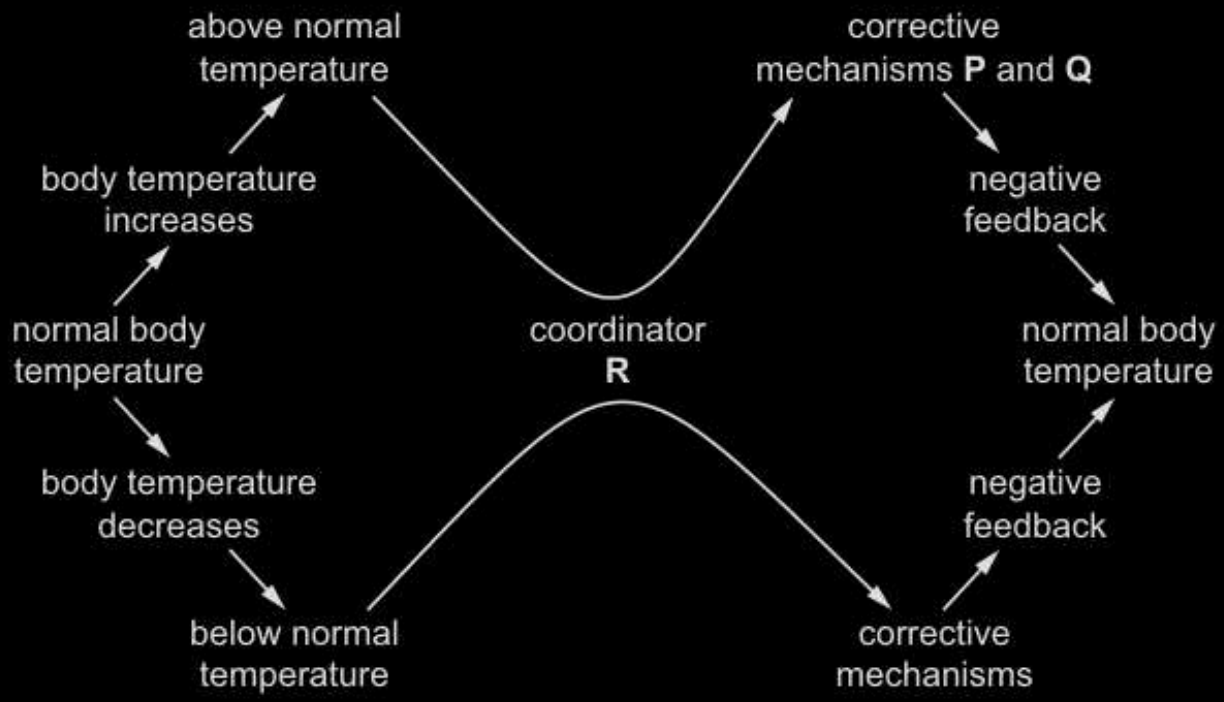
B It is less than that passing through Z.

C It is much less than that passing through X but slightly greater than that passing through Z.

D It is slightly less than that passing through X but much greater than that passing through Z.

Correct answer: A

20 The diagram shows the homeostatic control of body temperature.



What are corrective mechanisms **P** and **Q** and coordinator **R**?

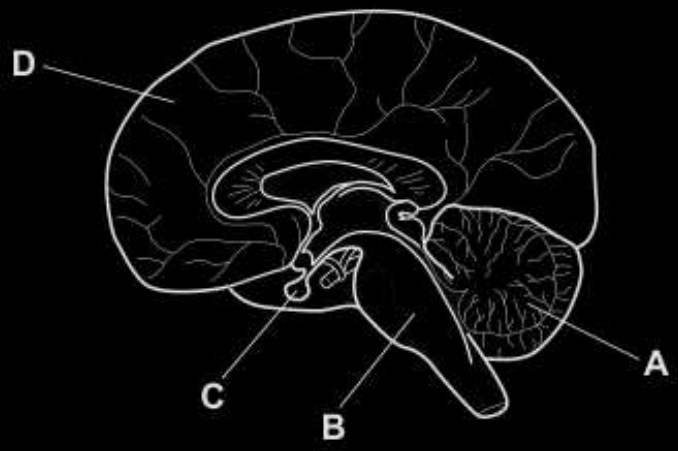
	corrective mechanisms		coordinator **R**
	P	**Q**	
A	sweating	more blood flows close to the skin surface	brain
B	sweating	more blood flows close to the skin surface	skin
C	shivering	less blood flows close to the skin surface	brain
D	shivering	less blood flows close to the skin surface	skin

May/June 2021 · Paper 1 · Variant 1 · Question 21

Correct answer: C

21 The diagram shows a section through the brain.

Which part of the brain secretes hormones?



May/June 2021 · Paper 1 · Variant 1 · Question 22

Correct answer: C

22 Which changes occur in the eye when viewing a distant object?

1 The ciliary muscles contract.

2 The ciliary muscles relax.

3 The suspensory ligaments are pulled tight.

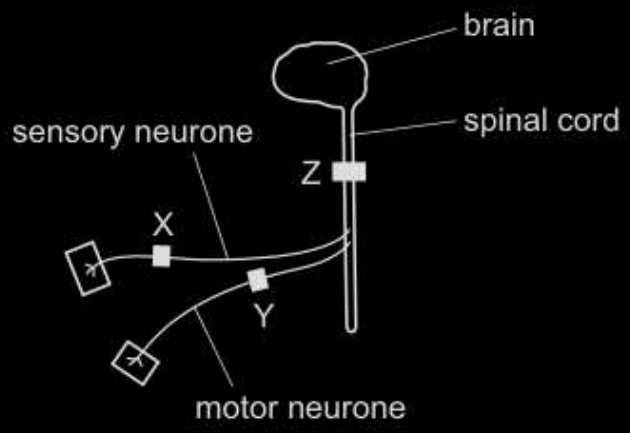
4 The suspensory ligaments are slackened.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

May/June 2021 · Paper 1 · Variant 1 · Question 23

Correct answer: C

23 The diagram shows three positions, X, Y and Z, where nerve impulses are blocked by a drug.



Which person listed below moves their leg in response to a pinprick, but does **not** feel it?

A a person with a block at X

B a person with a block at Y

C a person with a block at Z

D a person with no block

May/June 2021 · Paper 1 · Variant 1 · Question 25

Correct answer: B

25 What are the effects of drinking a large quantity of alcohol on the human body?

	effects on the body		
	acts as a	reaction time	damages the
A	depressant	decreases	kidney
B	depressant	increases	liver
C	stimulant	increases	kidney
D	stimulant	decreases	liver

Correct answer: D

26 The diagram shows the structure of a bacterium.

In what way does this differ from a cell of a fungus?

A The bacterium has a cell membrane.

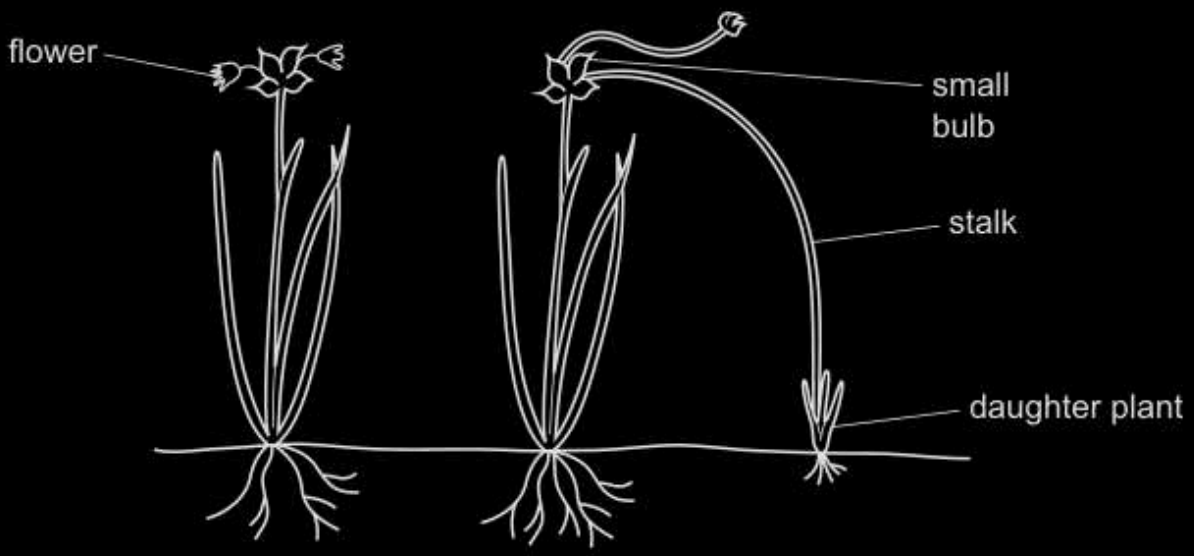
B The bacterium has a cell wall.

C The bacterium has cytoplasm.

D The bacterium has no true nucleus.

Correct answer: C

33 The diagram shows two onion plants.



Using the information in the diagram, which statement about these onion plants is correct?

A Daughter plants are produced from the small bulb by meiosis.

B Daughter plants produced from the flower are genetically identical to the parent plant.

C The plants can reproduce sexually and asexually.

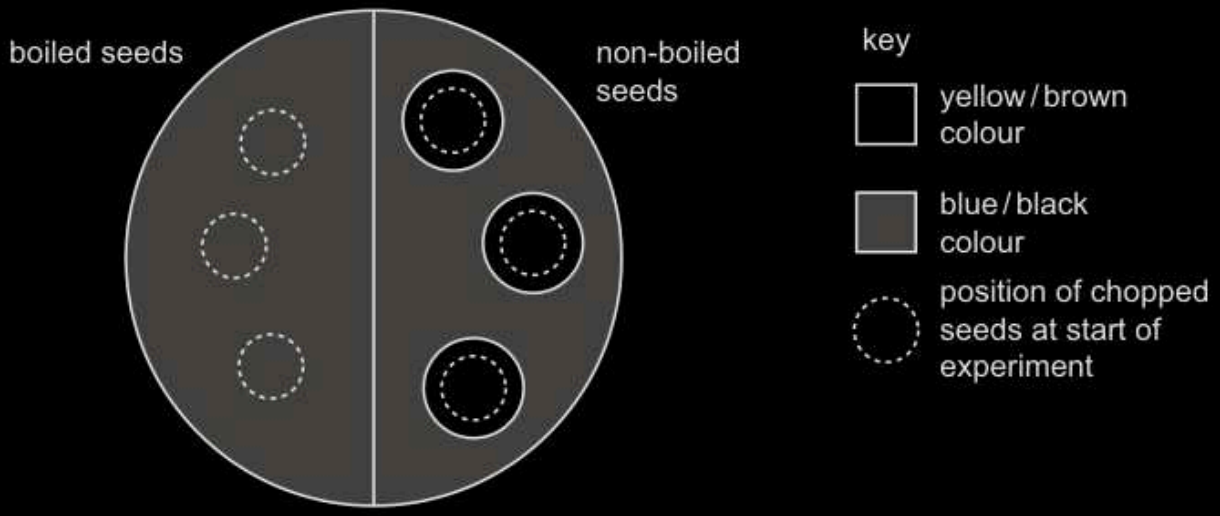
D Two parent plants are required for reproduction.

Correct answer: D

34 Six bean seeds were soaked in cold water. Three of them were boiled and cooled. The boiled and the non-boiled seeds were chopped up and then placed on the surface of agar jelly containing starch.

After two days, all the seeds were removed and the jelly was tested with iodine solution.

The diagram shows the result of the experiment.



What is the explanation for the results with the non-boiled bean seeds?

A They absorb iodine.

B They absorb starch.

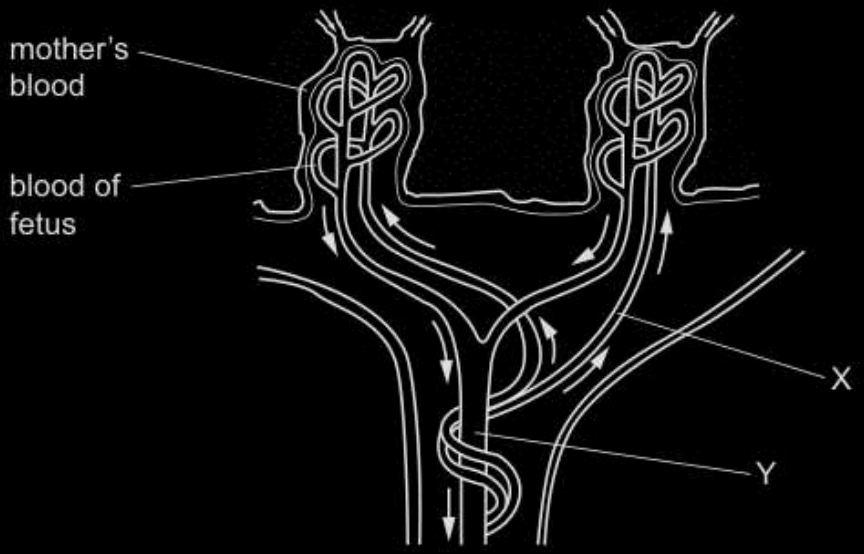
C They contain acid.

D They contain amylase.

May/June 2021 · Paper 1 · Variant 1 · Question 35

Correct answer: B

35 The diagram shows how the blood of a human fetus flows close to the mother's blood in the placenta.



Which substances are present at X in higher concentrations than at Y?

A carbon dioxide and glucose

B carbon dioxide and urea

C glucose and oxygen

D glucose and urea

May/June 2021 · Paper 1 · Variant 1 · Question 36

Correct answer: B

36 Why will antibiotics **not** cure a person with HIV infection?

A Viruses are much smaller than bacteria.

B Viruses are not affected by antibiotics.

C Viruses are made from protein and DNA.

D Viruses are not pathogens.

May/June 2021 · Paper 1 · Variant 1 · Question 37

Correct answer: A

37 Chromosomes, DNA molecules and genes are pieces of inherited material with different sizes.

Which statement describes the relationship between these pieces of inherited material?

A A chromosome is larger than a gene.

B A gene is larger than a chromosome.

C Multiple chromosomes make up a DNA molecule.

D Multiple DNA molecules make up a gene.

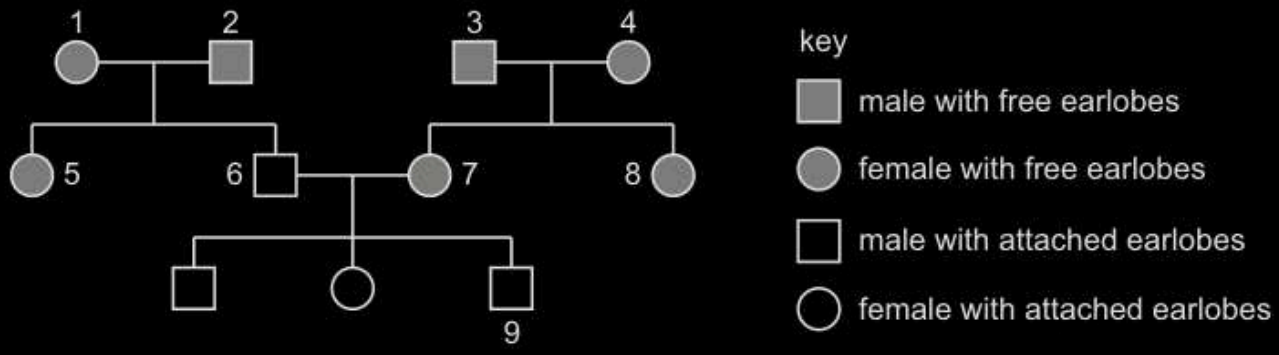
May/June 2021 · Paper 1 · Variant 1 · Question 38

Correct answer: A

38 Earlobes can either be attached to the cheek or be 'free' (unattached). This characteristic is controlled by a single gene.

The allele for attached earlobes is recessive.

The diagram shows the inheritance of attached earlobes in one family.



Which two individuals **must** be heterozygous for earlobe attachment?

A 1 and 7 **B** 3 and 4 **C** 5 and 8 **D** 6 and 9

May/June 2021 · Paper 1 · Variant 1 · Question 39

Correct answer: B

39 Which statements about natural selection are correct?

	natural selection can lead to better adapted species surviving	natural selection can lead to extinction of a species	natural selection can lead to gene mutations occurring
A	true	true	true
B	true	true	false
C	true	false	true
D	false	true	true

May/June 2021 · Paper 1 · Variant 1 · Question 40

Correct answer: A

40 Which feature of a cereal crop **cannot** be improved by selective breeding alone?

A ability to fix nitrogen

B resistance to drought

C tolerance to low temperatures

D number of seeds produced

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

May/June 2021 · Paper 1 · Variant 2 · Question 1

Correct answer: D

A 1 and 3 **B** 1 and 5 **C** 2 and 4 only **D** 2, 4 and 5

4 Which word describes an enzyme that has its active site changed by high temperature?

A denatured

B destroyed

C deleted

D distorted

May/June 2021 · Paper 1 · Variant 2 · Question 2

Correct answer: C

2 A piece of plant tissue is transferred from a beaker of water into a concentrated sugar solution.

Which row describes what happens?

	movement of water	volume of tissue cells
A	into the cells	decreases
B	into the cells	increases
C	out of the cells	decreases
D	out of the cells	increases

May/June 2021 · Paper 1 · Variant 2 · Question 3

Correct answer: B

3 The following statements refer to the movement of substances into and out of cells.

1 Energy from respiration is required.

2 Energy from respiration is not required.

3 Movement occurs down a concentration gradient.

4 Net movement is from a region of higher concentration to a region of lower concentration.

5 Net movement is from a region of lower concentration to a region of higher concentration.

Which statements apply to **active transport**?

A 1 and 3 **B** 1 and 5 **C** 2 and 4 only **D** 2, 4 and 5

May/June 2021 · Paper 1 · Variant 2 · Question 4

Correct answer: A

4 Which word describes an enzyme that has its active site changed by high temperature?

A denatured

B destroyed

C deleted

D distorted

May/June 2021 · Paper 1 · Variant 2 · Question 6

Correct answer: D

6 What describes the upper cuticle of a leaf?

A a permeable layer allowing water to enter the leaf

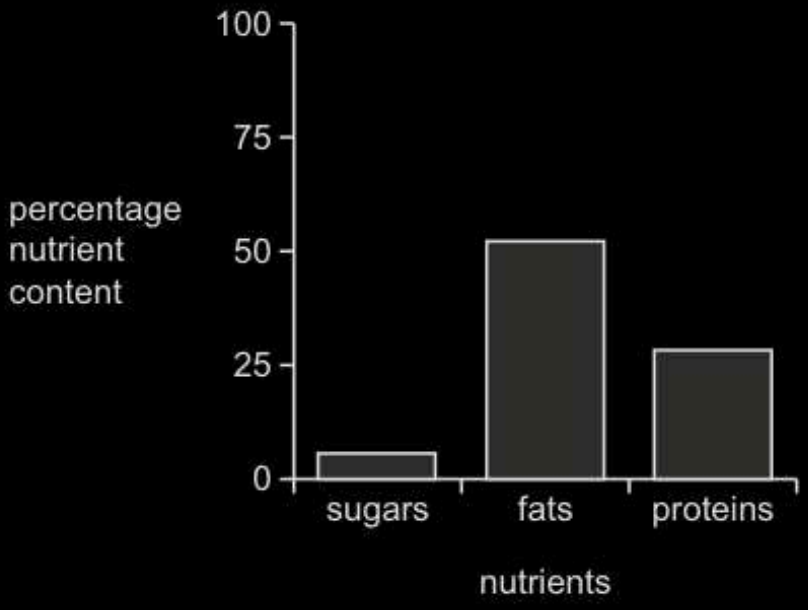
B a single layer of cells containing many chloroplasts

C a single layer of transparent cells allowing light to enter the leaf

D a thin non-cellular layer preventing water loss from the leaf

Correct answer: A

8 The graph shows some of the nutrients found in peanuts.



A student tested peanuts with four food tests.

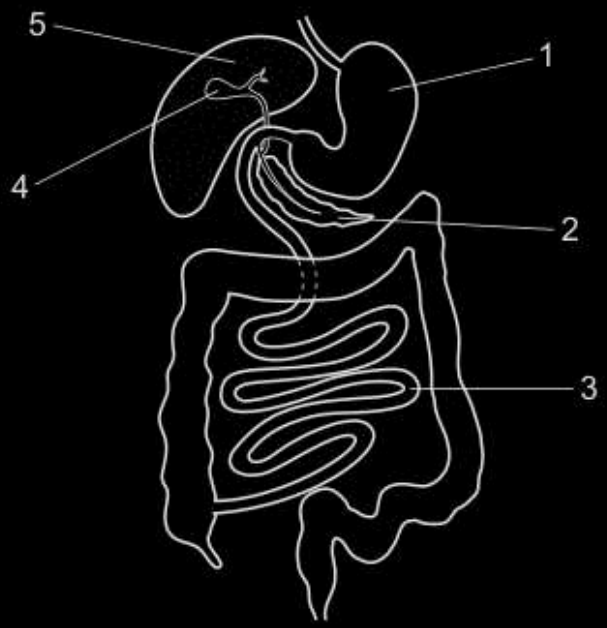
Which row shows the results?

	Benedict's test	biuret test	ethanol emulsion test	iodine test
A	yellow	purple	white	brown
B	black	blue	white	black
C	red	purple	clear	blue
D	blue	blue	clear	brown

May/June 2021 · Paper 1 · Variant 2 · Question 10

Correct answer: A

10 The diagram shows part of the human alimentary canal and associated organs.



Which row describes the functions of parts shown in the diagram?

	structure	function	structure	function
A	1	digestion of protein	3	absorption of the products of digestion
B	2	emulsifying fats	3	absorption of amino acids and glucose
C	4	production of bile	5	making digestive enzymes
D	4	storing digestive enzymes	2	making digestive enzymes

May/June 2021 · Paper 1 · Variant 2 · Question 11

Correct answer: B

11 How does water enter root hair cells?

A from high to low water potential by active transport

B from high to low water potential by osmosis

C from low to high water potential by active transport

D from low to high water potential by osmosis

May/June 2021 · Paper 1 · Variant 2 · Question 13

Correct answer: D

13 Which row describes the functions of the blood components?

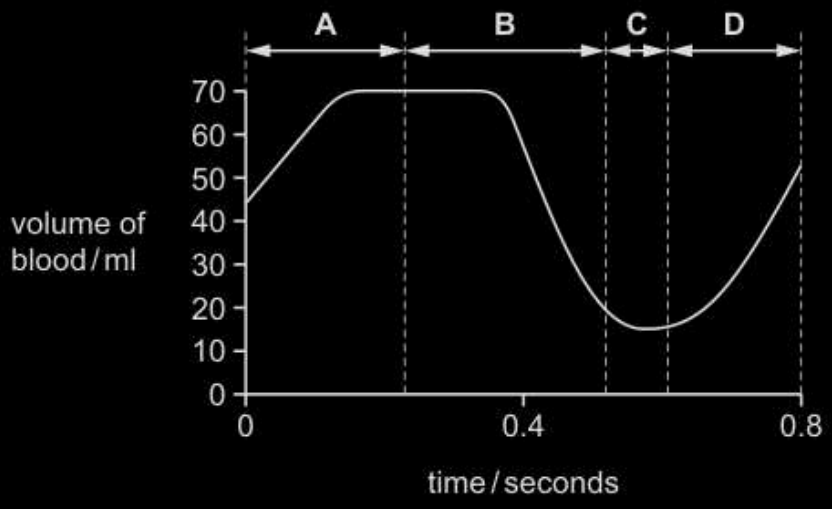
	plasma	platelets	white blood cells
A	antibody formation	clotting	transport of nutrients
B	clotting	transport of nutrients	antibody formation
C	clotting	antibody formation	transport of nutrients
D	transport of nutrients	clotting	antibody formation

May/June 2021 · Paper 1 · Variant 2 · Question 14

Correct answer: B

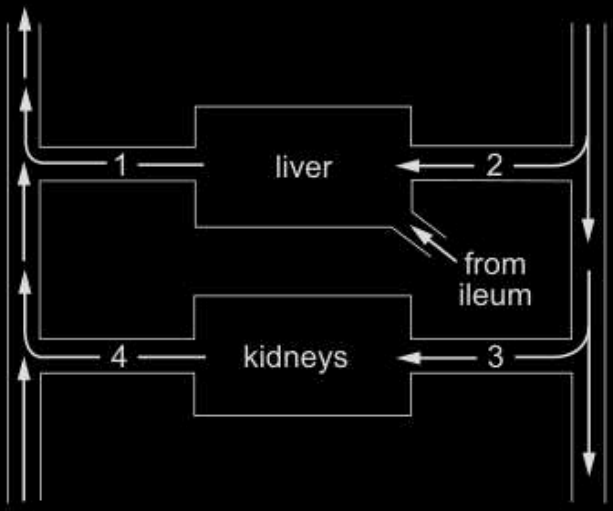
14 The diagram shows how the volume of blood in the left ventricle of a human heart changes during one heart beat.

When is the left ventricle contracting?



Correct answer: B

15 The diagram represents the blood supply to the liver and to the kidneys.



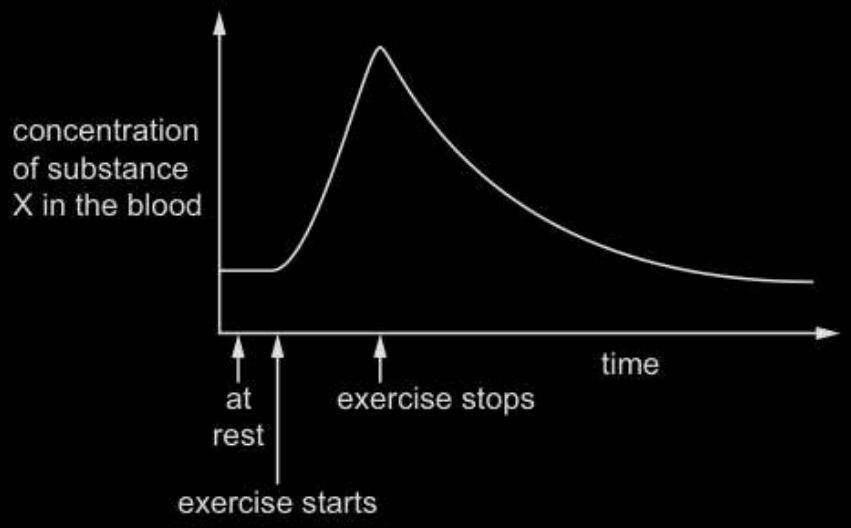
Which vessels contain blood with the highest and lowest concentrations of urea?

	highest	lowest
A	1	2
B	1	4
C	3	2
D	3	4

May/June 2021 · Paper 1 · Variant 2 · Question 17

Correct answer: C

17 The graph shows the concentration of substance X in a person's blood, before, during and after exercise.



What is substance X?

A alcohol

B glycogen

C lactic acid

D urea

May/June 2021 · Paper 1 · Variant 2 · Question 18

Correct answer: A

18 During exercise, a person breathes more deeply and more frequently.

Which is a result of this?

A Carbon dioxide is removed more rapidly.

B Heat loss is decreased.

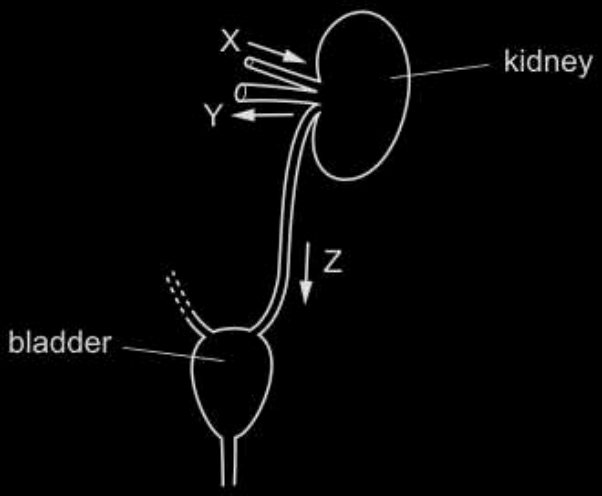
C Less glucose is delivered to the working muscles.

D Less oxygen diffuses into the blood.

May/June 2021 · Paper 1 · Variant 2 · Question 19

Correct answer: D

19 The diagram shows structures in a human which form and store urine. Liquids pass through tubes X, Y and Z in the directions shown by the arrows.



Which statement about the volume of liquid passing through Y in one day is correct?

A It is greater than that passing through X.

B It is less than that passing through Z.

C It is much less than that passing through X but slightly greater than that passing through Z.

D It is slightly less than that passing through X but much greater than that passing through Z.

May/June 2021 · Paper 1 · Variant 2 · Question 20

Correct answer: C

20 Which response will happen if a person walks into a cold environment?

A Hair erector muscles cause the body hairs to lay flat on the skin.

B More blood flows in the capillaries near the surface of the skin.

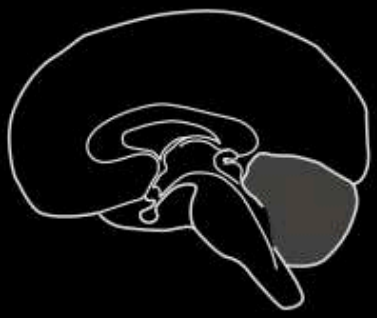
C Large body muscles contract and relax rapidly.

D The sweat glands produce a larger amount of sweat.

May/June 2021 · Paper 1 · Variant 2 · Question 21

Correct answer: A

21 The diagram shows the human brain.



What is the name of the shaded part of the brain?

A cerebellum

B cerebrum

C hypothalamus

D medulla

May/June 2021 · Paper 1 · Variant 2 · Question 22

Correct answer: C

22 What happens in the eye to produce the pupil reflex to bright light?

1 ciliary muscle contraction

2 circular muscle contraction

3 radial muscle relaxation

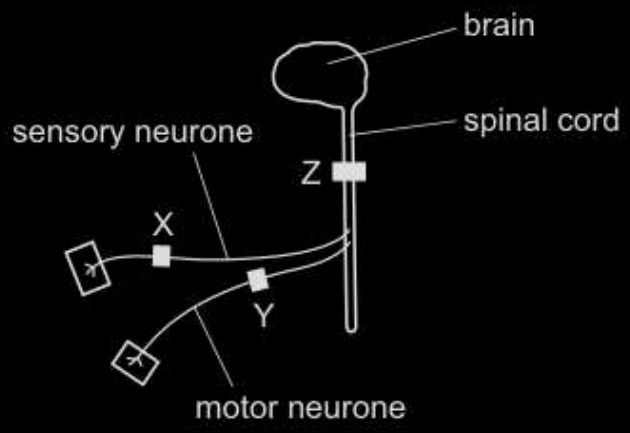
4 suspensory ligaments pulled tight

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

May/June 2021 · Paper 1 · Variant 2 · Question 23

Correct answer: C

23 The diagram shows three positions, X, Y and Z, where nerve impulses are blocked by a drug.



Which person listed below moves their leg in response to a pinprick, but does **not** feel it?

A a person with a block at X

B a person with a block at Y

C a person with a block at Z

D a person with no block

May/June 2021 · Paper 1 · Variant 2 · Question 25

Correct answer: A

25 Electronic cigarettes are a cigarette substitute allowing nicotine to be inhaled without the other components found in cigarette smoke.

Which effects of cigarette smoking will also be effects of using electronic cigarettes?

A Smoking increases heart rate, narrows the arterioles and increases blood pressure.

B Smoking irritates the airways and increases mucus production.

C Smoking reduces oxygen carried by red blood cells and reduces oxygen delivered to the tissues.

D Smoking narrows the airways and increases coughing.

Correct answer: D

26 The diagram shows the structure of a bacterium.

In what way does this differ from a cell of a fungus?

A The bacterium has a cell membrane.

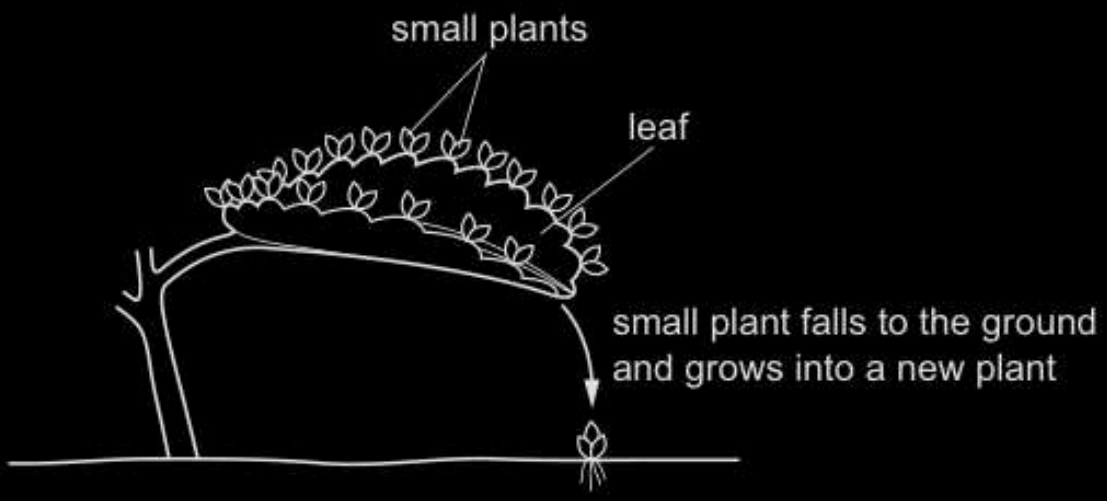
B The bacterium has a cell wall.

C The bacterium has cytoplasm.

D The bacterium has no true nucleus.

Correct answer: B

33 The diagram shows part of a *Bryophyllum* plant.



Which best describes how the new plants are produced?

	type of reproduction		type of cell division	
	asexual reproduction	sexual reproduction	meiosis	mitosis
A	✓	✗	✓	✗
B	✓	✗	✗	✓
C	✗	✓	✓	✗
D	✗	✓	✗	✓

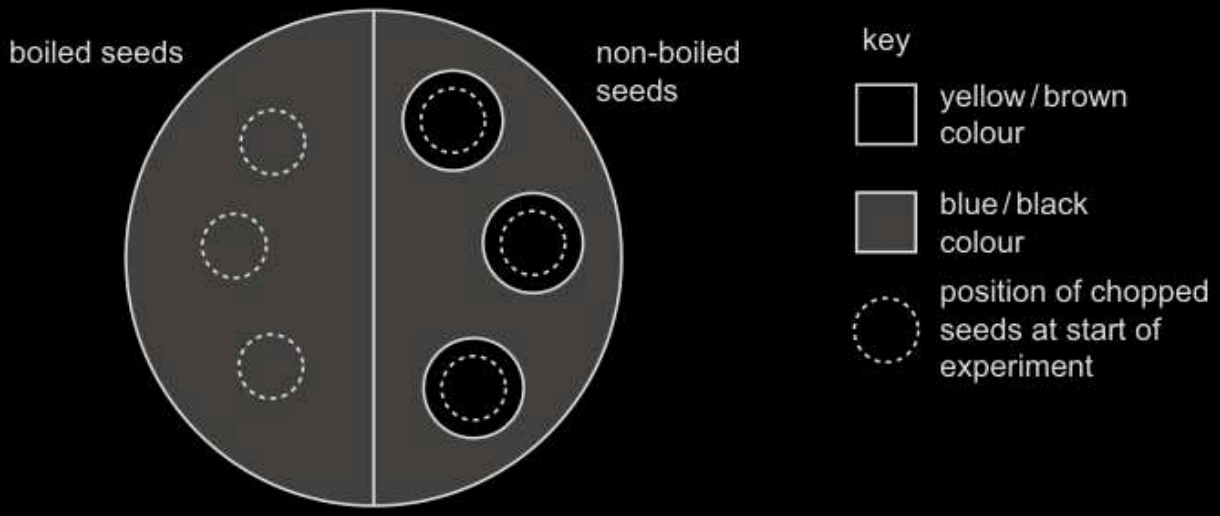
key
✓ = yes
✗ = no

Correct answer: D

34 Six bean seeds were soaked in cold water. Three of them were boiled and cooled. The boiled and the non-boiled seeds were chopped up and then placed on the surface of agar jelly containing starch.

After two days, all the seeds were removed and the jelly was tested with iodine solution.

The diagram shows the result of the experiment.



What is the explanation for the results with the non-boiled bean seeds?

A They absorb iodine.

B They absorb starch.

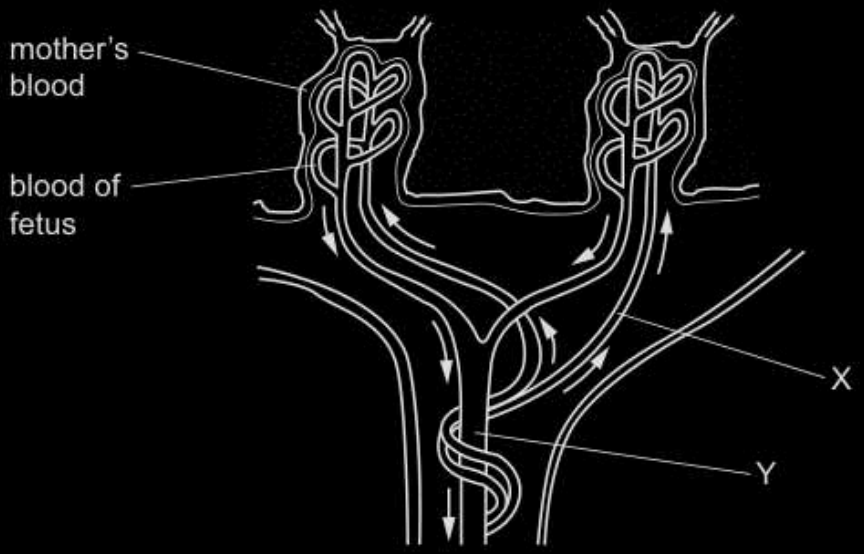
C They contain acid.

D They contain amylase.

May/June 2021 · Paper 1 · Variant 2 · Question 35

Correct answer: B

35 The diagram shows how the blood of a human fetus flows close to the mother's blood in the placenta.



Which substances are present at X in higher concentrations than at Y?

A carbon dioxide and glucose

B carbon dioxide and urea

C glucose and oxygen

D glucose and urea

May/June 2021 · Paper 1 · Variant 2 · Question 37

Correct answer: C

37 Which sex chromosomes are present in every sperm cell?

A X chromosome only

B Y chromosome only

C X or Y chromosome only

D X and Y chromosomes

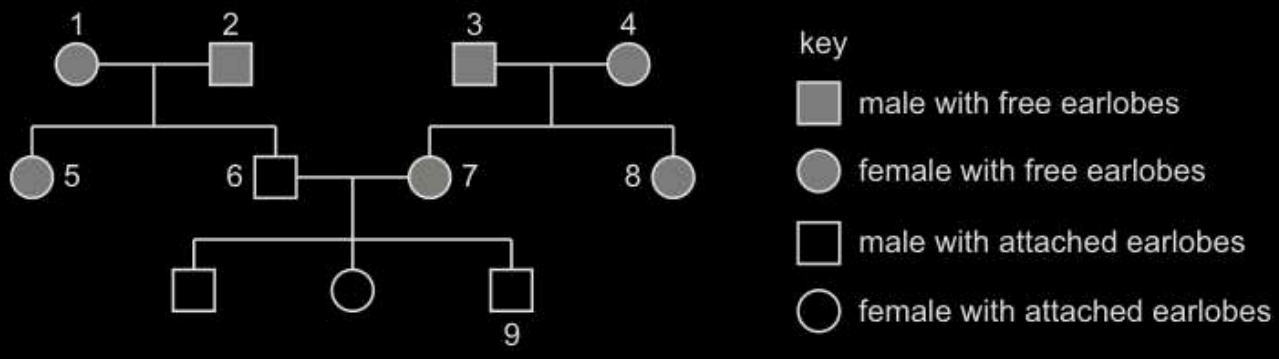
May/June 2021 · Paper 1 · Variant 2 · Question 38

Correct answer: A

38 Earlobes can either be attached to the cheek or be 'free' (unattached). This characteristic is controlled by a single gene.

The allele for attached earlobes is recessive.

The diagram shows the inheritance of attached earlobes in one family.



Which two individuals **must** be heterozygous for earlobe attachment?

A 1 and 7 **B** 3 and 4 **C** 5 and 8 **D** 6 and 9

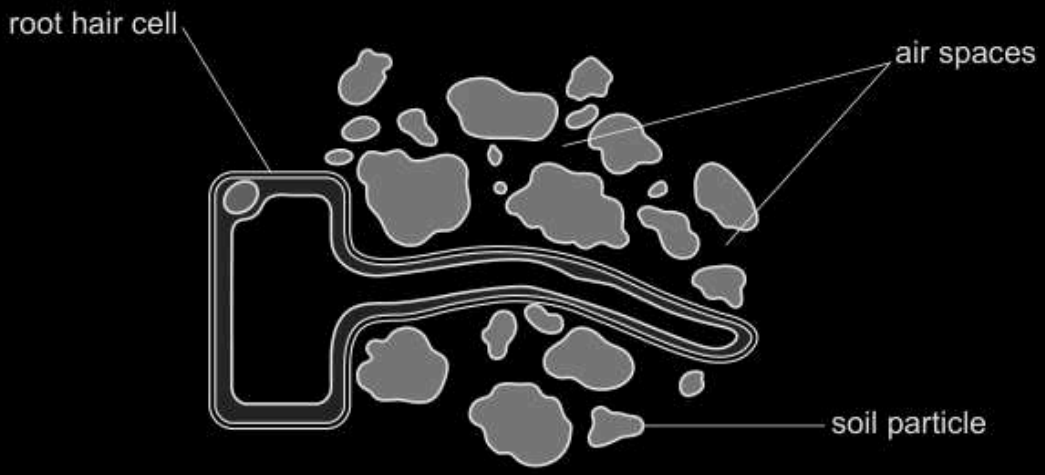
May/June 2021 · Paper 1 · Variant 2 · Question 39

Correct answer: B

39 Which statements about natural selection are correct?

	natural selection can lead to better adapted species surviving	natural selection can lead to extinction of a species	natural selection can lead to gene mutations occurring
A	true	true	true
B	true	true	false
C	true	false	true
D	false	true	true

1 The diagram shows a root hair cell from a lemon tree and some soil particles.



(a) (i) State **one** feature, visible in the diagram, which shows that a root hair cell is a plant cell.

..... [1]

(ii) Describe how mineral ions pass from the soil into the root hair cell.

.....

.....

.....

.....

.....

..... [4]

(b) The lemon tree has grown in the same soil for many years so the soil now lacks magnesium.

Describe and explain the effect on the lemon tree of a lack of magnesium.

.....

.....

..... [2]

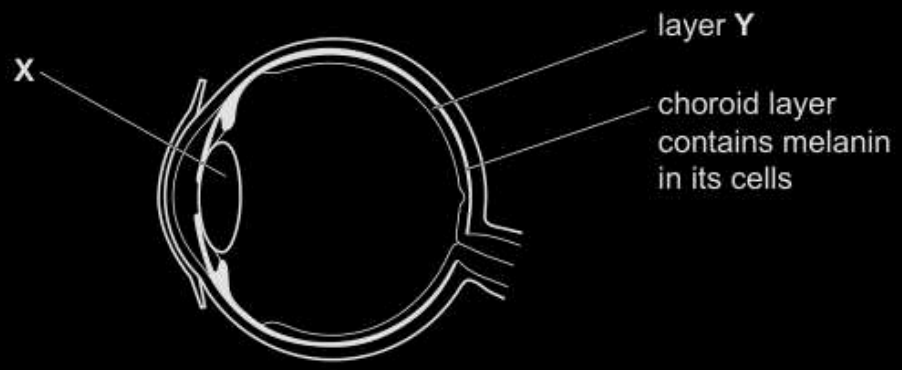
[Total: 7]

1(a)(i)	cell wall or large / central / sap **+** vacuole ;	**1**	
1(a)(ii)	*any four from:* dissolved in water ; through the cell wall ; carriers in membrane ; through the cell / partially permeable **AW +** membrane ; by active transport / diffusion ; correct reference to concentration gradient ; using energy (for active transport) ; from (aerobic) respiration (for active transport) ;	**4**	
1(b)	yellowing / chlorosis / reduced growth / cannot photosynthesise / make glucose / starch / sugar / carbohydrate ; deficiency of chlorophyll ;	**2**	

© UCLES 2021

Question	Answer	Marks	Guidance

2 A person visits an eye doctor to have an eye test. On the wall of the doctor's room is a diagram of an eye.



(a) (i) Name the part labelled **X**.

.....

[1]

(ii) Describe **one** way in which the structure of **X** is related to its function.

.....

.....

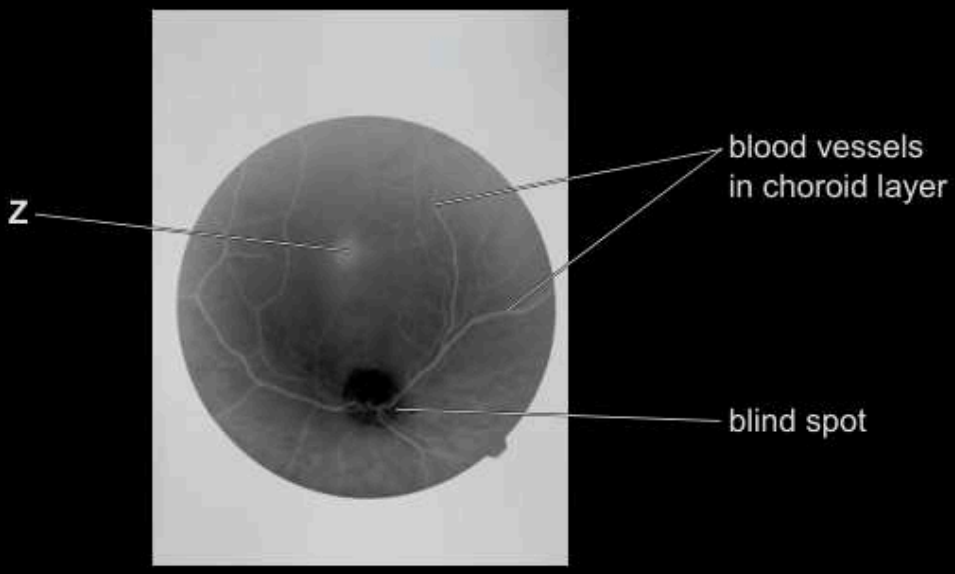
..... [2]

(iii) Name layer **Y**.

.....

[1]

(b) As part of the eye test, the doctor shines a bright light into the eye and takes a picture. The picture shows layer **Y** and the choroid layer behind it.



(i) Name structure **Z**.

.....

[1]

(ii) Layer **Y** does not contain blood vessels.
Explain how the blood vessels in the choroid layer are important for layer **Y**.

.....

.....

.....

.....

.....

..... [4]

(c) When the bright light is shone into the eye there is a pupil reflex which causes the pupil to become smaller.

Explain how named parts of the eye and of the nervous system are involved in this pupil reflex.

.....

.....

.....

.....

.....

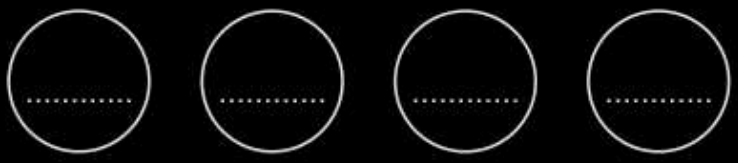
..... [4]

(d) Some people inherit a condition which prevents the production of the dark pigment melanin in their choroid and skin cells. The gene involved has two alleles, **A** (dominant) and **a** (recessive). Only people with two recessive alleles have this condition.

Two people, each with a heterozygous genotype, have children together.

(i) Complete the genetic diagram to show the possible genotypes and phenotypes of the children.

genotypes of parents ..... × .....

gametes 

possible genotypes of children ....

possible phenotypes of children ....

[4]

(ii) State the probability of these two people having a child who does **not** produce melanin.

..... [1]

[Total: 18]

2(a)(i)	lens ;	**1**	
2(a)(ii)	*any two from:* transparent / spherical / curved sides ; allows light to pass through / to focus / refract light ; **or** flexible / elastic / can change shape **AW** ; accommodation / focus objects ;	**2**	1 mark for structural feature and 1 mark for function
2(a)(iii)	retina ;	**1**	
2(b)(i)	macula / yellow spot / fovea ;	**1**	
2(b)(ii)	*any four from:* bring glucose / amino acids / minerals / vitamins ; for energy / repair of cells ; bring oxygen ; for (aerobic) respiration ; take urea / carbon dioxide away ; diffusion ; capillaries / veins / arteries mentioned in correct context ; bring white blood cells ; to fight infection ;	**4**	

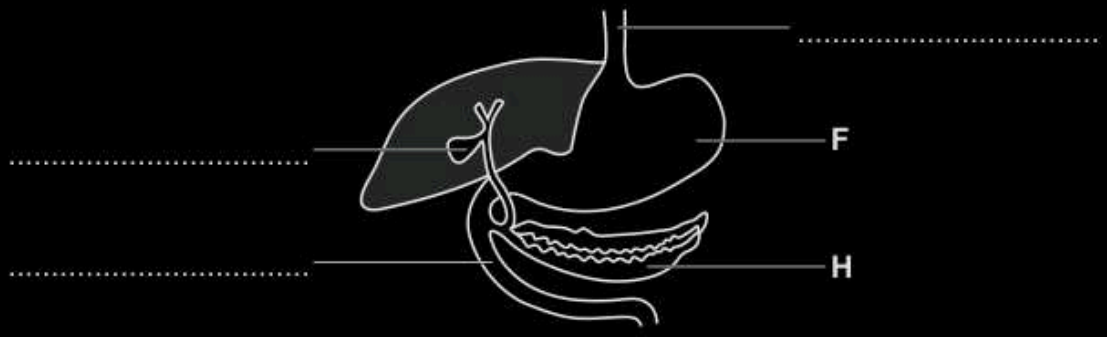
© UCLES 2021

Question	Answer	Marks	Guidance
2(c)	*any four from:* retina / rods detect extra light (intensity) ; impulse **+** to CNS / brain ; via optic nerve / sensory neurone ; back to eye / iris / effector / muscle **+** motor neurone ; radial muscles relax ; circular muscles contract ; iris becomes bigger / elongates ; reflex arc ;	**4**	
2(d)(i)	parental genotypes = Aa **+** Aa ; gametes a and A ; offspring genotypes AA, Aa, Aa and aa ; correctly linked phenotypes: AA & Aa = produce melanin, aa don't produce melanin ;	**4**	
2(d)(ii)	25% / ¼ / 0.25 / l in 4 / 1:3 ;	**1**	

© UCLES 2021

Question	Answer	Marks	Guidance

4 The diagram shows part of the human alimentary canal and its associated organs.



(a) Complete the three labels on the diagram by naming the structures. [3]

(b) Describe the functions of the structure labelled **F**.

.....

.....

.....

..... [3]

(c) Name structure **H** and describe its role in homeostasis.

name .....

role.....

.....

.....

[2]

[Total: 8]

4(a)	*clockwise from top:* oesophagus / gullet ; duodenum / small intestine ; gall bladder ;	**3**	 **R** ileum
4(b)	*any three from:* digestion / break down food or protein / protein to polypeptides / protein to amino acids ; (mechanical) muscle / contractions / movement **+** make food particles smaller / chyme ; (chemical) enzymes / proteases / pepsin **+** proteins ; hydrochloric acid **+** optimum pH **AW** ; hydrochloric acid **+** destroy microbes / defence against microbes ; mucus **+** protects lining / lubrication ;	**3**	
4(c)	pancreas ; control of blood glucose concentration / level **or** produces / releases **+** insulin / glucagon ;	**2**	

© UCLES 2021

Question	Answer	Marks	Guidance

May/June 2021 · Paper 2 · Variant 1 · Question 6

6 (a) Yoghurt is a milk product.
Outline the role of microorganisms in the process of yoghurt production.

.....

.....

.....

.....

.....

.....

..... [5]

(b) The diagram shows a food label from a pot of yoghurt.

contents	nutritional values in 100 g of yoghurt
energy	344 kJ
fat	4.5 g
total carbohydrates	5.5 g
sugar	5.5 g
fibre	0.0 g
protein	4.2 g
salt	0.1 g
calcium	0.125 g
water	81%

Yoghurt can contribute to a balanced diet but adult humans cannot survive by eating only yoghurt.

Explain this statement using the information on the label.

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

6(a)	*any five from:* heating / pasteurisation at start of process or after fermentation ; denaturation of milk protein / kill unwanted bacteria ; starter culture / bacteria ; *Lactobacillus (bulgaricus)* / *Streptococcus (thermophiles)* ; fermentation / anaerobic respiration ; warm temperature ; lactose / milk sugar ; lactic acid produced ; decreases pH ; coagulation / thickening ;	**5**	
6(b)	*any five from:* *(yoghurt contributes to a balanced diet)* carbohydrates for energy ; proteins for growth / repair ; water for chemical reactions / cytoplasm / transport ; fats for energy / insulation ; calcium for bones / prevent rickets ; *(yoghurt is not suitable on its own)* no **+** fibre **+** so poor bowel health / constipation ; no **+** iron ; no **+** vitamin C or other named vitamin ;	**5**	

© UCLES 2021

Question	Answer	Marks	Guidance

7 **(a)** Excess amino acids are broken down in liver cells to form molecules of urea.

Describe the pathway taken by a molecule of urea, from the liver of a mammal until it reaches the soil.

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

(b) Urea is a molecule which contains nitrogen.
Suggest why it is important that molecules that contain nitrogen are added to the soil.

.....

.....

.....

.....

..... [3]

[Total: 10]

Section C

Answer **either** Question 8 **or** Question 9.

Write your answers in the spaces provided.

7(a)	*any seven from:* transported in blood / plasma ; dissolved ; hepatic vein to vena cava / heart ; renal artery (to kidney) ; to kidney ; urea removed / filtered **+** from blood ; part of urine ; ureter (to bladder) ; bladder ; (bladder to) urethra ;	**7**	
7(b)	*any three from:* cycling / replenishing nitrogen ; used / converted by nitrifying bacteria / nitrogen fixing bacteria ; form **+** nitrates ; nitrates **+** taken up by plants ; production of proteins / amino acids ; increased soil fertility (for crops) / plants grow faster / bigger / **AW** ; animals / herbivores obtain protein / nitrogen from plants ;	**3**	

© UCLES 2021

Question	Answer	Marks	Guidance

8 **(a)** Rice, maize and wheat plants are the main carbohydrate source for more than 60% of the human population.

Describe how plants like these produce carbohydrates using materials from the environment.

.....

.....

.....

.....

.....

.....

..... [5]

(b) Many scientists predict that, as the world's human population increases, there will be a shortage of food.

Suggest and explain why an increase in human population is predicted to lead to a global human food shortage.

.....

.....

.....

.....

.....

.....

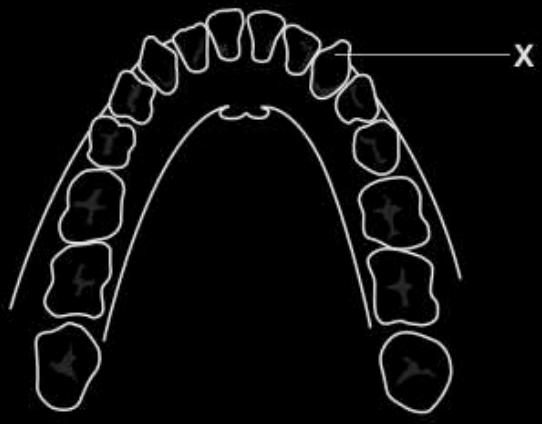
..... [5]

[Total: 10]

8(a)	*any five from:* photosynthesis ; in leaves / green parts of plants / in mesophyll / chloroplasts / using chlorophyll ; using light energy ; water **+** soil / irrigation / roots / root hair cells ; carbon dioxide **+** air / through stomata ; produce glucose / $C_6H_{12}O_6$; starch ; magnesium **+** from soil **+** to make chlorophyll ;	**5**	
8(b)	*any five from:* greater demand for food ; not enough land available for farming ; land being used for e.g. building / biofuels / deforestation ; increase in greenhouse gases ; (leads to) climate change ; droughts / floods / salinisation **AW** ; lack of nutrients in soils monoculture / overcrowding **AW** ;	**5**	

May/June 2021 · Paper 2 · Variant 2 · Question 1

1 The diagram shows the teeth in the lower jaw of a human.



(a) Name the type of tooth labelled **X** and describe **one** function of this type of tooth.

type of tooth .....

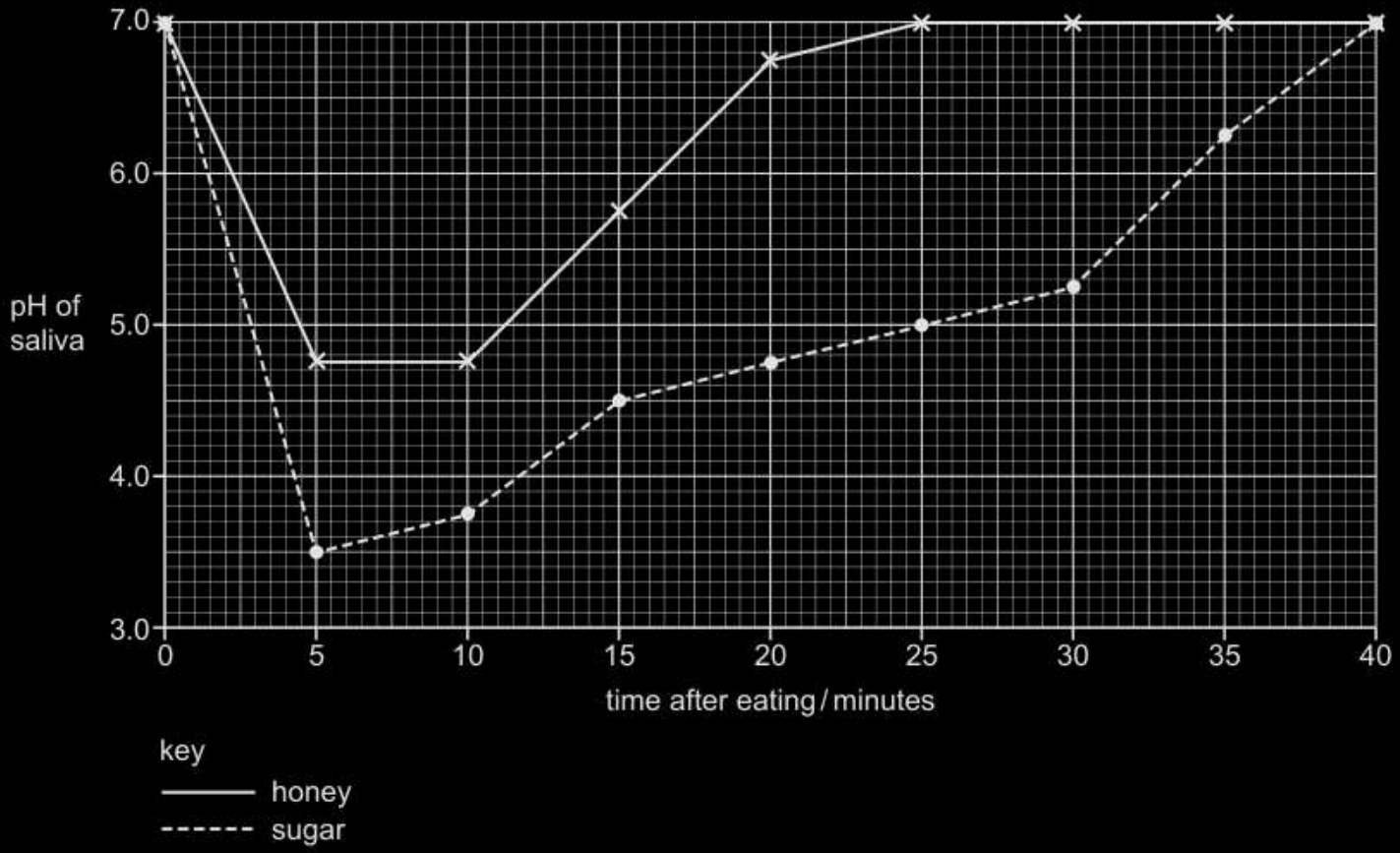
function .....

.....

[2]

(b) Food can be sweetened using honey or sugar.

The graph shows how the pH of saliva in the mouth changes with time after eating food sweetened with honey or sugar.



(i) State the lowest pH of saliva in the mouth after eating food sweetened with honey.

..... [1]

(ii) It takes more time for saliva to return to pH7.0 after eating food sweetened with sugar than after eating food sweetened with honey.

State how much more time it takes for the pH to return to 7.0.

..... [2]

(c) Dental decay is likely to occur when the pH of saliva falls below 5.5.

(i) Use information from the graph to explain whether sweetening food with honey or with sugar is more likely to lead to dental decay.

.....

.....

.....

.....

.....

..... [2]

(ii) State **two** ways to prevent dental decay.

1 .....

2 .....

[2]

(iii) A person with dental decay may also have gums that bleed. This makes it more likely that bacteria found in the mouth will enter the circulatory system.

Bacteria in the circulatory system can cause the blood to clot.

Small blood clots may move through the circulatory system to the coronary arteries.

Suggest and explain possible health problems that this may cause.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 13]

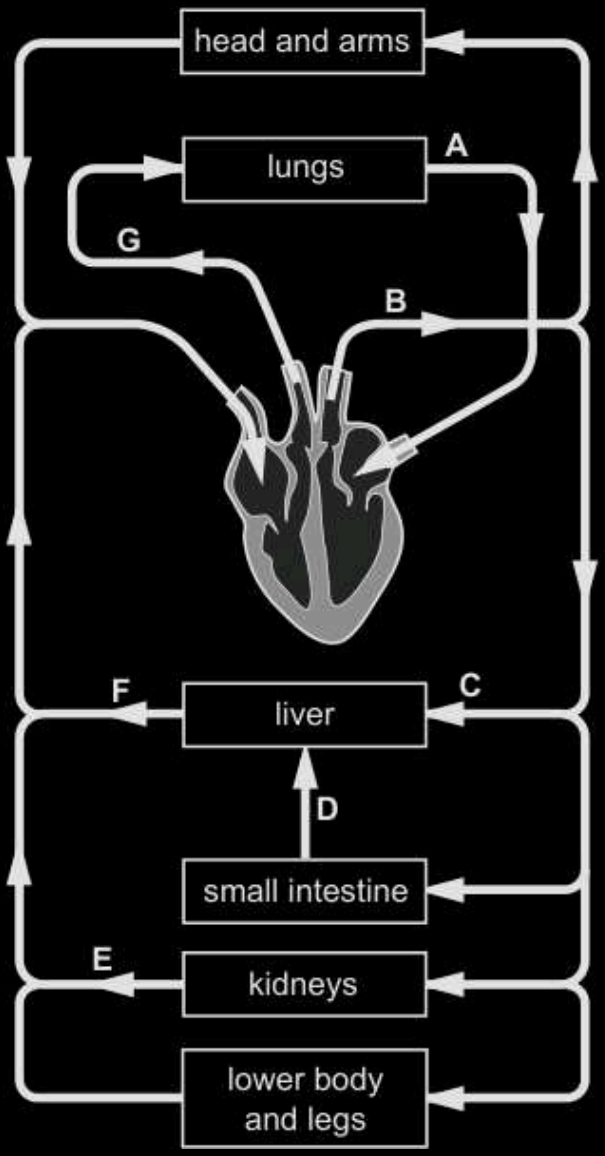
1(a)(i)	*(type of tooth)* **1** canine ; *(function)* **2** tearing / ripping / piercing ;	**2**	**Ig** chewing
1(b)(i)	4.75 ;	**1**	
1(b)(ii)	**1** 15 ; **2** minutes / min ;	**2**	
1(c)(i)	**1** lower pH / more acidic / any pH between 3.5 and 4.74 ; **2** longer time / any time between 25 and 35 minutes ;	**2**	
1(c)(ii)	**1** brushing / cleaning / toothpaste ; **2** flossing / mouthwash ; **3** fluoride ; **4** avoid sugar **AW** / use artificial sweetener ; **5** use honey rather than sugar ;	**2**	**Ig** dental check-ups
1(c)(iii)	**1** heart attack / angina **AW** / breathlessness ; **2** less ability **AW** of heart to **+** contract / pump ; **3** less blood to **+** body / tissues / organs or any named ; **4** less oxygen / less glucose to **+** body / tissues / organs or any named ; **5** anaerobic respiration / less aerobic respiration ; **6** lactic acid ; **7** less ability **AW** to carry out / fatigue **+** physical activity ;	**4**	

© UCLES 2021

Question	Answer	Marks	Guidance

2 The diagram shows the arrangement of major blood vessels in the human circulatory system.

Arrows show the direction that blood moves in each vessel.



(a) Seven of the blood vessels have been labelled on the diagram using the letters **A** to **G**.

Complete the table by writing **one** letter, **A** to **G**, for each name or description of a blood vessel.

The first row has been completed for you.

name or description	blood vessel
the aorta	**B**
carries blood containing the lowest concentration of urea	
one vessel that carries oxygenated blood	
carries blood at the highest pressure	
carries blood containing the highest concentration of glucose	
the hepatic vein	
an artery that carries deoxygenated blood	

[6]

(b) Explain why the movement of blood through the circulatory system of a human is described as a double circulation.

.....

.....

.....

.....

.....

..... [3]

[Total: 9]

2(a)	<table><tr><th>name or description</th><th>blood vessel</th></tr><tr><td>the aorta</td><td></td></tr><tr><td>carries blood containing the lowest concentration of urea</td><td>**E ;**</td></tr><tr><td>one vessel that carries oxygenated blood</td><td>**A / B / C ;**</td></tr><tr><td>carries blood at the highest pressure</td><td>**B ;**</td></tr><tr><td>carries blood containing the highest concentration of glucose</td><td>**D / F ;**</td></tr><tr><td>the hepatic</td><td>**F ;**</td></tr><tr><td>an artery that carries deoxygenated blood</td><td>**G ;**</td></tr></table>	**6**	
2(b)	**1** blood through heart twice **;** **2** heart / blood **+** to lungs / pulmonary **;** **3** heart / blood **+** to body / systemic **;**	**3**	

Question	Answer	Marks	Guidance

3 The diagram shows an elm tree.

(a) Elm trees may be infected by a fungus that causes a disease.

State **two** characteristics of a fungus.

1 .....

2 .....

[2]

(b) The fungus moves through an infected tree inside the xylem vessels.

The tree responds to stop the movement of the fungus by producing substances that block the xylem vessels. This causes the leaves of the tree to wilt and to become yellow.

Explain how the response of the tree to infection leads to these symptoms of disease.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

(c) An insect called the elm bark beetle is the vector of this disease.

Describe what is meant by the term **vector**.

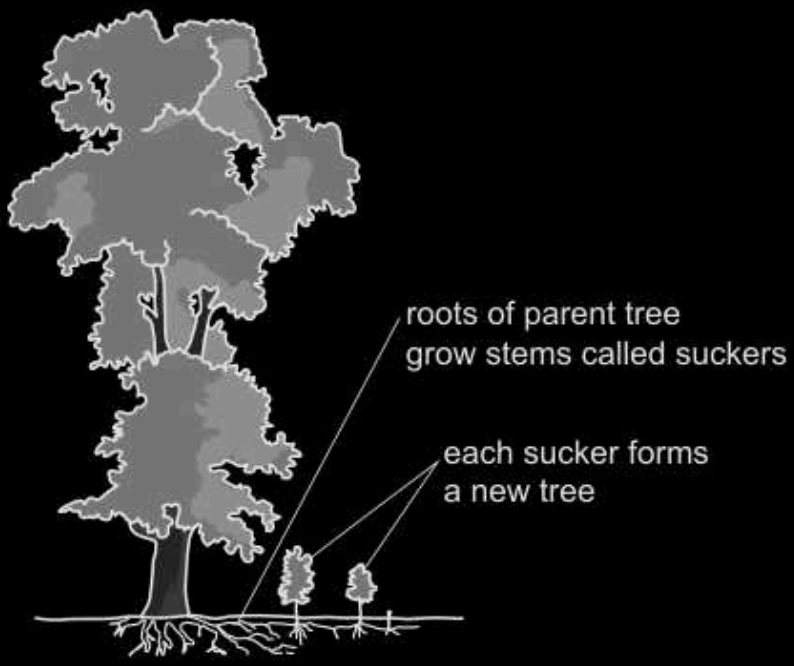
.....

.....

.....

..... [2]

(d) Elm trees can reproduce when the roots from one parent tree spread out near the surface of the soil. This is shown in the diagram below.



(i) Name the type of reproduction shown in the diagram.

..... [1]

(ii) Suggest why this type of reproduction makes it difficult to control the spread of the disease caused by the fungus.

.....

.....

.....

.....

.....

..... [3]

[Total: 13]

3(a)	**1** hyphae / mycelium ; **2** saprotrophic / saprophytic / feeds on dead matter **AW** / decomposer ; **3** spores ; **4** cell wall **+** chitin ; **5** more than one nucleus / multinucleate ;	**2**	

Question	Answer	Marks	Guidance
3(b)	**1** less / no **+** water transported / water absorbed / transpiration ; **2** water loss greater than water uptake ; **3** less **+** water in cells ; **4** flaccid / loss of turgor / plasmolysed ; **5** magnesium / Mg ; **6** less / no **+** chlorophyll ;	**5**	
3(c)	**1** transfer / carries / spreads **AW +** pathogen / fungus / parasite ; **2** one organism **AW** to another ; **3** not itself **+** affected / infected ;	**2**	
3(d)(i)	asexual ;	**1**	
3(d)(ii)	**1** close together ; **2** connected **AW** ; **3** genetically identical / all genetically susceptible / clones ;	**3**	

Question	Answer	Marks	Guidance

4 Cardiopulmonary resuscitation (CPR) is a first aid procedure.

Air is forced into a patient's lungs by another person. The heart is made to pump blood by repeated compression of the patient's chest over the heart.

A person performing CPR on a patient is shown in the diagrams.

air forced into lungs

repeated compression of chest

(a) (i) State what will be seen to happen to the patient's chest when air is forced into the lungs.

..... [1]

(ii) Name, in the correct order, the tubes through which air will travel from the patient's mouth to the alveoli of the lungs.

[2]

(iii) The air forced into the patient's lungs has different concentrations of gases compared to the air the patient would normally breathe in.

State **two** differences in the concentrations of gases.

.....

.....

..... [2]

(b) Explain how each action of the CPR procedure will benefit the patient:

air forced into the patient's lungs

.....

.....

.....

.....

repeated compression of the patient's chest.

.....

.....

.....

.....

[4]

[Total: 9]

4(a)(i)	rises / expands **AW** ;	**1**	
4(a)(ii)	ALL THREE from the list below = **2 marks** ;; **OR** ANY ONE or TWO from the list below = **1 mark** ; left box = larynx / trachea middle box = bronchi / bronchus right box = bronchiole(s)	**2**	left box **Ig** windpipe
4(a)(iii)	*air forced into the lungs has ...* **1** less oxygen ; **2** more carbon dioxide ; **3** more water vapour ;	**2**	

Question	Answer	Marks	Guidance
4(b)	**1** oxygen **+** lungs / alveoli / air sacs ; **2** oxygen **+** blood / red blood cells / RBC / erythrocytes / haemoglobin ; **3** diffusion ; **4** high to low concentration / down concentration gradient ; **5** blood from **+** left ventricle ; **6** blood to **+** organs / tissues / cells ; **7** aerobic respiration ; **8** re-start the heart ;	**4**	

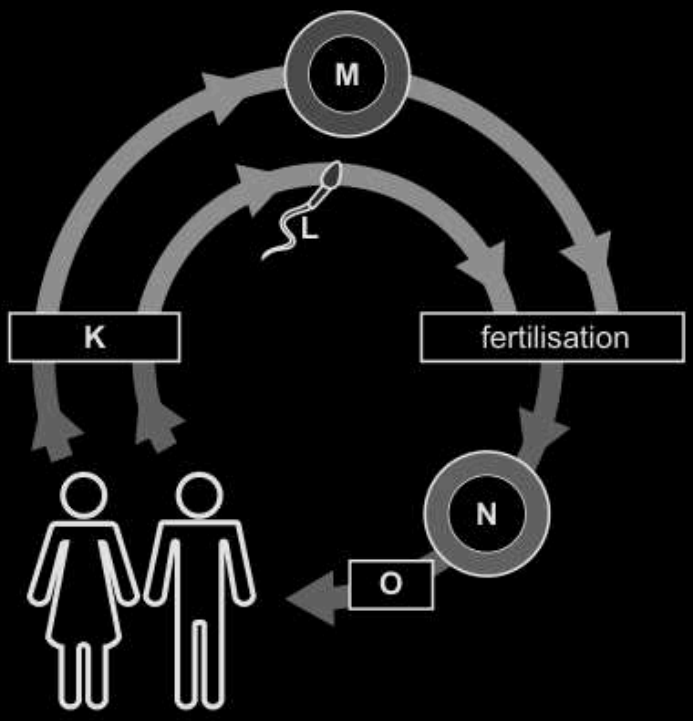
Question	Answer	Marks	Guidance

6 The diagram shows the human life cycle.

The components of the diagram are **not** drawn to the same scale.

The diagram shows:

- cells **L**, **M** and **N**
- two types of cell division, **K** and **O**.



(a) **L** and **M** are both the same type of cell.
Name this type of cell.

.....

[1]

(b) Outline differences between the nucleus of cell **M** and the nucleus of cell **N**.

.....

.....

.....

..... [3]

(c) Name and compare the types of cell division **K** and **O**.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

[Total: 10]

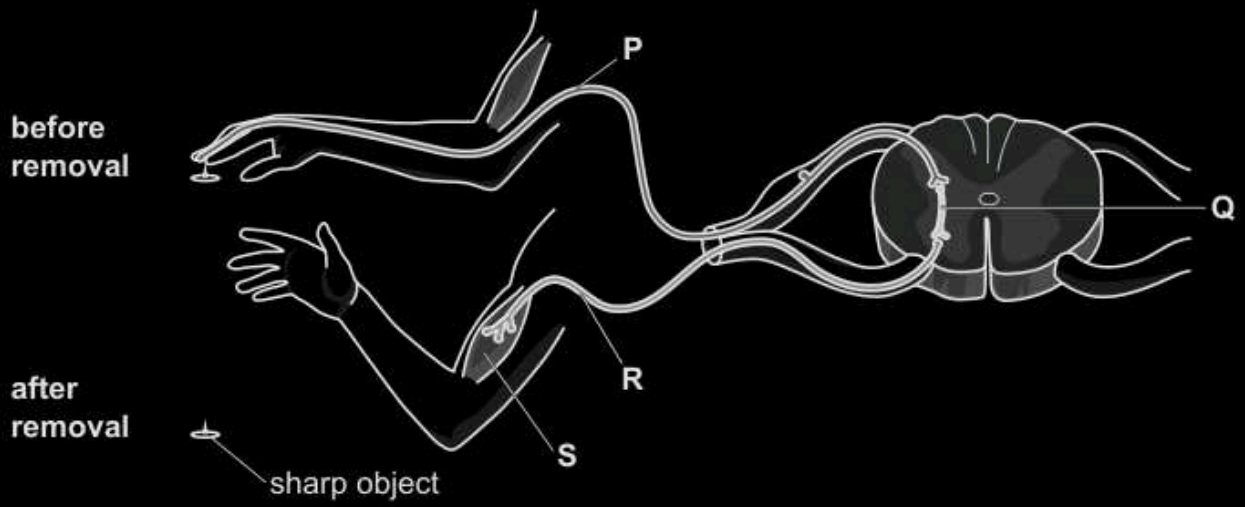
6(a)	gamete(s) ;	**1**	**Ig** sex cell
6(b)	**1** **M +** haploid / 23 chromosomes ; **2** **M +** only X chromosome ; **3** **N +** diploid / 46 chromosomes / 23 pairs of chromosomes ; **4** **N +** either XX or XY chromosomes ;	**3**	

© UCLES 2021

Question	Answer	Marks	Guidance
6(c)	**1** **K +** meiosis ;	**1**	
	2 **O +** mitosis ;	**1**	
	*(marking points **3–8** must be linked to meiosis or to **K** if meiosis not named)* **3** reduction division ; **4** chromosome **+** number halved / 46 to 23 ; **5** diploid **+** to haploid ; **6** four daughter **AW** cells ; **7** genetically different ; **8** takes place in **+** testes / ovaries / gonads ; *(marking points **9–13** must be linked to mitosis or to **O** if mitosis not named)* **9** chromosome **+** number maintained / 46 to 46 ; **10** diploid **+** to diploid ; **11** genetically identical / clones ; **12** growth ; **13** two daughter **AW** cells ;	**4**	

Question	Answer	Marks	Guidance

7 The diagram shows structures involved in a reflex action that results in removal of a person's hand from a sharp object.



(a) (i) Name and state the function of:

structure **P**

.....

.....

.....

structure **R**.

.....

.....

.....

[4]

(ii) Describe the location of structure **Q** in the nervous system.

.....

..... [2]

(b) Explain how the action of structure **S** results in the movement of bones at a **named** type of joint.

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

Section C

Answer **either** Question 8 **or** Question 9.

Write your answers in the spaces provided.

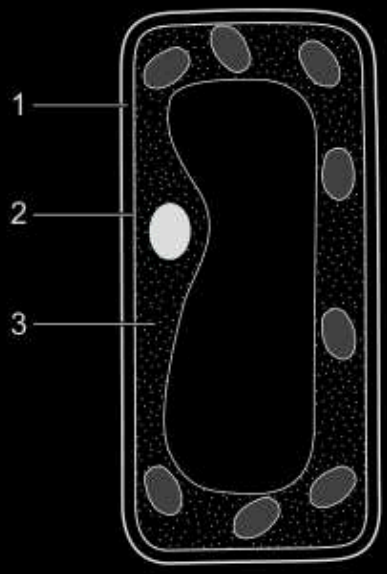
7(a)(i)	*(structure **P**)* **1** sensory / afferent **+** neurone / nerve ; **2** from **+** receptor ; *(structure **R**)* **3** motor / efferent **+** neurone / nerve ; **4** to **+** effector / muscle / biceps / **S** ; **5** impulse ;	**4**	
7(a)(ii)	**1** CNS / central nervous system / grey matter ; **2** spinal cord ;	**2**	

Question	Answer	Marks	Guidance
7(b)	**1** hinge ;	**1**	
	any three from: **2** contracts ; **3** pulls / jerks ; **4** radius **+** ulna ; **5** up / towards humerus / away from / flexion ;	**3**	

October/November 2020 · Paper 1 · Variant 1 · Question 1

Correct answer: D

1 The diagram shows the structure of a cell from a leaf of a green plant.



What are the numbered parts of the cell?

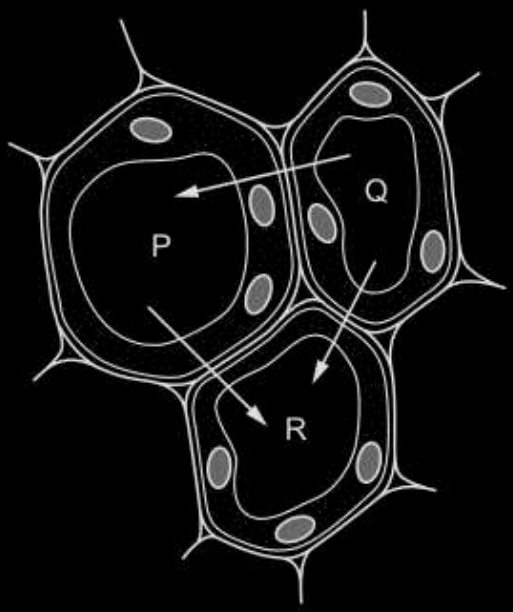
	1	2	3
A	cell membrane	cell wall	cytoplasm
B	cell membrane	cytoplasm	cell wall
C	cell wall	cytoplasm	cell membrane
D	cell wall	cell membrane	cytoplasm

© UCLES 2020

October/November 2020 · Paper 1 · Variant 1 · Question 2

Correct answer: C

2 The diagram shows three plant cells labelled P, Q and R. The arrows show the direction of water movement by osmosis.

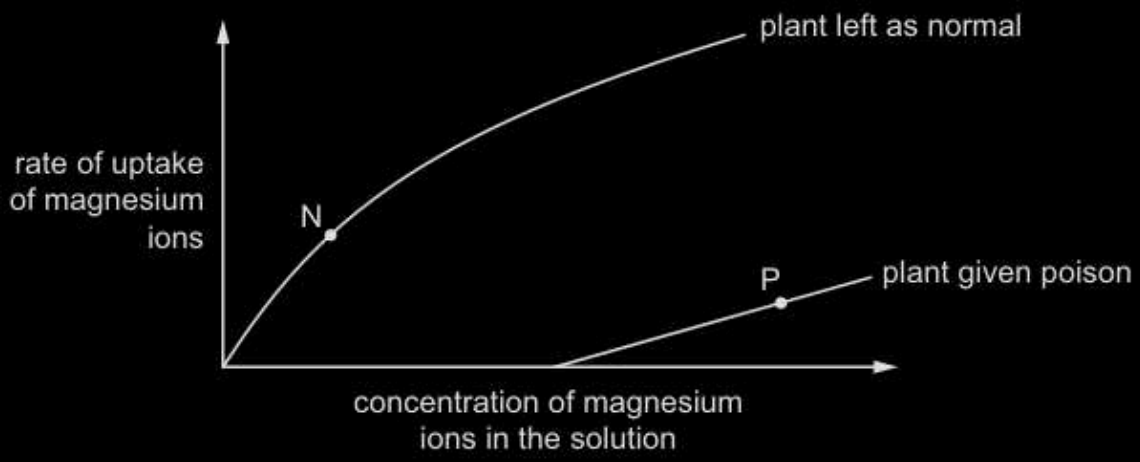


What is the correct order of water potential in the cells, from the highest to the lowest?

	highest	middle	lowest
A	P	Q	R
B	P	R	Q
C	Q	P	R
D	R	P	Q

Correct answer: B

3 An experiment measured the rate at which plants take up magnesium ions from solution. One plant was given a poison that stops respiration. Another plant was left as normal. The graph shows the results.



How are the magnesium ions being absorbed by the plants at points N and P?

	point N	point P
A	active transport	active transport
B	active transport	diffusion
C	diffusion	active transport
D	diffusion	diffusion

October/November 2020 · Paper 1 · Variant 1 · Question 4

Correct answer: C

4 Starch digestion occurs in the mouth cavity and in the duodenum but it stops in the stomach.

Why is this?

A All the starch has been digested before it reaches the stomach.

B Cells in the stomach do not produce amylase.

C The pH in the stomach changes the shape of the amylase.

D The temperature in the stomach is too high for amylase to work.

© UCLES 2020

October/November 2020 · Paper 1 · Variant 1 · Question 6

Correct answer: B

6 How do carbon dioxide and water enter a leaf?

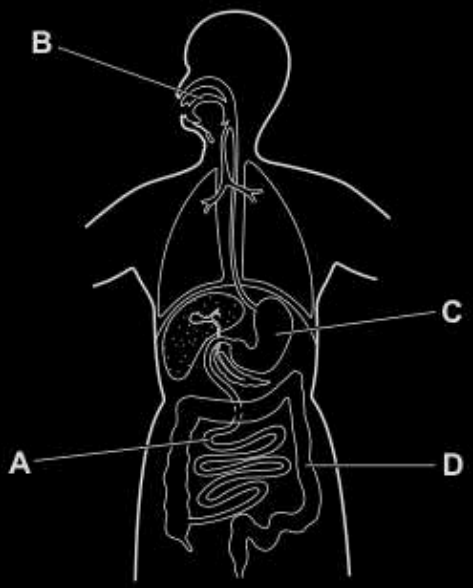
	carbon dioxide	water
A	diffusion	active transport
B	diffusion	transpiration pull
C	osmosis	active transport
D	osmosis	transpiration pull

October/November 2020 · Paper 1 · Variant 1 · Question 8

Correct answer: C

8 The diagram shows the human alimentary canal.

In which part does protein digestion begin?



October/November 2020 · Paper 1 · Variant 1 · Question 9

Correct answer: C

9 Blood flows away from the small intestine in the hepatic portal vein.

Where does the blood go next, and what may happen there?

A the kidneys, delivering glucose for respiration

B the kidneys, for removal of urea

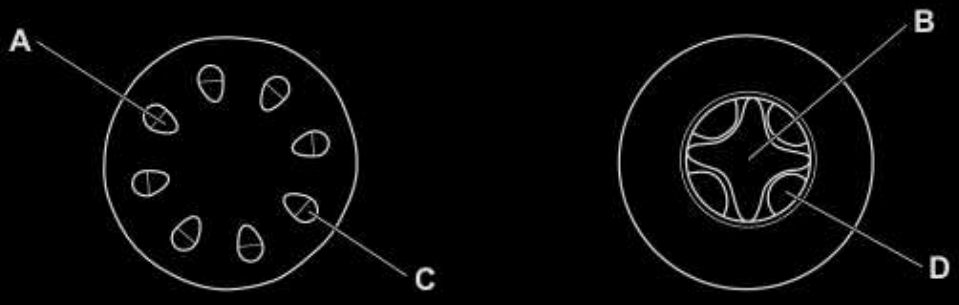
C the liver, where alcohol is broken down

D the liver, where bile is added to the blood

October/November 2020 · Paper 1 · Variant 1 · Question 11

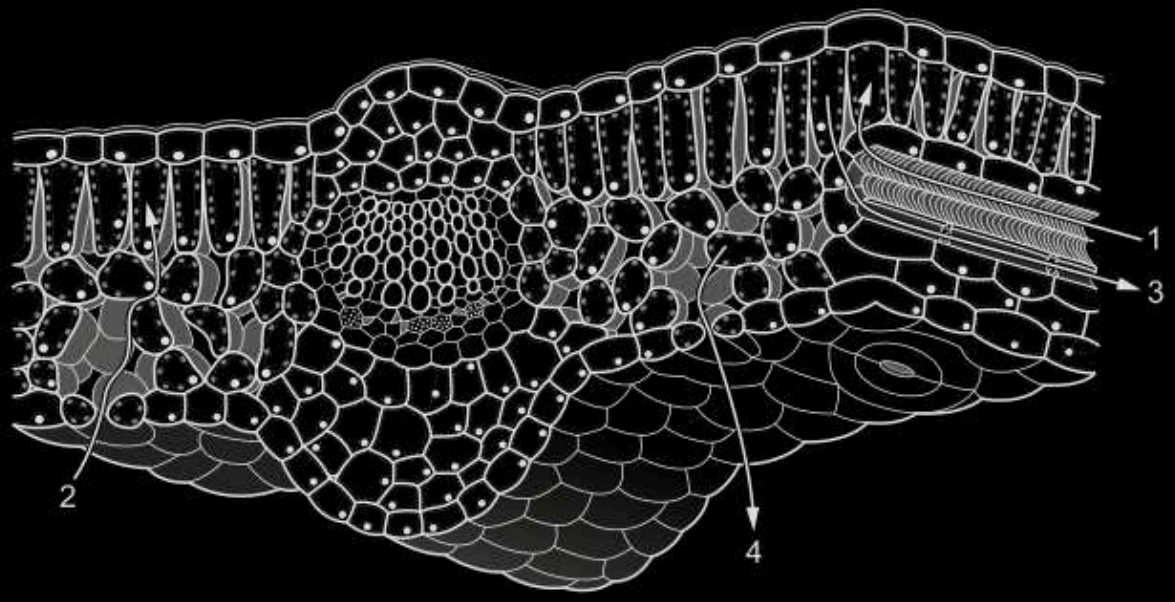
Correct answer: D

11 Which diagram shows the position of phloem tissue in a herbaceous, dicotyledonous root?



Correct answer: C

12 The diagram represents part of a leaf.



Which arrows show translocation and transpiration?

	translocation	transpiration
A	1	2
B	2	3
C	3	4
D	4	1

© UCLES 2020

October/November 2020 · Paper 1 · Variant 1 · Question 13

Correct answer: B

13 In the process of blood clotting, the ……1…… help to convert ……2…… to ……3…… which forms a clot.

Which words complete the description of blood clotting?

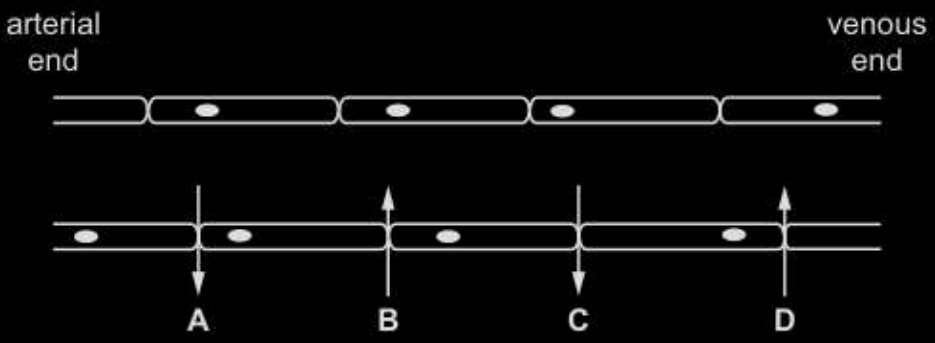
	1	2	3
A	platelets	fibrin	fibrinogen
B	platelets	fibrinogen	fibrin
C	red blood cells	fibrin	fibrinogen
D	red blood cells	fibrinogen	fibrin

October/November 2020 · Paper 1 · Variant 1 · Question 14

Correct answer: A

14 The diagram shows a section of capillary.

Which arrow represents tissue fluid formation?



October/November 2020 · Paper 1 · Variant 1 · Question 15

Correct answer: B

15 Which row shows possible causes of coronary heart disease?

	diet high in saturated fats	diet low in salt	regular exercise
A	✓	✓	✗
B	✓	✗	✗
C	✗	✓	✓
D	✗	✗	✓

key
✓ = yes
✗ = no

October/November 2020 · Paper 1 · Variant 1 · Question 19

Correct answer: B

19 During dialysis, substances such as glucose, mineral salts, urea and water move between the patient's blood and the dialysis fluid.

Which processes are involved in these movements?

	active transport	diffusion
A	yes	yes
B	no	yes
C	yes	no
D	no	no

© UCLES 2020

October/November 2020 · Paper 1 · Variant 1 · Question 20

Correct answer: A

20 Which organs in humans are involved in maintaining body temperature?

	brain	skin
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

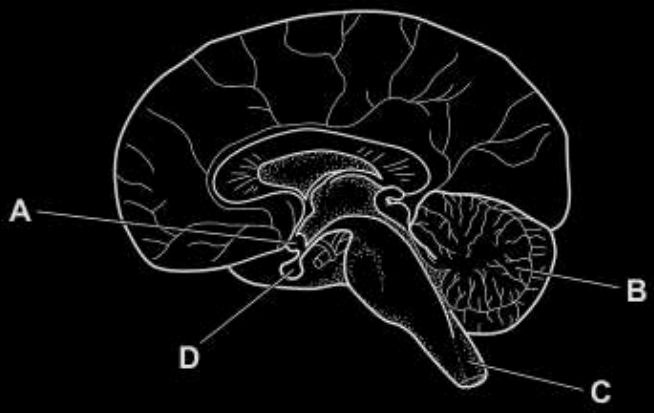
✓ = yes

✗ = no

Correct answer: A

21 The diagram shows a section through the brain.

Which part is the hypothalamus?

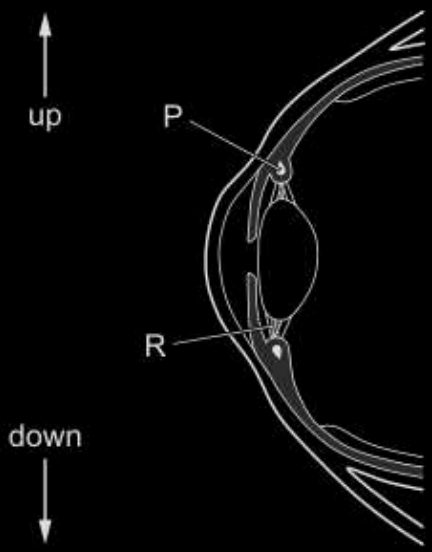


© UCLES 2020

[Turn over

Correct answer: A

22 The diagram shows a section through the eye.



When the person looks down and focuses on their mobile phone, what is the state of the structures P and R?

	P	R
A	contracted	loose
B	contracted	tight
C	relaxed	loose
D	relaxed	tight

October/November 2020 · Paper 1 · Variant 1 · Question 23

Correct answer: B

23 What is a sign of diabetes mellitus?

A glucose in the blood

B glucose in the urine

C insulin in the blood

D insulin in the urine

© UCLES 2020

October/November 2020 · Paper 1 · Variant 1 · Question 26

Correct answer: A

26 The table shows the characteristics of four microorganisms.

Which one could be a virus?

	contains one or more cells	contains one or more cell nuclei	produces spores
A	✗	✗	✗
B	✓	✗	✗
C	✓	✓	✗
D	✓	✓	✓

key

✓ = true

✗ = false

© UCLES 2020

[Turn over

October/November 2020 · Paper 1 · Variant 1 · Question 27

Correct answer: C

27 Which statement comparing the production of single cell protein and the production of protein in meat is **not** correct?

A Single cell protein can be produced more quickly.

B Production of single cell protein is independent of climatic changes.

C Production of single cell protein requires more space.

D Single cell protein contains less fat and more dietary fibre.

October/November 2020 · Paper 1 · Variant 1 · Question 35

Correct answer: B

35 What is an advantage to a baby of breast-feeding compared with bottle feeding?

A Only breast milk contains all the vitamins a growing baby needs.

B Only breast milk contains antibodies.

C Only breast milk is sterile.

D Only breast milk provides the correct volume of milk for the baby.

© UCLES 2020

Correct answer: A

37 In a species of mouse, fur colour can be black or white. Two black female mice were allowed to mate with the same black male. One female had nine young, all of which were black. The other female had seven young, five black and two white.

One of these white mice was male, and is allowed to mate with a heterozygous female.

What is the expected ratio of phenotypes of their offspring?

A 1 black : 1 white

B 1 black : 3 white

C 1 black : 2 grey : 1 white

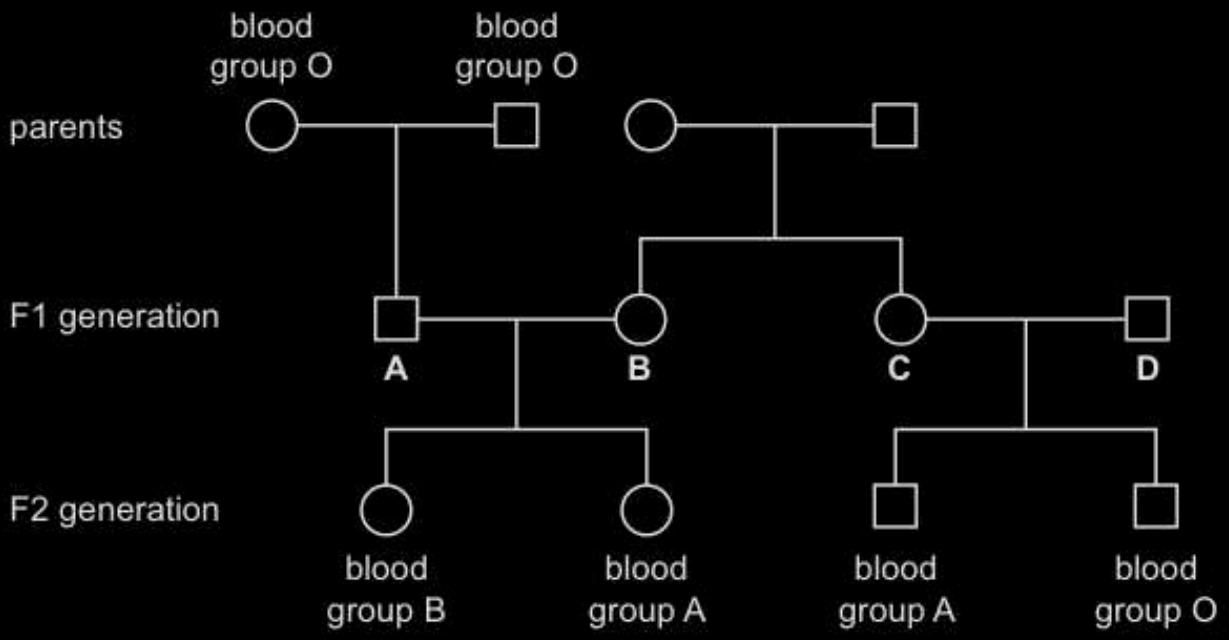
D 3 black : 1 white

© UCLES 2020

Correct answer: B

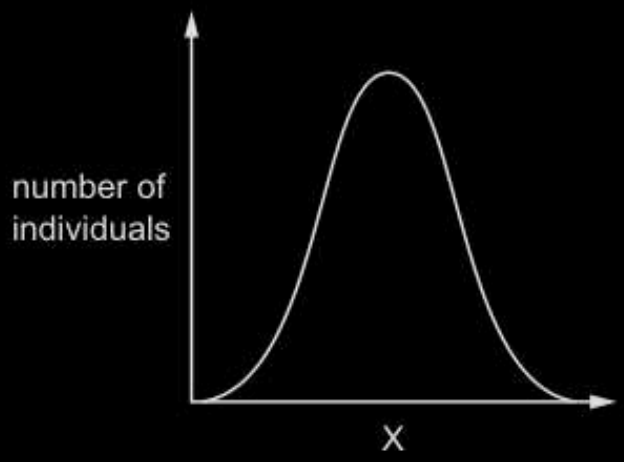
38 The diagram shows the blood group phenotypes of some members of a family.

Which member of the F1 generation must be heterozygous with the codominant alleles?



Correct answer: C

39 The diagram shows the distribution of a human characteristic, X, in the population.



What is the characteristic X?

A blood group

B eye colour

C height

D sex

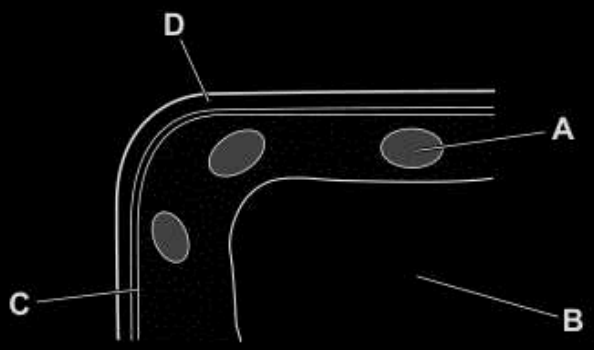
© UCLES 2020

October/November 2020 · Paper 1 · Variant 2 · Question 1

Correct answer: C

1 The diagram shows part of a plant cell.

Which part controls the entry of substances into the cell?



October/November 2020 · Paper 1 · Variant 2 · Question 2

Correct answer: A

2 Which statements about diffusion are correct?

1 Molecules move at random.

2 Molecules move down a concentration gradient.

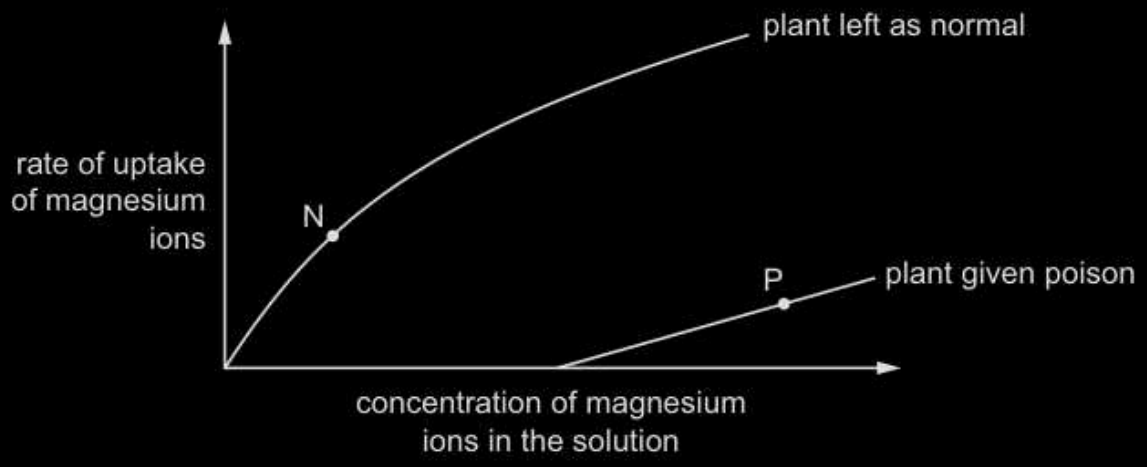
3 Molecules may move through a partially permeable membrane.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

October/November 2020 · Paper 1 · Variant 2 · Question 3

Correct answer: B

3 An experiment measured the rate at which plants take up magnesium ions from solution. One plant was given a poison that stops respiration. Another plant was left as normal. The graph shows the results.



How are the magnesium ions being absorbed by the plants at points N and P?

	point N	point P
A	active transport	active transport
B	active transport	diffusion
C	diffusion	active transport
D	diffusion	diffusion

October/November 2020 · Paper 1 · Variant 2 · Question 4

Correct answer: C

4 Starch digestion occurs in the mouth cavity and in the duodenum but it stops in the stomach.

Why is this?

A All the starch has been digested before it reaches the stomach.

B Cells in the stomach do not produce amylase.

C The pH in the stomach changes the shape of the amylase.

D The temperature in the stomach is too high for amylase to work.

October/November 2020 · Paper 1 · Variant 2 · Question 5

Correct answer: B

5 Which part of a leaf absorbs and uses carbon dioxide from the air?

A cuticle

B mesophyll

C phloem

D xylem

October/November 2020 · Paper 1 · Variant 2 · Question 6

Correct answer: B

6 How do carbon dioxide and water enter a leaf?

	carbon dioxide	water
A	diffusion	active transport
B	diffusion	transpiration pull
C	osmosis	active transport
D	osmosis	transpiration pull

October/November 2020 · Paper 1 · Variant 2 · Question 8

Correct answer: C

8 What are the substrates and end products for amylase and lipase?

	amylase		lipase	
	substrate	end product	substrate	end product
A	protein	amino acids	fat	fatty acids and glycerol
B	protein	amino acids	glycogen	glucose
C	starch	maltose	fat	fatty acids and glycerol
D	starch	maltose	protein	amino acids

October/November 2020 · Paper 1 · Variant 2 · Question 9

Correct answer: A

9 What is the main function of the villi in the ileum?

A to increase the surface area for absorption

B to move digested food along the intestine by peristalsis

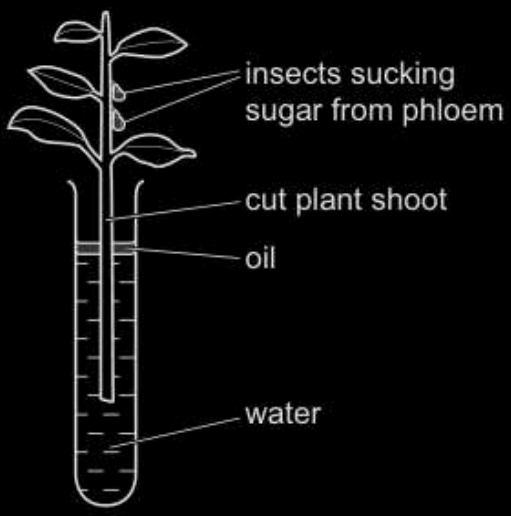
C to produce mucus so that materials can move smoothly

D to secrete amylase to digest starch in the food

October/November 2020 · Paper 1 · Variant 2 · Question 12

Correct answer: C

12 The diagram shows a cut plant shoot in a container of water.



What will stop the movement of water up this stem?

A a fall in the humidity of the air

B a rise in the air temperature

C a blockage in the xylem

D an insect sucking sugar from the phloem

October/November 2020 · Paper 1 · Variant 2 · Question 13

Correct answer: D

13 Which pathway does oxygenated blood follow in the human body?

A pulmonary artery → left and right atria → left and right ventricles

B pulmonary artery → left atrium → left ventricle → aorta

C pulmonary vein → left and right atria → left and right ventricles

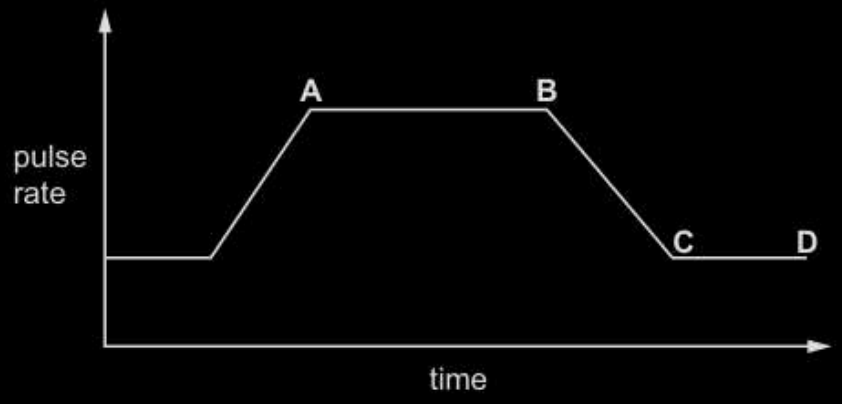
D pulmonary vein → left atrium → left ventricle → aorta

October/November 2020 · Paper 1 · Variant 2 · Question 14

Correct answer: B

14 The graph shows a person's pulse rate over a period of time during which they rest, pedal an exercise bike and then rest again.

At which point did the person stop pedalling?



October/November 2020 · Paper 1 · Variant 2 · Question 15

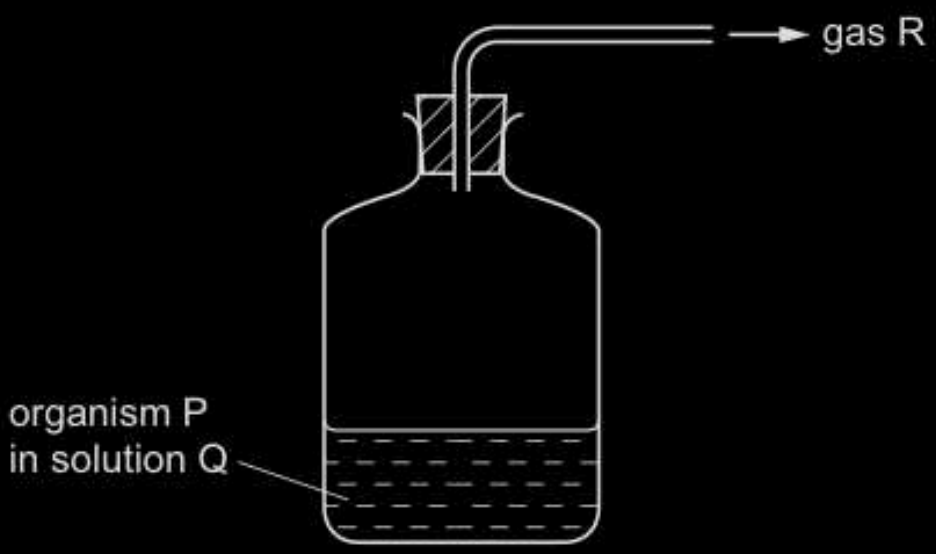
Correct answer: D

15 Which row shows the blood pressures in an artery, a capillary and a vein?

	pressure / kPa		
	artery	capillary	vein
A	0.6	4.0	13.0
B	4.0	0.6	13.0
C	13.0	0.6	4.0
D	13.0	4.0	0.6

Correct answer: C

17 The apparatus shown is used to make alcohol by anaerobic respiration.



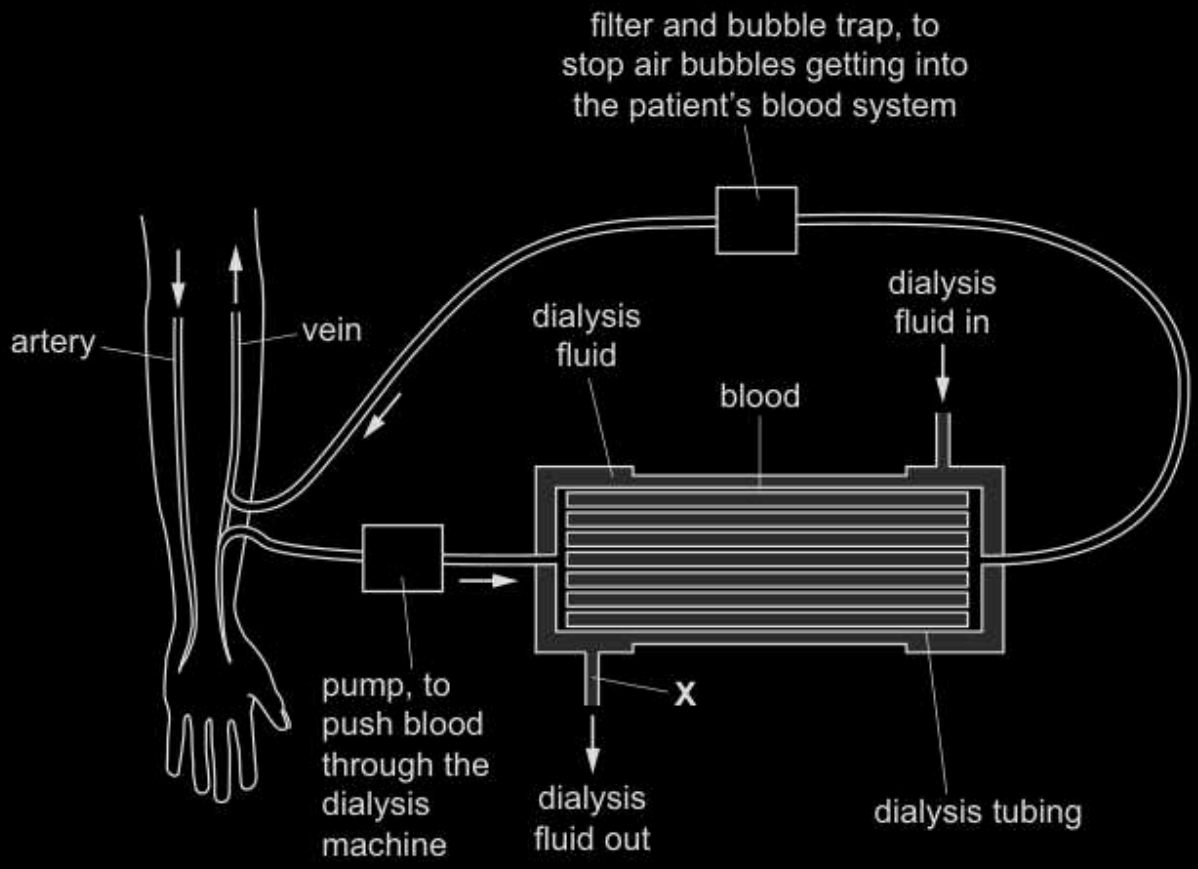
What are P, Q and R?

	P	Q	R
A	amylase	starch	oxygen
B	bacterium	amino acids	nitrogen
C	fungus	glucose	carbon dioxide
D	virus	sucrose	water vapour

October/November 2020 · Paper 1 · Variant 2 · Question 19

Correct answer: C

19 The diagram shows a dialysis machine.



Which substance would **not** be present in a sample of the dialysis fluid taken from point **X**?

A glucose molecules

B salt ions

C plasma proteins

D urea molecules

October/November 2020 · Paper 1 · Variant 2 · Question 20

Correct answer: B

20 Which part of the skin has a major role in insulating the human body?

A blood vessels

B fatty tissue

C hair follicles

D sweat glands

October/November 2020 · Paper 1 · Variant 2 · Question 21

Correct answer: A

21 What describes a reflex action?

A It is a rapid response to a stimulus, involving the spinal cord.

B It is a rapid response detected by motor neurones.

C It is a slow response to a stimulus, involving the spinal cord.

D It is a slow response detected by motor neurones.

October/November 2020 · Paper 1 · Variant 2 · Question 23

Correct answer: B

23 What is a sign of diabetes mellitus?

A glucose in the blood

B glucose in the urine

C insulin in the blood

D insulin in the urine

October/November 2020 · Paper 1 · Variant 2 · Question 26

Correct answer: A

26 The table shows the characteristics of four microorganisms.

Which one could be a virus?

	contains one or more cells	contains one or more cell nuclei	produces spores
A	✗	✗	✗
B	✓	✗	✗
C	✓	✓	✗
D	✓	✓	✓

key

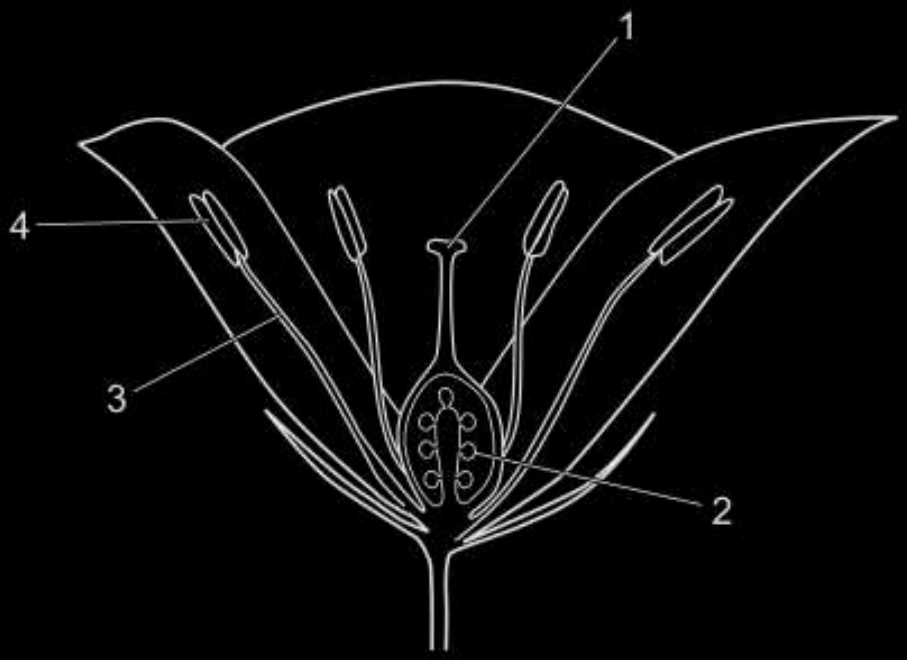
✓ = true

✗ = false

October/November 2020 · Paper 1 · Variant 2 · Question 33

Correct answer: C

33 The diagram shows a flower cut in half.



Which **two** parts of the flower produce haploid gametes?

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

October/November 2020 · Paper 1 · Variant 2 · Question 37

Correct answer: A

37 In a species of mouse, fur colour can be black or white. Two black female mice were allowed to mate with the same black male. One female had nine young, all of which were black. The other female had seven young, five black and two white.

One of these white mice was male, and is allowed to mate with a heterozygous female.

What is the expected ratio of phenotypes of their offspring?

A 1 black : 1 white

B 1 black : 3 white

C 1 black : 2 grey : 1 white

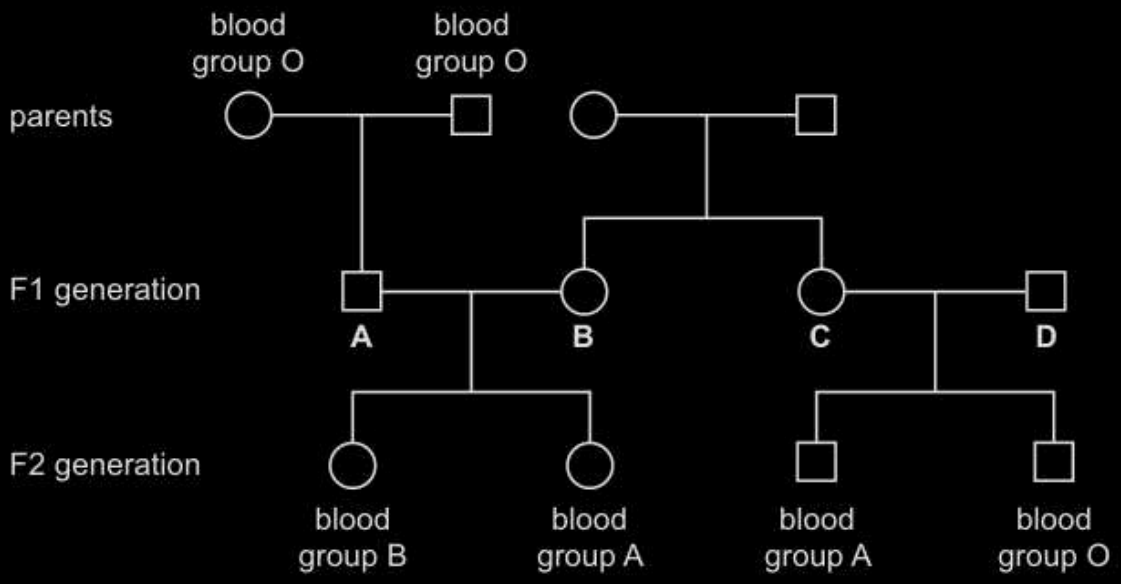
D 3 black : 1 white

October/November 2020 · Paper 1 · Variant 2 · Question 38

Correct answer: B

38 The diagram shows the blood group phenotypes of some members of a family.

Which member of the F1 generation must be heterozygous with the codominant alleles?



October/November 2020 · Paper 1 · Variant 2 · Question 39

Correct answer: D

39 Which human characteristic shows discontinuous variation?

A body mass

B heart rate

C height

D sickle cell anaemia

Correct answer: C

40 A farmer wants to produce extra-large, sweet oranges, by selective breeding.

Using information from the table, which plant types should the farmer select for breeding?

plant type	size of orange / cm			percentage sugar content	
	6–9	9–12	12–15	10	20
1		✗		✗	
2			✗	✗	
3	✗				✗
4		✗			✗
5	✗			✗	

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 4 and 5

2 Corals are a collection of small animals living together in the sea. Their skeletons produce coral reefs. The cells of these animals contain microscopic, green, plant-like organisms called algae.

(a) Suggest the name of the process by which the algae make their food and explain your answer.

process .....

explanation .....

..... [2]

(b) The corals and the algae depend on each other to survive.

Suggest

- how the algae benefit from the corals

.....

.....

.....

- how the corals benefit from the algae.

.....

.....

.....

[5]

(c) An increase in the average temperature of the sea over the last 20 years has caused the algae to manufacture chemicals that are poisonous to the corals.

Suggest and explain the long-term effect on the corals and on the algae of this rise in temperature.

.....

.....

.....

..... [3]

[Total: 10]

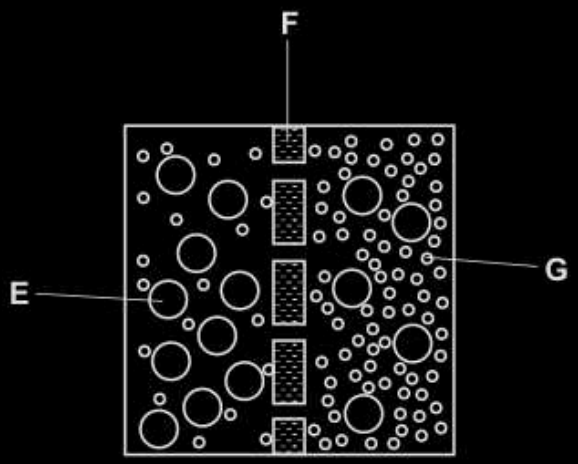
2(a)	photosynthesis ; green / plant **+** chlorophyll / chloroplasts ;	**2**	
2(b)	*algae benefit:* gain carbon dioxide ; protected / sheltered / habitat ; *coral benefit:* gain oxygen ; corals feed on **+** glucose / carbohydrate / protein / food **+** produced by algae ; *algae or coral:* respiration **+** gas produced or required ; photosynthesis **+** gas produced or required ;	**5**	
2(c)	coral dies / killed / decreases / extinction **AW** ; algae die / killed / decrease / extinction **AW** ; algae no longer **+** protected / cells to live in / get CO_2 ; coral / algae decompose ;	**3**	

Question	Answer	Marks	Guidance

5 **(a)** Name the substance absorbed by plants using the process of osmosis.

..... [1]

(b) The diagram represents osmosis taking place between two sucrose solutions of different concentrations.



Identify **E**, **F** and **G**.

molecule **E** .....

structure **F** .....

molecule **G** .....

[3]

(c) Explain, with reference to the diagram, how osmosis takes place.

.....

.....

.....

.....

.....

..... [5]

[Total: 9]

Section B

Answer **both** questions in this section.

Write your answers in the spaces provided.

5(a)	water ;	1	
5(b)	**(E)** sucrose ; **(F)** partially permeable membrane **AW** ; **(G)** water ;	3	
5(c)	movement of solvent / water / **G** ; water **+** from high to low water potential / **G** concentration **OR** water **+** from dilute to concentrated sucrose / **E** solution **OR** water **+** down its concentration gradient **AW** ; water / **G** **+** to left / from right ; through partially permeable membrane / **F** ; sucrose / solute / **E** too large **OR** water / **G** small enough to pass through ; diffusion **+** water **G** ; stops at equilibrium / concentration same on both sides **AW** ;	5	**A** selectively permeable membrane / Visking tubing / dialysis tubing

© UCLES 2020

Question	Answer	Marks	Guidance

6 **(a)** State the functions of the aorta and the vena cava and describe how each is adapted for its functions.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

(b) Some people are born with a hole in their heart that allows blood to pass between the left and the right sides of the heart. Suggest how this condition may affect the efficiency of their circulatory system.

.....

.....

.....

.....

..... [3]

[Total: 10]

6(a)	*aorta 4 max:* oxygenated blood ; from left ventricle **+** to rest of body ; thick walls ; elastic **+** muscle ; elastic recoil **OR** lumen can stretch **AW** ; high **+** pressure ; valve **+** prevent blood returning to heart **AW** ; *vena cava 4 max:* deoxygenated blood ; from the body / head **+** to the right atrium ; thin walls ; wide lumen ; low **+** pressure ; valves **+** to prevent backflow / one way flow ;	**7**	*maximum of 4 marks for either section*
6(b)	mixing / non-separation **+** oxy- and deoxygenated blood ; more oxygen in blood sent to the lungs ; less oxygen in blood sent to body / organs / tissues / cells **AW** ; reduced (aerobic) respiration / reduced energy / more anaerobic respiration / more lactic acid ; heart beats faster / harder **+** to meet oxygen demand ; (blood) pressure reduced ;	**3**	

© UCLES 2020

Question	Answer	Marks	Guidance

9 **(a)** Describe, with examples, what is meant by the term *mutation*. Name factors which may increase the rate of mutation.

.....

.....

.....

.....

.....

.....

..... [5]

(b) Artificial Insemination is a method of breeding farm animals in which the semen (liquid containing sperm) from a selected male animal is sent to a farmer to fertilise females of the same species.

Suggest possible advantages of this method over natural methods of breeding farm animals.

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

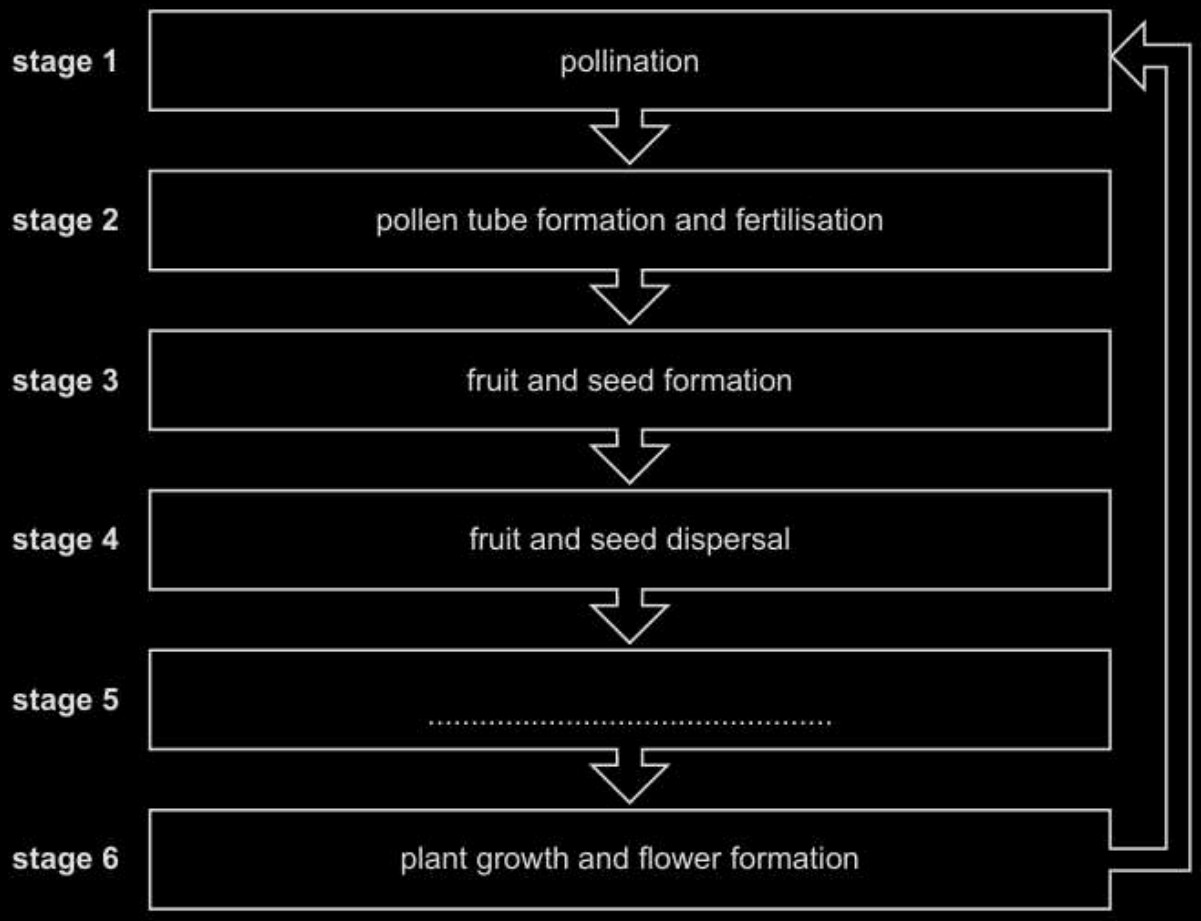
9(a)	change in gene / change in DNA ; change in chromosome number ; sickle-cell anaemia + change in gene ; Down's syndrome + increase in chromosome number / 47 chromosomes ; radiation ; chemicals / example ;	**5**	maximum of 1 mark for 2 examples alone, or if incorrectly linked to mutation

© UCLES 2020

Question	Answer	Marks	Guidance
9(b)	farmer does not need to keep male animals ; promotes cross-breeding / prevents inbreeding ; sperm transported from anywhere / far away ; artificial selection / selective breeding ; desirable features / characteristics ; increased chances of successful fertilisation **AW** ; quicker / easier / more convenient ; economic benefit **+** explanation ;	**5**	

© UCLES 2020

1 The flow chart shows stages in the life cycle of a flowering plant.



(a) The name of **stage 5** of the life cycle has not been included in the flow chart.

During this stage a radicle and a plumule are formed from a seed.

(i) State, in the correct place on the flow chart, the name of **stage 5**. [1]

(ii) Describe the role of enzymes in this stage of the plant's life cycle.

.....

.....

.....

.....

.....

..... [3]

(b) **Two** of the stages in the life cycle of this flowering plant **may** depend on the action of an animal.

Identify each of these stages **and** for each stage describe how the action of an animal may be required.

stage .....

action of an animal .....

.....

.....

.....

stage .....

action of an animal .....

.....

.....

.....

[6]

[Total: 10]

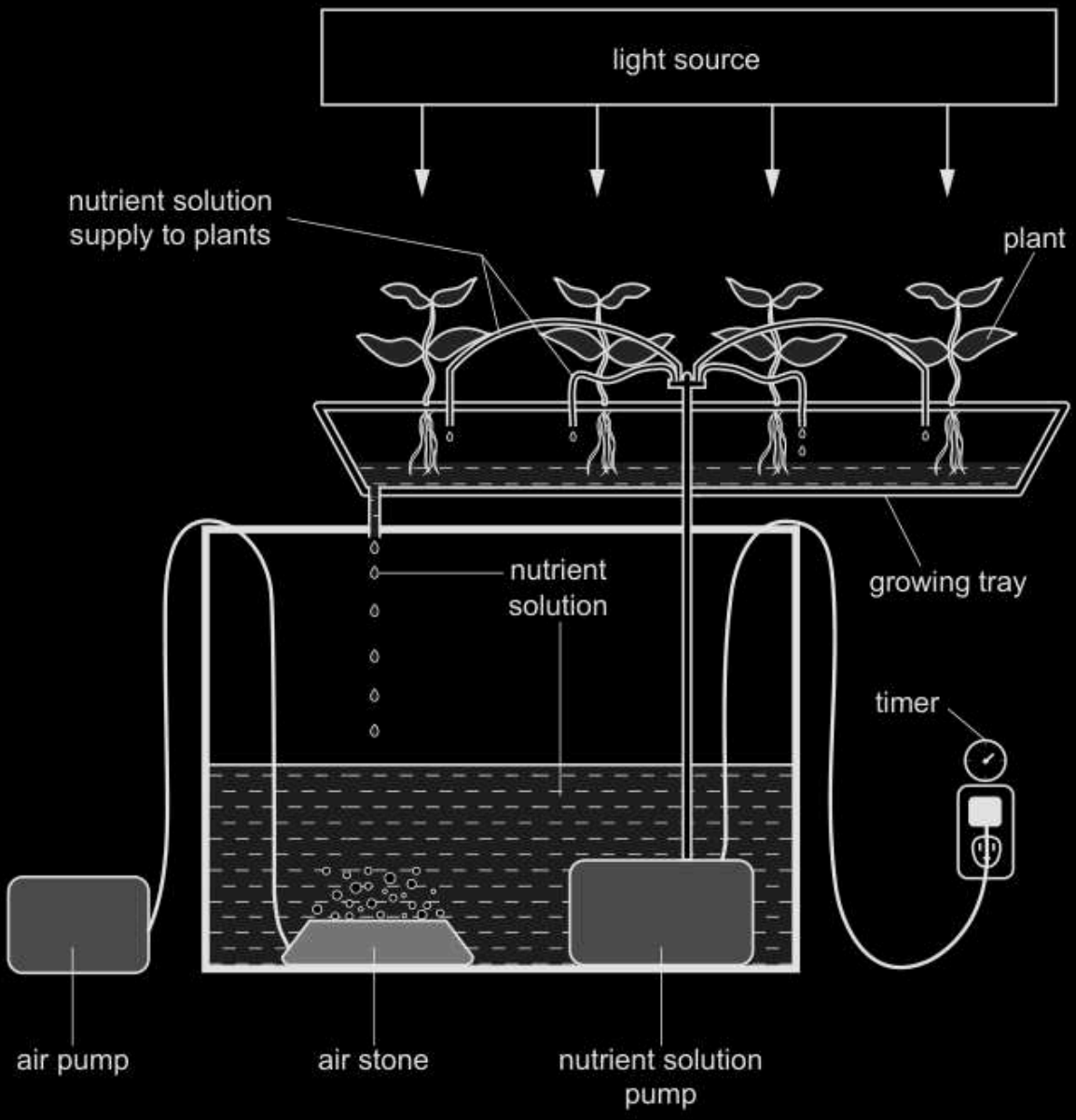
1(a)(i)	germination ;	**1**	
1(a)(ii)	**1** break down **AW** / digest / hydrolyse ; **2** amylase / maltase / lipase / other named digestive enzyme ; **3** named substrate **+** named product ; **4** respiration / protein synthesis ;	**3**	
1(b)	**response groups may be in either order**		
	response group 1 **1** (stage) 1 / one / pollination ; **ONLY if marking point 1 awarded then go on to award any two from ...** **2** pollen **+** movement / attachment ; **3** from anther **+** to stigma ; **4** of different flowers ;	**3**	
	response group 2 **5** (stage) 4 / four / fruit dispersal / seed dispersal ; **ONLY if marking point 5 awarded then go on to award any two from ...** **6** eats / consumes / undigested / attachment **AW** ; **7** movement **AW** ; **8** egestion / faeces / removal **AW** (e.g. falls off / spat out / thrown) ;	**3**	**6 7 8 R** if any reference to pollen / pollination **8** **R** excretion

Question	Answer	Marks	Guidance

October/November 2020 · Paper 2 · Variant 2 · Question 4

4 Hydroponics is a technique used to grow plants without soil.

The diagram shows plants being grown using hydroponics.



(a) The leaves of the plants are provided with a source of light for photosynthesis.

State, in **either** words **or** symbols, the equation for photosynthesis.

..... [2]

(b) The roots of the plants are provided with a nutrient solution. The nutrient solution contains magnesium ions.

Air is pumped through the nutrient solution using an air stone that contains many very small holes.

(i) Explain the advantage to the cells of the plant roots of pumping air through the nutrient solution using the air stone.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) State the type of cell in plant roots that absorbs magnesium ions from the solution.

..... [1]

(iii) State the process that transports magnesium ions against a concentration gradient into this type of cell.

..... [1]

(c) Farmers grew groups of 20 tomato plants for the same length of time in nutrient solutions with a range of different concentrations of magnesium ions.

They determined the best concentration of magnesium ions for growth by measuring the following for each group of 20 tomato plants:

- the mean height of the tomato plant stems (cm)
- the mean mass of the tomato plant stems and leaves (g).

The results of the investigation are shown in the table.

concentration of magnesium ions / parts per million	**mean height of tomato plant stems / cm**	**mean mass of tomato plant stems and leaves / g**
0	20.0	6.3
50	30.3	11.7
125	32.6	12.3
250	23.4	10.4
500	20.4	8.0

(i) State the best concentration of magnesium ions for growth.

..... parts per million [1]

(ii) Explain your answer using the results of the investigation **and** your knowledge of the importance of magnesium ions to plants.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 13]

Question 5 starts on the next page.

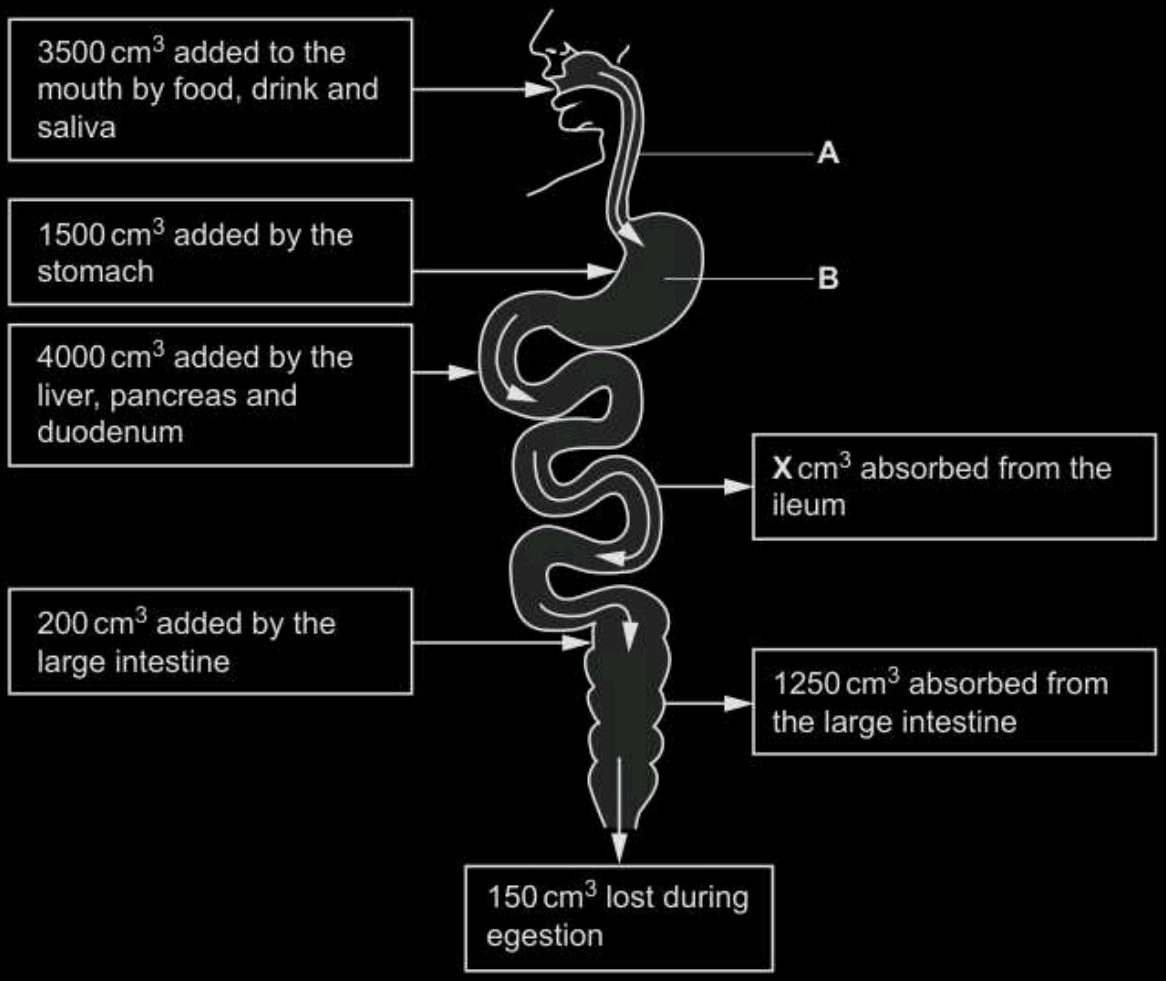
4(a)	**1** carbon dioxide / $6CO_2$ **+** water / $6H_2O$ **(⟶) ;** **2** **(⟶)** glucose / $C_6H_{12}O_6$ **+** oxygen / $6O_2$ **;**	**2**	
4(b)(i)	**1** increased / more / greater / higher / high **;** **any three from ...** **2** bubbles / surface area of air **;** **3** oxygen **;** **4** aerobic **+** respiration **;** **5** energy **;** **6** growth **;** **7** absorption **AW** / active transport **+** ions / salts / minerals / magnesium **;**	**4**	**1 Allow** only if linked to any of points **2** to **7** **5** **Reject** energy <u>produced</u>
4(b)(ii)	hair **;**	**1**	

© UCLES 2020

Question	Answer	Marks	Guidance
4(b)(iii)	active transport **;**	**1**	
4(c)(i)	125 **;**	**1**	
4(c)(ii)	**1** largest **AW +** height and mass **;** **2** magnesium **+** chlorophyll **;** **3** more **AW +** photosynthesis **;** **4** production of **+** glucose / starch / carbohydrate / sugar **;**	**4**	

Question	Answer	Marks	Guidance

5 The diagram shows the total volume of water entering and leaving organs of the human alimentary canal during one day.



(a) (i) Name the organs labelled **A** and **B**.

A .....

B .....

[2]

(ii) Calculate the volume of water, **X**, absorbed from the ileum.

..... cm^3 [1]

(iii) Describe the absorption of water from the ileum.

.....

.....

.....

.....

.....

..... [3]

(b) State how a **named** organ causes some of the water absorbed from the ileum to later become water in the urine.

.....

.....

.....

..... [2]

[Total: 8]

Section B

Answer **both** questions in this section.

Write your answers in the spaces provided.

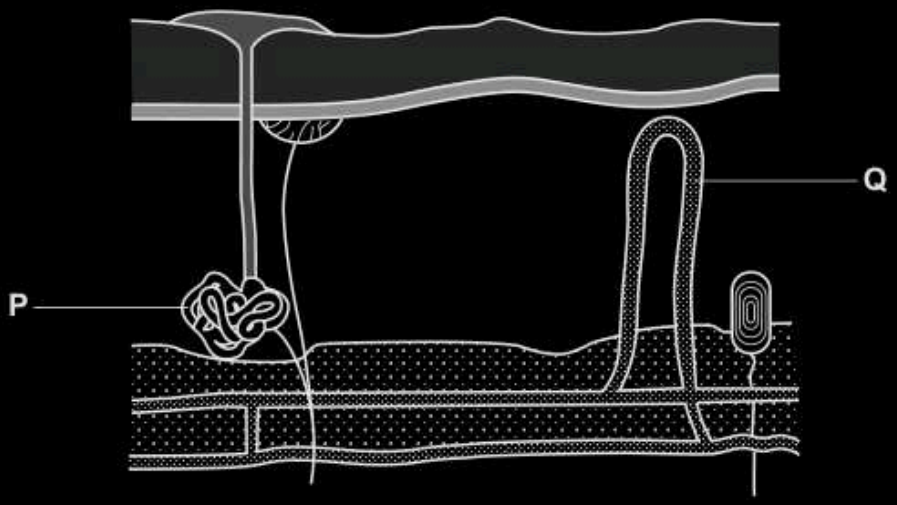
5(a)(i)	(**A**) oesophagus / gullet ; (**B**) stomach ;	**2**	
5(a)(ii)	7800 ;	**1**	
5(a)(iii)	**1** osmosis / diffusion ; **2** down **+** water concentration gradient / water potential gradient ; **3** villi / microvilli ; **4** into **+** blood / plasma / capillaries ;	**3**	
5(b)	**1** kidney ; **2** filters / removes water from **+** blood ;	**2**	

© UCLES 2020

Question	Answer	Marks	Guidance

October/November 2020 · Paper 2 · Variant 2 · Question 6

6 The diagram shows a vertical section through human skin in a hot environment.



(a) Name **and** outline the role of **P** and **Q** in the control of body temperature when a person is in a hot environment.

P .....

.....

.....

.....

.....

.....

Q .....

.....

.....

.....

.....

.....

[7]

(b) Explain how the process of temperature control in humans is an example of:

homeostasis .....

.....

negative feedback. .....

.....

.....

[3]

[Total: 10]

6(a)	(allow once in **either** section **P or** section **Q**) **1** removal / loss of **+** heat **;** (allow anywhere in section **P**) **2** sweat gland **;** **3** produce / release **+** sweat **;** **4** evaporates / vapour **;** (allow anywhere in section **Q**) **5** capillary **;** **6** increased **AW +** blood flow **;** **7** blood near **+** surface **AW ;**	**7**	
6(b)	(homeostasis) **1** constant / same **+** internal environment **AW /** temperature **;** (negative feedback) **2** change / increase / decrease **;** **3** correction / reversal / back (e.g. down / up / to normal) **AW ;**	**3**	

Question	Answer	Marks	Guidance

7 The kidneys are organs of excretion.

(a) Define the term *excretion*.

.....

.....

.....

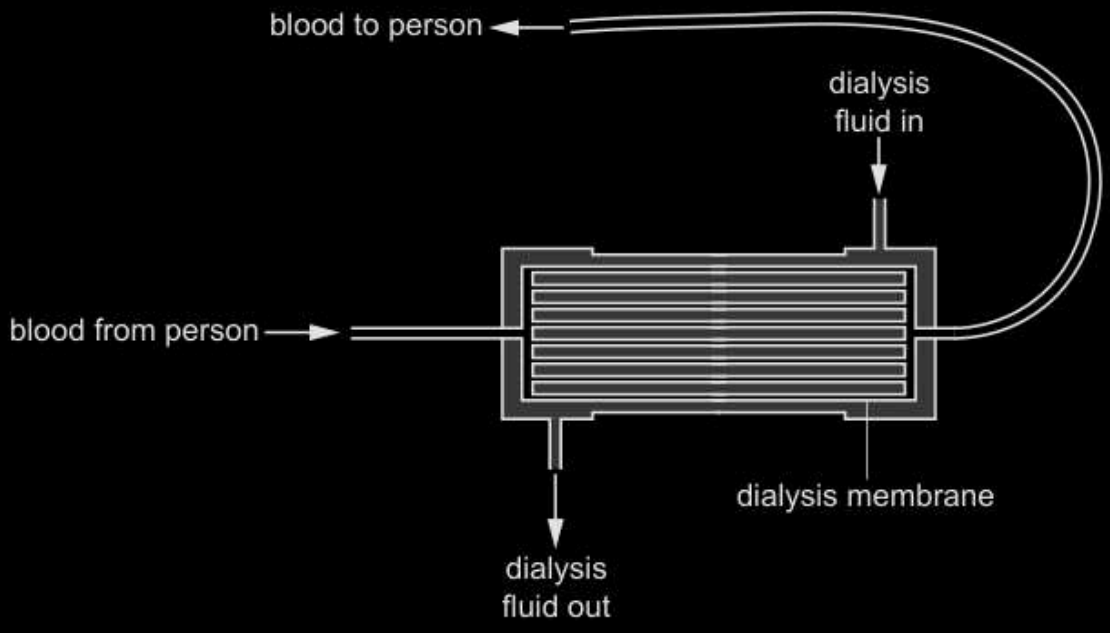
.....

..... [3]

(b) A person with kidney damage requires the process of dialysis to take place several times each week.

Dialysis is carried out using a kidney machine connected to the blood supply of the person.

The diagram shows a section through the internal structure of a kidney machine.



(i) Describe the structure of the dialysis membrane **and** explain how it functions.

.....

.....

.....

.....

.....

.....

.....

..... [5]

(ii) Explain why the dialysis fluid in the kidney machine must be continuously replaced.

.....

.....

.....

..... [2]

[Total: 10]

Section C

Answer **either** Question 8 **or** Question 9.

Write your answers in the spaces provided.

7(a)	**1** removal **AW** ; **2** toxins / toxic ; **3** waste products **AW** / urea / carbon dioxide ;	**3**	**3** **R** egestion	
7(b)(i)	**1** multiple tubes / multiple sheets / coiled / folded / long ; **2** large **AW** + surface area ; **3** selectively / partially / semi + permeable ; **4** diffusion / osmosis ; **5** small / waste products **AW** / urea / ions / water / toxins + do pass **AW** ; **6** large / protein / blood cells + do not pass **AW** ;	**5**		

© UCLES 2020

Question	Answer	Marks	Guidance
7(b)(ii)	**1** maintain gradient **AW** ; **2** prevent build up / ensure removal + waste products **AW** / urea / toxins ; **3** continued / more / maximum + diffusion / movement / exchange ;	**2**	

Question	Answer	Marks	Guidance

May/June 2020 · Paper 1 · Variant 1 · Question 1

Correct answer: C

1 Which structure is present in a liver cell and in a leaf cell?

A cell wall

B chloroplast

C cytoplasm

D sap vacuole

May/June 2020 · Paper 1 · Variant 1 · Question 2

Correct answer: B

2 By which process does water vapour pass out of a leaf?

A active transport

B diffusion

C osmosis

D translocation

May/June 2020 · Paper 1 · Variant 1 · Question 3

Correct answer: A

3 A student takes a potato and cuts three pieces from it. Each piece is 5 cm × 0.5 cm × 0.5 cm. He places the three potato pieces into three different concentrations of sugar solution.

After two hours, he removes the potato pieces from the sugar solutions and measures their lengths.

The results are shown in the table.

solution	length of potato piece after two hours / cm
X	5.2
Y	4.7
Z	5.3

What can be concluded from these results?

A Solution Y has a lower water potential than the potato cells.

B Solution Z has the lowest water potential.

C The potato piece in solution X increases in length because it takes up sugar.

D The potato piece in solution Y decreases in length because it loses sugar.

May/June 2020 · Paper 1 · Variant 1 · Question 4

Correct answer: D

4 Which property of enzymes is explained by the lock and key hypothesis?

A All enzymes are proteins.

B Enzymes are inactive at very low temperatures.

C Human enzymes are most active just below 40 °C.

D Most enzymes can catalyse only one reaction.

May/June 2020 · Paper 1 · Variant 1 · Question 8

Correct answer: D

8 Which organ produces an acid which kills the bacteria in ingested food?

A liver

B pancreas

C salivary gland

D stomach

May/June 2020 · Paper 1 · Variant 1 · Question 9

Correct answer: B

9 A patient has her gall bladder surgically removed.

How will this affect the functioning of her body?

A reducing the absorption of carbohydrates

B reducing the digestion of fats

C reducing the liver's ability to convert glucose to glycogen

D reducing the volume of stored urine

May/June 2020 · Paper 1 · Variant 1 · Question 11

Correct answer: D

11 Which feature of root hairs suggests that they take up ions from the soil by active transport?

A Their cell membranes are partially permeable.

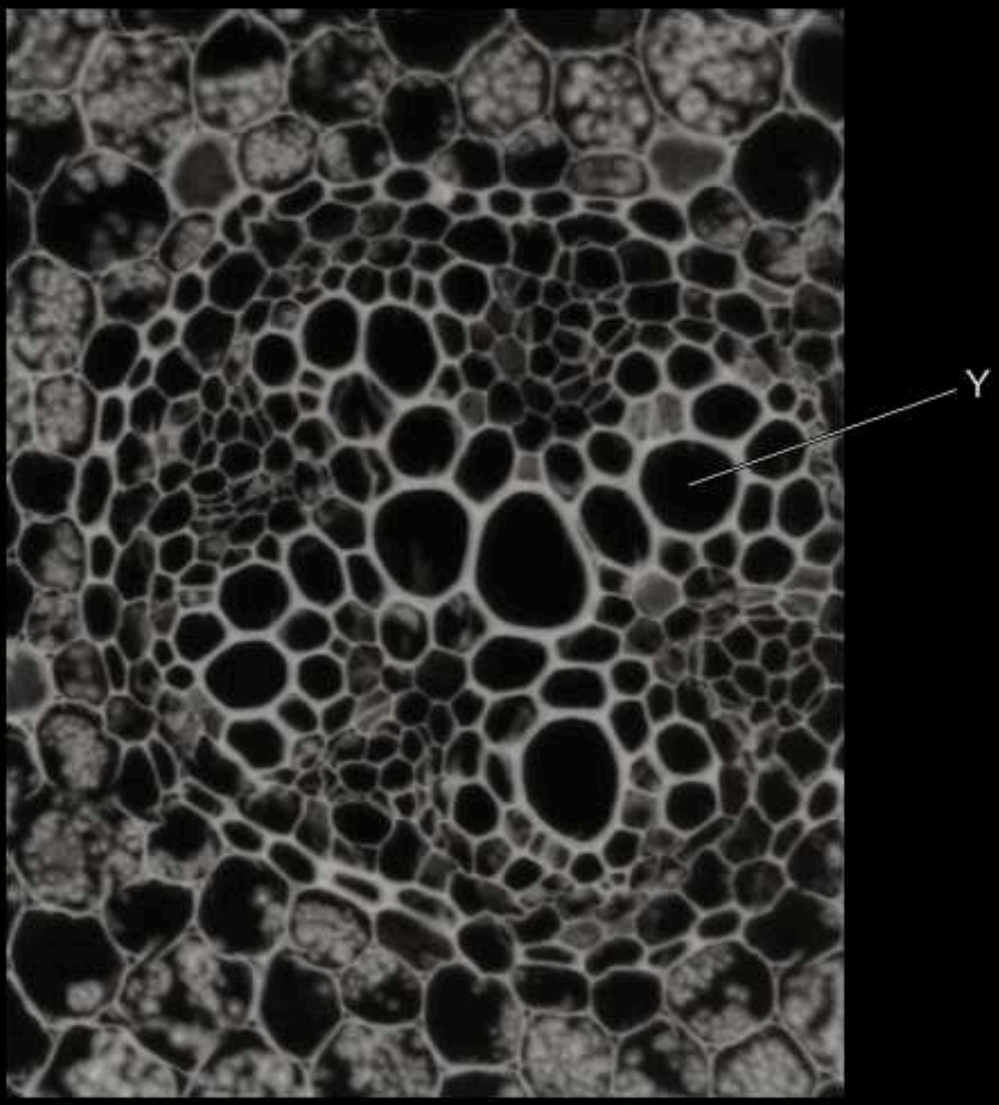
B They have a large surface area.

C They have a lower water potential than the soil.

D They take up ions more slowly in low oxygen concentrations.

Correct answer: D

12 The photomicrograph shows part of a section through a root.



The contents of Y are tested with Benedict's solution and with iodine solution.

Which results are expected?

	Benedict's solution	iodine solution
A	+	+
B	+	–
C	–	+
D	–	–

key

+ = positive result

– = negative result

May/June 2020 · Paper 1 · Variant 1 · Question 13

Correct answer: A

13 Which blood vessels carry blood to the liver?

	hepatic artery	hepatic portal vein	hepatic vein	vena cava
A	yes	yes	no	no
B	yes	no	yes	no
C	no	yes	no	yes
D	no	no	yes	yes

May/June 2020 · Paper 1 · Variant 1 · Question 14

Correct answer: B

14 Which blood vessels contain valves?

	capillary	renal artery	renal vein
A	✓	✗	✗
B	✗	✗	✓
C	✓	✓	✗
D	✗	✓	✓

key

✓ = contains valves

✗ = does not contain valves

May/June 2020 · Paper 1 · Variant 1 · Question 15

Correct answer: A

15 Which chamber of the heart would be the first to receive nicotine absorbed into the blood in the lungs of a cigarette smoker?

A left atrium

B left ventricle

C right atrium

D right ventricle

May/June 2020 · Paper 1 · Variant 1 · Question 17

Correct answer: B

17 What lines the walls of alveoli?

A a carpet of cilia

B a film of moisture

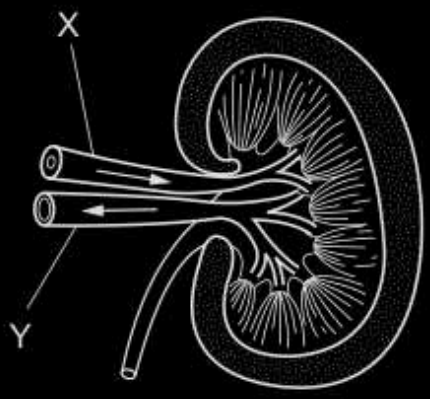
C a network of veins

D a thin sheet of muscle

May/June 2020 · Paper 1 · Variant 1 · Question 19

Correct answer: B

19 The diagram shows a kidney and blood vessels associated with it.



Which statements are correct?

1 Vessel X contains more urea than vessel Y.

2 Vessel Y contains urine.

3 Vessel X contains more oxygen than vessel Y.

A 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only

May/June 2020 · Paper 1 · Variant 1 · Question 21

Correct answer: C

21 Which statements describe the pupil reflex in bright light?

1 ciliary muscles contract
2 ciliary muscles relax
3 circular iris muscles contract
4 circular iris muscles relax
5 lens becomes rounder
6 lens becomes thinner
7 pupil constricts
8 pupil dilates
9 radial iris muscles contract
10 radial iris muscles relax

A 1, 6 and 9 **B** 2, 5 and 10 **C** 3, 7 and 10 **D** 4, 8 and 9

May/June 2020 · Paper 1 · Variant 1 · Question 22

Correct answer: C

22 Which chemical produced by the body alters the activity of a target organ and is destroyed by the liver?

A bile
B enzyme
C hormone
D saliva

May/June 2020 · Paper 1 · Variant 1 · Question 23

Correct answer: A

23 The diagram shows a section through the human brain.

Which labelled part contains the temperature regulation centre?

May/June 2020 · Paper 1 · Variant 1 · Question 37

Correct answer: D

37 Natural selection plays a role in evolution.

What describes features of natural selection that can affect evolution?

	better adapted individuals have a greater chance of surviving and breeding	survivors may transfer an advantageous feature to their offspring
A	no	no
B	no	yes
C	yes	no
D	yes	yes

May/June 2020 · Paper 1 · Variant 1 · Question 38

Correct answer: D

38 In guinea-pigs, the allele for black fur (B) is dominant to the allele for brown fur (b). A breeder can sell a brown guinea-pig for more money than a black guinea-pig.

Which cross will produce the most money?

A BB × BB **B** BB × Bb **C** Bb × Bb **D** Bb × bb

May/June 2020 · Paper 1 · Variant 1 · Question 39

Correct answer: B

39 The inheritance of the ABO blood groups in humans is controlled by three alleles (I^A, I^B and I^O), only two of which can be present in one individual.

What are the possible blood groups of children born to a homozygous group A woman and a heterozygous group B man?

A AB and B only

B AB and A only

C A, B and AB only

D A, B, AB and O

May/June 2020 · Paper 1 · Variant 1 · Question 40

Correct answer: C

40 Which statement is **always** true of dominant alleles?

A They cannot undergo mutation.

B They give a greater chance of survival than recessive alleles.

C They give the same phenotype in heterozygotes and homozygotes.

D They occur less frequently in the population than recessive alleles.

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

May/June 2020 · Paper 1 · Variant 2 · Question 1

Correct answer: C

1 Starting with the smallest, and ending with the largest, what is the correct sequence of these parts of an organism?

A chromosome → gene → nucleus → cell → tissue

B chromosome → gene → nucleus → tissue → cell

C gene → chromosome → nucleus → cell → tissue

D gene → chromosome → nucleus → tissue → cell

May/June 2020 · Paper 1 · Variant 2 · Question 2

Correct answer: C

2 Which movement of a substance in plants is an energy-consuming process?

A absorption of carbon dioxide by a palisade cell

B absorption of oxygen by a mesophyll cell

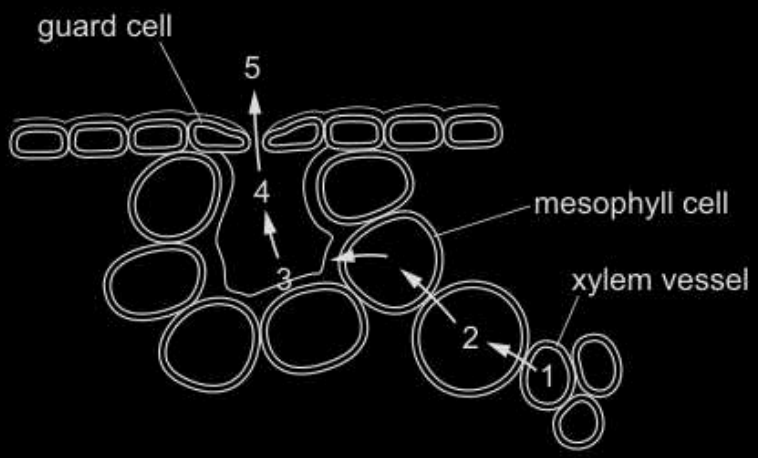
C nitrate uptake by root hair cells

D transport of water up through the xylem

May/June 2020 · Paper 1 · Variant 2 · Question 3

Correct answer: D

3 The diagram shows the movement of water through part of a leaf.



Which processes are involved in the movement of water at these stages?

	1–2	3–4	4–5
A	diffusion	evaporation	osmosis
B	evaporation	diffusion	osmosis
C	osmosis	diffusion	evaporation
D	osmosis	evaporation	diffusion

May/June 2020 · Paper 1 · Variant 2 · Question 4

Correct answer: D

4 Which substance is an enzyme?

A chlorophyll

B fibrinogen

C insulin

D lipase

May/June 2020 · Paper 1 · Variant 2 · Question 8

Correct answer: C

8 Which process happens in the duodenum?

A bile is formed

B faeces are formed

C fats are digested by lipase

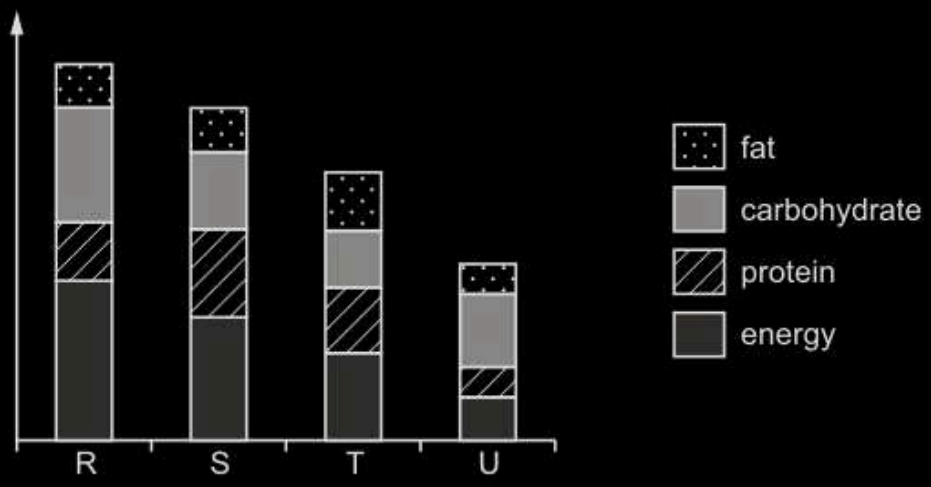
D pancreatic juice lowers the pH

May/June 2020 · Paper 1 · Variant 2 · Question 10

Correct answer: C

10 The bar chart shows the recommended daily food intake of four people:

- an 80 year-old retired man
- a male manual worker
- a 6 year-old child
- a breast-feeding mother.



Which row correctly identifies two of these four people?

A child is U; retired man is R

B manual worker is T; child is S

C mother is S; manual worker is R

D retired man is R; mother is U

May/June 2020 · Paper 1 · Variant 2 · Question 11

Correct answer: A

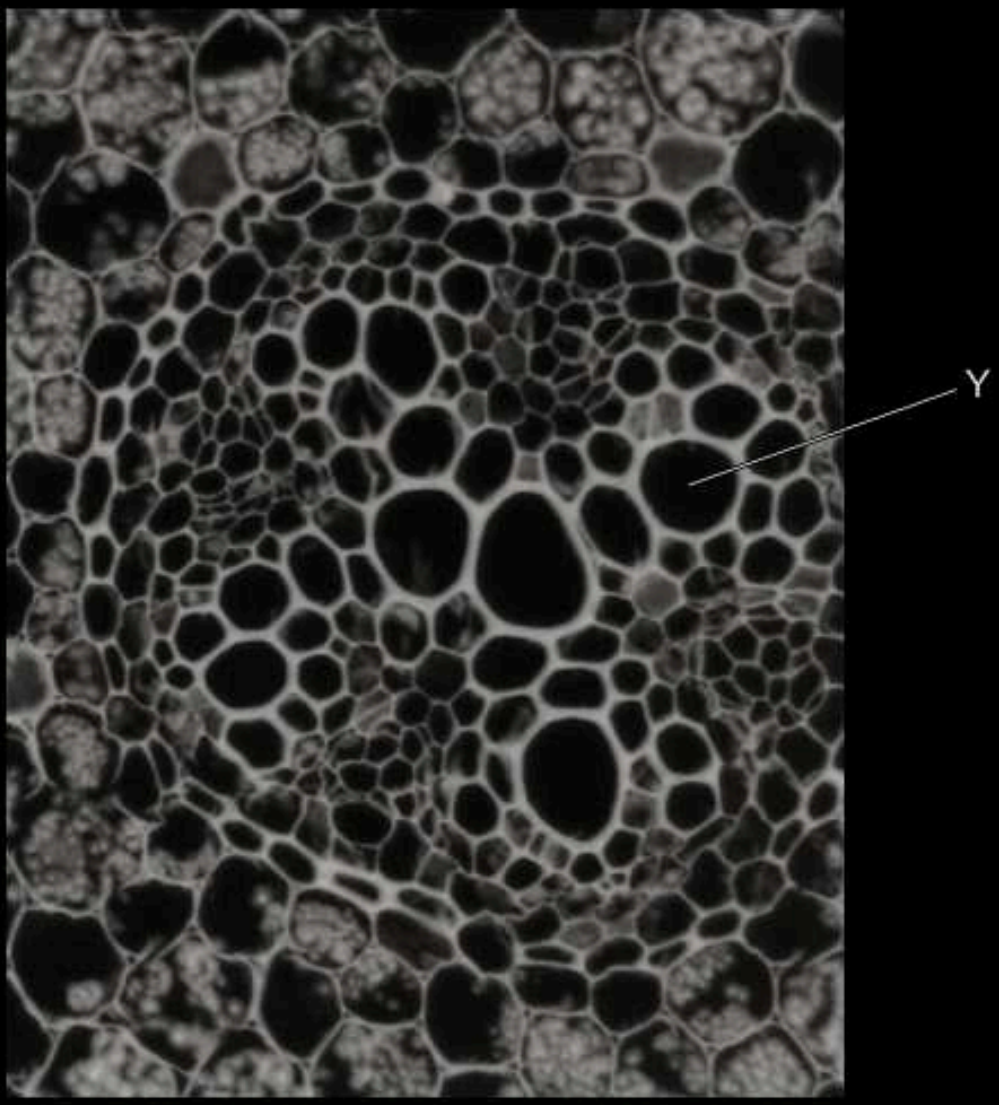
11 Water and ions can reach the xylem of a plant root through cell walls, without passing through a cell membrane.

How do these substances move through the cell walls?

	water	ions
A	diffusion	diffusion
B	diffusion	osmosis
C	osmosis	diffusion
D	osmosis	osmosis

Correct answer: D

12 The photomicrograph shows part of a section through a root.



The contents of Y are tested with Benedict's solution and with iodine solution.

Which results are expected?

	Benedict's solution	iodine solution
A	+	+
B	+	–
C	–	+
D	–	–

key

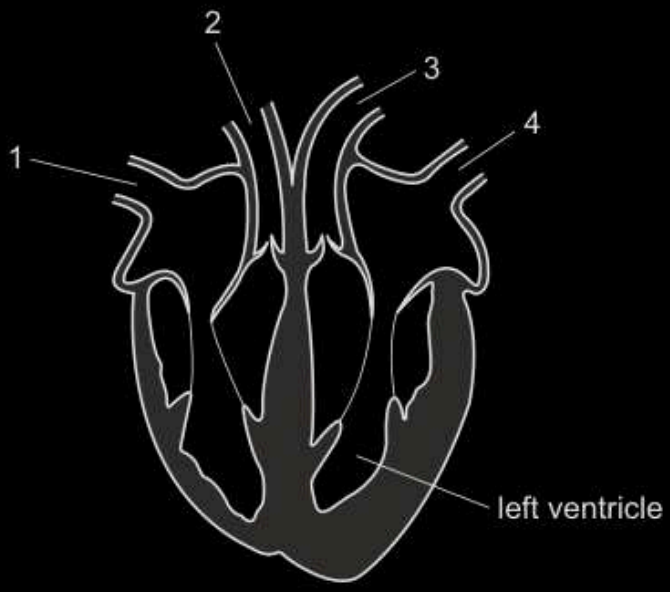
+ = positive result

– = negative result

May/June 2020 · Paper 1 · Variant 2 · Question 13

Correct answer: D

13 The diagram shows a section through the heart.



Which two numbered parts contain oxygenated blood?

A 1 and 2 **B** 2 and 3 **C** 2 and 4 **D** 3 and 4

May/June 2020 · Paper 1 · Variant 2 · Question 14

Correct answer: B

14 Which blood vessels contain valves?

	capillary	renal artery	renal vein
A	✓	✗	✗
B	✗	✗	✓
C	✓	✓	✗
D	✗	✓	✓

key

✓ = contains valves

✗ = does not contain valves

May/June 2020 · Paper 1 · Variant 2 · Question 15

Correct answer: A

15 Which chamber of the heart would be the first to receive nicotine absorbed into the blood in the lungs of a cigarette smoker?

A left atrium

B left ventricle

C right atrium

D right ventricle

May/June 2020 · Paper 1 · Variant 2 · Question 17

Correct answer: B

17 Which structure is lined with cilia?

A alveolus

B bronchus

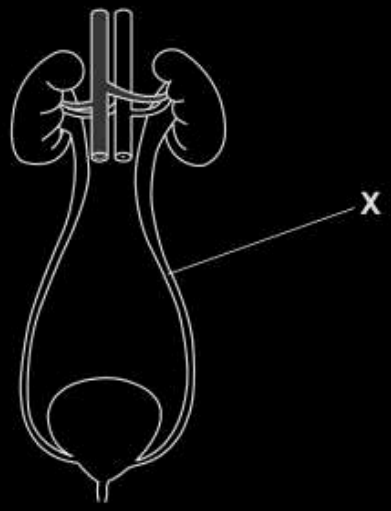
C oesophagus

D small intestine

May/June 2020 · Paper 1 · Variant 2 · Question 19

Correct answer: C

19 The diagram shows the human kidneys and their associated structures.



What is the structure labelled **X**?

A renal artery

B renal vein

C ureter

D urethra

May/June 2020 · Paper 1 · Variant 2 · Question 21

Correct answer: B

21 Which structures does light pass through when it is focused on the retina?

	cornea	lens	sclera
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key
✓ = yes
✗ = no

May/June 2020 · Paper 1 · Variant 2 · Question 23

Correct answer: A

23 The diagram shows a section through the human brain.

Which part senses and reacts to a change in blood temperature?

May/June 2020 · Paper 1 · Variant 2 · Question 34

Correct answer: B

34 Which statements about meiosis are correct?

	meiosis produces genetically identical nuclei	meiosis produces haploid nuclei
A	✗	✗
B	✗	✓
C	✓	✗
D	✓	✓

key
✓ = correct
✗ = not correct

May/June 2020 · Paper 1 · Variant 2 · Question 37

Correct answer: D

37 What describes features of natural selection?

	causes mutations to happen	leads to the evolution of new species	results in existing species becoming extinct
A	yes	yes	yes
B	yes	yes	no
C	yes	no	yes
D	no	yes	yes

May/June 2020 · Paper 1 · Variant 2 · Question 38

Correct answer: D

38 A farmer uses sperm from a black male sheep to artificially inseminate (fertilise) 20 light brown female sheep.

All of the offspring produced were black.

Which statement explains these results?

A The alleles for light brown colour are dominant.

B The male sheep is heterozygous and the allele for black colour is dominant.

C The male sheep is homozygous and the allele for black colour is codominant.

D The male sheep is homozygous and the allele for black colour is dominant.

May/June 2020 · Paper 1 · Variant 2 · Question 39

Correct answer: B

39 The inheritance of the ABO blood groups in humans is controlled by three alleles (I^A, I^B and I^O), only two of which can be present in one individual.

What are the possible blood groups of children born to a homozygous group A woman and a heterozygous group B man?

A AB and B only

B AB and A only

C A, B and AB only

D A, B, AB and O

May/June 2020 · Paper 1 · Variant 2 · Question 40

Correct answer: C

40 Which statement is **always** true of dominant alleles?

A They cannot undergo mutation.

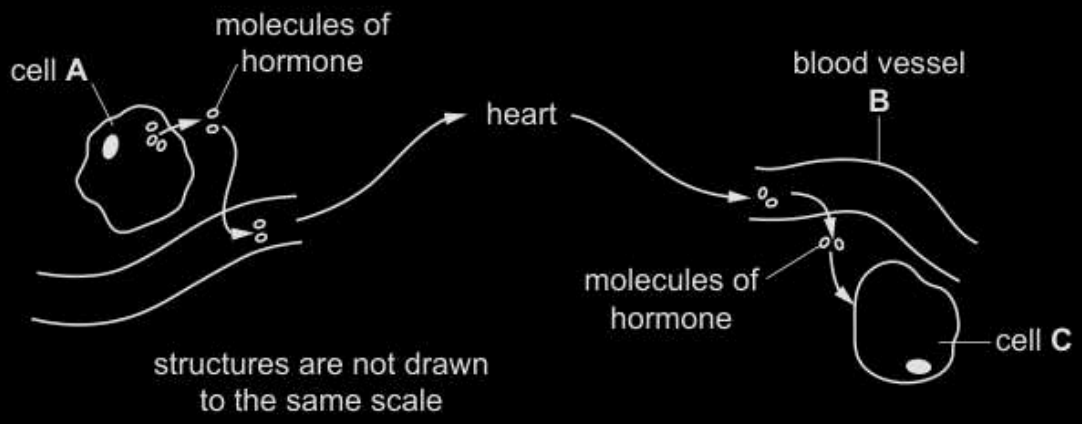
B They give a greater chance of survival than recessive alleles.

C They give the same phenotype in heterozygotes and homozygotes.

D They occur less frequently in the population than recessive alleles.

May/June 2020 · Paper 2 · Variant 1 · Question 1

1 The diagram shows the pathway taken by a hormone from cell **A** where it is produced to cell **C** that it affects.



(a) (i) Identify the type of blood vessel **B**. ..... [1]

(ii) Explain the ways in which this type of blood vessel is adapted for the transport of hormone molecules.

.....

.....

.....

.....

..... [4]

(iii) State how molecules of the hormone are transported through blood vessel **B**.

.....

..... [2]

(b) For a **named** hormone, identify the gland containing cell **A**, an organ containing cell **C** and a function or effect of the hormone you have named.

- named hormone .....
- the gland containing cell **A** .....
- an organ containing cell **C** .....
- a function or effect of the named hormone

.....

.....

[4]

(c) Cell **C** is one of a number of similar cells making up an organ.

State the name given to organs affected by hormones.

..... [1]

[Total: 12]

1(a)(i)	capillary ;	1
1(a)(ii)	very small **AW** / narrow ; thin / one cell thick / leaky **+** walls ; runs close to the cells ; very numerous / large surface area ; maximum uptake of hormone / maximum offloading of hormone **AW** ; by diffusion ;	4
1(a)(iii)	in solution ; in blood ; plasma ;	2
1(b)	(*for example*) insulin ; pancreas ; liver / muscles ; glucose to glycogen / glycogen storage ;	4
1(c)	target (organ) ;	1

Question	Answer	Marks

3 The photograph shows an insect called a leafhopper. When it feeds, its mouthparts penetrate the xylem of the plant on which it lives.

Leafhoppers carry disease-causing bacteria from plant to plant.

(a) (i) State the term used for a disease-causing organism.

..... [1]

(ii) State the term used for an animal, such as an insect, that carries disease-causing organisms.

..... [1]

(b) The bacteria carried by the leafhopper reproduce inside the plant stem and block the xylem.

(i) Suggest and explain why this can cause the leaves of the plant to wilt.

.....

.....

.....

..... [4]

(ii) The leaves of plants affected by these bacteria may also develop white patches. Suggest reasons for this.

.....

.....

(iii) Suggest an explanation for the fact that the fruits of plants affected by these bacteria are small and poorly developed.

.....

.....

.....

.....

..... [4]

[Total: 12]

3(a)(i)	pathogen ;	**1**
3(a)(ii)	vector ;	**1**
3(b)(i)	water lost from leaves ; by evaporation / transpiration ; cannot be replaced / cannot pass up stem ; water lost from cells ; lose their turgor / turgidity ;	**4**
3(b)(ii)	can't make chlorophyll ; reference to magnesium / zinc not getting to leaves ;	**2**
3(b)(iii)	reduced photosynthesis ; less sugar / starch / protein made ; less water available to fruit ; fewer nutrients or named nutrient sent to fruit ; less growth or development (linked to protein) ;	**4**

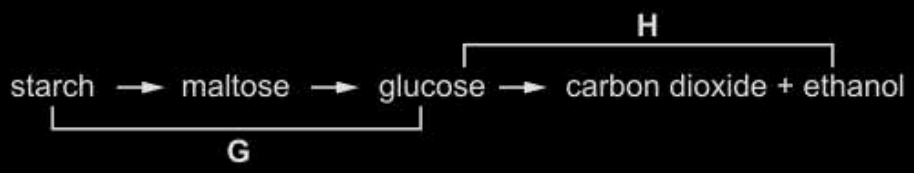
Question	Answer	Marks

4 **(a)** Explain the importance of the production of carbon dioxide in bread-making.

.....

..... [1]

(b) The diagram shows a flow chart for some of the chemical reactions that occur during bread-making.



Name the processes occurring at **G** and **H**.

G .....

H .....

[3]

(c) Explain what is causing the changes at **G**.

.....

.....

..... [3]

(d) **(i)** State the name of the microorganism used in bread-making and the group of organisms to which it belongs.

name .....

group .....

[2]

(ii) State **two** characteristics of organisms in this group.

1 .....

2 .....

[2]

[Total: 11]

4(a)	rise / texture ;	**1**
4(b)	**G** digestion ; **H** respiration ; anaerobic ;	**3**

© UCLES 2020

Question	Answer	Marks
4(c)	amylase ; starch to maltose (linked to amylase) ; maltase ; maltose to glucose (linked to maltase) ;	3
4(d)(i)	yeast ; fungi ;	2
4(d)(ii)	cell wall ; cytoplasm ; hyphae / mycelium ; cell nuclei ;	2

Question	Answer	Marks

5 **(a)** The passage describes the nucleus of a cell with missing words replaced by the letters **J**, **K**, **L** and **M**. Read the passage, then select words from the list below the passage to replace letters **J**, **K**, **L** and **M**.

> In the nucleus of a body cell from a person with Down's syndrome, there are **J** thread-like structures called **K**. These are made up of many units called **L** that instruct the cell to produce a particular **M**.

carbohydrate	**chromosomes**	**DNA**
fat	**forty-seven**	**genes**
protein	**twenty-four**	**twenty-three**

J .....

K .....

L .....

M .....

[4]

(b) When suddenly exposed to bright light, some people automatically sneeze. This is known as the photic sneeze reflex.

The photic sneeze reflex is the result of the possession of a dominant allele **A**. Use a genetic diagram to show how parents, both with the photic sneeze reflex, can have a child who is **not** affected by the photic sneeze reflex.

[3]

[Total: 7]

Section B

Answer **both** questions in this section.

Write your answers in the spaces provided.

5(a)	**J** 47 ; **K** chromosomes ; **L** genes ; **M** protein ;	**4**
5(b)	Aa x Aa ; aa correctly derived and identified ; correct use of the words parents **+** gametes / genotype **+** phenotype ;	**3**

Question	Answer	Marks

May/June 2020 · Paper 2 · Variant 1 · Question 7

7 **(a)** Describe how changes in gene structure may be caused.

.....

.....

.....

..... [3]

(b) Explain how changes in gene structure may eventually lead to the evolution of a species.

.....

.....

.....

.....

.....

.....

.....

..... [7]

[Total: 10]

Section C

Answer **either** Question 8 **or** Question 9.

Write your answers in the spaces provided.

7(a)	radiation ; chemicals ; mutation ;	3
7(b)	cause changes in appearance / phenotype ; variation ; change in environment ; advantage ; survival ; natural selection ; reproduction ; inheritance ; over a period of time ; breed only with similar type / separate species **AW** ;	7

© UCLES 2020

Question	Answer	Marks

May/June 2020 · Paper 2 · Variant 1 · Question 8

8 **(a)** List the structures within a plant leaf that **do not** possess chloroplasts.

.....

.....

.....

..... [4]

(b) Explain the importance to a plant of

(i) guard cells

.....

.....

.....

.....

..... [4]

(ii) root hair cells.

.....

.....

..... [2]

[Total: 10]

8(a)	cuticle ; epidermis ; xylem ; phloem ;	**4**
8(b)(i)	control the size / opening and closing ; of stomata ; (linked to) water loss / transpiration ; (allow) exchange of O_2 and CO_2 ; for respiration (or process described) ; and photosynthesis (or process described) ;	**4**
8(b)(ii)	for uptake of water ; ions ; oxygen ; large surface area ;	**2**

May/June 2020 · Paper 2 · Variant 1 · Question 9

9 **(a)** Describe the similarities and differences in the **functions** of a motor and a sensory neurone.

similarities .....

.....

.....

differences .....

.....

.....

[5]

(b) Describe the functions of the following parts of the brain.

(i) the cerebrum

.....

.....

.....

.....

..... [3]

(ii) the hypothalamus

.....

.....

.....

..... [2]

[Total: 10]

9(a)	*similarities*: both carry impulses ; quickly ; over long distances ; during voluntary / involuntary / reflex actions ; *differences* : *motor* – from CNS / spinal cord / brain ; to effector / named effector ; *sensory* – from receptor / named receptor ; to CNS / spinal cord / brain ;	**5**
9(b)(i)	reasoning / planning / problem solving / memory / emotions ; speech ; hearing ; movement ; recognition ; sight ;	**3**

Question	Answer	Marks
9(b)(ii)	homeostasis / maintenance of internal environment ; controls pituitary secretions ; reference to hormones / nervous system ;	**2**

1 In the diagram below the boxes on the left contain the names of organs of the human body.

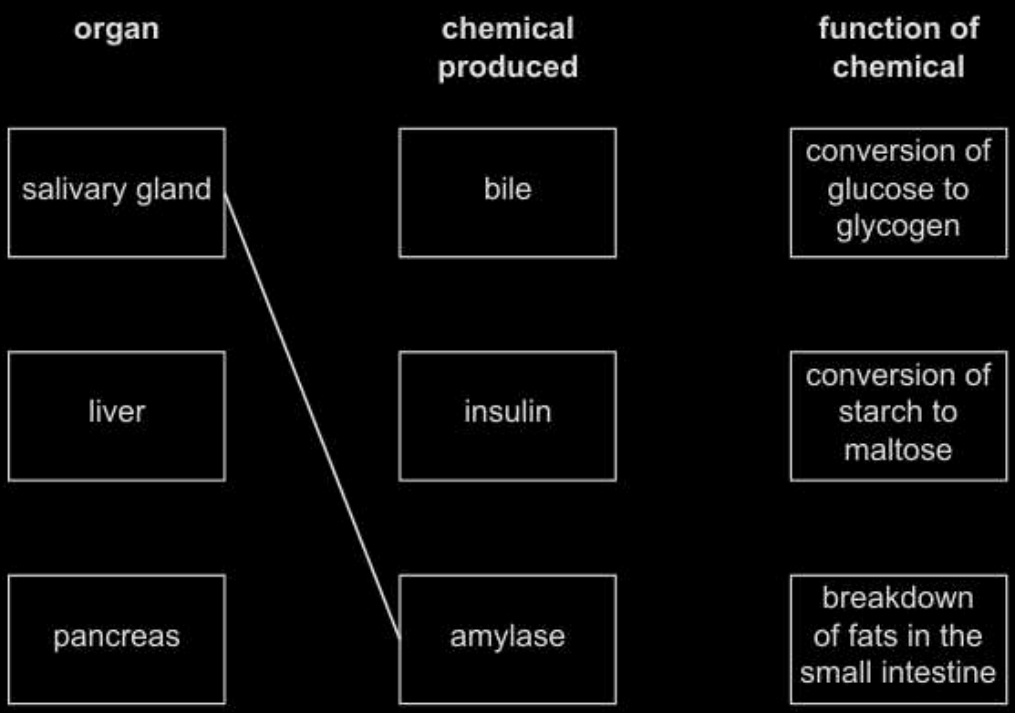
The boxes in the middle contain the names of chemicals produced by these organs.

The boxes on the right contain descriptions of one function of each of these chemicals.

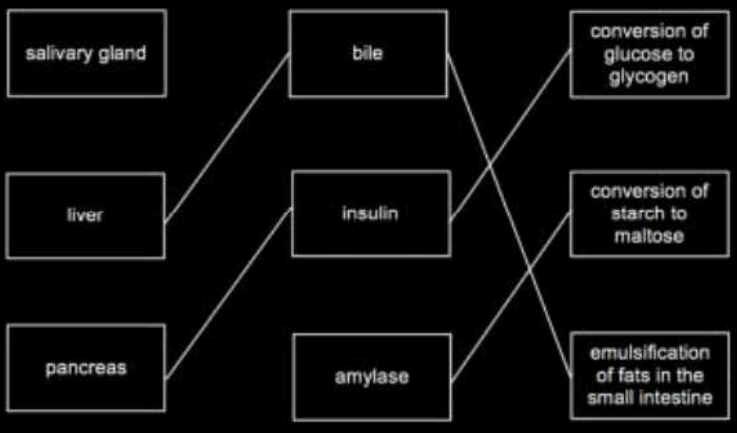
Draw lines to link each organ with the chemical it produces **and** to link each chemical with the description of its function.

One line has been drawn for you.

Draw **five more** lines.



[5]

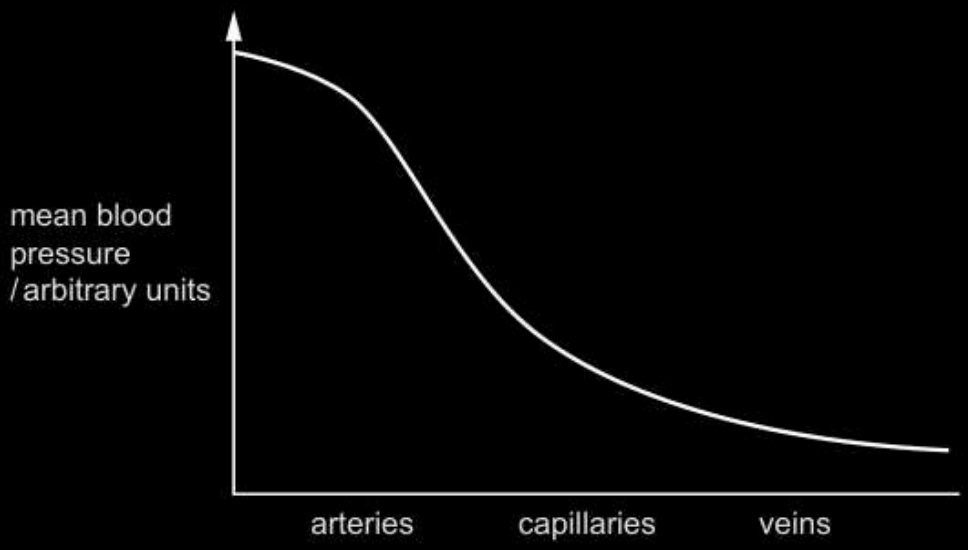
1	one mark for each correct line  	5

Question	Answer	Marks

4 The human double circulation has both

- a circulation taking blood to and from the body tissues, and
- a circulation taking blood to and from the tissues of the lungs.

The graph shows the change in mean blood pressure in the blood vessels as blood passes to and from the **body tissues**.



(a) Name, for the circulation shown in the graph

(i) the first artery that blood enters with the highest mean blood pressure

..... [1]

(ii) the last vein that blood enters with the lowest mean blood pressure.

..... [1]

(b) Explain, using your knowledge of the structures of arteries and veins, the change in mean blood pressure shown in the graph.

.....

.....

.....

.....

.....

..... [3]

(c) Describe, with reference to the movement of **named** substances, the process that takes place as blood passes through the capillaries of the body tissues.

.....

.....

.....

.....

.....

..... [3]

(d) Draw on the graph on page 8 a line to show the change in the mean blood pressure as blood passes to and from the **tissues of the lungs**.

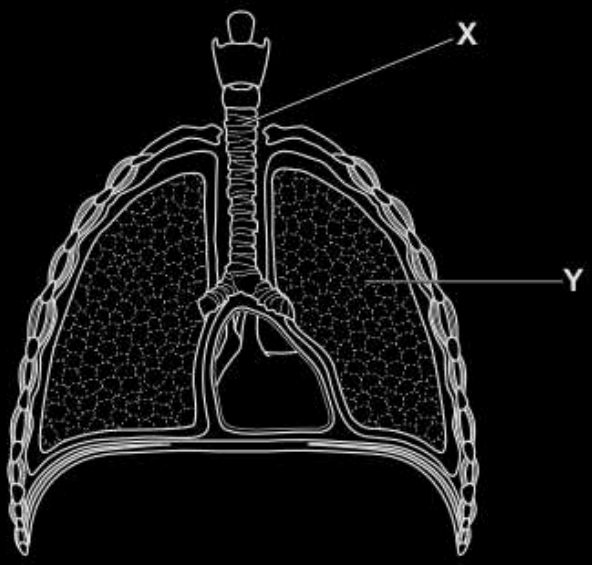
[2]

[Total: 10]

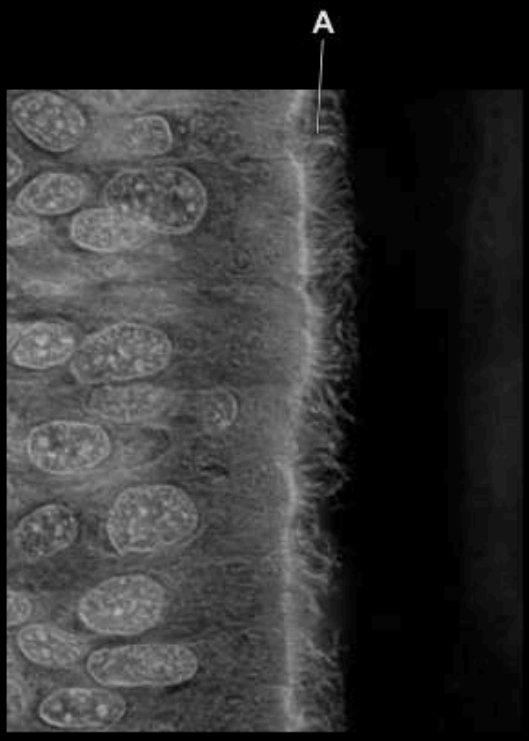
4(a)(i)	aorta ;	**1**
4(a)(ii)	vena cava ;	**1**
4(b)	pressure reduces (from artery to vein) ; vein **+** wider lumen **AW ;** vein **+** thinner layer of muscle / elastic / wall ;	**3**
4(c)	diffusion ; oxygen **+** out of capillary / blood ; carbon dioxide **+** into capillary / blood ; other named chemical **+** correct direction of movement ;	**3**
4(d)	line touches y-axis **+** line extends to end of x-axis ; curved shape to match **+** below that for systemic circuit at all points ;	**2**

Question	Answers	Marks

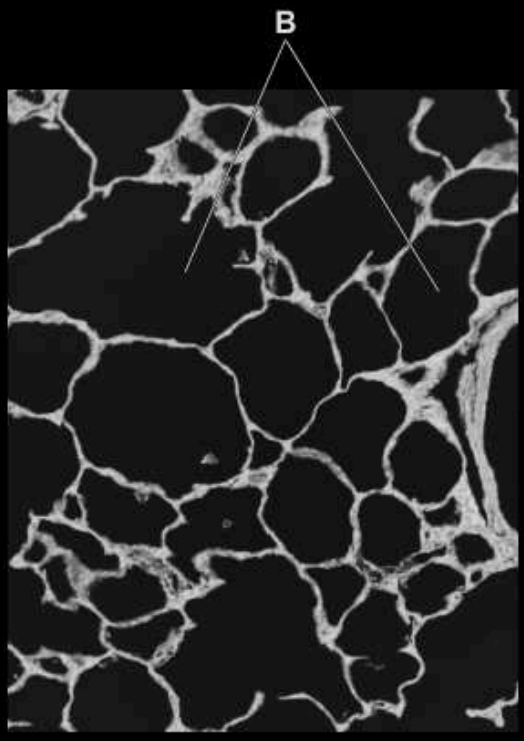
5 The diagram shows the human breathing system.



The photomicrographs below show further details of the two components labelled **X** and **Y**, from the breathing system of a non-smoker seen using a light microscope.



magnification of **X**



magnification of **Y**

(a) (i) Name the structures labelled **A**.

.....

[1]

(ii) Describe the role of the structures labelled **A**.

.....

.....

.....

..... [2]

(b) (i) Name the structures labelled **B**.

.....

[1]

(ii) Describe the changes to the structures labelled **B** that would be caused by tobacco smoke **and** explain the possible effects on a smoker of these changes.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(iii) Outline **two** reasons why many people regard smoking as socially unacceptable.

1 .....

.....

2 .....

.....

[2]

[Total: 10]

Section B

Answer **both** questions in this section.

Write your answers in the spaces provided.

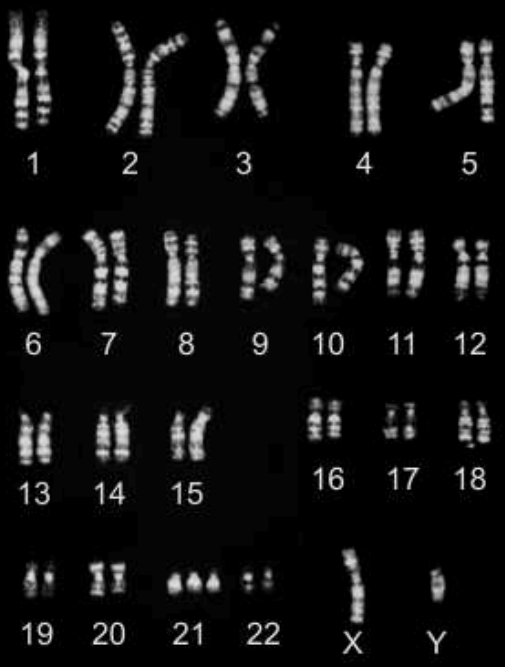
5(a)(i)	cilia ;	**1**
5(a)(ii)	move **+** up / away from lungs ; mucus ; (containing trapped) bacteria / pathogens / dust ;	**2**
5(b)(i)	alveoli / air sac ;	**1**

© UCLES 2020

Question	Answer	Marks
5(b)(ii)	walls broken down **AW** ; less surface area ; less gas exchange ; reference to diffusion ; emphysema ; shortness of breath / difficulty breathing **AW** ; less oxygen **+** to cells / muscles ; less aerobic **+** respiration ; physical activity difficult **AW** ;	**4**
5(b)(iii)	reference to passive smoking **AW** / effect on person other than the smoker ; smell ;	**2**

Question	Answer	Marks

6 The diagram shows a magnified image of the chromosomes from one skin cell of a person.



(a) State **two** conclusions that can be made about this person using **only** information in the diagram **and** explain each conclusion.

1 .....

.....

.....

.....

.....

2 .....

.....

.....

.....

.....

[5]

(b) Name the type of gamete produced by a human male.

.....

Outline the process for producing this type of gamete.

.....

.....

.....

.....

.....

.....

.....

[5]

[Total: 10]

6(a)	male ; X + Y / non-matching sex chromosomes ; Down's syndrome ; 47 + not 46 (chromosomes) ; three / extra + (chromosome) 21 ;	**5**
6(b)	*type of gamete:* sperm ; *process:* testes ; meiosis / reduction division ; chromosome number halved ; haploid ;	**5**

© UCLES 2020

Question	Answer	Marks

8 **(a)** Describe how a plant takes in carbon dioxide from the air and mineral ions from the soil.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

(b) Outline how a **named** mineral ion taken in by a plant from the soil is made available for use in the leaves of the plant to produce chlorophyll.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

8(a)	*carbon dioxide:* diffusion ; down concentration gradient **AW** ; through stomata ; into air spaces (in leaf) ;	**7**

© UCLES 2020

Question	Answer	Marks
8(a)	*mineral ions:* root hair ; through **+** cell wall ; through **+** cell membrane ; active transport ; against concentration gradient **AW** ; reference to use of energy ;	
8(b)	magnesium / nitrates ; dissolve / solution ; in water ; xylem ; reference to transpiration ; into palisade mesophyll ; into chloroplast ;	**3**

Question	Answer	Marks

October/November 2019 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 During a lesson about animal and plant cells, a student reads out a number of statements about cell structure. Only three of his statements are correct.

1 All cells have a cell wall.

2 Cell walls are made of cellulose.

3 Chromosomes carry DNA.

4 Cell walls contain starch.

5 All cells have a cell membrane.

6 A sap vacuole helps an animal cell maintain its turgor.

7 Chromosomes are found in the cytoplasm.

Which three statements are correct?

A 1, 3 and 7 **B** 2, 3 and 5 **C** 2, 4 and 6 **D** 4, 5 and 7

October/November 2019 · Paper 1 · Variant 1 · Question 2

Correct answer: B

2 Which statement is always correct for both diffusion and osmosis?

A They depend on energy from cells.

B They involve movement of particles down a concentration gradient.

C They involve the movement of water molecules.

D They take place only in liquids.

October/November 2019 · Paper 1 · Variant 1 · Question 3

Correct answer: A

3 Which process occurs by active transport?

A uptake of glucose by villi

B uptake of oxygen by root hair cells

C uptake of starch by villi

D uptake of water by root hair cells

October/November 2019 · Paper 1 · Variant 1 · Question 4

Correct answer: D

4 Which statement about enzymes is correct?

A Enzymes are made of lipid molecules and change the rate of a reaction.

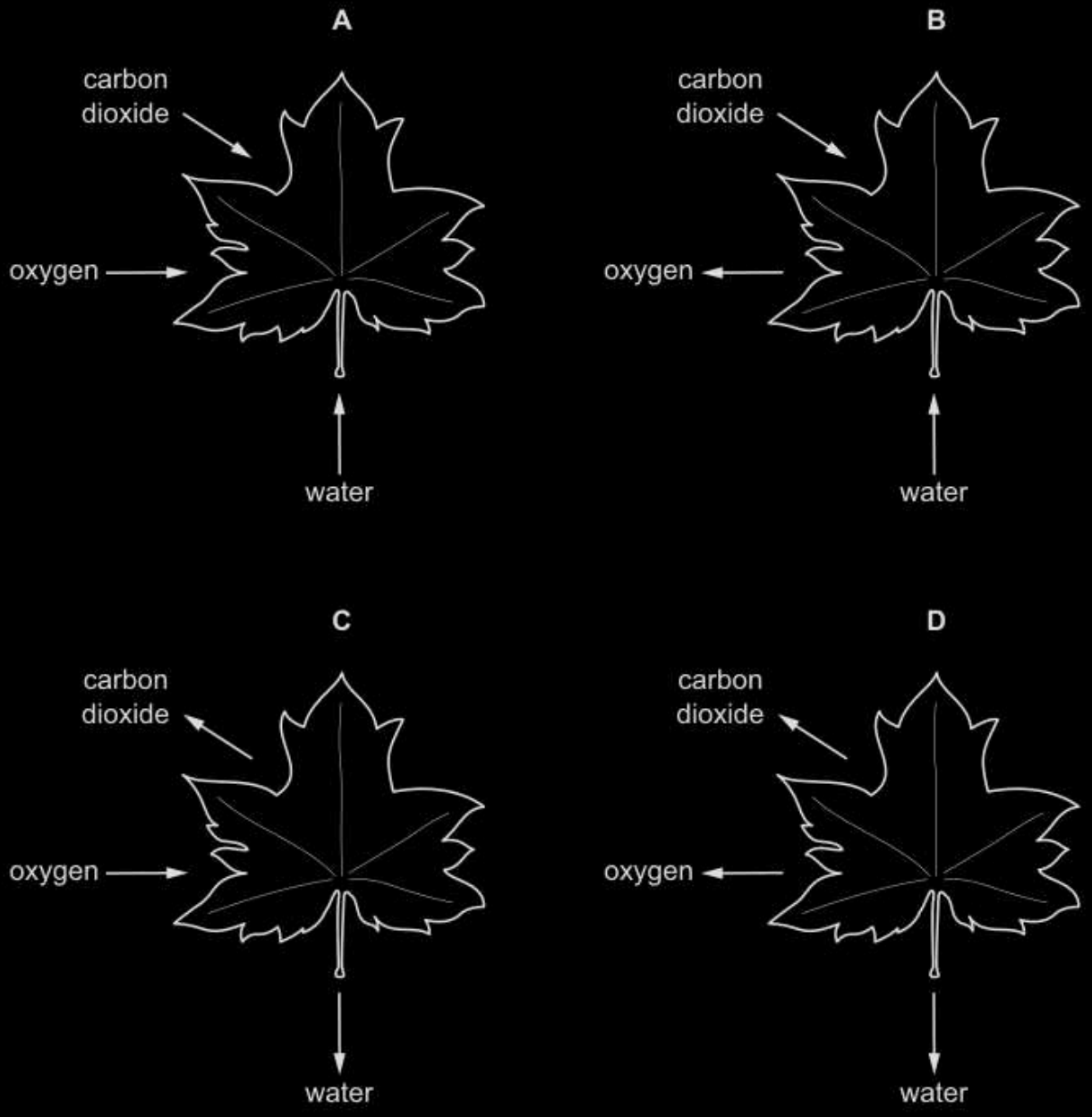
B Enzymes are permanently denatured by high and low temperatures.

C Enzymes can only operate within a living organism.

D Enzymes have individual shapes that lock onto specific molecules.

Correct answer: B

5 Which diagram shows substances going into and out of a leaf when glucose is being formed?



October/November 2019 · Paper 1 · Variant 1 · Question 10

Correct answer: C

10 The table shows the rates of absorption of two different sugars, arabinose and glucose, in living and dead intestines. The concentrations of the sugars inside the intestines were the same in each case.

	rate of absorption / arbitrary units	
	arabinose	glucose
living intestine	31	102
dead intestine	31	34

What are the main methods of absorption of arabinose and glucose in living intestine?

	arabinose	glucose
A	active transport	active transport
B	active transport	diffusion
C	diffusion	active transport
D	diffusion	diffusion

October/November 2019 · Paper 1 · Variant 1 · Question 12

Correct answer: C

12 What is the main cause of water moving up to the leaves in xylem vessels?

A active transport

B evaporation from the epidermis of the leaf

C evaporation from the walls of the mesophyll cells

D use of water in photosynthesis

October/November 2019 · Paper 1 · Variant 1 · Question 13

Correct answer: D

13 What is meant by double circulation?

A Blood travels down the left-hand side of the body and up the right.

B Blood travels away from the heart in the arteries and back to the heart in the veins.

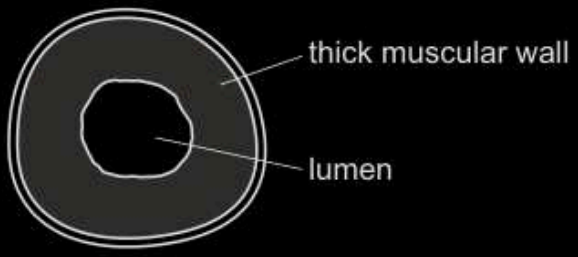
C There are two separate circulatory systems in the body.

D There is a low pressure circulation to the lungs and a high pressure circulation to body tissues.

October/November 2019 · Paper 1 · Variant 1 · Question 14

Correct answer: A

14 The diagram shows a section through a vessel carrying oxygenated blood, as seen through a microscope.



What is this vessel?

A coronary artery

B hepatic portal vein

C pulmonary artery

D renal vein

October/November 2019 · Paper 1 · Variant 1 · Question 15

Correct answer: A

15 Hypoplastic left heart failure is a rare heart condition in which the lower left chamber of the heart has not developed properly and is much smaller than normal.

The immediate result of this condition is to cause lower than normal blood flow into which blood vessel?

A aorta

B pulmonary artery

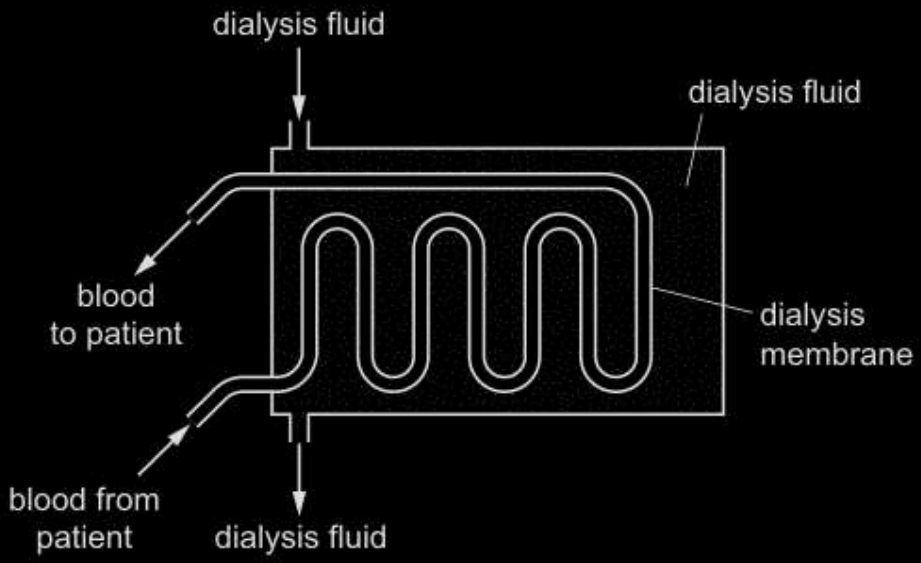
C pulmonary vein

D vena cava

October/November 2019 · Paper 1 · Variant 1 · Question 19

Correct answer: B

19 In a kidney dialysis machine, which substance **cannot** diffuse through the dialysis membrane?



A glucose

B insulin

C sodium

D urea

October/November 2019 · Paper 1 · Variant 1 · Question 21

Correct answer: B

21 Which structure in the brain registers changes in the blood concentration?

A cerebellum

B hypothalamus

C medulla

D pituitary gland

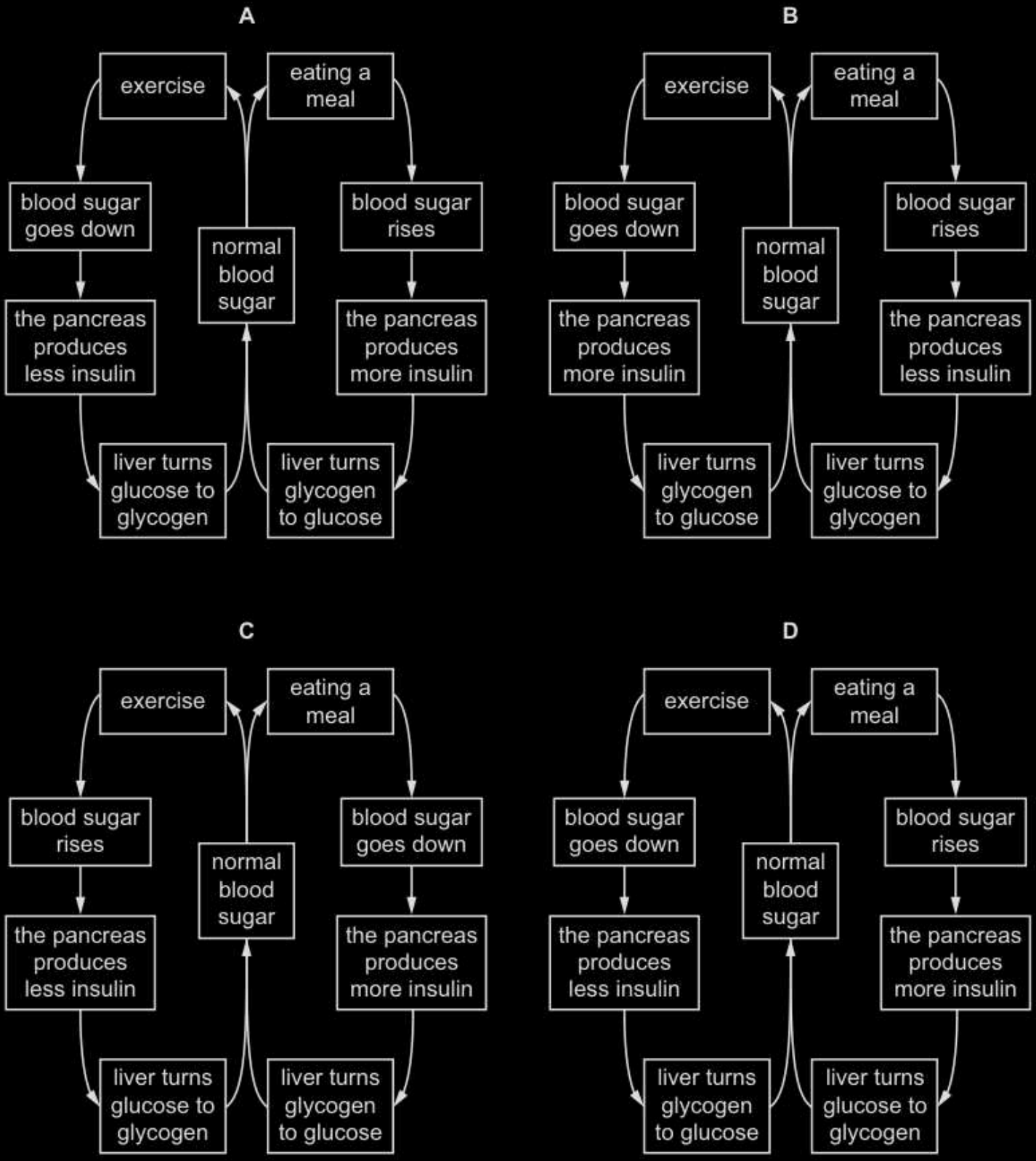
Correct answer: D

22 The diagram shows a section through a human eye.

Where are the receptors for the pupil (iris) reflex?

Correct answer: D

23 Which flow diagram shows the regulation of blood sugar levels?



October/November 2019 · Paper 1 · Variant 1 · Question 25

Correct answer: C

25 Which risk is greatly increased by the nicotine in cigarette smoke?

A bronchitis and emphysema

B cancer of the airways

C high blood pressure

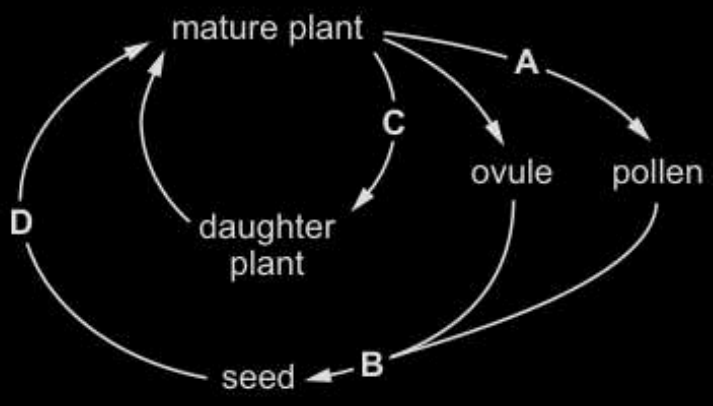
D reduced birth weight of babies

October/November 2019 · Paper 1 · Variant 1 · Question 34

Correct answer: A

34 The diagram shows the life cycle of a species of plant.

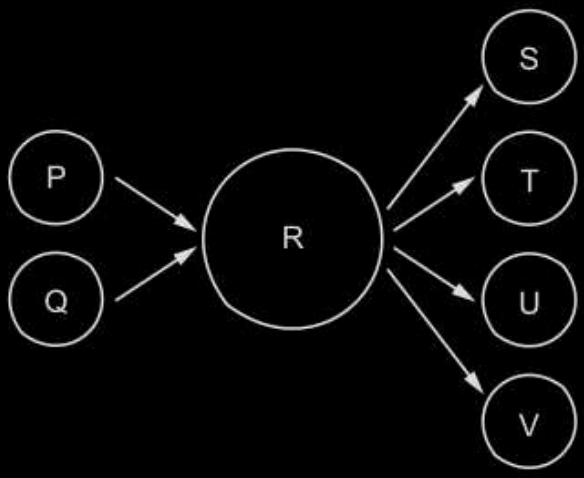
During which stage does meiosis (reduction division) occur?



October/November 2019 · Paper 1 · Variant 1 · Question 36

Correct answer: B

36 The diagram represents gametes P and Q fusing to give cell R. Cell R then produces gametes S, T, U and V.



Which statement about the numbers of chromosomes in the cells and gametes is correct?

A The numbers of chromosomes in P and Q are different.

B The numbers of chromosomes in P and S are the same.

C The number of chromosomes in S is one quarter of the number of chromosomes in R.

D The number of chromosomes in T is half the number of chromosomes in Q.

October/November 2019 · Paper 1 · Variant 1 · Question 37

Correct answer: D

37 Which statements about DNA are correct?

1 An allele carries molecules of DNA.

2 A chromosome includes a molecule of DNA.

3 A molecule of DNA is divided into genes.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

Correct answer: D

40 Which statement about dominant and recessive alleles is **not** correct?

A A dominant characteristic is seen in the phenotype of a heterozygote.

B A homozygous genotype may be either dominant or recessive.

C Recessive phenotypes always have two recessive alleles.

D The phenotype of a homozygote is always dominant.

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

October/November 2019 · Paper 1 · Variant 2 · Question 1

Correct answer: A

1 A human cheek cell and a spongy mesophyll cell from a leaf are examined under a microscope.

Which structures are seen in both cells?

A cell membrane, nucleus and cytoplasm

B cell wall, cell membrane and nucleus

C cytoplasm, cell wall and cell membrane

D nucleus, cytoplasm and cell wall

October/November 2019 · Paper 1 · Variant 2 · Question 2

Correct answer: A

2 Which statements about active transport in root hair cells are correct?

1 Energy is required.

2 Substances may be absorbed as ions.

3 The membrane is partially permeable.

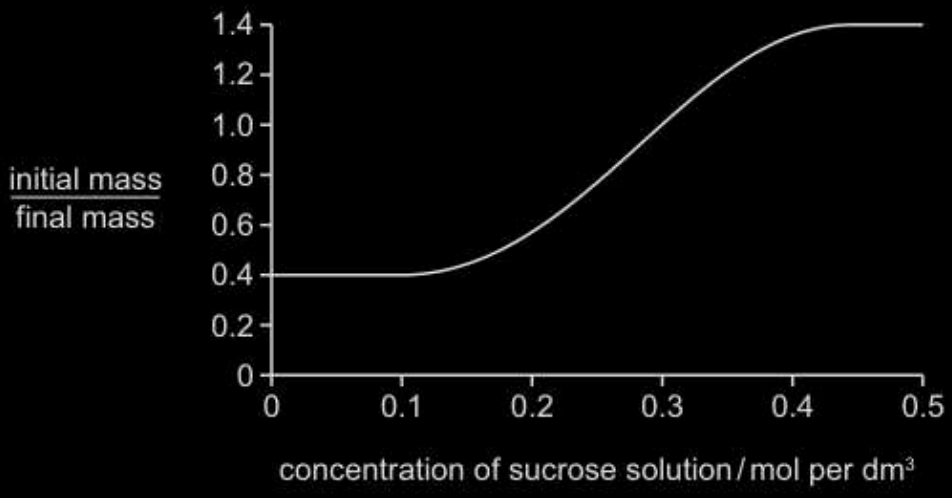
A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

October/November 2019 · Paper 1 · Variant 2 · Question 3

Correct answer: B

3 Discs of raw potato tissue are weighed (initial mass). They are then placed in sucrose solutions of different concentrations. After one hour their masses are measured again (final mass).

The graph shows how the ratio of initial mass : final mass changes with concentration of sucrose solution.



Which concentration of sucrose solution has the same water potential as that of the cells in the potato discs?

A 0.1 mol per dm^3

B 0.3 mol per dm^3

C 0.4 mol per dm^3

D 0.5 mol per dm^3

© UCLES 2019

October/November 2019 · Paper 1 · Variant 2 · Question 4

Correct answer: D

4 What happens to an enzyme after it has catalysed a reaction?

A It can no longer fit the substrate molecules.

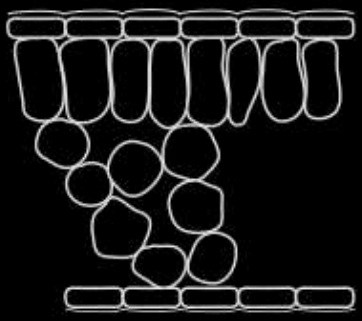
B It can now catalyse several different reactions.

C It has become part of the product molecules.

D It is available to catalyse the same reaction again.

Correct answer: C

6 The diagram shows some of the structures seen in a section through a dicotyledonous leaf.



Which two features are missing from this diagram?

A cuticle and epidermis

B epidermis and guard cells

C guard cells and vascular bundle

D vascular bundle and cuticle

© UCLES 2019

Correct answer: B

9 Which structure has villi to increase its surface area?

A alveolus

B ileum

C root hair

D trachea

© UCLES 2019

October/November 2019 · Paper 1 · Variant 2 · Question 10

Correct answer: C

10 The table shows the rates of absorption of two different sugars, arabinose and glucose, in living and dead intestines. The concentrations of the sugars inside the intestines were the same in each case.

	rate of absorption / arbitrary units	
	arabinose	glucose
living intestine	31	102
dead intestine	31	34

What are the main methods of absorption of arabinose and glucose in living intestine?

	arabinose	glucose
A	active transport	active transport
B	active transport	diffusion
C	diffusion	active transport
D	diffusion	diffusion

October/November 2019 · Paper 1 · Variant 2 · Question 11

Correct answer: A

11 Which statements about root hairs are correct?

1 The cell membrane can allow movement of water and ions into and out of the cell.

2 The cell vacuole extends into the root hair.

3 They are formed as an extension of the outer layer of root cells.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

October/November 2019 · Paper 1 · Variant 2 · Question 12

Correct answer: C

12 What is the main cause of water moving up to the leaves in xylem vessels?

A active transport

B evaporation from the epidermis of the leaf

C evaporation from the walls of the mesophyll cells

D use of water in photosynthesis

October/November 2019 · Paper 1 · Variant 2 · Question 13

Correct answer: B

13 Which blood vessel will have the highest urea concentration after a protein-rich meal?

A hepatic portal vein

B hepatic vein

C renal artery

D renal vein

October/November 2019 · Paper 1 · Variant 2 · Question 15

Correct answer: D

15 Which statements about blood vessels are correct?

	arteries always carry oxygenated blood	the pulmonary vein carries deoxygenated blood
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = yes

✗ = no

Correct answer: A

17 Which process does **not** use energy released from respiration?

A diffusion of substances down a concentration gradient

B muscle contraction

C production of protease enzymes in the pancreas

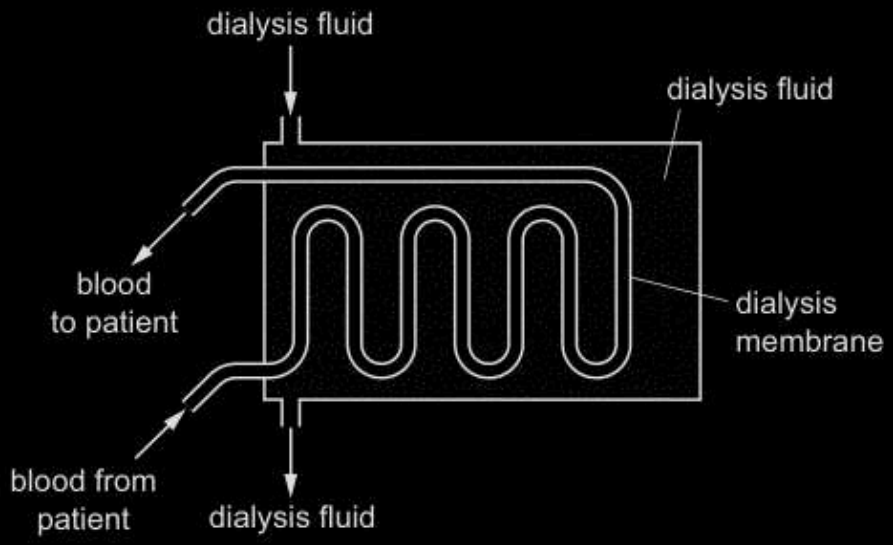
D replacing heat lost from the body

© UCLES 2019

October/November 2019 · Paper 1 · Variant 2 · Question 19

Correct answer: B

19 In a kidney dialysis machine, which substance **cannot** diffuse through the dialysis membrane?



A glucose

B insulin

C sodium

D urea

October/November 2019 · Paper 1 · Variant 2 · Question 20

Correct answer: A

20 On a hot day, the cheeks of a light-skinned person may appear red in colour.

What is the cause of this?

A Arterioles supplying the skin have dilated.

B Capillaries in the skin have moved closer to the surface.

C The person has started to sweat.

D Temperature receptors have ceased to function.

October/November 2019 · Paper 1 · Variant 2 · Question 21

Correct answer: B

21 Which structure in the brain registers changes in the blood concentration?

A cerebellum

B hypothalamus

C medulla

D pituitary gland

October/November 2019 · Paper 1 · Variant 2 · Question 23

Correct answer: B

23 In which situation would the amount of insulin secreted into a person's blood be highest?

A during a game of football

B having a meal of rice

C waking up in the early morning

D working outside on a cold day

October/November 2019 · Paper 1 · Variant 2 · Question 25

Correct answer: C

25 Which risk is greatly increased by the nicotine in cigarette smoke?

A bronchitis and emphysema

B cancer of the airways

C high blood pressure

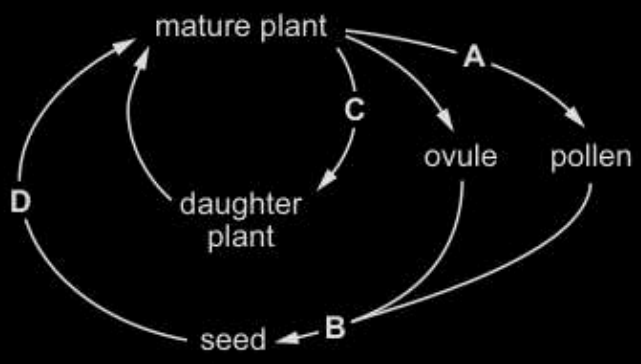
D reduced birth weight of babies

October/November 2019 · Paper 1 · Variant 2 · Question 34

Correct answer: A

34 The diagram shows the life cycle of a species of plant.

During which stage does meiosis (reduction division) occur?



October/November 2019 · Paper 1 · Variant 2 · Question 37

Correct answer: D

37 A student wrote some statements about chromosomes, but made a number of mistakes.

1 There are 46 pairs of chromosomes in a human body cell.

2 In gametes, chromosomes are found in pairs.

3 Males have one X and one Y chromosome in each body cell.

4 Chromosomes contain a long DNA molecule divided into sections called genes.

5 Chromosomes include genes which are divided into sections called DNA molecules.

Which two statements are correct?

A 1 and 2 **B** 1 and 3 **C** 2 and 5 **D** 3 and 4

October/November 2019 · Paper 1 · Variant 2 · Question 38

Correct answer: A

38 Elephants are hunted by poachers for their tusks, which are a source of ivory. In 1930, in a National Park, 1% of the elephants were born without any tusks. Lack of tusks was the result of a gene mutation. Currently, in the same area, 15% of the females and 9% of the males do not have tusks.

Which statements can explain this increase in the number of tuskless elephants?

1 Selection against tusked elephants is occurring.

2 Tuskless elephants are less likely to be killed by poachers.

3 Tuskless elephants reproduce and pass the allele to their offspring.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 only **D** 2 and 3 only

October/November 2019 · Paper 1 · Variant 2 · Question 40

Correct answer: D

40 Which statement about dominant and recessive alleles is **not** correct?

A A dominant characteristic is seen in the phenotype of a heterozygote.

B A homozygous genotype may be either dominant or recessive.

C Recessive phenotypes always have two recessive alleles.

D The phenotype of a homozygote is always dominant.

1 The table shows some of the components found in 100 cm^3 of cow's milk, breast milk and breast milk substitute (formula milk).

component	cow's milk	breast milk	breast milk substitute
protein/g	3.3	1.2	1.3
sugar/g	4.2	6.4	7.0
fat/g	3.0	4.0	1.4
calcium/mg	120.0	120.0	49.0
iron/mg	0.1	0.1	0.5
vitamin C/mg	1.0	2.0	8.3
vitamin D/μg	20.0	200.0	1.2

(a) Name **two** main components of a normal healthy diet that do **not** appear in the table.

1 .....

2 .....

[2]

(b) State which type of milk would be **least** likely to ensure the development of healthy bones and teeth, and explain your answer.

type of milk .....

explanation .....

.....

.....

[3]

(c) State which type of milk would provide a baby with the greatest amount of energy. Give your reasons.

type of milk .....

reasons .....

.....

.....

[3]

(d) Suggest why babies fed on breast milk may have more resistance to disease than those fed on any other type of milk.

.....

.....

.....

..... [2]

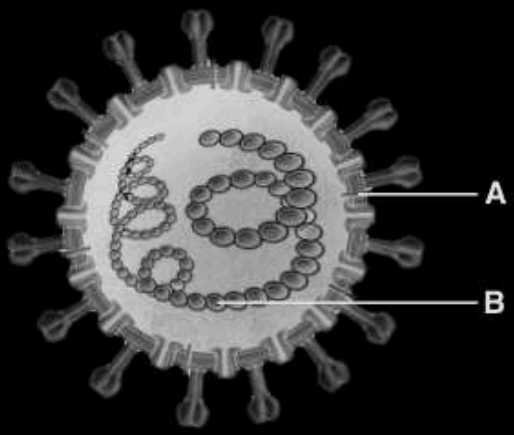
[Total: 10]

1(a)	water ; fibre / roughage ; starch ; other named ion ; other named vitamin ;	**2**	
1(b)	substitute (breast milk) ;	**1**	**A** formula
	lacking or low in vitamin D ; lacking or low in calcium ; reference to rickets ;	**2**	
1(c)	breast milk ;	**1**	
	highest fat / highest fat and sugar ; high sugar ; comparative data quote ; fat yields highest energy (per g) **AW** ;	**2**	
1(d)	antibodies ; (antibodies) made by mother / from mother ; colostrum ; passive immunity ;	**2**	

© UCLES 2019

Question	Answer	Marks	Guidance

2 The diagram shows the structure of a virus.



(a) Identify the structures labelled **A** and **B** on the diagram.

A .....

B ..... [2]

(b) Name a disease caused by

a virus .....

a bacterium .....
[2]

(c) **(i)** State **two** characteristics of bacteria that are **not** characteristics of viruses.

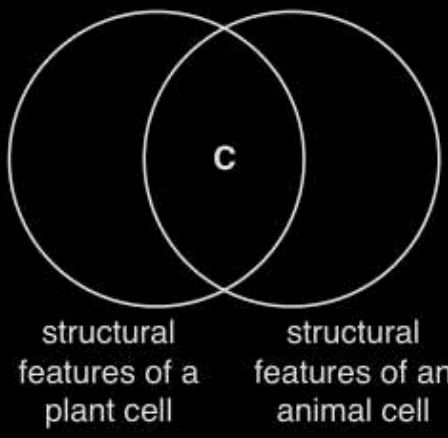
1 .....

2 .....
[2]

(ii) State **one** structural feature **absent** in a bacterial cell that is always present in a plant cell.

..... [1]

(d) The diagram shows how the structural features found in plant cells and in animal cells can be grouped together.



List **three** features that would be grouped in region **C** on the diagram.

1 .....

2 .....

3 .....

[3]

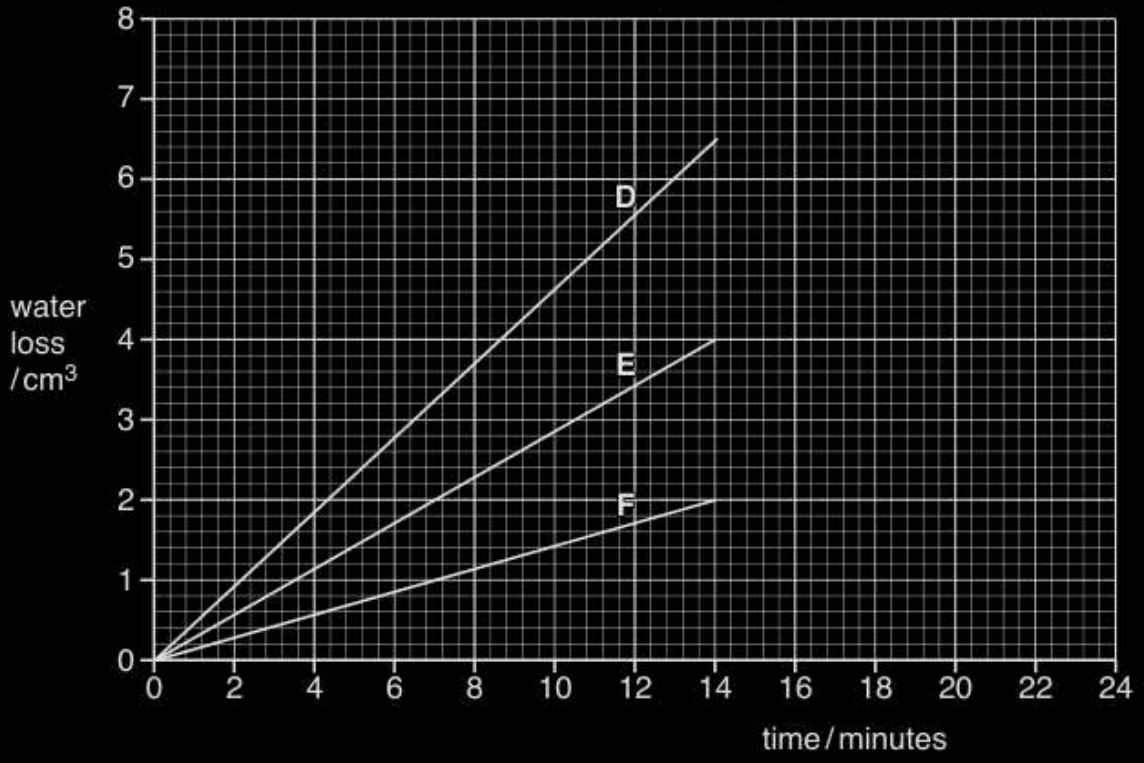
[Total: 10]

2(a)	**(A)** protein (coat) / capsid ; **(B)** RNA or DNA ;	**2**	
2(b)	any viral disease ; any bacterial disease ;	**2**	e.g. AIDS **A** HIV, 'flu' e.g. syphilis / cholera
2(c)(i)	cell / unicellular ; cytoplasm ; plasmid ; cell membrane ; cell wall ; ribosomes ; flagellum ; capsule ; reproduce without host / asexual reproduction / binary fission ; respiration ; method of nutrition ;	**2**	
2(c)(ii)	nucleus ; cellulose cell wall ; mitochondria ; large vacuole / cell sap ;	**1**	
2(d)	cytoplasm ; nucleus ; cell membrane ; ribosomes ; mitochondria ; vacuole ;	**3**	**R** cell sap or large vacuole

© UCLES 2019

Question	Answer	Marks	Guidance

3 The graph shows the rates of water loss for three plants, **D**, **E** and **F**, during the first 14 minutes of an experiment. The plants are of different species and are growing in identical conditions.



(a) Name the process by which plants lose water to the atmosphere.

..... [1]

(b) Suggest **three** reasons for the differences in rates of water loss shown by the three plant species.

1 .....

2 .....

3 .....

[3]

(c) In the experiment, after 14 minutes, air is blown across plant **E** for 2 minutes and a black bag is placed over plant **F** for the remaining 10 minutes.

(i) Continue the lines on the graph to show what would happen to the rates of water loss for plants **E** and **F**. [2]

(ii) Explain the shape of each line you have drawn.

plant **E** .....

.....

.....

plant **F** .....

.....

.....

[5]

3(a)	transpiration / evaporation / diffusion ;	**1**	
3(b)	numbers / size / surface area **+** leaves ; numbers of stomata ; sizes of stomata / degree to which stomata are open ; rates of water uptake / number of root hairs ; sizes of plant ; thickness of cuticle ; hairs / sunken stomata ;	**3**	
3(c)(i)	**E** (drawing) gradient increases for minimum of 2 minutes ; **F** line with a reduced gradient / horizontal line to 24 minutes ;	**2**	
3(c)(ii)	**E** increase **+** water loss / transpiration / evaporation / diffusion ; water (vapour) removed / humidity decreases ; increased concentration gradient ; **F** decrease **+** water loss / transpiration / evaporation / diffusion ; photosynthesis decreases / stops ; guard cells lose their turgidity ; stomata close / stomata not open ; humidity increases ; decreased concentration gradient ;	**5**	

© UCLES 2019

Question	Answer	Marks	Guidance

4 **(a)** Some people find that certain vegetables, such as Brussels sprouts, taste bitter and are unpleasant to eat. Scientists believe that one dominant allele (**T**) of a particular gene gives people the ability to detect the bitter taste.

Two parents find that Brussels sprouts taste bitter. In the space below, draw a genetic diagram to show how these parents can have a child who does **not** find that Brussels sprouts taste bitter.

[5]

(b) Genes are also important in the clotting of blood. Blood will clot normally if a person possesses at least one dominant allele (**B**). This gene is carried only on the X chromosome.

Suggest why more males than females are likely to suffer from a disease in which blood does **not** clot normally.

.....

.....

.....

.....

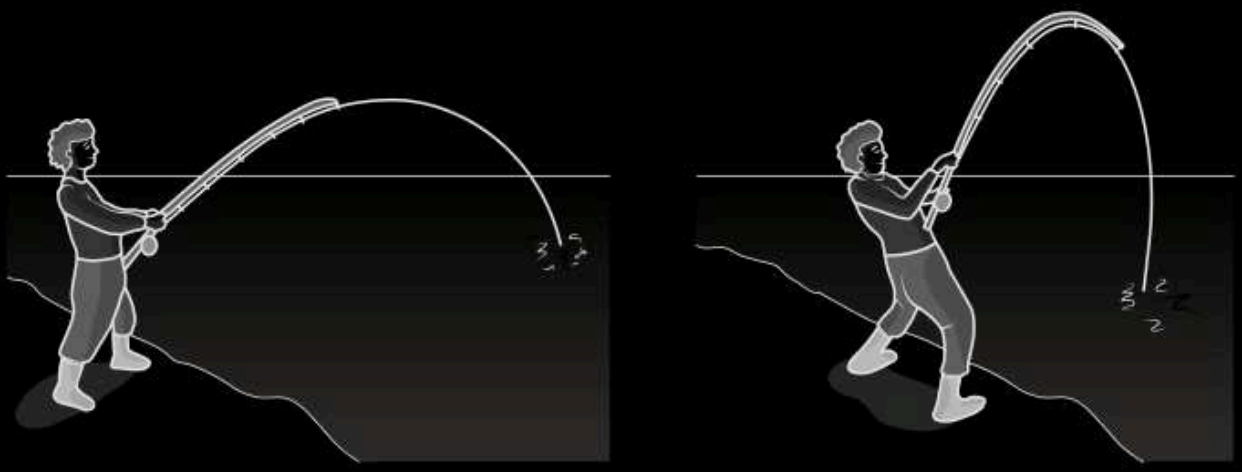
..... [4]

[Total: 9]

4(a)	Tt × Tt ; all four gametes correctly shown (T, t, T, t,) ; diagram labelled with parents / gametes / offspring ; four genotypes shown correctly (TT, Tt ,Tt, tt) ; identification of tt as child who doesn't find sprouts bitter ;	**5**	
4(b)	males are XY ; females are XX ; idea that males only one allele / B / b **AW** ; idea that females have two alleles/ B / b **AW** ; greater chance that males will not have the dominant allele (**B**) **ORA** ;	**4**	

Question	Answer	Marks	Guidance

6 The diagrams show a man fishing.



K L

(a) In diagram **K**, his fishing rod bends downwards when he catches a fish. Explain how his nervous system is made aware that the rod is bending.

.....

.....

.....

.....

.....

..... [5]

(b) Diagram **L** shows him raising the rod further after catching the fish.

Explain how **named** structures in his right arm have caused the arm to bend in the way shown.

.....

.....

.....

.....

.....

..... [5]

6(a)	stimulus ; receptors / named receptor ; in skin / eyes / retina ; impulses ; sensory / afferent neurone ; synapse ; relay / inter/ intermediate ; spinal cord / brain / CNS ;	**5**	
6(b)	biceps ; contracts ; triceps ; relaxes ; antagonistic ; pulls on ulna / ulna moves closer to humerus ; hinge joint + elbow ;	**5**	

© UCLES 2019

Question	Answer	Marks	Guidance

7 **(a)** Explain how energy from the Sun can eventually be used for active transport in the alimentary canal.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

(b) Solar energy is increasingly being used to replace fossil fuels to generate electricity. Explain the **disadvantages** of continuing to use fossil fuels to generate electricity.

.....

.....

.....

.....

..... [3]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7(a)	photosynthesis ; light energy ; starch / carbohydrate ; plant / producer eaten ; digested ; enzyme ; glucose ; respiration ; energy released ; for absorption **AW** ; movement against a concentration gradient ;	**7**	
7(b)	carbon dioxide ; greenhouse effect / global warming/climate change ; sulphur dioxide / NO_x ; acid rain ; conservation of resources / fossil fuels not renewable ;	**3**	

Question	Answer	Marks	Guidance

8 **(a)** Describe and explain how a protein molecule that is eaten becomes molecules of urea that are excreted.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [8]

(b) Explain why carbon dioxide does not normally pass out of the leaves of a plant during the day.

.....

.....

..... [2]

[Total: 10]

8(a)	digested / broken down /hydrolysis ; into amino acids ; by protease or named enzyme ; absorbed in small intestine, duodenum / ileum ; in villi / by capillaries / blood ; hepatic portal vein ; to liver ; deamination or described ; urea + in renal artery ; urea + taken to kidney ; blood filtered / urea leaves blood ; forms urine ; bladder / urethra ;	**8**	

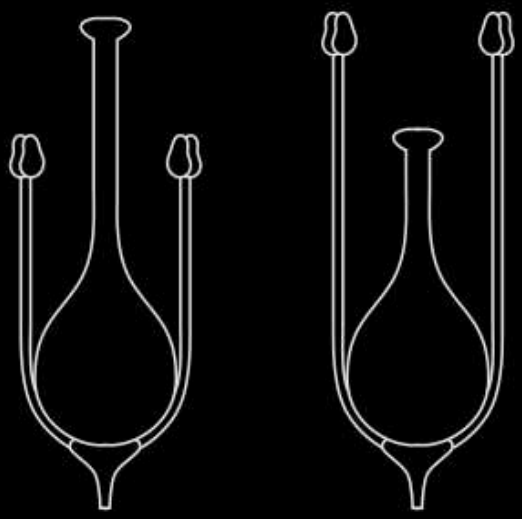
© UCLES 2019

Question	Answer	Marks	Guidance
8(b)	photosynthesis ; uses up CO_2 ; faster than is made during respiration ;	2	

Question	Answer	Marks	Guidance

October/November 2019 · Paper 2 · Variant 2 · Question 1

1 A species of flowering plant has flowers of two types. Some of the plants of this species have flowers called **pin** and some of the plants have flowers called **thrum**. The diagrams show sections through parts of these two types of flower.



pin flower

thrum flower

(a) Describe how the structure of the **pin** type of flower differs from that of the **thrum** type of flower.

.....

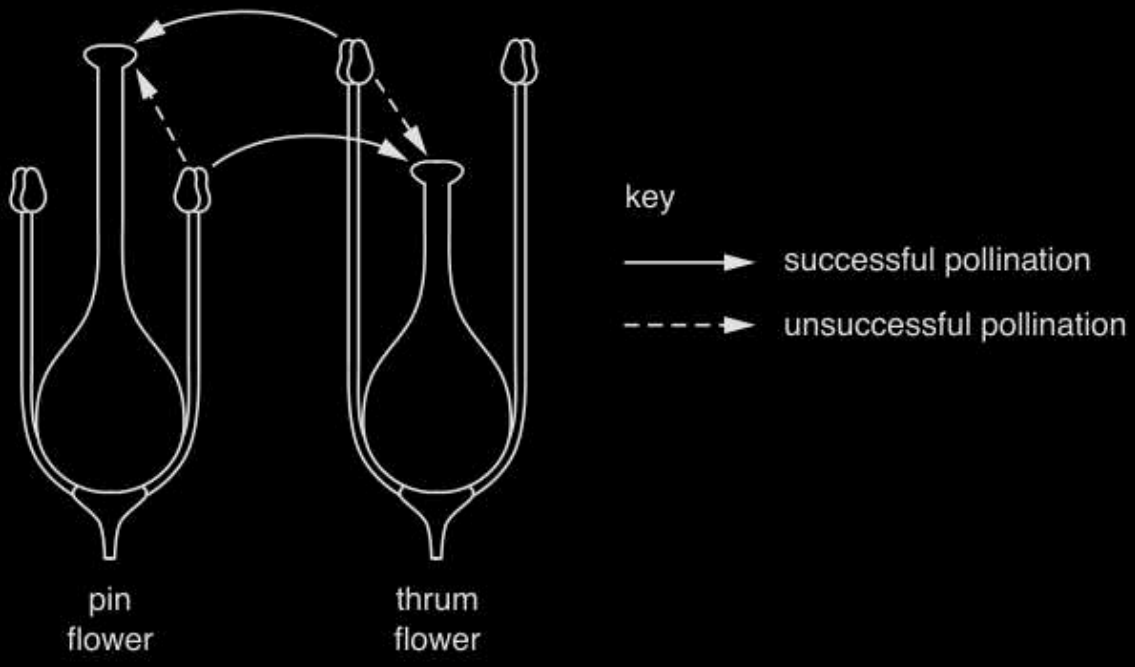
.....

.....

.....

..... [3]

(b) Pollination in this species of plant is not always successful in leading to fertilisation. Examples of successful pollination and of unsuccessful pollination are shown in the diagram below.



Use information from the diagram to name the type of pollination that is successful **and** suggest the advantage to the species of this type of pollination.

type of pollination .....

advantage .....

.....

.....

[3]

(c) The gene responsible for the production of the different types of flower has two alleles. One allele (**T**) is dominant to the other allele (**t**).

A plant with flowers of the **pin** type has a homozygous recessive genotype.

A plant with flowers of the **thrum** type has a heterozygous genotype.

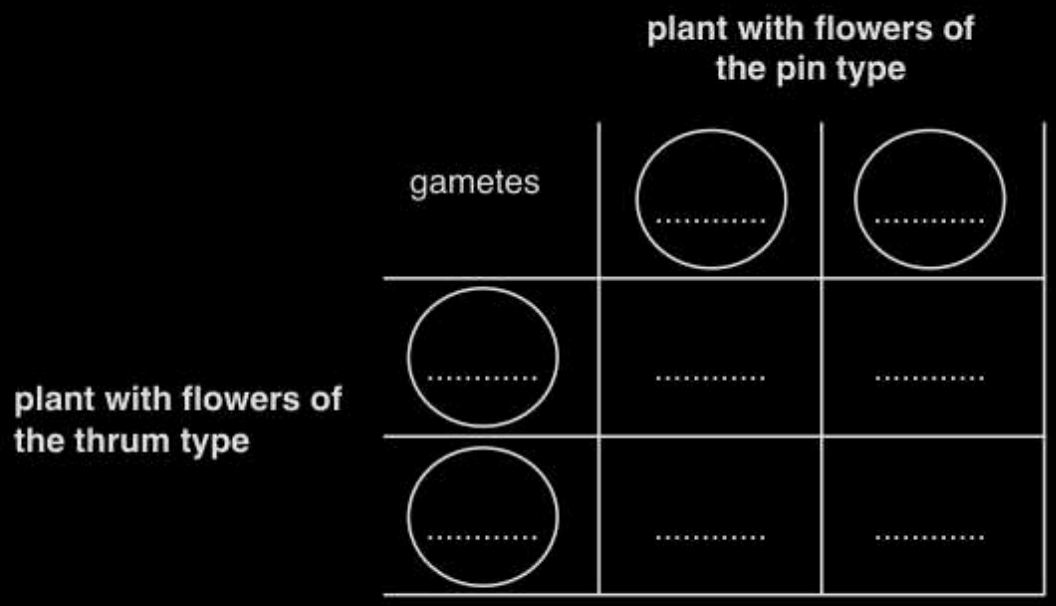
(i) Use the letters **T** and **t** to show the genotype of each of the following:

a plant with flowers of the **pin** type .....

a plant with flowers of the **thrum** type ..... [2]

(ii) A plant with flowers of the **pin** type was successfully pollinated by a plant with flowers of the **thrum** type.

Complete the genetic diagram shown below **and** state the likely ratio of new plants being produced with flowers of the **pin** and **thrum** types.



ratio of new plants with flowers of the **pin** and **thrum** types:

..... [3]

[Total: 11]

1(a)	stamen / anther / filament / androecium ; carpel / pistil / stigma / style / gynoecium ; correct length or height comparison between **pin** and **thrum** ;	**3**	
1(b)	*type of pollination* cross ; *advantage* genetic / genes / alleles ; variation / different ;	**3**	
1(c)(i)	*pin* tt ; *thrum* Tt ;	**2**	**A**Tt genotype to be written as tT
1(c)(ii)	*pin gametes* *thrum gametes* ; *offspring genotypes* Tt + Tt + tt + tt **OR** Tt + Tt + Tt + Tt ; *ratio* 2:2 / 1:1 / $\frac{1}{2}$ / 2 out of 4 / 50% / half / 50:50 **OR** 100% / all + thrum ;	**3**	

© UCLES 2019

Question	Answer	Marks	Guidance

2 Plants absorb light energy to use in the process of photosynthesis.

(a) **(i)** Name the part of a plant cell where photosynthesis takes place.

..... [1]

(ii) State the equation, in words or symbols, for photosynthesis.

..... [2]

(b) Photosynthesis also takes place in some bacteria. Suggest **and** explain the colour of these bacteria.

.....

.....

..... [2]

(c) These bacteria are able to absorb more light than plants because they have additional chemicals that absorb light. Scientists have suggested that it may be possible to transfer a gene that codes for one of these additional chemicals into the DNA of plants used to produce crops that humans eat.

(i) Name the process that scientists would use to transfer the gene from these bacteria into the DNA of a plant.

..... [1]

(ii) Suggest **and** explain the possible advantages of transferring this gene into the DNA of a plant used to produce crops that humans eat.

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

2(a)(i)	chloroplast ;	1	
2(a)(ii)	carbon dioxide / $6CO_2$ + water / $6H_20$ (⟶) ; (⟶) glucose / $C_6H_{12}O_6$ + oxygen / $6O_2$;	2	
2(b)	green ; chlorophyll ;	2	
2(c)(i)	genetic + engineering / modification ;	1	**Ig** biotechnology
2(c)(ii)	increased / more / greater / higher / high ; *any three from ...* light + absorbed **AW** ; photosynthesis ; production **AW** of + glucose / starch / carbohydrate / sugar ; growth / yield / protein synthesis / production of protoplasm ; profit **AW** ;	1 **max 3**	

Question	Answer	Marks	Guidance

October/November 2019 · Paper 2 · Variant 2 · Question 4

4 Proteins are an important component of the human diet. Protein molecules are digested and the products of digestion are absorbed into the blood.

(a) (i) State **one** principal source of protein in the diet.

.....

[1]

(ii) Describe the chemical digestion of proteins in **named** parts of the alimentary canal.

.....

.....

.....

.....

.....

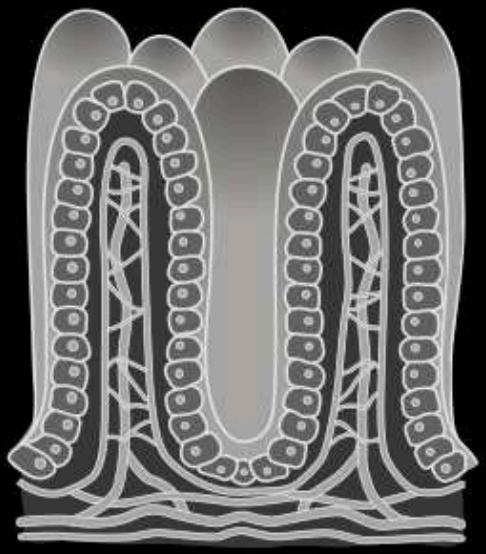
..... [4]

(iii) State where in the alimentary canal the products of digestion are absorbed into the blood.

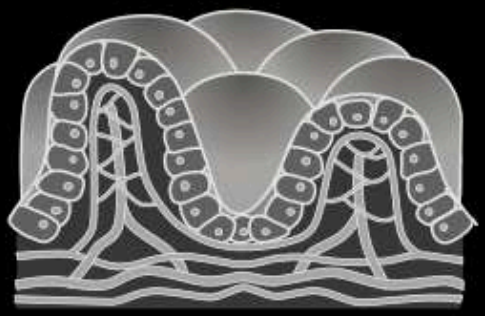
.....

[1]

(b) In a disease that affects a small number of people, the surface through which the products of digestion are absorbed into the blood changes. This change is shown in the diagrams below. Both diagrams are drawn to the same scale.



absorption surface in a person **not affected** by the disease



absorption surface in a person **affected** by the disease

Describe the change in the absorption surface in a person affected by the disease **and** suggest ways in which the person may be affected by this change.

change in absorption surface

.....

.....

.....

.....

ways in which person may be affected

.....

.....

.....

.....

[4]

[Total: 10]

4(a)(i)	meat / fish / eggs / dairy products / nuts / pulses ;	**1**	
4(a)(ii)	stomach ; small intestine / duodenum / ileum ; protease / pepsin / trypsin / erepsin / rennin / peptidases ; into **AW +** amino acids / polypeptides / peptides / peptones ;	**4**	
4(b)	less / reduced / smaller / small / shorter / short ; area ; less **+** diffusion / active transport ; of **+** glucose / amino acids / fatty acids / glycerol / vitamins / minerals ; less **+** assimilation ; result of deficiency in a component named in **4** ; more food required in diet **AW** ;	**4**	

© UCLES 2019

Question	Answer	Marks	Guidance

5 The statements **E** to **L** relate to the process of cell division.

E is used in the repair of damaged tissues

F results in the production of diploid cells

G results in the production of gametes

H results in the chromosome number being halved

J gives rise to genetically identical cells

K takes place in both animals and plants

L is used in the process of asexual reproduction

Complete the table by writing each statement letter in the correct box to show whether the statement relates to mitosis only, meiosis only, or to both.

The first letter has been written in the correct box for you. [6]

mitosis only	meiosis only	both mitosis and meiosis
E		

Section B

Answer **both** questions in this section.

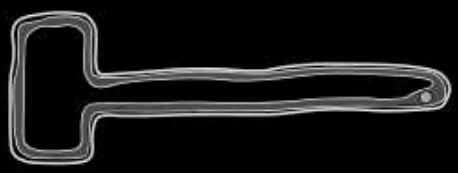
Write your answers in the spaces provided.

Question	Answer	Marks	Guidance
5	**(E no mark)** **F** left column **only ;** **J** left column **only ;** **L** left column **only ;** **G** middle column **only ;** **H** middle column **only ;** **K** right column **only ;**	6	

6 The diagrams show two types of cell specialised to carry out particular functions.

Name each type of cell shown and state the relationship between cell structure and cell function.

(a) cell from a plant



name of cell type .....

relationship between structure and function

.....

.....

.....

.....

.....

.....

[4]

(b) cell from human blood

name of cell type .....

relationship between structure and function

.....

.....

.....

.....

.....

.....

[6]

[Total: 10]

6(a)	root hair ; long / extension / projection / finger-like **AW** ; increased / large **+** surface area ; uptake **AW** of **+** water ; osmosis / diffusion ; uptake **AW** of **+** ions / named ion / minerals / salts ; active transport ;	**4**	

Question	Answer	Marks	Guidance
6(b)	red blood / erythrocyte ; biconcave ; increased / large **+** surface area ; oxygen **+** in / out / contained / carried **AW** ; oxygen **+** diffusion ; no nucleus / anucleate ; increased / large **+** volume / space ; haemoglobin ; oxygen **+** binding / attachment / combine **OR** reference to oxyhaemoglobin ; flexible / elastic / change shape / can squeeze ; into / through **+** capillaries ;	**6**	

Question	Answer	Marks	Guidance

9 Following sexual intercourse in humans, fertilisation may take place.

(a) Describe the process of fertilisation and the early development of a ball of cells.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) The ball of cells may then become implanted in the wall of the uterus and form a developing fetus.

Describe how a developing fetus obtains and uses the **named** substances it needs and removes its **named** waste products.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

[Total: 10]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

9(a)	gamete(s) ; egg(s) / ovum / ova + sperm ; fuse / join / combine **AW** ; zygote ; cell division ; mitosis ; embryo / blastula / blastocyst / morula ;	4	
9(b)	oxygen + respiration ; glucose + respiration ; amino acids + protein synthesis / growth / cell division ; ions / vitamins / minerals / salts / antibodies + for named use ; removal **AW** of + carbon dioxide / urea ; placenta ; umbilical cord ; diffusion ; capillaries ; mother's blood + fetus' **AW** blood ;		

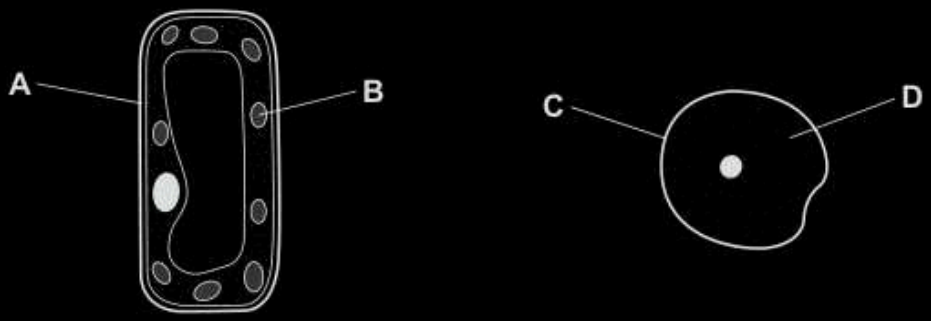
© UCLES 2019

May/June 2019 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 The diagram shows a cell from a plant leaf and a cell from an animal's skin.

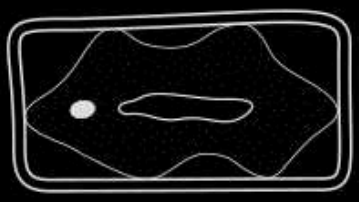
Which part will stain blue-black with iodine solution?



May/June 2019 · Paper 1 · Variant 1 · Question 2

Correct answer: D

2 The diagram shows a plant cell after being placed in a solution for an hour.



What provides the explanation for the appearance of this cell?

A The cell has gained water and is turgid.

B The cell has gained water and is no longer turgid.

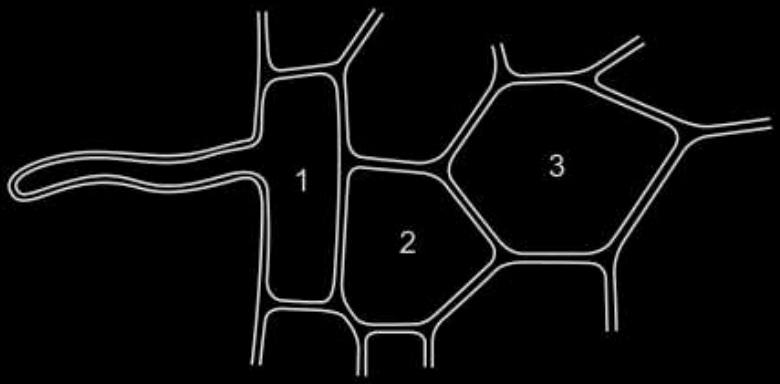
C The cell has lost water and is turgid.

D The cell has lost water and is no longer turgid.

May/June 2019 · Paper 1 · Variant 1 · Question 3

Correct answer: C

3 The diagram shows some cells in the root of a plant that is absorbing water from the soil.



How does the water potential of the cell marked 2 differ from the water potentials of the cells marked 1 and 3?

A higher than cell 1 and higher than cell 3

B higher than cell 1 and lower than cell 3

C lower than cell 1 and higher than cell 3

D lower than cell 1 and lower than cell 3

May/June 2019 · Paper 1 · Variant 1 · Question 4

Correct answer: C

4 Which statement about enzymes is **not** correct?

A All enzymes are made of protein.

B Enzymes are biological catalysts.

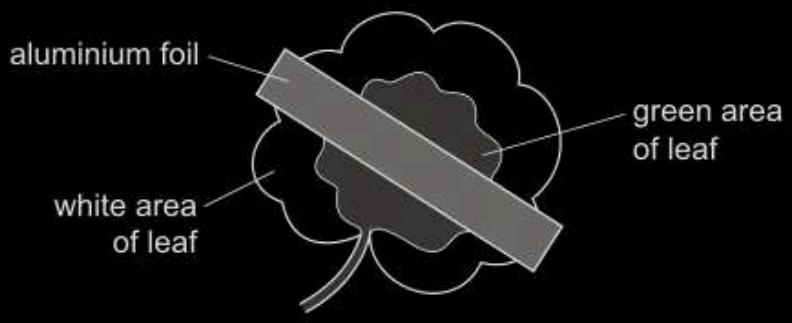
C Enzymes are destroyed during the reaction they are involved in.

D Most human enzymes work best at temperatures between 35 °C and 40 °C.

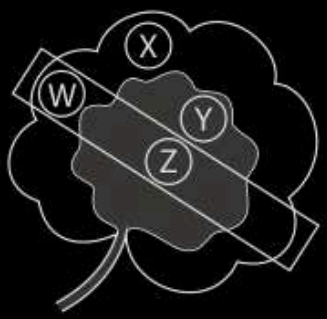
May/June 2019 · Paper 1 · Variant 1 · Question 7

Correct answer: D

7 A plant has leaves with both green and white areas. One of its leaves is partly covered with aluminium foil which blocks light.



The plant is then placed under a lamp for 24 hours. After this time, discs are cut from the areas of the leaf shown, and tested with iodine solution.



Which of the leaf discs will give a blue-black colour when tested with iodine solution?

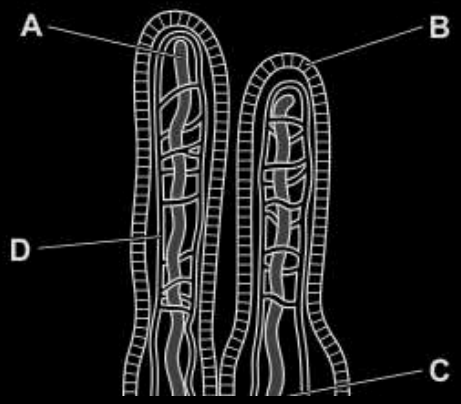
A W and X **B** X only **C** Y and Z **D** Y only

May/June 2019 · Paper 1 · Variant 1 · Question 8

Correct answer: A

8 The diagram shows a section of the wall of the small intestine.

Into which structure are most of the lipid components of the diet absorbed for transport to the rest of the body?



May/June 2019 · Paper 1 · Variant 1 · Question 10

Correct answer: D

10 Which statement describes the function of the stomach?

A It absorbs indigestible food material.

B It is the place where protease and bile work.

C It is the first place in the alimentary canal where starch digestion happens.

D It secretes protease and strong acid.

May/June 2019 · Paper 1 · Variant 1 · Question 11

Correct answer: A

11 Which statement best describes the movement of water through the stem of a plant?

A Water moves up the stem as it evaporates from the leaves.

B Water moves up the stem as it is pushed by water entering the roots.

C Water moves up the stem because it is used in photosynthesis.

D Water moves up the stem by osmosis.

May/June 2019 · Paper 1 · Variant 1 · Question 13

Correct answer: C

13 In a muscle, which two substances show net movement from the plasma **into** the tissue fluid?

A carbon dioxide and glucose

B carbon dioxide and lactic acid

C glucose and oxygen

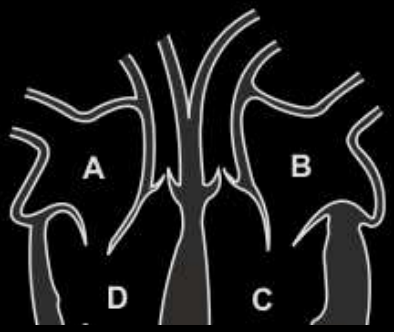
D lactic acid and oxygen

May/June 2019 · Paper 1 · Variant 1 · Question 14

Correct answer: B

14 A patient inhales a drug.

Which chamber of the patient's heart would be the first to receive blood containing the drug?



Correct answer: C

15 Which table shows vessels carrying blood to and from the heart, liver and lungs?

A

blood to organ	organ	blood away from organ
pulmonary artery	heart	pulmonary vein
hepatic portal vein	liver	hepatic vein
pulmonary vein	lungs	pulmonary artery

B

blood to organ	organ	blood away from organ
pulmonary artery	heart	pulmonary vein
hepatic vein	liver	hepatic portal vein
pulmonary vein	lungs	pulmonary artery

C

blood to organ	organ	blood away from organ
pulmonary vein	heart	pulmonary artery
hepatic portal vein	liver	hepatic vein
pulmonary artery	lungs	pulmonary vein

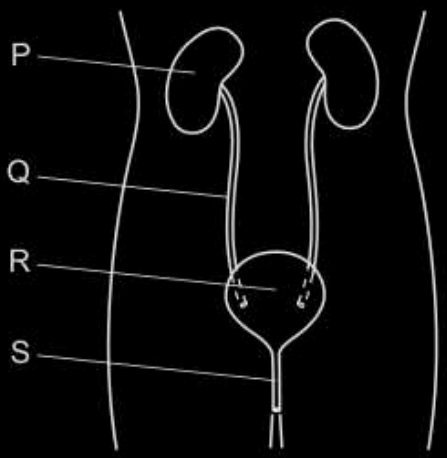
D

blood to organ	organ	blood away from organ
pulmonary vein	heart	pulmonary artery
hepatic vein	liver	hepatic portal vein
pulmonary artery	lungs	pulmonary vein

May/June 2019 · Paper 1 · Variant 1 · Question 19

Correct answer: C

19 The diagram shows the kidneys and their associated structures.



Which row identifies P, Q, R and S?

	P	Q	R	S
A	bladder	ureter	urethra	kidney
B	bladder	urethra	kidney	ureter
C	kidney	ureter	bladder	urethra
D	kidney	urethra	bladder	ureter

May/June 2019 · Paper 1 · Variant 1 · Question 20

Correct answer: D

20 By which mechanism would the body be able to lose most heat on a hot dry day, if the surrounding air temperature is higher than body temperature?

A Blood vessels in the skin become narrower.

B Hair erector muscles relax so hairs lay flat.

C Shivering by body muscles ceases.

D Sweat glands become more active.

May/June 2019 · Paper 1 · Variant 1 · Question 21

Correct answer: D

21 What are the short-term effects of adrenaline on the body?

1 decreased blood sugar concentration

2 decreased heart rate

3 increased blood sugar concentration

4 increased heart rate

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

May/June 2019 · Paper 1 · Variant 1 · Question 22

Correct answer: C

22 How does hormonal control usually differ from nervous control?

	speed of message	duration of effect
A	faster	long-term
B	faster	short-term
C	slower	long-term
D	slower	short-term

May/June 2019 · Paper 1 · Variant 1 · Question 23

Correct answer: D

23 A girl is waiting at a bus stop and is watching a bus travelling towards her.

What happens to her eyes as the bus gets closer?

	ciliary muscle	change in shape of lens	change in suspensory ligaments
A	contracts less	less rounded	less tension
B	contracts less	more rounded	more tension
C	contracts more	less rounded	more tension
D	contracts more	more rounded	less tension

May/June 2019 · Paper 1 · Variant 1 · Question 26

Correct answer: B

26 What are characteristics of bacteria?

	contain cytoplasm	have DNA	nuclear membrane present
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key

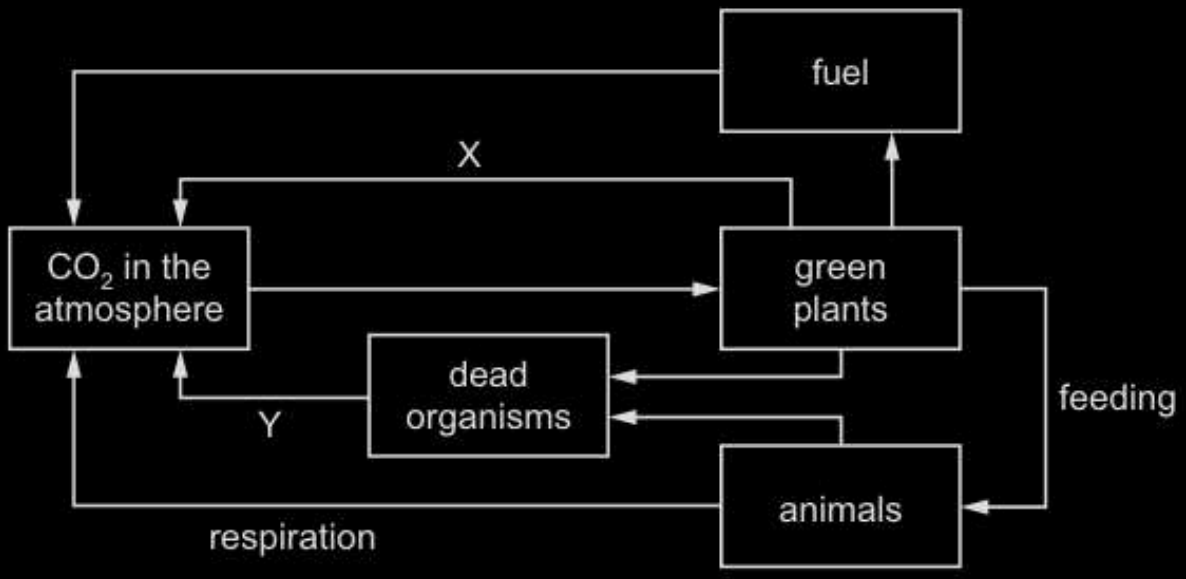
✓ = yes

✗ = no

May/June 2019 · Paper 1 · Variant 1 · Question 30

Correct answer: D

30 The diagram shows part of the carbon cycle.



What are the processes X and Y?

	X	Y
A	photosynthesis	decomposition
B	photosynthesis	excretion
C	respiration	photosynthesis
D	respiration	respiration

May/June 2019 · Paper 1 · Variant 1 · Question 37

Correct answer: A

37 Two heterozygous plants are crossed.

What is the likely ratio of homozygous genotypes to heterozygous genotypes in the offspring?

	homozygous genotypes	:	heterozygous genotypes
A	1	:	1
B	1	:	2
C	1	:	3
D	3	:	1

May/June 2019 · Paper 1 · Variant 1 · Question 38

Correct answer: B

38 Which row shows the sex chromosomes when parents have a baby boy?

	in the egg cell	in the sperm cell	in the zygote
A	X	X	XX
B	X	Y	XY
C	Y	X	XY
D	Y	Y	YY

May/June 2019 · Paper 1 · Variant 1 · Question 40

Correct answer: C

40 Which term describes reproduction between those members of a species that are best fitted to their environment?

A discontinuous variation

B gene mutation

C natural selection

D survival of the fittest

May/June 2019 · Paper 1 · Variant 2 · Question 1

Correct answer: B

1 Which two structures are found in all plant epidermal cells?

A cell wall and chloroplasts

B cell wall and nucleus

C chloroplasts and starch grains

D nucleus and starch grains

May/June 2019 · Paper 1 · Variant 2 · Question 2

Correct answer: C

2 Which process only occurs by diffusion?

A movement of sugars in the phloem

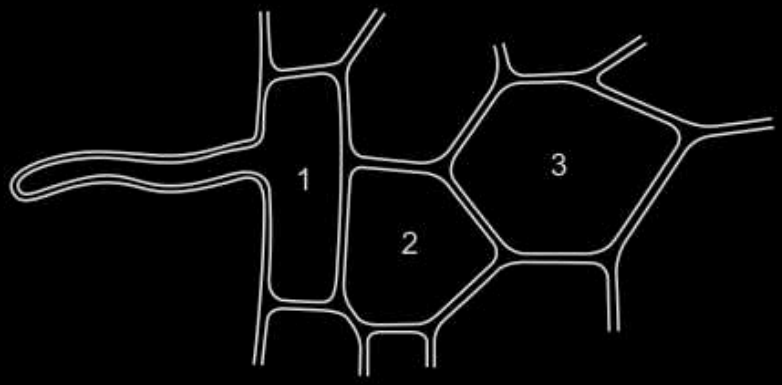
B movement of water in the xylem

C uptake of carbon dioxide by the blood

D uptake of glucose by the intestinal villi

Correct answer: C

3 The diagram shows some cells in the root of a plant that is absorbing water from the soil.



How does the water potential of the cell marked 2 differ from the water potentials of the cells marked 1 and 3?

A higher than cell 1 and higher than cell 3

B higher than cell 1 and lower than cell 3

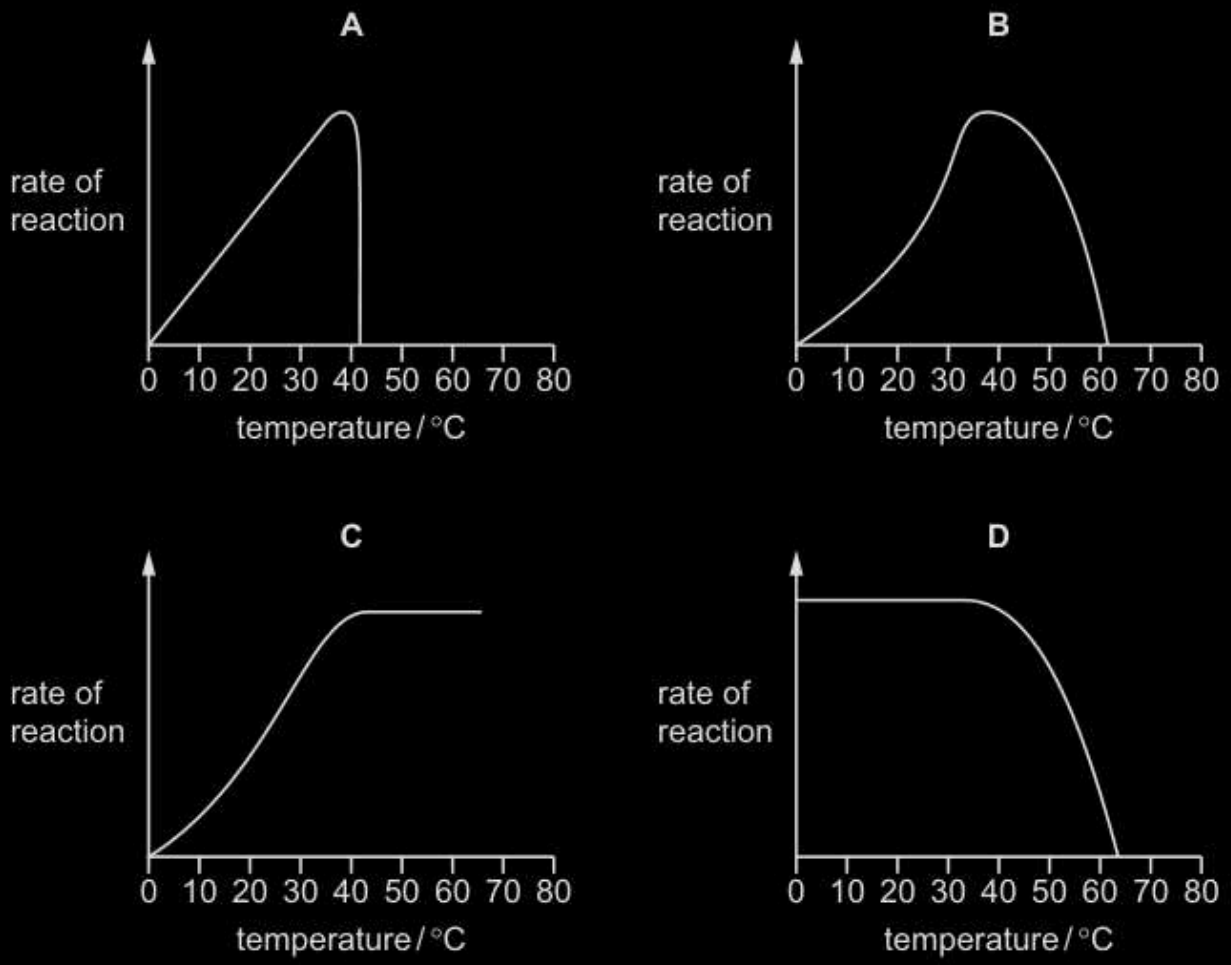
C lower than cell 1 and higher than cell 3

D lower than cell 1 and lower than cell 3

May/June 2019 · Paper 1 · Variant 2 · Question 4

Correct answer: B

4 Which graph shows the effect of temperature on the rate of a reaction catalysed by amylase?

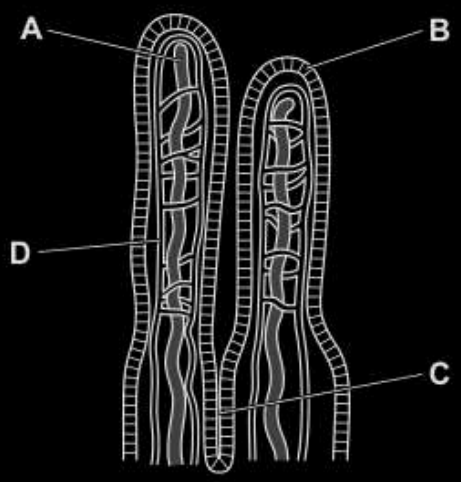


May/June 2019 · Paper 1 · Variant 2 · Question 8

Correct answer: A

8 The diagram shows a section of the wall of the small intestine.

Into which structure are most of the lipid components of the diet absorbed for transport to the rest of the body?



May/June 2019 · Paper 1 · Variant 2 · Question 9

Correct answer: C

9 A surgical procedure involves the removal of the gall bladder.

What are patients who have undergone this procedure **not** able to do?

A mix food with gastric juice

B produce bile

C store bile

D store urine

May/June 2019 · Paper 1 · Variant 2 · Question 11

Correct answer: A

11 Which statement best describes the movement of water through the stem of a plant?

A Water moves up the stem as it evaporates from the leaves.

B Water moves up the stem as it is pushed by water entering the roots.

C Water moves up the stem because it is used in photosynthesis.

D Water moves up the stem by osmosis.

May/June 2019 · Paper 1 · Variant 2 · Question 13

Correct answer: C

13 In a muscle, which two substances show net movement from the plasma **into** the tissue fluid?

A carbon dioxide and glucose

B carbon dioxide and lactic acid

C glucose and oxygen

D lactic acid and oxygen

May/June 2019 · Paper 1 · Variant 2 · Question 14

Correct answer: D

14 Through which valve does the blood pass as it leaves the right ventricle and which structure does it next reach?

	valve	structure
A	atrioventricular (bicuspid)	left atrium
B	atrioventricular (tricuspid)	right atrium
C	semi-lunar	aorta
D	semi-lunar	pulmonary artery

May/June 2019 · Paper 1 · Variant 2 · Question 15

Correct answer: A

15 Which statement about white blood cells is correct?

A They can ingest bacteria by a process called phagocytosis.

B They can produce antigens in response to foreign antibodies.

C They convert fibrinogen to fibrin when bleeding happens.

D They transport toxins from body tissues to the liver.

May/June 2019 · Paper 1 · Variant 2 · Question 19

Correct answer: B

19 How does the composition of the blood in the renal vein differ from that in the renal artery?

	carbon dioxide concentration	urea concentration
A	higher	higher
B	higher	lower
C	lower	higher
D	lower	lower

May/June 2019 · Paper 1 · Variant 2 · Question 21

Correct answer: B

21 Which statement describes a simple reflex action involving an arm or leg?

A It does not involve the brain or spinal cord.

B It involves the spinal cord, and the brain is informed as to what has happened.

C It involves the spinal cord, without any information passing to the brain.

D The response is caused by nerve impulses from the brain.

May/June 2019 · Paper 1 · Variant 2 · Question 22

Correct answer: B

22 What is correct for the formation of urea?

	formed from	formed in
A	amino acids	kidney
B	amino acids	liver
C	carbohydrate	kidney
D	carbohydrate	liver

May/June 2019 · Paper 1 · Variant 2 · Question 23

Correct answer: A

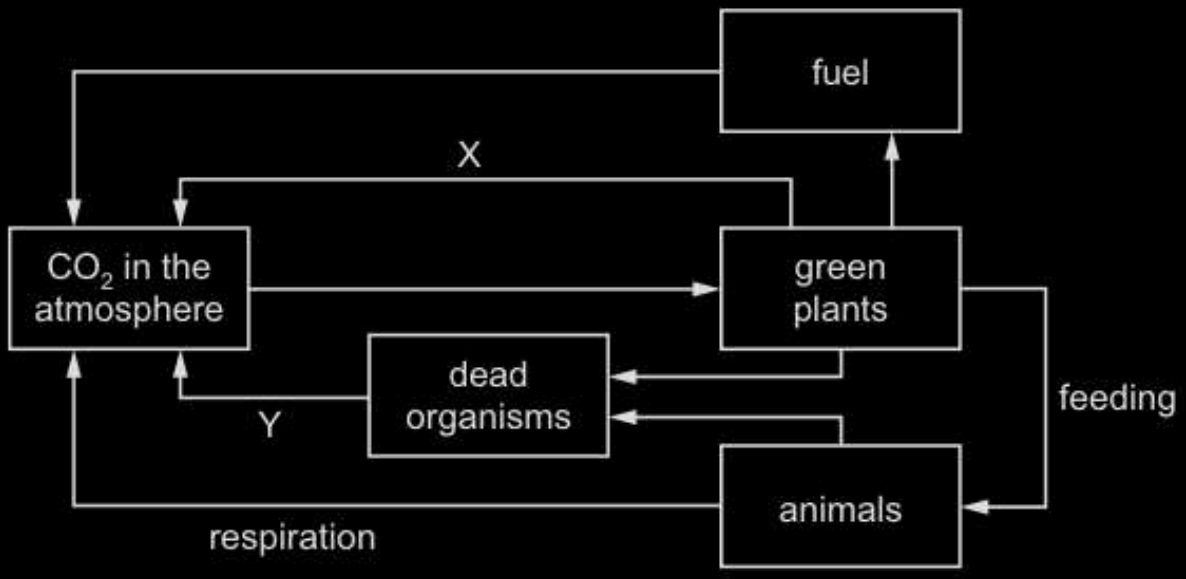
23 What happens when a person focuses on a near object?

	ciliary muscles	lens	suspensory ligaments
A	contract	rounded	loose
B	contract	thin	taut
C	relax	rounded	loose
D	relax	thin	taut

May/June 2019 · Paper 1 · Variant 2 · Question 30

Correct answer: D

30 The diagram shows part of the carbon cycle.



What are the processes X and Y?

	X	Y
A	photosynthesis	decomposition
B	photosynthesis	excretion
C	respiration	photosynthesis
D	respiration	respiration

May/June 2019 · Paper 1 · Variant 2 · Question 35

Correct answer: C

35 Which sign of syphilis may appear many years after infection, if treatment is **not** given?

A a discharge from the penis or vagina

B a hard red ulcer appearing where infection happened

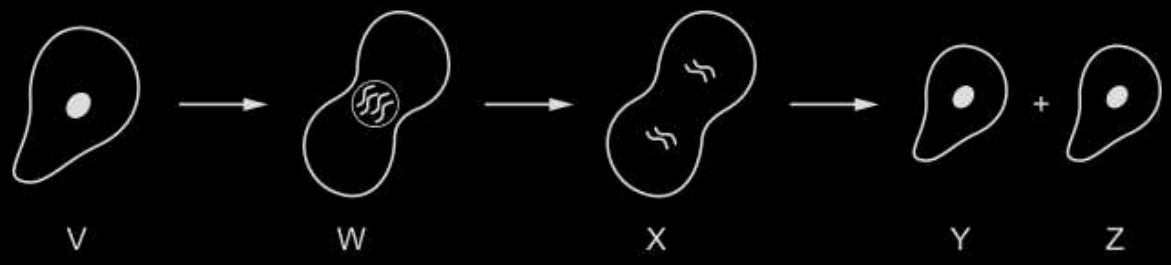
C damage to the central nervous system

D developing a red rash in many areas of the body

May/June 2019 · Paper 1 · Variant 2 · Question 36

Correct answer: C

36 The diagram shows some stages during the asexual reproduction of a single-celled organism.



Which row correctly shows the relative amounts of DNA in each of the cells V, W, X, Y and Z?

	V	W	X	Y	Z
A	1	1	1	2	2
B	1	1	2	1	1
C	1	2	2	1	1
D	2	1	1	2	2

May/June 2019 · Paper 1 · Variant 2 · Question 37

Correct answer: A

37 Two heterozygous plants are crossed.

What is the likely ratio of homozygous genotypes to heterozygous genotypes in the offspring?

	homozygous genotypes	:	heterozygous genotypes
A	1	:	1
B	1	:	2
C	1	:	3
D	3	:	1

May/June 2019 · Paper 1 · Variant 2 · Question 40

Correct answer: C

40 Which term describes reproduction between those members of a species that are best fitted to their environment?

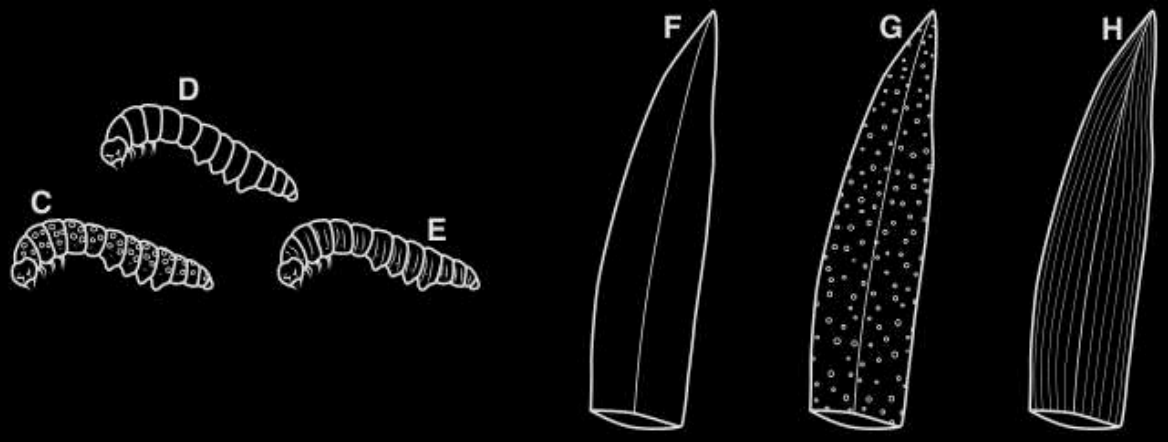
A discontinuous variation

B gene mutation

C natural selection

D survival of the fittest

2 The diagram shows three varieties, **C**, **D** and **E**, of the same species of insect, and three different leaf patterns, **F**, **G** and **H**, of the same species of plant on which the insect feeds.



(a) Suggest which variety of the insect is likely to be found in the greatest numbers in areas where leaf pattern **G** of the plant is found, and explain your answer.

variety of insect .....

explanation .....

.....

.....

[2]

(b) Over a period of many years, as they grow, the plants lose the dots and stripes on their leaves which become plain.

Suggest and explain what is likely to happen to the numbers of the different varieties of insect in an area where the majority of plants are old.

.....

.....

.....

.....

.....

..... [4]

(c) Two alleles, **T** and **t**, control the body pattern of the insects.
Insects with dots (**C**) are homozygous dominant.
Insects with stripes (**E**) are homozygous recessive.
Plain insects (**D**) are heterozygous.

Explain why all three varieties of insect will continue to be produced even in areas where all the plants have plain leaves.

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

2(a)	**C /** insect with dots / spots **AW ;**	**1**	
	less easily seen / camouflaged / markings same as leaf / not eaten **;**	**1**	
2(b)	**C** and **E** decline **AW ;** **C** and **E +** easily seen / eaten / not camouflaged / markings different to leaves **AW ;** **D +** increases / breeds / reproduces **;** **D +** less easily seen / camouflaged / markings same as leaf / not eaten **;** hands on gene / allele(s) **;** natural selection / evolution **;**	**4**	

© UCLES 2019

Question	Answer	Marks	Guidance
2(c)	**D +** Tt **;** Tt × Tt / heterozygous **+** cross / mate / reproduce **;** reference to handing on **both** T and t alleles / gametes **;** produces TT **+** dots **;** tt **+** stripes **;**	**4**	

Question	Answer	Marks	Guidance

6 **(a)** Define osmosis and explain why osmosis is a special form of diffusion.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

(b) Explain the importance of active transport in the human alimentary canal.

.....

.....

.....

..... [3]

[Total: 10]

6(a)	movement + molecules ; water ; water potential ; high to low ; until concentrations equal either side / equilibrium / force applied to prevent it ; (osmosis) partially permeable **AW** membrane ; (diffusion) membrane not required ; energy not required **AW** ;	**7**	
6(b)	absorption / uptake ; by villi ; of digested carbohydrate / glucose / amino acids / fatty acids / glycerol / minerals / ions **or** salts **or** named ; faster than diffusion ; against a concentration gradient ; to prevent wastage / egestion **AW** ;	**3**	

© UCLES 2019

Question	Answer	Marks	Guidance

8 Describe and compare the following:

(a) gene and allele

.....

.....

.....

.....

.....

.....

.....

..... [5]

(b) continuous variation and discontinuous variation.

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

8(a)	made of DNA ; part of a chromosome ; copied / passed on / inherited / hereditary ; one + character(istic) / trait ; one + protein / polypeptide ; (allele) a form / type / version of a gene ; may be dominant / recessive / co-dominant ;	**5**	

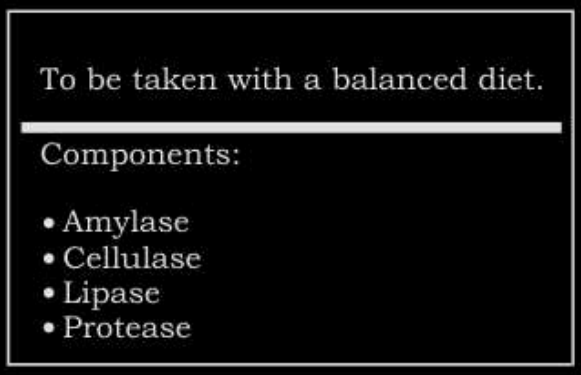
© UCLES 2019

Question	Answer	Marks	Guidance
8(b)	*continuous* over a range / many phenotypes / intermediates ; the result of interaction between genes and the environment ; valid example (skin colour, height, weight) ; *discontinuous* very few forms / few phenotypes / no intermediates ; the result of genes only ; valid example (blood group, eye colour, sex) ; inherited / passed on to next generation ;	5	

Question	Answer	Marks	Guidance

1 Food supplements are substances that are often added to a person's diet. These supplements may be in the form of a tablet.

The diagram shows the label from a food supplement.



This supplement may be taken by a person whose pancreas is not producing enough of some of the components shown on the label.

(a) (i) State the type of chemical that the supplement contains.

.....

[1]

(ii) Each food group in the diet is the substrate for a specific component of the supplement.

Complete the table by writing the most appropriate word or words in each box.

component of supplement	substrate	end product or products
amylase		
protease		
lipase		

[3]

(iii) Cellulase is a component of the supplement that is **not** usually found in the human body.

Use your knowledge of the structure of a plant cell to suggest the function of the cellulase component of the supplement.

.....

(b) The supplement is taken in the form of a tablet that is swallowed.

Each tablet is covered in a substance that breaks down **only** in alkaline conditions to release the components of the supplement.

Suggest why this covering is important if the supplement is to work effectively in the alimentary canal.

.....

.....

.....

.....

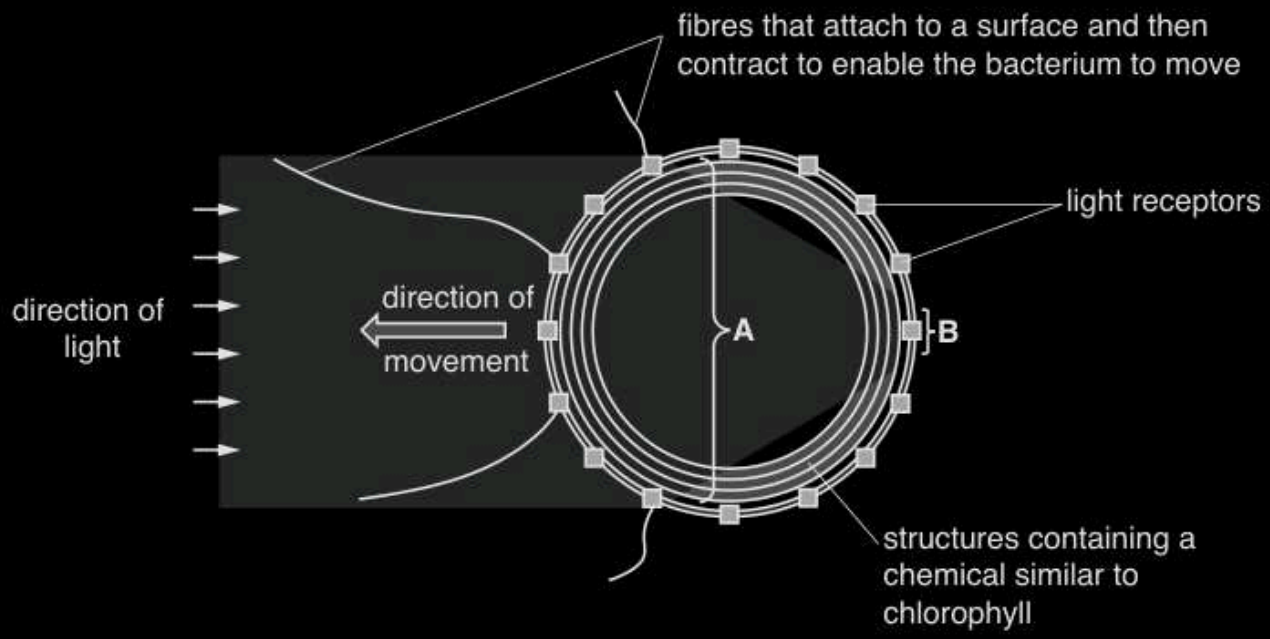
..... [3]

[Total: 9]

1(a)(i)	enzyme(s) / protein(s) / catalyst(s) ;	1	
1(a)(ii)	starch + maltose / glucose ; protein / polypeptide + amino acids / polypeptides / peptides / peptones ; fat / oil / lipid / triglyceride + fatty acids **and** glycerol ;	3	
1(a)(iii)	break down / digest / hydrolyse ; cellulose ;	2	
1(b)	stomach + acid / pH below 7.0 ; tablet does not break down / contents not released + stomach ; prevents enzyme + denaturation / destruction / inactivity ; small intestine / duodenum + alkali / pH above 7.0 ; tablet breaks down / contents released + small intestine / duodenum ; optimum **AW OR** fastest rate **AW** ;	3	

Question	Answer	Marks	Guidance

2 The diagram shows the structure of a type of bacterium and details of the response of the bacterium to light.



(a) Use information in the diagram to suggest:

(i) the sequence of events that takes place when the bacterium is provided with light from the direction shown

.....

.....

.....

.....

..... [3]

(ii) the advantage to the bacterium of responding to light in this way.

.....

.....

..... [2]

(b) There are similarities between the functions of parts of the bacterium and the functions of structures found in the human eye.

Suggest which structure in the human eye has a function similar to each of the following regions of the bacterium.

A .....

B .....

[2]

(c) The type of bacterium shown in the diagram fixes nitrogen.

Describe the role of bacteria in nitrogen fixation.

.....

.....

.....

..... [2]

(d) The enzyme that these bacteria use to fix nitrogen is destroyed in the presence of oxygen.

(i) Suggest why destruction of this enzyme is likely to occur in the type of bacterium in the diagram.

.....

.....

.....

..... [2]

(ii) Suggest **one** adaptation that such bacteria may have developed to prevent the destruction of this enzyme.

.....

.....

..... [1]

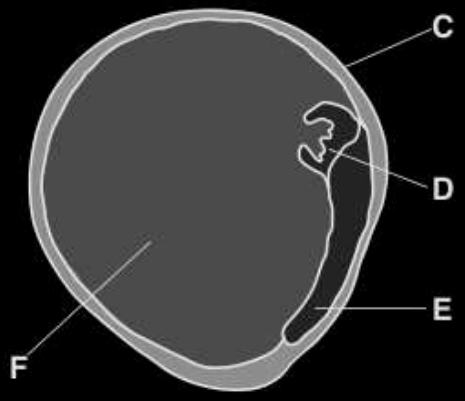
[Total: 12]

2(a)(i)	**1** (light) enters / refracted **AW** / converged / travels through (bacterium) ; **2** (light) detected / absorbed / received **AW +** by receptors ; **3** fibres contract ; **4** movement towards light ; **5** **sequence mark**: if point(s) **1** / **2** appear before point(s) **3** / **4** ;	**3**	
2(a)(ii)	more **AW +** light ; chlorophyll ; photosynthesis ; produce **AW +** glucose / starch / carbohydrate / sugar ;	**2**	
2(b)	(**A**) cornea / lens / aqueous humour / vitreous humour ; (**B**) retina / photoreceptors / fovea / yellow spot / rods / cones ;	**2**	
2(c)	(uses) nitrogen **+** gas / in air / atmospheric ; produce **AW +** ammonium ;	**2**	

Question	Answer	Marks	Guidance
2(d)(i)	photosynthesis ; oxygen + produced / released **AW** ;	2	
2(d)(ii)	any suggested method of having less oxygen present ; any suggested method of keeping oxygen away from enzyme ; mutation ;	1	

Question	Answer	Marks	Guidance

3 The diagram shows the internal structure of a seed from a pea plant.



Seeds from a pea plant are non-endospermic.

(a) Identify each of parts **C** to **F**.

C .....

D .....

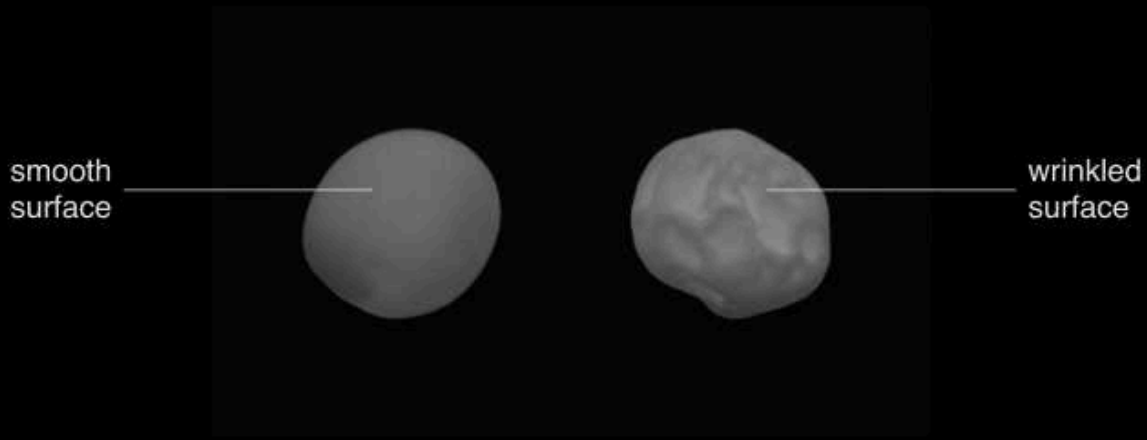
E .....

F .....

[4]

(b) The diagram below shows two seeds, one from each of two pea plants of the same species.

Seeds from this species of plant have either a smooth surface or a wrinkled surface.



The type of surface of a seed is controlled by a single gene. The allele of the gene that gives the seed a smooth surface (**R**) is dominant to the allele that gives the seed a wrinkled surface (**r**).

(i) Name the part of a cell that contains the genes.

.....

[1]

(ii) Define each of the following terms:

gene .....

.....

.....

dominant allele .....

.....

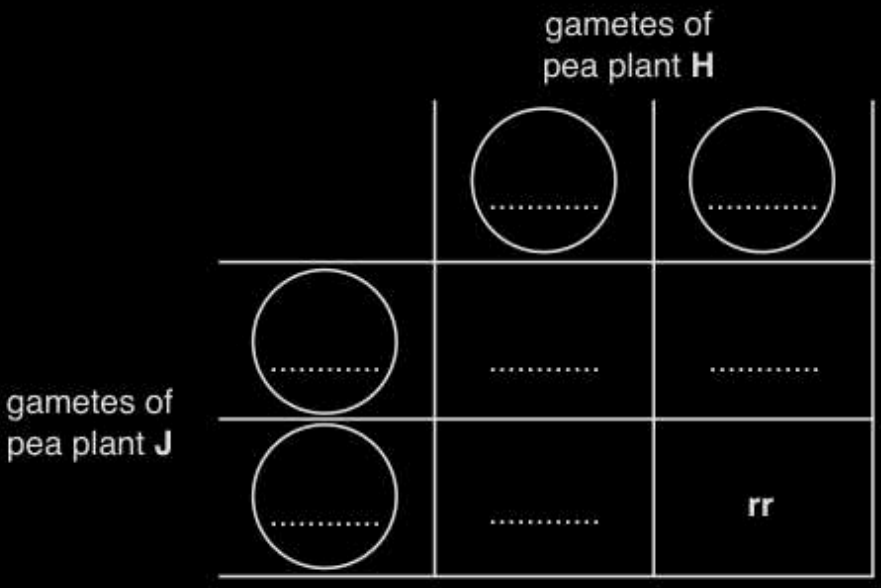
.....

[4]

(iii) One pea plant, **H**, has a homozygous genotype that develops seeds with a wrinkled surface. Another pea plant, **J**, has an unknown genotype.

When these two plants were cross-pollinated, approximately half of the new plants produced had peas with a wrinkled surface.

Complete the genetic diagram below for this cross **and** state the genotype of pea plant **J**.

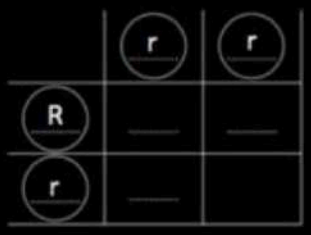
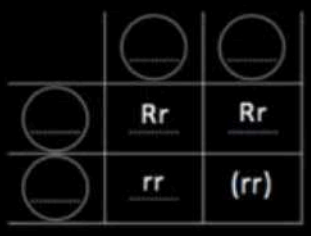


genotype of pea plant **J**: .....

[3]

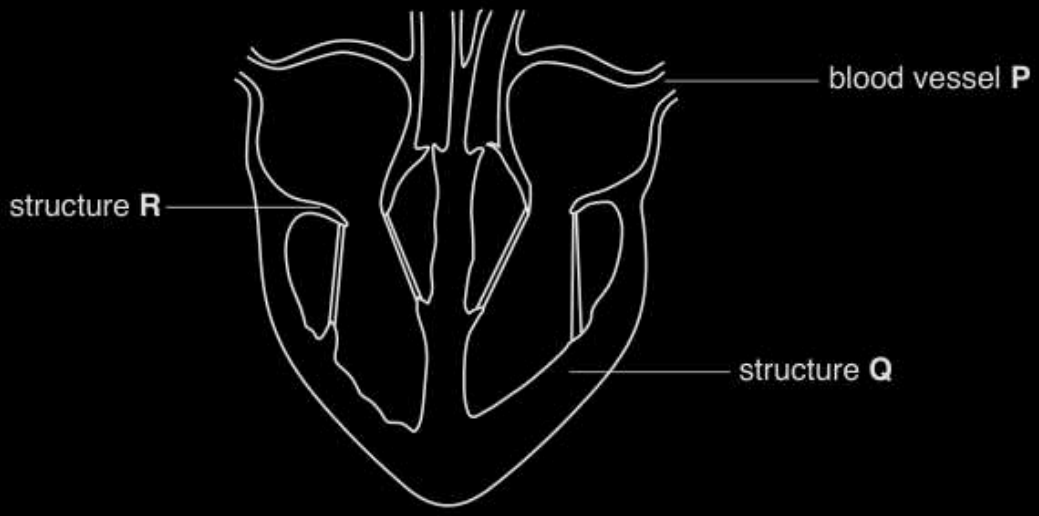
[Total: 12]

3(a)	(**C**) testa / seed coat ; (**D**) plumule ; (**E**) radicle ; (**F**) cotyledon(s) ;	**4**	
3(b)(i)	nucleus / chromosome(s) / DNA ;	**1**	
3(b)(ii)	*gene* made of DNA / section of a chromosome ; copied / passed on / inherited / hereditary ; one **AW +** characteristic / trait / protein / polypeptide ; *dominant allele* version / form / type **+** of a gene ; expressed **AW +** always / if only one copy present / by heterozygote / over another allele ;	**4**	

Question	Answer	Marks	Guidance
3(b)(iii)	**1**  ; **2**  ; **3** (genotype of **J**) Rr / heterozygous ;	**3**	**2,3 A** Rr genotypes to be written as rR

Question	Answer	Marks	Guidance

6 The diagram shows the internal structure of a human heart viewed from the front and its associated blood vessels.



Name and describe the function of each of the following:

(a) blood vessel **P**

.....

.....

.....

..... [3]

(b) structure **Q**

.....

.....

.....

..... [3]

(c) structure **R**.

.....

.....

.....

.....

..... [4]

6(a)	*blood vessel* ***P*** pulmonary vein ; oxygenated blood ; from lungs / to heart / to left atrium ;	**3**	
6(b)	*structure* ***Q*** left ventricle / muscle ; pumps / contracts / pushes blood ; high pressure ; blood to **+** body / aorta ;	**3**	
6(c)	*structure* ***R*** valve ; atrioventricular / tricuspid ; opens **+** blood into right ventricle ; closes **+** blood into pulmonary artery / blood to lungs ; prevent backflow **AW +** of blood ; (prevents backflow) into right atrium ;	**4**	

© UCLES 2019

Question	Answer	Marks	Guidance

7 The nervous system coordinates body functions.

(a) **(i)** Name the type of coordinated response that takes place as a result of a specific stimulus, such as the withdrawal of the hand from a hot object touching the skin.

..... [1]

(ii) Outline, **in the correct sequence**, the functions of each **named** type of neurone involved in producing this coordinated response.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

(b) Suggest why damage to a person's spinal cord may result in the person not being able to produce coordinated responses.

.....

.....

..... [2]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7(a)(i)	reflex / involuntary ;	1	
7(a)(ii)	**1** impulse ; **2** synapse ; **3** from **+** receptor ; **4** sensory / afferent ; **5** relay / inter / intermediate ; **6** CNS / spinal cord ; **7** motor / efferent ; **8** to **+** effector ; **9** muscle / named muscle correct for withdrawal reflex ; **10** **sequence mark**: point(s) **3**/**4**/**5**/**6** must be ticked before point(s) **7**/**8**/**9** ;	7	
7(b)	impulse **+** not transmitted **AW** ; effector / named effector **+** not activated **AW** ;	2	

Question	Answer	Marks	Guidance

9 **(a)** Describe the pathway of water molecules from the soil, through a plant and into the atmosphere.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

(b) Explain how variation in **one named** environmental factor would lead to an **increase** in the rate of water loss from a plant.

.....

.....

.....

.....

..... [3]

[Total: 10]

9(a)	root hair ; cell wall ; membrane ; partially **AW +** permeable ; osmosis / diffusion ; stem ; xylem ; leaf / mesophyll ; water film ; air / intercellular **AW +** spaces ; out of / through **+** stomata / guard cells ; transpiration / evaporation / water vapour ;	**7**	
9(b)		**3**	

factor ;	*variation* ;	*explanation* ;
temperature / heat	high / higher **AW**	increased **AW +** evaporation / water vapour
wind / air movement	fast / faster **AW**	fewer water molecules outside / increased concentration gradient
light	high / higher **AW**	stomata / guard cells **+** open
humidity / moisture in air	low / lower **AW**	fewer water molecules outside / increased concentration gradient

© UCLES 2019

October/November 2018 · Paper 1 · Variant 1 · Question 1

Correct answer: C

1 Four types of cell found in the leaf of a green plant are listed below.

1 epidermal cells (not including guard cells)

2 guard cells

3 palisade mesophyll cells

4 spongy mesophyll cells

Which cells contain chloroplasts?

A 1, 2, 3 and 4

B 1 and 2 only

C 2, 3 and 4 only

D 2 and 3 only

October/November 2018 · Paper 1 · Variant 1 · Question 2

Correct answer: D

2 Which statements about osmosis are correct?

1 Glucose molecules can move by osmosis.

2 Molecules pass through a partially permeable membrane.

3 Molecules move down a water potential gradient.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

October/November 2018 · Paper 1 · Variant 1 · Question 3

Correct answer: A

3 A seedling is placed, with its roots, in a solution containing potassium ions and left for 48 hours. The relative concentrations of the potassium ions are shown in the table.

	concentration of potassium ions in the solution	concentration of potassium ions in the root hair
start of experiment	20	15
after 48 hours	18	35

How did the potassium ions enter the cell?

	active transport	diffusion	osmosis
A	✓	✓	✗
B	✓	✓	✓
C	✓	✗	✗
D	✗	✗	✓

key

✓ = yes

✗ = no

October/November 2018 · Paper 1 · Variant 1 · Question 4

Correct answer: B

4 Enzyme action can be explained by the lock and key hypothesis.

Where is the active site and does the substrate act as the lock or as the key?

	active site on enzyme	active site on substrate	substrate acts as a lock	substrate acts as a key
A	✓	✗	✓	✗
B	✓	✗	✗	✓
C	✗	✓	✓	✗
D	✗	✓	✗	✓

key
✓ = yes
✗ = no

October/November 2018 · Paper 1 · Variant 1 · Question 5

Correct answer: C

5 Some organisms live at the bottom of the sea where it is very dark. To synthesise glucose, they use energy from chemicals in the very hot water that comes out of volcanoes.

What is a distinguishing feature of these organisms?

A Their enzymes are easily denatured by heat.

B They do not need carbon dioxide.

C They do not need to be green.

D They obtain energy only as carnivores.

October/November 2018 · Paper 1 · Variant 1 · Question 8

Correct answer: B

8 What is the main function of the colon?

A It absorbs digested food material.

B It absorbs water.

C It gets rid of undigested food material from the body.

D It produces enzymes to digest food.

October/November 2018 · Paper 1 · Variant 1 · Question 10

Correct answer: C

10 A person who has coeliac disease has flattened villi in their ileum. If a person with coeliac disease eats the same quantity and types of food as a disease-free person, they will have lower levels of glucose and amino acids in their hepatic portal vein.

Which row explains this?

	capillaries do not absorb the glucose molecules	the surface area of the ileum is reduced
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key
✓ = yes
✗ = no

October/November 2018 · Paper 1 · Variant 1 · Question 11

Correct answer: A

11 What is **least** likely to increase the transpiration rate in a plant?

A increased humidity

B increased sunlight

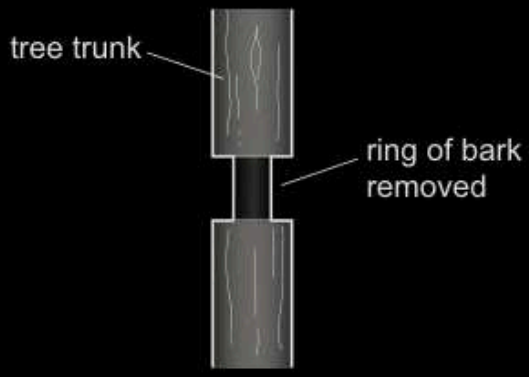
C increased temperature

D increased wind speed

October/November 2018 · Paper 1 · Variant 1 · Question 12

Correct answer: B

12 The diagram shows part of a tree trunk. A ring of bark including the phloem has been removed.



The tree will eventually die because removing the bark stops the transport of

A mineral salts to the leaves.

B organic nutrients to the roots.

C oxygen to the roots.

D water to the leaves.

October/November 2018 · Paper 1 · Variant 1 · Question 13

Correct answer: C

13 Which of these organs will receive blood from both an artery **and** a vein?

	kidney	liver	lung
A	✓	✓	✗
B	✓	✗	✓
C	✗	✓	✗
D	✗	✗	✓

key

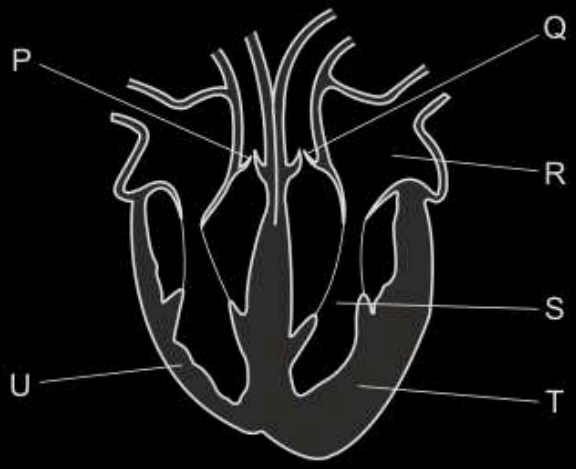
✓ = yes

✗ = no

October/November 2018 · Paper 1 · Variant 1 · Question 14

Correct answer: D

14 The diagram shows a section through the human heart.



Which suggests that blood leaves the heart at different pressures, when going to the lungs and to the body?

A Chambers R and S have different volumes.

B The walls of the atria are thinner than the walls of the ventricles.

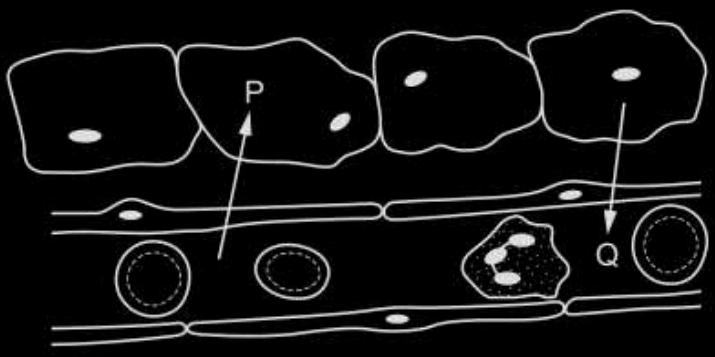
C Valve P is stronger than valve Q.

D Wall T is more muscular than wall U.

October/November 2018 · Paper 1 · Variant 1 · Question 15

Correct answer: D

15 The diagram shows chemicals being exchanged between some cells and a blood capillary.



What could be the identities of chemicals P and Q?

	P	Q
A	amino acids and oxygen	carbon dioxide and maltose
B	carbon dioxide and glucose	alcohol and oxygen
C	carbon dioxide and urea	oxygen and protein
D	glucose and oxygen	carbon dioxide and water

October/November 2018 · Paper 1 · Variant 1 · Question 17

Correct answer: B

17 Which process does **not** require energy from respiration?

A converting amino acids into proteins

B diffusion of oxygen into the blood

C keeping the body temperature at 37 °C

D mitosis

October/November 2018 · Paper 1 · Variant 1 · Question 18

Correct answer: D

18 By which route does oxygen pass into the blood?

A alveoli → capillaries → trachea → bronchioles → bronchi

B bronchi → alveoli → capillaries → trachea → bronchioles

C capillaries → bronchioles → alveoli → bronchi → trachea

D trachea → bronchi → bronchioles → alveoli → capillaries

October/November 2018 · Paper 1 · Variant 1 · Question 19

Correct answer: B

19 Which organ excretes most carbon dioxide from the human body?

A kidney

B lung

C rectum

D skin

October/November 2018 · Paper 1 · Variant 1 · Question 20

Correct answer: B

20 Which structures are **all** involved in controlling human body temperature?

A blood vessels near the skin surface, the cerebellum and sweat glands

B blood vessels near the skin surface, the hypothalamus and skeletal muscles

C kidneys, the cerebellum and sweat glands

D kidneys, the hypothalamus and skeletal muscles

October/November 2018 · Paper 1 · Variant 1 · Question 21

Correct answer: D

21 What happens to the ciliary muscle and what happens to the lens when the eye focuses on a distant object?

	ciliary muscle	lens
A	contracts	gets thicker
B	contracts	gets thinner
C	relaxes	gets thicker
D	relaxes	gets thinner

October/November 2018 · Paper 1 · Variant 1 · Question 22

Correct answer: C

22 Which statement about brain structure and function is **not** correct?

A The cerebellum controls balance.

B The cerebrum stores memories.

C The hypothalamus controls movement.

D The medulla controls heart rate.

October/November 2018 · Paper 1 · Variant 1 · Question 23

Correct answer: D

23 What is **not** an example of a simple reflex action?

A blinking when sand blows in your eyes

B jumping when hearing a loud noise

C removing your hand from a hot object

D steering a car to avoid hitting something

October/November 2018 · Paper 1 · Variant 1 · Question 26

Correct answer: C

26 Which statement is correct for all viruses but **not** for all bacteria and fungi?

A Most of the cell is cytoplasm.

B Their outer layer is cellulose.

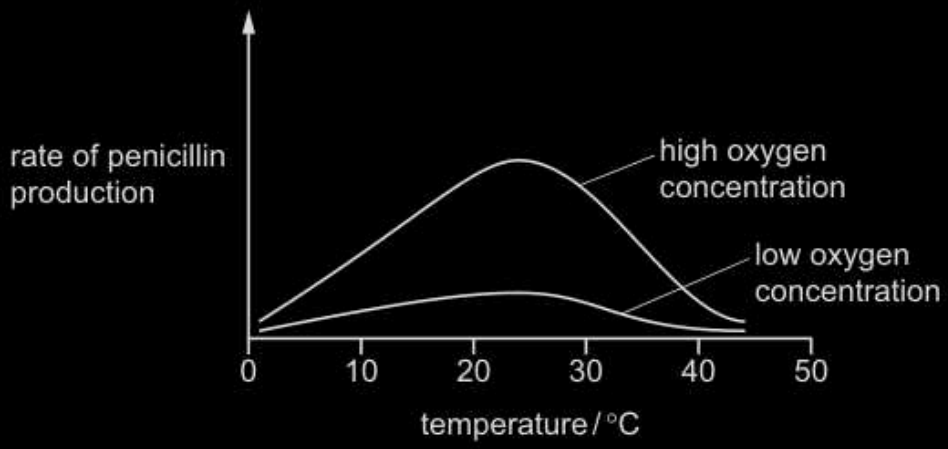
C They are all parasites.

D They have a nucleus containing DNA or RNA.

October/November 2018 · Paper 1 · Variant 1 · Question 27

Correct answer: C

27 The graph shows how the rate of penicillin production from the fungus *Penicillium* varies with temperature at two different oxygen concentrations.



What can be deduced from the graph?

1 The fungus *Penicillium* respires aerobically.

2 The fungus *Penicillium* respires anaerobically.

3 The optimum temperature for penicillin production is approximately 24 °C.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

October/November 2018 · Paper 1 · Variant 1 · Question 37

Correct answer: B

37 What is the primary function of DNA?

A It controls the absorption of nutrients.

B It controls the production of protein.

C It controls the rate of mutation.

D It controls the rate of reproduction.

October/November 2018 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 In the ABO blood group system, which genotype is homozygous dominant?

A $I^A\ I^O$ **B** $I^A\ I^B$ **C** $I^B\ I^B$ **D** $I^O\ I^O$

October/November 2018 · Paper 1 · Variant 1 · Question 39

Correct answer: C

39 In a species of plant, the allele for yellow flowers is dominant to the allele for white flowers.

Which offspring is it possible to produce from a cross between two plants heterozygous for flower colour?

A heterozygous yellow and heterozygous white

B heterozygous yellow only

C heterozygous yellow, homozygous yellow and homozygous white

D homozygous yellow only

October/November 2018 · Paper 1 · Variant 1 · Question 40

Correct answer: D

40 Which statement is correct?

A Evolution is natural selection.

B Evolution results in natural selection.

C Natural selection and evolution are independent of each other.

D Natural selection results in evolution.

October/November 2018 · Paper 1 · Variant 2 · Question 1

Correct answer: A

1 A cell is observed under a microscope.

Which feature identifies it as a plant cell?

A The cell contains a single large sap vacuole.

B The cell contains glucose and amino acids.

C The cell contains stored fat.

D The cell surface membrane is partially permeable.

October/November 2018 · Paper 1 · Variant 2 · Question 2

Correct answer: B

2 Which statements about diffusion are correct?

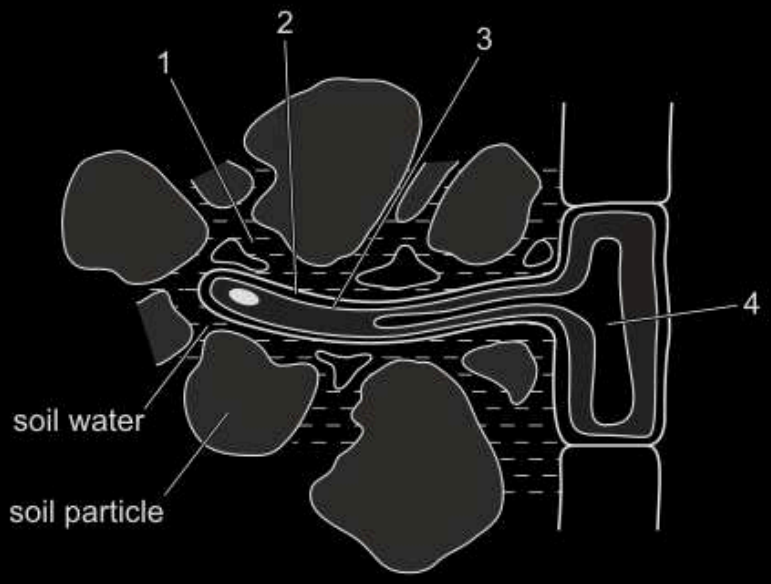
	molecules move from a higher to a lower concentration	only occurs within living systems	rate slows down as the temperature increases
A	✓	✓	✗
B	✓	✗	✗
C	✗	✓	✓
D	✗	✗	✓

key
✓ = yes
✗ = no

October/November 2018 · Paper 1 · Variant 2 · Question 3

Correct answer: B

3 The diagram shows a root hair cell and surrounding soil particles.



Osmosis occurs when regions of higher and lower concentration of water molecules are separated by a partially permeable membrane.

On the diagram, what are these regions?

	higher concentration of water molecules	partially permeable membrane	lower concentration of water molecules
A	1	2	4
B	1	3	4
C	4	2	1
D	4	3	1

October/November 2018 · Paper 1 · Variant 2 · Question 4

Correct answer: C

4 What are enzymes?

A fats that are secreted by glands in the digestive system

B fats that have a characteristic molecular shape

C proteins that act as biological catalysts

D proteins that are unaffected by temperature

October/November 2018 · Paper 1 · Variant 2 · Question 5

Correct answer: C

5 Some organisms live at the bottom of the sea where it is very dark. To synthesise glucose, they use energy from chemicals in the very hot water that comes out of volcanoes.

What is a distinguishing feature of these organisms?

A Their enzymes are easily denatured by heat.

B They do not need carbon dioxide.

C They do not need to be green.

D They obtain energy only as carnivores.

October/November 2018 · Paper 1 · Variant 2 · Question 8

Correct answer: C

8 What are the smaller basic units that make up glycogen and protein?

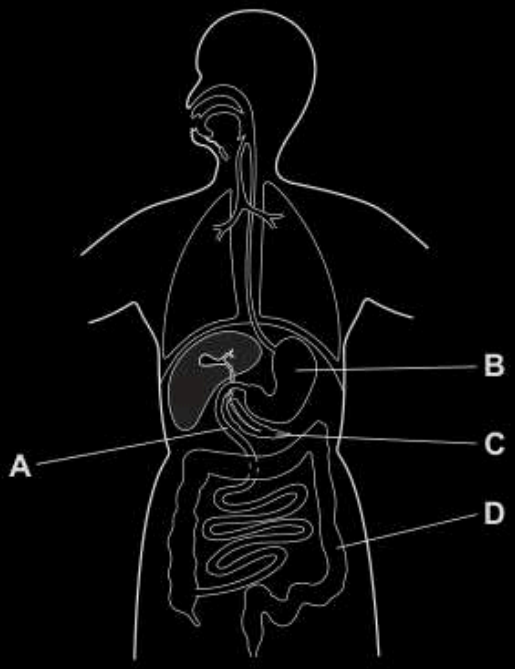
	glycogen	protein
A	amino acids	glucose
B	fatty acids	glucose
C	glucose	amino acids
D	glucose	fatty acids

October/November 2018 · Paper 1 · Variant 2 · Question 10

Correct answer: A

10 A gland secretes different types of enzymes into part of the alimentary canal which allows digestion of proteins and starch. It also receives alkaline secretions from another gland and contains muscles that are used in peristalsis.

Which part of the alimentary canal is being described?



October/November 2018 · Paper 1 · Variant 2 · Question 11

Correct answer: D

11 What is **not** a function of transpiration?

A It causes minerals to travel up through the plant.

B It causes water to move from the roots to the leaves.

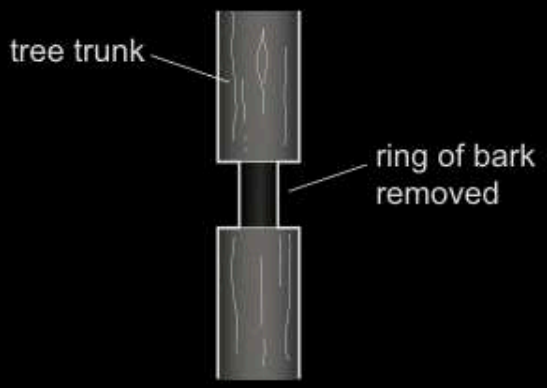
C It helps to keep the plant cool.

D It helps to release energy in cells.

October/November 2018 · Paper 1 · Variant 2 · Question 12

Correct answer: B

12 The diagram shows part of a tree trunk. A ring of bark including the phloem has been removed.



The tree will eventually die because removing the bark stops the transport of

A mineral salts to the leaves.

B organic nutrients to the roots.

C oxygen to the roots.

D water to the leaves.

October/November 2018 · Paper 1 · Variant 2 · Question 13

Correct answer: C

13 Which of these organs will receive blood from both an artery **and** a vein?

	kidney	liver	lung
A	✓	✓	✗
B	✓	✗	✓
C	✗	✓	✗
D	✗	✗	✓

key

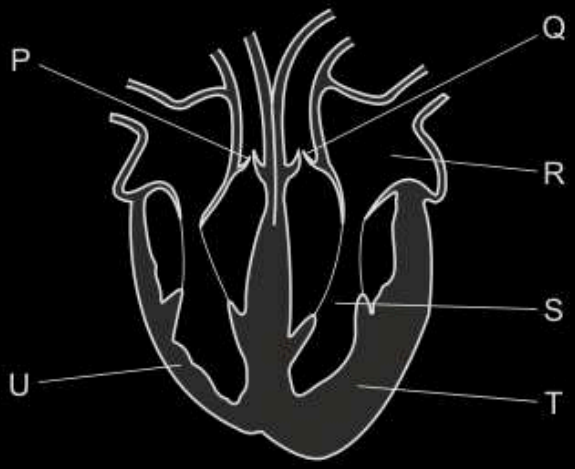
✓ = yes

✗ = no

October/November 2018 · Paper 1 · Variant 2 · Question 14

Correct answer: D

14 The diagram shows a section through the human heart.



Which suggests that blood leaves the heart at different pressures, when going to the lungs and to the body?

A Chambers R and S have different volumes.

B The walls of the atria are thinner than the walls of the ventricles.

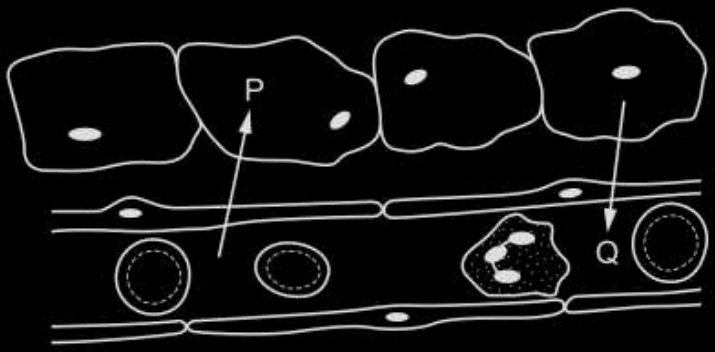
C Valve P is stronger than valve Q.

D Wall T is more muscular than wall U.

October/November 2018 · Paper 1 · Variant 2 · Question 15

Correct answer: D

15 The diagram shows chemicals being exchanged between some cells and a blood capillary.



What could be the identities of chemicals P and Q?

	P	Q
A	amino acids and oxygen	carbon dioxide and maltose
B	carbon dioxide and glucose	alcohol and oxygen
C	carbon dioxide and urea	oxygen and protein
D	glucose and oxygen	carbon dioxide and water

October/November 2018 · Paper 1 · Variant 2 · Question 20

Correct answer: B

20 Which structures are **all** involved in controlling human body temperature?

A blood vessels near the skin surface, the cerebellum and sweat glands

B blood vessels near the skin surface, the hypothalamus and skeletal muscles

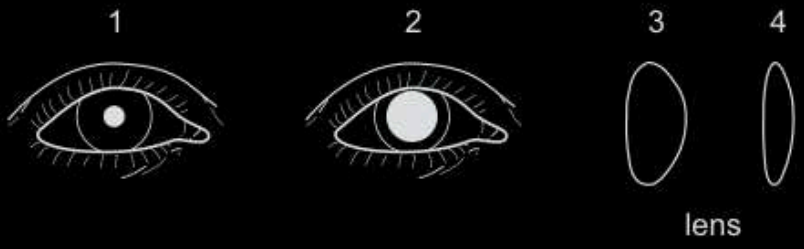
C kidneys, the cerebellum and sweat glands

D kidneys, the hypothalamus and skeletal muscles

October/November 2018 · Paper 1 · Variant 2 · Question 21

Correct answer: D

21 The diagrams show the eye viewed from the front and its lens in cross-section.



Which diagrams show the appearance of the pupil and the shape of the lens when looking up at the sky at night?

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

October/November 2018 · Paper 1 · Variant 2 · Question 22

Correct answer: B

22 What is a major function of the medulla in the brain?

A balance and coordination

B controlling contractions of the diaphragm

C reasoning and thinking

D secretion of hormones

October/November 2018 · Paper 1 · Variant 2 · Question 23

Correct answer: B

23 Which statement about insulin is correct?

A It converts glycogen to glucose.

B It is broken down by the liver.

C It is secreted by the liver.

D Its release increases blood glucose levels.

October/November 2018 · Paper 1 · Variant 2 · Question 37

Correct answer: B

37 What is the primary function of DNA?

A It controls the absorption of nutrients.

B It controls the production of protein.

C It controls the rate of mutation.

D It controls the rate of reproduction.

October/November 2018 · Paper 1 · Variant 2 · Question 38

Correct answer: C

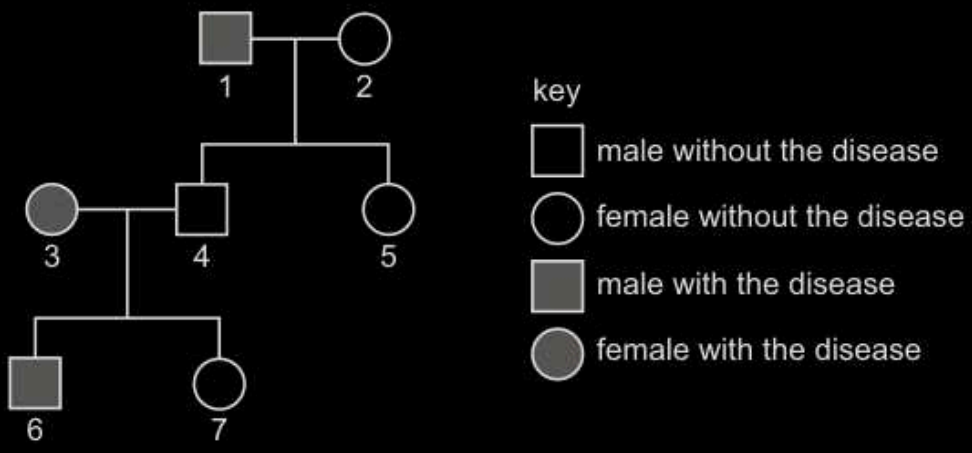
38 In the ABO blood group system, which genotype is homozygous dominant?

A $I^A\ I^O$ **B** $I^A\ I^B$ **C** $I^B\ I^B$ **D** $I^O\ I^O$

October/November 2018 · Paper 1 · Variant 2 · Question 39

Correct answer: A

39 The diagram shows a family in which some members suffer from a disease caused by a recessive allele.



Which are two members of the family who **must** be heterozygous for the gene?

A 5 and 7 **B** 3 and 6 **C** 2 and 5 **D** 1 and 4

Correct answer: D

40 Which statement is correct?

A Evolution is natural selection.

B Evolution results in natural selection.

C Natural selection and evolution are independent of each other.

D Natural selection results in evolution.

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

2 The diagram shows a seed (Fig. 1) and a plant developing from this seed (Fig. 2).

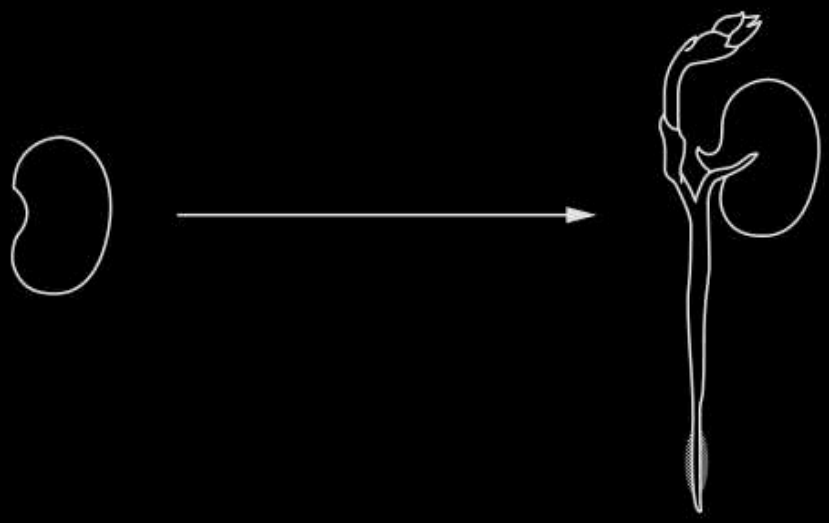


Fig. 1 **Fig. 2**

(a) **(i)** Name **two** structures visible in Fig. 2 that are **not** visible in Fig. 1.

1 .....

2 ..... [2]

(ii) Name the process represented by the arrow drawn between Fig. 1 and Fig. 2.
Describe the use of enzymes in this process.

name of process .....

use of enzymes .....

.....

.....

.....

[4]

(b) In an experiment, the growth of a plant from a seed is found to be faster as temperatures increase up to 35 °C.

Explain how the development of a plant from a seed would be affected by a temperature much higher than 35 °C.

.....

.....

..... [2]

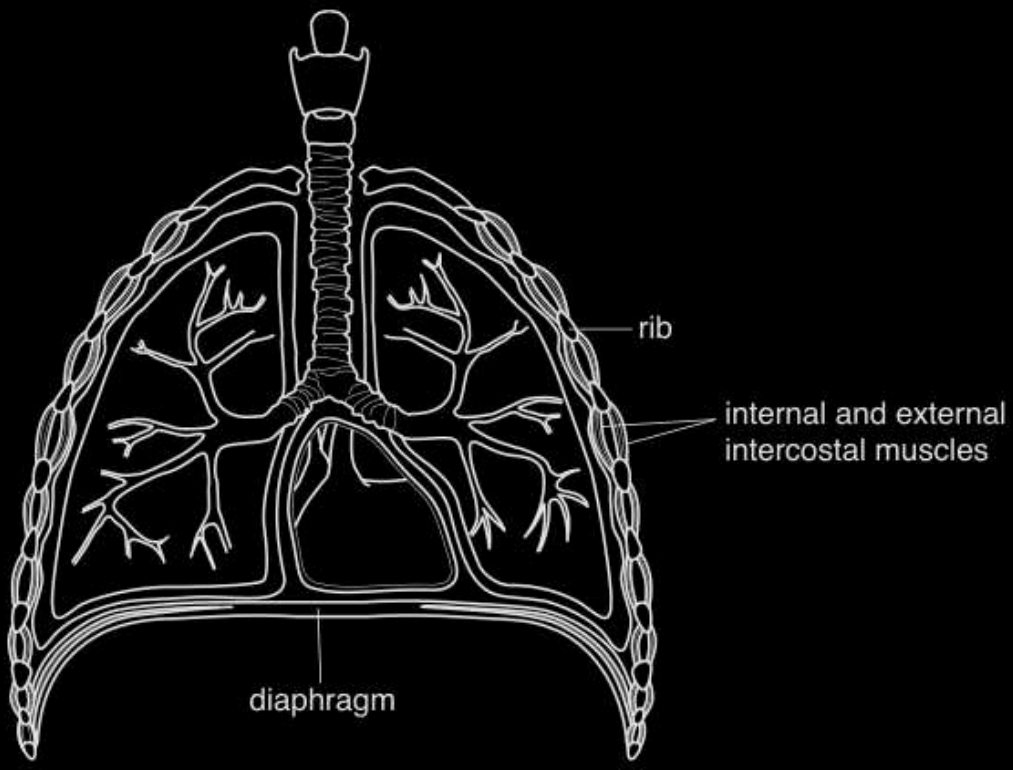
[Total: 8]

2(a)(i)	radicle / root ; root hairs ; plumule / shoot / stem ; leaves ; hypocotyl ;	**2**	
2(a)(ii)	*(name of process)* germination / growth ; *(use of enzymes) maximum 3 marks from* digest / breakdown / hydrolysis ; amylase ; starch **+** to maltose / glucose ; protease ; protein **+** to amino acids ; lipase ; fats **+** to fatty acids **and** glycerol ;	**4**	
2(b)	development **AW** slower / would not germinate / would not grow ; denature ; reactions / digestion / hydrolysis / breakdown **+** slow / don't occur ;	**2**	

© UCLES 2018

Question	Answer	Marks	Guidance

4 The diagram shows the human thorax.



(a) (i) Describe how each of the structures named in the diagram is involved when a person takes a single breath **in**.

.....

.....

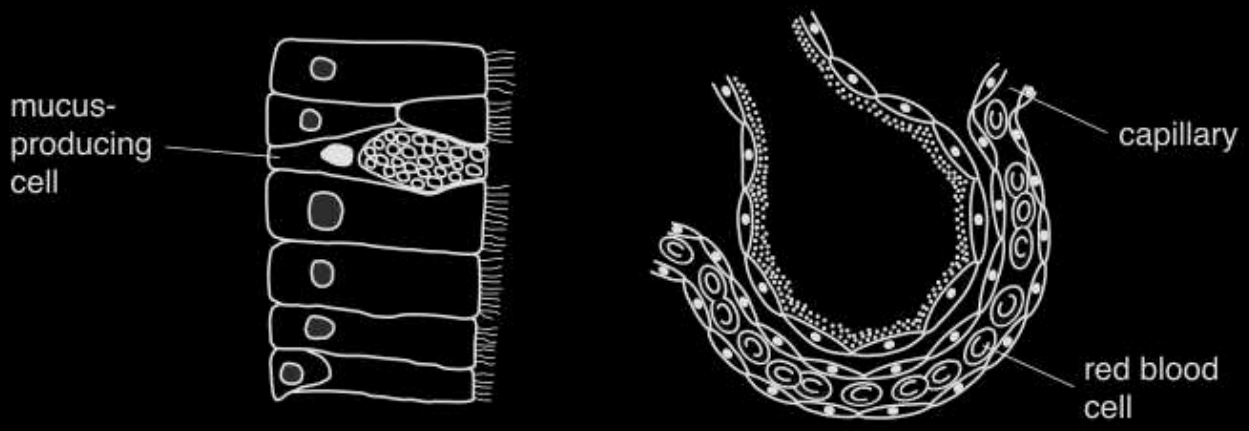
.....

.....

.....

..... [3]

(ii) The diagrams below show two magnified structures, **D** and **E**, from the thorax.



structure D **structure E**

Draw lines labelled **D** and **E** on the diagram of the thorax on page 8 to indicate the positions of **structure D** and **structure E**. [2]

(b) Describe how structure is related to function for each of the following:

a capillary, ...

.....

.....

a red blood cell. .....

.....

..... [5]

[Total: 10]

4(a)(i)	diaphragm **+** contracts ; diaphragm **+** moves down ; external intercostal muscles **+** contract ; internal intercostal muscles **+** relax ; ribs move **+** up / out ;	**3**	**Ig** references to volume / pressure
4(a)(ii)	**structure D** correctly identified on trachea / bronchus ; **structure E** identified at end of a bronchiole ;	**2**	**A** any method of correct identification
4(b)	*(capillary) max 3 marks from* wall **+** one cell thick ; diffusion ; example of **named** substance **+** in / out / through wall ; branching / network / contacts many cells / large surface area ; *(red blood cell) max 3 marks from* biconcave **AW** ; no nucleus ; haemoglobin ; large surface area ; oxygen in / out / carriage ; able to squeeze through capillaries / flexible ;	**5**	

© UCLES 2018

Question	Answer	Marks	Guidance

5 Fig. 1 shows cells from a plant tissue which have been mounted on a slide with distilled water and viewed using a microscope.

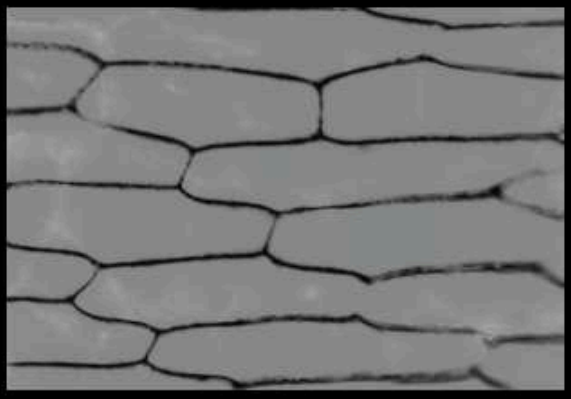


Fig. 1

Fig. 2 shows cells taken from the same plant tissue when mounted on a slide with concentrated salt solution.

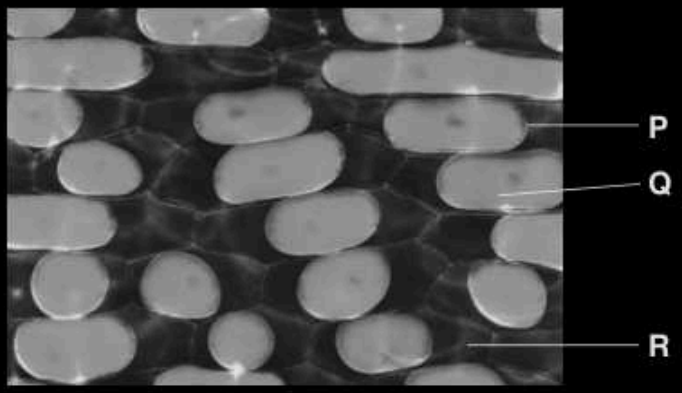


Fig. 2

(a) Explain the appearance of the cells in Fig. 2.

.....

.....

.....

.....

.....

..... [4]

(b) **(i)** Identify structures **P** and **Q** in Fig. 2.

P .....

Q ..... [2]

(ii) State the contents of location **R** in Fig. 2.

..... [1]

(iii) The concentrations of substances in structure **Q** are different from those in location **R**.

Explain how the properties of structure **P** result in differences in concentrations of substances in **Q** and **R**.

.....

.....

.....

.....

..... [3]

[Total: 10]

Section B

Answer **both** questions in this section.

Write your answers in the spaces provided.

5(a)	membrane or **P** / cytoplasm or **Q** / pulling away from wall ; plasmolysed / plasmolysis / flaccid ; water potential / concentration **+** lower outside than inside cells **ORA** ; diffusion / osmosis ; water **+** out of cells ; through **+** partially **AW** permeable membrane ;	**4**	
5(b)(i)	**(P)** membrane ; **(Q)** cytoplasm ;	**2**	
5(b)(ii)	salt **+** water / solution ;	**1**	
5(b)(iii)	partially **AW** permeable ; water can pass through ; other substances (e.g. salt) **+** cannot pass through ;	**3**	

© UCLES 2018

Question	Answer	Marks	Guidance

October/November 2018 · Paper 2 · Variant 1 · Question 8

8 **(a)** Describe and explain how each of the following affects enzyme activity:

pH, ...

.....

.....

.....

.....

.....

.....

.....

temperature. .....

.....

.....

.....

.....

.....

.....

..... [7]

(b) Name **one** enzyme that acts in a **named** part of the alimentary canal and describe the role of this enzyme in digestion.

.....

.....

.....

.....

..... [3]

[Total: 10]

8(a)	*(pH or temperature)* best / optimum / fastest ; rate slower both sides of the optimum ; active site ; change in shape ; substrate no longer fits ; shape of curve described ; denaturation **+** pH not optimum / high temperature **AW** ; *(temperature only)* Increase in temperature increases rate of reaction ; heat increases rate of molecular movement / kinetic energy ; more collisions (at higher temperature) ;	**7**	
8(b)	named enzyme from alimentary canal ; correct named location for action of enzyme named ; correct named substrate **+** product(s) for substrate named ;	**3**	

© UCLES 2018

Question	Answer	Marks	Guidance

1 Fig. 1 and Fig. 2 show transverse sections of two types of blood vessel.

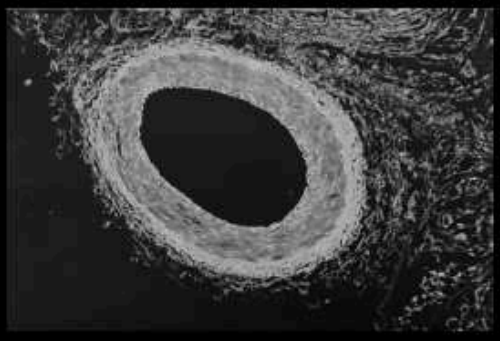


Fig. 1

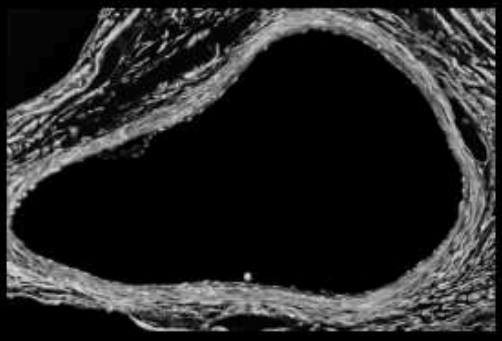


Fig. 2

(a) (i) Name the type of blood vessel shown in:

Fig. 1 .....

Fig. 2 ..... [2]

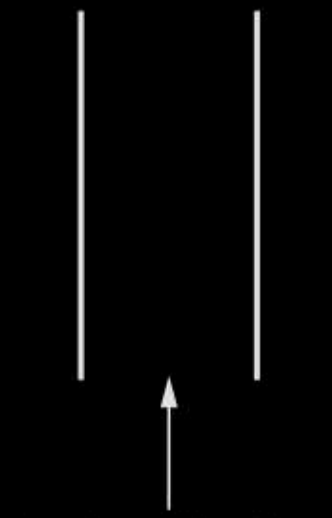
(ii) Describe the differences in the structures shown in Fig. 1 and Fig. 2 that helped you to identify these blood vessels.

.....

.....

.....[2]

(b) The diagram below is of a simplified, incomplete longitudinal section from the type of blood vessel shown in Fig. 2. The direction of blood flow is shown.



(i) Complete the diagram by drawing **one** set of valves.

[2]

(ii) Describe the function of these valves.

.....

.....

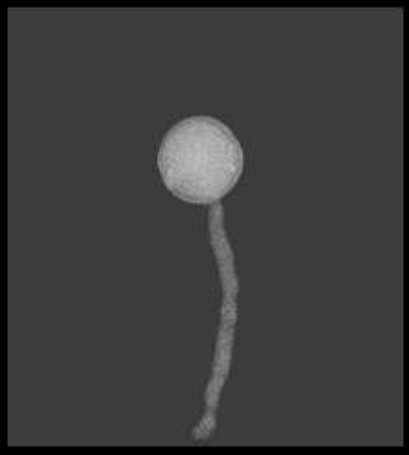
.....[2]

[Total: 8]

1(a)(i)	(Fig. 1) artery / arteriole / named artery ; (Fig. 2) vein / venule / named vein ;	**2**	
1(a)(ii)	reference to size / shape **+** lumen **AW** ; reference to size of wall / muscle / elastic ;	**2**	
1(b)(i)	two structures drawn toward lumen from similar height on opposite walls ; all structures drawn point upwards ;	**2**	
1(b)(ii)	open / close **OR** action of valve flaps described ; prevent back-flow of blood **OR** blood in one direction ; from lower regions of body ; low pressure ;	**2**	

Question	Answer	Marks	Guidance

3 The diagram shows a pollen grain with a pollen tube growing from it.



Pollen grains from the same type of plant were placed in sucrose solutions of different concentrations for a fixed amount of time. After this time, the pollen grains and tubes were examined using a microscope. The following observations were made for each concentration of sucrose:

- the number of pollen grains that had germinated to produce a pollen tube,
- the length of each pollen tube.

The table shows the results of the investigation.

% sucrose concentration	% of pollen grains germinated	mean pollen tube length/mm
1	6	0.005
2	13	0.008
4	25	0.015
8	56	0.040
10	31	0.030
20	25	0.018
40	13	0.006

(a) (i) A total of 12 pollen grains were placed in the 20% sucrose solution.

Use the information in the table to calculate the number of pollen grains that germinated to produce a pollen tube in the 20% sucrose solution.

..... [1]

(ii) Suggest why the **mean** pollen tube length was calculated for each sucrose concentration.

.....

.....[1]

(iii) Use the information in the table to suggest the optimum (best) concentration of sucrose solution for pollen tube germination and growth.

..... %

Explain how the information in the table enabled you to reach this conclusion.

.....

.....

.....

[3]

(iv) The germination of a pollen grain to form a pollen tube requires the movement of water into the pollen grain from its surroundings.

Suggest why placing a pollen grain in a solution with a higher sucrose concentration than in your answer to **(a)(iii)** may result in a lower percentage of germination.

.....

.....

.....

.....

.....[3]

(b) Describe the route taken by a growing pollen tube in a plant and explain its importance in plant reproduction.

.....

.....

.....

.....

.....

.....

.....[4]

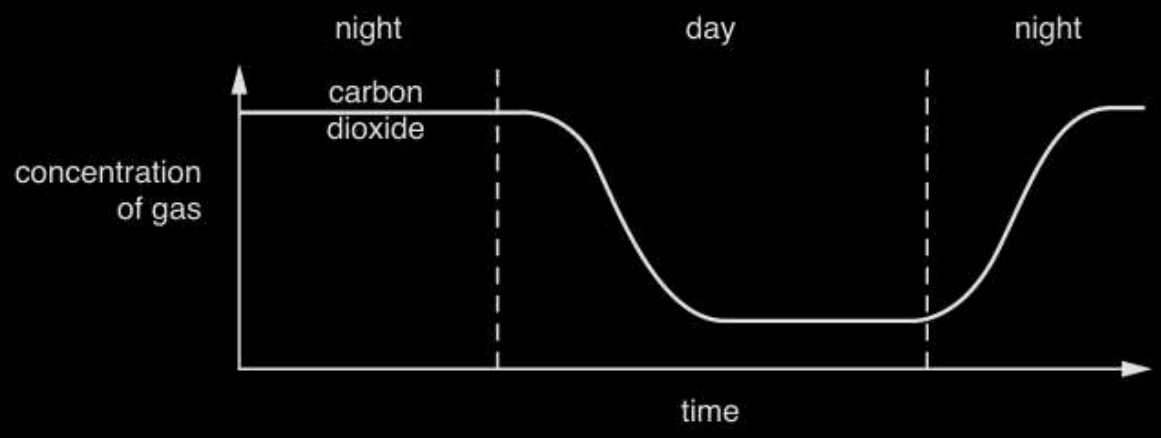
[Total: 12]

3(a)(i)	3 ;	1	**R** 3%
3(a)(ii)	some may not germinate / some may not produce a pollen tube ; accuracy / precision / reliability / validity / reduce error ;	1	
3(a)(iii)	8 ; highest **AW +** % / number ; highest **AW +** length **OR** longest ;	3	
3(a)(iv)	water **+** concentration / potential ; less / no **+** water enters **OR** water leaves ; osmosis / diffusion ;	3	
3(b)	stigma ; style ; ovary / ovule / ovum / embryo sac / female gamete ; movement of **AW +** male gamete / male nucleus / pollen nucleus ; fertilisation / fusion of gametes ;	4	

Question	Answer	Marks	Guidance

October/November 2018 · Paper 2 · Variant 2 · Question 4

4 The graph shows the variation in the concentration of dissolved carbon dioxide in the water of a pond over a period of 24 hours.



(a) (i) Explain how plants that live in the water contribute to the changes in concentration of dissolved carbon dioxide shown in the graph.

.....

.....

.....

.....

.....

.....

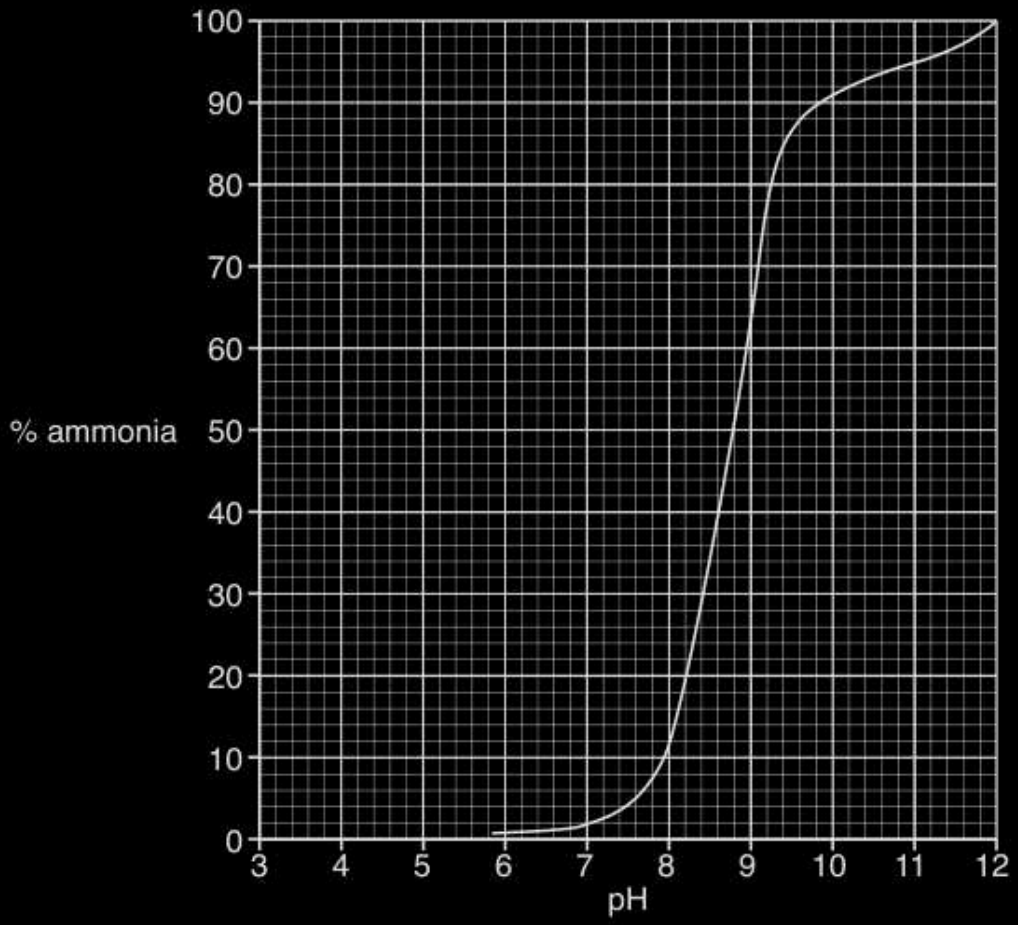
.....[4]

(ii) Draw a line on the graph above to show how the concentration of dissolved oxygen in the water of the same pond is likely to change over the same period of time. [2]

(b) Carbon dioxide dissolves in water to form an acidic solution. The higher the concentration of dissolved carbon dioxide, the lower the pH.

Ammonia is a toxic chemical found in ponds, that may harm fish.

The graph shows how the concentration of dissolved ammonia in the water of a pond changes as the pH changes.



Use this graph and the graph on page 8 to suggest and explain at which time of day fish that live in the pond are **most likely** to be affected by the toxic effects of ammonia.

Draw a (ring) around the correct time of day.

time of day **sunrise** **midday** **sunset**

explanation .....

.....

.....

.....

.....

[4]

[Total: 10]

4(a)(i)	carbon dioxide **+** falls / lower **+** during day ; carbon dioxide **+** rises / higher **+** at night ; respiration **+** during day and night ; respiration **+** produces / releases **+** carbon dioxide ; photosynthesis **+** during day / not at night ; photosynthesis **+** uses / absorbs **+** carbon dioxide ; photosynthesis faster than respiration during day ;	**4**	
4(a)(ii)	line starts at y-axis **+** spans all time axis ; any line lower **+** then rising above **+** then falling below existing line ;	**2**	

Question	Answer	Marks	Guidance
4(b)	midday circled ; less / low **+** carbon dioxide ; pH **+** increases / high / above 7 **OR** alkali ; more ammonia **AW** ;	**4**	

Question	Answer	Marks	Guidance

8 **(a)** Explain what is meant by each of the following terms and describe **one** example of each:

discontinuous variation, ...

.....

.....

.....

.....

.....

continuous variation. .....

.....

.....

.....

.....

.....

[7]

(b) Name the molecule that controls production of proteins in each body cell of a human and describe its importance in inheritance.

.....

.....

.....

.....[3]

[Total: 10]

8(a)	*(discontinuous)* idea of few **+** phenotypes / categories **AW OR** no intermediates ; inherited / passed to next generation ; genes / genetic / genotype / alleles ; correct example (e.g. blood group, eye colour, sex, ear lobes, tongue roll) ; bar chart / bar graph ; *(continuous)* idea of many **+** phenotypes / categories **AW OR** intermediates / range ; genes / genetic / genotype / alleles **+** environment ; any named environmental factor (e.g. diet / sunlight exposure) ; correct example (e.g. skin colour / hair colour / height / weight) ; normal distribution ;	**7**	
8(b)	DNA / deoxyribonucleic acid ; chromosomes / genes / alleles ; copied / replicated ; passed to next generation / from parents ; gametes / egg / sperm ;	**3**	

© UCLES 2018

Question	Answer	Marks	Guidance

9 **(a)** Explain what is meant by each of the following terms and describe **one** example of each in either a plant **or** an animal:

diffusion, ...

.....

.....

.....

.....

.....

active transport. .....

.....

.....

.....

.....

.....

[7]

(b) Starch is a carbohydrate stored inside plant cells.

Explain why starch is a more suitable storage substance than glucose.

.....

.....

.....

.....[3]

[Total: 10]

9(a)	*(diffusion)* **1** movement **+** particles / molecules / ions **;** **2** down concentration gradient / from high to low concentration **;** **3** named molecule or ion **;** **4** **(ONLY if point 3 awarded)** context for example named in point **3** **;** *(active transport)* **5** movement **+** particles / molecules / ions **;** **6** against concentration gradient / from low to high concentration **;** **7** energy **;** **8** respiration **;** **9** membrane / carrier proteins **;** **10** named molecule or ion moved by active transport **;** **11** **(ONLY if point 10 awarded)** context for example named in point **10** **;**	**7**	e.g. oxygen / carbon dioxide / glucose e.g. out of blood / into leaf / into villi e.g. nitrate ion / glucose e.g. into root hair / into villi
9(b)	*points referring to why starch is more suitable:* insoluble **;** molecule **+** large / compact / complex **OR** polysaccharide **;** remains inside cell / cannot cross membrane / grains **;** does not change water potential inside cell **;** prevents gain of water by cells **;** osmosis **OR** diffusion **+** water **;**	**3**	**A** reverse argument for all points if reference made to why glucose would be less suitable

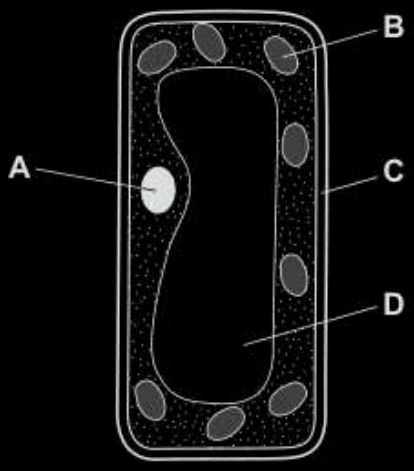
© UCLES 2018

May/June 2018 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 The diagram shows a palisade cell.

Which structure is the site of photosynthesis?



May/June 2018 · Paper 1 · Variant 1 · Question 3

Correct answer: B

3 A plant is growing in a fertile, well-watered soil. It is absorbing nitrate ions by active transport and water by osmosis.

How do the concentrations of nitrate ions and water in the plant's roots compare to the concentrations in the soil?

	concentration of nitrate ions in the roots	concentration of water in the roots
A	higher	higher
B	higher	lower
C	lower	higher
D	lower	lower

May/June 2018 · Paper 1 · Variant 1 · Question 4

Correct answer: A

4 In an enzyme's action, where is the active site and where are the lock and the key?

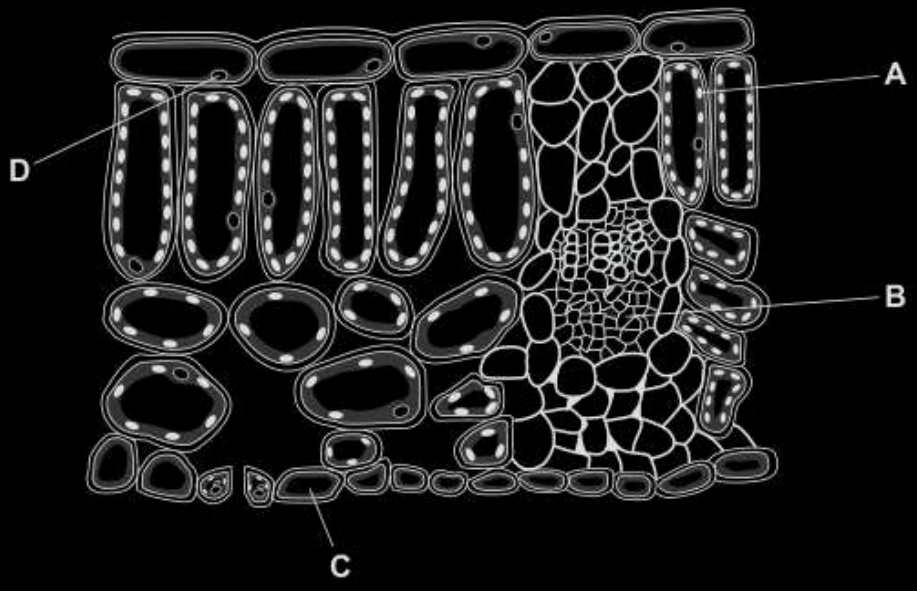
	active site	lock	key
A	on the enzyme	on the enzyme	on the substrate
B	on the enzyme	on the substrate	on the enzyme
C	on the substrate	on the enzyme	on the substrate
D	on the substrate	on the substrate	on the enzyme

May/June 2018 · Paper 1 · Variant 1 · Question 5

Correct answer: B

5 The diagram represents a cross-section of part of a leaf as seen using a microscope.

Where does translocation (movement of sucrose and amino acids) occur?

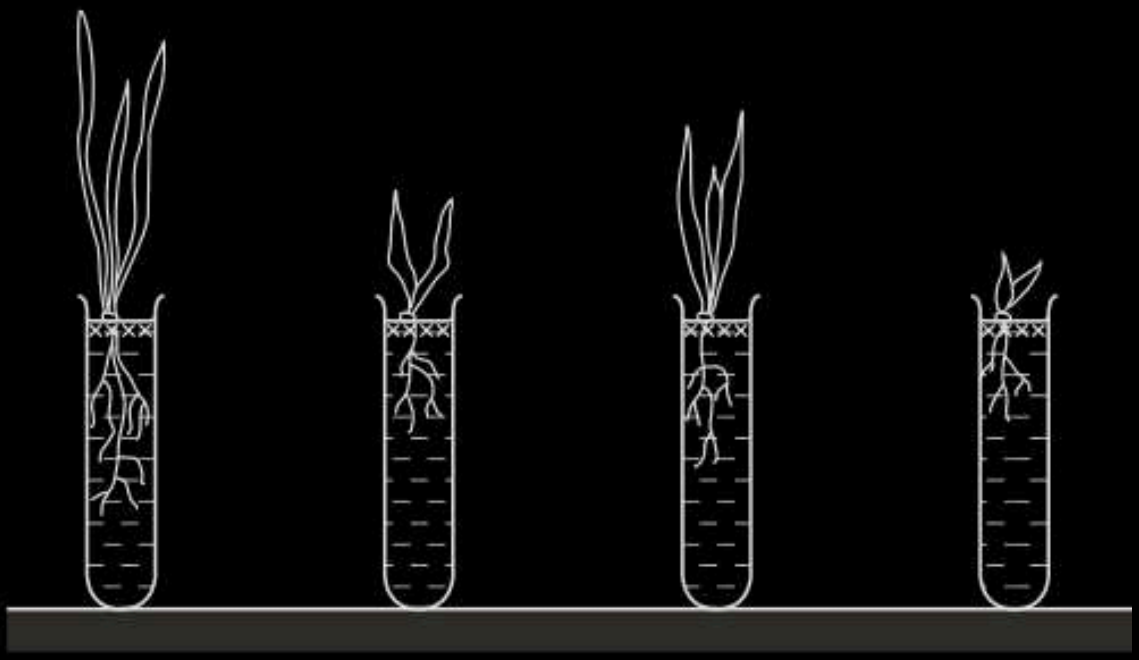


Correct answer: C

7 A student grows seedlings in four different test-tubes.

Tube W contains all the mineral ions needed for healthy plant growth.

The diagram shows the appearance of these seedlings after two weeks.



What do tubes X, Y and Z contain?

	X	Y	Z
A	all minerals except magnesium ions	all minerals except nitrate ions	water
B	all minerals except magnesium ions	water	all minerals except nitrate ions
C	all minerals except nitrate ions	all minerals except magnesium ions	water
D	water	all minerals except magnesium ions	all minerals except nitrate ions

May/June 2018 · Paper 1 · Variant 1 · Question 8

Correct answer: C

8 Which substance is built up from amino acids?

A glucose

B glycogen

C protein

D urea

May/June 2018 · Paper 1 · Variant 1 · Question 10

Correct answer: D

10 The surface area of the small intestine is increased by the villi in the intestine wall.

How does the increased surface area help absorption of digested materials?

A It makes peristalsis more efficient.

B More mucus is produced for lubrication.

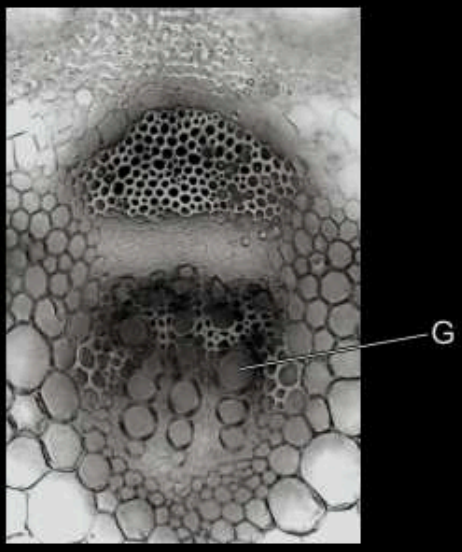
C More starch and protein can be absorbed.

D There is a greater chance of food molecules diffusing into the blood.

May/June 2018 · Paper 1 · Variant 1 · Question 11

Correct answer: D

11 The photograph shows the cells of a vascular bundle from a stem, as seen using a microscope.



What is tissue G, and what is its function?

	tissue	function
A	phloem	transports sugars to the roots
B	phloem	transports water to the leaves
C	xylem	transports sugars to the roots
D	xylem	transports water to the leaves

May/June 2018 · Paper 1 · Variant 1 · Question 13

Correct answer: A

13 Which blood vessel transports blood into the liver?

A hepatic portal vein

B hepatic vein

C pulmonary vein

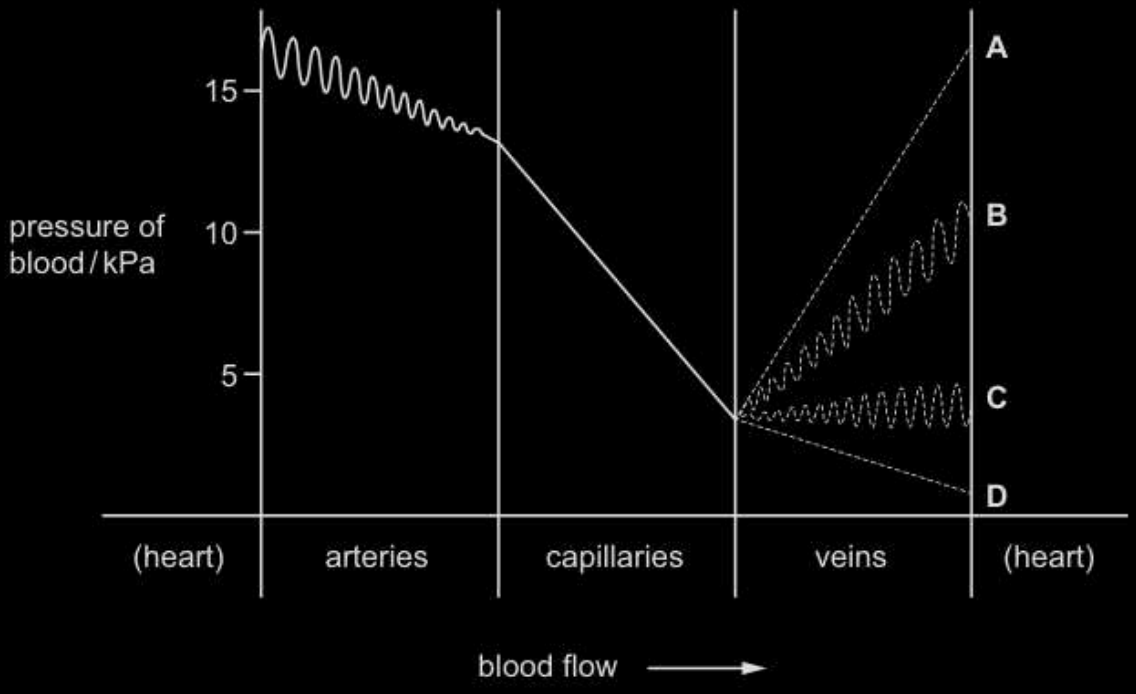
D renal vein

May/June 2018 · Paper 1 · Variant 1 · Question 14

Correct answer: D

14 The diagram shows the blood pressure of a person at rest as the blood leaves the heart and passes through arteries and then capillaries.

Which line shows the pressure of blood as it flows through veins before returning to the heart?



May/June 2018 · Paper 1 · Variant 1 · Question 15

Correct answer: A

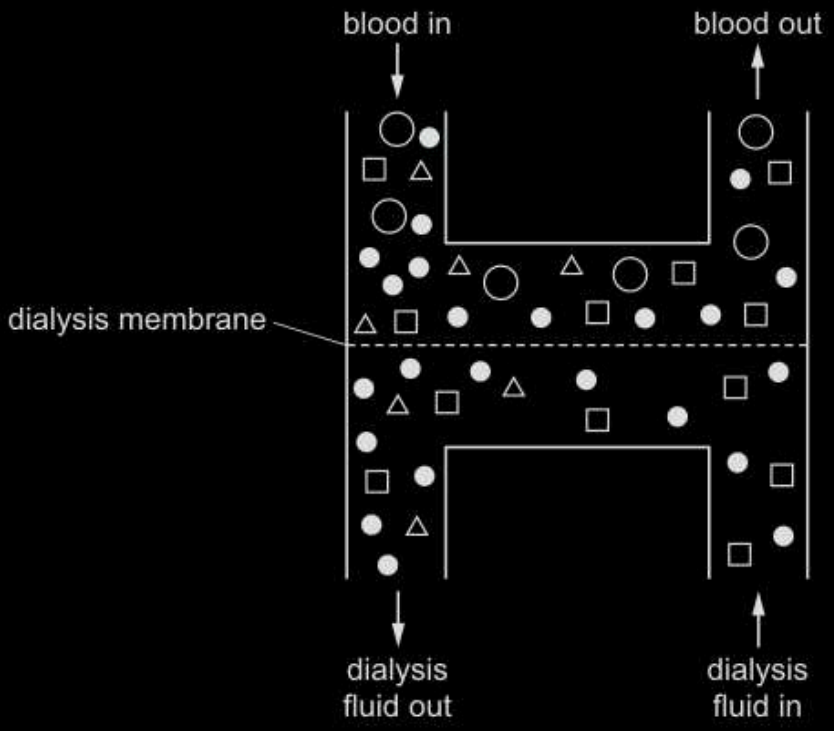
15 The diagram shows a section through part of a vein.

What could be the first organs found in the directions 1 and 2?

	1	2
A	heart	brain
B	intestine	liver
C	kidney	heart
D	lung	heart

Correct answer: D

19 The diagram shows how a kidney dialysis machine works. Each shape represents a molecule found in blood or dialysis fluid.



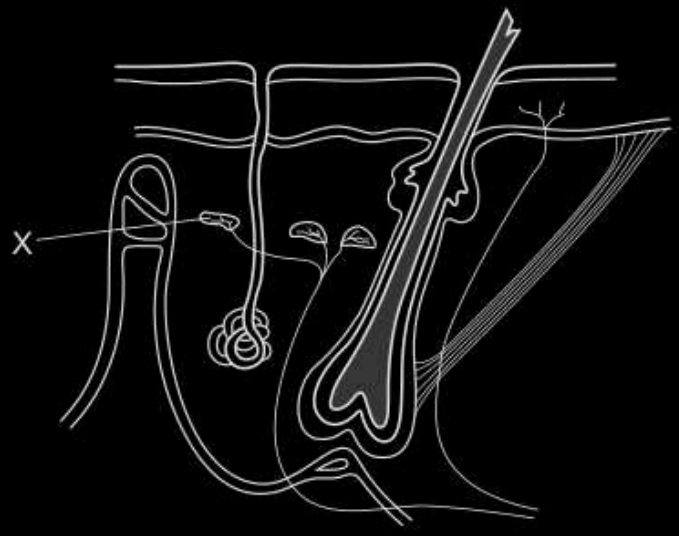
Which shape represents urea?

A ◯ **B** ● **C** □ **D** △

May/June 2018 · Paper 1 · Variant 1 · Question 20

Correct answer: B

20 The diagram shows some of the structures seen in a section through human skin.



What is the function of structure X?

A to cause capillaries to constrict

B to detect changes in temperature

C to receive impulses from the central nervous system

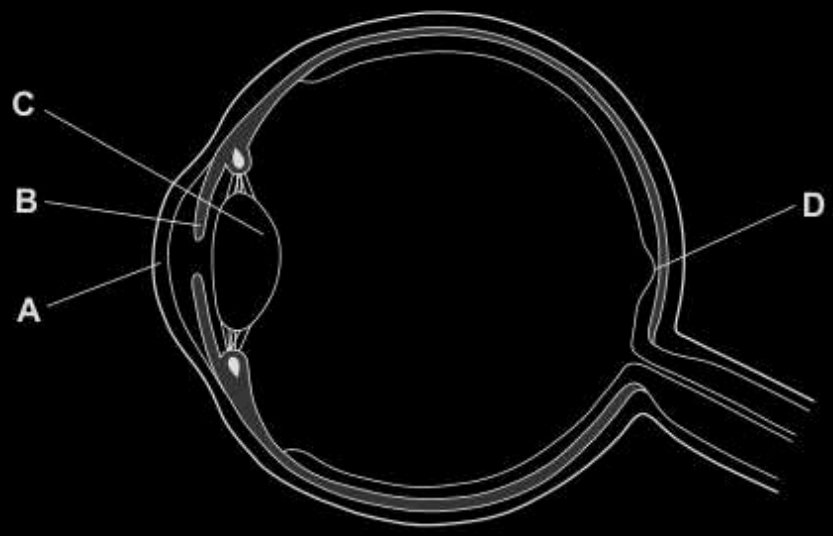
D to stimulate sweat glands to release sweat

May/June 2018 · Paper 1 · Variant 1 · Question 21

Correct answer: D

21 The diagram shows a section through an eye.

Which part is the receptor for the stimulus that results in a change in the size of the pupil?



May/June 2018 · Paper 1 · Variant 1 · Question 23

Correct answer: C

23 Which of these is a reflex action?

A increasing the blood glucose level by eating rice

B lifting a book off the table by contracting your arm muscles

C preventing an insect from flying into your eye by blinking

D using your brain to work out the answer to a problem

May/June 2018 · Paper 1 · Variant 1 · Question 27

Correct answer: D

27 Which table shows the features of bacteria, fungi and viruses?

A	feature	bacteria	fungi	viruses
	cell membrane	✓	✓	✓
	cell nucleus	✓	✗	✗
	cell wall	✓	✓	✗

key
✓ = yes
✗ = no

B	feature	bacteria	fungi	viruses
	cell membrane	✓	✓	✗
	cell nucleus	✓	✗	✗
	cell wall	✓	✗	✗

C	feature	bacteria	fungi	viruses
	cell membrane	✓	✗	✓
	cell nucleus	✗	✓	✗
	cell wall	✓	✓	✓

D	feature	bacteria	fungi	viruses
	cell membrane	✓	✓	✗
	cell nucleus	✗	✓	✗
	cell wall	✓	✓	✗

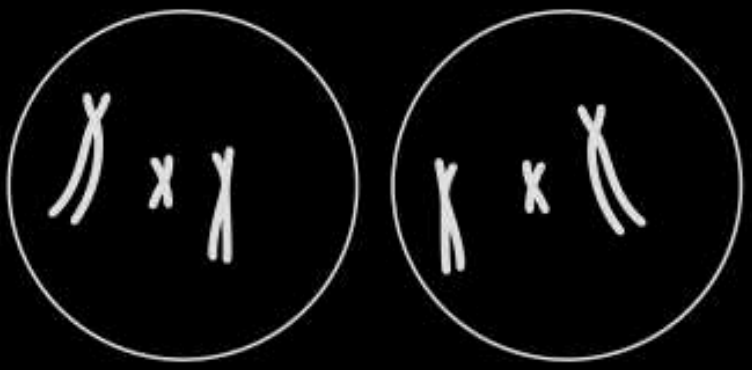
May/June 2018 · Paper 1 · Variant 1 · Question 33

Correct answer: B

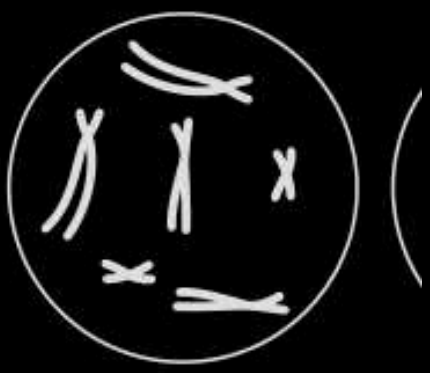
33 The diagram shows the chromosomes in a cell.

Which diagram shows the product of **one** division of the cell by mitosis?

A



B



C

D

May/June 2018 · Paper 1 · Variant 1 · Question 35

Correct answer: C

35 In which part of the human female reproductive system does a zygote start to divide to form a ball of cells?

A cervix

B ovary

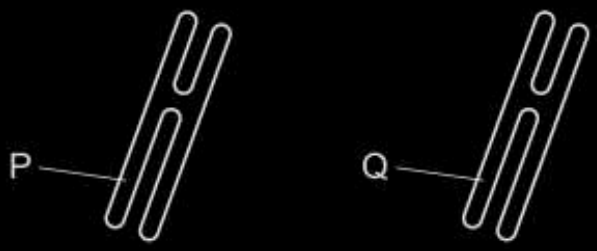
C oviduct

D uterus

May/June 2018 · Paper 1 · Variant 1 · Question 37

Correct answer: B

37 The diagram shows a pair of chromosomes from the same cell.



A gene is found at the point labelled P.

In a heterozygous individual, what will be found at the equivalent position labelled Q?

A a different allele of a different gene

B a different allele of the same gene

C a different gene of the same allele

D the same gene of the same allele

May/June 2018 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 Which characteristics show continuous and discontinuous variation in humans?

	continuous variation	discontinuous variation
A	blood groups	skin colour
B	free or attached ear lobes	blood groups
C	height	free or attached ear lobes
D	intelligence	hair colour

May/June 2018 · Paper 1 · Variant 1 · Question 39

Correct answer: B

39 Which statements about natural selection are correct?

	natural selection can lead to better adapted species surviving	natural selection can lead to extinction of a species	natural selection can lead to gene mutations occurring
A	true	true	true
B	true	true	false
C	true	false	true
D	false	true	true

May/June 2018 · Paper 1 · Variant 2 · Question 1

Correct answer: B

1 Which feature indicates that a root cell is from a plant and **not** an animal?

A cell membrane

B cell wall

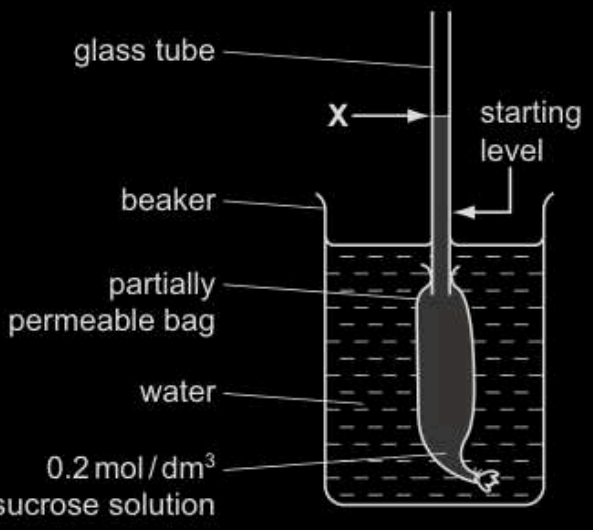
C chloroplast

D cytoplasm

May/June 2018 · Paper 1 · Variant 2 · Question 3

Correct answer: D

3 The diagram shows the result of an experiment. The liquid in the glass tube had risen to point **X** after three hours.



In a second experiment, which change could be made to cause the liquid to rise higher than **X**?

A a larger beaker

B a smaller bag

C water in the bag

D 0.4 mol / dm^3 sucrose solution in the bag

May/June 2018 · Paper 1 · Variant 2 · Question 4

Correct answer: A

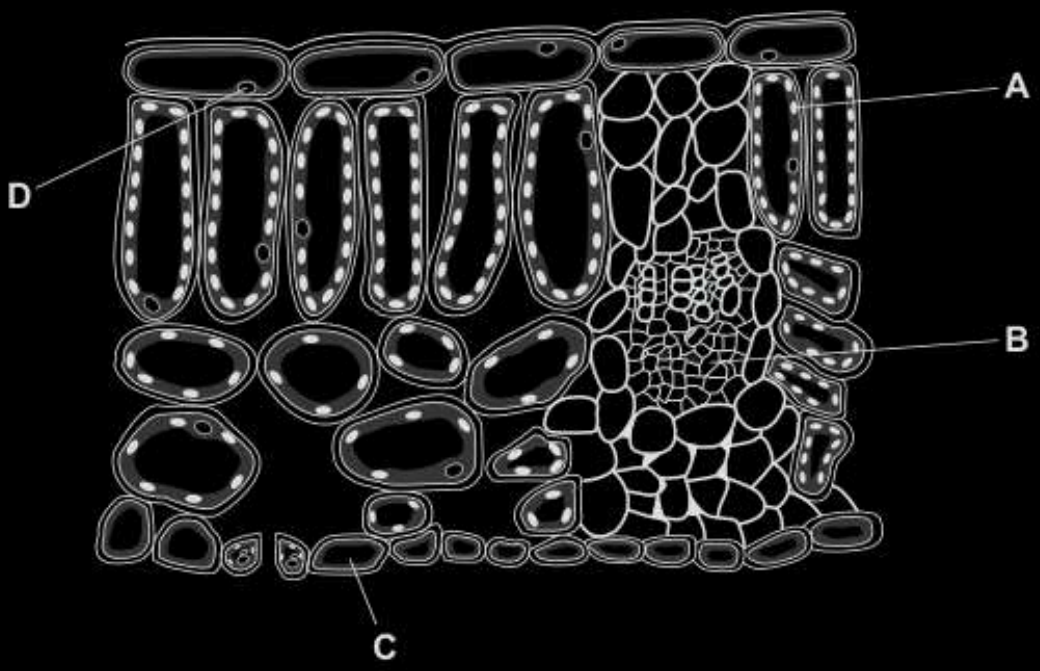
4 In an enzyme's action, where is the active site and where are the lock and the key?

	active site	lock	key
A	on the enzyme	on the enzyme	on the substrate
B	on the enzyme	on the substrate	on the enzyme
C	on the substrate	on the enzyme	on the substrate
D	on the substrate	on the substrate	on the enzyme

Correct answer: B

5 The diagram represents a cross-section of part of a leaf as seen using a microscope.

Where does translocation (movement of sucrose and amino acids) occur?

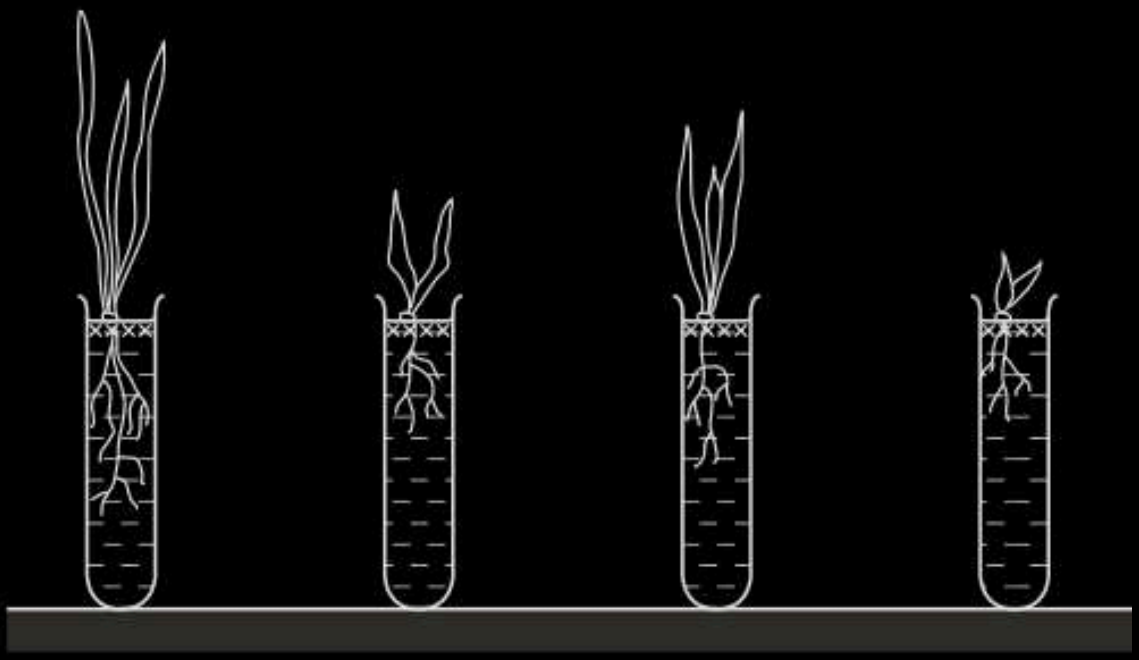


Correct answer: C

7 A student grows seedlings in four different test-tubes.

Tube W contains all the mineral ions needed for healthy plant growth.

The diagram shows the appearance of these seedlings after two weeks.



What do tubes X, Y and Z contain?

	X	Y	Z
A	all minerals except magnesium ions	all minerals except nitrate ions	water
B	all minerals except magnesium ions	water	all minerals except nitrate ions
C	all minerals except nitrate ions	all minerals except magnesium ions	water
D	water	all minerals except magnesium ions	all minerals except nitrate ions

May/June 2018 · Paper 1 · Variant 2 · Question 8

Correct answer: C

8 Which type of molecule is used to synthesise glycogen?

A amino acid

B fatty acid

C simple sugar

D starch

May/June 2018 · Paper 1 · Variant 2 · Question 11

Correct answer: C

11 The following paragraph describes water uptake by roots.

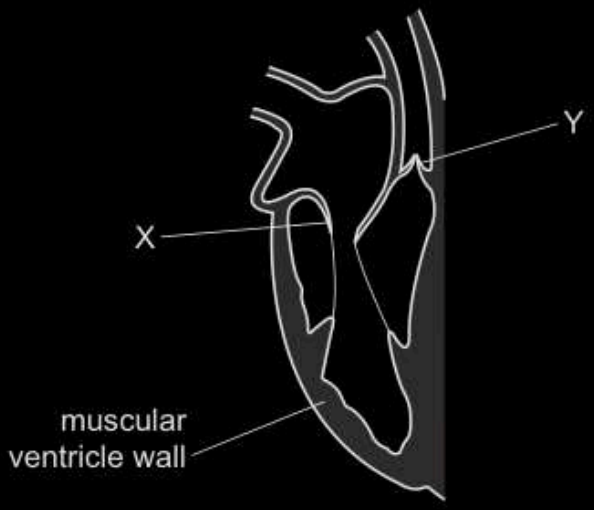
Root hair cells have ……1…… cell walls to allow them to absorb water. Cell cytoplasm has a ……2…… water potential than soil water so water moves by ……3…… from the soil to the root hair cell.

Which words correctly complete gaps 1, 2 and 3?

	1	2	3
A	large	larger	diffusion
B	permeable	higher	diffusion
C	permeable	lower	osmosis
D	rigid	higher	active transport

Correct answer: B

13 The diagram shows the right side of the human heart when the ventricle is relaxed.



Which row correctly describes the positions of valve X and valve Y when the ventricle contracts?

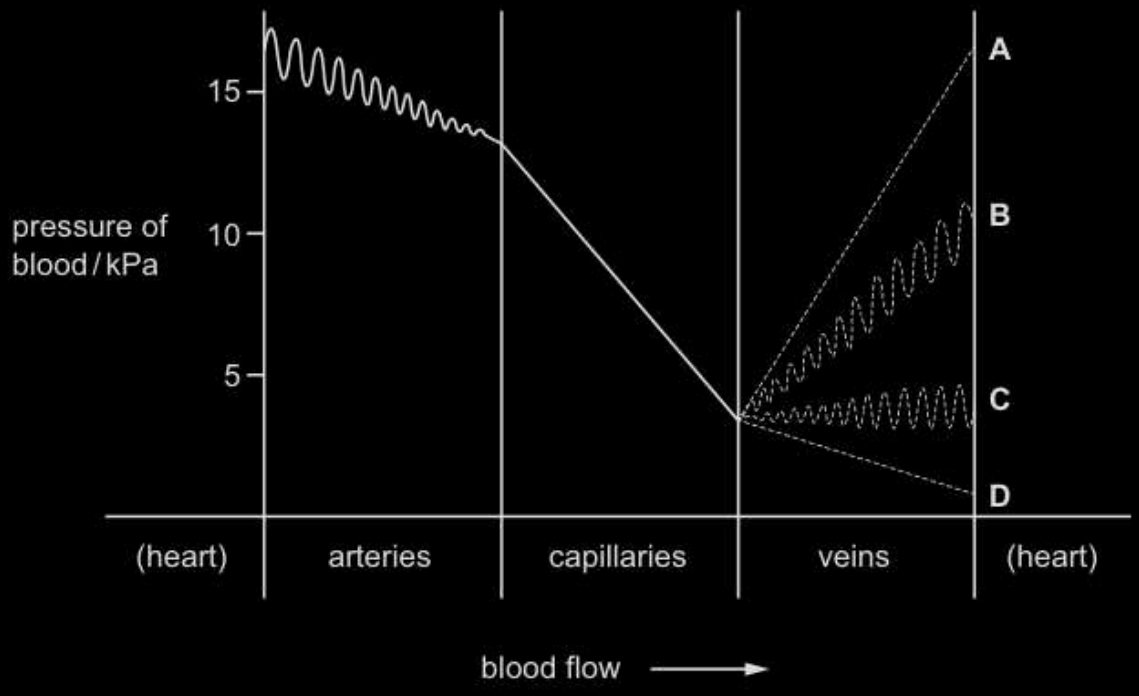
	valve X	valve Y
A	closed	closed
B	closed	open
C	open	closed
D	open	open

May/June 2018 · Paper 1 · Variant 2 · Question 14

Correct answer: D

14 The diagram shows the blood pressure of a person at rest as the blood leaves the heart and passes through arteries and then capillaries.

Which line shows the pressure of blood as it flows through veins before returning to the heart?



May/June 2018 · Paper 1 · Variant 2 · Question 15

Correct answer: A

15 The diagram shows a section through part of a vein.

What could be the first organs found in the directions 1 and 2?

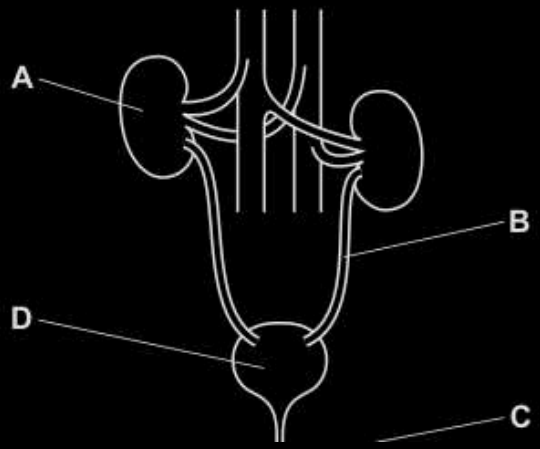
	1	2
A	heart	brain
B	intestine	liver
C	kidney	heart
D	lung	heart

May/June 2018 · Paper 1 · Variant 2 · Question 19

Correct answer: C

19 The diagram shows the human urinary system.

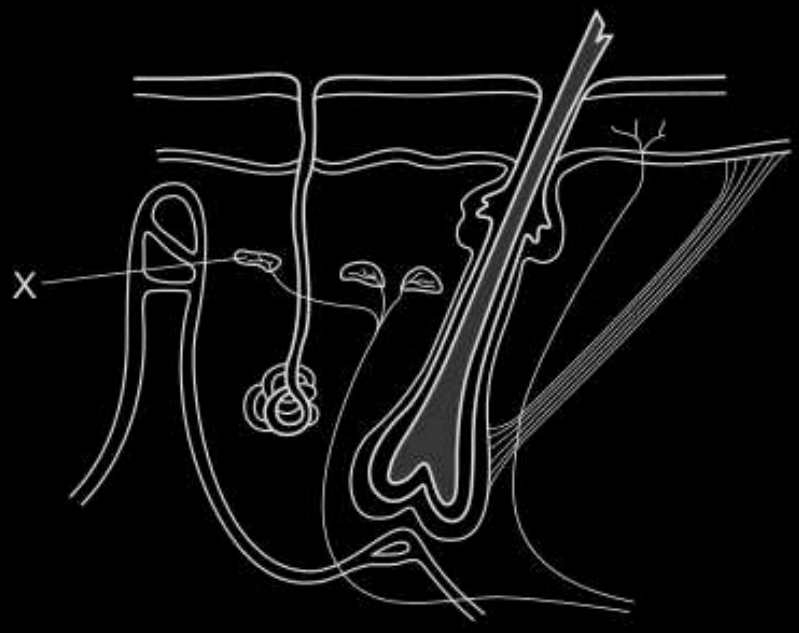
Which structure is the urethra?



May/June 2018 · Paper 1 · Variant 2 · Question 20

Correct answer: B

20 The diagram shows some of the structures seen in a section through human skin.



What is the function of structure X?

A to cause capillaries to constrict

B to detect changes in temperature

C to receive impulses from the central nervous system

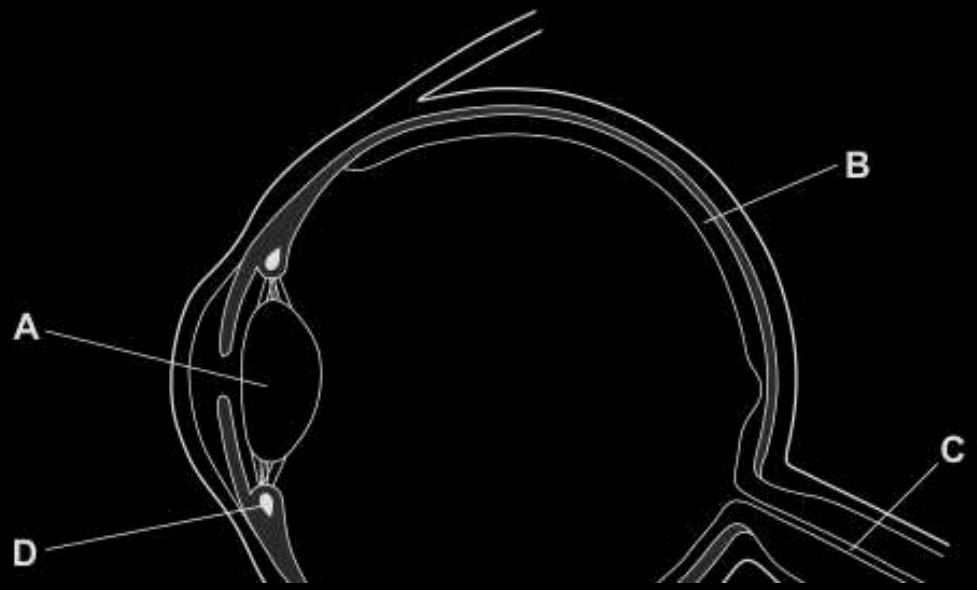
D to stimulate sweat glands to release sweat

May/June 2018 · Paper 1 · Variant 2 · Question 21

Correct answer: B

21 The diagram shows a horizontal section of a human eye.

Which part contains receptor cells sensitive to light?

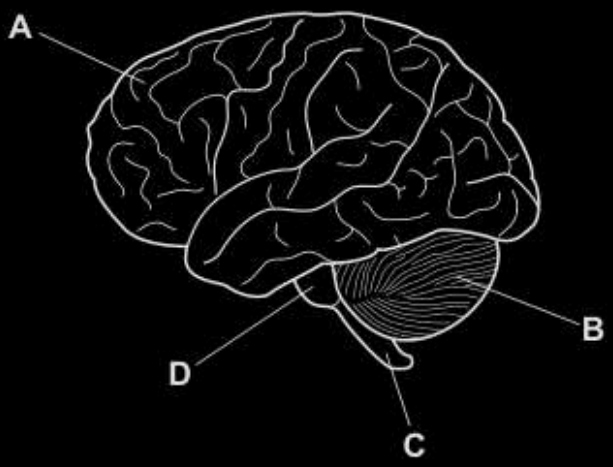


May/June 2018 · Paper 1 · Variant 2 · Question 22

Correct answer: A

22 The diagram shows the external view of a brain.

Which part of the brain is concerned with thinking and making decisions?



May/June 2018 · Paper 1 · Variant 2 · Question 23

Correct answer: B

23 In diabetes, glucose appears in the urine.

How does insulin prevent this?

A increasing the amount of water in the urine

B increasing the uptake of glucose into body cells

C preventing starch being digested into glucose

D raising blood pressure

May/June 2018 · Paper 1 · Variant 2 · Question 25

Correct answer: B

25 What are some effects of using heroin?

	ceasing its use causes withdrawal symptoms	injecting it can lead to AIDS	it stimulates the nervous system
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key

✓ = yes

✗ = no

May/June 2018 · Paper 1 · Variant 2 · Question 26

Correct answer: A

26 Antibiotics, such as penicillin, are produced in fermenters on a large scale.

Which substance is **not** needed for production of penicillin in this way?

A carbon dioxide

B mineral ions

C oxygen

D sugar

Correct answer: A

27 Some young plants were put into the soil and grew well for a few weeks. They then began to show signs of disease. Samples of the diseased leaves were examined using a microscope.

Which observations of the organism causing the disease show that it could be a fungus?

	long and thread-like structure	chloroplasts not present	cell walls present	nuclei surrounded by a membrane
A	✓	✓	✓	✓
B	✓	✓	✓	✗
C	✓	✗	✗	✓
D	✗	✓	✓	✓

key

✓ = yes

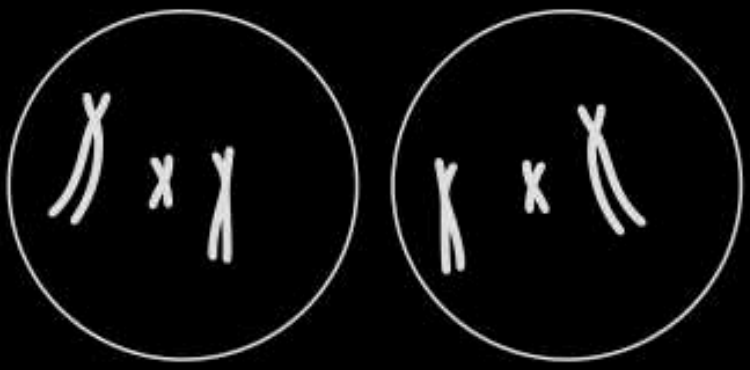
✗ = no

Correct answer: B

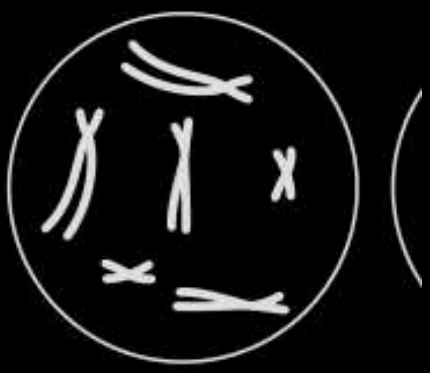
33 The diagram shows the chromosomes in a cell.

Which diagram shows the product of **one** division of the cell by mitosis?

A



B

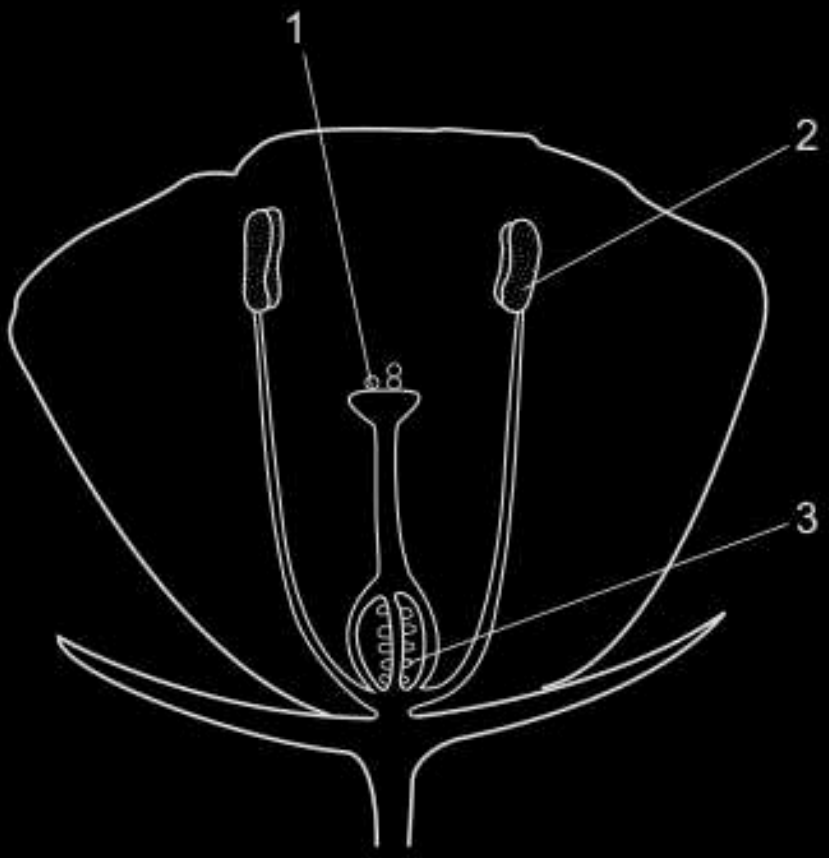


C

D

Correct answer: B

34 The diagram shows a section through a flower.



Which statement is correct?

A Fertilisation occurs at 1.

B Haploid gametes are produced at 2 and 3.

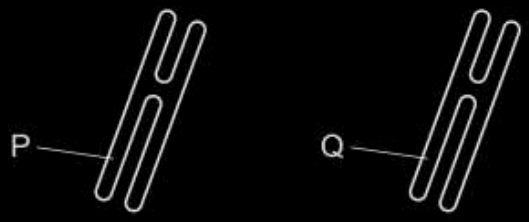
C Pollen is transferred by insects to 3.

D The pollen grain fuses with the female nucleus at 2.

May/June 2018 · Paper 1 · Variant 2 · Question 37

Correct answer: B

37 The diagram shows a pair of chromosomes from the same cell.



A gene is found at the point labelled P.

In a heterozygous individual, what will be found at the equivalent position labelled Q?

A a different allele of a different gene

B a different allele of the same gene

C a different gene of the same allele

D the same gene of the same allele

May/June 2018 · Paper 1 · Variant 2 · Question 38

Correct answer: A

38 The table shows the variation in foot length in a number of students.

foot length / cm	number of students
20.0–20.9	0
21.0–21.9	5
22.0–22.9	12
23.0–23.9	15
24.0–24.9	17
25.0–25.9	8
26.0–26.9	0

Which row identifies this type of variation and states its cause?

	type of variation	cause
A	continuous	genes and the environment
B	continuous	genes only
C	discontinuous	environment only
D	discontinuous	genes and the environment

Correct answer: D

39 The colour of the fruit of tomato plants is determined by alleles of the same gene. A tomato plant with red fruit was crossed with a tomato plant with yellow fruit. Of the offspring, 26 plants had red fruit and 24 had yellow fruit.

Three explanations were suggested.

1 Both parents were homozygous.

2 One parent had two recessive alleles.

3 One parent was heterozygous.

Which explanations are correct?

A 1 only **B** 3 only **C** 1 and 2 **D** 2 and 3

2 The diagram shows a male human gamete.

(a) (i) State the name of the male human gamete.

.....

[1]

The diagram below shows the male reproductive system and associated organs.

(ii) Label with a letter **X** on the diagram where the male gametes are produced.

[1]

(iii) The nucleus of the male gamete is different from the nuclei of other types of cell found at location **X**.

State the cause of this difference and explain its importance in reproduction.

cause of difference .....

.....

.....

explanation .....

.....

.....

.....

.....

[3]

(b) Describe the differences in size and mobility between the male human gamete and the female human gamete.

.....

.....

.....

.....[2]

(c) Complete the paragraph by writing the most appropriate word in each of the spaces.

The fusion of a male human gamete and a female human gamete to form a ..... is called ..... . A ball of cells is then formed that becomes implanted in the wall of the ..... .

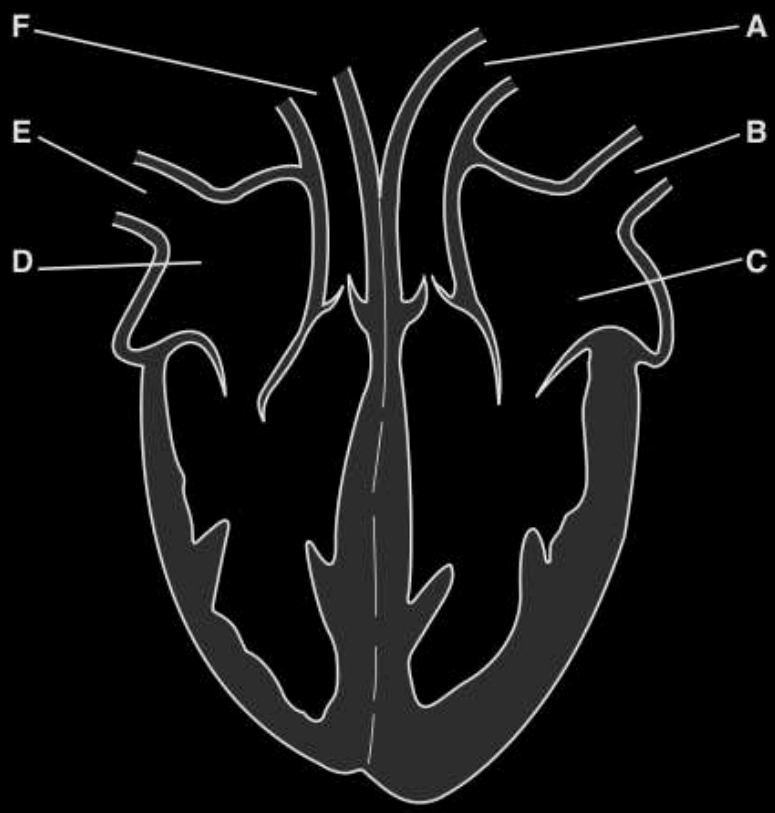
[3]

[Total: 10]

2(a)(i)	sperm ;	1	
2(a)(ii)	testis correctly labelled with **X** ;	1	
2(a)(iii)	sperm cells / gamete **+** meiosis **OR** other cells **+** mitosis ; different allele combinations on chromosomes ; reference to (nucleus of sperm) haploid / half number **AW** ; reference to restoration of diploid / full number **AW** ; (upon) fertilisation / fusion of gametes ;	3	Marks can be awarded under *cause of difference* or *explanation*
2(b)	male / sperm **+** small(er) ; male / sperm **+** motile / mobile / can swim **AW** ;	2	
2(c)	zygote ; fertilisation ; uterus / womb ;	3	

Question	Answer	Marks	Guidance

4 The diagram shows a human heart and associated blood vessels.



(a) Complete the table to show which of the parts **A** to **F** contain oxygenated blood and which contain deoxygenated blood.

Write **each** of the letters **A** to **F** in either the right or the left side of the table.

contain oxygenated blood	contain deoxygenated blood

[2]

(b) **(i)** Complete the table below to show which of **A** to **F** are involved in the circulation of blood to or from each of the following locations:

- the lungs,
- the body tissues.

Write **each** of the letters **A** to **F** in either the right or the left side of the table.

blood to or from the lungs	blood to or from the body tissues

[4]

(ii) Compare the pressure of blood in the circulation to the body tissues and the pressure of blood in the circulation to the lungs.

.....

.....[1]

(iii) Explain how the structure of the heart produces this difference in blood pressure.

.....

.....

.....

.....[3]

[Total: 10]

4(a)	contain oxygenated blood: **A + B + C ;** / contain deoxygenated blood: **D + E + F ;**	**2**	**A** one or two correct letters in both boxes for 1 mark in total
4(b)(i)	blood to or from the lungs: any two of **F + B + C ;** three correct **F + B + C ;;** / blood to or from the body tissues: any two of **A + D + E ;** three correct **A + D + E ;;**	**4**	

© UCLES 2018

Question	Answer	Marks	Guidance
4(b)(ii)	higher (to the body tissues) **AW ;**	1	
4(b)(iii)	left ventricle ; thicker **AW ;** muscle ; greater **+** contraction / force (applied to blood) **AW ;**	3	**ORA** for right ventricle for all points

© UCLES 2018

Question	Answer	Marks	Guidance

5 **(a)** **(i)** Describe the main characteristics of a virus.

.....

.....

.....

.....[3]

(ii) The human immunodeficiency virus (HIV) reproduces inside white blood cells and destroys them.

Use your knowledge of the functions of white blood cells to suggest why the virus is named the **immunodeficiency** virus.

.....

.....

.....[2]

(b) **(i)** HIV causes a disease called AIDS. The virus may be transmitted during sexual intercourse.

State **two** methods by which the spread of HIV by sexual intercourse may be controlled.

1 .....

.....

2 .....

.....

[2]

(ii) HIV may be transmitted in other ways.

State **two** ways, other than during sexual intercourse, by which HIV may be transmitted.

1 .....

.....

2 .....

.....

[2]

[Total: 9]

Section B

Answer **both** questions in this section.

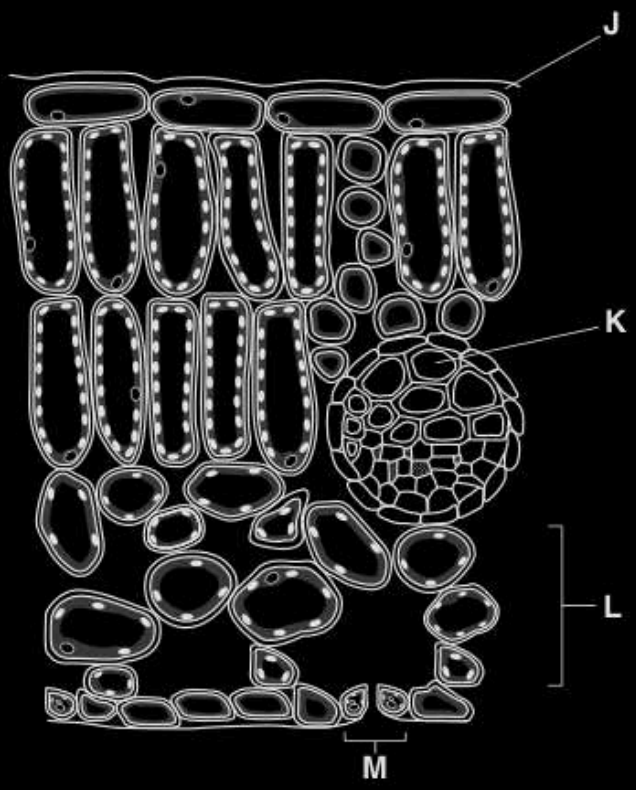
Write your answers in the spaces provided.

5(a)(i)	very **AW** small ; protein coat / capsid ; no membrane **/** no cytoplasm / no nucleus / non-cellular **/** no organelles ; nucleic acid / DNA / RNA ; parasitic / (may be) pathogen(ic) / cause disease / non-living **AW** ; unaffected by antibiotics ;	**3**	
5(a)(ii)	reference to immune / immunity (system or affected) ; reference to reduced **AW +** antibody production ; reference to reduced **AW +** phagocytic action ;	**2**	
5(b)(i)	barrier contraception / correct named contraceptive method e.g. condom ; abstinence ; only one partner **AW** ; reference to use of anti-retroviral drugs ;	**2**	**A** avoid unprotected sex
5(b)(ii)	sharing **+** needles / sharp objects / toothbrushes ; blood transfusion / contact ; from mother to baby / across placenta / during birth / during breast-feeding ;	**2**	

© UCLES 2018

Question	Answer	Marks	Guidance

6 The diagram shows a magnified transverse section through a leaf.



Name each of **J**, **K**, **L** and **M** and explain the importance of each in the process of transpiration.

J .....

.....

.....

.....

K .....

.....

.....

.....

.....

L .....

.....

.....

.....

6	*(J)* cuticle ; waterproof / waxy **AW** ; prevents / reduces **+** water loss / transpiration / evaporation ; *(K)* xylem ; transport **+** water ; from named location to named higher location ; *(L)* spongy ; air / intercellular **+** spaces ; water film **AW** ; evaporation / water vapour ; diffusion ; *(M)* stoma / stomata ; guard cell ; open / close ; reference to water vapour **+** exits ; diffusion ;	**10**	**Note** marking points must **not** be transferred between the sections **R** 'phloem' for xylem

Question	Answer	Marks	Guidance

7 **(a)** Define the term *allele*.

.....

.....

.....[2]

(b) In mice, the allele for black fur (**B**) is dominant to the allele for white fur (**b**).

Two mice with black fur were mated to produce offspring, all of which were black.

One of these black offspring was then mated with a mouse with white fur. Equal numbers of mice with black fur and with white fur were produced.

Complete the genetic diagram to show the pattern of inheritance in each cross.

phenotypes of first parents black fur × black fur

genotypes of first parents ..... × .....

gametes ....

genotypes ....

phenotype all have black fur

phenotypes of second parents black fur × white fur

genotypes of second parents ..... × .....

gametes ....

genotypes ....

phenotypes black fur white fur

[8]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7(a)	form / version ; gene ; reference to codes for / specifies production of **+** protein ; reference to dominant / codominant / recessive ;	**2**	
7(b)	BB ; x Bb ; B **+** B **+** B **+** b ; BB **+** Bb ; Bb (black) ; x bb (white) ; B **+** b **+** b **+** b ; Bb **+** bb ;	**8**	**Allow** any order within each marking point

© UCLES 2018

Question	Answer	Marks	Guidance

8 **(a)** Describe **one** example of a simple reflex action and explain the importance to the body of this action.

.....

.....

.....

.....

.....

.....

.....

.....[4]

(b) Describe how **named** components of the nervous system are involved in producing a reflex action.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 10]

8(a)	named stimulus / trigger for a specific reflex action ; correct named receptor for stimulus given ; action described correct for example given ; importance of specific action explained ;	**4**	
8(b)	synapse (anywhere in sequence) ; impulse / electrical pulse (anywhere in sequence) ; receptor ; detection of stimulus ; **then** sensory neurone ; relay / inter(mediate) / connector neurone ; reference to CNS / brain / spinal cord ; **then** motor neurone ; effector / named effector ; action of effector or described ;	**6**	**Ig** signal / message

Question	Answer	Marks	Guidance

9 **(a)** Explain the concept of control by negative feedback.

.....

.....

.....

.....

.....

.....

.....

.....[4]

(b) Describe how **two named** components of the skin are involved in regulating body temperature in **hot** conditions.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 10]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

9(a)	parameter / condition e.g. temperature ; change from set point / norm **AW** ; detected ; reference to communication or named method (e.g.nerve / impulse / hormone) ; reference to control centre / coordinator / hypothalamus / brain ; response / corrective mechanism(s) ; reversal /correction of initial change / return to set point or norm ;	**4**	

© UCLES 2018

Question	Answer	Marks	Guidance
9(b)	any two from nerve ending / blood vessels / sweat gland / hair ;;	2	
	(nerve ending) detects / receptor ; change / increase in temperature ; impulse / (message) to brain ; *(blood vessels)* dilate ; more blood to surface of skin ; reference to capillaries ; reference to increased radiation / heat loss ; *(sweat gland / duct)* secretion / release / skin surface **+** sweat ; reference to evaporation ; *(hair)* lowers ; less air trapped / loss of insulation ;	4	**A** if with reference to capillaries

© UCLES 2018

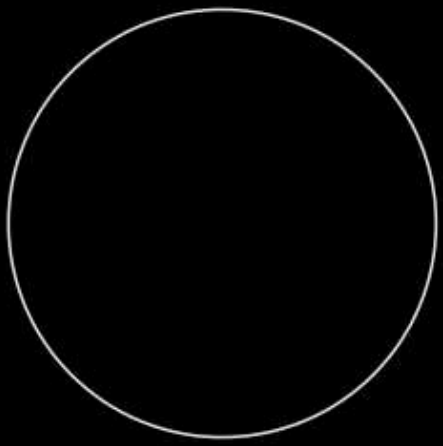
1 The diagrams show the bud of an insect-pollinated flower and a magnified transverse section through the same flower bud. The transverse section was taken at the position shown by the dotted line.

(a) Complete the table to show the name of each of structures **A** to **D** and to state **one** function of each structure.

letter	name of structure	function
A		
B		
C		
D		

[8]

(b) The diagram shows an incomplete transverse section through the stem of this plant.



(i) Complete the diagram by drawing and labelling the positions of each of the following tissues:

- xylem,
- phloem. [2]

(ii) State **one** function of xylem tissue.

.....

.....[1]

[Total: 11]

1(a)		8	

letter	name of structure	function
(A)	sepal / calyx ;	protect ;
(B)	petal / corolla ;	attract / landing platform ;
(C)	stamen / anther ; **or** filament ;	produces / contains **AW +** pollen / male gamete / male nucleus ; support anther ;
(D)	ovary / carpel / pistil ;	produces / contains **AW +** ovum / ovule / egg / female gamete ; **or** forms **AW** fruit / site of fertilisation ;

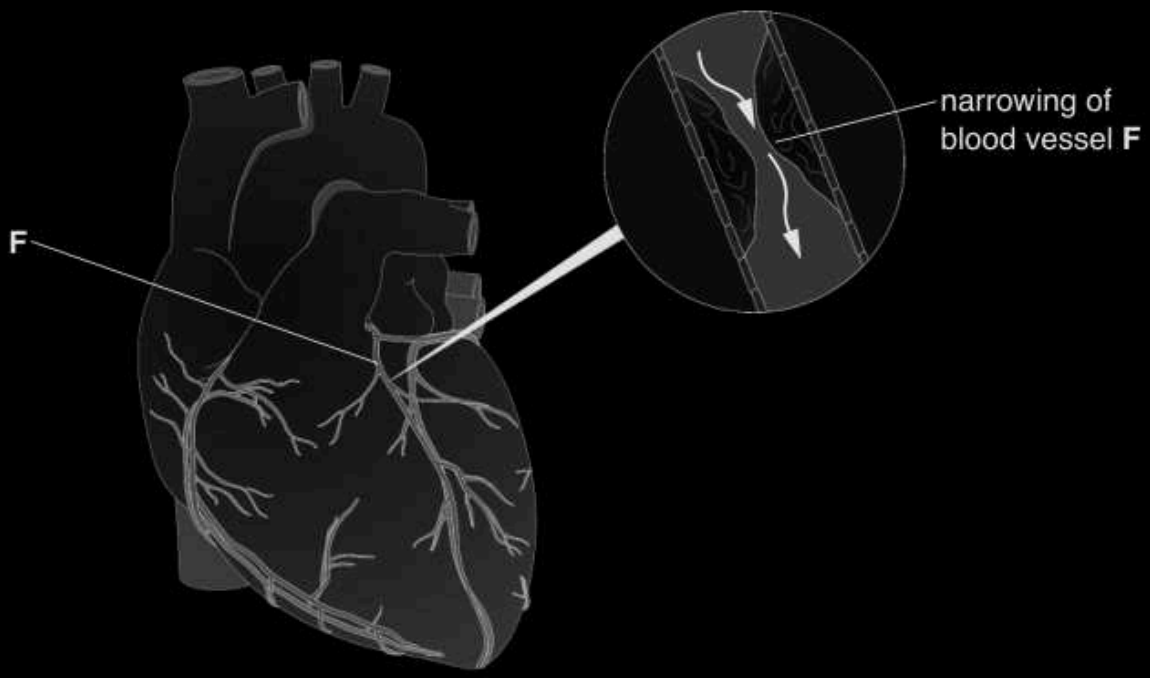
1(b)(i)	**1** one or more vascular bundles **+** each oval shaped **+** location correct ; **2** 'xylem' labelled on inside **+** 'phloem' labelled separately on outside of at least one oval vascular bundle ;	**2**	
1(b)(ii)	**1** transports / carries **AW +** water / ions / minerals ; **2** support ;	**1**	**A** named ion

© UCLES 2018

Question	Answer	Marks	Guidance

2 The diagram shows the human heart.

The blood vessel labelled **F** may become narrowed as shown.



(a) (i) Name blood vessel **F**.

.....

[1]

(ii) Name the disease caused by the narrowing of this blood vessel.

.....

[1]

(iii) State **three** factors that may lead to the narrowing of this blood vessel.

1 .....

2 .....

3 .....

[3]

(iv) Suggest and explain how a person might be affected by the disease caused by the narrowing of blood vessel **F**.

.....

.....

.....

.....

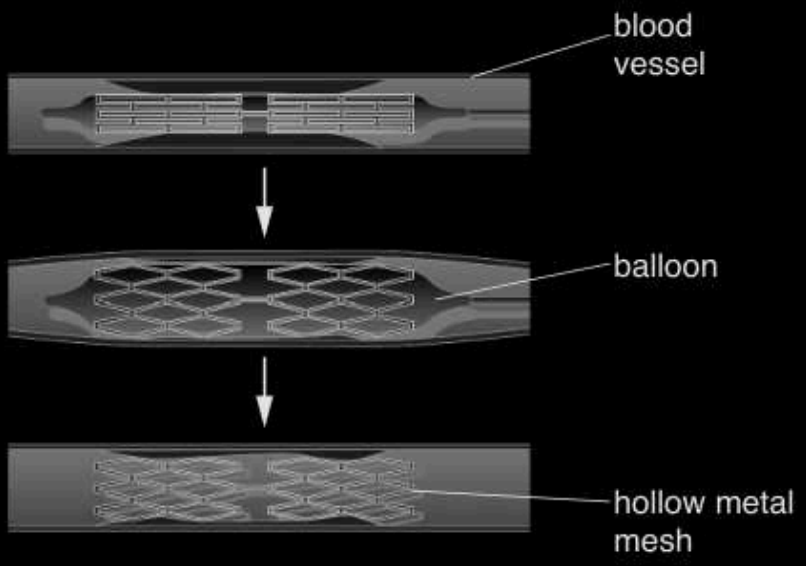
.....

.....[5]

(b) It is possible to treat the disease caused by the narrowing of blood vessel **F**, by carrying out an operation.

In the operation:

- a balloon surrounded by a metal mesh is inserted into the blood vessel and inflated,
- the balloon is then deflated and removed, leaving the metal mesh in place.



Suggest the purpose of each of the following:

inflating the balloon,

.....

.....

.....

leaving the hollow metal mesh in the blood vessel.

.....

.....

.....

[3]

[Total: 13]

2(a)(i)	coronary artery ;	1	
2(a)(ii)	coronary heart disease / heart disease / CHD / atherosclerosis / cardiac disease / arteriosclerosis / angina ;	1	
2(a)(iii)	**1** fat / oil / cholesterol **+** diet **AW** ; **2** stress **AW** ; **3** smoking ; **4** genetics / inheritance **AW** ; **5** lack of exercise ; **6** old age ; **7** obesity ;	3	**A** 'anxiety' / 'depression' for point **2**
2(a)(iv)	**1** heart attack **AW** / heart stops / angina **AW** / breathlessness ; **2** less blood **+** to body / tissues / organs or any named ; **3** less oxygen/ glucose **+** to body / tissues / organs or any named ; **4** less aerobic respiration **or** more anaerobic respiration ; **5** production of lactic acid ; **6** less ability **AW** **+** of heart to contract / pump blood ; **7** less ability **AW** **+** to carry out physical activity ;	5	

© UCLES 2018

Question	Answer	Marks	Guidance
2(b)	*(inflating the balloon)* **1** opens metal mesh **AW** ; **2** push / compress **+** blockage / fat ; **3** widen **AW** **+** blood vessel / lumen **AW** ; *(leaving the hollow metal mesh in the blood vessel)* **4** maintain **AW** **+** wider lumen **AW** ; **5** increase **AW** **+** blood flow ;	**3**	

© UCLES 2018

Question	Answer	Marks	Guidance

3 The table shows how the thickness of the lens of the eye changes when focussing on an object at different distances from the front of the eye.

distance from eye/cm	thickness of lens/mm
10	4.0
20	3.6
30	3.2
50	2.9
100	2.7
150	2.6
200	2.6

(a) (i) Describe the pattern shown by the data in the table.

.....

.....

.....[2]

(ii) Explain how **named** components of the eye change the thickness of the lens when focussing on an object as it moves further from the front of the eye.

.....

.....

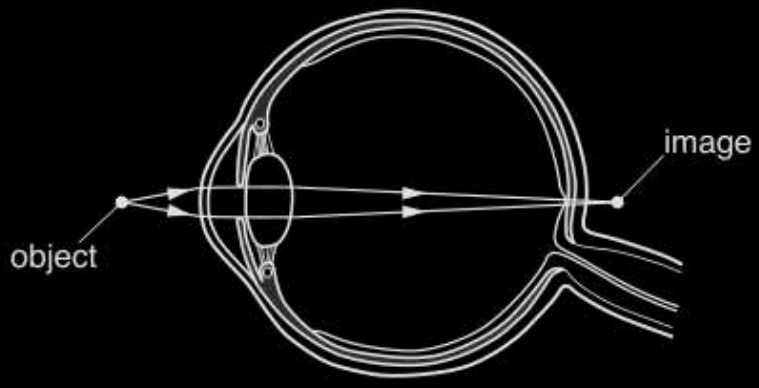
.....

.....

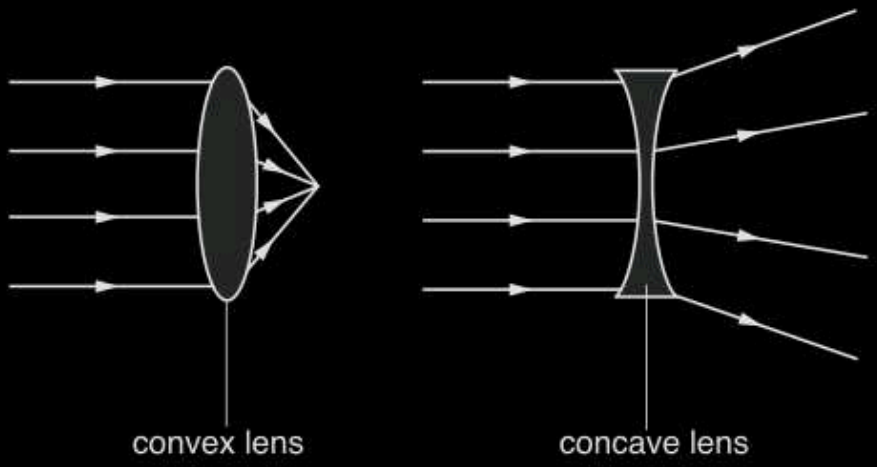
.....

.....[4]

(b) As a person becomes older, the lens of the eye becomes harder and less elastic. This results in the person seeing an image of a close object that is out of focus.



The diagram below shows how light is refracted by two types of artificial lens that may be used to manufacture spectacles (glasses).



Suggest and explain which type of lens would be most suitable for this older person.

.....

.....

.....

.....

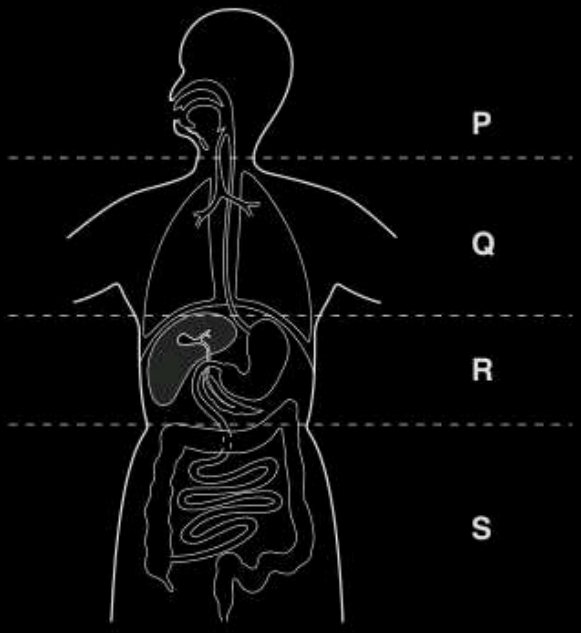
.....[3]

[Total: 9]

3(a)(i)	**1** distance from eye increases **+** thickness / width of lens decreases **or** inverse ; **2** up to 150 **+** cm **or** up to 2.6 **+** mm **or** then constant **AW** ;	**2**	**A** distance from eye decreases **+** thickness / width of lens increases
3(a)(ii)	**1** ciliary ; **2** muscles **+** relax ; **3** suspensory ligaments ; **4** ligaments **+** tighten / taut **AW** ; **5** lens **+** pulled / stretched ;	**4**	
3(b)	(*if convex stated*) **no mark for convex** **1** light **+** rays ; **2** increased / more **+** refraction / bending ; **3** before entering eye ; **4** converge / meet **+** on retina / fovea ; **5** object / image **+** clear / in focus ; **OR** (*if concave, i.e. incorrect, **or** no type of lens stated*) **6** reference to light rays **+** refraction **AW +** before entering eye ;	**3**	

Question	Answer	Marks	Guidance

5 The diagram shows a section through the human body divided into regions **P**, **Q**, **R** and **S**.



Complete the table below by matching the letters from the diagram to the statements in the table. There may be one or more than one letter for each statement.

contains an organ which	region or regions
… produces an acidic secretion	**R**
… contains villi	
… digests protein	
… produces insulin	
… contains bronchi	
… secretes amylase	
… ingests food	

[6]

[Total: 6]

Question 6 starts on the next page.

Section B

Answer **both** questions in this section.

Write your answers in the spaces provided.

Question	Answer	Marks	Guidance
5			

contains an organ which	region or regions
… produces an acidic secretion	(R)
… contains villi	R / S ;
… digests protein	R / S ;
… produces insulin	R ;
… contains bronchi	Q ;
… secretes amylase	P + R ;
… ingests food	P ;

6			**R** incorrect letter in any box

7 **(a)** Describe the cause of each of the following:

Down's syndrome,

.....

.....

.....

.....

sickle cell anaemia.

.....

.....

[3]

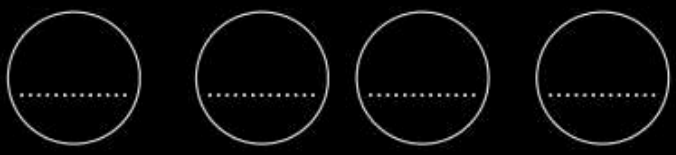
(b) **(i)** Parents, one with blood group B and the other with blood group A, have a child with blood group O.

The parents decide to have another child.

Complete the genetic diagram to show the possible blood groups for the second child of these parents.

genotypes of parents ..... x .....

gametes



possible genotypes of child ....

possible blood groups of child ....

[4]

(ii) State the probability of each of the following for the second child of these parents:

being the same sex as the first child .....

having the same blood group as the first child ..... [2]

(iii) Name the type of inheritance shown by the alleles that produce the blood group AB.

..... [1]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7(a)	**1** inherited / genetic **AW** ; *(Down's syndrome)* **2** mutation **+** chromosome ; **3** reference to chromosome 21 ; **4** one extra **or** 3 instead of 2 **or** 47 not 46 **or** 24 not 23 in **+** gamete/egg/sperm ; *(sickle cell anaemia)* **5** mutation **+** gene ; **6** homozygous recessive / two recessive alleles ;	**3**	
7(b)(i)	**1** I^BI^O **+** I^AI^O **or** BO **+** AO ; **2** I^B **+** I^O **+** I^A **+** I^O **or** B **+** O **+** A **+** O **or** gametes correct for stated parent genotypes ; **3** I^AI^B **+** I^BI^O **+** I^AI^O **+** I^OI^O **or** AB **+** BO **+** AO **+** OO **or** possible genotypes of child correct for stated gametes ; **4** AB **+** B **+** A **+** O **or** possible blood groups of child correct for stated genotypes ;	**4**	**Allow** any order within each marking point
7(b)(ii)	*(same sex)* 50% / half / ½ / 0.5 / 2 in 4 / 1 in 2 / 1:1 ; *(same blood group)* 25% / quarter / ¼ / 0.25 / 1 in 4 / 1:3 ;	**2**	
7(b)(iii)	codominance / codominant ;	**1**	**R** incomplete dominance

© UCLES 2018

Question	Answer	Marks	Guidance

8 **(a)** Outline the role of a **named type** of microorganism in the production of each of the following products:

yoghurt .....

.....

.....

.....

.....

bread .....

.....

.....

.....

.....

[6]

(b) Describe how a **named type** of microorganism can be used to produce human insulin on a commercial scale.

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

8(a)	**1** enzymes ; *(yoghurt)* **2** bacteria / *Lactobacillus* / *Streptococcus* ; **3** sugar / lactose **+** milk ; **4** production **AW +** of acid **or** reduction in pH ; **5** thickens / curdles / coagulates / reference to taste ; *(bread)* **6** fungus/ yeast / *Saccharomyces* ; **7** anaerobic **+** respiration **or** fermentation ; **8** production **AW +** carbon dioxide ; **9** rising ;	**6**	**A** once for **either** *yoghurt* **or** *bread*
8(b)	**1** genetic **+** engineering / modification ; **2** bacteria / fungus / named bacteria / named fungus ; **3** reference to insulin gene ; **4** from human **+** DNA / chromosome / genome ; **5** to bacterial / fungal **+** DNA / chromosome / genome / plasmid ; **6** fermenter ; **7** reproduce / multiply / divide / mitosis / binary fission ;	**4**	

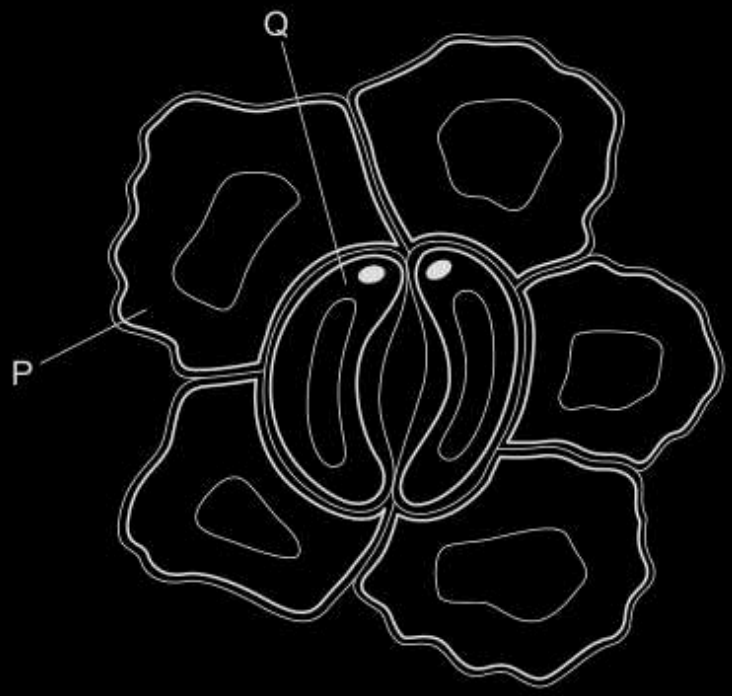
© UCLES 2018

Question	Answer	Marks	Guidance

October/November 2017 · Paper 1 · Variant 1 · Question 1

Correct answer: C

1 The diagram shows cells in the epidermis of a leaf.



To complete the diagram, which structural features should be added to the cells P and Q?

	P		Q	
	chloroplasts	nucleus	chloroplasts	nucleus
A	✓	✓	✗	✗
B	✓	✗	✓	✓
C	✗	✓	✓	✗
D	✗	✗	✗	✓

key
✓ = yes
✗ = no

October/November 2017 · Paper 1 · Variant 1 · Question 2

Correct answer: C

2 Which movement of a substance in plants is an energy-consuming process?

A absorption of carbon dioxide by a palisade cell

B loss of water into the air spaces of a leaf

C nitrate uptake by root hair cells

D transport of water up through the xylem

October/November 2017 · Paper 1 · Variant 1 · Question 3

Correct answer: B

3 The diagram shows the direction of water movement from one cell to another in a plant root.

Which cell has the higher water potential, and how does the water move?

	cell with higher water potential	process by which water moves
A	P	active transport
B	P	diffusion and osmosis
C	Q	active transport
D	Q	diffusion and osmosis

October/November 2017 · Paper 1 · Variant 1 · Question 4

Correct answer: D

4 According to the lock and key hypothesis, what is the lock and what is the key for the enzyme lipase?

	lock	key
A	lipids	fatty acids
B	fatty acids	lipase
C	lipids	lipase
D	lipase	lipids

October/November 2017 · Paper 1 · Variant 1 · Question 5

Correct answer: D

5 Which characteristic is the result of a deficiency of magnesium in plants?

A large green leaves

B purple spots on leaves

C reduced root growth

D yellow areas between leaf veins

October/November 2017 · Paper 1 · Variant 1 · Question 7

Correct answer: D

7 What describes the upper cuticle of a leaf?

A a permeable layer allowing water to enter the leaf

B a single layer of cells containing many chloroplasts

C a single layer of transparent cells allowing light to enter the leaf

D a thin non-cellular layer preventing water loss from the leaf

October/November 2017 · Paper 1 · Variant 1 · Question 8

Correct answer: A

8 Which is essential for the healthy growth of teeth in a baby?

A calcium

B iron

C nitrate

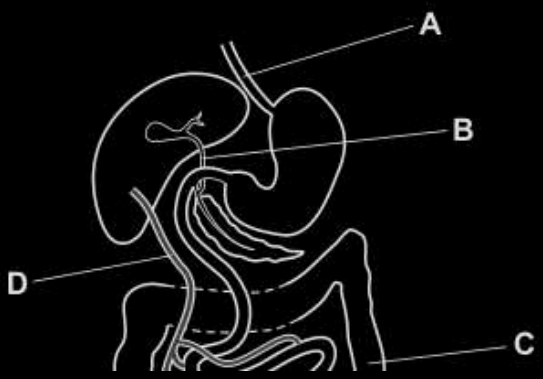
D urea

October/November 2017 · Paper 1 · Variant 1 · Question 9

Correct answer: D

9 The diagram shows part of the alimentary canal and associated organs.

Which part would contain high concentrations of glucose and amino acids, four hours after eating a meal?



October/November 2017 · Paper 1 · Variant 1 · Question 10

Correct answer: A

10 After eating, the pH in the mouth decreases.

Which statement explains this decrease?

A Bacteria release acids.

B Enzymes in saliva release acids.

C Salivary glands release acids.

D Sensory neurones in the tongue release acids.

October/November 2017 · Paper 1 · Variant 1 · Question 11

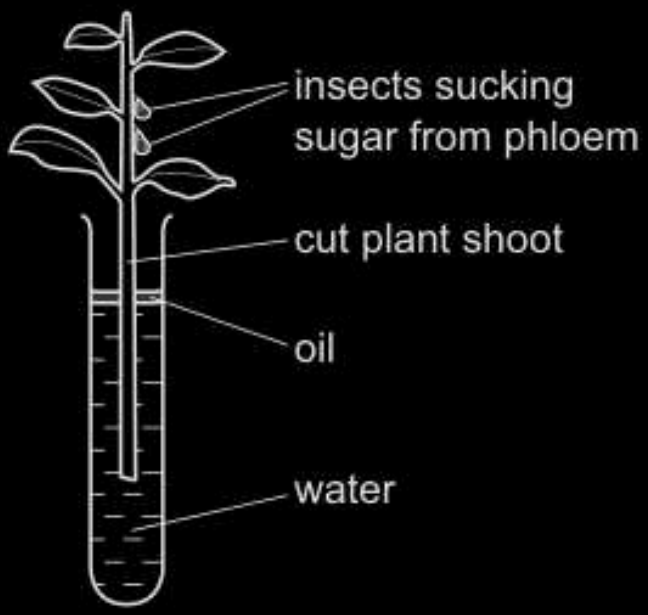
Correct answer: B

11 Which conditions result in the highest rate of transpiration from a plant?

	percentage humidity	temperature / °C
A	60	15
B	60	25
C	100	15
D	100	25

Correct answer: C

12 The diagram shows a cut plant shoot in a container of water.



What will stop the movement of water up this stem?

A a fall in the humidity of the air

B a rise in the air temperature

C air bubbles in the xylem

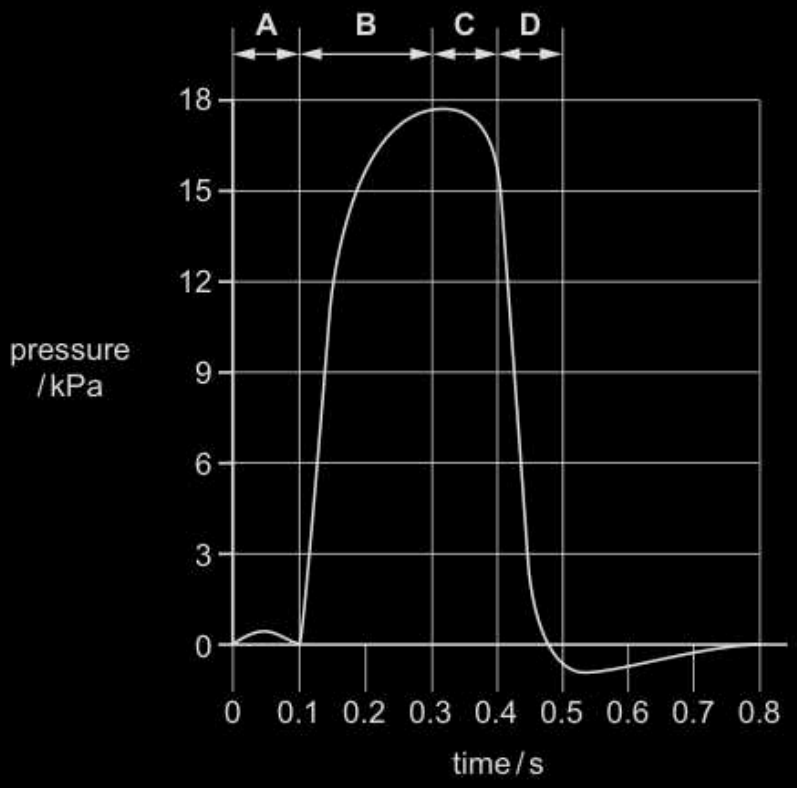
D an insect sucking sugar from the phloem

October/November 2017 · Paper 1 · Variant 1 · Question 13

Correct answer: A

13 The graph shows changes in the blood pressure in the left ventricle of the heart.

During which period is the left **atrium** contracting?



October/November 2017 · Paper 1 · Variant 1 · Question 14

Correct answer: D

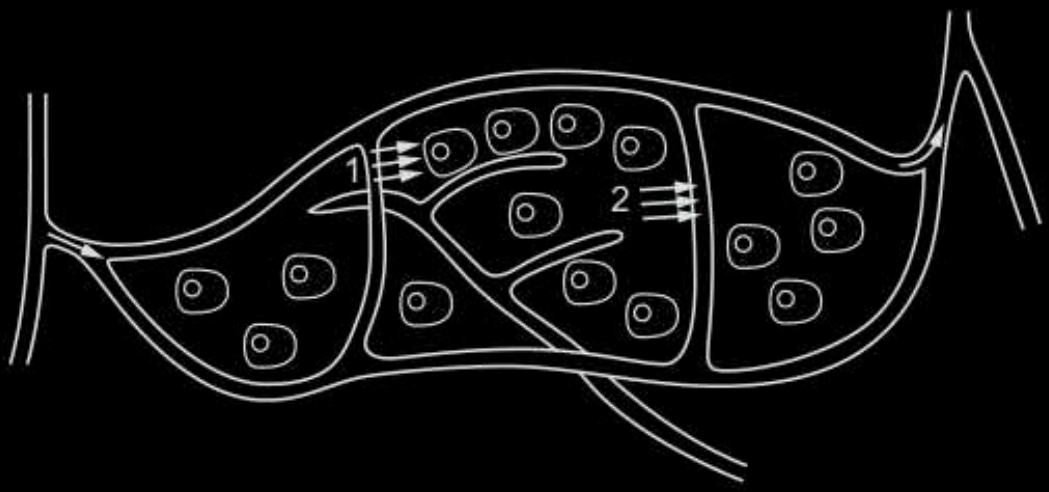
14 What are the effects of the following activities on pulse rate, when compared to resting pulse rate?

	slow walking	fast running
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

October/November 2017 · Paper 1 · Variant 1 · Question 15

Correct answer: C

15 The diagram shows capillaries with the direction of movement of materials.



What is happening at points 1 and 2?

	1	2
A	carbon dioxide leaves the blood	urea enters the blood
B	oxygen diffuses	red blood cells return to the blood
C	tissue fluid leaves the capillaries	waste products enter the blood
D	red blood cells move into the tissues	carbon dioxide diffuses

October/November 2017 · Paper 1 · Variant 1 · Question 17

Correct answer: B

17 Which structure is lined with cilia?

A alveolus

B bronchus

C oesophagus

D small intestine

October/November 2017 · Paper 1 · Variant 1 · Question 19

Correct answer: B

19 Men sometimes develop an enlarged prostate gland.

How might this affect urination?

A by blocking the ureter

B by preventing the exit of urine from the bladder

C by stopping the bladder filling with urine

D by stopping urine production in the kidneys

October/November 2017 · Paper 1 · Variant 1 · Question 20

Correct answer: C

20 What helps heat retention in the human body?

A actively secreting sweat glands

B dilated blood vessels near skin surface

C fat in and under the skin

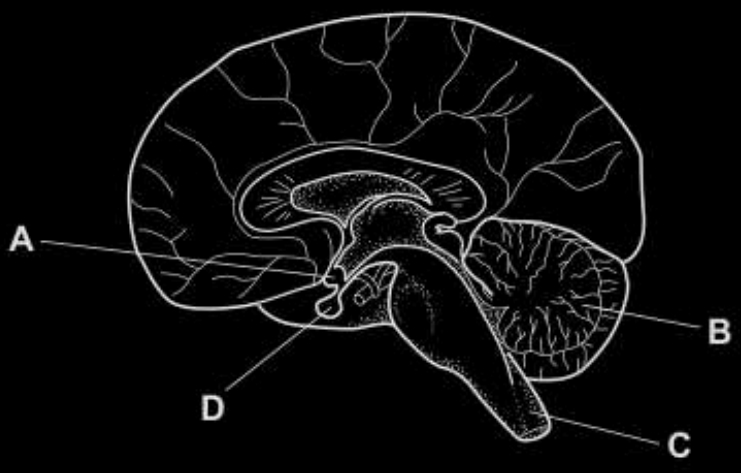
D relaxed hair erector muscles

October/November 2017 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 The diagram shows the brain in vertical section.

Which part is the hypothalamus?



October/November 2017 · Paper 1 · Variant 1 · Question 22

Correct answer: C

22 Which statements describe the pupil reflex in bright light?

1 ciliary muscles contract
2 ciliary muscles relax
3 circular muscles contract
4 circular muscles relax
5 lens becomes rounder
6 lens becomes thinner
7 pupil constricts
8 pupil dilates
9 radial muscles contract
10 radial muscles relax

A 1, 6 and 9 **B** 2, 5 and 10 **C** 3, 7 and 10 **D** 4, 8 and 9

October/November 2017 · Paper 1 · Variant 1 · Question 24

Correct answer: A

24 Which structure does **not** have its muscles arranged antagonistically?

A the heart ventricle
B the iris
C the oesophagus walls
D the upper arm

October/November 2017 · Paper 1 · Variant 1 · Question 37

Correct answer: C

37 A gene is a unit of inheritance that controls the production of

A a chromosome.
B an allele.
C a protein.
D DNA.

October/November 2017 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 The gene for insulin production can be removed from human pancreatic cells and added to the genetic material of a harmless bacterium.

What would be the next stage in using this process to treat diabetes?

A Add the altered bacterium to human food.

B Inject the altered bacterium into the blood of a person with diabetes.

C Put the bacterium into a fermenter to multiply rapidly.

D Use the altered bacterium in a nasal spray.

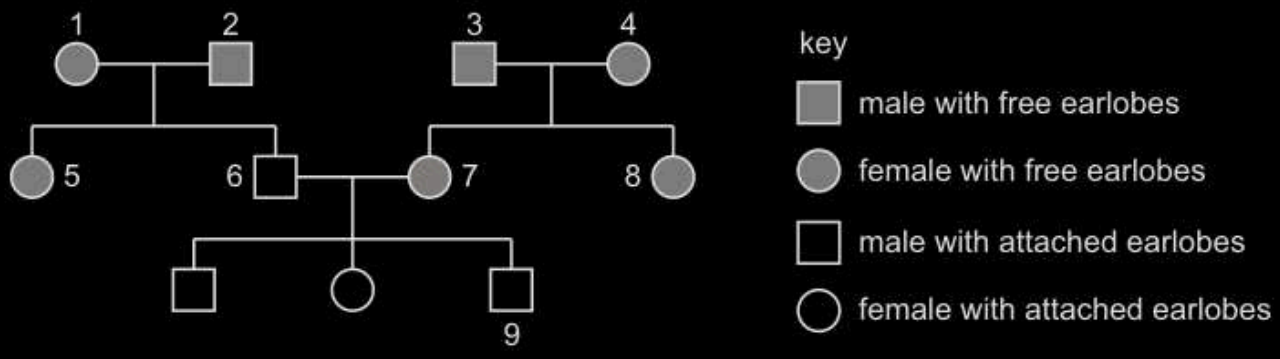
October/November 2017 · Paper 1 · Variant 1 · Question 39

Correct answer: A

39 Earlobes can either be attached to the cheek or 'free' (unattached). This characteristic is controlled by a single gene.

The allele for attached earlobes is recessive.

The diagram shows the inheritance of earlobe attachment in one family.



Which two individuals **must** be heterozygous for earlobe attachment?

A 1 and 7 **B** 3 and 4 **C** 5 and 8 **D** 6 and 9

Correct answer: C

40 The table shows the genotypes and phenotypes for hair colour for the members of a family, but **one** phenotype is shown incorrectly.

	genotype		phenotype
family member	allele 1	allele 2	hair colour
mother	a	A	brown
father	A	A	brown
son 1	a	A	blonde
daughter 1	a	a	blonde
son 2	A	A	brown
daughter 2	A	a	brown

Which family member has the **incorrect** phenotype?

A daughter 1

B daughter 2

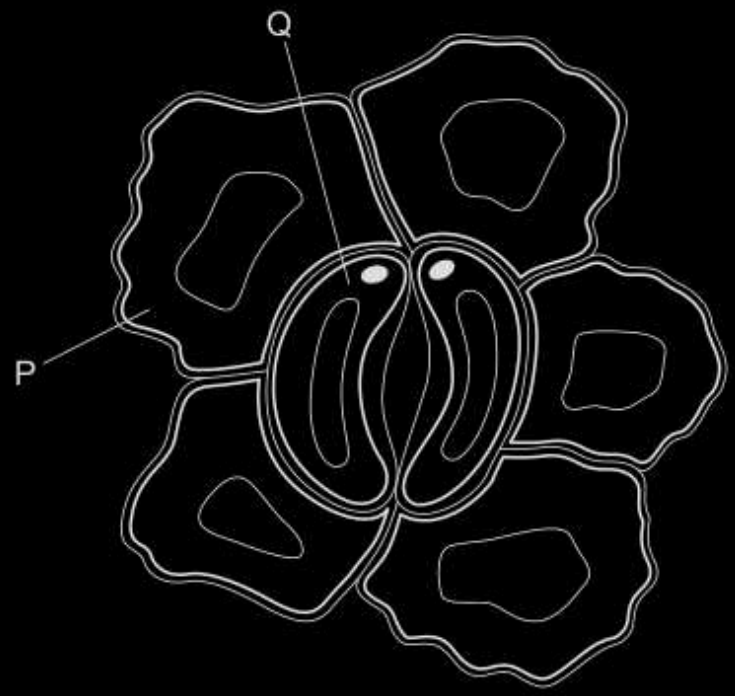
C son 1

D son 2

October/November 2017 · Paper 1 · Variant 2 · Question 1

Correct answer: C

1 The diagram shows cells in the epidermis of a leaf.



To complete the diagram, which structural features should be added to the cells P and Q?

	P		Q	
	chloroplasts	nucleus	chloroplasts	nucleus
A	✓	✓	✗	✗
B	✓	✗	✓	✓
C	✗	✓	✓	✗
D	✗	✗	✗	✓

key
✓ = yes
✗ = no

October/November 2017 · Paper 1 · Variant 2 · Question 2

Correct answer: B

2 By which process does water vapour pass out of a leaf?

A active transport

B diffusion

C osmosis

D translocation

October/November 2017 · Paper 1 · Variant 2 · Question 3

Correct answer: A

3 A student takes a potato and cuts three pieces from it. Each piece is 5 cm × 0.5 cm × 0.5 cm. He places the three potato pieces into three different sugar solutions.

After two hours, he removes the potato pieces from the sugar solutions and measures their lengths.

The results are shown in the table.

solution	length of potato piece after two hours / cm
X	5.2
Y	4.7
Z	5.3

What can be concluded from these results?

A Solution Y has a lower water potential than the potato.

B Solution Z has the lowest water potential.

C The potato piece in solution X increases in length because it takes up sugar.

D The potato piece in solution Y decreases in length because it loses sugar.

October/November 2017 · Paper 1 · Variant 2 · Question 4

Correct answer: D

4 According to the lock and key hypothesis, what is the lock and what is the key for the enzyme lipase?

	lock	key
A	lipids	fatty acids
B	fatty acids	lipase
C	lipids	lipase
D	lipase	lipids

October/November 2017 · Paper 1 · Variant 2 · Question 7

Correct answer: D

7 What describes the upper cuticle of a leaf?

A a permeable layer allowing water to enter the leaf

B a single layer of cells containing many chloroplasts

C a single layer of transparent cells allowing light to enter the leaf

D a thin non-cellular layer preventing water loss from the leaf

October/November 2017 · Paper 1 · Variant 2 · Question 8

Correct answer: B

8 Blood from the ileum is carried in the hepatic portal vein to the liver.

Why is this an advantage to the body?

A Amino acids can be converted to urea before they enter the general circulation.

B Excess glucose can be converted to glycogen for storage and not excreted.

C It ensures that fat products pass through the liver before they reach the heart.

D Toxic materials can be destroyed before they reach any body cells.

October/November 2017 · Paper 1 · Variant 2 · Question 9

Correct answer: B

9 The table shows the compositions of four foods.

Which food provides the most energy per gram?

	carbohydrate %	fat %	protein %	water %
A	1	16	28	55
B	2	83	2	13
C	5	4	3	88
D	12	20	23	45

October/November 2017 · Paper 1 · Variant 2 · Question 10

Correct answer: A

10 After eating, the pH in the mouth decreases.

Which statement explains this decrease?

A Bacteria release acids.

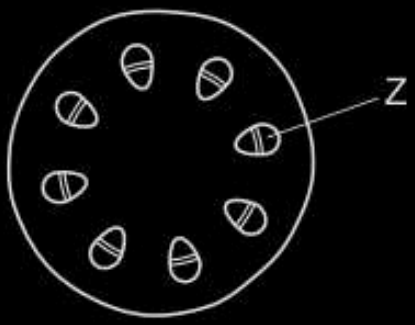
B Enzymes in saliva release acids.

C Salivary glands release acids.

D Sensory neurones in the tongue release acids.

Correct answer: C

12 The diagram shows a section through a plant stem.



Which process is responsible for moving substances in region Z?

A diffusion

B osmosis

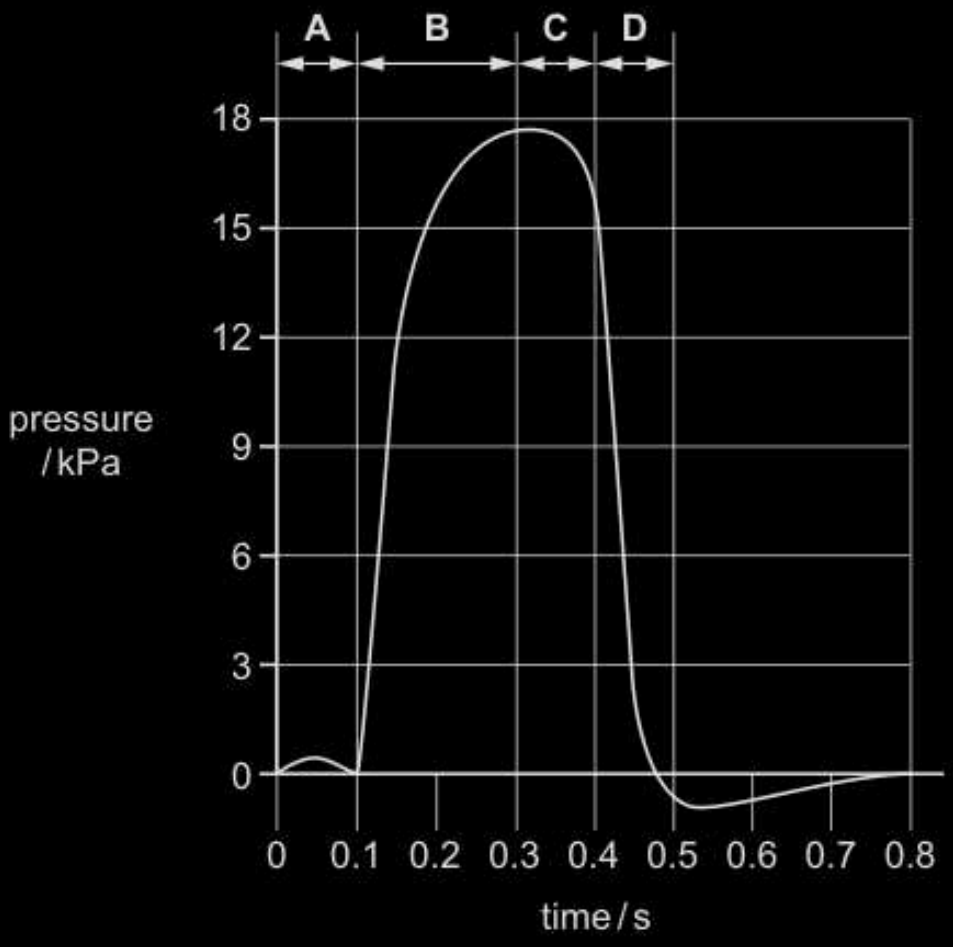
C translocation

D transpiration

Correct answer: A

13 The graph shows changes in the blood pressure in the left ventricle of the heart.

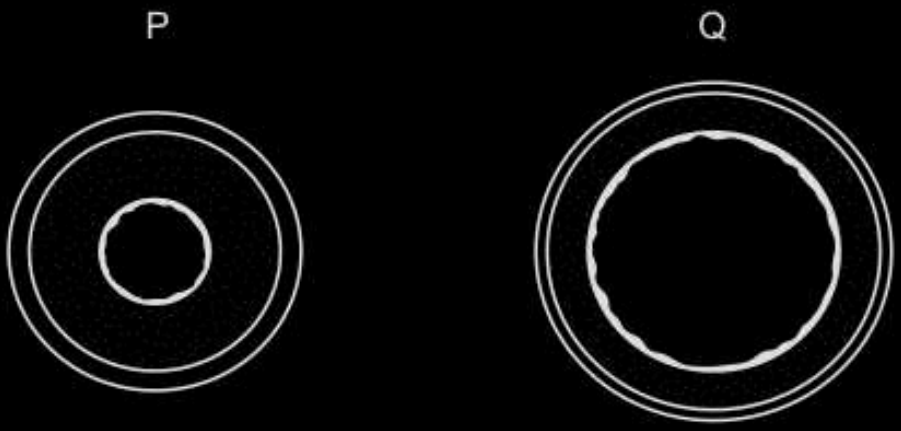
During which period is the left **atrium** contracting?



October/November 2017 · Paper 1 · Variant 2 · Question 14

Correct answer: B

14 The diagram shows cross-sections of two blood vessels, P and Q.



What types of blood vessel are they?

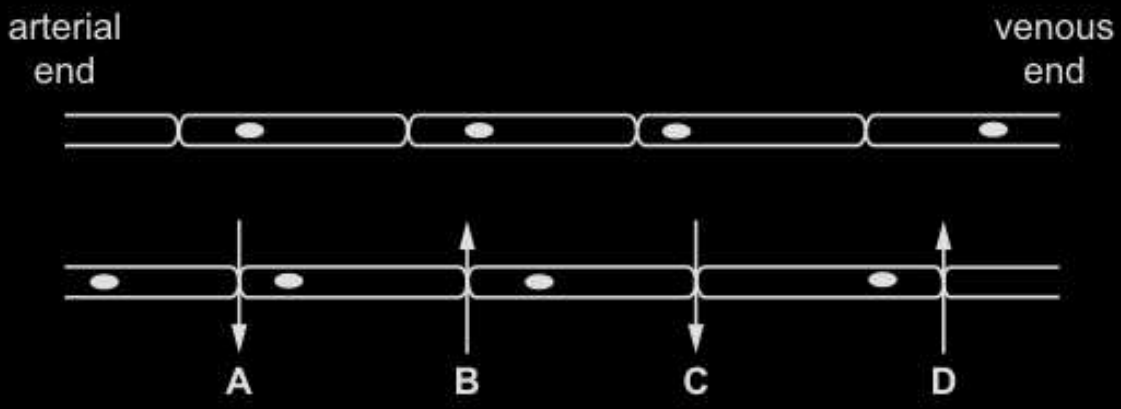
	P	Q
A	artery	capillary
B	artery	vein
C	vein	artery
D	vein	capillary

October/November 2017 · Paper 1 · Variant 2 · Question 15

Correct answer: A

15 The diagram shows a section of capillary.

Which arrow represents tissue fluid formation?



October/November 2017 · Paper 1 · Variant 2 · Question 16

Correct answer: C

16 What is **not** a feature of alveoli?

A a large surface area

B a moist surface

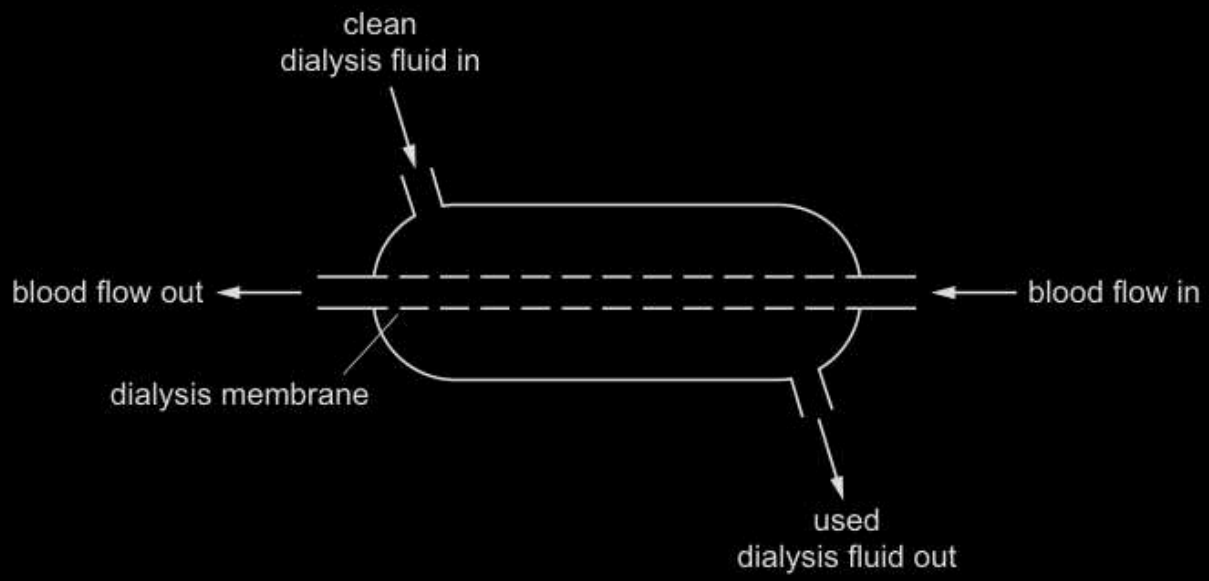
C cilia

D walls one cell thick

October/November 2017 · Paper 1 · Variant 2 · Question 19

Correct answer: B

19 The diagram shows a simplified dialysis machine.



How do the urea concentrations in the blood and dialysis fluid change as they pass through the apparatus?

	urea concentration in blood	urea concentration in dialysis fluid
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

October/November 2017 · Paper 1 · Variant 2 · Question 20

Correct answer: C

20 What helps heat retention in the human body?

A actively secreting sweat glands

B dilated blood vessels near skin surface

C fat in and under the skin

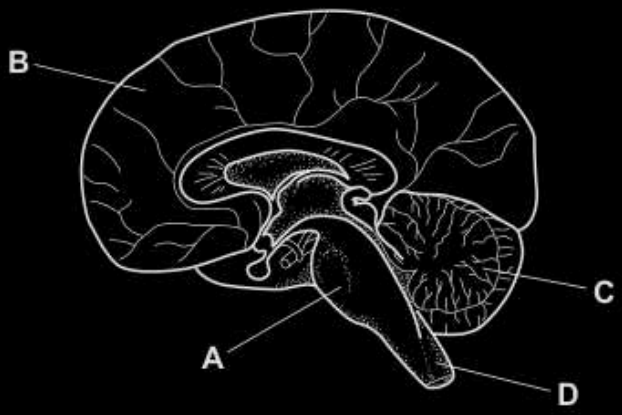
D relaxed hair erector muscles

October/November 2017 · Paper 1 · Variant 2 · Question 21

Correct answer: C

21 The diagram shows a section through the human brain.

Which part is the cerebellum?



October/November 2017 · Paper 1 · Variant 2 · Question 22

Correct answer: B

22 A person is reading a book in dim light.

Which row shows the state of the eye muscles?

	ciliary muscles	iris circular muscles	iris radial muscles
A	contracted	contracted	relaxed
B	contracted	relaxed	contracted
C	relaxed	contracted	relaxed
D	relaxed	relaxed	contracted

October/November 2017 · Paper 1 · Variant 2 · Question 23

Correct answer: C

23 Which chemical produced by the body alters the activity of a target organ and is destroyed by the liver?

A bile

B enzyme

C hormone

D saliva

October/November 2017 · Paper 1 · Variant 2 · Question 25

Correct answer: B

25 Which of these structures would be the first to receive nicotine absorbed into the blood from tobacco smoke?

A brain

B left atrium

C liver

D right atrium

October/November 2017 · Paper 1 · Variant 2 · Question 26

Correct answer: A

26 Some characteristics of microorganisms are listed.

1 can be killed by antibiotics
2 cell wall made of chitin
3 genetic material is DNA or RNA
4 have hyphae
5 make spores
6 may have flagella
7 have a protein coat
8 release enzymes to digest food
9 reproduce inside a host cell

Which row matches the features of bacteria, fungi and viruses?

	bacteria	fungi	viruses
A	1 and 6	4 and 5	7 and 9
B	2 and 4	3 and 8	1 and 5
C	6 and 9	1 and 7	2 and 3
D	7 and 8	2 and 3	6 and 9

October/November 2017 · Paper 1 · Variant 2 · Question 33

Correct answer: B

33 Which statements about meiosis are correct?

	meiosis produces genetically identical nuclei	meiosis produces haploid nuclei
A	false	false
B	false	true
C	true	false
D	true	true

October/November 2017 · Paper 1 · Variant 2 · Question 37

Correct answer: C

37 A gene is a unit of inheritance that controls the production of

A a chromosome.

B an allele.

C a protein.

D DNA.

October/November 2017 · Paper 1 · Variant 2 · Question 38

Correct answer: C

38 Two brothers had different blood groups. One was blood group A and the other was group B.

What can be concluded about their parents' blood group genotypes?

A Both of the parents must be heterozygous.

B Both of the parents must be homozygous.

C One parent (at least) must be heterozygous.

D One parent (at least) must be homozygous.

Correct answer: C

40 The table shows the genotypes and phenotypes for hair colour for the members of a family, but **one** phenotype is shown incorrectly.

	genotype		phenotype
family member	allele 1	allele 2	hair colour
mother	a	A	brown
father	A	A	brown
son 1	a	A	blonde
daughter 1	a	a	blonde
son 2	A	A	brown
daughter 2	A	a	brown

Which family member has the **incorrect** phenotype?

A daughter 1

B daughter 2

C son 1

D son 2

2 Fig. 2.1 shows a section through a human kidney and its associated structures.

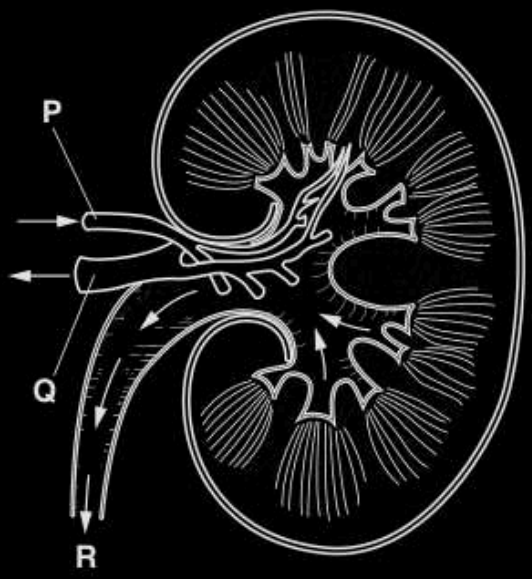


Fig. 2.1

(a) (i) Name the blood vessels **P** and **Q** in Fig. 2.1.

P .....

Q ..... [2]

(ii) Name the fluid **R** in Fig. 2.1 and describe the route taken by this fluid from leaving the kidney to being removed from the body.

name of fluid .....

route taken by fluid .....

.....

.....

.....

[4]

(b) A person with kidney damage requires the process of dialysis to take place several times each week.

Before dialysis begins, a surgical procedure is used to create a connection, called an AV fistula, between two blood vessels in the arm.

Fig. 2.2 shows the AV fistula and the connections between the blood vessels and a kidney machine.

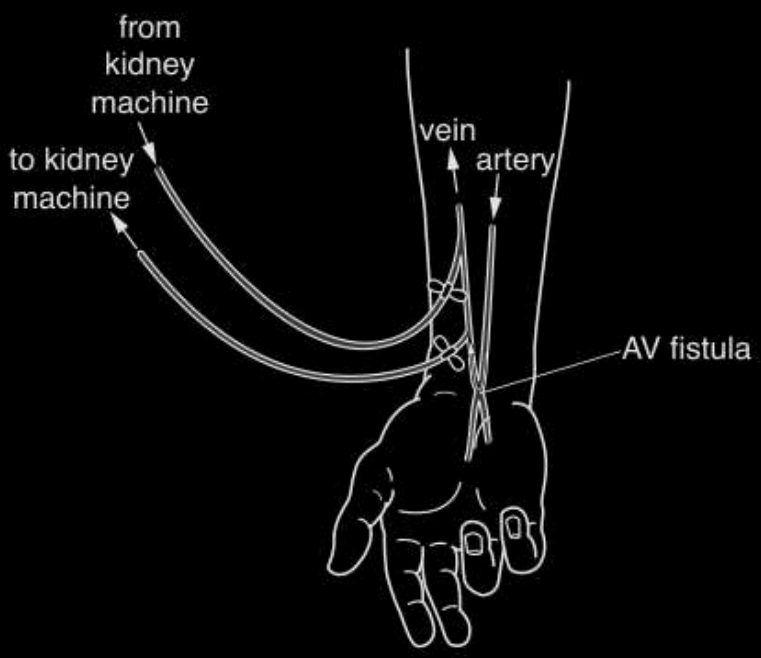


Fig. 2.2

(i) Suggest why the connection made between blood vessels is called an **AV** fistula.

.....[1]

(ii) State **one** way in which blood returning from the kidney machine will differ from blood leaving the body to enter the kidney machine.

.....

.....[1]

(c) Describe the role in dialysis of each of these components of a kidney machine:

artificial membrane

.....

.....

dialysis fluid

.....

.....

.....

[5]

[Total: 13]

2(a)(i)	**P** renal artery ; **Q** renal vein ;	**2**	
2(a)(ii)	urine ; ureter ; bladder ; urethra ;	**4**	**R** urea
2(b)(i)	reference to artery **+** vein ;	**1**	
2(b)(ii)	lower pressure ; less / no urea ; reference to increase / decrease **+** concentration of ions **AW** ;	**1**	
2(c)	(*artificial membrane*) partially permeable **AW** ; small molecules or named **+** pass through (membrane) ; correct reference to diffusion / osmosis ; proteins do not pass through (membrane) ; *(dialysis fluid)* no / low urea **OR** removes urea ; reference to concentration gradient ; same glucose concentration **+** as normal blood ; same ion **AW** concentration **+** as normal blood ;	**5**	

© UCLES 2017

Question	Answer	Marks	Guidance

4 Fig. 4.1 shows a human cell.

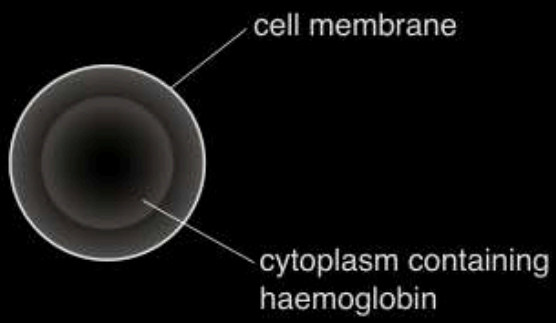


Fig. 4.1

(a) Name and state the main function of the type of cell shown in Fig. 4.1.

name .....

function .....

.....

[2]

(b) Suggest and explain what symptoms might be experienced by a person with an unusually low number of this type of cell.

.....

.....

.....

.....

.....

.....

.....[4]

(c) Explain what would happen to the cell shown in Fig. 4.1 if placed in pure water.

.....

.....

.....

.....

.....[3]

[Total: 9]

4(a)	red blood (cell) / erythrocyte ; oxygen **+** carriage / transport / absorption **AW** ;	**2**	
4(b)	tiredness / fatigue / weak / dizzy / faint ; inability to exercise / exert / inactive ; breathing problems ; reduced oxygen transport ; reduced respiration ; reduced energy (release) ; irregular menstruation **AW** ;	**4**	
4(c)	low <u>water</u> potential / concentration inside cell ; water enters ; by osmosis / diffusion ; cell expands ; cell bursts ; reference to no cell wall ;	**3**	

© UCLES 2017

Question	Answer	Marks	Guidance

6 Fig. 6.1a shows the right eye of a person **before** moving into an area of bright light.

Fig. 6.1a

Fig. 6.1b

(a) (i) Complete Fig. 6.1b to show the appearance of the right eye of the person shortly **after** moving into an area of bright light. [1]

(ii) With reference to **named** structures within the eye, describe the changes that take place when a person moves into an area of bright light.

.....

.....

.....

.....

.....[5]

(b) Name the type of action that occurs to make the changes that you have described and suggest why it is important that these changes take place.

type of action .....

why the changes take place .....

.....

.....

.....

[4]

[Total: 10]

6(a)(i)	shaded circle drawn in iris + smaller than in 6.1(a) ;	1	
6(a)(ii)	iris + muscle ; circular + contract ; radial + relax ; pupil ; constricts **AW** ;	5	
6(b)	*(type of action)* reflex / involuntary ; *(why the changes take place)* protection / prevent damage ; reduces light (entering) ; retina / rods / cones / photoreceptors ;	4	

© UCLES 2017

Question	Answer	Marks	Guidance

9 **(a)** Describe how the nervous system is involved in the maintenance of a constant body temperature.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

(b) Describe the role of the hormone adrenaline and give **one** example of a situation in which adrenaline may be released.

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

9(a)	hypothalamus ; reference to detecting blood temperature ; receptors ; in skin ; detect temperature **+** of surroundings **AW** ; impulse ; sensory neurone ; to central nervous system / brain / spinal cord ; motor neurone ; effector ; named effector ; shiver / vasoconstriction or vasodilation / sweat increase or decrease ; reference to negative feedback ;	**6**	**A** muscle or named effector / sweat gland
9(b)	correct named situation ; target organ / named ; *(any two effects from)* glycogen to glucose / sugar ; increased (heart) rate ; pupils dilate ; dry mouth ; breathing rate change ; diversion of blood to muscle / from gut ;	**4**	increased heart rate = 2 marks (named target organ **+** effect)

© UCLES 2017

1 Fig. 1.1 shows stages in the development of human twins.

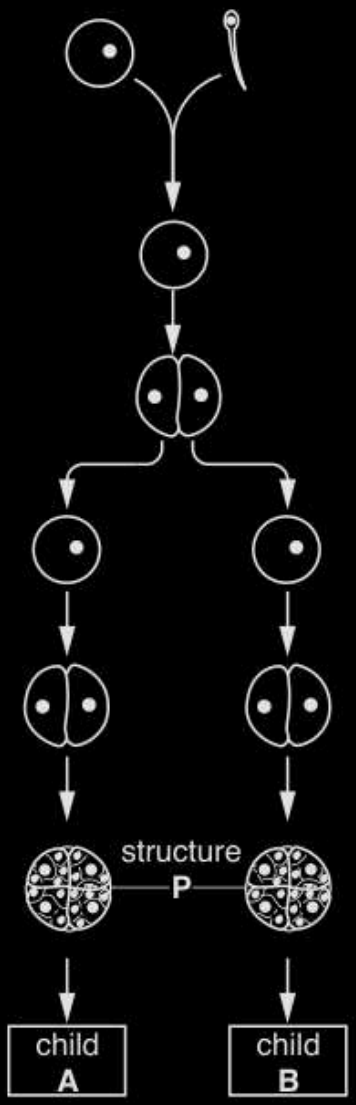


Fig. 1.1

(a) On Fig. 1.1, label and name each of the following:

- a gamete,
- a zygote. [2]

(b) Name the part of the female reproductive system that structure **P** enters.

.....[1]

(c) If the sex chromosome in the sperm is a Y chromosome, and in the ovum (egg) is an X chromosome, state the sex of child **A** and of child **B**. Explain your answer.

child **A** .....

child **B** .....

explanation .....

.....

.....

.....

.....

[4]

(d) Explain how a woman's body prevents further ova (eggs) from being released until the end of her pregnancy.

.....

.....

.....

.....

.....[4]

[Total: 11]

1(a)	gamete / egg / ovum / sperm correctly named and labelled ; zygote correctly named and labelled ;	**2**	
1(b)	uterus / womb / endometrium ;	**1**	
1(c)	*(child **A**)* male ; *(child **B**)* male ; *(explanation)* **1** zygote **+** X **+** Y ; **2** mitosis ; **3** identical **+** cells / twins ;	**4**	**A** boy / son for both
1(d)	**1** yellow body / follicle / corpus luteum / ovary ; **2** placenta ; **3** progesterone ; **4** inhibited / less / no **+** FSH / LH **+** production / release ; **5** FSH / LH **+** pituitary ; **6** FSH **+** ovum / egg **+** maturation / development ; **7** LH **+** ovum / egg **+** release ;	**4**	

© UCLES 2017

Question	Answer	Marks	Guidance

2 Fig. 2.1 shows changes in the body temperature of a person.

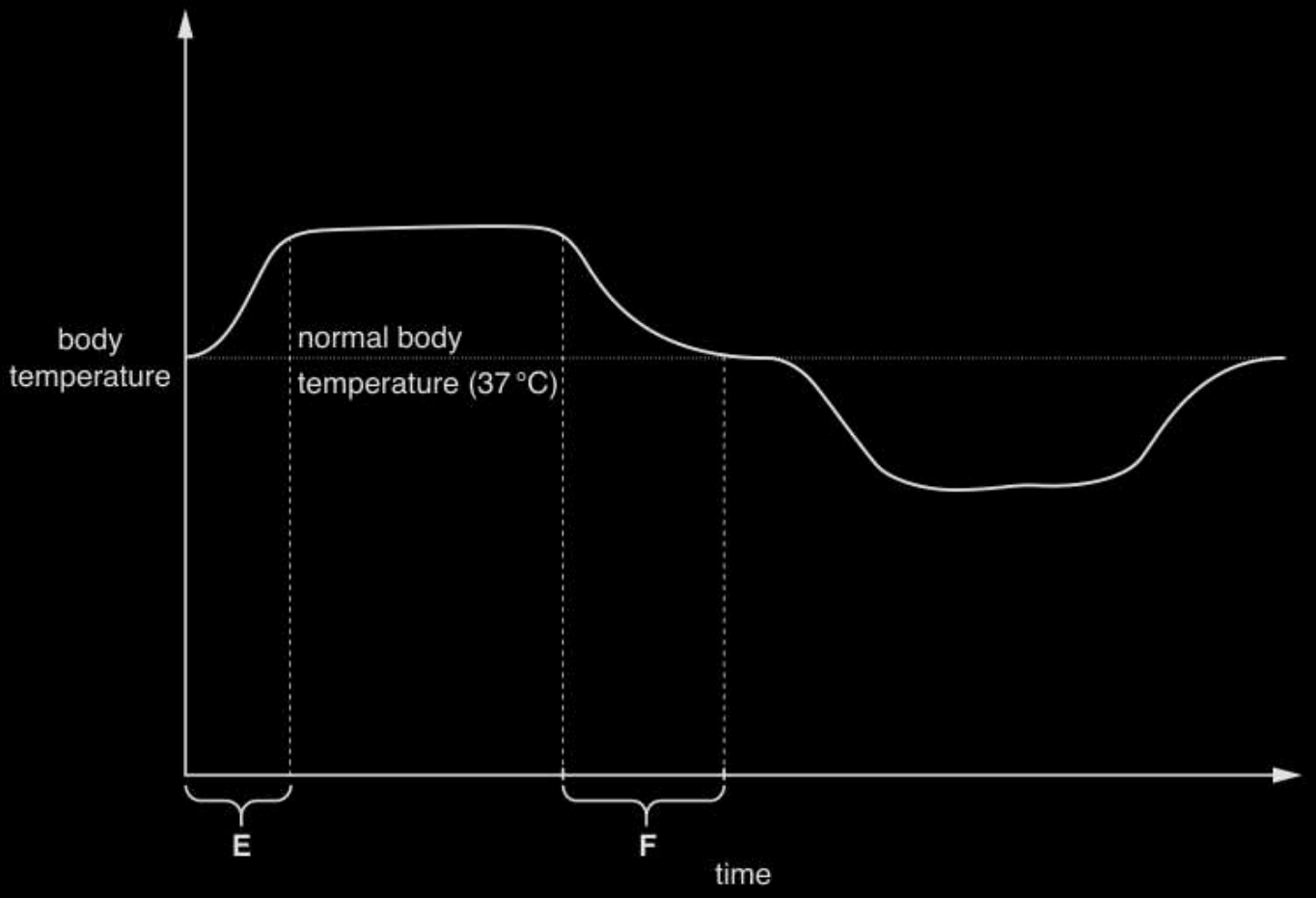


Fig. 2.1

(a) State the term for maintaining constant conditions, such as temperature, in the body.

.....[1]

(b) Suggest **three** things that could happen to account for the shape of the curve during time **E**.

1 .....

2 .....

3 .....

[3]

(c) Explain what is happening in the body to cause the change in body temperature at time **F**.

.....

.....

.....

.....

.....[3]

(d) Name each of the following:

(i) the process by which the return to a set point, as illustrated by Fig. 2.1, is achieved,

.....[1]

(ii) the part of the brain that controls this process.

.....[1]

[Total: 9]

2(a)	homeostasis ;	1	
2(b)	1 body temperature increased ; 2 exercise **AW** ; 3 increased / high **+** surrounding **AW** temperature ; 4 increased / high **+** surrounding **AW** humidity ; 5 hot **+** food / drink ; 6 increased / high **+** metabolism / respiration ; 7 fever / illness ; 8 ovulation ; 9 wear more clothes **AW** ;	3	
2(c)	1 sweat ; 2 evaporation ; 3 vasodilation **AW** ; 4 reduced **+** metabolism / respiration ; 5 more blood **+** to skin ; 6 heat loss / radiation ;	3	
2(d)(i)	negative feedback ;	1	
2(d)(ii)	hypothalamus ;	1	

© UCLES 2017

Question	Answer	Marks	Guidance

3 An experiment was carried out to investigate the inheritance of flower colour.

In cross 1, a plant with blue flowers (plant **G**) was pollinated by another plant with blue flowers (plant **H**). The resulting seeds were collected and labelled 'batch 1'.

In cross 2, plant **G** was pollinated by a plant with white flowers (plant **J**). The resulting seeds were collected and labelled 'batch 2'.

All the plants were of the same species.

Fifty seeds from each of batch 1 and batch 2 were grown and the number of plants with white flowers was counted.

The results are shown in Table 3.1.

Table 3.1

batch	number of plants with white flowers
1 (from cross 1)	12
2 (from cross 2)	27

(a) Name the dominant phenotype in this experiment .....[1]

The alleles controlling flower colour in this plant are **B** (dominant) and **b** (recessive).

(b) **(i)** State the genotypes of

plant **G** .....

plant **H** .....

plant **J** ..... [3]

(ii) State which of these plants is heterozygous .....[1]

Question 3 continues on page 7.

(c) In the box below, draw a genetic diagram for cross 1 (between plants **G** and **H**).

[5]

[Total: 10]

3(a)	blue ;	1	
3(b)(i)	(**G**) Bb / heterozygous ; (**H**) Bb / heterozygous ; (**J**) bb / homozygous **+** recessive ;	3	
3(b)(ii)	**G + H** ;	1	
3(c)	**1** parent genotypes / gametes correct from **G and H** in **(b)(i)** ; **2** offspring genotypes correct for parent genotypes / gametes ; **3** offspring phenotypes correct for stated genotypes ; **4** genotype / phenotype ratio correct ; **5** correct use of the terms gamete **+** genotype / phenotype ;	5	

© UCLES 2017

Question	Answer	Marks	Guidance

5 Muscles are arranged around joints in pairs. One muscle contracts to bend a limb at a joint, and the other contracts to straighten it.

(a) (i) State the term for muscles that act in this way .....[1]

(ii) Name the muscle in your arm that contracts to move your hand away from your nose after smelling a flower.

.....[1]

(b) Fig. 5.1 shows how muscles are arranged in the human leg and pelvis, and also shows the leg in two different positions, **R** and **S**.

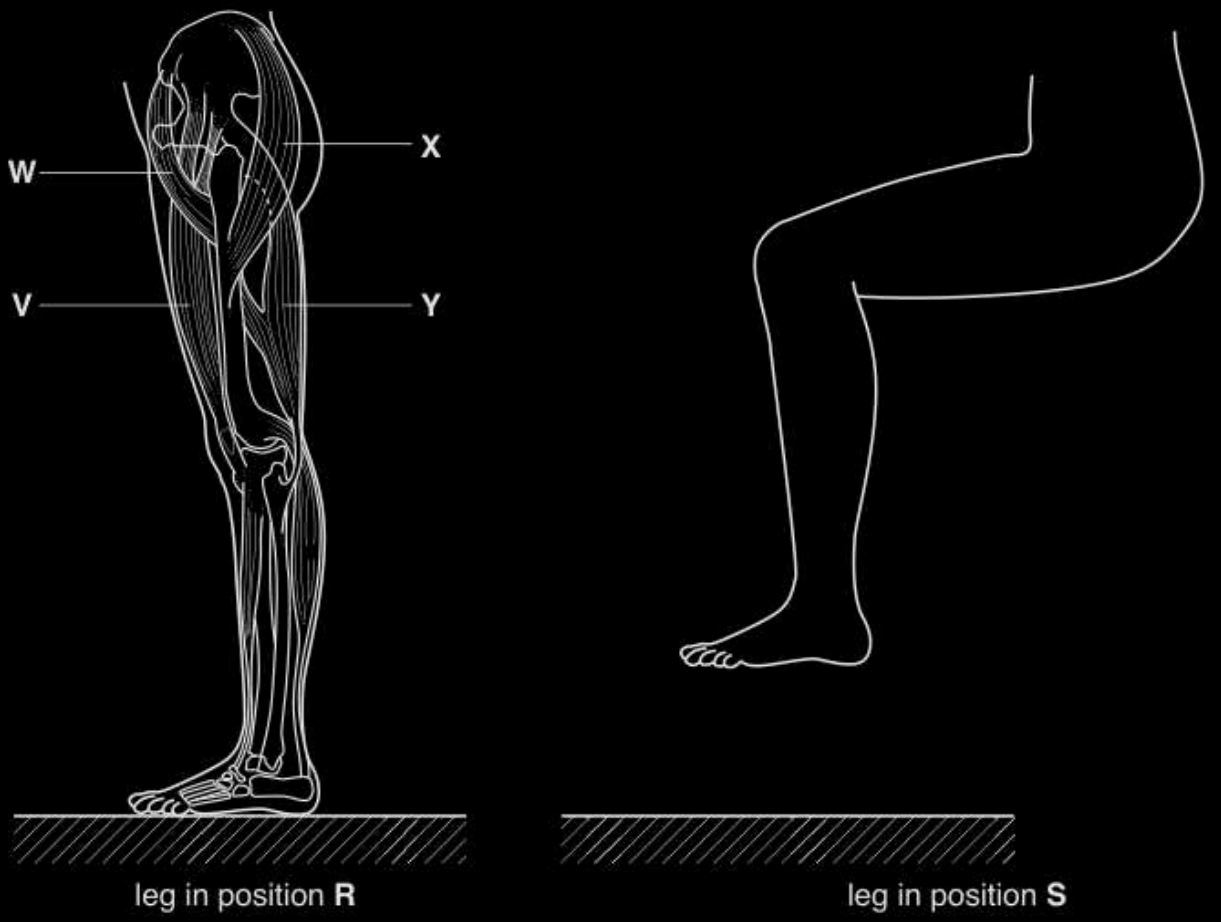


Fig. 5.1

(i) Using the letters in Fig. 5.1, identify the two muscles that contract to move the leg off the ground from position **R** to position **S**.

..... and ..... [2]

(ii) State what happens to the other muscles in Fig. 5.1 during this action.

.....[1]

(c) Describe how a similar arrangement of muscles in the eye helps vision in dim light.

.....

.....

.....

.....

.....

.....[3]

[Total: 8]

Section B

Answer **both** questions in this section.

5(a)(i)	antagonistic ;	**1**	
5(a)(ii)	triceps ;	**1**	
5(b)(i)	W ; Y ;	**2**	
5(b)(ii)	relax ;	**1**	
5(c)	**1** (muscles) in iris ; **2** radial **+** contract ; **3** circular **+** relax ; **4** pupil **+** wider / dilates / expands **AW** ; **5** more light ; **6** retina / rods / cones / photoreceptors **AW** ;	**3**	

© UCLES 2017

Question	Answer	Marks	Guidance

7 **(a)** Describe the causes and symptoms of each of the following:

(i) rickets,

.....

.....

.....

.....

.....

.....[4]

(ii) scurvy.

.....

.....

.....

.....

.....

.....[4]

(b) Suggest why the alimentary canal does not produce any enzymes to work on vitamins.

.....

.....

.....[2]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

7(a)(i)	*(causes)* **1** children ; **2** lack / deficiency **AW +** calcium ; **3** lack / deficiency **AW +** vitamin D ; **4** lack of any reasonable named food containing calcium / vitamin D ; **5** lack of sunlight ; *(symptoms)* **6** bones **+** soft / weak / bent / brittle / likely to fracture / do not develop properly ; **7** teeth **+** soft / decay / do not develop properly **AW** ; **8** bowed legs ;	**4**	
7(a)(ii)	*(causes)* **1** lack / deficiency **AW +** vitamin C ; **2** lack of any reasonable named food containing vitamin C ; *(symptoms)* **3** gums bleed ; **4** teeth loose / fall out ; **5** wound healing slow / poor ; **6** joint pain ; **7** shortness of breath ; **8** rough **AW** skin / weak connective tissue / damaged **AW** epithelium ;	**4**	
7(b)	**1** small / soluble / dissolved ; **2** not digested / not broken down / already in simplest form ; **3** diffusion ; **4** through **+** villi / capillary / cell membrane / partially **AW** permeable membrane ;	**2**	

© UCLES 2017

Question	Answer	Marks	Guidance

9 **(a)** Describe and explain the features of a gas exchange surface.

.....

.....

.....

.....

.....

.....[4]

(b) Explain the effect of exercise on the breathing rate of a person.

.....

.....

.....

.....

.....

.....[4]

(c) Living at high altitude increases the number of red blood cells in a person's blood.

Suggest why athletes sometimes train at high altitude.

.....

.....

.....

.....[2]

[Total: 10]

9(a)	1 2 3 4 5 6	thin / one cell thick / short distance ; moist ; large surface area ; permeable ; maximum / more / quick / efficient **+** diffusion / absorption / exchange ; reference to dissolving / in solution ;	4	
9(b)	1 2 3 4 5 6 7 8 9	faster / deeper **+** breathing ; more **+** oxygen ; (oxygen) in lungs / inhaled / in blood ; muscle ; faster / increased **+** respiration / oxidation of glucose ; aerobic ; more energy **+** required / released ; more **+** carbon dioxide removed **AW** ; delays / prevents **AW +** anaerobic respiration ;	4	
9(c)	1 2 3 4	less oxygen in atmosphere (at high altitude) ; more **+** haemoglobin ; oxyhaemoglobin **OR** more **+** oxygen carried / supplied ; competitive advantage **AW** ;	2	

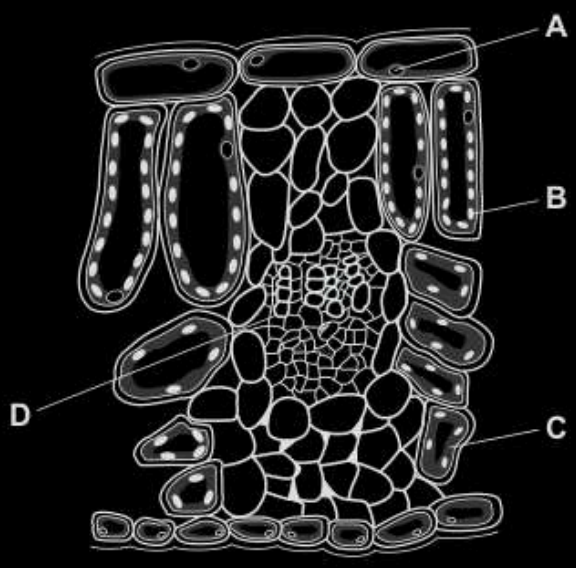
© UCLES 2017

May/June 2017 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 The diagram shows cells from a plant leaf.

Which structure contains a high concentration of magnesium?



May/June 2017 · Paper 1 · Variant 1 · Question 2

Correct answer: C

2 Which of these processes will always occur down a concentration gradient?

	active transport	diffusion	osmosis
A	✓	✓	✓
B	✓	✓	✗
C	✗	✓	✓
D	✗	✗	✗

key
✓ = yes
✗ = no

May/June 2017 · Paper 1 · Variant 1 · Question 3

Correct answer: B

3 A student cuts out four cylinders of potato. Each cylinder is 30 mm long. The cylinders are all of the same diameter.

The potato cylinders are placed in sugar solutions of different concentrations. After one hour, the lengths of the cylinders are measured again. The results are shown in the table.

Which sugar solution has a water potential closest to that of the potato cells?

	concentration of sugar / mol per dm^3	starting length / mm	length after one hour / mm
A	0.1	30	33
B	0.3	30	31
C	0.4	30	27
D	0.5	30	26

May/June 2017 · Paper 1 · Variant 1 · Question 4

Correct answer: A

4 Which are properties of enzymes?

	reusable many times	specific in their action	unaffected by temperature
A	✓	✓	✗
B	✗	✗	✗
C	✓	✗	✓
D	✗	✓	✓

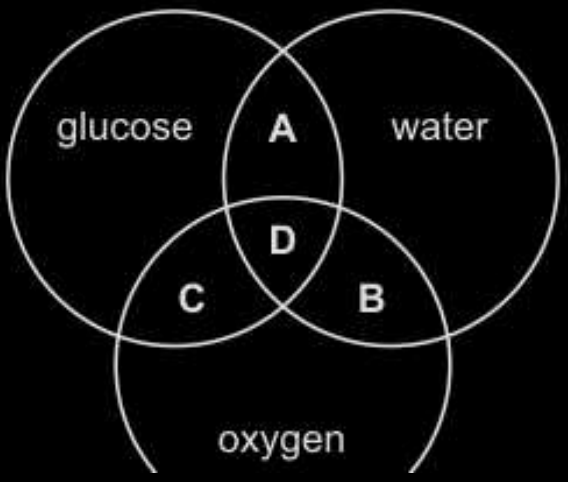
key
✓ = yes
✗ = no

May/June 2017 · Paper 1 · Variant 1 · Question 5

Correct answer: C

5 The diagram refers to some substances found in plant cells.

Which area of the diagram represents the end products of photosynthesis?



May/June 2017 · Paper 1 · Variant 1 · Question 9

Correct answer: C

9 Which is **not** a function of the liver?

A conversion of glucose to glycogen

B storage of glycogen

C secretion of insulin

D synthesis of proteins from amino acids

May/June 2017 · Paper 1 · Variant 1 · Question 10

Correct answer: C

10 A person has a blockage in their bile duct.

What will be a result of this?

A increased concentration of glucose in the blood

B increased concentration of urea in the blood

C slower digestion of fats

D slower digestion of starch

May/June 2017 · Paper 1 · Variant 1 · Question 11

Correct answer: A

11 Water and ions can reach the xylem of a plant root through cell walls, without passing through a cell membrane.

How do these substances move through the cell walls?

	water	ions
A	diffusion	diffusion
B	diffusion	osmosis
C	osmosis	diffusion
D	osmosis	osmosis

May/June 2017 · Paper 1 · Variant 1 · Question 13

Correct answer: B

13 Which chamber of the heart has the thickest walls?

A left atrium

B left ventricle

C right atrium

D right ventricle

May/June 2017 · Paper 1 · Variant 1 · Question 14

Correct answer: D

14 Which row shows the blood pressures in an artery, a capillary and a vein?

	pressure / kPa		
	artery	capillary	vein
A	0.6	4.0	13.0
B	4.0	0.6	13.0
C	13.0	0.6	4.0
D	13.0	4.0	0.6

May/June 2017 · Paper 1 · Variant 1 · Question 15

Correct answer: C

15 What is a difference between plasma and tissue fluid?

	plasma	tissue fluid
A	dissolved glucose	no dissolved glucose
B	less dissolved glucose	more dissolved glucose
C	more protein molecules	fewer protein molecules
D	no white blood cells	white blood cells

May/June 2017 · Paper 1 · Variant 1 · Question 20

Correct answer: A

20 Which parts of the skin are involved in the control of body temperature?

	sweat glands	temperature receptors	blood vessels
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

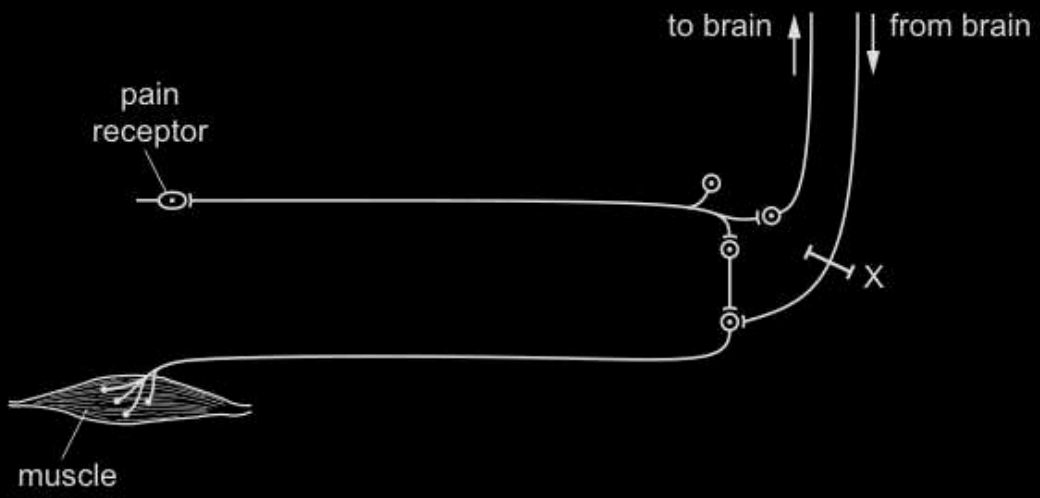
key

✓ = yes

✗ = no

Correct answer: B

23 The diagram shows some of the nerve pathways associated with a reflex action.



If the pathway at X is damaged, how does this affect the reflex?

A The person will not be aware that the reflex is occurring.

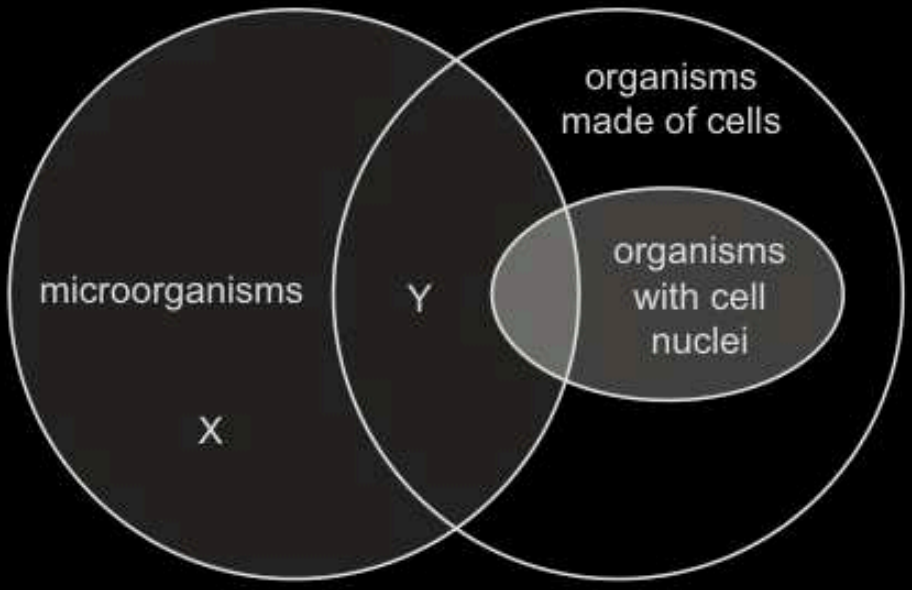
B The reflex cannot be controlled consciously.

C The response will occur without any stimulus.

D There is no response to the stimulus.

Correct answer: D

26 The diagram shows some characteristics of different organisms.



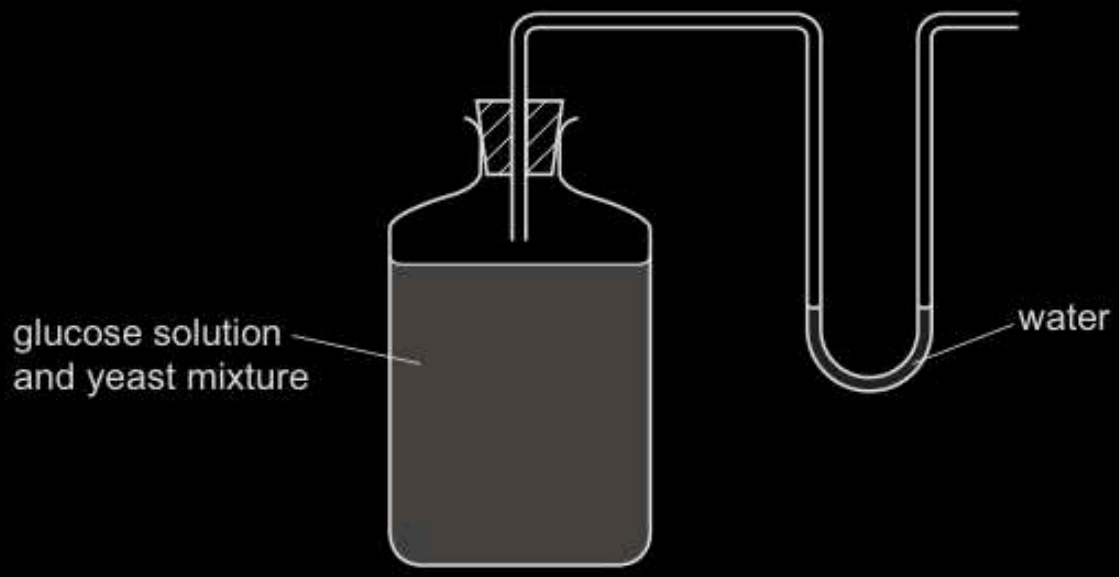
What do areas X and Y represent?

	X	Y
A	bacteria	fungi
B	bacteria	viruses
C	fungi	viruses
D	viruses	bacteria

May/June 2017 · Paper 1 · Variant 1 · Question 27

Correct answer: D

27 The diagram shows a simple apparatus that could be used to produce alcohol.



What is the role of the U-shaped tube containing water?

A to keep bacteria in and oxygen in

B to keep bacteria in and oxygen out

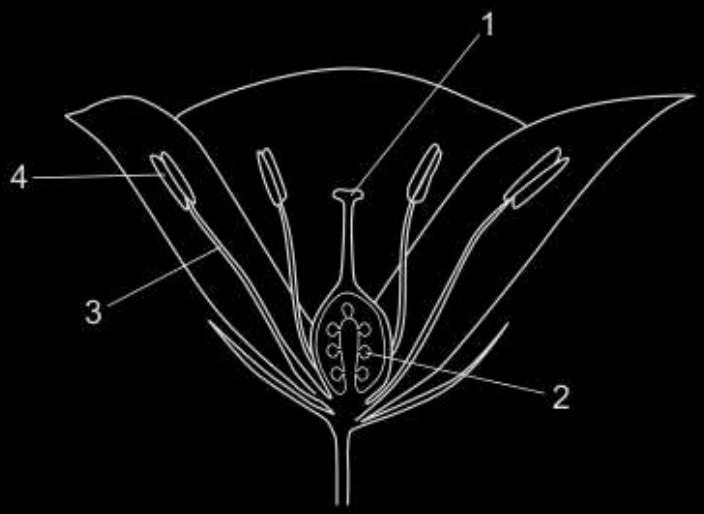
C to keep bacteria out and oxygen in

D to keep bacteria out and oxygen out

May/June 2017 · Paper 1 · Variant 1 · Question 33

Correct answer: C

33 The diagram shows a flower cut in half.



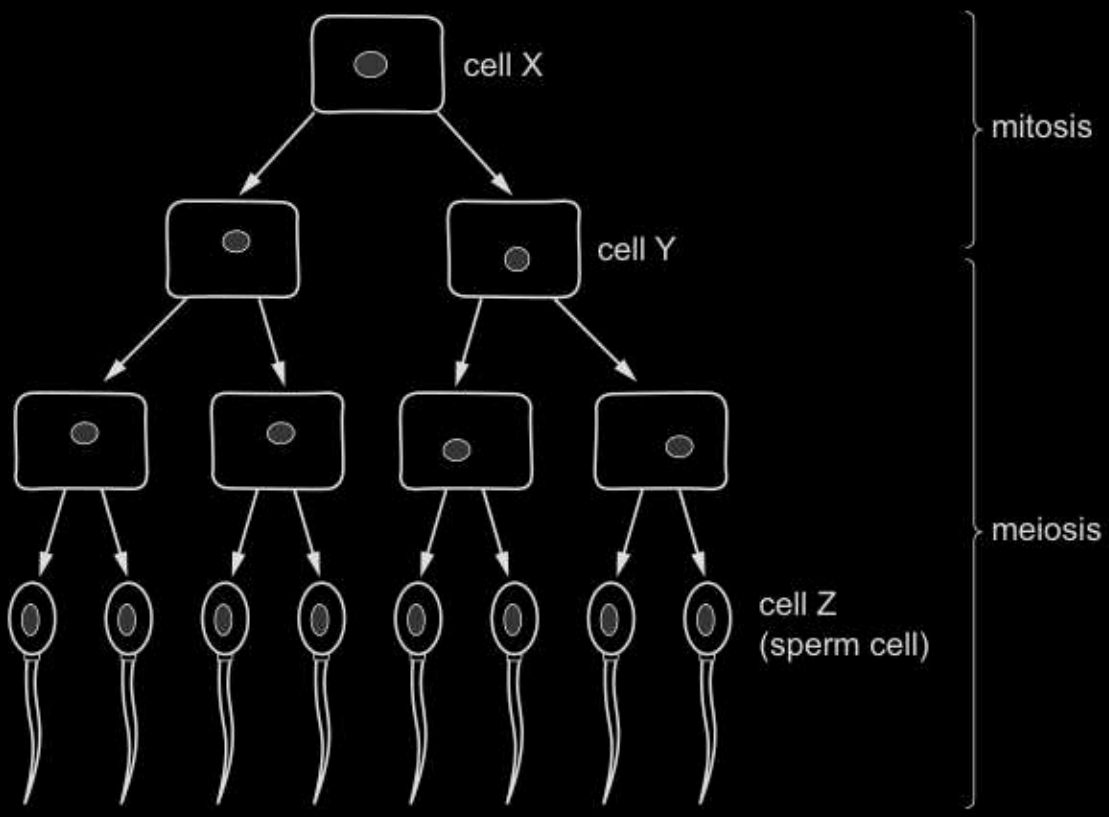
Which **two** parts of the flower produce haploid gametes?

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

May/June 2017 · Paper 1 · Variant 1 · Question 34

Correct answer: C

34 The diagram shows some stages in cell division in a fruit fly.



Cell X contains 8 chromosomes.

How many chromosomes are in cell Y and in cell Z?

	cell Y	cell Z
A	4	4
B	4	8
C	8	4
D	8	8

May/June 2017 · Paper 1 · Variant 1 · Question 35

Correct answer: C

35 What is the path taken by sperm cells during ejaculation from the male reproductive system?

A sperm duct → testis → urethra

B sperm duct → urethra → testis

C testis → sperm duct → urethra

D testis → urethra → sperm duct

May/June 2017 · Paper 1 · Variant 1 · Question 37

Correct answer: A

37 Which statements about genes and chromosomes are correct?

	A chromosome carries a molecule of DNA.	A gene is a section of DNA.
A	true	true
B	true	false
C	false	true
D	false	false

May/June 2017 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 Which feature of bacteria shows discontinuous variation?

A the diameters of their cells

B the masses of their cytoplasm

C the numbers of their flagella

D the thicknesses of their cell walls

May/June 2017 · Paper 1 · Variant 1 · Question 40

Correct answer: C

40 In fruit flies, the allele for an ebony coloured body is recessive to the allele for a grey coloured body. In an investigation, an ebony-bodied fly is crossed with a grey-bodied fly.

What will be the body colour of the offspring if the grey-bodied fly is heterozygous?

A all ebony

B all grey

C half ebony and half grey

D three-quarters grey and one-quarter ebony

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

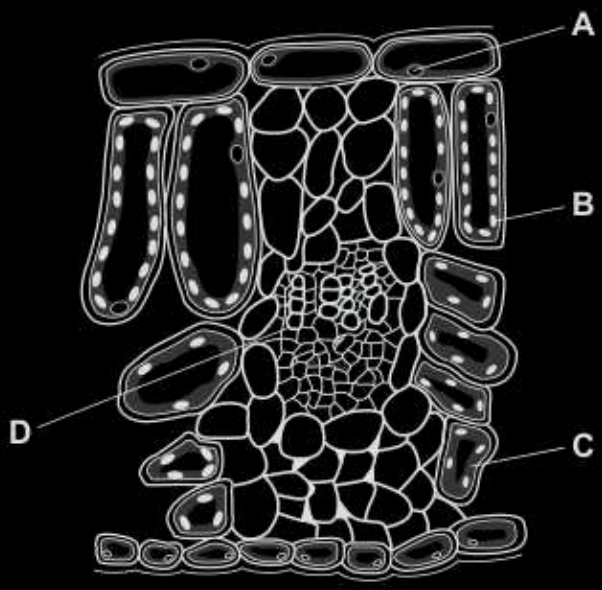
Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

May/June 2017 · Paper 1 · Variant 2 · Question 1

Correct answer: B

1 The diagram shows cells from a plant leaf.

Which structure contains a high concentration of magnesium?



May/June 2017 · Paper 1 · Variant 2 · Question 2

Correct answer: A

2 Which statements about diffusion are correct?

1 Molecules move at random.

2 Molecules move down a concentration gradient.

3 Molecules may move through a partially permeable membrane.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

May/June 2017 · Paper 1 · Variant 2 · Question 3

Correct answer: D

3 Which row defines active transport?

	cell membrane needed	movement of ions	respiration needed
A	no	down a concentration gradient	yes
B	no	up a concentration gradient	no
C	yes	down a concentration gradient	no
D	yes	up a concentration gradient	yes

May/June 2017 · Paper 1 · Variant 2 · Question 4

Correct answer: D

4 Which property of enzymes is explained by the lock and key hypothesis?

A All enzymes are proteins.

B Enzymes are inactive at very low temperatures.

C Human enzymes are most active just below 40 °C.

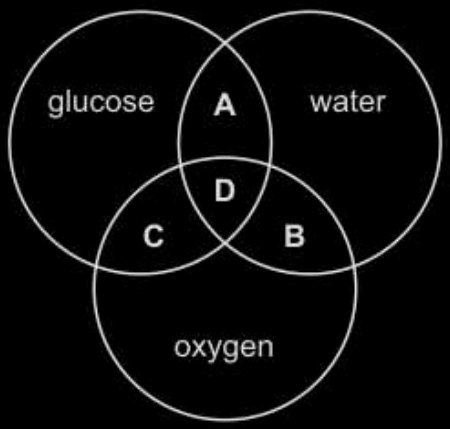
D Most enzymes can only catalyse one reaction.

May/June 2017 · Paper 1 · Variant 2 · Question 5

Correct answer: C

5 The diagram refers to some substances found in plant cells.

Which area of the diagram represents the end products of photosynthesis?



May/June 2017 · Paper 1 · Variant 2 · Question 8

Correct answer: D

8 Which symptom of malnutrition can be treated by an increased amount of protein in the diet?

A constipation

B heart disease

C obesity

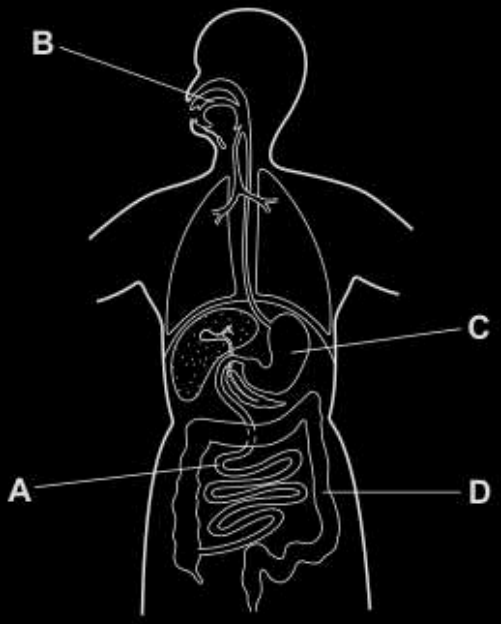
D stunted growth

May/June 2017 · Paper 1 · Variant 2 · Question 9

Correct answer: C

9 The diagram shows the human alimentary canal.

In which part does protein digestion begin?



May/June 2017 · Paper 1 · Variant 2 · Question 10

Correct answer: C

10 In which blood vessel does the concentration of glucose vary the most?

A aorta

B hepatic artery

C hepatic portal vein

D vena cava

May/June 2017 · Paper 1 · Variant 2 · Question 11

Correct answer: D

11 Which feature of root hairs suggests that they take up ions from the soil by active transport?

A Their cell membranes are partially permeable.

B They have a large surface area.

C They have a lower water potential than the soil.

D They take up ions more slowly in low oxygen concentrations.

May/June 2017 · Paper 1 · Variant 2 · Question 13

Correct answer: C

13 Why is a rise in pulse rate important for an athlete during a 200 m race?

A to increase the digestion of carbohydrates

B to increase ventilation of the lungs with fresh air

C to reduce the need for anaerobic respiration

D to reduce the need for sweating to cool the body

May/June 2017 · Paper 1 · Variant 2 · Question 14

Correct answer: B

14 Which row shows what may cause coronary heart disease?

	diet high in saturated fats	diet low in salt	regular exercise
A	✓	✓	✗
B	✓	✗	✗
C	✗	✓	✓
D	✗	✗	✓

key
✓ = yes
✗ = no

May/June 2017 · Paper 1 · Variant 2 · Question 15

Correct answer: C

15 What is a difference between plasma and tissue fluid?

	plasma	tissue fluid
A	dissolved glucose	no dissolved glucose
B	less dissolved glucose	more dissolved glucose
C	more protein molecules	fewer protein molecules
D	no white blood cells	white blood cells

May/June 2017 · Paper 1 · Variant 2 · Question 16

Correct answer: B

16 What lines the walls of alveoli?

A a carpet of cilia

B a film of moisture

C a network of veins

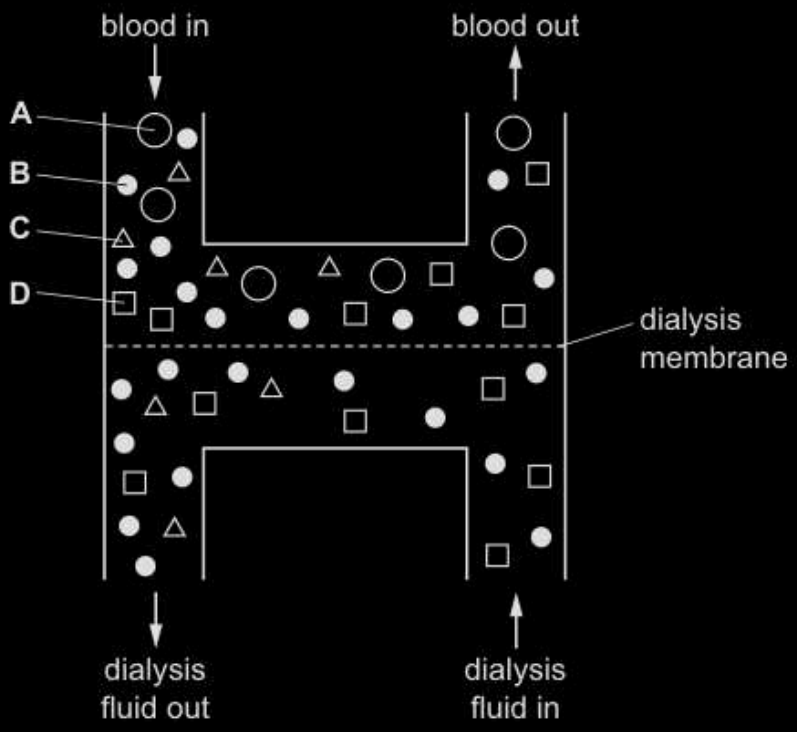
D a thin sheet of muscle

May/June 2017 · Paper 1 · Variant 2 · Question 19

Correct answer: A

19 The diagram represents dialysis in a kidney machine. Each shape represents a molecule found in blood or dialysis fluid.

Which shape represents protein?



May/June 2017 · Paper 1 · Variant 2 · Question 20

Correct answer: A

20 Which parts of the skin are involved in the control of body temperature?

	sweat glands	temperature receptors	blood vessels
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key

✓ = yes

✗ = no

May/June 2017 · Paper 1 · Variant 2 · Question 21

Correct answer: B

21 Which structures does light pass through when it is focused on the retina?

	cornea	lens	sclera
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key

✓ = yes

✗ = no

May/June 2017 · Paper 1 · Variant 2 · Question 22

Correct answer: B

22 Which row shows what causes adrenaline or insulin to be released, and the effect of this hormone on blood glucose concentration?

	hormone	when released into blood	effect on blood glucose concentration
A	adrenaline	after meals	decreases
B	adrenaline	when anxious	increases
C	insulin	after meals	increases
D	insulin	when fasting	decreases

May/June 2017 · Paper 1 · Variant 2 · Question 23

Correct answer: D

23 A man suffers a head injury and his cerebellum is damaged.

What is the result of this?

A inability to recognise visual stimuli

B irregular heartbeat

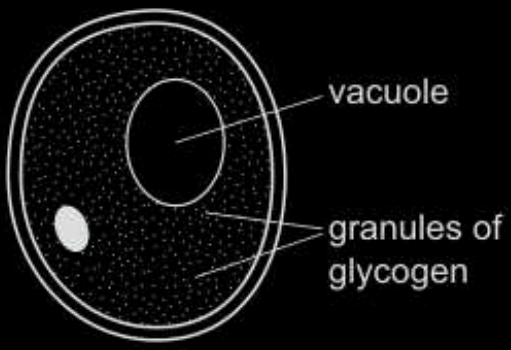
C lack of temperature control

D poor coordination of movements

May/June 2017 · Paper 1 · Variant 2 · Question 27

Correct answer: A

27 The diagram shows a cell and some of its features.



From where has this cell been taken?

A a fermenting alcoholic drink

B pus from a syphilitic sore

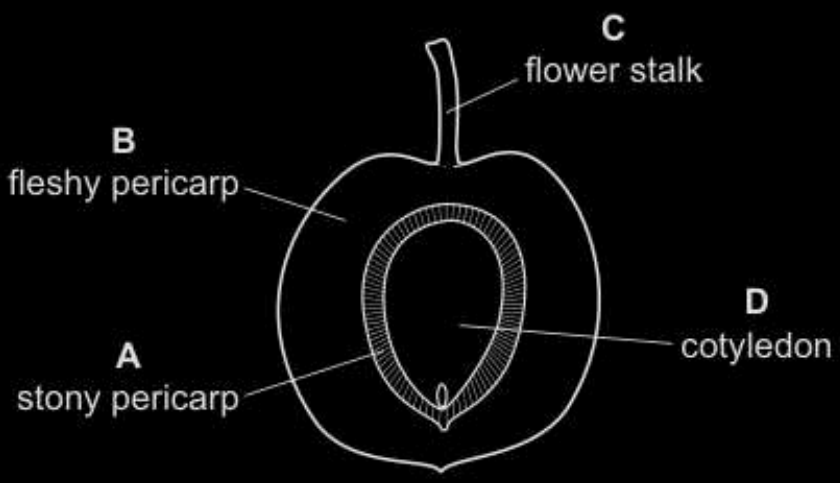
C the liver of an animal

D the mesophyll of a leaf

Correct answer: D

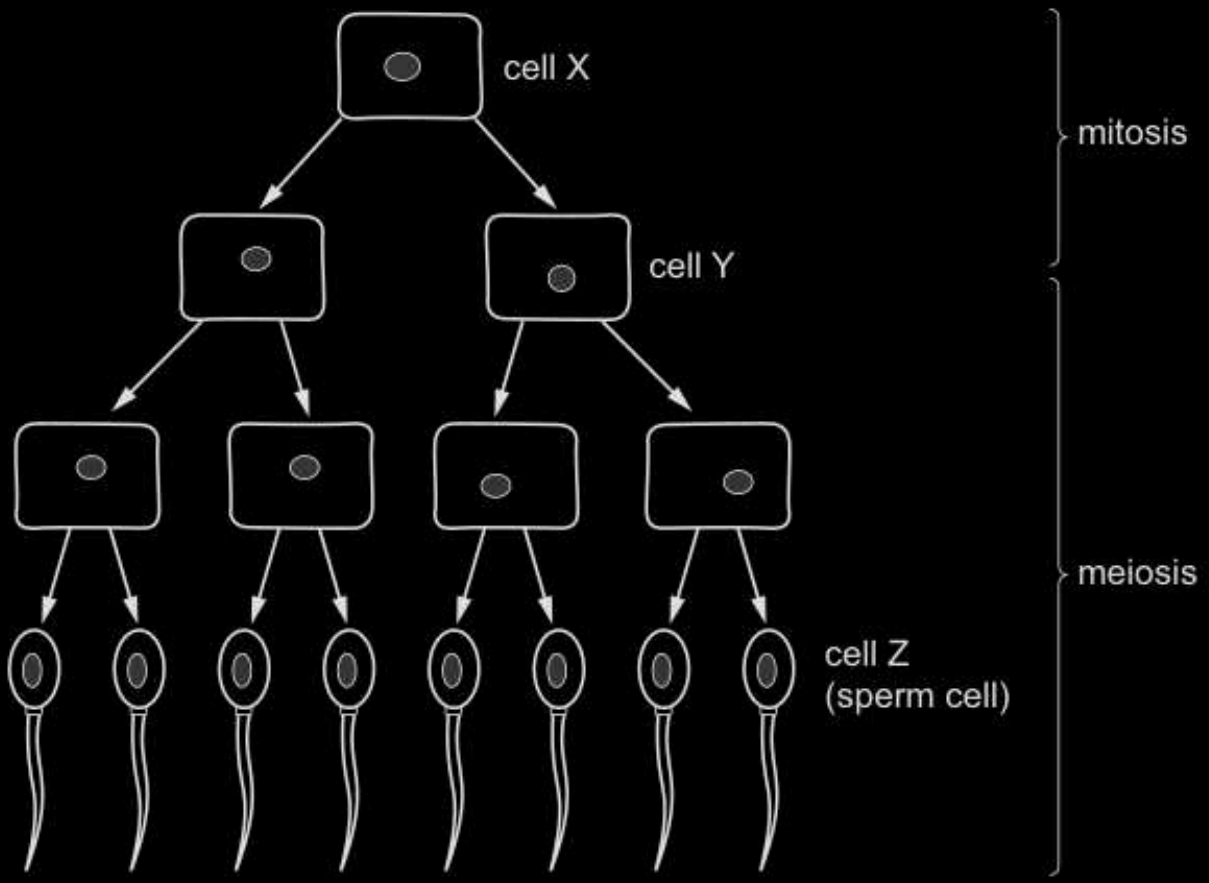
33 The diagram shows a section through a plum fruit.

Which structure has a genotype different to the other three?



Correct answer: C

34 The diagram shows some stages in cell division in a fruit fly.



Cell X contains 8 chromosomes.

How many chromosomes are in cell Y and in cell Z?

	cell Y	cell Z
A	4	4
B	4	8
C	8	4
D	8	8

May/June 2017 · Paper 1 · Variant 2 · Question 35

Correct answer: C

35 What is the path taken by sperm cells during ejaculation from the male reproductive system?

A sperm duct → testis → urethra

B sperm duct → urethra → testis

C testis → sperm duct → urethra

D testis → urethra → sperm duct

May/June 2017 · Paper 1 · Variant 2 · Question 37

Correct answer: A

37 What is essential for natural selection to occur?

	competition	variation
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = yes

✗ = no

May/June 2017 · Paper 1 · Variant 2 · Question 39

Correct answer: B

39 Which statement about chromosomes is correct?

A A chromosome is part of a DNA molecule.

B A chromosome carries a long molecule of DNA.

C Each chromosome controls the inheritance of one body feature.

D Genes are made up of a chain of chromosomes.

May/June 2017 · Paper 1 · Variant 2 · Question 40

Correct answer: C

40 In fruit flies, the allele for an ebony coloured body is recessive to the allele for a grey coloured body. In an investigation, an ebony-bodied fly is crossed with a grey-bodied fly.

What will be the body colour of the offspring if the grey-bodied fly is heterozygous?

A all ebony

B all grey

C half ebony and half grey

D three-quarters grey and one-quarter ebony

1 *Spirulina* is classified in the group bacteria. *Spirulina* is green in colour and is able to synthesise its own food.

(a) List **three** characteristics of bacteria.

1 .....

2 .....

3 ..... [3]

(b) **(i)** Suggest the name of the chemical that gives *Spirulina* its green colour.

..... [1]

(ii) Write, in words or symbols, the equation for the process by which *Spirulina* synthesises its own food.

..... → ..... [2]

(c) *Spirulina* can be eaten by humans.

The United Nations World Health Organization (WHO) made the following statement about *Spirulina*:

'For WHO Spirulina represents an interesting food for multiple reasons. Rich in iron and protein, it can be given to children without any risk. We at WHO consider it to be a very suitable food.'

Use your knowledge of animal nutrition to suggest and explain why WHO considers *Spirulina* to be a 'very suitable food'.

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

1(a)	single-celled ; no nucleus / nucleus not membrane bound ; no organelles ; cell wall ; (cell wall) not cellulose ; cell membrane / cytoplasm ; flagella / flagellum ; DNA circular / loop **OR** plasmid / single chromosome ; smaller than animal / plant cells **OR** 1–2 μm ;	**3**	
1(b)(i)	chlorophyll ;	**1**	
1(b)(ii)	carbon dioxide / $6CO_2$ **+** water / $6H_2O$; glucose / $C_6H_{12}O_6$ **+** oxygen / $6O_2$;	**2**	
1(c)	iron **+** haemoglobin / red blood cells ; prevent anaemia ; protein **+** growth / repair ; protein **+** (production of) enzymes / antibodies ; carbohydrates / glucose / starch / vitamins / ions ; reference to a component of balanced diet / dietary supplement **OR** to alleviate malnutrition / famine ;	**4**	
	Total:	**10**	

© UCLES 2017

Question	Answer	Marks	Guidance

2 A potometer is used to measure water uptake by a plant. Fig. 2.1 shows the stem and flower of a plant in a potometer. As water is taken up, the bubble moves in the direction shown.

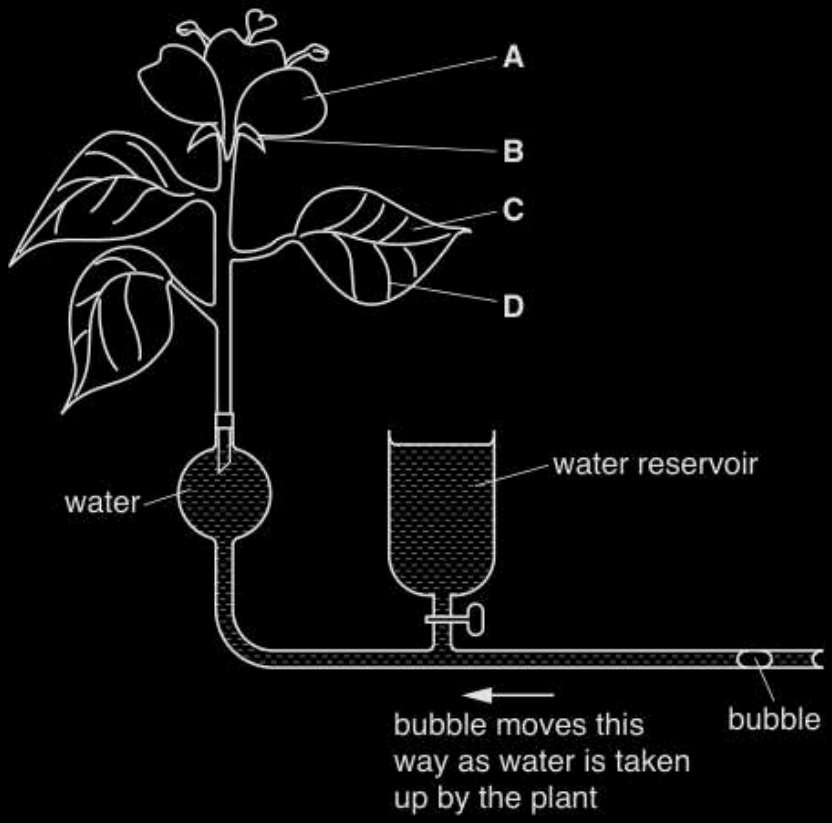


Fig. 2.1

(a) Name the parts **A**, **B**, **C** and **D** in Fig. 2.1.

A .....

B .....

C .....

D ..... [4]

(b) Describe the pathway taken by water as it moves from the potometer, through the plant stem and into the surrounding air.

.....

.....

.....

.....

.....[3]

(c) **(i)** In an experiment, the bubble moved a distance of 60 mm in 10 minutes.
Calculate the average rate at which the bubble moved in mm per min.

..... mm per min
[1]

(ii) The experiment was repeated in an area of lower light intensity.

Predict and explain what would happen to the rate at which the bubble moved.

prediction.....

.....

explanation.....

.....

.....

.....[3]

[Total: 11]

2(a)	(**A**) petal ; (**B**) sepal ; (**C**) leaf / lamina / cuticle ; (**D**) vein / midrib / vascular (tissue) **OR** xylem **+** phloem ;	**4**	
2(b)	xylem ; to leaves / flower(s) ; between leaf cells ; surface of mesophyll cells ; (leaf) air spaces ; (through) stomata / guard cells ;	**3**	
2(c)(i)	6 ;	**1**	
2(c)(ii)	reduces / lower(s) **AW** ; less photosynthesis ; less water needed ; stomata / guard cells **+** open less / are closed ; less diffusion ; less evaporation / transpiration ;	**3**	
	Total:	**11**	

© UCLES 2017

Question	Answer	Marks	Guidance

4 Fig. 4.1 shows a Bengal tiger.

Fig. 4.1

Fur colour in the Bengal tiger is controlled by a single gene. The dominant allele of the gene results in orange fur. A single change in this gene produces a recessive allele, which results in white fur in tigers with the homozygous recessive genotype.

(a) (i) Define the term *gene*.

.....

.....

.....

.....[3]

(ii) State the term used to describe a change in a gene.

..... [1]

(b) Using the letters **T** (orange) and **t** (white) to represent the alleles that control fur colour, draw a labelled genetic diagram to show how two tigers with orange fur may give rise to offspring with white fur.

[5]

(c) Bengal tigers have dark stripes on their fur.

Suggest why each of the following is true for the pattern of dark stripes:

it is **not** affected by whether the fur is orange or white

.....

.....

.....

it is unique to each individual

.....

.....

.....[3]

[Total: 12]

4(a)(i)	section of / made of / piece of **+** DNA / chromosome ; controls production of <u>one</u> protein ; may be copied ; unit of inheritance / passed on to next generation ;	**3**	
4(a)(ii)	<u>mutation</u> ;	**1**	
4(b)	Tt **+** Tt ; t **+** t ; tt ; tt offspring clearly indicated as white ; 3 × labels on genetic diagram correct ;	**5**	**A** parent / offspring / gamete / genotype / phenotype
4(c)	*(unaffected by fur colour)* (controlled by) different gene(s) ; *(unique to each individual)* alleles ; (as a result of) mutation(s) ; meiosis AW ; different combinations (of alleles) ; reference to some may be homozygous / heterozygous ; reference to codominance ;	**1** **2**	
	Total:	**12**	

Question	Answer	Marks	Guidance

5 The statements **E** to **K** relate to the process of reproduction.

E produces genetically identical offspring

F produces more individuals of the same species

G involves only one parent

H involves fusion of nuclei

I requires gametes

J forms a diploid zygote

K involves only cell division by mitosis

Table 5.1 shows a comparison of sexual and asexual reproduction.

Table 5.1

sexual reproduction only	asexual reproduction only	both sexual and asexual reproduction
	E	

Complete Table 5.1 by writing each letter in the correct box to match it to sexual reproduction only, asexual reproduction only, or to both.

The first letter has been written in the correct box for you. Use each letter **once** only. [6]

[Total: 6]

Section B

Answer **both** questions in this section.

5	*(sexual only)* **H ;** **I ;** **J ;**	**3**	**R** letters in more than one box **Ig** letters in incorrect box
	(asexual only) (**E**) **G ;** **K ;**	**2**	
	(both) **F ;**	**1**	
	Total:	**6**	

© UCLES 2017

Question	Answer	Marks	Guidance

6 **(a)** Describe the role of the cilia in the trachea.

.....

.....

.....

.....

.....[3]

(b) Fig. 6.1 shows components of the human gas exchange surface and an associated blood vessel.

Fig. 6.1

State the characteristics, and describe the roles, of each of the components shown in Fig. 6.1. You should make reference to **named** structures in your answer.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[7]

[Total: 10]

6(a)	move mucus ; (containing) bacteria / pathogens / dust ; (moves) up / away from lungs **AW** ; prevent infection ;	**3**	
6(b)	reference to diffusion ; (for) O_2 / CO_2 **+** exchange **AW** ; alveolus / air sac ; large surface area ; one cell thick **+** wall ; moist **AW** / mucus ; (gases) to dissolve ; capillary ; one cell thick **+** wall ; connect **AW** arteries **+** veins ; blood **+** moving ; red blood cells / erythrocytes ; no nucleus ; biconcave ; contain haemoglobin ; live for 90 / 120 days **OR** 3 / 4 months ; carry oxygen ; plasma ; carriage of carbon dioxide ;	**7**	
	Total:	**10**	

Question	Answer	Marks	Guidance

May/June 2017 · Paper 2 · Variant 1 · Question 7

7 Fig. 7.1 shows bacteria growing on the surface of a dish containing nutrient jelly. Paper discs, such as **M** and **N**, were soaked in solutions of different antibiotics and placed on top of the growing bacteria. A clear area on the jelly indicates that bacteria in that area have been killed.

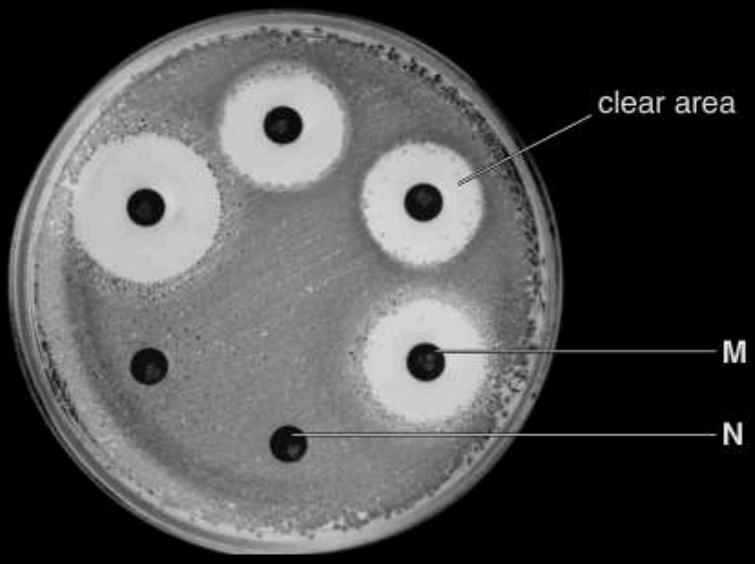


Fig. 7.1

(a) Use the information above, and your knowledge of the process of natural selection, to describe and explain the difference in appearance of the jelly surrounding discs **M** and **N**.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

(b) Describe how the process of artificial selection differs from that of natural selection. Include reference to the production of **one named** economically important plant or animal in your answer.

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

7(a)	**M** has clear area **+ N** no clear area ; *(for disc **M**)* reference to (bacteria) killed around **M** / not killed around **N** ; gene ; mutation ; resistant (to antibiotic) ; (resistant) survive ; (resistant) reproduce ; pass on resistance to next generation / offspring ; *(for disc **N**)* antibiotic (solution) not strong / concentrated enough **OR** incorrect antibiotic (for the bacteria) ;	**6**	 **ORA** for disc **N**
7(b)	named example of (artificially selected) animal / plant ; reason named example is economically important ; reference to human / farmer **AW** ; reference to breed together / cross breed **AW** ; over several generations / repeated ;	**4**	
	Total:	**10**	

Question	Answer	Marks	Guidance

8 **(a)** With reference to **named** components, describe how the structure of one animal cell (for example from fresh liver) would appear different from a plant cell (for example from an onion epidermis).

.....

.....

.....

.....

.....

.....

.....[4]

(b) State the relationship between structure and function for **both** of the following:

xylem vessels

.....

.....

.....

.....

red blood cells

.....

.....

.....

.....[6]

[Total: 10]

8(a)	reference to shape difference **+** plant **and** animal cell named ; *(plant cell)* cell wall ; nucleus pushed to outside / not towards centre ; vacuole ; chloroplast ; starch grains ;	**4**	 **ORA** for animal cell
8(b)	*(xylem vessels)* (S) hollow / dead ; (S) strengthened / lignification ; (S) extend from root to stem / leaves ; (S) narrow / tubes ; (F) conduction / transport ; (F) capillarity ; (F) of water ; (F) of ions ; (F) support ;	**3**	each section must refer to at least **one** structure (S) marking point and **one** function (F) marking point to score maximum 3 marks
	(red blood cells) (S) haemoglobin ; (S) no nucleus ; (S) biconcave ; (S) able to change shape **AW** ; (S) increased / large **+** surface area ; (F) squeeze **AW** through **+** capillaries ; (F) absorption / transport of oxygen ;	**3**	
	Total:	**10**	

© UCLES 2017

Question	Answer	Marks	Guidance

9 Compare each of the following processes:

(a) aerobic respiration and anaerobic respiration

.....

.....

.....

.....

.....[3]

(b) anaerobic respiration in muscles and anaerobic respiration in yeast

.....

.....

.....

.....

.....[3]

(c) diffusion and active transport

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

9(a)	glucose **+** required for both ; complete or incomplete breakdown (of glucose) ; ref. oxygen requirement ; amount of energy released ;	**3**	**A** each point **only** if linked to **either** 'aerobic' **or** 'anaerobic' respiration
9(b)	glucose **+** required for both ; reference to oxygen debt **AW** ; lactic acid ; carbon dioxide ; alcohol / ethanol ;	**3**	**A** each point **only** if linked to **either** 'muscles **or** 'yeast'
9(c)	movement of particles / molecules / named molecule ; concentration gradient ; membrane requirement ; living cell requirement ; energy requirement ; (energy from) respiration ; correct example ;	**4**	**A** each point **only** if linked to **either** 'diffusion' **or** 'active transport'
	Total:	**10**	

© UCLES 2017

1 **(a)** Fig. 1.1 shows the front view of a person's chest and abdomen.

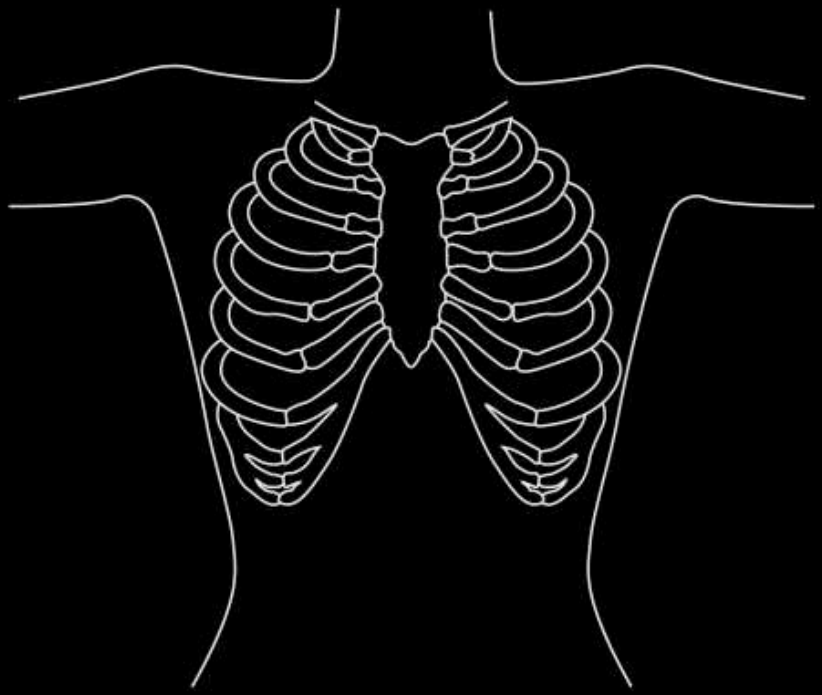


Fig. 1.1

On Fig. 1.1 draw:

- the diaphragm as it would appear immediately after breathing in,
- a circle (**O**) to show the position of the heart,
- a cross (**X**) to show the position of the liver. [3]

(b) Fig. 1.2 shows a person about to lift the handle of a bucket from position **A** to position **B**.

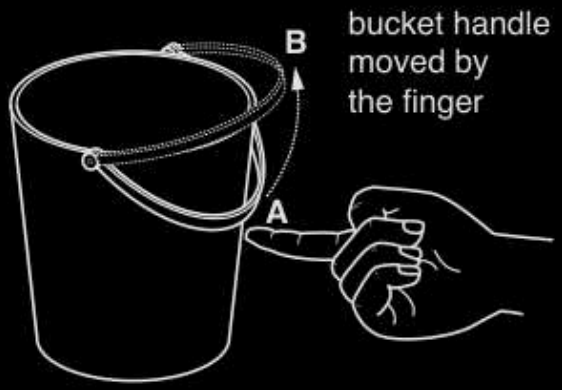


Fig. 1.2

The movement of the bucket handle, as shown, illustrates some features of the movement of a person's chest while breathing in.

(i) State **two similarities** between the movement of a person's chest while breathing in and the movement of the handle.

1 .....

2 .....

[2]

(ii) Explain the **differences** between the movement of a person's chest and the movement of the handle.

.....

.....

.....

.....

.....

.....

.....

.....[5]

[Total: 10]

1(a)	**1** line starts and ends at body wall **+** crosses abdomen below ribs ; **2** circle with all or part of it touching / on the sternum ; **3** middle of cross below diaphragm ;	**3**	
1(b)(i)	**1** any reference to up / rises / raised ; **2** out / forwards ; **3** reference to muscle ; **4** (muscle) contract(ion) ; **5** reference to (requires) energy ;	**2**	
1(b)(ii)	(movement of person's chest) **1** involuntary **AW** ; **2** intercostal (muscles) ; **3** (muscles) between the ribs / in the chest (wall) ; **4** (move) bone / ribs / ribcage ; **5** attached / hinged **+** to vertebrae / backbone / at back ; **6** *leads to increase in volume / decrease in pressure ; (movement of bucket handle) **7** voluntary **AW** ; **8** muscle in arm / finger **OR** reference to bicep(s) ; **9** external to / not part of **+** the bucket / handle ; **10** (move) metal / plastic **OR** reference to a single handle ; **11** attached / hinged **+** to bucket / at side ; **12** *does not lead to change in volume / pressure ;	**5**	* **A** once only for either chest or handle * **A** once only for either chest or handle

© UCLES 2017

Question	Answer	Marks	Guidance

2 Four of the stages associated with human nutrition are:

absorption **digestion** **egestion** **ingestion**

(a) (i) State the stage in nutrition that has **not** been included in the list above.

.....[1]

Fig. 2.1 shows the human alimentary canal.

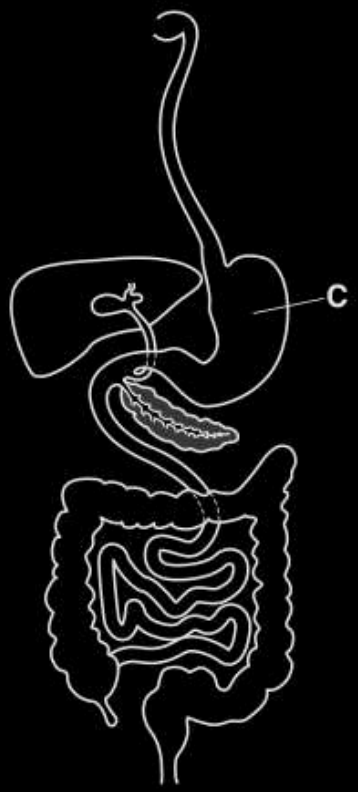


Fig. 2.1

(ii) On Fig. 2.1, use lines labelled **A** and **B** to indicate where egestion (**A**) and ingestion (**B**) occur. [2]

(b) State a component of the diet that is digested in region **C** in Fig. 2.1, and where the products are then absorbed into the blood.

component .....

where products absorbed .....
[2]

(c) Suggest the likely effect on digestion of removing a person's gall bladder.

.....

.....

.....

.....

.....

.....

.....

.....[5]

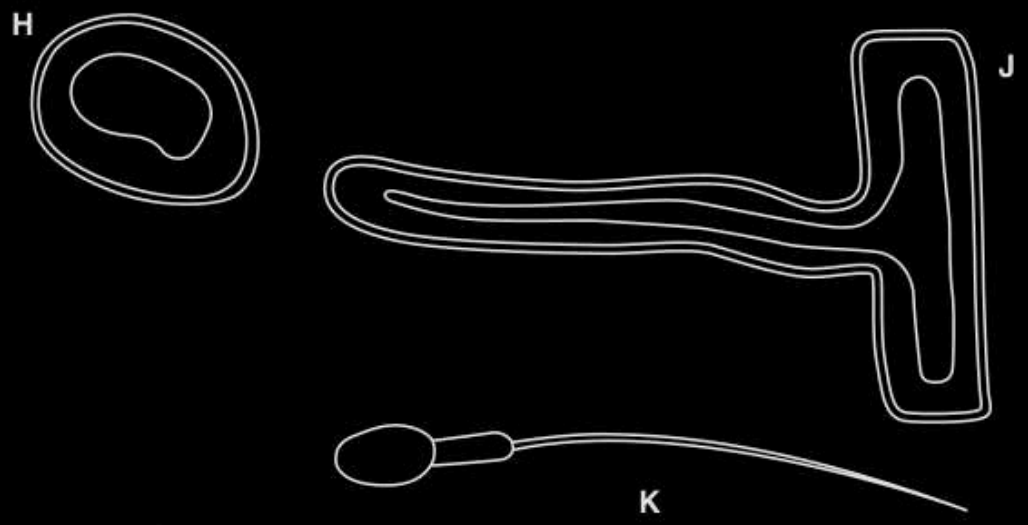
[Total: 10]

2(a)(i)	assimilation ;	**1**	
2(a)(ii)	(egestion) line ending / letter written below rectum **+** labelled '**A**' / 'egestion' ; (ingestion) line ending / letter written to left of or inside mouth **+** labelled '**B**' / 'ingestion' ;	**2**	
2(b)	(component) protein ; (where absorbed) ileum / small intestine / villi / capillaries ;	**2**	**R** amino acids **Ig** any reference to fat **A** jejunum
2(c)	**1** reference to bile (salts) ; **2** (bile) not stored ; **3** less / slower **+** release (of bile) ; **4** less / slower **+** emulsification **AW** ; **5** less / slower / not optimum **+** digestion (of fats) / (lipase) enzyme action ; **6** fewer fatty acids / less glycerol **+** produced / absorbed ; **7** incorrect pH / more acidic / less alkaline **+** in small intestine / duodenum ;	**5**	

© UCLES 2017

Question	Answer	Marks	Guidance

5 Fig. 5.1 shows cells with some of their structures missing.



not to scale

Fig. 5.1

(a) On the cells in Fig. 5.1, draw **and** label nuclei and chloroplasts in their appropriate positions. [4]

(b) State which cell has been produced by meiosis and name the organ in which it has been produced.

cell .....

organ ..... [2]

(c) Explain how energy plays an important part in the function of each of these cells.

cell **H**

.....

.....

.....

cell **J**

.....

.....

.....

cell **K**

.....

.....

.....

[6]

[Total: 12]

Section B

Answer **both** questions in this section.

5(a)	(**H**) nucleus drawn in cytoplasm **+** labelled ; (**J**) nucleus drawn in cytoplasm **+** labelled ; (**K**) nucleus drawn in cytoplasm **+** labelled ; chloroplast(s) drawn <u>only</u> in (**H**) **+** at least one labelled ;	**4**	
5(b)	(cell) sperm / male gamete / (cell) **K** ; (organ) testis / testicle / testes ;	**2**	
5(c)	(cell **H**) **1** traps / absorbs / needs / uses / converts **+** <u>light</u> (energy) ; **2** reference to photosynthesis **OR** production of glucose / starch / carbohydrate **OR** into chemical energy **OR** cell division / mitosis **OR** growth **OR** protein synthesis ; (cell **J**) **3** active transport / against a concentration gradient **AW** ; **4** intake **AW** of **+** ions / minerals / salts ; (cell **K**) **5** movement / use / action **+** of tail / flagellum ; **6** movement / motility / mobility / locomotion / swimming **AW** ;	**6**	

© UCLES 2017

Question	Answer	Marks	Guidance

6 **(a)** Describe the journey made by a molecule of carbon dioxide after its production by respiration in a body cell, to the moment it enters the air in the lungs.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[7]

(b) Explain how blood is kept flowing from the foot to the heart.

.....

.....

.....

.....

.....

.....[3]

[Total: 10]

6(a)	1 2 3 4 5 6 7 8 9 10 11 12	correct reference to (CO_2) diffusion ; cytoplasm ; (across) cell membrane ; tissue fluid ; * capillary ; plasma / (red) blood (cell) ; vein / vena cava ; heart + right (side of heart) ; reference to atrium / auricle + followed by ventricle ; pulmonary artery / arch ; * capillary ; alveolus / alveoli ;	7	* **A** once only either here or below **Ig** inferior / superior vena cava * **A** once only either here or above **Ig** reference to air sac
6(b)	1 2 3 4 5	(blood is under) pressure ; (moved) by (contraction of skeletal) muscles ; reference to veins / correct named vein ; reference to valves (in veins) ; (valves) prevent backflow of blood **AW** ;	3	

Question	Answer	Marks	Guidance

October/November 2016 · Paper 1 · Variant 1 · Question 1

Correct answer: A

1 Which row correctly shows the function of a cell structure and its location?

	cell structure	function	location
A	cell membrane	controls the passage of substances into and out of cells	both plant and animal cells
B	cell membrane	maintains turgor	both plant and animal cells
C	cell wall	controls the passage of substances into and out of cells	plant cells only
D	cell wall	maintains turgor	animal cells only

October/November 2016 · Paper 1 · Variant 1 · Question 2

Correct answer: C

2 Which row describes diffusion into a cell?

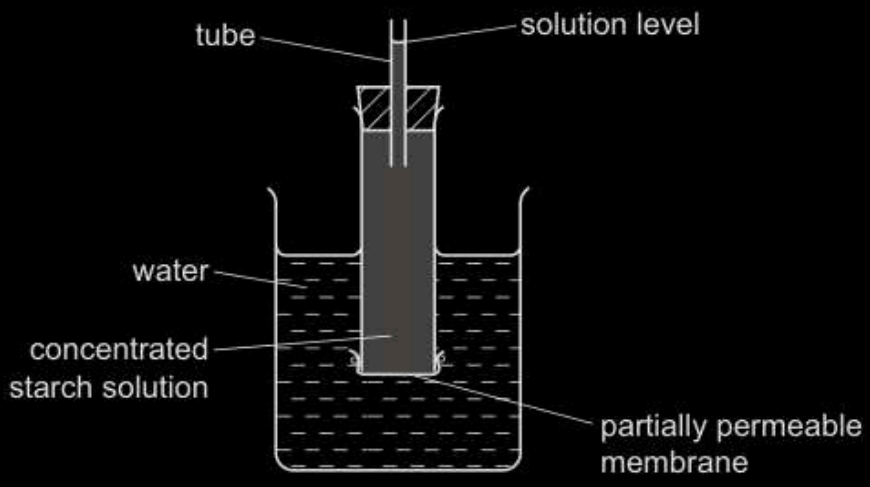
	energy required from the cell	particles move down a concentration gradient
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key
✓ = yes
✗ = no

October/November 2016 · Paper 1 · Variant 1 · Question 3

Correct answer: D

3 The diagram represents apparatus used to investigate osmosis.



Which molecules will move across the partially permeable membrane and which change will occur in the solution level?

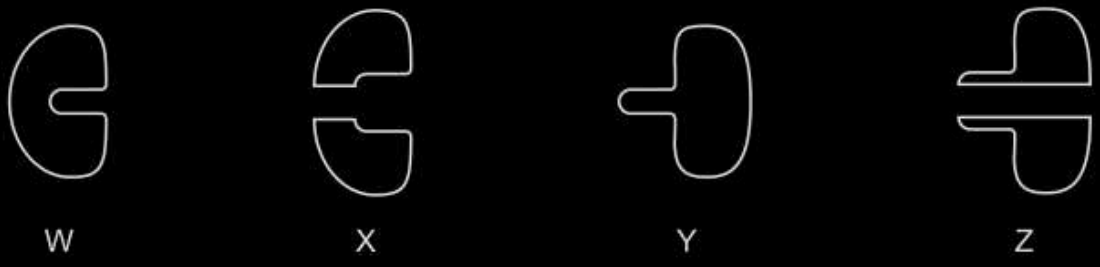
	molecules	solution level
A	starch	fall
B	starch	rise
C	water	fall
D	water	rise

October/November 2016 · Paper 1 · Variant 1 · Question 4

Correct answer: D

4 The diagram represents an enzyme molecule.

What could be substrates for this enzyme?



A W only **B** W or X **C** Y only **D** Y or Z

October/November 2016 · Paper 1 · Variant 1 · Question 5

Correct answer: B

5 How do carbon dioxide and water enter a leaf?

	carbon dioxide	water
A	diffusion	active transport
B	diffusion	transpiration
C	osmosis	diffusion
D	osmosis	transpiration

October/November 2016 · Paper 1 · Variant 1 · Question 8

Correct answer: D

8 Which function is **not** carried out by the liver?

A breakdown of alcohol

B conversion of glycogen to glucose

C formation of urea

D secretion of digestive enzymes

October/November 2016 · Paper 1 · Variant 1 · Question 9

Correct answer: A

9 Which row shows substances mainly absorbed into capillaries and substances mainly absorbed into lacteals, in the villi of the small intestine?

	mainly absorbed into capillaries	mainly absorbed into lacteals
A	amino acids	lipids
B	emulsified fat	glucose
C	glucose	amino acids
D	lipids	emulsified fat

October/November 2016 · Paper 1 · Variant 1 · Question 10

Correct answer: D

10 Which of these food substances increases the risk of obesity?

1 fruit and vegetables

2 processed food

3 salt

4 saturated fat

A 1, 2 and 3 **B** 1, 2 and 4 **C** 1 and 3 only **D** 2 and 4 only

October/November 2016 · Paper 1 · Variant 1 · Question 11

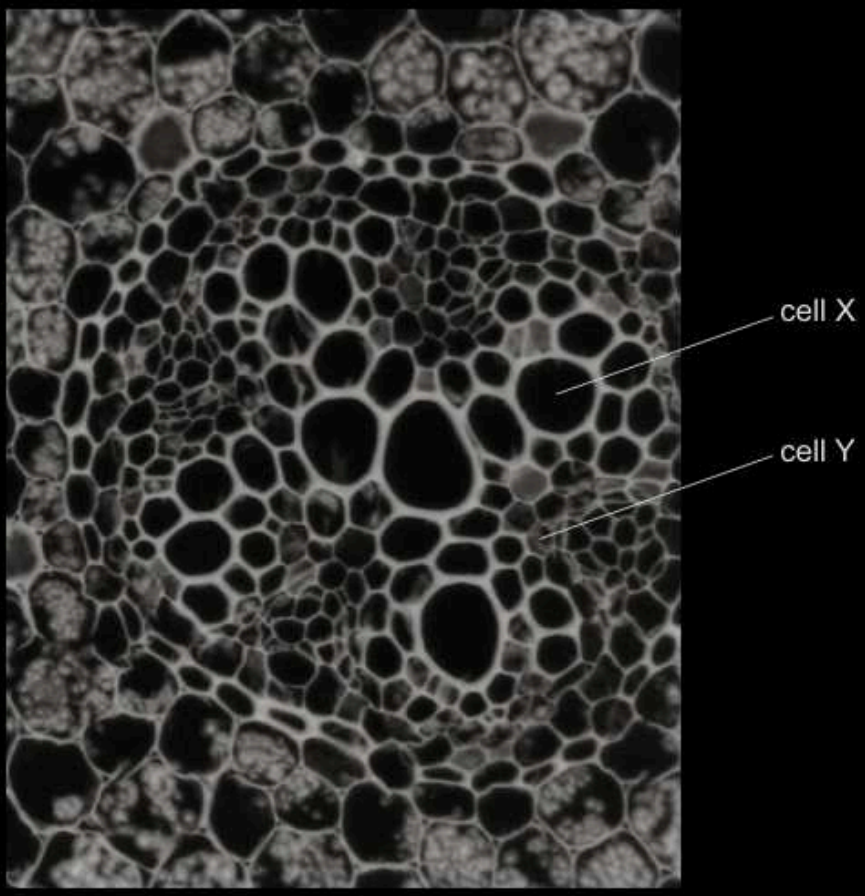
Correct answer: D

11 In which direction do water molecules move in the phloem and in the xylem of a plant stem?

	phloem	xylem
A	down only	up only
B	up only	down only
C	up only	both up and down
D	both up and down	up only

Correct answer: D

12 The photomicrograph shows part of a section through a root.



The contents of cell X and the contents of cell Y are each tested with Benedict's solution and with iodine solution.

Which results are expected?

	cell X		cell Y	
	Benedict's solution	iodine solution	Benedict's solution	iodine solution
A	+	+	–	–
B	+	–	+	+
C	–	+	–	+
D	–	–	+	–

key

+ = positive result

– = negative result

October/November 2016 · Paper 1 · Variant 1 · Question 13

Correct answer: C

13 Which states the sequence of contractions in one heart beat?

A left atrium and left ventricle together, pause, right atrium and right ventricle together

B left atrium, left ventricle, right atrium, right ventricle, pause

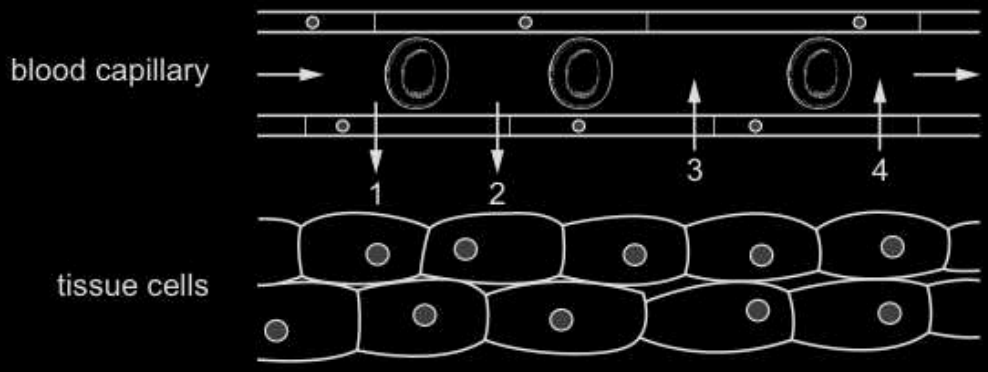
C right atrium and left atrium together, right ventricle and left ventricle together, pause

D right atrium, right ventricle, left atrium, left ventricle, pause

October/November 2016 · Paper 1 · Variant 1 · Question 14

Correct answer: C

14 The diagram shows the transfer of materials between capillaries and tissue fluid.



What could the numbered substances be?

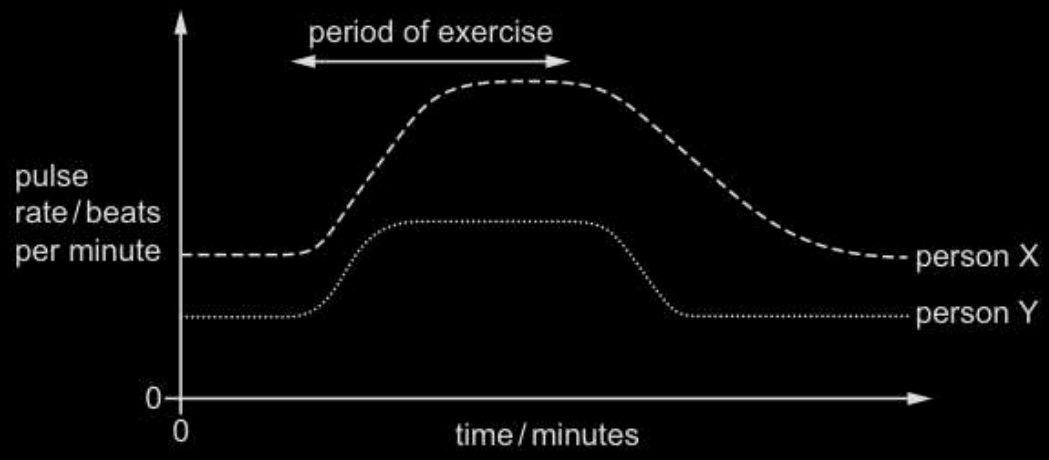
	1	2	3	4
A	carbon dioxide	oxygen	urea	glucose
B	carbon dioxide	urea	oxygen	glucose
C	oxygen	glucose	carbon dioxide	urea
D	oxygen	urea	carbon dioxide	glucose

October/November 2016 · Paper 1 · Variant 1 · Question 15

Correct answer: D

15 Two people carried out a short period of exercise on a treadmill. Each of the two people ran on the treadmill for the same amount of time and at the same speed.

The graph shows the pulse rates of these two people before, during and after the period of exercise.



Which interpretation of the graph is correct?

A Person X is fitter, and their heart rate increases more during exercise.

B Person X is fitter, and their heart rate recovers more quickly after exercise.

C Person Y is fitter, and their heart rate increases more during exercise.

D Person Y is fitter, and their heart rate recovers more quickly after exercise.

October/November 2016 · Paper 1 · Variant 1 · Question 20

Correct answer: B

20 What is a function of a part of the urinary system?

A Ureter carries urine from the bladder.

B Ureter carries urine from the kidneys.

C Urethra carries urine to the bladder.

D Urethra carries urine to the kidneys.

October/November 2016 · Paper 1 · Variant 1 · Question 21

Correct answer: B

21 Which part of the skin has a major role in insulating the human body?

A blood vessels

B fatty tissue

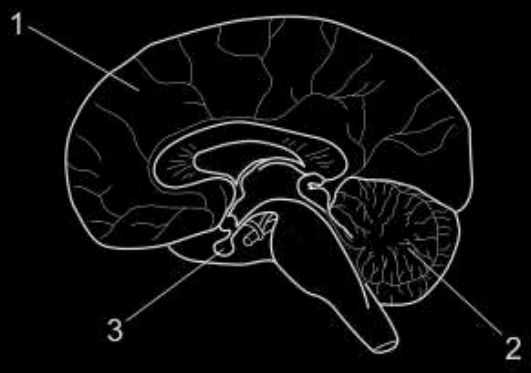
C hair follicles

D sweat glands

October/November 2016 · Paper 1 · Variant 1 · Question 22

Correct answer: A

22 The diagram shows a section of the human brain.



What are the numbered structures?

	1	2	3
A	cerebrum	cerebellum	pituitary gland
B	cerebrum	hypothalamus	medulla
C	cerebellum	cerebrum	pituitary gland
D	cerebellum	hypothalamus	medulla

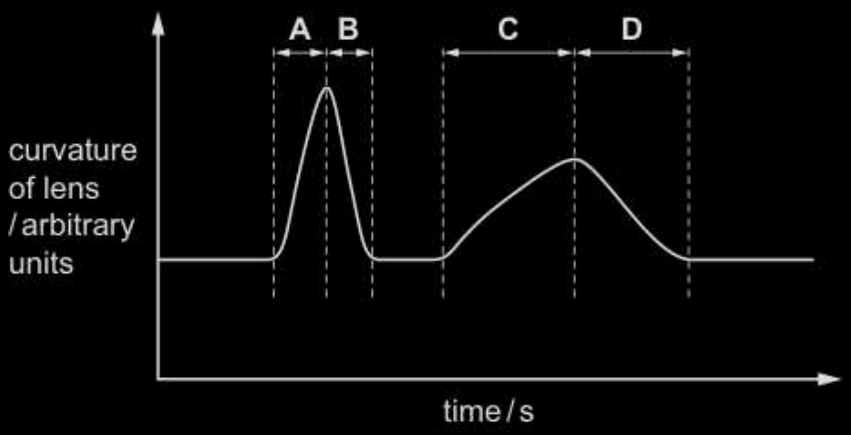
October/November 2016 · Paper 1 · Variant 1 · Question 23

Correct answer: A

23 The diagram shows the curvature of the lens in a person's eye.

The shape of the lens changes as the person watches two motorbikes go past at different speeds.

During which period was a motorbike moving towards the person at the higher speed?



October/November 2016 · Paper 1 · Variant 1 · Question 24

Correct answer: D

24 Nerve impulses in neurones can travel

1 away from the central nervous system

2 towards the central nervous system

3 within the central nervous system

In which direction do impulses in sensory and relay neurones travel?

	sensory neurone	relay neurone
A	1	2
B	1	3
C	2	1
D	2	3

October/November 2016 · Paper 1 · Variant 1 · Question 25

Correct answer: B

25 What are the effects on the body of drinking a large amount of alcohol?

	effect on brain	reaction time
A	depressed	decreases
B	depressed	increases
C	stimulated	decreases
D	stimulated	increases

October/November 2016 · Paper 1 · Variant 1 · Question 26

Correct answer: B

26 A new organism is discovered. It contains DNA in a cellular structure.

To which group of organisms could it belong and to which group could it **not** belong?

	could belong to	could **not** belong to
A	bacteria	fungi
B	bacteria	viruses
C	fungi	bacteria
D	viruses	bacteria

October/November 2016 · Paper 1 · Variant 1 · Question 34

Correct answer: B

34 Small pieces of root tissue are taken from an oil palm tree and placed in a nutrient medium. Each small piece produces a new oil palm tree.

Which type of reproduction is this and how do the genotypes of the new trees compare with that of the parent tree?

	type of reproduction	genotype
A	asexual	different
B	asexual	identical
C	sexual	different
D	sexual	identical

October/November 2016 · Paper 1 · Variant 1 · Question 36

Correct answer: A

36 What is an advantage of human breast milk compared with bottle milk?

A It contains antibodies from the mother.

B It contains fats that cannot be digested.

C It contains red blood cells from the mother.

D It has a higher iron content.

October/November 2016 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 The inheritance of the ABO blood groups depends on three alleles I^A, I^B and I^O.

What are the possible genotypes for a person of blood group A?

A I^AI^A only

B I^AI^A, and I^AI^B only

C I^AI^A, and I^AI^O only

D I^AI^A, I^AI^O and I^AI^B

October/November 2016 · Paper 1 · Variant 1 · Question 39

Correct answer: D

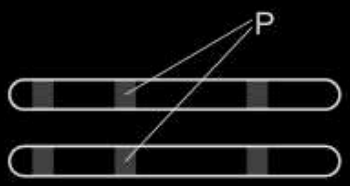
39 Two heterozygous individuals are crossed. Some of the offspring show the recessive characteristic.

What is the probability that one of these offspring that shows the recessive characteristic is homozygous?

A 0.00 **B** 0.25 **C** 0.50 **D** 1.00

Correct answer: D

40 The diagram shows a pair of chromosomes from the same cell.



What do the lines labelled P point to?

A the site of alleles made up of two or more genes which are always the same

B the site of alleles made up of two or more genes which might be different

C the site of genes made up of two or more alleles which are always the same

D the site of genes made up of two or more alleles which might be different

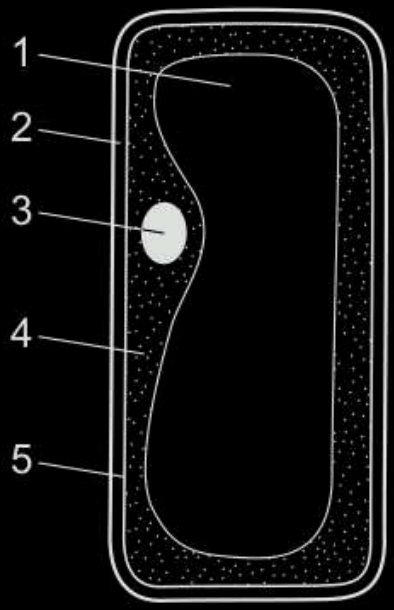
Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

Correct answer: A

1 The diagram shows a plant cell.



Which two structures are **not** found in animal cells?

A 1 and 2 **B** 2 and 3 **C** 3 and 4 **D** 4 and 5

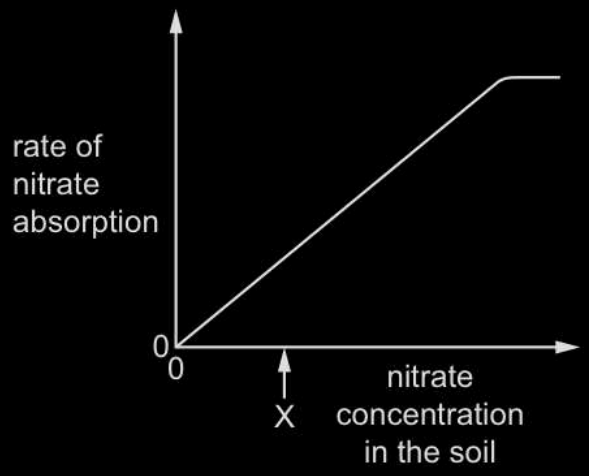
Correct answer: A

2 The rate of nitrate ion absorption by a root hair cell was measured at different soil nitrate concentrations.

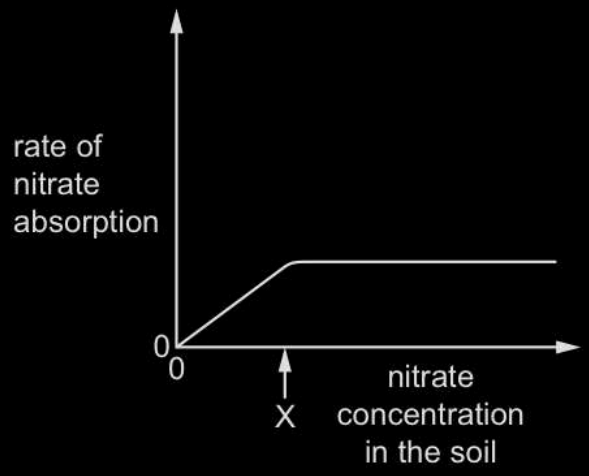
At X, the concentration of nitrate in the soil is the same as in the cell.

Which graph shows how the rate of absorption varies with nitrate concentration in the soil?

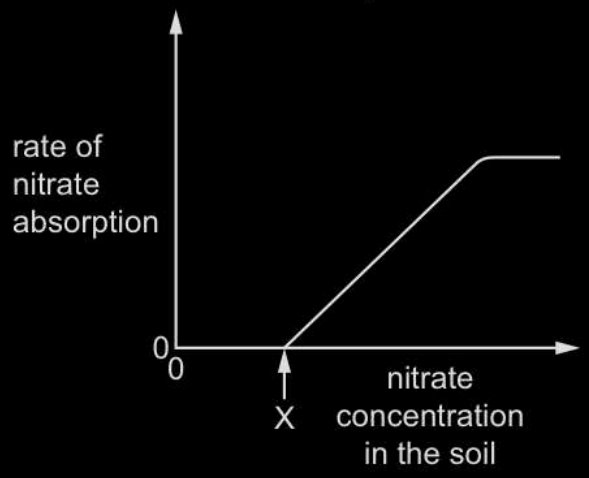
A



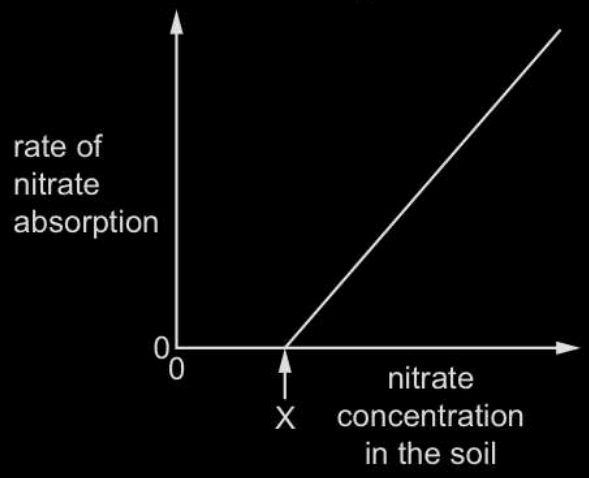
B



C



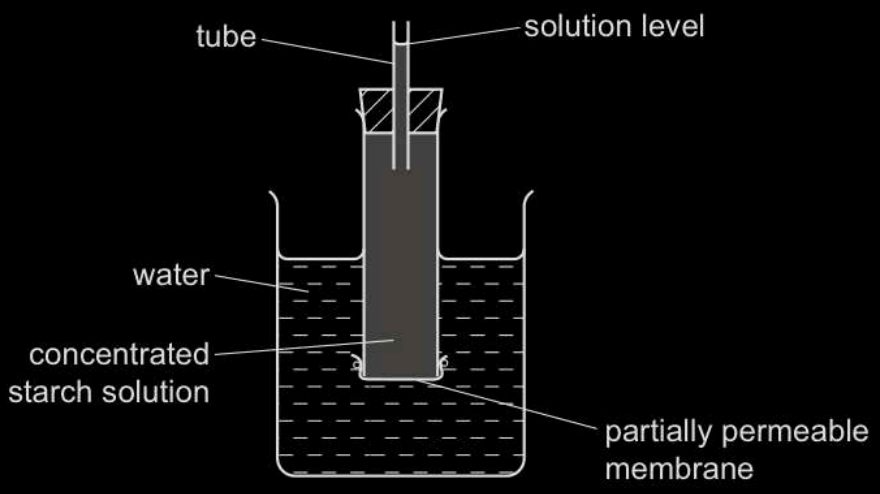
D



October/November 2016 · Paper 1 · Variant 2 · Question 3

Correct answer: D

3 The diagram represents apparatus used to investigate osmosis.



Which molecules will move across the partially permeable membrane and which change will occur in the solution level?

	molecules	solution level
A	starch	fall
B	starch	rise
C	water	fall
D	water	rise

October/November 2016 · Paper 1 · Variant 2 · Question 4

Correct answer: C

4 Starch digestion occurs in the mouth cavity and in the duodenum but it stops in the stomach.

Why is this?

A All the starch has been digested before it reaches the stomach.

B Cells in the stomach do not produce amylase.

C The pH in the stomach alters the shape of the amylase.

D The temperature in the stomach is too high for amylase to work.

October/November 2016 · Paper 1 · Variant 2 · Question 5

Correct answer: C

5 One of the tissues in a leaf was described as consisting of 'loosely packed cells with a layer of moisture on their surface to dissolve gases so that the gases can diffuse into the cells'.

To which tissue is this description referring?

A epidermis

B palisade mesophyll

C spongy mesophyll

D vascular bundle

October/November 2016 · Paper 1 · Variant 2 · Question 6

Correct answer: B

6 A growing plant is regularly watered with a solution. The composition of the solution is changed after which the plant's leaves become yellow in colour.

How can this problem be corrected?

A adding amino acids to the solution

B adding magnesium ions to the solution

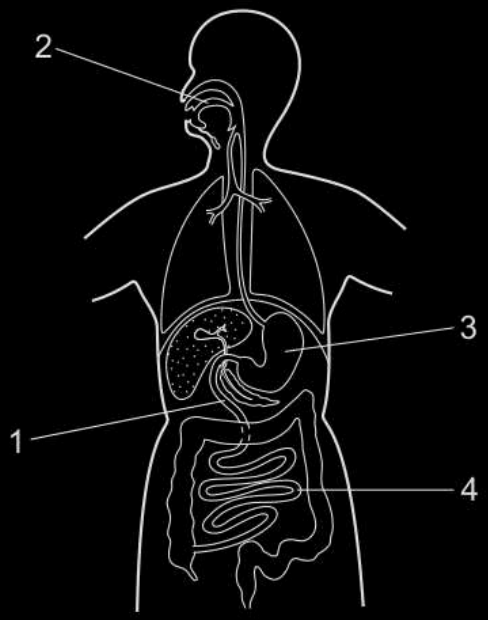
C bubbling carbon dioxide through the solution

D bubbling oxygen through the solution

October/November 2016 · Paper 1 · Variant 2 · Question 8

Correct answer: A

8 The diagram shows the alimentary canal and some associated organs.



Which row shows where amylase is released?

	1	2	3	4
A	✓	✓	✗	✗
B	✓	✗	✓	✗
C	✗	✓	✓	✗
D	✗	✓	✗	✓

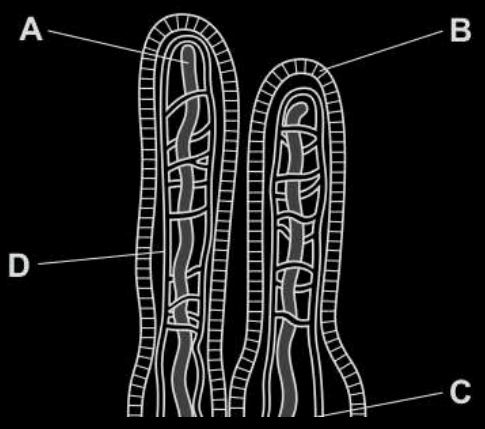
key
✓ = yes
✗ = no

October/November 2016 · Paper 1 · Variant 2 · Question 9

Correct answer: A

9 The diagram shows a section of the wall of the small intestine.

Into which structure are most lipid components absorbed for transport to the rest of the body?



October/November 2016 · Paper 1 · Variant 2 · Question 10

Correct answer: B

10 Which chemical change does **not** occur in the liver?

A amino acids to glucose

B glucose to amino acids

C glucose to glycogen

D glycogen to glucose

October/November 2016 · Paper 1 · Variant 2 · Question 11

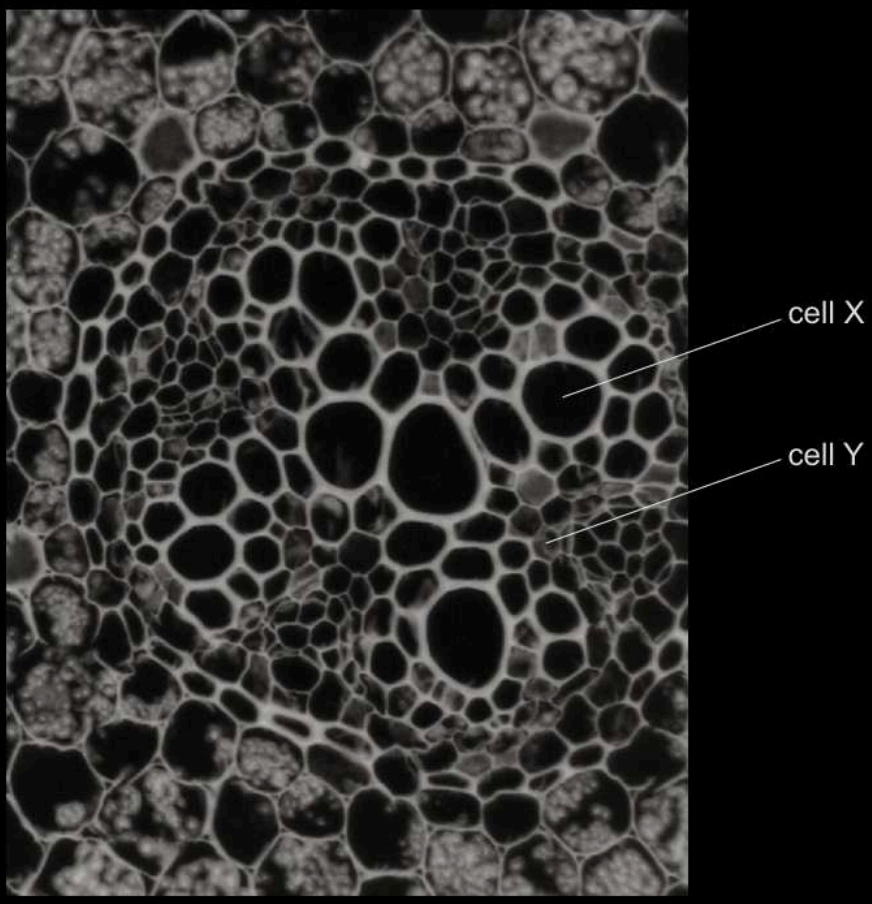
Correct answer: D

11 In which direction do water molecules move in the phloem and in the xylem of a plant stem?

	phloem	xylem
A	down only	up only
B	up only	down only
C	up only	both up and down
D	both up and down	up only

Correct answer: D

12 The photomicrograph shows part of a section through a root.



The contents of cell X and the contents of cell Y are each tested with Benedict's solution and with iodine solution.

Which results are expected?

	cell X		cell Y	
	Benedict's solution	iodine solution	Benedict's solution	iodine solution
A	+	+	–	–
B	+	–	+	+
C	–	+	–	+
D	–	–	+	–

key

+ = positive result

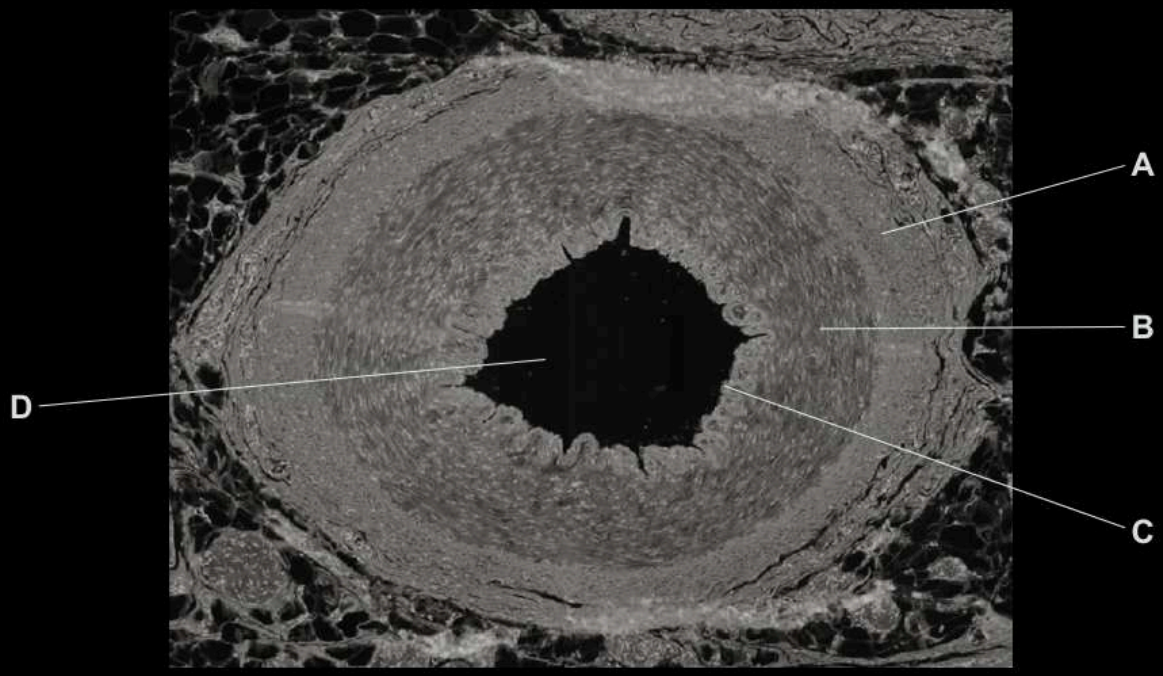
– = negative result

October/November 2016 · Paper 1 · Variant 2 · Question 13

Correct answer: C

13 The photograph shows a cross-section of an artery.

Which labelled part would be of the same thickness in a vein?



October/November 2016 · Paper 1 · Variant 2 · Question 14

Correct answer: C

14 In a muscle, which two substances show net movement from the plasma **into** the tissue fluid?

A carbon dioxide and glucose

B carbon dioxide and lactic acid

C glucose and oxygen

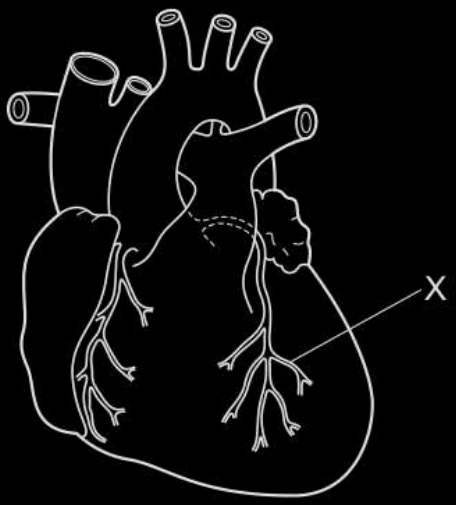
D lactic acid and oxygen

October/November 2016 · Paper 1 · Variant 2 · Question 15

Correct answer: B

15 The diagram shows an external view of the heart.

The left coronary artery is blocked at the point labelled X.



How would the blockage first affect the heart?

A Blood is prevented from entering the heart.

B Cells in the wall of the left ventricle die.

C The rate of contraction of the heart increases.

D The valves of the heart allow blood to flow backwards.

October/November 2016 · Paper 1 · Variant 2 · Question 20

Correct answer: C

20 Which row shows the parts of the urinary system that carry out different functions?

	makes urine	carries urine	holds urine	removes urine from body
A	bladder	ureter	urethra	kidney
B	bladder	urethra	ureter	kidney
C	kidney	ureter	bladder	urethra
D	kidney	urethra	bladder	ureter

October/November 2016 · Paper 1 · Variant 2 · Question 21

Correct answer: D

21 On a hot day, how does the skin react to lower the internal temperature of the body?

A arterioles constrict, sweat production decreases, hairs stand on end

B arterioles constrict, sweat production increases, hairs lie flat against the skin

C arterioles dilate, sweat production decreases, hairs lie flat against the skin

D arterioles dilate, sweat production increases, hairs lie flat against the skin

October/November 2016 · Paper 1 · Variant 2 · Question 22

Correct answer: D

22 The eye changes focus from looking at a wristwatch to looking at an aeroplane flying overhead.

Which changes occur inside the eye?

	shape of lens	suspensory ligaments	ciliary muscles
A	thicker	slack	contract
B	thicker	tight	relax
C	thinner	slack	contract
D	thinner	tight	relax

October/November 2016 · Paper 1 · Variant 2 · Question 24

Correct answer: D

24 Nerve impulses in neurones can travel

1 away from the central nervous system

2 towards the central nervous system

3 within the central nervous system

In which direction do impulses in sensory and relay neurones travel?

	sensory neurone	relay neurone
A	1	2
B	1	3
C	2	1
D	2	3

October/November 2016 · Paper 1 · Variant 2 · Question 26

Correct answer: B

26 A new organism is discovered. It contains DNA in a cellular structure.

To which group of organisms could it belong and to which group could it **not** belong?

	could belong to	could **not** belong to
A	bacteria	fungi
B	bacteria	viruses
C	fungi	bacteria
D	viruses	bacteria

October/November 2016 · Paper 1 · Variant 2 · Question 30

Correct answer: B

30 In which process is atmospheric nitrogen absorbed in root nodules and combined with other compounds?

A nitrification

B nitrogen fixation

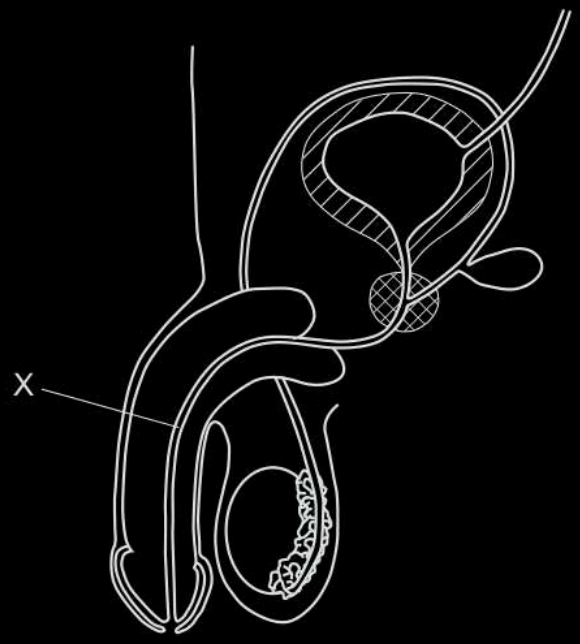
C putrefaction

D diffusion

October/November 2016 · Paper 1 · Variant 2 · Question 35

Correct answer: D

35 The diagram shows part of the male reproductive and urinary systems.



What is X?

A prostate gland

B sperm duct

C ureter

D urethra

October/November 2016 · Paper 1 · Variant 2 · Question 36

Correct answer: B

36 What are the functions of the amniotic sac and the amniotic fluid?

A act as a barrier to the passage of bacteria and allow the transfer of some antibodies to the fetus

B protect the fetus from physical shock and allow the fetus to move freely

C provide soluble nutrients such as glucose and mineral ions for the fetus

D supply oxygen to, and remove carbon dioxide from, the fetus

October/November 2016 · Paper 1 · Variant 2 · Question 38

Correct answer: C

38 The inheritance of the ABO blood groups depends on three alleles I^A, I^B and I^O.

What are the possible genotypes for a person of blood group A?

A I^AI^A only

B I^AI^A, and I^AI^B only

C I^AI^A, and I^AI^O only

D I^AI^A, I^AI^O and I^AI^B

October/November 2016 · Paper 1 · Variant 2 · Question 39

Correct answer: D

39 Two heterozygous individuals are crossed. Some of the offspring show the recessive characteristic.

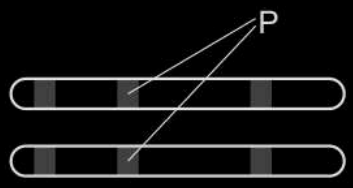
What is the probability that one of these offspring that shows the recessive characteristic is homozygous?

A 0.00 **B** 0.25 **C** 0.50 **D** 1.00

October/November 2016 · Paper 1 · Variant 2 · Question 40

Correct answer: D

40 The diagram shows a pair of chromosomes from the same cell.



What do the lines labelled P point to?

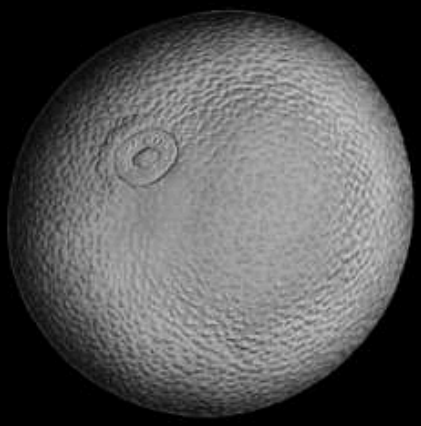
A the site of alleles made up of two or more genes which are always the same

B the site of alleles made up of two or more genes which might be different

C the site of genes made up of two or more alleles which are always the same

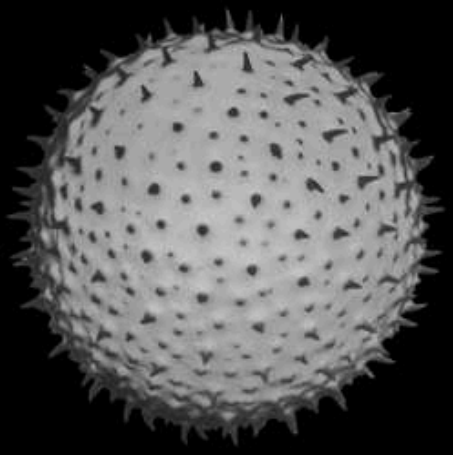
D the site of genes made up of two or more alleles which might be different

2 Fig. 2.1 shows a magnified pollen grain from each of two plant species, **A** and **B**.



magnification ×830

pollen grain from plant species **A**



magnification ×200

pollen grain from plant species **B**

Fig. 2.1

(a) (i) Use the information in Fig. 2.1 and your biological knowledge to describe **three** differences between the pollen grains from species **A** and species **B**.

Write your answers in Table 2.1.

Table 2.1

pollen grain from species **A**	pollen grain from species **B**

[3]

(ii) Cross-pollination takes place in both species **A** and species **B**.

Describe what is meant by the term *cross-pollination*.

.....

.....

.....

.....

.....

.....

..... [3]

(iii) Using the information provided by Fig. 2.1, describe how cross-pollination is most likely to occur in species **A** and in species **B**.

species **A**

.....

.....

.....

.....

species **B**

.....

.....

.....

.....[3]

(b) Fig. 2.2 is a photograph of the flowers of species **A**.

Fig. 2.2

List **two** features that would be present in the flowers of species **B** that are **not** present in those of species **A**.

1 .....

2 ..... [2]

[Total: 11]

2(a)(i)	**A** small + **B** large / **A** smaller / **B** larger ; **A** smooth + **B** spikes **AW** ; **A** wind dispersal / **B** animal dispersal ; **A** light / **B** heavy ;	**3**	
2(a)(ii)	transfer / movement + of pollen ; from anther ; to <u>stigma</u> ; of another plant (of same species) ;	**3**	 **A** stamen / pollen sac **R** another species

© UCLES 2016

Question	Answer	Mark	Guidance
2(a)(iii)	(plant species **A**) wind ; reference to light / small / carried (by wind) ; (plant species **B**) insect ; reference to attachment (to insect) / carried (by insect) moving between flowers ;	3	
2(b)	large (petals) ; (brightly) coloured / patterned (petals) ; anthers enclosed in flower / not hanging out / erect ; scent ; nectar ;	2	
	Total:	11	

© UCLES 2016

Question	Answer	Mark	Guidance

3 Fig. 3.1a is a diagram of the human skeleton. Fig. 3.1b is an X-ray that shows damage to part of a person's skeleton.

Fig. 3.1a

Fig. 3.1b

(a) (i) Draw the letter **X** on Fig. 3.1a to show the location of the damage to the person's skeleton that is shown in Fig. 3.1b.

[1]

(ii) Describe the damage to the person's skeleton shown in Fig. 3.1b.

.....

.....

.....

..... [2]

(b) Fig. 3.2 shows the arrangement of bones and muscles in part of a person's body.

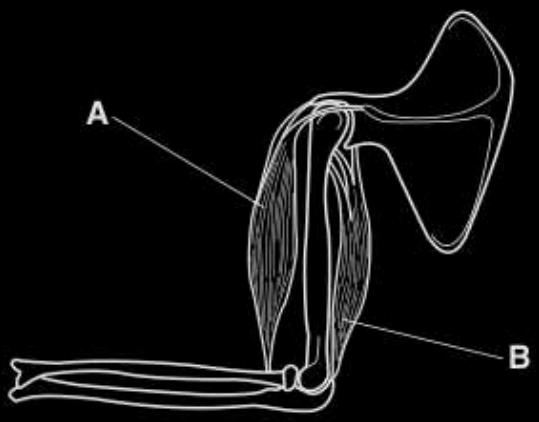


Fig. 3.2

(i) Name the muscle labelled **B** in Fig. 3.2.

……. [1]

(ii) Draw an arrow on Fig. 3.2 to show the direction of movement that will be caused when the muscle labelled **B** contracts.

[1]

(iii) When the muscle labelled **B** contracts, the muscle labelled **A** relaxes. State the term that refers to a pair of muscles that act in this way.

……. [1]

(c) Fig. 3.3 shows the approximate percentage of body mass that is due to the skeleton in birds and in humans.

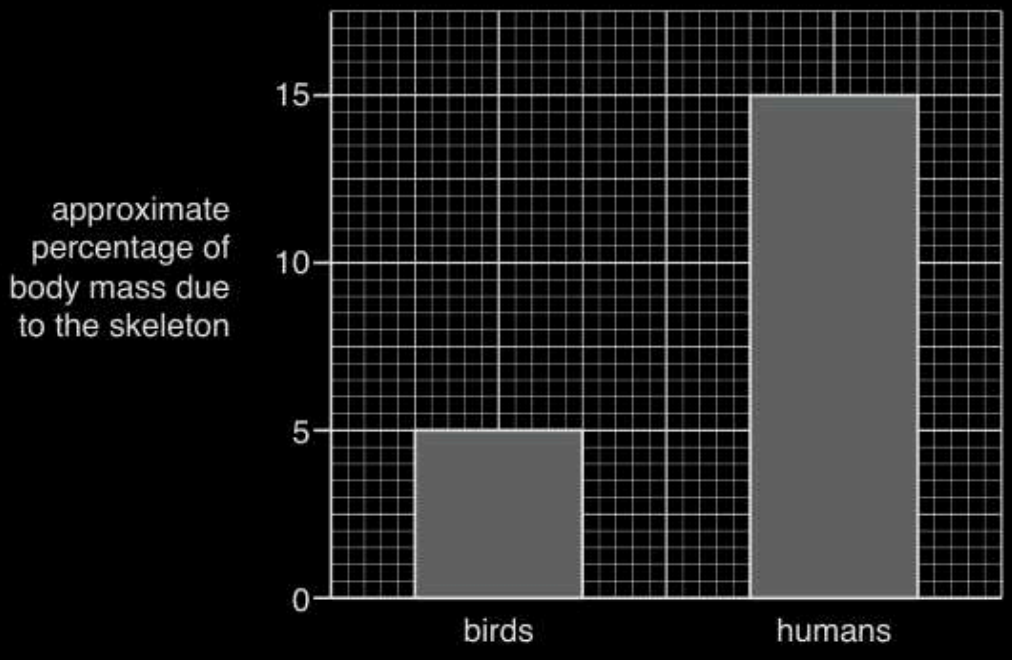


Fig. 3.3

(i) Use the information in Fig. 3.3 to compare the approximate percentage of body mass that is due to the skeleton in birds and in humans.

.....

..... [1]

(ii) One reason for this difference is that the bones of a bird contain air spaces. These air spaces are connected to the lungs of the bird.

The blood of birds contains more haemoglobin per unit volume than that of humans.

Suggest how these adaptations, and that shown in Fig. 3.3, help a bird to move by flying.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 11]

3(a)(i)	**X** drawn on left / right humerus of skeleton ;	**1**	
3(a)(ii)	humerus / bone / arm ; broken **AW** ;	**2**	
3(b)(i)	triceps / extensor ;	**1**	
3(b)(ii)	arrow down from above / below lower arm bones i.e. to left of elbow ;	**1**	
3(b)(iii)	antagonistic ;	**1**	
3(c)(i)	3 × larger + humans / 3 × smaller + birds / ratio 1:3 / ratio 5:15 ;	**1**	must have correct reference to birds and / or humans
3(c)(ii)	reduced weight ; fly further / faster ; more energy efficient **AW** ; more / sufficient oxygen available / carried by the blood / in the bones ; as oxyhaemoglobin ; (for aerobic) respiration ; (release) more / enough energy (for it to fly) ; (to use for) muscle + contraction ;	**4**	**R** anaerobic (respiration)
	Total:	**11**	

© UCLES 2016

Question	Answer	

4 Table 4.1a shows some of the recommended dietary allowances for a child under the age of six months.

Table 4.1b shows part of the composition of bottle milk that may be fed to a child under the age of six months.

Table 4.1a

component	recommended dietary allowance
energy	2770 kJ
protein	13 g
vitamin C	30 mg
iron	6 mg

Table 4.1b

component	amount per 100 cm^3 of bottle milk
energy	277 kJ
fat	3.6 g
carbohydrate	7.3 g
protein	1.3 g
vitamin C	3.0 mg
iron	0.6 mg

(a) (i) Calculate the volume of bottle milk that a child under the age of six months should be fed each day to obtain the recommended dietary allowance of each component listed in Table 4.1a.

You may use the space below to work out your answer.

..... [2]

(ii) State and explain the health risks to a child who is fed less bottle milk each day than the volume you have calculated.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) (i) Many mothers choose to feed their child breast milk rather than bottle milk.

Describe some advantages of breast milk compared with bottle milk.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) Suggest **one** reason why a mother may choose to feed her child bottle milk.

.....

.....

..... [1]

[Total: 11]

4(a)(i)	1000 / 1 ; cm^3 / dm^3 ;	
4(a)(ii)	malnutrition / deficiency disease ; lack of balanced diet ; reference to low / lack of energy (intake into body) ; (leads to) general lethargy ; reference to lack of fat or carbohydrate (for energy) ; reference to specified problem ; kwashiorkor / named symptom + lack of protein ; scurvy / named symptom + lack of vitamin C ; anaemia or described + lack of iron ;	

© UCL

Mark	Guidance
2	**A** 1 litre / 1 dm^3 for 2 marks **A** ml in place of cm^3, unit must match figure for second mark
4	**A** muscle contraction / protein synthesis / cell division / active transport / growth / passage of nerve impulses / maintenance of a constant body temperature / mend or repair tissues

Question	Answer	Mark	Guidance
4(b)(i)	sterile / not contaminated / no additives ; antibodies / hormones ; no need to warm / at correct temperature ; bonding between mother and child ; better proportions of nutrients **AW** ; does not cost money ; supply / availability on demand ;	**4**	
4(b)(ii)	lack of awareness of the benefits ; mother unable / chooses not to breast-feed ; reference to advertising by bottle milk providers ; not available to do so (e.g. at work / child in care) ; infection / drug addiction / damaged nipples ; mother does not produce (enough) milk ;	**1**	
	Total:	**11**	

© UCLES 2016

Question	Answer	Mark	Guidance

5 Complete the paragraph below by writing the most appropriate word in each of the spaces.

A ..... includes a long molecule of DNA that is divided into sections called genes. Genes may be copied and passed on to the next generation. Each gene may have two or more alternative forms called ..... . One of these forms may be ..... and the other recessive. A change in the structure of a gene or in the number of chromosomes is called a ..... . Chemicals and ..... may increase the rate at which these changes take place.

[5]

[Total: 5]

Section B

Answer **both** questions in this section.

5	chromosome ; allele(s) ; dominant ; mutation ; radiation / named radiation / radioactivity ;	**5**	**A** chromatid **A** carcinogens
	Total:	**5**	

Question	Answer	Mark	Guidance

7 **(a)** Define the term *hormone*.

.....

.....

.....

.....

.....

..... [3]

(b) **(i)** State the role of the hormone insulin in controlling blood sugar concentration.

.....

.....

.....

.....

.....

..... [3]

(ii) Name the condition caused if a person is unable to produce sufficient amounts of the hormone insulin.

Describe the signs of the condition that you have named, and its treatment.

name of condition

.....

description of signs and treatment

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

7(a)	chemical (substance) ; produced by (endocrine / ductless) gland ; (secreted) into / carried by blood ; (effect on) target organ ; destroyed by liver ;	**3**	**Ig** named examples of hormones
7(b)(i)	insulin is secreted when blood sugar / glucose is (too) high / lowers blood sugar / glucose levels ; (increased) sugar / glucose into cells ; reference to liver / muscle (cells) ; (converted) to glycogen ; (glycogen) for storage / is insoluble ;	**3**	

© UCLES 2016

Question	Answer	Mark	Guidance
7(b)(ii)	diabetes ; increased blood sugar / glucose concentration ; sugar / glucose in urine ; frequent urination ; tired / thirsty / weight loss ; insulin (by injection or tablet) ; dietary control ;	**4**	**A** sweet-smelling breath / urine **R** injection alone as a treatment, insulin required for mark
	Total:	**10**	

© UCLES 2016

Question	Answer	Mark	Guidance

8 **(a)** Describe the double circulation of blood in the human circulatory system and the different functions of the two circuits.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) Describe the structure of a capillary and the transfer of a **named** material between capillaries and tissue fluid.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

8(a)	through heart twice (during one circuit) **AW** ; circuit + to lungs / pulmonary ; blood to become oxygenated / pick up oxygen ; blood to lose carbon dioxide ; circuit + to body tissues / systemic ; to deliver oxygen / glucose (to body tissues) ; collect carbon dioxide ; reference to part of heart + named vessel ; low pressure lungs / high pressure body ;	**6**	**A** heart – body – heart – lungs

© UCLES 2016

Question	Answer	Mark	Guidance
8(b)	one cell thick / thin / 1μm + wall ; small diameter of lumen / RBC single file / width of RBC **AW** ; no muscle / no elastic tissue (compared with artery / vein) ; leaky walls **AW** ; diffusion ; reference to named material ;	4	**A** epithelium
	Total:	10	

© UCLES 2016

Question	Answer	Mark	Guidance

2 Fig. 2.1 shows part of the flowering head of a small tree that grows in tropical rainforests.

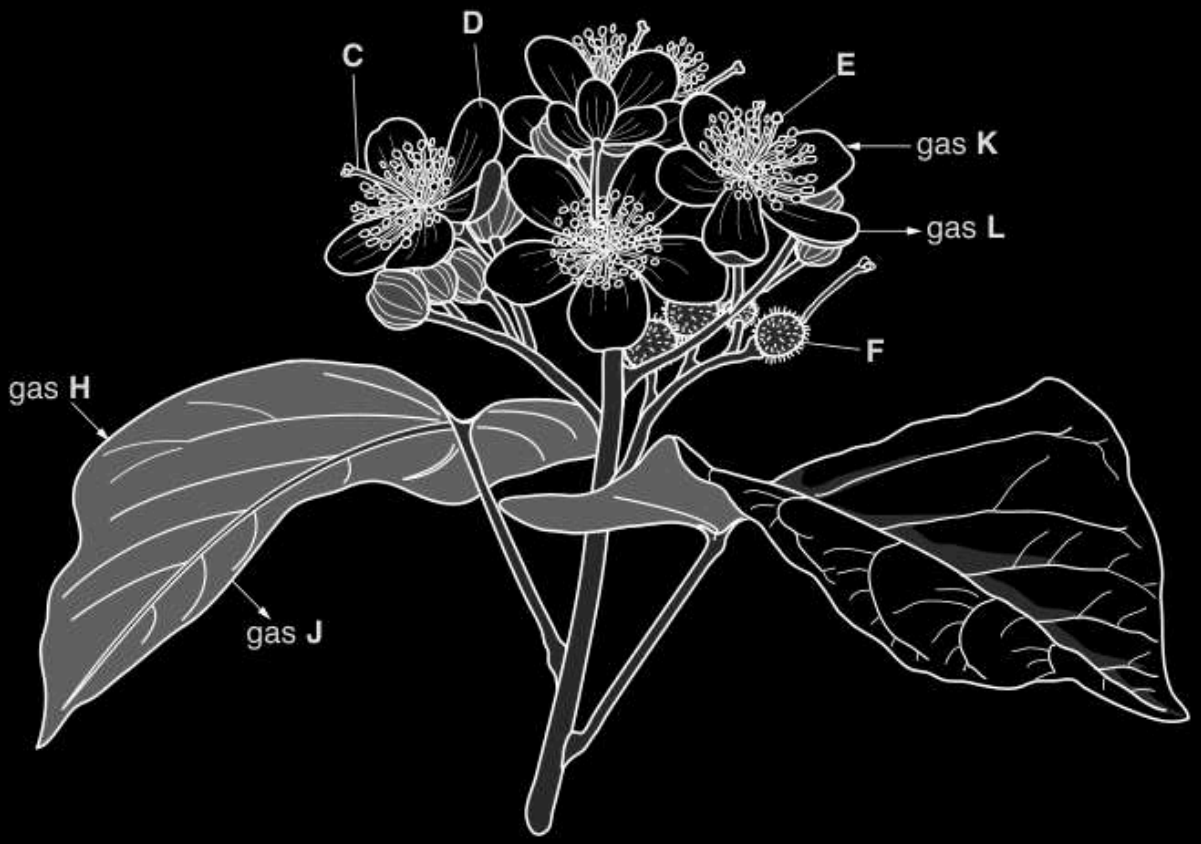


Fig. 2.1

(a) Identify the structures **C**, **D**, **E** and **F** shown on Fig. 2.1.

C .....

D .....

E .....

F ..... [4]

(b) **(i)** Structure **D** contains stomata similar to those found in the leaves.

H, **J**, **K** and **L** represent gases that pass into and out of the leaves and flowers during daylight hours.
Identify the gases **H** and **J**.

H .....

J ..... [2]

(ii) The flowers are very pale pink in colour. Suggest the identity of gases **K** and **L**.
Give an explanation for your answer.

K .....

L .....

explanation .....

.....

.....

.....[4]

(c) Extracts from the tree have many uses in medicine. Some of the extracts are alkaline and have anti-bacterial properties.

Some stomach problems are caused by excess acid production.
Suggest why these tree extracts are sometimes used to treat such stomach problems.

.....

.....

.....

.....[2]

[Total: 12]

2(a)	C – style / pistil / carpel / gynaecium ; D – petal / corolla ; E – anther / stamen / androecium ; F – fruit / pericarp ;	4	
2(b)(i)	H – CO_2 / carbon dioxide ; J – O_2 / oxygen / water <u>vapour</u> ;	2	

© UCLES 2016

Question	Answer	Mark	Guidance
2(b)(ii)	K – O_2 / oxygen ; L – CO_2 / carbon dioxide / water <u>vapour</u> ; no chlorophyll / no chloroplasts ; no photosynthesis ; respiration ; **A** transpiration if water vapour given for L ;	2 2	
2(c)	neutralises (stomach acid) / raises pH **AW** ; may kill / destroy (harmful) bacteria ;	2	
	Total:	12	

Question	Answer	Mark	Guidance

4 Fig. 4.1 shows a section through a heart connected to what is described as an LV Assist Device.

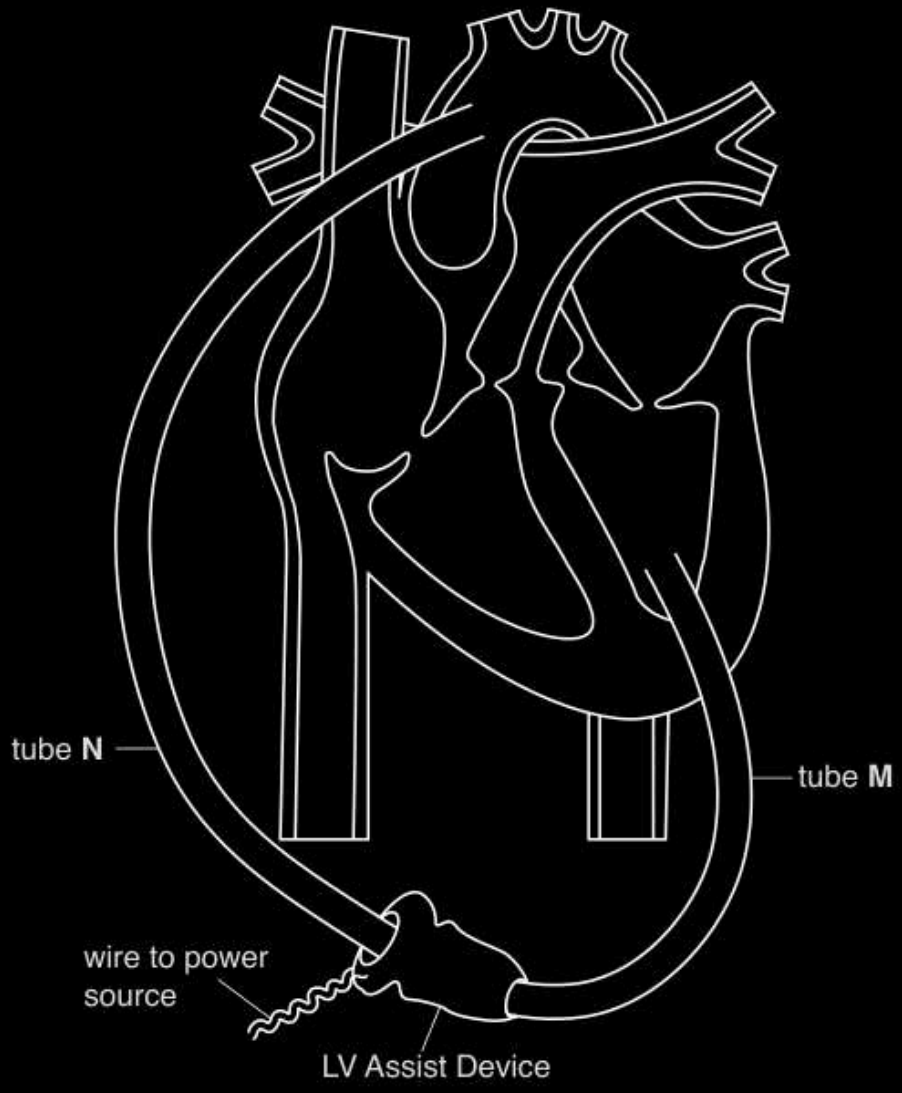


Fig. 4.1

(a) (i) Use Fig. 4.1 to suggest why the device is called an LV Assist Device.

.....

.....[1]

(ii) Name the type of tissue through which tube **M** passes.

..... [1]

(iii) Name the blood vessel into which tube **N** is inserted.

..... [1]

(b) Draw arrows on Fig. 4.1 to show

(i) the direction of blood flow in the blood vessels that carry blood into the heart. [2]

(ii) the direction of blood flow through the LV Assist Device. [1]

(c) Name the valve that is bypassed by blood flowing through the LV Assist Device.

.....[1]

(d) Sometimes the pulmonary circulation requires artificial assistance. Suggest and describe where, under these circumstances, an Assist Device would be fitted.

.....

.....

.....

.....[2]

[Total: 9]

4(a)(i)	Left Ventricle / helps blood flow or pressure ;	1	
4(a)(ii)	muscle ;	1	
4(a)(iii)	aorta ;	1	
4(b)(i)	arrow in or towards heart in vena(e) cava(e) ; arrow towards heart in pulmonary vein(s) ;	2	**A** only one vena cava / pulmonary vein annotated **R** if arrows in these vessels contradict
4(b)(ii)	arrow right to left through LVAssist Device / in tube M / N ;	1	**R** if arrows contradict
4(c)	aortic / semi-lunar ;	1	

© UCLES 2016

Question	Answer	Mark	Guidance
4(d)	right ventricle ; pulmonary artery ;	2	
	Total:	9	

Question	Answer	Mark	Guidance

5 Sweet clover is a member of the pea and bean family (leguminous plants) that grows amongst grass in fields used for cattle feed.

(a) Explain how sweet clover can improve the soil in which it grows.

.....

.....

.....

.....[2]

(b) As sweet clover dries, fungi that grow on it produce a chemical (dicoumarol) that prevents blood from clotting.

Suggest a protein that cattle which eat sweet clover affected by such fungi might be unable to produce.

..... [1]

(c) Dicoumarol has been used for many years as a constituent in a form of rat poison.
Fig. 5.1 shows the change in the LD_{50} for dicoumarol in rats from 2000 to 2015.
The LD_{50} is a measure of the amount of poison that will kill 50% of the rats that consume it.

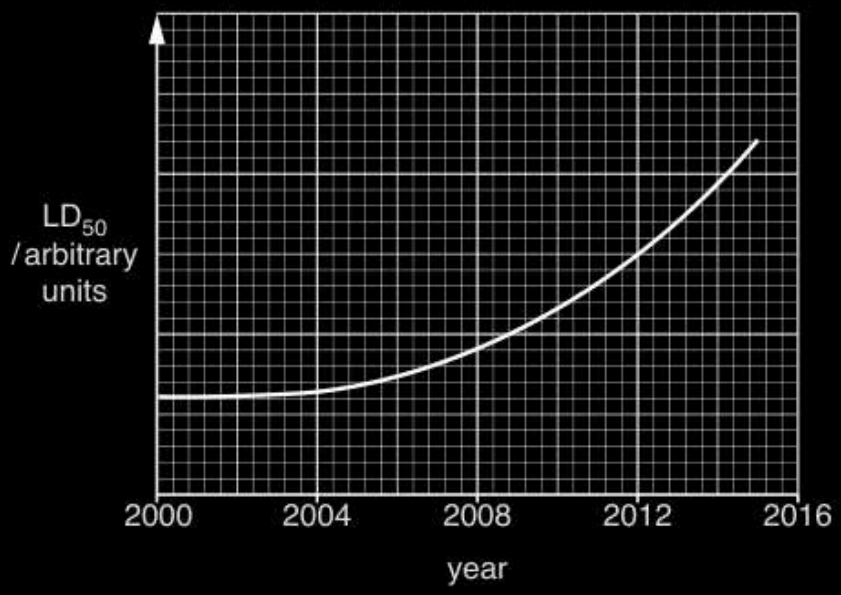


Fig. 5.1

(i) Describe the change in the LD_{50} for dicoumarol in rats from 2000 to 2015.

.....

.....

.....

.....[2]

(ii) Suggest an explanation for the change you have described.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 9]

Section B

Answer **both** questions in this section.

5(a)	1 bacteria (or correctly named) + roots / nodules ; 2 fix / convert / turn + nitrogen ; 3 ploughing / digging into soil ; 4 decay / decompose ; 5 (which produces) nitrates / nitrites / ammonium ;	2	**Ig** ammonia
5(b)	fibrinogen / fibrin ;	1	**A** thrombin / prothrombin / (pro)thrombokinase / factor VIII
5(c)(i)	increase ; no change + (from 2000) to 2002 / 3 / 4 / first 2 years / first 3 years / first four years ; exponential or described / more quickly **AW** ;	2	
5(c)(ii)	1 mutation / change / variation ; 2 reference to genes / DNA ; 3 resistance (to poison) / better adapted ; 4 survival ; 5 breeding / reproduction / produce offspring ; 6 natural selection / evolution ; 7 greater amount (of dicoumarol) now required to kill / used to kill / the LD_{50} must be increased ;	4	
	Total:	9	

© UCLES 2016

Question	Answer	Mark	Guidance
Section B			

6 **(a)** As a person gets older, the lens of the eye loses its elasticity.
Describe and explain how light rays from a near object pass through the eye but may not form a sharply focused image for an older person.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[7]

(b) Suggest how reading glasses may bring the image into focus.

.....

.....

.....

.....

.....

.....

.....[3]

[Total: 10]

6(a)	1	light / rays + bent / refracted / converged ; following points must be related to, or imply, the idea of light / rays (accept image) passing through:	**7**	
	2	cornea ;		
	3	pupil ;		
	4	aqueous / vitreous humour / bodies ;		
	5	lens ;		
	6	(lens has) less ability to change shape **AW** / accommodate ;		
	7	(therefore lens unable to bulge) fully / enough ;		
	8	(lens) cannot refract sufficiently / decrease focal length sufficiently **AW** ;		
	9	(cannot focus on the) retina / fovea / yellow spot ;		
6(b)	1	convex (reference to spectacle lens) **AW** ;	**3**	
	2	(light / rays) bend / refract / converge ;		
	3	before entering eye / (eye) lens ;		
	4	compensate for thinness / lack of convexity or fatness in eye lens ;		
	5	(to focus on the) retina / fovea / yellow spot ;		
		Total:	**10**	

8 **(a)** Describe the similarities and differences between the absorption of ions in plants and the uptake of glucose in humans.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

(b) Explain why humans absorb amino acids but plants do not.

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

8(a)	(similarities) 1 large surface area / both elongated structures **AW** ; 2 active transport / require energy / against concentration gradient ; 3 both in solution ; 4 both require a membrane ; (differences) 5 root hairs ; 6 villi ; 7 root hairs are single-celled ; 8 villi are multicellular **AW** ; 9 through cell walls in plants **ORA** ; 10 ions absorbed from the soil ; 11 glucose absorbed from / in the intestines / digestive system ;	**max 6**	
8(b)	1 plants use nitrates ; 2 (and) carbohydrates / products of photosynthesis ; 3 to make their own amino acids / **ORA** ; 4 humans rely on proteins + from other organisms ; 5 have to be ingested / eaten / consumed ; 6 have to be digested to amino acids ;	**4**	
	Total:	**10**	

© UCLES 2016

Question	Answer	Mark	Guidance

May/June 2016 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 Which two structures are found in all plant epidermal cells?

A cell wall and chloroplasts

B cell wall and nucleus

C chloroplasts and starch grains

D nucleus and starch grains

May/June 2016 · Paper 1 · Variant 1 · Question 2

Correct answer: C

2 Which processes can **only** occur through a membrane?

	active transport	diffusion	osmosis
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key
✓ = yes
✗ = no

May/June 2016 · Paper 1 · Variant 1 · Question 3

Correct answer: A

3 The small intestine of a person contains a lower concentration of glucose than is present in the blood.

The cells of the villi absorb glucose.

By which process is the glucose absorbed?

A by active transport against the concentration gradient

B by active transport down the concentration gradient

C by diffusion against the concentration gradient

D by diffusion down the concentration gradient

May/June 2016 · Paper 1 · Variant 1 · Question 4

Correct answer: D

4 X and Y are the reactants in a chemical reaction for which E is the enzyme. The product is Z.

The first three stages in the reaction are shown.

$$X + E \rightarrow XE$$

$$Y + XE \rightarrow XYE$$

$$XYE \rightarrow ZE$$

What is the fourth, and final, stage of this reaction?

A $X + Y + E \rightarrow Z$

B $X + YE \rightarrow Z$

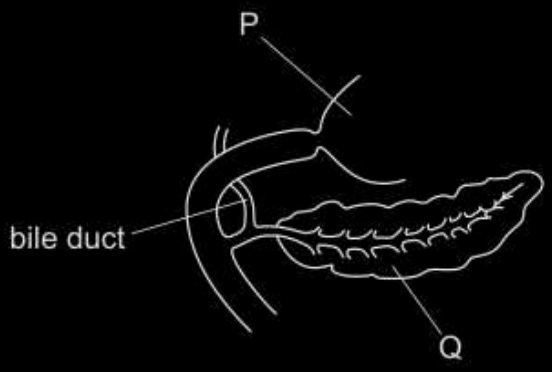
C $Y + ZE \rightarrow YZ + E$

D $ZE \rightarrow Z + E$

May/June 2016 · Paper 1 · Variant 1 · Question 8

Correct answer: D

8 The diagram shows part of the alimentary canal and associated structures.



What are organs P and Q?

	P	Q
A	ileum	gall bladder
B	ileum	pancreas
C	stomach	gall bladder
D	stomach	pancreas

May/June 2016 · Paper 1 · Variant 1 · Question 9

Correct answer: C

9 Which statement about chemical digestion in the human alimentary canal is correct?

A Digestion of carbohydrates is completed in the colon.

B Enzymes are secreted to break down cellulose in the duodenum.

C Protein digestion is completed in the ileum.

D The stomach secretes enzymes to break down starch.

May/June 2016 · Paper 1 · Variant 1 · Question 11

Correct answer: A

11 Which statement best describes the movement of water through the stem of a plant?

A Water moves up the stem as it evaporates from the leaves.

B Water moves up the stem as it is pushed by water entering the roots.

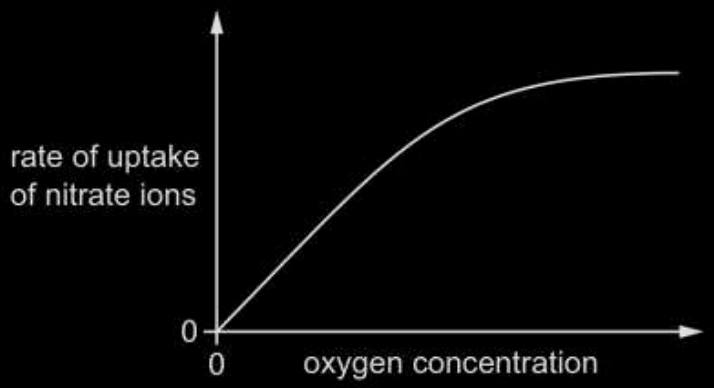
C Water moves up the stem because it is used in photosynthesis.

D Water moves up the stem by osmosis.

May/June 2016 · Paper 1 · Variant 1 · Question 12

Correct answer: A

12 The graph shows the effect of oxygen concentration on the uptake of nitrate ions from the soil into a root hair cell.



What can be concluded from this information?

A Nitrate ions enter the root hair cell by active transport.

B Nitrate ions enter the root hair cell by osmosis.

C Nitrate ions leave the root hair cell by diffusion.

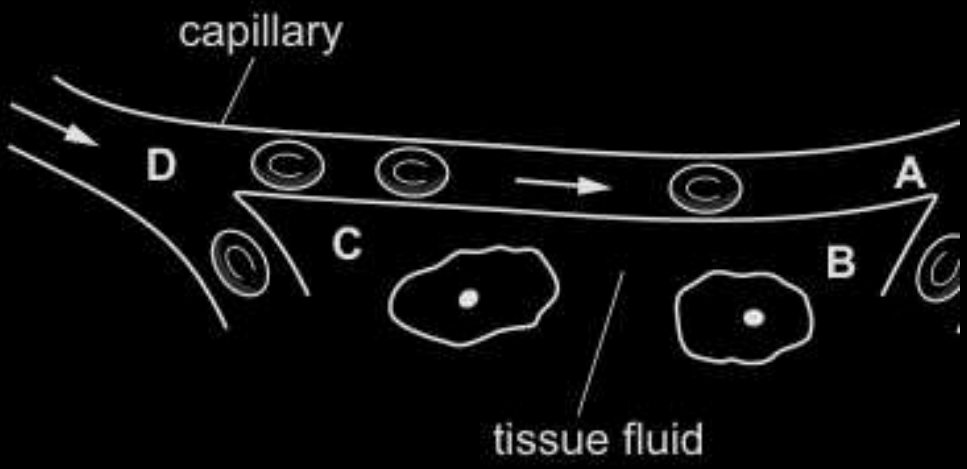
D Nitrate ions leave the root hair cell in low oxygen concentrations.

May/June 2016 · Paper 1 · Variant 1 · Question 13

Correct answer: D

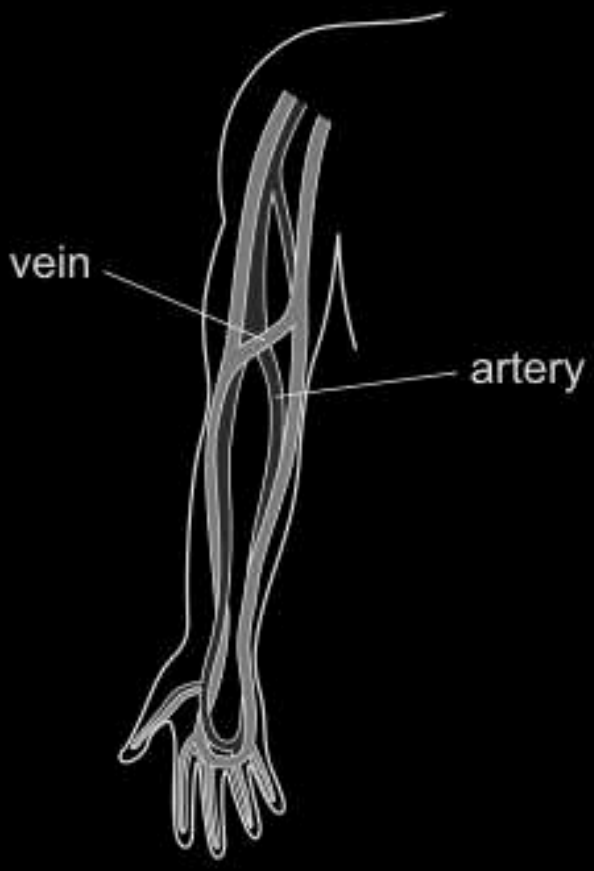
13 The diagram shows the movement of blood through a tissue.

At which labelled point is the pressure highest?



Correct answer: C

14 The diagram shows arteries and veins in the human forearm.



Which statements apply to a vein in the human forearm?

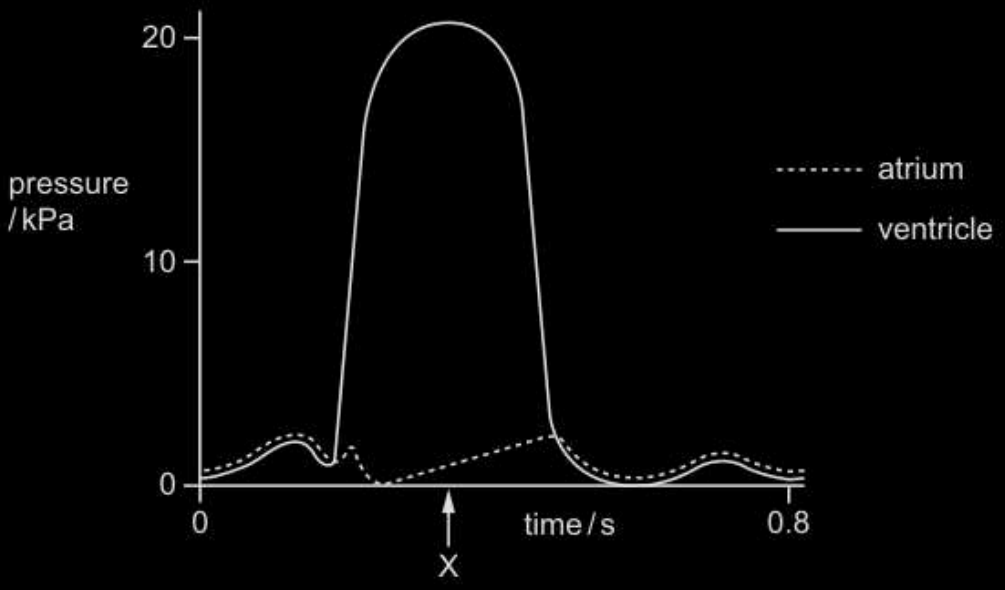
	carries oxygenated blood	has valves	blood is returning to the heart
A	✓	✓	✗
B	✓	✗	✗
C	✗	✓	✓
D	✗	✗	✓

key
✓ = yes
✗ = no

May/June 2016 · Paper 1 · Variant 1 · Question 15

Correct answer: B

15 The graph shows pressure changes in the left atrium and in the left ventricle during one heartbeat.



What is the state of the valves in the heart at time X?

	left atrio-ventricular valve (bicuspid)	semi-lunar valve (in aorta)
A	closed	closed
B	closed	open
C	open	closed
D	open	open

May/June 2016 · Paper 1 · Variant 1 · Question 20

Correct answer: C

20 During a sunny day, which substance is excreted from a leaf?

A carbon dioxide

B nitrogen

C oxygen

D sucrose

May/June 2016 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 Which process is **not** a result of negative feedback?

A A decrease in the surrounding temperature leads to a decrease in respiration rate.

B A decrease in the surrounding temperature leads to a decrease in sweating.

C A decrease in the surrounding temperature leads to a decrease in blood flow through the skin surface.

D A decrease in the surrounding temperature leads to shivering.

May/June 2016 · Paper 1 · Variant 1 · Question 22

Correct answer: D

22 What is an example of a reflex arc?

A retina → motor neurone → relay neurone → sensory neurone → iris muscle

B retina → motor neurone → sensory neurone → relay neurone → iris muscle

C retina → sensory neurone → motor neurone → relay neurone → iris muscle

D retina → sensory neurone → relay neurone → motor neurone → iris muscle

May/June 2016 · Paper 1 · Variant 1 · Question 23

Correct answer: A

23 Which structures cover the pupil at the front of a human eye?

A conjunctiva and cornea

B conjunctiva and sclera

C cornea and retina

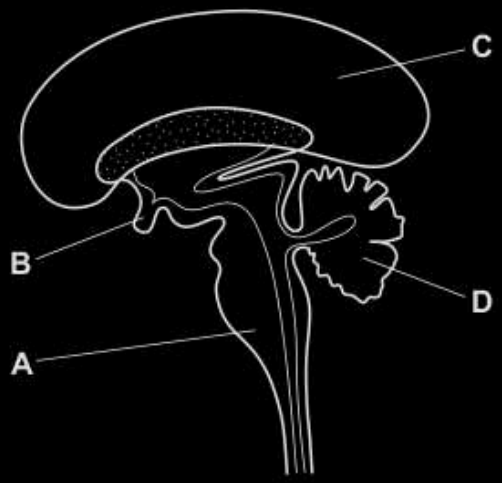
D retina and sclera

May/June 2016 · Paper 1 · Variant 1 · Question 24

Correct answer: D

24 The diagram shows a vertical section through the human brain.

Which structure controls balance and muscular coordination?



May/June 2016 · Paper 1 · Variant 1 · Question 26

Correct answer: A

26 Where is penicillin produced?

A in a fungus

B in bacteria

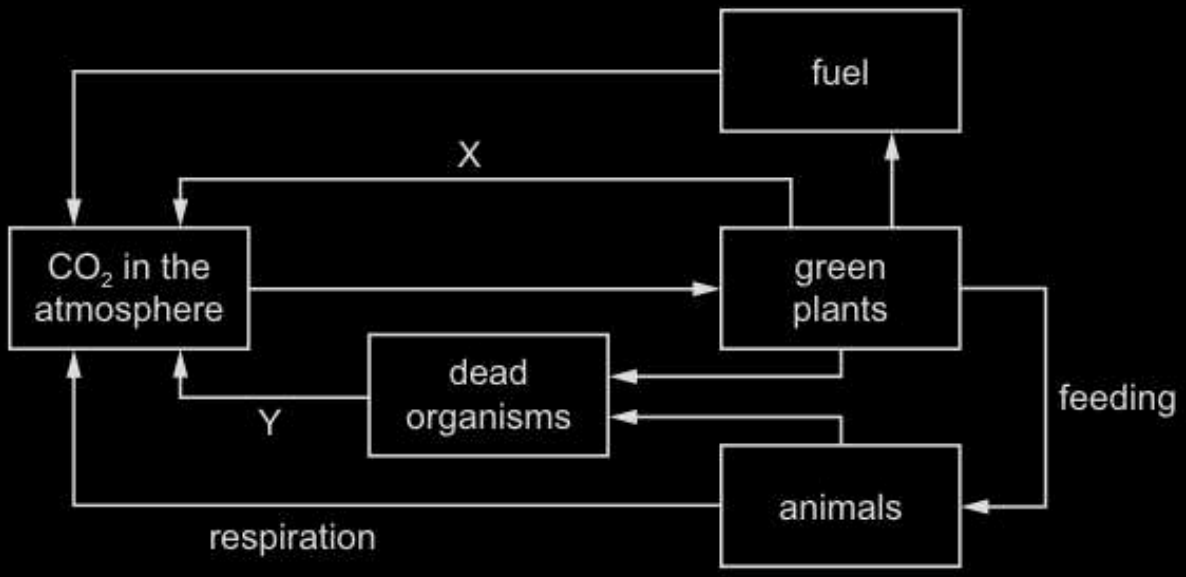
C in the brain

D in the pancreas

May/June 2016 · Paper 1 · Variant 1 · Question 30

Correct answer: D

30 The diagram shows part of the carbon cycle.



What are the processes X and Y?

	X	Y
A	photosynthesis	decomposition
B	photosynthesis	excretion
C	respiration	photosynthesis
D	respiration	respiration

May/June 2016 · Paper 1 · Variant 1 · Question 34

Correct answer: B

34 A diploid cell produces further cells with a haploid number of chromosomes.

Which type of cell division is involved and what is the purpose of the cells produced?

	type of cell division	purpose of cells
A	meiosis	asexual reproduction
B	meiosis	sexual reproduction
C	mitosis	asexual reproduction
D	mitosis	sexual reproduction

May/June 2016 · Paper 1 · Variant 1 · Question 38

Correct answer: A

38 In the ABO blood group system, which alleles are codominant?

A I^A and I^B

B I^A and I^o

C I^B and I^o

D I^A, I^B and I^o

May/June 2016 · Paper 1 · Variant 1 · Question 39

Correct answer: D

39 Two black female mice were mated with the same black male. One female had nine offspring, all of which were black. The other female had six black and two white offspring.

Which cross represents the parents of the all black family?

	female	male
A	bb	Bb
B	Bb	Bb
C	Bb	BB
D	BB	Bb

key

B = allele for black

b = allele for white

May/June 2016 · Paper 1 · Variant 1 · Question 40

Correct answer: A

40 Which statement is evidence that genes are copied and passed on to the next generation?

A Asexual reproduction produces genetically identical offspring.

B Different alleles of a gene can produce variation in phenotype.

C Each species of a plant or animal has a fixed number of chromosomes.

D Sexual reproduction produces genetically different offspring.

May/June 2016 · Paper 1 · Variant 2 · Question 1

Correct answer: A

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

May/June 2016 · Paper 1 · Variant 2 · Question 2

Correct answer: C

2 Which processes can **only** occur through a membrane?

	active transport	diffusion	osmosis
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key
✓ = yes
✗ = no

May/June 2016 · Paper 1 · Variant 2 · Question 3

Correct answer: B

3 The diagrams show a cylindrical net packed with rubber balloons full of air. The structure is used by a teacher to explain wilting.

all the balloons fully inflated

the same number of balloons with some of the air let out

What is represented by the parts of the structure shown?

	air	balloons	net	rubber
A	cells	cell sap	cell walls	epidermis
B	cell sap	cells	epidermis	cell walls
C	cell walls	epidermis	cell sap	cells
D	epidermis	cell walls	cells	cell sap

May/June 2016 · Paper 1 · Variant 2 · Question 4

Correct answer: C

4 Amylase solution is tested with Benedict's solution, biuret solution and iodine solution.

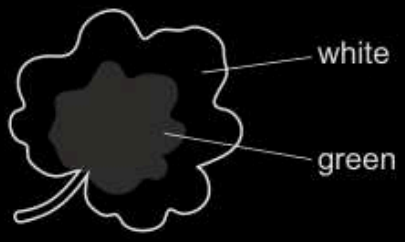
Which colours are obtained?

	Benedict's solution	biuret solution	iodine solution
A	blue	blue	blue-black
B	blue	blue	brown
C	blue	purple	brown
D	red	purple	blue-black

May/June 2016 · Paper 1 · Variant 2 · Question 6

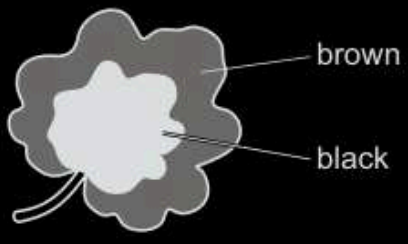
Correct answer: A

6 A leaf was picked from a plant that had been in sunlight for eight hours.

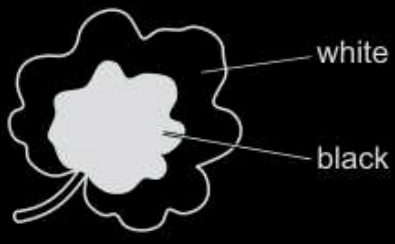


What does this leaf look like after it is tested with iodine solution?

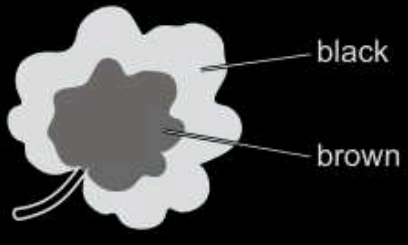
A



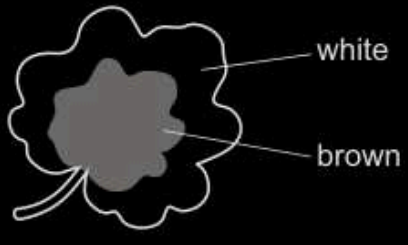
B



C



D



May/June 2016 · Paper 1 · Variant 2 · Question 8

Correct answer: B

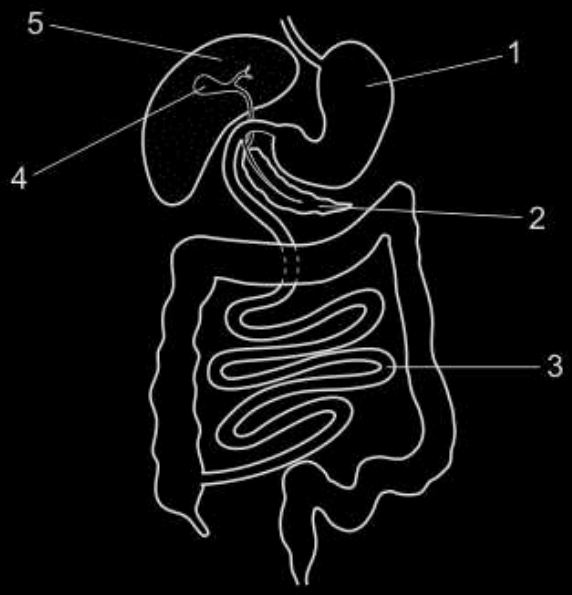
8 What is correct for the formation of urea?

	formed from	formed in
A	amino acids	kidney
B	amino acids	liver
C	carbohydrate	kidney
D	carbohydrate	liver

May/June 2016 · Paper 1 · Variant 2 · Question 9

Correct answer: A

9 The diagram shows part of the alimentary canal and associated organs.



Which row correctly describes the functions of parts shown in the diagram?

	structure	function	structure	function
A	1	digestion of protein	3	absorption of the products of digestion
B	2	emulsifying fats	3	absorption of amino acids and glucose
C	4	production of bile	5	making digestive enzymes
D	4	storing digestive enzymes	2	making digestive enzymes

May/June 2016 · Paper 1 · Variant 2 · Question 11

Correct answer: A

11 What shows how the rate of transpiration changes when conditions in the atmosphere change?

	reduced wind	increased humidity
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

May/June 2016 · Paper 1 · Variant 2 · Question 12

Correct answer: C

12 What is the main cause of water moving up to the leaves in xylem vessels?

A active transport

B evaporation from the epidermis of the leaf

C evaporation from the walls of the mesophyll cells

D use of water in photosynthesis

May/June 2016 · Paper 1 · Variant 2 · Question 13

Correct answer: D

13 Some factors associated with coronary heart disease are listed.

1 high blood pressure

2 high intake of fruit and vegetables

3 high intake of saturated fats

4 low blood cholesterol

5 low intake of processed foods

Which factors **decrease** the risk of coronary heart disease?

A 1, 2 and 3 **B** 1, 3 and 5 **C** 2, 3 and 4 **D** 2, 4 and 5

May/June 2016 · Paper 1 · Variant 2 · Question 14

Correct answer: C

14 Which statements about arteries are correct?

1 All arteries carry oxygenated blood.

2 Arteries carry blood at high pressure.

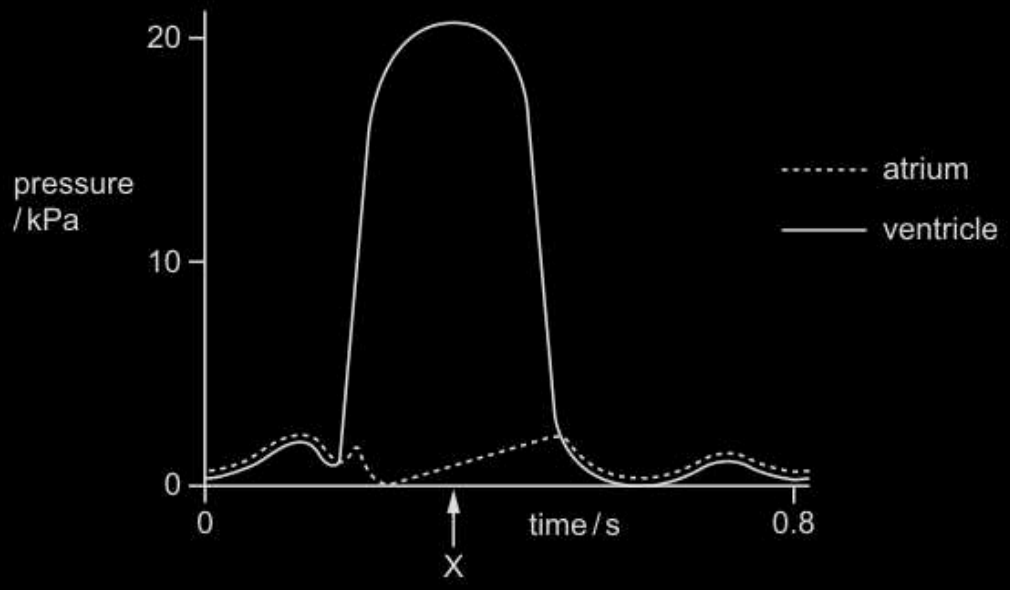
3 All arteries carry blood away from the heart.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

May/June 2016 · Paper 1 · Variant 2 · Question 15

Correct answer: B

15 The graph shows pressure changes in the left atrium and in the left ventricle during one heartbeat.



What is the state of the valves in the heart at time X?

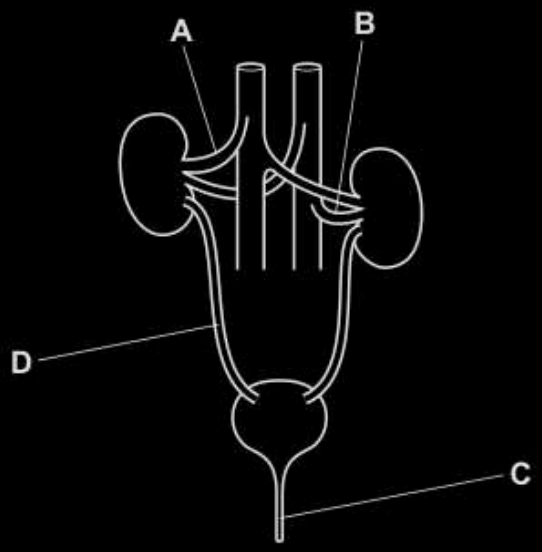
	left atrio-ventricular valve (bicuspid)	semi-lunar valve (in aorta)
A	closed	closed
B	closed	open
C	open	closed
D	open	open

May/June 2016 · Paper 1 · Variant 2 · Question 20

Correct answer: D

20 The diagram shows the human urinary system.

Which labelled structure is the ureter?



May/June 2016 · Paper 1 · Variant 2 · Question 21

Correct answer: A

21 Which process is **not** a result of negative feedback?

A A decrease in the surrounding temperature leads to a decrease in respiration rate.

B A decrease in the surrounding temperature leads to a decrease in sweating.

C A decrease in the surrounding temperature leads to a decrease in blood flow through the skin surface.

D A decrease in the surrounding temperature leads to shivering.

May/June 2016 · Paper 1 · Variant 2 · Question 22

Correct answer: D

22 What is an example of a reflex arc?

A retina → motor neurone → relay neurone → sensory neurone → iris muscle

B retina → motor neurone → sensory neurone → relay neurone → iris muscle

C retina → sensory neurone → motor neurone → relay neurone → iris muscle

D retina → sensory neurone → relay neurone → motor neurone → iris muscle

May/June 2016 · Paper 1 · Variant 2 · Question 23

Correct answer: D

23 What is a role of adrenaline, and where is it destroyed in the body?

	role of adrenaline	destroyed by
A	glycerol changed to glucose	kidney
B	glycerol changed to glucose	liver
C	glycogen changed to glucose	kidney
D	glycogen changed to glucose	liver

May/June 2016 · Paper 1 · Variant 2 · Question 24

Correct answer: D

24 The diagram shows a vertical section through the human brain.

Which structure controls balance and muscular coordination?

May/June 2016 · Paper 1 · Variant 2 · Question 37

Correct answer: A

37 Which statements describe an allele?

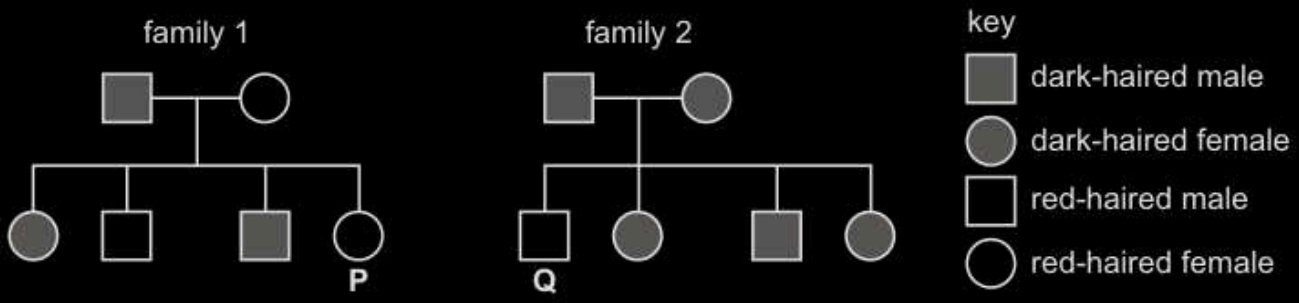
	an alternative form of a gene	copied during cell division	part of a DNA molecule
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key
✓ = yes
✗ = no

May/June 2016 · Paper 1 · Variant 2 · Question 39

Correct answer: B

39 The diagram shows the pattern of inheritance of dark hair and red hair in two families.



If individuals **P** and **Q** have children together, which prediction can be made about the hair colour of these children?

A All the children will have dark hair.

B All the children will have red hair.

C Half the children will have dark hair.

D Three-quarters of the children will have dark hair.

May/June 2016 · Paper 1 · Variant 2 · Question 40

Correct answer: A

40 Which statement is evidence that genes are copied and passed on to the next generation?

A Asexual reproduction produces genetically identical offspring.

B Different alleles of a gene can produce variation in phenotype.

C Each species of a plant or animal has a fixed number of chromosomes.

D Sexual reproduction produces genetically different offspring.

1 Fig. 1.1 shows a plant with a magnified transverse section (TS) of the root and an incomplete, magnified transverse section (TS) of the stem.

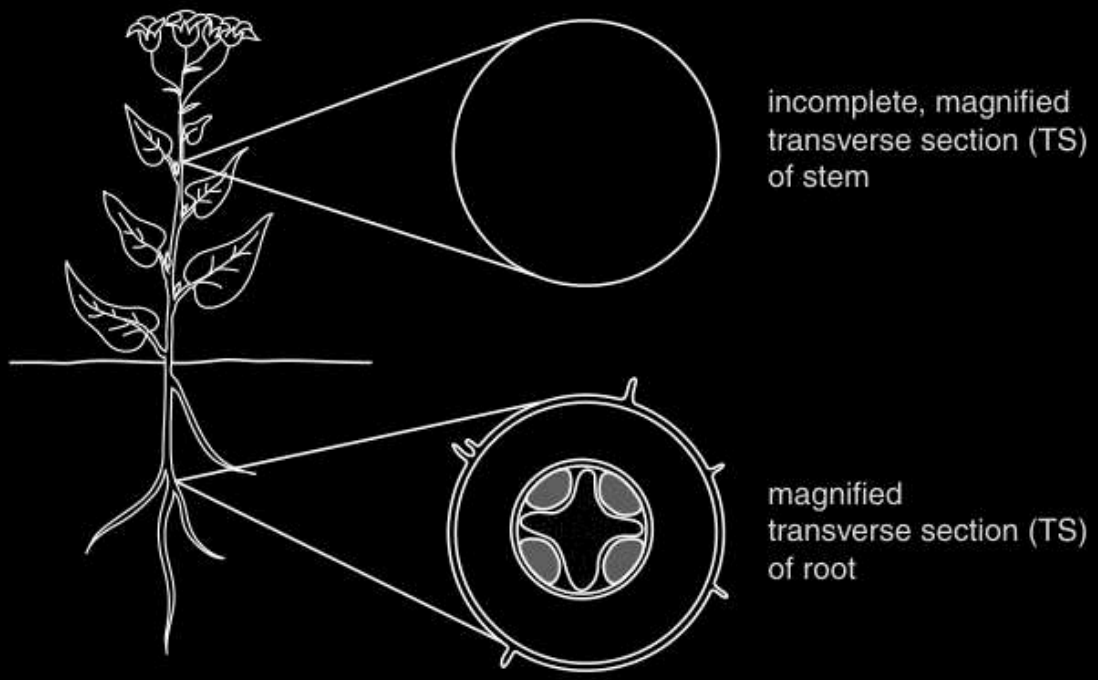


Fig. 1.1

(a) The transverse section of the root in Fig. 1.1 shows the distribution of xylem and phloem tissue.

(i) Using a labelling line and the letter **X**, show, on Fig. 1.1, the location of the xylem tissue in the transverse section (TS) of the root. [1]

(ii) Complete the transverse section (TS) of the stem in Fig. 1.1 to show the distribution of xylem and phloem tissue. [2]

(iii) Using a label line and the letter **P**, show the location of the phloem tissue that you have drawn. [1]

(b) State the functions of phloem tissue.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 7]

1 (a) (i)	Label line **X** on xylem on transverse section of root ;		[1]
(ii)	vascular bundle drawn towards outside of transverse section of stem ; vascular bundle divided into two sections ; three or more vascular bundles drawn ;		[max 2]
(iii)	label line **P** drawn in correct location (outer side of vascular bundle) ;		[1]
(b)	translocation ; transport ; dissolved / in solution ; sugar(s) / sucrose / products of photosynthesis ; amino acids ; reference from source **+** to sink / leaves to roots **AW** ;		[max 3]
			[Total: 7]

Question	Expected Answer	Additional Guidance	Mark

3 Digestion in the human alimentary canal is carried out by the action of enzymes. Each food group is the substrate for a specific enzyme.

(a) Fig. 3.1 shows diagrams that each represent the action of a specific enzyme to break down a substrate into one or more end products.

Diagram **F** in Fig. 3.1 has been completed for you.

Complete diagrams **G** and **H** in Fig. 3.1.

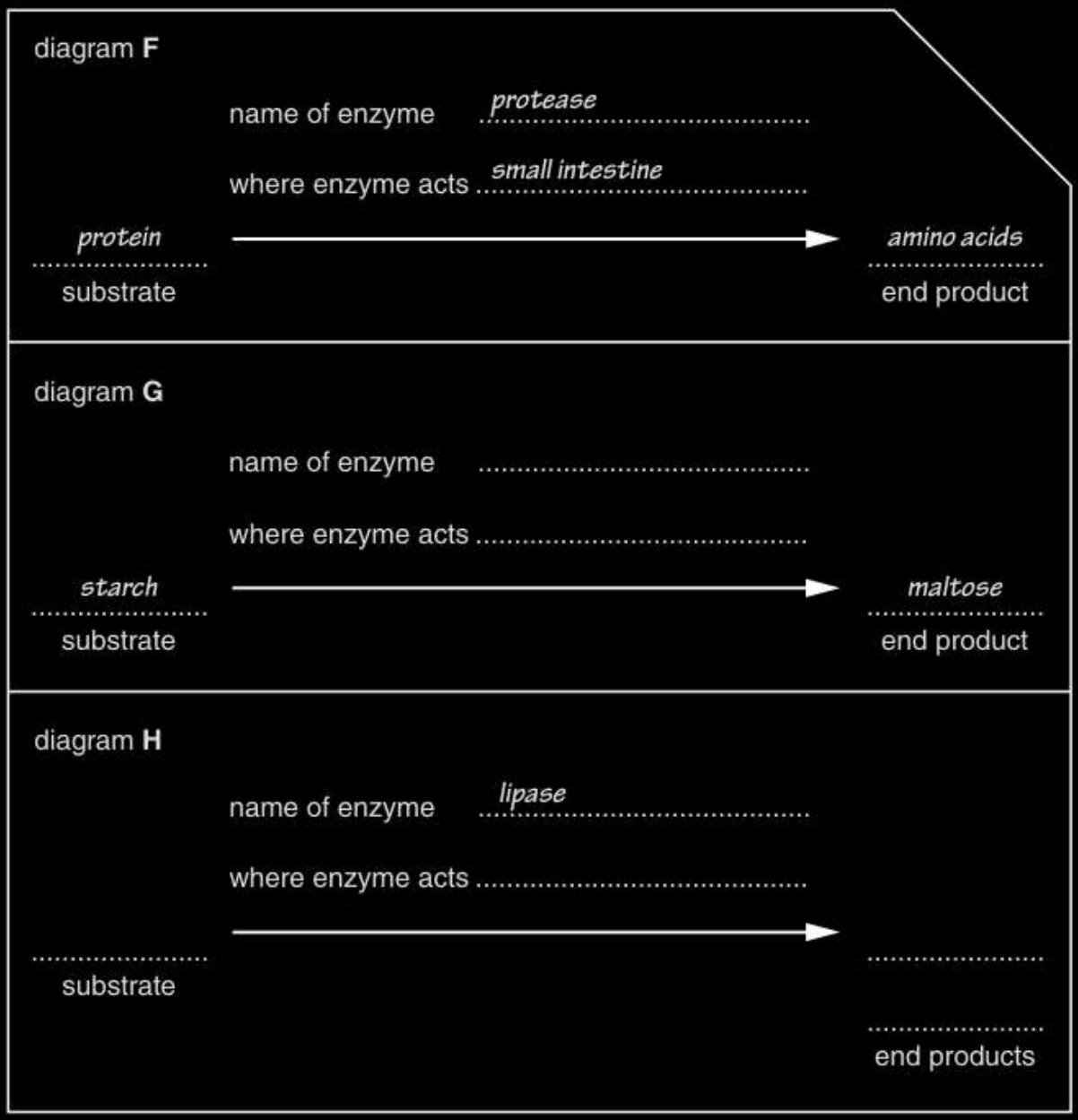


Fig. 3.1

[6]

(b) Amino acids are the end products of protein digestion. Describe what happens to these amino acids from the point of digestion until they reach the liver.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

3	(a)	*diagram* ***G:*** amylase ; mouth / small intestine / duodenum ; *diagram* ***H:*** small intestine / duodenum ; fat / lipid / oil ; fatty acid (s) ; glycerol ;		[6]
	(b)	reference to absorption / diffusion ; villi ; capillary ; blood / plasma ; reference to hepatic portal vein ;		[max 4]
				[Total: 10]

Question	Expected Answer	Additional Guidance	Mark

6 Enzyme **Q** is active in the human alimentary canal. Fig. 6.1 shows the effect of pH on the rate of reaction of enzyme **Q**.

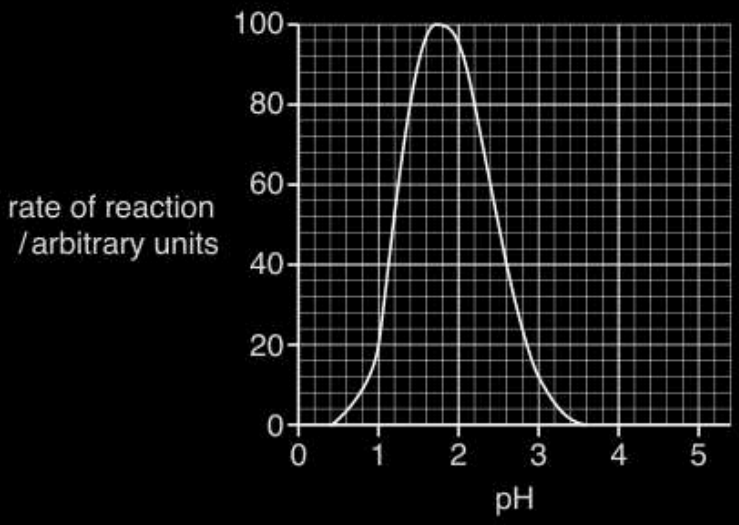


Fig. 6.1

(a) (i) Use the information in Fig. 6.1 to name the region of the alimentary canal where enzyme **Q** is active.

…….. [1]

(ii) Use your knowledge of the 'lock and key' hypothesis of enzyme action to explain why enzyme **Q** is active only in this region of the alimentary canal and not in any other region.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) All enzymes contain carbon. List **three** other chemical elements that must be present in each molecule of an enzyme.

1 .....

2 .....

3 ..... [3]

[Total: 10]

6	**(a)** **(i)**	stomach ;		[1]
	(ii)	enzyme and substrate are lock and key ; reference to optimum pH / (hydrochloric) acid (in stomach) ; active site ; complementary **AW** ; substrate ; if pH changes / alkaline ; enzyme denatured / changes shape **AW** ; (substrate) no longer fits ;		[max 6]
	(b)	hydrogen ; oxygen ; nitrogen ;		[3]
				[Total: 10]

Question	Expected Answer	Additional Guidance	Mark

8 The brain is one component of the nervous system.

(a) State **two** other components of the nervous system.

1 .....

2 .....[2]

(b) Fig. 8.1 shows the human brain.

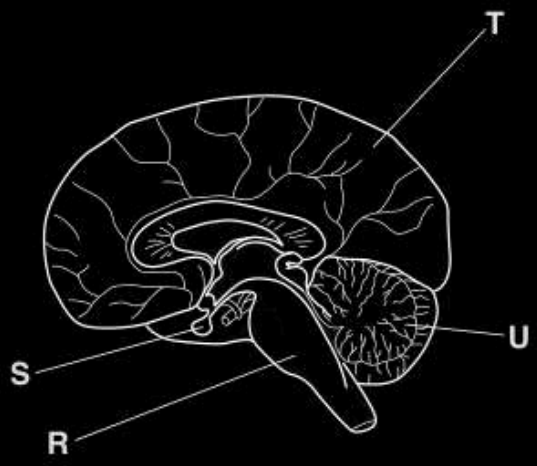


Fig. 8.1

Name and describe the functions of parts **R**, **S**, **T** and **U**, labelled in Fig. 8.1, in terms of coordinating and regulating bodily functions.

R .....

.....

.....

.....

.....

S .....

.....

.....

.....

.....

T .....

.....

.....

.....

.....

U .....

.....

.....

.....

..... [8]

[Total: 10]

8 (a)	spinal cord ; nerves / neurones ; named receptor / sense organ ;		[2]
(b)	**R +** medulla ; control of heart rate / breathing / involuntary actions ; relay of impulses between brain and spinal cord ; **S +** pituitary gland ; secretes / releases hormone(s) ; example of named hormone released **+** correct function ; **T +** cerebrum / cerebral hemispheres ; voluntary movement or example of ; thought / memory / any other acceptable function ; **U +** cerebellum ; coordination of movement ; maintenance of posture / balance ;	**A** functions only if linked to correct named part of the brain e.g. TSH released which stimulates the thyroid gland to secrete thyroxine.	[max 8]
			[Total: 10]

Question	Expected Answer	Additional Guidance	Mark

2 Fig. 2.1 shows a fetus developing inside its mother.

Fig. 2.1

(a) Complete Table 2.1, using letters from Fig. 2.1, to identify each of the following:

Table 2.1

	letter
a structure that contains urea released by the fetus	.....
a region that contains cells, almost all of which could be used for determining the sex of the fetus	.....
a structure that is used to expel the fetus at birth	.....
a temporary structure that would have been expelled during menstruation if pregnancy had not occurred	.....

[4]

(b) Every day, babies are born suffering from severe withdrawal symptoms as a result of their mothers taking drugs during pregnancy.

(i) Name a drug which the mother may have taken during pregnancy that might cause these withdrawal symptoms.

..... [1]

(ii) Suggest how drugs taken by the mother have been able to affect her developing fetus.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 9]

2	**(a)**	C / D / E ; C / D ; F ; E ;		[4]
	(b) **(i)**	heroin / alcohol / nicotine / named addictive drug ;		[1]
	(ii)	**Ig** reference to named drug / substance (e.g. CO, tar) mother's blood ; diffusion ; across or through placenta ; fetus / baby / embryo + blood ; umbilical cord / umbilical vein ;	**R** umbilical artery	[max 4]
				[Total 9]

3 Table 3.1 is a list of some of the constituents of a healthy human diet.

Table 3.1

constituent
carbohydrates
fats
proteins
mineral salts
fibre / roughage

(a) State one of these constituents that

(i) does not require digestion, ...[1]

(ii) is the body's main storage substance. .....[1]

(b) Name **two** dietary constituents that are missing from the list in Table 3.1 and for each state **two** reasons for its importance in the diet.

constituent: .....

1 .....

.....

2 .....

.....

constituent: .....

1 .....

.....

2 .....

.....[6]

(c) The chimpanzee is an animal whose metabolism is very similar to that of a human, but one type of chimpanzee lacks the gene responsible for the manufacture of amylase.

Suggest how this will affect the diet of this type of chimpanzee.

.....

.....

.....[2]

[Total: 10]

3 (a) (i)	mineral salts / fibre / roughage ;		[1]
(ii)	fat / lipid ;		[1]

Question	Expected answers	Additional guidance	Marks
(b)	*constituent:* <u>water</u> ; *reasons for importance:* solvent / constituent of urine / reference to chemical reactions / constituent cells OR cyto-, proto-plasm OR blood / temperature regulator OR sweating / transporter / osmoregulator AW / prevents dehydration / lubrication / prevents constipation ;;		[1] [max 2]
	constituent: (named) vitamin ; *reasons for importance:* prevent deficiency disease AW / reference to correct specified benefit of any vitamin ;;		[1] [max 2]
(c)	little or no <u>starch</u> in diet / cannot digest <u>starch</u> AW ; relies on sugar or named sugar AW ; (needs to) eat fruit / animals / fewer plants ; may rely on fat / protein (for energy) ;		[max 2]
			[Total 10]

5 Table 5.1 shows a number of processes that occur in the lungs and thorax (chest cavity).

Table 5.1

process involved	✓
diffusion into red blood cells	
diffusion into the alveoli	
the diaphragm contracts	
the diaphragm relaxes	
the external intercostal muscles contract	
the internal intercostal muscles relax	
the ribs rise	
the ribs fall	
pressure in the thorax increases	
pressure in the thorax decreases	

(a) Carbon dioxide arrives at the lungs in capillaries. Place a tick (✓) in each box against a process involved in removing carbon dioxide from the blood and expelling it to the atmosphere. [4]

(b) Table 5.2 shows the percentage of oxygen in inspired and expired air of three people, J, K and L.

Table 5.2

person	% of oxygen in inspired air	% of oxygen in expired air
J	20.5	15.0
K	20.5	16.5
L	20.5	18.5

Describe the differences shown in Table 5.2 and suggest reasons for them.

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 8]

Section B

Answer **both** questions in this section.

5 (a)	✓ diffusion into the alveoli *(box 2)* ; ✓ the diaphragm relaxes *(box 4)* ; ✓ the ribs fall *(box 8)* ; ✓ pressure in the thorax increases *(box 9)* ;		[4]
(b)	1. any stated difference between inspired and expired % O_2 ; 2. O_2 used in respiration ; 3. **Person J** – the most / more (than normal) O_2 absorbed / used ; 4. **Person J** - active / taking exercise / athlete / pregnant / high respiratory rate / high blood cell or red blood cell count AW ; 5. **Person K** – moderate activity / normal ; 6. **Person L** – low O_2 absorption / use ; 7. **Person L** – (named) lung disease / anaemia / smoker / inactive / sleeping / elderly / dying / low respiratory rate / reference to low red blood cell count / carboxyhaemoglobin ;	**A** any disease that would restrict O_2 uptake	[4]
			[Total 8]

Question	Expected answers	Additional guidance	Marks

6 **(a)** Describe the differences in structure and function between a cell wall and a cell membrane.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

(b) Explain, with examples, the relationship between cells, tissues and organs.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

6 (a)	*structure:* 1. cell wall thick / cell membrane thin ; 2. cellulose in cell wall ; 3. fat / protein in cell membrane ; 4. cell wall surrounds OR protects cell membrane **ORA** ; *function:* 5. wall <u>permeable</u> + membrane semi-<u>permeable</u> AW ; 6. cell wall no control / cell membrane has control over what enters cell ; 7. cell wall involved in turgor / support / protection / shape / prevents bursting ; 8. osmosis (only) through membrane ; 9. active transport (only) through membrane ;		[max 6]
(b)	1. (cell) the unit of life AW ; 2. tissues are made up of cells AW ; 3. cells in tissues have common / specific function ; 4. organs are made up of tissues AW ; 5. working / combining together ; 6. one example each of a named <u>cell</u> identified as such + a named <u>organ</u> identified as such ;	**A for tissue examples including:** blood, muscle, nervous, epithelial, connective, xylem, phloem, palisade, epidermis **A for organ examples including:** muscle, heart, leaf, flower, root, stem	[max 3] [1]
			[Total 10]

Question	Expected answers	Additional guidance	Marks

8 **(a)** Describe the industrial manufacture of single cell protein.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[8]

(b) Suggest problems of using viruses in biotechnology.

.....

.....

.....

.....[2]

[Total: 10]

8 (a)	1. sterilised / aseptic technique ; 2. <u>fermenter</u> ; 3. bacterium / fungus / algae / named example ; 4. substrate / cultured medium or named / broth / any two chemicals in the medium ; 5. oxygen / air ; 6. bubbles / sparger / aerator / paddle / stirrer AW ; 7. temperature regulation / control / cooling jacket ; 8. pH ref ; 9. optimum or best for growth or reproduction of organism ; 10. filtration / collection / harvesting / separating (the product) ; 11. name / use of product e.g. mycoprotein / meat substitute / cattle feed ;		[max 8]
(b)	1. size reference / extremely small AW ; 2. reproduce only in living cells / pathogenic AW / parasitic ; 3. specific ; 4. may need to separate them from living tissue / difficult to isolate ;	**A** named viral disease	[max 2]
			[Total 10]

9 **(a)** With reference to the inheritance of blood groups, explain what is meant by each of the following terms:

(i) *dominance*

.....

.....

.....

.....

.....

.....

.....[3]

(ii) *codominance.*

.....

.....

.....

.....

.....

.....

.....[3]

(b) Explain the difference between the number of chromosomes in a gamete and in a body cell.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

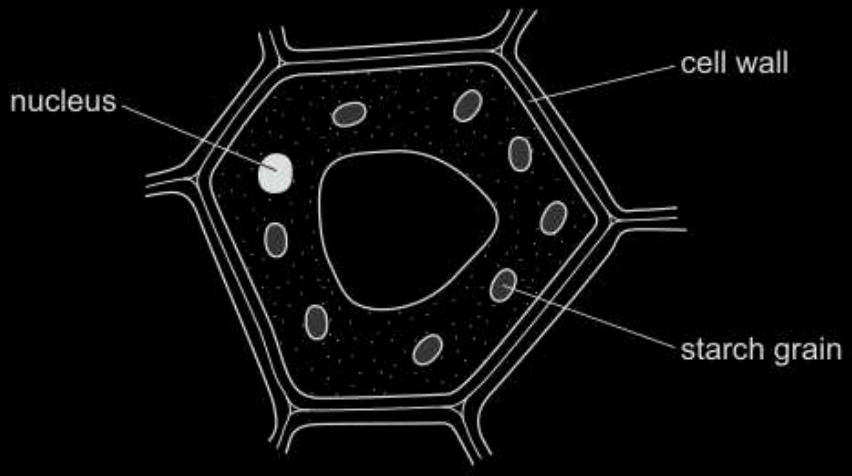
9	(a) (i)	1. allele(s) ; 2. responsible for a character(istic) / trait ; 3. only one needs to be present / reference to heterozygote AW ; 4. for the character to appear / be expressed (in phenotype) AW ; 5. e.g. I^AI^O + group A / I^BI^O + group B / I^AI^B + dominant to I^O ;	**A** *A, B, O instead of* I^A, I^B, I^O	[max 3]
	(ii)	1. <u>two</u> alleles ; 2. neither being recessive / (equally) dominant ; 3. both have an effect / are expressed / phenotype intermediate ; 4. reference to heterozygote AW ; 5. I^AI^B / *AB* ;		[max 3]

Question	Expected answers	Additional guidance	Marks
(b)	1. number of chromosomes in a gamete is half those in a somatic or body cell **ORA** / one versus two sets of chromosomes ; 2. correct use of the terms haploid + diploid ; 3. 46 v. 23 ; 4. gametes / haploid cells are the result of meiosis / reduction division ; 5. somatic / body cells occur in / produced by mitosis ; 6. (diploid) number restored + at fertilisation ; 7. each parent has equal share in genotype of offspring AW ;	**A** sex cell = gamete / one named gamete	[max 4]
			[Total 10]

October/November 2015 · Paper 1 · Variant 1 · Question 1

Correct answer: C

1 The diagram shows a plant cell. The cell is stained with iodine solution.



After staining with iodine solution, what are the colours of the cell wall and the starch grain?

	cell wall	starch grain
A	blue-black	blue-black
B	blue-black	orange-brown
C	orange-brown	blue-black
D	orange-brown	orange-brown

October/November 2015 · Paper 1 · Variant 1 · Question 2

Correct answer: A

2 What describes the diffusion of molecules?

A movement from a region of their higher concentration to a region of their lower concentration down a concentration gradient

B movement from a region of their higher concentration to a region of their lower concentration up a concentration gradient

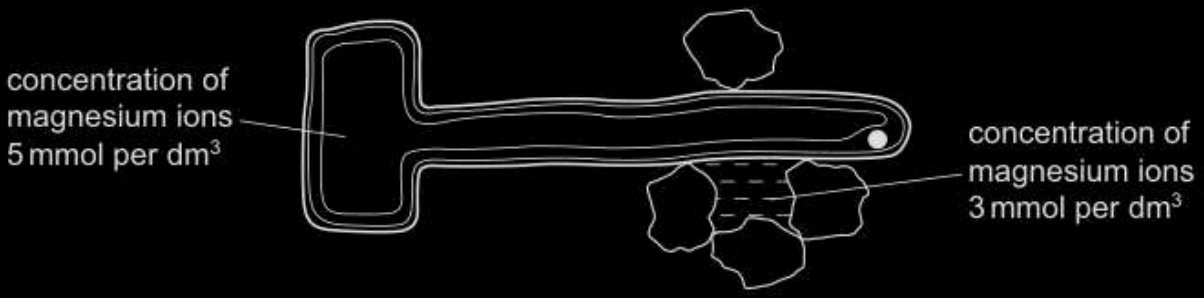
C movement from a region of their lower concentration to a region of their higher concentration down a concentration gradient

D movement from a region of their lower concentration to a region of their higher concentration up a concentration gradient

October/November 2015 · Paper 1 · Variant 1 · Question 3

Correct answer: A

3 The diagram shows the concentration of magnesium ions in a healthy root hair cell of a plant and in the soil water surrounding it.



For the plant to remain healthy, how will the magnesium ions move?

A into the cell by active transport

B into the cell by diffusion

C out of the cell by active transport

D out of the cell by diffusion

October/November 2015 · Paper 1 · Variant 1 · Question 4

Correct answer: D

4 What is the enzyme that controls a reaction in which both the enzyme and the substrate can denature at high temperatures?

A amylase

B insulin

C lipase

D protease

October/November 2015 · Paper 1 · Variant 1 · Question 5

Correct answer: B

5 Which part of a leaf contains cells, but with the fewest chloroplasts?

A cuticle

B epidermis

C spongy mesophyll

D stomata

Correct answer: A

6 Which features of the mesophyll layer in a leaf aid rapid diffusion of carbon dioxide into the cells?

	air spaces	layer of moisture on the outside of the cell walls
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = aids diffusion

✗ = does not aid diffusion

Correct answer: C

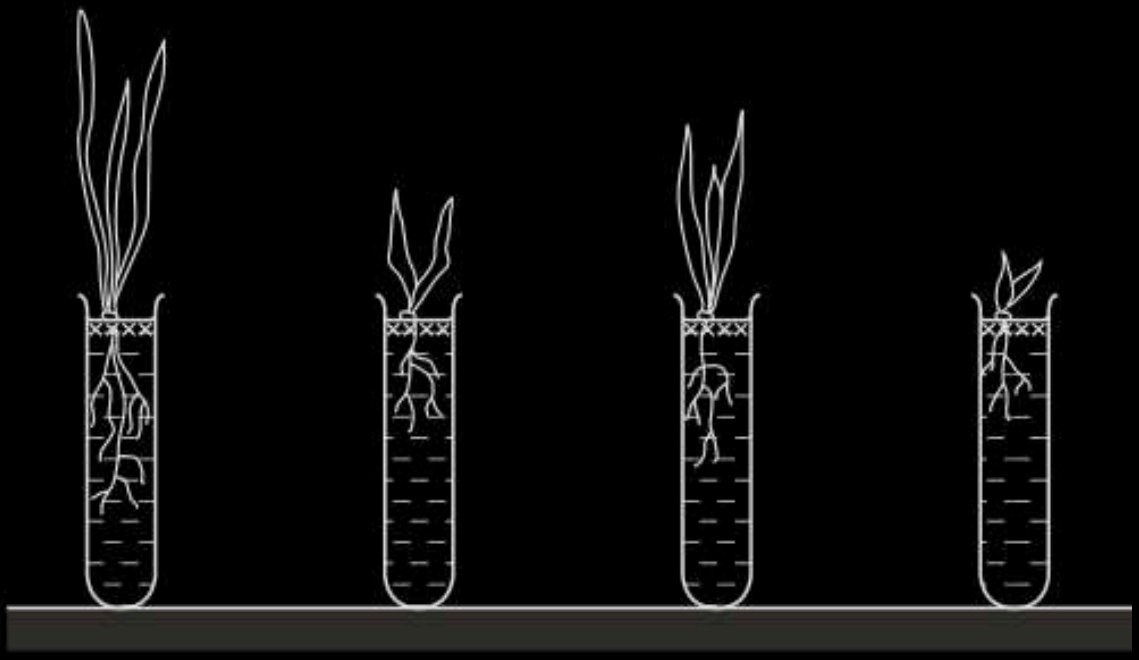
7 A student grows seedlings in four different test-tubes.

Tube W contains all the mineral ions needed for healthy plant growth.

The other three tubes contain

- water
- all minerals except magnesium ions
- all minerals except nitrates.

The diagram shows the appearance of these seedlings after two weeks.

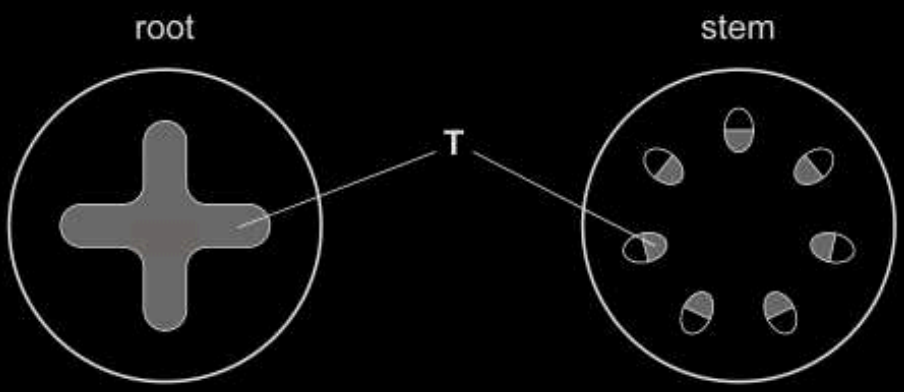


What do tubes X, Y and Z contain?

	X	Y	Z
A	all minerals except magnesium ions	all minerals except nitrates	water
B	all minerals except magnesium ions	water	all minerals except nitrates
C	all minerals except nitrates	all minerals except magnesium ions	water
D	water	all minerals except magnesium ions	all minerals except nitrates

Correct answer: D

11 The diagram shows cross-sections through the root and stem of the same plant.



What is tissue **T**?

A epidermis

B mesophyll

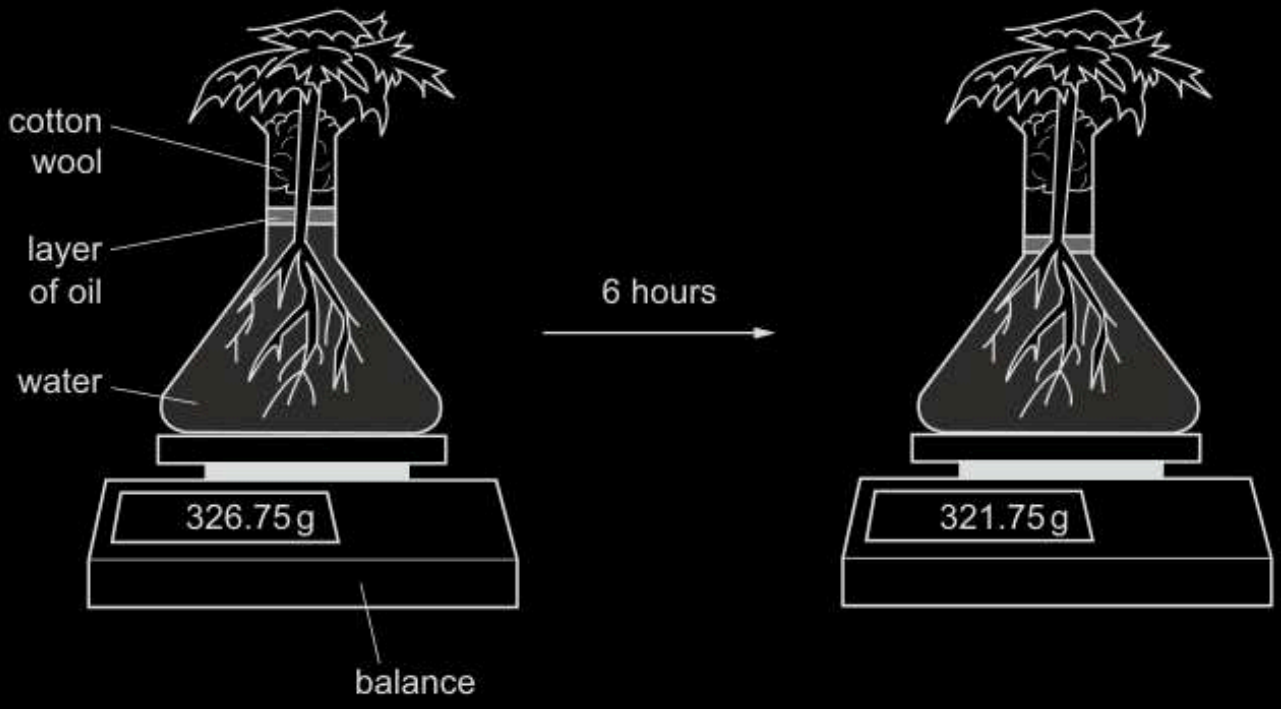
C phloem

D xylem

October/November 2015 · Paper 1 · Variant 1 · Question 12

Correct answer: D

12 The diagrams show a plant in a flask of water. It is left in the light at 16 °C for six hours.



What explains the change in mass after six hours?

A absorption of water into the root hairs

B evaporation of water from the flask

C photosynthesis in the leaves of the plant

D transpiration from the leaves of the plant

October/November 2015 · Paper 1 · Variant 1 · Question 13

Correct answer: C

13 Which organ receives blood from two major vessels and all of this blood leaves it in a single major vessel?

A heart

B kidney

C liver

D lung

October/November 2015 · Paper 1 · Variant 1 · Question 14

Correct answer: B

14 In the human circulatory system, what causes the transfer of materials from the capillaries to the tissue fluid?

A active transport

B blood pressure

C capillarity

D osmosis

October/November 2015 · Paper 1 · Variant 1 · Question 15

Correct answer: A

15 Which chamber of the heart would be the first to receive nicotine absorbed into the blood in the lungs of a cigarette smoker?

A left atrium

B left ventricle

C right atrium

D right ventricle

October/November 2015 · Paper 1 · Variant 1 · Question 19

Correct answer: B

19 A person drinks a large amount of water.

How does this affect the water potentials of the liquids in the renal arteries, the renal veins and the ureters?

	water potential in renal arteries	water potential in renal veins	water potential in ureters
A	higher	higher	higher
B	higher	little change	higher
C	lower	lower	little change
D	lower	lower	lower

October/November 2015 · Paper 1 · Variant 1 · Question 20

Correct answer: A

20 What are the responses to decreased body temperature?

A shivering and vasoconstriction in blood vessels near the skin surface

B shivering and vasodilation in blood vessels near the skin surface

C sweating and vasoconstriction in blood vessels near the skin surface

D sweating and vasodilation in blood vessels near the skin surface

October/November 2015 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 What are the symptoms of diabetes?

	concentration of glucose	
	in blood	in urine
A	+	+
B	+	–
C	–	+
D	–	–

key

+ = increased

– = decreased

October/November 2015 · Paper 1 · Variant 1 · Question 22

Correct answer: D

22 Which row correctly describes the connections of the neurones in a reflex action?

	receptor to central nervous system	between neurones	central nervous system to effector
A	motor neurone	relay neurone	sensory neurone
B	motor neurone	sensory neurone	relay neurone
C	sensory neurone	motor neurone	relay neurone
D	sensory neurone	relay neurone	motor neurone

October/November 2015 · Paper 1 · Variant 1 · Question 23

Correct answer: A

23 What does the hypothalamus control?

A body temperature

B heart rate

C insulin secretion

D rate of breathing

October/November 2015 · Paper 1 · Variant 1 · Question 34

Correct answer: B

34 What passes down a pollen tube after pollination?

A female nucleus

B male nucleus

C ovule

D pollen grain

October/November 2015 · Paper 1 · Variant 1 · Question 37

Correct answer: C

37 In horses, red hair is dominant to brown.

A breeder crosses a number of heterozygous red-haired horses.

Approximately what percentage of the offspring will be red-haired?

A 25% **B** 50% **C** 75% **D** 100%

October/November 2015 · Paper 1 · Variant 1 · Question 38

Correct answer: B

38 Which row in the table is correct?

	chromosome in mother's ovum	chromosome in father's sperm	sex of baby
A	X	X	male
B	X	Y	male
C	Y	X	female
D	X	Y	female

October/November 2015 · Paper 1 · Variant 1 · Question 39

Correct answer: D

39 Which statement is correct?

A Evolution is natural selection.

B Evolution results in natural selection.

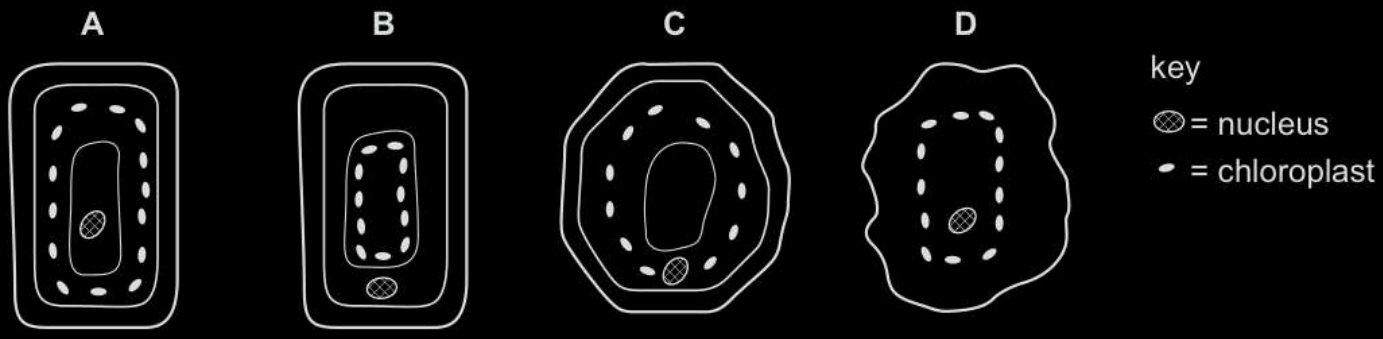
C Natural selection and evolution are independent of each other.

D Natural selection results in evolution.

October/November 2015 · Paper 1 · Variant 2 · Question 1

Correct answer: C

1 Which diagram shows the positions of the chloroplasts and the nucleus in a cell?



October/November 2015 · Paper 1 · Variant 2 · Question 2

Correct answer: A

2 How do plants absorb nitrate ions by active transport?

A against a concentration gradient using energy

B against a concentration gradient without using energy

C down a concentration gradient using energy

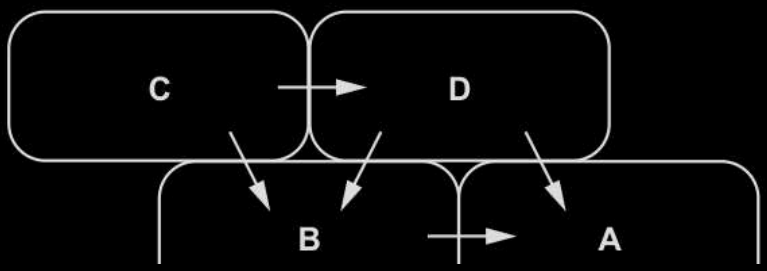
D down a concentration gradient without using energy

October/November 2015 · Paper 1 · Variant 2 · Question 3

Correct answer: C

3 The diagram shows the net movement of water by osmosis between four adjacent cells.

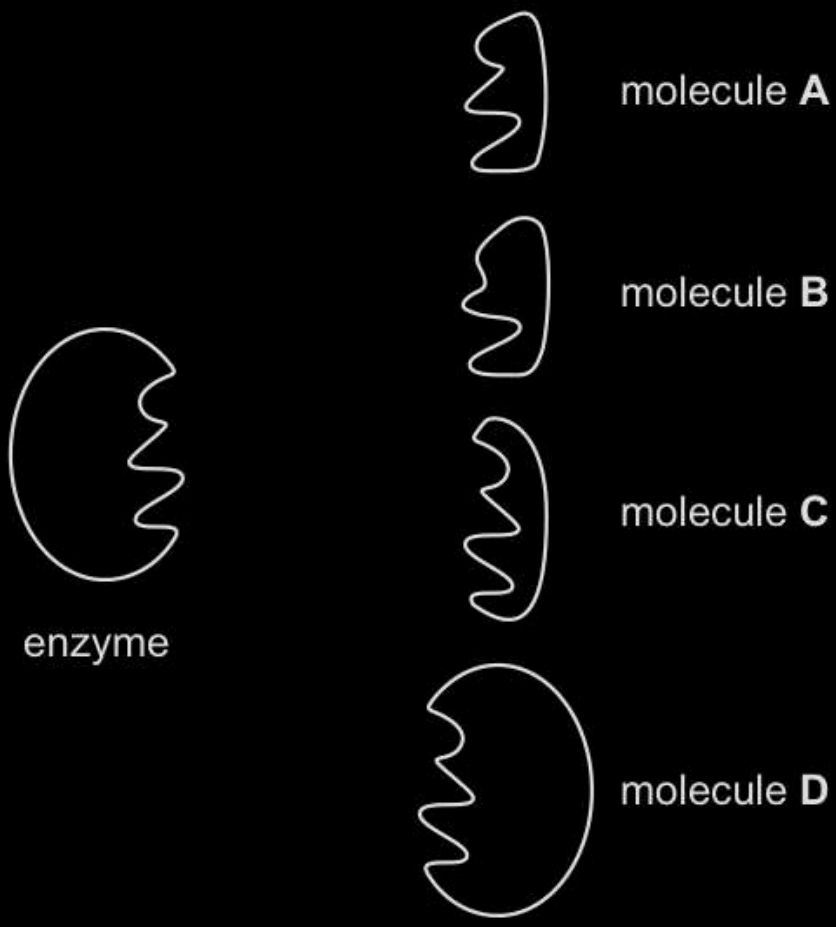
Which cell has the highest water potential?



Correct answer: A

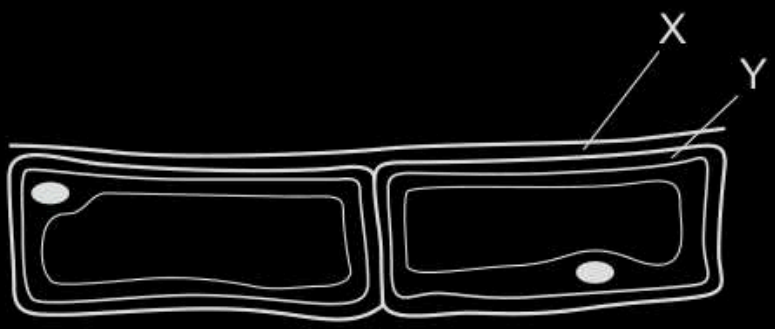
4 The diagram represents an enzyme and four molecules, **A**, **B**, **C** and **D**.

Which molecule is the substrate of this enzyme?



Correct answer: D

5 The diagram shows the upper layers of a leaf.



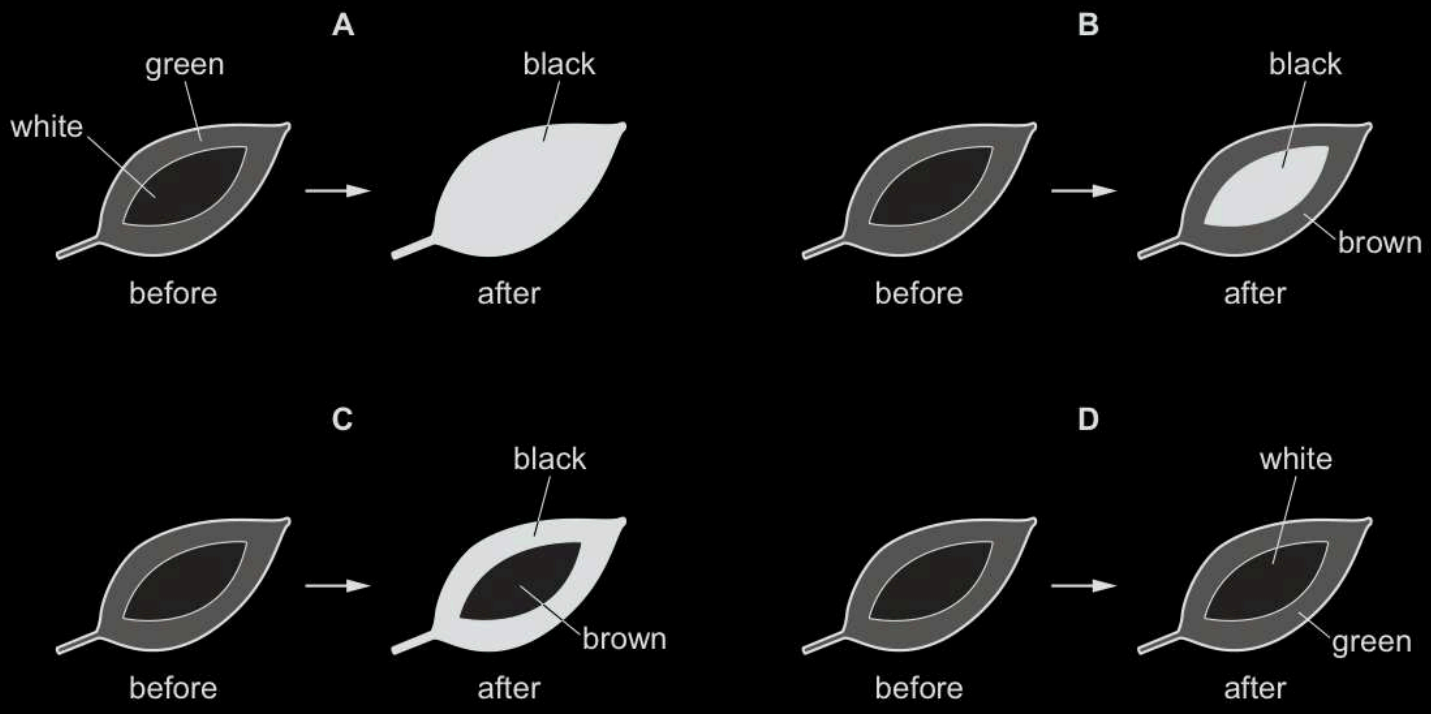
What are the structures labelled X and Y?

	X	Y
A	cell membrane	cell wall
B	cell wall	cell membrane
C	cell wall	cuticle
D	cuticle	cell wall

Correct answer: C

6 A variegated plant is destarched and then placed in the sunlight for several hours. A leaf is then taken and tested for starch using iodine solution.

Which diagram shows the correct results?

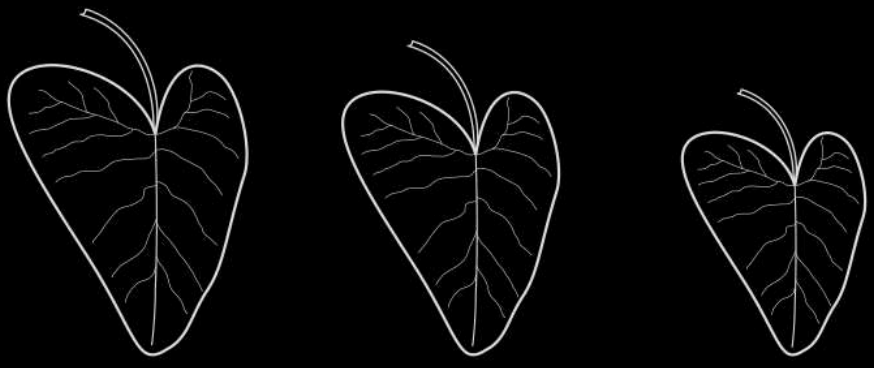


October/November 2015 · Paper 1 · Variant 2 · Question 7

Correct answer: B

7 A container is filled with water from a lake and placed on an outside table in the sunlight. Healthy plant seedlings are placed in the container and left for 3 months. Distilled water is added to the container as required. At the end of each month a typical mature leaf grown during the previous 30 days is removed and drawn to the same scale.

The diagram shows the three drawings that are made.



month 1 month 2 month 3

Why is the leaf from month 3 the smallest?

A insufficient light for photosynthesis

B nitrates from the water had all been absorbed

C phloem vessels unable to transport water

D plants cannot grow without soil

October/November 2015 · Paper 1 · Variant 2 · Question 8

Correct answer: A

8 Which is a large molecule built up from amino acids?

A amylase

B glucose

C glycogen

D urea

October/November 2015 · Paper 1 · Variant 2 · Question 9

Correct answer: A

9 Each part of the alimentary canal has the optimum pH for the enzymes that digest food there.

What is the optimum pH for an enzyme that works in the stomach?

A 2.0

B 6.5 – 7.5

C 7.5 – 8.5

D 12.0

October/November 2015 · Paper 1 · Variant 2 · Question 11

Correct answer: D

11 Samples of four different tissues are removed from the same plant.

The concentration of water in each of the four tissues is measured.

Which tissue is likely to have the highest water concentration?

A leaf epithelium

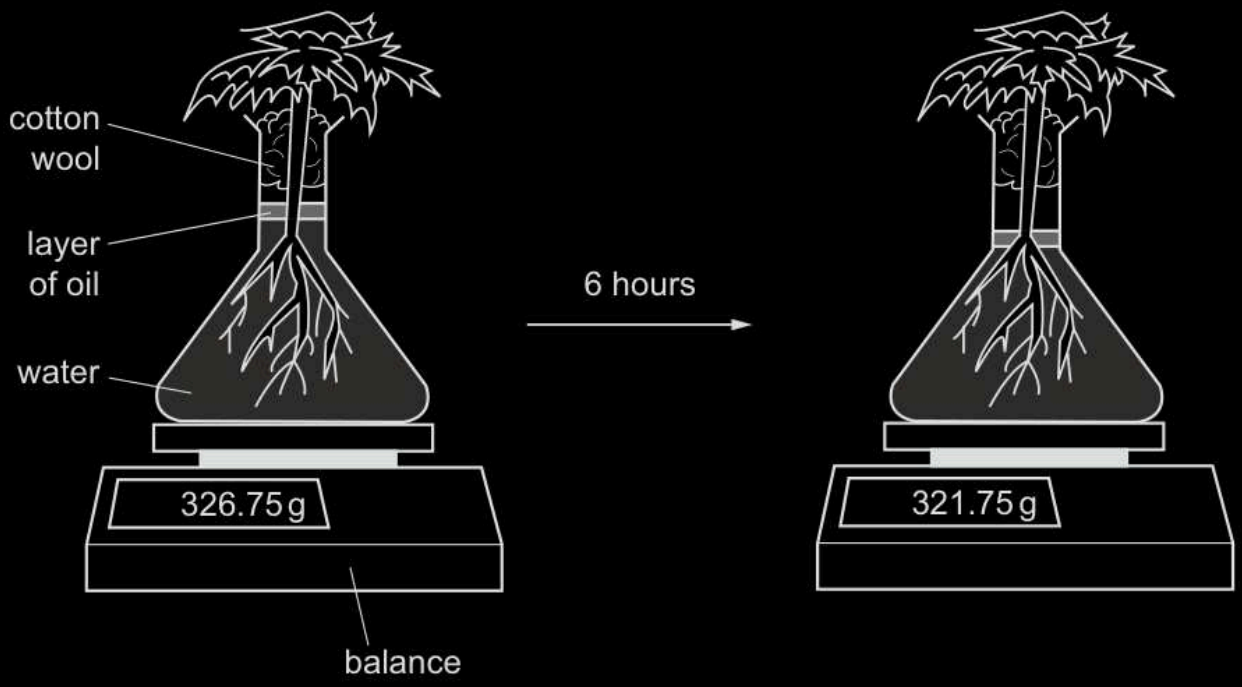
B leaf mesophyll

C root epidermis

D xylem

Correct answer: D

12 The diagrams show a plant in a flask of water. It is left in the light at 16 °C for six hours.



What explains the change in mass after six hours?

A absorption of water into the root hairs

B evaporation of water from the flask

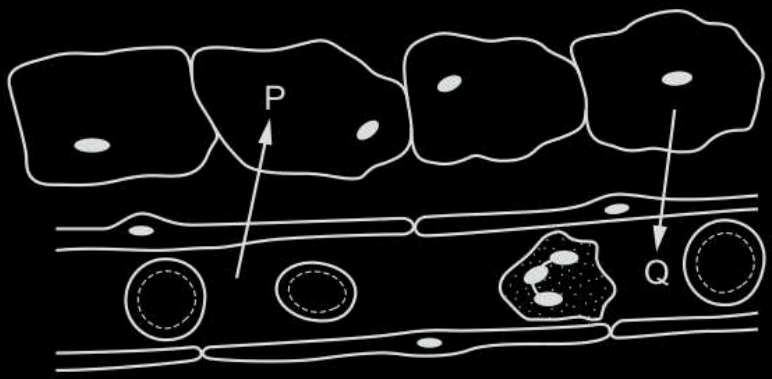
C photosynthesis in the leaves of the plant

D transpiration from the leaves of the plant

October/November 2015 · Paper 1 · Variant 2 · Question 13

Correct answer: D

13 The diagram shows chemicals being exchanged between some cells and a blood capillary.



What could be the identities of chemicals P and Q?

	P	Q
A	amino acids and oxygen	carbon dioxide and maltose
B	carbon dioxide and glucose	alcohol and oxygen
C	carbon dioxide and urea	oxygen and protein
D	glucose and oxygen	carbon dioxide and water

October/November 2015 · Paper 1 · Variant 2 · Question 14

Correct answer: B

14 In the human circulatory system, what causes the transfer of materials from the capillaries to the tissue fluid?

A active transport

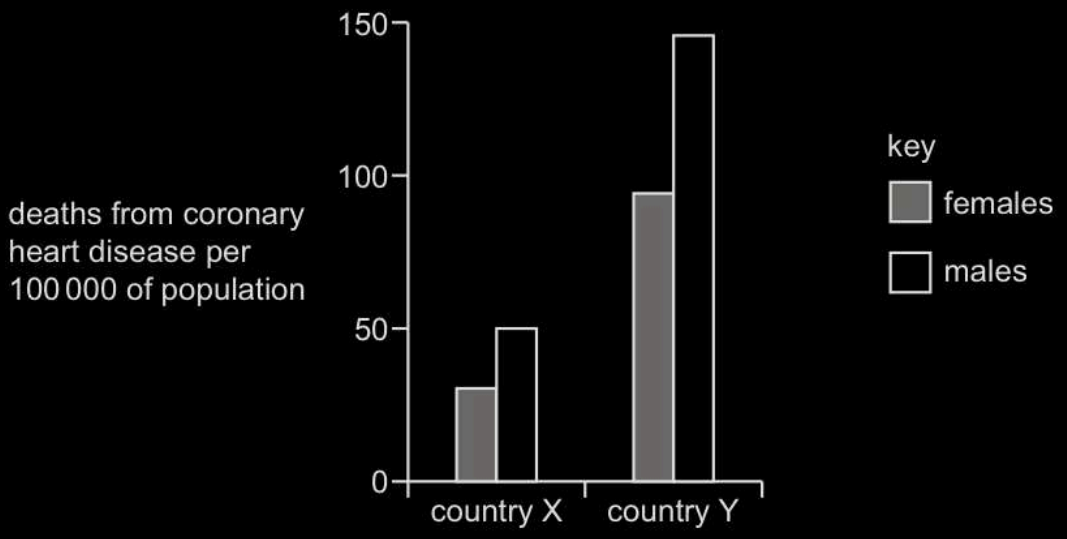
B blood pressure

C capillarity

D osmosis

Correct answer: C

15 The graph shows the death rates from coronary heart disease in two different countries.



What could **not** explain the difference between the two countries?

A less smoking in country X

B lower average age in country X

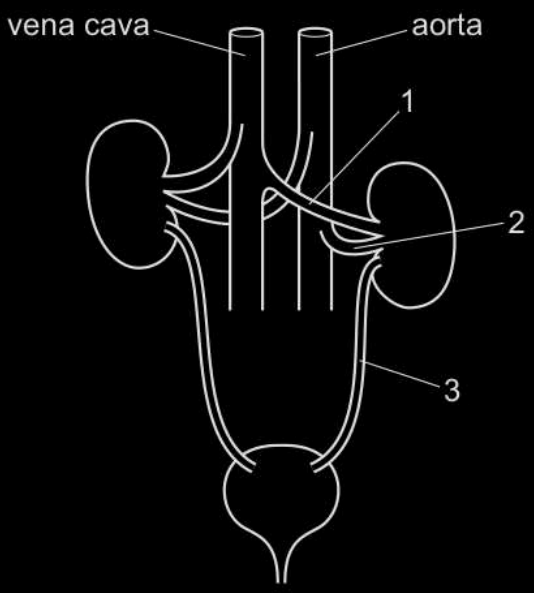
C more males in country Y

D more saturated fats eaten in country Y

October/November 2015 · Paper 1 · Variant 2 · Question 19

Correct answer: A

19 The diagram shows the kidneys and associated structures.



What are the contents of the labelled structures?

	1	2	3
A	deoxygenated blood	oxygenated blood	urine
B	oxygenated blood	deoxygenated blood	urine
C	oxygenated blood	urine	deoxygenated blood
D	urine	oxygenated blood	deoxygenated blood

October/November 2015 · Paper 1 · Variant 2 · Question 20

Correct answer: A

20 What are the responses to decreased body temperature?

A shivering and vasoconstriction in blood vessels near the skin surface

B shivering and vasodilation in blood vessels near the skin surface

C sweating and vasoconstriction in blood vessels near the skin surface

D sweating and vasodilation in blood vessels near the skin surface

October/November 2015 · Paper 1 · Variant 2 · Question 21

Correct answer: B

21 Which process keeps conditions such as temperature and glucose concentration constant inside the body?

A excretion

B homeostasis

C secretion

D vasodilation

October/November 2015 · Paper 1 · Variant 2 · Question 22

Correct answer: C

22 What is a function of sensory neurones?

A transmitting impulses from muscle to spinal cord

B transmitting impulses from receptor to muscle

C transmitting impulses from receptor to spinal cord

D transmitting impulses from spinal cord to muscle

October/November 2015 · Paper 1 · Variant 2 · Question 23

Correct answer: A

23 What does the hypothalamus control?

A body temperature

B heart rate

C insulin secretion

D rate of breathing

October/November 2015 · Paper 1 · Variant 2 · Question 25

Correct answer: B

25 What are the cause, signs and symptoms and treatment for diabetes mellitus?

	cause	signs and symptoms	treatment
A	damaged pancreas cells	excess sugar in blood	adrenaline injections
B	damaged pancreas cells	sugar in urine	insulin injections
C	sugar in urine	damaged pancreas cells	insulin injections
D	sugar in urine	damaged pancreas cells	adrenaline injections

October/November 2015 · Paper 1 · Variant 2 · Question 28

Correct answer: B

28 The diagram shows the label from a bottle of gin.

What will happen, during the next few hours, after a person drinks a large amount of gin?

A Their judgement of distance will improve.

B Their muscle control will be reduced.

C Their reaction time will decrease.

D Their urine output will decrease.

October/November 2015 · Paper 1 · Variant 2 · Question 34

Correct answer: B

34 What passes down a pollen tube after pollination?

A female nucleus

B male nucleus

C ovule

D pollen grain

October/November 2015 · Paper 1 · Variant 2 · Question 38

Correct answer: B

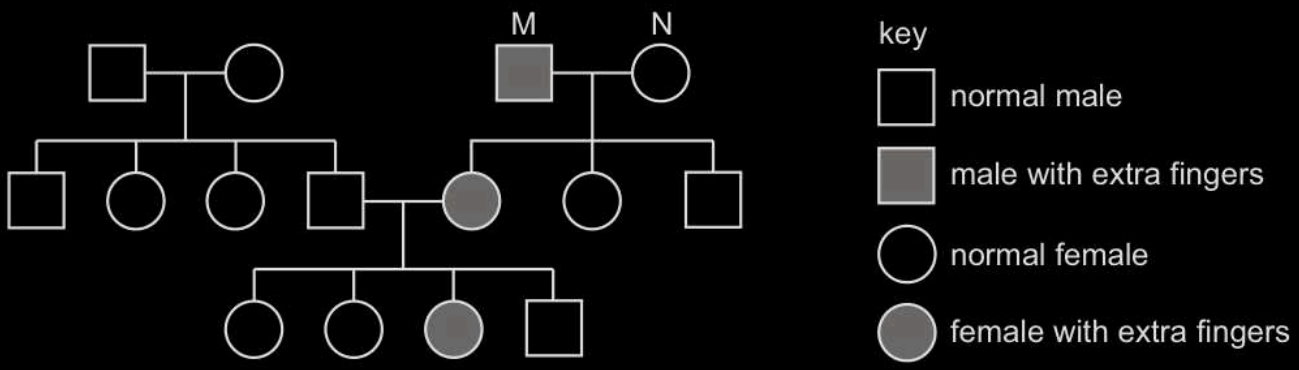
38 Which row in the table is correct?

	chromosome in mother's ovum	chromosome in father's sperm	sex of baby
A	X	X	male
B	X	Y	male
C	Y	X	female
D	X	Y	female

October/November 2015 · Paper 1 · Variant 2 · Question 39

Correct answer: C

39 The diagram shows the inheritance of the allele, E, for polydactyly (having extra fingers) which is dominant over the normal allele, e.



What are the genotypes of persons M and N?

	M	N
A	EE	ee
B	Ee	Ee
C	Ee	ee
D	ee	ee

1 Fig. 1.1 shows a plant cell before and after being placed in a concentrated salt solution.

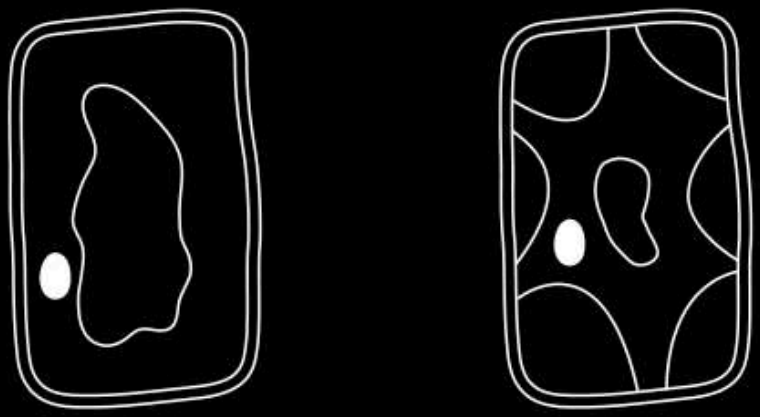


Fig. 1.1

(a) With reference to **named** parts of the cell, describe changes in the appearance of the cell after being placed in the concentrated salt solution.

.....

.....

.....

.....

.....

..... [3]

(b) Explain how the changes you have described in **(a)** have occurred.

.....

.....

.....

.....

.....

..... [3]

[Total: 6]

1 (a)	(cell) membrane / cytoplasm / nucleus ; pulled / moved away from (cell) wall / plasmolysed ; vacuole ; reduced in size / smaller / AW ;	[max. 3]	
(b)	ref. direction of water potential / concentration gradient / AW ; osmosis / diffusion ; (movement of) water out of cells ; ref. partial permeability AW of (cell) membrane ;	[max. 3]	
		[Total: 6]	

© Cambridge International Examinations 2015

Question	Expected Answer	Mark	Additional guidance

2 Fig. 2.1 shows a vertical section through a human heart viewed from the front.
Two chambers, **X** and **Y**, are labelled.

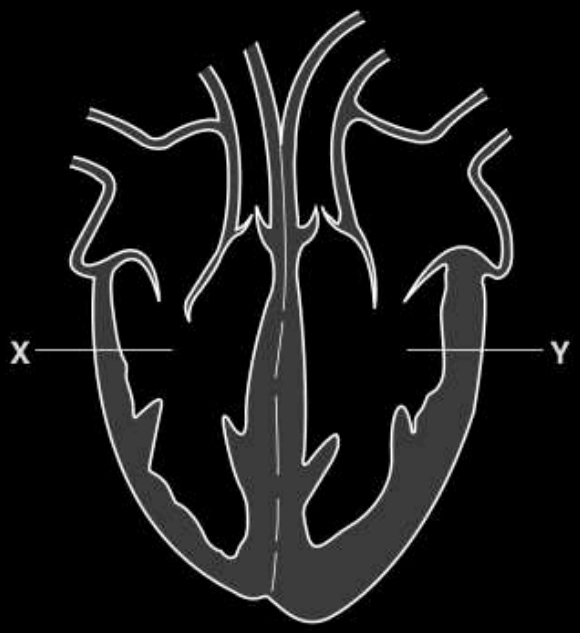


Fig. 2.1

(a) Use Fig. 2.1, and your knowledge of the circulatory system, to complete Table 2.1.

Table 2.1

chamber	name of chamber	name of blood vessel carrying blood from chamber
X		
Y		

[4]

(b) Fig. 2.2(a) shows how the mean blood pressure changes as blood flows through different types of blood vessel after leaving the heart.

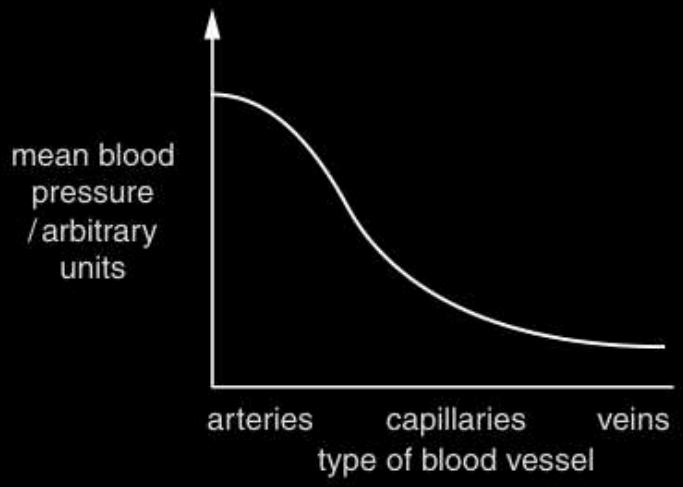


Fig. 2.2(a)

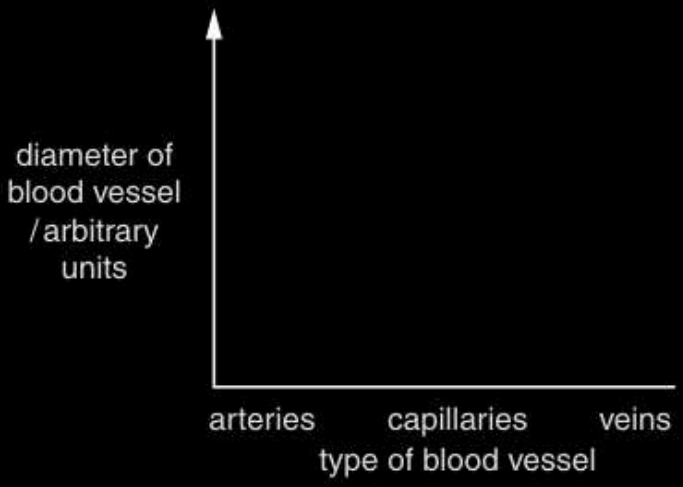


Fig. 2.2(b)

(i) Draw a line on Fig. 2.2(b) to show how the diameters of the vessels that blood flows through vary.

[2]

(ii) Use the line you have drawn on Fig. 2.2(b), and your biological knowledge, to explain why the mean blood pressure is higher in an artery than in a vein.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(c) Fig. 2.3 shows blood returning to the heart at low pressure through a vein in a leg.

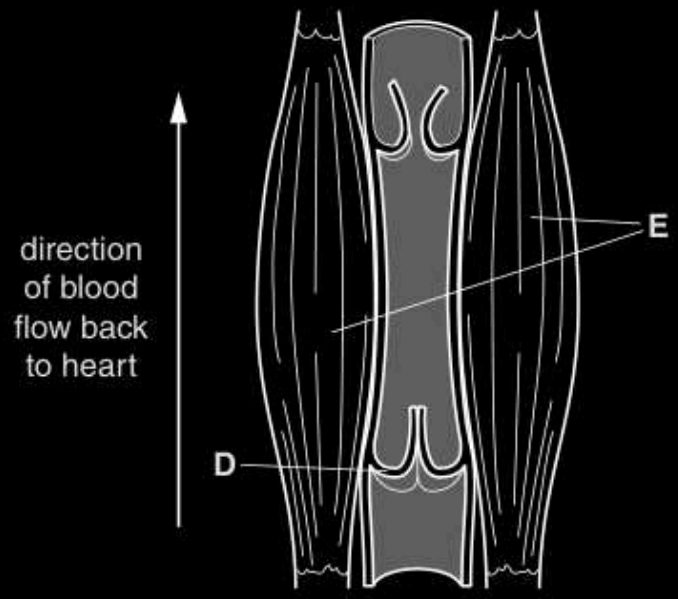


Fig. 2.3

Name part **D** in Fig. 2.3.

Explain how this part enables blood to return to the heart.

(i) name of part **D** .....

function .....

.....

..... [2]

(ii) Suggest how the parts labelled **E** in Fig. 2.3 help blood to return to the heart.

.....

.....

.....

.....

..... [2]

[Total: 14]

2 (a)	**X** right ventricle ; pulmonary artery ; **Y** left ventricle ; aorta ;	 [4]	
(b) (i)	line decreases (lowest at capillaries) followed by increase ; vein diameter drawn to be higher than artery ;	 [2]	
(ii)	ref. heart / ventricle + pump / source of pressure ; pressure relates to distance from heart / pump ; resistance / friction ; narrow lumen (in artery) ; thick / muscular / elastic <u>walls</u> (in artery) ; generation of <u>tissue fluid</u> in capillaries ;	 [max. 4]	**A** arteries take blood from / are close to heart **ORA** for vein
(c) (i)	<u>valve</u> ; prevents backflow of blood / allows flow in one direction only ;	 [2]	

© Cambridge International Examinations 2015

(ii)	muscle(s) ; contract ; put pressure + wall of vein / blood in vein ;	[max. 2]	
		[Total: 14]	

© Cambridge International Examinations 2015

Question	Expected Answer	Mark	Additional guidance

3 **(a)** Fig. 3.1 shows the distribution of blood groups in the population of a country.

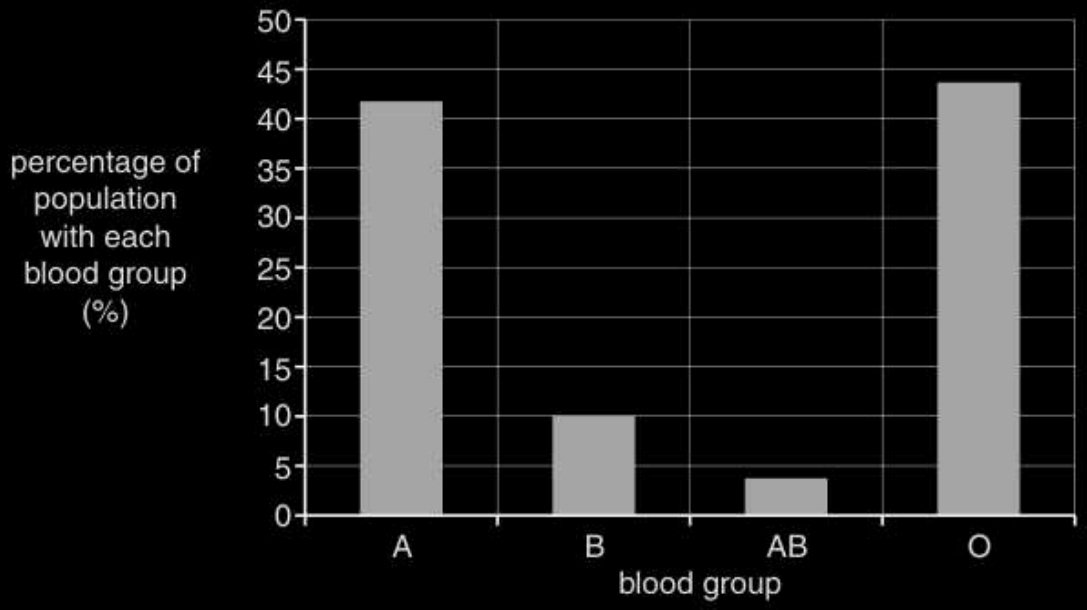


Fig. 3.1

(i) State the type of variation shown in Fig. 3.1.

.....

[1]

(ii) Give a reason for your answer to part **(a)(i)**.

.....

.....

[1]

(iii) The population of this country is approximately 63 million people. Use the information in Fig. 3.1 to calculate the approximate number of people in the country that have **blood group B**.

Show your working in the space below.

.....

[2]

(b) Table 3.1 shows the distribution of blood groups in the populations of four countries.

Table 3.1

country	percentage of population with each blood group (%)			
	A	B	AB	O
S	23	38	10	29
T	42	10	4	44
U	26	18	.....	52
V	36	14	4	46

(i) Calculate the percentage of the population of country **U** that has blood group AB.

Write your answer in the space provided in Table 3.1. [1]

(ii) Suggest why the percentage of the population with each blood group varies between the countries listed.

.....

.....

.....

.....

..... [2]

(c) Suggest why it might be necessary to know a person's blood group.

.....

.....

.....

.....

..... [2]

(d) A child's mother has blood group AB and the child's father has blood group O.

Draw a (ring) around each possible genotype and blood group of the child.

genotypes	I^AI^A	I^Ai^o	I^BI^B	I^Bi^o	I^AI^B	i^oi^o
blood groups	**A**	**B**	**AB**	**O**		

You may use the space below to work out your answer.

[2]

[Total: 11]

3 (a) (i)	discontinuous ;	[1]	
(ii)	no intermediate values / distinct / separate categories ;	[1]	
(iii)	(63 / 100) × 10 ; 6.3 million / 6 300 000 / 6.3×10^6 ;	[2]	award 2 marks for correct answer alone
(b) (i)	4 (%) ;	[1]	
(ii)	ref. alleles / genes ; each person has two (for blood group) ; different (allele / gene) frequency in different populations ; ref. inbreeding (of separate groups) AW ;	[max. 2]	**A** example of inbreeding e.g. geographical isolation / culture
(c)	blood transfusion / tissue (**A** blood donor) / organ transplant ; clotting with ref. to blood / prevent rejection ; to check paternity ;	[max. 2]	**A** if ref. is to donor or recipient **A** ORA
(d)	$I^AI^o + I^BI^o$; A + B ;	[2]	
		[Total: 11]	

© Cambridge International Examinations 2015

Question	Expected Answer	Mark	Additional guidance

6 **(a)** Explain why most foods must be digested.

.....

.....

.....

.....

.....

..... [3]

(b) Describe the digestion of fats.

You should include reference to the following in your answer:

- named regions of the alimentary canal and associated organs
- named chemicals, including the end products of fat digestion.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

[Total: 10]

6 (a)	molecules large + must be converted to smaller ; insoluble + need to be converted to soluble / non-diffusible to diffusible AW ; to be absorbed ; from (small) intestine / ileum ; into blood / capillaries / lymph / lacteals ;	[max. 3]	**A** across intestine wall / into villi
(b)	bile from + liver / gall bladder / bile duct ; ref. emulsification ; increased surface area ; ref. provision of optimum pH / alkaline conditions ; small intestine / duodenum ; lipase ; from pancreas ; fatty acids ; glycerol ;	[max. 7]	
		[Total: 10]	

© Cambridge International Examinations 2015

Question	Expected Answer	Mark	Additional guidance

7 **(a)** A person looks up from focusing on a near object to focus on an object further away.

Describe how changes that take place in **named** components of the person's eye produce a focused image of the distant object.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) Suggest why these changes that take place in the eye are controlled by the nervous system, rather than by a hormone.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7 (a)	ciliary muscles ; relax ; suspensory ligaments ; tighten / become taut / pulled AW ; increased tension / pull on lens ; lens long + thin / flatter / less fat / convex ; light refracted / bent (by lens) ; (light) converges / (produces image) on retina / fovea ;	[max. 6]	**R** bent more
(b)	nervous system quicker AW / hormones slower ; example of relevant situation ; electrical / impulses ; via neurones / nerve cells ; hormones are chemical ; carried / transported via blood ;	[max. 4]	
		[Total: 10]	

© Cambridge International Examinations 2015

Question	Expected Answer	Mark	Additional guidance

9 **(a)** Define the term *excretion*.

.....

.....

.....

.....

.....

..... [3]

(b) Explain how a kidney machine carries out the functions of a kidney for a person with kidney disease.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

[Total: 10]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

9	(a)	removal from organism / body ; toxic / poisonous ; waste (products) ; from metabolism ;	[max. 3]	
	(b)	1 blood from patient to machine / returned to patient ; 2 using partially permeable membrane / tubes AW ; 3 ref. dialysis fluid ; 4 ref. relative concentrations AW ; 5 filters / removes substances from + blood / unwanted out / wanted in ; 6 by diffusion ; 7 urea / nitrogenous products / toxins / poisons ; 8 salt(s) / ions / small molecules ; 9 excess water / ref. osmoregulation ; 10 large molecules / proteins stay in blood ; 11 practical aspect e.g. bubble trap / repetition / frequency / duration of procedure / dialysis fluid frequently changed / temperature ref / counter flow ;	[max. 7]	
			[Total: 10]	

© Cambridge International Examinations 2015

1 Fig. 1.1 shows a carpel of a flower with two pollen grains on top.

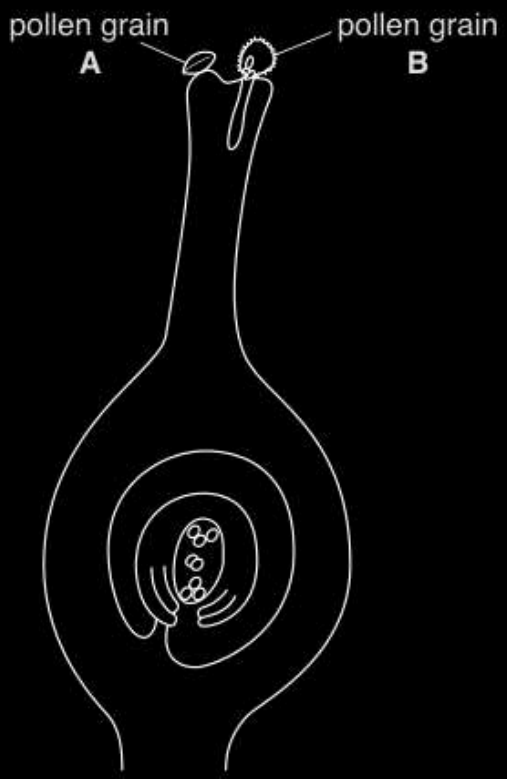


Fig. 1.1

(a) Name each of the following:

(i) the process that has brought the pollen grains to the carpel

.....[1]

(ii) the part of the carpel to which the pollen grains are attached.

.....[1]

(b) On Fig. 1.1, draw a line to show the route taken by the pollen tube up to the point at which male and female gametes fuse. [3]

(c) Name each of the following:

(i) a part of the carpel where the nuclei are all **diploid**

.....[1]

(ii) a part of the carpel that contains **haploid** nuclei.

.....[1]

(d) Suggest why pollen grain **A** has not developed a pollen tube.

.....

.....[1]

[Total: 8]

1 (a) (i)	pollination ;	[1]	**I** insect / wind / cross / self
(ii)	stigma ;	[1]	
(b)	line (continuing) down style ; entering micropyle ; touching / entering embryo sac ;	[3]	
(c) (i)	stigma / style / ovary wall ;	[1]	
(ii)	ovary / ovule / embryo sac / ovum / ova / egg (cell) / gamete ;	[1]	
(d)	from a different species / type / kind of plant ; ref. chemicals on stigma unsuitable ; couldn't germinate / not viable / infertile AW ; not there long enough / grain immature / wrong enzymes ;	[max 1]	**A** e.g. wrong sugar concentration, toxic chemical
		[Total: 8]	

© Cambridge International Examinations 2015

Question	Expected Answer	Mark	Additional guidance

3 Fig. 3.1 shows blood pressure changes as blood flows through part of the circulatory system, beginning at the right atrium, travelling to the lungs, and ending in the pulmonary vein.

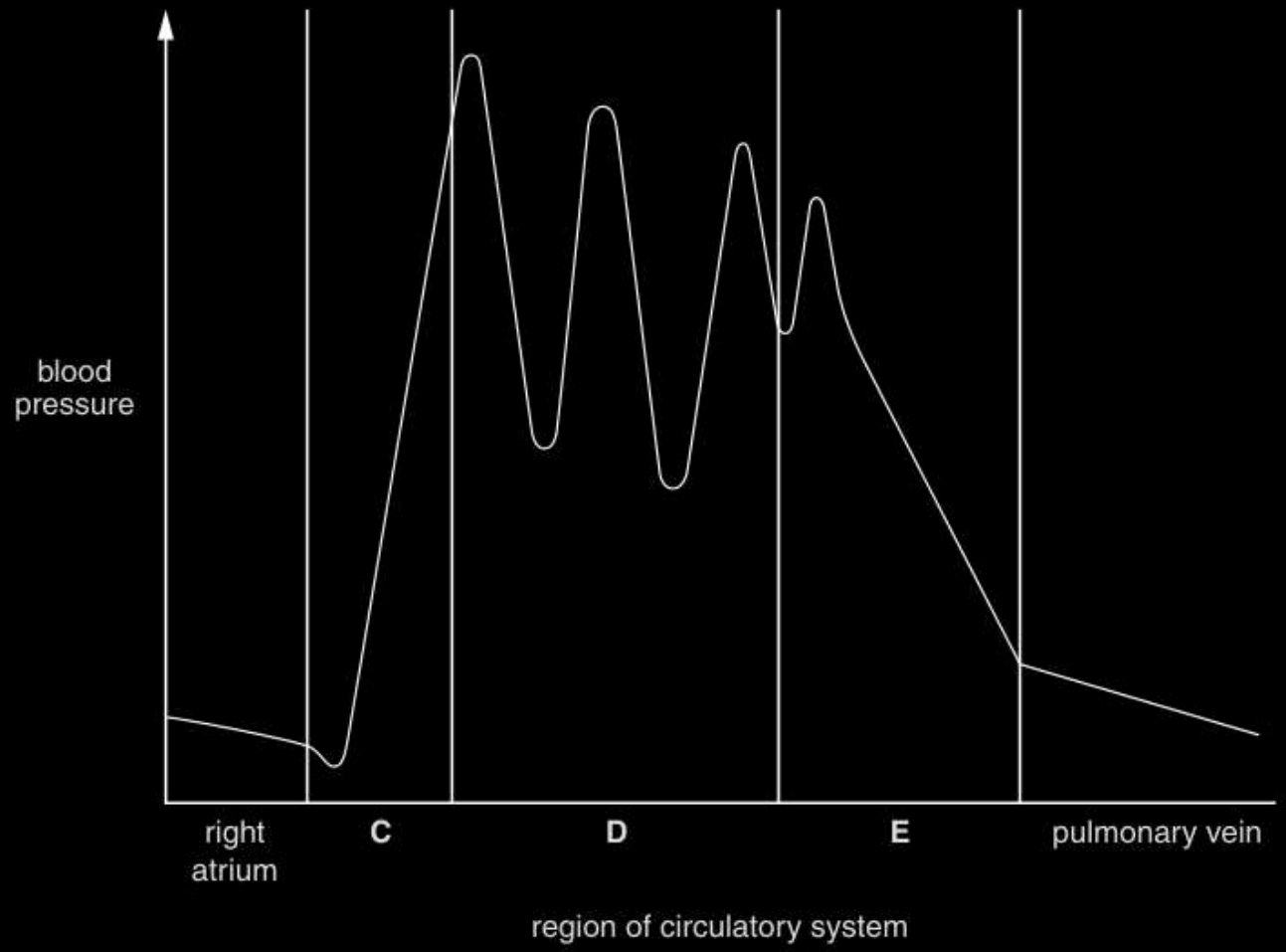


Fig. 3.1

(a) State which chamber of the heart is represented by **C**. Explain your answer.

chamber **C** .....

explanation .....

.....[2]

(b) Explain the reasons for the regular changes in blood pressure in region **D**.

.....

.....

.....

.....[2]

(c) Important chemical changes occur in the blood as it passes through region **E**.

(i) Identify region **E**.

.....[1]

(ii) Describe and explain the chemical changes that occur.

.....

.....

.....

.....

.....

.....

.....[3]

(d) Describe and explain how the shape of a graph drawn to show blood pressure changes as blood flows from the heart to the rest of the body and back again would differ from Fig. 3.1.

.....

.....

.....

.....

.....

.....

.....[3]

[Total: 11]

3 (a)	right ventricle ; blood flows from right atrium to right ventricle / pressure increases ;	[2]	**A** pressure increases with incorrect chamber for 1 mark
(b)	increases when heart / ventricle + contracts / pumps ; decreases when heart / ventricle + relaxes / doesn't pump ;	[2]	**A** for 1 mark : ref. heart beat / pulse
(c) (i)	capillaries / lung /alveoli / air sac / venule ;	[1]	
(ii)	CO_2 lost * ; O_2 gained / oxygenated * ; haemoglobin / oxyhaemoglobin ; change in pH / less acidic ; ref. to diffusion gradient / concentration gradient ;	[max 3]	*OR **A** gas exchange for 1 mark
(d)	wider variations in pressure AW ; pressure high(er) AW ; left ventricle wall large / thick / more muscular ; blood has further to travel / takes longer ; more or many fluctuations / undulations AW ;	[max 3]	**I** increasing pressure
		[Total: 11]	

© Cambridge International Examinations 2015

Question	Expected Answer	Mark	Additional guidance

4 Fig. 4.1 shows a section through the leaf of a plant.

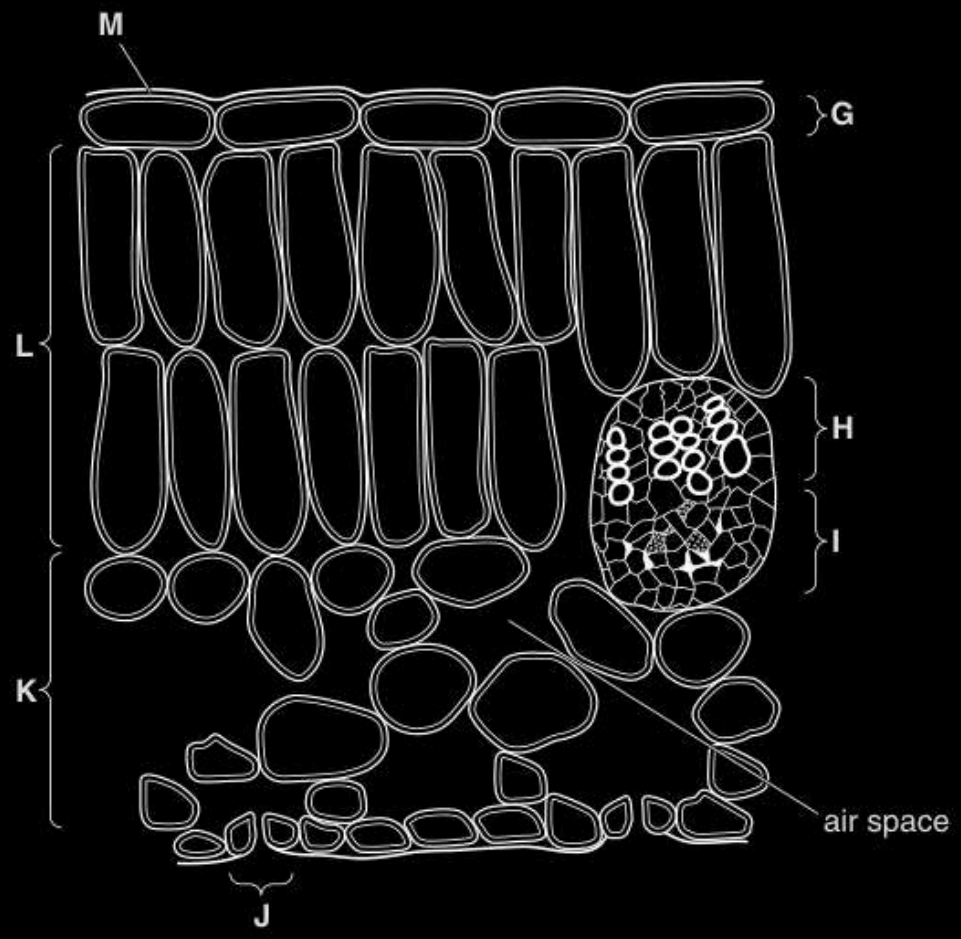


Fig. 4.1

(a) State the letters that identify each of the following:

(i) the region containing cells with the greatest number of chloroplasts

.....[1]

(ii) two regions that contain cells with **no** chloroplasts.

..... and .....[1]

(b) Explain why Fig. 4.1 shows the leaf during daylight hours.

.....

.....[1]

(c) **(i)** Identify the tissues labelled **H** and **I**.

H .....

I .....[2]

(ii) Describe the functions of these two tissues.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

(d) Name a gas that gradually increases in concentration in the air space during the hours of darkness. Give an explanation for your answer.

gas .....

explanation .....

.....

.....

.....[3]

[Total: 12]

4	(a) (i)	**L** ;	[1]	
	(ii)	any 2 from : **G** , **H** , **I** ;	[1]	**M** incorrect as doesn't contain cells
	(b)	stomata / **J** / guard cells open **ORA** ;	[1]	
	(c) (i)	**H** – xylem ; **I** – phloem ;	[2]	
	(ii)	(**H**) carries water ; minerals / salts / ions ; (provides) support ; (**I**) carries sucrose / sugar ; amino acids ; to / from leaf (phloem only) / rest of plant / through plant ;	[max 4]	
	(d)	carbon dioxide ; from respiration ; OR water vapour ; from transpiration (stream) ; cannot escape / stomata or guard cells closed ;	[3]	gas stated must be CO_2 OR water vapour 2 marks for explanation
			[Total:12]	

© Cambridge International Examinations 2015

Question	Expected Answer	Mark	Additional guidance

7 **(a)** **(i)** Explain how wilting occurs in a plant.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

(ii) Explain why mature plants do not wilt as much as newly-transplanted seedlings, even when both are exposed to similar conditions.

.....

.....

.....

.....

.....

.....

.....[3]

(b) Suggest an explanation for what would happen to a plant if it is accidentally 'watered' with a concentrated salt solution.

.....

.....

.....

.....

.....

.....

.....[3]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7 (a) (i)	water lost / plant dehydrated ; by transpiration / evaporation / through stomata ; faster than it is gained ; cells / tissues lose turgor / become flaccid / plasmolysed AW ; no longer support plant ; any condition leading to wilting ;	[max 4]	**A** lack of water, low humidity, high light intensity, high temperature, wind
(ii)	(mature plant) root system undamaged **ORA** ; ref. root hairs ; water uptake sufficient for plant **ORA** ; plant cells / tissues turgid **ORA** ; xylem supports older plants / ref. lignin ;	[max 3]	**I** ref. more / less water uptake
(b)	plant / leaves will wilt ; water lost ; from cells / tissues ; by osmosis / diffusion ; lower water potential in soil water / ref. correct water potential gradient AW ;	[max 3]	
		[Total: 10]	

© Cambridge International Examinations 2015

Question	Expected Answer	Mark	Additional guidance

9 **(a)** Outline the processes of *mitosis* and *meiosis*, including references to where they occur.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[7]

(b) Describe the advantages of asexual reproduction in plants with reference to a **named** commercially important application.

.....

.....

.....

.....

.....

.....

.....

.....[3]

[Total: 10]

9	(a)	both cell <u>division</u> ; (mitosis) in body cells or named ; for growth / repair / replacement ; chromosome number retained AW ; ref. asexual reproduction / single parent ; produces copy / identical / similar ; produces two cells ; (meiosis) in ovaries* / ovule ; testes* / anthers ; ref. gamete / sex cell production ; chromosome number halved / cells become haploid ; sexual reproduction ; <u>genetically</u> different ; produces four cells ;	[max 7]	max 4 marks for mitosis or meiosis **A** in zygote **I** growth / repair *within* a cell * OR **A** gonads for one mark **I** fertilisation
	(b)	name of correct organism / crop / technique ; always an exact / similar copy ; known characteristics / flavour AW ; more certain outcome / can be conducted in controlled conditions; not reliant on pollination (agent) ; only one parent needed ; quick(er) ; less expensive / greater profit / higher yield / more offspring ;	[1] [max 2] [3]	**A** cutting, micro-propagation, tubers, bulbs, layering, runners
			[Total: 10]	

© Cambridge International Examinations 2015

May/June 2015 · Paper 1 · Variant 1 · Question 1

Correct answer: D

1 The cell wall of a plant cell is removed using an enzyme.

What would happen if this cell is then placed in distilled water?

A It would take longer for the cell to become turgid.

B Proteins in the cytoplasm would leave through the cell membrane.

C The cell would become smaller as water passes out.

D The cell would burst as water moves into it.

May/June 2015 · Paper 1 · Variant 1 · Question 2

Correct answer: A

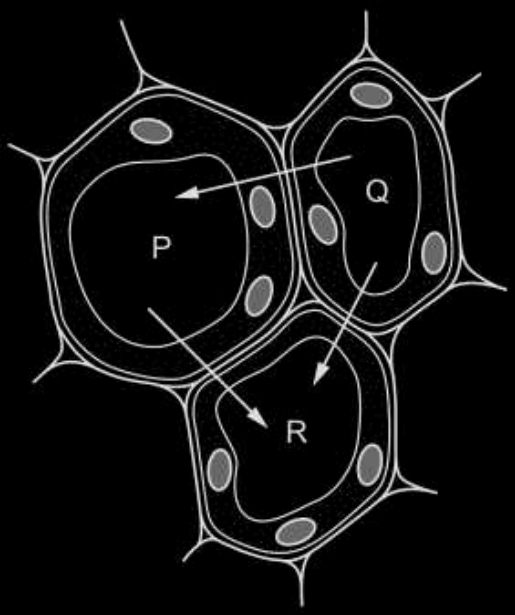
2 Which processes are responsible for the uptake of ions from the soil by a plant and the uptake of glucose into the villi of a human?

	uptake of ions by a plant	uptake of glucose into the villi
A	active transport	active transport
B	active transport	osmosis
C	diffusion	osmosis
D	osmosis	active transport

May/June 2015 · Paper 1 · Variant 1 · Question 3

Correct answer: C

3 The diagram shows three plant cells labelled P, Q and R. The arrows show the direction of water movement by osmosis.



What is the correct order of water potential in the cells, from the highest to the lowest?

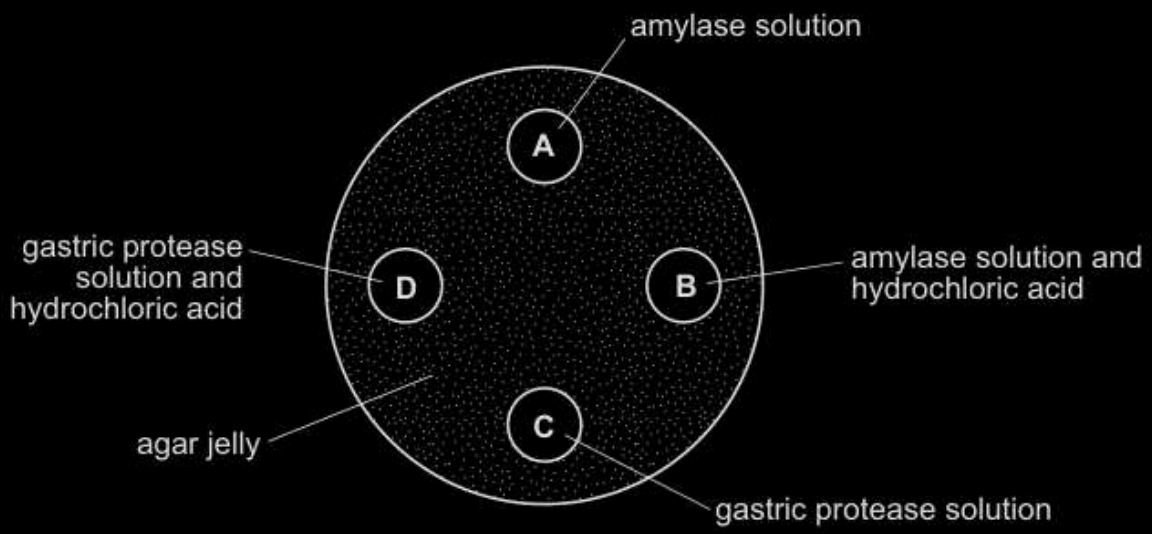
	highest	middle	lowest
A	P	Q	R
B	P	R	Q
C	Q	P	R
D	R	P	Q

May/June 2015 · Paper 1 · Variant 1 · Question 4

Correct answer: A

4 A dish is filled with agar jelly containing starch. Four holes are cut in the jelly and each hole is filled as shown.

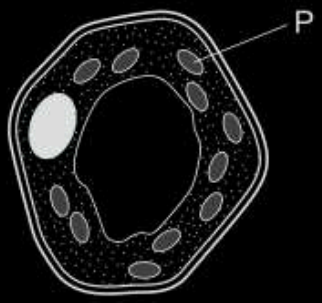
After 30 minutes, which hole will be surrounded by the largest area without starch?



May/June 2015 · Paper 1 · Variant 1 · Question 5

Correct answer: A

5 The diagram shows a plant cell.



Compared with the rest of the cell, which row describes the concentrations of oxygen and magnesium inside structure P during the daytime?

	oxygen	magnesium
A	high	high
B	high	low
C	low	high
D	low	low

May/June 2015 · Paper 1 · Variant 1 · Question 6

Correct answer: C

6 Some organisms live at the bottom of the seas where it is very dark. To synthesise glucose, they use energy from chemicals in the very hot water that comes out of volcanoes.

What is a distinguishing feature of these organisms?

A Their enzymes are easily denatured by heat.

B They do not need carbon dioxide.

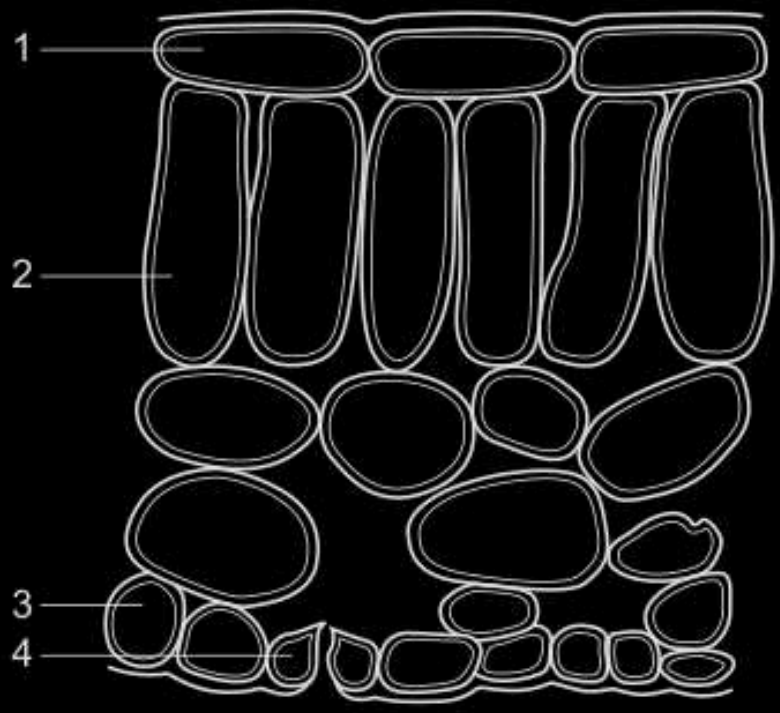
C They do not need to be green.

D They obtain energy only as carnivores.

May/June 2015 · Paper 1 · Variant 1 · Question 7

Correct answer: D

7 The diagram shows cells in a section through a leaf of a typical green plant. (No cell contents are shown.)



Which cells usually contain chloroplasts?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

May/June 2015 · Paper 1 · Variant 1 · Question 8

Correct answer: B

8 The element nitrogen is needed to form

A fat.

B protein.

C starch.

D sugar.

May/June 2015 · Paper 1 · Variant 1 · Question 9

Correct answer: B

9 Which process involves the use of nutrients inside cells?

A absorption

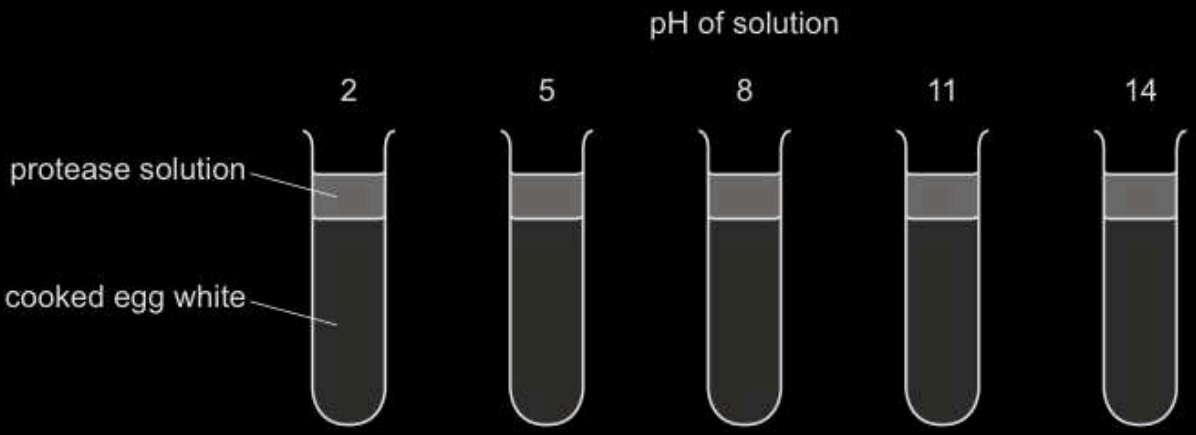
B assimilation

C digestion

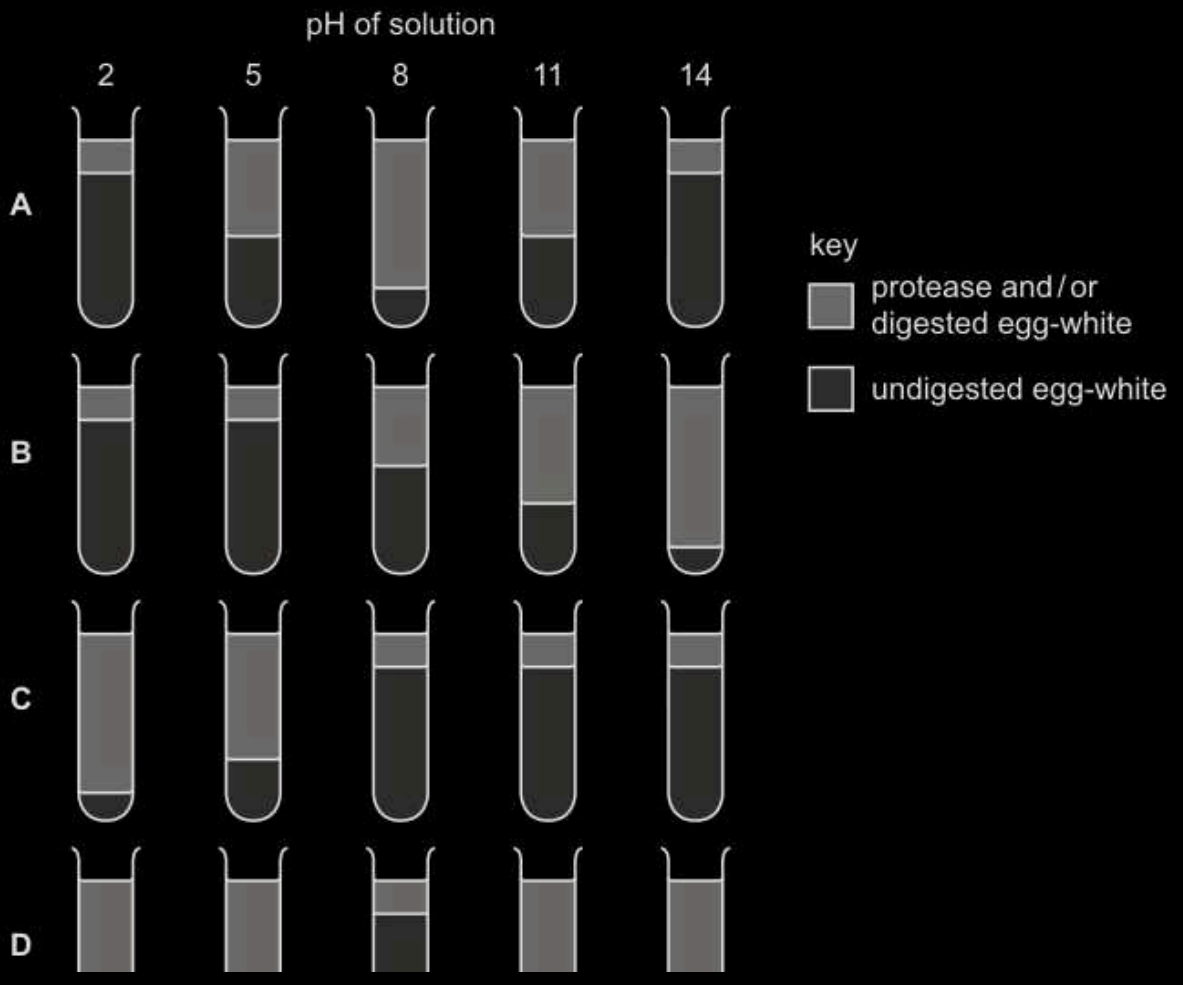
D ingestion

Correct answer: C

10 Five tubes containing cooked egg-white are set up as shown. Protease solutions of different pH are added to each tube.



Which diagram shows the results of this experiment for a protease from the stomach?



May/June 2015 · Paper 1 · Variant 1 · Question 11

Correct answer: C

11 Under which set of conditions will the transpiration rate of a well-watered plant be fastest?

A a cool, dry, windless day

B a cool, rainy, windy day

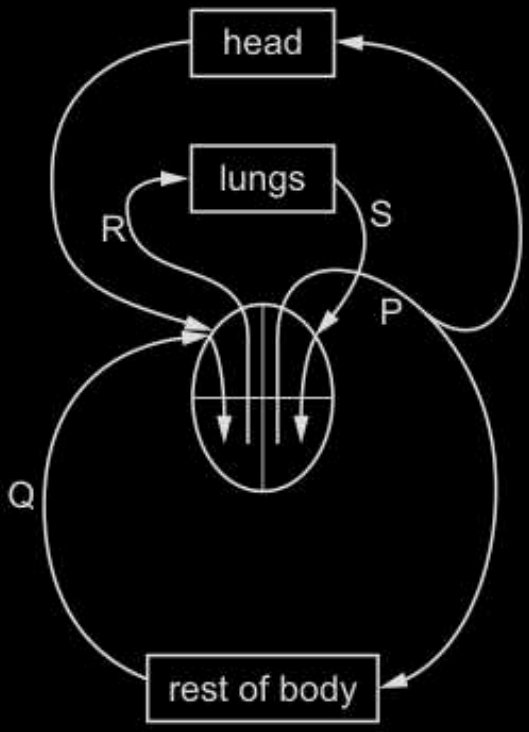
C a hot, dry, windy day

D a hot, rainy, windy day

May/June 2015 · Paper 1 · Variant 1 · Question 13

Correct answer: C

13 The diagram represents the heart and some major blood vessels.

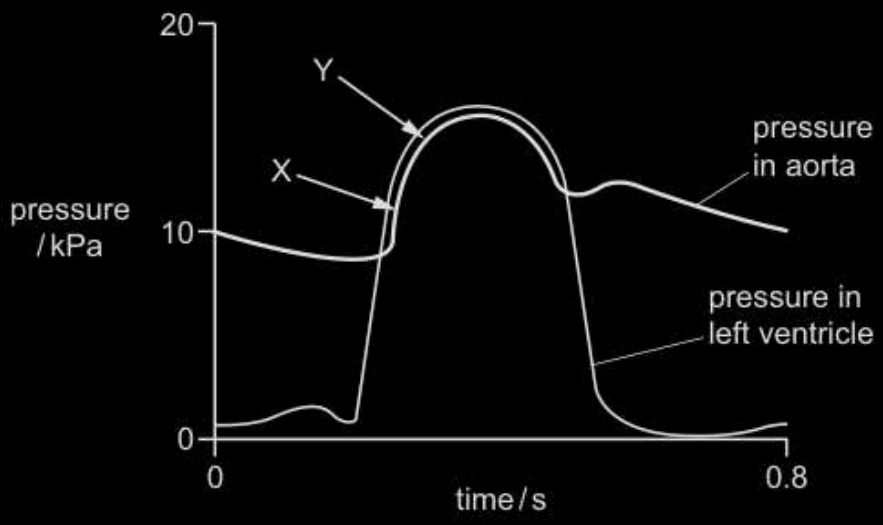


What are possible blood pressures (in kPa) for the vessels shown on the diagram?

	P	Q	R	S
A	1	4	2	16
B	4	16	2	1
C	16	2	4	1
D	16	4	1	2

Correct answer: B

14 The diagram shows changes in pressure in the aorta and the left ventricle during one complete heart beat.



What causes the increase in pressure between points X and Y?

A contraction of the left atrium

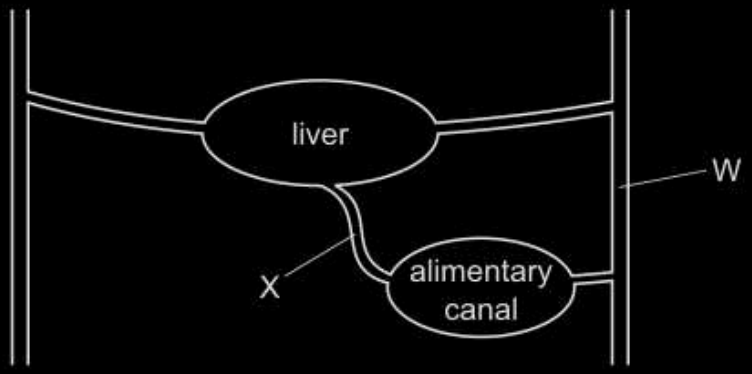
B contraction of the left ventricle

C relaxation of the left atrium

D relaxation of the left ventricle

Correct answer: B

15 The diagram shows the liver and its blood supply.



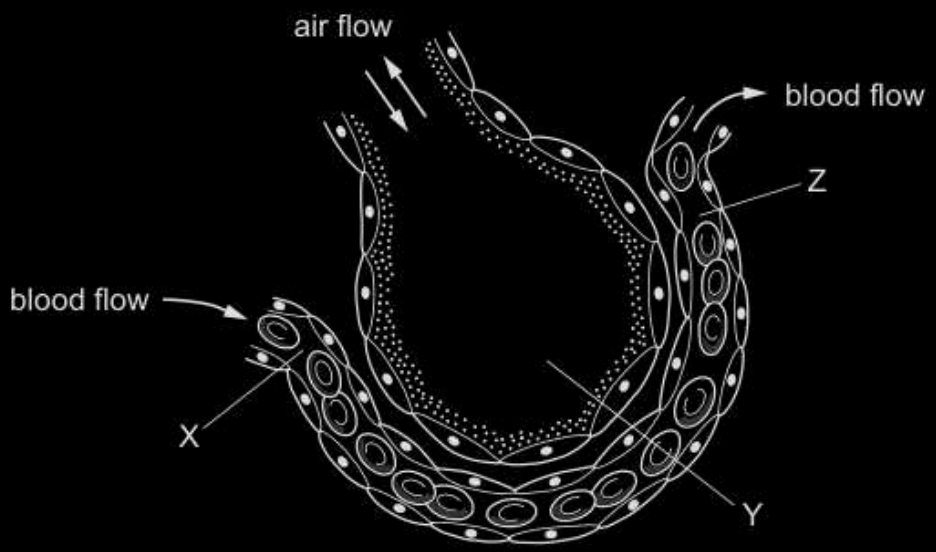
What is the name of blood vessel X, and which substance increases in concentration between vessel W and vessel X?

	name of X	substance
A	hepatic artery	carbon dioxide
B	hepatic portal vein	carbon dioxide
C	hepatic artery	oxygen
D	hepatic portal vein	oxygen

May/June 2015 · Paper 1 · Variant 1 · Question 18

Correct answer: B

18 The diagram shows a section of an alveolus and a capillary in a lung.



What are the relative concentrations of **carbon dioxide** at X, Y and Z?

	X	Y	Z
A	high	high	high
B	high	low	low
C	low	high	high
D	low	high	low

May/June 2015 · Paper 1 · Variant 1 · Question 19

Correct answer: A

19 Where are most nitrogen compounds excreted from humans?

A kidneys

B liver

C rectum

D skin

May/June 2015 · Paper 1 · Variant 1 · Question 20

Correct answer: D

20 When the temperature of the air is higher than body temperature, which of these control mechanisms can help to maintain a constant body temperature?

	constriction of blood vessels in skin	shivering	sweating
A	✓	✓	✓
B	✓	✗	✗
C	✗	✓	✗
D	✗	✗	✓

May/June 2015 · Paper 1 · Variant 1 · Question 21

Correct answer: D

21 A motor neurone transmits impulses from

A muscle to spinal cord.

B receptor to muscle.

C receptor to spinal cord.

D spinal cord to muscle.

May/June 2015 · Paper 1 · Variant 1 · Question 22

Correct answer: A

22 In which order does light pass through these structures in the eye?

A cornea → aqueous humour → lens → vitreous humour → retina

B cornea → vitreous humour → lens → aqueous humour → retina

C lens → aqueous humour → cornea → vitreous humour → retina

D lens → vitreous humour → cornea → aqueous humour → retina

May/June 2015 · Paper 1 · Variant 1 · Question 23

Correct answer: B

23 Why is glucose found in the urine of diabetics?

A increased uptake and use of glucose by the body cells

B not enough glucose in the blood is converted to glycogen

C stored fats in the body are being oxidized

D too much glucose is absorbed by the kidney cells

May/June 2015 · Paper 1 · Variant 1 · Question 26

Correct answer: B

26 In a practical lesson, a student makes the following observations about some organisms.

'They are spherical, unicellular organisms which are visible with the light microscope.

Diameter is 0.01 mm. Cytoplasm is present and, after staining, a nucleus can be seen.

A vacuole is present.'

What is being described?

A bacterium

B fungus

C plant

D virus

May/June 2015 · Paper 1 · Variant 1 · Question 27

Correct answer: D

27 The fungus *Penicillium* is grown in fermenters to make penicillin. The table shows the time taken for the mycelium of the fungus to double in size with different types of carbohydrate. All other conditions were kept constant.

type of carbohydrate	glucose	maltose	starch
time for mycelium of *Pencillium* to double in size / hours	15	20	30

What can be concluded from the data?

A Glucose molecules are larger than maltose molecules.

B *Penicillium* does not produce amylase.

C *Penicillium* does not produce protease.

D *Penicillium* takes longer to use starch than to use sugars.

May/June 2015 · Paper 1 · Variant 1 · Question 33

Correct answer: B

33 What describes the chromosome number of these human cells?

	egg-producing cell in ovary	sperm	zygote	cell in an embryo
A	diploid	diploid	haploid	haploid
B	diploid	haploid	diploid	diploid
C	diploid	haploid	haploid	diploid
D	haploid	haploid	diploid	diploid

May/June 2015 · Paper 1 · Variant 1 · Question 37

Correct answer: C

37 Which of these may be heterozygous?

A a haploid cell

B an allele of a gene

C an organism with a dominant phenotype

D an organism with a recessive genotype

May/June 2015 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 A red-flowered plant is crossed with a white-flowered plant. All the offspring have red flowers.

What is the genotype of these offspring?

A RR and Rr **B** RR only **C** Rr only **D** rr only

May/June 2015 · Paper 1 · Variant 1 · Question 39

Correct answer: A

39 A person with Down's syndrome is born with 47 chromosomes in each cell, instead of 46.

What could cause this?

A A mutation happened during the production of the egg cell.

B More than one sperm fused with the egg at fertilisation.

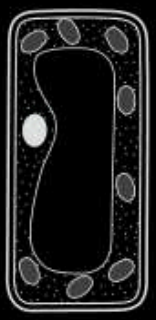
C Radiation caused a change in structure of a gene in the father's sperm.

D The mother was exposed to harmful chemicals while she was pregnant.

May/June 2015 · Paper 1 · Variant 2 · Question 1

Correct answer: B

1 The diagram shows a plant cell.



Which structure identifies this as a plant cell rather than an animal cell?

A cell membrane

B cell wall

C cytoplasm

D nucleus

May/June 2015 · Paper 1 · Variant 2 · Question 2

Correct answer: A

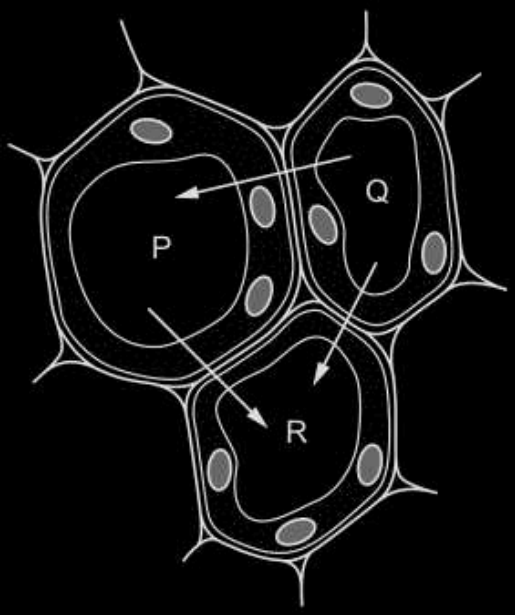
2 Which processes are responsible for the uptake of ions from the soil by a plant and the uptake of glucose into the villi of a human?

	uptake of ions by a plant	uptake of glucose into the villi
A	active transport	active transport
B	active transport	osmosis
C	diffusion	osmosis
D	osmosis	active transport

May/June 2015 · Paper 1 · Variant 2 · Question 3

Correct answer: C

3 The diagram shows three plant cells labelled P, Q and R. The arrows show the direction of water movement by osmosis.



What is the correct order of water potential in the cells, from the highest to the lowest?

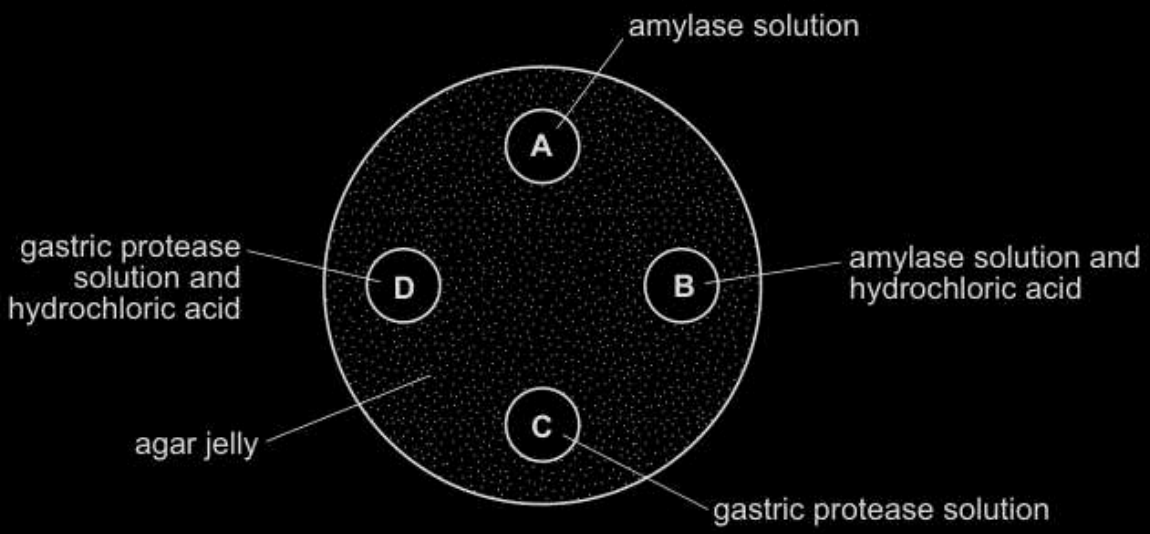
	highest	middle	lowest
A	P	Q	R
B	P	R	Q
C	Q	P	R
D	R	P	Q

May/June 2015 · Paper 1 · Variant 2 · Question 4

Correct answer: A

4 A dish is filled with agar jelly containing starch. Four holes are cut in the jelly and each hole is filled as shown.

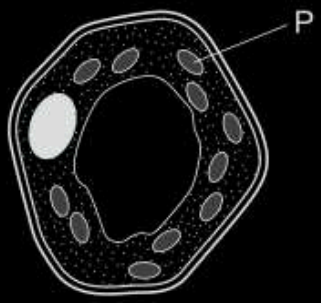
After 30 minutes, which hole will be surrounded by the largest area without starch?



May/June 2015 · Paper 1 · Variant 2 · Question 5

Correct answer: A

5 The diagram shows a plant cell.



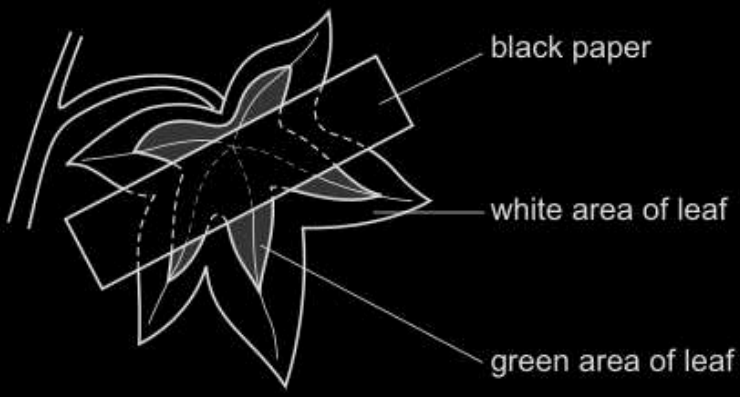
Compared with the rest of the cell, which row describes the concentrations of oxygen and magnesium inside structure P during the daytime?

	oxygen	magnesium
A	high	high
B	high	low
C	low	high
D	low	low

Correct answer: A

6 A plant has leaves that are green in the centre but white around the edges.

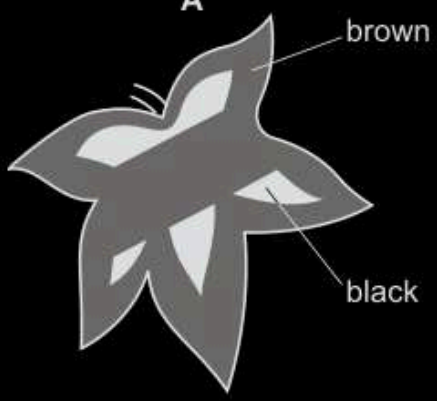
The plant is destarched and then has one of its leaves partly covered with black paper on both sides of the leaf, as shown.



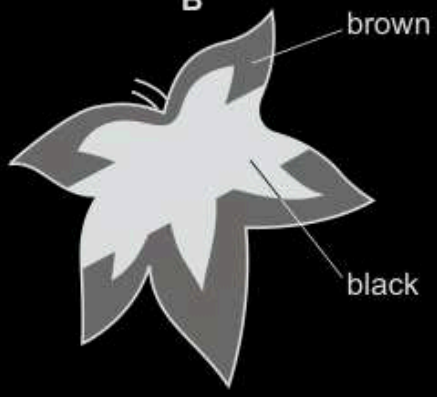
The plant is placed in bright light for 48 hours. The leaf is then tested for starch.

Which diagram shows the colours that are obtained?

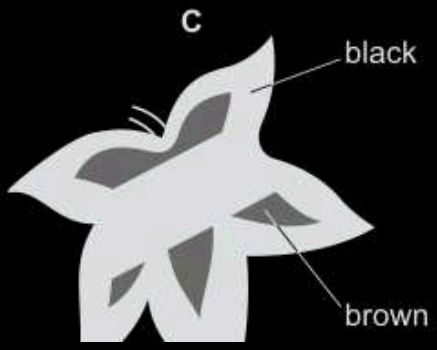
A



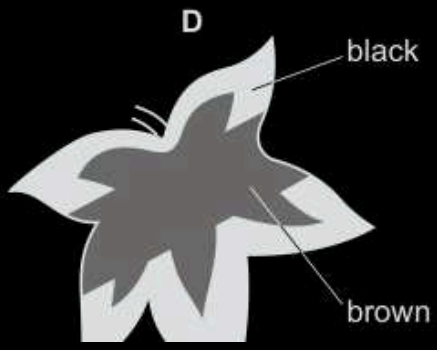
B



C



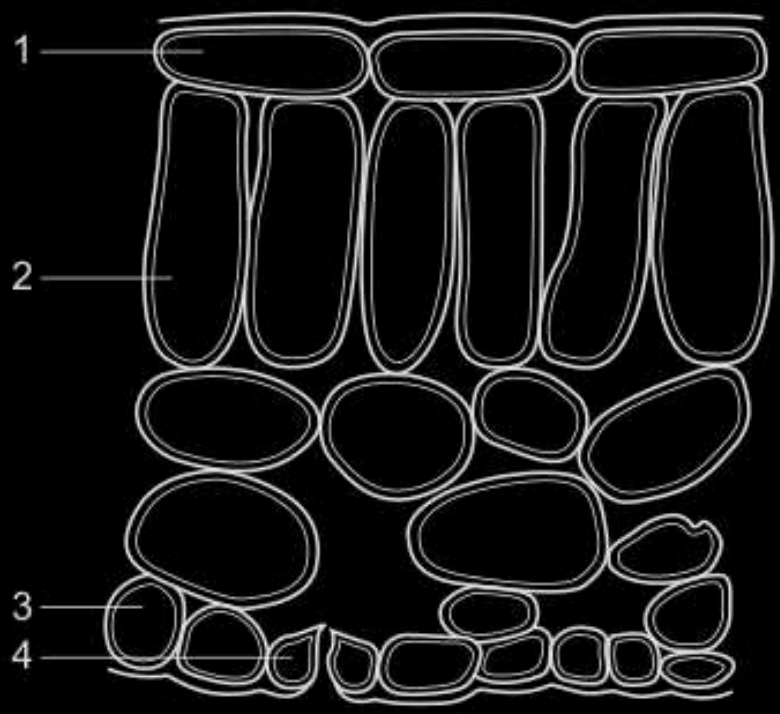
D



May/June 2015 · Paper 1 · Variant 2 · Question 7

Correct answer: D

7 The diagram shows cells in a section through a leaf of a typical green plant. (No cell contents are shown.)



Which cells usually contain chloroplasts?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

May/June 2015 · Paper 1 · Variant 2 · Question 8

Correct answer: C

8 What is an example of assimilation?

A absorption of glycerol into lacteals

B breakdown of alcohol in the liver

C building of proteins from amino acids

D release of a hormone from a gland

May/June 2015 · Paper 1 · Variant 2 · Question 9

Correct answer: C

9 What are the basic units from which glycogen is synthesised?

A amino acids

B fatty acids

C glucose

D glycerol

May/June 2015 · Paper 1 · Variant 2 · Question 10

Correct answer: C

10 A food is known to contain protein, but no reducing sugar.

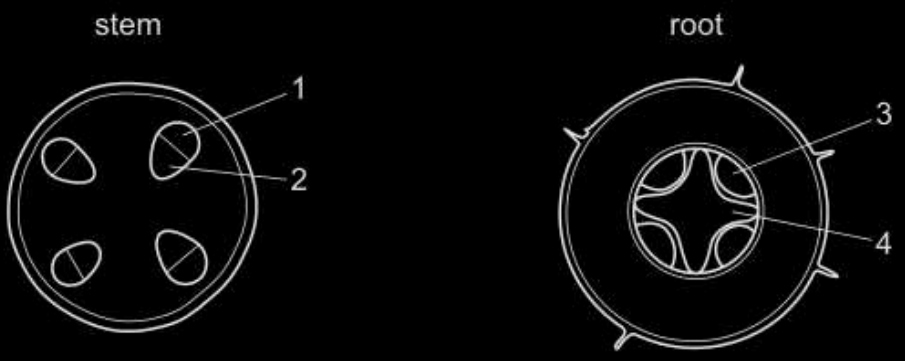
When this food is tested with biuret solution and Benedict's solution which colours would a student observe?

	biuret test	Benedict's test
A	blue	blue
B	blue	brick red
C	lilac / violet	blue
D	lilac / violet	brick red

May/June 2015 · Paper 1 · Variant 2 · Question 11

Correct answer: D

11 The diagrams show transverse sections of a plant stem and of a plant root.



Which regions contain xylem vessels?

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

Correct answer: A

12 In an experiment to investigate transpiration, the leaves of four identical shoots are treated as follows.

1 upper surfaces covered with waterproof jelly

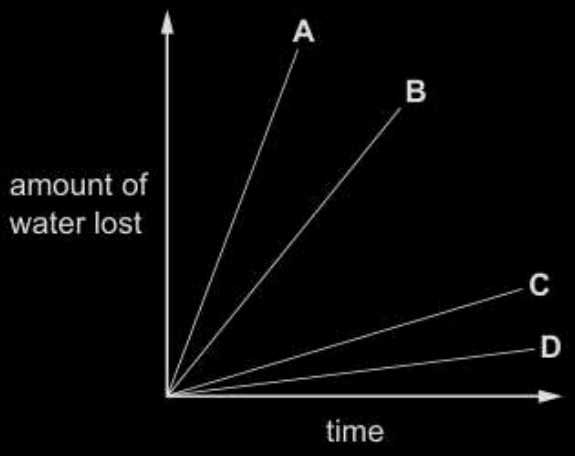
2 lower surfaces covered with waterproof jelly

3 upper and lower surfaces covered with waterproof jelly

4 untreated

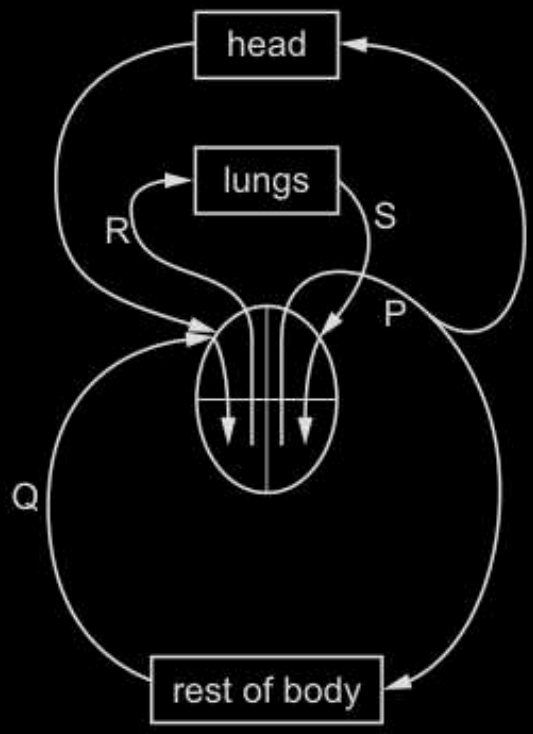
The graph shows the water lost by the four shoots.

Which line shows the result for shoot 4?



Correct answer: C

13 The diagram represents the heart and some major blood vessels.

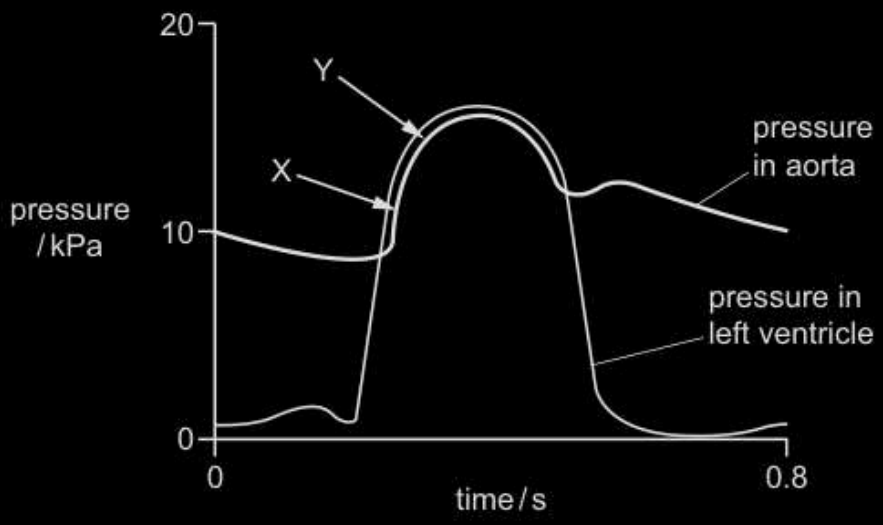


What are possible blood pressures (in kPa) for the vessels shown on the diagram?

	P	Q	R	S
A	1	4	2	16
B	4	16	2	1
C	16	2	4	1
D	16	4	1	2

Correct answer: B

14 The diagram shows changes in pressure in the aorta and the left ventricle during one complete heart beat.



What causes the increase in pressure between points X and Y?

A contraction of the left atrium

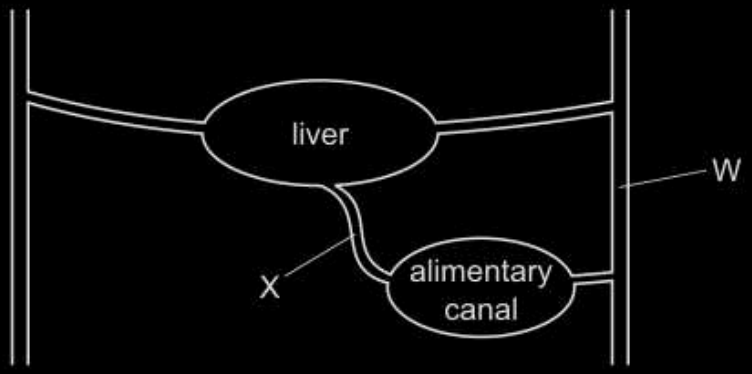
B contraction of the left ventricle

C relaxation of the left atrium

D relaxation of the left ventricle

Correct answer: B

15 The diagram shows the liver and its blood supply.

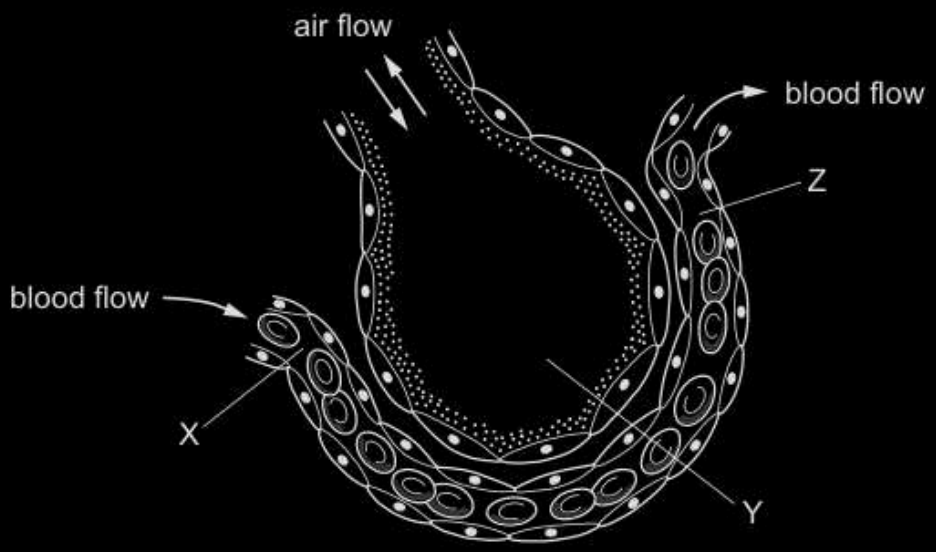


What is the name of blood vessel X, and which substance increases in concentration between vessel W and vessel X?

	name of X	substance
A	hepatic artery	carbon dioxide
B	hepatic portal vein	carbon dioxide
C	hepatic artery	oxygen
D	hepatic portal vein	oxygen

Correct answer: B

18 The diagram shows a section of an alveolus and a capillary in a lung.



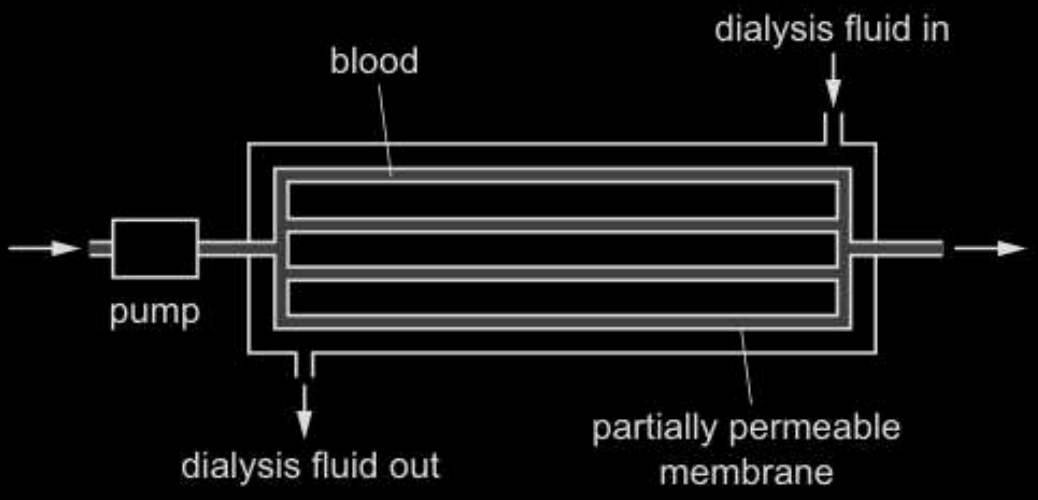
What are the relative concentrations of **carbon dioxide** at X, Y and Z?

	X	Y	Z
A	high	high	high
B	high	low	low
C	low	high	high
D	low	high	low

May/June 2015 · Paper 1 · Variant 2 · Question 19

Correct answer: B

19 The diagram represents a kidney machine.



Which two substances are both present in the dialysis fluid entering the machine?

A glucose and protein

B glucose and salts

C protein and urea

D urea and salts

May/June 2015 · Paper 1 · Variant 2 · Question 21

Correct answer: D

21 What are characteristics of all neurones?

	carry information within the brain	stimulate muscles or glands	transmit electrical impulses
A	✓	✓	✗
B	✓	✗	✓
C	✗	✓	✗
D	✗	✗	✓

May/June 2015 · Paper 1 · Variant 2 · Question 23

Correct answer: B

23 A patient complains to the doctor of unexplained weight loss, severe thirst and frequent need of urination.

A test shows high levels of glucose in the urine.

Which condition does the doctor diagnose?

A cardiovascular disease

B diabetes

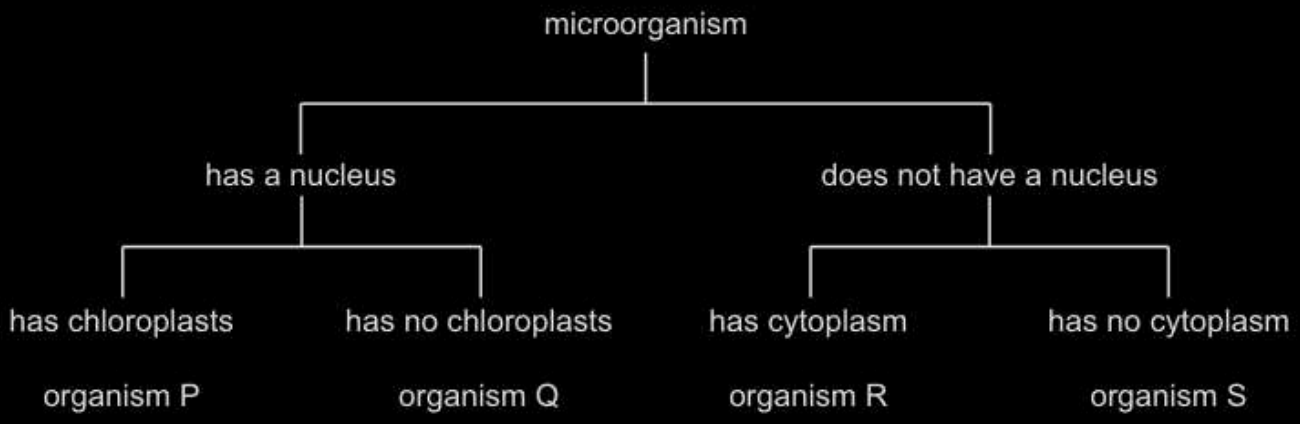
C kidney failure

D obesity

May/June 2015 · Paper 1 · Variant 2 · Question 26

Correct answer: C

26 A student examined four different microorganisms. She noted whether the microorganisms had a nucleus, chloroplasts and cytoplasm. She constructed the table below to identify the microorganisms.



What are organisms Q, R and S?

	Q	R	S
A	bacterium	virus	fungus
B	bacterium	virus	plant
C	fungus	bacterium	virus
D	fungus	bacterium	plant

May/June 2015 · Paper 1 · Variant 2 · Question 34

Correct answer: D

34 What is always essential for seeds to begin germinating?

A carbon dioxide

B light

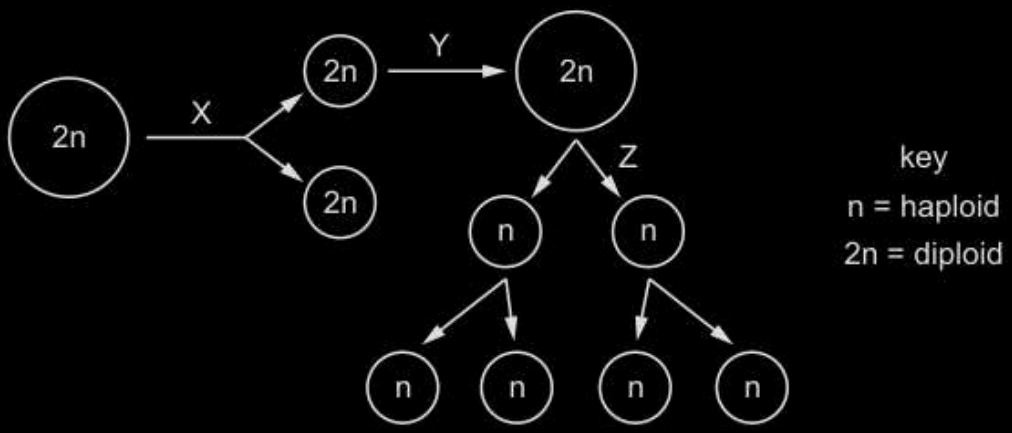
C mineral salts

D oxygen

May/June 2015 · Paper 1 · Variant 2 · Question 36

Correct answer: C

36 The diagram shows some cells in the life cycle of an animal.



Which processes are occurring at X, Y and Z?

	X	Y	Z
A	maturation	meiosis	mitosis
B	meiosis	maturation	mitosis
C	mitosis	maturation	meiosis
D	mitosis	meiosis	maturation

May/June 2015 · Paper 1 · Variant 2 · Question 37

Correct answer: C

37 Which of these may be heterozygous?

A a haploid cell

B an allele of a gene

C an organism with a dominant phenotype

D an organism with a recessive genotype

May/June 2015 · Paper 1 · Variant 2 · Question 39

Correct answer: A

39 A person with Down's syndrome is born with 47 chromosomes in each cell, instead of 46.

What could cause this?

A A mutation happened during the production of the egg cell.

B More than one sperm fused with the egg at fertilisation.

C Radiation caused a change in structure of a gene in the father's sperm.

D The mother was exposed to harmful chemicals while she was pregnant.

May/June 2015 · Paper 1 · Variant 2 · Question 40

Correct answer: A

40 Bacteria can be genetically engineered to produce human insulin.

Before this method was developed, the only insulin available was that from cattle or pigs. It was obtained from extracts of animal pancreas.

Which statements about the two methods are correct?

W Large numbers of bacteria can be cultured in a small space.

X Bacteria reproduce very quickly and make insulin quickly.

Y People sometimes develop diseases from insulin taken from cows or pigs.

Z The insulin produced in bacteria is not the same as that produced in the human pancreas.

A W, X and Y **B** W, X and Z **C** W, Y and Z **D** X, Y and Z

1 Fig. 1.1 shows a vertical section through the skin in two different environmental conditions, **A** and **B**.

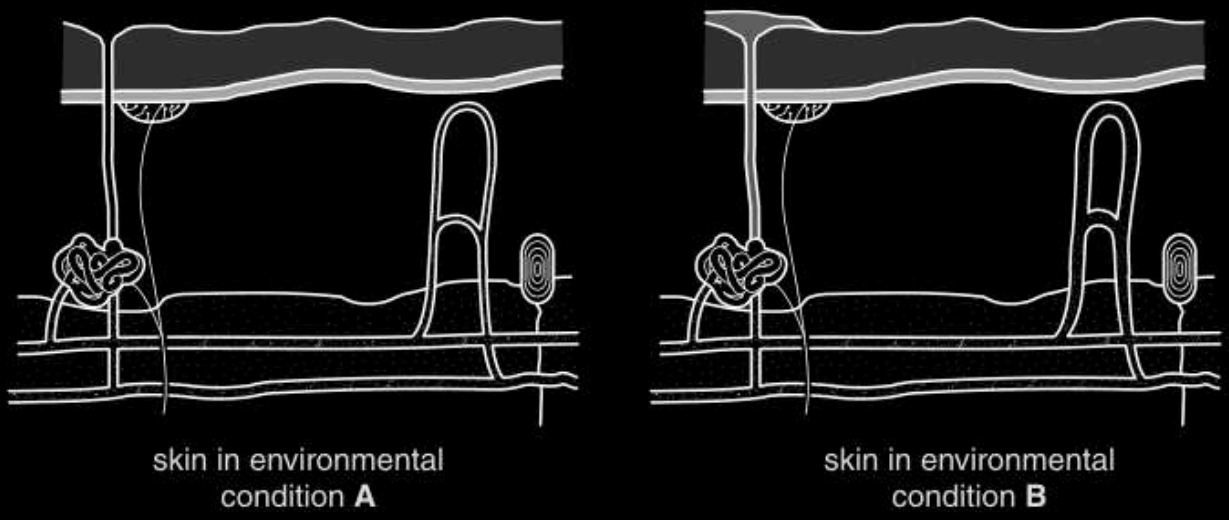


Fig. 1.1

(a) On Fig. 1.1, label each of the following:

- a sweat gland
- a capillary. [2]

(b) Use the information in Fig. 1.1 to suggest how environmental condition **B** is different from environmental condition **A**.

.....

..... [1]

(c) **(i)** State **two** differences between the skin in environmental condition **A** and the skin in environmental condition **B**.

1 .....

.....

2 .....

..... [2]

(ii) Explain the advantages to a person in environmental condition **B** of the **two** differences you have stated in your answer to **(c)(i)**.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 8]

1	**(a)** **(i)**	sweat gland labelled ; capillary labelled ;	**R** sweat duct	[2]
	(b)	warmer ;	must be a comparative statement	[1]
	(c) **(i)**	<u>sweat</u> (present in **B** / **ORA**) ; capillaries carry more blood (in **B** / **ORA**) ; capillaries wider / dilated (in **B** / **ORA**) ;	 **Ig** constrict / shrink	[max 2]
	(ii)	evaporation (of sweat) ; more blood near (surface of) skin ; increased heat loss / cooling ; allows regulation of temperature / prevents overheating ;		[max 3]
				[Total 8]

2 Table 2.1 shows the volume of blood supplied to parts of the body at rest and during strenuous exercise.

Table 2.1

part of body	volume of blood supplied in cm^3/min	
	at rest	during strenuous exercise
brain	750	750
heart	250	750
skeletal muscle	1200	12500
skin	500	1900
kidneys	1100	.....
digestive organs	1400	600
other	600	400
Total	5800	17500

(a) (i) Calculate the volume of blood that is supplied to the brain **at rest** as a percentage of the total supplied to the whole body.

Show your working in the space below.

..... % [2]

(ii) Name the blood vessels that supply each kidney with blood.

..... [1]

(iii) Using the information in Table 2.1, calculate the volume of blood supplied to the kidneys during strenuous exercise.

Write your answer in the space provided in Table 2.1. [1]

(b) Use the information in Table 2.1 to name **two** parts of the body that have an increased supply of blood during strenuous exercise.

Explain the advantage to the body of increasing the supply of blood to each of the parts you name.

name of part .....

advantage .....

.....

.....

.....

name of part .....

advantage .....

.....

.....

..... [4]

(c) Using the information in Table 2.1, suggest why eating immediately before exercise is not recommended.

.....

.....

.....

..... [2]

[Total: 10]

2	**(a)** **(i)**	(750/5800) *100 ; 12.9 / 13 (%) ;		[2]
	(ii)	renal artery ;		[1]
	(iii)	600 ;		[1]

Question	Expected answers	Additional guidance	Marks
(b)			[max 4]
	heart ;		[1]
	skeletal muscle ;		[1]
	for above named parts ref. supply of more oxygen / glucose ;		[max 2]
	ref. increased (aerobic) respiration / prevent anaerobic respiration ;		
	remove lactic acid (for skeletal muscle only) ;		
	ref. contract harder / faster ;		
	skin ;		
	increased heat loss ;		
(c)	less blood to digestive organs ;		[max 2]
	less digestion ;		
	less / slower absorption of products of digestion ;	**A** ref. active transport in digestive system	
			[Total 10]

3 Fig. 3.1 shows a fetus developing in the uterus of a mother. The fluid labelled **C** contains cells from the fetus.

A long, hollow needle may be used to withdraw some of the fluid into a syringe. The DNA from the cells in this fluid can then be analysed to find the sex of the fetus and to detect mutations.

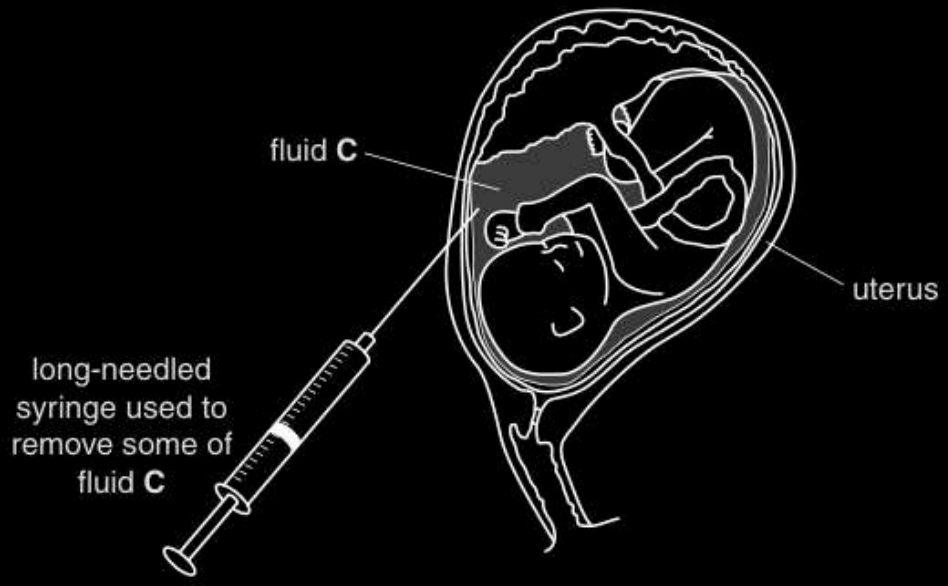


Fig. 3.1

(a) Name fluid **C** and state its function.

name of fluid .....

function .....

..... [2]

(b) **(i)** Label the placenta on Fig. 3.1 using a line and the letter **P**. [1]

(ii) State **two** functions of the placenta.

1 .....

.....

2 .....

..... [2]

(c) Fig. 3.2 shows the chromosomes found in the nucleus of one cell of a developing fetus.

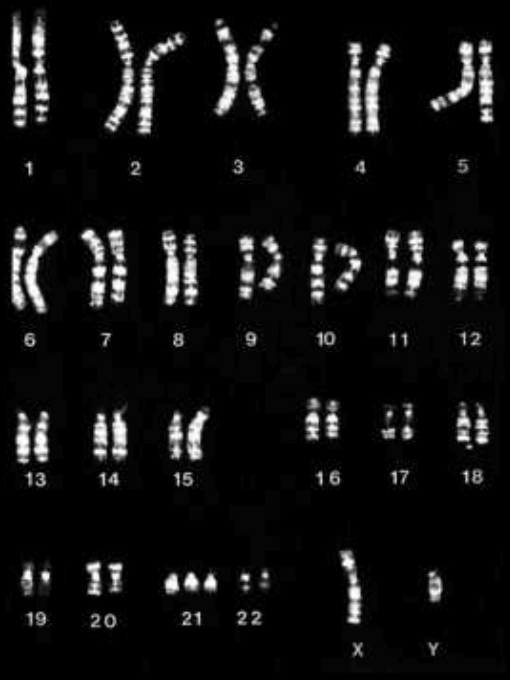


Fig. 3.2

State the sex of this fetus and explain your answer.

sex .....

explanation .....

..... [2]

(d) This fetus has a mutation.

(i) Describe the mutation shown in Fig. 3.2.

.....

.....

.....

..... [2]

(ii) Suggest the condition that this child could be born with as a result of this mutation.

..... [1]

[Total: 10]

3 (a)	amniotic ; prevents physical harm / damage to fetus ;		[2]
(b) (i)	placenta correctly labelled ;		[1]
(ii)	exchange / passage in correct direction of + nutrients / named ; + gases / named ; + excretory products / named ; + antibodies ; ref. prevent mixing of maternal and fetal blood ;		[max 2]

Question	Expected answers	Additional guidance	Marks
(c)	male ; presence of Y (chromosome) ;	 **R** gene	[2]
(d) (i)	one extra chromosome / trisomy / three not two / 47 not 46 ; position / pair 21 ;		[2]
(ii)	Down's syndrome ;		[1]
			[Total 10]

7 **(a)** Describe the significance of each of the following features of a dicotyledonous leaf in terms of the process named:

(i) the distribution of chloroplasts in the process of photosynthesis

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) stomata and mesophyll cells in the process of gas exchange.

.....

.....

.....

.....

.....

..... [3]

(b) Transverse sections were taken from the root and stem of a dicotyledonous plant.

Describe differences in how **two named** tissues involved in transport are arranged in each of these sections.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7	**(a)**	**(i)**	in palisade ; spongy ; ref. more in palisade ; none in epidermis / xylem / phloem / vein / transparent epidermis ; near leaf surface / (sun)light / to absorb more or most sunlight ; rapid **AW** rate of photosynthesis ; guard cells ; controlling stomata ;	 **A** exposed to **A** with ref. either distribution or number	[max 4]

Question	Expected answers	Additional guidance	Marks
(ii)	CO_2 ; O_2 ; correct ref. to photosynthesis or respiration ; diffusion ; *(stomata):* allow water vapour out / transpiration ; *(mesophyll cells):* moist surface ; large surface area ;		[max 3]
(b)	xylem + phloem ; around edge of stem ; centre of root / description of arrangement ; many xylem (stem) + one xylem (root) ; phloem and xylem paired in stem / unpaired in root ;	**A** in vascular bundles in stem	[max 3]
			[Total 10]

9 **(a)** Describe and explain how an **increase** in each of the following factors surrounding a plant affects the rate of transpiration:

- temperature

.....

.....

.....

.....

- light intensity

.....

.....

.....

.....

- humidity.

.....

.....

.....

..... [7]

(b) Suggest the importance of transpiration to a plant.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

9 (a)	correct ref. diffusion in any section ; *temperature* rate increases ; increased evaporation (of water) ; from surface of mesophyll cells ; ref. (diffusion) gradient* ; ref. kinetic energy / molecules move faster ; *light intensity* rate increases ; more stomata open / stomata open wider ; increased surface area for water loss ; *humidity* rate decreases ; ref. (diffusion) gradient* ;	 **R** guard cells open **Ig** water alone	[max 7]

Question	**Expected answers**	**Additional guidance**	**Marks**
(b)	to bring water / salts ; from roots to leaves / up the stem ; for photosynthesis ; cool leaf / plant ;	**A** nutrients	[max 3]
			[Total 10]

1 Fig. 1.1 shows the effect of temperature on the rate at which yeast cells in a nutrient solution produce bubbles of a gas.

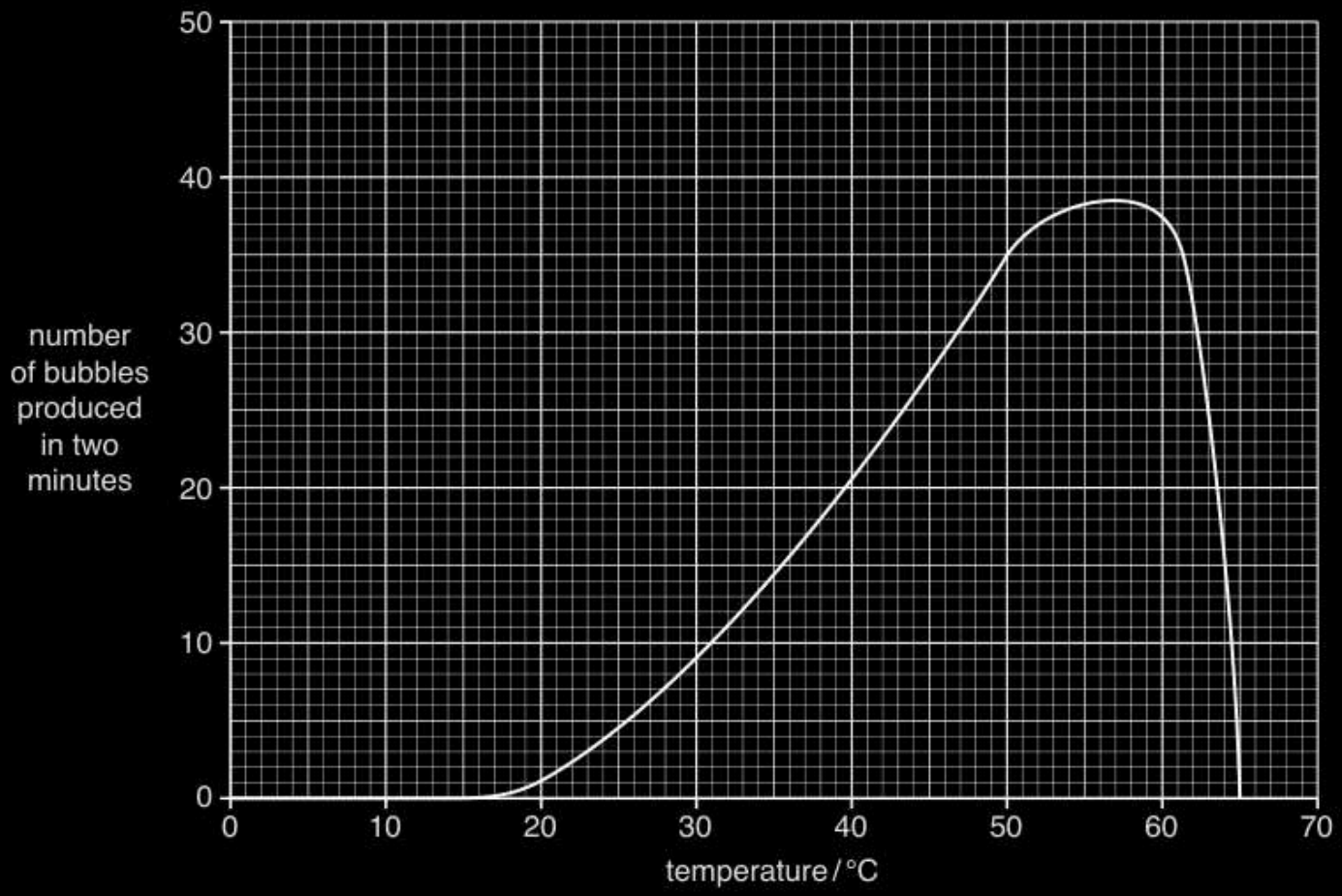


Fig. 1.1

(a) Name each of the following:

- the gas released

.....

- the metabolic process that releases it

.....

- the essential chemical constituents of the nutrient solution.

..... and ..... [3]

(b) Use Fig. 1.1 to find the optimum temperature for the metabolic process.

..... [1]

(c) Explain the shape of the curve after 60 °C.

.....

.....

.....

.....

.....

.....

..... [3]

(d) Explain what would happen to the rate at which bubbles of the gas are produced by the yeast if the temperature of the solution is then gradually reduced from 65 °C to 45 °C.

.....

.....

.....

..... [2]

[Total: 9]

1	(a)	carbon dioxide / CO_2 ; respiration / fermentation ; *any two for one mark:* sugar OR carbohydrates OR named / water / protein OR amino acids / oxygen / lipids OR fats / mineral ions OR named ;	**A** two named mineral ions for 1 mark	[3]
	(b)	any temperature in range 56 – 58 °C ;		[1]
	(c)	1. rate decreases **+** zero / stops ; 2. enzymes ; 3. denatured ; 4. references to active site / lock and key hypothesis ; 5. yeast dies / destroyed / killed ;	1. **A** number of bubbles per (two) minutes(s) 2. **R** denaturing of yeast / killing of enzymes	[max 3]
	(d)	1. rate remains at zero / stays the same / no change / no effect ; 2. denaturing is permanent / enzymes still denatured ; 3. death of yeast ;	1. **A** no gas production / no bubbles 2. **A** active site still changed	[max 2]
				[Total 9]

Question	Expected answers	Additional guidance	Marks

2 Fig. 2.1 shows a plant called the ghost plant and a magnified diagram of its flower. It is called a ghost plant because it is often completely white in colour. Cells of the ghost plant do not contain chloroplasts.

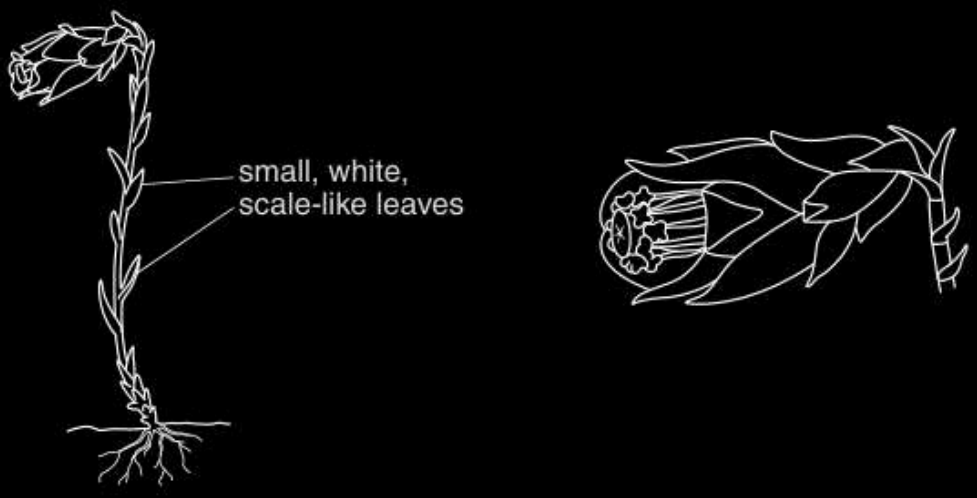


Fig. 2.1

(a) From the appearance of the flower, suggest how it is pollinated. Give a reason for your answer.

how the flower is pollinated .....

reason .....

..... [2]

Question 2 continues on page 6

Question 2 continues over the page

(b) The ghost plant has an unusual feeding relationship with an underground fungus and with a nearby tree, as shown in Fig. 2.2 (opposite).

(i) From the information given in Fig. 2.2, name the producer.

..... [1]

(ii) The flow of nutrients in the feeding relationship is shown by the arrow on Fig. 2.2. Suggest **three** nutrients that flow from the tree to the ghost plant.

1 .....

2 .....

3 ..... [3]

(c) Suggest why the leaves of the ghost plant do not possess stomata or large intercellular spaces.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 11]

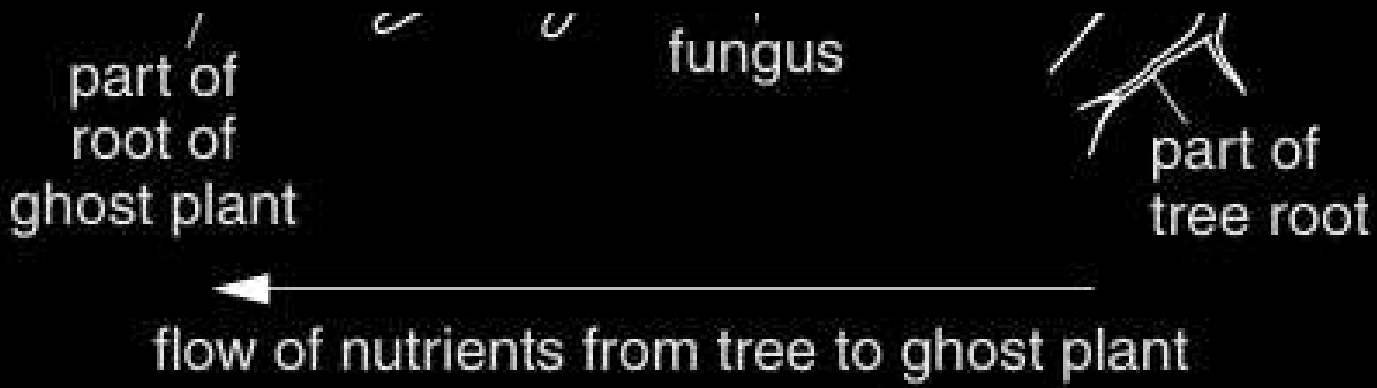

part of
root of
ghost plant
fungus
part of
tree root
flow of nutrients from tree to ghost plant

Fig. 2.2

2 (a)	*how the flower is pollinated:* insect / self ; *reason (insect):* 1. stigma / carpel **+** enclosed ; 2. anthers / stamens **+** enclosed ; 3. anthers / stamens **+** small ; 4. white colour may attract (insects) ; *reason (self):* 5. anthers and stigma close ; 6. white / not brightly coloured **+** doesn't attract (insects);	2. **A** not pendulous	[max 2] [1] [max 1]
(b) (i)	tree ;		[1]
(ii)	carbohydrate / sucrose / glucose / sugar ; amino acids ; mineral / ions / salts / nitrate ;	**lg** water **R** protein **A** phosphate / potassium **lg** magnesium	[3]
(c)	1. no chlorophyll ; 2. no photosynthesis ; 3. carbohydrates obtained from other organism / cannot make carbohydrates ; 4. no CO_2 requirement ; 5. reduced need for gas diffusion / gas exchange ; 6. ref. reduced / no transpiration / description ;	1. **lg** chloroplasts	[max 5]
			[Total 11]

3 Fig. 3.1 shows the blood supply to cells in the liver.

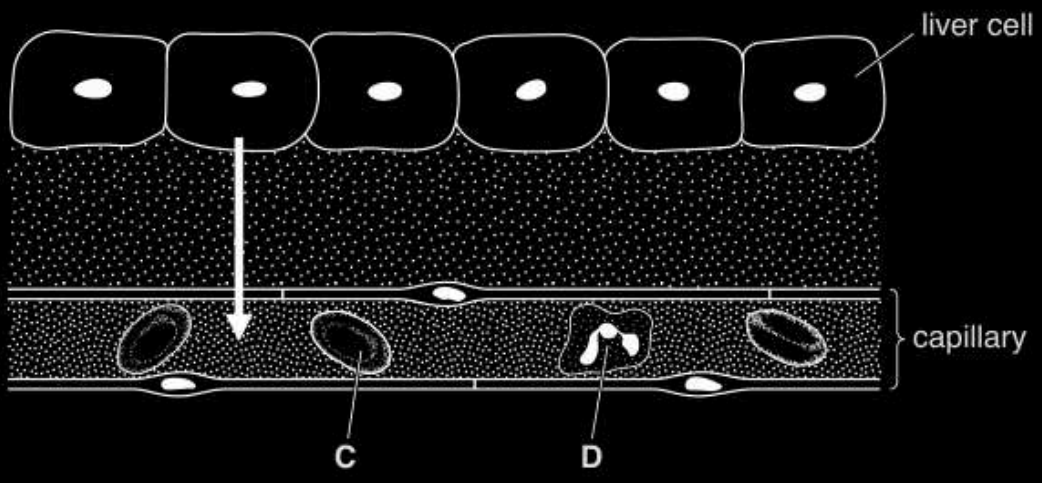


Fig. 3.1

(a) (i) Name the cells labelled **C** and **D** in Fig. 3.1.

C .....

D ..... [2]

(ii) The arrow in Fig. 3.1 shows the movement of substances from the liver cells into the capillary.

Name **three** substances that move in the direction shown.

1 .....

2 .....

3 ..... [3]

(b) (i) Describe the effect of adrenaline on liver cells.

.....

.....

.....

..... [2]

(ii) State a situation in which this might occur.

..... [1]

(c) Sometimes the liver is unable to remove glucose from the blood. This condition is called diabetes.

(i) State **two** symptoms of this condition.

1 .....

2 ..... [2]

(ii) State how this condition is treated.

.....

..... [1]

[Total: 11]

3	**(a)**	**(i)**	*C* – RBC / erythrocyte ; *D* – WBC / phagocyte / polymorph / neutrophil ;	**A** polymorph / neutrophil	[2]

Question	Expected answers	Additional guidance	Marks
(ii)	1. urea ; 2. CO_2 ; 3. iron ; 4. glucose ; 5. (named) vitamins ; 6. water ;	1. **A** uric acid / creatinine	[max 3]
(b) (i)	1. glycogen ; 2. (broken down to) glucose ; 3. (glucose) released / carried in blood / increases blood glucose level ;	**Ig** ref. to glucagon / pancreas	[max 2]
(ii)	fight / flight / fright (or described) ;		[1]
(c) (i)	increased / high **+** blood glucose / sugar level ; presence of glucose / sugar **+** in urine ;		[2]
(ii)	1. insulin ; 2. controlled CHO / sugar in diet ;		[max 1]
			[Total 11]

4 The dominant allele for the ability to smell the scent of a particular flower is represented by **A**. The recessive allele, which does not allow a person to smell the scent of the flower, is represented by **a**.

(a) Using these letters, indicate each of the following:

(i) the genotype of a woman who is unable to smell the flower

..... [1]

(ii) the possible alleles found in the gametes of a woman who can smell the flower.

..... and ..... [2]

(b) Fig. 4.1 represents some alleles on part of the sex chromosomes of a woman and of a man.

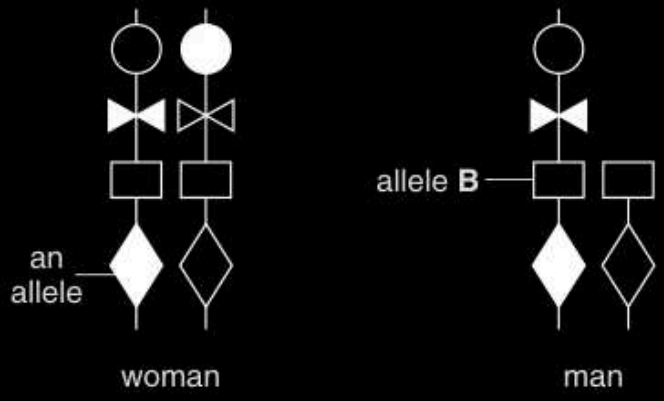


Fig. 4.1

In the space below, draw these alleles as they might appear in a sperm cell that carries the Y chromosome.

[2]

(c) Fig. 4.2 shows how the alleles on one of the chromosomes might appear in a cell taken from somewhere else in the man's body. Allele **B** shows a mutation.

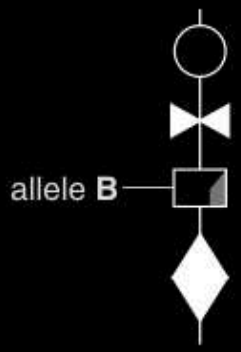


Fig. 4.2

Suggest **two** possible causes of the mutation.

1 .....

2 ..... [2]

(d) Mutated alleles such as that shown in Fig. 4.2 are usually recessive.

Use your knowledge of genetics to explain why society discourages marriage between closely-related people.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

4 (a) (i)	aa ;		[1]
(ii)	a / A ; A ;		[2]
(b)	chromosome with 2 alleles drawn ; the same two alleles as in Fig. 4.1 ;		[2]
(c)	 radiation or type named ; chemicals or mutagenic chemical named ;	**Ig** ref. to inheritance e.g. uv, gamma, alpha e.g. mustard gas, benzene, tar, etc.	[2]

Question	Expected answers	Additional guidance	Marks
(d)	 1. parents with normal phenotype / appear normal / do not have mutational condition ; 2. heterozygotes / carriers of recessive allele ; 3. <u>greater chance</u> (correctly qualified) ; 4. children + inheriting 2 recessive alleles / being homozygous ; 5. suffering from mutational condition ;	**A** reverse argument throughout **A** mutated as being equiv. to recessive throughout 5. **Ig** genetic disease	[max 3]
			[Total 10]

5 Table 5.1 shows the mean daily water intake and loss by a person.

Table 5.1

water intake / dm^3		water loss / dm^3	
drinks	1.50	faeces	0.10
food	0.75	sweat	0.52
		urine	1.50
		exhaled air	……..
Total	2.25	**Total**	2.50

(a) (i) Using the information in Table 5.1, calculate the daily loss of water in exhaled air.

..... [1]

(ii) Explain why exhaled air contains water.

.....

.....

.....

..... [2]

(b) Explain why, even though 2.25 dm^3 of water are taken in through the mouth, the faeces contain only 0.10 dm^3 of water.

.....

.....

.....

.....

.....

..... [3]

(c) Explain the importance of water in urine.

.....

.....

..... [2]

(d) The difference between water intake and water loss by a person is accounted for by water produced by a metabolic process in the body. Name this metabolic process.

..... [1]

[Total: 9]

Section B

Answer **both** questions in this section.

Write your answers in the spaces provided.

5 (a) (i)	0.38 **+** dm^3 ;		[1]
(ii)	evaporation / water vapour ; from (moist lining of) alveoli / lungs ;	 **Ig** other named parts of breathing system	[2]
(b)	1. water is absorbed **+** blood ; 2. (from) ileum / small intestine / colon / large intestine ; 3. ref need to prevent water loss / importance of water in body (e.g. solvent / transport) ; 4. water lost by other methods / ref. to any named other method of water loss ;		[max 3]
(c)	water is a solvent / dissolves ; any one named solute ;	 **A** salts	[2]
(d)	<u>respiration</u> ;	if qualified must be aerobic	[1]
			[Total 9]

Question	Expected answers	Additional guidance	Marks

7 **(a)** Describe each of the following processes:

(i) active transport

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) osmosis.

.....

.....

.....

.....

.....

..... [3]

(b) Explain what happens to a red blood cell when it is placed in pure water.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

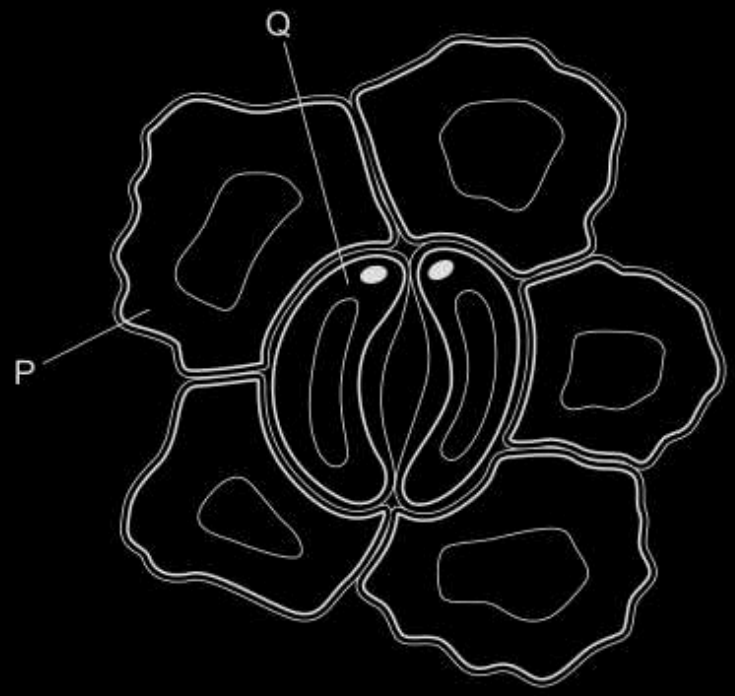
Write your answers in the spaces provided.

7 (a) (i)	1. requires energy ; 2. from respiration 3. movement of ions / molecules / substances / particles ; 4. (through) living / cell membrane ; 5. against concentration gradient / described ; 6. correct example, e.g. mineral ions and root hair cells / glucose and ileum	2. **R** produced from respiration 4. **Ig** partially permeable 6. **A** intestine for ileum	[max 4]
(ii)	1. water ; 2. down (water) concentration / water potential gradient ; 3. partially permeable membrane ; 4. by diffusion ; 5. passive / energy not required ;		[max 3]
(b)	1. water uptake ; 2. ref. water potential higher outside of cell / ref. concentration gradient ; 3. increase in volume / cell swells ; 4. membrane cannot withstand pressure ; 5. cell bursts ;	1. **Ig** osmosis 3. **Ig** turgidity 4. **A** no cell wall 5. **A** haemolysis	[max 3]
			[Total 10]

Question	Expected answers	Additional guidance	Marks

Correct answer: C

1 The diagram shows cells in the epidermis of a leaf.



To complete the diagram, which structural features should be added to the cells P and Q?

	P		Q	
	chloroplasts	nucleus	chloroplasts	nucleus
A	✓	✓	✗	✗
B	✓	✗	✓	✓
C	✗	✓	✓	✗
D	✗	✗	✗	✓

October/November 2014 · Paper 1 · Variant 1 · Question 2

Correct answer: B

2 The list shows the absorption of three different substances by organisms.

1 mineral ions entering a root hair cell from the soil

2 oxygen entering the blood from an alveolus

3 water entering a root hair cell from the soil

In each case, what is the mechanism of absorption?

	mineral ions	oxygen	water
A	active transport	active transport	diffusion
B	active transport	diffusion	osmosis
C	diffusion	active transport	diffusion
D	diffusion	osmosis	active transport

October/November 2014 · Paper 1 · Variant 1 · Question 3

Correct answer: D

3 A cube of fresh potato is weighed. It is then placed in a test-tube containing a dilute solution of sucrose. After an hour, its mass has increased.

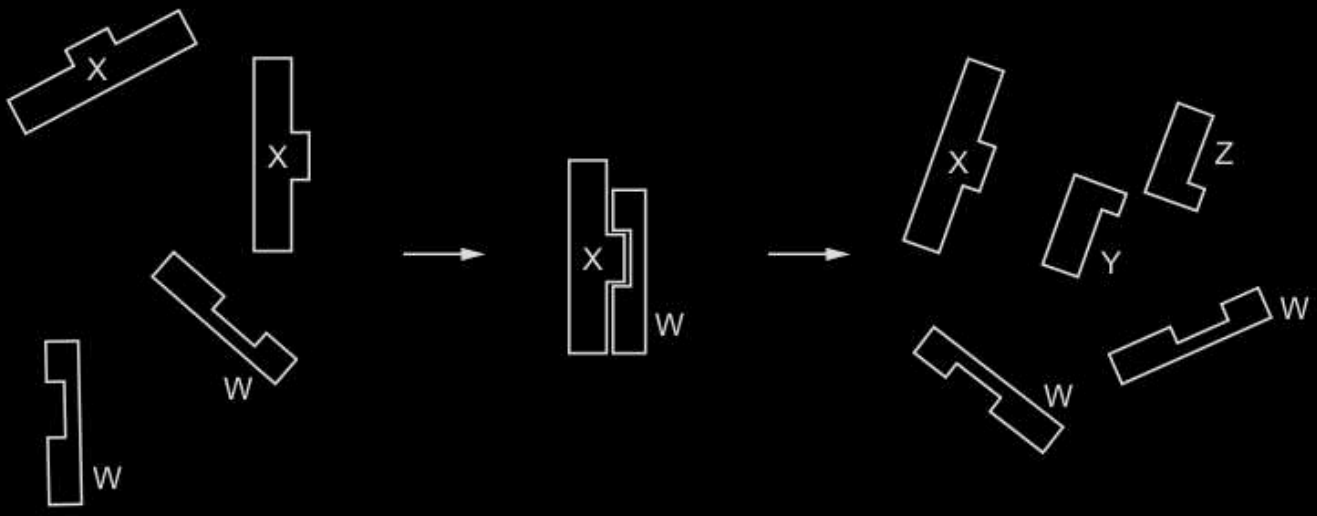
Which process has occurred and what has happened to the concentration of the sucrose in the solution in the test-tube?

	process	sucrose concentration
A	active transport	decreased
B	active transport	increased
C	osmosis	decreased
D	osmosis	increased

October/November 2014 · Paper 1 · Variant 1 · Question 4

Correct answer: B

4 The diagram illustrates the 'lock and key' hypothesis of enzyme action.



What are the enzyme, product and substrate in this reaction?

	enzyme	product	substrate
A	W	X	Y and Z
B	W	Y and Z	X
C	X	W	Y and Z
D	X	Y and Z	W

October/November 2014 · Paper 1 · Variant 1 · Question 6

Correct answer: A

6 What are the effects of a lack of magnesium ions and nitrate ions on plant growth?

	lack of magnesium ions	lack of nitrate ions
A	leaves go yellow between the veins	unable to form enzymes
B	roots lack root hairs	overall growth stunted
C	unable to form amino acids	unable to form glucose
D	unable to form cellulose	oldest leaves go yellow

October/November 2014 · Paper 1 · Variant 1 · Question 9

Correct answer: B

9 Which blood vessels carry absorbed nutrients and oxygen into the liver?

	carries absorbed nutrients	carries oxygen
A	hepatic artery	hepatic vein
B	hepatic portal vein	hepatic artery
C	hepatic portal vein	hepatic vein
D	hepatic vein	hepatic artery

October/November 2014 · Paper 1 · Variant 1 · Question 10

Correct answer: A

10 A piece of butter containing fat is eaten.

After the fat has been swallowed, which chemicals will enter the stomach and the ileum?

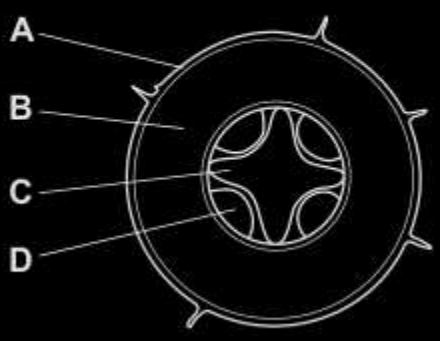
	stomach	ileum
A	fat	fatty acids and glycerol
B	fatty acids	glucose
C	fatty acids and glycerol	fatty acids and glycerol
D	glucose	fat

October/November 2014 · Paper 1 · Variant 1 · Question 11

Correct answer: D

11 The diagram shows a section through a root.

Which tissue transports amino acids?



October/November 2014 · Paper 1 · Variant 1 · Question 12

Correct answer: A

12 Which statement describes transpiration?

A evaporation of water from mesophyll cells and its loss through the stomata

B gaseous exchange between the leaves and the atmosphere

C movement of water by osmosis from the roots to the leaves

D movement of water up through the xylem and into the mesophyll cells

October/November 2014 · Paper 1 · Variant 1 · Question 13

Correct answer: B

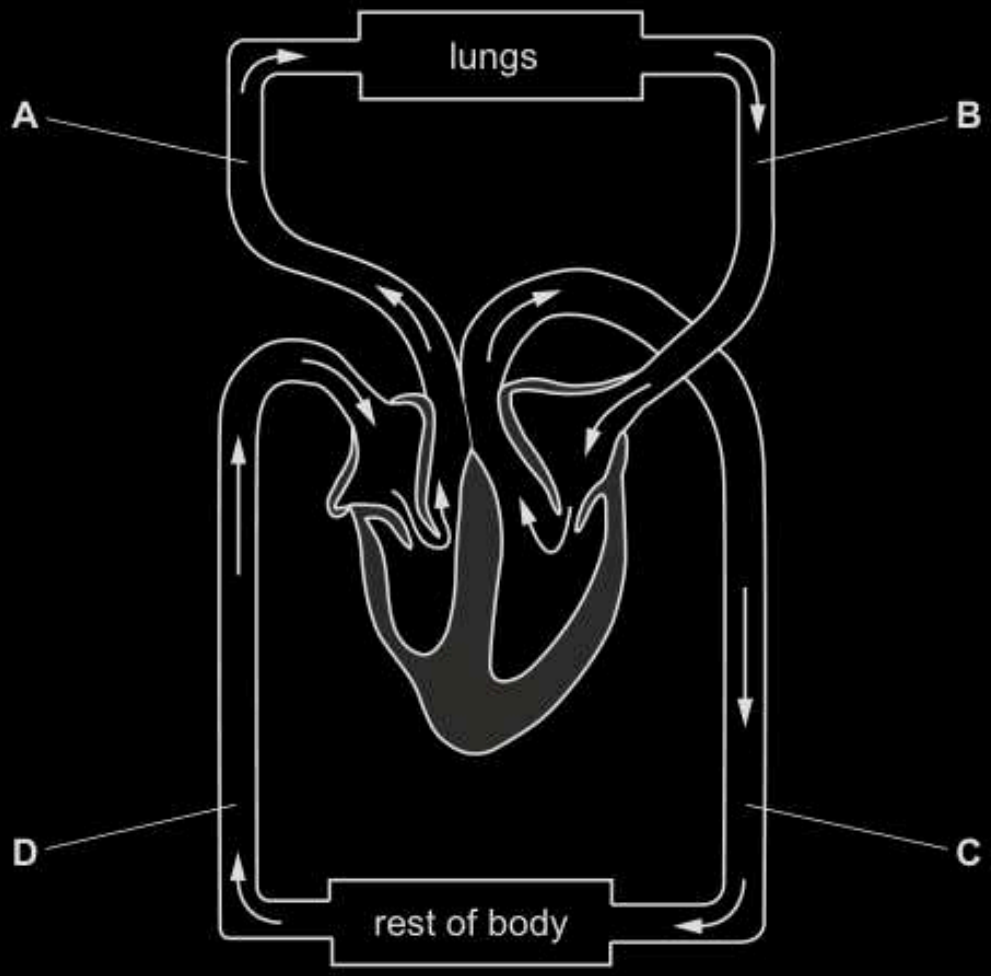
13 Which blood vessels contain valves?

	aorta	capillaries	renal artery	renal vein
A	✓	✓	✗	✗
B	✓	✗	✗	✓
C	✗	✓	✓	✗
D	✗	✗	✓	✓

Correct answer: C

14 The diagram shows the circulatory system.

In which vessel is the blood pressure highest?



October/November 2014 · Paper 1 · Variant 1 · Question 15

Correct answer: A

15 The table shows the characteristics of the blood in one blood vessel in the body.

oxygen concentration	carbon dioxide concentration	pressure
high	low	high

Which blood vessel contains blood with these characteristics?

A aorta

B pulmonary artery

C pulmonary vein

D vena cava

October/November 2014 · Paper 1 · Variant 1 · Question 16

Correct answer: C

16 Some characteristics of the human gas exchange system are listed.

1 a difference in temperature between the air in the alveoli and the blood

2 a large surface area of the alveoli and the blood capillaries

3 a short diffusion distance between the air and the blood

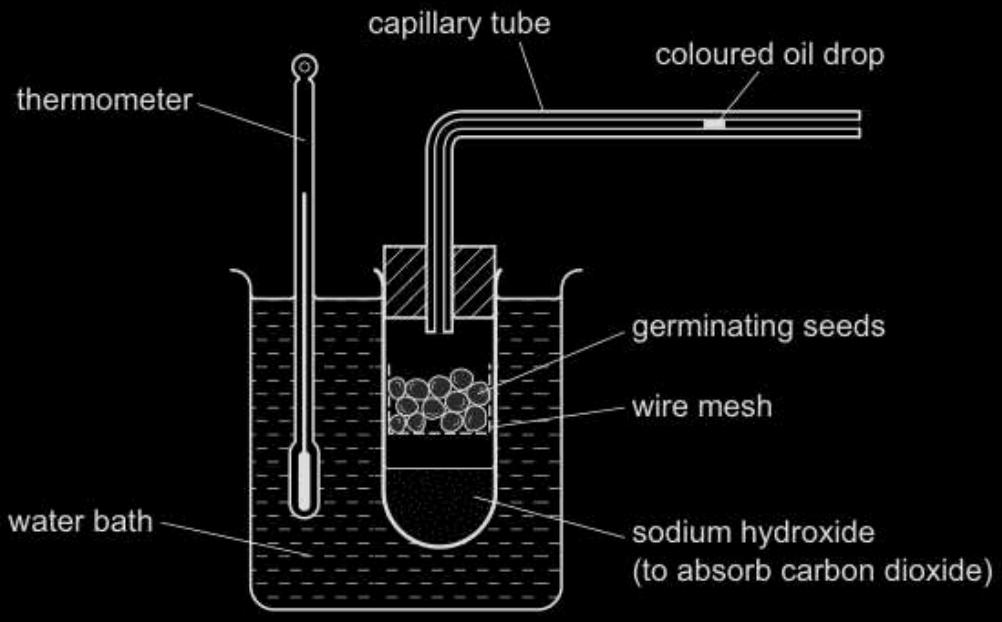
4 elastic fibres in the walls of the alveoli

Which factors affect the rate of oxygen uptake into the blood?

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 3 and 4

Correct answer: C

17 The diagram shows how some apparatus is set up to investigate respiration in germinating seeds.



The coloured oil drop moves along the capillary tube.

What causes the movement of the coloured oil drop?

A carbon dioxide released

B heat released

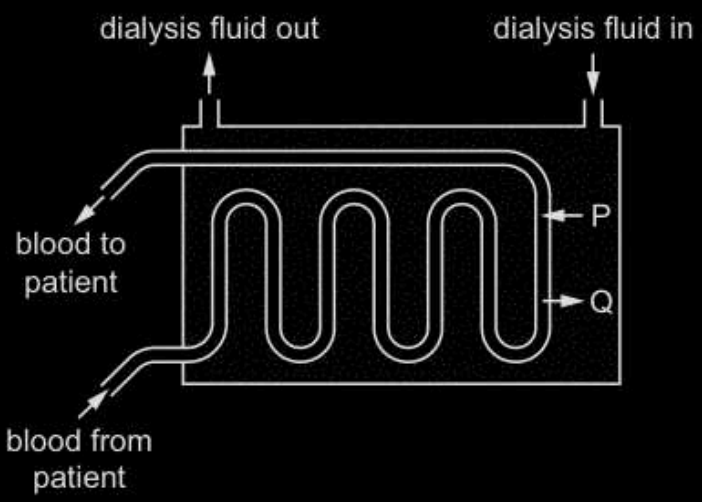
C oxygen used

D water used

October/November 2014 · Paper 1 · Variant 1 · Question 19

Correct answer: D

19 The diagram shows a dialysis machine.



In a person whose health problems affect only the kidneys, which substances will move as shown at P and Q?

	P	Q
A	glucose	protein
B	glucose	urea
C	some salts	protein
D	some salts	urea

October/November 2014 · Paper 1 · Variant 1 · Question 20

Correct answer: D

20 A person's kidneys remove waste from the blood and help to keep the composition of the blood constant.

Which word describes these functions?

A absorption

B assimilation

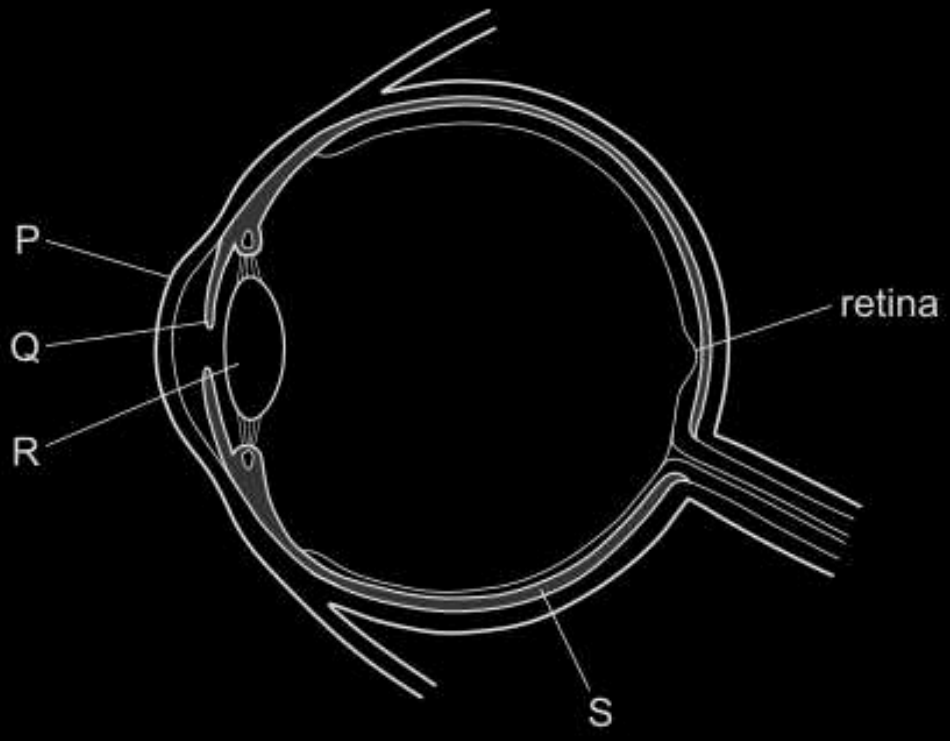
C diffusion

D homeostasis

October/November 2014 · Paper 1 · Variant 1 · Question 21

Correct answer: B

21 The diagram shows a section through the eye.



Which pair of structures focus light rays onto the retina?

A P and Q **B** P and R **C** Q and R **D** Q and S

October/November 2014 · Paper 1 · Variant 1 · Question 23

Correct answer: C

23 In what way is the pupil reflex different from some other reflexes?

A A change in the environment causes impulses in the sensory neurone.

B The function of the reflex is protective.

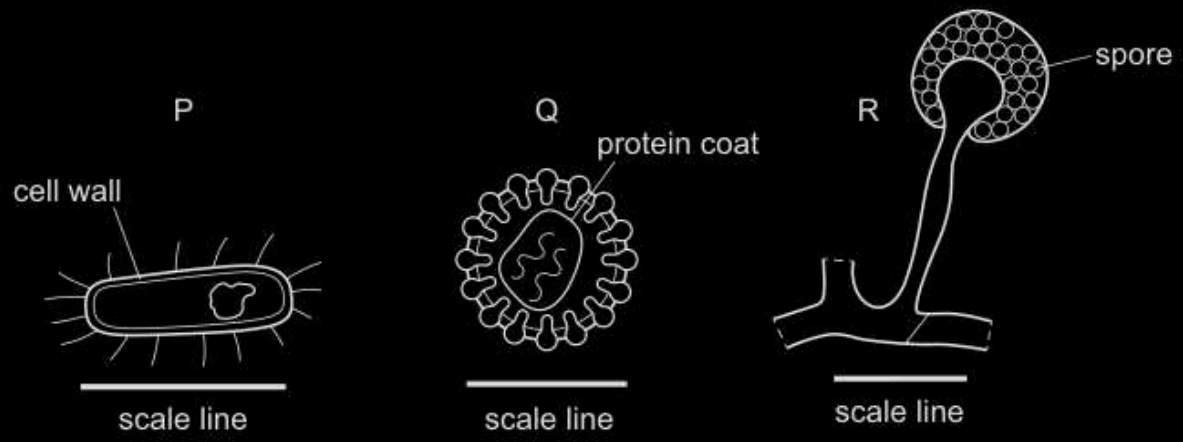
C The relay neurones are in the brain.

D The response occurs quickly.

October/November 2014 · Paper 1 · Variant 1 · Question 27

Correct answer: C

27 The diagrams show three organisms, **not** drawn to the same scale. Each diagram has its own scale line.



Using the same units (μm) in each diagram, what length is the scale line for each organism?

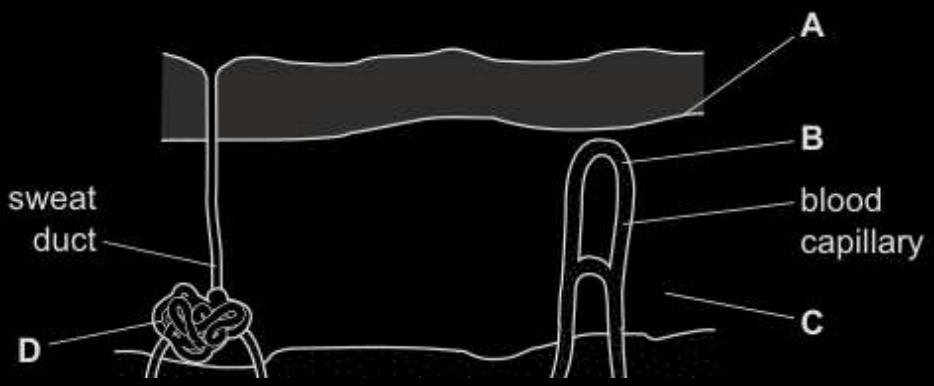
	P	Q	R
A	0.1	2.0	80–100
B	0.1	80–100	2.0
C	2.0	0.1	80–100
D	80–100	2.0	0.1

October/November 2014 · Paper 1 · Variant 1 · Question 33

Correct answer: A

33 The diagram represents a section through the human skin.

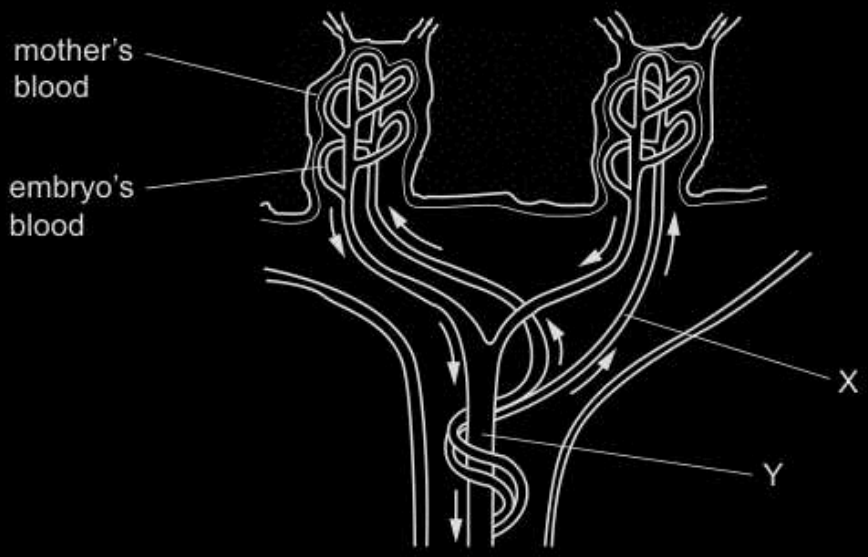
In which part is mitosis occurring most rapidly?



October/November 2014 · Paper 1 · Variant 1 · Question 36

Correct answer: B

36 The diagram shows how the blood of a human embryo flows close to the mother's blood in the placenta.



Which substances are present at X in higher concentrations than at Y?

A carbon dioxide and glucose

B carbon dioxide and urea

C glucose and oxygen

D glucose and urea

October/November 2014 · Paper 1 · Variant 1 · Question 37

Correct answer: B

37 Which characteristic shows continuous variation?

A blood group phenotypes

B body weight

C sex

D sickle cell anaemia

October/November 2014 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 In the ABO blood grouping system, which genotype is homozygous dominant?

A $I^A\ I^O$ **B** $I^A\ I^B$ **C** $I^B\ I^B$ **D** $I^O\ I^O$

October/November 2014 · Paper 1 · Variant 1 · Question 40

Correct answer: A

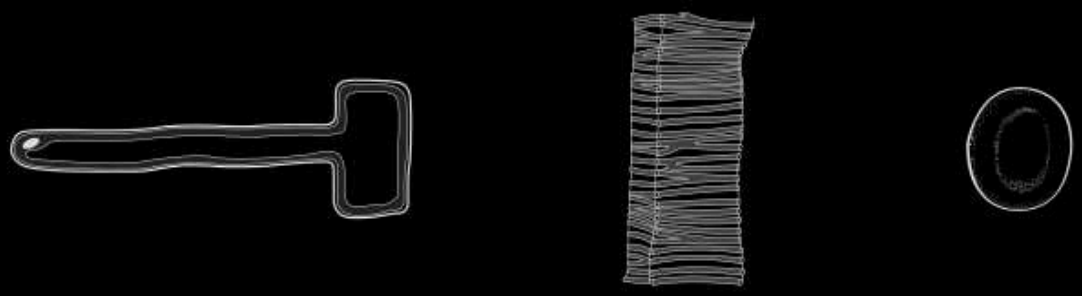
40 If a homozygous recessive is crossed with a heterozygous individual, what would be the ratio of dominant to recessive individuals in the offspring?

A 1:1 **B** 2:1 **C** 3:1 **D** 4:1

October/November 2014 · Paper 1 · Variant 2 · Question 1

Correct answer: C

1 The diagram shows three cellular structures.



1 2 3

Which statements about these cells are correct?

	1	2	3
A	adapted to carry oxygen	lacks a nucleus	adapted to carry water
B	adapted to carry water	lacks a nucleus	is in contact with the soil
C	is in contact with the soil	adapted to carry water	adapted to carry oxygen
D	is in contact with the soil	is in contact with the soil	lacks a nucleus

October/November 2014 · Paper 1 · Variant 2 · Question 2

Correct answer: C

2 Active transport, diffusion and osmosis are described below.

1 the movement of ions or molecules across the cell membrane against a concentration gradient using energy

2 the movement of ions or molecules from a region of high concentration to a region of low concentration down a concentration gradient

3 the movement of water molecules from a region of their higher concentration to a region of their lower concentration through a partially permeable membrane

What links the descriptions with their names?

	diffusion	osmosis	active transport
A	1	3	2
B	2	1	3
C	2	3	1
D	3	2	1

October/November 2014 · Paper 1 · Variant 2 · Question 3

Correct answer: D

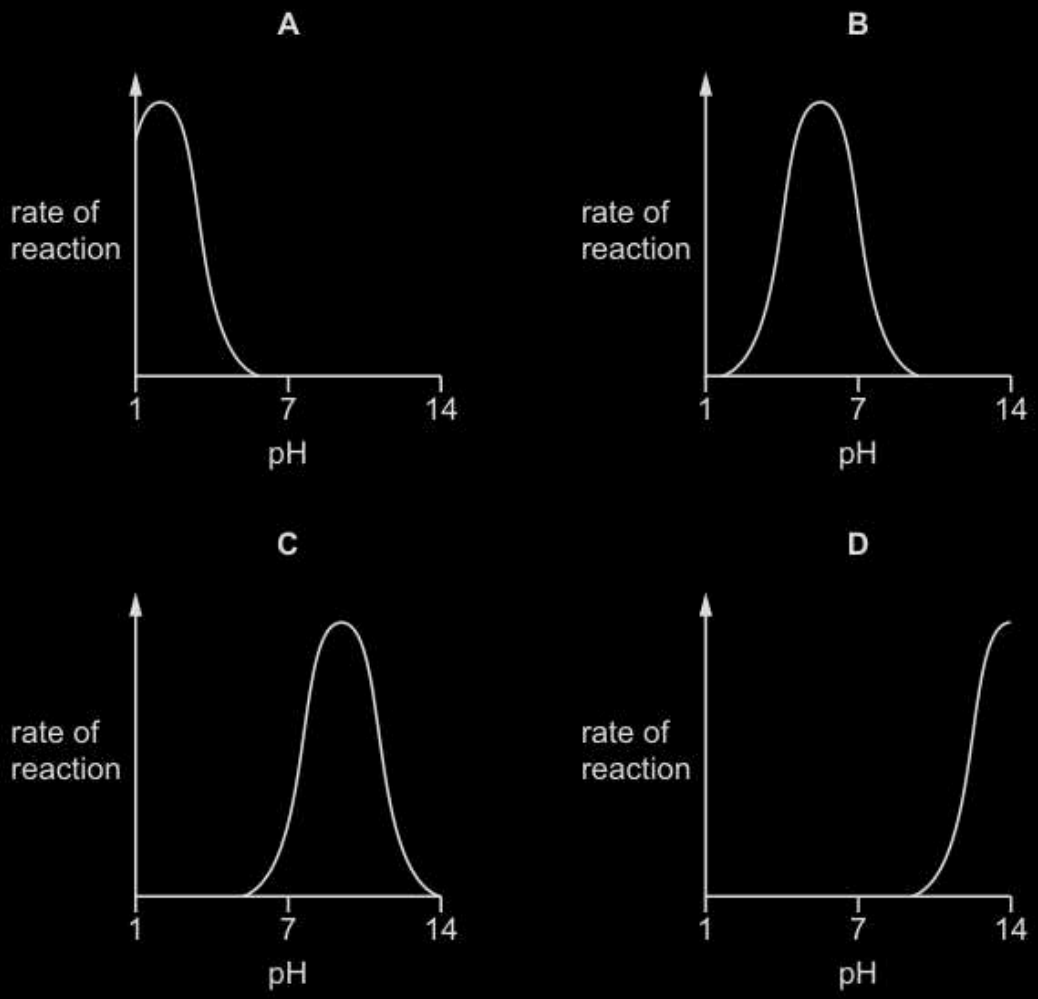
3 A cube of fresh potato is weighed. It is then placed in a test-tube containing a dilute solution of sucrose. After an hour, its mass has increased.

Which process has occurred and what has happened to the concentration of the sucrose in the solution in the test-tube?

	process	sucrose concentration
A	active transport	decreased
B	active transport	increased
C	osmosis	decreased
D	osmosis	increased

Correct answer: A

4 Which graph best represents the effect of pH on the action of gastric protease?

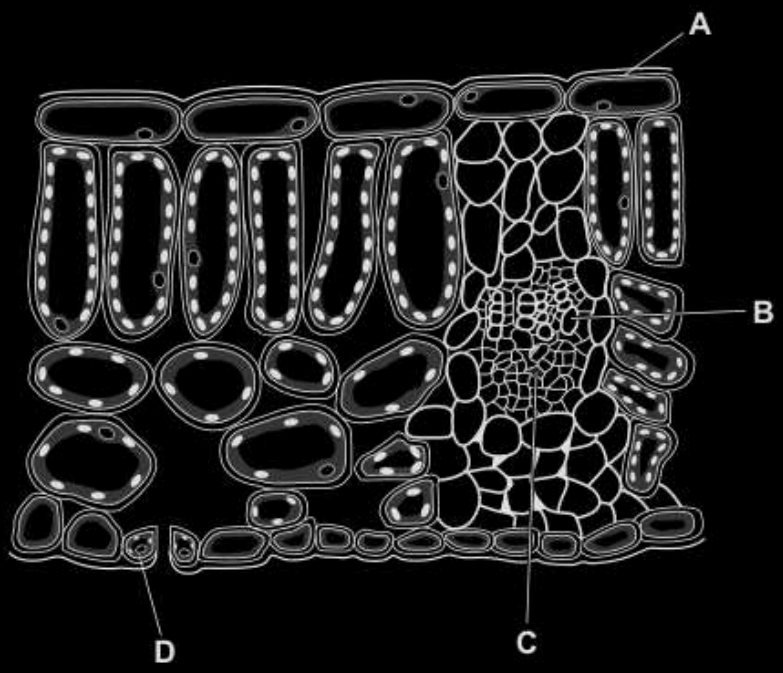


October/November 2014 · Paper 1 · Variant 2 · Question 5

Correct answer: B

5 The diagram shows a section through part of a dicotyledonous leaf, as seen under the microscope.

Where is the xylem?



October/November 2014 · Paper 1 · Variant 2 · Question 6

Correct answer: B

6 When a plant is photosynthesising, which gases enter and leave the stomata?

	carbon dioxide	oxygen	water vapour
A	enters	leaves	enters
B	enters	leaves	leaves
C	leaves	enters	enters
D	leaves	enters	leaves

October/November 2014 · Paper 1 · Variant 2 · Question 8

Correct answer: A

8 Which molecule is absorbed in the ileum?

A glycerol

B glycogen

C protein

D starch

Correct answer: B

9 Which table describes the elements present in carbohydrate, protein and fat?

A

	elements			
	C	N	O	H
carbohydrate	✓	✓	✓	✓
protein	✓	✗	✓	✓
fat	✓	✓	✗	✓

B

	elements			
	C	N	O	H
carbohydrate	✓	✗	✓	✓
protein	✓	✓	✓	✓
fat	✓	✗	✓	✓

C

	elements			
	C	N	O	H
carbohydrate	✓	✗	✗	✓
protein	✓	✓	✓	✓
fat	✓	✓	✗	✓

D

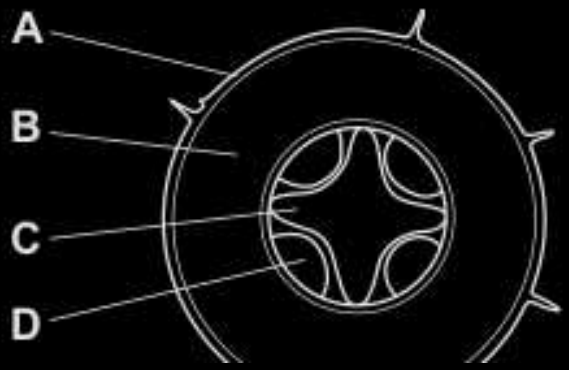
	elements			
	C	N	O	H
carbohydrate	✓	✗	✓	✓
protein	✓	✓	✗	✓
fat	✓	✗	✗	✓

October/November 2014 · Paper 1 · Variant 2 · Question 11

Correct answer: D

11 The diagram shows a section through a root.

Which tissue transports amino acids?

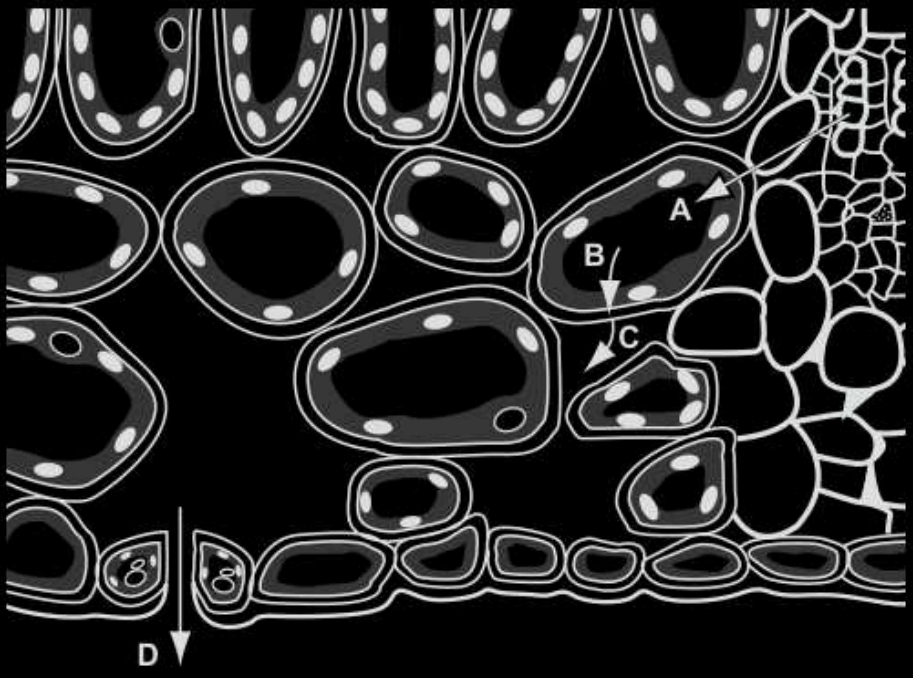


October/November 2014 · Paper 1 · Variant 2 · Question 12

Correct answer: C

12 The diagram shows the movement of water molecules in a leaf from the xylem to the atmosphere.

At which stage does evaporation occur?



October/November 2014 · Paper 1 · Variant 2 · Question 13

Correct answer: C

13 Which organs receive blood from both an artery **and** a vein?

	kidney	liver	lung
A	✓	✓	✗
B	✓	✗	✓
C	✗	✓	✗
D	✗	✗	✓

October/November 2014 · Paper 1 · Variant 2 · Question 14

Correct answer: C

14 The diagram shows a section through the human heart.

Why is the wall of the right ventricle thicker than the wall of the right atrium?

A The chambers have different oxygen concentrations.

B The right atrium must pump blood all the way around the body.

C The right ventricle must exert a greater force than the right atrium.

D The right ventricle receives more blood than the right atrium.

Correct answer: A

15 The table shows the characteristics of the blood in one blood vessel in the body.

oxygen concentration	carbon dioxide concentration	pressure
high	low	high

Which blood vessel contains blood with these characteristics?

A aorta

B pulmonary artery

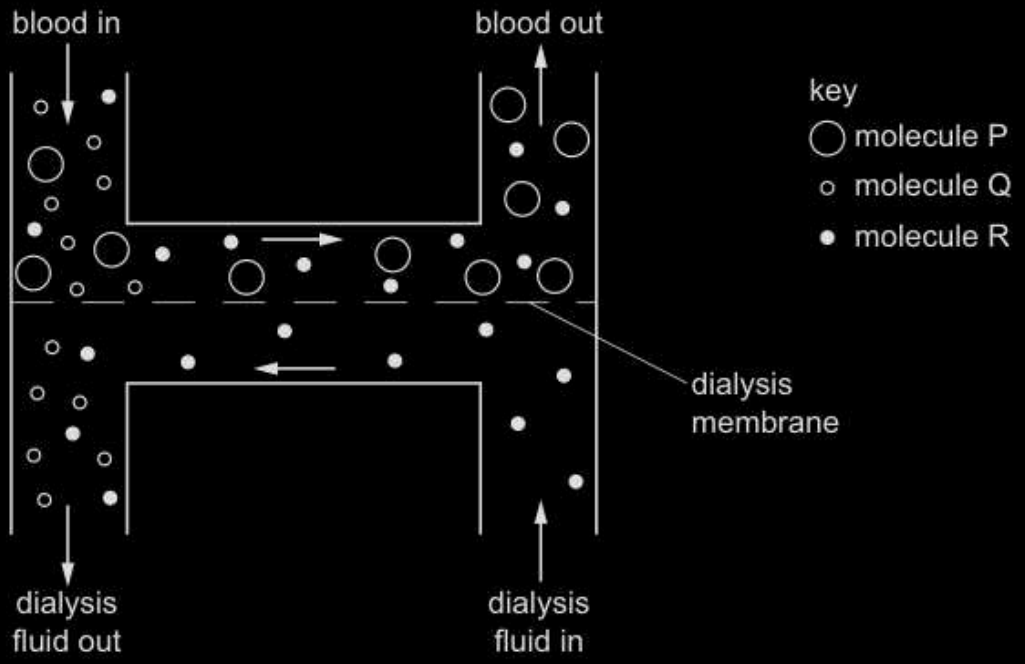
C pulmonary vein

D vena cava

October/November 2014 · Paper 1 · Variant 2 · Question 19

Correct answer: D

19 The diagram shows what happens to molecules of glucose, protein and urea as blood passes through a kidney dialysis machine.



What are molecules P, Q and R?

	molecule P	molecule Q	molecule R
A	glucose	protein	urea
B	glucose	urea	protein
C	protein	glucose	urea
D	protein	urea	glucose

October/November 2014 · Paper 1 · Variant 2 · Question 20

Correct answer: B

20 Which process is responsible for maintaining a person's body temperature when they are too cold?

A excretion

B homeostasis

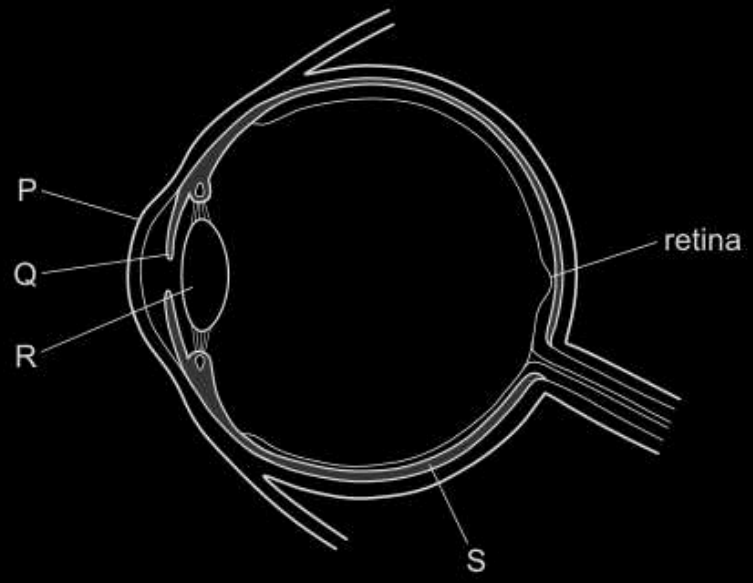
C sweating

D vasodilation

October/November 2014 · Paper 1 · Variant 2 · Question 21

Correct answer: B

21 The diagram shows a section through the eye.



Which pair of structures focus light rays onto the retina?

A P and Q **B** P and R **C** Q and R **D** Q and S

October/November 2014 · Paper 1 · Variant 2 · Question 25

Correct answer: C

25 Which risk is greatly increased by the nicotine in cigarette smoke?

A bronchitis and emphysema

B cancer of the airways

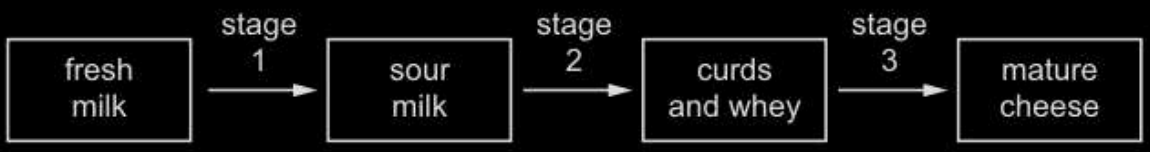
C high blood pressure

D reduced birth weight of babies

October/November 2014 · Paper 1 · Variant 2 · Question 27

Correct answer: C

27 The diagram shows three stages in the production of a blue cheese.



The three stages use micro-organisms and enzymes, as shown by P, Q, R and S.

P bacterial culture added

Q bacteria killed by pasteurisation

R rennin added

S fungal mould added

What is done at each of the stages?

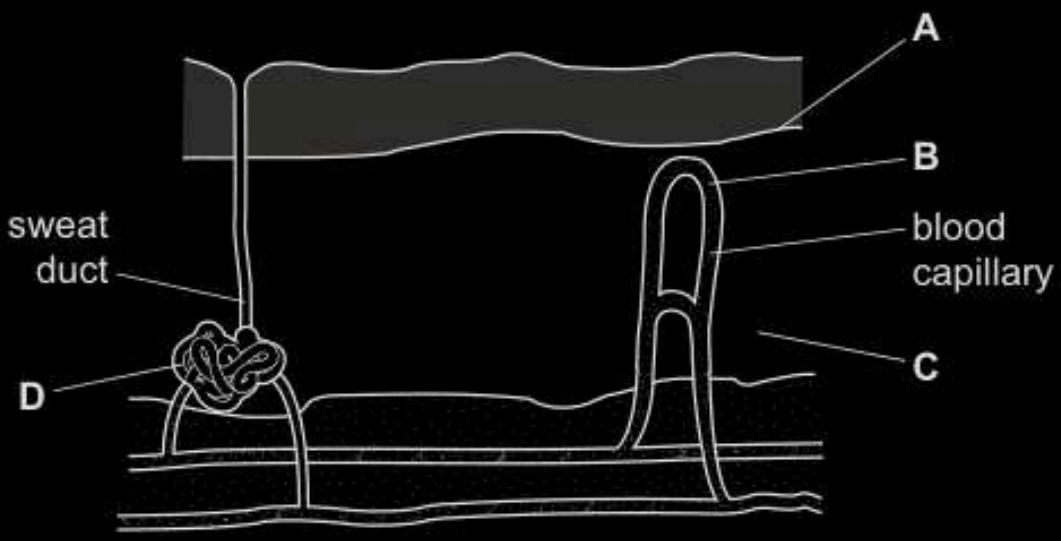
	stage 1	stage 2	stage 3
A	P and then Q	R	S
B	P and then Q	S	R
C	Q and then P	R	S
D	Q and then P	S	R

October/November 2014 · Paper 1 · Variant 2 · Question 33

Correct answer: A

33 The diagram represents a section through the human skin.

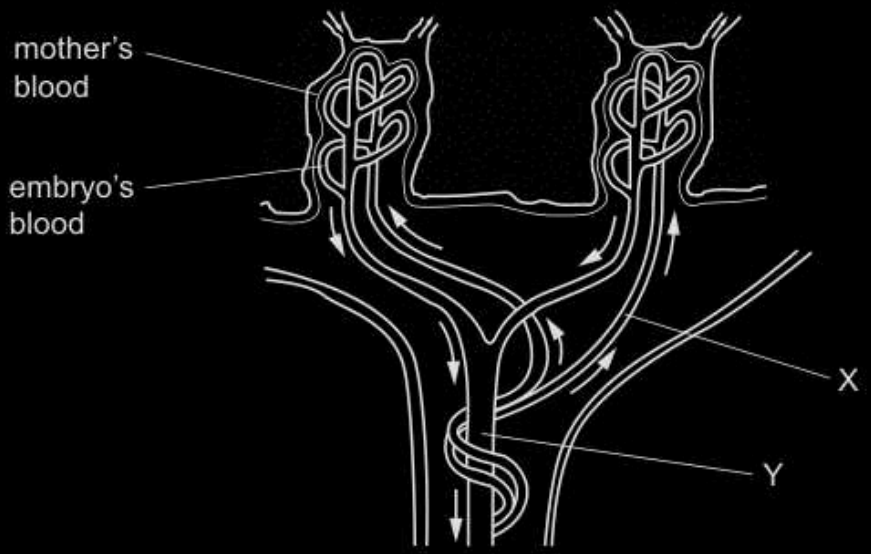
In which part is mitosis occurring most rapidly?



October/November 2014 · Paper 1 · Variant 2 · Question 36

Correct answer: B

36 The diagram shows how the blood of a human embryo flows close to the mother's blood in the placenta.



Which substances are present at X in higher concentrations than at Y?

A carbon dioxide and glucose

B carbon dioxide and urea

C glucose and oxygen

D glucose and urea

October/November 2014 · Paper 1 · Variant 2 · Question 37

Correct answer: C

37 Which term describes reproduction between those members of a species that are best fitted to their environment?

A discontinuous variation

B gene mutation

C natural selection

D survival of the fittest

October/November 2014 · Paper 1 · Variant 2 · Question 38

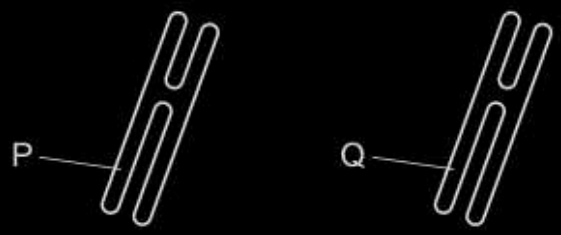
Correct answer: C

38 In the ABO blood grouping system, which genotype is homozygous dominant?

A $I^A\ I^O$ **B** $I^A\ I^B$ **C** $I^B\ I^B$ **D** $I^O\ I^O$

Correct answer: B

40 The diagram shows a pair of chromosomes from the same cell.



A gene is found at the point labelled P.

In a heterozygous individual, what will be found at the equivalent position labelled Q?

A a different allele of a different gene

B a different allele of the same gene

C a different gene of the same allele

D the same gene of the same allele

1 Fig. 1.1 shows a section through a plant organ with a very small pipette (micropipette) inserted into tissue **A**.

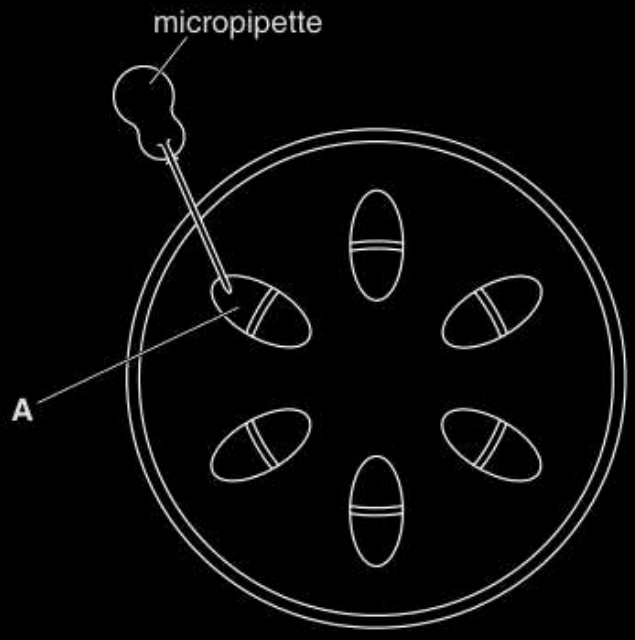


Fig. 1.1

(a) **(i)** Name the plant organ shown in Fig. 1.1. ..... [1]

(ii) Name the tissue labelled **A**. ..... [1]

(b) The micropipette is used to take samples from tissue **A** at regular intervals over a 24-hour period. These samples are then analysed for their content. Explain how this might provide information about the rate of photosynthesis in the plant.

.....

.....

.....

.....

.....

.....

..... [4]

(c) Some insects have mouthparts similar to the tube of the micropipette. Suggest reasons for these insects using tissue **A** to supply their food.

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

1 (a) (i)	stem ;	[1]		
(ii)	phloem ;	[1]		
(b)	(photosynthesis) produces glucose / sugar / starch / carbohydrates ; changed into sucrose ; passes down the phloem / tissue A ; concentration (of sucrose) varies ; highest when photosynthesis rate is highest / AW ;	[max. 4]		
(c)	(sucrose / sugar turned to) glucose ; used for respiration ; to release energy ; amino acids ; used to make protein ; for growth / repair ;	[max. 4]	**R** produce/make	
		[Total: 10]		

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance

2 Fig. 2.1 shows the digestion, absorption and assimilation of a chemical (**B**). It also shows the formation of its waste product, chemical **F**.

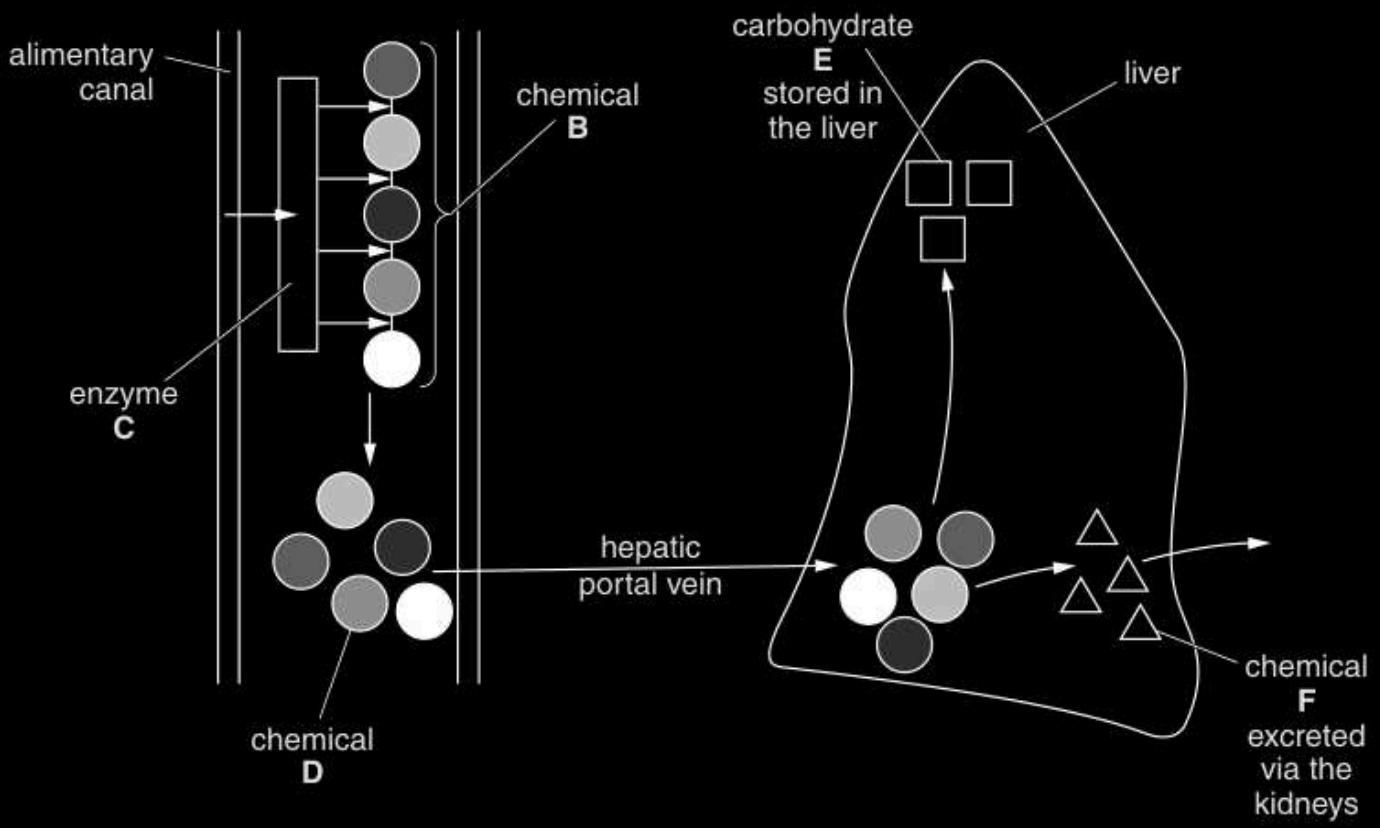


Fig. 2.1

(a) Name

chemical **B** .....

enzyme **C** .....

chemical **D** .....

carbohydrate **E** .....

chemical **F** ..... [5]

(b) **(i)** Describe what happens to carbohydrate **E** before it can be used in metabolic reactions in body cells.

.....

.....

..... [2]

(ii) Name the process in which the product of carbohydrate **E** is used within a body cell.

process ..... [1]

(iii) Describe how different types of body cells and tissues make use of this process.

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 12]

2	**(a)**	B = protein / (poly)peptide ; C = protease or named* ; D = amino acids* ; E = <u>glycogen</u> ; F = urea ;	[5]	*__A__ ecf for incorrect substrate *__A__ ecf for incorrect substrate __R__ urine
	(b) **(i)**	broken down / converted / changed ; (role of) glucagon / adrenaline ; to <u>glucose</u> ; made soluble ;	[max. 2]	
	(ii)	<u>respiration</u> ;	[1]	

© Cambridge International Examinations 2014

(iii)	to release energy ;	[1]	
	any 3 from: for growth ;	[max. 3]	**A** production of complex molecules
	mitosis / meiosis / cell division ;		
	active transport ;		
	impulse production ;		
	temperature regulation ;		
	muscular activity / movement ;	[Total: 4]	**A** named example
		[Total: 12]	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance

6 **(a)** Describe the two forms of variation found within a natural population.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

(b) Explain how variation is important for the long-term survival of the species.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

6 (a)	continuous variation ; gradual change / range ; between extremes ; genes + environment ; discontinuous variation ; few and distinct differences ; controlled by genes alone ; any correct example of one or the other correctly linked ;	[max. 5]	**R** if one e.g. correct, the other incorrect

© Cambridge International Examinations 2014

(b)	some variations are advantageous ; competition ; in the organism's habitat / environment ; organism survives / differential survival / ORA ; breeds / reproduces ; passes on the advantage / beneficial gene / allele ; over many generations / ref. time ; continuous adaptation to the changing environment ; evolution / natural selection / AW ; changes that result from the environment not so important ;	[max. 5]	
		[Total: 10]	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance

8 **(a)** **(i)** Explain how meiosis is involved in the formation of gametes.

.....

.....

.....

..... [2]

(ii) Explain how two plants, one with red flowers and one with white flowers, can produce seeds that will grow into plants with either red or white flowers in the ratio of 1:1.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) Suggest how two plants, of a different species, one with red and one with white flowers, could produce seeds that will grow into plants with pink flowers.

.....

.....

.....

.....

..... [2]

[Total: 10]

8 (a) (i)	in testes / anthers / ovaries ; cell division ; halving of chromosome numbers / haploid ; so that diploid number is restored on fertilisation ;	[max. 2]	
(ii)	one (either) colour is controlled by a dominant allele; one by a recessive allele ; one parent heterozygous – (or described, e.g. Rr) ; one is homozygous recessive (or described) ; correct ref. to gametes ; gametes correctly identified for both parents ; how gametes pair to produce offspring in 1:1 ratio ;	[max. 6]	(all points acceptable on an annotated genetic diagram)
(b)	mutation ; ref to a named mutagen ; possible co-dominance ; ref. to heterozygous plants having pink flowers ; ref. availability of certain ions (as in *Hydrangea*) ;	[max. 2]	
		[Total: 10]	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance

9 **(a)** Describe the path taken by a molecule of carbon dioxide, and the processes in which it is involved, from the moment it leaves the atmosphere until it forms part of a starch molecule in a leaf.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

(b) Explain how a lack of magnesium in the soil in which a plant is growing may reduce the formation of starch in a leaf.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

9	(a)	between guard cells / through stoma ; into intercellular / air space ; dissolving in water (film) ; diffusion ; through cell wall ; of mesophyll cell (or named) ; to chloroplast ; during photosynthesis ; links with water molecule ; glucose + forms starch ;	[max. 7]	
	(b)	little magnesium absorbed ; deficiency in chlorophyll ; less light energy trapped ; photosynthesis inhibited / limiting factor / AW ; less glucose / starch / carbohydrate formed ;	[max. 3]	
			[Total: 10]	

© Cambridge International Examinations 2014

1 Fig. 1.1 shows a kidney and its associated structures. The arrows show the direction of flow of fluids in these structures.

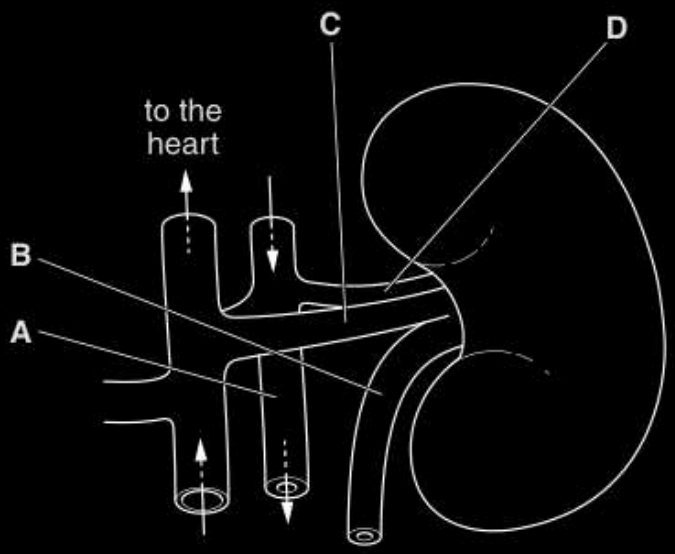


Fig. 1.1

(a) (i) Name the structure labelled **A** in Fig. 1.1. ..... [1]

(ii) Name the chamber of the heart through which blood in structure **A** last passed.

..... [1]

(b) Table 1.1 shows the relative concentrations of various substances in structures **B** and **C**. Complete the table to show the possible concentrations of these substances in structure **D**.

Table 1.1

substance	relative concentration in structure		
	B	**C**	**D**
amino acids	0.00	0.05	
glucose	0.00	0.10	
mineral ions	1.50	0.72	
proteins	0.00	8.00	
urea	2.00	0.03	

[5]

(c) Explain how the relative concentrations of glucose might change in structures **B**, **C** and **D** in a person with diabetes.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

1 (a) (i)	(dorsal) aorta ;	[1]	
(ii)	left ventricle ;	[1]	
(b)	(amino acids) 0.05 ; (glucose) 0.10 to 0.15 ; (mineral ions) 0.72 to 2.22 ; (proteins) 8.00 ; (urea) 0.03 to 2.03 ;	[5]	
(c)	B would contain some / more / high (glucose) / C would contain more / high (glucose) / D would contain more / high (glucose) ; lack of Insulin ; glucose would not be converted into glycogen ; kidney unable to/doesn't reabsorb all glucose ;	[max. 3]	
		[Total: 10]	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance

2 Fig. 2.1 shows parts of the alimentary canal that lie in the upper part of the human body.

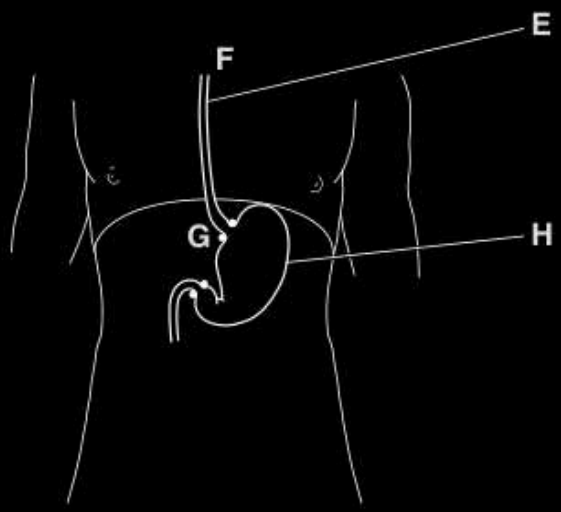


Fig. 2.1

(a) (i) Name the part labelled **E**. ..... [1]

(ii) Name the process that carries food from **F** to **G**. ..... [1]

(b) Suggest why the walls of part **H** are normally coated with mucus.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

(c) Sometimes, particularly when a person is lying flat, partly digested food returns into structure **E** through the valve at **G**. This can cause discomfort known as heartburn.

(i) Suggest why heartburn is not a biologically accurate name for this condition.

.....

..... [1]

(ii) Suggest and explain why medications for this condition are often alkaline in nature.

.....

.....

.....

..... [2]

[Total: 10]

2 (a) (i)	oesophagus / gullet ;	[1]	
(ii)	peristalsis ;	[1]	
(b)	ref. protection / barrier / prevents damage / breakdown / digestion ; of walls ; acid / HC*l* ; ref. protease ; walls are made of protein ; ref. lubrication ;	[max. 5]	
(c) (i)	heart not involved / no connection between E and the heart / AW ;	[1]	
(ii)	less mucus in E ; acid (from stomach) ; (acid) damages the cells / walls ; (acid) neutralised (by the medication) ;	[max. 2]	
		[Total: 10]	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance

4 Fig. 4.1 is an incomplete flow chart of the process of anaerobic respiration in yeast and in muscles.

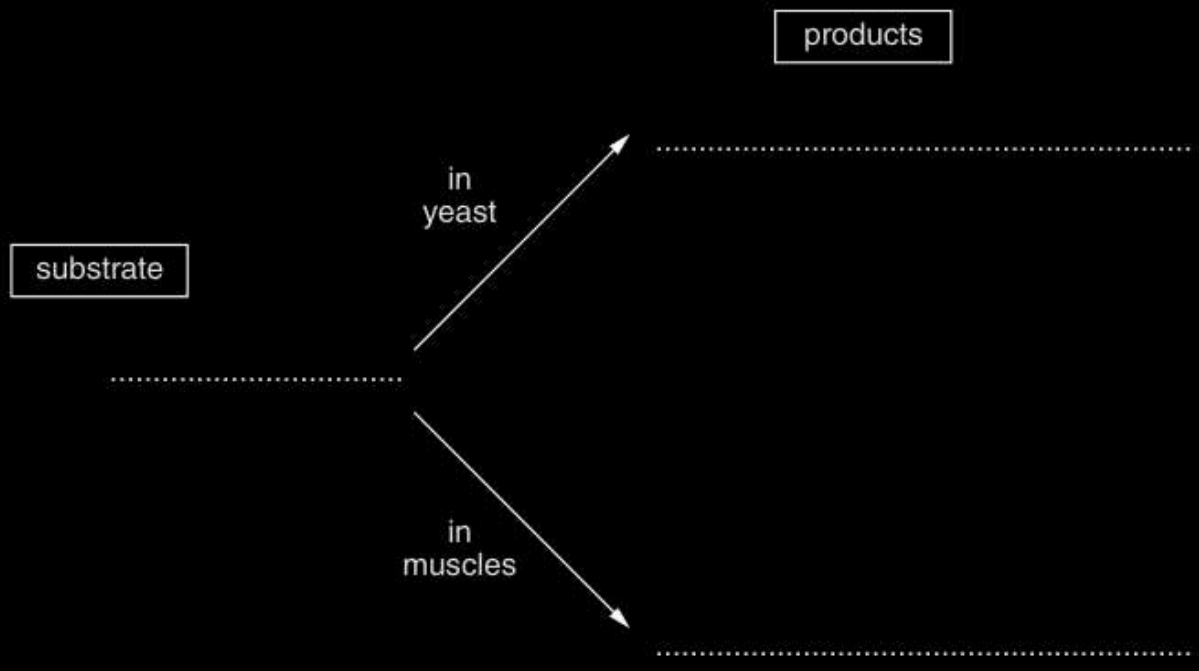


Fig. 4.1

(a) Complete Fig. 4.1 to show the substrate and products for anaerobic respiration in yeast and in muscles. [4]

(b) Yeast cells may be killed by their use in food biotechnology. Explain what causes the death of the yeast in each of the following.

(i) brewing

.....

.....

.....

..... [2]

(ii) making bread

.....

..... [1]

(c) Explain why muscle cells are not killed by anaerobic respiration in a healthy person.

.....

.....

.....

..... [2]

(d) Suggest why anaerobic respiration does not release as much energy as aerobic respiration for each molecule of the same substrate.

.....

.....

.....

..... [2]

[Total: 11]

4	**(a)**	glucose / $C_6H_{12}O_6$ (substrate) ; (yeast) alcohol / ethanol / C_2H_5OH ; (yeast) carbon dioxide / CO_2 ; (muscles) lactic acid / lactate / $C_3H_6O_3$;	[4]	
	(b) (i)	food / glucose deficiency / AW ; (killed) by alcohol ; poisoned by competing organisms (e.g. bacteria) ;	[max. 2]	
	(ii)	(killed) by heat / baking / high temperature ;	[1]	
	(c)	lactic acid removed / broken down / converted ; by circulation / blood / AW ; lactic acid not toxic (at concentrations experienced) ;	[max. 2]	
	(d)	substrate / glucose not completely broken down ; chemical energy ; still contained within product / lactic acid / alcohol ;	[max. 2]	
			[Total: 11]	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance

5 There are many different drugs available to treat high blood pressure. Fig. 5.1 shows the mean heart rates of two groups of people, **J** and **K**, over a five-year period.
From the start, and throughout the period, group **K** were treated with a drug called a beta-blocker.
Group **J** did not take any form of medication.

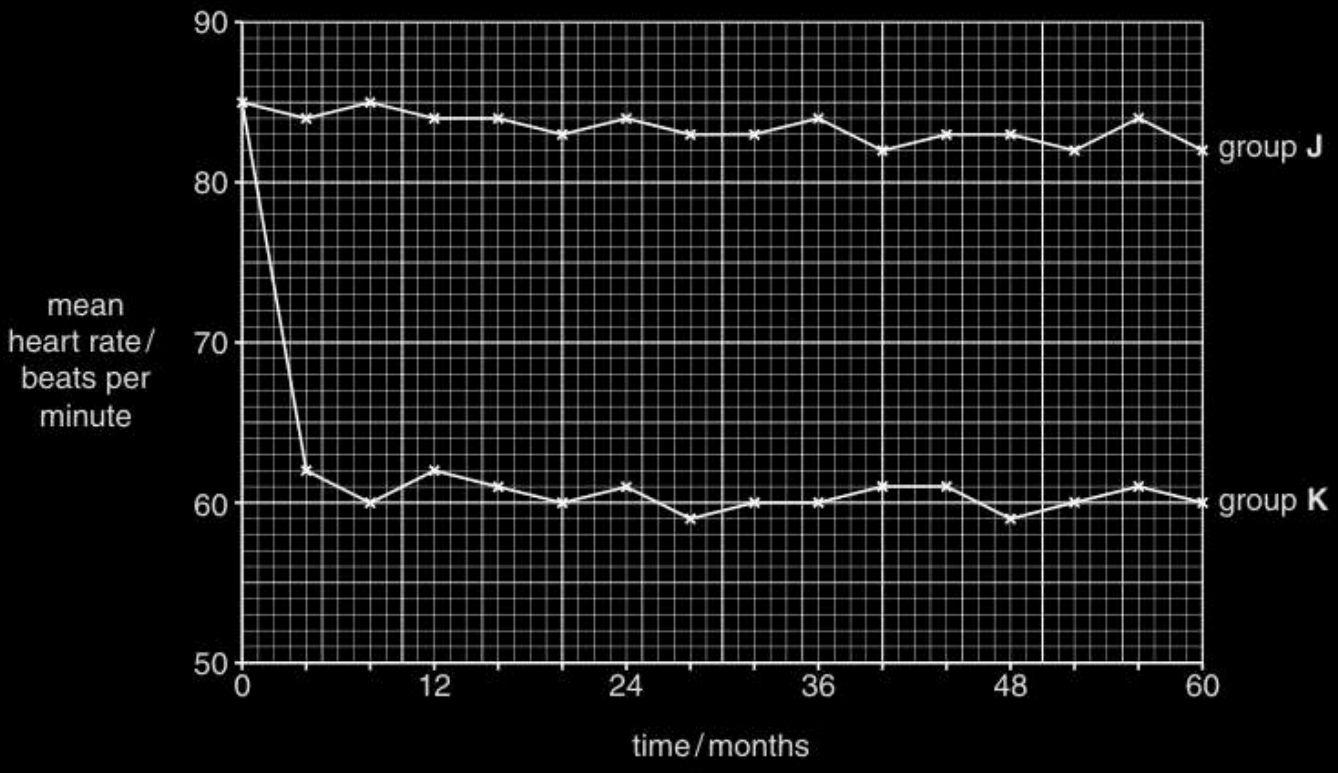


Fig. 5.1

(a) (i) State the term used to describe group **J**. ..... [1]

(ii) Using information from Fig. 5.1, describe the effect on the heart of taking beta-blockers.

.....

.....

.....

.....

.....

..... [3]

(b) On Fig. 5.1, draw a curve to show the expected effect on the mean heart rate of Group **J** if, after three years, half of them started to take beta-blockers. [3]

(c) Some other drugs reduce blood pressure by having an effect on blood vessels. Suggest how these drugs may cause a decrease in blood pressure.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 11]

Section B

Answer **all** questions in this section.

Write your answers in the spaces provided.

5	**(a) (i)**	control ;	[1]	
	(ii)	rate **+** drops / reduces / AW ; rapidly / AW / quoted figures (85 to 62 **+** bpm) ; during first 4 months ; remains (more or less) constant ; (constant at) 59 / 60 / 61 / 62 + bpm ;	[max. 3]	**A** stays low
	(b)	line / curve starts at 36 months ; drops with similar gradient to line K ; to between 70 and 74 bpm ; levels to run parallel with the J and K ;	[max. 3]	
	(c)	arteries / arterioles ; muscular wall ; relax / prevent constriction ; larger lumens / dilation / widens / AW ; ref. reduces **+** deposits / atheroma / AW ; less **+** resistance / friction / AW ;	[max. 4]	
			[Total: 11]	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance
(a)	*ovule*: in ovary ; contains female gamete ; ref. to haploid ;	[max. 2]	
	seed: ovule after fertilisation ; ref. diploid (or with ref. fruit) ; (grows) larger than ovule ; stores food / ref. cotyledons ; contains (rest of) embryo / radicle + plumule ; (surrounded by) testa ; ref. dispersal ;	[max. 3]	
	fruit: seed(s) + ovary (wall) / pericarp ; ref. dispersal ;	[max. 1] [total: 6]	

© Cambridge International Examinations 2014

(b)	dry ; light ; (may be) winged / hairy / feathery / helicopter / parachute ; large surface area ; to allow wind to detach it from parent plant ; to delay its descent ; (allow it to be carried) long distance AW / away from parent plant ;	[max. 4]	
		[Total: 10]	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance
7 (a)	fast / instant ; automatic / involuntary / spontaneous / unconscious / AW ; response / reaction ; to a stimulus ;	[max. 3]	**A** ref. to no decision involved
(b) (i)	named stimulus* ; correct receptor / named* ; impulse / electrical **+** pulse ; sensory / afferent / receptor + motor / efferent / effector neurones, in correct order ; CNS / spinal cord* ; correct or named effector* ; correct action (of effector)* ;	[max. 5]	**** description must match example*** **R** brain if it directs response
(ii)	automatic / innate / does not need to be learnt ; ref. protection / aids survival / damage limitation ;	[2]	**A** example, e.g. prevents too much light entering eye
		[Total: 10]	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance
8 (a) (i)	starting with a producer ; plausible food chain with 3 consumers **+** arrows in correct direction ;		**I** ref to the Sun
(ii)	labelled pyramid with organisms named in food chain ; in correct order with named producer labelled at bottom ; pyramid of correct proportions for given food chain ;	[max. 4]	
(b)	some organisms / parts remain uneaten ; energy lost in faeces / undigested food ; urine / excretory products / excretion ; respiration ; energy lost as heat ; homeostasis / named example ; in movement / muscular contraction (or any e.g. of same) ; in nervous impulses ; catabolic reactions / named ; active transport ; ref. decomposition / decay	[max. 6]	
		[Total: 10]	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance
9 (a)	blood goes through heart twice (in one complete circuit of the body) ; circulation to / from lungs / pulmonary ; circulation to / from (rest of) body / systemic ; lungs + low pressure ; body + high pressure ;	[max. 4]	
(b)	two sides to the heart / heart completely divided ; four chambers / two atria + two ventricles / all 4 named chambers ; beats continually ; right side / atrium + receives blood from body ; right side / ventricles + pumps blood to lungs ; left side / atrium receives blood from lungs ; left side / ventricle + pumps blood to (rest of) body ; left ventricle thicker-walled / more muscular + than right ventricle ; ventricles thicker-walled / more muscular + than atria ; further to pump blood / generate higher pressure ; ref. valves + one-way flow / prevent backflow ;	[max. 6]	
		[Total: 10]	

© Cambridge International Examinations 2014

7 **(a)** Explain what is meant by the term *reflex action*.

.....

.....

.....

.....

.....

..... [3]

(b) **(i)** Name a reflex action and explain how it is brought about.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

(ii) Explain its value to the human body.

.....

.....

.....

..... [2]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7 (a)	fast / instant ; automatic / involuntary / spontaneous / unconscious / AW ; response / reaction ; to a stimulus ;	[max. 3]	**A** ref. to no decision involved
(b) (i)	named stimulus* ; correct receptor / named* ; impulse / electrical **+** pulse ; sensory / afferent / receptor + motor / efferent / effector neurones, in correct order ; CNS / spinal cord* ; correct or named effector* ; correct action (of effector)* ;	[max. 5]	**** description must match example*** **R** brain if it directs response
(ii)	automatic / innate / does not need to be learnt ; ref. protection / aids survival / damage limitation ;	[2]	**A** example, e.g. prevents too much light entering eye
		[Total: 10]	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Additional Guidance

9 **(a)** Explain what is meant by the term *double circulation*.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Explain how the heart is adapted to keep blood flowing in a double circulation.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

[Total: 10]

9	(a)	blood goes through heart twice (in one complete circuit of the body) ; circulation to / from lungs / pulmonary ; circulation to / from (rest of) body / systemic ; lungs + low pressure ; body + high pressure ;	[max. 4]	
	(b)	two sides to the heart / heart completely divided ; four chambers / two atria + two ventricles / all 4 named chambers ; beats continually ; right side / atrium + receives blood from body ; right side / ventricles + pumps blood to lungs ; left side / atrium receives blood from lungs ; left side / ventricle + pumps blood to (rest of) body ; left ventricle thicker-walled / more muscular + than right ventricle ; ventricles thicker-walled / more muscular + than atria ; further to pump blood / generate higher pressure ; ref. valves + one-way flow / prevent backflow ;	[max. 6]	
			[Total: 10]	

© Cambridge International Examinations 2014

May/June 2014 · Paper 1 · Variant 1 · Question 1

Correct answer: C

1 Which of the following have both cytoplasm and cell walls?

A liver cells

B red blood cells

C root hair cells

D xylem vessels

May/June 2014 · Paper 1 · Variant 1 · Question 2

Correct answer: C

2 Plant cells are placed in a solution with a lower water potential than that of the cells.

Which row is correct?

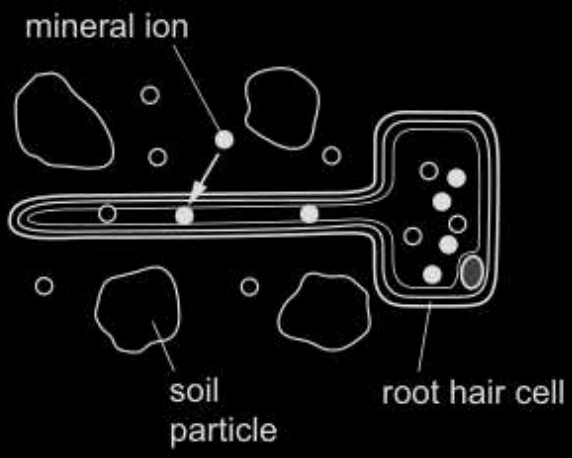
	movement of water by osmosis	volume of vacuole
A	enters cells	decreases
B	enters cells	increases
C	leaves cells	decreases
D	leaves cells	increases

May/June 2014 · Paper 1 · Variant 1 · Question 3

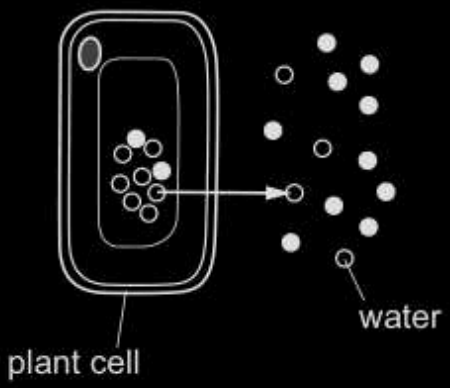
Correct answer: A

3 Which diagram illustrates the process of active transport?

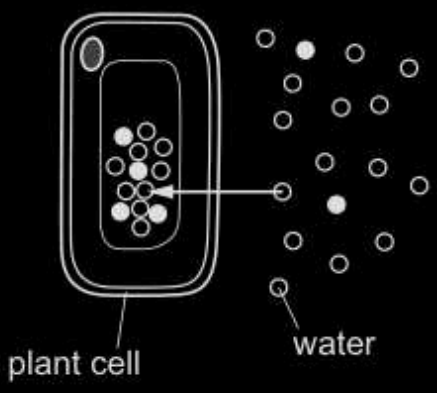
A



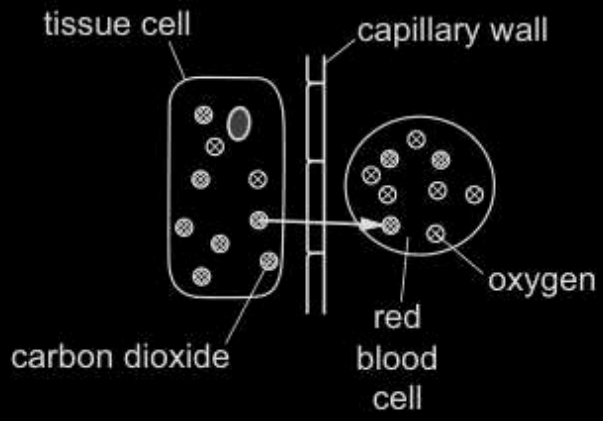
B



C



D

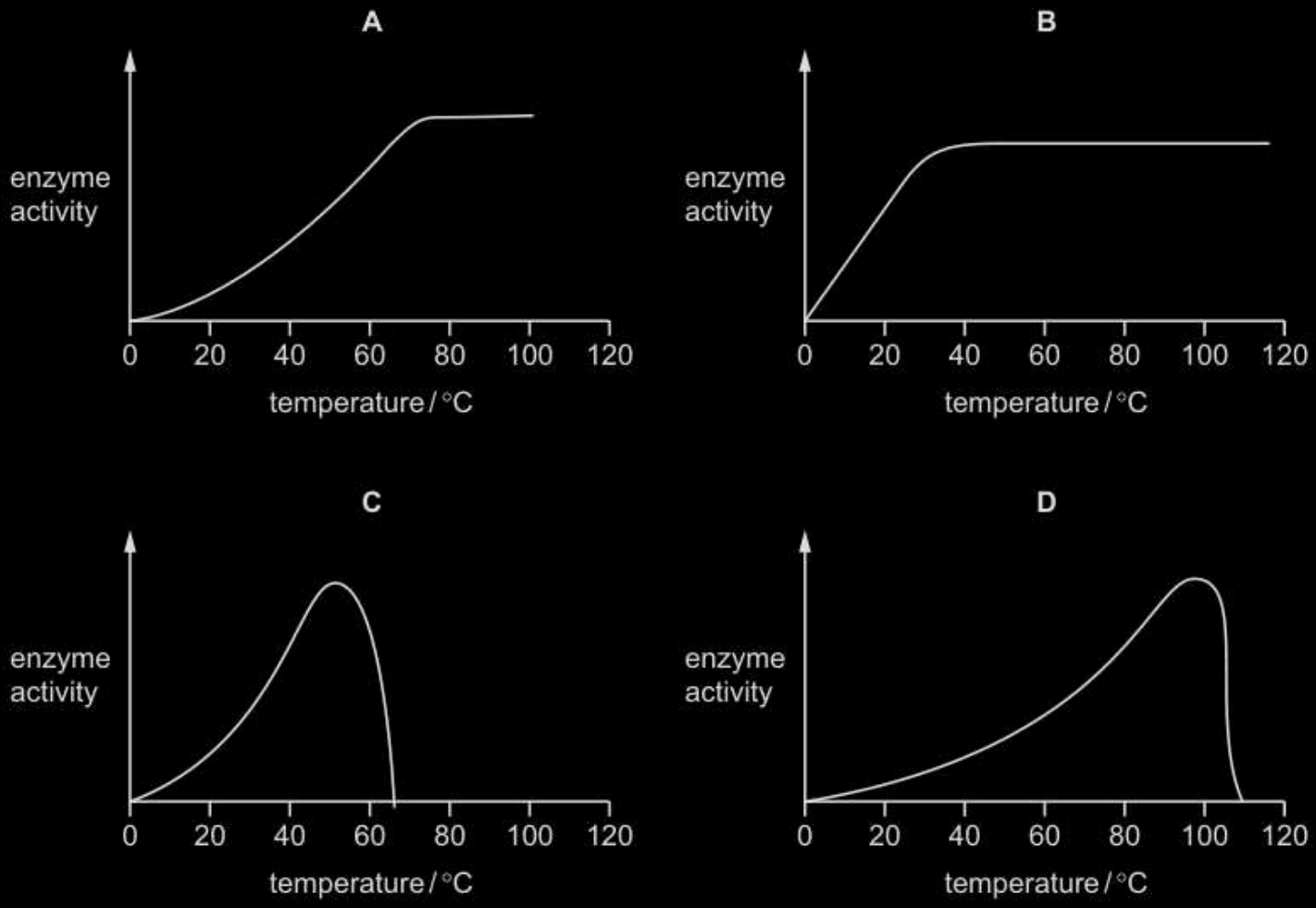


May/June 2014 · Paper 1 · Variant 1 · Question 4

Correct answer: D

4 A bacterium lives in hot springs at temperatures of 75 °C to 85 °C.

Which graph represents the activity of enzymes found in these bacteria?

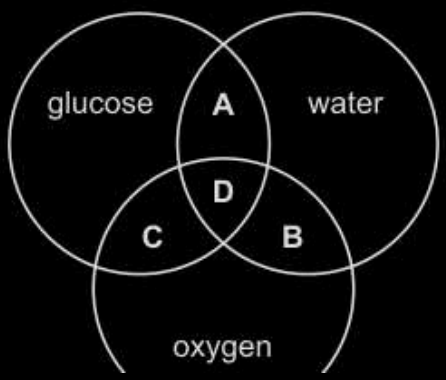


May/June 2014 · Paper 1 · Variant 1 · Question 6

Correct answer: C

6 The diagram refers to some substances found in plant cells.

Which area of the diagram represents the end products of photosynthesis?



May/June 2014 · Paper 1 · Variant 1 · Question 8

Correct answer: D

8 Which symptom of malnutrition can be treated by an increased amount of protein in the diet?

A constipation

B heart disease

C obesity

D stunted growth

May/June 2014 · Paper 1 · Variant 1 · Question 9

Correct answer: C

9 The table shows the rates of absorption of two different sugars, arabinose and glucose, in living and dead intestines. The concentrations of the sugars inside the intestines were the same in each case.

	rate of absorption (arbitrary units)	
	arabinose	glucose
living intestine	31	102
dead intestine	31	34

What are the main methods of absorption of arabinose and glucose in living intestine?

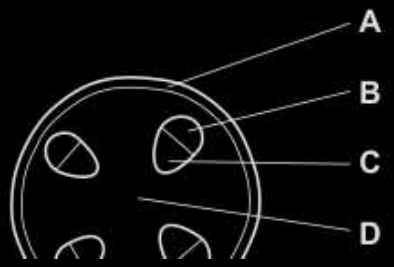
	arabinose	glucose
A	active transport	active transport
B	active transport	diffusion
C	diffusion	active transport
D	diffusion	diffusion

May/June 2014 · Paper 1 · Variant 1 · Question 11

Correct answer: B

11 The diagram shows a section through the stem of a dicotyledonous plant.

Which tissue transports sugars through the stem?



May/June 2014 · Paper 1 · Variant 1 · Question 12

Correct answer: C

12 Which effects will an increase in temperature and an increase in humidity have on the transpiration rate of a plant?

	transpiration rate	
	with increased temperature	with increased humidity
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

May/June 2014 · Paper 1 · Variant 1 · Question 13

Correct answer: A

13 Which sequence shows the shortest route taken by blood travelling from a leg to an arm in the human body?

A leg → heart → lungs → heart → arm

B leg → heart → lungs → kidney → arm

C leg → kidney → heart → lungs → arm

D leg → lungs → heart → alimentary canal → arm

May/June 2014 · Paper 1 · Variant 1 · Question 14

Correct answer: A

14 The diagram shows a section through part of a vein.

What could be the first organs found in the directions 1 and 2?

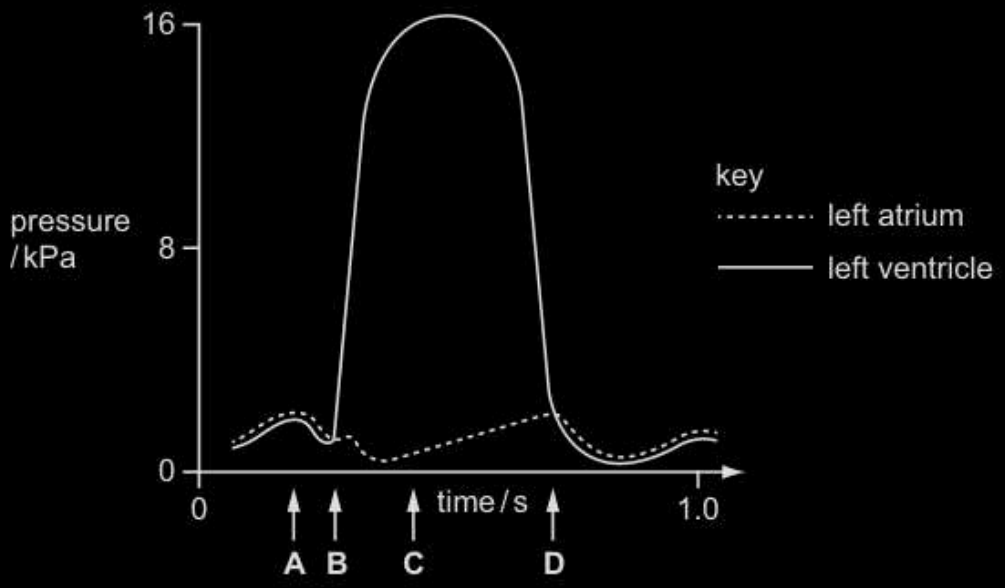
	1	2
A	heart	brain
B	intestine	liver
C	kidney	heart
D	lung	heart

May/June 2014 · Paper 1 · Variant 1 · Question 15

Correct answer: B

15 The graph shows the pressure changes in the left atrium and the left ventricle while the heart is beating.

When does the atrio-ventricular (bicuspid) valve close?

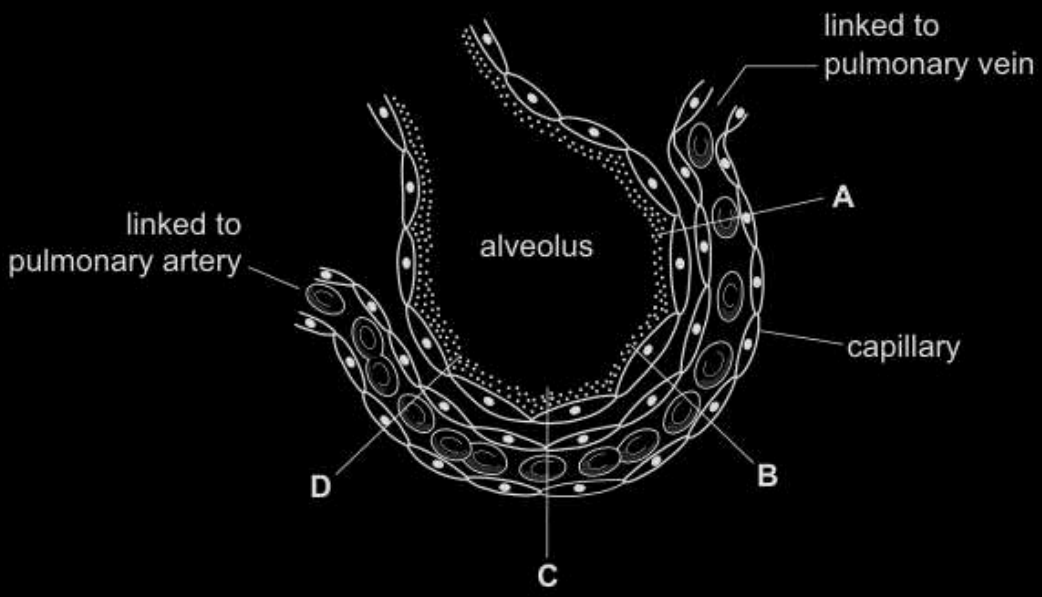


May/June 2014 · Paper 1 · Variant 1 · Question 17

Correct answer: D

17 The diagram shows an alveolus and an associated blood capillary.

At which point will the greatest rate of diffusion of carbon dioxide occur?



May/June 2014 · Paper 1 · Variant 1 · Question 19

Correct answer: B

19 Which process occurs in a kidney dialysis machine?

A Large protein molecules are removed from the blood plasma.

B Materials pass out of the blood down a concentration gradient.

C Oxygen is used up in removing materials from the blood.

D Pressure forces dialysis fluid into the blood.

May/June 2014 · Paper 1 · Variant 1 · Question 20

Correct answer: C

20 What happens when the body temperature rises above normal?

	arterioles near skin surface	sweat secreted
A	constricted	yes
B	constricted	no
C	dilated	yes
D	dilated	no

May/June 2014 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 A man stands 10 metres away from a sign and can see it clearly. He walks towards the sign and stops 0.5 metres from it.

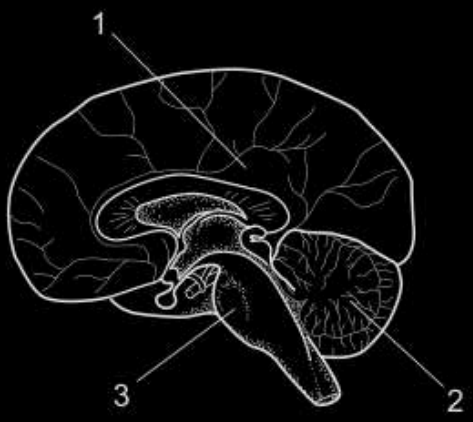
Which changes occur in his eyes so that the sign is still in focus?

	ciliary muscles	suspensory ligaments	lens becomes	result is light rays refracted
A	contract	slacken	thicker	more
B	contract	tighten	thinner	less
C	relax	slacken	thinner	less
D	relax	tighten	thicker	more

May/June 2014 · Paper 1 · Variant 1 · Question 22

Correct answer: C

22 The diagram shows a section through the brain.



What are some functions of the parts labelled 1, 2 and 3?

	1	2	3
A	centre for learning	forms visual images	controls digestion
B	controls blood pressure	centre for touch sensations	controls breathing rate
C	controls speech	controls smooth movements	controls heart rate
D	forms memory store	controls balance	determines intelligence

May/June 2014 · Paper 1 · Variant 1 · Question 23

Correct answer: D

23 How does adrenaline affect glucose uptake by muscle cells and carbohydrate conversion by liver cells?

	glucose uptake	carbohydrate conversion
A	decreases	glucose to glycogen
B	decreases	glycogen to glucose
C	increases	glucose to glycogen
D	increases	glycogen to glucose

May/June 2014 · Paper 1 · Variant 1 · Question 27

Correct answer: D

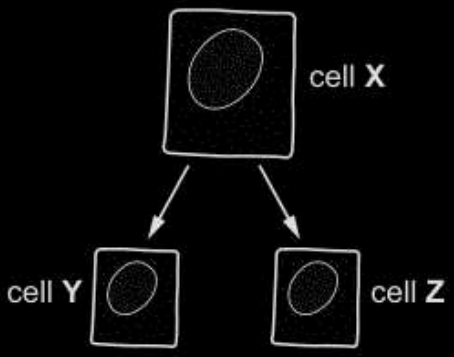
27 Using the key, which organism is a virus?

1	has a cell wall	go to 2
	does not have a cell wall	go to 3
2	cell wall is made of chitin	organism **A**
	cell wall is made of cellulose	organism **B**
3	has a cell membrane	organism **C**
	has a protein coat	organism **D**

May/June 2014 · Paper 1 · Variant 1 · Question 34

Correct answer: B

34 Cell **X** contains 24 chromosomes. It divides by mitosis to produce cells **Y** and **Z**.



How many chromosomes does cell **Z** contain?

A 12 **B** 24 **C** 46 **D** 48

May/June 2014 · Paper 1 · Variant 1 · Question 36

Correct answer: A

36 Which substances are present in breast milk but **not** in bottled milk made from milk powder?

A antibodies

B carbohydrates

C proteins

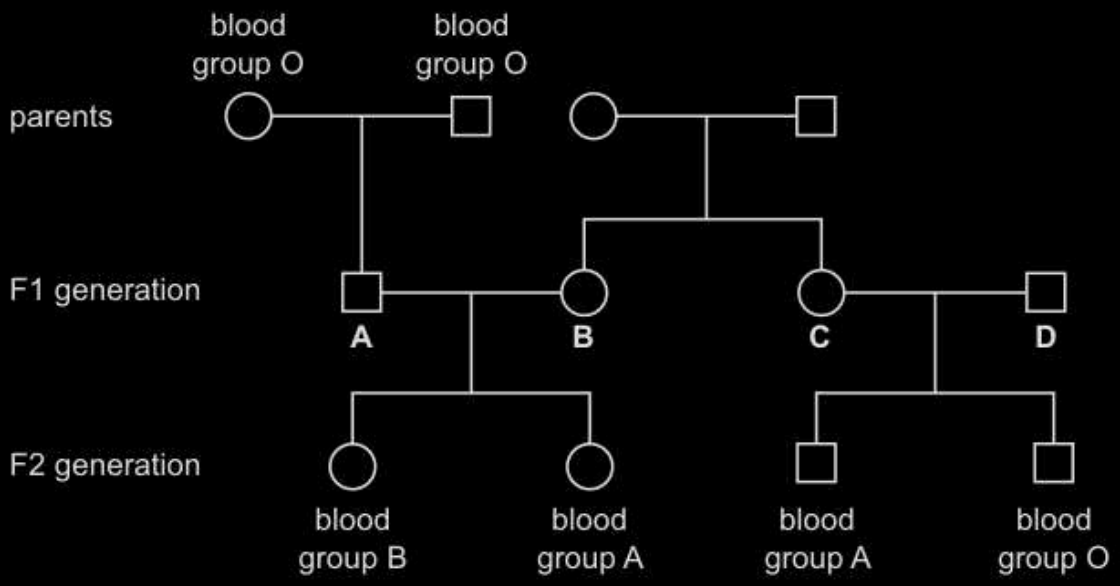
D vitamins

May/June 2014 · Paper 1 · Variant 1 · Question 38

Correct answer: B

38 The diagram shows the blood group phenotypes of some members of a family.

Which member of the F1 generation must be heterozygous, with the codominant alleles?



May/June 2014 · Paper 1 · Variant 1 · Question 39

Correct answer: B

39 Which statement about chromosomes is correct?

A Chromosomes are long DNA molecules called genes which are divided into sections.

B Chromosomes include a long molecule of DNA divided into sections called genes.

C Chromosomes include genes which are divided into sections called DNA molecules.

D Genes include long DNA molecules called chromosomes.

May/June 2014 · Paper 1 · Variant 1 · Question 40

Correct answer: C

40 Which statement is **always** true of dominant alleles?

A They cannot undergo mutation.

B They give a greater chance of survival than recessive alleles.

C They give the same phenotype in heterozygotes and homozygotes.

D They occur more frequently in the population than recessive alleles.

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

May/June 2014 · Paper 1 · Variant 2 · Question 1

Correct answer: D

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

May/June 2014 · Paper 1 · Variant 2 · Question 2

Correct answer: C

2 Which statement is always correct when oxygen is diffusing out of a plant cell?

A The concentration of carbon dioxide is higher inside the cell than outside.

B The concentration of carbon dioxide is higher outside the cell than inside.

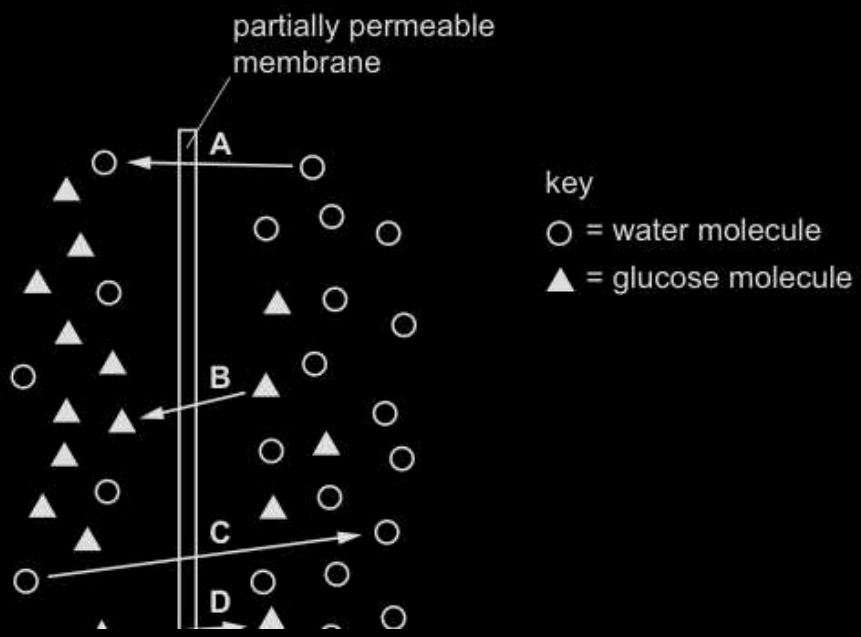
C The concentration of oxygen is higher inside the cell than outside.

D The concentration of oxygen is higher outside the cell than inside.

Correct answer: A

3 The diagram represents the passage of water molecules and glucose molecules across a partially permeable cell surface membrane.

Which arrow indicates osmosis?



Correct answer: C

4 The diagram represents the 'lock and key' mechanism of an enzyme that works best at pH 7.

What shows the enzyme and its substrate at pH 13?

	enzyme	substrate
A		
B		
C		
D		

May/June 2014 · Paper 1 · Variant 2 · Question 5

Correct answer: D

5 What are the signs of magnesium deficiency and nitrogen deficiency in leaves of plants?

1 purple spots on leaves

2 reduced growth of roots

3 reduced growth with loss of leaf colour

4 yellowing of leaves between the veins

	magnesium deficiency	nitrogen deficiency
A	2	1
B	2	3
C	4	1
D	4	3

May/June 2014 · Paper 1 · Variant 2 · Question 6

Correct answer: D

6 When carbon dioxide is being used in photosynthesis, what is its diffusion pathway in a leaf after entering a stoma?

1 cell membrane

2 cell wall

3 chloroplast membrane

4 cytoplasm

5 intercellular spaces

A $2 \rightarrow 1 \rightarrow 5 \rightarrow 4 \rightarrow 3$

B $2 \rightarrow 5 \rightarrow 1 \rightarrow 3 \rightarrow 4$

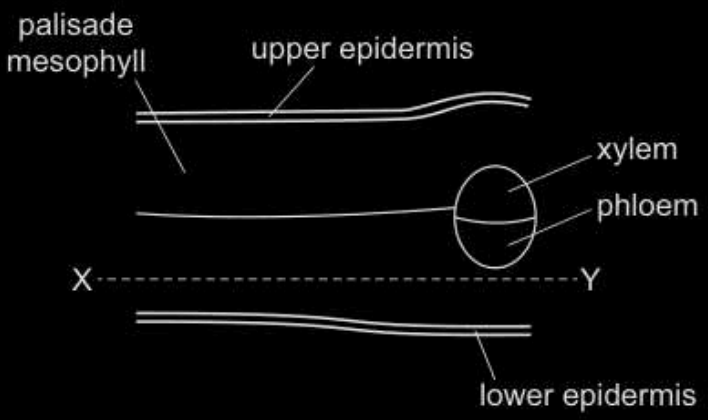
C $5 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

D $5 \rightarrow 2 \rightarrow 1 \rightarrow 4 \rightarrow 3$

May/June 2014 · Paper 1 · Variant 2 · Question 7

Correct answer: D

7 The diagram shows the arrangement of the tissues of a leaf as seen in cross-section under the microscope.



What is the arrangement of the cells in the section X-Y?

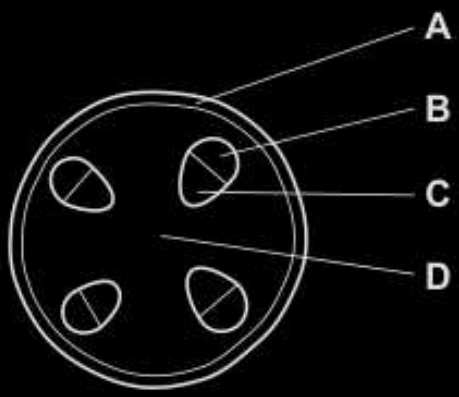
A B C D

May/June 2014 · Paper 1 · Variant 2 · Question 11

Correct answer: B

11 The diagram shows a section through the stem of a dicotyledonous plant.

Which tissue transports sugars through the stem?



May/June 2014 · Paper 1 · Variant 2 · Question 12

Correct answer: B

12 What is the main source of the energy that causes water to rise up a plant stem?

A difference in water potential between cell sap and soil water

B heat from the Sun

C light absorbed by chlorophyll

D respiration of sugars made in photosynthesis

May/June 2014 · Paper 1 · Variant 2 · Question 13

Correct answer: A

13 Which sequence shows the shortest route taken by blood travelling from a leg to an arm in the human body?

A leg → heart → lungs → heart → arm

B leg → heart → lungs → kidney → arm

C leg → kidney → heart → lungs → arm

D leg → lungs → heart → alimentary canal → arm

May/June 2014 · Paper 1 · Variant 2 · Question 14

Correct answer: A

14 The diagram shows a section through part of a vein.

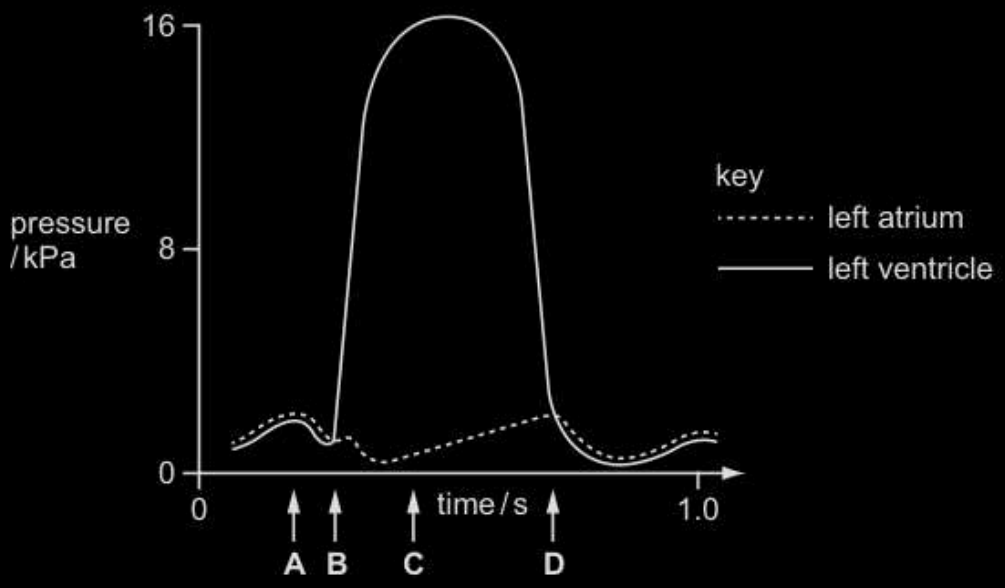
What could be the first organs found in the directions 1 and 2?

	1	2
A	heart	brain
B	intestine	liver
C	kidney	heart
D	lung	heart

Correct answer: B

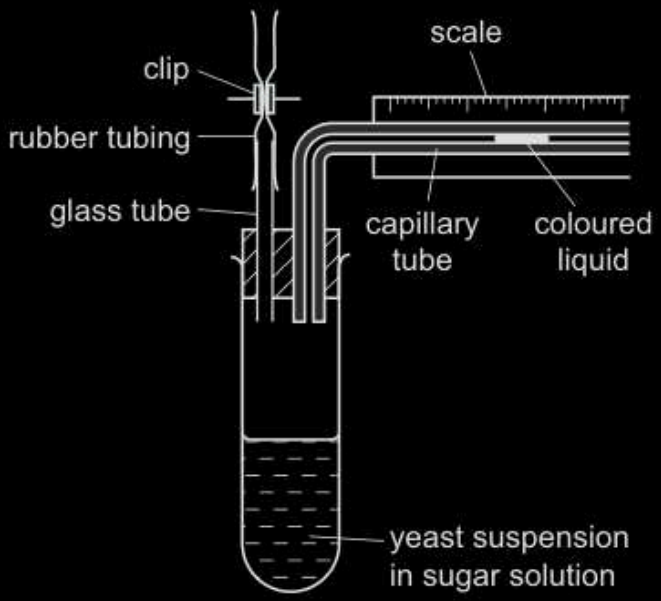
15 The graph shows the pressure changes in the left atrium and the left ventricle while the heart is beating.

When does the atrio-ventricular (bicuspid) valve close?



Correct answer: D

17 The diagram shows apparatus that can be used to measure the rate of anaerobic respiration of yeast.



Which precautions must be taken during the setting up of this apparatus?

P A layer of oil must be poured over the surface of the yeast suspension.

Q The clip must not be closed until the yeast suspension has had time to reach room temperature.

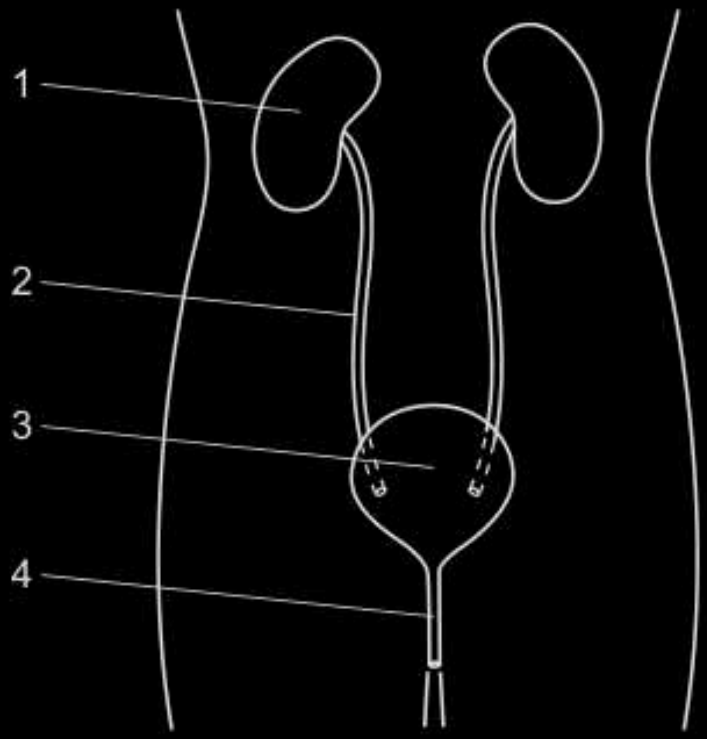
R The coloured liquid must be near the right hand end of the capillary tube.

S The yeast suspension must be made with boiled and cooled water.

A P, Q and R **B** Q, R and S **C** R, S and P **D** S, P and Q

Correct answer: C

19 The diagram shows the human urinary system.



What are the correct labels for structures 1, 2, 3 and 4?

	1	2	3	4
A	bladder	ureter	kidney	urethra
B	bladder	urethra	kidney	ureter
C	kidney	ureter	bladder	urethra
D	kidney	urethra	bladder	ureter

May/June 2014 · Paper 1 · Variant 2 · Question 20

Correct answer: C

20 Which part of the brain controls body temperature?

A cerebellum

B cerebrum

C hypothalamus

D medulla

May/June 2014 · Paper 1 · Variant 2 · Question 21

Correct answer: A

21 What is the role of motor neurones in reflex action?

A carrying nerve impulses from the central nervous system to an effector

B connecting a receptor to the central nervous system

C forming a synapse with a sensory neurone

D transferring energy from the stimulus to a nerve impulse

May/June 2014 · Paper 1 · Variant 2 · Question 22

Correct answer: D

22 Opticians sometimes place drops of a chemical in a patient's eye to keep the pupil wide open.

Which muscles contract when this chemical is used?

A ciliary muscles

B circular iris muscles

C muscles that move the eyeball

D radial iris muscles

May/June 2014 · Paper 1 · Variant 2 · Question 23

Correct answer: D

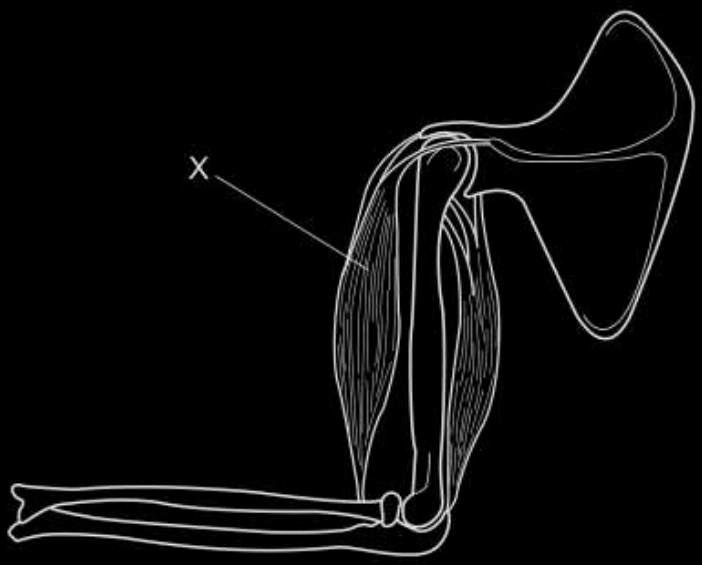
23 How does adrenaline affect glucose uptake by muscle cells and carbohydrate conversion by liver cells?

	glucose uptake	carbohydrate conversion
A	decreases	glucose to glycogen
B	decreases	glycogen to glucose
C	increases	glucose to glycogen
D	increases	glycogen to glucose

May/June 2014 · Paper 1 · Variant 2 · Question 24

Correct answer: A

24 The diagram shows parts of the human forelimb.



What will happen if a nerve impulse stimulates X?

	muscle contracted	effect
A	biceps	arm bends
B	biceps	arm straightens
C	triceps	arm bends
D	triceps	arm straightens

May/June 2014 · Paper 1 · Variant 2 · Question 26

Correct answer: B

26 Yeast is used in alcohol production.

Under which conditions will yeast produce most alcohol?

	glucose present	oxygen present
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

May/June 2014 · Paper 1 · Variant 2 · Question 27

Correct answer: D

27 Using the key, which organism is a virus?

1 has a cell wall ..... go to 2
does not have a cell wall ..... go to 3

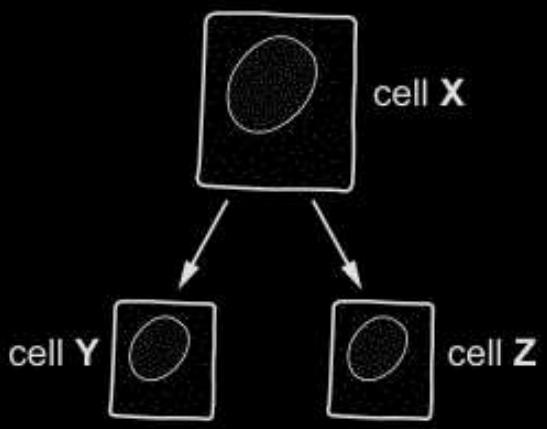
2 cell wall is made of chitin ..... organism **A**
cell wall is made of cellulose ..... organism **B**

3 has a cell membrane ..... organism **C**
has a protein coat ..... organism **D**

May/June 2014 · Paper 1 · Variant 2 · Question 34

Correct answer: B

34 Cell **X** contains 24 chromosomes. It divides by mitosis to produce cells **Y** and **Z**.



How many chromosomes does cell **Z** contain?

A 12 **B** 24 **C** 46 **D** 48

May/June 2014 · Paper 1 · Variant 2 · Question 35

Correct answer: A

35 What are the symptoms in the different stages of syphilis?

	first stage	second stage	third stage
A	a single sore on the skin	fever, widespread skin rash	heart disease, dementia
B	headaches, fever	widespread skin rash	hair loss, liver disease
C	slight fever	few or no symptoms	death from serious infectious disease
D	widespread skin rash	fever, sores on the skin	hair loss, liver disease

May/June 2014 · Paper 1 · Variant 2 · Question 36

Correct answer: A

36 Which substances are present in breast milk but **not** in bottled milk made from milk powder?

A antibodies

B carbohydrates

C proteins

D vitamins

May/June 2014 · Paper 1 · Variant 2 · Question 37

Correct answer: C

37 What is the effect of environment on discontinuous variation, and what is an example of this type of variation in humans?

	environmental effect	example
A	large	ABO blood group system
B	large	height
C	small	ABO blood group system
D	small	height

May/June 2014 · Paper 1 · Variant 2 · Question 38

Correct answer: B

38 In the inheritance of blood groups in humans, the MN system is controlled by a single gene. The gene has two alleles, M and N, that are co-dominant.

The offspring of two parents were two boys of blood group MN and M and a girl of blood group N.

What are possible genotypes of the parents?

	father's genotype	mother's genotype
A	MM	MN
B	MN	MN
C	NN	MM
D	NN	MN

May/June 2014 · Paper 1 · Variant 2 · Question 39

Correct answer: B

39 Which statement about chromosomes is correct?

A Chromosomes are long DNA molecules called genes which are divided into sections.

B Chromosomes include a long molecule of DNA divided into sections called genes.

C Chromosomes include genes which are divided into sections called DNA molecules.

D Genes include long DNA molecules called chromosomes.

May/June 2014 · Paper 1 · Variant 2 · Question 40

Correct answer: C

40 Which statement is **always** true of dominant alleles?

A They cannot undergo mutation.

B They give a greater chance of survival than recessive alleles.

C They give the same phenotype in heterozygotes and homozygotes.

D They occur more frequently in the population than recessive alleles.

1 Fig. 1.1 shows a sample of human blood seen using a microscope.

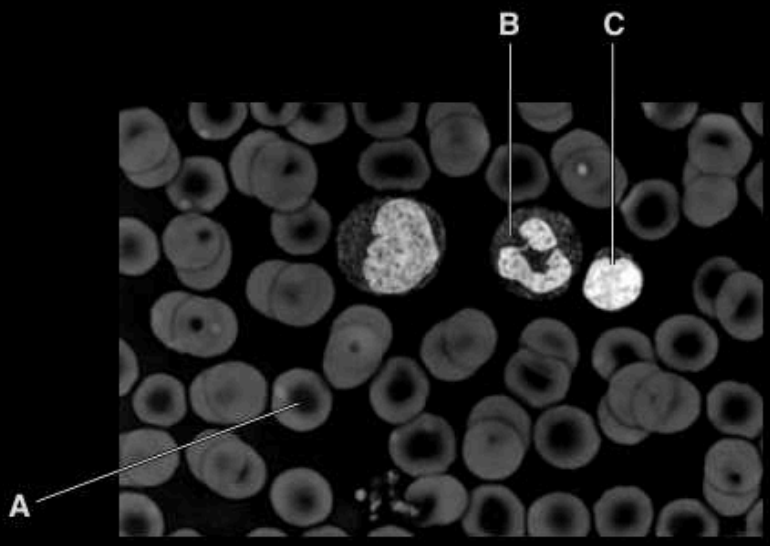


Fig. 1.1

(a) (i) Name the type of cell labelled **A** in Fig. 1.1. State the function of this type of cell.

type of cell .....

function

.....

[2]

(ii) Use your knowledge of the structure of this type of cell to suggest why the cell labelled **A** in Fig. 1.1 appears to be more lightly coloured at its centre than at its edge.

.....

.....

.....

.....

.....[3]

(b) (i) Name the type of cells labelled **B** and **C** in Fig. 1.1.

B ..... **C** ..... [1]

(ii) Some diseases can cause a person to have fewer of cells **B** and **C** in the blood. Use your knowledge of how cells **B** and **C** carry out their functions to suggest a problem this may cause for a person. Give an explanation for your answer.

problem

.....

explanation

.....

.....

.....

.....

.....

.....

[4]

[Total: 10]

1 (a) (i)	red (blood cell) ; absorb / carry / transport oxygen / transport CO_2 ;	[2]		**R** carry substances **Ig** contain haemoglobin
(ii)	thinner in middle / ref. biconcave ; ref. haemoglobin ; more (haemoglobin) at edges than at centre ; light more easily able to pass through centre ; lack of nucleus ;	[3]		
(b) (i)	**B** – white blood cell (phagocyte) **C** – white blood cell (lymphocyte) ;	[1]		

© Cambridge International Examinations 2014

(b) (ii)	**problem:** reduced immunity / immune response / less able to fight infection / kill microorganisms or pathogens / more likely to suffer (infectious) disease* ;	[1]	**A** named diseases **R** leukaemia **A** bacteria / virus / fungi **Ig** germs
	explanation: ref. phagocytosis ; ref. antibody production ; microorganisms / pathogens/bacteria/viruses / remain in blood / body / not destroyed ; more likely to succumb to (infectious) disease* AW ;	[3]	Max. 3 for explanations. * accept once only in either place
	Total	**[10]**	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Guidance

2 Fig. 2.1 shows the mean distance that molecules must travel during gas exchange between air in the lungs and blood in the circulatory system in birds and mammals.

This distance is known as the thickness of the blood-gas barrier.

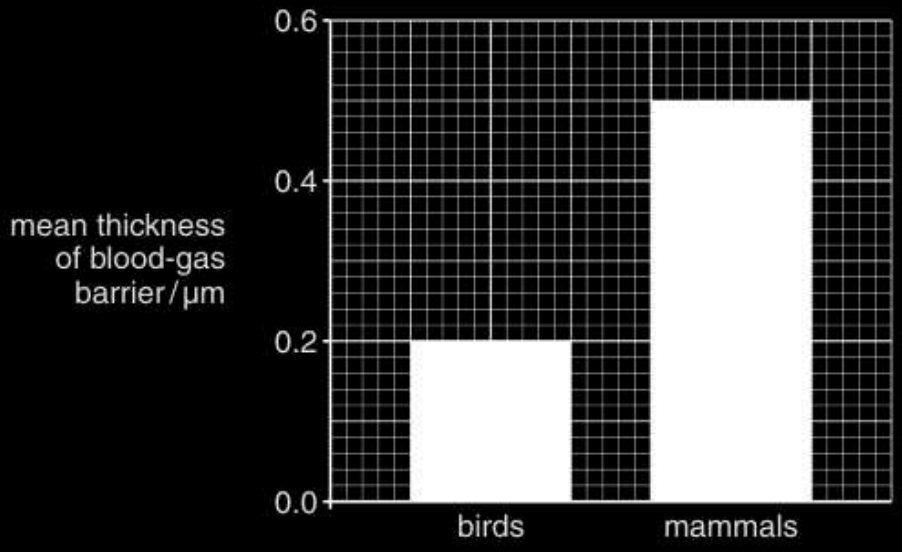


Fig. 2.1

(a) Name **two** gases that cross the blood-gas barrier during gas exchange.

1. .....

2. ..... [1]

(b) **(i)** Use information from Fig. 2.1 to compare the thickness of the blood-gas barrier in birds and mammals.

.....

.....

.....

.....[2]

(ii) Explain how the difference in thickness of the blood-gas barrier suggests that movement of a bird by flying requires more energy than movement by a mammal on land.

.....

.....

.....

.....

.....

.....[3]

[Total: 6]

2 (a)	carbon dioxide + oxygen ;	[1]	**A** CO_2 and O_2
(b) (i)	thicker in mammals / ora ; by 2.5 times / use of comparative figures ;	[2]	 **A** 0.2 µm and 0.5 µm
(ii)	(thinner barrier) results in faster/greater gas exchange / diffusion ; more oxygen supplied / more carbon dioxide removed ; ref. (aerobic) respiration ; ref. increased (respiration) ; ref. muscles ;	[3]	
	Total	**[6]**	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Guidance

3 Fig. 3.1 shows some parts of the human alimentary canal and associated organs.

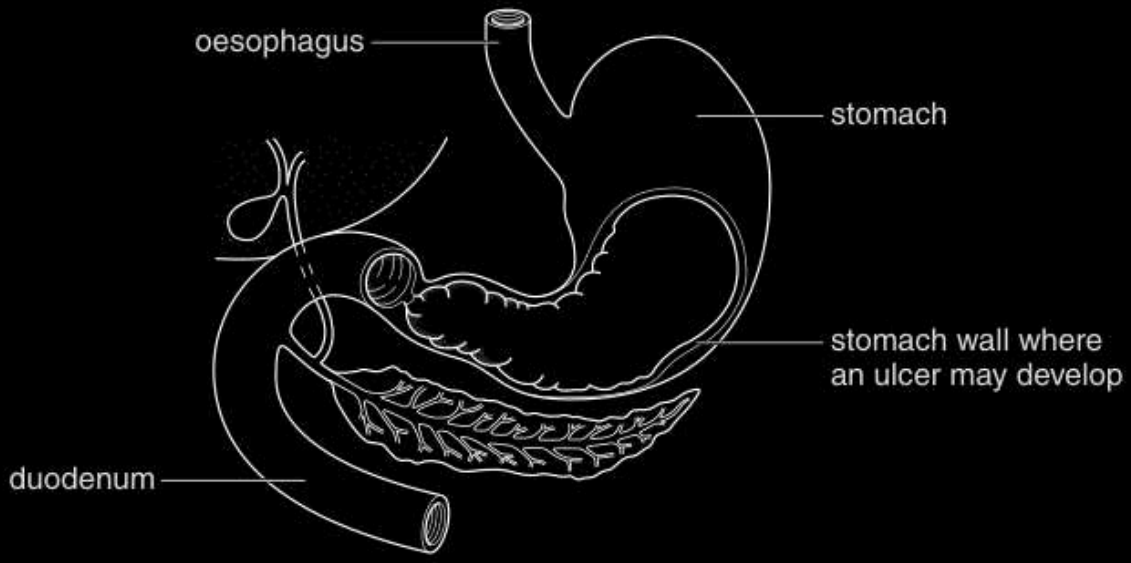


Fig. 3.1

(a) Name and describe the process that moves food down the oesophagus to the stomach.

name of process .....

description of process.....

.....

.....

.....

.....

.....

[4]

(b) A sore can develop on the wall of the stomach. This sore is called an ulcer, which can cause a person pain. The pain may be relieved by taking a drug that reduces the amount of acid produced by the cells in the stomach wall.

Suggest and explain how the processes taking place in the stomach may be affected in a person taking this drug.

.....

.....

.....

.....

.....

.....

.....[4]

(c) Infection with a type of bacterium in a person's stomach can increase the likelihood of an ulcer developing.

(i) Name a type of drug that may be taken to treat bacterial infection.

..... [1]

(ii) Suggest and explain what problems may occur if a person stops the treatment before all the bacteria are killed.

.....

.....

.....

.....

.....

.....[3]

[Total: 12]

Turn over for Question 4

3	(a)	peristalsis ; muscles ; circular ; contract ; behind food ; longitudinal ; relax + ref to food ; pushing (bolus / AW) ; wave action / rhythmic ;	[4]	
	(b)	less digestion ; of protein ; ref. protease / pepsin ; (enzyme) no / reduced activity ; (stomach) change in pH (if direction stated must be correct) / acidity / less acid ; ref. fewer microorganisms / bacteria / pathogens killed ;	[4]	**A** work best in acid conditions **R** drug neutralises the acid

© Cambridge International Examinations 2014

(c) (i)	antibiotic ;	[1]	**A** named antibiotic, e.g. metronidazole, tetracycline
(ii)	ref. resistant (bacteria) ; can multiply / reproduce/increase in number (again) ; pain / symptoms can recur / ulcer continues to grow AW ; ref. unable to kill / remove resistant bacteria using same antibiotic / bacteria tolerant / AW ;	[3]	**R** ref. immunity
	Total	**[12]**	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Guidance

4 Fig. 4.1 shows a type of plant cell.

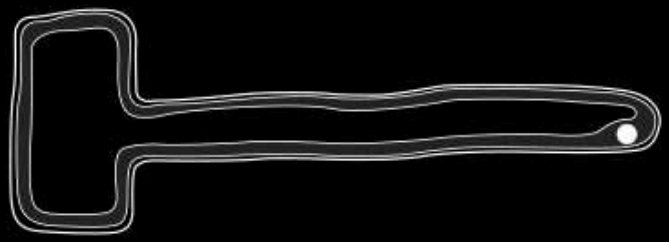


Fig. 4.1

(a) (i) Name the type of cell shown in Fig. 4.1.

..... [1]

(ii) Describe how water is taken up from the soil into the cytoplasm of the cell shown in Fig. 4.1.

.....

.....

.....

.....

.....

.....[3]

(b) The cell shown in Fig. 4.1 also takes up ions from the soil. Fig. 4.2 shows the relationship between the rate of ion uptake and the concentration of oxygen in the soil surrounding the cell.

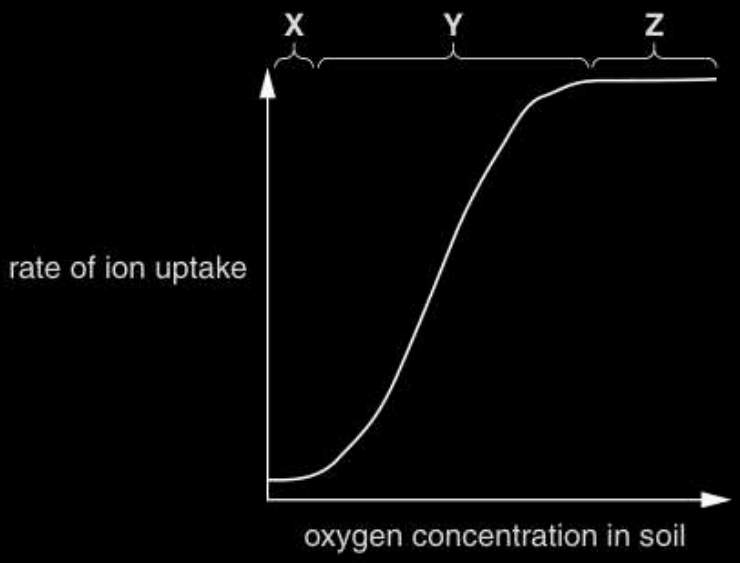


Fig. 4.2

(i) Using Fig. 4.2, describe the effect of increasing oxygen concentration in the soil on the rate of ion uptake.

.....

.....

.....

.....

.....[2]

(ii) Suggest how most ions are taken up during section **X** and section **Y** on Fig. 4.2. Give an explanation for your answers.

process during section **X** .....

process during section **Y** .....

explanation

.....

.....

.....

.....

.....

.....

.....

.....

[5]

(iii) Suggest a reason for the shape of the graph during section **Z** on Fig. 4.2.

.....

.....[1]

[Total: 12]

4 (a) (i)	root + hair / epidermis ;	[1]	
(ii)	osmosis / diffusion ; high <u>water</u> conc. to low(er) <u>water</u> conc. / down water potential gradient / hypotonic to hypertonic ; through / across partially permeable (AW) ; membrane ;	[3]	**R** active transport
(b) (i)	**(at Y)** (initial) increase in O_2 concentration leads to increase in uptake ; **(at X or Z)** (further) increase in O_2 concentration leads to no increase in uptake ;	[2]	
(ii)	**process during section X:** diffusion ; **process during section Y:** active transport / active uptake ; **explanation:** (during) X / diffusion + passive / not energy-requiring ; (during) Y / active transport + requires energy ; (energy for active transport from) respiration ; respiration requires oxygen ;	[5]	Max. 3 for explanations. **Ig** against concentration gradient
(iii)	rate of (ion) uptake dependent on another / other factor(s) / named factor e.g. temperature / ref. other limiting factor ;	1	
	Total	**[12]**	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Guidance

6 **(a)** Define the term *homeostasis.*

.....

.....

.....[2]

(b) Use the concept of control by negative feedback to explain what would happen in each of the following situations.

(i) A person goes outside wearing only short trousers and a short-sleeved shirt on a day when the air temperature is below 10 °C.

.....

.....

.....

.....

.....

.....

.....

.....[4]

(ii) A person drinks an excessive volume of water.

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

6 (a)	maintenance of / constant ; internal environment / conditions within the body ;	[2]	**A** regulating / control **Ig** specific examples
(b) (i)	low temperature detected ; by receptors / sensors on skin ; brain ; *any two corrective mechanisms from* shivering / hairs raised / sweating reduced/stopped / vasoconstriction of blood vessels (**R** of capillaries) ;; temperature rises again ;	[4]	 **A** hypothalamus
(ii)	(colon) water absorbed into blood ; rise / excess (in water content of blood) detected ; excess water excreted / removed / more urine produced AW ; by kidney ; correct ref. ureter / bladder / urethra in correct context ;	[4]	 **A** ref. hypothalamus **A** correct ref. to ADH **A** nephron
	Total	**[10]**	

© Cambridge International Examinations 2014

Question	Expected Answer	Mark	Guidance

7 **(a)** Compare the processes of mitosis and meiosis.

.....

.....

.....

.....

.....

.....

.....

.....[4]

(b) Use your knowledge of how sex is inherited to explain which parent determines the sex of a child.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9

Write your answers in the spaces provided.

7 (a)	<table><tr><th>Mitosis</th><th>Meiosis</th></tr><tr><td>2 daughter cells</td><td>4 daughter cells</td></tr><tr><td>haploid</td><td>diploid</td></tr><tr><td>chromosome number maintained</td><td>chromosome number halved</td></tr><tr><td>occurs in all organs / body cells</td><td>occurs in gamete producing organs / named</td></tr><tr><td>produces body cells / used in growth</td><td>produces gametes / named</td></tr><tr><td>ref. asexual reproduction</td><td>ref. sexual reproduction</td></tr><tr><td>no genetic variation in offspring AW</td><td>genetic variation in offspring AW</td></tr></table>	[4]	All points must be comparative to gain credit.
(b)	Father ; *Max 5 from the following:* father is XY / contains Y chromosomes ; mother is XX / does not contain Y chromosomes ; father produces sperm with either X or Y (chromosomes) ; mother produces eggs only with X (chromosomes) ; sex depends on which sperm fertilises the egg ; 1:1 male : female in offspring / AW ;	[1] [5]	**A** marks in Punnett Square
	Total	**[10]**	

© Cambridge International Examinations 2014

Section C			
Question	**Expected Answer**	**Mark**	**Guidance**

1 **(a)** State **three** substances found in the urine of a healthy person.

1 .....

2 .....

3 ..... [3]

(b) The concentration of a person's urine can vary according to their diet.

Explain how changes in a person's diet can affect the concentration of their urine.

.....

.....

.....

.....

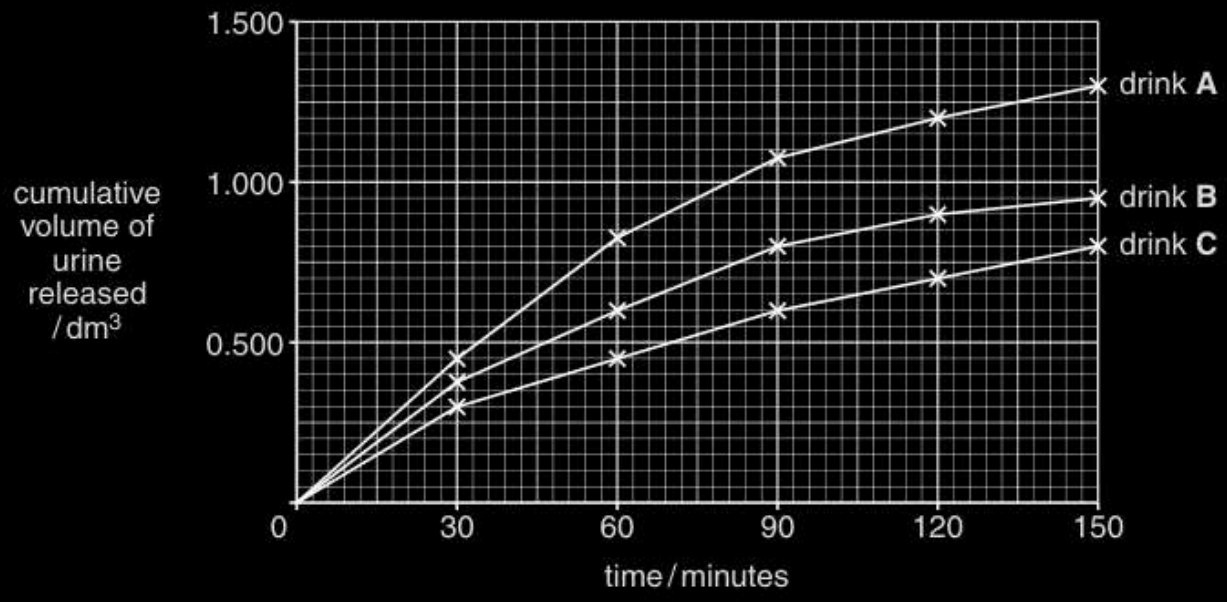
.....

.....

.....[4]

(c) An investigation was carried out into the effect of diet on the rate of production of urine. Three students each took 1.5 dm^3 of a different drink **A**, **B** or **C**.

Fig. 1.1 shows the volume of urine released by each student over the next two and a half hours.



Suggest which of the three drinks it would be better to avoid on a very hot day.

Give an explanation for your answer.

drink ..... [1]

explanation .....

.....

.....

.....

.....

.....[4]

[Total : 12]

1	(a)	3 named substances, e.g. water salts / ions / named, e.g. Na^+, Cl^-, NH_4^+, Ca^{2+} urea / nitrogenous waste / other named ;
	(b)	more protein / ORA ; correct ref. amino acids / ORA ; broken down in / converted by liver / deamination ; less water / more salts / ions + in diet / ORA ; (urine) more concentrated / more urea in (urine) / ORA ;

© Cambridge Internatio

	Syllabus	Paper
une 2014	5090	22

	Mark	Guidance
	[3]	**A** any other 3 correct substances, e.g. hormones, pigments, enzymes **R** sugar / glucose **A** any three named ions for 3 marks **A** any three named nitrogenous waste products for 3 marks, e.g. creatinine, uric acid
	[4]	**Ig** ref. to specific foods **A** ref. glucose / sugar only with ref. to diabetes

(c)	drink **A** ; increases volume of / more water in + urine / produces most / lot of / more urine ; water already being lost in sweat / AW ; (sweating) more than usual ; ref. temperature regulation / to reduce body temperature / keep cool / AW ; danger of dehydration / increases thirst / AW ;	[1] [4]	 Mark independently of drink named **Ig** ref. heat loss in urine
	Total	**[12]**	

© Cambridge International Examinations 2014

Expected Answer	Mark	Guidance

2 Two students performed an experiment to illustrate inheritance. They each made a 'spinner' similar to the one shown in Fig. 2.1. A result is recorded when a disc is spun and stops with one side nearest the surface.

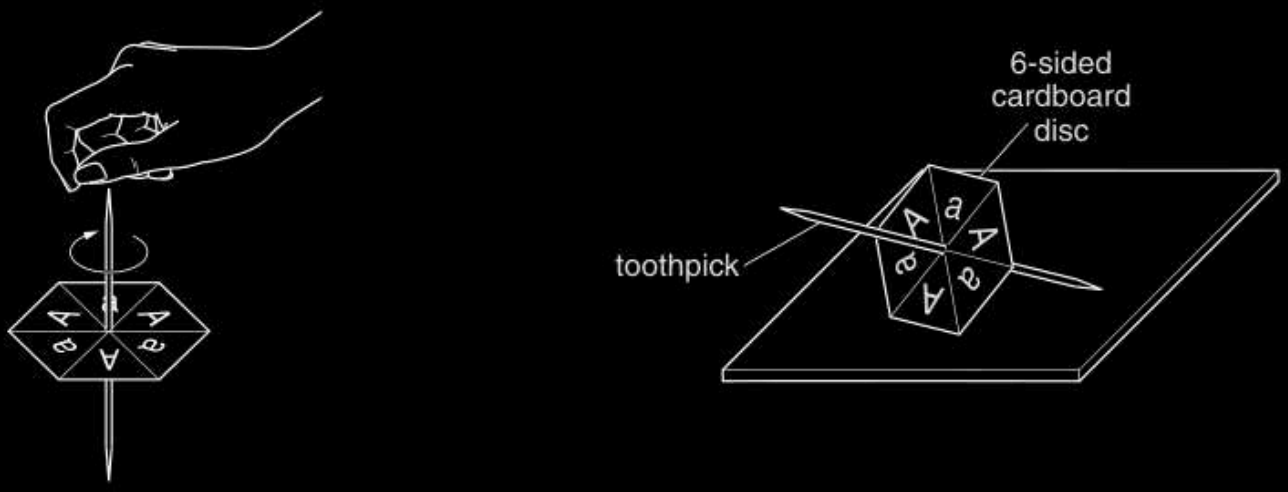


Fig. 2.1

(a) Complete Table 2.1, to show the expected results if the two students spin their discs, at the same time, on 80 separate occasions.

Table 2.1

combination of letters on each occasion	number of times each combination of letters is recorded
A and A	20
A and a	
a and a	

[1]

(b) Suggest **two** reasons why the results they obtained may have been different from the expected results.

.....

.....[2]

(c) Suggest the feature or stage in the process of inheritance represented by each of the following:

(i) the students .....

(ii) the spinning of the disc .....

(iii) the letters on the disc .....

[3]

(d) Describe how the students could modify their spinners and use them to illustrate the possible inheritance of blood groups by children of a father who has codominant blood group alleles and a mother who is heterozygous for blood group A.

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 11]

2 **(a)**	<table><tr><th>combination of letters on each occasion</th><th>number of times each combination of letters is recorded</th></tr><tr><td>A and A</td><td>20</td></tr><tr><td>A and a</td><td>40</td></tr><tr><td>a and a</td><td>20</td></tr></table>	[1]	
(b)	expected are theoretical or statistical / spinning is random or due to chance ; disc poorly made / toothpick doesn't pass through middle / AW; error in counting ;	[2]	**Ig** wind / force of spin **A** 'it' is random
(c) **(i)**	parents (cells) / genotypes / gonads or both named ;	[3]	
(ii)	meiosis / reduction division / gamete (formation) / fertilisation ;		
(iii)	genes / alleles / chromosomes / gametes ;		**R** genotype

© Cambridge International Examinations 2014

(d)	Correct alleles, A, B and O (with or without I) ; one disc with A and B and one disc with A and O ; each disc with correct numbers of alleles, i.e. 3 for discs given in question ; representing father and mother ; spin several / many times ; results recorded / counted ;	[5]	 e.c.f. with letters used in point 1 **R** if either parent has wrong alleles
	Total	**[11]**	

© Cambridge International Examinations 2014

Expected Answer	Mark	Guidance

3 Fig. 3.1 shows a flowering head of wheat, and individual flowers before and after opening.

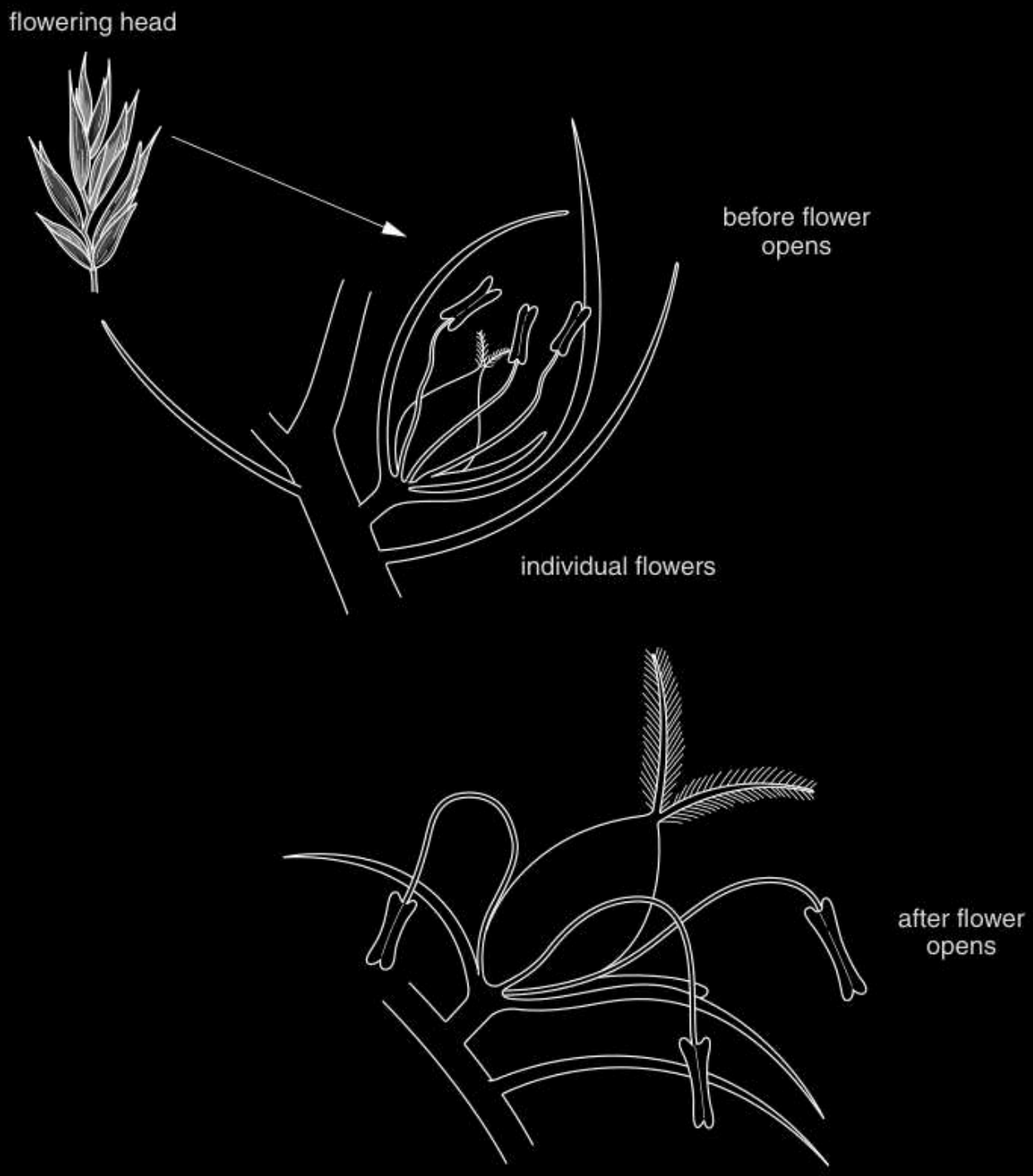


Fig. 3.1

The anthers release most of their pollen before the flower opens. The rest is released after the flower opens.

(a) Name the type of pollination found in the wheat plant before the flower opens.

.....

[1]

(b) **(i)** Using the information provided by Fig. 3.1, describe pollination in the wheat plant after the flower opens.

.....

.....

.....

.....

.....[3]

(ii) Wheat pollen is relatively heavy and is released for only a few hours after the flowers open.

Suggest **two** disadvantages of this.

.....

.....[2]

(c) Scientists are working to introduce genes into wheat plants to make them resistant to attack by insect pests (greenfly) and to encourage root nodule bacteria from pea and bean plants to live in their roots.

(i) Name the type of experimental work in which these scientists are involved.

.....

[1]

(ii) Suggest how the growth of root nodule bacteria on the roots of wheat plants could reduce the amount of fertiliser required by a growing wheat crop.

.....

.....

.....

.....

.....

.....

.....

.....

.....[5]

[Total:12]

3 (a)	self (–pollination) ;	[1]	
(b) (i)	(carried by) <u>wind</u> ; pollen to stigma ; of another (wheat) plant / flower ; correct ref. to cross-pollination (now being possible) ;	[3]	**Ig** ref. to animals
(ii)	wind can't carry / can't be carried far / reduced dispersal ; too much dependence on self-pollination / lack of (genetic) variation AW ; wind may not be blowing (over short time period) ; reduces chances of pollination / fertilisation ;	[2]	**R** if ref. to seed / fruit

© Cambridge International Examinations 2014

(c) (i)	genetic engineering / genetic modification ;	[1]	**Ig** gene transfer / biotechnology
(ii)	(bacteria) fix / convert / change / turn ; atmospheric / soil nitrogen ; (to) ammonium ; (to) nitrates ; (to make) amino acids / proteins ; (nitrates) absorbed / (amino acids or proteins) used by plants ;	[5]	**R** ammonia
	Total	**[12]**	

© Cambridge International Examinations 2014

Expected Answer	Mark	Guidance

4 Fig. 4.1(a) shows the reproductive organs of a man and Fig. 4.1(b) shows the reproductive organs of a woman.

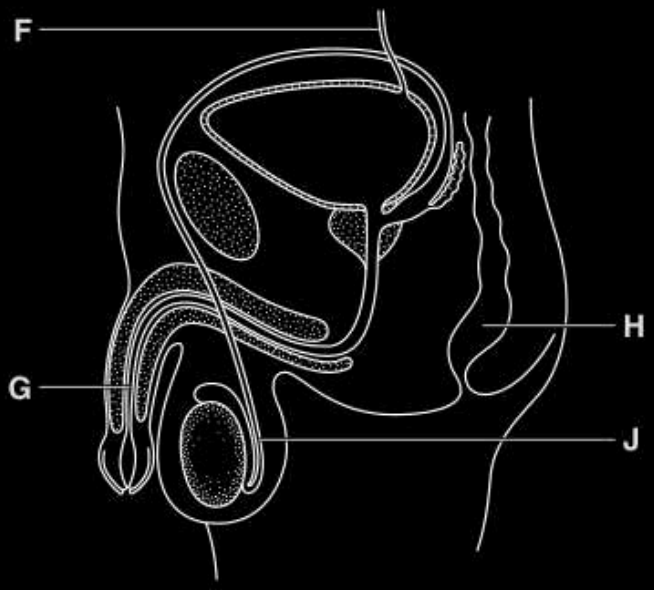


Fig 4.1(a)

Fig. 4.1(b)

(a) Complete Table 4.1, stating the names of the structures in Fig. 4.1(a) and indicating whether they carry urine and/or sperms using yes **or** no as appropriate.

Table 4.1

structure identified by letter	name of structure	carries urine (yes or no)	carries sperms (yes or no)
F			
G			
H			
J			

[4]

(b) Indicate by drawing a line across each of the relevant parts **in both Fig 4.1(a)** and **4.1(b)**, where a cut may be made in order to carry out a form of surgical contraception.

[2]

(c) In older men, the prostate gland tends to increase in size. Suggest an explanation for how this may affect urination.

.....

.....

.....

.....[2]

[Total : 8]

4 (a)	<table><tr><th>structure identified by letter</th><th>name of structure</th><th>carries urine (yes or no)</th><th>carries sperms (yes or no)</th></tr><tr><td>F</td><td>ureter</td><td>yes</td><td>no</td></tr><tr><td>G</td><td>urethra</td><td>yes</td><td>yes</td></tr><tr><td>H</td><td>rectum</td><td>no</td><td>no</td></tr><tr><td>J</td><td>vas deferens / sperm duct</td><td>no</td><td>yes</td></tr></table>	[4]	1 mark per correct row ; spelling of ureter and urethra **must** be correct
(b)	line drawn across sperm duct ; line drawn across oviduct ;	[2]	**R** if more than one line drawn on each Fig.– unless across same structure **R** if more than one structure cut **Ig** skin cuts
(c)	closes / restricts AW the urethra ; adverse effect on urination AW ;	[2]	**Ig** ref bladder **Ig** refs to pain on urination
	Total	**[8]**	

© Cambridge International Examinations 2014

Expected Answer	Mark	Guidance

8 **(a)** Describe how peristalsis causes food to be moved along the alimentary canal.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

(b) Explain why the heart muscle is **not** described as an antagonistic muscle.

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

8 (a)	muscles ; circular ; contract ; behind food ; longitudinal ; relax behind food / contract in front of food ; pushing / forcing / squeezing (bolus / AW) ; wave action / rhythmic ;	[6]	**R** if mention of parts outside of alimentary canal, e.g. trachea **R** if mention of contraction of longitudinal muscles behind food **Ig** moving
(b)	its muscles work on their own ; muscle not arranged in pairs / ORA ; no flexor / ORA ; no extensor / ORA ; no muscle relaxes when it contracts / ORA ; not attached to bones / ORA ; does not cause movement at a joint / ORA ;	[4]	 **A** ref. to one muscle
	Total	**[10]**	

© Cambridge International Examinations 2014

Expected Answer	Mark	Guidance

9 **(a)** Explain the health risks of each of the following:

(i) a high-fat diet

.....

.....

.....

.....

.....

.....

.....

.....

.....[5]

(ii) a low-protein diet

.....

.....

.....

.....

.....

.....[3]

(b) Explain why women may sometimes require iron supplements to their diet.

.....

.....

.....

.....[2]

[Total: 10]

9 (a) (i)	obesity ; strain on skeleton / effect on joints ; strain on heart / pumps harder / pumps faster ; breathing difficulties ; risk of diabetes ; social implications / example, e.g. bullying, clothing ; atheroma / AW ; high blood pressure ; heart disease / heart attack / other cardiovascular condition / AW ;	[5]	**R** ref. in veins / <u>on</u> arteries **Ig** blood vessels **A** cholesterol
(ii)	poor muscle development ; stunted / poor growth ; heart failure ; lack of / deficiency in one named protein, e.g. haemoglobin / antibodies / enzymes / hormones / thrombin ; AVP, e.g. reduced / deficient RBC production / poor wound healing / poor tissue / cell / organ repair / blood clotting / anaemia ;	[3]	Reference to a **negative** effect required.

© Cambridge International Examinations 2014

(b)	menstruation ; loss of blood ; haemoglobin ;	[2]	
	Total	**[10]**	

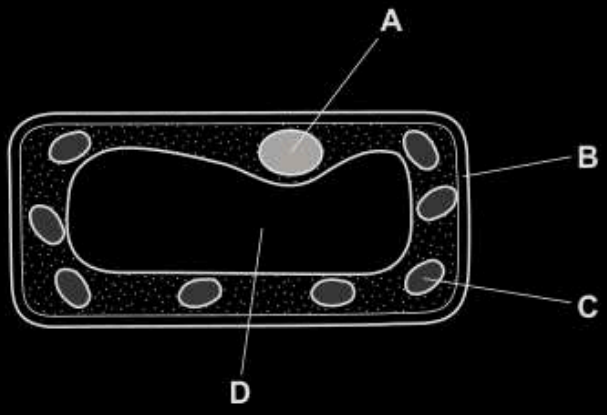
© Cambridge International Examinations 2014

October/November 2013 · Paper 1 · Variant 1 · Question 1

Correct answer: C

1 A plant is grown in bright sunlight. The diagram shows what is seen when a cell fro[illegible] placed under a microscope. After a few hours, a leaf from this plant is stained w[illegible] solution.

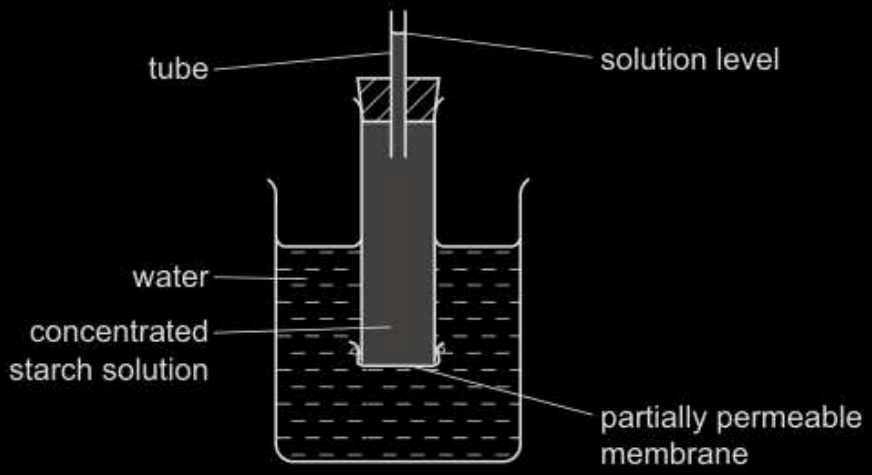
What will be stained blue / black?



October/November 2013 · Paper 1 · Variant 1 · Question 2

Correct answer: D

2 The diagram represents apparatus used to investigate osmosis.



Which molecules will move across the partially permeable membrane and which change will occur in the solution level?

	molecules	solution level
A	starch	fall
B	starch	rise
C	water	fall
D	water	rise

October/November 2013 · Paper 1 · Variant 1 · Question 3

Correct answer: A

3 The small intestine of a person contains a lower concentration of glucose than is p… blood.

The cells of the villi absorb glucose.

By which process is the glucose absorbed?

A by active transport against the concentration gradient

B by active transport with the concentration gradient

C by diffusion against the concentration gradient

D by diffusion with the concentration gradient

October/November 2013 · Paper 1 · Variant 1 · Question 4

Correct answer: A

4 Which statements are correct for all enzymes?

1 They are proteins.

2 They are secreted into the alimentary canal.

3 They speed up biochemical reactions.

4 None of them work at low pH.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

October/November 2013 · Paper 1 · Variant 1 · Question 6

Correct answer: A

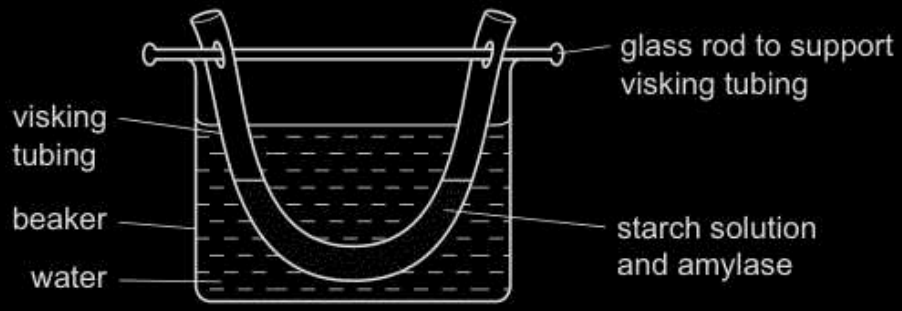
6 What is a function of each of these types of plant cell?

	phloem cells	root hair cells
A	sugar transport	ion uptake
B	sugar transport	transpiration
C	photosynthesis	ion uptake
D	photosynthesis	transpiration

October/November 2013 · Paper 1 · Variant 1 · Question 8

Correct answer: D

8 An investigation is carried out on digestion and absorption in the alimentary canal. [illegible] shows the apparatus used. The visking tubing is permeable to small molecules such as [illegible] but not to large molecules such as starch.



After one hour, samples of water in **the beaker** are tested with Benedict's solution and with iodine solution.

Which colours are obtained?

	colour obtained after heating with Benedict's solution	colour obtained after adding iodine solution
A	blue	blue-black
B	blue	yellow-brown
C	red	blue-black
D	red	yellow-brown

October/November 2013 · Paper 1 · Variant 1 · Question 9

Correct answer: C

9 What are the substrate and end-products of digestion by the enzyme lipase?

	substrate	end-product
A	carbohydrate	glucose
B	fat	amino acids
C	fat	fatty acids and glycerol
D	protein	fatty acids and glycerol

October/November 2013 · Paper 1 · Variant 1 · Question 10

Correct answer: A

10 Food tests were carried out on four different substances.

Which substance contained both protein and reducing sugar?

substance	Benedict's test	biuret test	emulsion test	iodine test
A	✓	✓	✗	✗
B	✓	✗	✓	✓
C	✗	✓	✓	✗
D	✗	✗	✗	✓

key
✓ = positive
✗ = negative

October/November 2013 · Paper 1 · Variant 1 · Question 11

Correct answer: D

11 In which direction do water molecules move in the phloem and in the xylem of a plant stem?

	phloem	xylem
A	down only	up only
B	up only	down only
C	up only	both up and down
D	both up and down	up only

October/November 2013 · Paper 1 · Variant 1 · Question 12

Correct answer: B

12 Four similar leafy shoots are exposed to different conditions. The rates of water uptake and the rates of water loss are measured.

The results are shown in the table.

Which shoot is most likely to wilt?

	water uptake / mm^3 per min	water loss / mm^3 per min
A	14	13
B	10	12
C	5	5
D	4	2

October/November 2013 · Paper 1 · Variant 1 · Question 13

Correct answer: D

13 What is the correct route for blood flow in a human?

A left atrium → left ventricle → lungs → right ventricle → right atrium

B left atrium → left ventricle → right ventricle → right atrium → lungs

C right atrium → right ventricle → left ventricle → left atrium → lungs

D right atrium → right ventricle → lungs → left atrium → left ventricle

October/November 2013 · Paper 1 · Variant 1 · Question 14

Correct answer: B

14 The table refers to blood vessels in the human body.

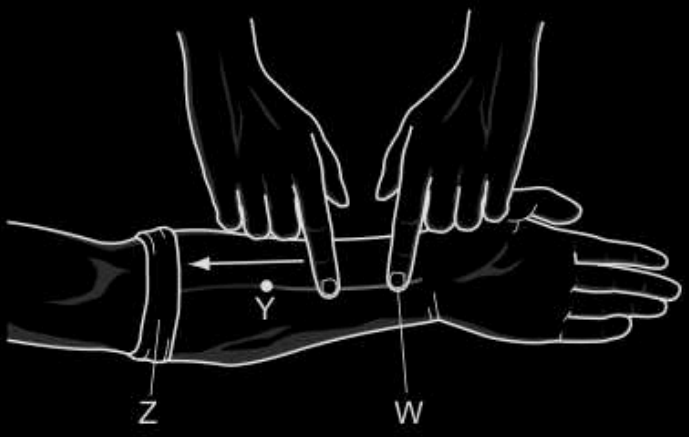
vessel	blood carried		oxygenated / deoxygenated
	from	to	
aorta	**P**	all organs except lungs	oxygenated
pulmonary vein	lungs	heart	**Q**
hepatic artery	aorta	**R**	oxygenated
hepatic portal vein	alimentary canal	liver	**S**

What are **P**, **Q**, **R** and **S**?

	P	Q	R	S
A	left ventricle	deoxygenated	kidney	deoxygenated
B	left ventricle	oxygenated	liver	deoxygenated
C	right ventricle	deoxygenated	kidney	oxygenated
D	right ventricle	oxygenated	liver	oxygenated

Correct answer: C

15 The diagram shows an investigation of blood flow in the veins of the lower arm.



A cloth is tightly wrapped round the arm at point Z and the veins stand out clearly. One finger presses firmly on the vein at W.

When another finger strokes the vein, as shown in the diagram, the vein lies flat between points W and Y.

Four reasons are given to explain why the vein lies flat.

1 The bandage at Z prevents backflow of blood.

2 The finger pressed at W prevents more blood entering the vein.

3 A valve at Y prevents backflow of blood.

4 A valve at Z prevents more blood from entering the vein.

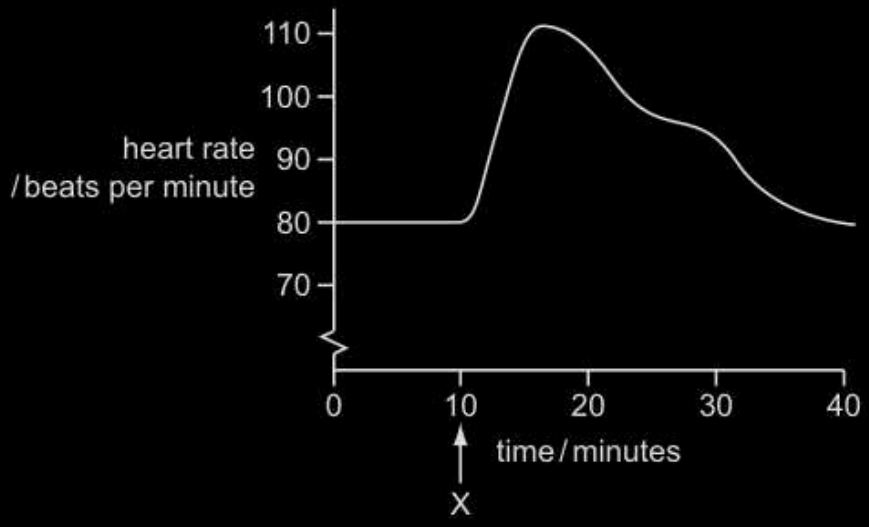
Which are correct?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

October/November 2013 · Paper 1 · Variant 1 · Question 18

Correct answer: B

18 A person begins to smoke a cigarette at time X. The graph shows how their heart rate



Which substance in cigarette smoke is the main cause of the change in heart rate between 10 and 18 minutes?

A carbon monoxide

B nicotine

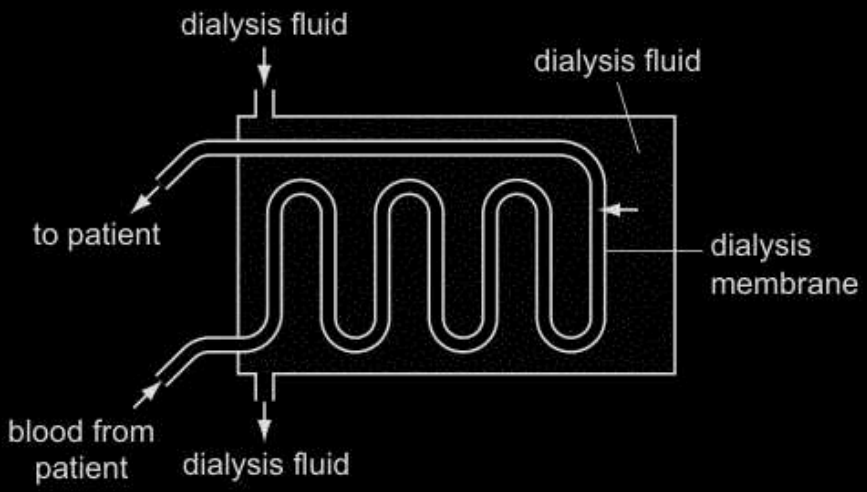
C smoke particles

D tar

October/November 2013 · Paper 1 · Variant 1 · Question 19

Correct answer: B

19 In a kidney dialysis machine, which substance **cannot** diffuse through the dialysis membrane?



A glucose

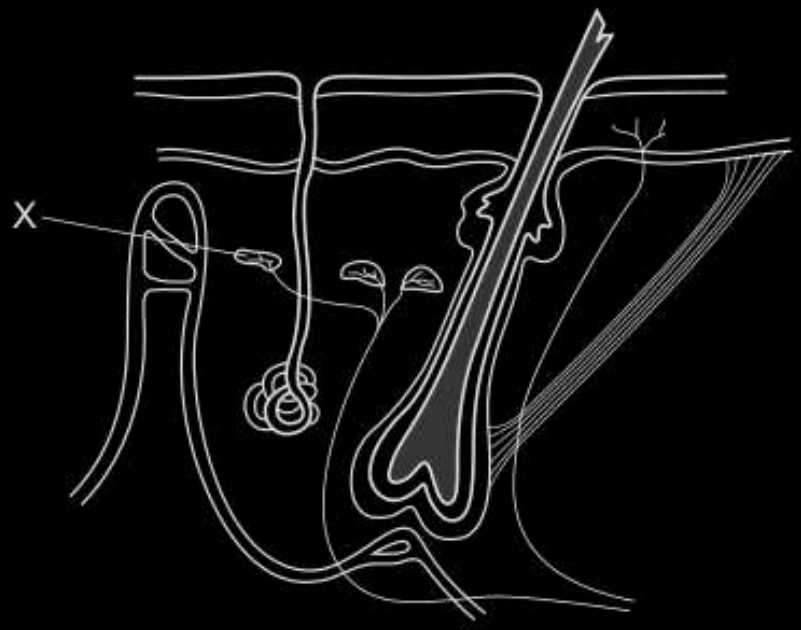
B insulin

C sodium

D urea

Correct answer: B

20 The diagram shows some of the structures seen in a section through human skin.



What is the function of structure X?

A to cause capillaries to constrict

B to detect changes in temperature

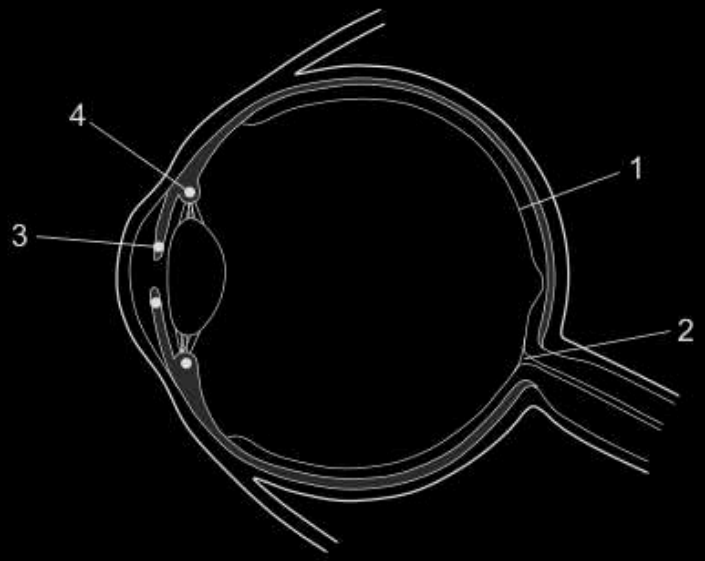
C to receive impulses from the central nervous system

D to stimulate sweat glands to release sweat

October/November 2013 · Paper 1 · Variant 1 · Question 22

Correct answer: A

22 The diagram shows a section through the eye.



In the pupil reflex, which row gives the sites of the effectors and receptors involved?

	effectors	receptors
A	3	1
B	3	2
C	4	1
D	4	2

October/November 2013 · Paper 1 · Variant 1 · Question 23

Correct answer: A

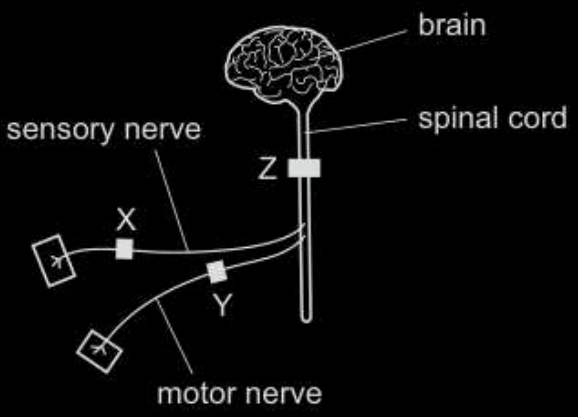
23 In a person suffering from diabetes mellitus, how do the concentrations of glucose in the blood and in the urine differ from those of a healthy person?

	concentration of glucose in blood	concentration of glucose in urine
A	higher	higher
B	higher	same
C	same	lower
D	lower	lower

October/November 2013 · Paper 1 · Variant 1 · Question 24

Correct answer: B

24 A local anaesthetic is a drug used to block nerve impulses. The diagram represent… nervous system. X,Y, and Z show sites where the anaesthetic can be injected.



In an experiment, one person can feel a pin prick their leg but cannot move their leg.

Where was the anaesthetic injected in this person?

A at X

B at Y

C at Z

D at X and at Y

October/November 2013 · Paper 1 · Variant 1 · Question 25

Correct answer: B

25 Chemicals in tobacco smoke lead to the breakdown of the elastic tissue in the walls of the alveoli.

What is the name of this condition?

A bronchitis

B emphysema

C lung cancer

D pneumonia

October/November 2013 · Paper 1 · Variant 1 · Question 26

Correct answer: A

26 The table shows the characteristics of four microorganisms.

Which one could be a virus?

	contains DNA	contains one or more cells	contains one or more cell nuclei	produces spores
A	✗	✗	✗	✗
B	✓	✓	✗	✗
C	✓	✓	✓	✗
D	✓	✓	✓	✓

key
✓ = true
✗ = false

October/November 2013 · Paper 1 · Variant 1 · Question 27

Correct answer: C

27 When bacteria act on milk, which row describes the formation of yoghurt?

	substrate	product	pH
A	glucose	starch	lower
B	lactic acid	lactose	higher
C	lactose	lactic acid	lower
D	starch	glucose	higher

October/November 2013 · Paper 1 · Variant 1 · Question 35

Correct answer: D

35 What is the result of cutting both the sperm ducts in a man?

A He is unable to develop sperms.

B He is unable to pass urine.

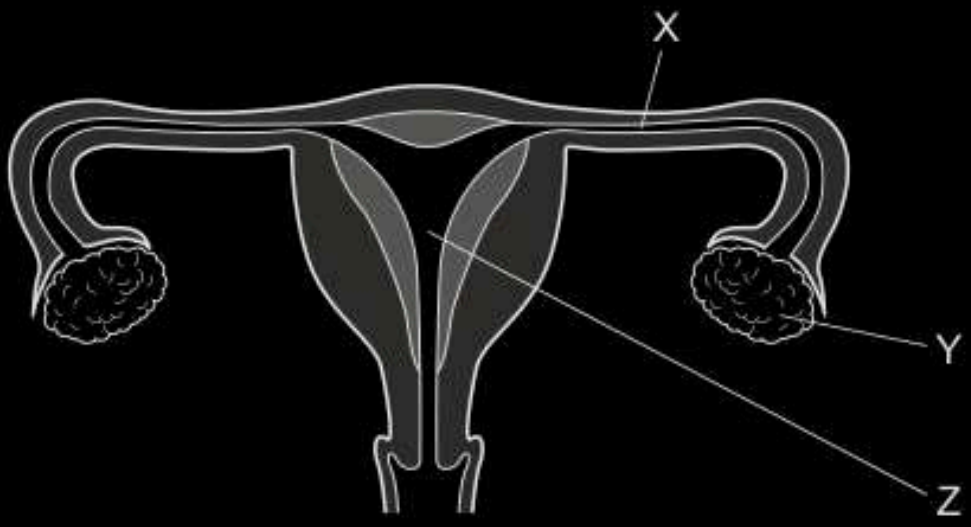
C Male sex hormones no longer circulate in his blood.

D Sperm are not emitted from the urethra.

October/November 2013 · Paper 1 · Variant 1 · Question 36

Correct answer: C

36 The diagram shows a section through the female reproductive system.



During pregnancy, where does mitosis occur in the cells of the embryo?

	X	Y	Z
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✗	✓

key

✓ = takes place

✗ = does not take place

October/November 2013 · Paper 1 · Variant 1 · Question 37

Correct answer: B

37 Some genotypes that occur in blood groups are given.

Which genotype results in a phenotype that shows co-dominance?

A $I^A I^A$

B $I^A I^B$

C $I^B I^O$

D $I^O I^O$

October/November 2013 · Paper 1 · Variant 1 · Question 38

Correct answer: A

38 A human cell contains all of the following.

Which is the smallest in size?

A gene

B nucleus

C X-chromosome

D Y-chromosome

October/November 2013 · Paper 1 · Variant 1 · Question 39

Correct answer: C

39 The allele for white flowers is recessive to the allele for red flowers.

Which statement is **not** correct?

A An allele for red in the genotype will always be seen in the phenotype.

B Crossing two heterozygotes will produce an approximate 3 : 1 ratio.

C Red flowers are always heterozygous.

D White flowers are always homozygous.

October/November 2013 · Paper 1 · Variant 1 · Question 40

Correct answer: C

40 What is a result of natural selection?

A dogs that are friendly to humans

B grapes that contain no seeds

C mosquitoes that are resistant to insecticides

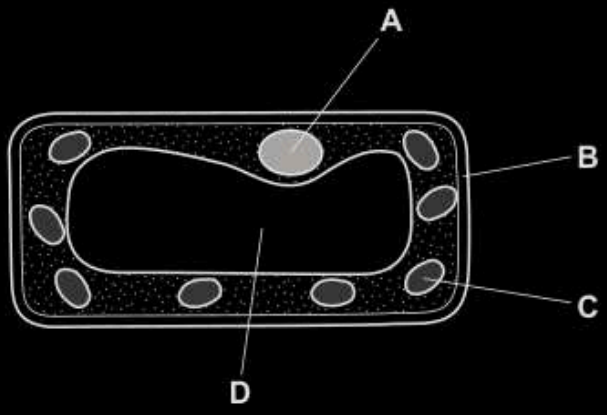
D onion crops that have a pleasant taste

October/November 2013 · Paper 1 · Variant 2 · Question 1

Correct answer: C

1 A plant is grown in bright sunlight. The diagram shows what is seen when a cell fro[illegible] placed under a microscope. After a few hours, a leaf from this plant is stained w[illegible] solution.

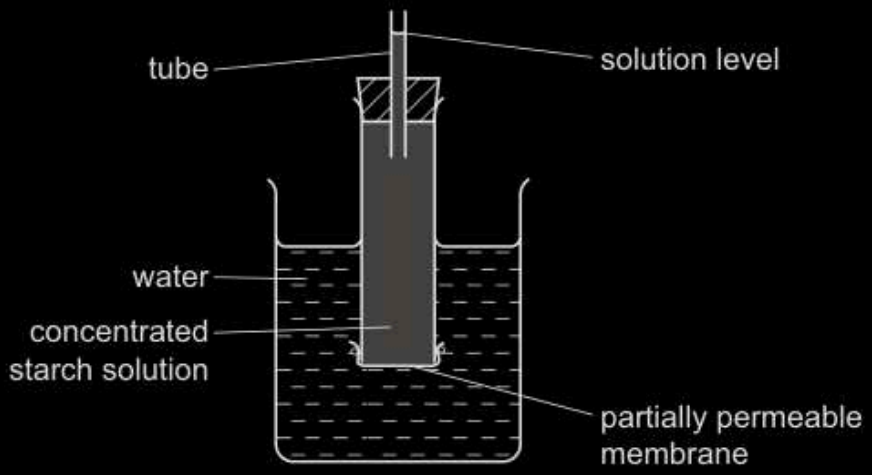
What will be stained blue / black?



October/November 2013 · Paper 1 · Variant 2 · Question 2

Correct answer: D

2 The diagram represents apparatus used to investigate osmosis.



Which molecules will move across the partially permeable membrane and which change will occur in the solution level?

	molecules	solution level
A	starch	fall
B	starch	rise
C	water	fall
D	water	rise

October/November 2013 · Paper 1 · Variant 2 · Question 3

Correct answer: A

3 The small intestine of a person contains a lower concentration of glucose than is p… blood.

The cells of the villi absorb glucose.

By which process is the glucose absorbed?

A by active transport against the concentration gradient

B by active transport with the concentration gradient

C by diffusion against the concentration gradient

D by diffusion with the concentration gradient

October/November 2013 · Paper 1 · Variant 2 · Question 4

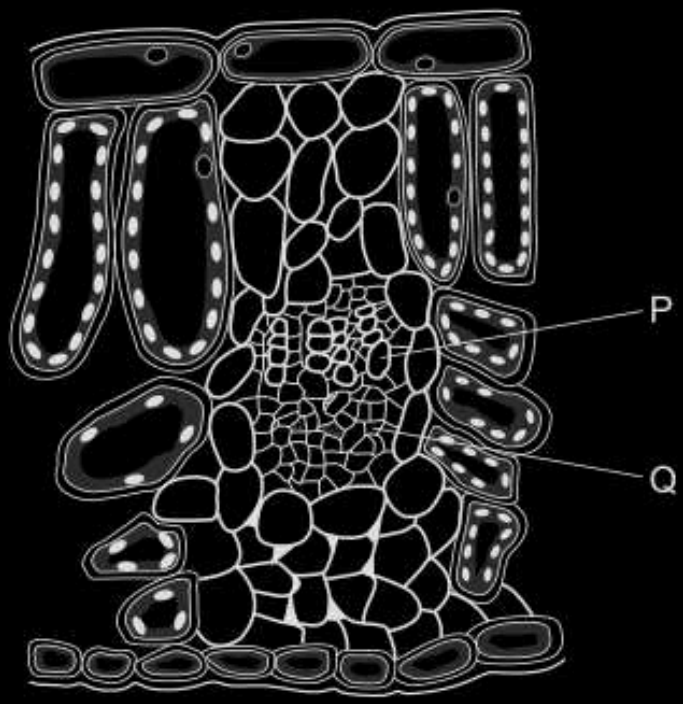
Correct answer: A

4 In an enzyme action, where is the active site and where are the lock and the key?

	active site	key	lock
A	on the enzyme	on the substrate	on the enzyme
B	on the enzyme	on the enzyme	on the substrate
C	on the substrate	on the enzyme	on the substrate
D	on the substrate	on the substrate	on the enzyme

Correct answer: C

7 The diagram represents a section through part of a leaf.



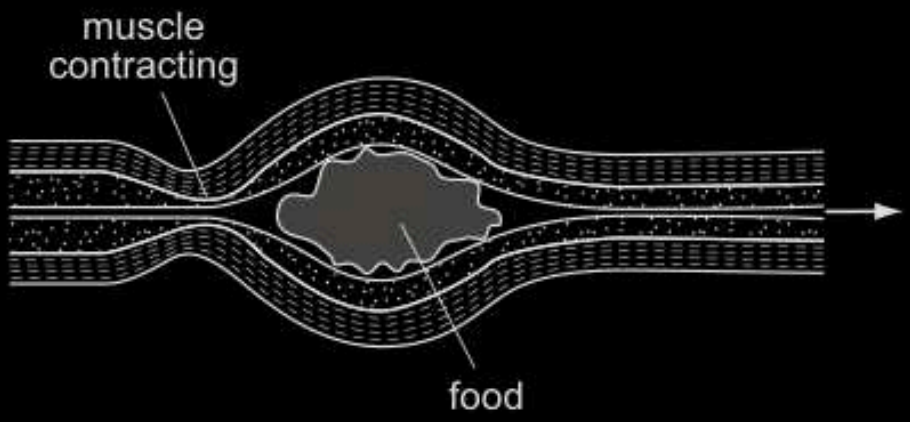
How do carbon dioxide and water enter the leaf?

	carbon dioxide	water
A	active transport	through tissue P
B	active transport	through tissue Q
C	diffusion	through tissue P
D	diffusion	through tissue Q

October/November 2013 · Paper 1 · Variant 2 · Question 8

Correct answer: D

8 The diagram shows some food moving through the digestive system.



Which process is shown?

A diffusion

B digestion

C ingestion

D peristalsis

October/November 2013 · Paper 1 · Variant 2 · Question 9

Correct answer: C

9 What are the substrate and end-products of digestion by the enzyme lipase?

	substrate	end product
A	carbohydrate	glucose
B	fat	amino acid
C	fat	fatty acids and glycerol
D	protein	fatty acids and glycerol

October/November 2013 · Paper 1 · Variant 2 · Question 10

Correct answer: A

10 The products of digested food are present in the ileum.

Which substances enter a blood capillary and a lacteal in a villus?

	blood capillary	lacteal
A	amino acids and glucose	fatty acids and glycerol
B	amino acids and glycerol	fatty acids and glucose
C	fatty acids and glucose	amino acids and glycerol
D	fatty acids and glycerol	amino acids and glucose

October/November 2013 · Paper 1 · Variant 2 · Question 12

Correct answer: B

12 Four similar leafy shoots are exposed to different conditions. The rates of water uptake and the rates of water loss are measured.

The results are shown in the table.

Which shoot is most likely to wilt?

	water uptake / mm^3 per min	water loss / mm^3 per min
A	14	13
B	10	12
C	5	5
D	4	2

October/November 2013 · Paper 1 · Variant 2 · Question 13

Correct answer: D

13 What is the correct route for blood flow in a human?

A left atrium → left ventricle → lungs → right ventricle → right atrium

B left atrium → left ventricle → right ventricle → right atrium → lungs

C right atrium → right ventricle → left ventricle → left atrium → lungs

D right atrium → right ventricle → lungs → left atrium → left ventricle

October/November 2013 · Paper 1 · Variant 2 · Question 14

Correct answer: D

14 Which row describes the functions of the blood components?

	plasma	platelets	white blood cells
A	antibody formation	clotting	transport of nutrients
B	clotting	transport of nutrients	antibody formation
C	clotting	antibody formation	transport of nutrients
D	transport of nutrients	clotting	antibody formation

October/November 2013 · Paper 1 · Variant 2 · Question 15

Correct answer: B

15 In the liver, which substances are present in higher concentration in the hepatic vein than in the hepatic artery?

A carbon dioxide and protein

B carbon dioxide and urea

C oxygen and protein

D oxygen and urea

October/November 2013 · Paper 1 · Variant 2 · Question 19

Correct answer: A

19 Which parts of the skin are involved in the control of body temperature?

	sweat glands	temperature receptors	blood vessels
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

October/November 2013 · Paper 1 · Variant 2 · Question 20

Correct answer: C

20 How is the shape of the lens changed when the eye focuses on a distant object?

	ciliary muscles	suspensory ligaments	lens
A	contract	pulled tight	pulled thin
B	contract	slackened	allowed to bulge
C	relax	pulled tight	pulled thin
D	relax	slackened	allowed to bulge

October/November 2013 · Paper 1 · Variant 2 · Question 21

Correct answer: A

21 In a kidney dialysis machine, which substance **will not** diffuse from the patient's blood into the dialysis fluid?

A protein

B salts

C urea

D water

October/November 2013 · Paper 1 · Variant 2 · Question 22

Correct answer: B

22 Which structure in the brain registers changes in the blood concentration?

A cerebellum

B hypothalamus

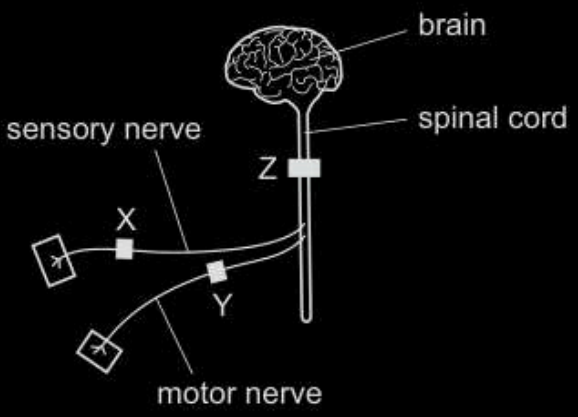
C medulla

D pituitary gland

October/November 2013 · Paper 1 · Variant 2 · Question 24

Correct answer: B

24 A local anaesthetic is a drug used to block nerve impulses. The diagram represent… nervous system. X,Y, and Z show sites where the anaesthetic can be injected.



In an experiment, one person can feel a pin prick their leg but cannot move their leg.

Where was the anaesthetic injected in this person?

A at X

B at Y

C at Z

D at X and at Y

October/November 2013 · Paper 1 · Variant 2 · Question 25

Correct answer: B

25 Chemicals in tobacco smoke lead to the breakdown of the elastic tissue in the walls of the alveoli.

What is the name of this condition?

A bronchitis

B emphysema

C lung cancer

D pneumonia

October/November 2013 · Paper 1 · Variant 2 · Question 26

Correct answer: A

26 The table shows the characteristics of four microorganisms.

Which one could be a virus?

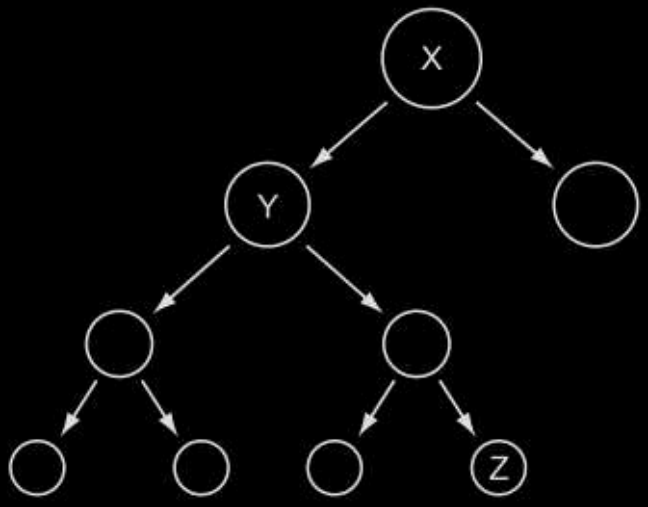
	contains DNA	contains one or more cells	contains one or more cell nuclei	produces spores
A	✗	✗	✗	✗
B	✓	✓	✗	✗
C	✓	✓	✓	✗
D	✓	✓	✓	✓

key
✓ = true
✗ = false

October/November 2013 · Paper 1 · Variant 2 · Question 34

Correct answer: D

34 Cell X first divides by mitosis and then cell Y divides by meiosis.



How does cell Z compare to cell X?

	cell Z	
	same number of chromosomes as cell X	genetically identical to cell X
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key
✓ = true
✗ = false

October/November 2013 · Paper 1 · Variant 2 · Question 35

Correct answer: D

35 What is the result of cutting both sperm ducts in a man?

A He is unable to develop sperms.

B He is unable to pass urine.

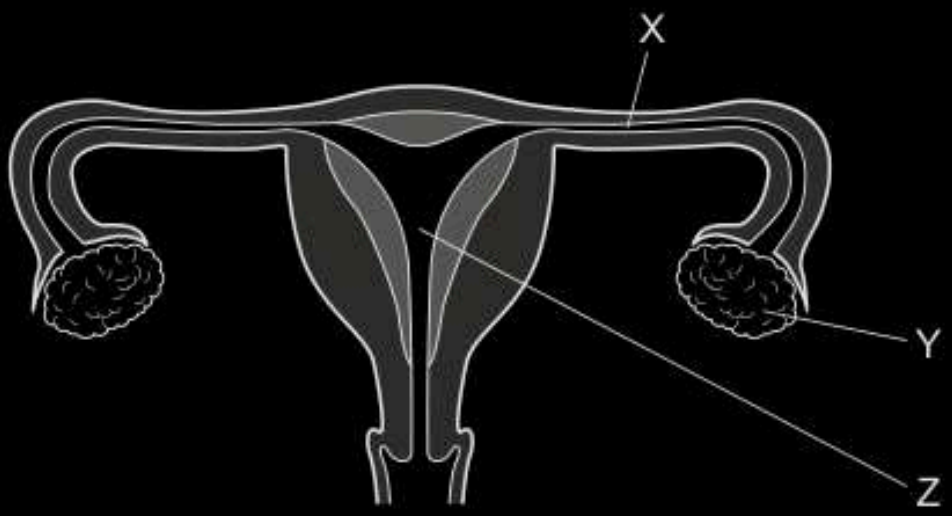
C Male sex hormones no longer circulate in his blood.

D Sperm are not emitted from the urethra.

October/November 2013 · Paper 1 · Variant 2 · Question 36

Correct answer: C

36 The diagram shows a section through the female reproductive system.



During pregnancy, where does mitosis occur in the cells of the embryo?

	X	Y	Z
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✗	✓

key

✓ = takes place

✗ = does not take place

October/November 2013 · Paper 1 · Variant 2 · Question 38

Correct answer: C

38 The table shows the genotypes and phenotypes for hair colour for the members of [illegible] **one** phenotype is shown incorrectly.

	genotype		phenotype
family member	allele 1	allele 2	hair colour
mother	a	A	brown
father	A	A	brown
son 1	a	A	blonde
daughter 1	a	a	blonde
son 2	A	A	brown
daughter 2	A	a	brown

Which family member has the **incorrect** phenotype?

A daughter 1

B daughter 2

C son 1

D son 2

October/November 2013 · Paper 1 · Variant 2 · Question 39

Correct answer: C

39 A gene is a unit of inheritance that controls the production of

A a chromosome.

B an allele.

C a protein.

D DNA.

October/November 2013 · Paper 1 · Variant 2 · Question 40

Correct answer: C

40 What is a result of natural selection?

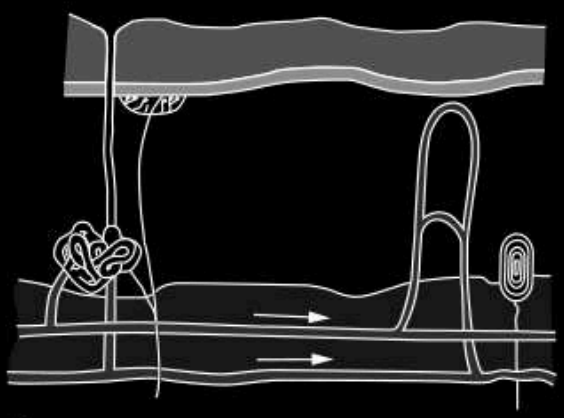
A dogs that are friendly to humans

B grapes that contain no seeds

C mosquitoes that are resistant to insecticides

D onion crops that have a pleasant taste

1 Fig. 1.1(a) and Fig. 1.1(b) show a section through the skin of a person at two different body temperatures.



Key

→ direction of blood flow

Fig. 1.1(a)

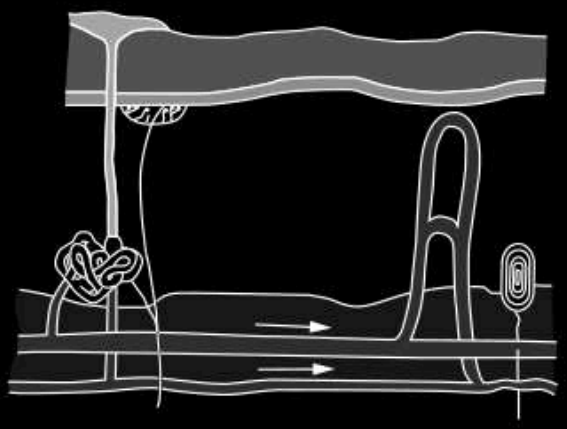


Fig. 1.1(b)

(a) On Fig. 1.1(a), identify and label each of the following.

(i) a sweat gland

(ii) a temperature receptor. [2]

(b) (i) State what has happened to the body temperature to cause the changes shown in Fig. 1.1(b) compared with Fig. 1.1(a).

.....[1]

(ii) State **two** reasons for the answer you have given in **(b)(i)**.

1 .....

2 .....[2]

(c) (i) On Fig. 1.1(a), label and name the type of blood vessel that constricts wh... body temperature falls.

(ii) Explain the effects caused by constriction of the blood vessel you have named in **(c)(i)**.

.....

.....

.....

.....[4]

[Total: 11]

...Cambridge.c...

1 (a) (i)	label line must touch the sweat gland;	[1]	**A** label on gland or duct **A (i) = gland**
(ii)	label line must touch some part of receptor under Malpighian layer;	[1]	**(ii)** = temp receptor
(b) (i)	(temperature / it) increases / rises;	[1]	**R** cooler / decreases OWTTE
(ii)	In either order: sweat; wider blood vessels / capillaries / more blood;	[2]	**R** veins
(c) (i)	arteriole; the upper horizontal blood vessel labelled;	[2]	**A** Artery (the word – ignore what is labelled) (ignore name given to it)
(ii)	less blood (flows); to capillaries (R capillaries constrict) to sweat glands; blood carries heat / less heat carried less heat lost by skin / radiation ; less sweat produced; ref to latent heat of vaporisation;	[max 4]	**A** less heat lost by blood
		[Total 11]	

3 **(a)** Name **two** constituents of a diet that are absorbed without digestion. Fo... constituent, name the part of the alimentary canal in which it is absorbed.

constituent 1 ..... absorbed in .....

constituent 2 ..... absorbed in ..... [2]

(b) Suggest a constituent of a healthy diet that is never absorbed. .....[1]

(c) Fig. 3.1 is a diagrammatic representation of the small intestine containing three types of food molecule, a fat, a carbohydrate and a protein, before they have been digested. Fig. 3.1 also shows a lacteal and a capillary. The different features in Fig. 3.1 have not been drawn to the same scale.

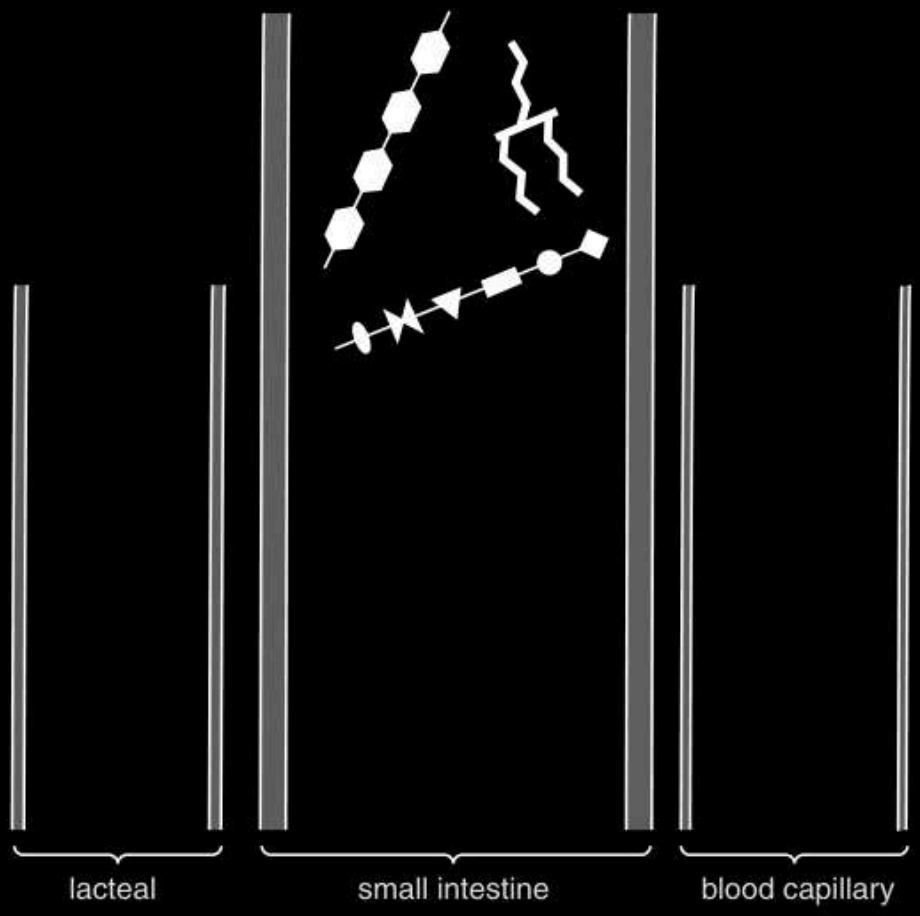


Fig. 3.1

On Fig. 3.1, draw **and** label the molecules as they would appear **after** they have been digested and then absorbed by the lacteal and by the capillary. [4]

(d) Some absorbed molecules travel directly to the liver.

(i) Name **three** of these molecules.

molecule 1 .....

molecule 2 .....

molecule 3 ..... [3]

(ii) Explain how these molecules are carried to the liver.

.....

.....

.....

.....[2]

[Total: 12]

3 **(a)**	water + any part of the alimentary canal after the oesophagus; ions / minerals / named + ileum; vitamins / named + ileum / colon glucose + ileum	[max 2]	**A** small intestine for ileum
(b)	fibre / roughage / cellulose / fruits / seeds / tomato skins / maize pericarps;	[1]	**A** any suitable named plant part
(c)	products drawn in correct absorptive vessel – All correct = 2 marks 2 correct = 1 mark ;; all molecules totally digested ; 2 products correctly named ;	[max 4]	**I** any diagrams in the intestine **R** any products drawn in both capillaries and lacteals All bonds must be broken – a minimum of 3 products of each type required.
(d) (i)	amino acids; glucose/ other named monosaccharide; vitamins / named; ions / salts / minerals / named; water;	[max 3]	
(ii)	in solution; in blood / plasma; by (hepatic) portal vein;	[max 2]	
		[Total 12]	

October/November 2013 · Paper 2 · Variant 1 · Question 6

6 **(a)** Describe the functions of xylem in a plant.

.....

.....

.....

.....

.....

.....[3]

(b) Explain how starch and protein stored in the roots of a plant during the day can be used in the growth of a shoot during the night.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[7]

[Total: 10]

6 (a)	support / rigidity; lignin; conduct/transport of water; ions / minerals / salts; ref to narrow/hollow/continuous / lumen; from roots(hair) + ref to any aerial part of plant;	[max 3]	**R** transport of amino acids / sugars
(b)	ref enzyme / named enzyme; action of enzyme ; named product of enzyme action; ref to solubility of product diffusion(from storage cells to phloem or vice –versa); to phloem; sucrose/amino acids (moved up phloem) ; ref translocation/transport up stem; no photosynthesis at night ; (energy) released during respiration; sugar provides energy for growth; amino acids make proteins; to make new cytoplasm / cells;	[max 7]	**A** amino acids / glucose / sugar **R** produced
		[Total 10]	

7 **(a)** Explain the importance of active transport in plants and in humans.

in plants .....

.....

.....

.....

in humans .....

.....

.....

.....[4]

(b) **(i)** Explain what is meant by the term *diffusion*.

.....

.....[2]

(ii) Explain why respiration depends on the process of diffusion.

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7 (a)	*ref. to <u>against</u> concentration gradient; **A** a description *(plants) any 2 from*: minerals / ions / salts; from soil; via root hairs; *(humans) any 2 from*: glucose; from gut; via villi ;	[max 4]	*credit once only, under plants or humans,
(b) (i)	movement of <u>molecules / particles / ions;</u> high concentration to low;	[2]	
(ii)	ref. crossing a membrane ; thin; oxygen in (to cell) ; glucose in (to cell) ; carbon dioxide out (of cell) ;	[max 4]	**A** ref to entering named plant or animal cell Allow 1 mark for O_2 in + Co_2 out - if only refer to gases in the lungs.
		[Total 10]	

8 **(a)** Explain the importance of *meiosis* in sexual reproduction.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

(b) Fruits and seeds may be dispersed by the wind.

Describe the features of a **named** wind-dispersed fruit or seed and explain the importance of these features in the process of dispersal.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 10]

8 (a)	cell division; ref. to chromosomes; diploid to haploid; gamete formation/ named gametes; all gametes genetically different; fertilisation restores chromosome number / AW; leads to variation in population;	[max 4]	**A** halving of chromosome number
(b)	name of fruit / seed (not linked to following marks); structural feature of fruit/seed that is a wind-dispersal adaptation; ref. large surface area / buoyancy; *wind removes fruit / seed from parent plant; *seed relatively light in mass; *to delay descent of seed AW; spread over large area / prevent overcrowding / competition ;	[max 6]	Accept * points for descriptions of pollen for max 3 e.g. floats for long time / increases air resistance
		[Total 10]	

1 Fig. 1.1 shows a demonstration related to blood circulation.

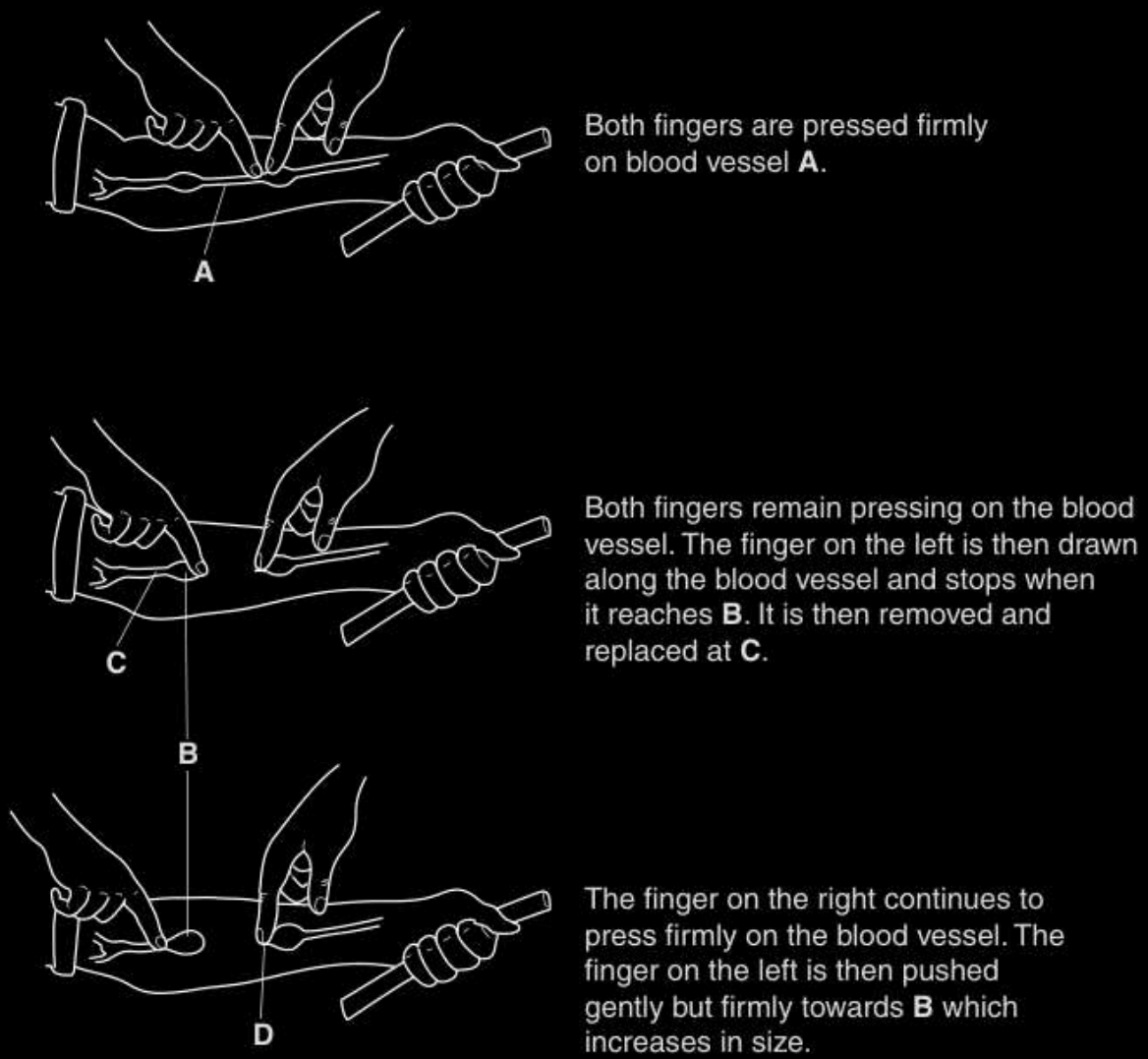


Fig. 1.1

(a) (i) Name the type of blood vessel labelled **A** in Fig. 1.1.

..... [1]

(ii) Name the structure leading to the effect shown in this blood vessel at position **B**.

..... [1]

(b) In the space below, draw a longitudinal section (cut along its length) through the vessel as it appears at position **B**.

[2]

(c) Suggest why, at the end of the demonstration, the blood vessel is no longer visible between positions **B** and **D**.

.....

.....

.....

.....

.....

.....

.....[4]

(d) Suggest why a rod was repeatedly gripped tightly before carrying out this demonstration.

.....

.....

.....

.....[2]

[Total: 10]

1 **(a)** **(i)** vein;

(ii) valve;

(b) a valve shown with flaps touching;

vessel widest at correct side of valve + walls shown both before and after valve; [2]

(c) (vein) empty / no blood flowing through;

blood has been pushed out of (vein);

* blood cannot flow back (due to valve / from B);

* finger on the right / at D + prevents blood flowing;

walls of vein thin thus not visible beneath skin **AW**; [max 4]

(d) muscle + contraction(s);

increase blood pressure;

increases circulation / blood flow / helps fill (vein) with blood; [2]

[Total 10]

2 Fig. 2.1 shows how an alcoholic drink is produced in some countries.

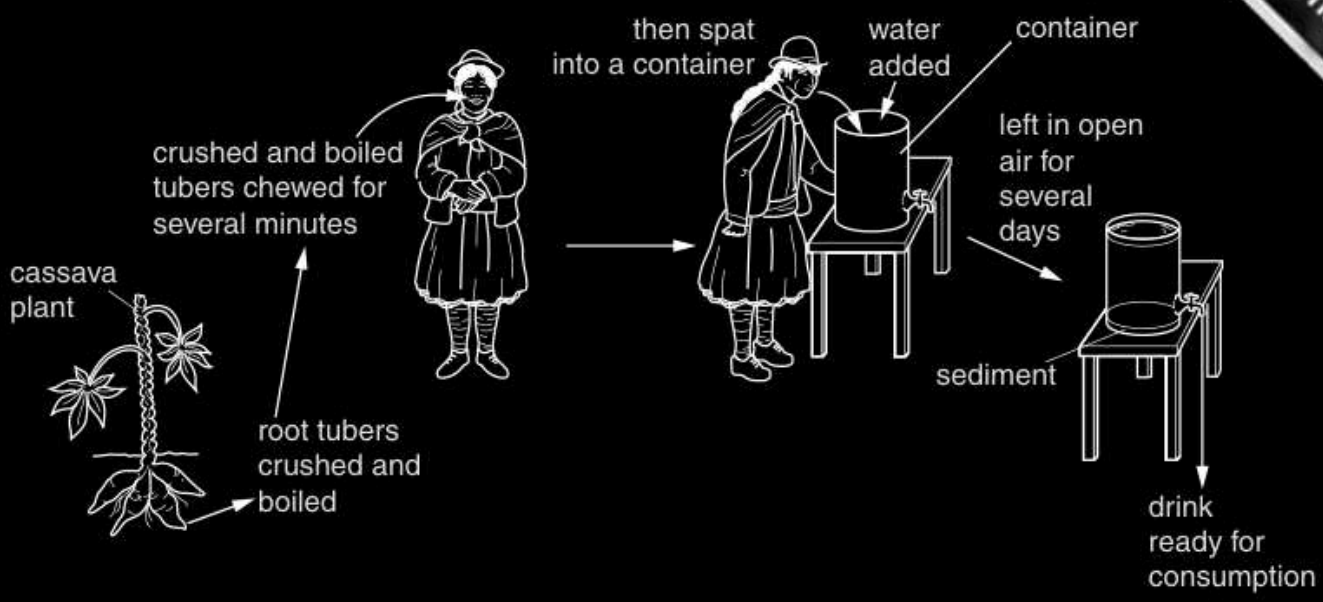


Fig. 2.1

(a) The root tubers of the cassava plant store starch. After removal from the plant, the tubers are crushed and boiled.

Suggest the effect that crushing and boiling will have on the cells of the tubers.

.....

..... [2]

(b) After they have been crushed and boiled, the cassava tubers are chewed for several minutes.

Explain what happens to the starch during this time.

.....

.....

.....

..... [3]

(c) **(i)** Name the process that must occur in the container to produce alcohol and ... space below, give an equation for this process.

process.....

equation

[3]

(ii) Name the type of organism, whose spores are found in soil and floating in the air, that will bring about the production of alcohol.

.....[1]

(d) Suggest why the sediment in the bottom of the container increases in quantity as the container is left to stand for several days.

.....

.....

.....

.....[2]

[Total: 11]

2 **(a)** damage / breaks **AW** (cells / tuber);

ref. cell wall / membrane;

releasing (cell) contents / starch;

kills cells / denatures enzymes / stops metabolic reactions; [max 2]

(b) amylase / carbohydrase;

digests / breaks down / hydrolyses + starch;

(to) <u>maltose</u>; [3]

(c) **(i)** fermentation / anaerobic respiration; [1]

left hand side of equation
glucose / $C_6H_{12}O_6$;

right hand side of equation
$2C_2H_5OH + 2CO_2$ / alcohol/ethanol + carbon dioxide; [2]

(ii) fungus / yeast ; [1]

(d) yeast (cells) + reproduce / increase in number / multiply;

(sediment contains) yeast (cells) / bacteria / chewed tuber **AW**

yeast (cells) dead + sugar / food used up / alcohol is toxic **AW**; [max 2]

[Total 11]

3 Fig. 3.1 shows six different animal and plant cells.

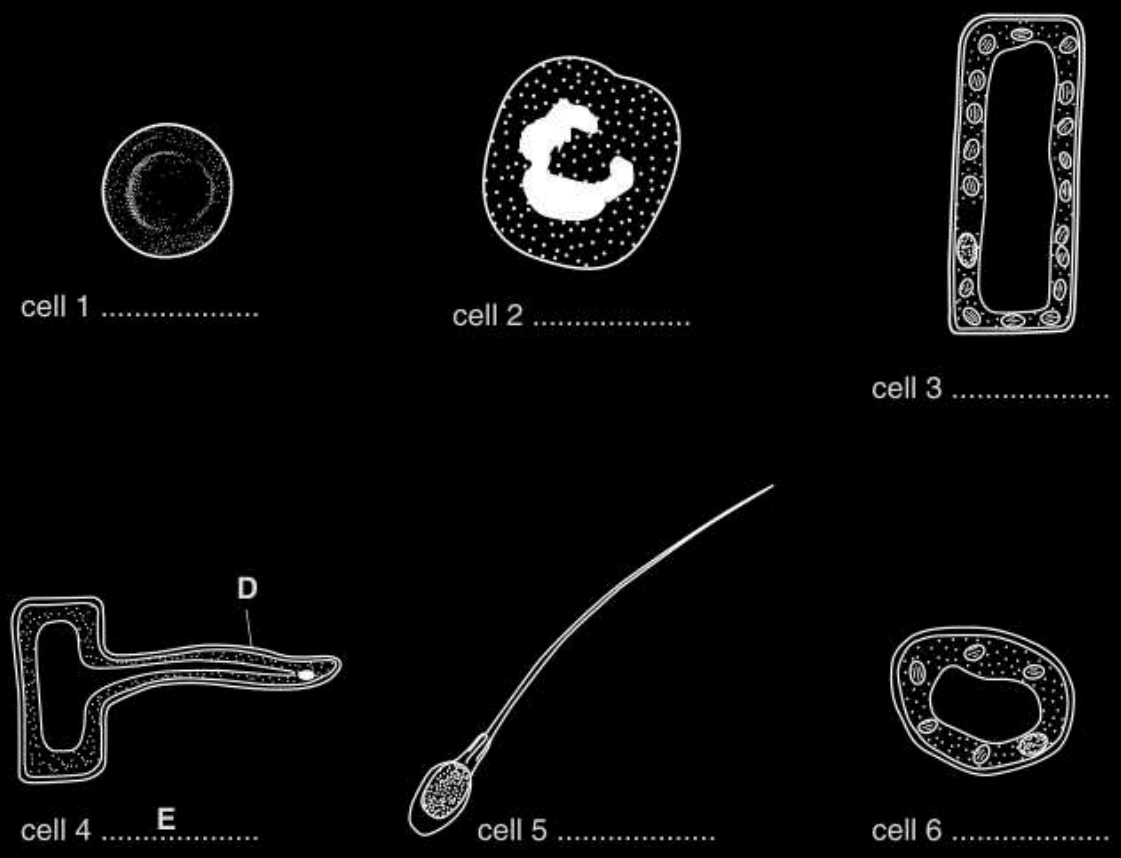


Fig. 3.1

(a) (i) Name cells 4 and 5. Write your answers below.

cell 4.....

cell 5..... [2]

(ii) Describe the function of structure **D** in cell 4.

.....

.....[3]

(b) You are now required to identify each cell **by letter**, following a series of instructi

When you have identified each cell, write the appropriate letter beneath the identifie cell in Fig. 3.1, in the space provided.

Cell **E** has already been identified for you.

Select a cell in Fig. 3.1, then ask yourself the following questions.

Q1 Does the cell have a cell wall?
if it does, go to Q2
if it does not, go to Q3

Q2 Does the cell have chloroplasts?
if it does, go to Q4
if it does not, then it is cell **E**

Q3 Does the cell have a nucleus?
if it does, go to Q5
if it does not, then it is cell **F**

Q4 Does the cell show more than 5 chloroplasts?
if it does, it is cell **G**
if it does not, then it is cell **H**

Q5 Does the nucleus occupy more than half of the cell's cytoplasm?
if it does, it is cell **J**
if it does not, then it is cell **K**

When you have completed this process for the cell you selected, repeat the process for another cell, and continue until all cells have been identified by letter. [5]

[Total: 10]

3 **(a)** **(i)** (cell 4) - root <u>hair</u>;

(cell 5) - sperm(atozoon) / male gamete;

(ii) (root hair) increases surface area;

(for) absorption / movement (A correct named method of molecular transport) into **AW**;

minerals / ions / salts (or named);

water;

oxygen;

(cell wall) gives mechanical support / allows turgor; [max 3]

(b) cell 1 (RBC) – **F**;

cell 2 (WBC) – **K**;

cell 3 (palisade) – **G**;

cell 5 (sperm) – **J**;

cell 6 (spongy) – **H** ; [5]

[Total 10]

4 Fig. 4.1 shows a large jar in which plants are growing.

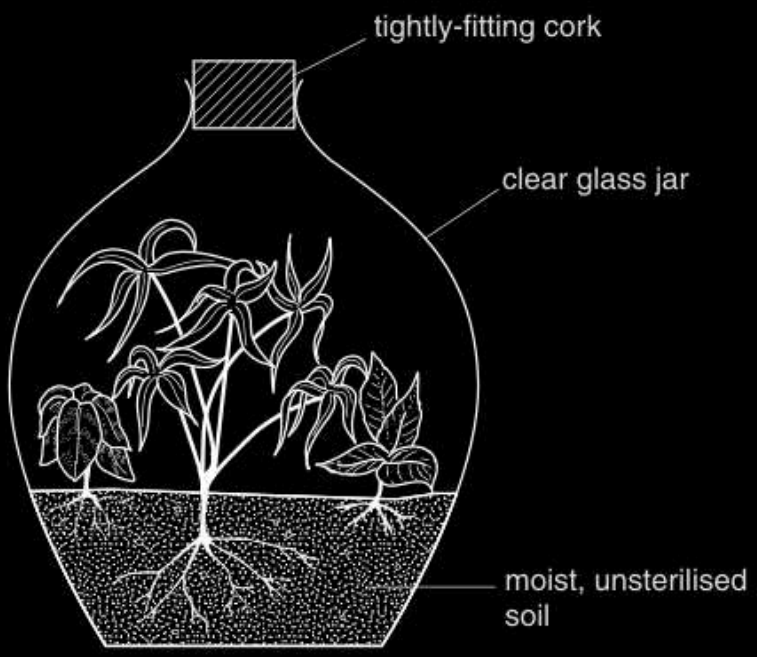


Fig. 4.1

This jar provides an environment in which plants can live for many months without adding water or removing the tightly-fitting cork to allow air to enter.

(a) State the reason for placing the jar where it can receive a supply of sunlight.

.....[1]

(b) Suggest why the plants in the jar show only very limited growth compared with similar plants growing under natural conditions.

.....

.....

.....

.....[3]

(c) The cork prevents atmospheric air from entering the jar. Explain how the plants are able to remain alive without a continuous supply of fresh air.

.....

.....

.....

.....

.....[4]

(d) Explain why no water needs to be added to the jar.

.....

.....

.....

.....[3]

[Total: 11]

4 **(a)** for photosynthesis / to make food/carbohydrate/starch/ glucose/sugar;

(b) * limited nitrates / nitrates not topped up;

limited protein production;

* limited magnesium + for chlorophyll;

limited carbon dioxide ;

limited photosynthesis / limited production of food/ carbohydrate/starch/glucose/sugar;

limited space / volume / area; [max 3]

(c) ref. respiration + photosynthesis (occurring in jar);

respiration + releases CO_2;

CO_2 + used for photosynthesis;

photosynthesis + releases O_2;

O_2 + used for respiration;

ref. microorganisms / bacteria / fungi / decomposers + in soil; [max 4]

(d) water from leaves/transpiration **AW** / water from soil evaporates;

respiration + produces/releases water;

*returned to soil / condenses;

*absorbed **AW** / used by plants; [max 3]

[Total 11]

6 **(a)** State the similarities between sweating and transpiration.

.....

.....

.....

.....[3]

(b) Describe the differences between sweating and transpiration.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[7]

[Total: 10]

6 **(a)** loss of water;

involves evaporation / as vapour;

to the atmosphere / surroundings;

cooling effect;

through pores **AW**;

in epidermis;

both affected by temp / humidity / wind (speed); [max 3]

(b) (Accept reverse arguments where relevant)

occurs in animals;

sweating ref. (loss of) urea / ions / salts / minerals;

under nervous control;

ref. homeostasis / determined by body temperature;

(sweat) glands / ducts;

sweat extracted from blood;

(from) skin;

transpiration from leaves / stems;

(helps to) bring ions up stem/xylem / to leaves/cells;

(helps to) bring water up stem/xylem / to leaves/cells;

consequence of stomata open for photosynthesis; [max 7]

[Total 10]

7 **(a)** Explain why the lungs are considered to be organs of excretion.

.....

.....

.....

.....

.....[3]

(b) Describe how the kidneys maintain blood at a constant concentration.

Explain the importance of this function.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[7]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7 **(a)** CO_2 / water;

from body **AW**;

waste product / toxic material;

respiration;

metabolic (process); [max 3]

(b) homeostasis / osmoregulation;

salts / ions / minerals / urea;

water;

from blood;

ref. reabsorption into blood;

in varying quantities / excess;

depending on food/water intake;

and on loss in sweating;

tissue fluid therefore at constant concentration;

water would otherwise enter or leave cells;

osmosis / diffusion;

effect on cell appearance (e.g. swell / burst / shrink);

affect on enzyme action / metabolism; [max 7]

[Total 10]

May/June 2013 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 Which structures are present in a root hair cell?

	nucleus	chloroplast
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = present

✗ = absent

May/June 2013 · Paper 1 · Variant 1 · Question 2

Correct answer: A

2 What is an example of active transport?

A movement of glucose molecules into the cells of the villi

B movement of glucose molecules down a concentration gradient

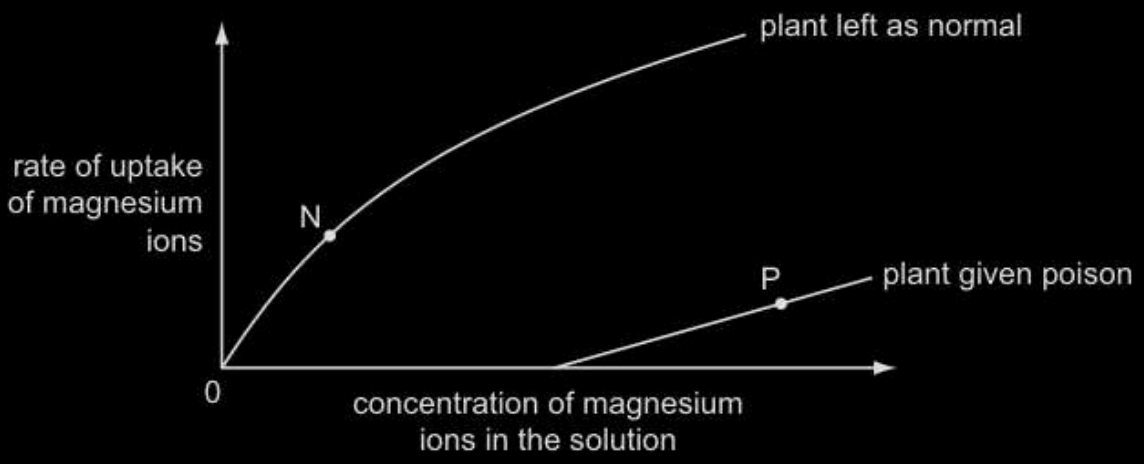
C movement of ions in blood plasma

D movement of water in the transpiration stream

May/June 2013 · Paper 1 · Variant 1 · Question 3

Correct answer: B

3 An experiment measured the rate at which plants take up magnesium ions from solution. One plant was given a poison that stops respiration. Another plant was left as normal. The graph shows the results.

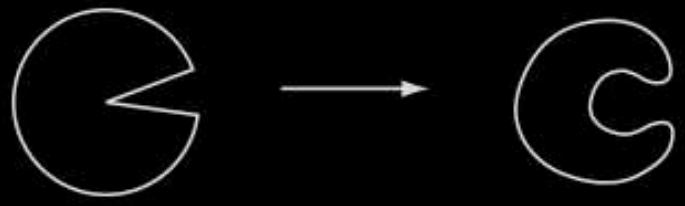


How are the magnesium ions being absorbed by the plants at points N and P?

	point N	point P
A	active transport	active transport
B	active transport	diffusion
C	diffusion	active transport
D	diffusion	diffusion

Correct answer: B

4 The diagram represents how an enzyme molecule changes in shape.



What explains this change?

A It has been cooled to 5 °C.

B It has been heated to 70 °C.

C It has been placed in a concentrated salt solution.

D It has been placed in a dilute salt solution.

May/June 2013 · Paper 1 · Variant 1 · Question 8

Correct answer: A

8 Which table states the tests for different food groups?

A

test	reagent	positive result
starch	iodine	blue-black
reducing sugars	Benedict's	orange
protein	biuret	purple
fats	ethanol	milky white

B

test	reagent	positive result
starch	biuret	blue-black
reducing sugars	Benedict's	orange
protein	iodine	purple
fats	ethanol	milky white

C

test	reagent	positive result
starch	iodine	blue-black
reducing sugars	Benedict's	purple
protein	biuret	milky white
fats	ethanol	orange

D

test	reagent	positive result
starch	biuret	blue-black
reducing sugars	Benedict's	purple
protein	iodine	milky white
fats	ethanol	orange

May/June 2013 · Paper 1 · Variant 1 · Question 9

Correct answer: A

9 What are the basic units from which molecules of glycogen and of protein are synthesised?

	basic units of glycogen	basic units of protein
A	glucose	amino acids
B	glucose	fatty acids
C	glycerol	amino acids
D	glycerol	fatty acids

May/June 2013 · Paper 1 · Variant 1 · Question 10

Correct answer: B

10 The table shows where different components of food are digested in the alimentary canal.

Which option states where **protein** is digested?

	mouth (buccal) cavity	stomach	duodenum	ileum
A	✓	✗	✓	✗
B	✗	✓	✓	✓
C	✗	✗	✓	✓
D	✗	✓	✗	✓

key

✓ = substance is digested

✗ = substance is not digested

May/June 2013 · Paper 1 · Variant 1 · Question 11

Correct answer: A

11 What contributes to the wilting of plant leaves?

A the mesophyll cells lose turgor

B the phloem stops translocating

C the stomata close

D the xylem fills with air

May/June 2013 · Paper 1 · Variant 1 · Question 13

Correct answer: C

13 After muscular exercise, which blood vessel carries the **lowest** concentration of carbon dioxide?

A hepatic vein

B pulmonary artery

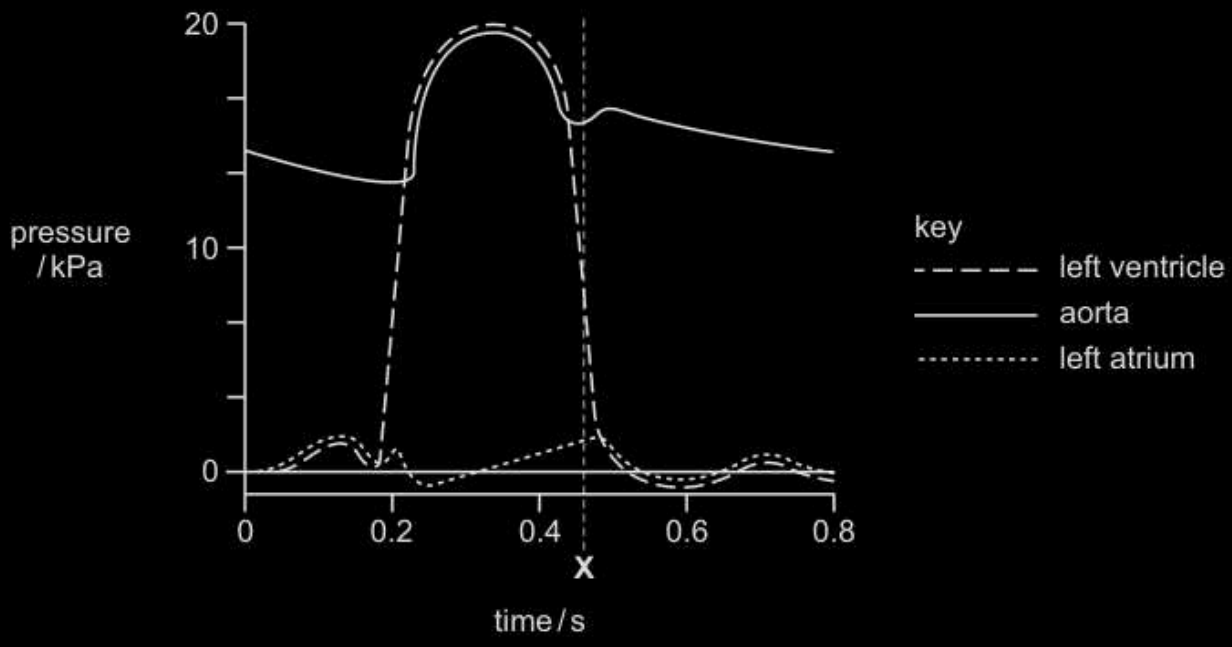
C pulmonary vein

D vena cava

May/June 2013 · Paper 1 · Variant 1 · Question 14

Correct answer: A

14 The diagram shows the pressures in the left side of the heart during one heart beat.



Which valves are open and which are closed at the time marked X?

	bicuspid	semi-lunar
A	closed	closed
B	closed	open
C	open	closed
D	open	open

May/June 2013 · Paper 1 · Variant 1 · Question 15

Correct answer: B

15 Two hours after eating a meal, which vessel contains blood with the highest concentration of glucose?

A aorta

B hepatic portal vein

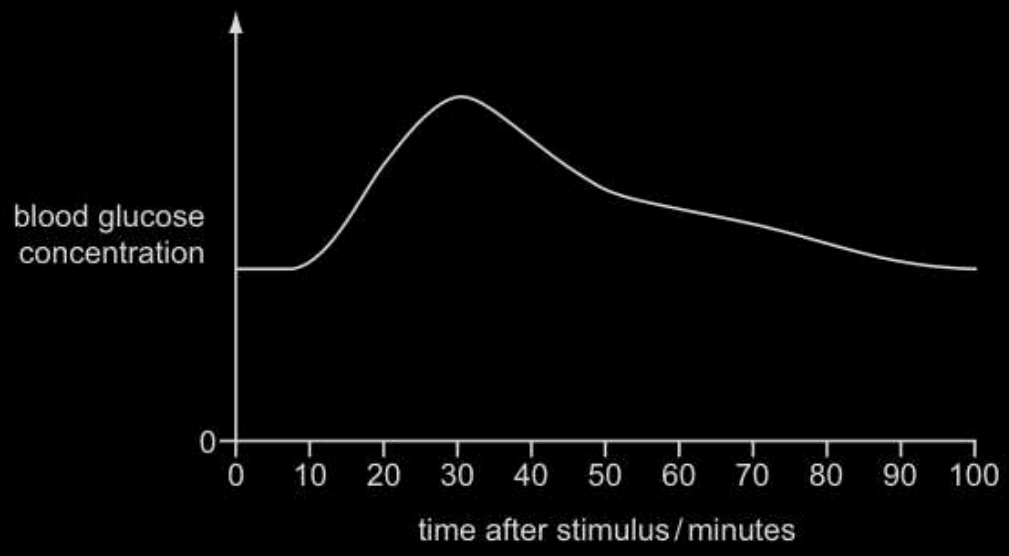
C pulmonary vein

D renal vein

May/June 2013 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 The graph shows changes in blood glucose concentration when a body responds to an external stimulus.



What is the hormone responsible for the change in blood glucose concentration in the first 30 minutes?

A adrenaline

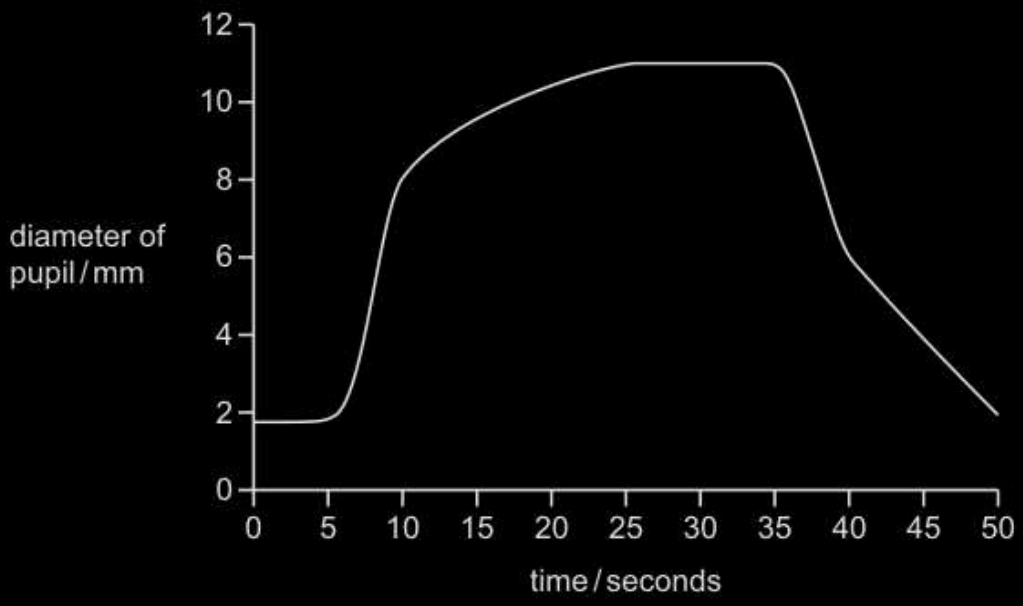
B fibrinogen

C glycogen

D insulin

Correct answer: D

22 The graph shows the changes in the size of the pupil of the eye as the light intensity of the surroundings is changed.



Between which times is the light intensity increasing?

A 5 to 10 seconds

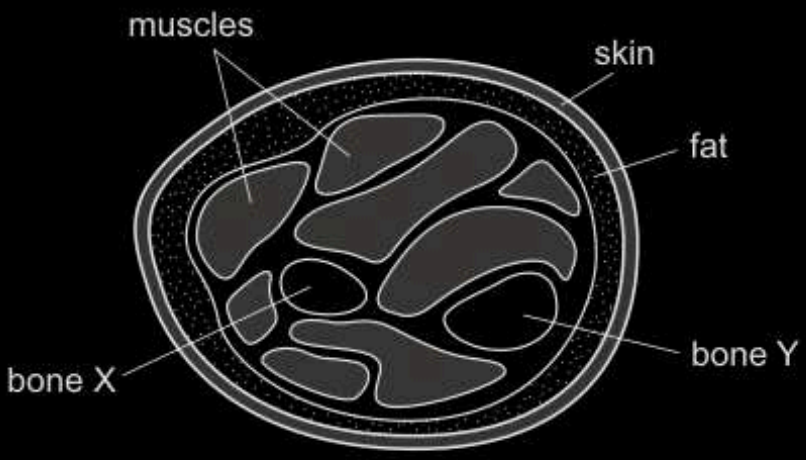
B 10 to 25 seconds

C 25 to 35 seconds

D 35 to 40 seconds

Correct answer: D

24 The diagram shows a section through the lower arm of a human.

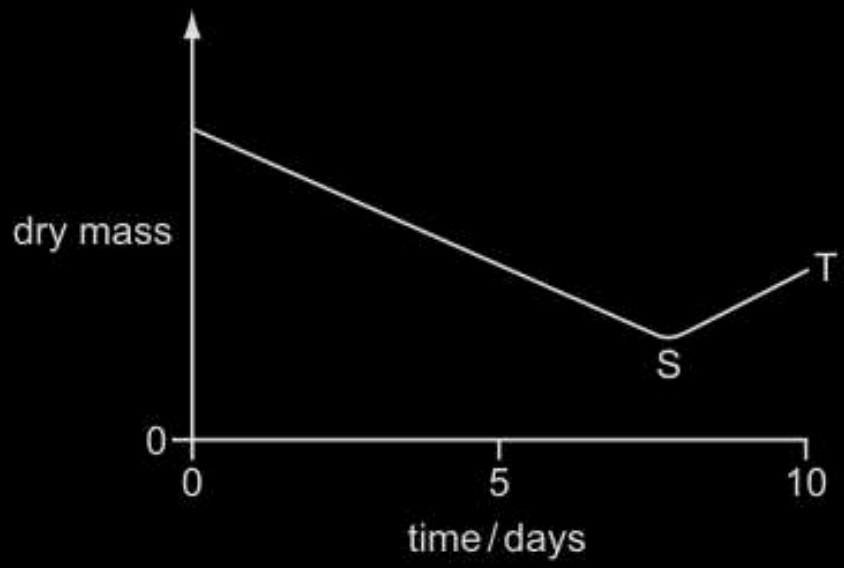


What are bones X and Y?

	X	Y
A	humerus	radius
B	humerus	ulna
C	radius	humerus
D	radius	ulna

Correct answer: B

33 The graph shows changes in the dry mass of a seed as it germinates.



What causes the change shown between points S and T?

A osmosis

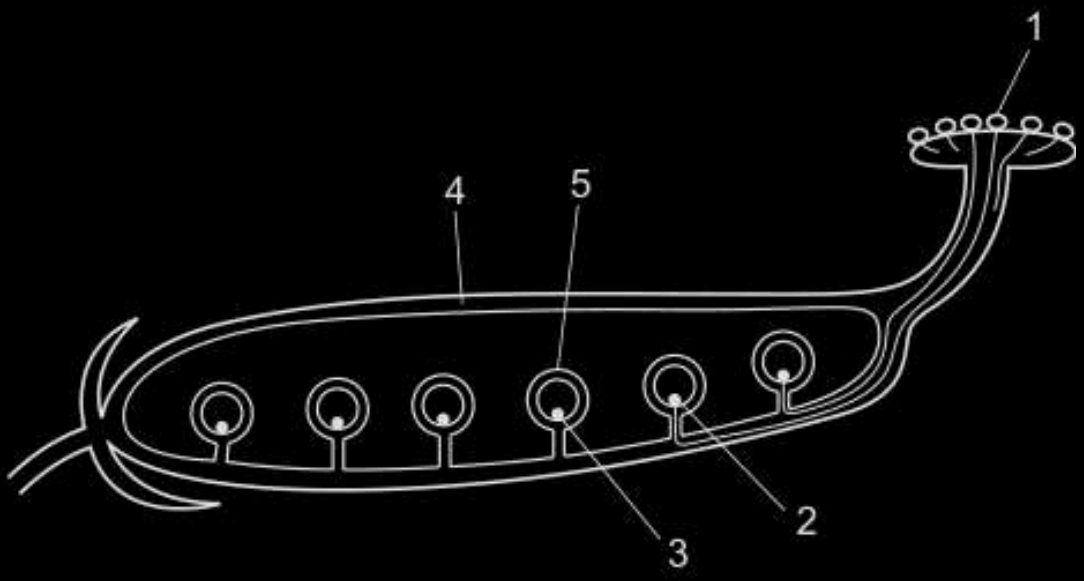
B photosynthesis

C respiration

D transpiration

Correct answer: B

34 The diagram shows part of a flower after it has been pollinated.



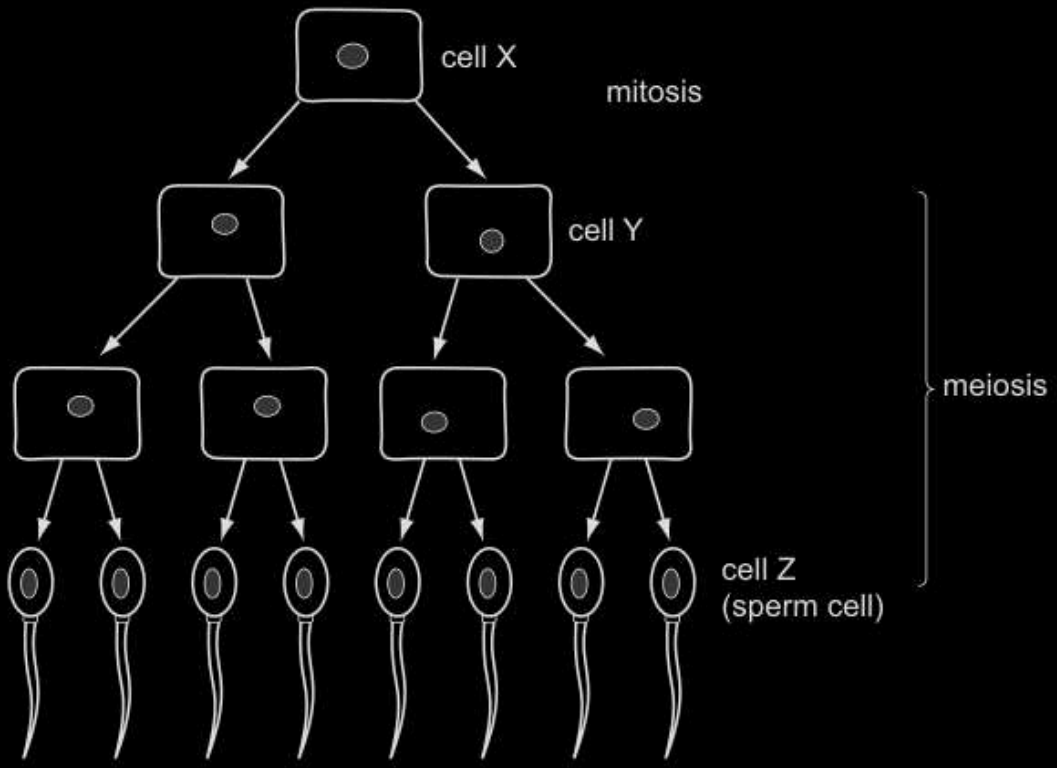
Which labelled structures are diploid and which are haploid?

	diploid	haploid
A	1	4
B	2	1
C	3	2
D	4	5

May/June 2013 · Paper 1 · Variant 1 · Question 36

Correct answer: C

36 The diagram shows some stages in cell division in a fruit fly.



Cell X contains 8 chromosomes.

How many chromosomes are in cell Y and in cell Z?

	cell Y	cell Z
A	4	4
B	4	8
C	8	4
D	8	8

May/June 2013 · Paper 1 · Variant 1 · Question 37

Correct answer: B

37 The genotype for the height of an organism is written as Tt.

What conclusion may be drawn?

A The allele for height has at least two different genes.

B There are at least two different alleles of the gene for height.

C There are two different genes for height, each having a single allele.

D There is one allele for height with two different forms.

May/June 2013 · Paper 1 · Variant 1 · Question 39

Correct answer: D

39 How many chromosomes are there in a zygote which develops into a baby with Down's syndrome?

A 23 **B** 24 **C** 46 **D** 47

May/June 2013 · Paper 1 · Variant 2 · Question 1

Correct answer: B

1 Which structures are present in a root hair cell?

	nucleus	chloroplast
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key
✓ = present
✗ = absent

May/June 2013 · Paper 1 · Variant 2 · Question 2

Correct answer: A

2 What is an example of active transport?

A movement of glucose molecules into the cells of the villi

B movement of glucose molecules down a concentration gradient

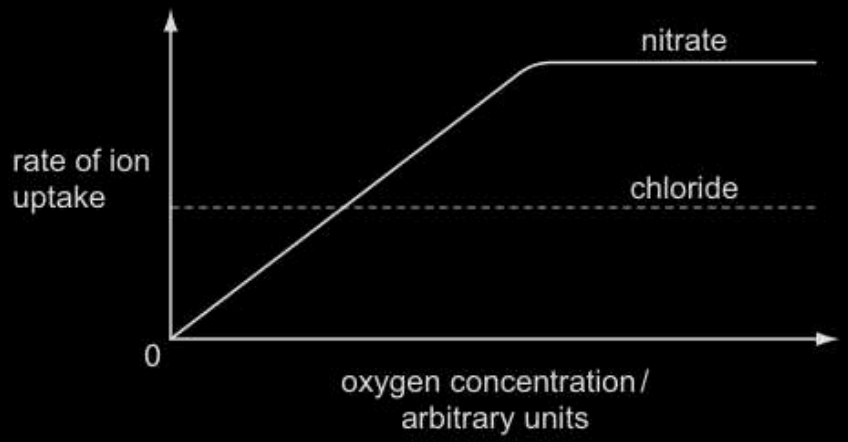
C movement of ions in blood plasma

D movement of water in the transpiration stream

Correct answer: C

3 The roots of a plant are placed in a dilute solution containing chloride and nitrate ions.

The graph shows how the rate of uptake of chloride and nitrate ions by the roots of the plant varies with oxygen concentration.



What can be concluded about how chloride and nitrate ions enter the roots?

	chloride	nitrate
A	active transport	active transport
B	active transport	diffusion
C	diffusion	active transport
D	diffusion	diffusion

May/June 2013 · Paper 1 · Variant 2 · Question 4

Correct answer: A

4 Four test tubes, each containing $2cm^3$ of amylase solution are treated as follows:

1 boiled, then cooled to 1 °C

2 boiled, then cooled to 25 °C

3 frozen, then warmed to 1 °C

4 frozen, then warmed to 25 °C

10 cm^3 of starch solution were then added to each tube and after 5 minutes, 2 drops of iodine solution were added to each tube.

Which row shows the results?

	1	2	3	4
A	black	black	black	yellow
B	black	yellow	black	yellow
C	yellow	black	yellow	black
D	yellow	yellow	yellow	black

May/June 2013 · Paper 1 · Variant 2 · Question 6

Correct answer: C

6 What describes the upper cuticle of a leaf?

A a single layer of cells containing many chloroplasts

B a single layer of transparent cells allowing light to enter the leaf

C a thin non-cellular layer preventing water loss from the leaf

D a permeable layer allowing water to enter the leaf

May/June 2013 · Paper 1 · Variant 2 · Question 8

Correct answer: A

8 Which classes of foods contain the elements shown?

	carbohydrates	fats	proteins
A	C,H,O	C,H,O	C,H,O and N
B	C,H,O	C,H,O and N	C,H,O
C	C,H,O	C,H,O and N	C,H,O and N
D	C,H,O and N	C,H,O	C,H,O

May/June 2013 · Paper 1 · Variant 2 · Question 9

Correct answer: B

9 A positive food test produces an orange precipitate.

Which test was performed?

A Amino acids were mixed with cold biuret solution.

B Glucose was heated with Benedict's solution.

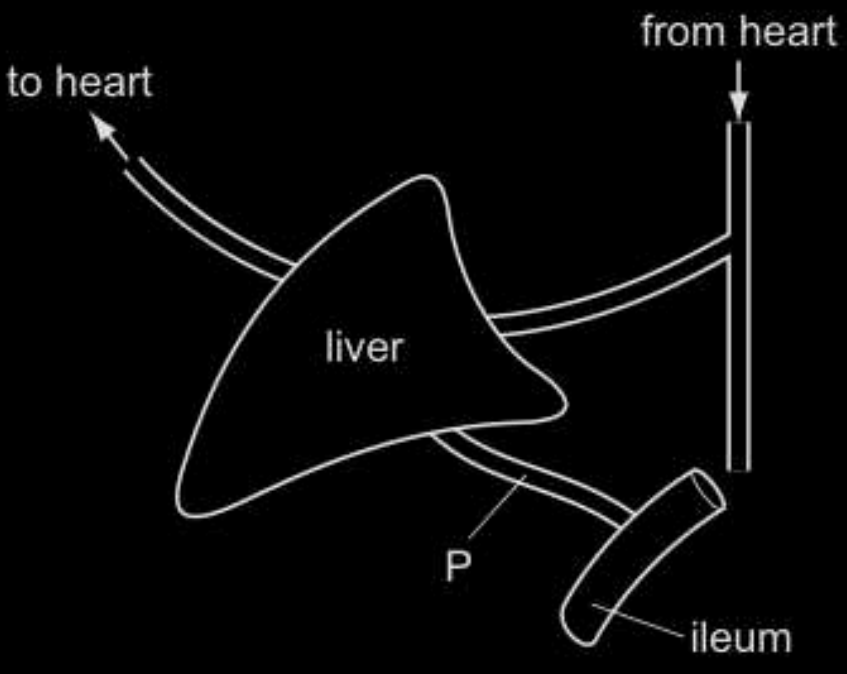
C Protein was heated with Benedict's solution.

D Protein was mixed with biuret solution and heated.

May/June 2013 · Paper 1 · Variant 2 · Question 10

Correct answer: B

10 The diagram represents the liver and some associated structures.



What does the blood in blood vessel P transport?

A bile to the ileum

B glucose to the liver

C glycogen to the liver

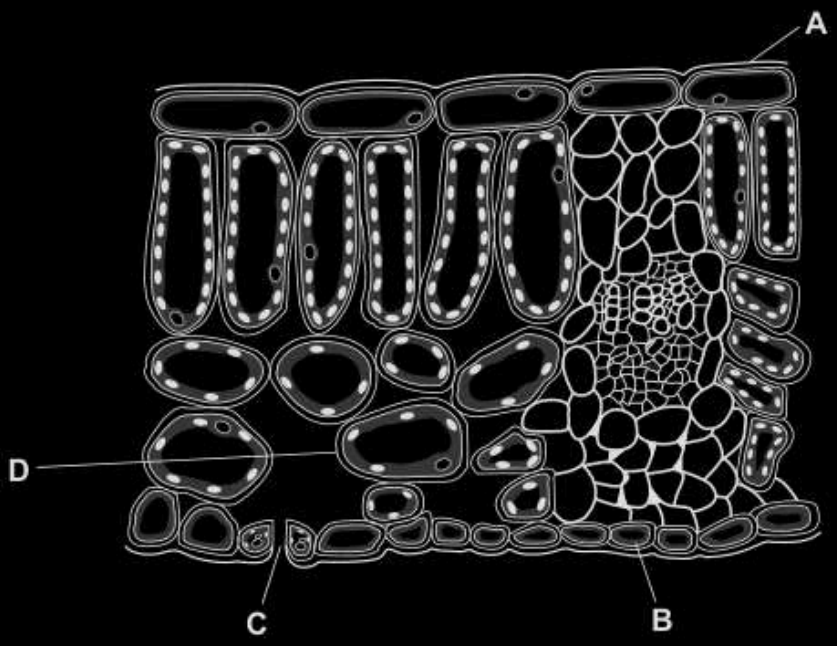
D urea to the ileum

May/June 2013 · Paper 1 · Variant 2 · Question 11

Correct answer: D

11 The diagram shows a cross-section of a leaf.

From which part does most of the water evaporate during transpiration?



May/June 2013 · Paper 1 · Variant 2 · Question 13

Correct answer: B

13 Compared to a vein, an artery has

A a thinner wall.

B a wall with more elastic tissue.

C a wider lumen.

D valves.

May/June 2013 · Paper 1 · Variant 2 · Question 14

Correct answer: C

14 Which row shows the blood vessels carrying blood to and from the organs listed?

	blood vessel carrying blood to the organ	organ	blood vessel carrying blood from the organ
A	aorta	heart	pulmonary vein
B	hepatic artery	liver	hepatic portal vein
C	pulmonary artery	lung	pulmonary vein
D	renal vein	kidney	renal artery

May/June 2013 · Paper 1 · Variant 2 · Question 15

Correct answer: B

15 Two hours after eating a meal, which vessel contains blood with the highest concentration of glucose?

A aorta

B hepatic portal vein

C pulmonary vein

D renal vein

May/June 2013 · Paper 1 · Variant 2 · Question 17

Correct answer: B

17 A new organism is discovered. It is found to be made of cells and contains DNA.

To which group of organisms could it belong to and to which group could it **not** belong?

	could be	could **not** be
A	bacteria	fungi
B	bacteria	viruses
C	fungi	bacteria
D	viruses	bacteria

May/June 2013 · Paper 1 · Variant 2 · Question 20

Correct answer: D

20 If the air temperature is higher than the core body temperature, which processes can increase heat loss from the body?

	shivering by muscles	evaporation of sweat	vasodilation in the skin
A	✓	✗	✓
B	✓	✗	✗
C	✗	✓	✓
D	✗	✓	✗

key

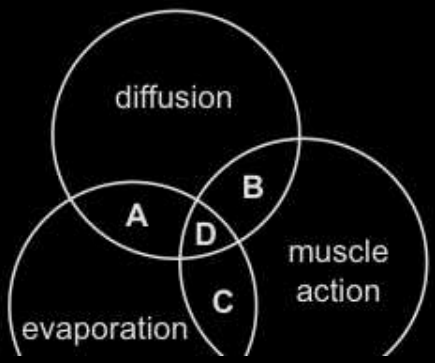
✓ = can increase heat loss

✗ = cannot increase heat loss

May/June 2013 · Paper 1 · Variant 2 · Question 22

Correct answer: B

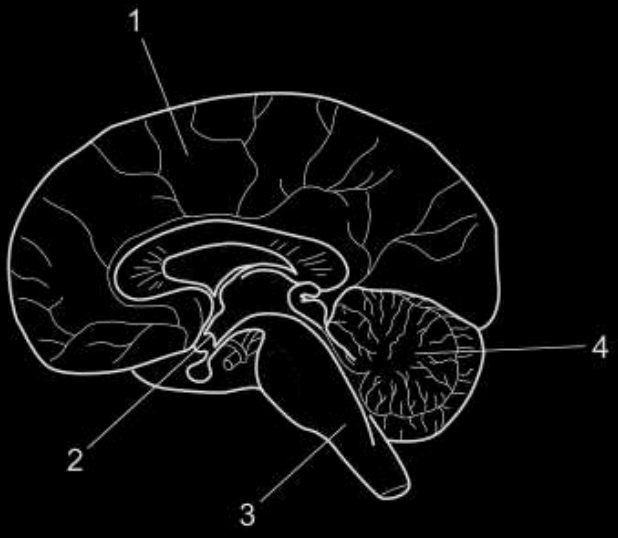
22 Which processes help the excretion of carbon dioxide from the lungs?



May/June 2013 · Paper 1 · Variant 2 · Question 23

Correct answer: B

23 The diagram represents a section through the human brain.



What are the numbered parts?

	cerebrum	cerebellum	medulla	hypothalamus
A	1	2	4	3
B	1	4	3	2
C	4	1	3	2
D	4	1	2	3

May/June 2013 · Paper 1 · Variant 2 · Question 24

Correct answer: D

24 Insulin is a hormone, synthesised in the pancreas and is distributed around the body by the blood.

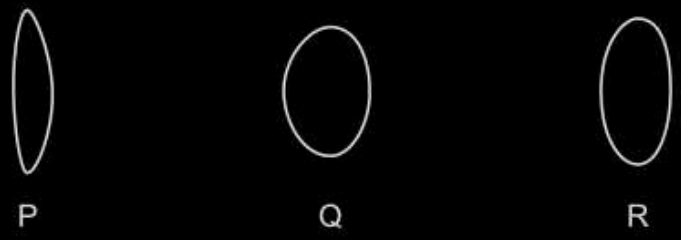
What describes its rate of secretion and its concentration in the blood?

	rate of secretion	concentration in the blood
A	constant	constant
B	constant	varied
C	varied	constant
D	varied	varied

May/June 2013 · Paper 1 · Variant 2 · Question 25

Correct answer: C

25 The diagram shows the appearance of the side view of the lens when it focuses on objects at different distances from the eye.



Which shows the appearance of the lens when a person focuses first on a near object and then on a distant object?

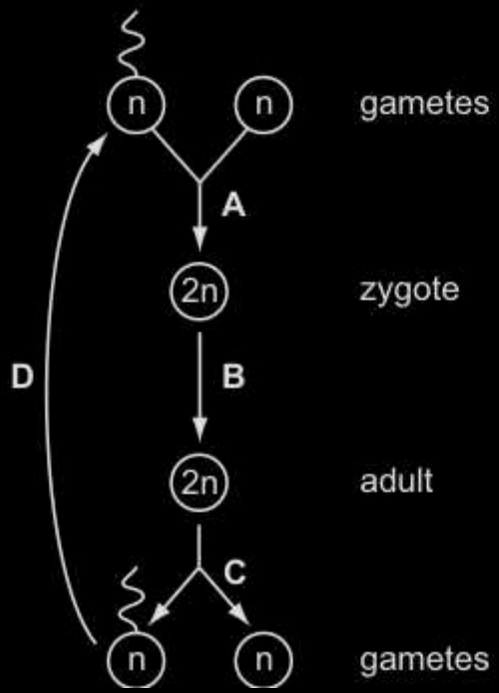
	near object	distant object
A	P	Q
B	P	R
C	Q	P
D	R	Q

May/June 2013 · Paper 1 · Variant 2 · Question 34

Correct answer: C

34 The diagram shows cells at different stages in the life cycle of an organism.

At which stage does meiosis occur?



May/June 2013 · Paper 1 · Variant 2 · Question 37

Correct answer: B

37 The genotype for the height of an organism is written as Tt.

What conclusion may be drawn?

A The allele for height has at least two different genes.

B There are at least two different alleles of the gene for height.

C There are two different genes for height, each having a single allele.

D There is one allele for height with two different forms.

May/June 2013 · Paper 1 · Variant 2 · Question 38

Correct answer: A

38 A scientist takes 4 pairs of samples from a wild cherry tree.

In which pair of samples could there be cells with different genotypes?

A two fruits

B two leaves

C two petals

D two root cuttings

May/June 2013 · Paper 1 · Variant 2 · Question 39

Correct answer: D

39 How many chromosomes are there in a zygote which develops into a baby with Down's syndrome?

A 23 **B** 24 **C** 46 **D** 47

1 **(a)** Table 1.1 lists the daily requirements for some of the components in the diet of a young child.

Table 1.1

diet component	daily requirement
energy	8 MJ
fat	50 g
protein	19 g
vitamin C	25 mg
vitamin D	0.005 mg
calcium	800 mg
iron	10 mg

State **two** components, other than those in Table 1.1, that are required in a balanced diet.

1. .....

2. .....

[2]

(b) Malnutrition is common in countries where there is famine.

List **two** problems that may contribute to famine.

1. .....

2. .....

[2]

(c) Fig. 1.1 shows the label from the packet of a type of food sometimes fed to young children suffering from severe malnutrition.

For Examiner's Use

Emergency Famine Food

92 g pack (provides 2.0 MJ)

Ready to use – does not require water or refrigeration

Contains: peanut paste, vegetable oil, powdered milk, powdered sugar, vitamins (including C and D), minerals (including iron and calcium)

Date of manufacture: June 2012 *Use within 2 years*

Fig. 1.1

(i) State **three** effects of malnutrition which may occur in young children.

1. .....

2. .....

3. .....

[3]

(ii) Explain how this emergency famine food helps to overcome the effects of malnutrition.

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 11]

1 (a)	Carbohydrate; Water; roughage / fibre; other named mineral / other named vitamin;	[max 2]	
(b)	**(Environmental):** Drought; Flooding; Infertile land e.g. poor soil structure; Natural disasters e.g. earthquakes/ very severe weather; **(Political):** Significant increase in population; unequal distribution / supply of food; poverty / economic breakdown; war / political considerations;	[max 2]	Accept rain if qualified to suggest that it is below the required level for crop growth e.g. inadeguate rainfall Ignore: tornadoes
(c) (i)	**Diseases:** Rickets; Scurvy; Kwashiorkor; Marasmus; beri-beri; **Symptoms:** stunted growth; low body weight /death; weak/brittle bones; lowers immunity; anaemia; bleeding membranes; fatigue; obesity;	[max 3]	**A** any other relevant descriptions of symptoms

© Cambridge International Examinations 2013

(ii)	Contains everything for a balanced diet / all nutrients in right proportions; Peanut / sugar / milk + energy / respiration; oil + correct use eg provide energy / synthesise cell membranes; peanut paste / powdered milk + protein; protein + for growth / new cells / new protoplasm / tissue repair / named protein / prevents kwashiorkor; vitamin C + wound healing / prevention of scurvy; vitamin D + bones / teeth / prevention of rickets; iron + for haemoglobin / blood / prevention of anaemia; calcium + bones / teeth / blood clotting / prevention of rickets;	[max 4]	
		[Total: 11]	

© Cambridge International Examinations 2013

For Examiner's Use

2 **(a)** Receptors receive stimuli and convert them into electrical impulses.

Fig. 2.1 shows the pathway taken by electrical impulses in a reflex action. Complete Fig. 2.1 by writing the name of the appropriate component on the dotted lines.

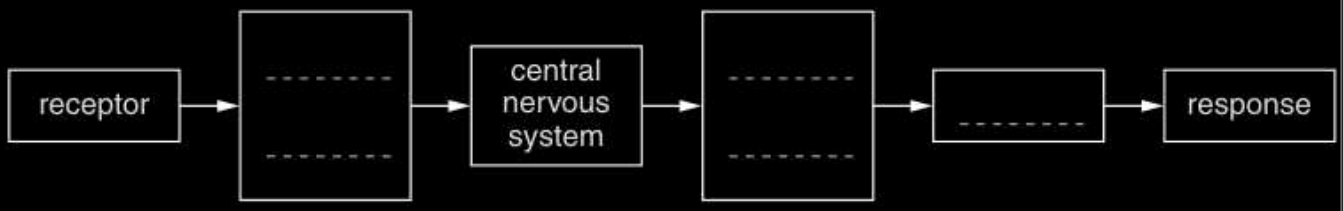


Fig. 2.1 [2]

(b) The brain is one part of the central nervous system. Fig. 2.2 is a diagram of the human brain.

Fig. 2.2

Damage to the brain can sometimes occur as the result of an accident.

Name the parts of the brain labelled **A** and **B** in Fig. 2.2 and suggest a problem that may be experienced by a person who has damage to that part of the brain.

part **A** .....

problem caused by damage

.....

.....

part **B** .....

problem caused by damage

.....

.....

[4]

(c) Scientists are able to treat people with some types of brain damage. They may do this by injecting the patient with cells taken from another person (donor). These cells then travel to the brain where they divide and specialise to become groups of fully functioning brain cells.

For Examiner's Use

(i) State the type of cell division that takes place when the cells that have been injected reach the patient's brain.

.....[1]

(ii) State the term used to describe a group of cells that are specialised to perform a specific function.

.....[1]

(d) Female patients were injected with cells from male donors. After a period of time, the scientists examined brain cells from these patients and looked for groups of brain cells containing the Y chromosome.

Explain why finding groups of brain cells containing the Y chromosome would suggest to the scientists that the treatment may have been successful.

.....

.....

.....

.....

.....

.....[3]

[Total: 11]

2 (a)	Box 1: sensory / afferent neurone Box 2: motor / efferent neurone /fibre Box 3: effector / muscle / gland / A named effector 3 correct = 2 marks , 1 or 2 correct = 1 mark	[2]	R nerve R unspecified organs
(b)	**(part A)** cerebrum / cerebral hemispheres / frontal lobes / fore brain; **(Problem)** loss of voluntary movement / speech difficulties / change in sensory perception / / learning difficulty / memory loss / change in emotional behaviour; **(part B)** Cerebellum / hind brain; **(Problem)** loss of control over movement / loss of learnt activity / loss of balance / coordination;	[max 4]	Require the effect of the damage. Need the idea of "loss"
(c) (i)	Mitosis;	[1]	
(ii)	Tissue;	[1]	
(d)	Y chromosome is only in males / female is XX; injected cells have travelled to brain; Replication / cell division has occurred / tissue has formed; Cells present after time period;	[max 3]	
		[Total: 11]	

© Cambridge International Examinations 2013

For Examiner's Use

3 **(a)** Fig. 3.1 shows how pollination takes place in two different species of plant, species **P** and species **Q**.

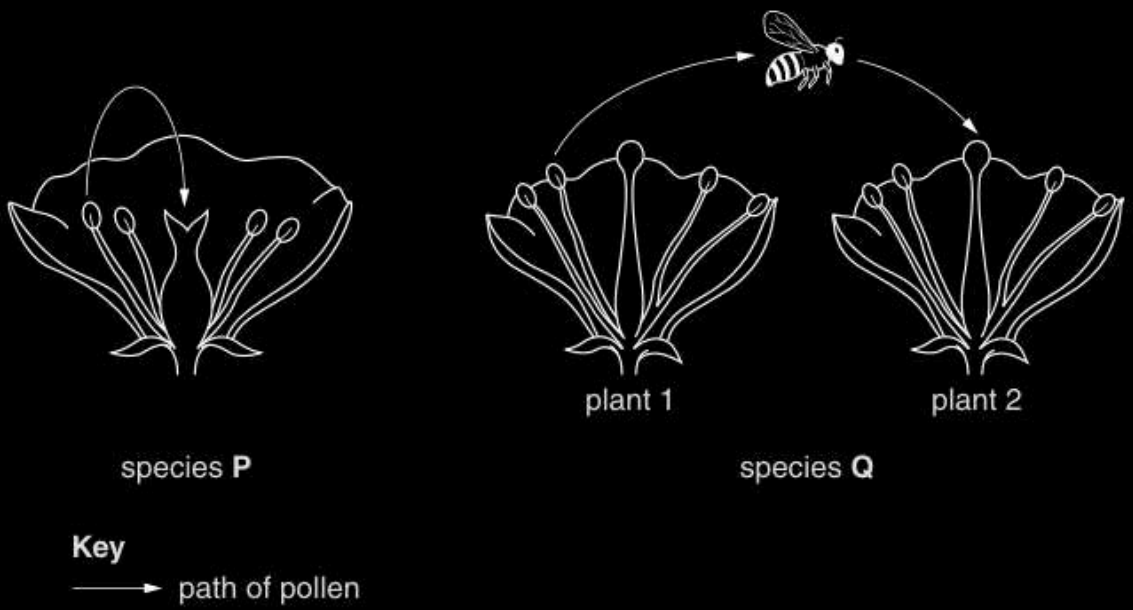


Fig. 3.1

(i) Using the information in Fig. 3.1, suggest and explain how each of these species of plant is pollinated.

species **P**

.....

.....

species **Q**

.....

.....

[4]

(ii) Suggest **two** ways in which a flower from a plant of species **Q** in Fig. 3.1 may be adapted to increase the likelihood of pollination taking place.

1. .....

2. .....

[2]

(b) State why species **Q** shows more variation in its phenotype than species **P**.

.....

.....[1]

(c) Describe the events that take place in a flower after pollination until fertilisation has taken place.

For Examiner's Use

.....

.....

.....

.....

.....

.....[3]

[Total: 10]

3	(a) (i)	**(species P)** self / insect / wind ; (pollen) from anther to stigma*; of same flower; **(species Q)** Cross; (by) insect (pollination); (pollen) from anther to stigma*; of different plant;	[max 4]	* accept once only in either P or Q Accept plausibly named insect eg bee, wasp
	(ii)	brightly coloured petals; large petals; scent; nectar; sticky pollen or stigma;	[max 2]	
	(b)	Cross pollination / has 2 parents / ref genes / alleles;	[1]	Accept "exchange of traits"
	(c)	pollen tube; grows / digests + down style; to ovule / embryo sac / micropyle; male + female gametes /egg / nuclei fuse;	[max 3]	
			[Total: 10]	

© Cambridge International Examinations 2013

5 **(a)** Define the term *gene*.

For Examiner's Use

.....

.....

..... [2]

(b) A gene mutation occurs when part of the DNA on a single chromosome is changed. State **two** factors that may increase the rate of gene mutation.

1. .....

2. .....

[2]

(c) Sickle cell anaemia is caused by a gene mutation. Fig. 5.1 shows red blood cells from a healthy person and from a person with sickle cell anaemia.

For Examiner's Use

- biconcave shape
- elastic
- non-sticky
- lifespan of 120 days

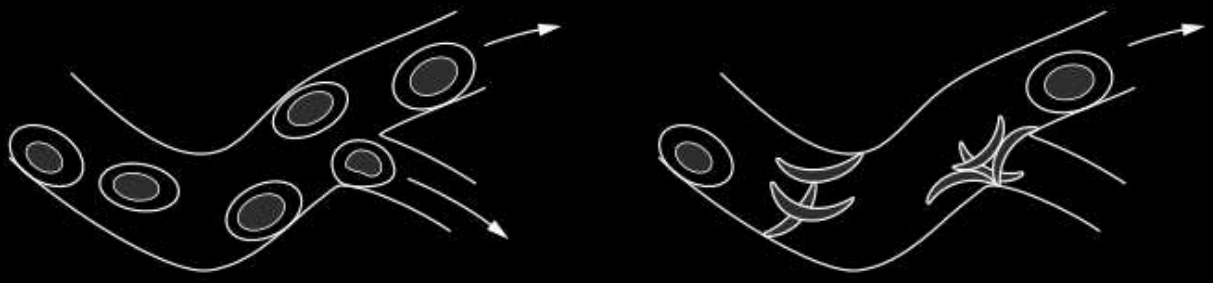
healthy person

- crescent shape
- non-elastic
- sticky
- lifespan of 10 to 20 days

person with sickle cell anaemia

Fig. 5.1

Fig. 5.2 shows the flow of red blood cells through a blood vessel in a healthy person and in a person with sickle cell anaemia.



healthy person

person with sickle cell anaemia

Fig. 5.2

Use information from Fig. 5.1 and Fig. 5.2 to suggest why the transport of oxygen to tissues is reduced in a person with sickle cell anaemia.

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 8]

Section B

Answer **both** questions in this section.

Write your answers in the spaces provided.

For Examiner's Use

5	(a)	unit of inheritance / give characteristics / traits; section of DNA; can be copied; (each) specifies production of one A/W protein / polypeptide;	[max 2]	Accept contains hereditary information
	(b)	radiation; chemicals; faulty cell division;	[max 2]	
	(c)	reduction in surface area / volume ; so less haemoglobin; *less AW oxygen + carriage / absorption / in blood (cell);* loss of elasticity + more difficult to move through blood vessels; sticky + may clump together / clot; (causing) blockage of blood vessel; reduction of blood flow (to tissue);	[max 4]	
			[Total: 8]	

© Cambridge International Examinations 2013

7 **(a)** Outline factors, other than alcohol, that may increase a person's risk of developing coronary heart disease.

For Examiner's Use

.....

.....

.....

.....

.....

.....

.....

.....[4]

(b) Describe the effects of excessive consumption of alcohol on the individual and society as a whole.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 10]

Section C

For Examiner's Use

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7	(a)	ref to high fat diet; obesity / overweight ; high blood pressure; ref to cholesterol / saturated fatty acids / animal fats + in smoking; lack of exercise; genetic / inherited / family history (of heart disease); age / time factor; stress; diabetes;
	(b)	**Effect on individual:** depressant / lowers blood pressure; reduced reaction times/ poor muscular co-ordination AW ; risk taking / delusions A/W; liver damage; tolerance to alcohol / increased dosage required; addiction / dependence; cost (to individual) financial / social; **Effect on Society :** anti social behaviour towards family A/W or example; anti social behaviour towards community A/W or examp financial cost to community; ref to drink driving;

© Cambridge Internatic

n diet / blood;	[max 4]	
ple;	[max 6]	Accept inability to take decisions AW A refs to smell / poor personal hygiene , low self esteem, lose friends etc. e.g. reduced income / physical harm e.g. stealing / wanton damage etc e.g. cost of damage / "welfare" for family / "rehabilitation cost of the individual"
	[Total: 10]	

May/June 2013 · Paper 2 · Variant 1 · Question 8

8 **(a)** Describe the function of the hepatic portal vein.

.....

.....

.....

.....

.....

.....[4]

(b) With reference to **named** molecules, describe the functions of the liver.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 10]

8 (a)	carries / transports + food molecules / named example; dissolved / in solution / digested; from small intestine; to liver; in blood;	[max 4]	
(b)	breakdown + amino acids / deamination; into urea; and a carbohydrate; (converts) glucose + to glycogen (A. reverse); ref. storage (of glycogen); ref. drugs / toxins / alcohol; ref. hormones (A. named hormone); ref. storage of iron / vitamins or named; breaks down haemoglobin / red blood cells; synthesises bile; Produces (plasma)proteins;	[max 6]	
		[Total: 10]	

© Cambridge International Examinations 2013

1 Fig. 1.1 shows a vertical section through a pair of guard cells and some other cells on the lower surface of a leaf.

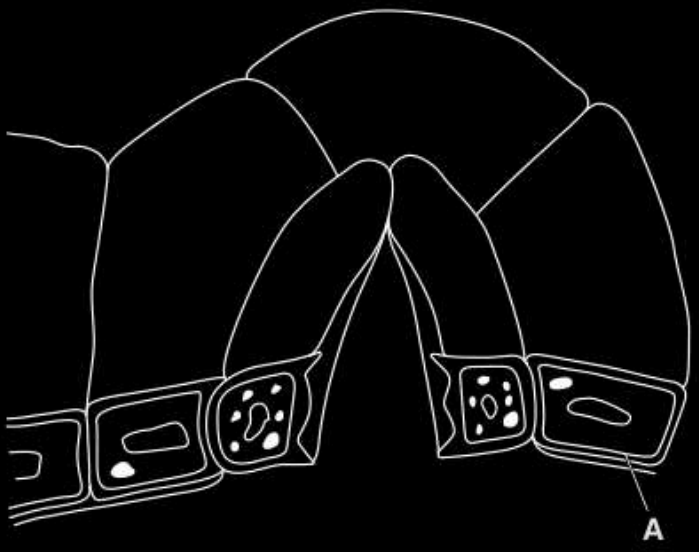


Fig. 1.1

(a) (i) Name the type of cell labelled **A** on Fig. 1.1. ..... [1]

(ii) Draw an arrow on Fig. 1.1 to show the position of the stoma. [1]

(b) Fig. 1.1 shows the guard cells as they appear at 1300 hours. In the space below, draw a diagram to show the guard cells, **in surface view**, as they would appear at 0100 hours.

On your diagram, draw and label the structural features of **one** of the guard cells.

[4]

(c) Explain the advantage to the plant of the difference in the guard cells at 1300 hours and at 0100 hours.

For Examiner's Use

1300 hours .....

.....

.....

0100 hours .....

.....

.....[5]

[Total: 11]

2 In 1822, a man, Alexis Bidagan, suffered an injury from a gun fired at close range. The injury was in the form of a hole about 10 cm in diameter, penetrating both his chest and stomach walls, below his diaphragm. When the wound healed, the edge of the hole in his stomach sealed itself with the edge of the hole in his chest wall. Fig. 2.1 shows the position of the opening that remained to Alexis's stomach until he died 58 years later.

For Examiner's Use

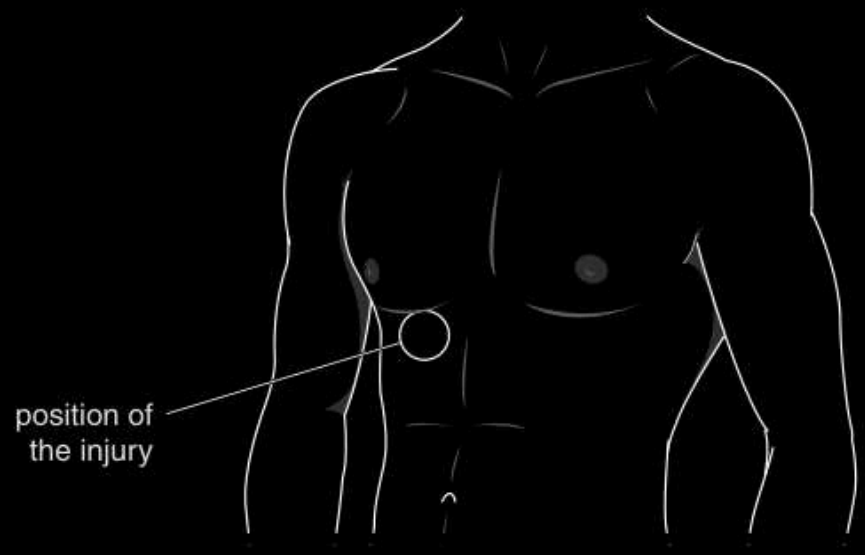


Fig. 2.1

(a) Name **two** organs, other than the stomach, that would have been exposed to infection through the hole **before** the wound healed.

1 .....

1 (a) (i)	epidermal / epidermis;	1	
(ii)	arrow shown clearly pointing to / or passing through stoma;	1	A arrow head on either end
(b)	stoma shown clearly more closed than in Fig. 1.1;	1	R any view other than surface view
	any 3 correctly identified and labelled features from: nucleus; vacuole / cell sap; cytoplasm; chloroplast; cell wall; cell membrane; vacuolar membrane / tonoplast;	Max 3	Ignore mitochondrion / ribosome

© Cambridge International Examinations 2013

(c)	Correct ref. to photosynthesis anywhere in (c); (1300 hrs) *allows CO_2 to enter; *allows O_2 to be released; (* allow ONE for ref. gaseous exchange); ref. to water loss / transpiration + cooling / bringing water or ions or minerals (to the leaf) / from the soil; (0100 hrs) prevents / reduces / stops + transpiration / loss of water; correct ref. wilting; O_2 entry + for respiration;	Max 5	R water loss for temperature regulation / control
		[Total: 11]	

© Cambridge International Examinations 2013

For Examiner's Use

4 Fig. 4.1 shows part of the circulatory system, and **some** of the structures associated with organ **B**.

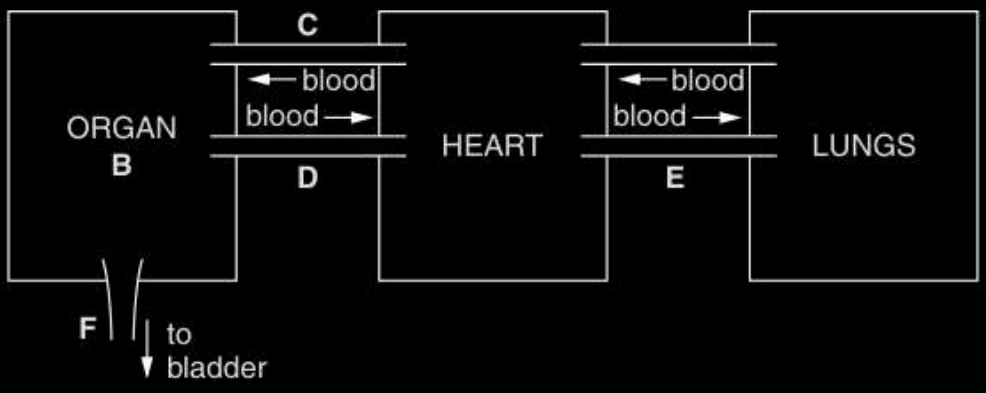


Fig. 4.1

(a) Name organ **B** in Fig. 4.1 ..... [1]

(b) **(i)** Name blood vessels **C** and **E** in Fig. 4.1.

C .....

E ..... [2]

(ii) Name the chambers of the heart, in the order in which blood passes through them from **D** to **E** in Fig. 4.1.

.... [2]

(c) Complete Table 4.1 to show **four** differences between the contents of **F** and the blood vessel, **C**, in a healthy person.

Table 4.1

difference	C	F
1		
2		
3		
4		

[4]

[Total: 9]

4 (a)	kidney;	1	
(b) (i)	C – renal artery / aorta; E – pulmonary artery;	2	No e.c.f. in this instance
(ii)	right atrium/auricle; right ventricle;	2	
(c)	**C** / **F** blood + urine; (a named) cells / platelets / plasma + no cells / platelets / plasma; protein/antibodies / amino acids / fats + none; lower urea concentration / higher urea concentration; glucose + no glucose; fewer salts / ions / less water / more / salts or ions / water; more hormones / vitamins / fewer hormones / vitamins;	Max 4	Ignore refs. to O_2 / CO_2 waste products Ignore minerals
		[Total: 9]	

© Cambridge International Examinations 2013

7 **(a)** List the components of a balanced diet. State **one** use of each component.

For Examiner's Use

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

(b) Explain how this diet should be modified for each of the following people:

(i) a person with diabetes

.....

.....

.....

.....[2]

(ii) a person with a history of heart problems.

.....

.....

.....

.....[2]

[Total: 10]

Section C

For Examiner's Use

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7 (a)	1. protein + for growth / repair / production of protoplasm or antibodies or enzymes or hormones; 2. carbohydrates (ignore names) + for energy; 3. fats + for energy / insulation / solvent (e.g. for some vitamins); 4. named mineral / ion + function*; 5. named vitamin + function*; 6. fibre / roughage + effective digestive transit AW; 7. water + solvent / other correct use;	Max 6	Ignore refs. energy do not penalise for refs to energy production *Disallow if function is incorrect for named component.
(b) (i)	(diabetic) reduced carbohydrate / sugar or named ; digestion / breakdown to glucose; lack of insulin / cells do not take up glucose / no glucose to glycogen; high blood sugar / glucose;	Max 2	Ignore ref. fats
(ii)	(heart patient) reduced fat; animal/saturated (fat); deposition on / in artery / atheroma / atherosclerosis; of heart / coronary; increased blood pressure;	Max 2	R no fat Ignore refs. to cholesterol Deposition may be of cholesterol Ignore refs. to heart attack
		[Total: 10]	

© Cambridge International Examinations 2013

8 **(a)** Define the term *hormone*.

.....

.....

.....

.....[3]

(b) State where male hormones and female hormones are produced in mammals and describe their roles in reproduction.

male hormones .....

.....

.....

.....

.....

female hormones .....

.....

.....

.....

.....

.....

.....

.....

.....

.....[7]

[Total: 10]

8 (a)	a chemical ; released into / carried by the blood; to affect a target organ; destroyed in the liver;	Max 3	
(b)	1. (male)testosterone androgens + testes; 2. for sperm / male gamete production; 3. 2ndry sexual characteristics (or one named); 4. (female) oestrogen + ovary; 5. development / release of an ovum / egg ; 6. 2ndry sexual characteristics (or one named); 7. repairs uterus lining / inhibits production of FSH; 8. progesterone + produced in ovary/ corpus luteum / placenta; 9. maintenance of uterus lining; 10. inhibition of ovulation AW; 11. LH / luteinising hormone + from pituitary; 12. triggers ovulation / development of corpus luteum; 13. FSH / follicle stimulating hormone + from pituitary;	Max 7	(male) Max 2 A production Function must be linked to correct hormone Ignore refs. uterine wall (x2) (female) Max 5
		[Total: 10]	

© Cambridge International Examinations 2013

9 **(a)** Describe the structure and functions of capillaries in the circulatory system.

For Examiner's Use

.....

.....

.....

.....

.....

.....

.....

.....

.....[5]

(b) Describe the functions of each of the following:

(i) white blood cells

.....

.....

.....

.....

.....

.....[3]

(ii) platelets.

.....

.....

.....

.....[2]

[Total: 10]

9 (a)	1. join arteries to veins; 2. walls + thin / one-cell thick / elastic; 3. allow passage of (tissue) fluid / plasma / permeable; 4. microscopic / pass easily between cells / large surface area / narrow lumen; 5. pressure reduction (along capillary); 6. ref. diffusion; 7. to / from + cells / tissues; 8. **any 2** of the following: (may be carried, passed in / out) glucose, amino acids, oxygen, CO_2 , hormones, urea, ions / salts,	Max 5	(Ignore 'They are one cell thick')
(b) (i)	(WBCs) phagocytes / phagocytosis or described ; antibodies / antitoxins; ref. bacteria / viruses / dead cells / pathogens / microorganisms / microbes; immune response / rejection AW;	Max 3	Ignore germs A ref. immune system / immunity
(ii)	(platelets) plug damaged vessels; fibrinogen; to fibrin; clotting; ref. antithrombin / prothrombin / thrombin / thromboplastin / thrombokinase;	Max 2	R fibres
		[Total: 10]	

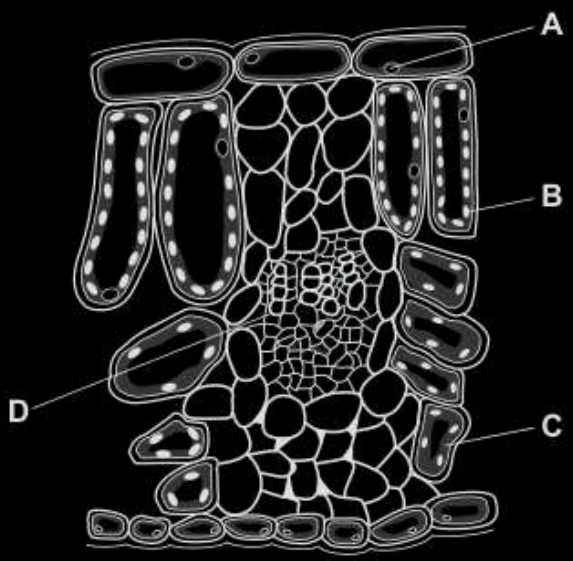
© Cambridge International Examinations 2013

October/November 2012 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 The diagram shows cells from a plant leaf.

Which structure contains a high concentration of magnesium?



October/November 2012 · Paper 1 · Variant 1 · Question 2

Correct answer: B

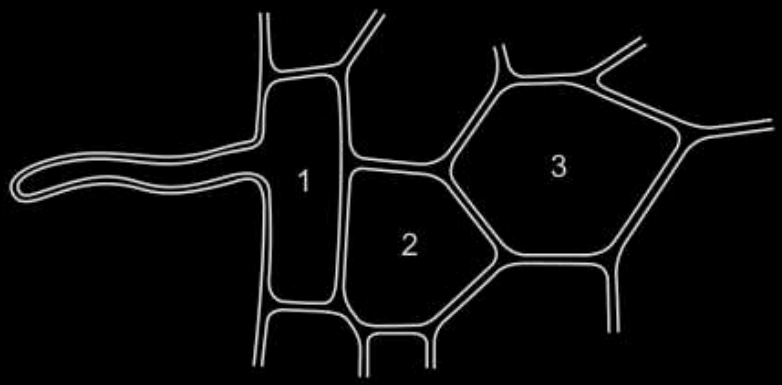
2 Which processes are responsible for the uptake of ions from the soil by a plant and the uptake of glucose into the villi of a human?

	uptake of ions into root hairs	uptake of glucose into the villi
A	active transport	osmosis
B	active transport	active transport
C	diffusion	osmosis
D	osmosis	active transport

October/November 2012 · Paper 1 · Variant 1 · Question 3

Correct answer: C

3 The diagram shows some cells in the root of a plant that is absorbing water from the soil.



How does the water potential of the cell marked 2 differ from the water potentials of the cells marked 1 and 3?

A higher than cell 1 and cell 3

B higher than cell 1 and lower than cell 3

C lower than cell 1 and higher than cell 3

D lower than cell 1 and lower than cell 3

October/November 2012 · Paper 1 · Variant 1 · Question 4

Correct answer: D

4 According to the lock and key hypothesis, which is the lock and which is the key for the enzyme lipase?

	key	lock
A	fatty acids	lipids
B	lipase	lipids
C	lipase	fatty acids
D	lipids	lipase

October/November 2012 · Paper 1 · Variant 1 · Question 8

Correct answer: C

8 A person tries eating a diet consisting only of lettuce leaves and water.

Which condition might develop?

A constipation

B heart disease

C rickets

D scurvy

October/November 2012 · Paper 1 · Variant 1 · Question 9

Correct answer: C

9 The table shows changes in the concentrations of blood components as the blood flows through an organ.

blood component	change in concentration
carbon dioxide	increased
glucose	increased
oxygen	reduced
urea	increased

Which organ has the blood passed through?

A brain

B kidney

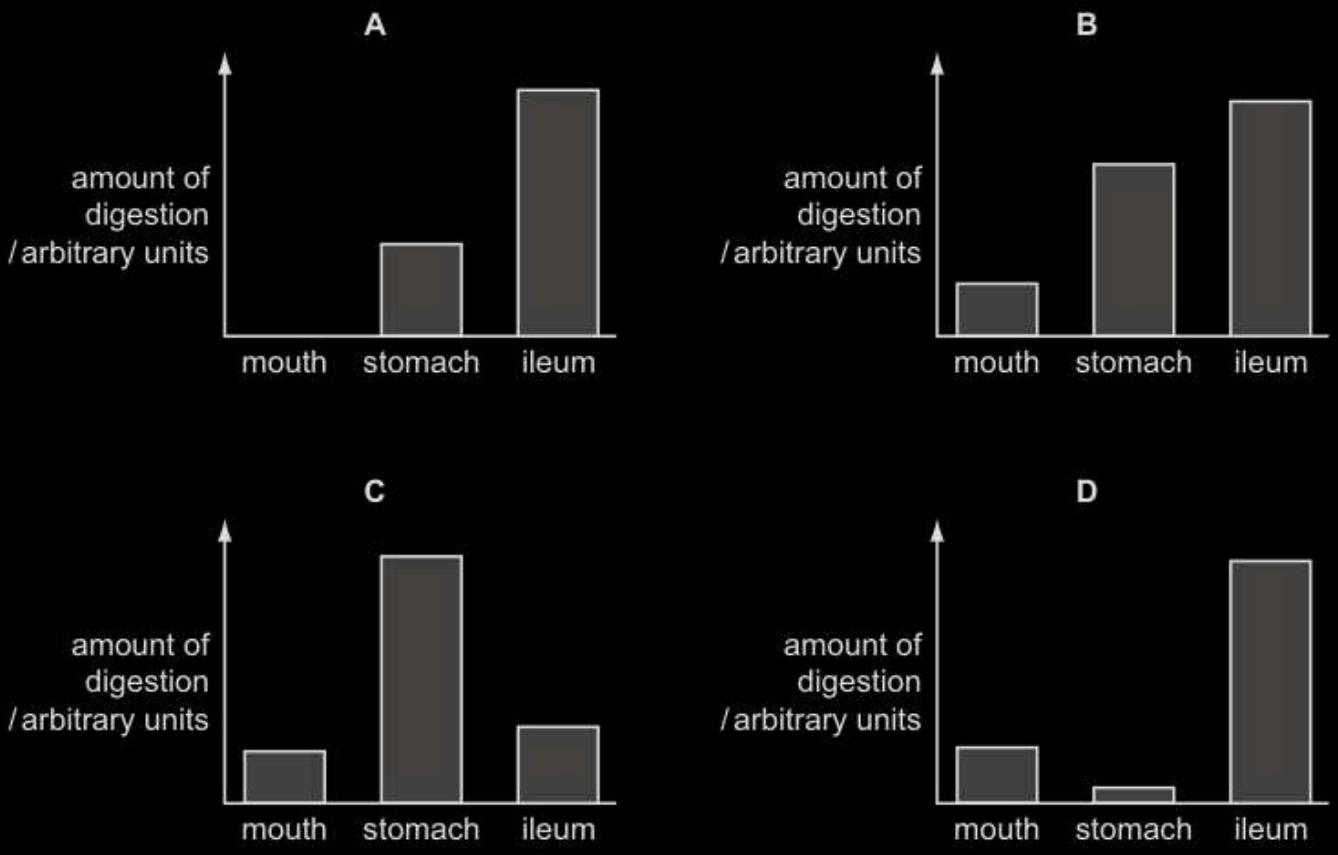
C liver

D stomach

October/November 2012 · Paper 1 · Variant 1 · Question 10

Correct answer: D

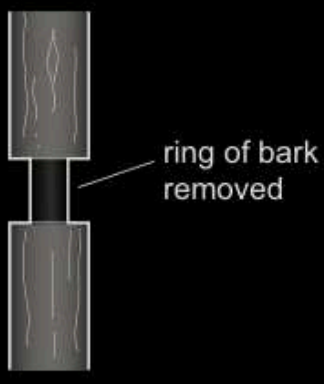
10 Which bar chart represents the amount of **starch** digested in the mouth, stomach and ileum of a human?



October/November 2012 · Paper 1 · Variant 1 · Question 11

Correct answer: B

11 The diagram shows a tree trunk, with a ring of bark, which includes the phloem, removed.



The tree will eventually die because this action cuts off the supply of

A mineral salts to the leaves.

B organic nutrients to the roots.

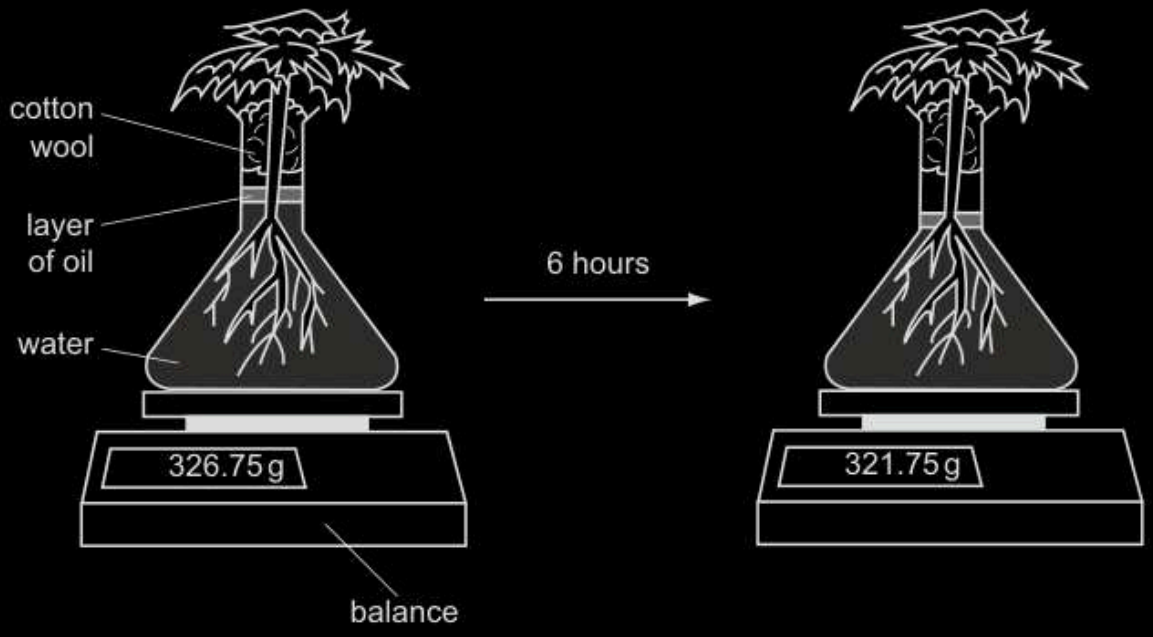
C oxygen to the roots.

D water to the leaves.

October/November 2012 · Paper 1 · Variant 1 · Question 12

Correct answer: D

12 The diagrams show a plant in a flask of water. It is left in the light at 16 °C for six hours.



What explains the change in mass after six hours?

A absorption of water into the root hairs

B evaporation of water from the flask

C photosynthesis in the leaves of the plant

D transpiration from the leaves of the plant

October/November 2012 · Paper 1 · Variant 1 · Question 13

Correct answer: B

13 In the human circulatory system, what causes the transfer of materials from the capillaries to the tissue fluid?

A active transport

B blood pressure

C capillarity

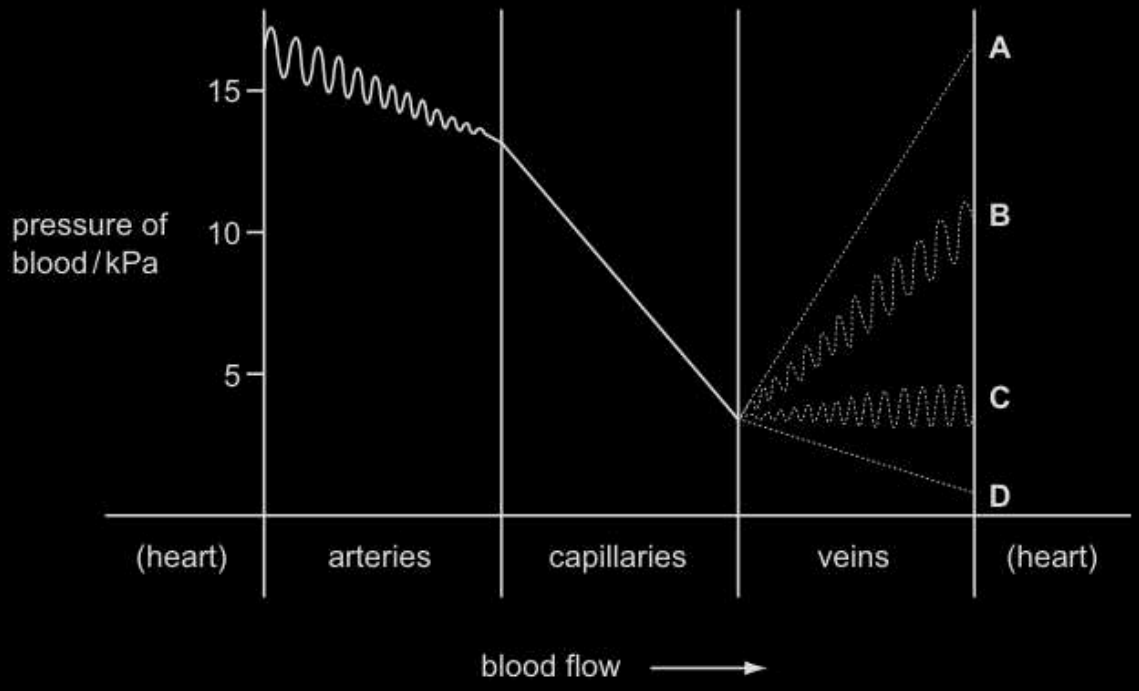
D osmosis

October/November 2012 · Paper 1 · Variant 1 · Question 14

Correct answer: D

14 The diagram shows the blood pressure of a person at rest as the blood leaves the heart and passes through arteries and then capillaries.

Which line shows the pressure of blood as it flows through veins before returning to the heart?

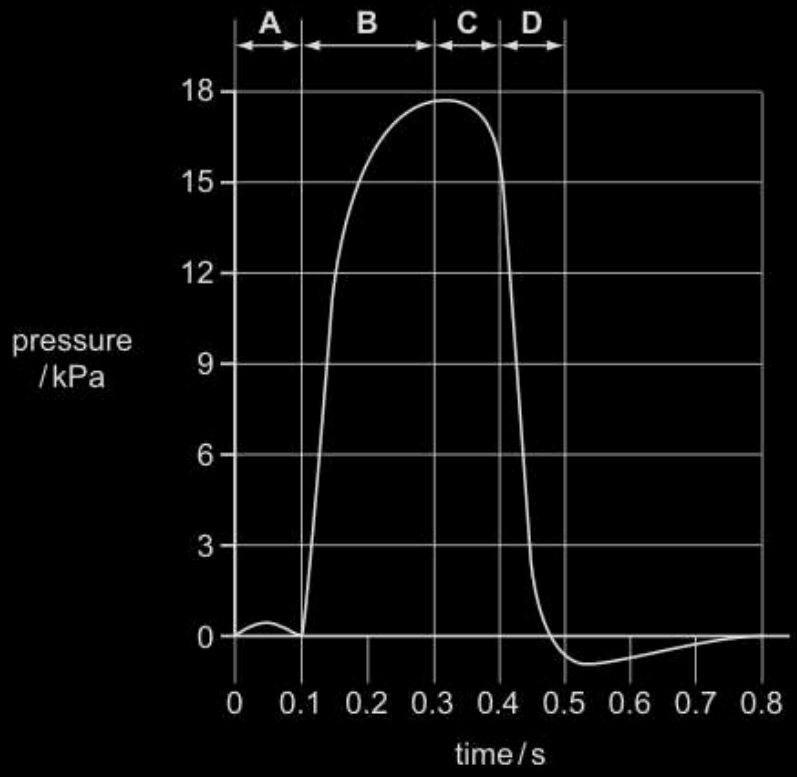


October/November 2012 · Paper 1 · Variant 1 · Question 15

Correct answer: A

15 The graph shows changes in the blood pressure in the left ventricle of the heart.

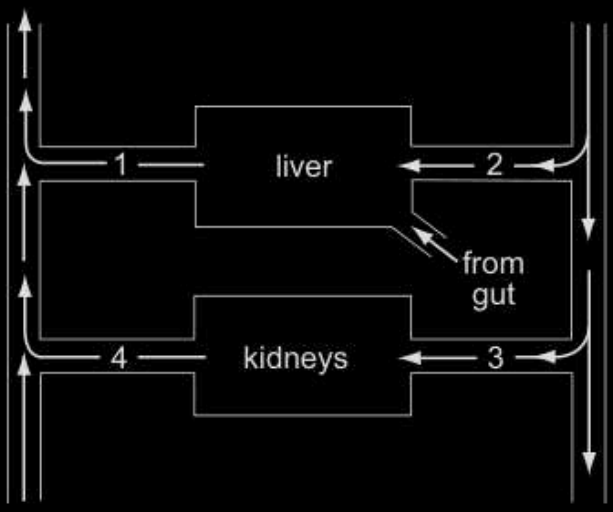
During which period is the left atrium contracting?



October/November 2012 · Paper 1 · Variant 1 · Question 20

Correct answer: B

20 The diagram represents the blood supply to the liver and to the kidneys.



Which vessels contain blood with the highest and lowest concentrations of urea?

	highest	lowest
A	1	2
B	1	4
C	3	2
D	3	4

October/November 2012 · Paper 1 · Variant 1 · Question 22

Correct answer: C

22 What happens in the eye when a person walks from a dark room into sunlight?

	circular muscles of the iris	radial muscles of the iris	pupil size
A	relax	contract	decreases
B	relax	contract	increases
C	contract	relax	decreases
D	contract	relax	increases

October/November 2012 · Paper 1 · Variant 1 · Question 23

Correct answer: B

23 The diagram shows a section of the brain.

What could be a result of damage to the part labelled X?

A difficulty in breathing

B inability to control body temperature

C may cause rickets

D raised insulin levels

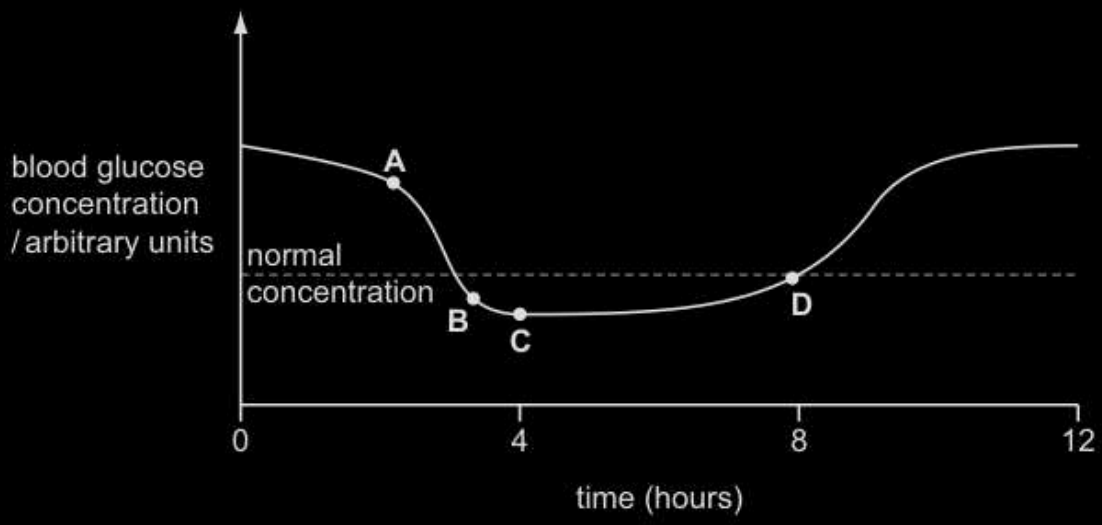
October/November 2012 · Paper 1 · Variant 1 · Question 24

Correct answer: A

24 A person with diabetes mellitus is receiving treatment with insulin injections.

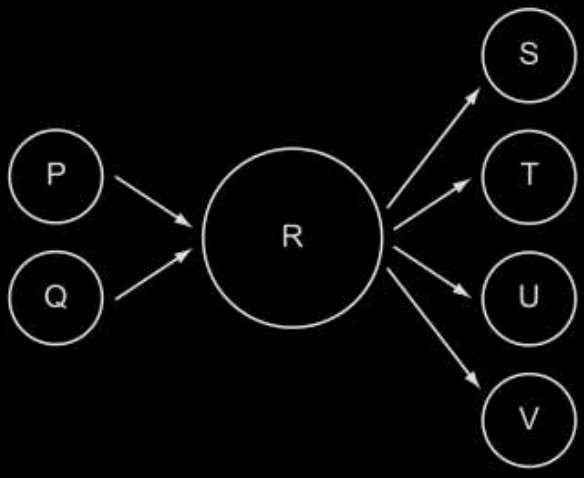
The graph shows how this person's blood glucose concentration changed during part of one day.

At what point was an insulin injection given?



Correct answer: B

33 The diagram represents gametes P and Q fusing to give cell R. Cell R then produces gametes S, T, U and V.



Which statement about the numbers of chromosomes in the cells and gametes is correct?

A The numbers of chromosomes in P and Q are different.

B The numbers of chromosomes in P and S are the same.

C The number of chromosomes in S is one quarter of the number of chromosomes in R.

D The number of chromosomes in T is half the number of chromosomes in Q.

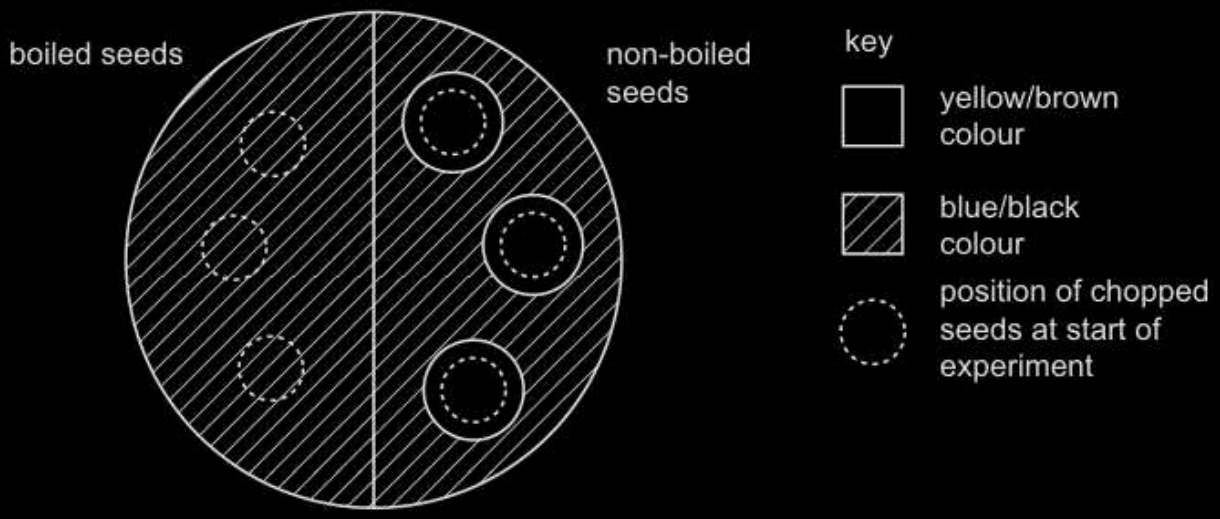
October/November 2012 · Paper 1 · Variant 1 · Question 34

Correct answer: D

34 Six bean seeds were soaked in cold water. Three of them were boiled and cooled. The boiled and the non-boiled seeds were chopped up and then placed on the surface of agar jelly containing starch.

After two days, all the seeds were removed and the jelly was tested with iodine solution.

The diagram shows the result of the experiment.



What is the explanation for the results with the non-boiled bean seeds?

A They absorb iodine.

B They absorb starch.

C They contain acid.

D They contain amylase.

October/November 2012 · Paper 1 · Variant 1 · Question 35

Correct answer: D

35 Which row describes the exchange of substances at the placenta?

	passing from mother to fetus	passing from fetus to mother
A	carbon dioxide, glucose and alcohol	oxygen and urea
B	carbon dioxide, nicotine and glucose	oxygen and urea
C	oxygen and urea	carbon dioxide and glucose
D	oxygen, glucose and antibodies	carbon dioxide and urea

October/November 2012 · Paper 1 · Variant 1 · Question 37

Correct answer: C

37 Which method could **not** be used to produce human insulin from genetically engineered bacteria?

A Bacteria are ground up and used as a source of insulin.

B Insulin is extracted from gases given off from the fermenter in which bacteria are grown.

C Insulin is extracted from homogenised bacteria.

D Insulin is extracted from the nutrient medium from a fermenter in which bacteria have been grown.

October/November 2012 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 One gene has two codominant alleles, A^E and A^F, and one recessive allele, A^G.

How many different genotypes and phenotypes are possible?

	genotypes	phenotypes
A	3	3
B	4	6
C	6	4
D	6	6

October/November 2012 · Paper 1 · Variant 1 · Question 39

Correct answer: D

39 Some normal fruit flies are subjected to radiation in a laboratory. As a result, they produce offspring with unusual characteristics, such as white eyes.

What causes this?

A continuous variation

B discontinuous variation

C dominance

D mutation

October/November 2012 · Paper 1 · Variant 1 · Question 40

Correct answer: C

40 What would be the genotype(s) of the offspring from a cross between two organisms with the genotype Tt?

A all Tt

B half TT, half tt

C quarter TT, half Tt, quarter tt

D three quarters TT, quarter Tt

October/November 2012 · Paper 1 · Variant 2 · Question 2

Correct answer: B

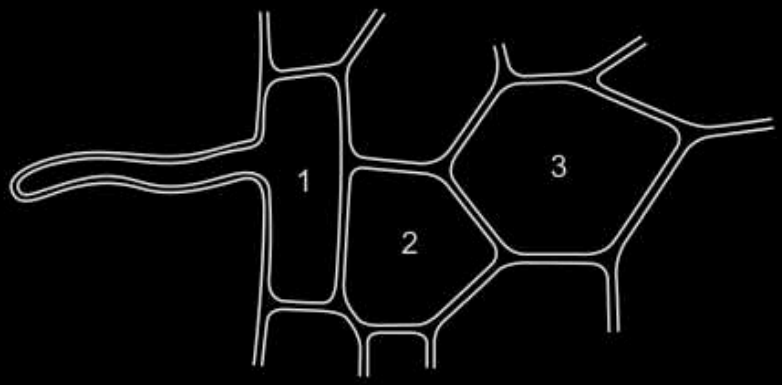
2 Through which organs do carbon dioxide molecules, magnesium ions and nitrate ions enter plants?

	carbon dioxide molecules	magnesium ions	nitrate ions
A	leaves	roots	leaves
B	leaves	roots	roots
C	roots	leaves	leaves
D	roots	leaves	roots

October/November 2012 · Paper 1 · Variant 2 · Question 3

Correct answer: C

3 The diagram shows some cells in the root of a plant that is absorbing water from the soil.



How does the water potential of the cell marked 2 differ from the water potentials of the cells marked 1 and 3?

A higher than cell 1 and cell 3

B higher than cell 1 and lower than cell 3

C lower than cell 1 and higher than cell 3

D lower than cell 1 and lower than cell 3

October/November 2012 · Paper 1 · Variant 2 · Question 4

Correct answer: D

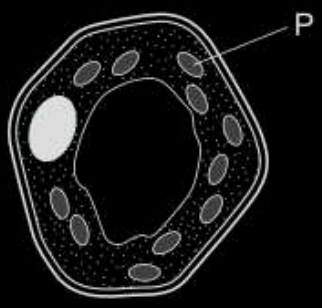
4 According to the lock and key hypothesis, which is the lock and which is the key for the enzyme lipase?

	key	lock
A	fatty acids	lipids
B	lipase	lipids
C	lipase	fatty acids
D	lipids	lipase

October/November 2012 · Paper 1 · Variant 2 · Question 6

Correct answer: A

6 The diagram shows a plant cell.



Compared with the rest of the cell, which row describes the concentrations of oxygen and magnesium inside structure P during the daytime?

	oxygen	magnesium
A	high	high
B	high	low
C	low	high
D	low	low

October/November 2012 · Paper 1 · Variant 2 · Question 7

Correct answer: D

7 For which processes do plants need either nitrate ions or magnesium ions?

	synthesis of cellulose	synthesis of chlorophyll	synthesis of proteins
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✗
D	✗	✓	✓

key

✓ = nitrate ions or magnesium ions needed

✗ = nitrate ions or magnesium ions not needed

October/November 2012 · Paper 1 · Variant 2 · Question 8

Correct answer: C

8 A person tries eating a diet consisting only of lettuce leaves and water.

Which condition might develop?

A constipation

B heart disease

C rickets

D scurvy

October/November 2012 · Paper 1 · Variant 2 · Question 9

Correct answer: D

9 When a person eats some egg white, protein and water enter the stomach. Which substances are found leaving the stomach and leaving the small intestine?

	leaving the stomach	leaving the small intestine
A	amino acids and water	amino acids and water
B	fatty acids, glycerol and water	fatty acids, glycerol and water
C	protein and water	fatty acids and glycerol
D	protein, amino acids and water	water

October/November 2012 · Paper 1 · Variant 2 · Question 10

Correct answer: B

10 The table shows the compositions of four foods.

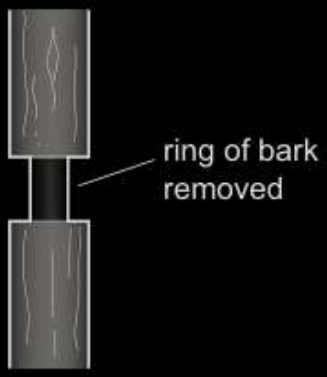
Which food provides the most energy per gram?

	carbohydrate %	fat %	protein %	water %
A	1	16	28	55
B	2	83	2	13
C	5	4	3	88
D	25	20	23	35

October/November 2012 · Paper 1 · Variant 2 · Question 11

Correct answer: B

11 The diagram shows a tree trunk, with a ring of bark, which includes the phloem, removed.



The tree will eventually die because this action cuts off the supply of

A mineral salts to the leaves.

B organic nutrients to the roots.

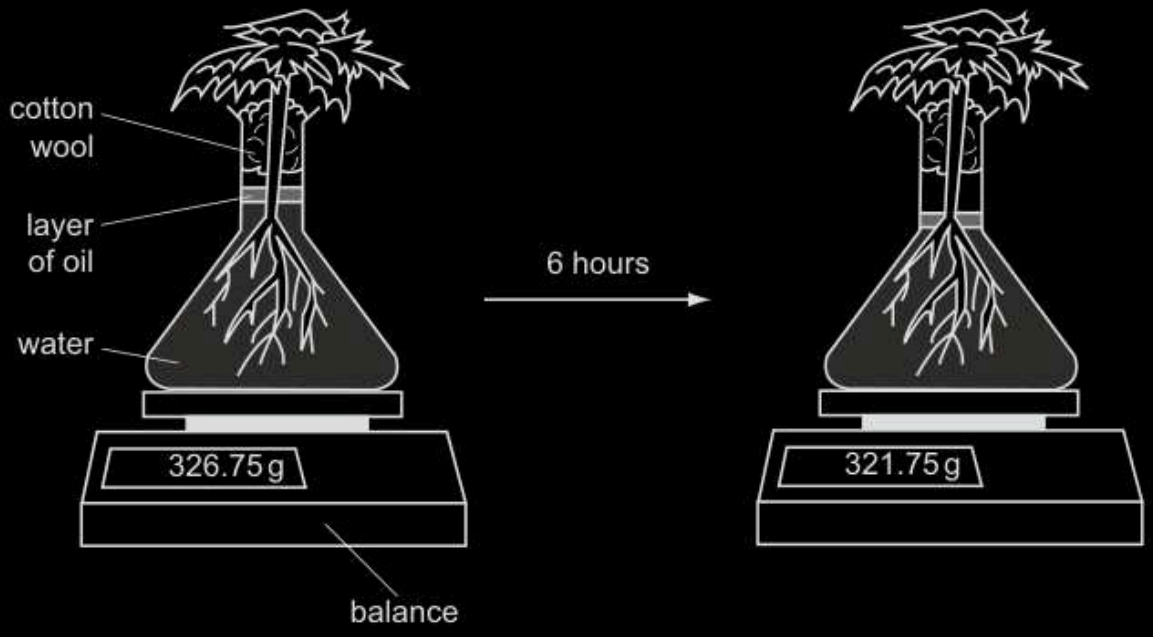
C oxygen to the roots.

D water to the leaves.

October/November 2012 · Paper 1 · Variant 2 · Question 12

Correct answer: D

12 The diagrams show a plant in a flask of water. It is left in the light at 16 °C for six hours.



What explains the change in mass after six hours?

A absorption of water into the root hairs

B evaporation of water from the flask

C photosynthesis in the leaves of the plant

D transpiration from the leaves of the plant

October/November 2012 · Paper 1 · Variant 2 · Question 13

Correct answer: B

13 In the human circulatory system, what causes the transfer of materials from the capillaries to the tissue fluid?

A active transport

B blood pressure

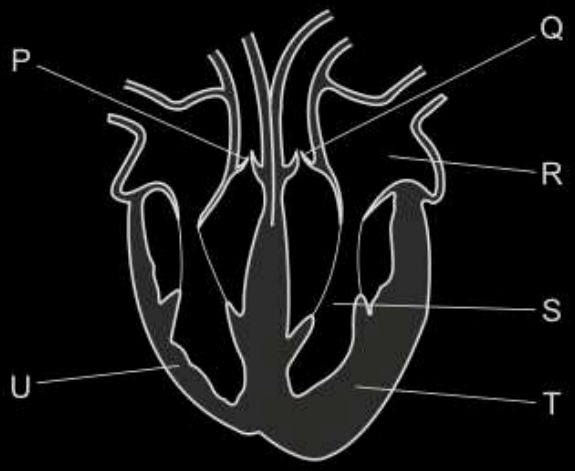
C capillarity

D osmosis

October/November 2012 · Paper 1 · Variant 2 · Question 14

Correct answer: D

14 The diagram shows a section through the human heart.



Which labelled features suggest that blood leaves the heart at different pressures, when going to the lungs and to the body?

A Chambers R and S have different volumes.

B The walls of the atria are thinner than the walls of the ventricles.

C Valve P is stronger than valve Q.

D Wall T is more muscular than wall U.

October/November 2012 · Paper 1 · Variant 2 · Question 15

Correct answer: B

15 Which blood vessel carries blood with the highest concentration of urea?

A hepatic artery

B hepatic vein

C renal artery

D renal vein

October/November 2012 · Paper 1 · Variant 2 · Question 17

Correct answer: C

17 What properties make the alveoli walls efficient at gas exchange?

	elastic tissue in walls	equal oxygen concentrations inside and out	walls one cell thick
A	✓	✓	✗
B	✓	✓	✓
C	✓	✗	✓
D	✗	✓	✓

October/November 2012 · Paper 1 · Variant 2 · Question 19

Correct answer: B

19 What is **not** an excretory product of mammals?

A carbon dioxide in expired air

B undigested food in faeces

C urea in sweat

D urea in urine

October/November 2012 · Paper 1 · Variant 2 · Question 20

Correct answer: A

20 Which process is **not** a result of negative feedback?

A A decrease in the surrounding temperature leads to a decrease in respiration rate.

B A decrease in the surrounding temperature leads to a decrease in sweating.

C A decrease in the surrounding temperature leads to a decrease in blood flow through the skin surface.

D A decrease in the surrounding temperature leads to shivering.

October/November 2012 · Paper 1 · Variant 2 · Question 21

Correct answer: D

21 Three directions in which nerve impulses in neurones can travel in the nervous system are listed.

1 away from the central nervous system
2 towards the central nervous system
3 within the central nervous system

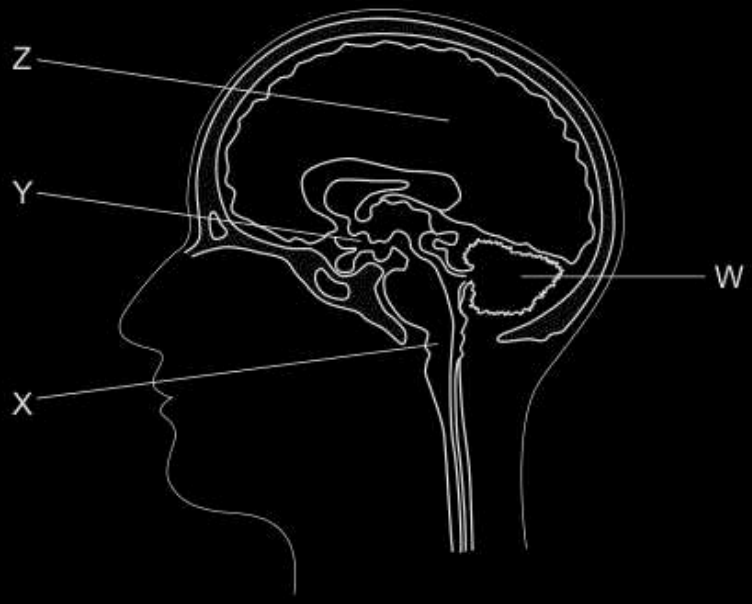
In which direction do impulses in sensory and relay (intermediate) neurones travel?

	sensory neurone	relay neurone
A	1	2
B	1	3
C	2	1
D	2	3

October/November 2012 · Paper 1 · Variant 2 · Question 22

Correct answer: A

22 The diagram shows a section through the brain.



Which regions control balance, breathing and memory?

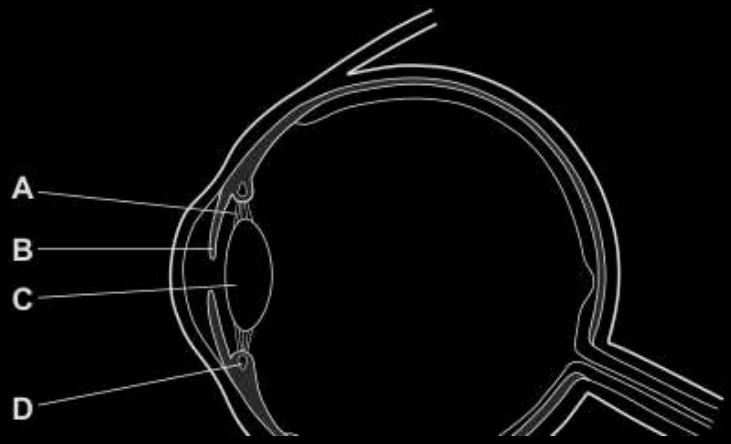
	balance	breathing rate	memory
A	W	X	Z
B	X	Y	Z
C	Y	Z	W
D	Z	W	Y

October/November 2012 · Paper 1 · Variant 2 · Question 24

Correct answer: B

24 The diagram shows a section through a human eye.

Which structure contains muscle fibres that contract in response to sudden changes in light intensity?



October/November 2012 · Paper 1 · Variant 2 · Question 30

Correct answer: C

30 Nitrogen is cycled in ecosystems. In what form is most of the nitrogen in animals?

A amino acid

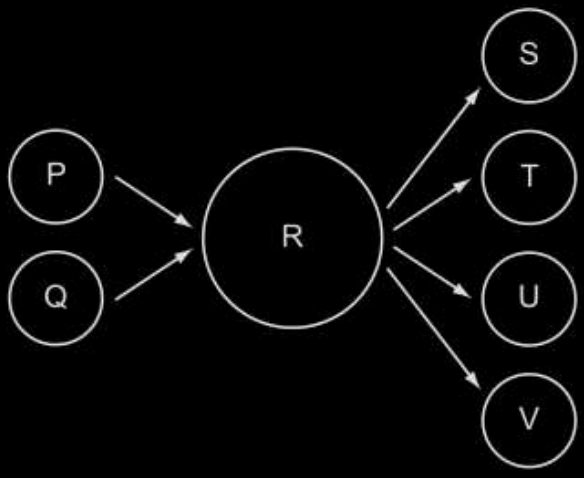
B nitrogen gas

C proteins

D urea

Correct answer: B

33 The diagram represents gametes P and Q fusing to give cell R. Cell R then produces gametes S, T, U and V.



Which statement about the numbers of chromosomes in the cells and gametes is correct?

A The numbers of chromosomes in P and Q are different.

B The numbers of chromosomes in P and S are the same.

C The number of chromosomes in S is one quarter of the number of chromosomes in R.

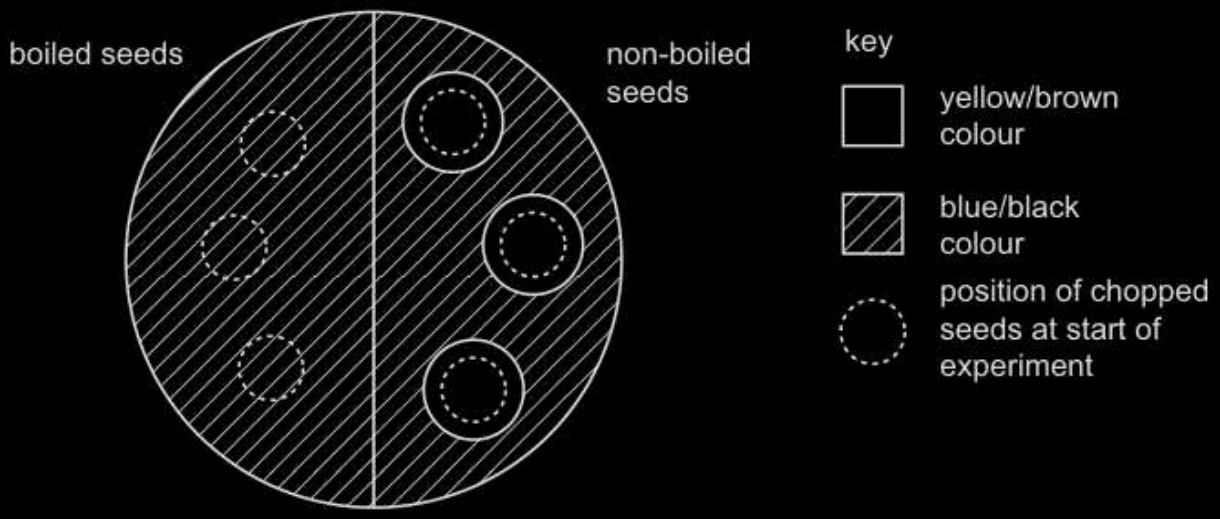
D The number of chromosomes in T is half the number of chromosomes in Q.

Correct answer: D

34 Six bean seeds were soaked in cold water. Three of them were boiled and cooled. The boiled and the non-boiled seeds were chopped up and then placed on the surface of agar jelly containing starch.

After two days, all the seeds were removed and the jelly was tested with iodine solution.

The diagram shows the result of the experiment.



What is the explanation for the results with the non-boiled bean seeds?

A They absorb iodine.

B They absorb starch.

C They contain acid.

D They contain amylase.

October/November 2012 · Paper 1 · Variant 2 · Question 35

Correct answer: A

35 The diagram shows a developing fetus in the uterus.

What is the liquid at X called?

A amniotic fluid

B blood

C urine

D water

October/November 2012 · Paper 1 · Variant 2 · Question 36

Correct answer: C

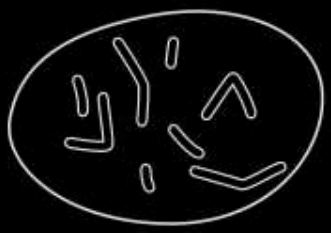
36 Which statement correctly describes how a method of birth control works?

	method	mechanism
A	chemical	prevents eggs from passing down the fallopian tubes
B	hormonal	paralyses sperm after ejaculation
C	mechanical	inserts a barrier between egg and sperm
D	surgical	cuts the urethra so no sperm can emerge

October/November 2012 · Paper 1 · Variant 2 · Question 37

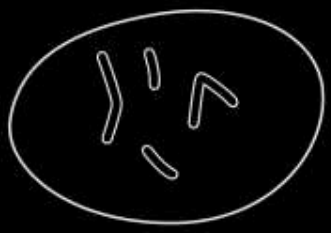
Correct answer: C

37 The diagram shows the chromosomes in the nucleus of a cell that divides by mitosis.

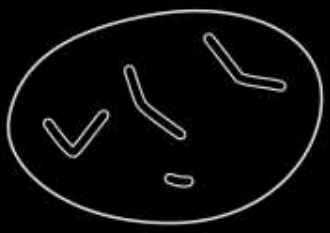


Which diagram shows the chromosomes in the nucleus of one of the daughter cells produced?

A



B



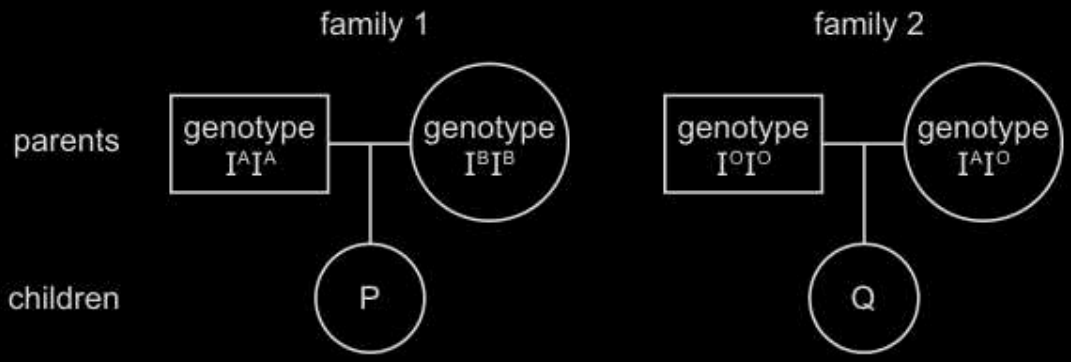
C

D

October/November 2012 · Paper 1 · Variant 2 · Question 38

Correct answer: C

38 The diagram shows inheritance of blood groups in two different families.



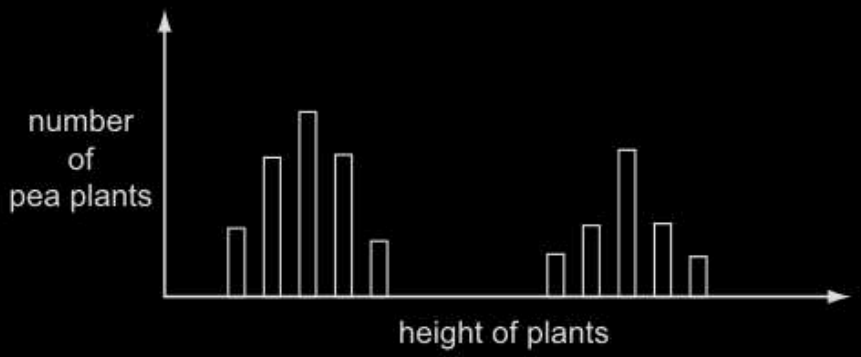
For each of P and Q, what is the probability that they have a co-dominant genotype?

	P	Q
A	0.0	1.0
B	0.5	1.0
C	1.0	0.0
D	1.0	0.5

October/November 2012 · Paper 1 · Variant 2 · Question 39

Correct answer: C

39 The bar chart shows the heights of pea plants grown from 500 pea seeds.



What variation do the plants show?

A continuous variation only

B discontinuous variation only

C both continuous variation and discontinuous variation

D neither continuous variation nor discontinuous variation

October/November 2012 · Paper 1 · Variant 2 · Question 40

Correct answer: B

40 A man of blood group A, and his wife of blood group O, had two children, both of blood group A. The man concluded that he must be homozygous for the allele I^A, since he thought half his children would be of group O if he were heterozygous.

Why was his conclusion unsound?

A Blood group mutations are common.

B Genetic ratios are unreliable for small numbers.

C His wife might have been heterozygous.

D The expected ratio for a heterozygous father and group O mother is 3 group A: 1 group O.

1 The urinary systems of a dog and of a human being are similar in structure and function. Fig. 1.1 shows the urinary system of a dog.

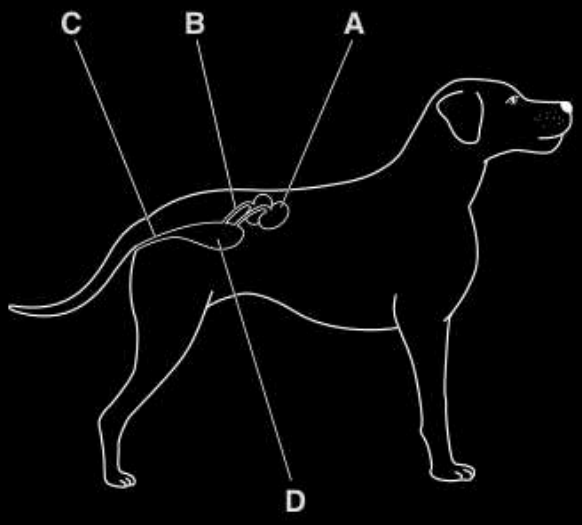


Fig. 1.1

(a) (i) Name the parts **A**, **B** and **C** in Fig. 1.1.

A .....

B .....

C ..... [3]

(ii) State the function of **D**.

..... [1]

(b) Fig. 1.2 shows the relationship between an alveolus and a blood capillary in the lung of a mammal. The arrows show the path taken by oxygen from inhaled air entering the blood during gas exchange.

For Examiner's Use

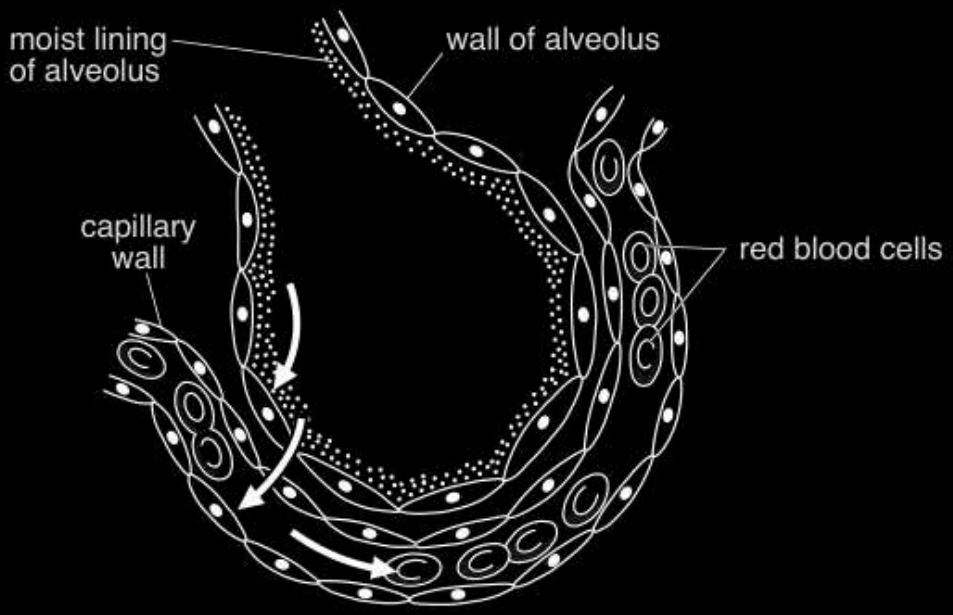


Fig. 1.2

(i) Draw **two** arrows on Fig. 1.2 to show the path taken by most carbon dioxide leaving the blood.

.....

.....

..... [2]

(ii) Explain why this movement of carbon dioxide is considered to be part of excretion.

.....

.....

..... [3]

(c) Hormones are also substances that need to be excreted from the body after they have carried out their functions.
Explain how this is done.

.....

.....

..... [2]

[Total: 11]

1 (a) (i) **A** kidney;
B ureter;
C urethra; [3]

(ii) **D** storage AW + urine; [1]

(b) (i) arrow shown leaving plasma;
arrow from moisture to alveolar air;
One arrow covering the entire journey = 1 mark [2]

(ii) removal + from body / organism;
waste;
of respiration / metabolism; [3]

(c) destroyed / broken down **AW**;
by liver;
products removed / reabsorbed; [max 2]

[Total: 11]

For Examiner's Use

2 **(a)** The graphs in Fig. 2.1 show the effect of changing pH on the rate of reactions controlled by two different enzymes found in the alimentary canal of a mammal.

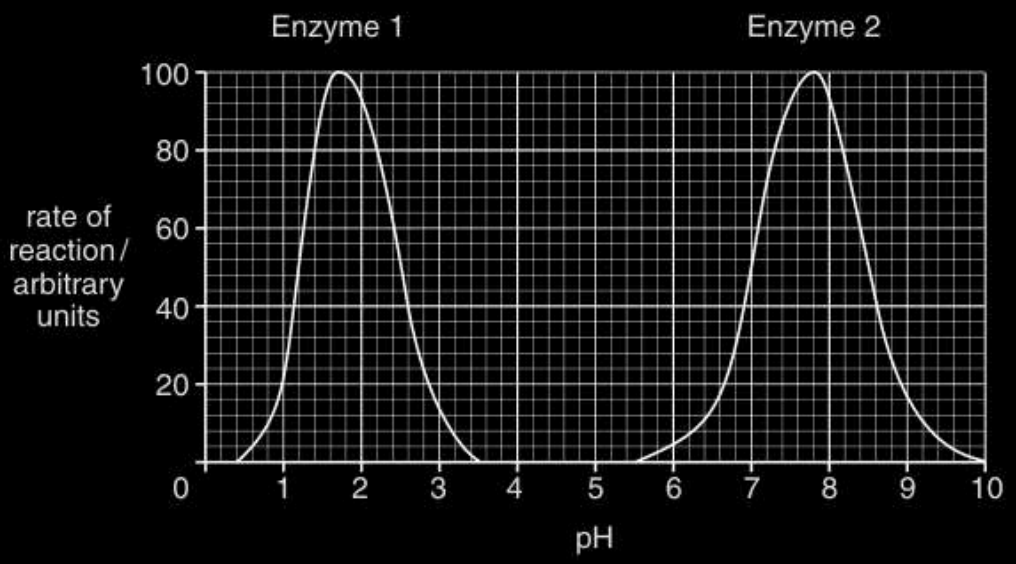


Fig. 2.1

(i) Explain what is meant by the term pH.

.....

..... [2]

(ii) Using the information in Fig. 2.1, state the rate of reactions controlled by Enzyme 2 at pH 7.

..... [1]

(iii) Suggest in which part of the alimentary canal Enzyme 1 is likely be active.

..... [1]

Give reasons for your suggestion.

.....

..... [2]

(b) Enzymes in the alimentary canal are involved in reactions that break down large molecules into small molecules, for example, starch into maltose.
Give an example in another, named part of the body, where small molecules are built up into larger ones.

named part of the body .....

small molecules .....

larger molecules .....

[3]

[Total: 9]

2 **(a)** **(i)** measure AW of;
acidity / alkalinity (A H^+ concentration); [2]

(ii) 48 – 52 + (arbitrary) units / %; [1]

(iii) stomach; [1]
optimum AW pH of the enzyme is (approx) 2;
acidic;
due to HC*l;* [max 2]

(b) in liver;
glucose;
to glycogen;
A other examples muscle / aa / protein; skin / glyc + f.a. / fat etc [3]

[Total: 9]

3 Atmospheric air contains oxygen and carbon dioxide.

For Examiner's Use

(a) Complete table 3.1 to show the percentage of oxygen and carbon dioxide in inhaled and exhaled air.

Table 3.1

gas	% gas in air	
	inhaled air	exhaled air
oxygen		
carbon dioxide		

[2]

(b) **(i)** Explain how oxygen is used by a muscle cell.

.....

.....

..... [3]

(ii) Explain what happens in a muscle cell when oxygen is in short supply.

.....

..... [2]

(c) At high altitudes, oxygen is less available than it is at low altitudes.
Suggest modifications of the circulatory and respiratory systems that might help people that live for many years at high altitude.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

3 **(a)** oxygen 19 – 21% + 14 – 16%;
carbon dioxide 0.03 – 0.045% + 3 – 4.5%; [2]

(b) **(i)** (aerobic) respiration;
release energy;
from glucose;
for contraction;
lactic acid +ref. oxygen debt **AW**;
R produce AW energy
A give **AW**; [max 3]

(ii) anaerobic (respiration);
less energy released;
(produces) lactic acid;
(muscle) becomes fatigued / tired / ref. cramp / pain; [max 2]

(c) rapid breathing mechanism / deeper breathing;
modified lung structure or described;
more (efficient) haemoglobin;
more efficient blood supply to organs / tissues or e.g.; larger heart;
more red blood cells;
faster heart rate / faster circulation of blood; [max 3]

[Total: 10]

5 Fig. 5.1 shows a palisade cell from the leaf of a flowering plant.

For Examiner's Use

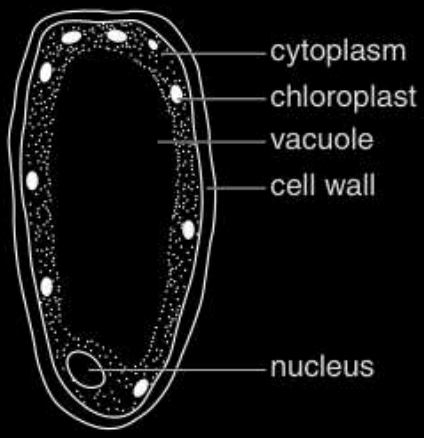


Fig. 5.1

(a) State **three** ways in which this cell differs from a fungal hypha.

palisade cell	fungal hypha
1.	.....
.....	.....
2.	.....
.....	.....
3.	.....
.....	.....

[3]

Fig. 5.2 shows the apparatus used to produce 5 dm^3 of red wine from grape juice.

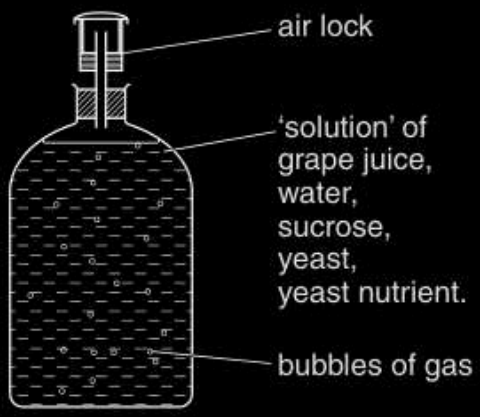


Fig. 5.2

(b) State an equation for the chemical process by which the alcohol is produced.

..... [2]

(c) Suggest a suitable temperature to ensure a good yield of alcohol.

For Examiner's Use

..... [1]

Table 5.1 shows the rate of bubble release and the concentrations of sugar and alcohol in the grape juice over the first 10 days.

Table 5.1

time /days	number of bubbles per hour	concentration of sugar/grams per dm^3	concentration of alcohol/%
1	60	200	0.0
2	40	150	0.0
3	20	100	0.0
7	2	50	0.5
10	1	30	2.0

(d) (i) State the process that was occurring from day 1–3 to produce the bubbles of gas.

..... [1]

(ii) Explain why alcohol started to be produced only after several days.

.....

.....

..... [2]

(iii) When the alcohol content reached 15%, no further bubbles were produced. Suggest why the alcohol content could not increase further.

.....

..... [1]

[Total: 10]

Section B

For Examiner's Use

Answer **all** the questions in this section.

Write your answers in the spaces provided.

5 **(a)** one nucleus per cell in palisade v hypha – coenocytic / several nuclei;
separate cells each with wall v not separate cells;
large central vacuole v several small vacuoles;
stores starch v stores glycogen;
chloroplasts / chlorophyll present v absent;
walls are made of different materials (chitin for hypha); [max 3]

(b) $C_6H_{12}0_6 \rightarrow 2\ C_2H_5OH + 2C0_2$;
glucose / sucrose; → alcohol / ethanol + carbon dioxide;
A a word or chemical equation, 1 mark each side, but if chemical, must balance. [2]

(c) 25 – 40 °C; [1]

(d) **(i)** respiration;
R anaerobic / fermentation [1]

(ii) the oxygen has been used (up);
and no more can enter ;
the yeast starts to respire anaerobically; [max 2]

(iii) yeast has died;
depletion of substrate or named;
respiration / fermentation ceases; [max 1]

[Total: 10]

6 Aphids are insects that feed by piercing the phloem of soft, young shoots. Aphids then use their sharp, hollow mouthparts for sucking up their nutrition from the plant.

(a) Explain why phloem is more suitable for this purpose than any other plant tissue.

.....

.....

.....

..... [3]

(b) Suggest and explain the effect on the plant shoot of a large number of aphids feeding at the same time over a period of several weeks.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

[Total: 10]

6 **(a)** within reach;
contains sucrose / sugar / carbohydrate;
contains amino acids;
in solution / water;
not lignified / is softer **AW**;
R starch / glucose / mineral ions
Ig nutrients [3]

(b) impaired growth **AW** / death;
amino acids used to make protein;
less / no new cytoplasm;
less sucrose or carbohydrate;
less glucose;
for use in respiration;
energy used;
for any named purpose;
may introduce viruses / disease;
damage to plant **AW;**
Ig wilting [max 7]

[Total: 10]

7 **(a)** Explain the differences between *mitosis* and *meiosis*. Your explanation should include references to where the processes occur and why they occur.

For Examiner's Use

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

(b) Explain how it is possible that parents with different blood groups are able to produce **only** children with blood groups **different** from themselves.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

Section C

For Examiner's Use

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7 **(a)** (mitosis) chromosome number maintained / diploid **AW**;
identical offspring / clones;
e.g. of where it occurs – plant or animal / in bacteria;
growth / repair;
asexual reproduction;
2 new cells produced;
(meiosis) chromosome number halved / haploid **AW**;
in gonads / testes / ovaries / anthers;
to produce gametes / sex cells;
any gamete correctly named;
sexual reproduction;
4 new cells produced;
R growth / repair of cells [max 5]

(b) correct parental blood groups identified (AB and O, **or** A and B);
(in text or diagram) parents' genotypes identified as AB and OO or AA and BB;
*correct gametes clearly identified for both parents;
*the word gametes correctly used;
genotype of offspring (AO + BO **or** AB) shown;
blood groups of offspring identified as Group A and B **or** AB;
*available with wrong genotypes, so long as they match [max 5]

[Total: 10]

8 **(a)** Explain what is meant by *double circulation*.

.....

.....

.....

..... [3]

(b) Describe the composition and the importance of plasma in the circulatory system.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

[Total: 10]

8 **(a)** blood passes through heart twice;
lungs + to rest of body;
lower pressure in pulmonary circulation ORA;
correct ref oxygenated blood / deoxygenated blood; [max 3]

(b) water;
solvent / carries dissolved / in solution;
any two from:
for salts or ions / glucose /amino acids / vitamins / fat or fatty acids + glycerol;;
urea;
plasma or blood proteins or named;
hormones;
transport of blood cells / platelets;
heat;
carbon dioxide;
to service body cells / target organs; [max 7]

[Total: 10]

2 Species **D** lives in a forest. Fig. 2.1 shows what can happen, over the same period of time, if some members of species **D** migrate from the forest to a desert region.

For Examiner's Use

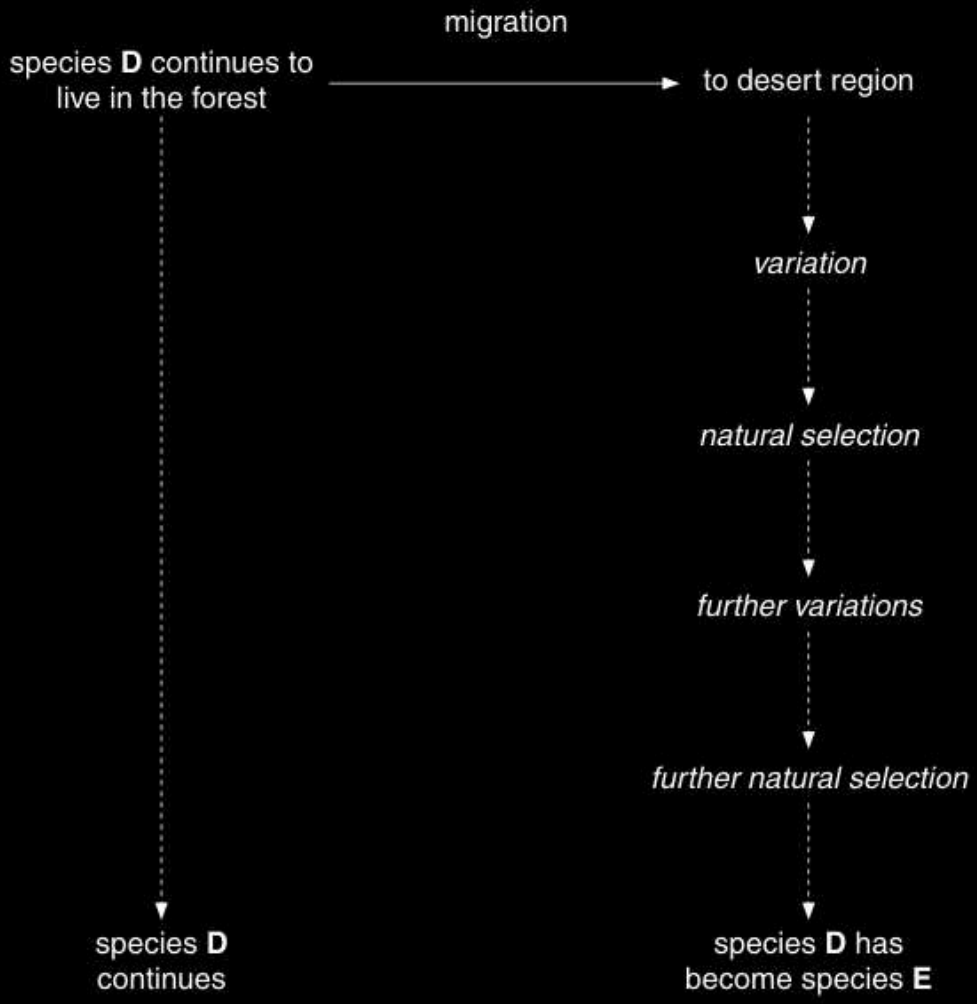


Fig. 2.1

(a) Name the **two** main causes of variation in organisms.

1. .....

2. ..... [2]

(b) Name the process involving variation and natural selection that has led to the development of species **E**.

..... [1]

(c) Suggest how the processes of variation and natural selection may have acted to produce the new species **E** in the desert.

For Examiner's Use

.....

.....

.....

.....

.....

.....

.....

.....[4]

(d) Members of species **D** are no longer able to breed successfully with members of new species **E**. Suggest reasons for this.

.....

.....

.....[2]

[Total: 9]

2 **(a)** genes / mutation / named common mutagen;

environment / habitat / named environmental factor; [2]

(b) evolution / speciation; [1]

(c) different/changed environment;

mutation(s);

variations + advantageous **AW** / better adapted;

survive;

reproduce / passed on;

cumulative effect / over many generations;

leading to change in phenotype / appearance / or e.g.; [Max 4]

(d) difference in genes/DNA;

difference in chromosomes;

ref to problems with fertilisation;

no sexual attraction / incompatible;

geographical separation; [Max 2]

[Total: 9]

3 Fig. 3.1 shows a plant and the position in which it grows in the dead leaves that collect where two branches of a tree join.

For Examiner's Use

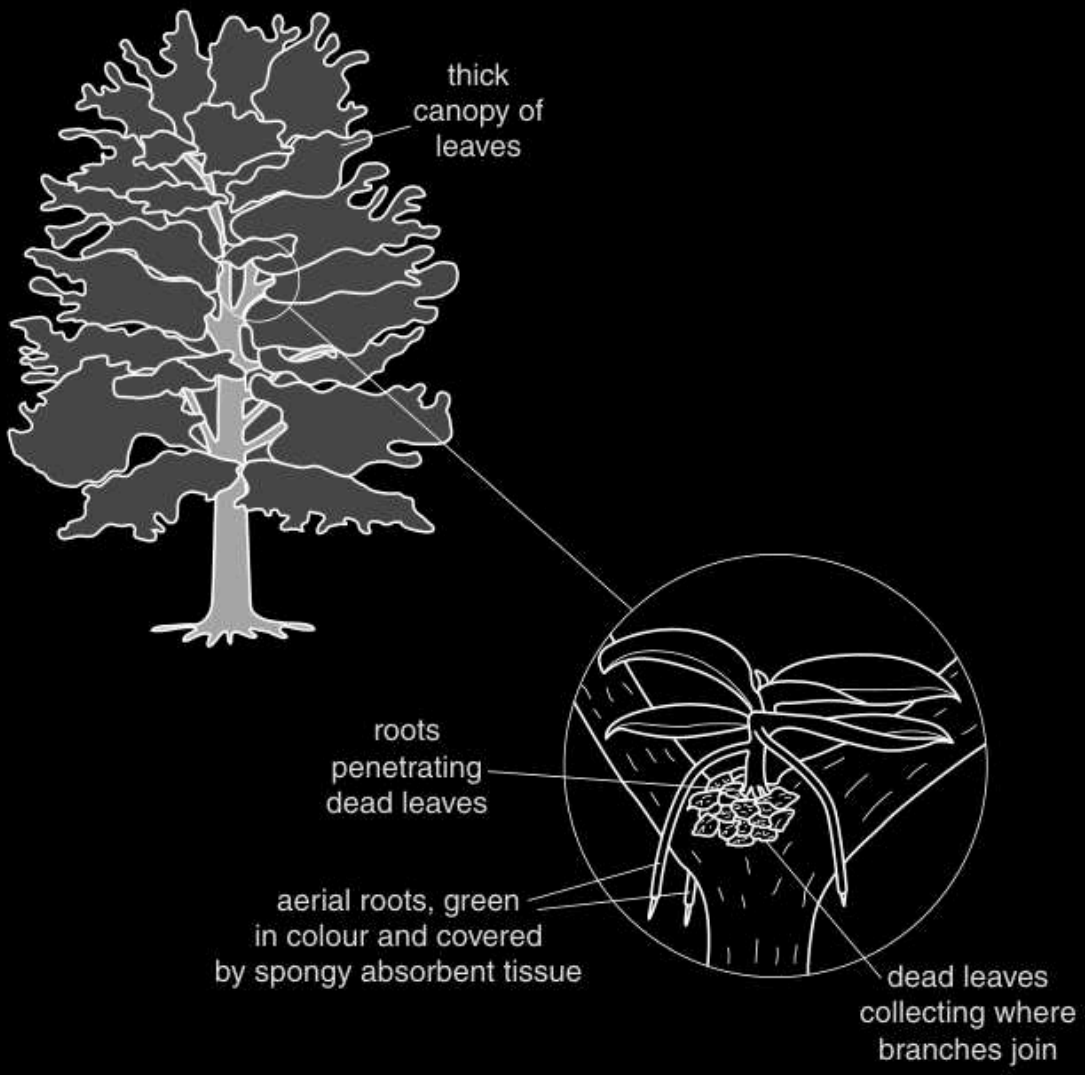


Fig. 3.1

(a) State **two** ions that the roots of the plant will absorb from the dead leaves. For each ion, state its importance to the plant.

ion 1 .....

importance .....

ion 2 .....

importance .....[4]

(b) Explain how the ions you have mentioned in **(a)** are made available to the plant from the dead leaves.

For Examiner's Use

.....

.....

.....

.....

.....[4]

(c) Suggest and explain why the aerial roots of the plant are green and covered with a spongy absorbent tissue.

green

.....

.....

covered with a spongy absorbent tissue

.....

.....[3]

[Total: 11]

3 **(a)** any 2 correct ions;;

any correct function for each ion;;

e.g.

nitrate;

protein / amino acid production / named protein / DNA;

magnesium;

chlorophyll; [Max 4]

(b) decomposition/decay/putrefaction/enzyme action;

*by bacteria;

*fungi;

of named chemical in plant leaves;

nitrification (or described); [Max 4]

(c) chlorophyll/chloroplasts + absorbs light/photosynthesis;

leaves do not receive enough light/in shade **AW**;

to absorb water/moisture/water vapour;

water store/retains water/dead leaves lack water; [Max 3]

[Total: 11]

4 Fig. 4.1 shows a person sitting on a chair with his legs crossed, watching the television. A friend notices that the person's foot is making very slight regular kicking movements as indicated by the arrow on Fig. 4.1.

For Examiner's Use

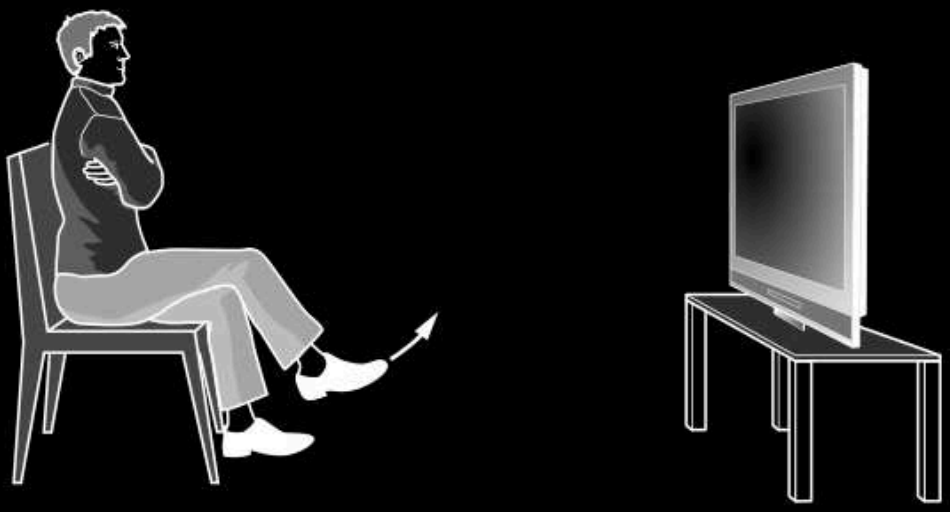


Fig. 4.1

The friend times the movements with her watch and announces that the person's heart is beating 70 times per minute.

(a) (i) Explain how the friend was able to make this deduction.

.....

.....

.....

.....

.....[4]

(ii) Explain why the person's leg kicks at a faster rate when there is an exciting programme on the television.

.....

.....[1]

Fig. 4.2(a) shows a chair leaning against a wall, carefully balanced and with a scale pan hanging from the front of the seat. Fig. 4.2(b) shows what happens when a mass is then added to the scale.

For Examiner's Use

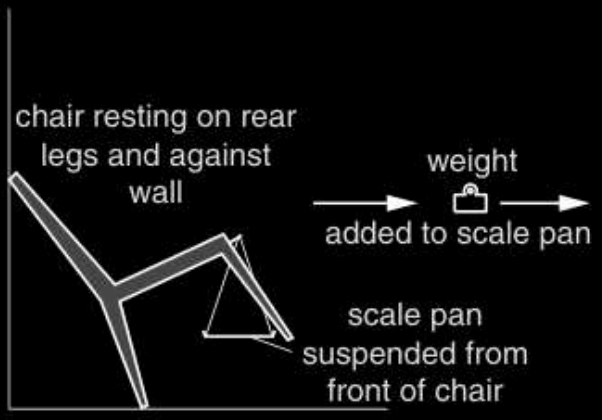


Fig. 4.2(a)

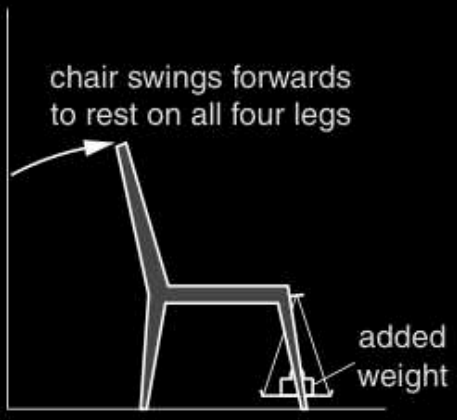


Fig. 4.2(b)

A student sits in a chair, leaning against a wall as shown in Fig. 4.2(a), and **remains perfectly still**.

(b) Use your knowledge of how blood moves through the circulatory system to suggest an explanation for why, after a few minutes, the chair falls forwards similar to the chair in Fig. 4.2(b).

.....

.....

.....

.....

.....[4]

[Total: 9]

4 **(a)** **(i)** pulse (beat);

in artery in leg;

increased pressure;

ref. one pulse beat/kick for every heart beat; [Max 4]

(ii) adrenaline/heart beats faster; [1]

(b) blood + legs/feet;

in veins;

no use of leg muscles ;

blood not pushed from one set of valves to the next;

increases mass/weight of the (lower) leg; [Max 4]

[Total: 9]

5 The graph in Fig. 5.1 shows the concentration of nicotine in the blood of a person before, during and after smoking a cigarette.

For Examiner's Use

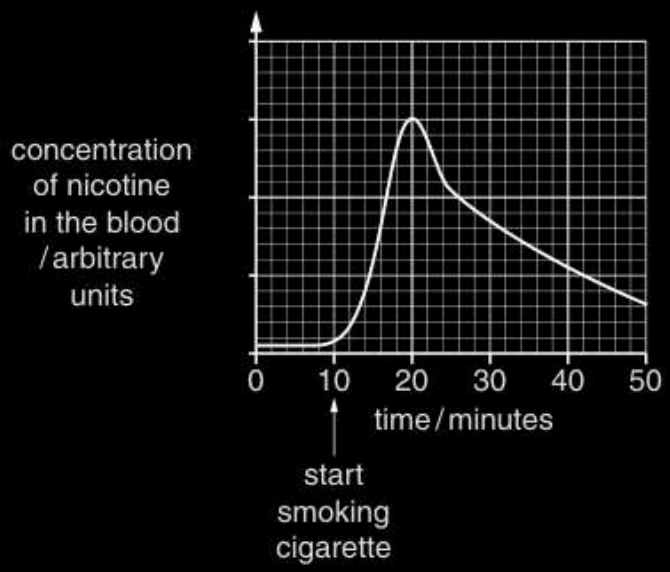


Fig. 5.1

(a) Use Fig. 5.1 to find the time taken for the person to smoke the cigarette.

..... [1]

(b) Suggest why the concentration of nicotine in the person's blood was not zero before starting to smoke the cigarette.

.....

.....[1]

(c) Name two components of the cigarette smoke, other than nicotine, that may be harmful to the person. For each component, state the harm that it may cause.

component 1 .....

harm it may cause .....

.....

component 2 .....

harm it may cause .....

.....[4]

(d) Fig. 5.2 shows the blood pressure of the same person over the same time period.

For Examiner's Use

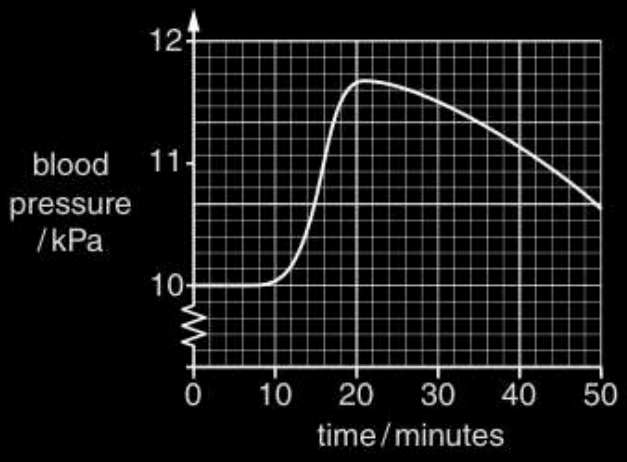


Fig. 5.2

(i) Use Fig. 5.2 to describe the effects of smoking the cigarette on this person's blood pressure.

.....

.....[2]

(ii) Suggest how smoking a cigarette resulted in this effect on blood pressure.

.....

.....[1]

(iii) Suggest possible harmful effects of this change in blood pressure.

.....

.....[2]

[Total: 11]

Section B

For Examiner's Use

Answer **all** the questions in this section.

Write your answers in the spaces provided.

5 **(a)** 10/11 minutes; [1]

(b) smoker/has recently smoked/passive smoking; [1]

(c) Any 2 from:

carbon monoxide/CO;

affect on O_2 carriage/fatty deposits in walls of bvs;

carbon dixide/CO_2;

prevents loss of CO_2 from blood;

tar;

carcinogenic properties/lung cancer/inhibits gaseous diffusion/damages alveolar walls or cilia; [4]

(d) **(i)** increase;

followed by decrease;

reading from graph with units; [Max 2]

(ii) arterial constriction, fat deposits or diameter reduction/heart rate increases/heart pumps harder or faster AW; [1]

(iii) prolonged raised pressure/cumulative effect;

damage to capillaries/ref. thin walls of capillaries;

any relevant effect e.g. damage to kidneys/brain/heart/blood vessels; [Max 2]

[Total: 11]

7 **(a)** Define the term, *excretion*.

For Examiner's Use

.....

.....

.....[3]

(b) Describe how a kidney machine carries out the functions of a kidney for a person suffering from kidney disease.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[7]

[Total: 10]

Section C

For Examiner's Use

Answer **either** question 8 **or** question 9.

Write your answer in the spaces provided.

7 **(a)** removal from organism/body;

toxic/poisonous;

waste (products);

from metabolism or described; [Max 3]

(b) filters/removes substances from + blood;

using partially permeable membrane **AW;**

ref. dialysis fluid;

urea/nitrogenous products;

salt(s)/ions/small molecules;

toxins/poisons;

ref. relative concentrations **AW**;

excess water/ref. osmoregulation;

large molecules stay in blood;

such as proteins;

ref. diffusion; [Max 7]

[Total: 10]

8 **(a)** Explain what is meant by the *lock and key* hypothesis for enzyme action.

.....

.....

.....

.....

.....

.....

.....

.....

.....[5]

(b) Describe how enzyme action is affected by an increase in temperature.

.....

.....

.....

.....

.....

.....

.....

.....

.....[5]

[Total: 10]

8 **(a)** active site (of enzyme);

of particular/special/complementary/exact + shape;

fits substrate molecule/ref enzyme-substrate complex;

splitting or joining of substrate molecule(s)/products formed;

(the idea) molecule(s) or product released;

enzyme ready to be used again/unchanged;

ref. specificity; [Max 5]

(b) *reaction becomes faster with higher temperature;

faster movement of molecules/more collisions;

*ref. maximum/optimum;

*slows rapidly;

*stops;

(heat has) changed/destroyed (shape) of active site;

denatured/lost 3D structure;

substrate no longer fits; [Max 5]

[Total: 10]

May/June 2012 · Paper 1 · Variant 1 · Question 1

Correct answer: D

1 What is found in both animal and plant cells?

A cellulose cell wall

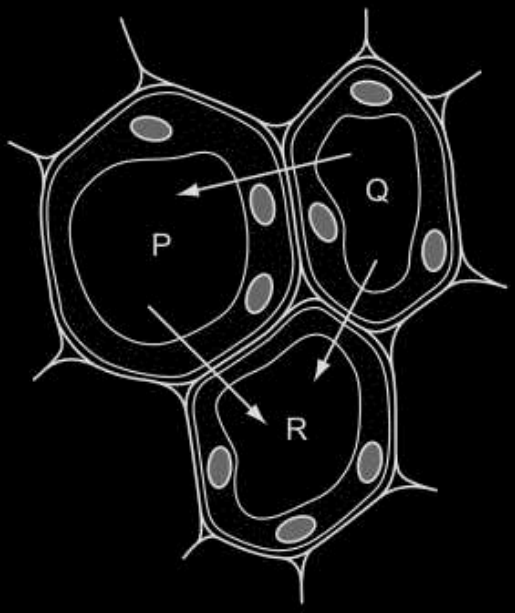
B chloroplast

C starch grain

D vacuole or vacuoles

Correct answer: C

2 The diagram shows part of a leaf with three spongy mesophyll cells labelled P, Q and R. The arrows show the direction of water movement by osmosis.



Which is the correct order of water potential in the cells from the highest to the lowest?

	highest	middle	lowest
A	P	Q	R
B	P	R	Q
C	Q	P	R
D	R	P	Q

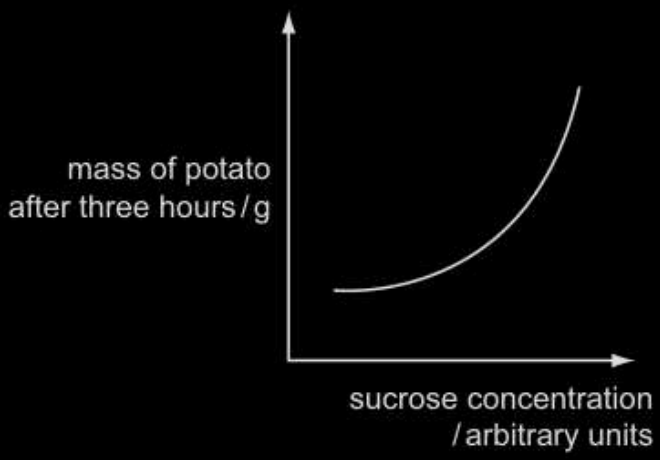
May/June 2012 · Paper 1 · Variant 1 · Question 3

Correct answer: D

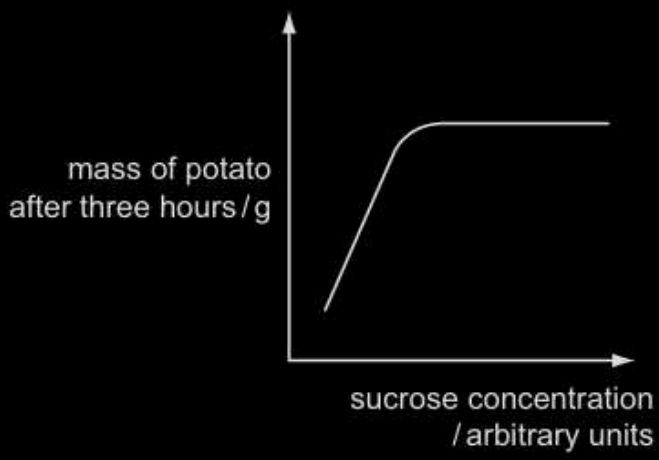
3 Identical pieces of potato are placed in sucrose solutions of different concentrations. After three hours, the mass of each potato piece is measured.

Which graph shows the results of this experiment?

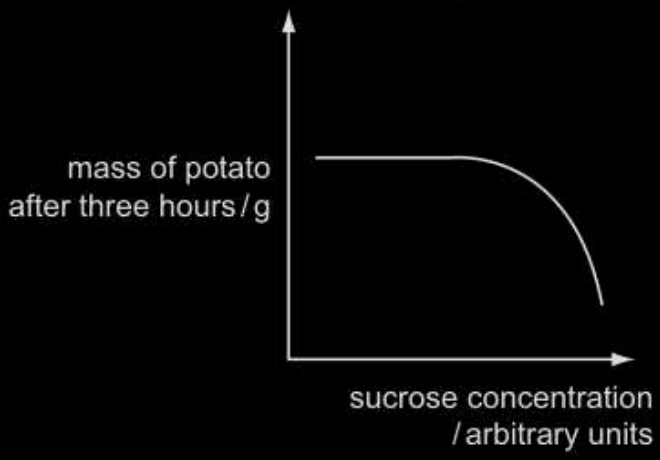
A



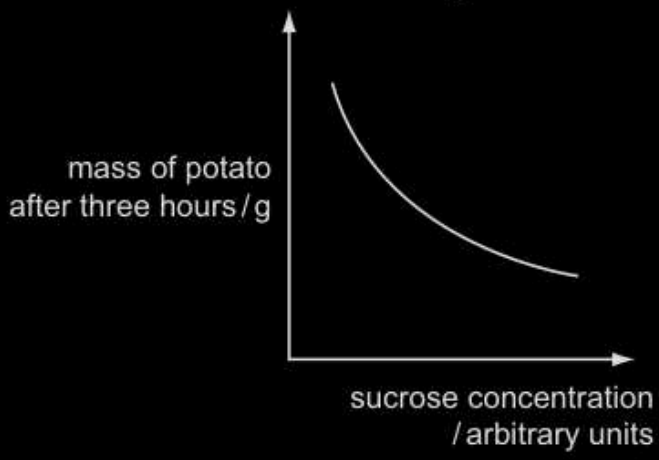
B



C



D



May/June 2012 · Paper 1 · Variant 1 · Question 4

Correct answer: C

4 Starch is digested to maltose by the enzyme amylase.

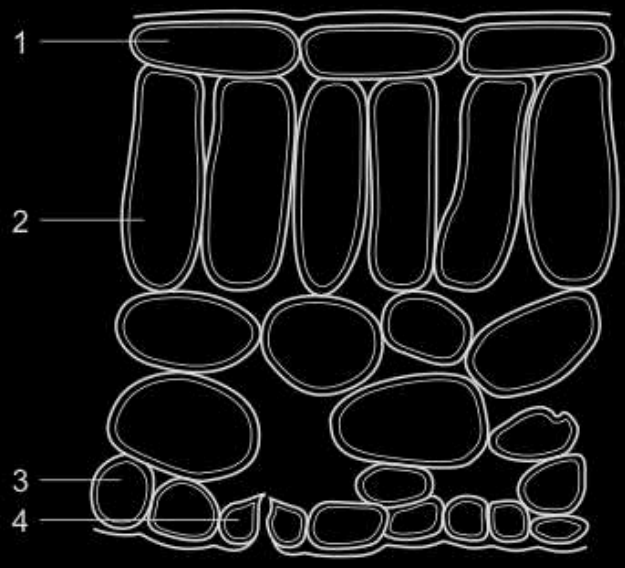
According to the 'lock and key' hypothesis, which is the 'key' and which is the 'lock'?

	'key'	'lock'
A	amylase	maltose
B	amylase	starch
C	starch	amylase
D	starch	maltose

May/June 2012 · Paper 1 · Variant 1 · Question 6

Correct answer: D

6 The diagram shows cells in a section through a leaf of a green plant.
(No cell contents are shown.)



Which cells contain chloroplasts?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

May/June 2012 · Paper 1 · Variant 1 · Question 8

Correct answer: B

8 Which chemical elements are present in **both** fats and proteins?

A carbon, hydrogen, oxygen and nitrogen

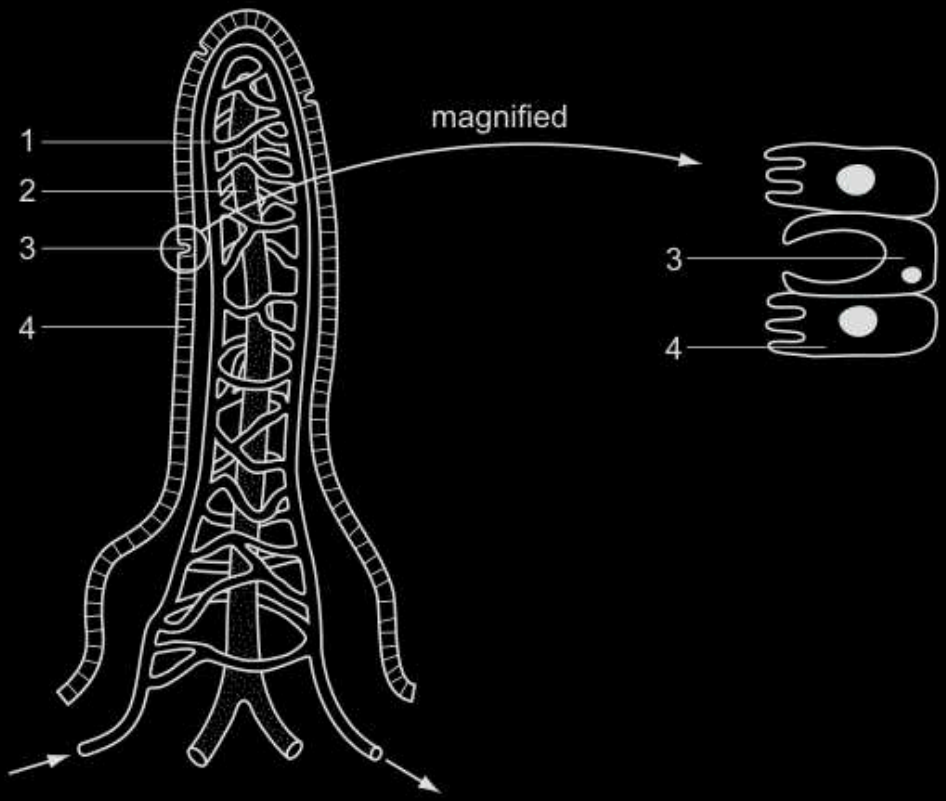
B carbon, hydrogen and oxygen only

C carbon, hydrogen and nitrogen only

D carbon, oxygen and nitrogen only

Correct answer: D

10 The diagram shows a section through a villus.

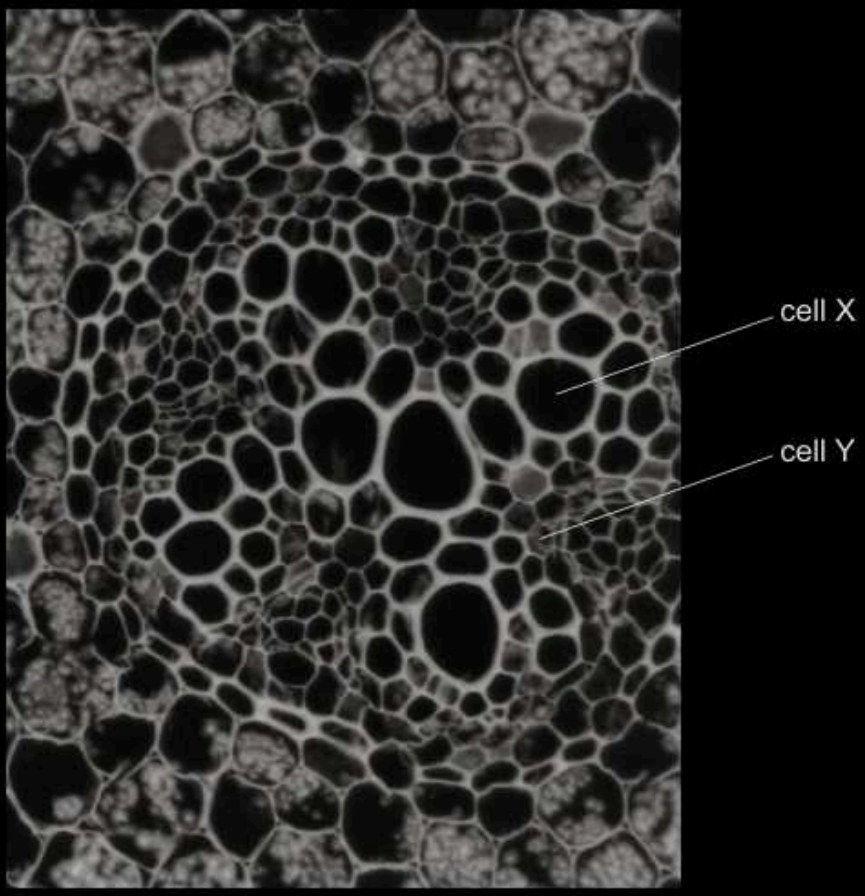


Which sequence correctly describes the functions of the numbered parts?

	1	2	3	4
A	transports digested fats	transports glucose	absorbs digested food	produces mucus
B	transports digested fats	transports glucose	produces mucus	absorbs digested food
C	transports glucose	transports digested fats	absorbs digested food	produces mucus
D	transports glucose	transports digested fats	produces mucus	absorbs digested food

Correct answer: D

11 The photomicrograph shows part of a section through a stem.



The contents of cell X and the contents of cell Y are each tested with Benedict's reagent and with iodine solution.

What results are expected?

	cell X		cell Y	
	Benedict's reagent	iodine solution	Benedict's reagent	iodine solution
A	+	+	–	–
B	+	–	+	+
C	–	+	–	+
D	–	–	+	–

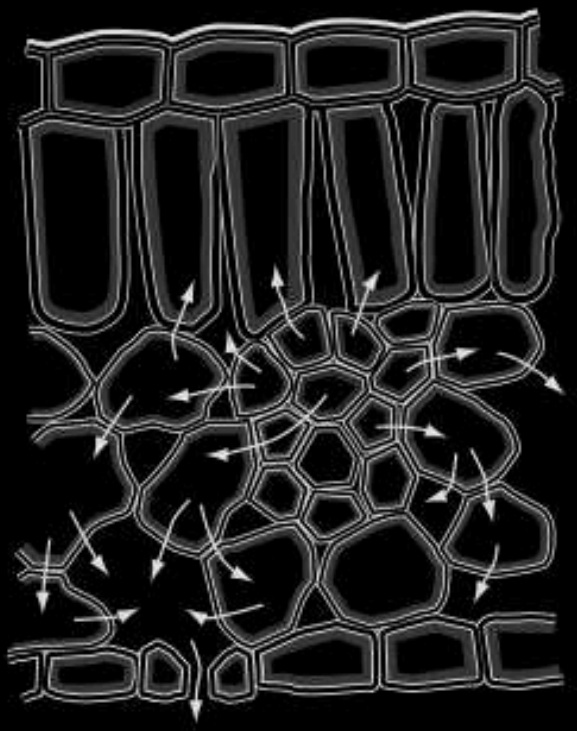
key

+ = positive result

– = negative result

Correct answer: D

12 The diagram shows a section through a green leaf.



The arrows represent the movement of

A carbon dioxide during respiration.

B oxygen during photosynthesis.

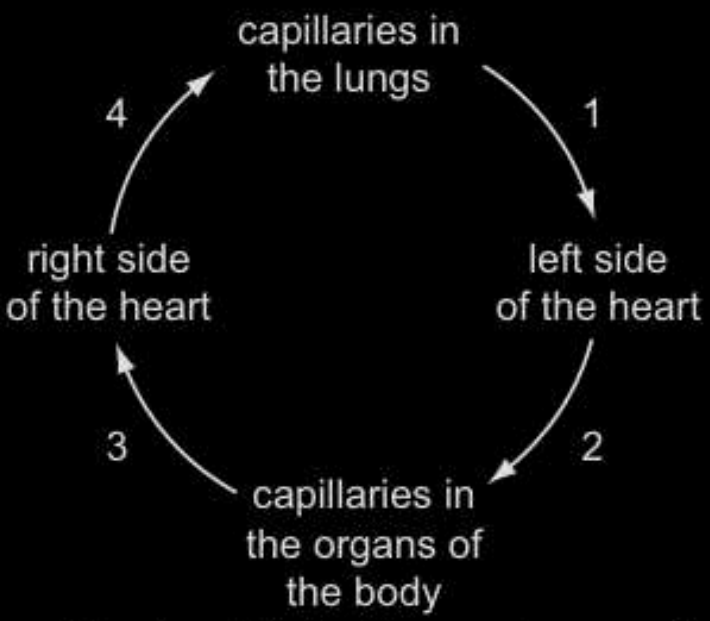
C sugars during translocation.

D water during transpiration.

May/June 2012 · Paper 1 · Variant 1 · Question 13

Correct answer: A

13 The diagram shows the direction of blood flow in the human body.



Which numbered stages have blood containing the most oxygen?

A 1 and 2 **B** 2 and 3 **C** 3 and 4 **D** 4 and 1

May/June 2012 · Paper 1 · Variant 1 · Question 14

Correct answer: D

14 Which row in the table describes the features of the pulmonary vein?

	feature of pulmonary vein		
	blood	lumen	muscle layer
A	deoxygenated	narrow	thin
B	deoxygenated	wide	thick
C	oxygenated	narrow	thick
D	oxygenated	wide	thin

May/June 2012 · Paper 1 · Variant 1 · Question 15

Correct answer: C

15 Which is a difference between plasma and tissue fluid?

	plasma	tissue fluid
A	less dissolved glucose	more dissolved glucose
B	dissolved glucose	no dissolved glucose
C	more protein molecules	fewer protein molecules
D	no white blood cells	white blood cells

May/June 2012 · Paper 1 · Variant 1 · Question 18

Correct answer: D

18 Which process does **not** require energy?

A focussing of eye

B growth of hair

C secretion of sweat

D tissue respiration

May/June 2012 · Paper 1 · Variant 1 · Question 20

Correct answer: A

20 In a kidney machine, protein molecules are not lost from the blood.

How is loss of protein prevented?

A Membranes prevent protein molecules diffusing out of the blood.

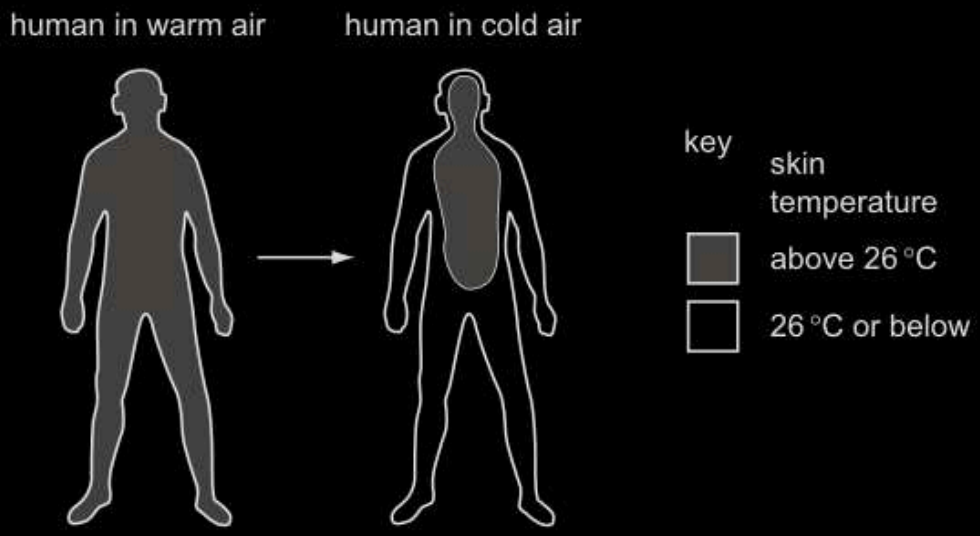
B Proteins are actively transported back into the blood.

C Proteins do not enter the kidney machine.

D The dialysis fluid contains protein, so there is no diffusion gradient.

Correct answer: A

21 The diagram shows skin temperature of a human when exposed to warm air and then exposed to cold air.



What causes the observed change in skin temperature on exposure to cold air?

A less blood flowing just below the skin

B less blood going to the heart and lungs

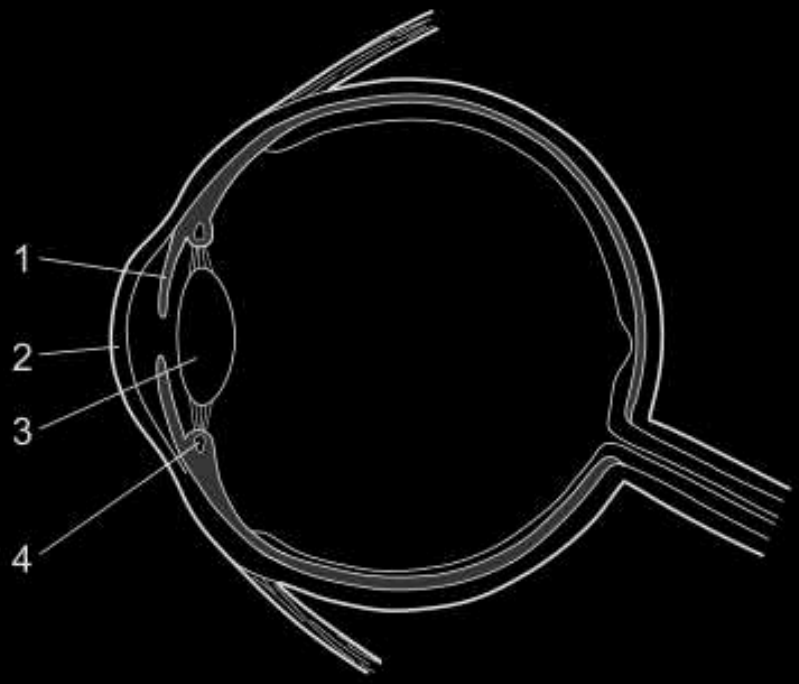
C more blood flowing just below the skin

D more blood going to the heart and lungs

May/June 2012 · Paper 1 · Variant 1 · Question 23

Correct answer: C

23 The diagram shows a section through an eye.



Which parts help to focus an image on the retina?

A 1 and 3 only **B** 1 and 4 only **C** 2, 3 and 4 **D** 3 and 4 only

May/June 2012 · Paper 1 · Variant 1 · Question 24

Correct answer: D

24 What happens when the body temperature of a human is controlled?

A body temperature is made more different from environmental temperature

B body temperature is made more different from ideal body temperature

C body temperature is made more similar to environmental temperature

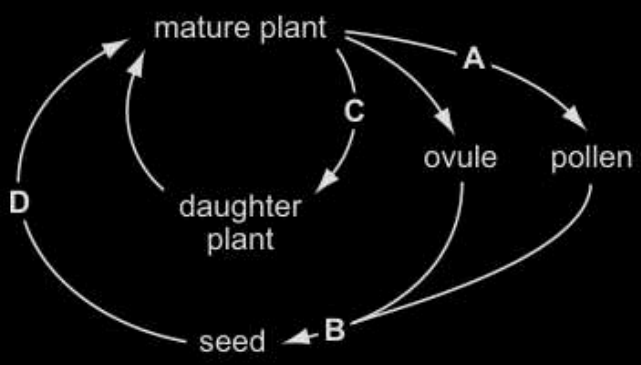
D body temperature is made more similar to ideal body temperature

May/June 2012 · Paper 1 · Variant 1 · Question 34

Correct answer: A

34 The diagram shows the life cycle of a species of plant.

During which stage does meiosis (reduction division) occur?



May/June 2012 · Paper 1 · Variant 1 · Question 38

Correct answer: A

38 A person has blood group A.

Which statement about his genotype is correct?

A At least one of his alleles is dominant.

B He must be heterozygous.

C He must be homozygous.

D His alleles are codominant.

May/June 2012 · Paper 1 · Variant 1 · Question 39

Correct answer: C

39 In fruit flies, the allele for an ebony coloured body is recessive to the allele for a grey coloured body. In an investigation, an ebony-bodied fly was crossed with a grey-bodied fly.

What will be the body colour of the offspring if the grey-bodied fly was heterozygous?

A all ebony

B all grey

C half ebony and half grey

D three-quarters grey and one-quarter ebony

May/June 2012 · Paper 1 · Variant 1 · Question 40

Correct answer: D

40 Which statement about dominant and recessive alleles is **not** correct?

A A dominant characteristic is seen in the phenotype of a heterozygote.

B A homozygous genotype may be either dominant or recessive.

C Recessive phenotypes always have two recessive alleles.

D The phenotype of a homozygote is always dominant.

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

May/June 2012 · Paper 1 · Variant 2 · Question 1

Correct answer: D

1 Some structures found in cells are listed.

1 cell wall

2 cell membrane

3 chloroplast

4 cytoplasm

5 nucleus

Which structures are found in both animal cells and plant cells?

A	1	2	4
B	1	2	3
C	2	3	5
D	2	4	5

May/June 2012 · Paper 1 · Variant 2 · Question 2

Correct answer: C

2 Which processes can **only** occur through a membrane?

	active transport	diffusion	osmosis
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

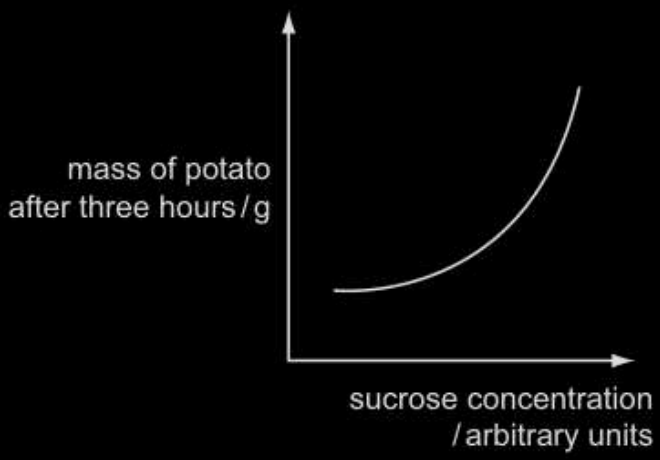
May/June 2012 · Paper 1 · Variant 2 · Question 3

Correct answer: D

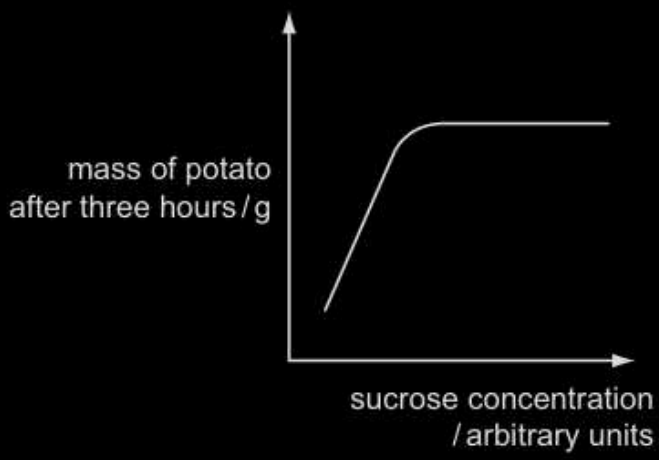
3 Identical pieces of potato are placed in sucrose solutions of different concentrations. After three hours, the mass of each potato piece is measured.

Which graph shows the results of this experiment?

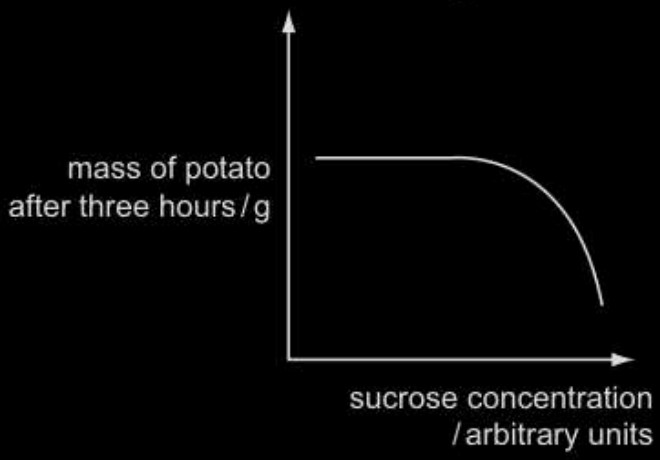
A



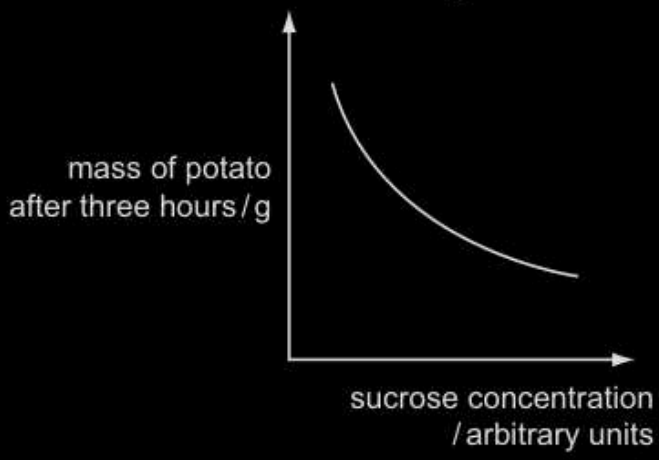
B



C



D



May/June 2012 · Paper 1 · Variant 2 · Question 4

Correct answer: C

4 Starch is digested to maltose by the enzyme amylase.

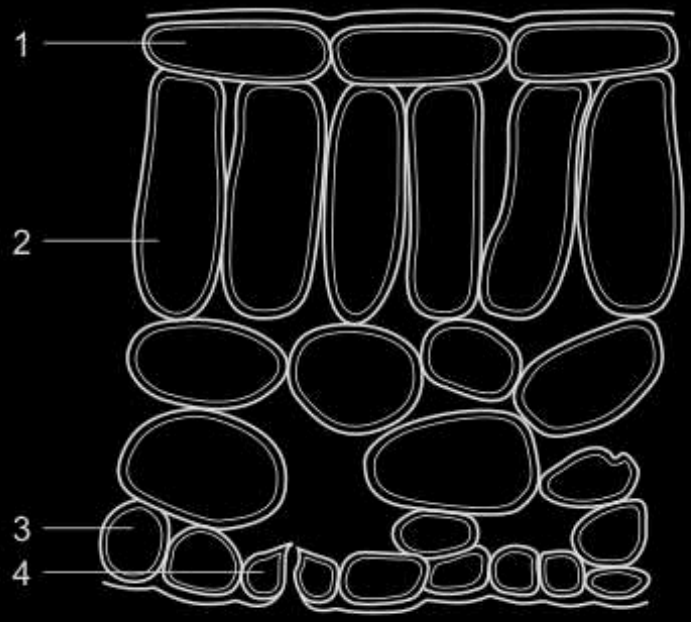
According to the 'lock and key' hypothesis, which is the 'key' and which is the 'lock'?

	'key'	'lock'
A	amylase	maltose
B	amylase	starch
C	starch	amylase
D	starch	maltose

May/June 2012 · Paper 1 · Variant 2 · Question 6

Correct answer: D

6 The diagram shows cells in a section through a leaf of a green plant. (No cell contents are shown.)



Which cells contain chloroplasts?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

May/June 2012 · Paper 1 · Variant 2 · Question 8

Correct answer: A

8 Which process is stimulated by adrenaline in the cells of the liver?

A breakdown of glycogen, increasing the blood glucose level

B breakdown of excess amino acids, forming urea

C breakdown of proteins, releasing amino acids into the blood

D conversion of excess blood glucose to glycogen

May/June 2012 · Paper 1 · Variant 2 · Question 9

Correct answer: D

9 Which fact about proteins makes them essential components of the human diet?

A They are an energy source.

B They are used in synthesising hormones.

C They can be stored as a food reserve.

D They provide amino acids for cell growth.

May/June 2012 · Paper 1 · Variant 2 · Question 10

Correct answer: B

10 Transpiration increases when conditions become

A less humid and less windy.

B less humid and more windy.

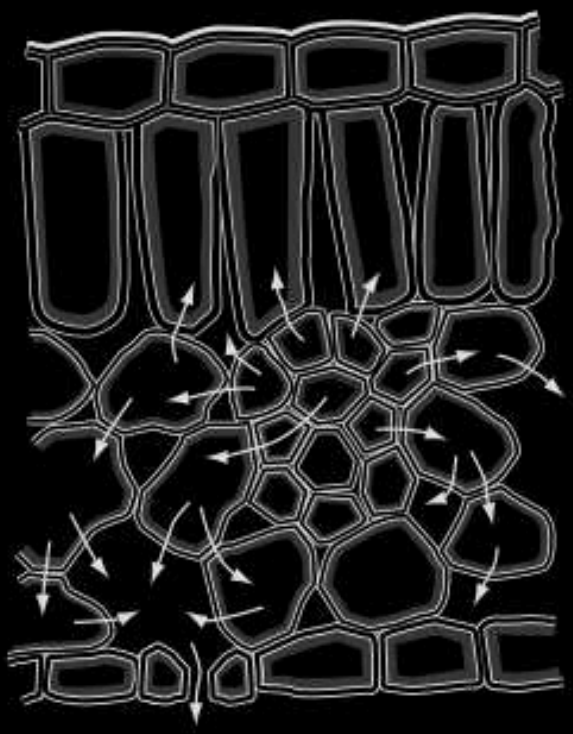
C more humid and less windy.

D more humid and more windy.

May/June 2012 · Paper 1 · Variant 2 · Question 11

Correct answer: D

11 The diagram shows a section through a green leaf.



The arrows represent the movement of

A carbon dioxide during respiration.

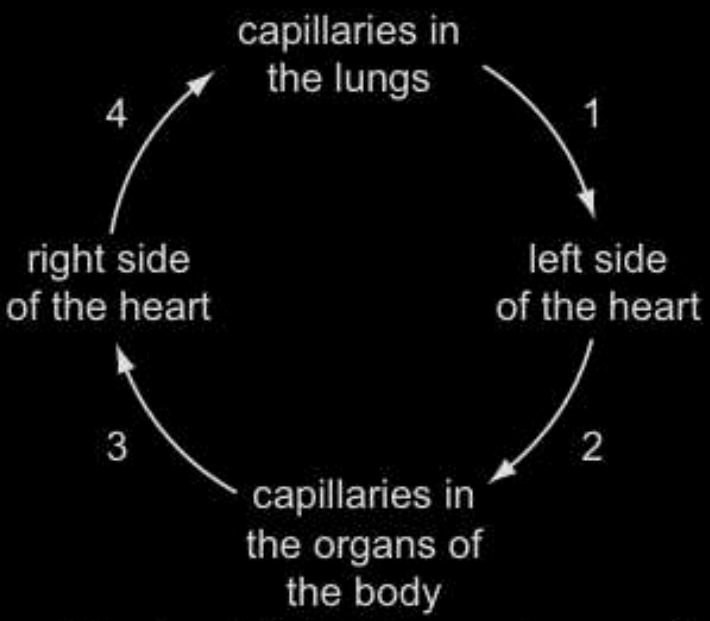
B oxygen during photosynthesis.

C sugars during translocation.

D water during transpiration.

Correct answer: A

12 The diagram shows the direction of blood flow in the human body.



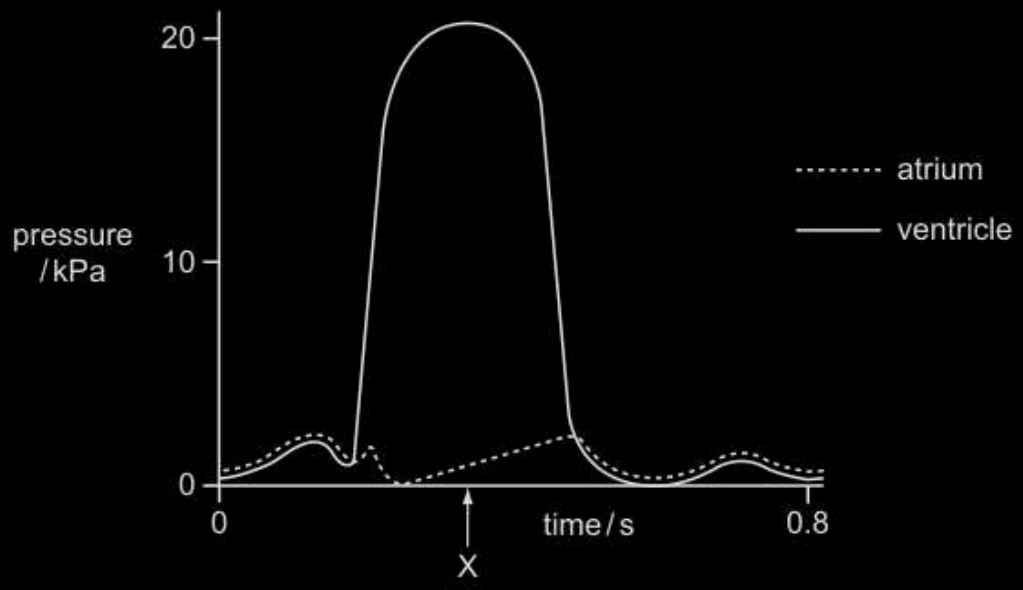
Which numbered stages have blood containing the most oxygen?

A 1 and 2 **B** 2 and 3 **C** 3 and 4 **D** 4 and 1

May/June 2012 · Paper 1 · Variant 2 · Question 13

Correct answer: B

13 The graph shows pressure changes in the left atrium and in the left ventricle during one heartbeat.



What is the state of the valves in the heart at time X?

	atrio-ventricular valve (bicuspid)	semi-lunar valve (in aorta)
A	closed	closed
B	closed	open
C	open	closed
D	open	open

May/June 2012 · Paper 1 · Variant 2 · Question 14

Correct answer: C

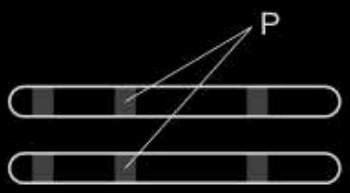
14 What is a difference between plasma and tissue fluid?

	plasma	tissue fluid
A	less dissolved glucose	more dissolved glucose
B	dissolved glucose	no dissolved glucose
C	more protein molecules	fewer protein molecules
D	no white blood cells	white blood cells

May/June 2012 · Paper 1 · Variant 2 · Question 19

Correct answer: D

19 The diagram shows a pair of chromosomes from the same cell.



What do the lines labelled P point to?

A the site of an allele made up of two or more genes which are always the same

B the site of an allele made up of two or more genes which might be different

C the site of a gene made up of two or more alleles which are always the same

D the site of a gene made up of two or more alleles which might be different

May/June 2012 · Paper 1 · Variant 2 · Question 20

Correct answer: D

20 Which substances can be removed from the blood by kidney machines?

A glucose, glycogen and proteins

B glucose, proteins and urea

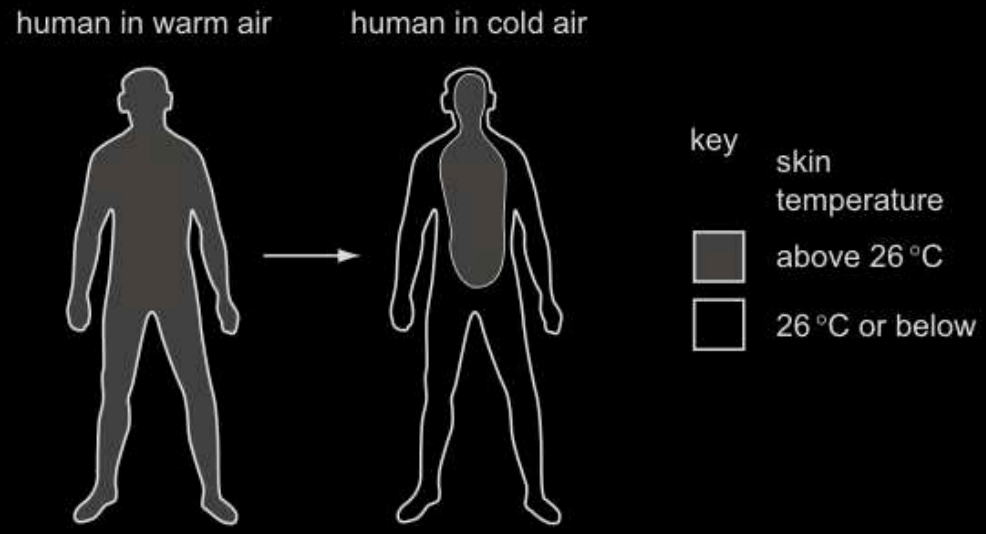
C glycogen, urea and some salts

D urea and some salts only

May/June 2012 · Paper 1 · Variant 2 · Question 21

Correct answer: A

21 The diagram shows skin temperature of a human when exposed to warm air and then exposed to cold air.



What causes the observed change in skin temperature on exposure to cold air?

A less blood flowing just below the skin

B less blood going to the heart and lungs

C more blood flowing just below the skin

D more blood going to the heart and lungs

May/June 2012 · Paper 1 · Variant 2 · Question 22

Correct answer: D

22 A child is frightened by a loud noise and shouts for help.

In which order are the different types of neurone involved in this response?

	involved first	→	involved last
A	motor neurone	relay neurone	sensory neurone
B	motor neurone	sensory neurone	relay neurone
C	sensory neurone	motor neurone	relay neurone
D	sensory neurone	relay neurone	motor neurone

May/June 2012 · Paper 1 · Variant 2 · Question 23

Correct answer: B

23 Which of these is a sign of diabetes mellitus?

A lack of haemoglobin

B raised blood glucose levels

C reduced urine production

D too much insulin

May/June 2012 · Paper 1 · Variant 2 · Question 25

Correct answer: A

25 Which listed substances are **all** broken down by the liver?

A alcohol, drugs and progesterone

B drugs, adrenaline and urea

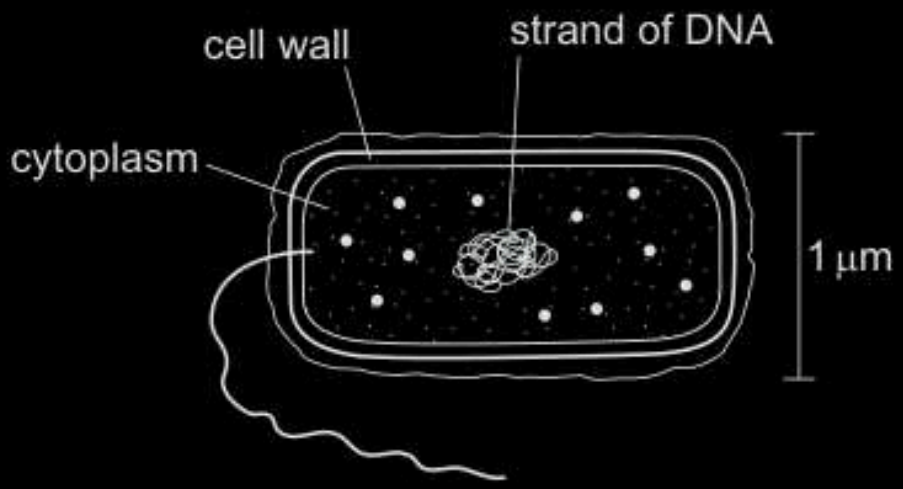
C oestrogen, water and drugs

D urea, alcohol and drugs

May/June 2012 · Paper 1 · Variant 2 · Question 27

Correct answer: A

27 The diagram shows a cell.



To which group of organisms does the cell belong?

A bacteria

B fungi

C plants

D viruses

May/June 2012 · Paper 1 · Variant 2 · Question 28

Correct answer: B

28 By which process do producers obtain energy?

A digestion

B photosynthesis

C translocation

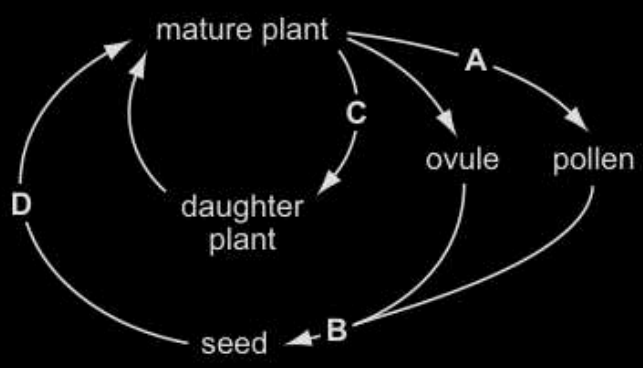
D transpiration

May/June 2012 · Paper 1 · Variant 2 · Question 34

Correct answer: A

34 The diagram shows the life cycle of a species of plant.

During which stage does meiosis (reduction division) occur?



May/June 2012 · Paper 1 · Variant 2 · Question 35

Correct answer: A

35 Pregnant women are advised to eat a diet with enough protein, calcium and iron for the developing baby.

Which needs are supplied by these constituents of food for the baby's bones, growing tissues and red blood cells?

	bones	growing tissues	red blood cells
A	calcium	protein	iron
B	carbohydrate	calcium	protein
C	iron	carbohydrate	calcium
D	protein	iron	carbohydrate

May/June 2012 · Paper 1 · Variant 2 · Question 36

Correct answer: D

36 How are sperm cells different from egg cells in size and in number?

	size of sperm cell	number of sperm cells
A	larger	fewer
B	larger	more
C	smaller	fewer
D	smaller	more

May/June 2012 · Paper 1 · Variant 2 · Question 37

Correct answer: D

37 The grids show the alleles and offspring of four pairs of parents.

Which grid shows codominance amongst the offspring?

A

♂ \ ♀	I^A	I^O
I^A	I^AI^A	I^AI^O
I^A	I^AI^A	I^AI^O

B

♂ \ ♀	I^A	I^O
I^A	I^AI^A	I^AI^O
I^O	I^AI^O	I^OI^O

C

♂ \ ♀	I^O	I^O
I^A	I^AI^O	I^AI^O
I^B	I^BI^O	I^BI^O

D

♂ \ ♀	I^A	I^B
I^A	I^AI^A	I^AI^B
I^O	I^AI^O	I^BI^O

May/June 2012 · Paper 1 · Variant 2 · Question 38

Correct answer: B

38 What is the primary function of DNA?

A controls the absorption of nutrients

B controls the production of protein

C controls the rate of reproduction

D controls the rate of mutation

May/June 2012 · Paper 1 · Variant 2 · Question 39

Correct answer: A

39 Two individuals are heterozygous for a particular gene.

Which statements about their offspring are correct?

1 Their offspring may show two phenotypes as a result of this gene.

2 Offspring with the recessive phenotype will be homozygous for this gene.

3 Their offspring have a 50 % chance of being heterozygous for this gene.

A 1, 2 and 3

B 1 and 2 only

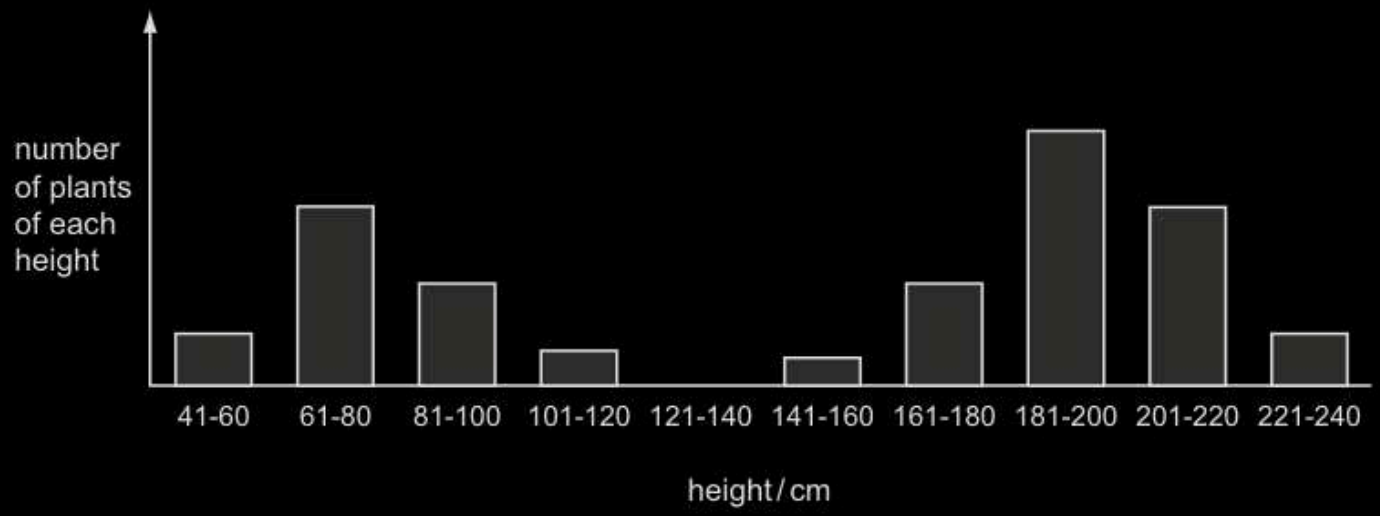
C 1 and 3 only

D 2 and 3 only

May/June 2012 · Paper 1 · Variant 2 · Question 40

Correct answer: C

40 The heights of 500 pea plants of the same age were measured to the nearest 20 cm. The results are shown in the chart below.



Variation in height of these pea plants shows

A continuous variation only.

B discontinuous variation only.

C both continuous and discontinuous variation.

D neither continuous nor discontinuous variation.

1 **(a)** **(i)** Name the process by which plants manufacture carbohydrates from raw materials.

..... [1]

Starch is an insoluble carbohydrate stored inside plant cells.

(ii) Explain why starch is a more suitable storage substance than the soluble sugar glucose.

.....

.....

..... [2]

(b) Before a plant can use it, the stored starch must first be broken down by an enzyme. Fig. 1.1 shows the '*lock and key*' hypothesis of how enzymes work.

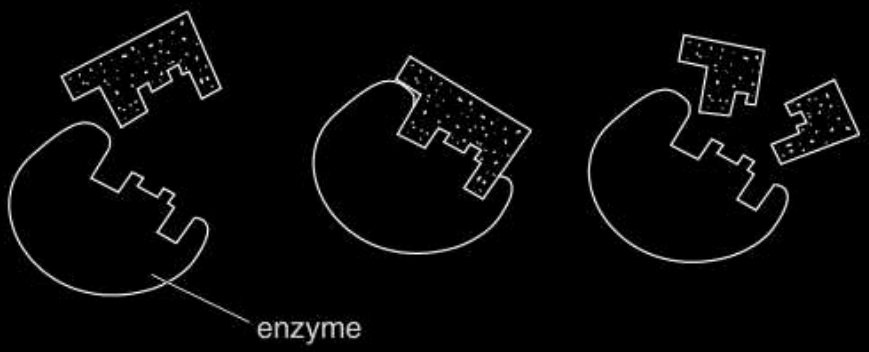


Fig. 1.1

Describe the '*lock and key*' hypothesis of enzyme action shown in Fig. 1.1.

.....

.....

.....

..... [3]

(c) In an investigation, two plants were grown in a solution containing mineral ions including nitrate and magnesium. Plant **A** was provided with air containing oxygen and plant **B** was provided with air from which the oxygen had been removed. Fig. 1.2 shows the plants after a period of growth in these conditions.

For Examiner's Use

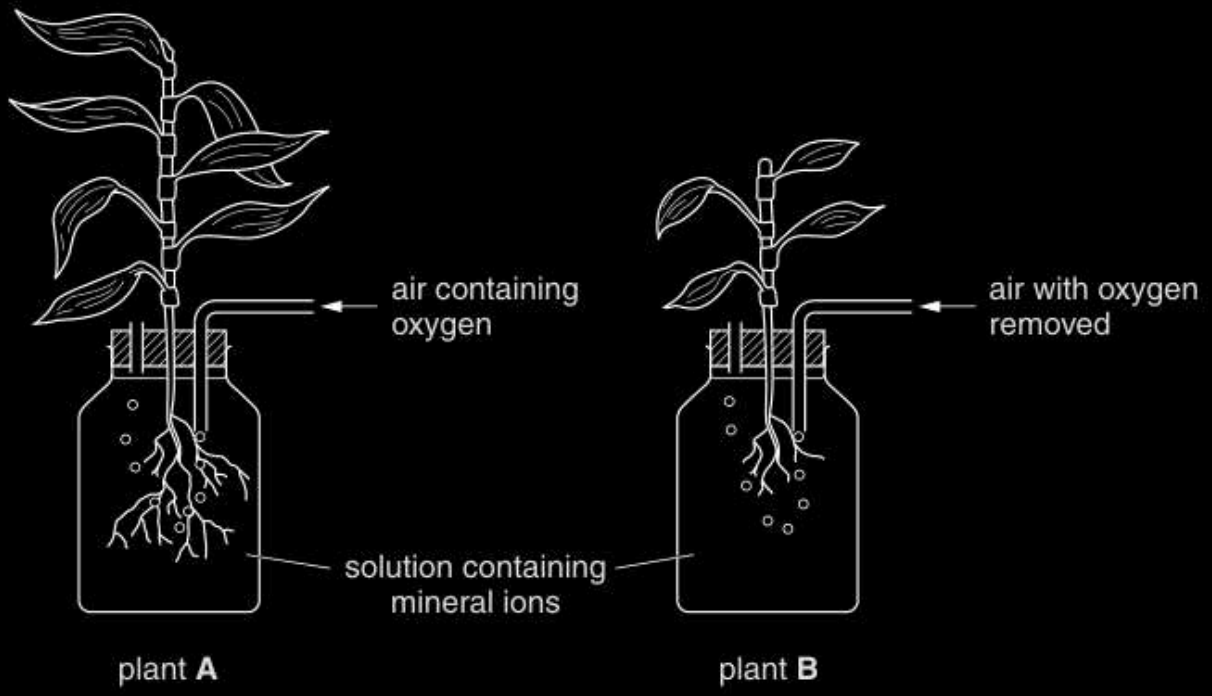


Fig. 1.2

Suggest reasons for the increased growth of the plant in the solution supplied with air containing oxygen.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

1 **(a)** **(i)** photosynthesis; [1]

(ii) starch stays inside cell / glucose can move R glucose used up;
does not lower water potential inside cell;
prevents water gain by cells;
correct ref. osmosis / diffusion; [2]

(b) correct ref. active site;

substrate;
perfect fit / complementary shape / specificity;
products / named suitable product / small molecules;
products leave + active site / enzyme re-useable / unchanged; [3]

(c) respiration;
root (cells / hairs);
increased uptake of ions;
by active transport;
ref. energy requirement of active transport;
(increased) chlorophyll production + magnesium;
(increased) photosynthesis (since more chlorophyll);
(increased) amino acid / protein production + nitrates; [4]

[Total: 10]

3 Cystic fibrosis is a genetic condition in humans that results from a failure to inherit a particular dominant allele of a gene.

For Examiner's Use

(a) State where genes are found in a cell.

..... [1]

(b) **(i)** Use a fully labelled genetic diagram to show how cystic fibrosis is inherited by the children of two heterozygous parents. Use the letter **D** to represent the dominant allele and **d** to represent the recessive allele.

[3]

(ii) State the expected ratio of phenotypes in the children.

..... [1]

Fig. 3.1 shows some of the main regions of the alimentary canal in a person.

For Examiner's Use

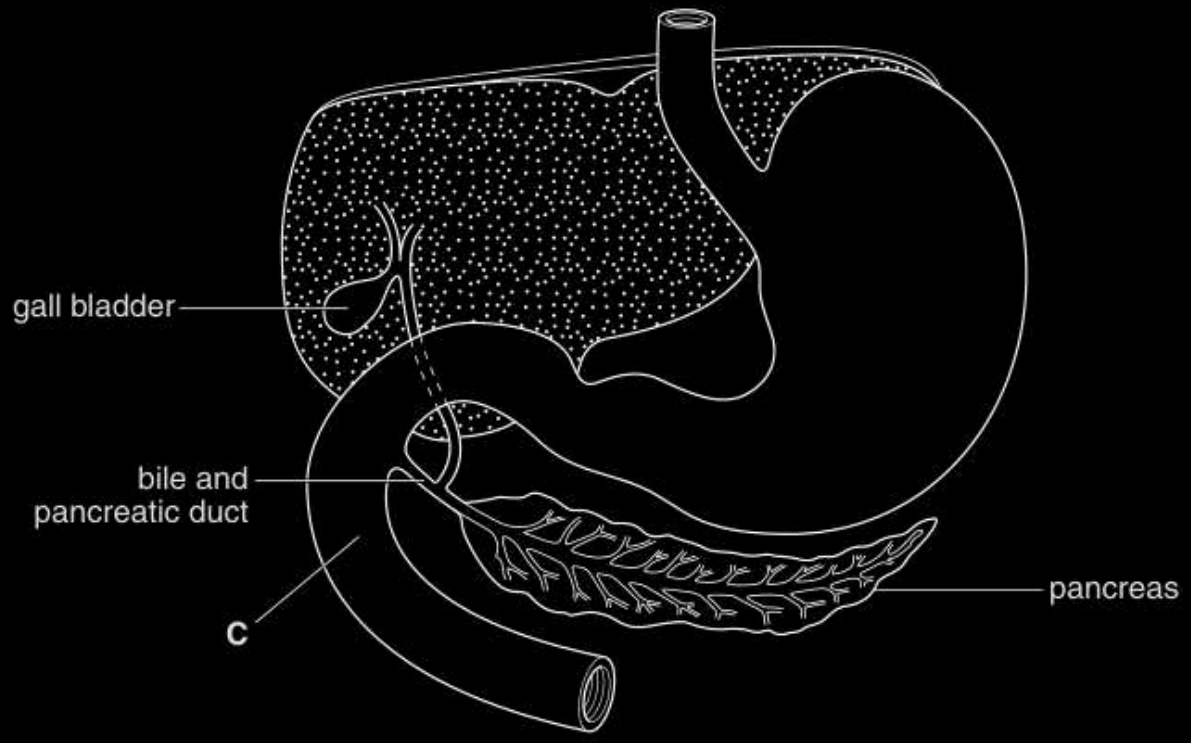


Fig. 3.1

(c) State the name of region **C**.

.....

[1]

(d) One effect of cystic fibrosis is that the bile and pancreatic duct becomes blocked with mucus. Suggest why a person whose bile and pancreatic duct is blocked may find it difficult to gain weight despite eating a balanced diet.

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

3 **(a)** nucleus / chromosomes; [1]

(b) **(i)** Dd + Dd;

correctly shown gametes;

correctly drawn and completed punnett square or gamete linkage; [3]

(ii) phenotype ratio correctly expressed and identified
(A. ratio / percentage / words); [1]

(c) duodenum / small intestine; [1]

(d) reduction in enzymes / pancreatic juice entering duodenum;
correct ref. to (reduced) bile action;
less digestion / emulsification AW;
especially of fats;
fewer molecules to absorb / less absorption qualified;
and use for assimilation into larger molecules;
used in growth;
ref. less fat stored /AW; [4]

[Total: 10]

4 Fig. 4.1 shows a horizontal section of the human eye and the pathway taken by light rays as they leave an object.

For Examiner's Use

(a) Complete the diagram by continuing the lines from the object to show how the light rays produce a focussed image on the retina.

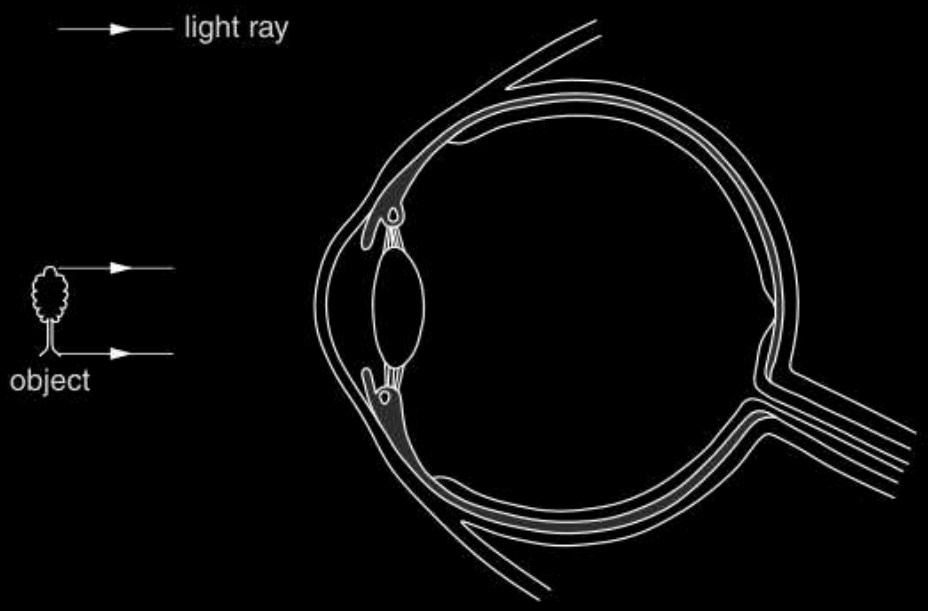


Fig. 4.1

[3]

(b) **(i)** State how the appearance of the pupil in the eye will change when a person moves from dim light into an area of bright light.

..... [1]

(ii) Explain how this change is brought about by structures in the eye.

.....

.....

..... [2]

(c) The change in appearance of the pupil when entering an area of bright light is a reflex action.

(i) Define the term *reflex action*.

.....

..... [2]

(ii) Suggest why drugs that prevent this reflex action from occurring should be avoided.

.....

.....

..... [2]

[Total: 10]

4 **(a)** rays continue parallel until hit cornea;
rays converge at cornea;
rays converge at lens;
meet before retina + continue to hit retina; [3]

(b) **(i)** narrows / decreases in size or diameter / constricts (R contracts); [1]

(ii) iris + muscles;
circular + contract; [2]

(c) **(i)** fast / rapid / quick;

reaction / response + (to) stimulus;

automatic / involuntary / no involvement of conscious thought AW; / cannot be controlled
R no involvement of brain [2]

(ii) too much light allowed to enter AW;
damage to retina / rods / cones / light-sensitive cells; [2]

[Total: 10]

For Examiner's Use

5 **(a)** Fig. 5.1 shows the arrangement of teeth in the lower jaw of an adult person.

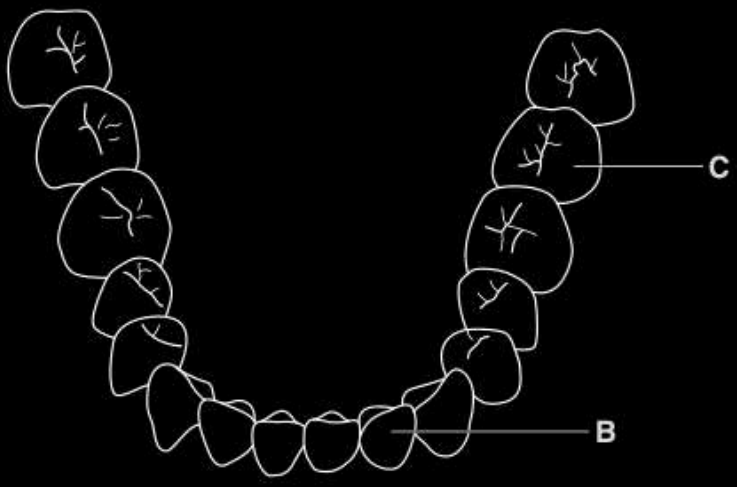


Fig. 5.1

Identify the types of teeth labelled **B** and **C** and state one function of each.

type **B** .....

function ..... [2]

type **C** .....

function ..... [2]

(b) During a single day two people ate the same amount of food containing a large amount of carbohydrate.

Person D ate the food in three equal portions at 7.00 am, 1.00 pm and 8.00 pm, following which he brushed his teeth using toothpaste before going to bed.

Person E ate the food in smaller portions more frequently during the day and did not brush her teeth before going to bed.

(i) List the chemical elements that make up carbohydrates.

..... [1]

(ii) If persons **D** and **E** continue their eating habits for several years, suggest in which person dental decay will occur first. Explain your reasoning.

person .....

explanation .....

.....

.....

.....

..... [3]

[Total: 8]

Section B

For Examiner's Use

Answer **both** questions in this section.

Write your answers in the spaces provided.

5 **(a)** **B** incisor;

cutting / biting / nibbling R holding; [2]

C molar (R pre-molar / wisdom);

grinding / crushing / chewing R shearing; [2]

(b) **(i)** carbon + hydrogen + oxygen (A C H O); [1]

(ii) (person) **E**; [1]
(more frequent meals) allows more sugar build up on teeth;
no brushing to remove bacteria / sugar / plaque;
more acid contact with teeth;
ref. toothpaste is alkaline;
ref no / less neutralisation of acid; [2]

[Total: 8]

[Total: 50]

Section B

6 **(a)** State the function of phloem in a plant.

.....

.....

.....

..... [2]

(b) Describe how a molecule of water moves through a plant from the soil until it enters the atmosphere through the leaves.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [8]

[Total: 10]

6 **(a)** transport sucrose / sugar / amino acids (R glucose / food);
in solution;
ref. direction of movement (A around the plant); [2]

(b) osmosis;
through cell membrane;
of root hair;
ref. movement from cell to cell;
ref. movement through or between cell walls;
enters xylem;
transpiration pull / stream / capillarity / molecular cohesion / root pressure;
evaporation / ref. water vapour;
from mesophyll cells / into air spaces;
ref. water potential gradient;
diffusion;
through stomata; [8]

[Total: 10]

7 **(a)** **(i)** State **one similarity** and **one difference** in the functions of the urethra in a male and in a female adult person.

For Examiner's Use

similarity .....

.....

difference .....

..... [2]

(ii) State the **differences** between male and female human gametes in terms of size, numbers and mobility.

.....

.....

.....

..... [3]

(b) Describe the **advantages** and **disadvantages** of surgical methods of birth control.

advantages .....

.....

.....

.....

.....

disadvantages .....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

Section C

For Examiner's Use

Answer **either** question 8 **or** question 9.

Write your answers in the spaces provided.

7 **(a)** **(i)** carries urine (R just urea) in male and female;
carries semen / sperms / gametes AW in male; [2]

(ii) ref. size comparison;
ref. qualified numbers comparison;
ref. mobility comparison; [3]

(b) (*advantages*)
only needs to be undergone once AW (e.g. less trouble);
high reliability / effectiveness AW; [2]

(*disadvantages*)
does not protect against sexually transmitted diseases;
ref. risks of surgery / anaesthetic;
not a temporary solution;
difficult / expensive to reverse;
need access to medical services AW; [4 max]

[Total: 10]

Section C

1 Fig. 1.1 shows the human eye in horizontal section.

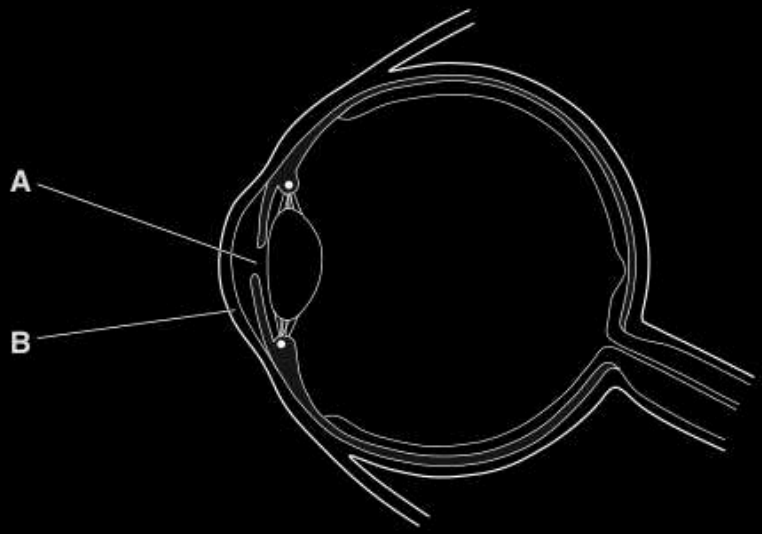


Fig. 1.1

(a) (i) Identify **A** and **B** that are labelled on Fig. 1.1.

A .....

B ..... [2]

(ii) Describe what happens to **A** when light entering the eye becomes less intense.

.....

..... [1]

(iii) Place a letter **Z** on Fig. 1.1 where a response occurs as a result of a reflex action. [1]

(b) In some people's eyes, the retina becomes completely detached from the tissues beneath. Explain how this will affect their ability to see.

.....

.....

.....

.....

..... [3]

(c) As people get older, cloudy (opaque) patches sometimes form in the lens of the eye. These are called cataracts.
Suggest how cataracts might affect the ability of the lens to carry out its function.

For Examiner's Use

.....

.....

.....

..... [3]

[Total: 10]

1 **(a)** **(i)** A pupil / aqueous humour / anterior chamber;
B cornea; [2]

(ii) dilates / gets big / enlarges / opens / gets wide / expands; [1]

(iii) **Z** on ciliary muscle / iris / external muscle; [1]

(b) reasonable ref. light receptors / sensitive cells / rods / cones;

not in contact with optic nerve;

no impulses;

to brain;

no picture formed / unable to see / blindness;
(R blurred vision or reduced visionary powers)

less / no nutrition for retina; [max. 3]

(c) failure to focus (all) light (rays);

blurred image AW;

any reference to the passage of light rays being impaired
(e.g. reflection / refraction / deflection / absorption / convergence);

ref. possible change in elasticity / ability to accommodate;

faded colour vision; [max. 3]

2 Fig. 2.1 shows some of the information on the packets of two breakfast cereals.

For Examiner's Use

Cereal C

Nutrition Information

Typical value per 100 g

ENERGY	1623 kJ	
PROTEIN	13 g	
CARBOHYDRATE	78 g	
of which sugars	24 g	
starch	54 g	
FAT	1.5 g	
of which saturates	0.5 g	
FIBRE	2.5 g	
SODIUM	0.4 g	
SALT	1 g	
VITAMINS:	**(% GDA)**	
VITAMIN D	7.4 µg	(147)
VITAMIN C	88 mg	(147)
THIAMIN (B_1)	2.1 mg	(147)
RIBOFLAVIN (B_2)	2.4 mg	(147)
NIACIN	26.5 mg	(147)
VITAMIN B_6	2.9 mg	(147)
FOLIC ACID	294 µg	(147)
VITAMIN B_{12}	1.47 µg	(147)
MINERALS:		
IRON	16.2 mg	(73)

Cereal D

Nutrition Information

Typical value per 100 g

ENERGY	1600 kJ	
PROTEIN	10 g	
CARBOHYDRATE	68 g	
of which sugars	20 g	
starch	48 g	
FAT	5 g	
of which saturates	0.9 g	
FIBRE	9 g	
SODIUM	0.01 g	
SALT	0.03 g	
VITAMINS:	**(% GDA)**	
THIAMIN (B_1)	1 mg	(73)
RIBOFLAVIN (B_2)	2.3 mg	(145)
NIACIN	13.1 mg	(73)
VITAMIN B_6	2.9 mg	(145)
FOLIC ACID	290 µg	(145)
VITAMIN B_{12}	0.73 µg	(73)
MINERALS:		
IRON	10.2 mg	(73)

Fig. 2.1

(a) The Guideline Daily Amount (GDA) of energy for an average adult is 8400 kJ.

(i) Calculate the percentage of this GDA a person would obtain by eating **one 25 g** serving of Cereal **D**. Show your working in the space provided.

Answer .....% [3]

(ii) State how the daily energy requirement of a hard-working farmer would differ from the GDA described above.

..... [1]

(b) Cereal **D** is considered to be better for people suffering from constipation than Cereal **C**. Suggest a reason for this.

For Examiner's Use

..... [1]

(c) Rickets is a condition that affects some children.

(i) Describe the symptoms of rickets.

.....

..... [2]

(ii) State which cereal, **C** or **D**, should be eaten by children to prevent rickets and explain your answer.

cereal .....

explanation .....

.....

..... [2]

[Total: 9]

2 **(a)** **(i)** Mark as follows:
Award 2 marks for a correct answer with no working;;
Award 3 marks for a correct answer with any working;;;
Award 2 marks max. for correct working with no or wrong answer;;

Answer: 4.7(6)(%) (A 4.7 / 4.8);;

Examples of correct working are as shown below :

$\frac{1600}{4}$ / OR $1600 \times \frac{25}{100}$ / OR 400 (;)

$\frac{400}{8400}$ (;) [3]

(ii) higher (R ref. to higher GDA); [1]

(b) more or a high amount AW fibre / roughage; [1]

(c) **(i)** bones;
soft / weak / deformed / bent / ref. bandy legs ; [2]
(R bent 'legs' – that is the effect of the knee joint.)
(Mark the first effect in a list)

(ii) C;

vitamin D / uptake of calcium / calcium used by bones AW; [max. 2]

3 In Fig. 3.1, the line drawn represents the cell membrane of a plant cell.

For Examiner's Use

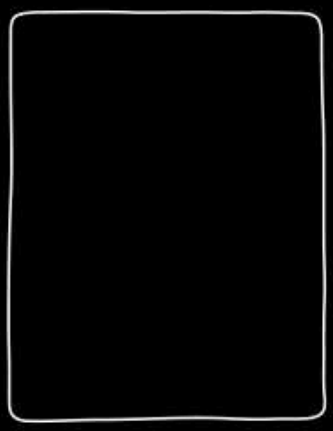


Fig. 3.1

(a) On Fig. 3.1 draw, name and label

(i) a structure that gives the cell its rigid shape,

(ii) a structure that contains chromosomes,

(iii) a structure that contains varying amounts of water, ions and sugars. [3]

(b) List three structural changes that must occur in young, unmodified plant cells as they develop into xylem tissue.

1 .....

2 .....

3 ..... [3]

(c) A small, leafy branch is cut from a tree. After some hours, the stem of the branch remains firm but the leaves become limp. Suggest an explanation for this.

stem remains firm .....

.....

.....

leaves become limp .....

.....

.....

..... [5]

[Total: 11]

3 **(a)** **(i)** cell wall + correct drawing (outside existing line);

(ii) nucleus + correct drawing (must be in cytoplasm);

(iii) vacuole / cytoplasm + correct drawing (if vacuole, must be larger than the nucleus.); [3]

If no marks scored through unacceptable drawings, allow max. 1 if all 3 are shown in the correct positions – vacuole in the middle, cell wall on the outside and nucleus between.

(b) elongation of cell (R cylindrical);

loss of cell contents / or one named content (A ref. death of cell / hollow)
(R chloroplasts / dissolving of cell contents.);

loss of end walls;

addition of thickening / strengthening material / lignin; [max. 3]

(c) (*stem*) contains xylem;

(xylem is) thickened / strengthened / contains lignin;

provides support;

(*leaf*) xylem less important for support;

loss of water / transpiration;

water not replaced / doesn't reach leaves;

cells become flaccid / lose turgor AW (R plasmolysed);

cells unable to support weight of leaf / wilts; [max. 5]

4 Fig. 4.1 shows a section through a ripe fruit of a tomato plant.

For Examiner's Use

Fig. 4.1

(a) Suggest and describe how seeds in this fruit may be dispersed.

.....

.....

.....

.....

..... [5]

(b) Normal tomato plants are diploid (2n), having two sets of chromosomes in each cell. However, some tomato plants are triploid, having three sets of chromosomes (3n) in each cell. This can be an advantage as these plants produce larger fruits.

(i) Suggest a reason why it may be a **disadvantage** to have three sets of chromosomes.

..... [1]

(ii) Suggest why the triploid condition is more common in plants than in animals.

.....

..... [2]

(c) Explain how two parents who do not have Down's syndrome can produce a child who has the syndrome.

.....

.....

.....

.....

..... [3]

[Total:11]

4 **(a)** animal dispersal (A plausible named animal);

attraction + colour / succulence / smell;

(fruit) eaten / ref. food;

time for digestive processes AW;

seeds are **not** digested;

pass out with faeces / egested AW;

ref. seeds discarded elsewhere; [max. 5]

(b) **(i)** ref. to interference with the sexual reproductive process / infertility;

meiosis / reduction division not possible;

gamete formation / seed production impaired AW; [max. 1]

(ii) (A reverse argument re. diploid animals)

many plants can reproduce asexually / propagate vegetatively;

mitosis unaffected / meiosis not involved;

triploids more hardy / survive better AW; [max. 2]

(c) chromosomes;

mutation / ref. to meiosis AW ;

gamete + one extra / 24 (chromosomes);

one extra (chromosome) / 47 + inherited / handed on; [max. 3]
(mark not awarded for saying offspring have 47 chromosomes)

5 **(a)** Fig. 5.1 shows the effect of temperature on the activity of enzyme **E**.

For Examiner's Use

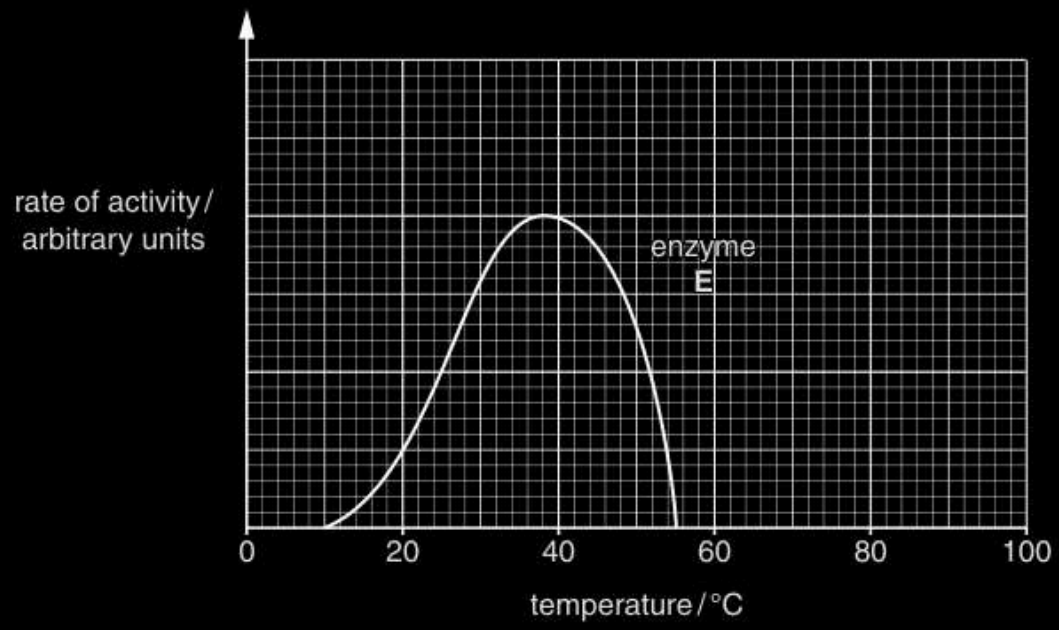


Fig. 5.1

(i) State the optimum temperature for enzyme **E**. ..... [1]

(ii) Suggest a possible identity for enzyme **E**, where it is found, and its function.

identity of enzyme **E** .....

where it is found .....

function .....

[3]

(b) Fig. 5.2 shows the effect of temperature on the activity of another enzyme, **F**.

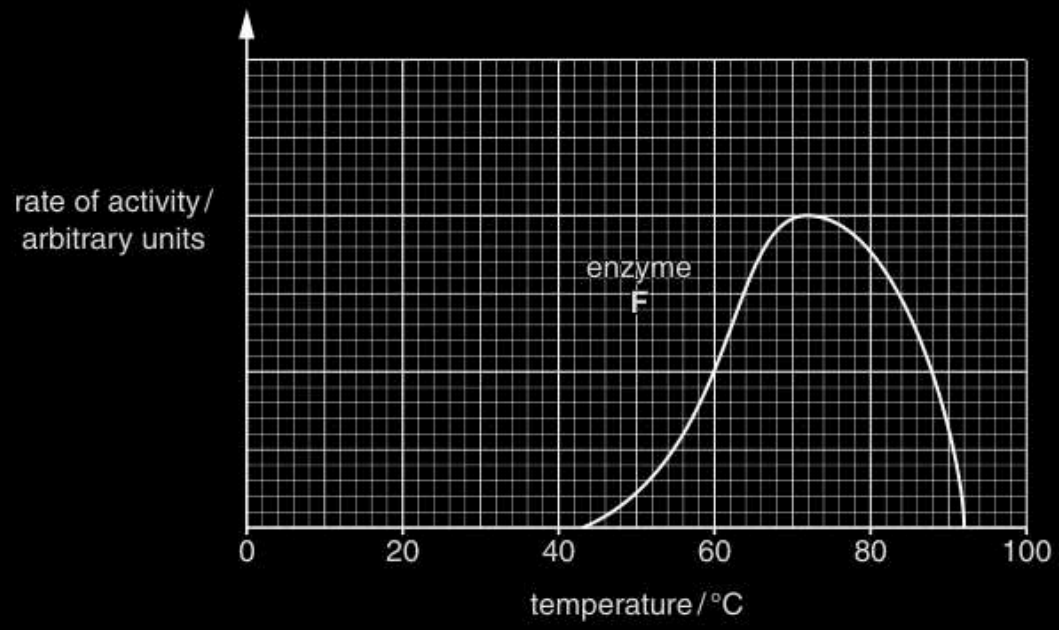


Fig. 5.2

State and explain what would happen to the activity of enzyme **E** at the optimum temperature for enzyme **F**.

For Examiner's Use

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 9]

Section B

For Examiner's Use

Answer **both** questions in this section.

Write your answers in the spaces provided.

5 (a) (i) 37–39 °C (A range or any temperature within the range); [1]

(ii) (If first line blank, then Max. 1 for location if correct for function)

A any human enzyme or enzyme from biotechnology ;

location (specific rather than general) correct for enzyme;

function correct for named enzyme (substrate + product); [3]

(b) denatured / destroyed;

stops working / no product made / cannot be used again;

relevant ref. active site;

substrate no longer fits / no formation enzyme-substrate complex AW;

correct use of lock and key idea;

possible effect of heat on shape of substrate molecule; [max. 5]

[Total: 50]

Section B

8 **(a)** **(i)** Explain the term *osmosis*.

.....

.....

.....

.....

.....

..... [4]

(ii) Explain the ways in which *active transport* is different from *osmosis*.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Explain how it is possible for oxygen in the lungs to diffuse rapidly into the blood.

.....

.....

.....

.....

.....

..... [2]

[Total: 10]

8 **(a) (i)** (osmosis) ref. diffusion;

of water ;

from high to low (water) molecular concentration or water potential / low to high solute concentration.;

Ref. to partially permeable membrane AW;

passive / no energy required ;

e.g. of an application; [max. 4]

(ii) (active transport) movement of ions / molecules / substances;

against a concentration gradient;

needs living or cell membrane;

energy required;

from respiration;

e.g. of an application; [max. 4]

(b) thin walls / one cell thick + alveoli / air sacs / capillaries;

haemoglobin + oxygen absorption;

ref. relatively high concentration of oxygen in the lungs / oxygen regularly replenished / more oxygen in the air than in the blood;

continuous movement of blood;

large surface area / alveoli have rich supply / surrounded by capillaries; [max. 2]

9 **(a)** Describe the removal of excretory products from the human body.

For Examiner's Use

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [8]

(b) Explain why the removal of faeces from the body is **not** regarded as excretion.

.....

.....

.....

..... [2]

[Total: 10]

9 **(a)** toxic / poisonous materials (R harmful);

waste products + metabolism AW;
(R refs. to salts)

urea;

liver / blood + kidneys;

urine;

urethra;

sweat;

skin / (epi)dermis;

carbon dioxide;

from cells / blood + alveoli;

exhaled / breathed out / expelled or removed from lungs AW; [max. 8]

(b) undigested food;

cellulose / fibre / lignin / roughage;

not from metabolism AW; [max. 2]

October/November 2011 · Paper 1 · Variant 1 · Question 1

Correct answer: A

1 A human cheek cell and a spongy mesophyll cell from a leaf are examined under a microscope.

Which structures are seen in both cells?

A cell membrane, nucleus and cytoplasm

B cell wall, cell membrane and nucleus

C cytoplasm, cell wall and cell membrane

D nucleus, cytoplasm and cell wall

Correct answer: D

3 The mass of a cube of fresh potato is found. It is then placed in a test-tube containing a dilute solution of sucrose. After an hour, its mass has increased.

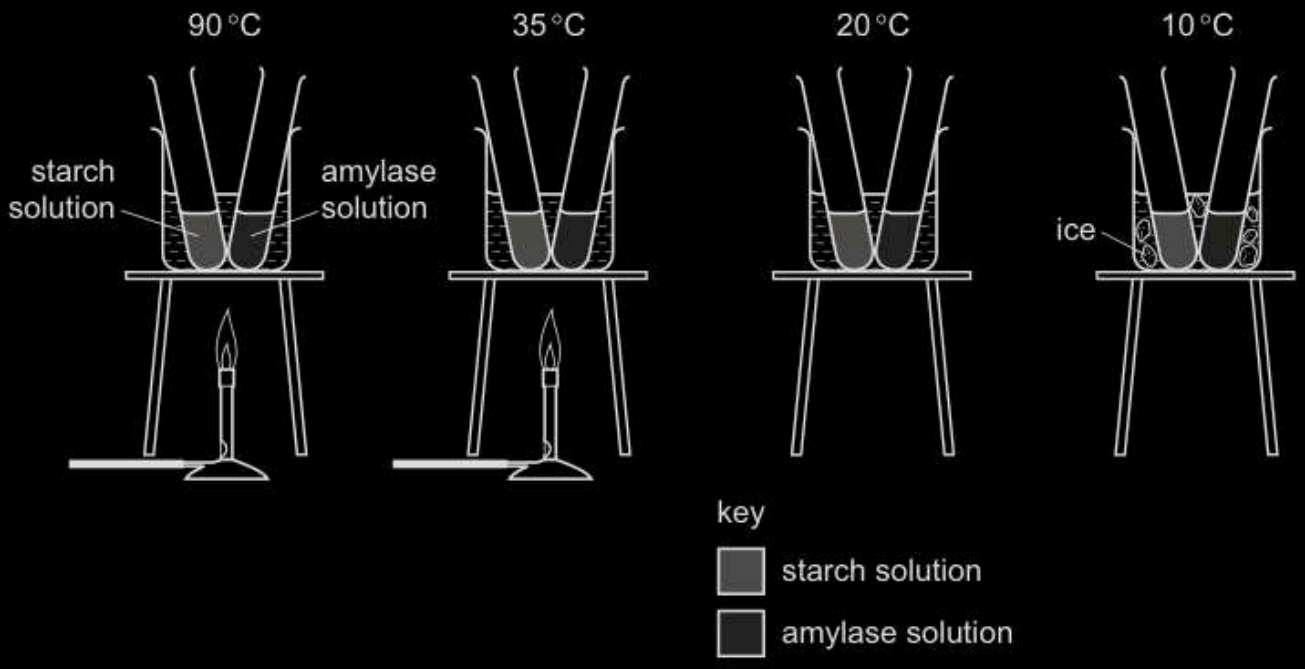
Which process has occurred and what has happened to the concentration of the sucrose in the solution in the test-tube?

	process	sucrose concentration
A	active transport	decreased
B	active transport	increased
C	osmosis	decreased
D	osmosis	increased

Correct answer: A

4 The diagram shows an experiment on amylase.

Each beaker contains water at the temperature shown.



After five minutes, each test-tube of amylase is poured into the test-tube of starch solution in the same beaker.

After leaving the tubes for 5 minutes, samples of the mixture are tested with iodine solution and are then tested again at 5 minute intervals.

Which results are expected?

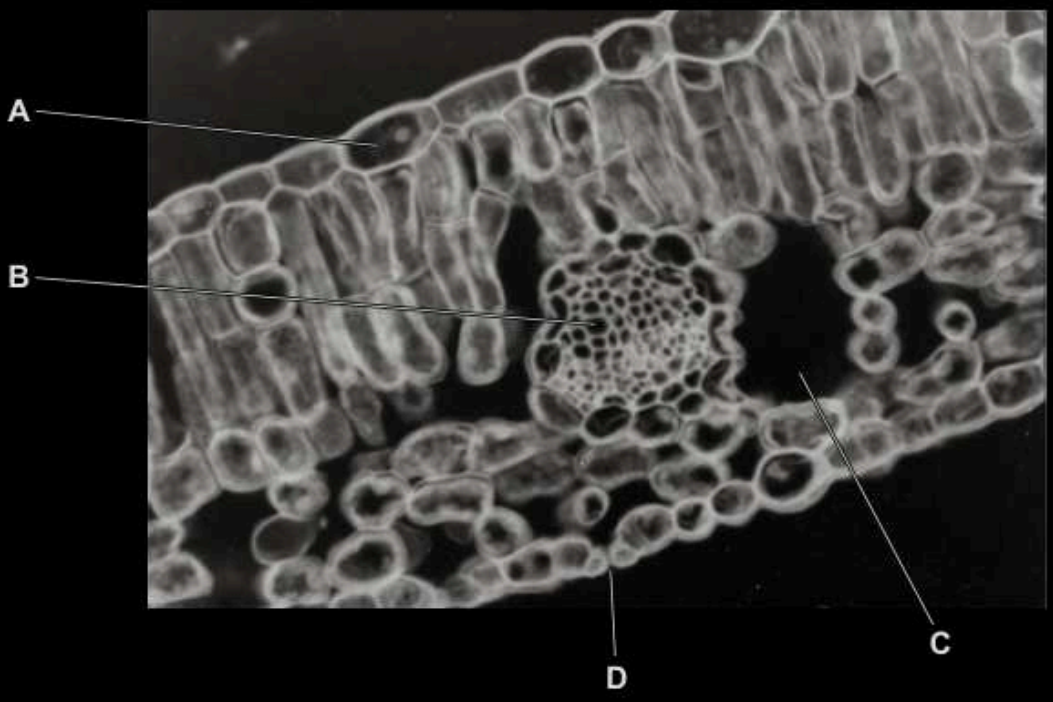
	90 °C	35 °C	20 °C	10 °C
A	blue-black after 30 minutes	goes yellow-brown immediately	goes yellow-brown after 5 minutes	blue-black after 30 minutes
B	blue-black after 30 minutes	goes yellow-brown after 5 minutes	goes yellow-brown immediately	blue-black after 30 minutes
C	goes yellow-brown immediately	goes yellow-brown after 5 minutes	goes yellow-brown after 5 minutes	blue-black after 30 minutes
D	goes yellow-brown after 5 minutes	blue-black after 30 minutes	blue-black after 30 minutes	goes yellow-brown after 5 minutes

October/November 2011 · Paper 1 · Variant 1 · Question 5

Correct answer: D

5 The photomicrograph shows a transverse section of a leaf.

Where does carbon dioxide enter the leaf?



October/November 2011 · Paper 1 · Variant 1 · Question 8

Correct answer: A

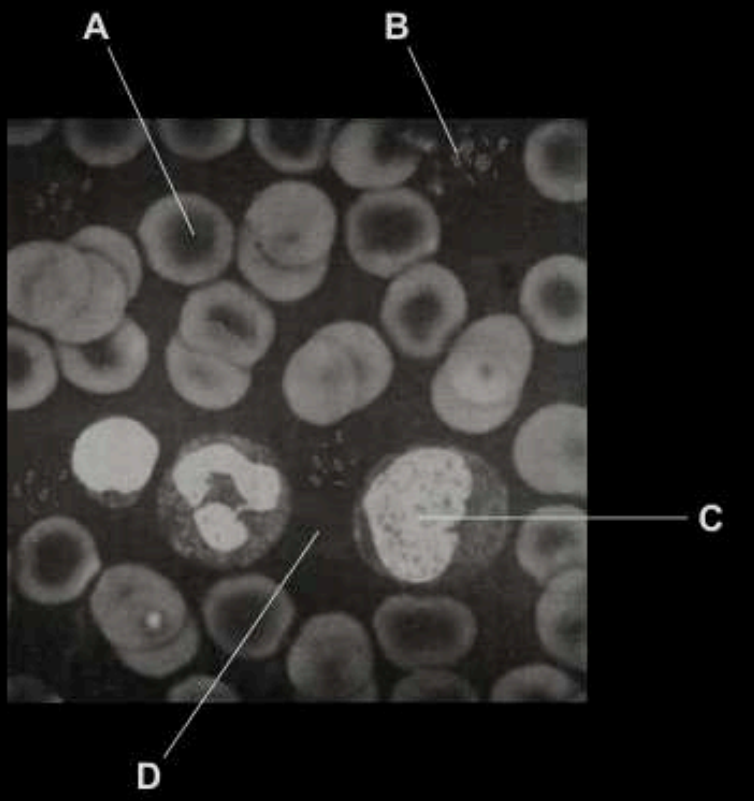
8 Which chemical elements are found in carbohydrates, fats and proteins?

	carbohydrates	fats	proteins
A	carbon, hydrogen and oxygen	carbon, hydrogen and oxygen	carbon, hydrogen, oxygen and nitrogen
B	carbon, hydrogen and oxygen	carbon, hydrogen, oxygen and nitrogen	carbon, hydrogen and oxygen
C	carbon, hydrogen, oxygen and nitrogen	carbon, hydrogen and oxygen	carbon, hydrogen and oxygen
D	carbon, hydrogen, oxygen and nitrogen	carbon, hydrogen and oxygen	carbon, hydrogen, oxygen and nitrogen

Correct answer: A

10 The photomicrograph shows human blood.

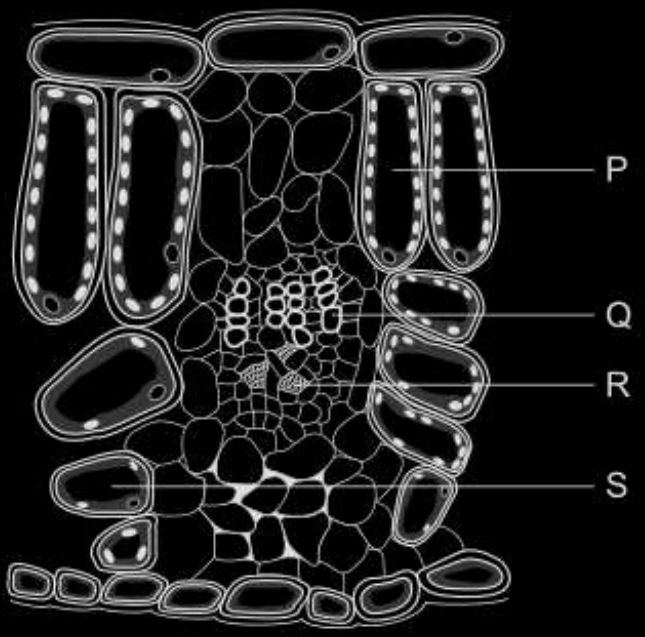
Which component cannot function effectively if a person's diet lacks iron?



magnification ×1000

Correct answer: C

11 The diagram shows part of a transverse section of a leaf.



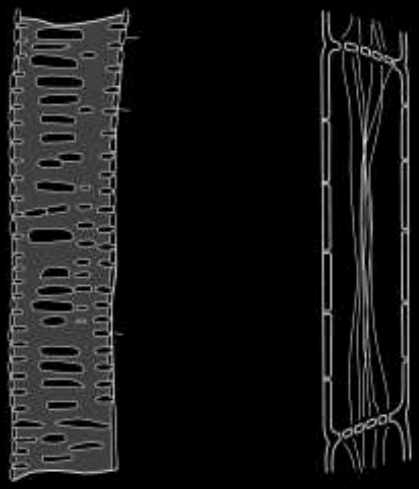
Which cells conduct water into the leaf and which cells conduct sugars out of the leaf?

	conduct water	conduct sugars
A	P	Q
B	Q	P
C	Q	R
D	R	Q

October/November 2011 · Paper 1 · Variant 1 · Question 12

Correct answer: D

12 The diagram shows two plant cells, X and Y, drawn to different scales.



X Y

Samples of the contents of X and Y were tested for nutrients.

What results are expected?

	X		Y	
	Benedict's reagent	iodine in potassium iodide solution	Benedict's reagent	iodine in potassium iodide solution
A	+	+	–	+
B	+	–	+	–
C	–	+	–	+
D	–	–	+	–

key

– = negative result

+ = positive result

October/November 2011 · Paper 1 · Variant 1 · Question 13

Correct answer: B

13 Which blood vessel carries absorbed food material from the small intestine to the liver?

A coronary artery

B hepatic portal vein

C pulmonary artery

D renal vein

Correct answer: A

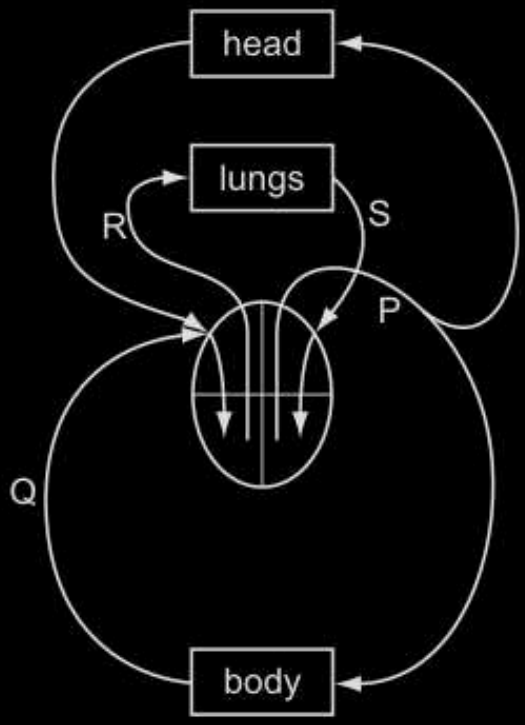
14 The diagram shows a section through part of a blood vessel.

What could be the first organs found in the directions 1 and 2?

	1	2
A	heart	brain
B	intestine	liver
C	kidney	heart
D	lung	heart

Correct answer: C

15 The diagram represents the heart and some major blood vessels.



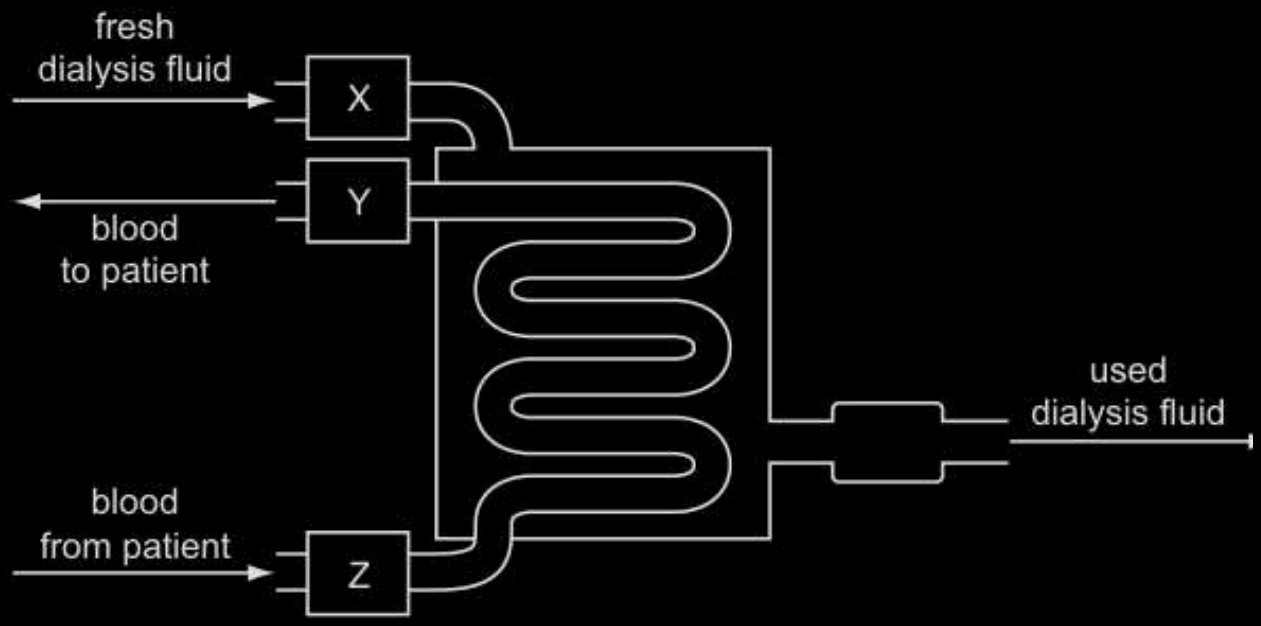
Which are possible blood pressures (in kPa) for the vessels shown on the diagram?

	P	Q	R	S
A	1	4	2	16
B	4	16	2	1
C	16	2	4	1
D	16	4	1	2

October/November 2011 · Paper 1 · Variant 1 · Question 20

Correct answer: C

20 The diagram represents a kidney dialysis machine.



What are the parts labelled X, Y and Z?

	bubble trap	roller pump	water bath for temperature control
A	X	Y	Z
B	Y	X	Z
C	Y	Z	X
D	Z	Y	X

October/November 2011 · Paper 1 · Variant 1 · Question 21

Correct answer: C

21 What helps heat retention in the human body?

A actively secreting sweat glands

B dilated skin blood vessels

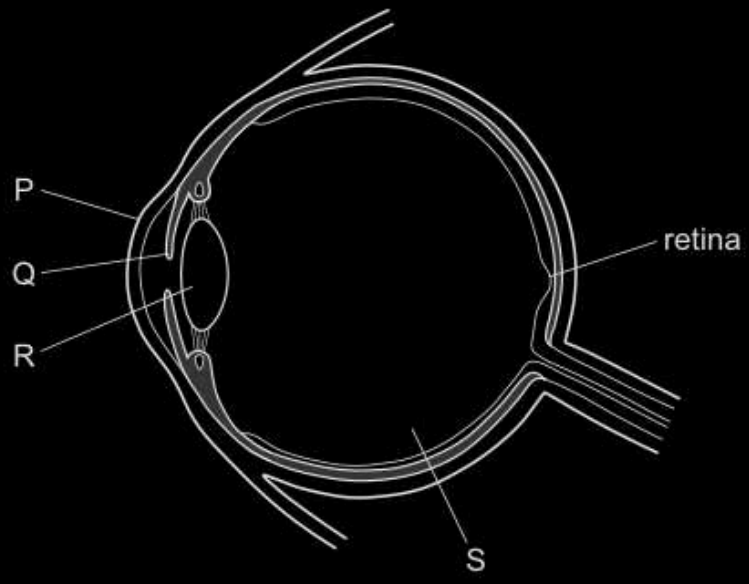
C fat in and under the skin

D relaxed hair erector muscles

October/November 2011 · Paper 1 · Variant 1 · Question 22

Correct answer: B

22 The diagram shows a section through the eye.



Which pair of structures focus light rays onto the retina?

A P and Q **B** P and R **C** Q and R **D** Q and S

October/November 2011 · Paper 1 · Variant 1 · Question 23

Correct answer: C

23 Which part of the central nervous system controls the body's water balance?

A cerebellum

B cerebrum

C hypothalamus

D medulla

October/November 2011 · Paper 1 · Variant 1 · Question 24

Correct answer: D

24 How does adrenaline affect glucose uptake by muscle cells and carbohydrate conversion by liver cells?

	glucose uptake	carbohydrate conversion
A	decreases	glucose to glycogen
B	decreases	glycogen to glucose
C	increases	glucose to glycogen
D	increases	glycogen to glucose

October/November 2011 · Paper 1 · Variant 1 · Question 25

Correct answer: D

25 What is a major effect of nicotine in tobacco smoke?

A It causes lung cancer.

B It destroys cilia in the trachea.

C It increases mucus production in the trachea.

D It increases the desire to smoke.

October/November 2011 · Paper 1 · Variant 1 · Question 34

Correct answer: C

34 What is a major advantage of feeding breast milk rather than milk made up from milk powder to a baby?

A It contains a higher percentage of calcium for growth of the baby's bones.

B It contains all the carbohydrates, proteins and vitamins needed by the baby.

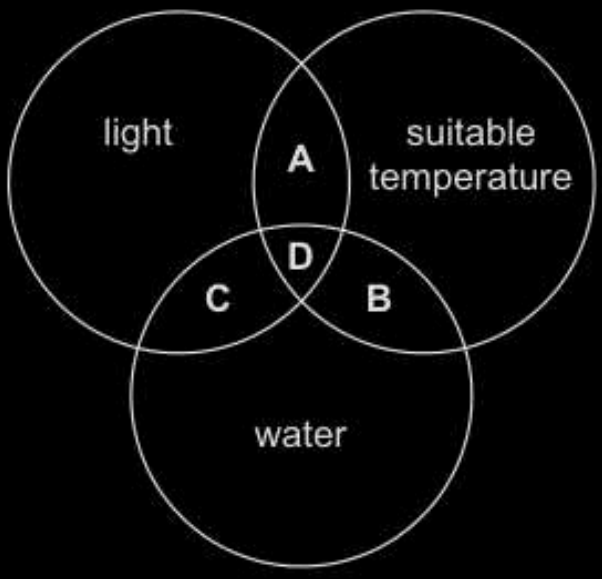
C It contains antibodies from the mother, which protect the baby from infectious diseases.

D It contains less protein, sugar and fat, which prevent the baby from becoming obese.

October/November 2011 · Paper 1 · Variant 1 · Question 36

Correct answer: B

36 Which conditions are necessary to activate enzymes when a seed germinates?



October/November 2011 · Paper 1 · Variant 1 · Question 37

Correct answer: C

37 A recessive homozygote is crossed with a heterozygote of the same gene.

What will be the phenotypes of the F_1 generation?

A all dominant

B 75 % dominant 25 % recessive

C 50 % dominant 50 % recessive

D 25 % dominant 50 % heterozygous 25 % recessive

October/November 2011 · Paper 1 · Variant 1 · Question 40

Correct answer: D

40 Two heterozygotes are crossed. Some of the offspring show the recessive characteristic.

What is the probability that one of these offspring that shows the recessive characteristic is homozygous?

A 0.00 **B** 0.25 **C** 0.50 **D** 1.00

October/November 2011 · Paper 1 · Variant 2 · Question 1

Correct answer: A

1 A human cheek cell and a spongy mesophyll cell from a leaf are examined under a microscope.

Which structures are seen in both cells?

A cell membrane, nucleus and cytoplasm

B cell wall, cell membrane and nucleus

C cytoplasm, cell wall and cell membrane

D nucleus, cytoplasm and cell wall

October/November 2011 · Paper 1 · Variant 2 · Question 3

Correct answer: D

3 The mass of a cube of fresh potato is found. It is then placed in a test-tube containing a dilute solution of sucrose. After an hour, its mass has increased.

Which process has occurred and what has happened to the concentration of the sucrose in the solution in the test-tube?

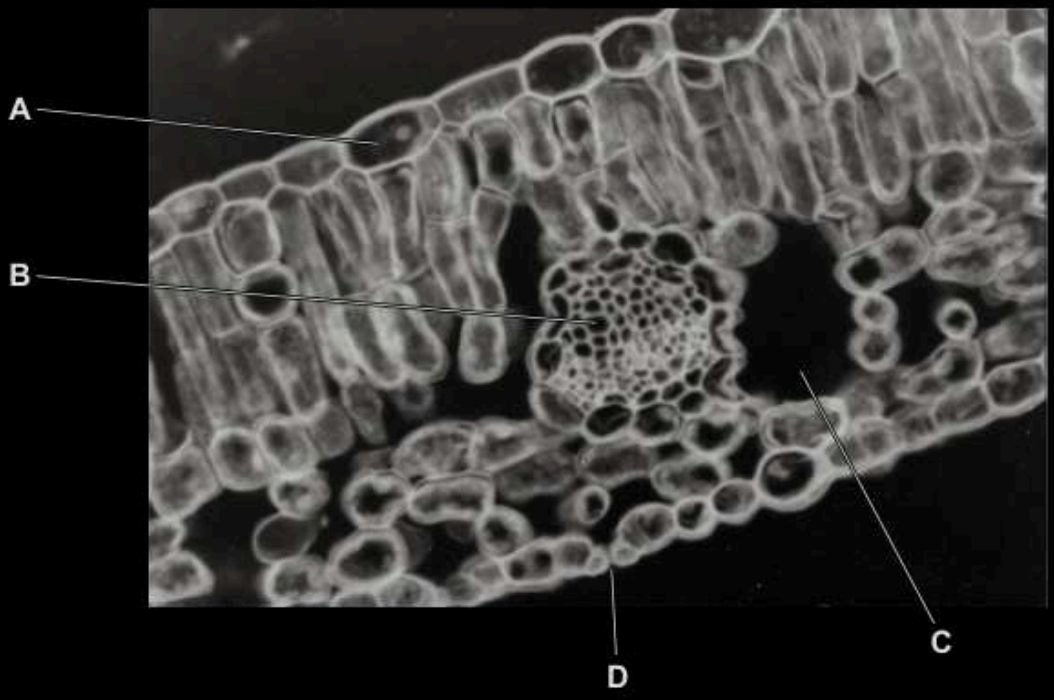
	process	sucrose concentration
A	active transport	decreased
B	active transport	increased
C	osmosis	decreased
D	osmosis	increased

October/November 2011 · Paper 1 · Variant 2 · Question 4

Correct answer: D

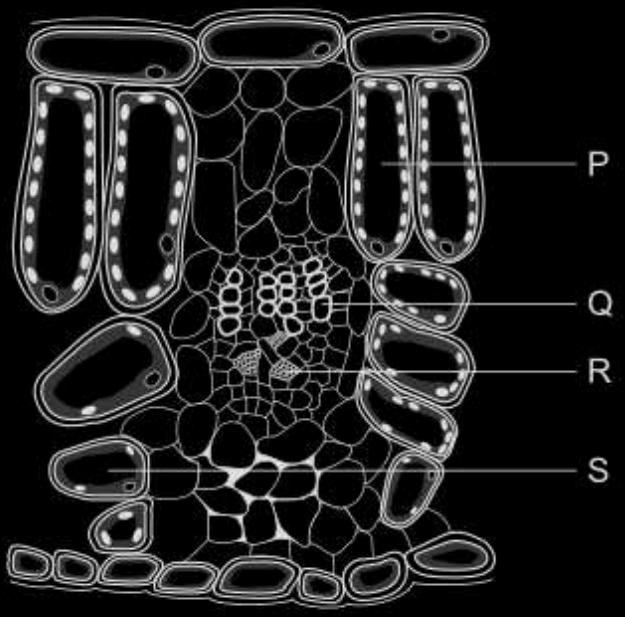
4 The photomicrograph shows a transverse section of a leaf.

Where does carbon dioxide enter the leaf?



Correct answer: C

7 The diagram shows part of a transverse section of a leaf.



Which cells conduct water into the leaf and which cells conduct sugars out of the leaf?

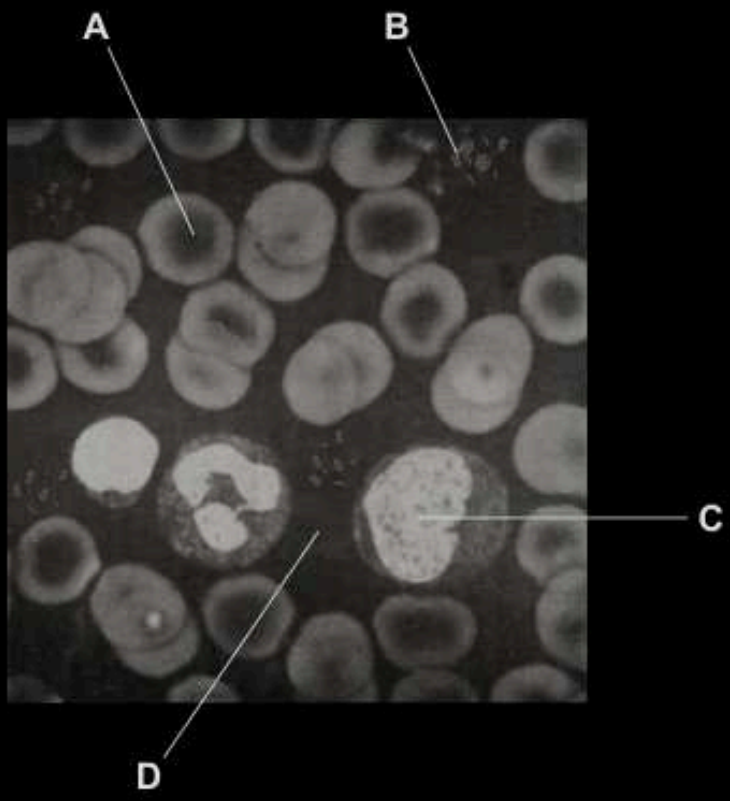
	conduct water	conduct sugars
A	P	Q
B	Q	P
C	Q	R
D	R	Q

October/November 2011 · Paper 1 · Variant 2 · Question 9

Correct answer: A

9 The photomicrograph shows human blood.

Which component cannot function effectively if a person's diet lacks iron?



magnification ×1000

October/November 2011 · Paper 1 · Variant 2 · Question 10

Correct answer: A

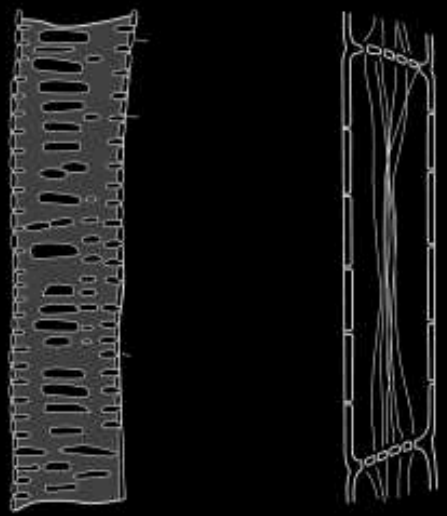
10 Which chemical elements are found in carbohydrates, fats and proteins?

	carbohydrates	fats	proteins
A	carbon, hydrogen and oxygen	carbon, hydrogen and oxygen	carbon, hydrogen, oxygen and nitrogen
B	carbon, hydrogen and oxygen	carbon, hydrogen, oxygen and nitrogen	carbon, hydrogen and oxygen
C	carbon, hydrogen, oxygen and nitrogen	carbon, hydrogen and oxygen	carbon, hydrogen and oxygen
D	carbon, hydrogen, oxygen and nitrogen	carbon, hydrogen and oxygen	carbon, hydrogen, oxygen and nitrogen

October/November 2011 · Paper 1 · Variant 2 · Question 11

Correct answer: D

11 The diagram shows two plant cells, X and Y, drawn to different scales.



X Y

Samples of the contents of X and Y were tested for nutrients.

What results are expected?

	X		Y	
	Benedict's reagent	iodine in potassium iodide solution	Benedict's reagent	iodine in potassium iodide solution
A	+	+	–	+
B	+	–	+	–
C	–	+	–	+
D	–	–	+	–

key

– = negative result

+ = positive result

October/November 2011 · Paper 1 · Variant 2 · Question 12

Correct answer: B

12 Which blood vessel carries absorbed food material from the small intestine to the liver?

A coronary artery

B hepatic portal vein

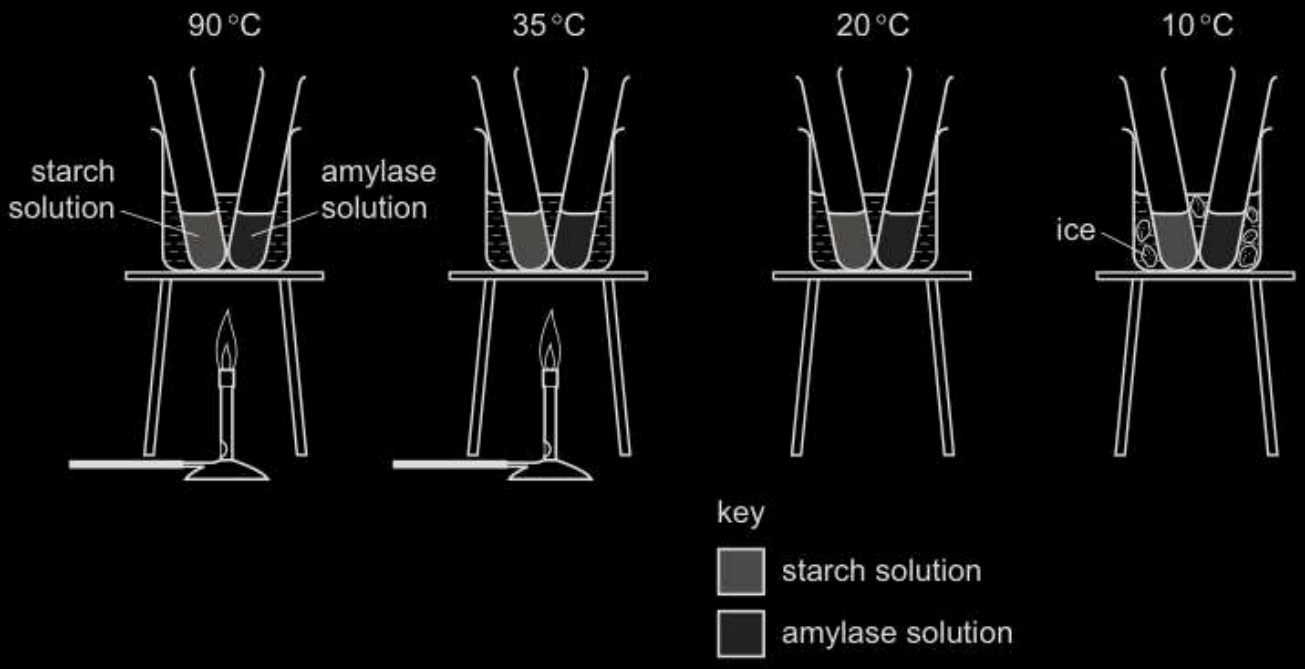
C pulmonary artery

D renal vein

Correct answer: A

13 The diagram shows an experiment on amylase.

Each beaker contains water at the temperature shown.



After five minutes, each test-tube of amylase is poured into the test-tube of starch solution in the same beaker.

After leaving the tubes for 5 minutes, samples of the mixture are then tested with iodine solution and then tested again at 5 minute intervals.

Which results are expected?

	90 °C	35 °C	20 °C	10 °C
A	blue-black after 30 minutes	goes yellow-brown immediately	goes yellow-brown after 5 minutes	blue-black after 30 minutes
B	blue-black after 30 minutes	goes yellow-brown after 5 minutes	goes yellow-brown immediately	blue-black after 30 minutes
C	goes yellow-brown immediately	goes yellow-brown after 5 minutes	goes yellow-brown after 5 minutes	blue-black after 30 minutes
D	goes yellow-brown after 5 minutes	blue-black after 30 minutes	blue-black after 30 minutes	goes yellow-brown after 5 minutes

Correct answer: A

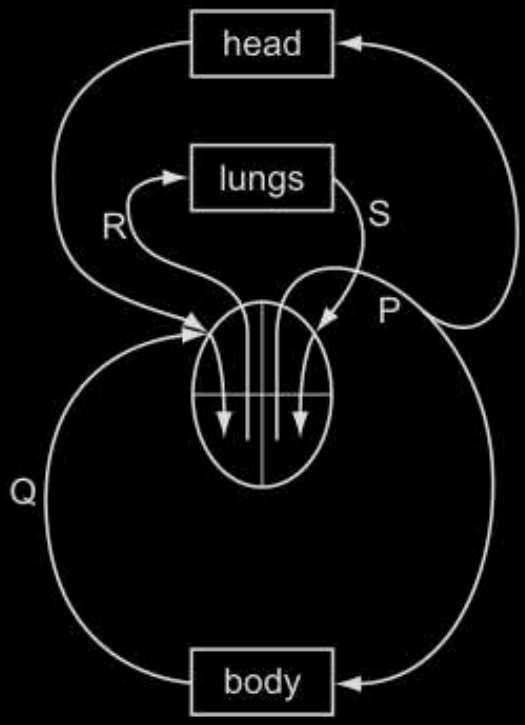
16 The diagram shows a section through part of a blood vessel.

What could be the first organs found in the directions 1 and 2?

	1	2
A	heart	brain
B	intestine	liver
C	kidney	heart
D	lung	heart

Correct answer: C

17 The diagram represents the heart and some major blood vessels.



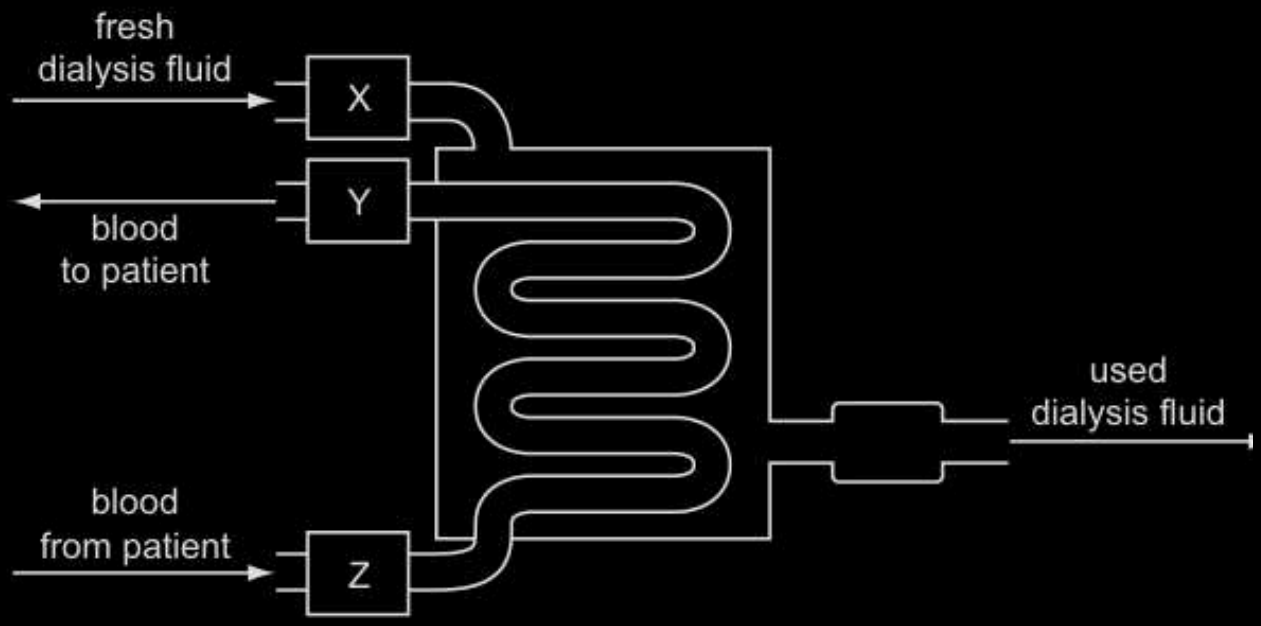
Which are possible blood pressures (in kPa) for the vessels shown on the diagram?

	P	Q	R	S
A	1	4	2	16
B	4	16	2	1
C	16	2	4	1
D	16	4	1	2

October/November 2011 · Paper 1 · Variant 2 · Question 18

Correct answer: C

18 The diagram represents a kidney dialysis machine.



What are the parts labelled X, Y and Z?

	bubble trap	roller pump	water bath for temperature control
A	X	Y	Z
B	Y	X	Z
C	Y	Z	X
D	Z	Y	X

October/November 2011 · Paper 1 · Variant 2 · Question 19

Correct answer: C

19 What helps heat retention in the human body?

A actively secreting sweat glands

B dilated skin blood vessels

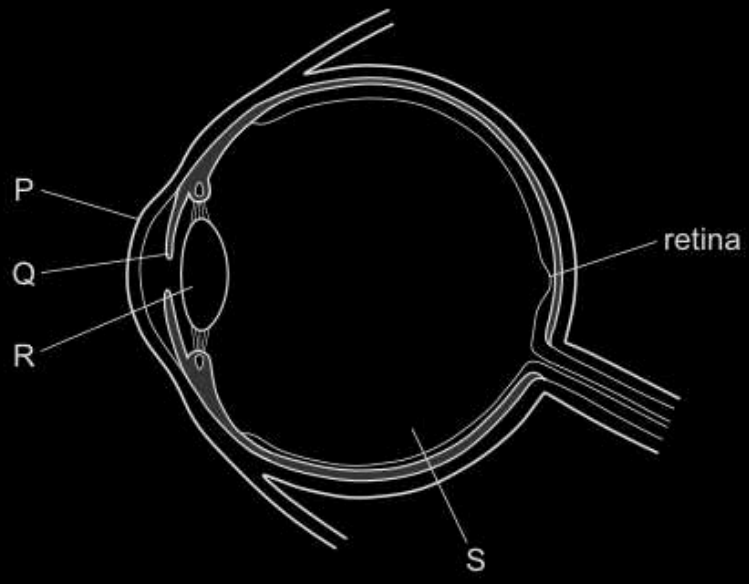
C fat in and under the skin

D relaxed hair erector muscles

October/November 2011 · Paper 1 · Variant 2 · Question 25

Correct answer: B

25 The diagram shows a section through the eye.



Which pair of structures focus light rays onto the retina?

A P and Q **B** P and R **C** Q and R **D** Q and S

October/November 2011 · Paper 1 · Variant 2 · Question 26

Correct answer: C

26 Which part of the central nervous system controls the body's water balance?

A cerebellum

B cerebrum

C hypothalamus

D medulla

October/November 2011 · Paper 1 · Variant 2 · Question 27

Correct answer: D

27 How does adrenaline affect glucose uptake by muscle cells and carbohydrate conversion by liver cells?

	glucose uptake	carbohydrate conversion
A	decreases	glucose to glycogen
B	decreases	glycogen to glucose
C	increases	glucose to glycogen
D	increases	glycogen to glucose

October/November 2011 · Paper 1 · Variant 2 · Question 28

Correct answer: D

28 What is a major effect of nicotine in tobacco smoke?

A It causes lung cancer.

B It destroys cilia in the trachea.

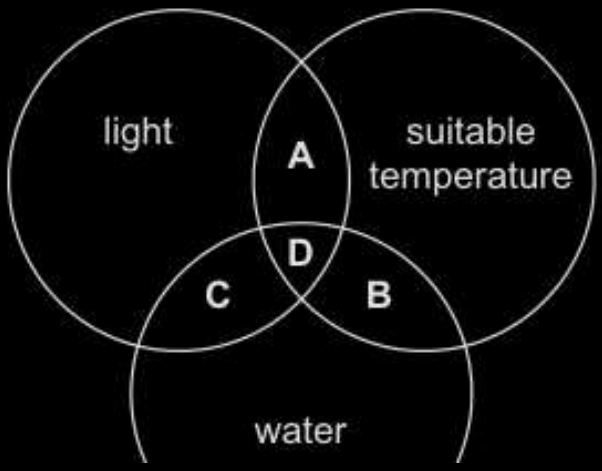
C It increases mucus production in the trachea.

D It increases the desire to smoke.

October/November 2011 · Paper 1 · Variant 2 · Question 33

Correct answer: B

33 Which conditions are necessary to activate enzymes when a seed germinates?



October/November 2011 · Paper 1 · Variant 2 · Question 34

Correct answer: C

34 A recessive homozygote is crossed with a heterozygote of the same gene.

What will be the phenotypes of the F_1 generation?

A all dominant

B 75 % dominant 25 % recessive

C 50 % dominant 50 % recessive

D 25 % dominant 50 % heterozygous 25 % recessive

October/November 2011 · Paper 1 · Variant 2 · Question 38

Correct answer: C

38 What is a major advantage of feeding breast milk rather than milk made up from milk powder to a baby?

A It contains a higher percentage of calcium for growth of the baby's bones.

B It contains all the carbohydrates, proteins and vitamins needed by the baby.

C It contains antibodies from the mother, which protect the baby from infectious diseases.

D It contains less protein, sugar and fat, which prevent the baby from becoming obese.

October/November 2011 · Paper 1 · Variant 2 · Question 40

Correct answer: D

40 Two heterozygotes are crossed. Some of the offspring show the recessive characteristic.

What is the probability that one of these offspring that shows the recessive characteristic is homozygous?

A 0.00 **B** 0.25 **C** 0.50 **D** 1.00

1 Fig. 1.1 shows cells from three different types of organism (not drawn to the same scale).

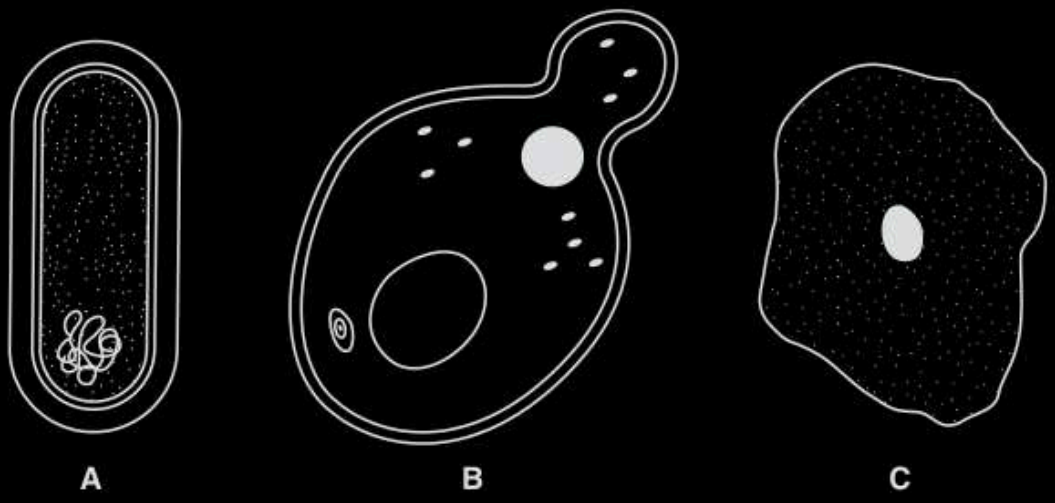


Fig. 1.1

(a) Name the type of organism represented by each of the cells and in each case give a reason for your answer.

*organism **A*** .....

reason .....

*organism **B*** .....

reason .....

*organism **C*** .....

reason .....[3]

(b) Fig. 1.2 shows a one-celled organism that has both plant and animal characteristics.

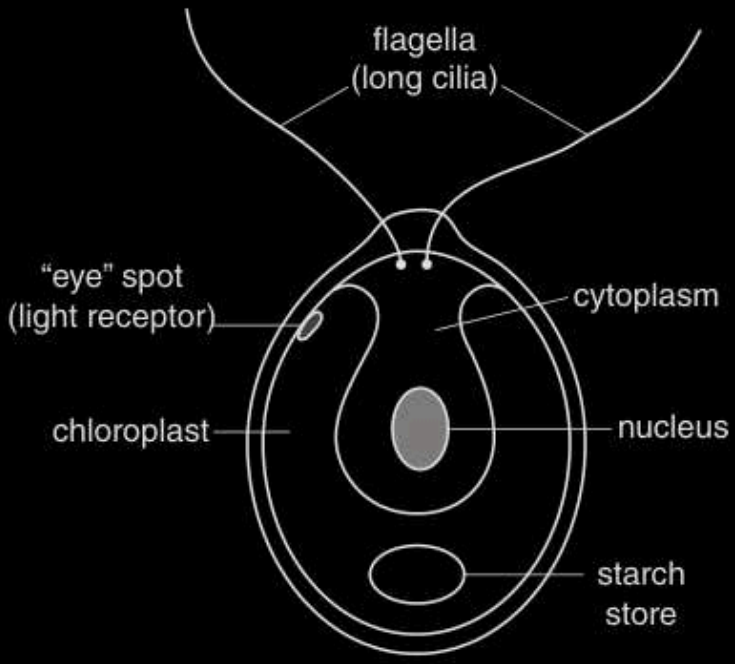


Fig. 1.2

State two reasons in each case why the organism might be identified as

(i) an animal

1. .....

2. .....[2]

(ii) a plant

1. .....

2. .....[2]

[Total: 7]

1 **(a)** **A** – bacterium (or named) + no nucleus/wall + no vacuole/slime capsule (A ref. nuclear strand AW/no nuclear membrane AW);

B – fungus/yeast + not angular/no central or large vacuole/budding;

C – animal or named + no cell wall/*only* cell membrane (A *Amoeba*/protozoan*)* (R protoctist) (R named cells); [3]

(b) **(i)** 2 from: eye/light receptor, cilia/flagella, locomotion;; [2]

(ii) 2 from: cell wall, starch, chloroplast/chlorophyll;; [2]

[Total: 7]

2 Cultivated banana plants produce fruits with seeds that are infertile (unable to develop). Wild banana plants produce fruits with large, fertile seeds.

(a) Suggest the type of reproduction usually found in

- cultivated bananas .....
- wild bananas .....[2]

(b) State two commercial advantages that result from the type of reproduction found in cultivated bananas.

1. .....

2. .....[2]

'Black sigatoka' is a fungal disease of banana leaves. The hyphae of the fungus spread through the intercellular spaces as shown in Fig. 2.1.

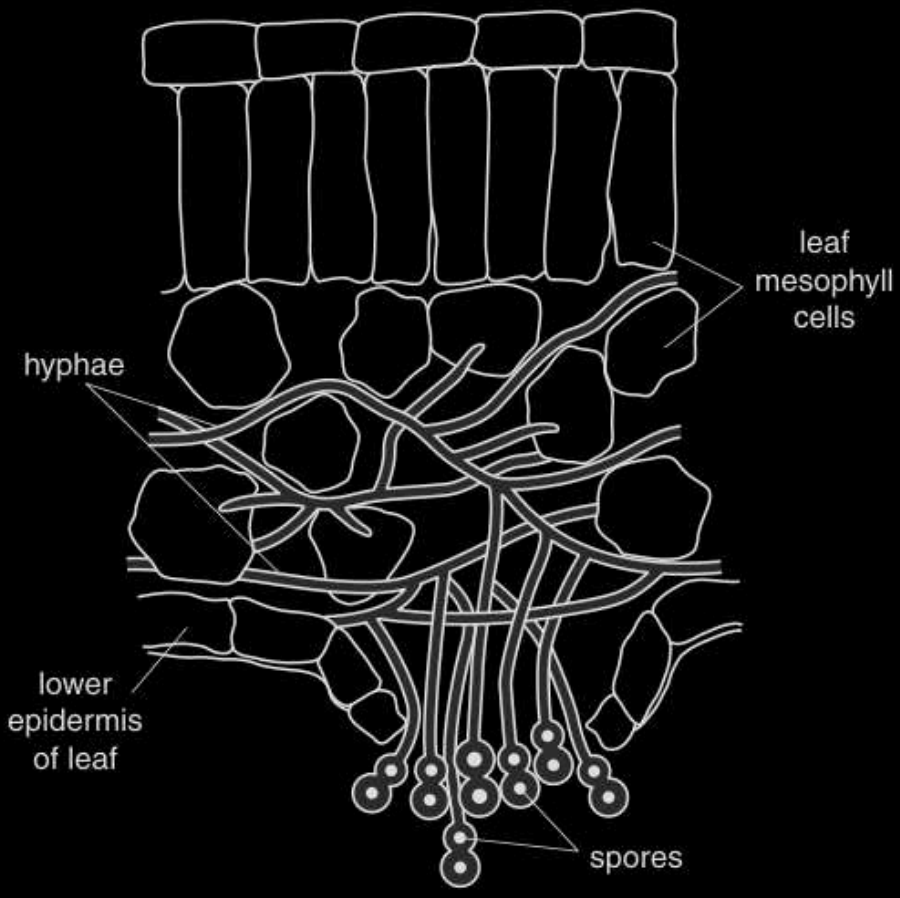


Fig. 2.1

(c) Suggest ways by which the fungus would eventually cause the leaf to die.

.....

.....

.....

.....

.....

.....

.....

.....[4]

(d) Explain why fungal diseases tend to kill a higher percentage of cultivated than of wild banana plants.

.....

.....

.....

.....

.....

.....[3]

[Total:11]

2 **(a)** asexual/vegetative;
sexual (Ignore asexual); [2]

(b) 2 from:
more certain, known quality/quantity of fruit or described*,
favourable conditions, greater % of fruit is flesh, faster,
greater profit/higher yield, (*allow ecf if wrong type of reproduction);; [2]

(c) interferes with movement of gases/blocks stomata;
interference with transpiration;
digests cell contents/ref. enzymes/separates cells;
takes nutrients from the plant;
kills cells/protective toxins released by cells;
no/less photosynthesis;
blocks veins/vascular bundles/phloem/xylem; [max 4]

(d) (A reverse argument) plants close together;
genetically identical;
little variation/mutation;
all/very large numbers lack resistance; [max 3]

[Total: 11]

3 Fig. 3.1 shows a section through human skin.

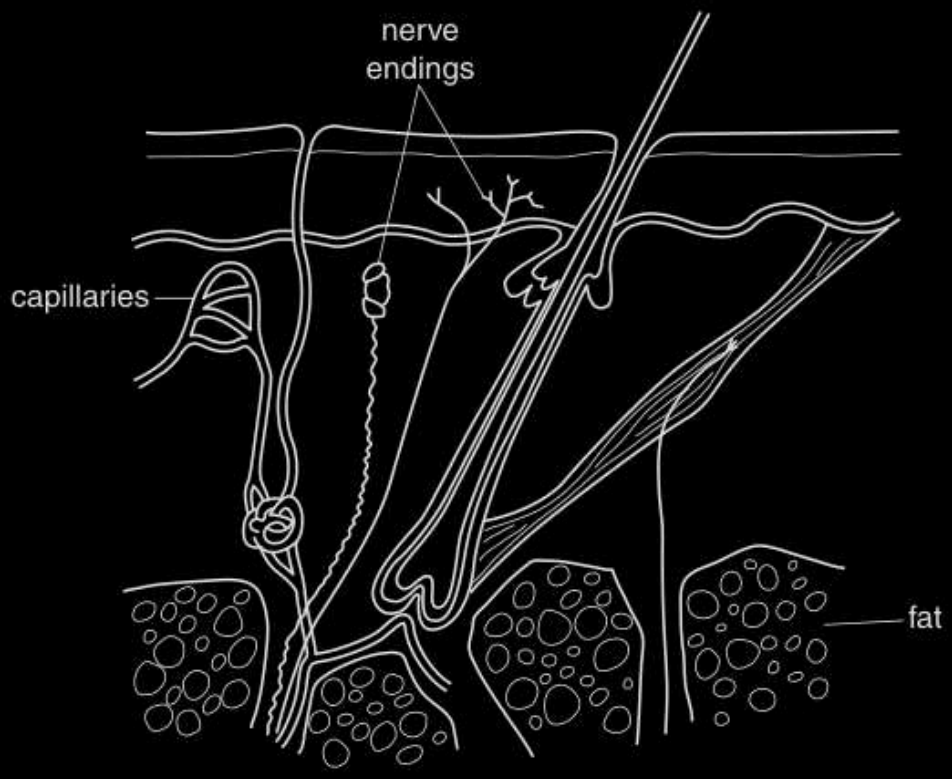


Fig. 3.1

(a) Suggest two possible functions of the nerve endings shown in Fig. 3.1.

1. .....

2. .....[2]

(b) Explain how the capillaries are involved in the loss of heat from the body during exercise.

.....

.....

.....

.....

.....

.....[3]

(c) Fig. 3.2 shows a yak. The yak is a large animal that lives at high altitudes (up to 5 500 m).

Fig. 3.2

Suggest why

(i) the sweat glands of this animal are largely non-functional;

.....

.....

.....

.....[2]

(ii) during those times of the year when food is plentiful, the yak stores a thick layer of fat beneath its skin;

.....

.....

.....

.....[2]

(iii) the yak has a compact body with small ears and short tail.

.....

.....

.....

.....[2]

[Total: 11]

3 **(a)** one per line, mark the first, any 2 from:
detection of pressure, temperature, pain, touch;; [2]
(A for ONE mark max. a reference to the detection of stimuli)

(b) dilation;
more blood;
blood carries heat;
heat lost from + body surface/skin/named heat transfer method;
capillaries supply sweat glands; [max 3]

(c) **(i)** (A reverse argument) very little sweat lost;
no need to sweat/sweating would be detrimental AW;
fur would inhibit evaporation;
less heat lost;
*ref. low external temperature; [max 2]

(ii) stores energy;
supplies energy/heat;
insulates (against heat loss);
*ref. low external temperature; [max 2]

(iii) (A reverse argument)
(for ears/tail) reduced surface/small surface area;
ref. small surface area to volume ratio for the whole animal (Assume that 'it' = the yak);
from which heat can be lost;
*ref. low external temperature; [max 2]
(n.b. * = once only)

[Total: 11]

5 **(a)** **(i)** Name the structures within the nucleus that are responsible for inheritance.

.....[1]

(ii) Name the chemical in these structures that makes inheritance possible.

.....[1]

A fly was homozygous dominant for each of **two** different characters. It was crossed with a fly showing the recessive phenotypes for both characters. Two of the offspring were then crossed. The result of this cross is shown in Fig. 5.1.

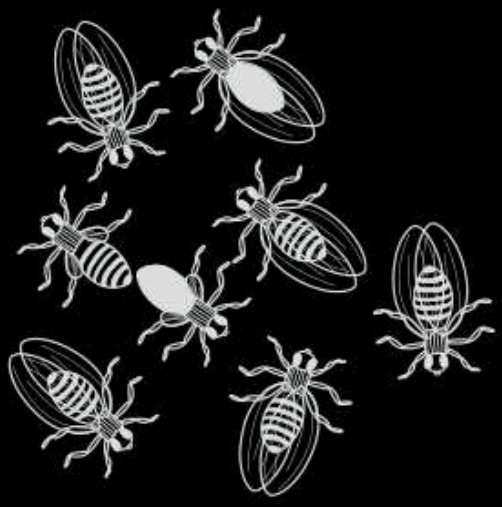


Fig. 5.1

(b) State the two pairs of contrasting characters shown by these flies.

1. .....

2. .....[2]

(c) **(i)** State the recessive phenotype of one of the pairs of characters you have identified in **(b)**.

Give a reason for your answer.

recessive phenotype .....

reason .....[2]

(ii) Draw fully-labelled genetic diagrams to show the possible outcomes of crossing a fly showing the recessive phenotype, that you have identified in **c(i)**, with one showing the dominant phenotype.

[6]

[Total: 12]

Section B

Answer **all** the questions in this section.

Write your answers in the spaces provided.

5 **(a)** **(i)** chromosomes/genes; [1]

(ii) DNA (mark the first); [1]

(b) 1 and 2 in either order: stripey + black (abdomens) AW;
long + short wings (A no wings); [2]

(c) **(i)** black body/short wings;
hybrid cross yields 1:3/1 in 4/fewer of recessive phenotype AW; [2]

(ii) rr × RR (any matching upper and lower case letters);
r (r) R (R) + gametes*/G/g/or encircled;
Rr;
long wings/stripey + Rr (A anywhere);
all of this genotype/phenotype AW (A 'all the same');

rr	×	Rr;
r (r)		R r + gametes* (* = once only);
Rr	+	rr;
long/stripey		short/black;
1:1	/	50/50;

[max 6]

[Total: 12]

Section B

6 **(a)** Describe and explain how microorganisms are used to produce a hormone commercially.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[7]

(b) State the advantages of obtaining hormones by this method.

.....

.....

.....

.....

.....

.....[3]

[Total:10]

6 **(a)** human/named donor animal/named cell;
gene or DNA for hormone/insulin;
cut/removed from chromosome;
ref. use of enzymes;
inserted into plasmid/DNA;
of bacterium;
culture medium AW (R agar plate);
oxygen supplied/aeration;
suitable temperature/pH/sterility;
bacteria divide/reproduce;
the gene makes insulin/hormone;
separated from infusion; [max 7]

(b) conditions (or named) can be controlled;
for maximum yield/large amounts;
no harm to human;
no harm to animal/sheep AW;
insulin is (exact) match of human insulin– not of another animal AW;
cheaper AW/higher profits/safer/no transmission of disease; [max 3]

[Total: 10]

October/November 2011 · Paper 2 · Variant 2 · Question 1

1 A student was asked to make a model of a plant cell. She took a length of tubing made from a substance that allows only water molecules to pass through and enclosed it in a flexible permeable membrane as shown in Fig. 1.1.

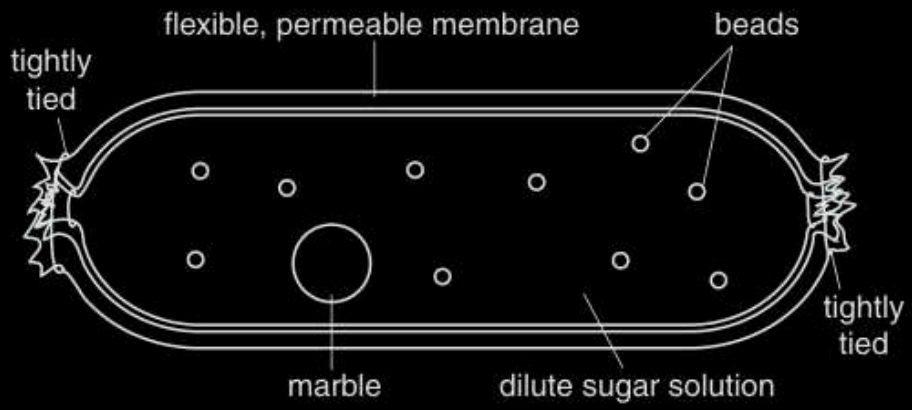


Fig. 1.1

(a) Name the structures found in a plant mesophyll cell that are best represented by the following items used in her model:

the marble .....

the beads .....

the tubing .....[3]

(b) State a structure found in a plant cell that is **not** represented in the student's model.

.....[1]

(c) The student then placed her model into a concentrated sugar solution for half an hour.

State and explain the similarities between what might happen to her model during that time and what would happen to a plant cell placed in the same solution.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 10]

1 **(a)** nucleus;
chloroplasts;
(cell) membrane; [3]

(b) vacuole / ribosomes / starch grains / mitochondria / tonoplast; [1]

(c) water leaves;
by osmosis / diffusion;
ref. water potential (A any ref. to differential in concentration);
cell / cytoplasm / protoplasm / model shrinks (R ref. vacuole);
ref. decreased pressure / turgidity / firmness / flaccidity;
tubing pulls away from permeable membrane OR cytoplasm or cell membrane pulls away from (cell) wall;
ref. plasmolysis;
no sugar leaves cell / enters space between tubing and permeable membrane OR between cell membrane and wall; [max 6]

[Total: 10]

2 Fig. 2.1 shows the chromosomes present in the nucleus of a human cell.

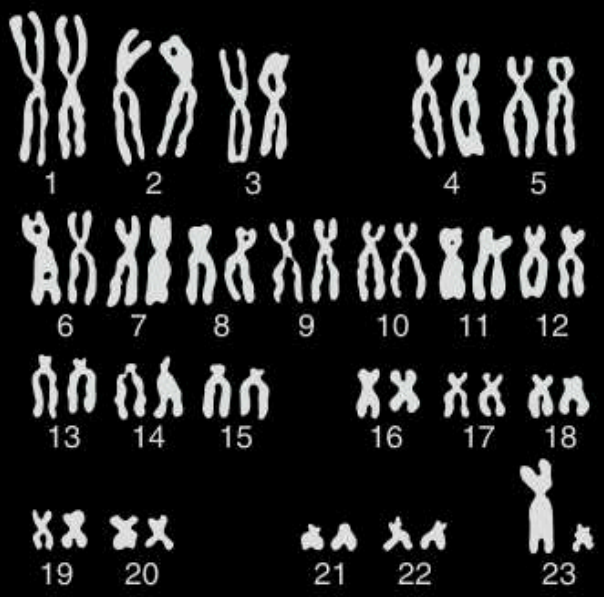


Fig. 2.1

(a) Name the chemical contained within a chromosome that is responsible for inheritance.

.....[1]

(b) **(i)** Name the type of cell division that occurs only in the sex organs to produce gametes.

.....[1]

(ii) State how many chromosomes would be present in a gamete from this person.

..... [1]

Fig. 2.2 shows the chromosomes present in the nucleus of a cell from a different person.

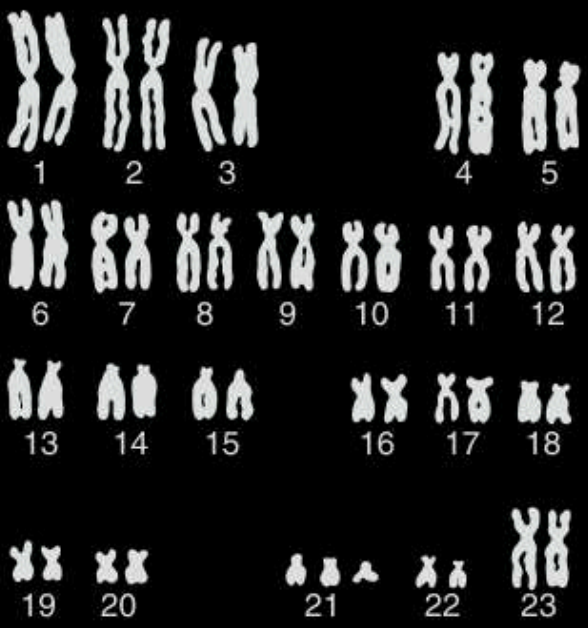


Fig. 2.2

(c) State the sex of this person and explain your answer.

sex .....

explanation .....[2]

(d) Suggest a condition which the person whose chromosomes are shown in Fig. 2.2 has and explain your answer.

condition .....

explanation .....

.....[3]

(e) Suggest why offspring that are produced by crossing two different species are usually infertile.

.....

.....

.....

.....[2]

[Total: 10]

2 **(a)** DNA; [1]

(b) **(i)** meiosis (R if a 't' appears in the word) / reduction division; [1]

(ii) 23; [1]

(c) female;
2 similar sex, 23rd or final (pair) / 2 X / no Y / only X [chromosome(s)]; [2]

(d) Down's syndrome / polysomy;
1 extra (chromosome) / 47 (chromosomes) / 3 / trisomy;
ref. chromosome no. 21; [3]

(e) chromosomes / genes (R mention of alleles);
do not match / pair / are different / different number of;
unable to produce gametes / meiosis not possible; [max 2]

[Total: 10]

3 The following is a list of terms associated with a person's responses. Each term is identified by a letter.

A – brain
B – contraction
C – gland
D – hormone
E – impulse
F – motor neurone
G – muscle
H – receptor
I – sensory neurone
J – spinal cord

(a) **Using their identifying letters only**, place the terms in the order in which they are involved in the following:

(i) Peeling an onion causes a student's eyes to water.

.....[2]

(ii) A student decides to open a book.

.....[1]

(iii) A student hears a sudden loud noise and, shortly afterwards, his heart beats faster.

.....

.....[2]

(b) Name the type of response involved in **(a)(i)** and **(a)(ii)**.

(a)(i) .....

(a)(ii) .....[2]

(c) Describe how responses involving the nervous system differ from those that involve the effects of hormones.

.....

.....

.....

.....[3]

[Total: 10]

3 **(a)** **(i)** **H**, **E**, **I** (or **I**, **E**), **A**, **F**, **C** all correct = 2, 3 in correct consecutive sequence =1 (Ignore any other '**E**'s);; [2]

(ii) **A**, **E**, **J**, **F**, **G**, **B** (Ignore any other **E**'s); [1]

(iii) **H**, **E**, **I** (or **I**, **E**), **A**, **F**, **C**, **D**, **G** (**B**) (4 in correct consecutive sequence = 1) (Ignore any other '**E**'s) (which can include the final **B**);; [2]

(b) reflex / involuntary / unintentional (R *spinal* reflex);
voluntary / deliberate / intentional; [2]

(c) faster / fast v. slow;
electrical / impulses v. chemical / blood;
neurones v. blood / chemical;
(R nerves)
muscles v. glands / organs;
temporary AW v. more lasting AW;
sometimes deliberate AW v. less controlled AW;
specific organs v more general effect; [max 3]

[Total: 10]

6 Explain the importance to a plant of

(a) the distribution of its chloroplasts

.....

.....

.....

.....

.....

.....

.....

.....[4]

(b) the presence of stomata

.....

.....

.....

.....

.....

.....[3]

(c) the position of its vascular bundles

.....

.....

.....

.....

.....

.....[3]

[Total: 10]

6 **(a)** in palisade*;
spongy*;
(*or for ONE mark, mesophyll)
none in epidermis / transparent epidermis;
near leaf surface / (sun)light / to absorb more or most sunlight (A exposed to);
chloroplasts able to move within cell to be nearer to leaf surface AW;
rapid AW rate of photosynthesis (A with ref. either distribution or number);
guard cells;
controlling stomata; [max 4]

(b) allow *CO_2 in (to leaf) (R cells);
*O_2 out;
(*Allow ONE mark only for ref. gas exchange)
for photosynthesis / to make CHOs;
allow water vapour out / transpiration;
to bring water / salts to leaves / cool leaf / plant; [max 3]

(c) in centre of root;
round outer part of / in a circle in stems;
ref. distribution in leaves (depending on mono /dicot);
xylem + strengthened / lignified;
support AW / resist bending ;
resists pulling in root / anchorage / reduces compression; [max 3]

[Total: 10]

7 Describe how blood is made to flow in a continuous circulation around the body with reference to

(a) the heart

.....

.....

.....

.....

.....

.....

.....

.....[4]

(b) the arteries

.....

.....

.....

.....

.....

.....[3]

(c) the veins

.....

.....

.....

.....

.....

.....[3]

[Total: 10]

Section C

Answer **either** question 8 **or** question 9.

Write your answer in the spaces provided.

7 **(a)** valves;
prevent backflow;
muscular + contraction;
ventricles + thick walls / thick muscles / powerful contraction (A if there is ref. only to left ventricle);
pump / push / squeeze;
(creates) pressure in blood system;
never tires or suffers from cramp / rhythmic; [max 4]

(b) always carrying blood away from heart / under pressure;
thick walled;
muscular;
narrow lumen AW;
allows recoil / maintains pulse beat (A elasticity);
link to capillaries / veins; [max 3]

(c) return to heart / low pressure / large lumen;
have valves;
at intervals along their length;
prevent backflow;
thin-walled;
allows (skeletal) muscular contraction to 'knead' the blood AW (R ref. to muscular walls); [max 3]

[Total: 10]

Section C

9 **(a)** Describe how energy enters and then flows through a biological system.

.....

.....

.....

.....

.....

.....

.....

.....[4]

(b) **(i)** Explain how energy is made available in cells.

.....

.....

.....

.....[2]

(ii) State the uses of energy in the human body.

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

9 **(a)** from sun / light + plants / producers;
(used for) photosynthesis;
to make *carbohydrates (or named);
*containing chemical energy;
*eaten by consumers / herbivore / carnivore / animals or correctly named;
decay / decomposers or named;
ref. non-cyclical / lost in metabolic processes AW (A respiration) (A ref. to ATP); [max 4]
(* These marks available for those who interpret the question with ref. to food flowing through the gut.)

(b) **(i)** respiration;
breaks down carbohydrates (or named);
releases energy (**only if linked** to CHO breakdown or respiration) (A liberates AW, R makes / produces); [max 2]
(R equation alone)

(ii) for growth / repair / cell division;
manufacturing other molecules / chemical substances / ref. anabolic reactions AW (A reasonable ref. to enzyme action, R metabolic reactions);
nervous impulses;
movement / locomotion / muscle contraction;
maintaining body temperature / keeping warm;
active transport / uptake AW; [max 4]

[Total: 10]

May/June 2011 · Paper 1 · Variant 1 · Question 1

Correct answer: B

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 2 and 5

May/June 2011 · Paper 1 · Variant 1 · Question 2

Correct answer: C

2 The sentence describes the uptake of water by a plant.

Water moves into the root hairs of a plant by osmosis through a .....1..... permeable cell membrane, ……2…… a water potential gradient.

Which words correctly complete gaps 1 and 2?

	1	2
A	fully	down
B	fully	up
C	partially	down
D	partially	up

May/June 2011 · Paper 1 · Variant 1 · Question 3

Correct answer: A

3 Below is a series of cell processes.

1 mineral ions entering root hair cells

2 glucose uptake by villus cells

3 water entering root hair cells

Which of these involve active transport?

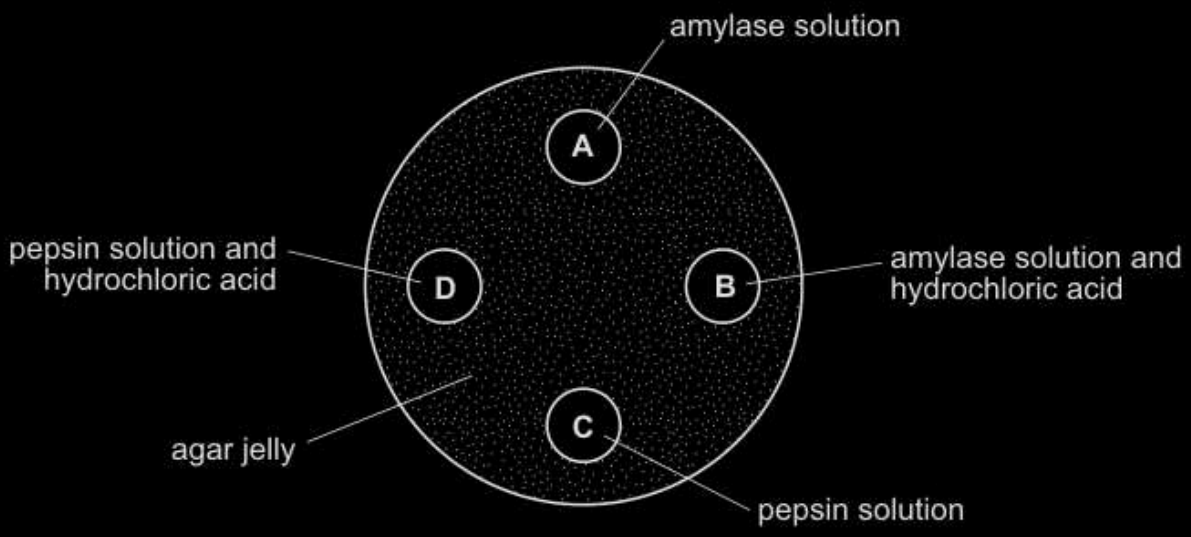
A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

May/June 2011 · Paper 1 · Variant 1 · Question 4

Correct answer: A

4 A dish is filled with agar jelly containing starch. Four holes are cut in the jelly and each hole is filled as shown.

After 30 minutes, which hole will be surrounded by the largest area without starch?



May/June 2011 · Paper 1 · Variant 1 · Question 5

Correct answer: B

5 Magnesium is an essential element for

A the formation of cell walls.

B the formation of chlorophyll.

C the formation of proteins.

D the process of cell division.

May/June 2011 · Paper 1 · Variant 1 · Question 10

Correct answer: C

10 A protein solution is tested using three different reagents.

Which set of results is obtained?

	iodine solution	Benedict's solution	biuret solution
A	black	blue	blue
B	black	red	blue
C	brown	blue	purple
D	brown	red	purple

Correct answer: C

12 The photomicrograph shows part of a sunflower stem.

Which tissue transports water and mineral salts and which tissue transports sucrose?

	water and mineral salts	sucrose
A	1	4
B	2	3
C	3	2
D	4	1

May/June 2011 · Paper 1 · Variant 1 · Question 13

Correct answer: D

13 How do veins differ from arteries?

	width of lumen	wall thickness	elastic fibres	muscles in wall
A	narrower	thicker	more	less
B	narrower	thinner	less	more
C	wider	thicker	more	more
D	wider	thinner	less	less

May/June 2011 · Paper 1 · Variant 1 · Question 14

Correct answer: A

14 The table shows the characteristics of the blood in one blood vessel in the body.

oxygen concentration	carbon dioxide concentration	pressure
high	low	high

Which blood vessel contains blood with these characteristics?

A aorta

B pulmonary artery

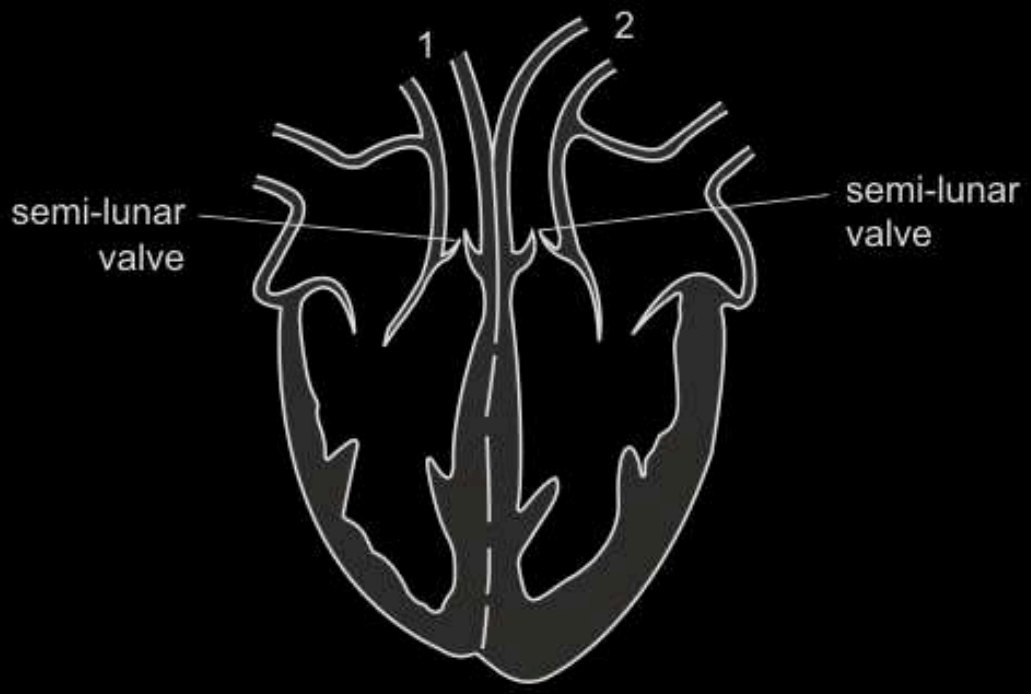
C pulmonary vein

D vena cava

May/June 2011 · Paper 1 · Variant 1 · Question 15

Correct answer: A

15 The diagram shows a section through the human heart.



What happens as blood is being pumped out of the heart?

	semi-lunar valves	vessel through which blood passes to the lungs
A	open	1
B	open	2
C	closed	1
D	closed	2

May/June 2011 · Paper 1 · Variant 1 · Question 17

Correct answer: C

17 Which process does **not** depend on respiration?

A active uptake of ions

B conduction of nervous impulses

C diffusion of glucose

D muscle contraction

May/June 2011 · Paper 1 · Variant 1 · Question 20

Correct answer: B

20 What is **not** an excretory product of mammals?

A carbon dioxide in expired air

B undigested food in faeces

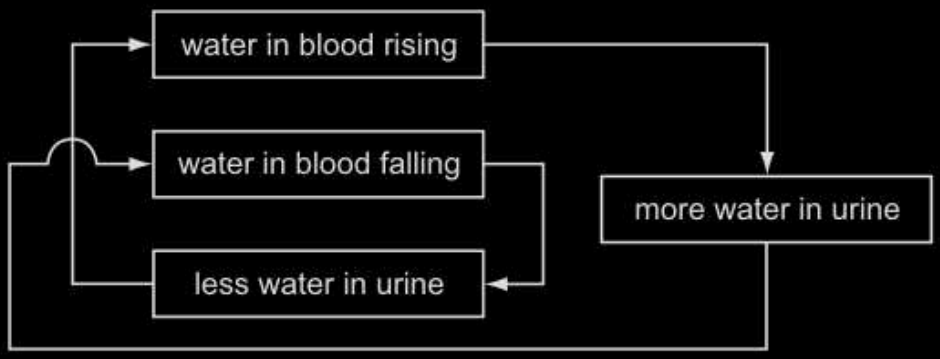
C urea in sweat

D urea in urine

May/June 2011 · Paper 1 · Variant 1 · Question 21

Correct answer: D

21 The diagram refers to the control of water concentration in the blood.



Why is this a negative feedback system?

A It decreases the amount of water in the blood.

B It increases any change in the amount of water in the blood.

C It increases the amount of water in the blood.

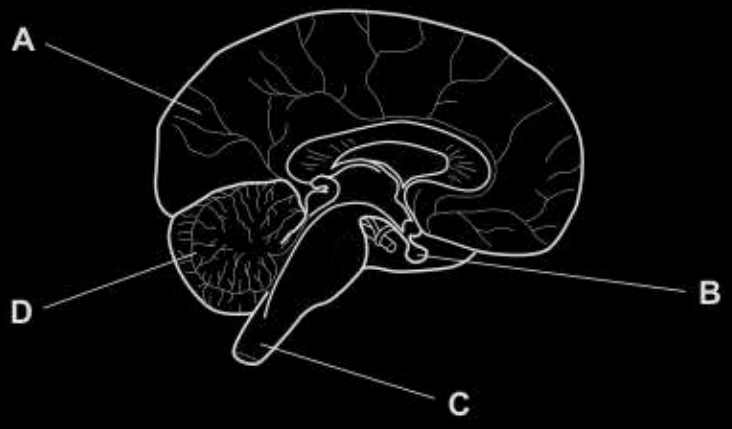
D It reverses any change in the amount of water in the blood.

May/June 2011 · Paper 1 · Variant 1 · Question 22

Correct answer: D

22 The diagram represents a section through the brain and part of the spinal cord.

Which part of the brain controls balance?



May/June 2011 · Paper 1 · Variant 1 · Question 23

Correct answer: A

23 What are the functions of the relay, motor and sensory neurones in a reflex response?

	relay neurone	motor neurone	sensory neurone
A	to connect neurones within the central nervous system	to conduct impulses to the effector from the central nervous system	to conduct impulses from the receptor to the central nervous system
B	to conduct impulses to the effector	to connect neurones within the central nervous system	to receive the stimulus
C	to conduct impulses from the central nervous system to the effector	to conduct impulses from the receptor to the central nervous system	to connect neurones within the central nervous system
D	to conduct impulses from the receptor to the central nervous system	to conduct impulses from the receptor to the central nervous system	to conduct impulses to the effector

May/June 2011 · Paper 1 · Variant 1 · Question 24

Correct answer: B

24 What **decreases** as a result of adrenaline secretion?

A blood sugar level

B digestive activity

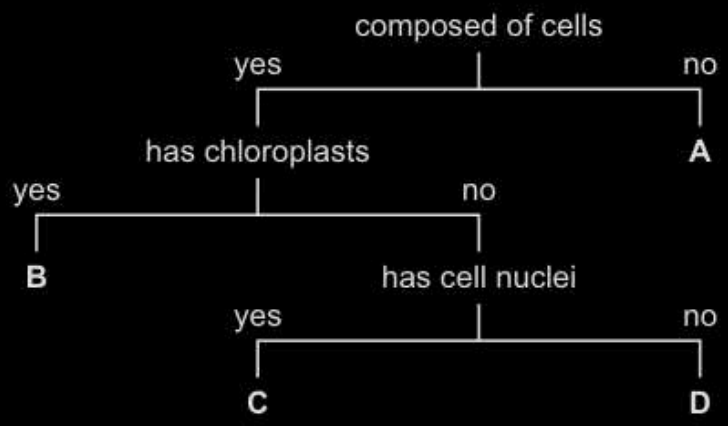
C heart rate

D size of the pupils of the eyes

May/June 2011 · Paper 1 · Variant 1 · Question 26

Correct answer: C

26 Using the flow diagram, which organism is a fungus?



May/June 2011 · Paper 1 · Variant 1 · Question 28

Correct answer: A

28 On decomposition, which substance in decaying plants is the major source of carbon for carbon dioxide production by the micro-organisms involved?

A cellulose

B glucose

C starch

D sucrose

May/June 2011 · Paper 1 · Variant 1 · Question 34

Correct answer: B

34 Which statement correctly describes advantages or disadvantages of self-pollination to a plant?

A It needs a lot of pollen but can happen when a plant is on its own.

B It needs little pollen but there is a high chance of pollination.

C It needs no agent to transfer pollen but pollination is unlikely.

D It needs two plants of the same species but there is little variation in the offspring.

May/June 2011 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 Which of these may be heterozygous?

A a haploid cell

B an allele of a gene

C an organism with a dominant phenotype

D an organism with a recessive genotype

May/June 2011 · Paper 1 · Variant 1 · Question 39

Correct answer: C

39 Flower colour is controlled by a single pair of alleles. The allele for red flowers is dominant to the allele for white flowers.

A plant homozygous for red flowers is crossed with a plant homozygous for white flowers. All the resulting plants have red flowers (F_1 generation).

When the F_1 generation are crossed with each other, 18 plants are obtained. 12 plants have red flowers and 6 have white flowers (F_2 generation).

What ratio is expected in the F_2 generation and what ratio has been obtained?

	expected ratio red to white	obtained ratio red to white
A	1 : 1	2 : 1
B	1 : 1	3 : 1
C	3 : 1	2 : 1
D	3 : 1	3 : 1

May/June 2011 · Paper 1 · Variant 2 · Question 1

Correct answer: B

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 2 and 5

May/June 2011 · Paper 1 · Variant 2 · Question 2

Correct answer: A

2 Below is a series of cell processes.

1 mineral ions entering root hair cells

2 glucose uptake by villus cells

3 water entering root hair cells

Which of these involve active transport?

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

May/June 2011 · Paper 1 · Variant 2 · Question 3

Correct answer: C

3 The sentence describes the uptake of water by a plant.

Water moves into the root hairs of a plant by osmosis through a .....1..... permeable cell membrane, ...2..... a water potential gradient.

Which words correctly complete gaps 1 and 2?

	1	2
A	fully	down
B	fully	up
C	partially	down
D	partially	up

May/June 2011 · Paper 1 · Variant 2 · Question 4

Correct answer: B

4 Magnesium is an essential element for

A the formation of cell walls.

B the formation of chlorophyll.

C the formation of proteins.

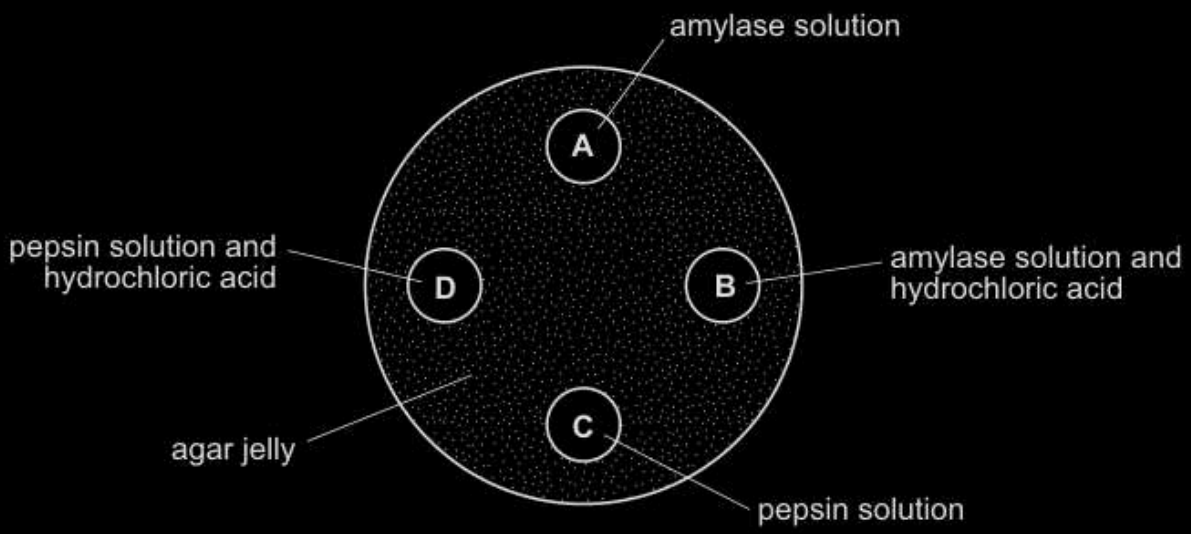
D the process of cell division.

May/June 2011 · Paper 1 · Variant 2 · Question 5

Correct answer: A

5 A dish is filled with agar jelly containing starch. Four holes are cut in the jelly and each hole is filled as shown.

After 30 minutes, which hole will be surrounded by the largest area without starch?



May/June 2011 · Paper 1 · Variant 2 · Question 8

Correct answer: C

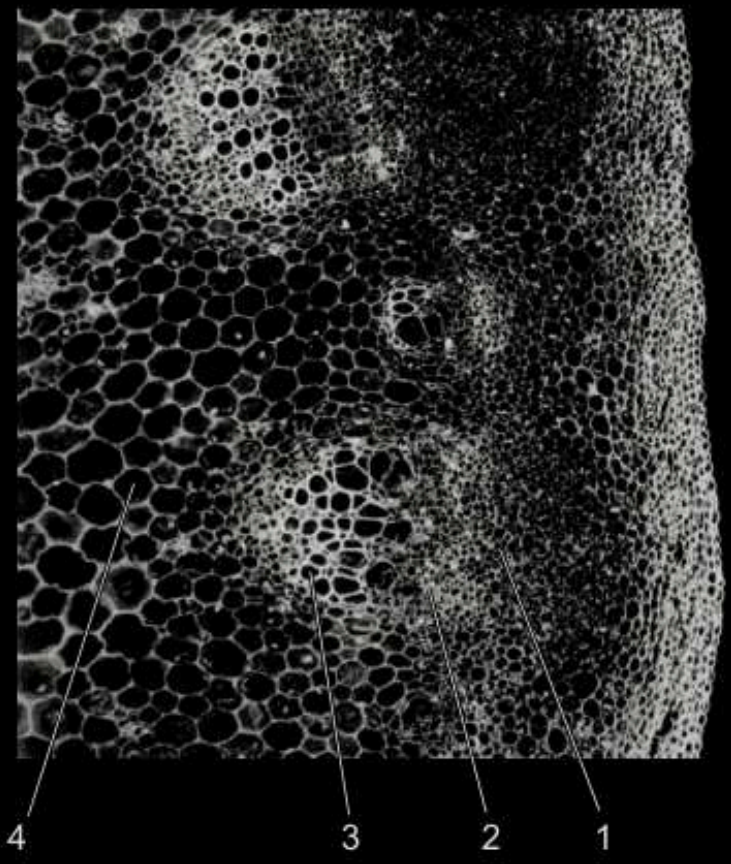
8 A protein solution is tested using three different reagents.

Which set of results is obtained?

	iodine solution	Benedict's solution	biuret solution
A	black	blue	blue
B	black	red	blue
C	brown	blue	purple
D	brown	red	purple

Correct answer: C

11 The photomicrograph shows part of a sunflower stem.



Which tissue transports water and mineral salts and which tissue transports sucrose?

	water and mineral salts	sucrose
A	1	4
B	2	3
C	3	2
D	4	1

May/June 2011 · Paper 1 · Variant 2 · Question 13

Correct answer: A

13 The table shows the characteristics of the blood in one blood vessel in the body.

oxygen concentration	carbon dioxide concentration	pressure
high	low	high

Which blood vessel contains blood with these characteristics?

A aorta

B pulmonary artery

C pulmonary vein

D vena cava

May/June 2011 · Paper 1 · Variant 2 · Question 14

Correct answer: D

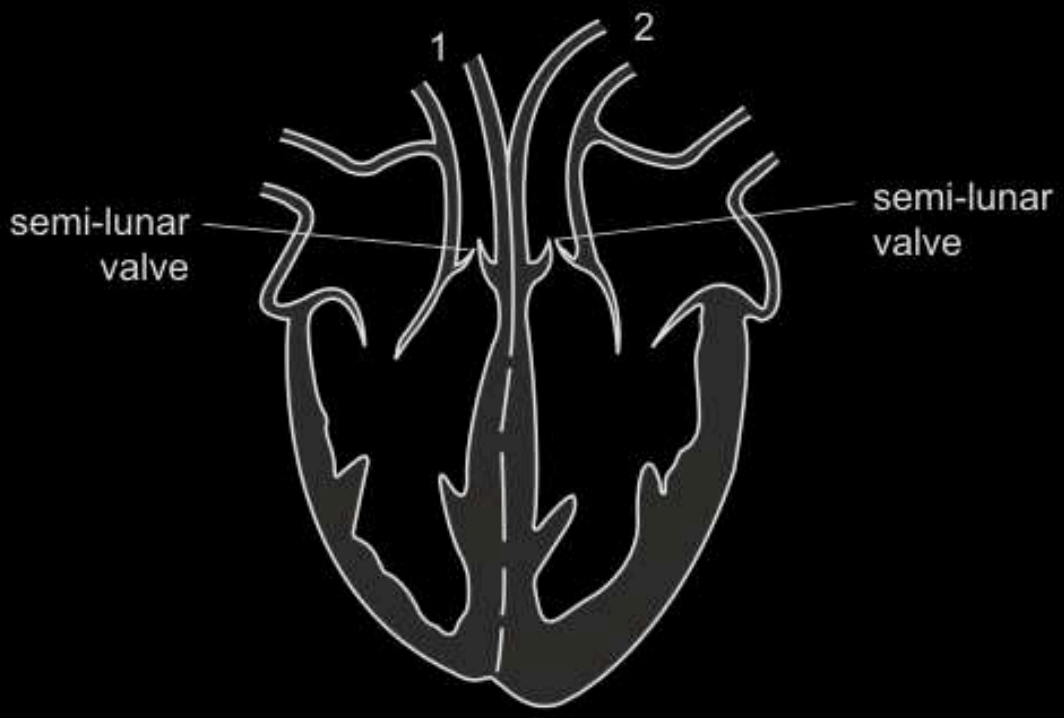
14 How do veins differ from arteries?

	width of lumen	wall thickness	elastic fibres	muscles in wall
A	narrower	thicker	more	less
B	narrower	thinner	less	more
C	wider	thicker	more	more
D	wider	thinner	less	less

May/June 2011 · Paper 1 · Variant 2 · Question 15

Correct answer: A

15 The diagram shows a section through the human heart.



What happens as blood is being pumped out of the heart?

	semi-lunar valves	vessel through which blood passes to the lungs
A	open	1
B	open	2
C	closed	1
D	closed	2

May/June 2011 · Paper 1 · Variant 2 · Question 17

Correct answer: C

17 Which process does **not** depend on respiration?

A active uptake of ions

B conduction of nervous impulses

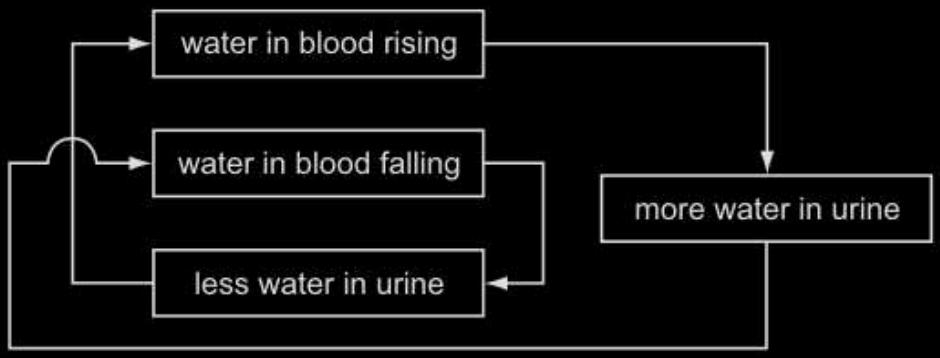
C diffusion of glucose

D muscle contraction

May/June 2011 · Paper 1 · Variant 2 · Question 19

Correct answer: D

19 The diagram refers to the control of water concentration in the blood.



Why is this a negative feedback system?

A It decreases the amount of water in the blood.

B It increases any change in the amount of water in the blood.

C It increases the amount of water in the blood.

D It reverses any change in the amount of water in the blood.

May/June 2011 · Paper 1 · Variant 2 · Question 20

Correct answer: B

20 What is **not** an excretory product of mammals?

A carbon dioxide in expired air

B undigested food in faeces

C urea in sweat

D urea in urine

May/June 2011 · Paper 1 · Variant 2 · Question 22

Correct answer: B

22 What **decreases** as a result of adrenaline secretion?

A blood sugar level

B digestive activity

C heart rate

D size of the pupils of the eyes

May/June 2011 · Paper 1 · Variant 2 · Question 23

Correct answer: A

23 What are the functions of the relay, motor and sensory neurones in a reflex response?

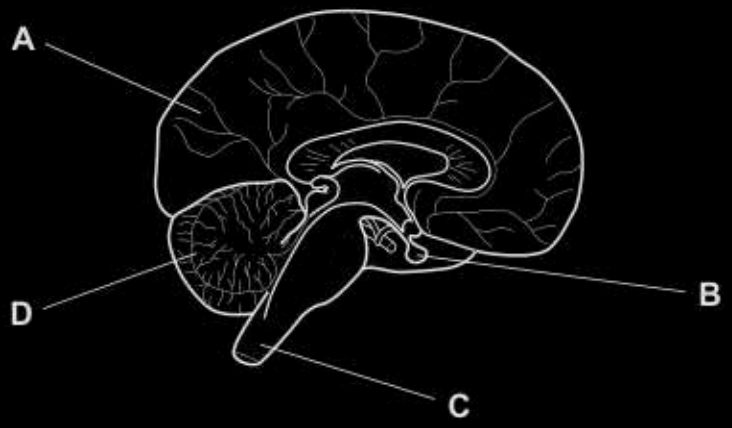
	relay neurone	motor neurone	sensory neurone
A	to connect neurones within the central nervous system	to conduct impulses to the effector from the central nervous system	to conduct impulses from the receptor to the central nervous system
B	to conduct impulses to the effector	to connect neurones within the central nervous system	to receive the stimulus
C	to conduct impulses from the central nervous system to the effector	to conduct impulses from the receptor to the central nervous system	to connect neurones within the central nervous system
D	to conduct impulses from the receptor to the central nervous system	to conduct impulses from the receptor to the central nervous system	to conduct impulses to the effector

May/June 2011 · Paper 1 · Variant 2 · Question 24

Correct answer: D

24 The diagram represents a section through the brain and part of the spinal cord.

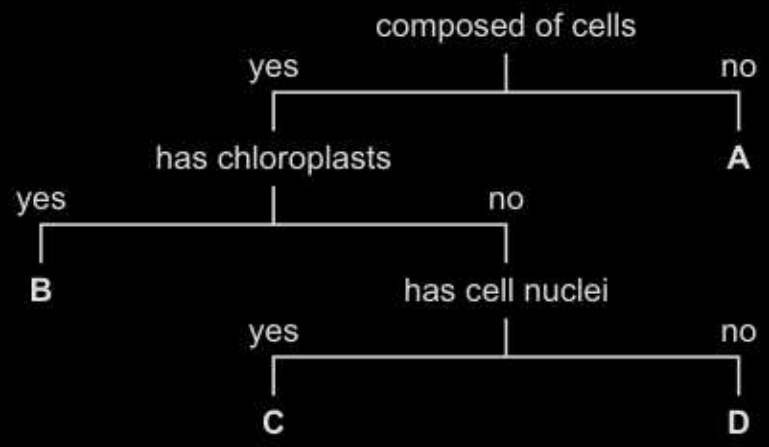
Which part of the brain controls balance?



May/June 2011 · Paper 1 · Variant 2 · Question 25

Correct answer: C

25 Using the flow diagram, which organism is a fungus?



May/June 2011 · Paper 1 · Variant 2 · Question 27

Correct answer: A

27 On decomposition, which substance in decaying plants is the major source of carbon for carbon dioxide production by the micro-organisms involved?

A cellulose

B glucose

C starch

D sucrose

May/June 2011 · Paper 1 · Variant 2 · Question 32

Correct answer: B

32 Which statement correctly describes advantages or disadvantages of self-pollination to a plant?

A It needs a lot of pollen but can happen when a plant is on its own.

B It needs little pollen but there is a high chance of pollination.

C It needs no agent to transfer pollen but pollination is unlikely.

D It needs two plants of the same species but there is little variation in the offspring.

May/June 2011 · Paper 1 · Variant 2 · Question 38

Correct answer: C

38 Flower colour is controlled by a single pair of alleles. The allele for red flowers is dominant to the allele for white flowers.

A plant homozygous for red flowers is crossed with a plant homozygous for white flowers. All the resulting plants have red flowers (F_1 generation).

When the F_1 generation are crossed with each other, 18 plants are obtained. 12 plants have red flowers and 6 have white flowers (F_2 generation).

What ratio is expected in the F_2 generation and what ratio has been obtained?

	expected ratio red to white	obtained ratio red to white
A	1 : 1	2 : 1
B	1 : 1	3 : 1
C	3 : 1	2 : 1
D	3 : 1	3 : 1

May/June 2011 · Paper 1 · Variant 2 · Question 40

Correct answer: C

40 Which of these may be heterozygous?

A a haploid cell

B an allele of a gene

C an organism with a dominant phenotype

D an organism with a recessive genotype

2 Fig. 2.1 shows an experiment on the uptake of water by a leafy shoot placed in a light, breezy environment.

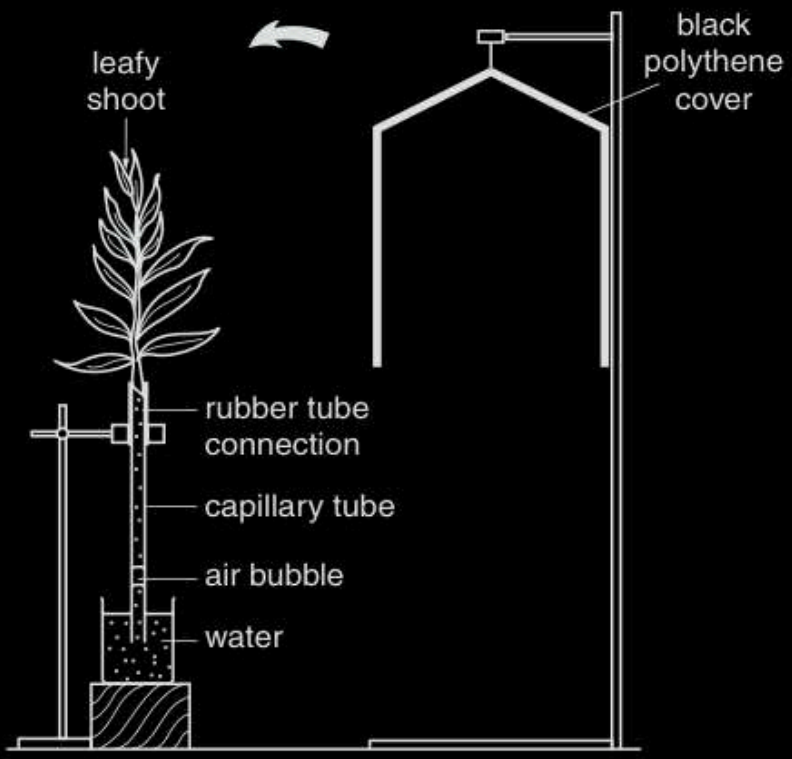


Fig. 2.1

(a) Explain what is likely to happen to the air bubble in the capillary tube.

.....

.....

.....[3]

The black polythene cover is then placed over the plant.

(b) **(i)** State the effect this is likely to have on what happens to the bubble.

.....[2]

(ii) Explain your answer to **b(i)**.

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 11]

2 **(a)** moves towards plant/upwards;
as water is used for photosynthesis;
lost during transpiration/evaporation from leaves;
ref. transpiration pull; [max 3]

(b) **(i)** over period of time/gradually;
(bubble) slows down;
(might) stop; [max 2]

(ii) vapour builds up around/stays close to leaf;
cover prevents air current/wind reaching leaf;
increased humidity;
slower rate of water loss/transpiration/evaporation;
diffusion gradient less steep AW;
photosynthesis stops;
stomata close;
less water used by shoot; [max 6]

[Total: 11]

3 Fig. 3.1 shows some stages in human reproduction.

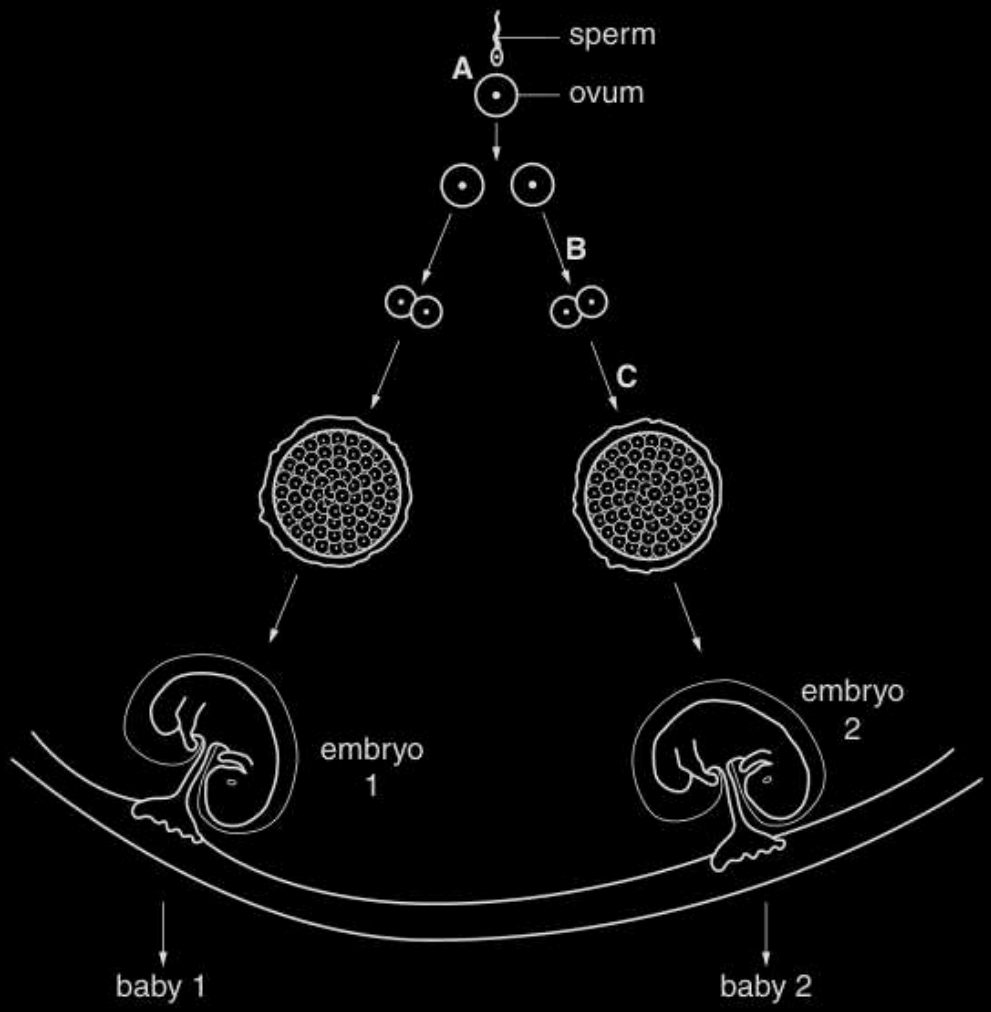


Fig. 3.1

(a) Name the process about to occur at **A** and state where in the reproductive system it normally occurs.

process .....

where it occurs .....[2]

(b) Name the type of cell division taking place at **B** and **C**.

.....[1]

The two embryos develop, are born and grow to become adults.

(c) Suggest two ways in which these two adults must be **similar** to each other, and explain your answers.

1. .....

2. .....

explanation

.....

.....[4]

(d) Suggest two ways in which they may **differ** from one another, and explain your answers.

1. .....

2. .....

explanation

.....

.....[4]

[Total: 11]

3 **(a)** fertilisation;
oviduct / Fallopian tube; [2]

(b) mitosis; [1]

(c) any 2 *genetic* similarities
(e.g. sex/eye colour/blood groups/shape of ears/nose etc.)/same genes* once only/
R look alike/same age/same chromosomes;;
characters caused by genes only;
have both inherited identical/same genes* once only;
they both arise from the same zygote/one sperm + one egg; [max 4]

(d) any two *environmentally affected* characters (*e.g.* size/weight/hair colour)
R different traits unqualified/intelligence;;
ref. to environment/genes only partly responsible;
named relevant environmental factor e.g. amount of sun light/lack of food etc.; [4]

[Total: 11]

4 Fig. 4.1 shows a section through a flower. A pollen grain has been deposited on the carpel.

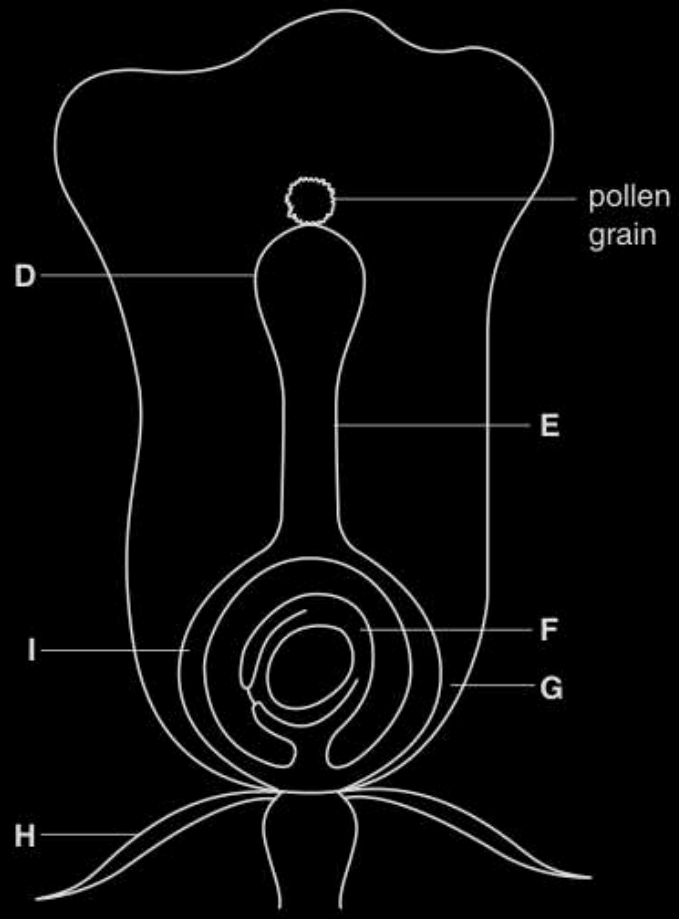


Fig. 4.1

(a) **(i)** Name the process that has occurred in bringing the pollen grain to the carpel.

..... [1]

(ii) State how this process has been brought about in this flower and give two reasons for your answer.

how the process has been brought about .....

.....

reason 1 .....

reason 2 .....[3]

(b) On Fig. 4.1, draw and label the pollen tube and the nuclei as they would appear just before the nuclei fuse. [3]

(c) Changes occur in the flower after the nuclei fuse. Use the letters in Fig. 4.1 to identify

- the parts of the flower that will die as the fruit is formed .....
- the part of the flower that will form the testa .....
- the part of the flower that will form the fruit wall. .....[4]

[Total: 11]

4 **(a)** **(i)** pollination; [1]

(ii) insect (or named);

any two from: large petals/stigma not feathery/stigma protected/
large or rough pollen grains
(If self pollination – no marks);; [3]
(If 'wind' given for (ii) – no mark, but allow 'no nectary' for *one* mark)
(If cross pollination/pollen transferred from anther to stigma given for (ii)
allow two reason marks A no anthers for a valid reason)

(b) pollen tube drawn down style and entering ovule;
through micropyle;
(female) nucleus drawn and labelled in ovule/embryo sac;
(male) nucleus shown in bottom half of pollen tube and labelled;
female and male nucleus identified correctly; [max 3]

(c) D, E, G (and H) – 3 for 2 marks, 2 for 1 mark;;
F;
I; [4]

[Total: 11]

5 Fig. 5.1 shows the percentage of undigested molecules of starch as they pass through the alimentary canal.

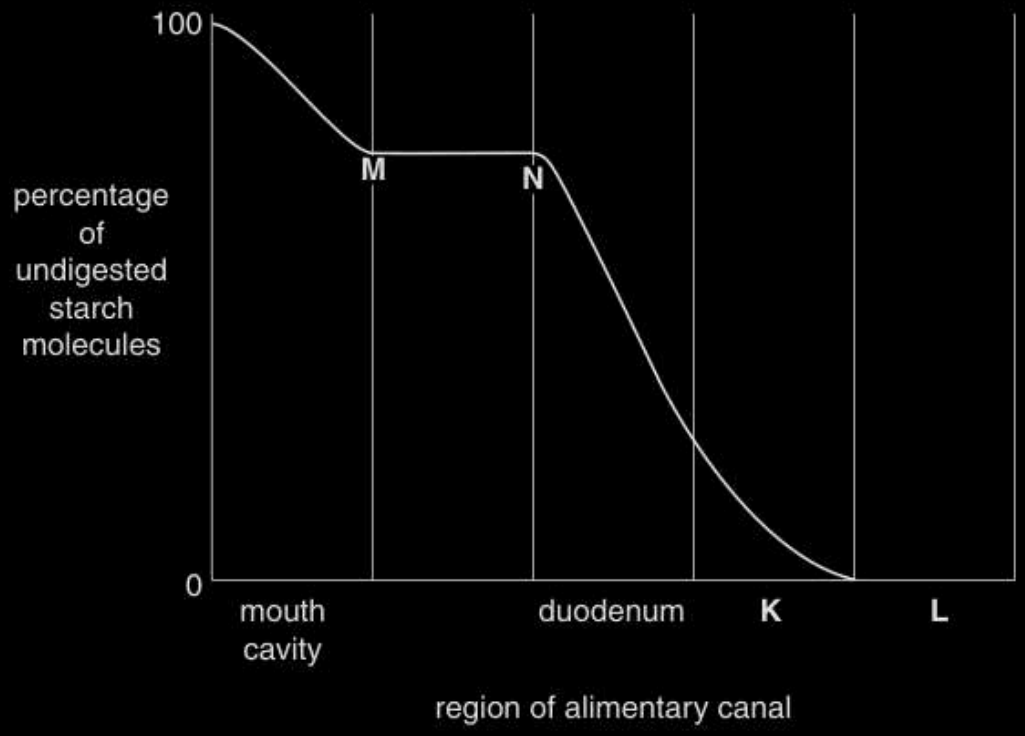


Fig. 5.1

(a) Name the parts of the alimentary canal represented by **K** and **L**.

K .....

L ..... [2]

(b) State and explain what happens to the digestion of starch between **M** and **N**.

.....

.....

.....

.....[4]

(c) Explain why the curve is not shown extending into region **L** of the graph.

.....

.....

.....

.....[2]

[Total: 8]

Section B

Answer **all** questions in this section.

Write your answers in the spaces provided.

5 **(a)** K – ileum/jejunum R small intestine;
L – colon/large intestine; [2]

(b) digestion or enzyme activity stops/amylase doesn't work;
pH incorrect/too acidic;
no suitable amylase produced;
in the stomach; [4]

(c) no starch;
all digested/only water absorption in colon; [2]

[Total: 8]

Section B

6 **(a)** With reference to the blood vessels involved, describe

- how the liver is supplied with the requirements for its functions
- how its products are taken away.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[8]

(b) Explain the importance of capillaries in excretory organs.

.....

.....

.....

.....[2]

[Total: 10]

6 **(a)** hepatic artery;
carries oxygen;
hormones/insulin/adrenalin;
hepatic portal vein;
from gut/villi/small intestine;
glucose (A with ref to artery) R refs glucose – glycogen;
amino acids (A with ref to artery);
hepatic vein;
removes carbon dioxide;
urea;
any named other product of the liver; [max 8]

(b) thin walls/one cell thick;
allow diffusion;
short distance (to diffuse)/close to cells;
of urea/salts etc. into kidneys;
of carbon dioxide into alveoli/lungs;
oxygen to excretory tissues; [max 2]

[Total: 10]

8 **(a)** Describe the functions of the cell membrane.

.....

.....

.....

.....

.....

.....

.....[5]

(b) Explain the advantages to a plant of having its cell membranes surrounded by cell walls.

.....

.....

.....

.....

.....

.....

.....[5]

[Total: 10]

8 **(a)** selectively permeable AW;
osmosis/diffusion;
active transport;
against conc. gradient;
uses energy;
only required substances enter/controls passage of substances;
manufactured substances exported A named examples;
holds/forms boundary for cytoplasm; [max 5]

(b) water enters;
osmosis/diffusion;
ref. concentration gradient/more concentrated cell sap;
pressure within cell/turgor;
cell wall tough/flexible/made of cellulose;
prevents cell bursting/membrane tearing;
supplies support to plant/stem/leaves;
gives shape/rigidity/support to cell; [max 5]

[Total: 10]

9 **(a)** For each of the following, state where, in the reflex arc, they are found and state their functions.

- sensory neurones

.....

.....

- relay neurones

.....

.....

- motor neurones

.....

.....[5]

(b) Explain why, just after hearing a sudden loud noise, a person's heart beats faster.

.....

.....

.....

.....

.....

.....

.....

.....[5]

[Total: 10]

9 **(a)** impulses anywhere;
Accept specific examples for next 6 mps
(sensory) from sense organ/receptor;
to CNS/spinal cord/brain;
(relay) within CNS;
connecting sensory to motor neurones;
(motor) from CNS/spinal cord/brain;
to effector/muscle/gland; [max 5]

(b) sound + stimulus;
ears AW;
one correct reference to a neurone;
brain;
gland/adrenal/suprarenal;
adrenaline/epinephrine;
blood;
heart muscles;
ref. fight/flight/fright etc. response – or described; [max 5]

[Total: 10]

2 Table 2.1 shows the masses of various substances in 100 g of banana fruit and in 100 g of avocado fruit.

Table 2.2 shows a person's recommended daily intake for calcium, iron and vitamin C.

Table 2.1

substance	banana	avocado
fat	0.33 g	14.60 g
cholesterol	0.00 g	0.00 g
carbohydrate	22.80 g	8.53 g
fibre	2.60 g	6.70 g
protein	1.10 g	2.00 g
calcium	5.00 mg	12.00 mg
iron	0.26 mg	0.55 mg
vitamin C	8.70 mg	10.00 mg

Table 2.2

calcium	500 mg
iron	13 mg
vitamin C	62 mg

Bananas form a significant part of the diet of some communities.

(a) Explain why bananas can be a valuable part of a person's diet.

.....

.....

.....

.....[4]

(b) Explain why a diet of only bananas is not suitable for a growing child.

.....

.....

.....

.....

.....[3]

(c) Explain which of the two fruits, banana or avocado, it would be better to eat just before running a race.

.....

.....

.....

.....[3]

[Total: 10]

2 **(a)** CHO + for energy / respiration;
low fat + avoidance of obesity / vascular disease AW;
no cholesterol + avoidance of vascular disease AW;
vitamin C + wound healing / avoidance of scurvy or described;
contains fibre + effective digestive transit AW;
calcium + for bones / teeth;
iron + for blood / RBCs / haemoglobin;
protein + for growth / repair / enzymes; [max. 4]

(b) not enough protein + for growth / repair or described;
not enough calcium + for bone / tooth development;
(A lacks)
not enough iron + for blood / haemoglobin / RBCs; [3]

(c) (banana) carbohydrate (or named) + energy / respiration;
instant / more readily available / more quickly changed or digested;
(A quick)

glucose (any ref to its presence anywhere or to its production);
(could score with first marking point – for 2 marks)

avocado + high(er) in energy;

fat takes longer to be converted to glucose AW / less readily available; [max. 3]

[Total: 10]

3 Fig. 3.1 shows energy flow in a habitat.

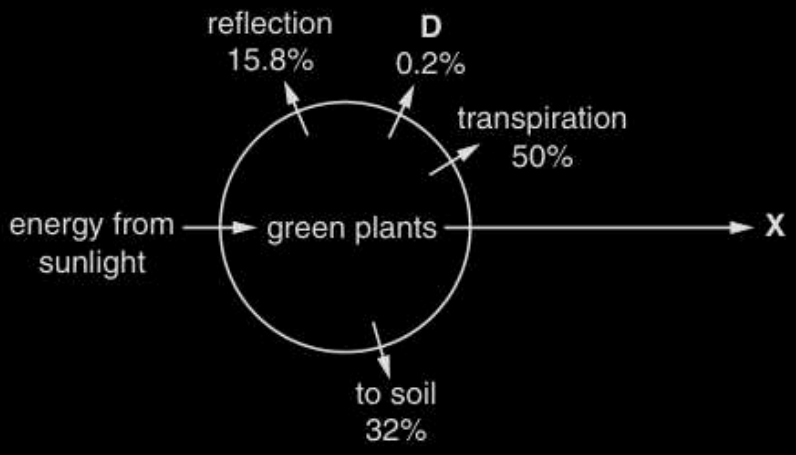


Fig. 3.1

(a) (i) Suggest the identity of **X** in Fig. 3.1. .....[1]

(ii) State the form in which energy passes from the green plants to **X**.

.....[1]

(iii) Calculate the percentage of energy passing to **X**.

.....[1]

(b) State the name of the process **D** in Fig. 3.1 .....[1]

(c) (i) Describe how energy passes into the soil and is then released.

.....

.....

.....

.....

.....[3]

(ii) Suggest how the release of this energy is useful to seeds in the soil.

.....

.....

.....

.....[2]

[Total: 9]

3 **(a)** **(i)** herbivore / named herbivore / consumer / omnivore; [1]

(ii) chemical (energy); [1]

(iii) 2 (%); [1]

(b) respiration; [1]

(c) **(i)** respiration;

bacteria / fungi / decomposers (N.B. look for idea of organism);
(R the process)
organic or dead matter / organic molecules or named;

energy released + when broken down / decomposed (A acted on by decomposers) / decayed AW / when fossil fuel burnt; [max. 3]

(ii) makes soil warmer / heat released;
for germination;
increases AW rate of enzyme action;
faster AW growth (of radical / plumule); [max. 2]
(R 'for' growth / helps growth)
(R any reference to a seedling / plant)

[Total: 9]

4 Fig. 4.1 shows a plant cell.

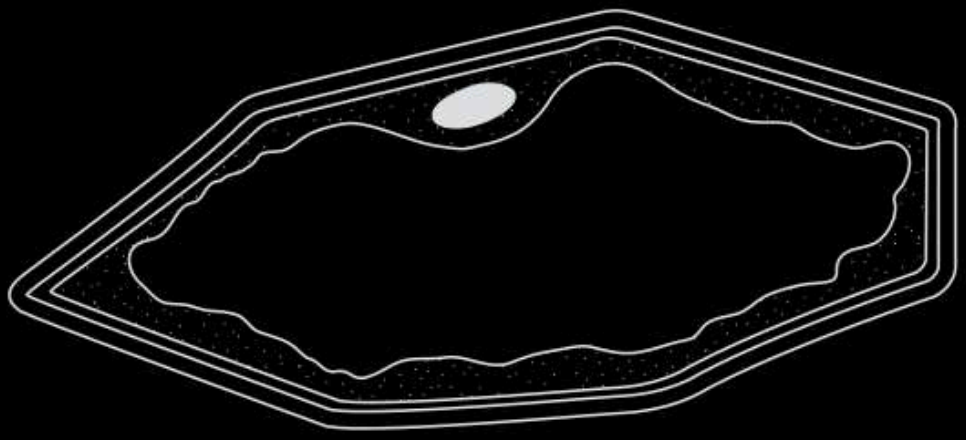


Fig. 4.1

(a) On Fig. 4.1, label a partially permeable membrane. [1]

(b) It is possible to remove the cell wall of a plant cell by using an enzyme. Suggest and explain what would happen if a cell that had been treated in this way, and also an untreated cell, were both put into distilled water for an hour.

treated cell .....

.....

.....

untreated cell .....

.....

.....[5]

(c) Cells are taken from one plant, treated with the enzyme described in **(b)**, and then grown on a culture medium. There, each of these cells divides to form a group of cells. Each group of cells then develops into a new plant.

(i) State the type of division that the nuclei of these cells have undergone.

.....[1]

(ii) Suggest and explain a possible commercial use of this procedure.

.....

.....

.....

.....

.....[3]

[Total: 10]

4 **(a)** either cell or tonoplast membrane labelled (line must end on membrane); [1]

(b) (treated) water enters* (* = or below);
<u>osmosis</u> / <u>diffusion</u> *;
cell swells * ;
bursts [N.B. This mark stands alone and is NOT included in the 4 max. for part **(b)**] ; [1]
(untreated) water enters *;
<u>osmosis</u> / <u>diffusion</u> *;
cell pressure increases / becomes turgid AW;
cell swells * ;
cell wall + withstands pressure / prevents it from bursting AW; [max. 4]

(c) **(i)** <u>mitosis</u>; [1]

(ii) *(Reject any answers relating to fungi or bacteria)*

many plants / greater profit / economic benefit / cheap;
(A higher / greater yield)
identical / clone (R similar, A same);

of known / desired characteristic;

grown under controlled conditions;

greater certainty; [max. 3]
(Ignore references to single parent required)

[Total: 10]

5 Fig. 5.1(a) and Fig. 5.1(b) show graphs of the pulse and breathing rates of two students, **E** and **F**, during and after one minute's vigorous exercise.

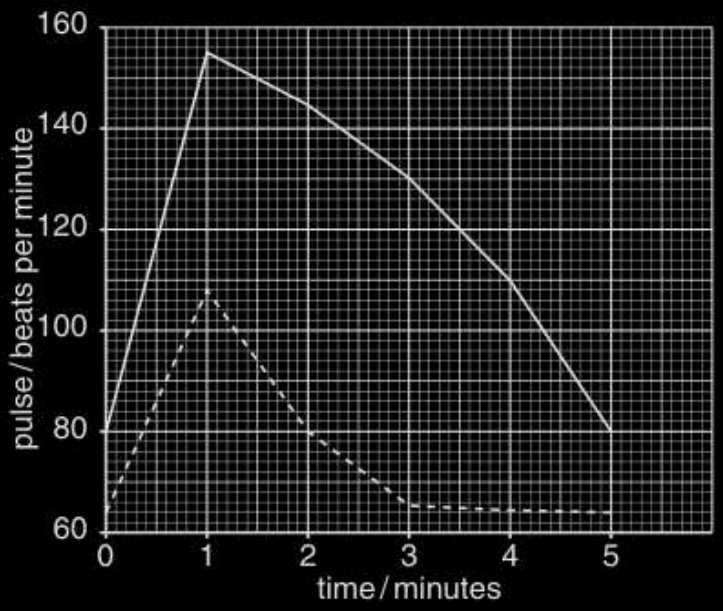


Fig. 5.1(a)

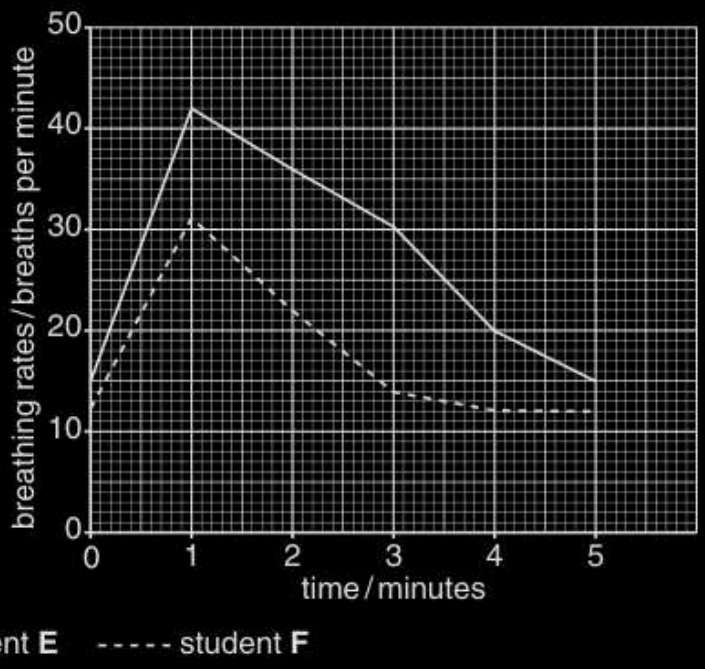


Fig. 5.1(b)

(a) State how long it took for student **F**'s pulse and breathing rates to return to their original levels after the student had finished exercising.

pulse rate .....

breathing rate ..... [2]

(b) Explain why the pulse and breathing rates of both students increased during exercise.

.....

.....

.....

.....

.....[4]

(c) Suggest and describe possible reasons for the differences in the effect of vigorous exercise on these two students.

.....

.....

.....

.....

.....[4]

[Total: 10]

Section B

Answer **all** questions in this section.

Write your answers in the spaces provided.

5 **(a)** 2.5 – 4 + minutes for pulse (A any within range);
3 – 4 + minutes for breathing (A anything within range); [2]

(b) muscles;

For the remaining marks in this part, there must be the use somewhere of a word that indicates enhancement of at least one of the factors (i.e. more / greater, faster etc.).
Thus, for example, 'more oxygen' scores, and so, then, would any ref. to CO_2 removal or energy, even if 'more' is not repeated.

more / faster blood;
more oxygen / glucose;
ref. more CO_2 removed / lactic acid;
more energy (R produced, made, manufactured, etc.); [max. 4]

(c) *(If they do not indicate which student they are talking about, mark up to **2** max.)*

*student F is fitter / exercises regularly AW;

*more efficient muscles / better breakdown of lactic acid;

*more efficient circulation / no or limited cholesterol in blood vessels;

*more efficient lungs / respiratory system;

*exercised less vigorously;

*more haemoglobin / more RBCs;
(*A reverse arguments for Student E)

student E was a smoker, F was not;

student E was obese / overweight AW (R large mass);

emphysema;

suffered from asthma / bronchitis / heart problems / lung infection AW; [max. 4]

[Total: 10]

6 **(a)** **(i)** State what is meant by *transpiration*.

.....

.....

.....[3]

(ii) Explain how the transpiration stream is important for the production of proteins in a plant.

.....

.....

.....[2]

(b) Describe and explain how tissues in the stem of a plant become stained when a plant is carefully removed from the soil and placed in a solution of dye for a few hours.

.....

.....

.....

.....

.....

.....

.....[5]

[Total: 10]

6 **(a)** **(i)** loss of water;
as vapour / evaporation;
from leaves / aerial parts of a plant / to the atmosphere;
through stomata (A between or through guard cells); [max. 3]

(ii) brings nitrates;
dissolved in water;
from soil / to combine with carbohydrates; [max. 2]

(b) dye diffuses (R ref. to osmosis);
through permeable (R semi- / partially- / selectively-);
(Ignore refs. to cell membrane)
(cell) wall;

of root (hair) cells;

xylem (any correct ref.);

ref. to travelling upwards (A refs. to transpiration pull / stream capillarity / root pressure AW);
[max. 5]

[Total: 10]

7 **(a)** Describe the functions of the kidneys.

.....

.....

.....

.....

.....

.....[4]

(b) Describe and explain the ways in which a person may be kept alive even when both of their kidneys have stopped functioning.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 10]

Section C

Answer **one** question from this section.

Write your answers in the spaces provided.

7 **(a)** excretion (A excretory);
removal of urea ;
salts / ref. osmoregulation / varying amounts of water;
toxins / poisons / hormones (R harmful);
filtration of / removal from + blood / plasma;
forming urine; [max. 4]

(b) ref dialysis;
blood along a partially permeable tube;
washing / bathing fluid (dialysis fluid could score 2 marks);
ref controlled contents of bathing fluid (i.e. concentrations as required in blood)
(Look for more than just a ref. to urea.);
diffusion (any correct ref.);
keeps blood at constant osmotic potential;
washing fluid changed;
temperature regulation;
urea removed;
blood from and to patient ;
kidney transplants / donation;
bubble trap / adding anticoagulants / counter flow / large surface area / pump ; [max. 6]
(A coiled tube)

[Total: 10]

Section C

9 **(a)** Explain why there are approximately equal numbers of babies born of each sex.

.....

.....

.....

.....[3]

(b) With the aid of a genetic diagram, describe the possible genotypes and phenotypes of the offspring of two parents who both have codominant alleles for a named character.

.....

.....

.....

.....

.....

.....

.....

.....

[7]

[Total: 10]

9 **(a)** ova / female gametes / eggs + carry X **;
sperms / male gametes + carry either X or Y**;
(** reject if the term alleles / genes is used)

in equal numbers;

(thus equal chances) of XX + girl or XY + boy; [max. 3]
(Punnett Square or genetic diagrams cannot score without explanation)

(b) (They are supposed to know AB blood group – but A any other e.g.)
N.B. There is a mark for each of the terms 'parents', 'gametes' and 'offspring' in the genetic diagram below.

parents / P; $I^AI^B \times I^AI^B$ (no I – penalise once only);
(A mother, father / male, female)

gametes / G; $I^A \quad I^B \quad I^A \quad I^B$;

offspring; $I^AI^A \quad I^AI^B \quad I^AI^B \quad I^BI^B$;
(A F_1 / children)

ratios / numbers of genotypes stated;

ratios / numbers of phenotypes stated;

phenotypes linked with genotypes; [max. 7]

N.B. Award marks up to a Max. 4 if
(i) both parents have at least one codominant allele in their genotypes e.g. I^AI^A or I^BI^o

(ii) if the answer is solely in text with no diagram.

[Total: 10]

October/November 2010 · Paper 1 · Variant 1 · Question 1

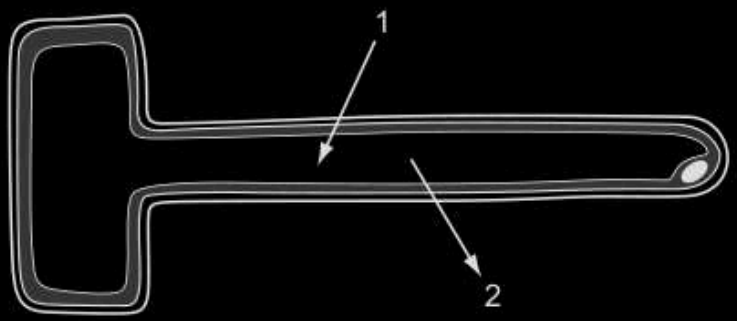
Correct answer: B

1 Which terms describe the parts of a plant?

	leaf	leaf mesophyll	leaf epidermis
A	organ	cell	tissue
B	organ	tissue	tissue
C	organ system	tissue	cell
D	tissue	cell	cell

Correct answer: B

2 The diagram shows a root hair.



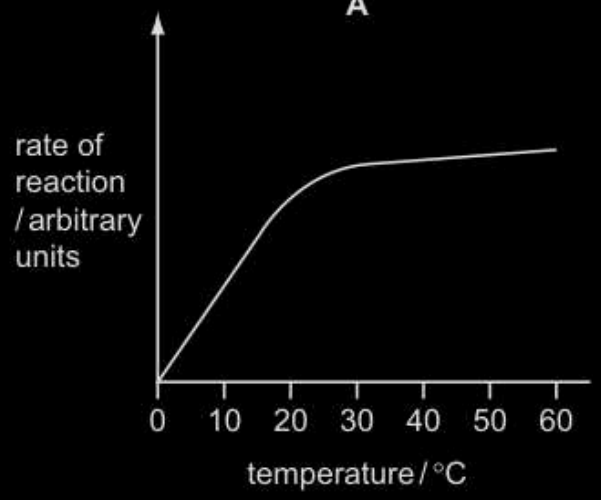
Which arrows show the direction in which it is **possible** for nitrate ions and water molecules to move?

	nitrate ions	water molecules
A	1 only	1 and 2
B	1 and 2	1 and 2
C	1 and 2	2 only
D	2 only	1 only

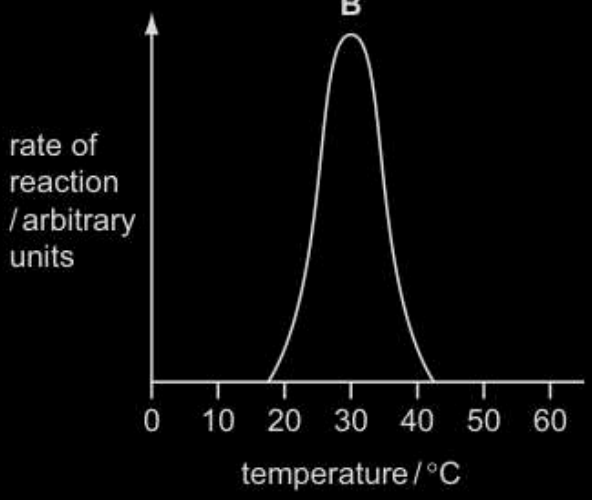
Correct answer: C

4 Which graph shows the effect of temperature on enzyme-controlled reactions?

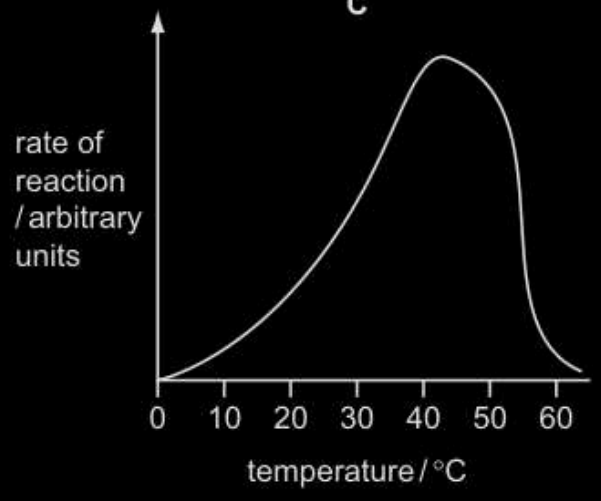
A



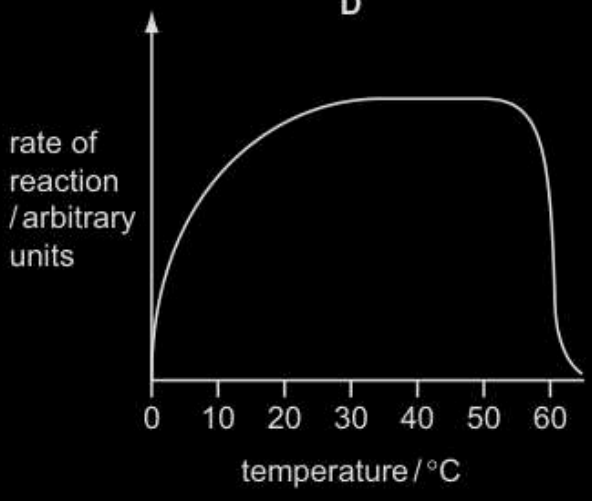
B



C



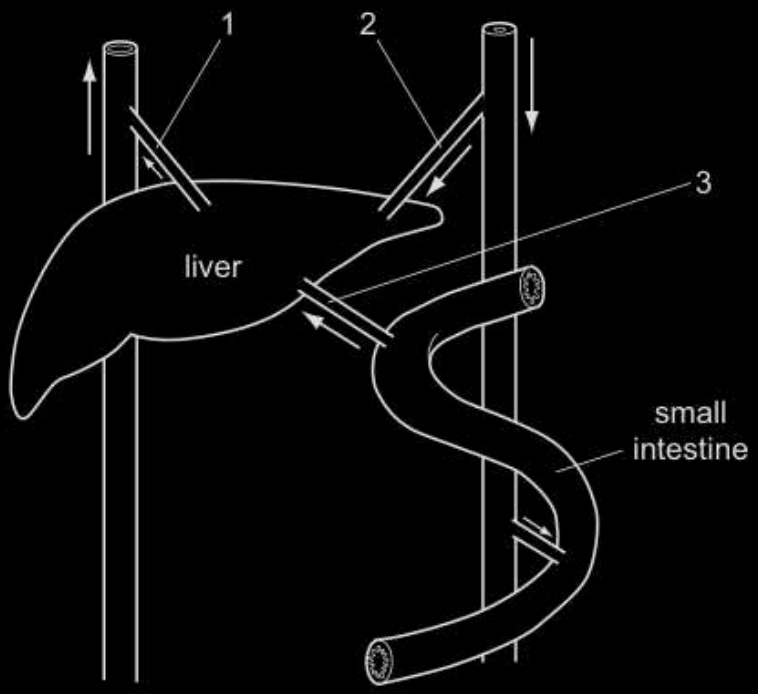
D



October/November 2010 · Paper 1 · Variant 1 · Question 8

Correct answer: A

8 The diagram represents the liver and associated blood vessels.



After a meal, how is the blood affected by the liver as it passes between these blood vessels?

	blood vessels	effect of liver
A	2 to 1	glucose added
B	2 to 1	urea removed
C	3 to 1	glucose added
D	3 to 1	urea removed

October/November 2010 · Paper 1 · Variant 1 · Question 9

Correct answer: B

9 Which chemical test shows the presence of an enzyme in a biological washing powder?

A Benedict's

B biuret

C ethanol emulsion

D iodine solution

October/November 2010 · Paper 1 · Variant 1 · Question 11

Correct answer: C

11 Which adaptations of a root hair cell make it suitable for water uptake?

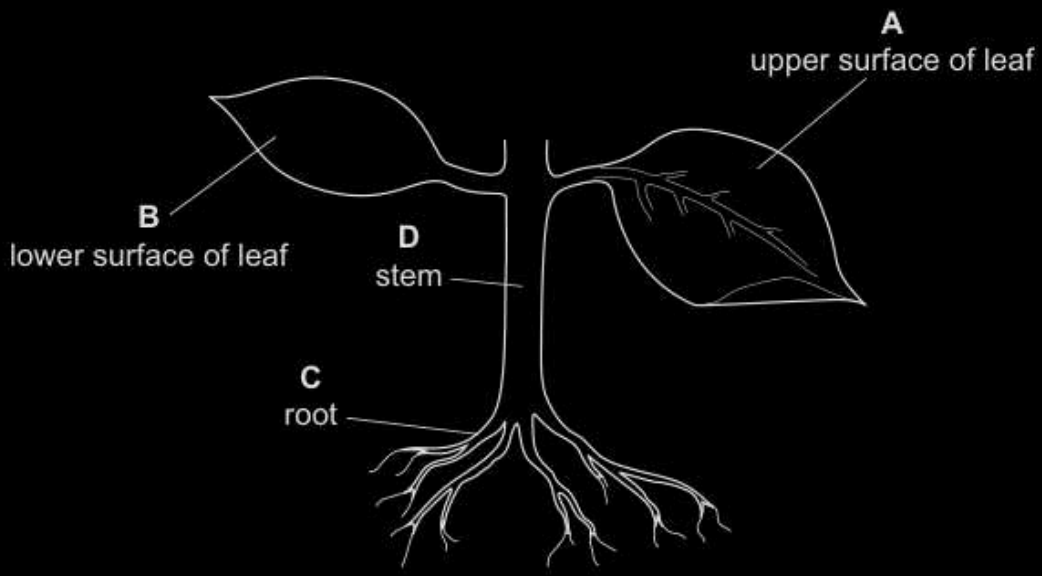
	partially permeable cell membrane	surface area to volume ratio of the cell
A	absent	high
B	absent	low
C	present	high
D	present	low

October/November 2010 · Paper 1 · Variant 1 · Question 12

Correct answer: B

12 The diagram shows part of a flowering plant.

Where does most transpiration take place?



October/November 2010 · Paper 1 · Variant 1 · Question 13

Correct answer: D

13 Vitamin C changes cholesterol into a form which is less likely to be deposited in artery walls.

Which dietary advice would help prevent coronary heart disease?

A Eat more bread and less meat.

B Eat more butter and less bread.

C Eat more meat and less oranges.

D Eat more oranges and less butter.

October/November 2010 · Paper 1 · Variant 1 · Question 14

Correct answer: A

14 Which sequence shows the shortest route taken by blood travelling from a leg to an arm in the human body?

A leg → heart → lungs → heart → arm

B leg → heart → lungs → kidney → arm

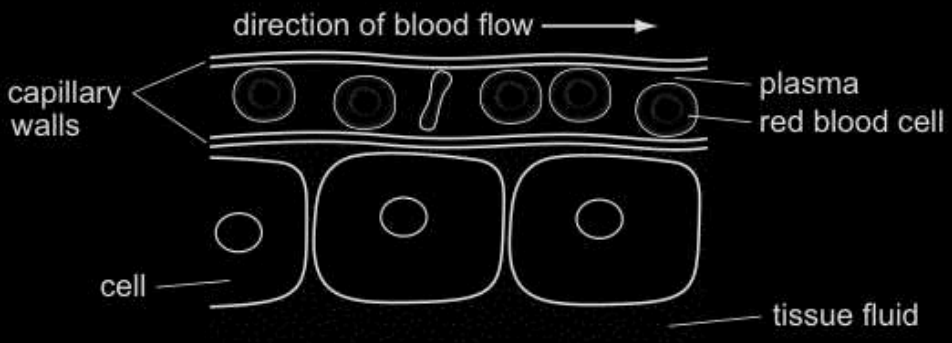
C leg → kidney → heart → lungs → arm

D leg → lungs → heart → gut → arm

October/November 2010 · Paper 1 · Variant 1 · Question 15

Correct answer: A

15 The diagram shows a blood capillary close to some tissue cells bathed in tissue fluid. Exchange of nutrients takes place here.

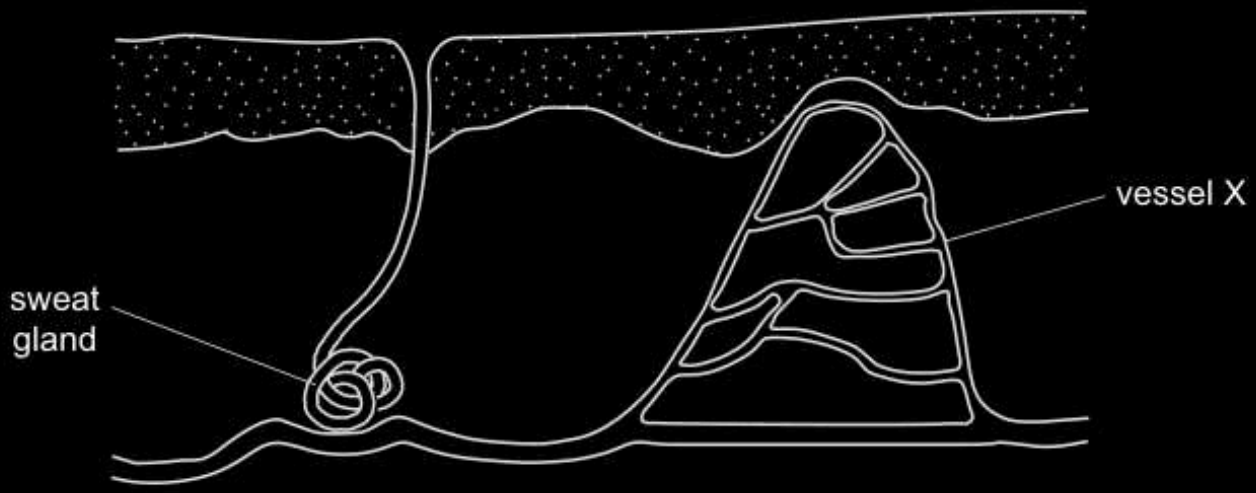


Which row shows the type of nutrient in the plasma and in the tissue fluid and the method of transfer between the two?

	plasma	tissue fluid	method of transfer
A	amino acid	amino acid	diffusion
B	amino acid	protein	osmosis
C	protein	amino acid	digestion
D	protein	protein	active uptake

Correct answer: C

20 The diagram shows a section through skin.

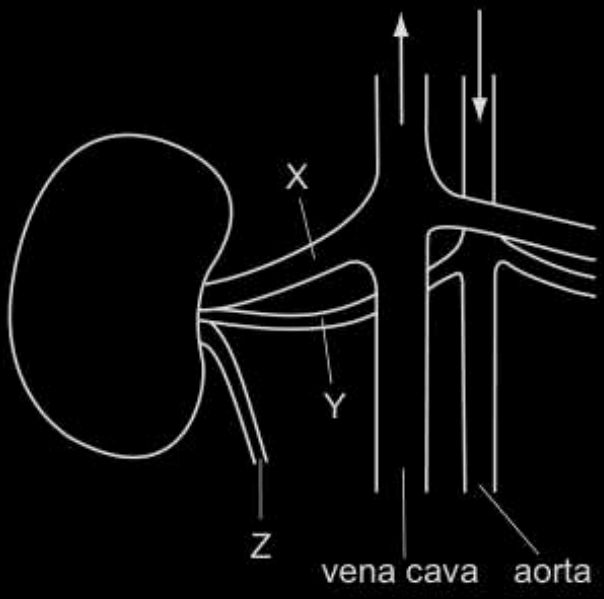


What happens if the body temperature starts to fall below normal?

	sweat glands	blood flow in vessel X
A	secrete sweat	decreases
B	secrete sweat	increases
C	stop secreting sweat	decreases
D	stop secreting sweat	increases

Correct answer: C

21 The diagram shows the structures associated with a human kidney.



What are the relative concentrations of urea in X, Y and Z?

A X is sometimes higher than Y

B Y is always higher than Z

C Y is always lower than Z

D Z is sometimes lower than X

Correct answer: A

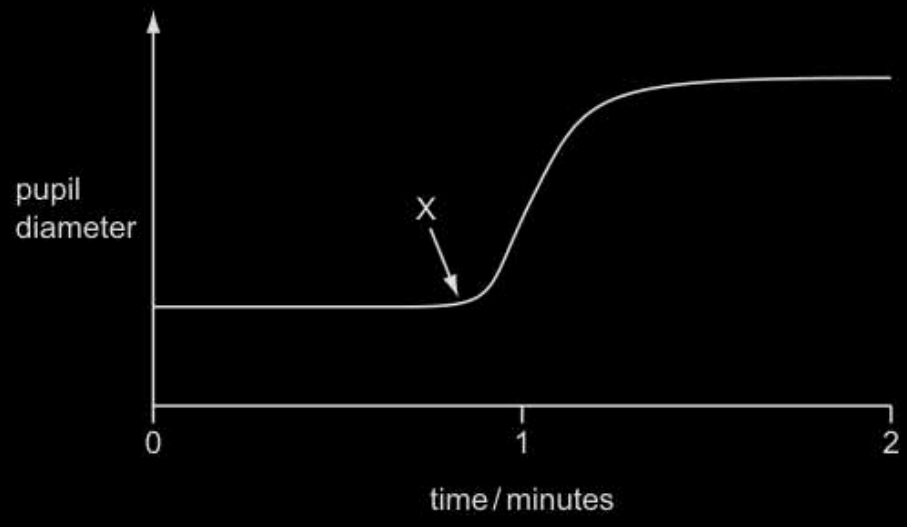
22 The diagram shows a section through the human brain.

What are the functions of the labelled regions?

	memory	balance
A	W	X
B	W	Y
C	X	W
D	X	Y

Correct answer: B

23 The graph shows how the diameter of the pupil of a person's eye changed during the course of two minutes.



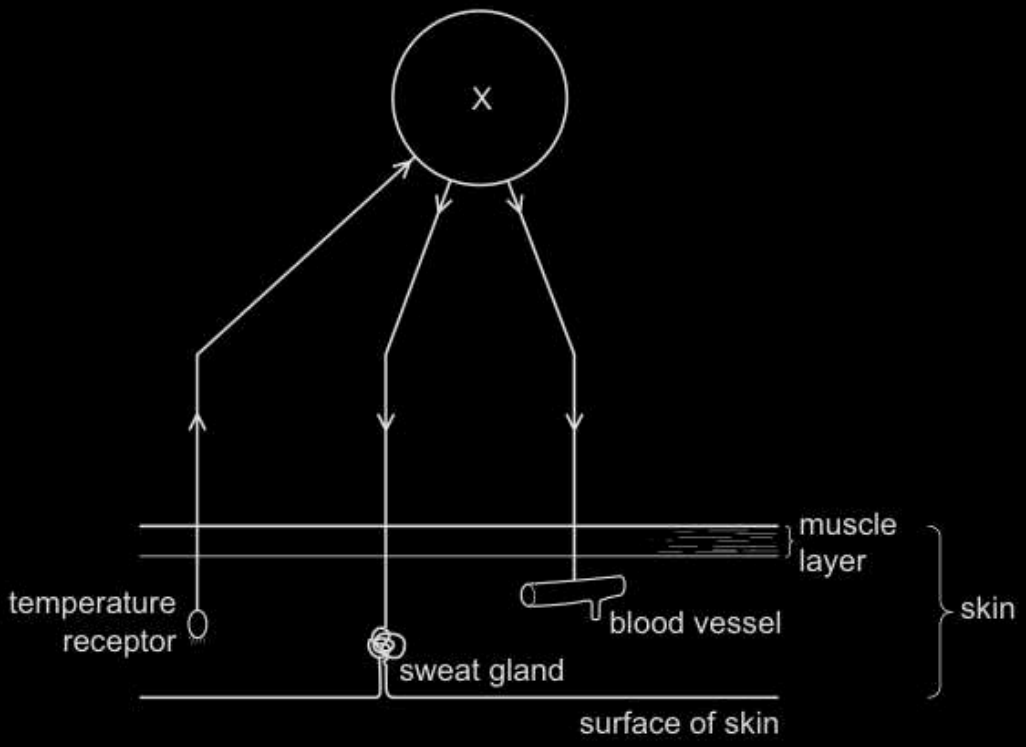
What happens to the light intensity at X and which muscles begin to contract?

	light intensity	iris muscles contracting
A	decreases	circular
B	decreases	radial
C	increases	circular
D	increases	radial

October/November 2010 · Paper 1 · Variant 1 · Question 26

Correct answer: B

26 The diagram shows some nerve pathways involved in temperature control of the human body.



Which part of the nervous system does X represent?

A cerebrum

B hypothalamus

C medulla

D pituitary gland

October/November 2010 · Paper 1 · Variant 1 · Question 33

Correct answer: B

33 What passes down a pollen tube after pollination?

A female nucleus

B male nucleus

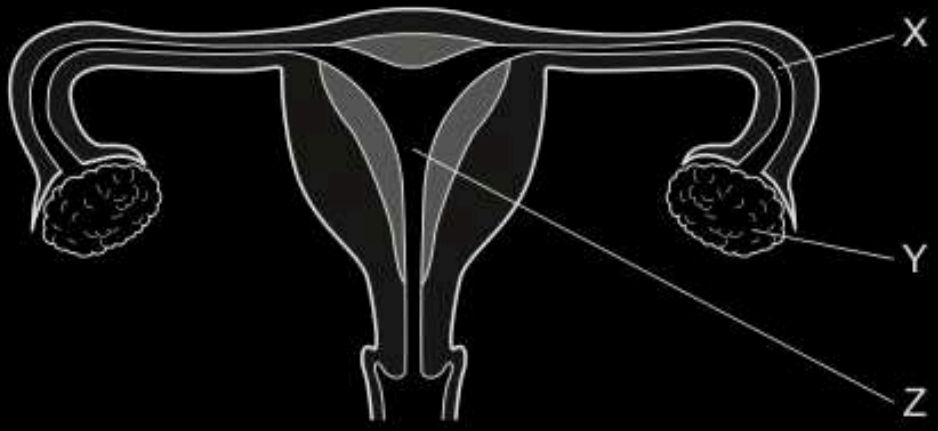
C ovule

D pollen grain

October/November 2010 · Paper 1 · Variant 1 · Question 35

Correct answer: C

35 The diagram shows a section through the female reproductive system.



During pregnancy, where does mitosis occur in the cells of the embryo?

	X	Y	Z
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✗	✓

key

✓ = takes place

✗ = does not take place

October/November 2010 · Paper 1 · Variant 1 · Question 36

Correct answer: A

36 What is a reason for breast milk being better for a baby than bottled milk?

A It contains antibodies for disease protection.

B It contains calcium ions for bone development.

C It contains protein for growth.

D It contains sugar for energy.

October/November 2010 · Paper 1 · Variant 1 · Question 37

Correct answer: A

37 Which statement is evidence that genes may be copied and passed on to the next generation?

A Asexual reproduction produces genetically identical offspring.

B Different alleles of a gene can produce variation in phenotype.

C Each species of a plant or animal has a fixed number of chromosomes.

D Sexual reproduction produces genetically different offspring.

October/November 2010 · Paper 1 · Variant 1 · Question 38

Correct answer: B

38 The genotype for the height of an organism is written as Tt.

What conclusion may be drawn?

A The allele for height has at least two different genes.

B There are at least two different alleles of the gene for height.

C There are two different genes for height, each having a single allele.

D There is one allele for height with two different forms.

October/November 2010 · Paper 1 · Variant 1 · Question 39

Correct answer: C

39 The body cells of a mammal contain two copies of 24 000 genes, making 48 000 genes in total.

Of these 48 000 genes in the body cells of a mammal, how many would have been inherited from its mother?

A 6 000

B 12 000

C 24 000

D 48 000

October/November 2010 · Paper 1 · Variant 1 · Question 40

Correct answer: B

40 The inheritance of the ABO blood groups in humans is controlled by three alleles (I^A, I^B and I^O), only two of which can be present in one individual.

What are the possible blood groups of children born to a homozygous group A woman and a heterozygous group B man?

A A and B only

B A and AB only

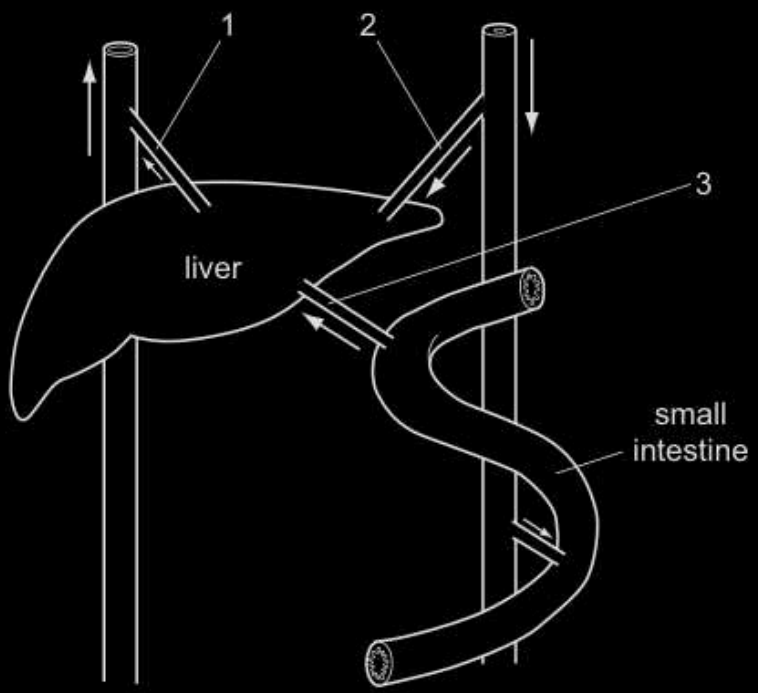
C A, B and AB only

D A, B, AB and O

October/November 2010 · Paper 1 · Variant 2 · Question 1

Correct answer: A

1 The diagram represents the liver and associated blood vessels.



After a meal, how is the blood affected by the liver as it passes between these blood vessels?

	blood vessels	effect of liver
A	2 to 1	glucose added
B	2 to 1	urea removed
C	3 to 1	glucose added
D	3 to 1	urea removed

October/November 2010 · Paper 1 · Variant 2 · Question 3

Correct answer: B

3 Which chemical test shows the presence of an enzyme in a biological washing powder?

A Benedict's

B biuret

C ethanol emulsion

D iodine solution

October/November 2010 · Paper 1 · Variant 2 · Question 4

Correct answer: D

4 Vitamin C changes cholesterol into a form which is less likely to be deposited in artery walls.

Which dietary advice would help prevent coronary heart disease?

A Eat more bread and less meat.

B Eat more butter and less bread.

C Eat more meat and less oranges.

D Eat more oranges and less butter.

October/November 2010 · Paper 1 · Variant 2 · Question 5

Correct answer: B

5 Which terms describe the parts of a plant?

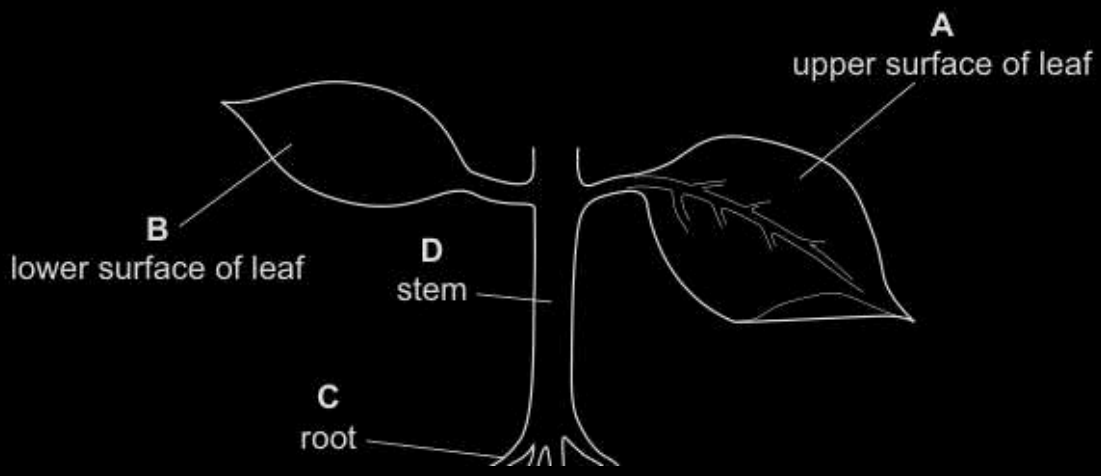
	leaf	leaf mesophyll	leaf epidermis
A	organ	cell	tissue
B	organ	tissue	tissue
C	organ system	tissue	cell
D	tissue	cell	cell

October/November 2010 · Paper 1 · Variant 2 · Question 6

Correct answer: B

6 The diagram shows part of a flowering plant.

Where does most transpiration take place?



October/November 2010 · Paper 1 · Variant 2 · Question 7

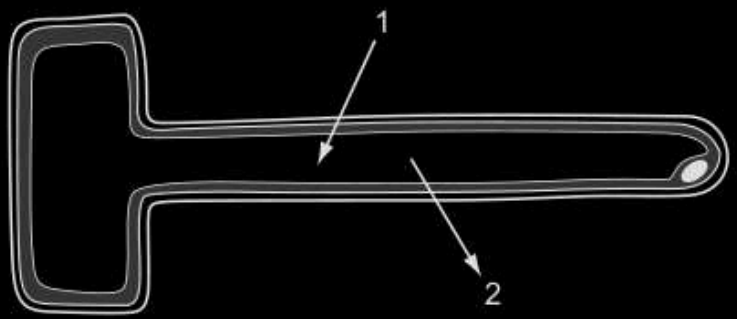
Correct answer: C

7 Which adaptations of a root hair cell make it suitable for water uptake?

	partially permeable cell membrane	surface area to volume ratio of the cell
A	absent	high
B	absent	low
C	present	high
D	present	low

Correct answer: B

8 The diagram shows a root hair.



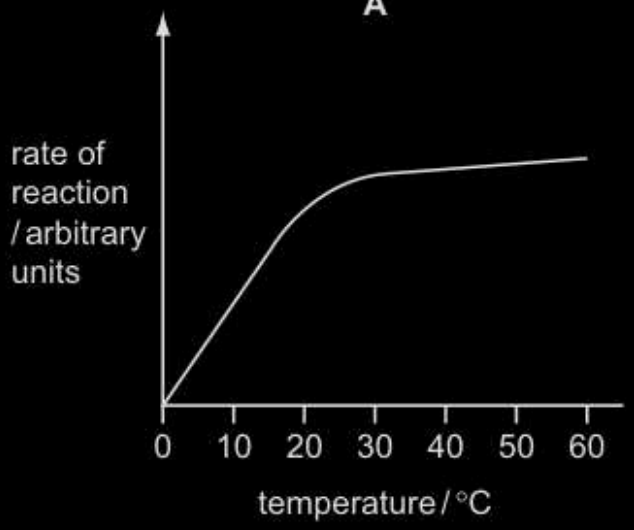
Which arrows show the direction in which it is **possible** for nitrate ions and water molecules to move?

	nitrate ions	water molecules
A	1 only	1 and 2
B	1 and 2	1 and 2
C	1 and 2	2 only
D	2 only	1 only

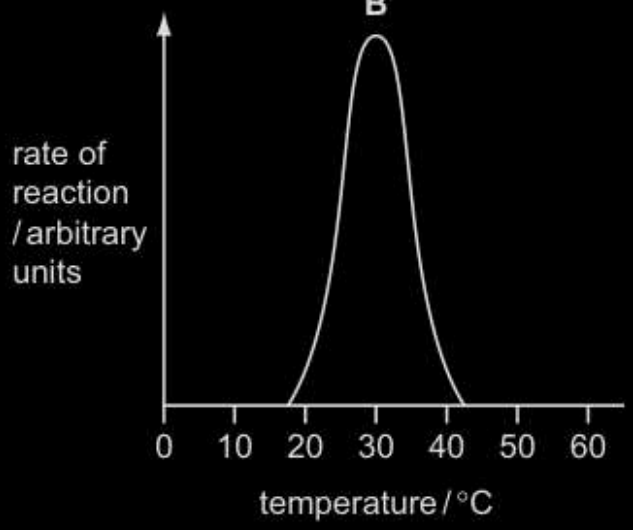
Correct answer: C

13 Which graph shows the effect of temperature on enzyme-controlled reactions?

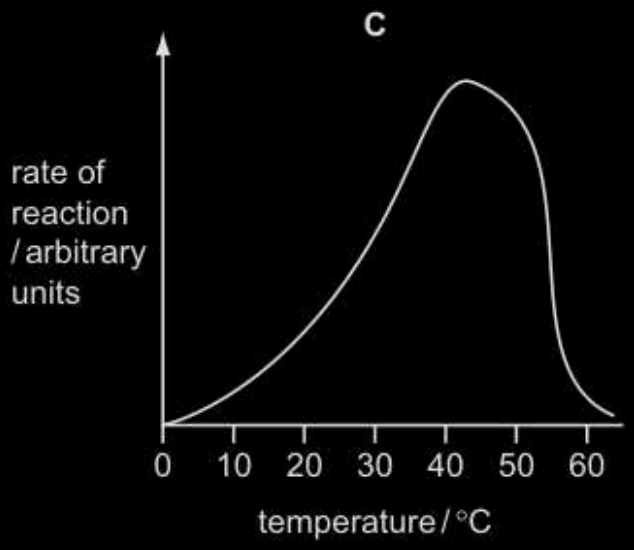
A



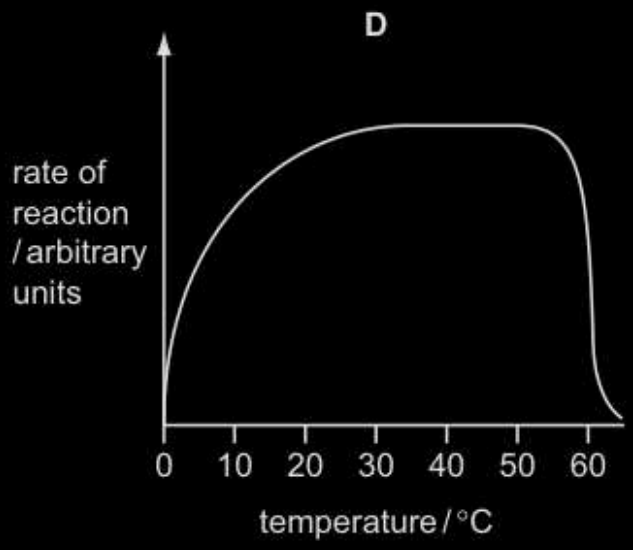
B



C



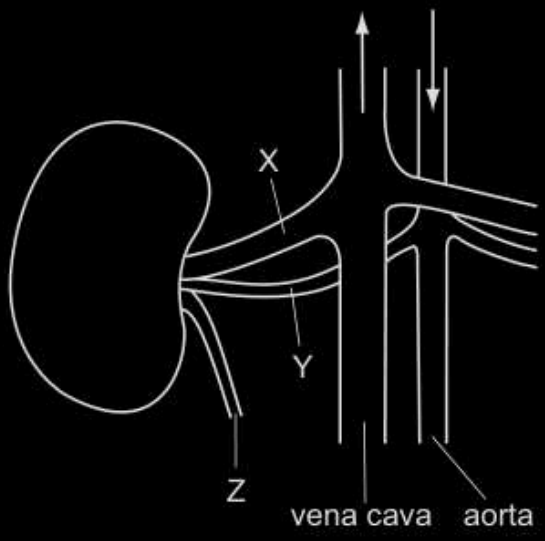
D



October/November 2010 · Paper 1 · Variant 2 · Question 15

Correct answer: C

15 The diagram shows the structures associated with a human kidney.



What are the relative concentrations of urea in X, Y and Z?

A X is sometimes higher than Y

B Y is always higher than Z

C Y is always lower than Z

D Z is sometimes lower than X

October/November 2010 · Paper 1 · Variant 2 · Question 16

Correct answer: A

16 Which sequence shows the shortest route taken by blood travelling from a leg to an arm in the human body?

A leg → heart → lungs → heart → arm

B leg → heart → lungs → kidney → arm

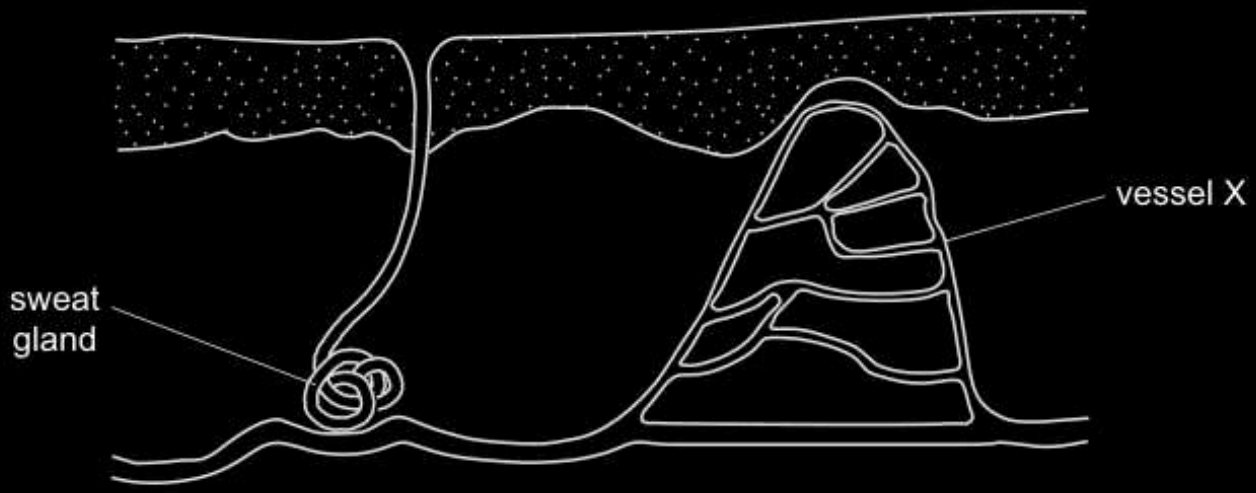
C leg → kidney → heart → lungs → arm

D leg → lungs → heart → gut → arm

October/November 2010 · Paper 1 · Variant 2 · Question 17

Correct answer: C

17 The diagram shows a section through skin.



What happens if the body temperature starts to fall below normal?

	sweat glands	blood flow in vessel X
A	secrete sweat	decreases
B	secrete sweat	increases
C	stop secreting sweat	decreases
D	stop secreting sweat	increases

October/November 2010 · Paper 1 · Variant 2 · Question 19

Correct answer: B

19 What passes down a pollen tube after pollination?

A female nucleus

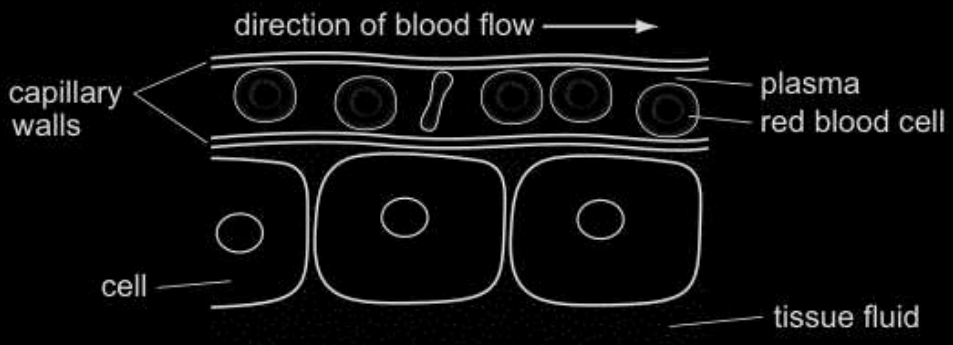
B male nucleus

C ovule

D pollen grain

Correct answer: A

23 The diagram shows a blood capillary close to some tissue cells bathed in tissue fluid. Exchange of nutrients takes place here.

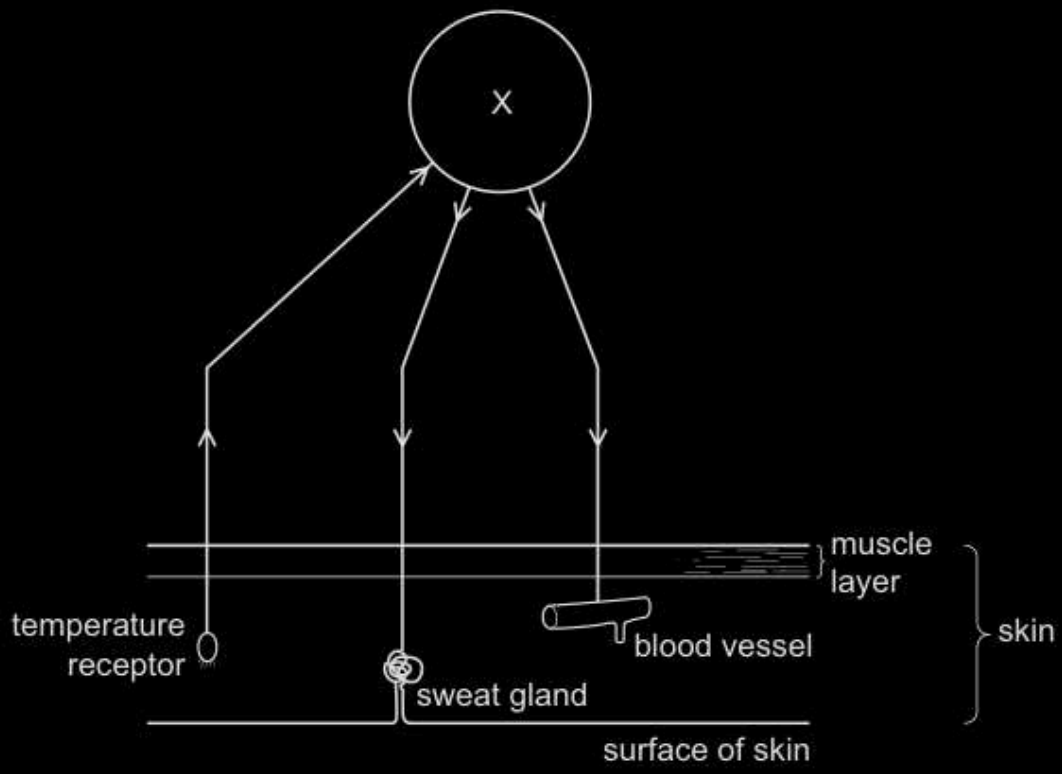


Which row shows the type of nutrient in the plasma and in the tissue fluid and the method of transfer between the two?

	plasma	tissue fluid	method of transfer
A	amino acid	amino acid	diffusion
B	amino acid	protein	osmosis
C	protein	amino acid	digestion
D	protein	protein	active uptake

Correct answer: B

24 The diagram shows some nerve pathways involved in temperature control of the human body.



Which part of the nervous system does X represent?

A cerebrum

B hypothalamus

C medulla

D pituitary gland

Correct answer: A

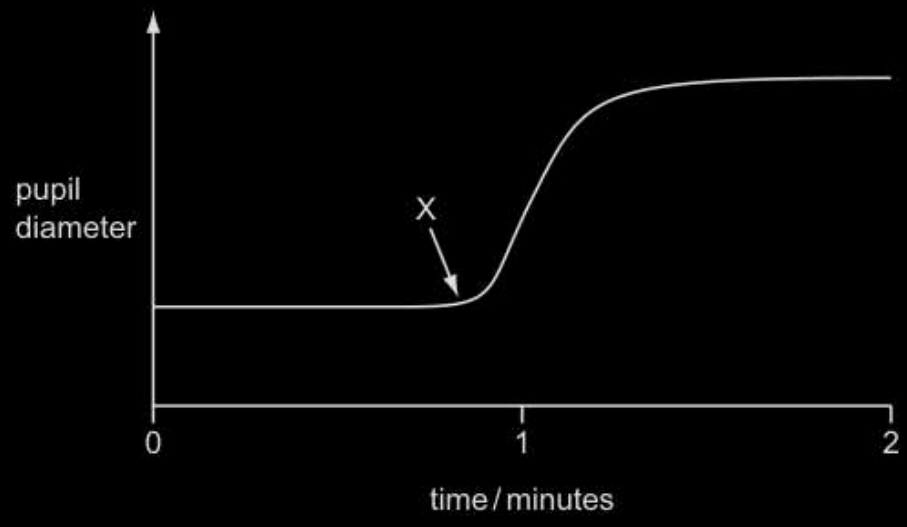
25 The diagram shows a section through the human brain.

What are the functions of the labelled regions?

	memory	balance
A	W	X
B	W	Y
C	X	W
D	X	Y

Correct answer: B

26 The graph shows how the diameter of the pupil of a person's eye changed during the course of two minutes.



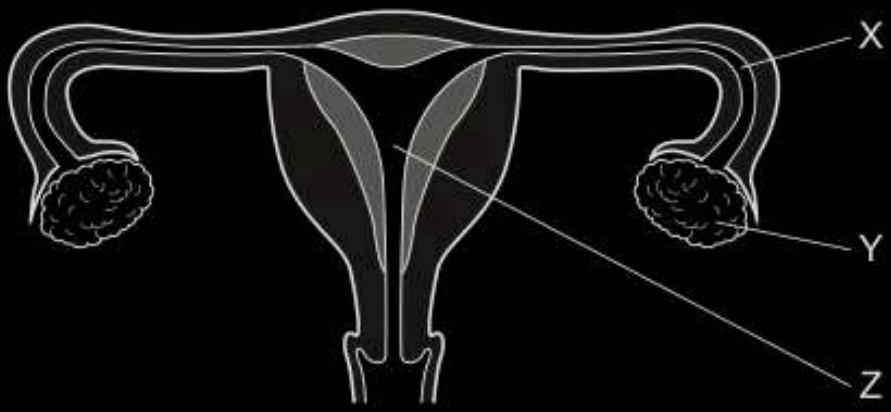
What happens to the light intensity at X and which muscles begin to contract?

	light intensity	iris muscles contracting
A	decreases	circular
B	decreases	radial
C	increases	circular
D	increases	radial

October/November 2010 · Paper 1 · Variant 2 · Question 27

Correct answer: C

27 The diagram shows a section through the female reproductive system.



During pregnancy, where does mitosis occur in the cells of the embryo?

	X	Y	Z
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✗	✓

key

✓ = takes place

✗ = does not take place

October/November 2010 · Paper 1 · Variant 2 · Question 28

Correct answer: A

28 What is a reason for breast milk being better for a baby than bottled milk?

A It contains antibodies for disease protection.

B It contains calcium ions for bone development.

C It contains protein for growth.

D It contains sugar for energy.

October/November 2010 · Paper 1 · Variant 2 · Question 35

Correct answer: B

35 The inheritance of the ABO blood groups in humans is controlled by three alleles (I^A, I^B and I^O), only two of which can be present in one individual.

What are the possible blood groups of children born to a homozygous group A woman and a heterozygous group B man?

A A and B only

B A and AB only

C A, B and AB only

D A, B, AB and O

October/November 2010 · Paper 1 · Variant 2 · Question 36

Correct answer: B

36 The genotype for the height of an organism is written as Tt.

What conclusion may be drawn?

A The allele for height has at least two different genes.

B There are at least two different alleles of the gene for height.

C There are two different genes for height, each having a single allele.

D There is one allele for height with two different forms.

October/November 2010 · Paper 1 · Variant 2 · Question 37

Correct answer: C

37 The body cells of a mammal contain two copies of 24 000 genes, making 48 000 genes in total.

Of these 48 000 genes in the body cells of a mammal, how many would have been inherited from its mother?

A 6 000

B 12 000

C 24 000

D 48 000

Correct answer: A

38 Which statement is evidence that genes may be copied and passed on to the next generation?

A Asexual reproduction produces genetically identical offspring.

B Different alleles of a gene can produce variation in phenotype.

C Each species of a plant or animal has a fixed number of chromosomes.

D Sexual reproduction produces genetically different offspring.

2 Fig. 2.1 shows components of the faeces of a cow, an herbivorous mammal.

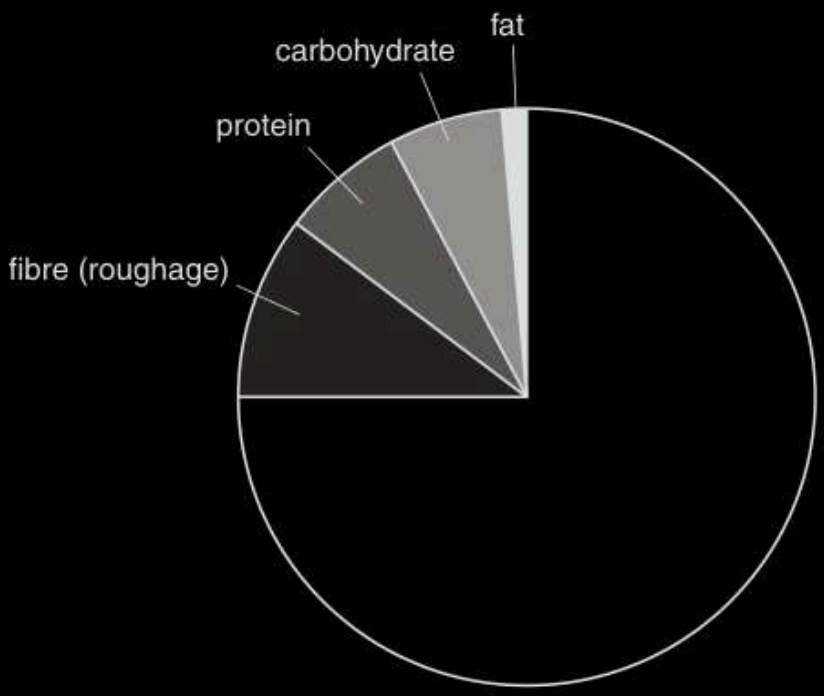


Fig. 2.1

(a) Name the component that makes up 75% of the faeces.

..... [1]

(b) Suggest what makes up a major part of the fibre in the faeces.

..... [1]

If used as a fertiliser, faeces will eventually increase the nitrates in the soil.

(c) Identify the component in faeces that is responsible for this increase in nitrates and describe its conversion into nitrates.

component ..... [1]

how it is converted .....

.....

.....

.....

.....

.....

..... [6]

(d) Explain why a mixture of faeces and urine is a better fertiliser than faeces alone.

.....

.....

..... [2]

[Total: 11]

2 **(a)** water; [1]

(b) cell walls / cellulose / undigested grass; [1]

(c) protein (A in how it is converted);
decomposition AW (A digests);
by bacteria / fungi / saprotrophs (or named);
to amino acids;
to ammonia/um (salts);
nitrites ;
nitrification ; [7]

(d) larger range of / more ions AW / more water;
urea / nitrogenous – or named;
easier / quicker to convert to nitrate AW; [max. 2]

[Total: 11]

3 Fig. 3.1 shows the human digestive system.

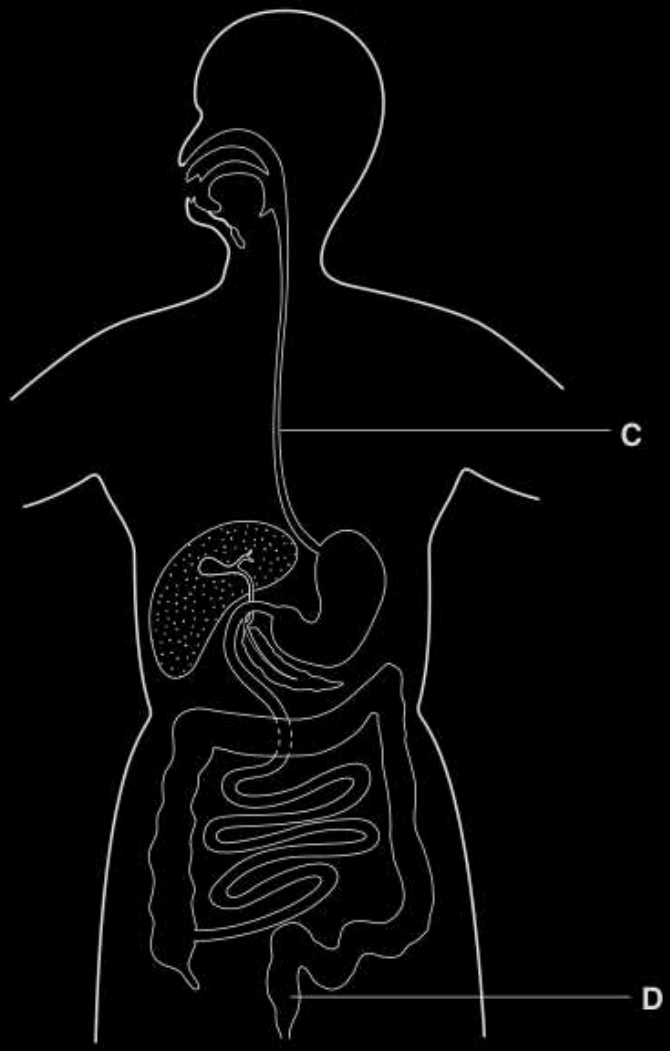


Fig. 3.1

(a) Identify parts **C** and **D** on Fig. 3.1.

C .....

D ..... [2]

(b) **(i)** Use a label line on Fig. 3.1, marked '**X**', to indicate where fat digestion begins. [1]

(ii) Explain your reasons for selecting this region.

.....

.....

.....

.....

..... [4]

(c) Adrenaline reduces the secretion of mucus in the body. Suggest why a person who suffers from stress may also suffer from damage to the lining of their stomach wall.

.....

.....

..... [2]

[Total: 9]

3 **(a)** C – oesophagus / gullet;
D – rectum; [2]

(b) **(i)** line to duodenum beyond entrance of pancreatic duct; [1]
(and before the point at which it crosses the colon)

(ii) lipase / enzyme for fat digestion;
not present until then / not present + in mouth / stomach;
released from pancreas / duodenal walls / in intestinal juice;
bile (salts);
released from gall bladder / bile duct / liver;
ref. correct pH / emulsification AW; [max. 4]

(c) wall protected by mucus;
wall (or stomach) made of protein;
protease / enzyme digests stomach wall ;
acid in contact with wall; [max. 2]

[Total: 9]

4 Fig. 4.1 shows the blood groups of the members of two families.

	Family 1	Family 2
blood groups of parents :	**B** and **A**	**O** and **B**

blood groups of children:	**A**	**O**	**AB**	**B**		**B**	**AB**	**B**
child number :	1	2	3	4	x	5	6	7
grandchild :					8			

Fig. 4.1

The alleles responsible for blood groups are I^A, I^B and I^o.

(a) State the term used to describe the relationship between alleles I^A and I^B.

..... [1]

(b) Identify, by number, which one of the children had been adopted by their family and could not be the genetic offspring of the parents. Explain your answer.

child number .....

explanation .....

.....

..... [2]

(c) When children 4 and 5 grow up, they have a child of their own, child 8, as shown in Fig. 4.1. Using a genetic diagram, explain the possible genotypes and phenotypes of child 8.

[4]

[Total: 7]

4 **(a)** incomplete / codominance; [1]

(b) child 6;
ref. to the group O parent unable to supply the I^A allele / there is no I^A in either parent AW; [2]

(c) I^BI^O × I^BI^O;

I^B, I^O, I^B, I^O;

I^BI^B I^BI^O I^OI^O;

group B group O; [4]
(failure to use prefix I, penalise first mark only. Allow gametes and genotypes of children if correct for wrong parents)

[Total: 7]

5 Fig. 5.1 shows the rate of water uptake and of water loss for a plant over a 24-hour period.

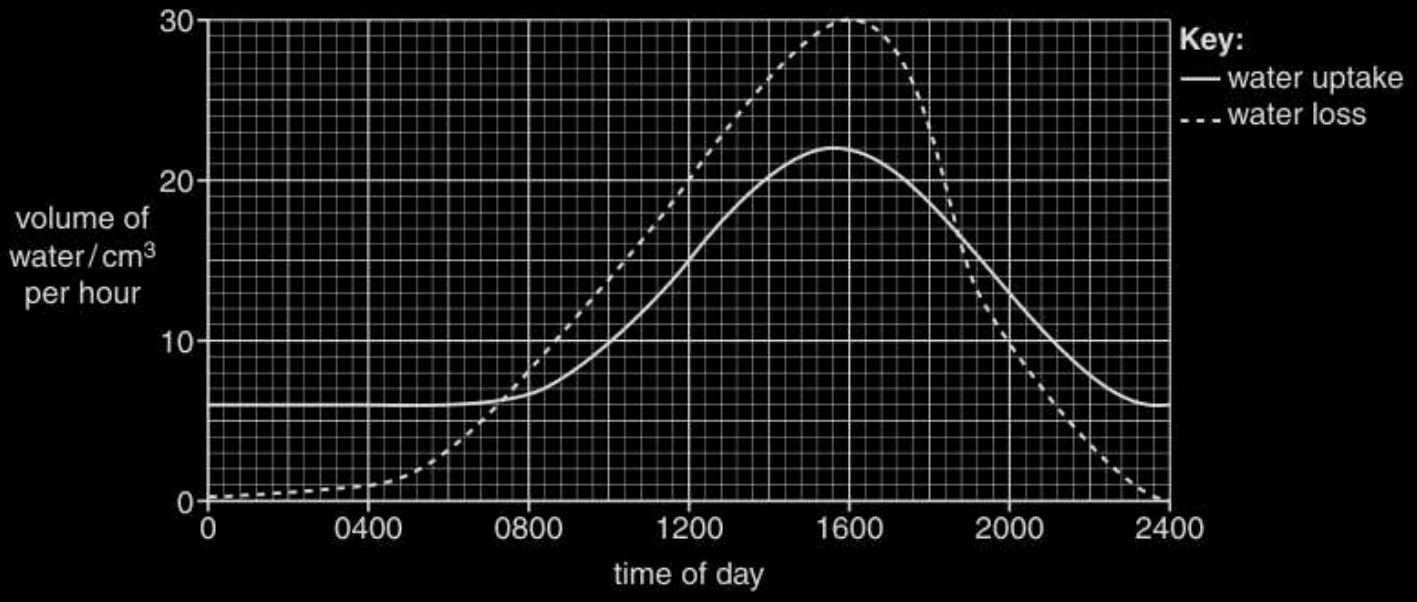


Fig. 5.1

(a) Determine the rate of water uptake at 1200 hours. ..... [1]

(b) **(i)** Name the cells through which water is absorbed from the soil.

..... [1]

(ii) Name the cells between which water vapour passes to the atmosphere.

..... [1]

(c) **(i)** State three uses of water within a plant between midnight and 0400 hours.

1. .....

2. .....

3. ..... [3]

(ii) State two additional uses of water between 0800 and 1900 hours.

1. .....

2. ..... [2]

(d) Explain what may happen to the plant between 1400 and 1800 hours.

.....

.....

..... [2]

[Total: 10]

Section B

Answer **all** the questions, including questions 6, 7 and either 8 **Either** or 8 **Or**.

Write your answers in the spaces provided.

5 **(a)** 15 cm^3 per hour; [1]

(b) **(i)** root hair (cells); [1]
guard cells (Ignore mesophyll) (R stomatal cells); [1]

(c) **(i)** any 3 from: solvent, transport medium, turgidity AW,
medium for enzyme action AW, hydrolysis / reactant AW,
elongation (R growth unqualified);;; [3]

(ii) (in either order) photosynthesis;
cooling / transpiration / any from **(i)** not mentioned; [2]

(d) loss of turgor AW;
wilting;
closing of stomata;
more transpiration / water loss than uptake / ref. overheating; [max. 2]

[Total: 10]

[Total for Section A: 50]

Section B

6 **(a)** Explain how chromosomes are involved in the process of inheritance.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

(b) Explain why a kidney transplant from a member of the same family is more likely to be successful than from someone who is unrelated.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

6 **(a)** made of DNA;
carries genes;
responsible for characteristics AW (A traits / enzymes);
(A named character or condition – e.g. eye colour / Down's syndrome)
passed from one generation to the next AW ;
during reproduction / ref. fertilisation;
correct ref. to chromosomes during cell (nuclear) division ;
sexual + variation / asexual + no variation; [max. 5]

(b) each person has unique set of genes AW;
genes make proteins;
therefore proteins different in different people;
relatives share genes / have common genes;
relatives have similar proteins;
relatives have similar tissues;
blood groups / types must be the same;
white blood cells / lymphocytes;
make antibodies;
against foreign protein / antigens;
greater chance of rejection AW if not related / less chance if related; [max. 5]

[Total: 10]

7 A person is stung on the hand by an insect and automatically withdraws their hand rapidly.

(a) Describe the part played by the nervous system in this action.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

(b) Explain why the heart then beats faster.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

7 **(a)** receptor or correct e.g.;
(converts) stimulus;
to impulse / electric pulse;
sensory + neurone / nerve fibre or cell;
synapse;
CNS / spinal cord (R brain);
(N.b. If brain directs the response i.e. any idea of a decision being made – stop marking)
relay neurone;
motor neurone;
flexor or named muscle;
effector; [max. 7]

(b) adrenaline;
in blood to heart;
defence mechanism / fright / prepare for action – or described;
(e.g. muscular action)
faster circulation of blood / faster delivery of O_2 or glucose; [max. 3]

[Total: 10]

1 Fig. 1.1 shows a vertical section through a human heart. The valves have not been shown.

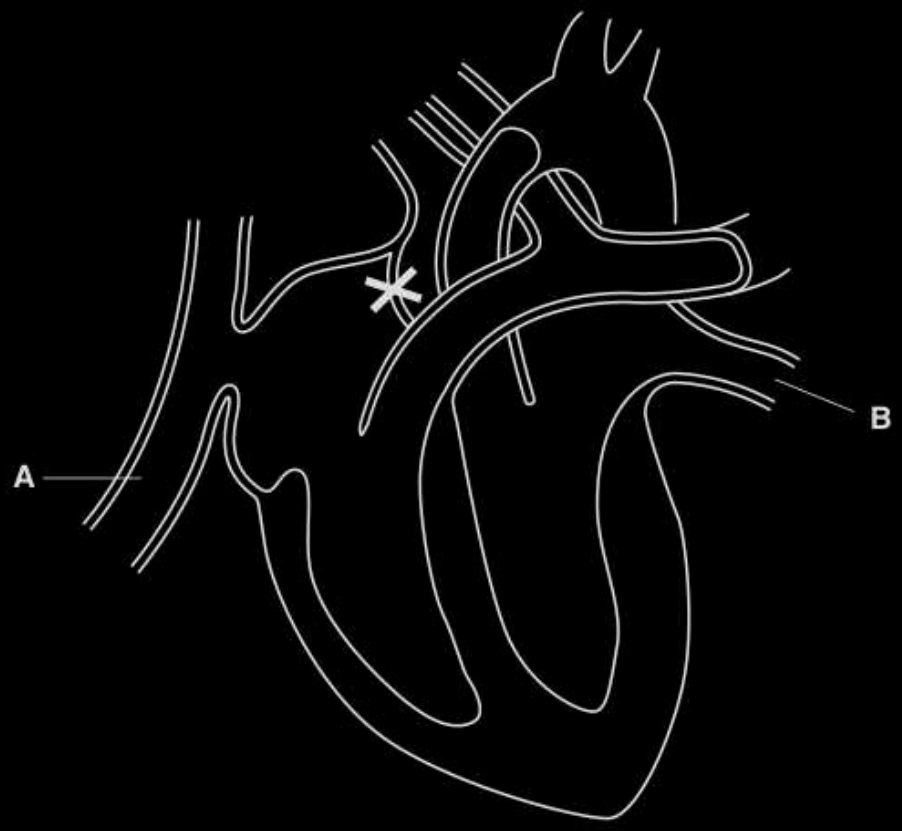


Fig. 1.1

(a) Identify structures **A** and **B** in Fig. 1.1.

A .....

B ..... [2]

(b) Complete Fig. 1.1 by drawing the valves to show how they control the direction of blood flow through the heart. [4]

(c) On Fig. 1.1, draw arrows to show the direction of blood flow into, through and out of the heart. [2]

Some people have a condition known as 'a hole in the heart'. This allows a connection between the left and right atrium at point **X** in Fig. 1.1.

(d) Suggest two problems this might cause.

1. .....

2. .....[2]

[Total: 10]

1 **(a)** A vena cava (I anterior / superior);

B pulmonary vein; [2]

(b) 2 sets of aortic valves shown;
2 sets of atrio-ventricular valves shown in correct places;
unattached end(s) of aortic valve(s) pointing upwards;
unattached end(s) of a-v valve(s) pointing downwards; [4]

(c) arrow(s) correctly into, through and out of right side of heart;
(i.e. look for evidence of all three sites)
arrow(s) correctly into, through and out of left side of the heart; [2]

(d) Any two (**mark the first, one per line unless a line is left blank) from:
mixing of blood, ref. possible effect on pressures of blood (but if stated, must be lower), ref. possible effect on volume of blood (leaving heart or one chamber), reduced oxygen carriage round the body, heart has to work harder, feeling tired AW, shortness of breath, fluid build-up, ;; [2]
(In fact, blood passes from left to right atrium – but do NOT penalise if wrong.)

[Total: 10]

3 Fig. 3.1 shows a water plant, bladderwort, that lives under water in a small lake. The bladderwort traps then digests small water animals such as water fleas.

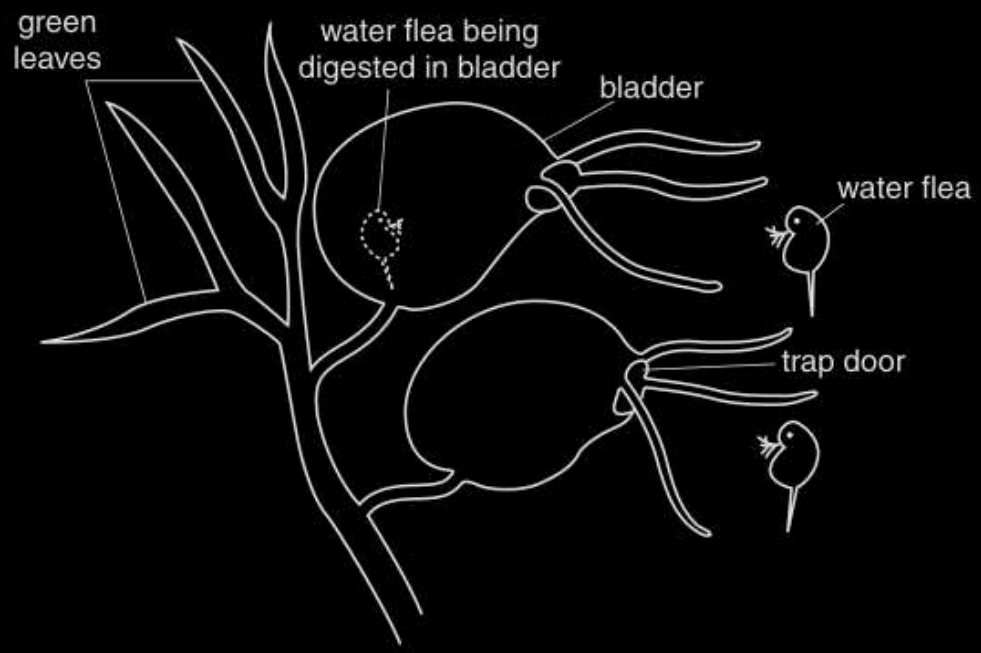


Fig. 3.1

(a) (i) State the process by which the plant is likely to obtain its carbohydrates.

.....[1]

(ii) Suggest how the raw materials for this process are made available to the plant.

.....

.....

.....

.....

.....[4]

The plant is able to supplement its nitrogen requirements by absorbing products from the digested water fleas.

(b) (i) Name an enzyme the plant must produce in its bladder in order to allow it to carry out this process.

.....[1]

(ii) Name the chemicals absorbed by the bladder after digestion and suggest how the plant uses them.

name .....

use .....

.....[3]

(c) State the form in which nitrogen-containing chemicals are absorbed by a plant growing on land.

.....[1]

[Total: 10]

3 (a) (i) photosynthesis; [1]

(ii) *CO_2;

(CO_2 or H_2O) from atmosphere / air / environment / lake;
(or described)
respiration;

(CO_2) in solution; [max 3]

*water (for photosynthesis stated or implied); [1]
(*R on equation alone with no explanation)

(b) (i) protease or named protease; [1]

(ii) amino acids;
to make proteins;
for growth / repair / (proto)cytoplasm; [3]

(c) nitrates; [1]

[Total: 10]

6 **(a)** Explain what is meant by a *hormone*.

.....

.....

.....

.....

.....

.....

.....

.....[4]

(b) Give an example of a hormone and describe how it is involved in maintaining constant conditions within the human body.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 10]

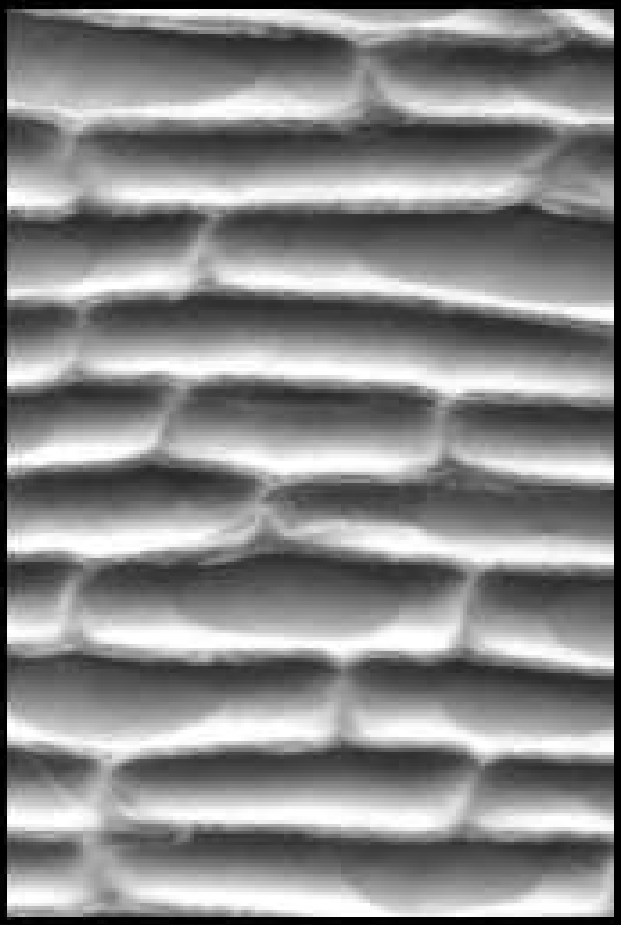
6 **(a)** chemical (R named hormone);
released / produced by a gland;
into blood;
target organ;
effect / alter the activity of;
destroyed by the liver; [max 4]

(b) *deviation from set point (or described);
(For insulin / ADH, set point must be ref. glucose / water in the blood)
named hormone
(for more than one hormone, mark the best account. R adrenaline / thyroxine);
named gland that produces it;
(R for any hormone that is not involved in homeostasis)
name of target organ;
effect (which must be homeostatic) correct for example;
*homeostasis;
*return to set point AW (A 'keeps constant', but R regulates / controls); [max 6]
(Need not refer to blood)
(*Only these marks available if hormone is incorrect.)

[Total: 10]

Correct answer: D

1 The photomicrograph shows onion epidermis.



Which term describes this onion epidermis?

A cell

B organ

C organ system

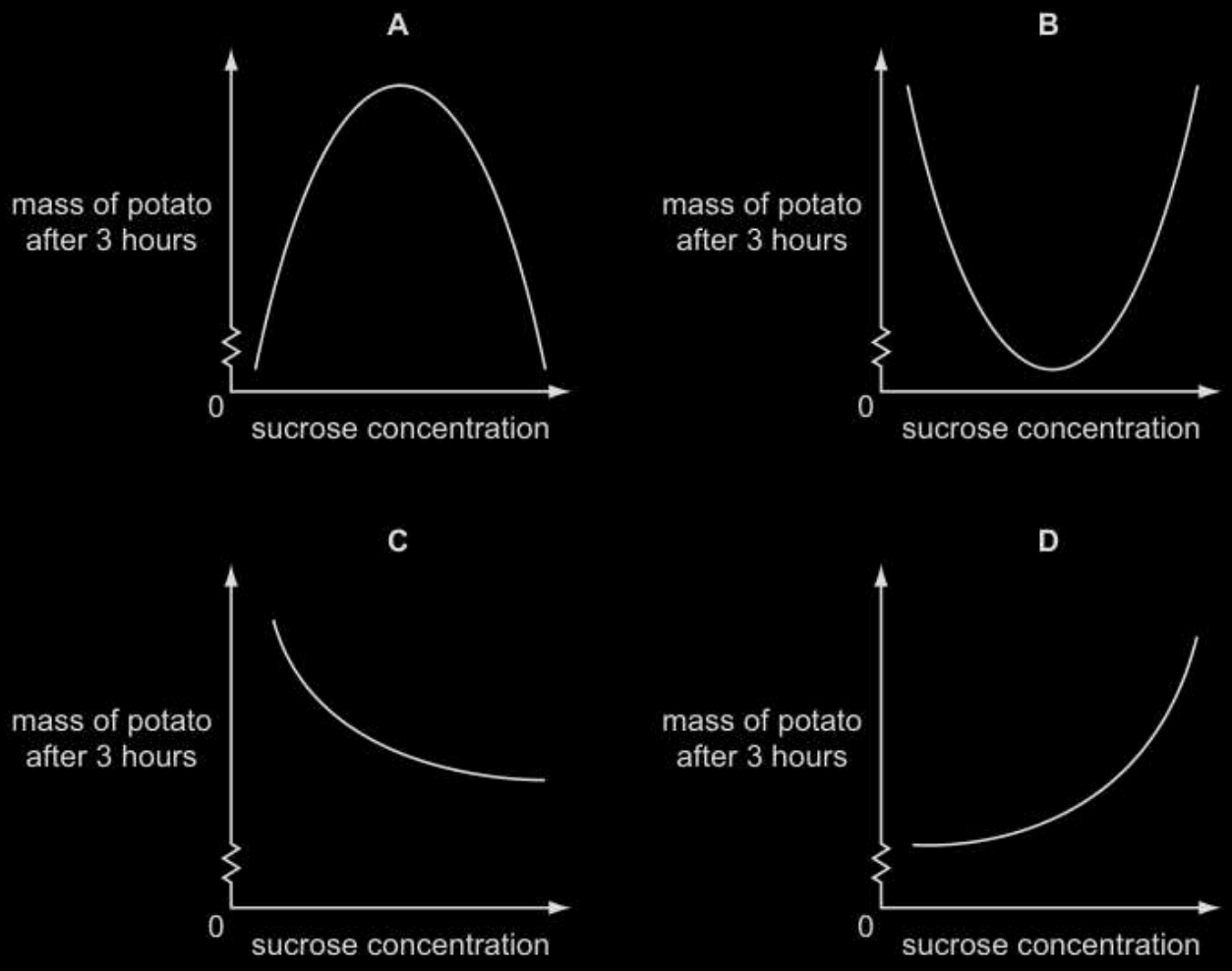
D tissue

May/June 2010 · Paper 1 · Variant 1 · Question 2

Correct answer: C

2 Identical pieces of potato are placed in sucrose solutions of different concentrations. After three hours, the mass of each potato piece is measured.

Which graph shows the results of this experiment?



May/June 2010 · Paper 1 · Variant 1 · Question 3

Correct answer: D

3 Which of these processes require energy from respiration?

	diffusion	osmosis
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = energy required

✗ = energy not required

May/June 2010 · Paper 1 · Variant 1 · Question 5

Correct answer: D

5 Which signs show that a plant has been grown in a culture solution deficient in magnesium?

A purple leaves and poor root growth

B small leaves and thin stem

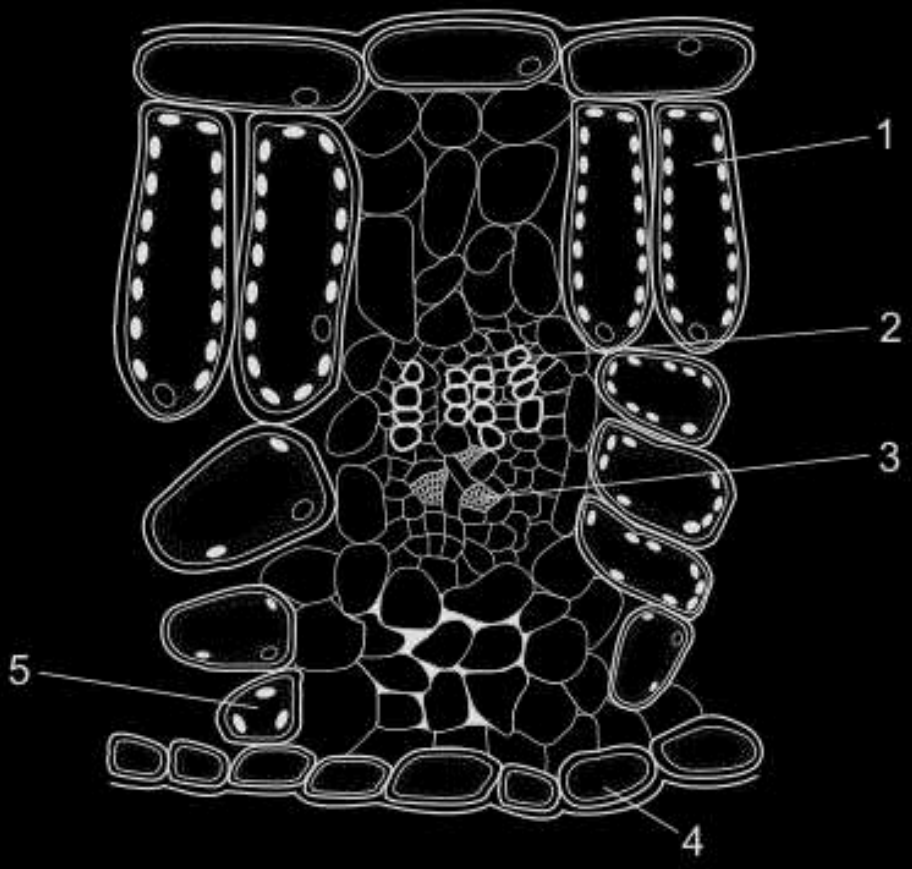
C white upper leaves and normal lower leaves

D yellow stem and yellow leaves

May/June 2010 · Paper 1 · Variant 1 · Question 7

Correct answer: A

7 The diagram shows a section through a leaf.



Which cells have the functions shown?

	photosynthesis	transport
A	1 and 5	2 and 3
B	2 and 4	1 and 4
C	3 and 4	2 and 5
D	4 and 5	3 and 4

May/June 2010 · Paper 1 · Variant 1 · Question 9

Correct answer: B

9 In which regions of the alimentary canal does amylase act?

A mouth cavity and pancreas

B mouth cavity and ileum

C stomach and pancreas

D stomach and ileum

May/June 2010 · Paper 1 · Variant 1 · Question 10

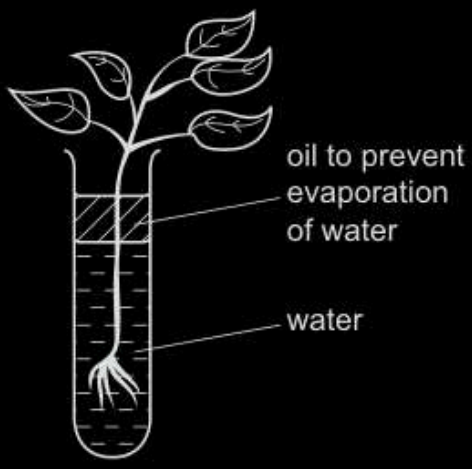
Correct answer: C

10 Which substances are transported in xylem and in phloem?

	xylem	phloem
A	amino acids and mineral ions	amino acids and water
B	mineral ions and sucrose	starch and mineral ions
C	mineral ions and water	sucrose and water
D	starch and water	sucrose and starch

Correct answer: B

12 Five similar plants are placed in test-tubes as shown.



Some of the plants have their leaves coated with grease to reduce transpiration. Each plant is weighed in its test-tube at the start of the experiment and again two days later.

The results are shown in the table.

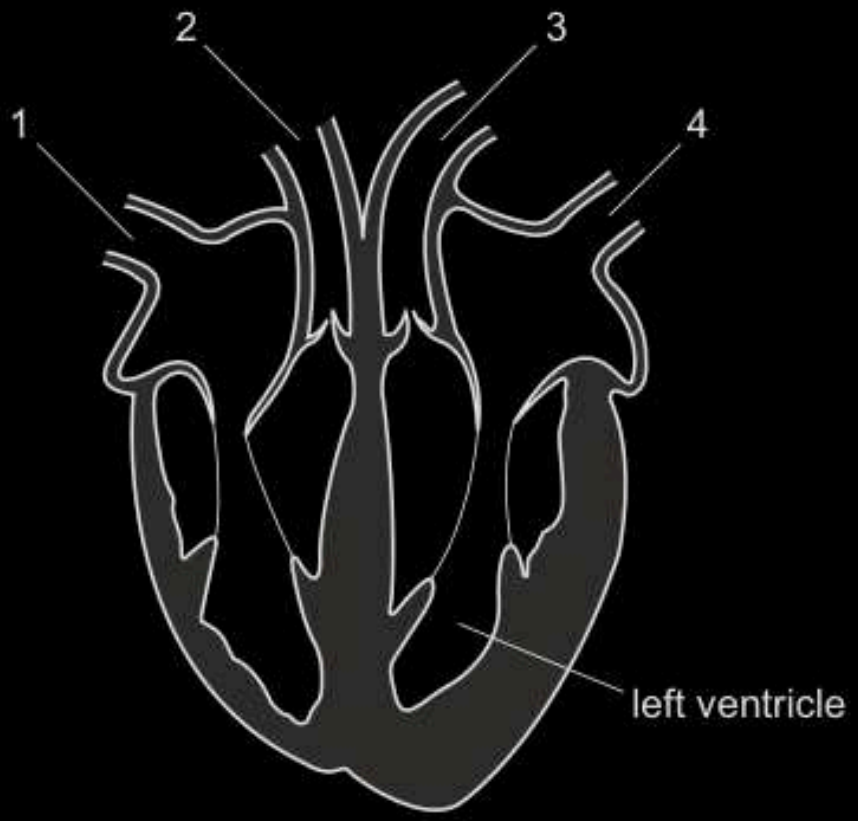
	mass / g				
	plant 1	plant 2	plant 3	plant 4	plant 5
at the start of experiment	105	121	107	111	119
after two days	103	97	84	110	93

Which plants had their leaves coated with grease?

A 1 and 2 **B** 1 and 4 **C** 2 and 5 **D** 2, 3 and 5

Correct answer: D

14 The diagram shows a vertical section through the heart.



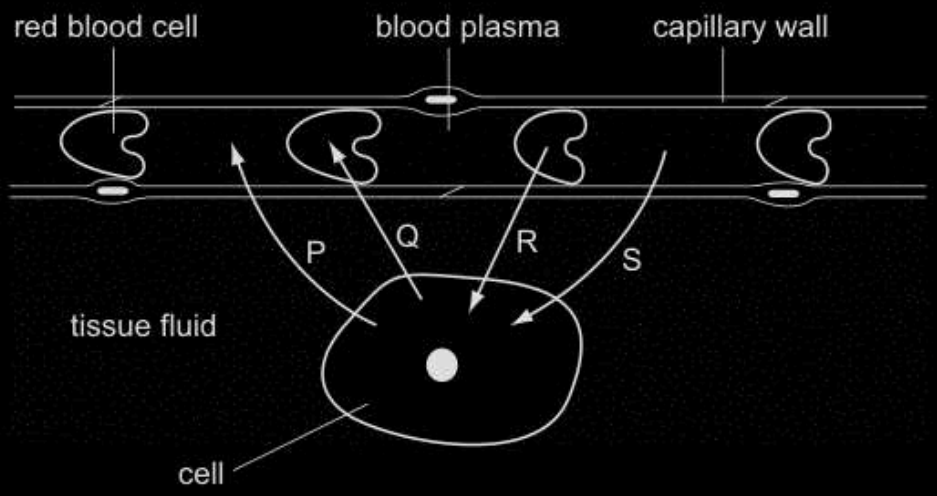
What are the functions of the numbered blood vessels?

	carries blood to body	carries blood to lungs	carries blood from lungs	carries blood from body
A	1	4	3	2
B	2	1	3	4
C	2	3	4	1
D	3	2	4	1

May/June 2010 · Paper 1 · Variant 1 · Question 15

Correct answer: D

15 The diagram represents a blood capillary with an adjacent cell. The arrows represent the transfer of substances between the capillary and the cell.



Which arrows represent glucose, carbon dioxide and oxygen?

	glucose	carbon dioxide	oxygen
A	P	R	Q
B	Q	S	P
C	R	Q	S
D	S	P	R

May/June 2010 · Paper 1 · Variant 1 · Question 17

Correct answer: C

17 What do the cilia do in the bronchi of the lungs?

	trap bacteria	move mucus out of the lungs
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = function of the cilia

✗ = not a function of the cilia

May/June 2010 · Paper 1 · Variant 1 · Question 19

Correct answer: B

19 Which **cannot** be an example of excretion?

A Carbon dioxide is breathed out from the lungs.

B Undigested food leaves the body through the anus.

C Urea leaves the body in urine.

D Water is removed through the kidneys.

May/June 2010 · Paper 1 · Variant 1 · Question 20

Correct answer: B

20 Which structures are **all** involved in controlling human body temperature?

A blood vessels near the skin surface, the cerebellum and sweat glands

B blood vessels near the skin surface, the hypothalamus and skeletal muscles

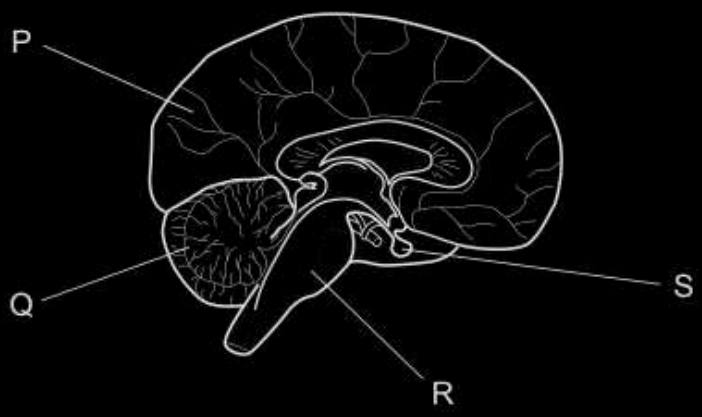
C kidneys, the cerebellum and sweat glands

D kidneys, the hypothalamus and skeletal muscles

May/June 2010 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 The diagram shows a section through the brain and part of the spinal cord.



Which row correctly identifies the cerebral hemisphere, cerebellum, medulla and pituitary gland?

	cerebral hemisphere	cerebellum	medulla	pituitary gland
A	P	Q	R	S
B	P	R	Q	S
C	Q	R	S	P
D	S	P	R	Q

May/June 2010 · Paper 1 · Variant 1 · Question 22

Correct answer: A

22 Which is **not** controlled directly by the nervous system?

A blood glucose concentration

B dilation of the blood vessels

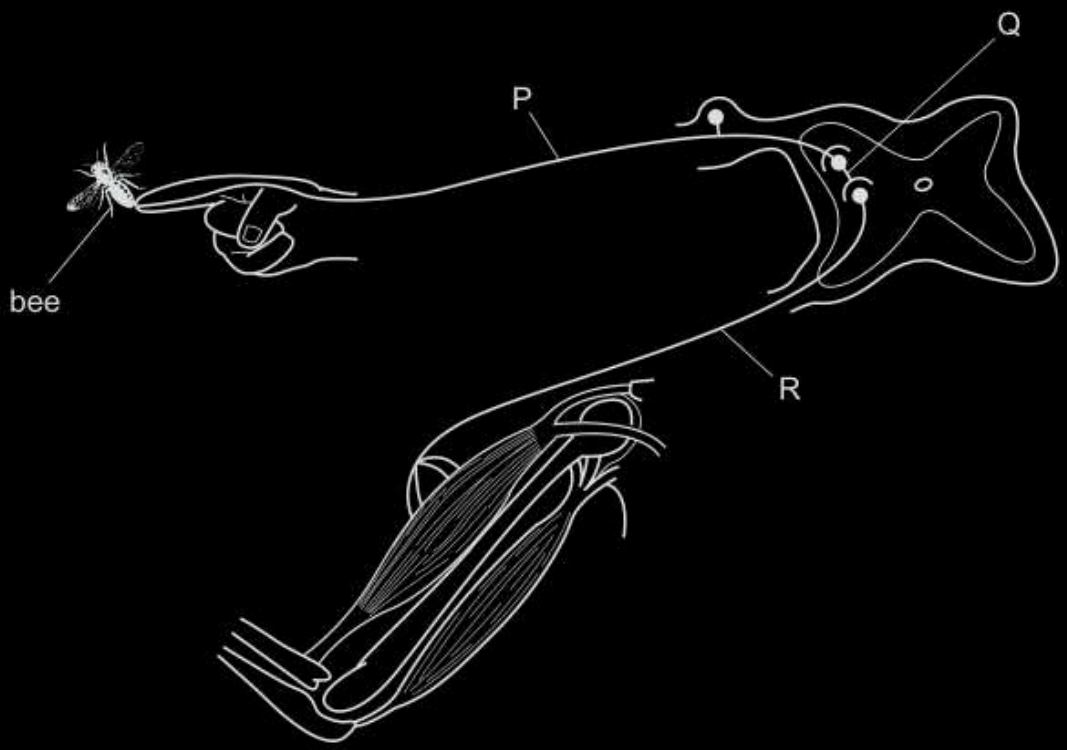
C heart rate

D pupil diameter in the eye

May/June 2010 · Paper 1 · Variant 1 · Question 23

Correct answer: B

23 The diagram shows a reflex arc in which a bee sting causes the arm to be moved quickly.



If the relay neurone at Q is damaged, how will the transmission of nerve impulses in the reflex arc be affected?

A They cannot pass from P to Q.

B They cannot pass from P to R.

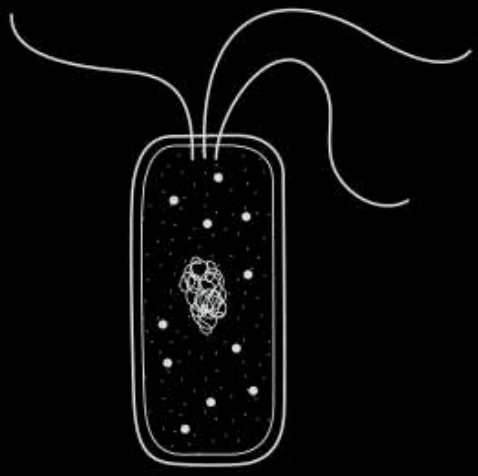
C They cannot pass from Q to P.

D They cannot pass from R to Q.

May/June 2010 · Paper 1 · Variant 1 · Question 26

Correct answer: D

26 The diagram shows the structure of a bacterium.



In what way does this differ from a cell of a fungus?

A The bacterium has a cell membrane.

B The bacterium has a cell wall.

C The bacterium has cytoplasm.

D The bacterium has no true nucleus.

May/June 2010 · Paper 1 · Variant 1 · Question 30

Correct answer: A

30 Which bacteria convert urea to ammonia?

A decay bacteria

B denitrifying bacteria

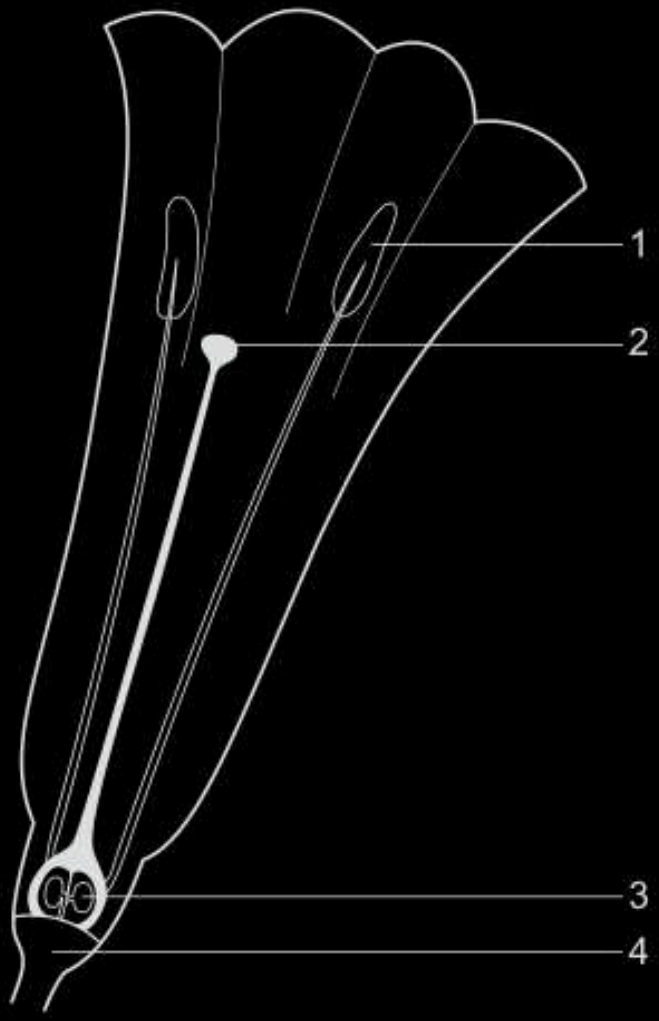
C nitrifying bacteria

D nitrogen-fixing bacteria

May/June 2010 · Paper 1 · Variant 1 · Question 33

Correct answer: A

33 The diagram shows a section through a flower.



In which structures are haploid nuclei formed by reduction division?

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

May/June 2010 · Paper 1 · Variant 1 · Question 35

Correct answer: D

35 After sexual intercourse, sperm can survive for 3 days in the uterus and oviducts. Ovulation can occur any time from day 13 to day 15 and an egg cell can live for 2 days after ovulation.

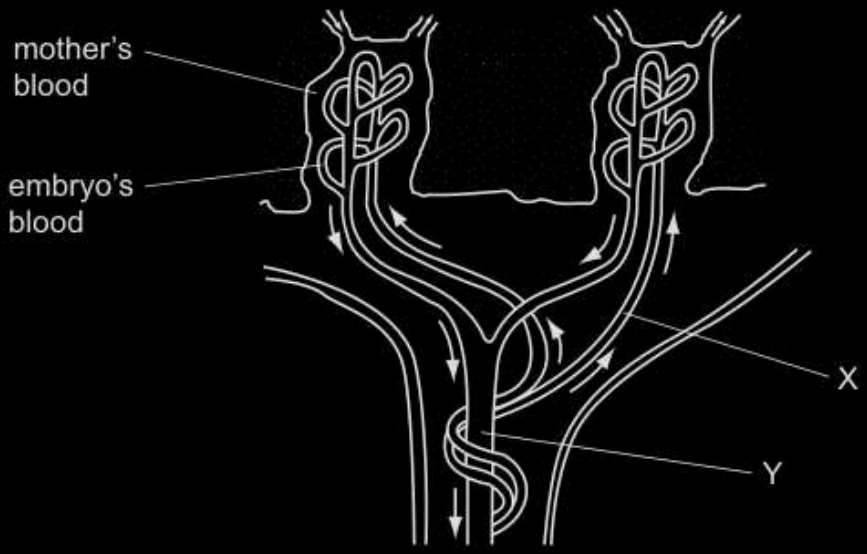
How long is the longest possible fertile phase of the menstrual cycle?

A 2 days **B** 3 days **C** 5 days **D** 7 days

May/June 2010 · Paper 1 · Variant 1 · Question 36

Correct answer: B

36 The diagram shows how the blood of a human embryo flows close to the mother's blood in the placenta.



Which substances are present at X in higher concentrations than at Y?

A carbon dioxide and glucose

B carbon dioxide and urea

C glucose and oxygen

D glucose and urea

May/June 2010 · Paper 1 · Variant 1 · Question 37

Correct answer: B

37 Which statement about human blood groups is correct?

A A person with the blood group A cannot have an I^O allele.

B A person with the blood group B may have either the genotype I^BI^B or I^BI^O.

C In a person with blood group AB, the I^B allele is recessive to the allele I^A.

D The alleles I^O and I^B are co-dominant and have equal effect on a phenotype.

May/June 2010 · Paper 1 · Variant 1 · Question 39

Correct answer: B

39 Which statement about chromosomes is correct?

A Chromosomes are long DNA molecules called genes which are divided into sections.

B Chromosomes include a long molecule of DNA divided into sections called genes.

C Chromosomes include genes which are divided into sections called DNA molecules.

D Genes include long DNA molecules called chromosomes.

May/June 2010 · Paper 1 · Variant 1 · Question 40

Correct answer: A

40 In maize, one allele of a particular gene allows chlorophyll production while the other allele prevents this, giving plants with cream coloured leaves.

Half the seeds from a cross between two green-leaved plants were sown in trays kept in the dark. The other half were sown in similar conditions except that they received light.

The table gives the results obtained.

numbers of seedlings			
kept in the dark		kept in the light	
green leaves	cream leaves	green leaves	cream leaves
?	405	320	110

What was the number of green-leaved plants formed from seeds germinating in the dark?

A 0 **B** 110 **C** 320 **D** 405

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

May/June 2010 · Paper 1 · Variant 2 · Question 3

Correct answer: C

3 Cubes of hard-boiled egg white are placed in test-tubes containing 5 cm^3 of water. Other substances are added to each tube as shown in the chart. The tubes are left for eight hours and then tested for amino acids.

tube	solution added	results of test for amino acids
1	pepsin	absent
2	pepsin + alkali	absent
3	none	absent
4	pepsin + acid	large amounts
5	boiled pepsin + acid	traces
6	acid	traces
7	alkali	absent

Which tubes show that pepsin is an enzyme?

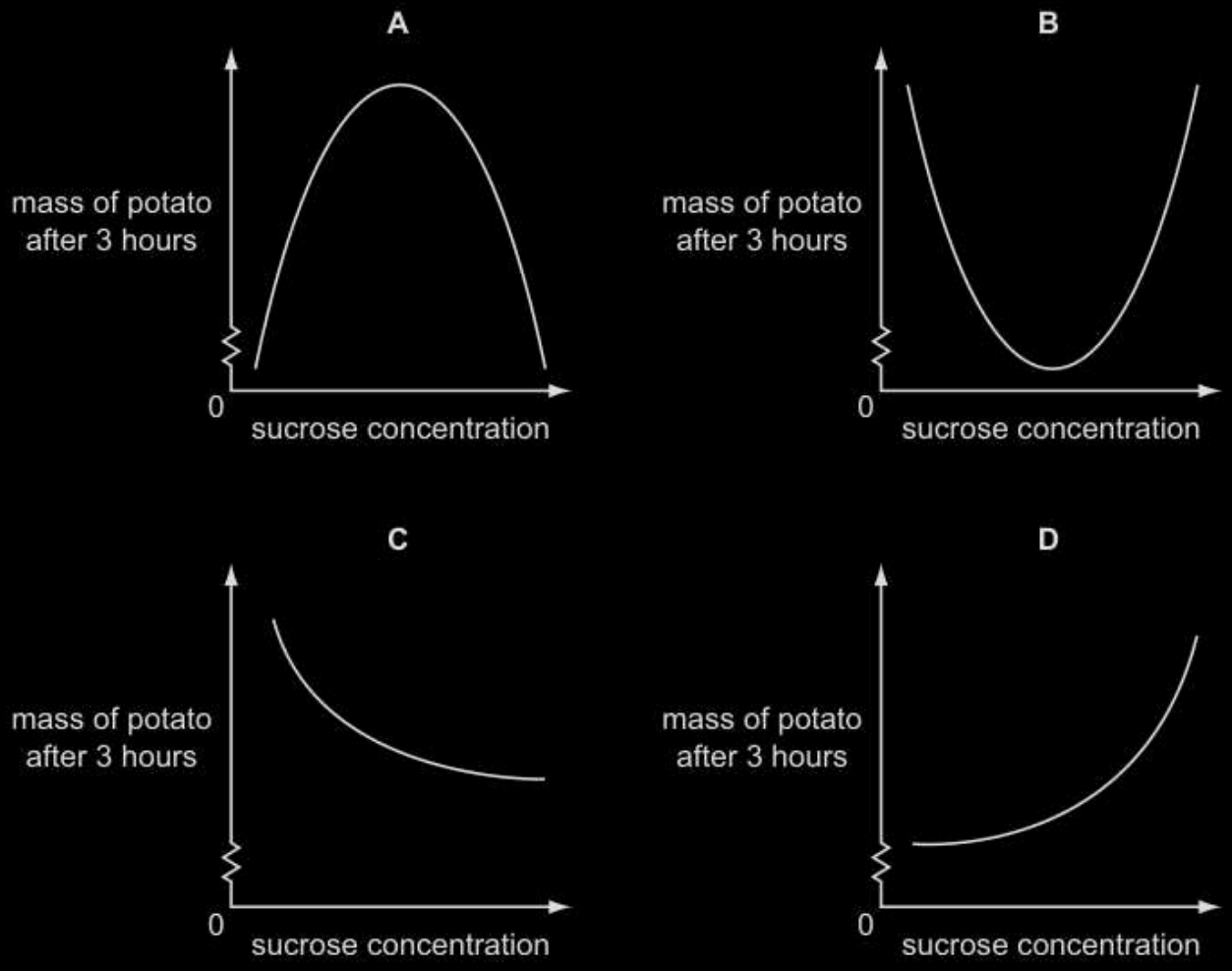
A 1 and 6 **B** 2 and 7 **C** 4 and 5 **D** 5 and 6

May/June 2010 · Paper 1 · Variant 2 · Question 4

Correct answer: C

4 Identical pieces of potato are placed in sucrose solutions of different concentrations. After three hours, the mass of each potato piece is measured.

Which graph shows the results of this experiment?



May/June 2010 · Paper 1 · Variant 2 · Question 5

Correct answer: D

5 Which of these processes require energy from respiration?

	diffusion	osmosis
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = energy required

✗ = energy not required

May/June 2010 · Paper 1 · Variant 2 · Question 6

Correct answer: B

6 In which regions of the alimentary canal does amylase act?

A mouth cavity and pancreas

B mouth cavity and ileum

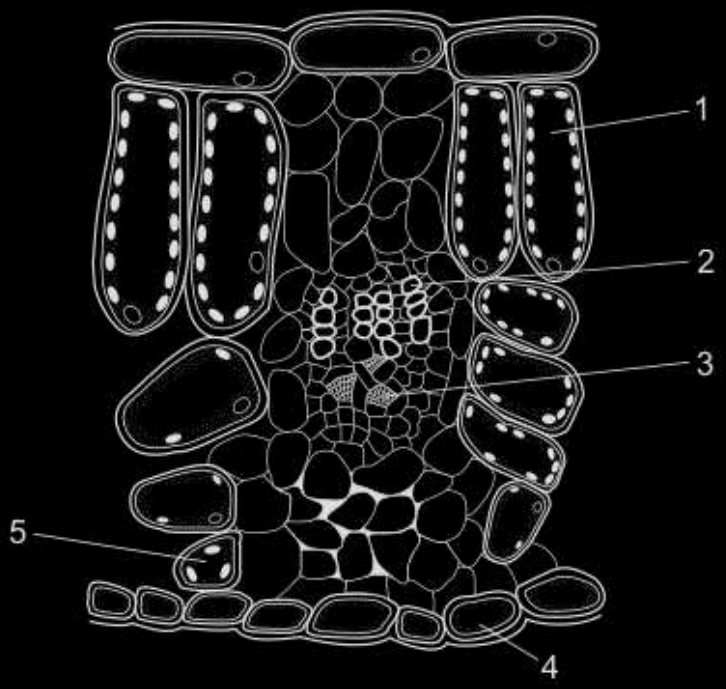
C stomach and pancreas

D stomach and ileum

May/June 2010 · Paper 1 · Variant 2 · Question 8

Correct answer: A

8 The diagram shows a section through a leaf.



Which cells have the functions shown?

	photosynthesis	transport
A	1 and 5	2 and 3
B	2 and 4	1 and 4
C	3 and 4	2 and 5
D	4 and 5	3 and 4

May/June 2010 · Paper 1 · Variant 2 · Question 9

Correct answer: C

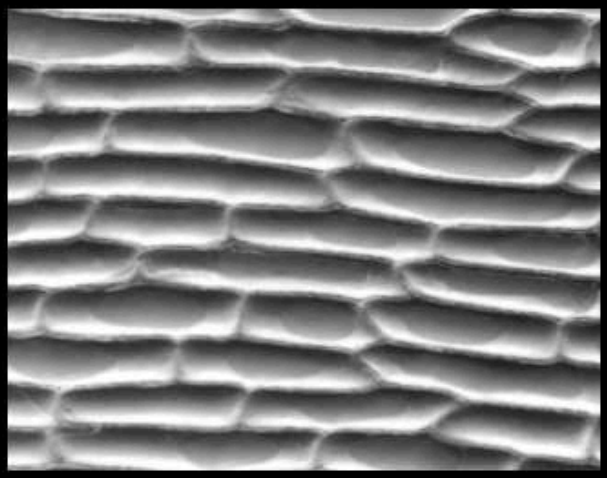
9 Which substances are transported in xylem and in phloem?

	xylem	phloem
A	amino acids and mineral ions	amino acids and water
B	mineral ions and sucrose	starch and mineral ions
C	mineral ions and water	sucrose and water
D	starch and water	sucrose and starch

May/June 2010 · Paper 1 · Variant 2 · Question 11

Correct answer: D

11 The photomicrograph shows onion epidermis.



Which term describes this onion epidermis?

A cell

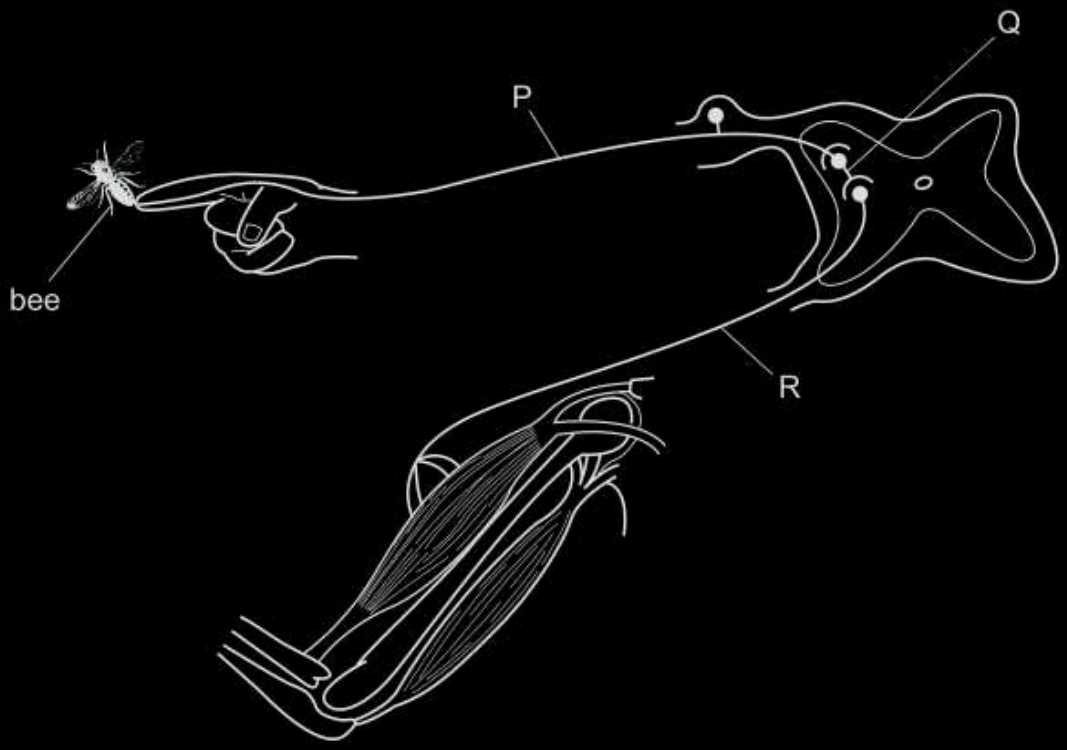
B organ

C organ system

D tissue

Correct answer: B

14 The diagram shows a reflex arc in which a bee sting causes the arm to be moved quickly.



If the relay neurone at Q is damaged, how will the transmission of nerve impulses in the reflex arc be affected?

A They cannot pass from P to Q.

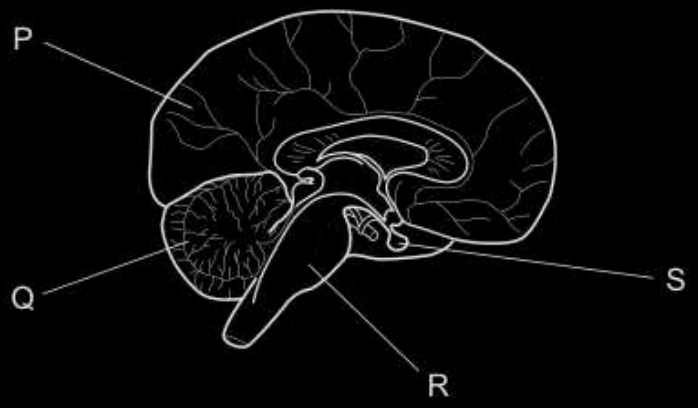
B They cannot pass from P to R.

C They cannot pass from Q to P.

D They cannot pass from R to Q.

Correct answer: A

15 The diagram shows a section through the brain and part of the spinal cord.

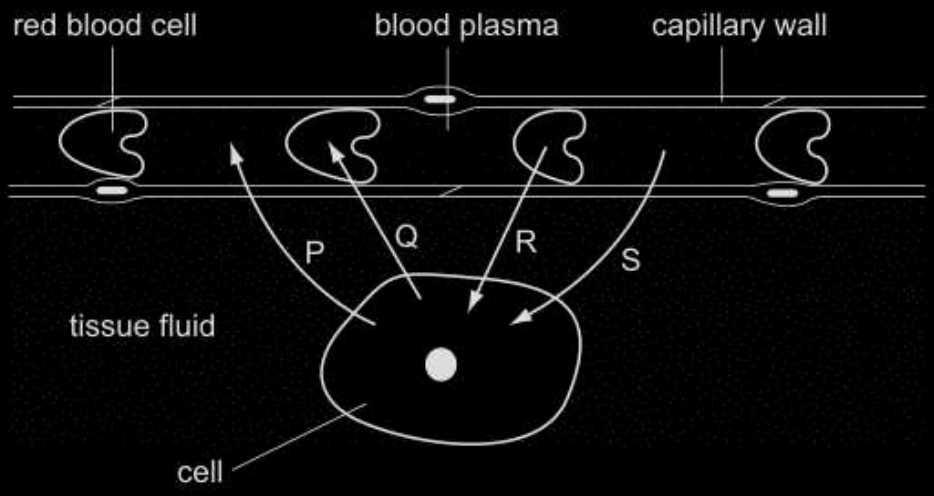


Which row correctly identifies the cerebral hemisphere, cerebellum, medulla and pituitary gland?

	cerebral hemisphere	cerebellum	medulla	pituitary gland
A	P	Q	R	S
B	P	R	Q	S
C	Q	R	S	P
D	S	P	R	Q

Correct answer: D

16 The diagram represents a blood capillary with an adjacent cell. The arrows represent the transfer of substances between the capillary and the cell.



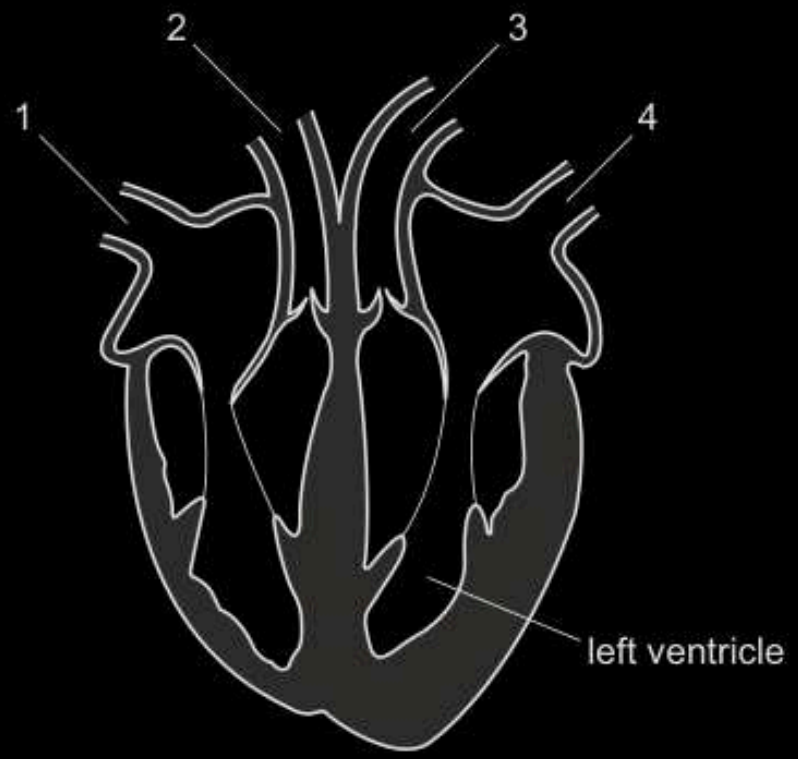
Which arrows represent glucose, carbon dioxide and oxygen?

	glucose	carbon dioxide	oxygen
A	P	R	Q
B	Q	S	P
C	R	Q	S
D	S	P	R

May/June 2010 · Paper 1 · Variant 2 · Question 17

Correct answer: D

17 The diagram shows a vertical section through the heart.



What are the functions of the numbered blood vessels?

	carries blood to body	carries blood to lungs	carries blood from lungs	carries blood from body
A	1	4	3	2
B	2	1	3	4
C	2	3	4	1
D	3	2	4	1

May/June 2010 · Paper 1 · Variant 2 · Question 18

Correct answer: B

18 Which structures are **all** involved in controlling human body temperature?

A blood vessels near the skin surface, the cerebellum and sweat glands

B blood vessels near the skin surface, the hypothalamus and skeletal muscles

C kidneys, the cerebellum and sweat glands

D kidneys, the hypothalamus and skeletal muscles

May/June 2010 · Paper 1 · Variant 2 · Question 20

Correct answer: C

20 What do the cilia do in the bronchi of the lungs?

	trap bacteria	move mucus out of the lungs
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = function of the cilia

✗ = not a function of the cilia

May/June 2010 · Paper 1 · Variant 2 · Question 22

Correct answer: A

22 Which is **not** controlled directly by the nervous system?

A blood glucose concentration

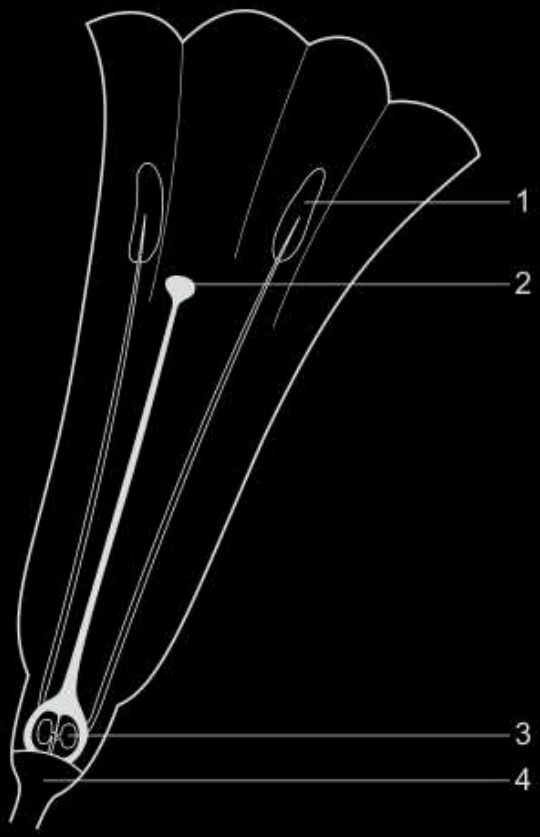
B dilation of the blood vessels

C heart rate

D pupil diameter in the eye

Correct answer: A

24 The diagram shows a section through a flower.



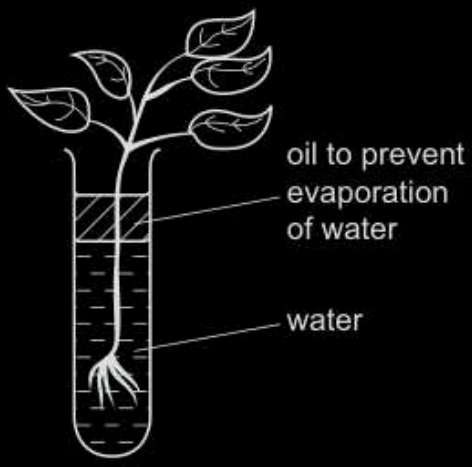
In which structures are haploid nuclei formed by reduction division?

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

May/June 2010 · Paper 1 · Variant 2 · Question 25

Correct answer: B

25 Five similar plants are placed in test-tubes as shown.



Some of the plants have their leaves coated with grease to reduce transpiration. Each plant is weighed in its test-tube at the start of the experiment and again two days later.

The results are shown in the table.

	mass / g				
	plant 1	plant 2	plant 3	plant 4	plant 5
at the start of experiment	105	121	107	111	119
after two days	103	97	84	110	93

Which plants had their leaves coated with grease?

A 1 and 2 **B** 1 and 4 **C** 2 and 5 **D** 2, 3 and 5

May/June 2010 · Paper 1 · Variant 2 · Question 26

Correct answer: B

26 Which statement about human blood groups is correct?

A A person with the blood group A cannot have an I^O allele.

B A person with the blood group B may have either the genotype I^BI^B or I^BI^O.

C In a person with blood group AB, the I^B allele is recessive to the allele I^A.

D The alleles I^O and I^B are co-dominant and have equal effect on a phenotype.

May/June 2010 · Paper 1 · Variant 2 · Question 27

Correct answer: B

27 Which statement about chromosomes is correct?

A Chromosomes are long DNA molecules called genes which are divided into sections.

B Chromosomes include a long molecule of DNA divided into sections called genes.

C Chromosomes include genes which are divided into sections called DNA molecules.

D Genes include long DNA molecules called chromosomes.

May/June 2010 · Paper 1 · Variant 2 · Question 28

Correct answer: A

28 In maize, one allele of a particular gene allows chlorophyll production while the other allele prevents this, giving plants with cream coloured leaves.

Half the seeds from a cross between two green-leaved plants were sown in trays kept in the dark. The other half were sown in similar conditions except that they received light.

The table gives the results obtained.

numbers of seedlings			
kept in the dark		kept in the light	
green leaves	cream leaves	green leaves	cream leaves
?	405	320	110

What was the number of green-leaved plants formed from seeds germinating in the dark?

A 0 **B** 110 **C** 320 **D** 405

Correct answer: D

30 The diagram shows the structure of a bacterium.

In what way does this differ from a cell of a fungus?

A The bacterium has a cell membrane.

B The bacterium has a cell wall.

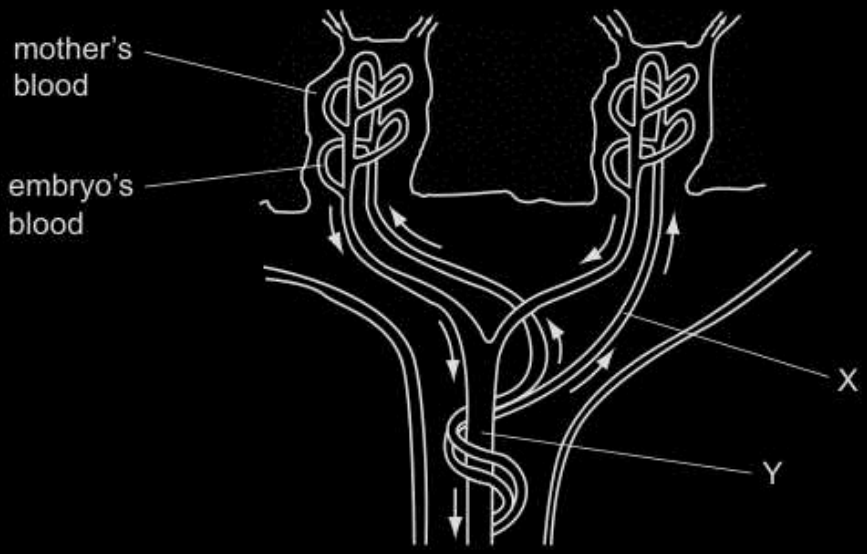
C The bacterium has cytoplasm.

D The bacterium has no true nucleus.

May/June 2010 · Paper 1 · Variant 2 · Question 32

Correct answer: B

32 The diagram shows how the blood of a human embryo flows close to the mother's blood in the placenta.



Which substances are present at X in higher concentrations than at Y?

A carbon dioxide and glucose

B carbon dioxide and urea

C glucose and oxygen

D glucose and urea

May/June 2010 · Paper 1 · Variant 2 · Question 33

Correct answer: D

33 After sexual intercourse, sperm can survive for 3 days in the uterus and oviducts. Ovulation can occur any time from day 13 to day 15 and an egg cell can live for 2 days after ovulation.

How long is the longest possible fertile phase of the menstrual cycle?

A 2 days **B** 3 days **C** 5 days **D** 7 days

May/June 2010 · Paper 1 · Variant 2 · Question 34

Correct answer: B

34 Which **cannot** be an example of excretion?

A Carbon dioxide is breathed out from the lungs.

B Undigested food leaves the body through the anus.

C Urea leaves the body in urine.

D Water is removed through the kidneys.

Correct answer: A

38 Which bacteria convert urea to ammonia?

A decay bacteria

B denitrifying bacteria

C nitrifying bacteria

D nitrogen-fixing bacteria

2 Table 2.1 shows some of the major constituents in a person's sweat on a warm day (in micrograms per 100 cm^3).

For Examiner's Use

Table 2.1

nitrogenous compounds (including urea, amino acids and broken-down hormones)	glucose	sodium chloride
31.5	2.5	3.5

(a) State and explain the effects of sweating on the urine produced during a hotter day.

.....

.....

.....

.....

.....[4]

To prevent sweating, some people use a spray (antiperspirant) that blocks the sweat ducts.

(b) Explain why it is important to use an antiperspirant **only** on those parts of the body, such as under the arms, that produce the most sweat.

.....

.....

.....

.....

.....[3]

People who sweat a lot and do not wash regularly may suffer from body odour.

(c) Suggest why the regular use of an antibacterial soap is better than an antiperspirant spray for controlling body odour.

.....

.....

.....[3]

[Total: 10]

2 **(a)** lower volume / less urine ;
less water / more concentrated ;
correct ref. to fewer nitrogenous compounds / salts to be removed ;
kidneys reabsorb more water ;
to keep blood concentration constant ; [4 max]

(b) sweating inhibited / AW ;
loss of ability to regulate temperature effectively ;
body would overheat / AW ORA ;
effect on metabolism / enzymes ; [3 max]

(c) kills / AW, bacteria (**R** germs) ;
responsible for decomposition / breakdown ;
of nitrogenous compounds / waste products ;
can be used all over body / does not block sweat ducts ;
temperature regulation not affected ; [3 max]

3 Fig. 3.1 shows a small, deep-rooted bush growing in a warm, dry climate.
Branches **B** and **C** have a similar number of leaves, but the leaves of branch **B** are enclosed in a transparent polythene bag that empties into a container.

For Examiner's Use

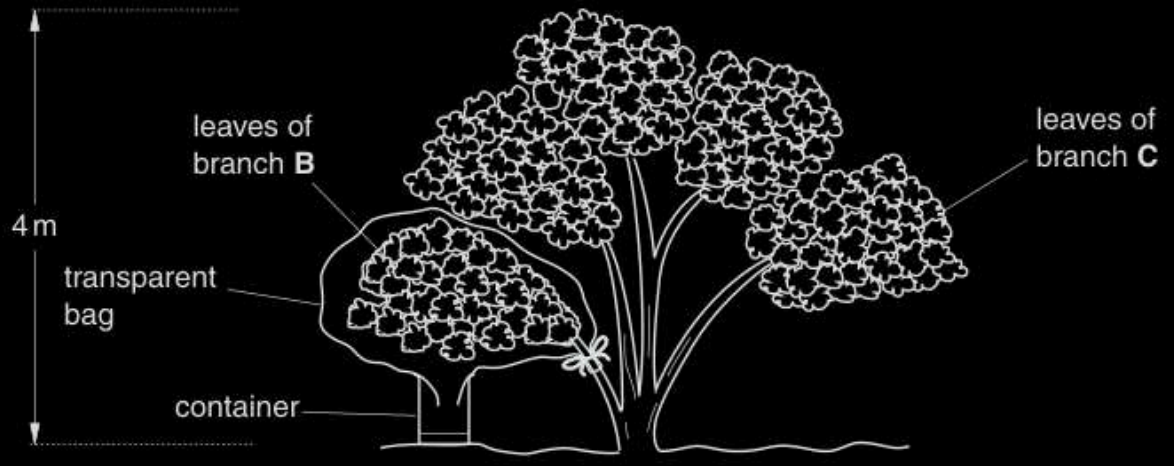


Fig. 3.1

Fig. 3.2 is a graph showing the total volume of water lost by the leaves of each of the two branches during the same day.

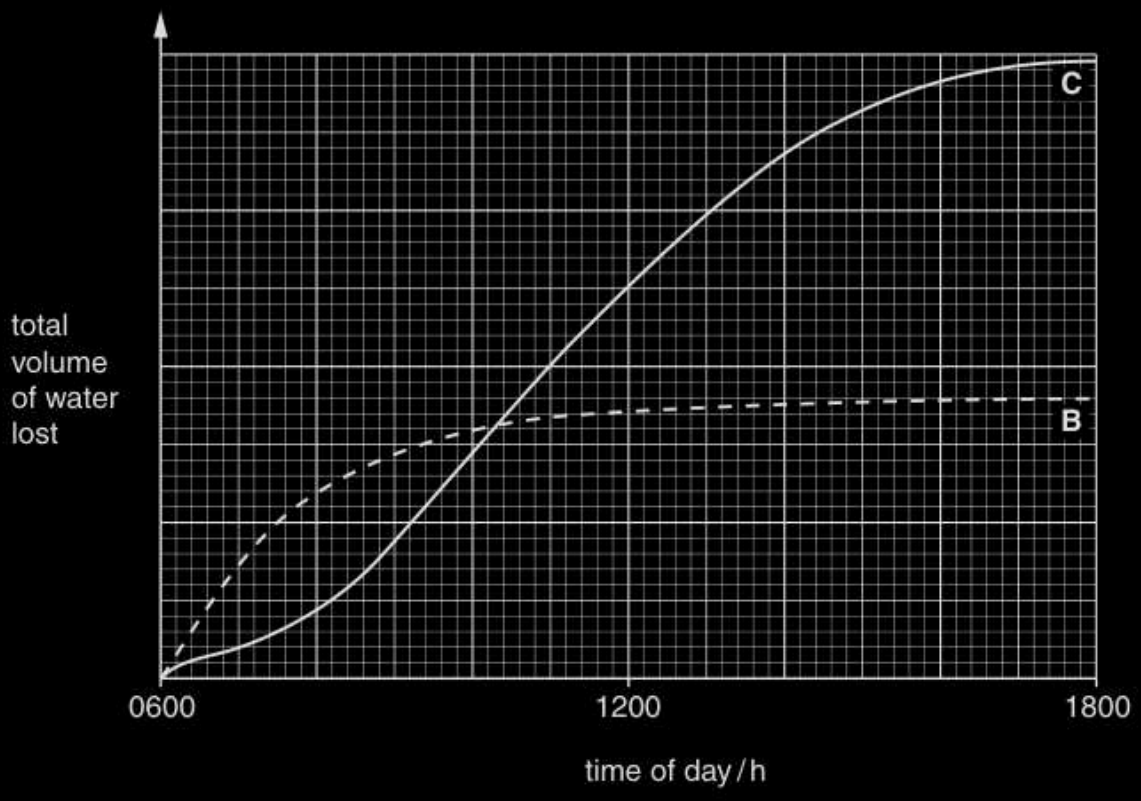


Fig. 3.2

(a) State two environmental factors responsible for the water loss during the day by branch **C**. For each factor, explain how it affects water loss.

For Examiner's Use

factor 1. .....

explanation

.....

.....

factor 2. .....

explanation

.....

.....[5]

(b) Explain how the volume of water lost from branch **B** is at first greater, then less than that lost from branch **C**.

.....

.....

.....[2]

(c) Suggest why, even for certain plants that are poisonous to humans, the container in Fig. 3.1 can supply travellers with safe drinking water.

.....

.....

.....

.....[3]

[Total: 10]

3 **(a)** (factors) – *any 2 from*: humidity / AW, temperature / heat, light, wind, amount of water in soil AW ;; [2]

(explanations): *(dry air ORA)* ref. concentration gradient ;
faster rate of evaporation / transpiration ;
faster rate of <u>diffusion</u> ;

(*higher temperature ORA*) faster rate of evaporation ;
faster molecular movement / ref. energy ;

(*bright light ORA*) stomata open ;
allows greater volume of vapour to be lost / AW ;

(*moving air ORA*) blows away water / vapour / moisture;
increases concentration gradient ;
faster rate of <u>diffusion</u> ;

(*soil water*) less water uptake ;
stomata close / AW ;
stops water loss ; [3 max]

(b) warms up more quickly at first / like glasshouse ;
humidity increases ;
becomes a limiting factor or described / less transpiration ; [2 max]

(c) water comes from the soil (not plant) ;
carried in xylem / xylem just hollow tubes ;
contains only dissolved salts / metabolites carried in phloem ;
poison remains in cells ;
it is only water that evaporates during transpiration ; [3 max]

4 Fig. 4.1(a) and Fig. 4.1(b) each shows cells from the lining of the trachea. One is from a smoker and one is from a non-smoker.

For Examiner's Use

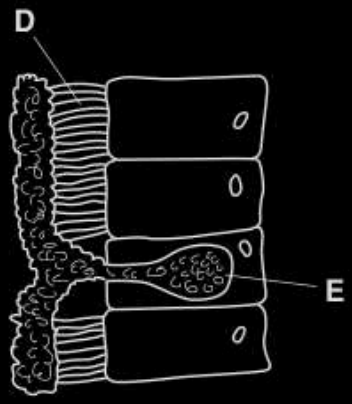


Fig. 4.1(a)

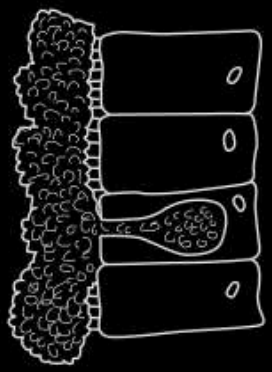


Fig. 4.1(b)

(a) (i) Identify **D** and **E** in Fig. 4.1(a).

D .....

E ..... [2]

(ii) Describe the function of **D**.

.....

..... [2]

Fig. 4.2(a) and Fig. 4.2(b) show cross-sections through the alveoli of a smoker and of a non-smoker.

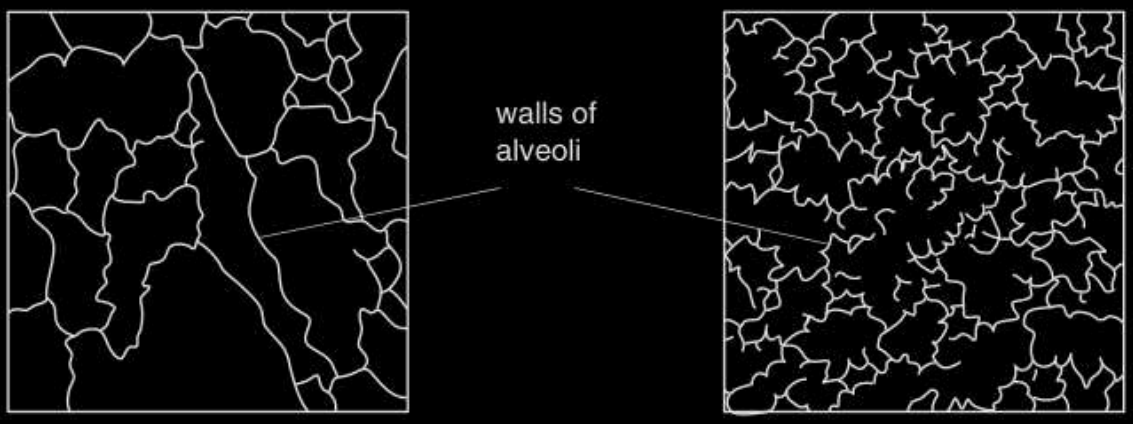


Fig. 4.2(a) **Fig. 4.2(b)**

(b) (i) Identify the figures on this page that show the trachea and alveoli of the smoker.

Fig. ..... and Fig. ..... [1]

(ii) Explain how the effect of smoking on the alveoli could affect the general health of a smoker.

For Examiner's Use

.....

.....

.....

.....

.....

.....

.....

.....[5]

[Total: 10]

4 **(a)** **(i)** **D** cilia ;
E goblet (cell) / mucus (-producing cell) / gland (cell) ; [2]

(ii) ref. beating / AW ;
moving mucus + towards throat / upwards / away from lungs ;
containing germs / dirt ; [2]

(b) **(i)** Fig. 4.1(b) + Fig. 4.2(a) (**A** in either order) ; [1]

(ii) carcinogenic / AW ;
tar + impervious to gases ;
emphysema / break down of alveoli walls ;
reduced surface area ;
less O_2 absorption / to red blood cells / body cells ;
named affected organ (e.g. extremities / brain / heart) ;
effect on (named) organ ;
tiredness / shortness of breath ; [5 max]

5 Fig. 5.1 shows some cells from a root of a plant.

For Examiner's Use

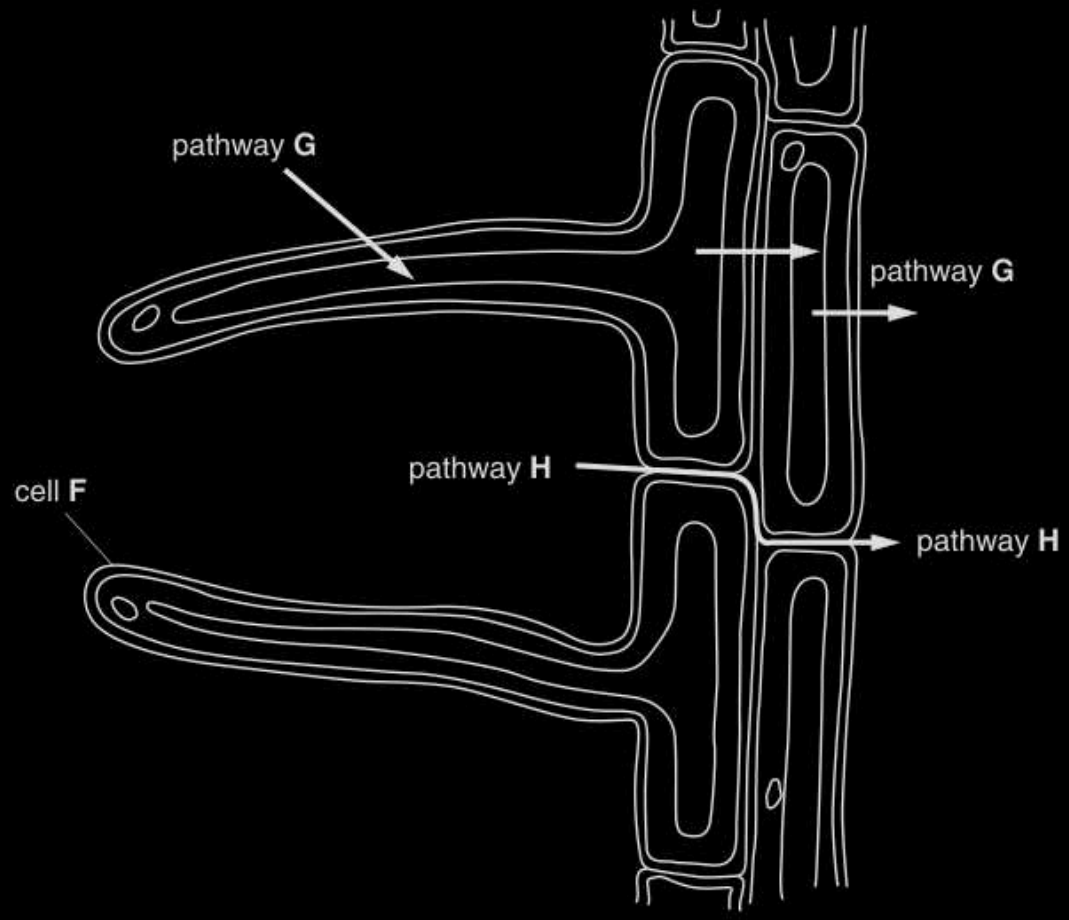


Fig. 5.1

(a) Identify cell **F** in Fig. 5.1.

F ..... [1]

(b) Name the mineral ions absorbed by roots that are essential components of

(i) chlorophyll, ...

(ii) amino acids. ..... [2]

(c) **G** and **H** show two different pathways for the uptake of mineral ions from the soil.

(i) Explain how ions are taken up via pathway **G**, even when their concentration in the surrounding soil is very low.

.....

.....

.....[3]

(ii) Suggest and explain why pathway **H** is more suited to the entry of ions that are in high concentration in the soil.

For Examiner's Use

.....

.....

.....

.....

.....

.....[4]

[Total: 10]

Section B

For Examiner's Use

Answer **all** the questions including questions 6, 7 and 8 **Either** or 8 **Or**.

Write your answers in the spaces provided.

5 **(a)** root hair ; [1]

(b) **(i)** magnesium / nitrates* ;

(ii) nitrates* (*once only) ; [2]

(c) **(i)** active transport / uptake ;
requires energy ;
from respiration / mitochondria ;
ref. living / cell + membrane ;
*ref. (against) concentration gradient ; [3 max]

(ii) *ref. concentration gradient ;
diffusion ;
cellulose / cell wall ;
fully permeable ;
direct pathway to xylem / no barrier ; [4 max]
[* = allow once only in **(i)** or **(ii)**]

[Total: 50]

Section B

8 Either **(a)** Describe how an amino acid molecule passes from the lumen of the ileum to the liver.

For Examiner's Use

.....

.....

.....

.....

.....

.....[3]

(b) Describe what could happen to an amino acid molecule from the time it enters the liver to the time its component elements leave the body.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[7]

[Total: 10]

8 **Or** **(a)** Define the term *mitosis*.

For Examiner's Use

.....

.....[2]

(b) Describe the role of mitosis in organisms.

.....

.....[2]

(c) Explain how the cells of offspring have the same number of chromosomes as the cells of their parents.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 10]

8 Either

(a) absorption / passes into ;
villus ;
capillary ;
blood plasma ;
in solution ;
(hepatic) portal vein ; [3 max]

(b) made into protein or named ;
deamination or described ;
carbohydrate production / glucose ;
ref. respiration / loss as CO_2 ;
storage as glycogen ;
urea ;
hepatic vein ;
renal artery ;
kidney ;
ureter ;
urine / sweat ;
bladder / urethra ; [7 max]

8 OR

(a) cell / nuclear division ;
producing genetically identical cells/ nuclei ;
maintaining chromosome number ; [2 max]

(b) growth ;
repair / replacement of cells / tissues ;
asexual / vegetative reproduction / cloning ; [2 max]

(c) in sexual reproduction ;
gametes ;
produced by reduction division / meiosis ;
have half the number of chromosome / haploid ;
fertilisation ;
fusion of nuclei / gametes ;
restoration of chromosome number / diploid ;
ref. asexual reproduction ;
mitosis ; [6 max]

3 *Skimming* is a process by which the constituents that collect at the top of fresh milk are removed. Skimmed milk has had all these constituents removed.

For Examiner's Use

Table 3.1 shows the nutritional contents of 100 cm^3 of skimmed milk and 100 cm^3 of unskimmed milk.

Table 3.1

	skimmed milk	unskimmed milk
energy	146 kj	267 kj
protein	3.4 g	3.4 g
carbohydrate	5.0 g	4.6 g
saturated fat	0.04 g	2.3 g
unsaturated fat	0.06 g	1.3 g
sodium	0.1 g	0.1 g
calcium	124 mg	119 mg

(a) Name two constituents of a healthy diet, not listed above, and state their importance to a person's health.

constituent 1 .....

importance

.....

.....

constituent 2 .....

importance

.....

..... [5]

(b) Using the information in Table 3.1, suggest why skimmed milk is considered to be better for the circulatory system than unskimmed milk.

..... [2]

(c) Explain why a mother's milk is better for her baby than milk from another species of mammal.

.....

.....

.....

..... [4]

[Total: 11]

3 **(a)** *Award **one each** for constituents.*

fibre / roughage ;

bulk / for muscles to push against AW / prevents constipation / prevents bowel cancer ;

peristalsis ;

---

water ;

prevents dehydration / ref. osmoregulation ;
medium for enzyme action /digestion /metabolic processes ;
solvent / transport / sweating ; [2 max]

(for importance)

vitamin C

(any two from) wound healing, anaemia, bleeding spots on the skin, loose teeth, bleeding gums, prevents scurvy ;;

---

vitamin D ;

uptake / storage + of calcium / phosphorus ;
healthy bones / teeth / anti-ricketic ;

Fe ;

haemoglobin ;
oxygen carriage / absorption ;

(Accept other vitamins / ions – 1 for name, 2 for importance ;;; [5 max]
A name of vitamin in 'importance' – In lists, mark first one only,
A 'vitamin**s**' (in the plural, and unspecified) for a mark, but importance must refer to at least **two** separate functions for **one mark**.

(b) less (overall) fat content ;
particularly saturated fat ;
ref. deposition in blood vessels / atheroma / raised blood pressure
heart disease or problem (**A** atherosclerosis) ; [2 max]

(c) correct constituents / AW ;
in correct proportions ;
antibodies ;
non-allergenic ;
readily available / cheap / correct temperature ;
unlikely to be sterile / uncontaminated (**A** more hygienic) ; [4 max]

4 Fig. 4.1 shows some structures in a section through human skin.

For Examiner's Use

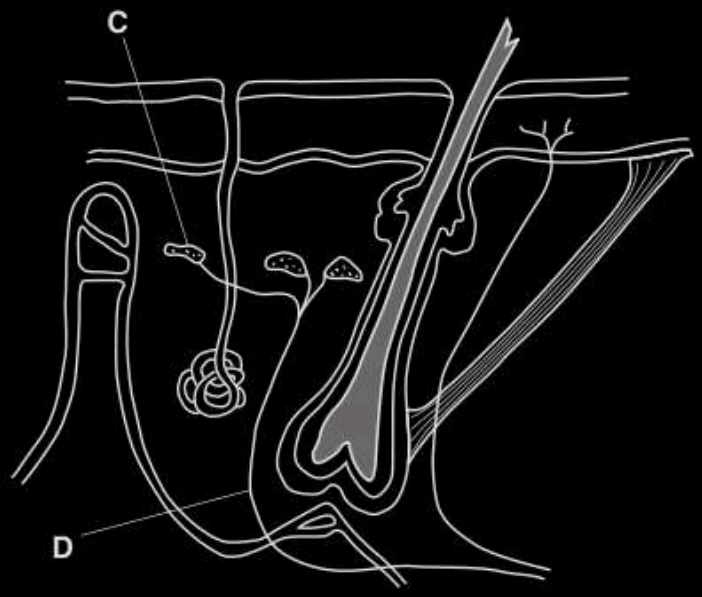


Fig. 4.1

(a) The body is able to maintain its internal environment within narrow limits. State the term for this process.

..... [1]

(b) Structures **C** and **D**, in Fig. 4.1, are involved in the process of temperature regulation.

Identify structures **C** and **D** and state the part they play in the process.

C .....

part played.....

.....

D .....

part played.....

..... [4]

(c) The consumption of alcohol causes the muscles in artery walls to relax.

Taking this into consideration, suggest why people who work in environmental temperatures below 5 °C might be advised not to drink alcohol before or during work.

.....

.....

.....

..... [3]

[Total: 8]

4 **(a)** homeostasis ; [1]

(b) **C** – receptor / sensor ;
detecting changes (in temperature) (**A** even if misidentified) ;
D – sensory / afferent + neurone / nerve cell or fibre (**R** nerve) ;
impulses to CNS / brain / spinal cord / (**A** even if misidentified) ; [4]

(c) more blood ;
to capillaries ;
(vaso)dilate (**A** with ref. to capillaries or arteri(ol)es) ;
blood carries heat ;
more heat lost ;
lowers body temperature (still) further / hypothermia / AW ; [3 max]

May/June 2010 · Paper 2 · Variant 2 · Question 5

5 Fig. 5.1 shows a section through a microscopic, one-celled organism often found growing on the bark of trees.

For Examiner's Use

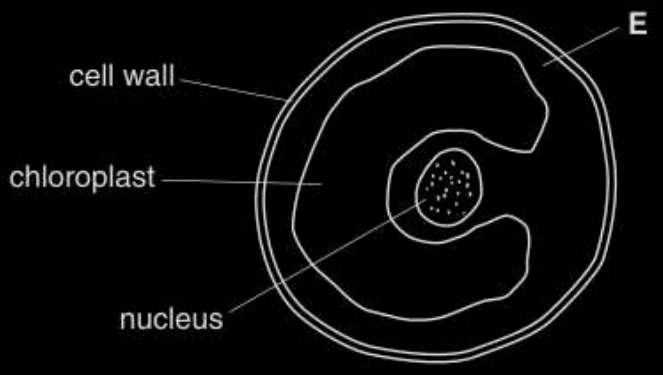


Fig. 5.1

(a) Suggest the identity of the region labelled **E** in Fig. 5.1.

..... [1]

(b) **(i)** List the structures visible in Fig. 5.1 which suggest that this organism might be a plant.

.....

..... [2]

(ii) List the ways in which this cell differs from a palisade cell in a plant leaf.

.....

.....

.....

..... [4]

Fig. 5.2 shows the same cell reproducing.

For Examiner's Use

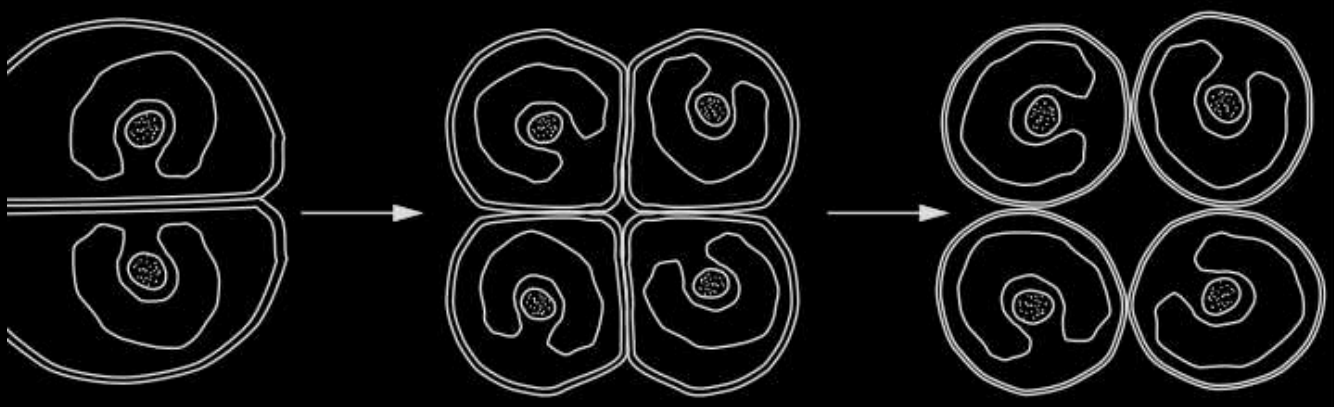


Fig 5.2

(c) Explain why successive generations of this organism have remained unchanged in appearance for many thousands of years.

.....

.....

..... [4]

[Total: 11]

Section B

For Examiner's Use

Answer **all** the questions including questions 6, 7 and 8 **Either** or 8 **Or**.

Write your answers in the spaces provided.

5 **(a)** cytoplasm ; [1]

(b) **(i)** *Mark the first two structures mentioned.*
chloroplast(s) ;
cell wall ; [2]

(ii) ***A*** *all converse points(ref. palisade cell) as long as cell type is clear.*
no vacuole / no cell sap ;
one chloroplast only ;
large chloroplast / ref shape of chloroplast ;
round / spherical shape of cell ;
no tonoplast / vacuolar membrane / AW ;
cell is entire organism / not part of a tissue ;
ref. position of nucleus ; [4 max]

(c) asexual (**A** no sexual) reproduction (**A** vegetative reproduction)
binary fission ;
mitosis ;
identical (**R** similar) offspring / no variation / clone ;
no (**A** limited) natural selection ;
no (**A** limited) evolution ;
no meiosis / no fertilisation / no gametes / only one parent
(**R** no mutation) ; [4 max]

[Total: 50]

Section B

6 **(a)** Describe how a molecule of carbon dioxide in the air becomes part of a carbohydrate molecule stored in a leaf of a plant.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [7]

(b) Describe how a carbohydrate molecule stored in a leaf of a plant can become a starch molecule stored in the root.

.....

.....

.....

.....

..... [3]

[Total: 10]

6 **(a)** stoma(ta) ;
Intercellular / air + space ;
diffusion (anywhere) ;
dissolves ;
mesophyll (cell) / named (any relevant ref.) ;

<u>chloroplast</u> ;
*water (as a reactant) ;
*light / photolysis ;

<u>photosynthesis</u> ;
*glucose / starch ; [7 max]
(***A** formulae and **A** on equation)

(b) changed to <u>sucrose</u> ;
suitable enzyme reference ;
in solution ;
translocated / carried + <u>phloem</u> ;
<u>from</u> cells (when made) / <u>into</u> cells (when stored) ; [3 max]

8 Either

A person cuts their hand on a piece of glass that they are picking up from the ground.

(a) Describe how the body's defence mechanisms protect the person from infection.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [8]

(b) Explain how the person would know if the cut had damaged an artery.

.....

.....

..... [2]

[Total: 10]

8 Or

For Examiner's Use

Potatoes contain starch. Describe and explain what happens to a piece of potato, that has been fried in oil, as it passes through the mouth (buccal cavity) and small intestine.

mouth (buccal cavity)

.....

.....

.....

.....

.....

small intestine

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[Total: 10]

8E (a) capillaries / blood vessels damaged ;
bleeding / blood flow ;
platelets / thrombokinase / prothrombin / thrombin ;
fibrinogen ;
fibrin ;
clotting ;
scab (or described) ;
new cell growth ;
re-establishment of bacteria-proofing / skin re-seals ;
white blood cells or named ;
antibodies / antitoxin ;
phagocytosis or described ; [8 max]

(b) bright red in colour / oxygenated blood ;
blood leaves in spurts / ref pulse in arteries ; [2]
(Ignore references to pressure)

8O (a) [pre]molar + grinding / [canine]incisor + cutting) / teeth + mechanical digestion
(**R** chewing) ;
saliva(ry) ;
starch to maltose* (**A** disaccharide, **R** sucrose) ;
amylase* ;
bolus (or described – **A** action of tongue / action of mucin) ;

(b) starch to maltose* [once only in **(a)** or **(b)**] ;
amylase* [once only in **(a)** or **(b)**] ;
maltose to glucose ;
lipase ;
fats to fatty acids and glycerol ;
absorbed by villi ;
capillaries + glucose / amino acids ;
lacteals / lymph for fatty acids + glycerol (**A** fat) ;
emulsification of fats AW ;
by bile ;
ref. to protein digestion ; [10 max]

October/November 2009 · Paper 1 · Variant 1 · Question 1

Correct answer: D

1 Which processes can take place in a root hair cell when oxygen is **not** available?

A active transport only

B diffusion only

C active transport and osmosis only

D diffusion and osmosis only

Correct answer: A

2 The diagram represents a cell as seen under the electron microscope.

What type of cell is this?

	type of cell	reason
A	animal cell	outer layer is the cell membrane
B	bacterium	no chromosomes are visible
C	plant cell	cytoplasm is visible
D	plant cell	cell wall is visible

October/November 2009 · Paper 1 · Variant 1 · Question 3

Correct answer: A

3 The table shows the concentrations of some mineral ions in the root hair of a plant and in the soil around it.

mineral ion	concentration in the root hair (arbitrary units)	concentration in the soil (arbitrary units)
magnesium	75	15
nitrate	126	47

How are these mineral ions absorbed from the soil by the plant?

	magnesium	nitrate
A	active transport	active transport
B	active transport	diffusion
C	diffusion	active transport
D	diffusion	diffusion

October/November 2009 · Paper 1 · Variant 1 · Question 4

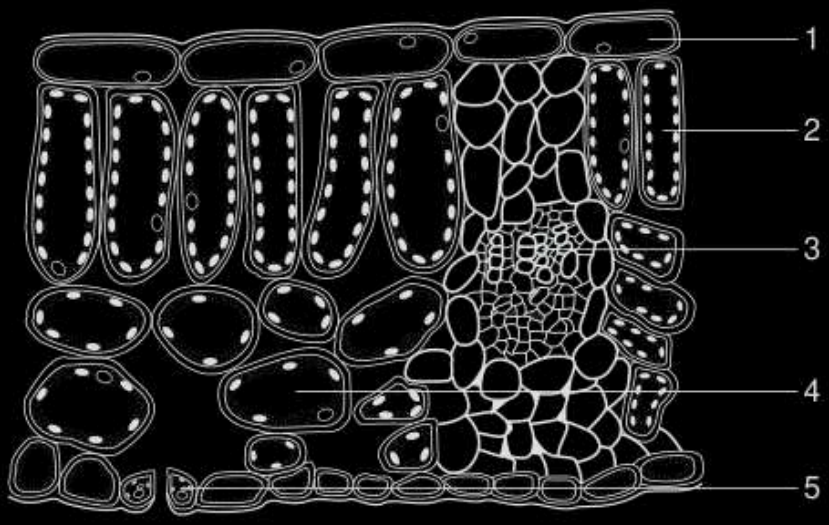
Correct answer: C

4 Protease breaks down proteins into amino acids.

In the 'lock and key' hypothesis, what is the lock and what is the key?

	lock	key
A	amino acid	protease
B	protease	amino acid
C	protease	protein
D	protein	protease

Correct answer: C

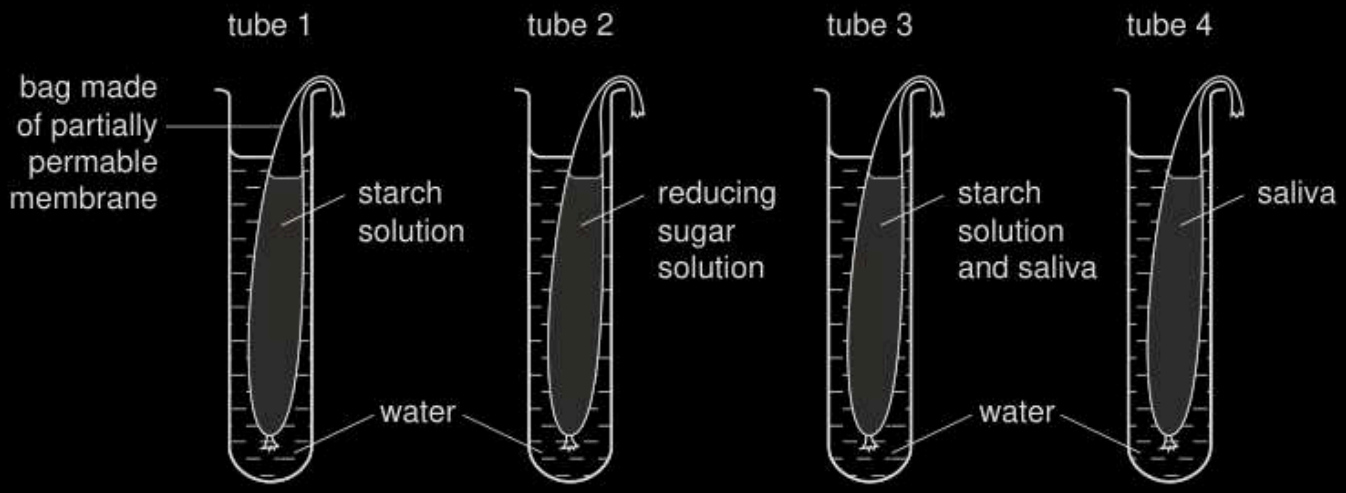


Which cells absorb carbon dioxide?

A 1, 2 and 3 **B** 2, 3 and 4 **C** 2, 4 and 5 **D** 1, 3 and 5

October/November 2009 · Paper 1 · Variant 1 · Question 9

Correct answer: B



After 20 minutes at 35 °C a sample of the water around the bag in each tube is boiled with Benedict's solution.

What are the expected results?

	tube 1	tube 2	tube 3	tube 4
A	blue	orange	blue	orange
B	blue	orange	orange	blue
C	orange	blue	orange	blue
D	orange	orange	blue	orange

October/November 2009 · Paper 1 · Variant 1 · Question 10

Correct answer: B

10 Four foods were analysed for protein, fat and carbohydrate.

Which food contains the most energy in a 100 g portion?

	protein %	fat %	carbohydrate %
A	20	7	1
B	10	9	20
C	4	5	6
D	0.5	0.5	5

October/November 2009 · Paper 1 · Variant 1 · Question 11

Correct answer: B

11 During translocation in plants, which substance is moved and in which direction?

	substance	from	to
A	sucrose	anthers	stigmas
B	sucrose	leaves	roots
C	water	roots	leaves
D	water	soil	root hairs

October/November 2009 · Paper 1 · Variant 1 · Question 12

Correct answer: C

12 Which substances are transported in the phloem and which in the xylem?

	phloem	xylem
A	nitrate ions	sugar
B	starch	nitrate ions
C	sugar	water
D	water	starch

October/November 2009 · Paper 1 · Variant 1 · Question 13

Correct answer: D

13 Blood enters the left atrium (left auricle) of the heart through

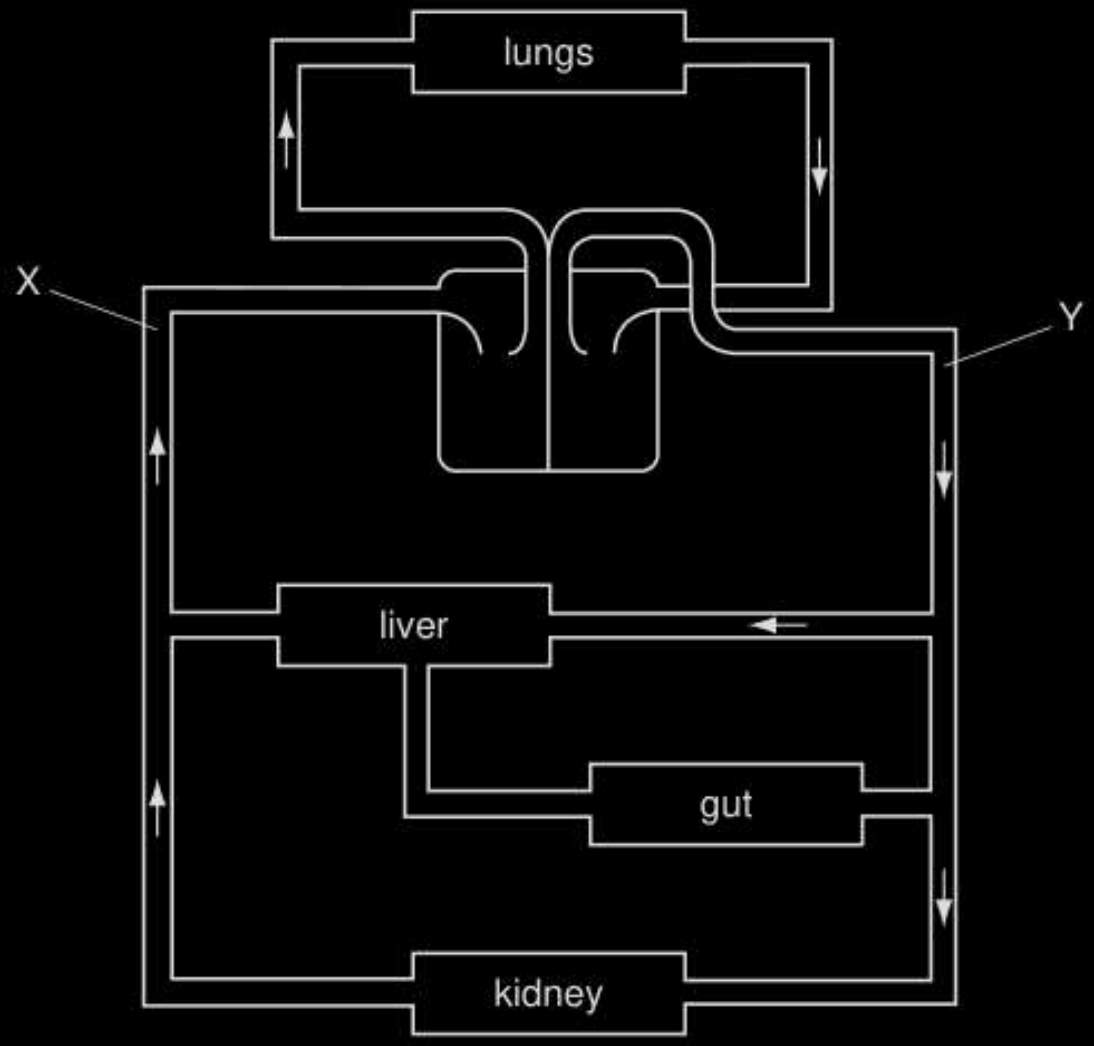
A the aorta.

B the bicuspid valve.

C the posterior vena cava.

D the pulmonary vein.

Correct answer: C



Which vessel **must** the blood pass through in flowing from X to Y?

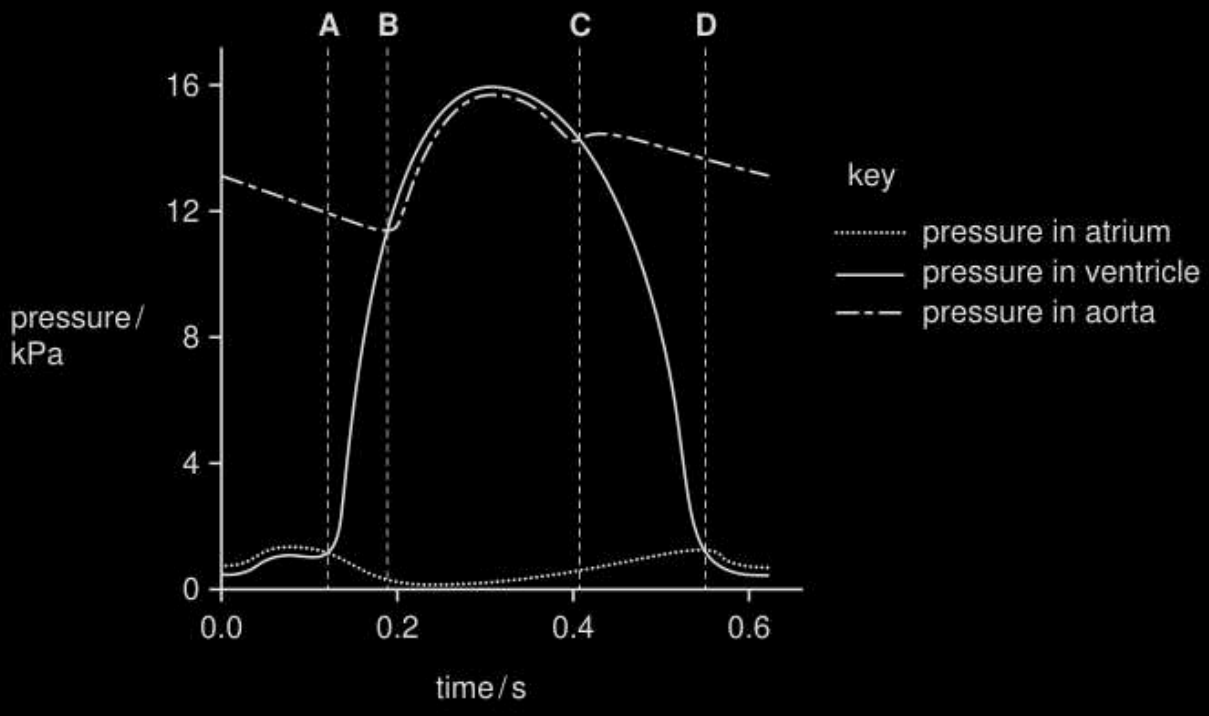
A hepatic artery

B hepatic portal vein

C pulmonary artery

D renal vein

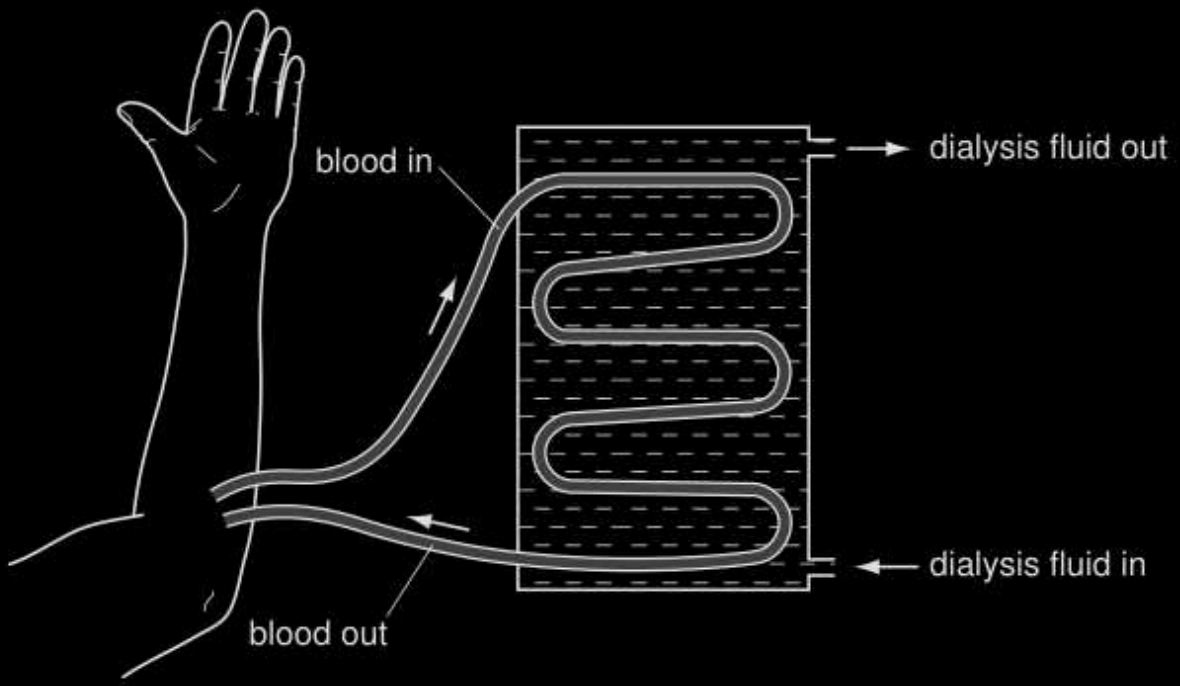
Correct answer: B



October/November 2009 · Paper 1 · Variant 1 · Question 20

Correct answer: A

20 The diagram represents part of a kidney machine.



Which substance must be at the same concentration in the dialysis fluid and in the blood?

A glucose

B salt

C urea

D water

October/November 2009 · Paper 1 · Variant 1 · Question 21

Correct answer: C

21 When a person moves from a cold room into a hot room, the following responses occur.

1 The brain co-ordinates the response.

2 The skin begins to secrete sweat.

3 Sweat evaporates from the skin surface.

4 Temperature receptors are stimulated in the skin.

What is the correct sequence of events?

A $3 \rightarrow 4 \rightarrow 1 \rightarrow 2$

B $2 \rightarrow 3 \rightarrow 4 \rightarrow 1$

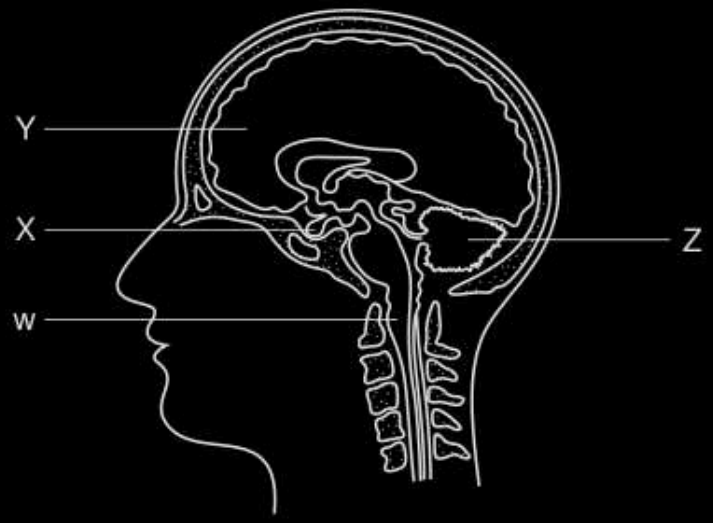
C $4 \rightarrow 1 \rightarrow 2 \rightarrow 3$

D $1 \rightarrow 4 \rightarrow 2 \rightarrow 3$

October/November 2009 · Paper 1 · Variant 1 · Question 22

Correct answer: D

22 The diagram is a section through the human head showing the brain and associated structures.



Where are the cerebellum and the pituitary gland?

	cerebellum	pituitary gland
A	W	Z
B	X	Y
C	Y	W
D	Z	X

October/November 2009 · Paper 1 · Variant 1 · Question 23

Correct answer: A

23 A person is sitting in a dark room.

What happens in the eye when a light is switched on?

	circular muscle of iris	size of pupil
A	contracts	decreases
B	contracts	increases
C	relaxes	decreases
D	relaxes	increases

October/November 2009 · Paper 1 · Variant 1 · Question 26

Correct answer: B

26 Two species of bacteria work together and make yoghurt.

What do they produce?

	ethanol	lactic acid	protease
A	✓	✓	✓
B	✗	✓	✓
C	✓	✗	✓
D	✓	✓	✗

October/November 2009 · Paper 1 · Variant 1 · Question 34

Correct answer: C

34 A plant P is pollinated. A seed from this plant develops into plant Q.

The cells of plant Q contain 14 chromosomes.

How many of these chromosomes are identical to chromosomes from plant P?

	if P was self pollinated	if P was cross pollinated
A	7	7
B	7	14
C	14	7
D	14	14

October/November 2009 · Paper 1 · Variant 1 · Question 35

Correct answer: D

35 What is the sequence of structures through which a mammalian sperm passes after leaving the penis and before fusing with an ovum?

A ureter → uterus→ vagina

B vagina → fallopian tube → ovary

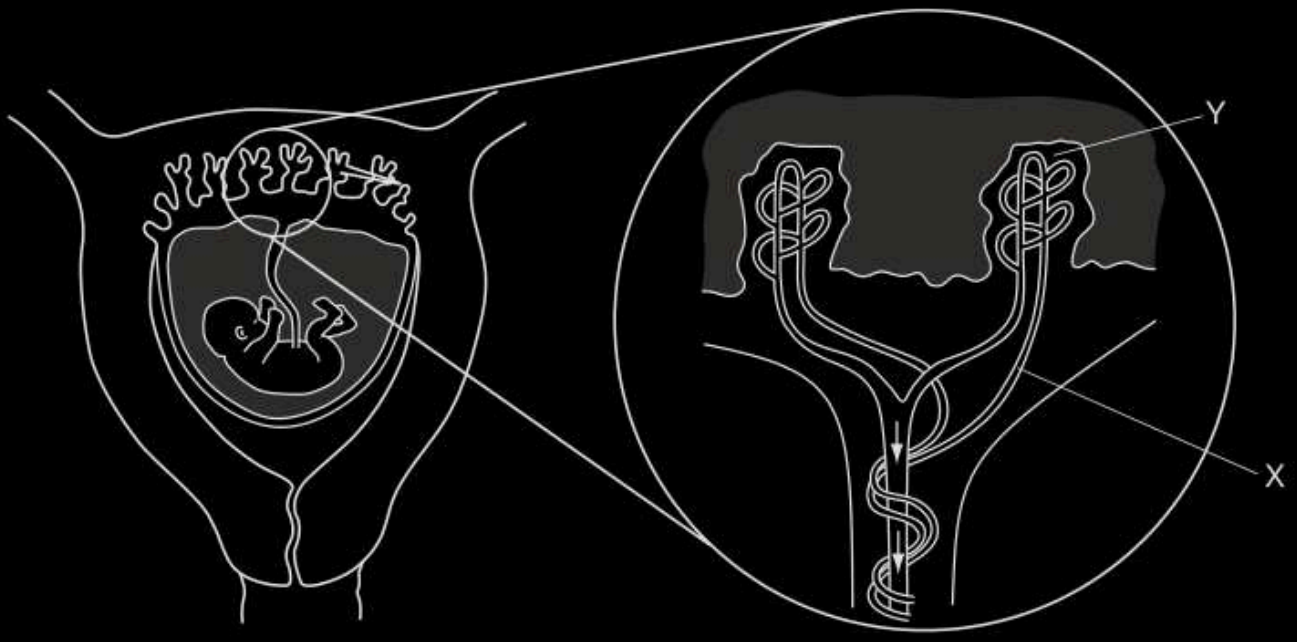
C vagina → urethra → fallopian tube

D vagina → uterus → fallopian tube

October/November 2009 · Paper 1 · Variant 1 · Question 36

Correct answer: C

36 The diagram shows a fetus in the uterus.



Which substances will be at a higher concentration at Y than at X?

A carbon dioxide and glucose

B carbon dioxide and urea

C glucose and oxygen

D oxygen and urea

October/November 2009 · Paper 1 · Variant 1 · Question 37

Correct answer: D

red form
with black spots

black form
with red spots

What kind of variation is shown by the beetle and why?

A continuous variation because it is controlled by genes

B continuous variation because there are two forms

C discontinuous variation because it is controlled by genes

D discontinuous variation because the two forms are distinct

October/November 2009 · Paper 1 · Variant 1 · Question 39

Correct answer: C

39 Which statement about the genotypes of organisms is correct?

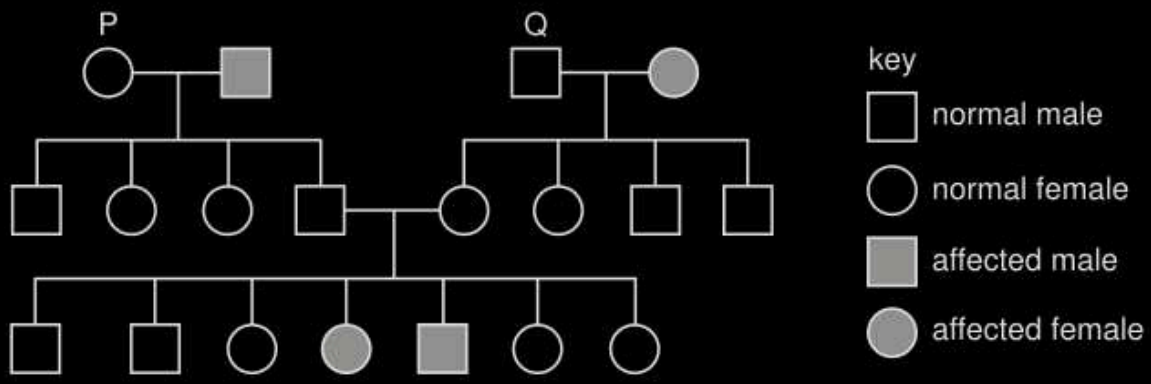
A Dominant alleles are only found in homozygotes.

B One recessive allele always causes a recessive phenotype.

C Recessive phenotypes must be homozygous.

D The dominant phenotype must be heterozygous.

Correct answer: D



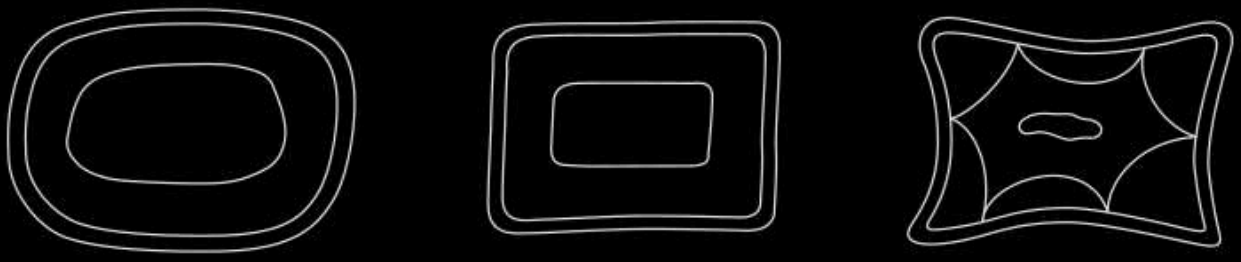
T represents the dominant allele and t represents the recessive allele.

What are the most likely genotypes of individuals P and Q?

	P	Q
A	Tt	Tt
B	Tt	TT
C	TT	Tt
D	TT	TT

2 Fig. 2.1 shows three similar plant cells 5 minutes after being placed in different solutions, **G**, **H** and **I**. One of these solutions has a composition and water potential exactly the same as that of the cell sap in the cells.

For Examiner's Use



cell in solution **G** | cell in solution **H** | cell in solution **I**

Fig. 2.1

(a) On **one** of the cells in Fig. 2.1, label clearly

(i) with the letter **J**, a structure made of cellulose,

(ii) with the letter **K**, the part of the cell that would contain the nucleus. [2]

(b) Identify the solution that has a **higher** water potential than the cell sap.
Explain what has happened to cause the appearance of the cell in that solution.

solution ..... [1]

explanation .....

.....

.....

..... [4]

(c) **(i)** On the cell in solution **I**, shade all the regions that would contain solution **I**. [1]

(ii) Explain your answer to **(i)**.

.....

.....

.....

..... [3]

[Total: 11]

2 **(a)** *For each letter, if more than one cell labelled, all labels must be correct*

(i) label line (or otherwise indicated) **J** to / in / on cellulose cell wall
(**R** label lines that terminate on inner line of cellulose cell wall) ;

(ii) label line (or otherwise) **K** to cytoplasm ; [2]

(b) solution G (**A** under explanation if solution line blank) ; [1]

water enters ;

*osmosis / diffusion ;

*cell sap more concentrated / diffusion or osmotic or water potential gradient ;

increase in volume AW / pressure / turgor ; [max 4]
(* available with wrong or no solution identified)

(c) (i) space between membrane and cell wall shaded on right hand diagram ; [1]
(ignore shaded cell wall)

(ii) (*These marks are available with no or wrong shading if they are in reasonable context*)

cell wall permeable ;

solution passes in ;

correct ref. to partial permeability of cell membrane ; [3]

[Total: 11]

For Examiner's Use

3 During heart transplant procedures, it is possible to keep the heart beating while it is being transported many miles from the donor to the patient. Fig. 3.1 shows diagrammatically how the heart is kept beating during transportation.

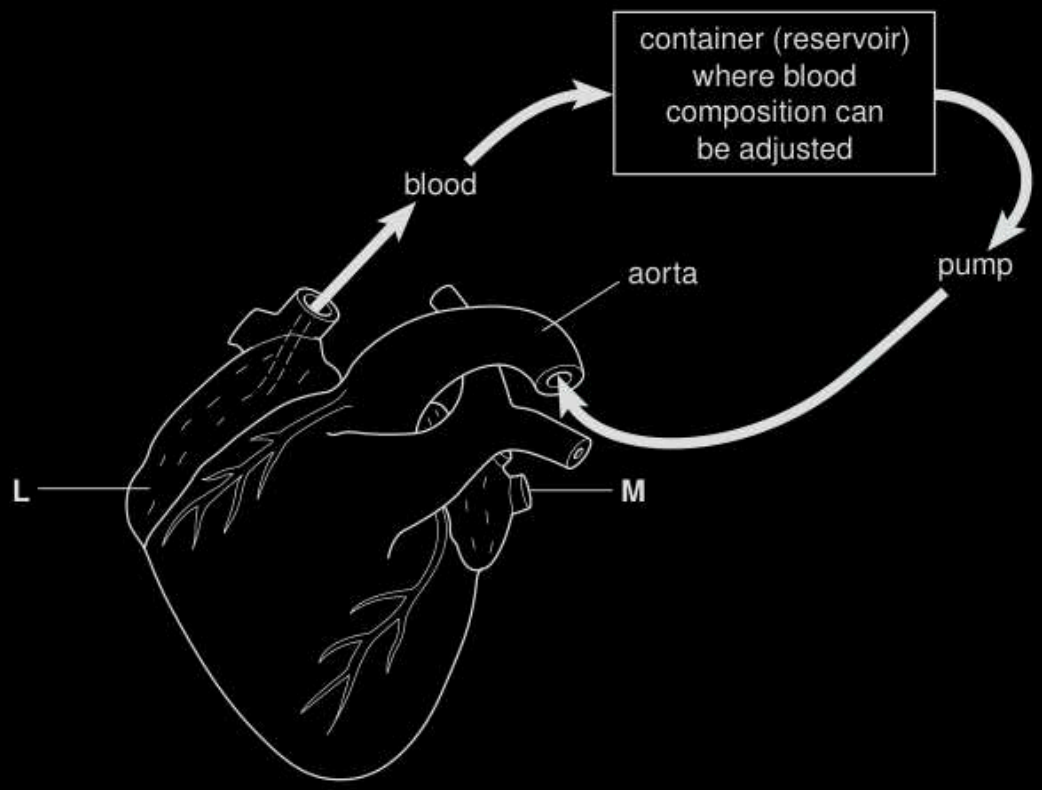


Fig. 3.1

(a) Identify structures **L** and **M** on Fig. 3.1.

L .....

M ..... [2]

(b) Suggest **two** chemicals which must be added to the blood in the reservoir, and explain your suggestions.

chemical 1 .....

explanation .....

.....

chemical 2 .....

explanation .....

..... [4]

(c) Using your knowledge of the structure of the heart, explain why blood entering the aorta from the reservoir flows into the heart muscle and **not** into the left ventricle.

For Examiner's Use

.....

.....

.....

.....

..... [3]

[Total: 9]

3 **(a)** **(L)** right + atrium / auricle ;

(M) pulmonary vein ; [2]

(b) (chemicals) *glucose ;

*oxygen ; [2]

(explanation max 2 – each mark available once only either under glucose or oxygen)

for energy (**R** production) ;

during respiration ;

heart <u>muscle / cells / tissue</u> ; (max 2) [4]

(***A** adrenalin; to stimulate heart muscle; insulin; for glucose absorption by heart cells; Heparin; to stop blood clotting)

(c) valve (if named, must be aortic / semi-lunar) ;

closes (available with wrongly-named valve) ;

sending blood through <u>coronary</u> blood vessel / artery(ies) (**R** vein) ; [3]

[Total: 9]

4 Fig. 4.1 shows two models of seeds which use the same type of seed dispersal.

For Examiner's Use

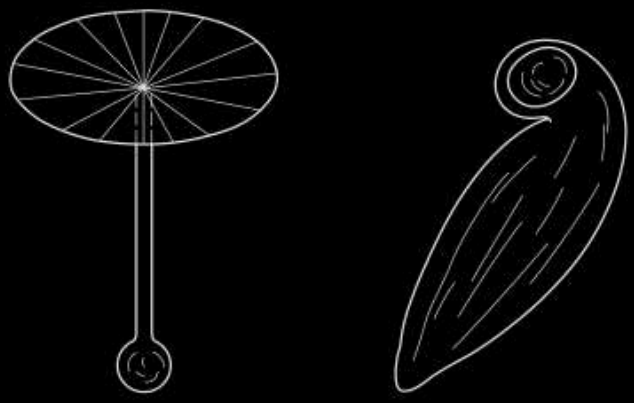


Fig. 4.1

(a) Name the structure within which a seed develops.

..... [1]

(b) Suggest the type of seed dispersal shown.
Explain how the structures in Fig. 4.1 are adapted for this type of dispersal.

type of dispersal .....

adaptation and explanation .....

..... [3]

Fig. 4.2 shows the percentage of seeds of a particular plant that germinate at different temperatures.

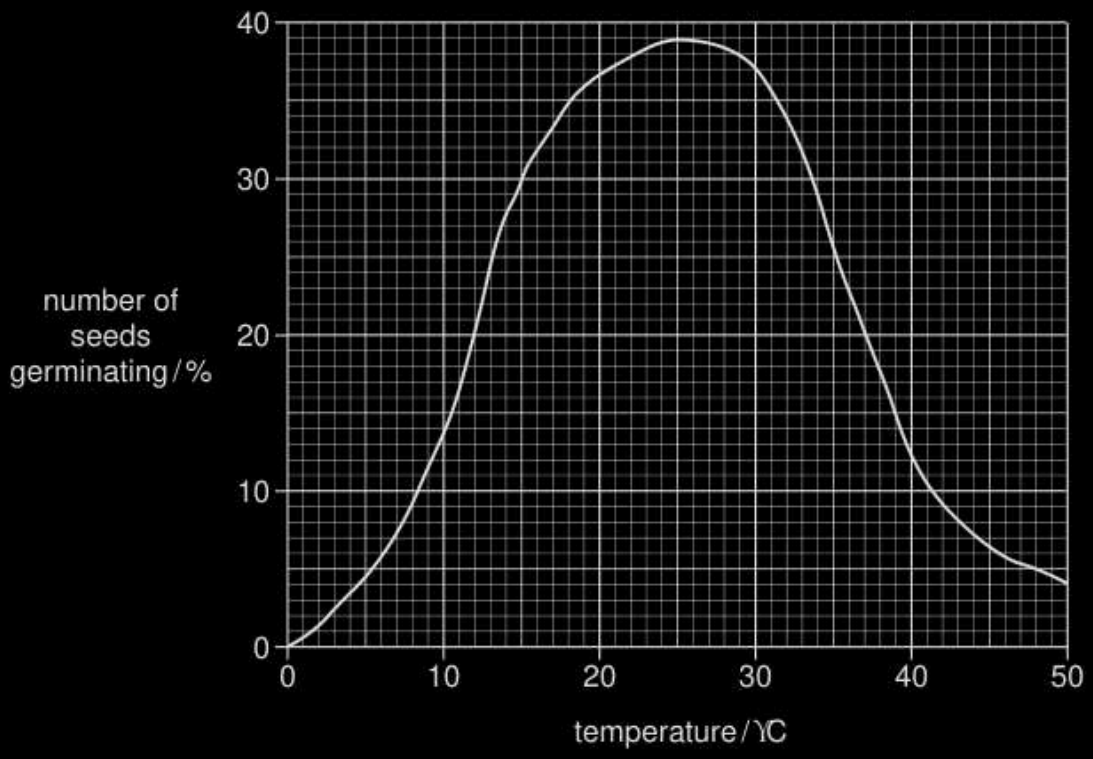


Fig. 4.2

(c) State two possible reasons why no more than 40% of the seeds germinate, even at the optimum temperature.

For Examiner's Use

1. .....

2. ..... [2]

(d) Suggest three reasons why, after germination, many seedlings do not live to become mature plants.

1. .....

2. .....

3. ..... [3]

[Total: 9]

4 **(a)** fruit / pericarp / ovary / testa ; [1]

(b) wind / <u>moving</u> air ; (1)

*(**R** any refs. to water dispersal)*
ref parachute / helicopter wing AW ;

large surface area / light ;

buoyancy / holds seed in air or floats ; (max 2) [3]

(c) lack of water AW ;

lack of O_2 (for whatever reason) ;

seed not viable / dormancy not complete AW / eaten ; [max 2]

(d) overcrowding / lack of light ;

unsuitable temperature ;

eaten ;

disease ;

unsuitable substrate / lack of ions or minerals / wrong pH ;

(**A,** here, ref. insufficient nutrients in soil = unsuitable substrate)

drought AW ; [max 3]

[Total: 9]

6 **(a)** Define a *hormone*.

.....

.....

.....

.....

.....

..... [4]

(b) Describe how the nervous system is involved in producing a **named** reflex action.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

[Total: 10]

6 **(a)** chemical (**R** named hormone) ;

produced by a gland (**R** specific named endocrine gland) ;

carried + by the blood ;

affects a target organ (**R** specific target organ) ;

destroyed by liver ; [max 4]

(b) stimulus or example ;

receptor / or described (**R** named sense organs) ;

impulse / electrical pulse AW ;

sensory neurone or sensory nerve cell (**R** nerve) ;

correct ref. to CNS for action described ;

relay / intermediate neurone or nerve cell (**A** 'interneurone') ;

motor or effector neurone or nerve cell ;

effector or described (muscle or gland) ;

correct response for stimulus given / named reflex ; [max 6]

[Total: 10]

7 **(a)** State where DNA is found and describe its importance in living organisms.

For Examiner's Use

.....

.....

.....

.....

.....

..... [3]

(b) Describe the causes of

(i) sickle cell anaemia,

.....

.....

(ii) Down's syndrome.

.....

.....

[3]

(c) Explain, with the use of a genetic diagram, how a child may be born with a blood group different from that of either parent.

[4]

[Total: 10]

Question 8 is in the form of an **Either**/**Or** question. Answer only question 8 **Either** or question 8 **Or**.

For Examiner's Use

7 **(a)** in nucleus / plasmids / chloroplasts / mitochondria ;

found in chromosomes / genes / contains genetic information ;
(**R** contains genes)

can be copied / inherited / ref responsible for characters (characteristics) ;

controls production of a protein ; [max 3]

(b) mutation (with ref. either sickle cell anemia or Down's) ;

(sickle cell) genetic or ref. gene / allele / DNA / change in Hb molecule ;

(Down's syndrome) one extra chromosome / 47 instead of 46 ; [3]

(c) correct plausible genotypes of parents (**A** A,B,O,I^A,I^B,I^o only) ;
(if they are linked by lines to the offspring, since they are then gametes and available below)

phenotypes / blood groups of both parents (**A** 'B' as Group B etc.) ;

gametes correctly shown – either separated / in circles etc. / punnet square / linked correctly by lines to offspring ;

offspring correctly identified + with different stated blood group / phenotype ;

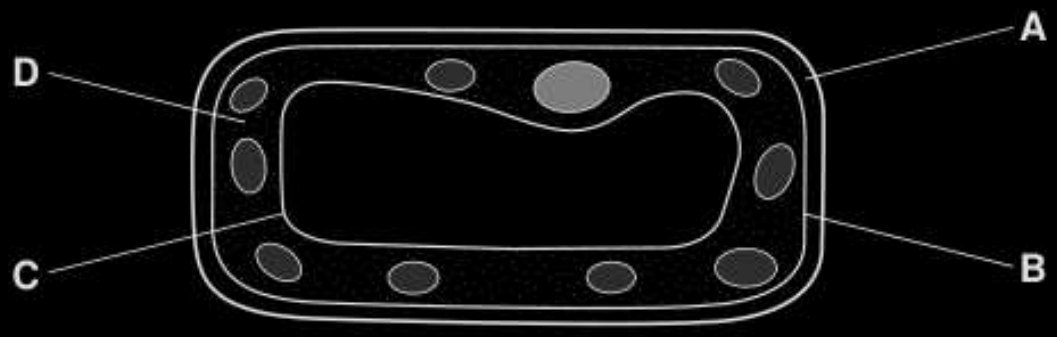
any correct descriptive term (gametes / parents / genotype / children) ; [max 4]
(**A** mother + father)

(*Allow all marks on a written description)*

[Total: 10]

May/June 2009 · Paper 1 · Variant 1 · Question 1

Correct answer: B



May/June 2009 · Paper 1 · Variant 1 · Question 2

Correct answer: A

2 Which liquid has the highest water potential at atmospheric pressure?

A distilled water

B leaf cytoplasm

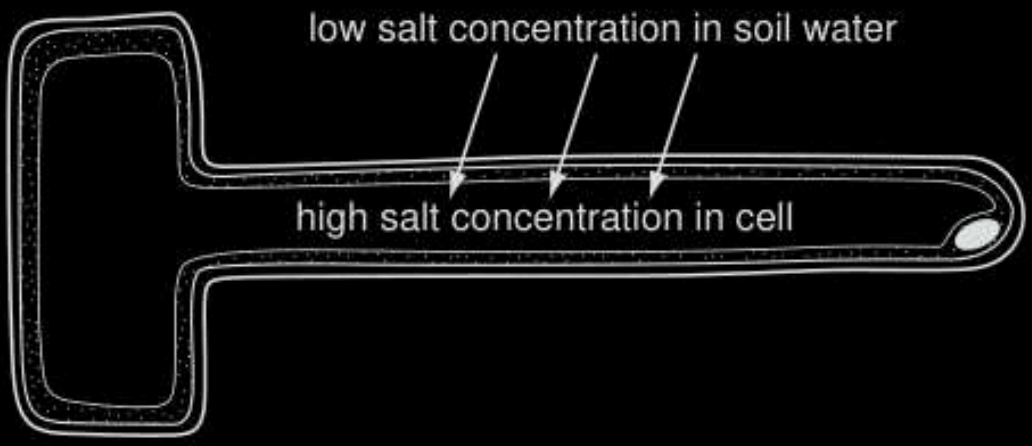
C root hair cell sap

D soil water

May/June 2009 · Paper 1 · Variant 1 · Question 3

Correct answer: A

3 The arrows represent the movement of salts into a root hair cell.



What describes the movement of the salts?

A active transport against the concentration gradient

B active transport down the concentration gradient

C diffusion against the concentration gradient

D diffusion down the concentration gradient

May/June 2009 · Paper 1 · Variant 1 · Question 4

Correct answer: B

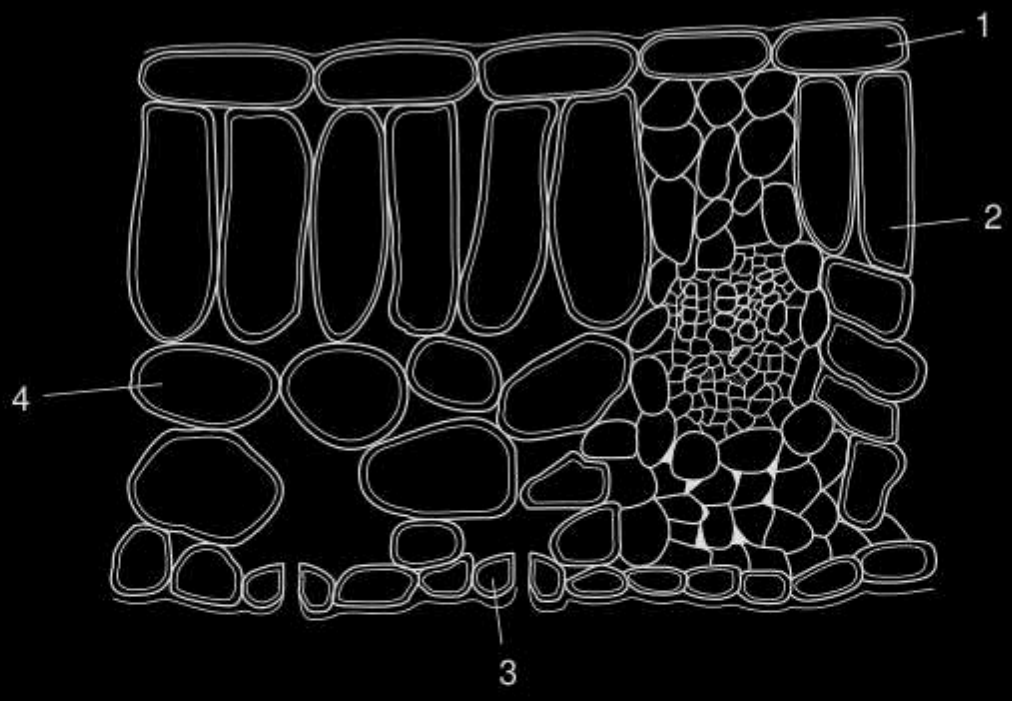
4 Four test-tubes containing egg white are incubated for ten minutes at 35 °C. Different substances are then added to the four test-tubes.

In which test-tube is the egg white first digested?

	substances added
A	1 cm^3 lipase solution and three drops of sodium hydroxide solution
B	1 cm^3 protease solution and three drops of dilute hydrochloric acid
C	1 cm^3 protease solution and three drops of water
D	1 cm^3 lipase solution and three drops of water

May/June 2009 · Paper 1 · Variant 1 · Question 7

Correct answer: D



Which cells contain chloroplasts?

A 1 and 2

B 2 only

C 2 and 3 only

D 2, 3 and 4

May/June 2009 · Paper 1 · Variant 1 · Question 8

Correct answer: A

8 Bread contains dietary fibre, fat, protein and starch.

Which substance contributes **least** to the energy obtained by a person eating the bread?

A dietary fibre

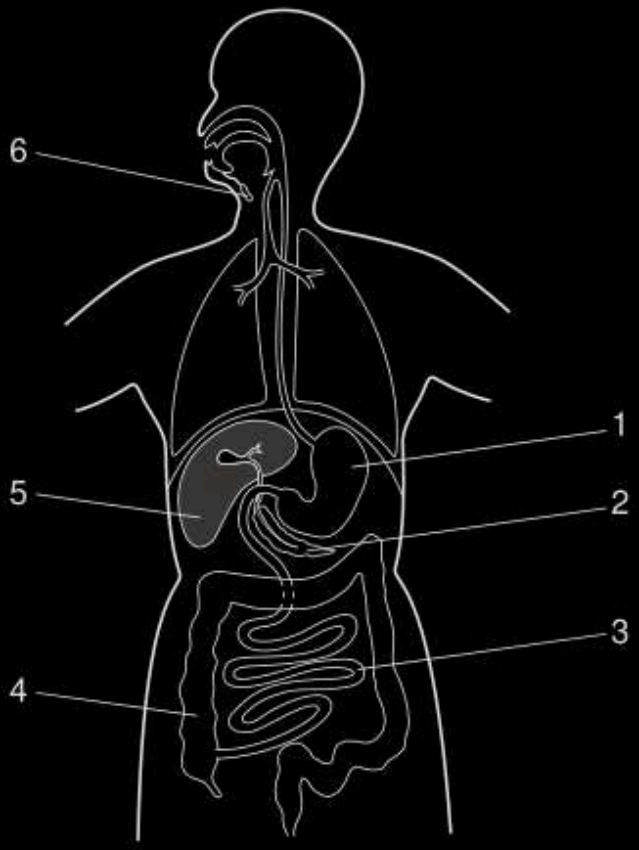
B fat

C protein

D starch

May/June 2009 · Paper 1 · Variant 1 · Question 10

Correct answer: B



Which numbered structures secrete digestive enzymes?

A 1, 2, 3 and 4

B 1, 2, 3 and 6

C 2, 3, 4 and 5

D 2, 3, 5 and 6

May/June 2009 · Paper 1 · Variant 1 · Question 11

Correct answer: D

11 What are the functions of the xylem?

	carrying sugars	carrying water	carrying mineral ions	giving support
A	✓	✗	✗	✓
B	✓	✓	✗	✗
C	✗	✓	✓	✗
D	✗	✓	✓	✓

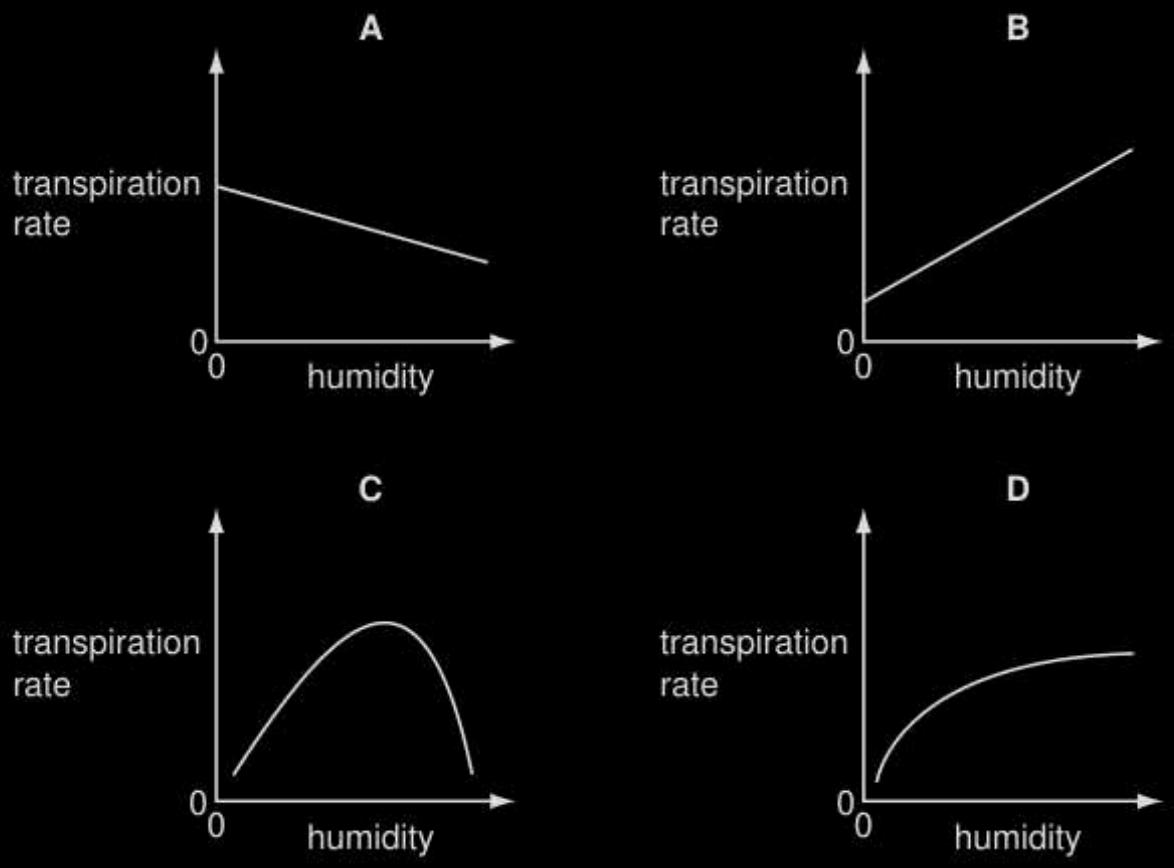
key

✓ = a function of xylem

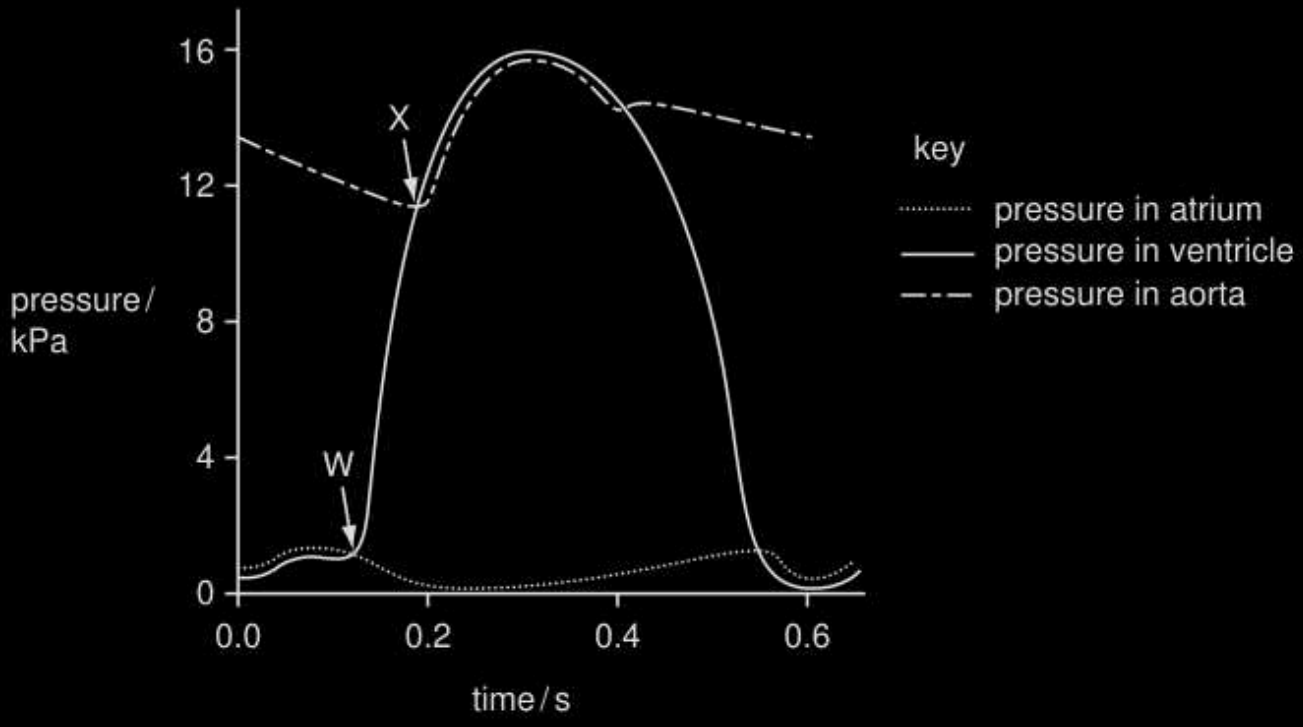
✗ = not a function of xylem

May/June 2009 · Paper 1 · Variant 1 · Question 12

Correct answer: A



Correct answer: A

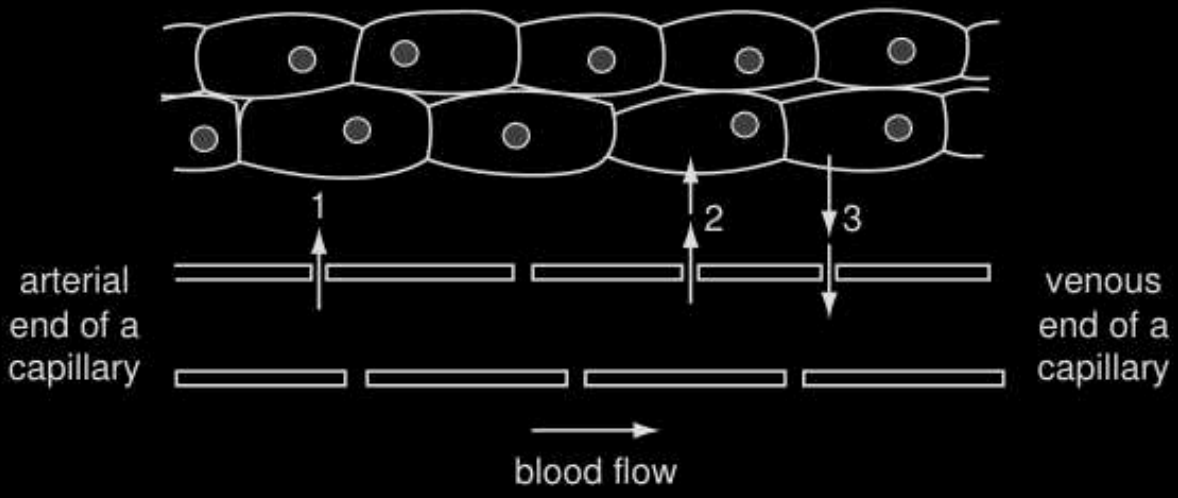


Between points W and X, are the following valves open or closed?

	atrio-ventricular	semi-lunar
A	closed	closed
B	closed	open
C	open	closed
D	open	open

May/June 2009 · Paper 1 · Variant 1 · Question 14

Correct answer: D



Which substances can 1, 2 and 3 represent?

	1	2	3
A	glucose	tissue fluid	carbon dioxide
B	oxygen	carbon dioxide	glucose
C	tissue fluid	glucose	oxygen
D	tissue fluid	oxygen	carbon dioxide

May/June 2009 · Paper 1 · Variant 1 · Question 15

Correct answer: B

15 A vein has a wider lumen than an artery.

What is the advantage of this?

A bringing blood into close contact with the tissues

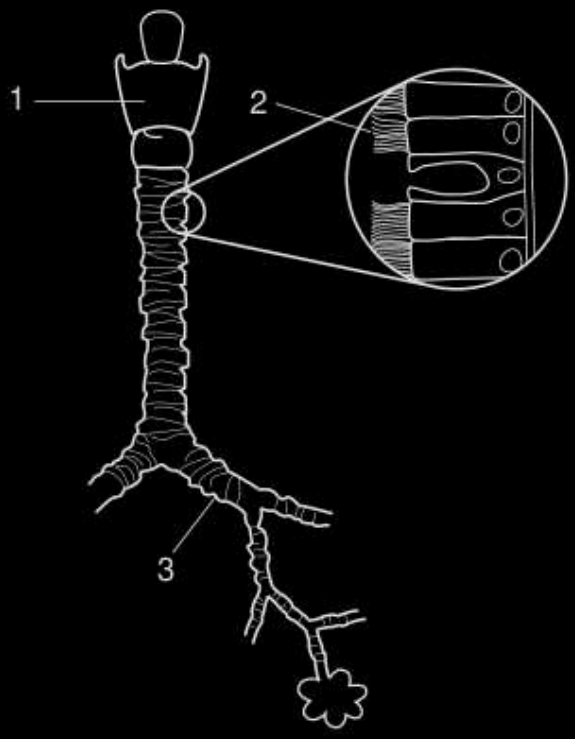
B offering less resistance to blood flow

C preventing back flow of blood

D resisting high blood pressure from the heart

May/June 2009 · Paper 1 · Variant 1 · Question 16

Correct answer: A

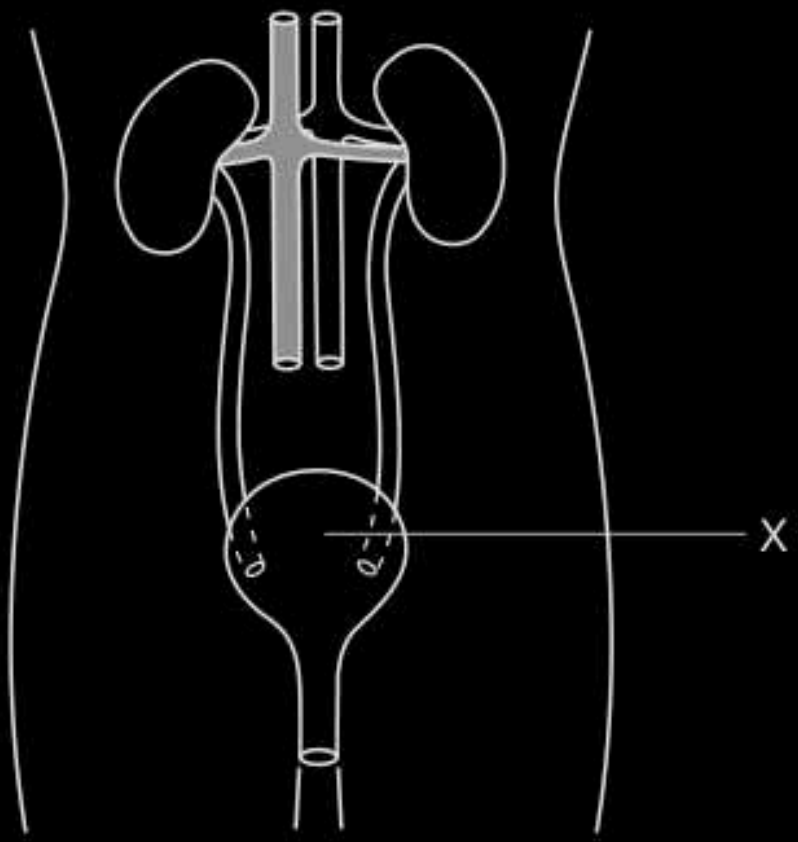


Which row in the table identifies the structures labelled 1, 2 and 3?

	1	2	3
A	larynx	cilia	bronchus
B	larynx	bronchus	bronchiole
C	trachea	bronchus	bronchiole
D	trachea	cilia	bronchus

May/June 2009 · Paper 1 · Variant 1 · Question 20

Correct answer: D



What is the function of X?

A to excrete urea

B to produce urea

C to produce urine

D to store urine

May/June 2009 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 A person walks from a warm room into a cold one.

Which of these reactions occurs first?

A cold receptors in the skin are stimulated

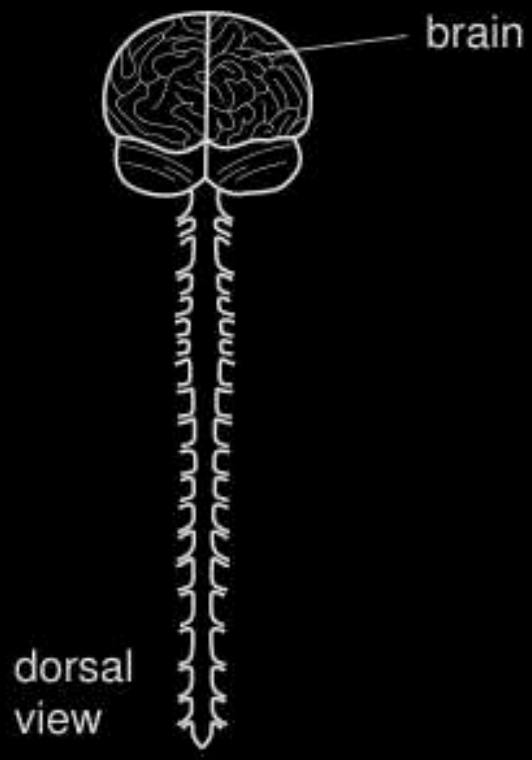
B cooler blood reaches the brain

C deep arterioles in the skin dilate

D shivering begins

May/June 2009 · Paper 1 · Variant 1 · Question 22

Correct answer: B



Which structures are visible?

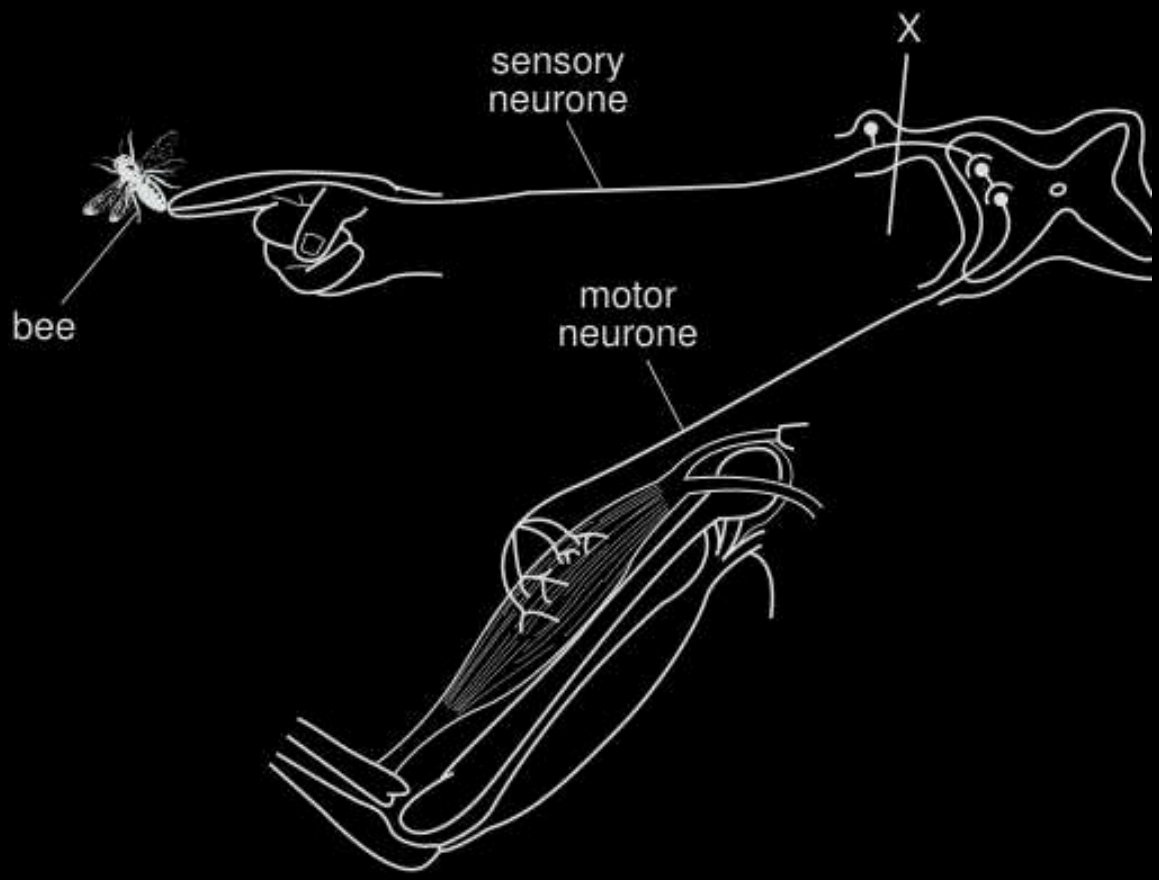
A cerebrum, cerebellum, hypothalamus, spinal cord

B cerebrum, spinal nerves, spinal cord, cerebellum

C spinal nerves, cerebrum, medulla, pituitary gland

D spinal cord, medulla, cerebrum, hypothalamus

Correct answer: A



A bee stings the finger, as shown.

What are the effects of this sting on the person?

	pain felt	arm moved
A	no	no
B	no	yes
C	yes	no
D	yes	yes

May/June 2009 · Paper 1 · Variant 1 · Question 24

Correct answer: D

24 What is an effect of insulin?

A decreased cell respiration

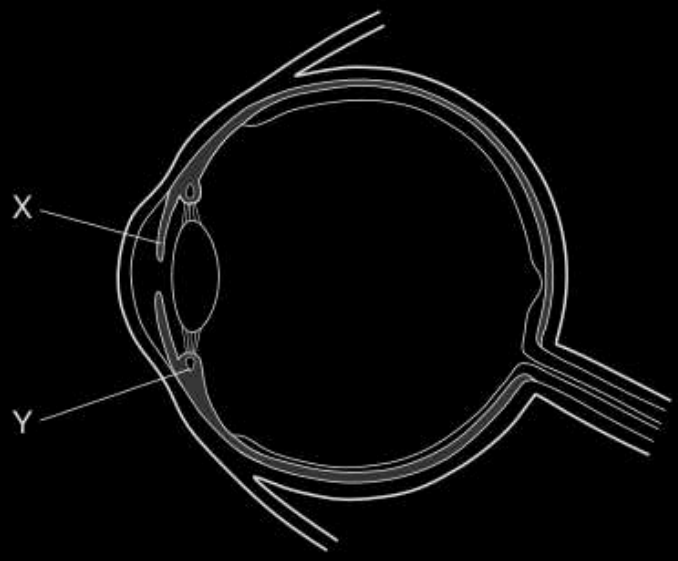
B decreased absorption of glucose by cells

C increased excretion of glucose

D increased synthesis of glycogen

May/June 2009 · Paper 1 · Variant 1 · Question 25

Correct answer: A



What happens to the circular muscles at X and Y when the eye focuses on a near object in bright light?

	X	Y
A	contract	contract
B	contract	relax
C	relax	contract
D	relax	relax

May/June 2009 · Paper 1 · Variant 1 · Question 27

Correct answer: D

27 The table shows some structures found in four different cells.

Which cell is from a fungus?

	cell wall	cell membrane	nuclear membrane	chloroplasts
A	✓	✓	✓	✓
B	✗	✓	✓	✓
C	✓	✓	✗	✗
D	✓	✓	✓	✗

key

✓ = present

✗ = absent

May/June 2009 · Paper 1 · Variant 1 · Question 32

Correct answer: D

32 Which term is used to describe both of these processes?

Process 1 In some towns, empty glass bottles are melted down. The new glass is used to make new bottles.

Process 2 Bacteria in the soil break down protein in dead animals and plants. The nitrates released by the bacteria are used by plants.

A conservation

B decomposition

C pollution

D recycling

May/June 2009 · Paper 1 · Variant 1 · Question 33

Correct answer: C

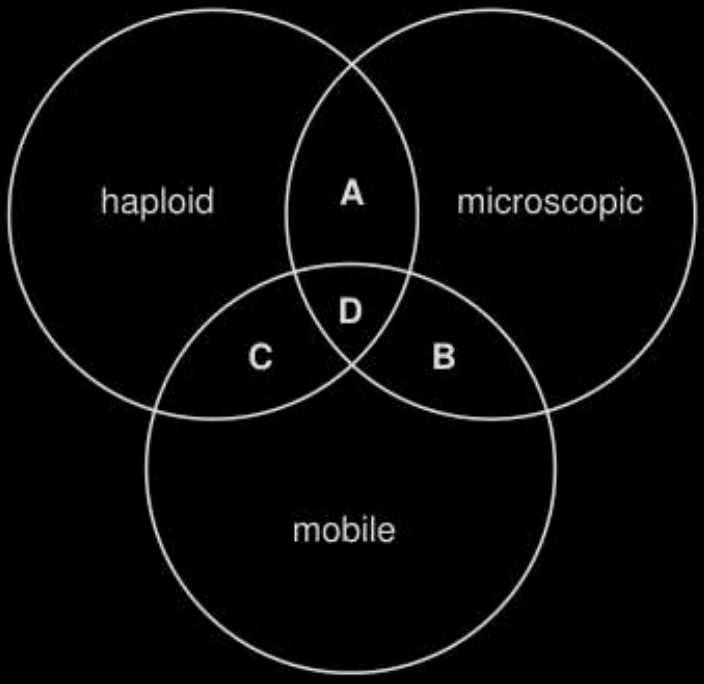
33 The table shows some features of cell division.

Which line correctly describes the division of a cell by mitosis?

	number of chromosomes	number of daughter cells	new cells genetically identical to parent cell	used in growth
A	halved	2	no	no
B	halved	4	yes	no
C	stays the same	2	yes	yes
D	stays the same	4	no	yes

May/June 2009 · Paper 1 · Variant 1 · Question 36

Correct answer: D



May/June 2009 · Paper 1 · Variant 1 · Question 37

Correct answer: B

37 Down's syndrome and sickle cell anaemia are both genetic diseases.

How are they caused?

	Down's syndrome	sickle cell anaemia
A	chromosomal mutation	chromosomal mutation
B	chromosomal mutation	gene mutation
C	gene mutation	chromosomal mutation
D	gene mutation	gene mutation

May/June 2009 · Paper 1 · Variant 1 · Question 38

Correct answer: B

38 A couple have three children. The table shows some of the children's characteristics.

child	sex	blood group	sickle-cell / normal
1	male	B	sickle-cell
2	female	AB	normal
3	male	O	normal

What do the characteristics show?

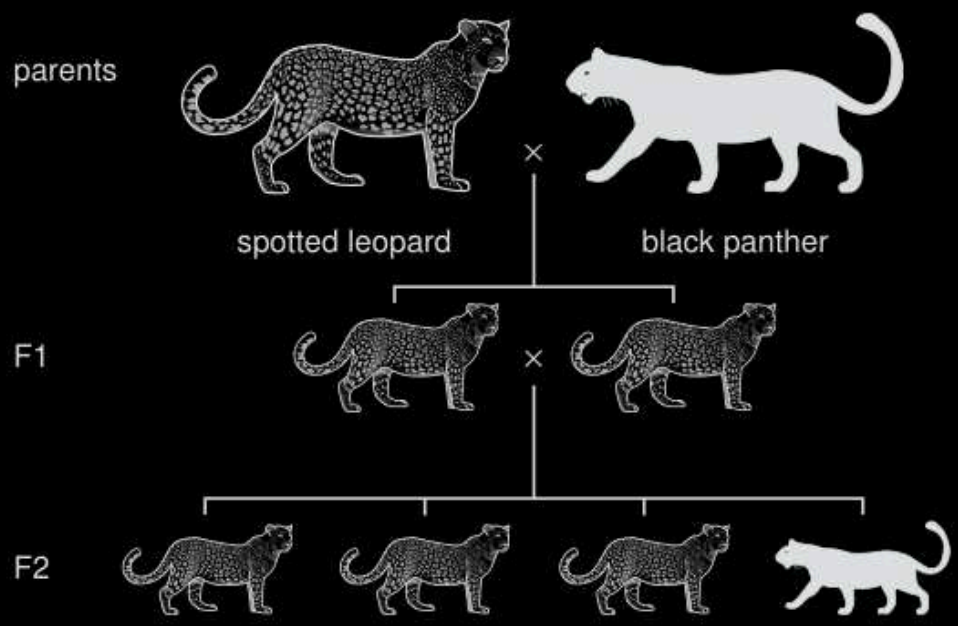
A continuous variation only

B discontinuous variation only

C both continuous and discontinuous variation

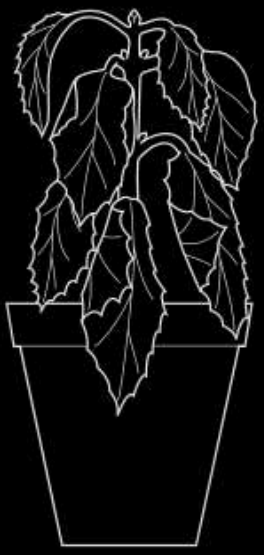
D neither continuous nor discontinuous variation

Correct answer: A



What are the genotypes of the original two parents?

	spotted leopard	black panther
A	HH	hh
B	HH	Hh
C	Hh	hh
D	hh	HH



(b) **(i)** State two environmental conditions that may have led to the change in appearance of the plant.

For Examiner's Use

1. .....

2. ..... [2]

(ii) Explain how environmental conditions have brought about this change.

.....

.....

.....

.....

..... [4]

(c) In the space below, draw and label guard cells and stomata as they would appear in the leaves of the plant in Fig. 1.1 and in Fig. 1.2.

in Fig. 1.1 **in Fig. 1.2**

[3]

[Total: 10]

1 **(a)** wilting/ed / flaccid (**R** plasmolysed) ; [1]

(b) **(i)** *Mark the first, one per line from:*
low humidity / dry air AW (**R** 'humidity') ;
wind ;
lack of (available) water / drought ;
high or raised temperature / hot / warm (**R** warmth, temp. unqualified) ; [max 2]

(ii) (**A** even if condition is inaccurate)
evaporation / (evapo) transpiration ;
water loss faster than rate of water uptake AW ;
loss of water from cells (**R** plant) ;
loss of turgor / flaccidity / ref. pressure AW (**R** plasmolysed) ;
loss of support (**R** droop / wilt) ; [max 4]

(c) (T.S. can score stoma size and labels only)
(labels – in either drawing) guard cell(s) + stoma(ta) ;
(drawings, must be 2) sausage shaped, touching at top and bottom in both ;
larger stoma in left-hand drawing ; [3]

2 In the inheritance of the colour of mouse fur, the allele for yellow fur (**D**) is dominant to the allele for grey fur (**d**).

For Examiner's Use

(a) Two heterozygous yellow-coloured mice produce offspring. Use a fully labelled genetic diagram to show how the colour of mouse fur is inherited by the offspring.

State the expected ratios of genotypes and phenotypes in the offspring.

[6]

A particular combination of these alleles is known as a 'lethal' combination. Young that inherit this combination die in the uterus during the very early stages of development. This results in a 2:1 ratio of fur colour in the surviving offspring.

(b) Identify the lethal combination of alleles and explain how you reached this answer.

lethal combination .....

explanation .....

.....

..... [3]

[Total: 9]

2 **(a)** Dd × Dd (**R** if wrong symbols used) ;
D d D d (*) ;
(* = **A** if correctly deduced from wrong cross)
DD Dd Dd dd (*) ;
1 : 2 : 1 (look for link with genotypes) ;
3 : 1
+ yellow : grey ;
correct ref. gametes (**A** even if qualifying incorrect cross) ; [6]

(b) DD (**A** e.c.f. for incorrect symbols) ;
ref. 1 in 4 would be DD ;
leaves ratio 2 yellow : 1 grey ; [3]
(**A** explanation on diagram – accept on **(a)** so long as linked)

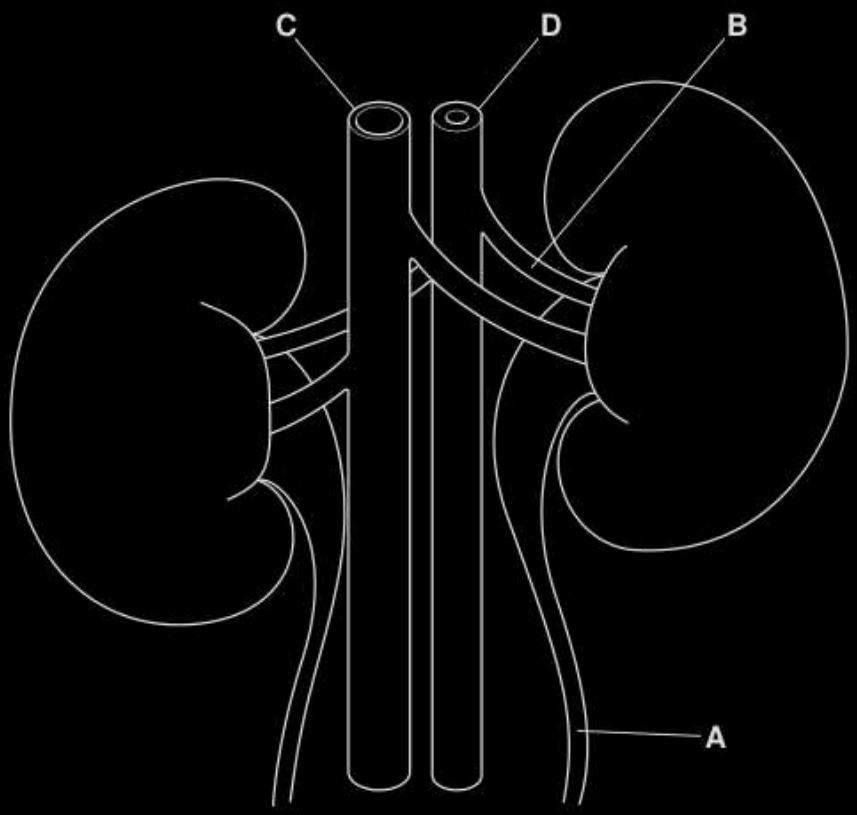


Fig. 4.1

(a) **(i)** Identify structure **A** in Fig. 4.1.

..... [1]

(ii) Peristalsis occurs continually in structure **A**. Describe and explain how this helps the structure to carry out its function.

.....

.....

..... [3]

(b) Identify structure **B** on Fig. 4.1 and state how the structural features of **C** and **D** enabled you to make your identification.

structure **B** .....

structural features of **C** and **D** .....

..... [3]

(c) On a hot day, a person consumed **only** meat before a day of energetic work. Explain the likely changes in the composition of the person's urine during the day.

For Examiner's Use

.....

.....

.....

..... [3]

[Total: 10]

4 **(a)** **(i)** ureter (*accurate spelling)* ; [1]

(ii) wave like / rhythmic AW ;
contraction of muscles (if named must be circular) ;
pushes urine (or description of) (**R** urea alone) ;
to bladder ; [max 3]

(b) renal artery ;
C has thinner walls (or described) than **D** (o. r. a.) ;
C has wider (lumen AW) than **D** (o. r. a.) ; [3]

(c) (water lost as) ref. sweat ;
more AW + urea (in urine) ;
blood concentration has to be maintained ;
more water (re)absorbed in kidneys / less water in urine / urine more concentrated ; [max 3]

7 **(a)** State the equation for anaerobic respiration in yeast.

For Examiner's Use

[2]

(b) **(i)** Describe and explain the changes that occur in breathing and heartbeat as a person climbs a mountain.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) Explain how these changes affect the working of the leg muscles during the climb.

For Examiner's Use

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]

Question 8 is in the form of an **Either**/**Or** question.

Answer only question 8 **Either** or question 8 **Or**.

For Examiner's Use

7 **(a)** $C_6H_{12}O_6$ / glucose / hexose / monosaccharide / simple sugar (**I** yeast) ;
$2C_2H_5OH + 2CO_2$ / alcohol or ethanol + carbon dioxide ; [2]
(**I** any refs. to energy)

(b) **(i)** (breathing) fast(er) ; | **A** 'breathe more' for *one* mark
deep(er) ; |
(heart beat) fast(er) ; | **A** 'more' for *one* mark
more powerfully / larger stroke volume AW ; |
(**A** ref. higher blood pressure)
faster circulation of blood ;
supplying more AW oxygen* / compensation for lower O_2 concentration ;
removing more AW carbon dioxide* ; [max 4]
[* or in **(ii)**]

(ii) (muscles) increased + supplies of glucose (to muscles) ;
increased + work-rate (person) / contraction (muscle) ;
faster + respiration (in muscle cells) ;
more + energy ;
increased supply of O_2 [* or in **(i)**] ;
increased removal of CO_2 [* or in **(i)**] ;
delays lactic acid production / removes lactic acid ; [max 4]

[Total: 10]

8 **Either** **(a)** State the effect on a plant of the lack of

(i) nitrates,

.....

.....

(ii) magnesium.

.....

.....

[3]

(b) Explain the significance of leaf structure in the process of photosynthesis.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[7]

[Total: 10]

8 Or (a) Explain how root hair cells are suited to the functions they perform.

For Examiner's Use

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Explain the significance of leaf structure in the process of transpiration.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

[Total: 10]

8E (a) (nitrates) reduced (to zero) protein / amino acid manufacture ;
poor / stunted / restricted AW + growth (**A** no) ;
(magnesium) yellow leaves / chlorosis ;
less / no <u>chlorophyll</u> ; [max 3]

(b) thin + short distance for gases to move ;
thin + ref. light penetration ;
flat / broad / large surface area / rt. angles to sun + <u>more</u> (AW) light absorption ;
(with large surface area, 'more' not required after +)
<u>chloroplasts</u> ;
in mesophyll (or named) ;
epidermis / cuticle + transparent for light entry ;
stomata / pores + gas movement (**I** water vapour) ;
air spaces + gaseous movement (**I** water vapour) ;
by <u>diffusion</u> ;
cell surfaces + large surface area for CO_2 entry ;
presence of vein / v.b. / xylem + to bring water /phloem to remove products ; [max 7]

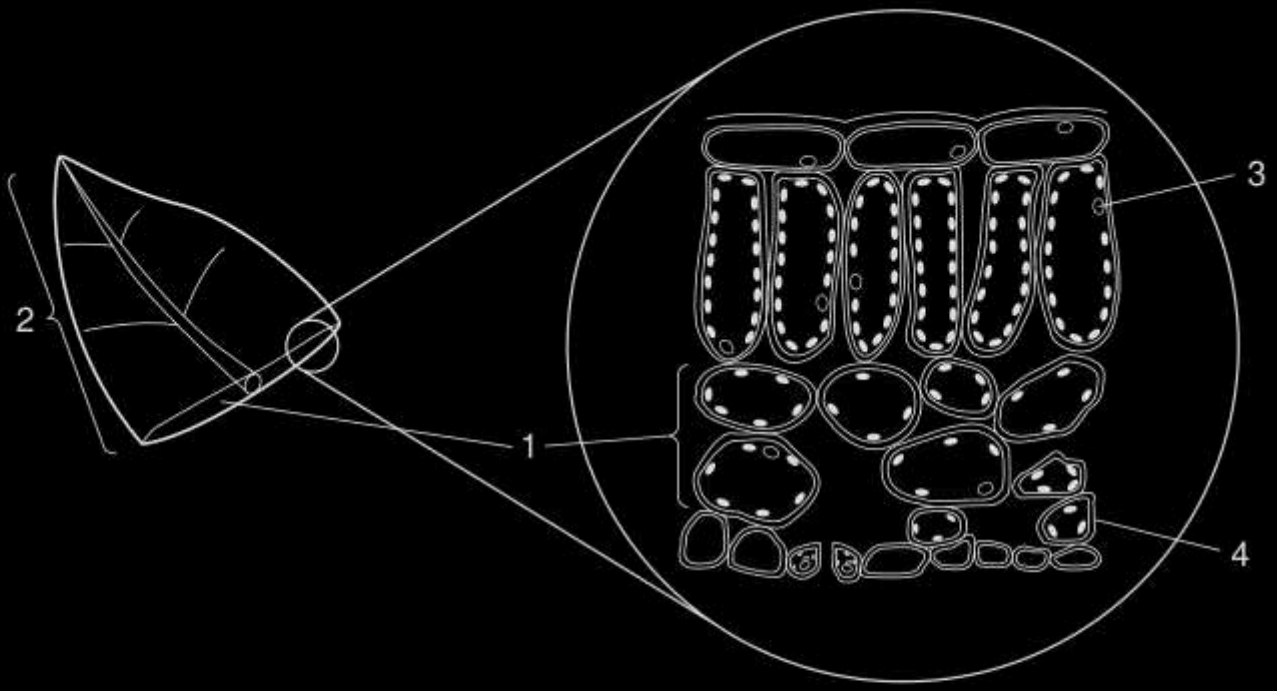
[Total: 10]

8O(a) long / microscopic or very small ;
large surface area ;
<u>increases / maximum</u> + uptake ;
water / ions / oxygen absorbed ;
in contact with soil water / between soil particles ; [max 4]

(b) xylem ;
brings water ;
leaf cell contents more concentrated ;
water leaves xylem by osmosis ;
water (film) on (surfaces of mesophyll) cells ;
evaporates + air spaces ;
increased / high humidity inside leaf ;
concentration gradient (or described) ;
<u>diffusion</u> ;
through stomata / pores ; [max 6]

[Total: 10]

Correct answer: C



Which letter identifies a cell, a tissue and an organ?

	cell	tissue	organ
A	3	2	4
B	1	4	3
C	4	1	2
D	2	3	1

October/November 2008 · Paper 1 · Variant 1 · Question 2

Correct answer: D

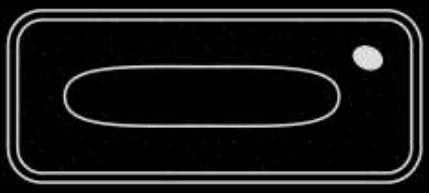


diagram 1

The cell is now placed in a concentrated sugar solution.

It changes to appear as in diagram 2.

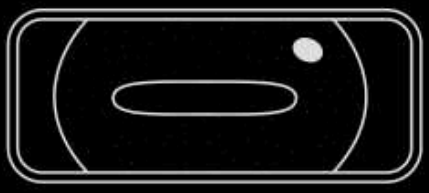


diagram 2

Which statement explains the change?

A Sugar has moved into the cell.

B Sugar has moved out of the cell.

C Water has moved into the cell.

D Water has moved out of the cell.

October/November 2008 · Paper 1 · Variant 1 · Question 3

Correct answer: B

3 Four tubes containing 10 cm^3 of 1 % starch solution were treated in different ways and then mixed with saliva. After 30 minutes, 1 cm^3 of iodine in potassium iodide solution was added to each tube.

In which tubes were the contents a yellow-brown colour?

	tube incubated at 35 °C	tube incubated at 75 °C	tube incubated at pH 2.5	tube incubated at pH 6.9
A	✓		✓	
B	✓			✓
C		✓		✓
D		✓	✓	

key

✓ = yellow-brown colour

October/November 2008 · Paper 1 · Variant 1 · Question 4

Correct answer: D

4 According to the lock and key hypothesis, which is the lock and which is the key for the enzyme lipase?

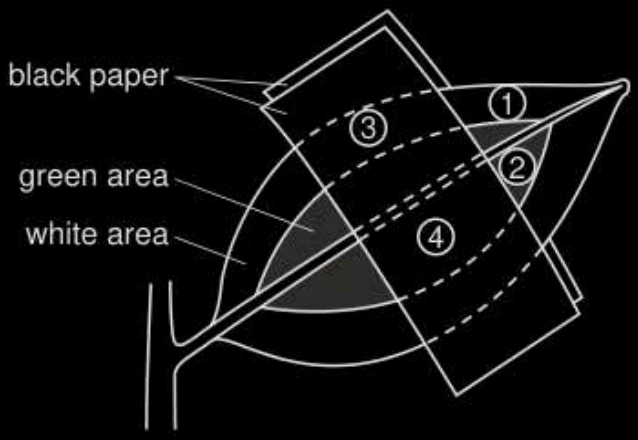
	key	lock
A	fatty acids	lipids
B	lipase	lipids
C	lipase	fatty acids
D	lipids	lipase

October/November 2008 · Paper 1 · Variant 1 · Question 7

Correct answer: C

7 A plant has leaves that are partly green and partly white. The plant is destarched and a leaf is partly covered by black paper.

The plant is placed in bright light for several hours. Four discs are then cut from the leaf in the positions shown and are tested for starch.



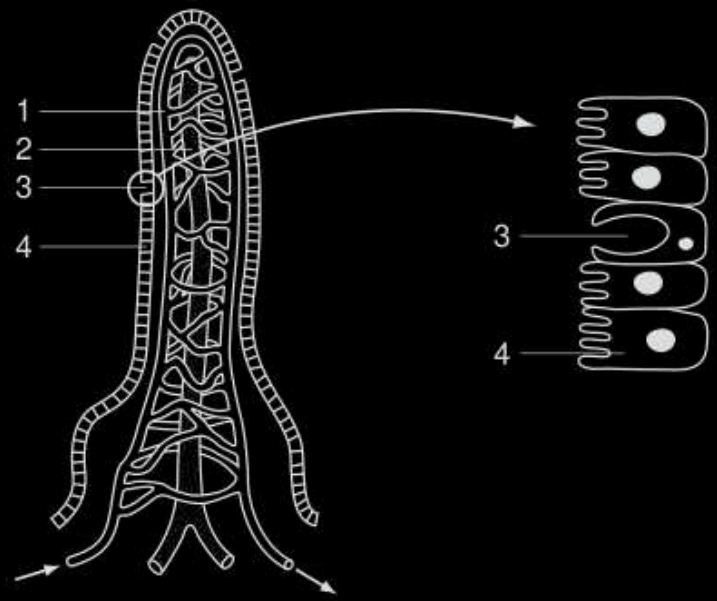
Which discs contain starch?

A 1 only **B** 1 and 2 **C** 2 only **D** 3 and 4

October/November 2008 · Paper 1 · Variant 1 · Question 9

Correct answer: D

9 The diagram represents a villus.



Which sequence correctly describes the functions of the numbered parts?

	1	2	3	4
A	absorbs digested fats	absorbs glucose	produces enzymes	produces mucus
B	absorbs digested fats	absorbs glucose	produces mucus	produces enzymes
C	absorbs glucose	absorbs digested fats	produces enzymes	produces mucus
D	absorbs glucose	absorbs digested fats	produces mucus	produces enzymes

October/November 2008 · Paper 1 · Variant 1 · Question 10

Correct answer: A

10 It has been shown that animals restricted to a diet of milk only, eventually suffer from blood disorders involving a lack of pigment in their red blood cells.

Which is the most likely explanation of this?

A Milk is deficient in iron.

B Milk contains more calcium than is required by most animals.

C Milk is deficient in vitamin A.

D Milk contains no roughage.

October/November 2008 · Paper 1 · Variant 1 · Question 11

Correct answer: D

11 Crop plants are grown in well-watered soils.

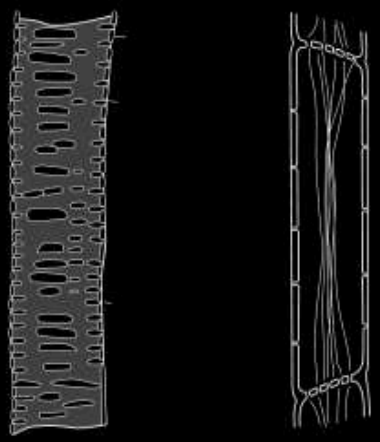
Their root hair cells have higher concentrations of mineral ions than in the soil surrounding them.

What processes are used by these plants to absorb water and mineral ions?

	water	mineral ions
A	active uptake	osmosis
B	diffusion	osmosis
C	osmosis	diffusion
D	osmosis	active uptake

October/November 2008 · Paper 1 · Variant 1 · Question 12

Correct answer: D



X Y

Samples of the contents of X and Y were tested.

What results are expected?

	X		Y	
	Benedict's reagent	iodine	Benedict's reagent	iodine
A	+	+	+	+
B	+	–	+	–
C	–	+	–	+
D	–	–	–	–

key

– = negative result

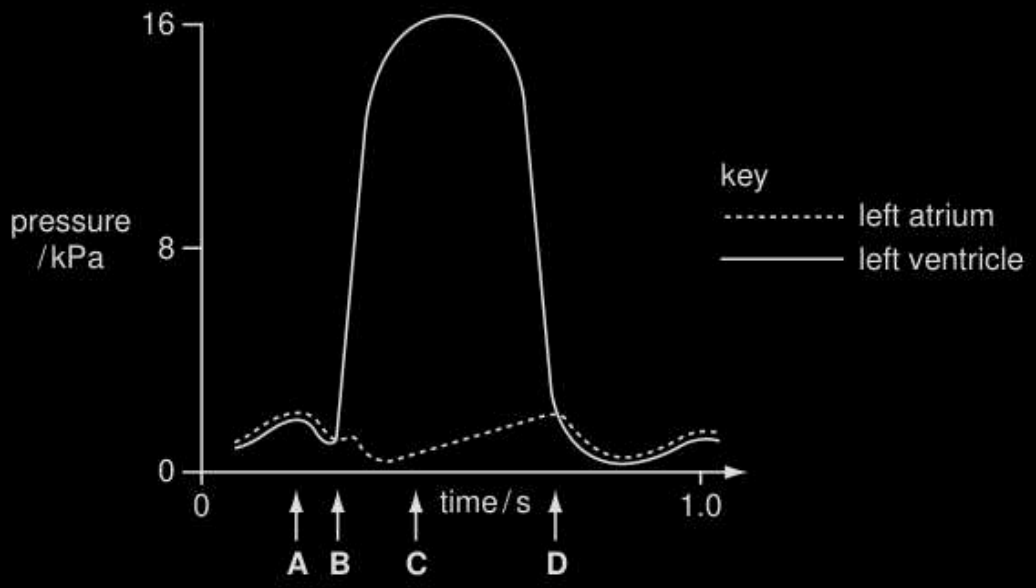
+ = positive result

October/November 2008 · Paper 1 · Variant 1 · Question 13

Correct answer: B

13 The graph shows the pressure changes in the left atrium and the left ventricle while the heart is beating.

At which point do the atrio-ventricular (bicuspid and tricuspid) valves close?



October/November 2008 · Paper 1 · Variant 1 · Question 14

Correct answer: C

14 The table shows substances that pass between capillaries and tissues in a part of the body.

substance	into the capillaries from the tissues	out of the capillaries into the tissues
oxygen		✓
carbon dioxide	✓	
amino acids		✓
urea	✓	

key

✓ = does pass

In which part of the body are these capillaries?

A between the alveoli

B in the kidney

C in the liver

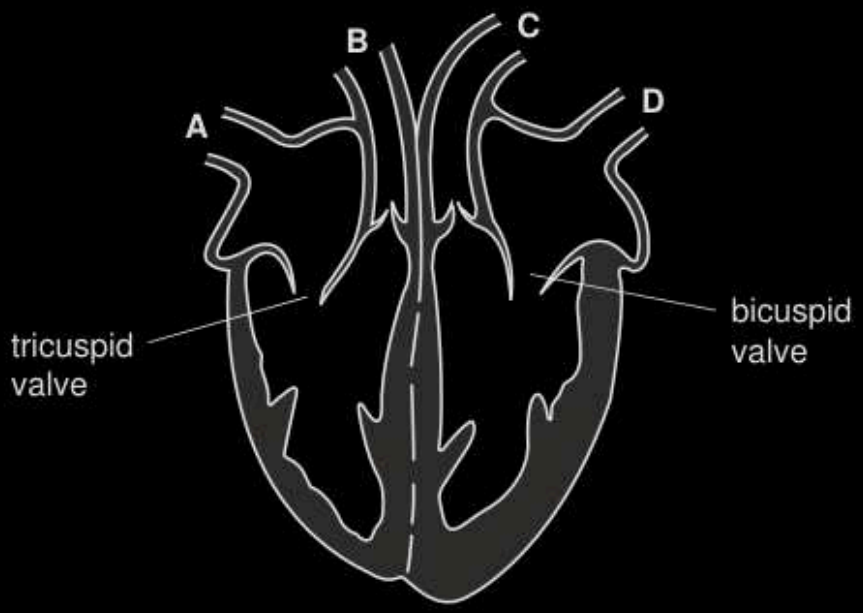
D in the villi

October/November 2008 · Paper 1 · Variant 1 · Question 15

Correct answer: D

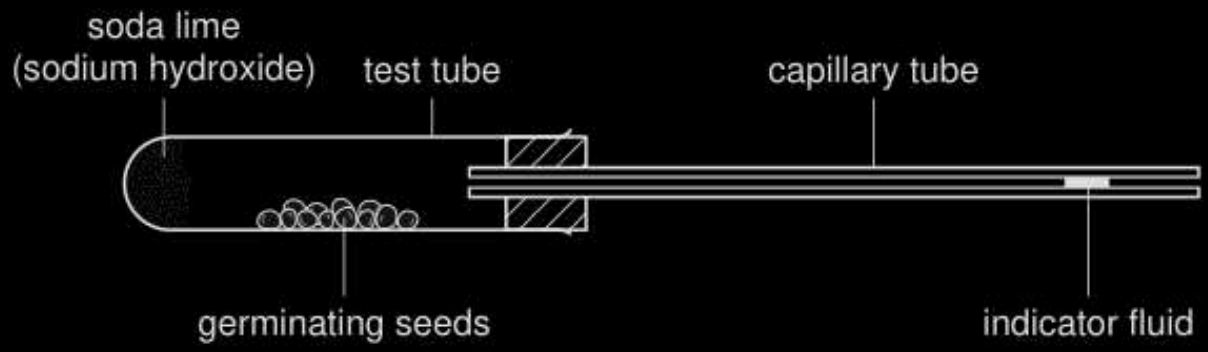
15 Look at the diagram of a heart.

Which vessel is a vein carrying oxygenated blood?



October/November 2008 · Paper 1 · Variant 1 · Question 18

Correct answer: C



The indicator fluid in the capillary tube shown in the diagram above will

A move away from the test-tube because of oxygen output by the seeds.

B move towards the test-tube because of carbon dioxide intake by the seeds.

C move towards the test-tube because of oxygen intake by the seeds.

D remain stationary, because carbon dioxide output and oxygen intake are equal.

October/November 2008 · Paper 1 · Variant 1 · Question 20

Correct answer: D

20 Urea is produced in one organ, filtered from the blood by a second organ and stored inside a third organ before being expelled from the body.

Which organs carry out these functions?

	production	filtration	storage
A	kidney	bladder	liver
B	kidney	liver	bladder
C	liver	bladder	kidney
D	liver	kidney	bladder

October/November 2008 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 What is meant by negative feedback?

A A change away from a set point causes a change back towards the set point.

B A change away from a set point causes further change away from the set point.

C A change towards a set point causes a change away from a set point.

D Changes away from a set point are prevented.

October/November 2008 · Paper 1 · Variant 1 · Question 22

Correct answer: B

22 What is a sign of diabetes mellitus?

A glucose in the blood

B glucose in the urine

C insulin in the blood

D insulin in the urine

October/November 2008 · Paper 1 · Variant 1 · Question 23

Correct answer: A

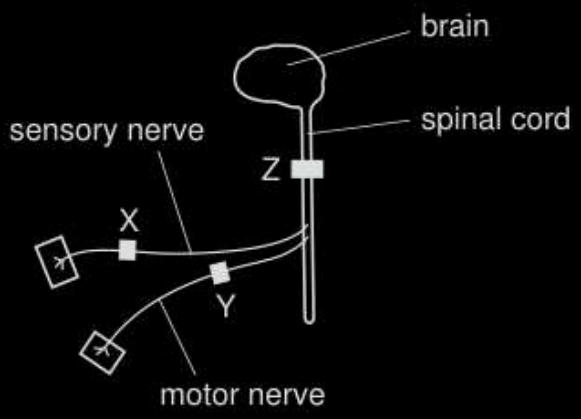
23 Which changes occur in the body when a person is shocked?

	increase in	decrease in
A	the diameter of the pupils in the eye	the speed of peristalsis
B	the rate of conversion of glycogen to glucose	the diameter of the pupils in the eye
C	the rate of urine formation	the rate of conversion of glycogen to glucose
D	the speed of peristalsis	the rate of urine formation

October/November 2008 · Paper 1 · Variant 1 · Question 24

Correct answer: B

24 The diagram represents a central nervous system. X,Y, and Z show possible sites where the system can be blocked by a local anaesthetic.



Of four men, one had no anaesthetic block and the other three had only one anaesthetic block at X, Y or Z.

One of the men can feel a pinprick on his leg but cannot move it.

Where is the anaesthetic block?

A block is at X

B block is at Y

C block is at Z

D no block

October/November 2008 · Paper 1 · Variant 1 · Question 26

Correct answer: A

26 The table shows the characteristics of four microbes.

Which one could be a virus?

	contains DNA	contains one or more cells	contains one or more cell nuclei	produces spores
A	✓	✗	✗	✗
B	✓	✓	✗	✗
C	✓	✓	✓	✗
D	✓	✓	✓	✓

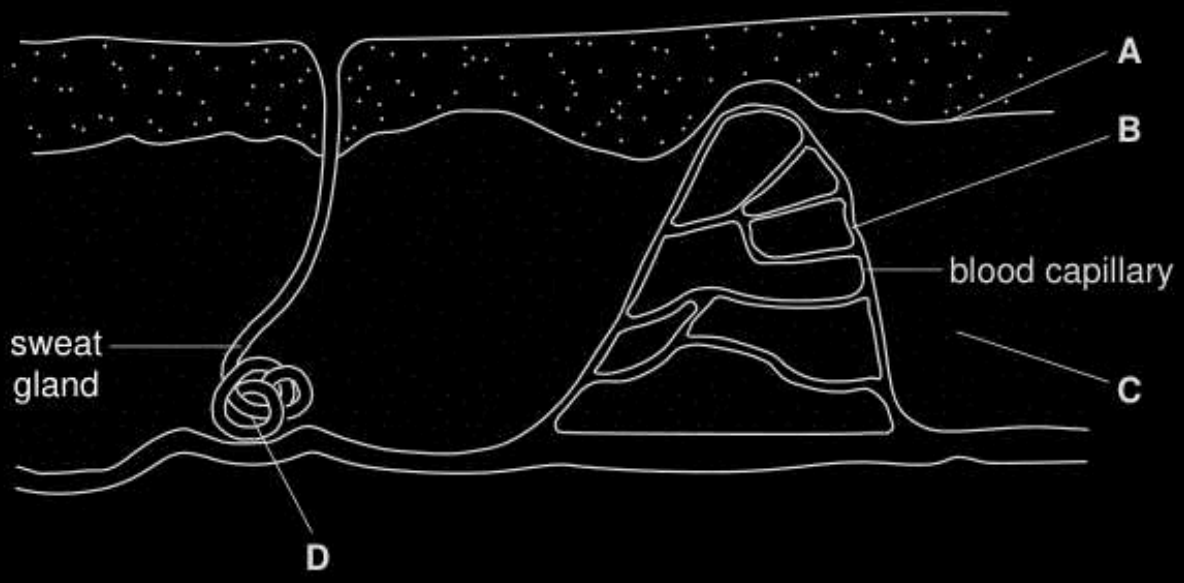
key
✓ = true
✗ = false

October/November 2008 · Paper 1 · Variant 1 · Question 34

Correct answer: A

34 The diagram represents a section through the human skin.

In which part is mitosis occurring most rapidly?



October/November 2008 · Paper 1 · Variant 1 · Question 35

Correct answer: C

35 What is an advantage of feeding a baby with bottled milk rather than breast milk?

A Breast milk contains antibodies.

B It is easier to keep the milk free of pathogens.

C The baby's intake of food can be measured.

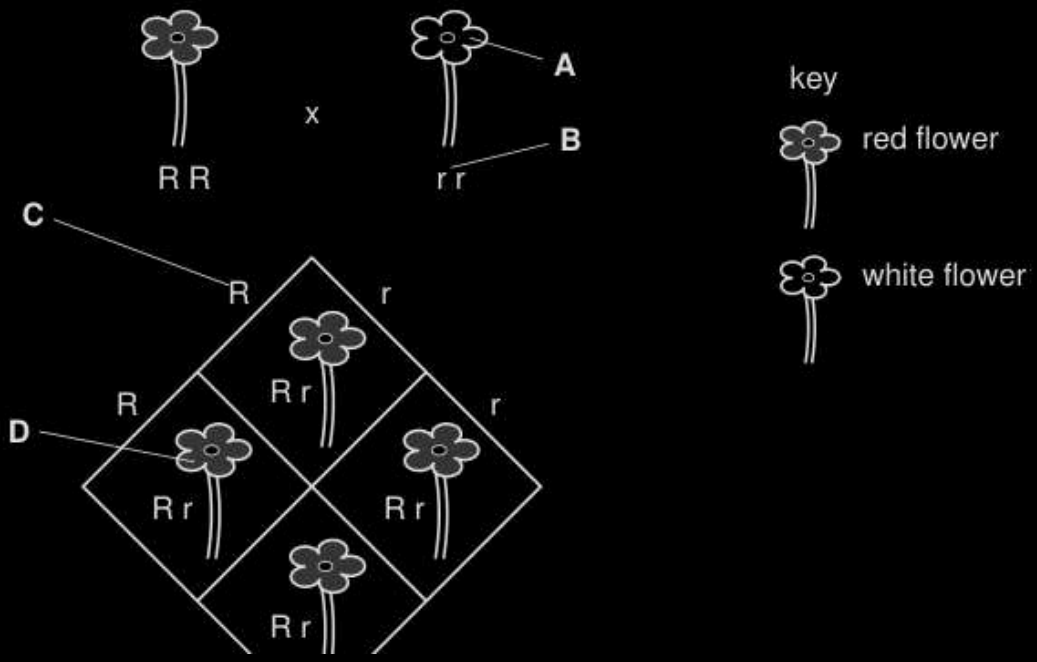
D The milk is always at the right temperature.

October/November 2008 · Paper 1 · Variant 1 · Question 38

Correct answer: D

38 The diagram shows a simple genetic cross between a red flower and a white flower.

Which represents the dominant phenotype?



October/November 2008 · Paper 1 · Variant 1 · Question 40

Correct answer: C

40 Which blood group genotype is homozygous dominant?

A I^A I^O

B I^A I^B

C I^B I^B

D I^O I^O

2 Fig. 2.1 shows the liver receiving chemicals from and sending chemicals to some other organs.

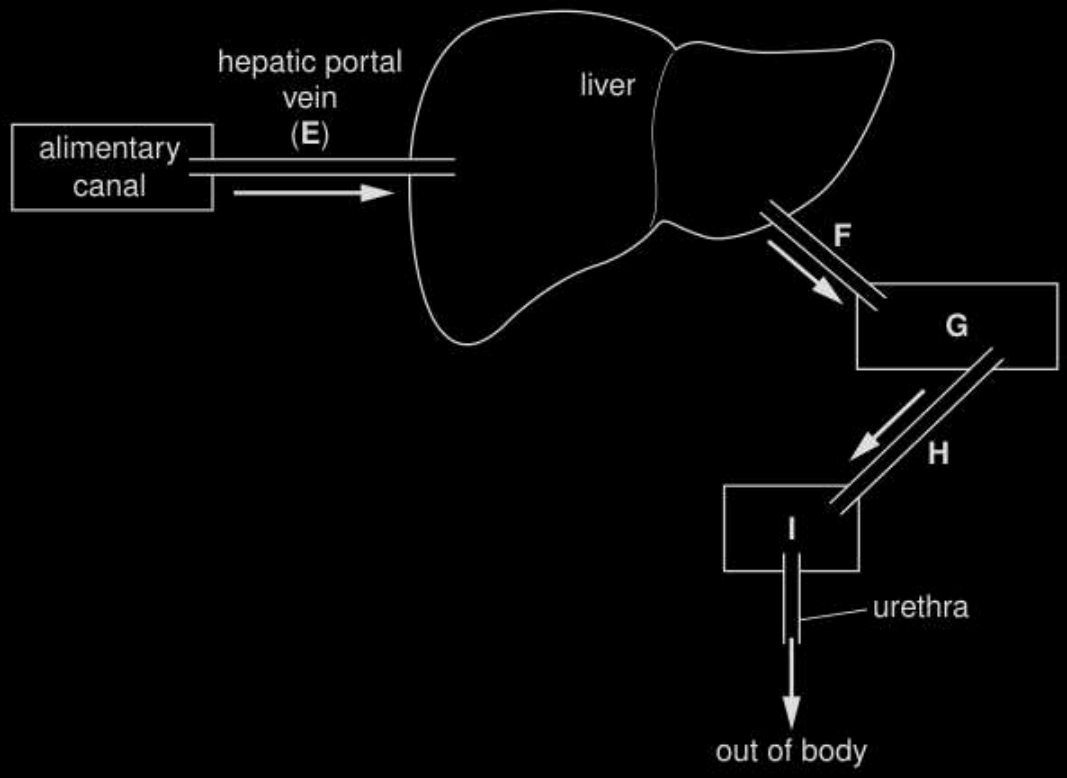


Fig. 2.1

(a) Identify organs **G** and **I**.

G .....

I ..... [2]

(b) Name the carbohydrate travelling in the hepatic portal vein (**E**), and explain how, on arrival in the liver, it is converted into a storage compound.

named carbohydrate .....

explanation .....

..... [4]

(c) Describe how the composition of the contents of **F** and **H** differ in a healthy person.

.....

.....

.....

..... [4]

[Total: 10]

2 **(a)** **(G)** kidney ;
(I) bladder (R gall bladder) ; [2]

(b) glucose ; [1]

insulin ;
from pancreas ;
in blood ;
glycogen ; [max 3]

(c) #**F** + blood/cells ;
Any **two** from: more urea in **H**, more toxins in **H**, glucose only in **F**, amino acids only in **F**, qualified salt concentration in either ;;
H + urine ;
Ref. O_2/CO_2 differences ; [max 4]
(# A reverse argument for alternative structure)

[Total: 10]

3 Fig. 3.1 shows a section through a leaf.

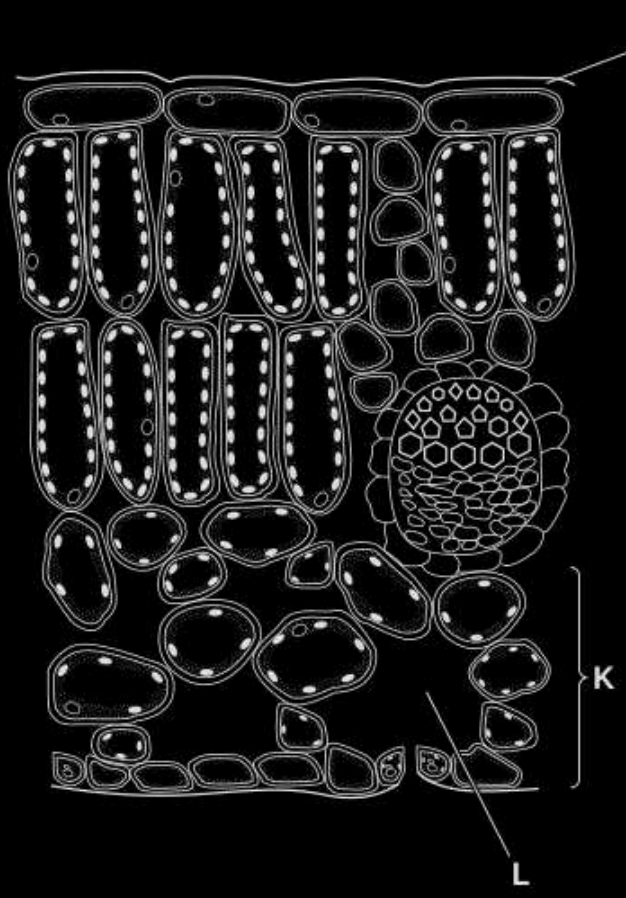


Fig. 3.1

(a) Identify structures **J** and **K**.

J .....

K ..... [2]

(b) **(i)** On Fig. 3.1, use arrows to show the pathway taken by water from its arrival in this part of the leaf until it is lost into the atmosphere. [3]

(ii) Place a large **X** so that its arms cross as closely as possible to the point at which evaporation of water is occurring. [1]

Fig. 3.2a shows the rates of transpiration for a particular species of plant at different air temperatures.

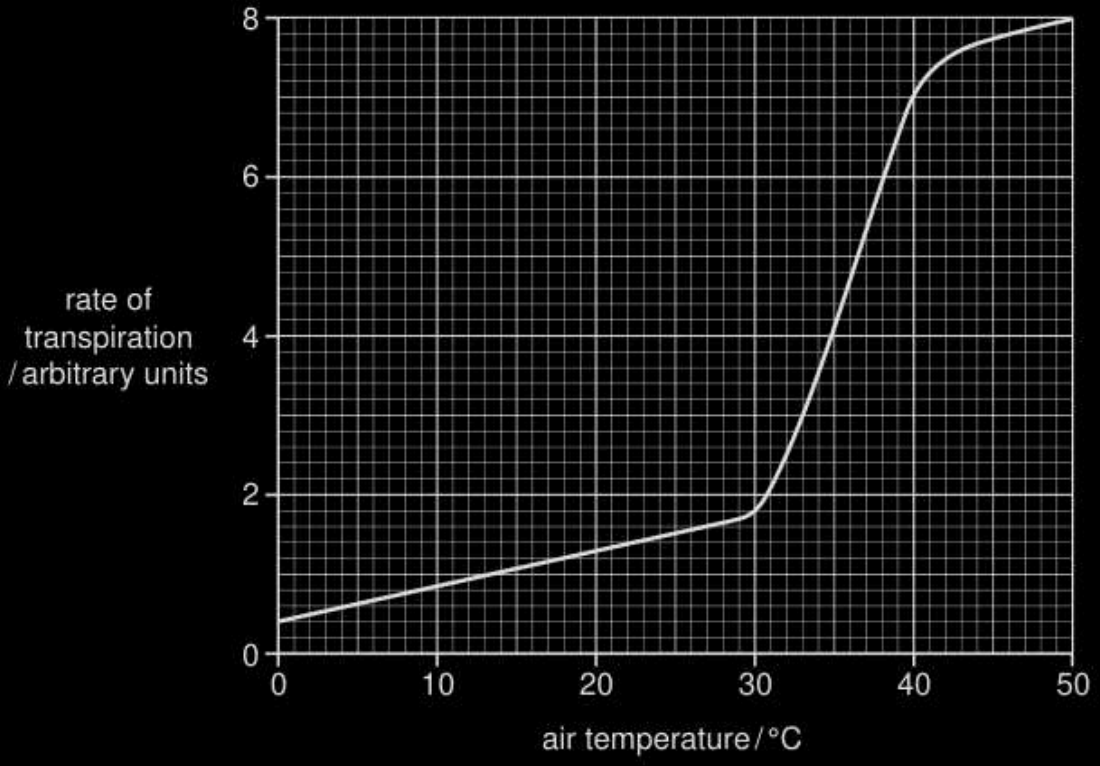


Fig. 3.2a

Fig. 3.2b shows the air temperatures and the temperatures inside the leaf at point **L** in Fig. 3.1 for the same plant during 24 hours.

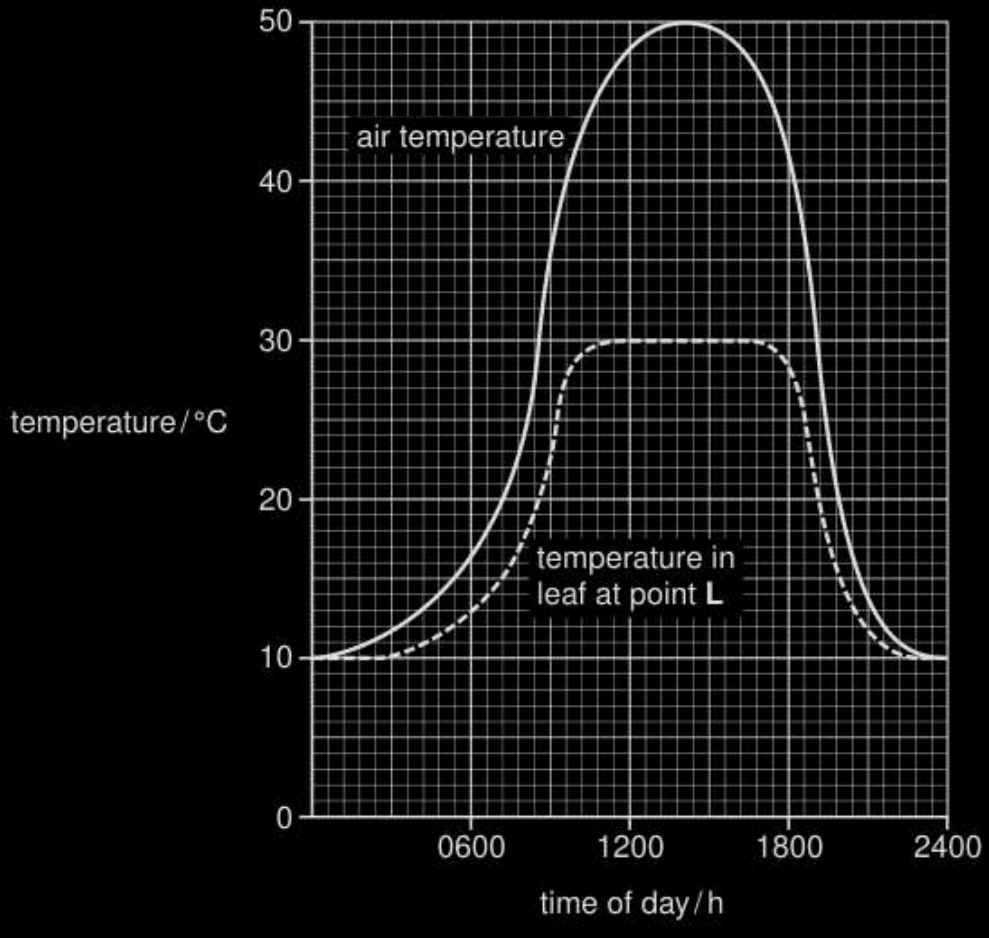


Fig. 3.2b

Use

Fig. 3.2c shows stomata as they appear in this leaf at three different times during the day.

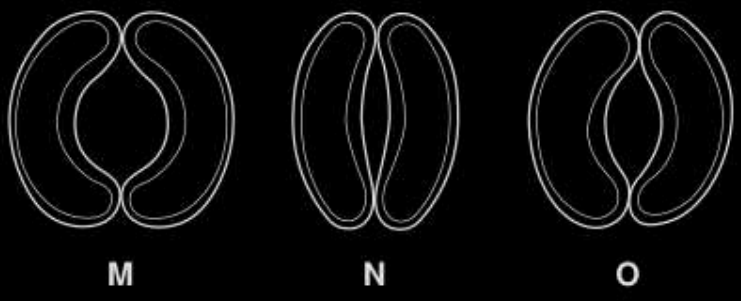


Fig. 3.2c

(c) Using information provided in Fig. 3.2a and Fig. 3.2b, state which of the stomatal pores, **M**, **N** and **O**, in Fig. 3.2c, shows their appearance at the following times of day.

(i) 03:00 .....

(ii) 19:30 .....

[2]

(d) Suggest why the temperature inside the leaf never rises above 30 °C, even though the air temperature rises much higher than this.

.....

.....

..... [2]

[Total: 10]

3 **(a)** **(J)** cuticle or described ;
(K) spongy (+ mesophyll) (ignore refs to lower epidermis) ; [2]

(b) **(i)** one arrow (somewhere) leaving xylem (R any that pass through phloem) ;
passing into any mesophyll cell ;
entering air space in spongy mesophyll ;
passing out of stoma ;
(sequence must be plausible) [max 3]

(ii) **X** placed where water enters air space/wall of mesophyll cell ; [1]
(R **X** on guard cell)

(c) **(i)** **N** ;

(ii) **O** ; [2]

(d) evaporation (R transpiration) ;
fast(er) in high temperatures ;
cools/removes heat ; [max 2]

[Total: 10]

4 Fig. 4.1 shows human blood containing pathogenic (disease-causing) organisms.

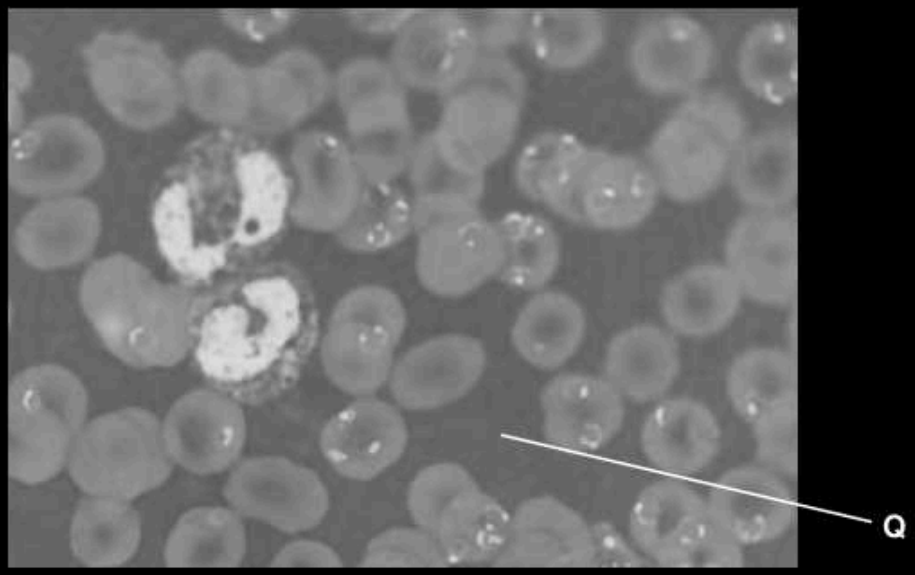


Fig. 4.1

(a) **(i)** Name the liquid labelled **Q** in Fig. 4.1. ..... [1]

(ii) Name two mineral ions which may be found in liquid **Q** and, for each ion, state its function in the body.

ion 1 ..... function .....

ion 2 ..... function .....

[3]

(b) On Fig. 4.1, label

(i) a white blood cell,

(ii) a red blood cell infected with the pathogenic organism.

[2]

(c) The pathogenic organisms were introduced into the blood by a mosquito while feeding. Suggest why the mosquito feeds from a capillary and not from an artery.

.....

.....

.....

..... [3]

[Total: 9]

4 **(a)** **(i)** **(Q)** plasma ; [1]

(ii) 2 named ions (iron and calcium on syllabus) ;
iron + red blood cells/haemoglobin ;
calcium + ref. bones or teeth/blood clotting ; [3]
(A any others correct with function e.g. Mg activates enzymes/for RBCs, Na/K for impulse transmission/ref. effect on cell membrane) (R N / I_2 / S & any other elements)

(b) WBC correctly labelled ;
infected RBC correctly labelled (If several labelled, all must be correct) ; [2]

(c) (capillary) close to surface ;
thin/one cell thick ;
low blood pressure ; [3]
(A reverse arguments for artery)

[Total: 9]

October/November 2008 · Paper 2 · Variant 1 · Question 7

7 **(a)** What is meant by the terms

(i) *discontinuous variation*,

(ii) *continuous variation*?

Describe **one** example of each type. [7]

(b) State the causes of

(i) sickle cell anaemia,

(ii) Down's syndrome. [3]

[Total: 10]

Question 8 is in the form of an **Either**/**Or** question. Answer only question 8 **Either** or question 8 **Or**.

7 **(a)** discontinuous – valid example (such as eye colour, tall + dwarf peas, red hair, albinism, sex)
(A labelled bar charts) ; [1]
continuous – valid example (A skin colour and labelled graph) ; [1]
(R eye colour)

(i) (discontinuous) few forms ;
distinct from one another/no intermediates AW ;
the result of inheritance ;
of genes ;

(ii) (continuous) many forms ;
small differences from one to the next/range ;
extremes at either end may show considerable difference ;
caused by genes + the environment ;
e.g. of environmental factor ;
[max 5]

(b) mutation (in either **(i)** or **(ii)**) ; [1]

(i) (sickle cell) of gene ;
affecting haemoglobin (formation) ; [max 1]

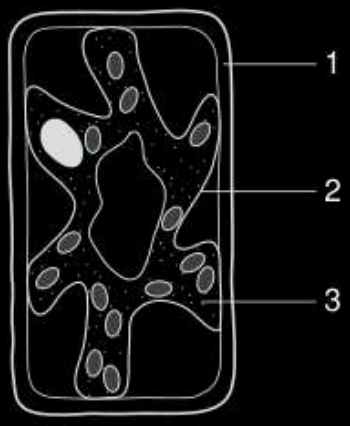
(ii) (Down's) of chromosome/one extra chromosome ; [max 1]

[Total: 10]

Either

May/June 2008 · Paper 1 · Variant 1 · Question 1

Correct answer: D



Which numbered structures are partially permeable?

A 1 and 2 only **B** 1 and 3 only **C** 1 only **D** 2 only

May/June 2008 · Paper 1 · Variant 1 · Question 2

Correct answer: A

2 The small intestine of a person contains a low concentration of glucose produced from the digestion of starch.

Glucose is taken up by the cells of the villi.

How does the glucose move?

A by active transport against the concentration gradient

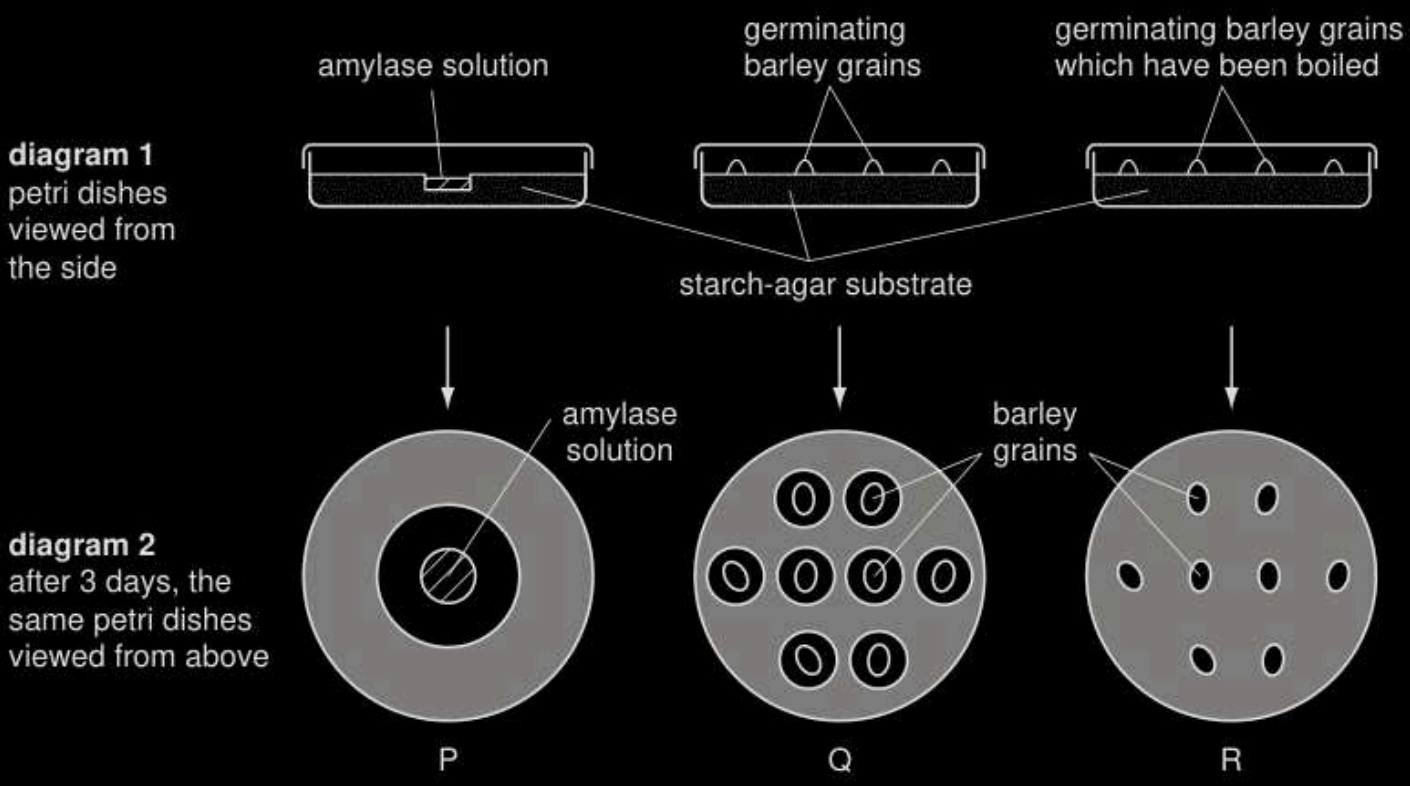
B by active transport with the concentration gradient

C by diffusion against the concentration gradient

D by diffusion with the concentration gradient

May/June 2008 · Paper 1 · Variant 1 · Question 3

Correct answer: B



Which is the **best** explanation of the results?

A Amylase is produced by barley grains that have been boiled.

B Amylase from barley grains is denatured when they are boiled.

C Germinating grains prevent iodine from staining starch blue/black.

D Starch from the substrate is used by the grains as an energy source.

May/June 2008 · Paper 1 · Variant 1 · Question 4

Correct answer: A

4 Enzyme action can be explained by the lock and key hypothesis.

Where is the active site and which acts as the lock or key?

	active site	lock / key
A	on the enzyme	substrate acts as a key
B	on the enzyme	substrate acts as a lock
C	on the substrate	enzyme acts as a key
D	on the substrate	enzyme acts as a lock

May/June 2008 · Paper 1 · Variant 1 · Question 5

Correct answer: D

5 Where and how does carbon dioxide enter a plant?

	where	how
A	root hair cells	active uptake
B	root hair cells	diffusion
C	stomata	active uptake
D	stomata	diffusion

May/June 2008 · Paper 1 · Variant 1 · Question 8

Correct answer: A

8 What is produced as a result of digestion by a protease?

A amino acids

B fatty acids

C glucose

D glycerol

May/June 2008 · Paper 1 · Variant 1 · Question 9

Correct answer: A

9 For which food test must the sample be dissolved in ethanol?

A fat

B protein

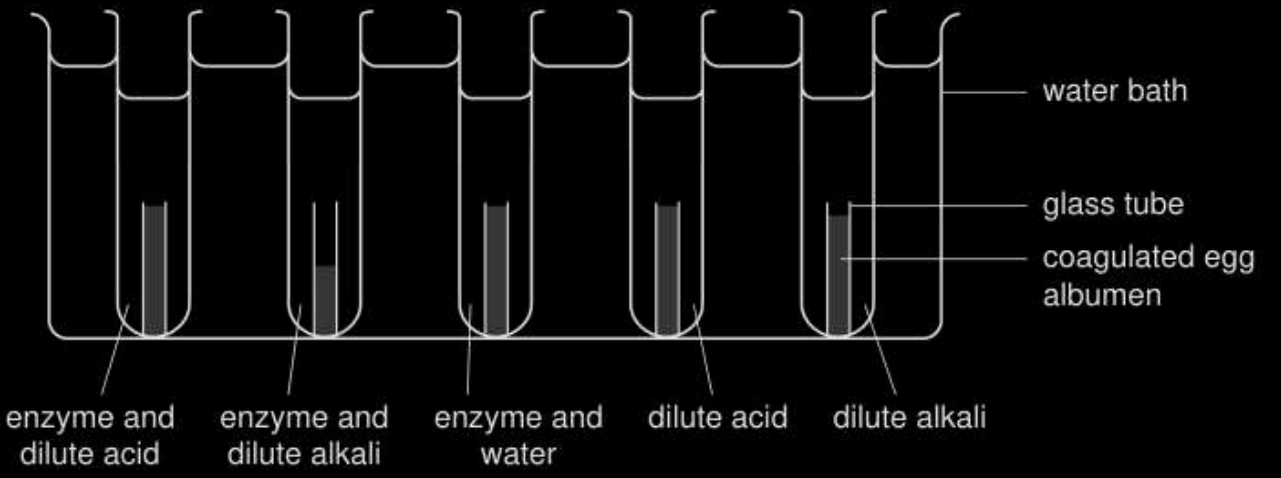
C reducing sugar

D starch

May/June 2008 · Paper 1 · Variant 1 · Question 10

Correct answer: B

10 An experiment was set up to investigate the effect of an enzyme on coagulated egg albumen. The results, after three hours, are shown in the diagram.



Where in the digestive system is the enzyme produced?

A liver

B pancreas

C salivary glands

D stomach wall

May/June 2008 · Paper 1 · Variant 1 · Question 11

Correct answer: B

11 When a complete ring of bark is removed from the trunk of a tree, it will eventually die because this action cuts off the supply of

A mineral salts to the leaves.

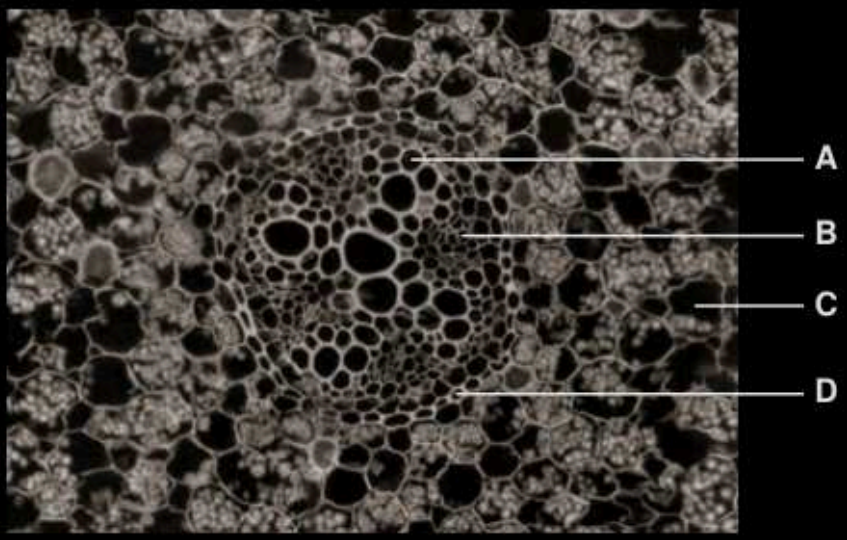
B nutrients to the roots.

C oxygen to the roots.

D water to the leaves.

May/June 2008 · Paper 1 · Variant 1 · Question 12

Correct answer: B

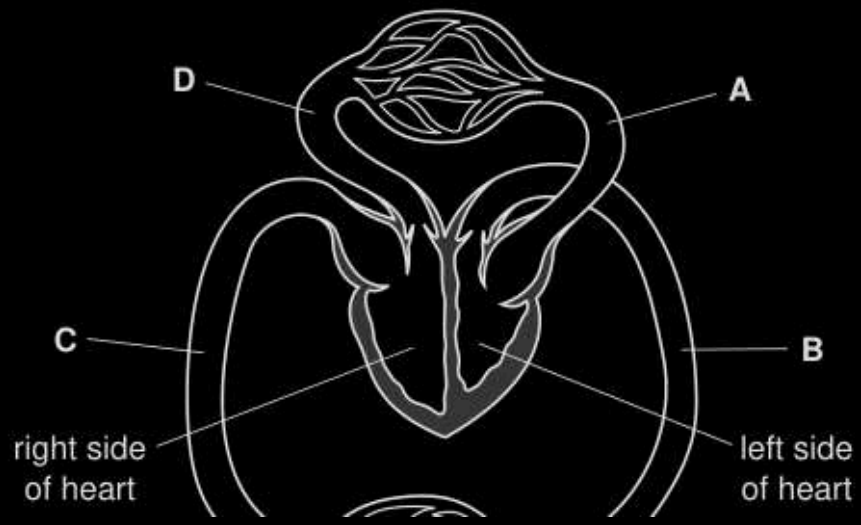


May/June 2008 · Paper 1 · Variant 1 · Question 13

Correct answer: B

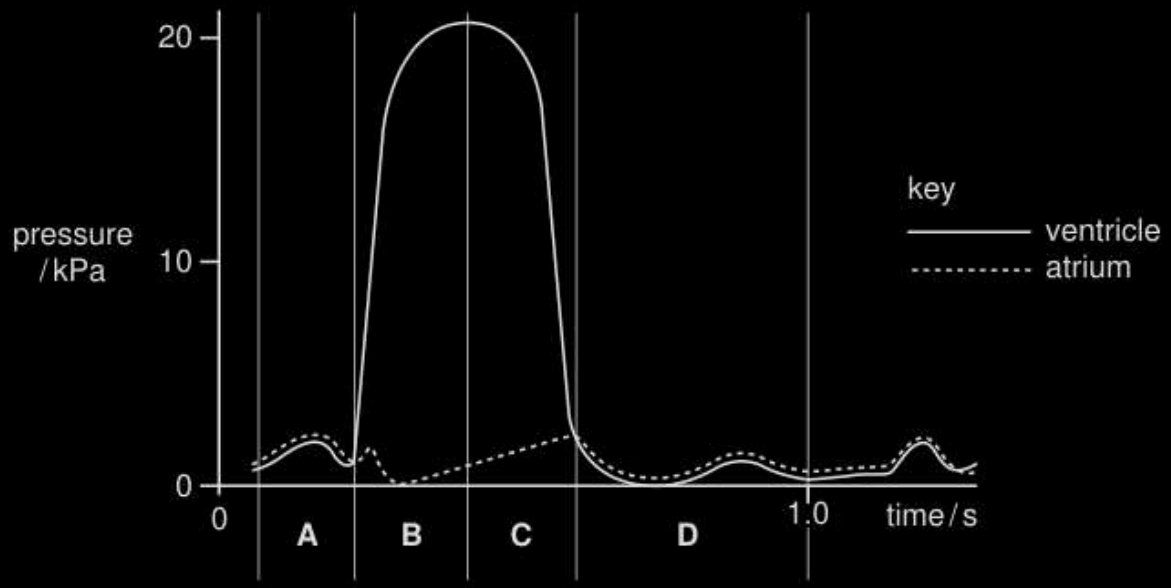
13 The diagram represents part of the human circulatory system.

Where is the blood pressure highest?



May/June 2008 · Paper 1 · Variant 1 · Question 14

Correct answer: B



May/June 2008 · Paper 1 · Variant 1 · Question 15

Correct answer: C

15 In the human body, blood circulating from the gut to the heart passes through the

A aorta.

B kidneys.

C liver.

D lungs.

May/June 2008 · Paper 1 · Variant 1 · Question 18

Correct answer: D

18 Which does **not** produce carbon dioxide?

A a muscle fibre

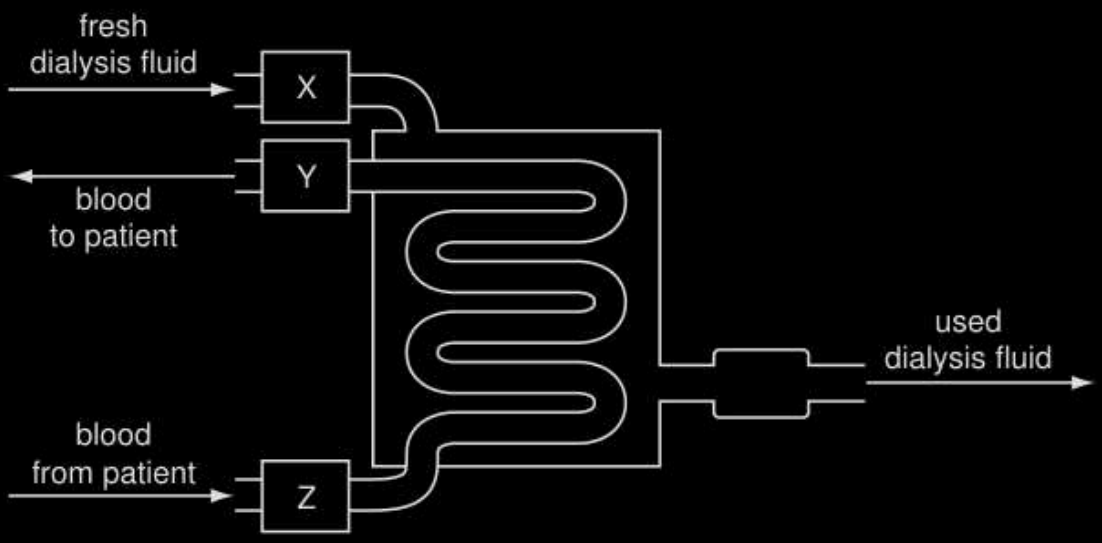
B a sensory neuron

C blood

D saliva

May/June 2008 · Paper 1 · Variant 1 · Question 20

Correct answer: B



What are the parts labelled X, Y and Z?

	bubble trap	roller pump	water bath for temperature control
A	X	Y	Z
B	Y	Z	X
C	Y	X	Z
D	Z	Y	X

May/June 2008 · Paper 1 · Variant 1 · Question 22

Correct answer: C

22 Adrenaline is secreted quickly when a person is frightened.

How does this affect the heart beat and the liver?

	heart beat	liver converts
A	decreases	glycogen to glucose
B	decreases	glucose to glycogen
C	increases	glycogen to glucose
D	increases	glucose to glycogen

May/June 2008 · Paper 1 · Variant 1 · Question 23

Correct answer: B

23 Which part of the nervous system is the centre for processes such as thought, memory and judgement?

A cerebellum

B cerebrum

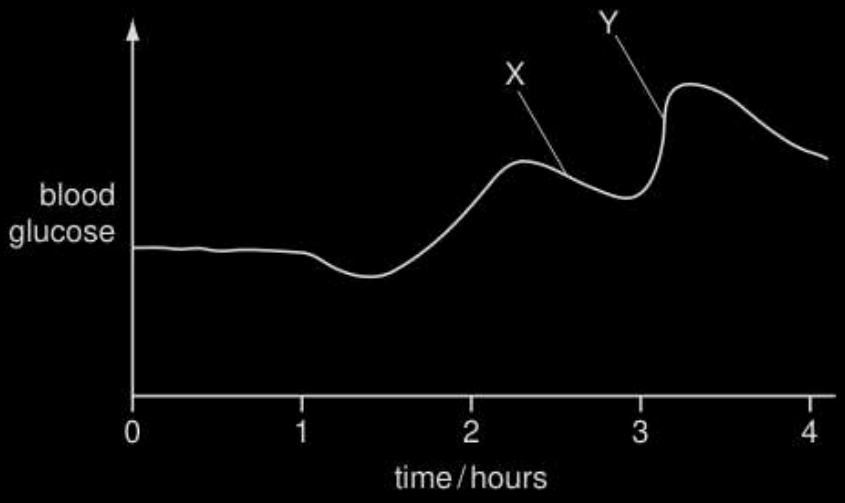
C medulla

D spinal cord

May/June 2008 · Paper 1 · Variant 1 · Question 24

Correct answer: D

24 The graph shows changes in a person's blood glucose concentration over four hours.



What might cause the changes at X and Y?

	X	Y
A	decreased insulin	decreased adrenaline
B	decreased insulin	increased adrenaline
C	increased adrenaline	increased insulin
D	increased insulin	increased adrenaline

May/June 2008 · Paper 1 · Variant 1 · Question 26

Correct answer: B

26 What is the role of the bacteria used in cheese and yoghurt production?

A to coagulate casein to form 'curds'

B to convert lactose to lactic acid

C to prevent the growth of pathogens

D to produce enzymes

May/June 2008 · Paper 1 · Variant 1 · Question 32

Correct answer: B

32 Cutting down large areas of tropical forest can lead to a reduction in rainfall.

What is the reason for the reduction in rainfall?

A a reduction in photosynthesis

B a reduction in transpiration

C an increase in flooding

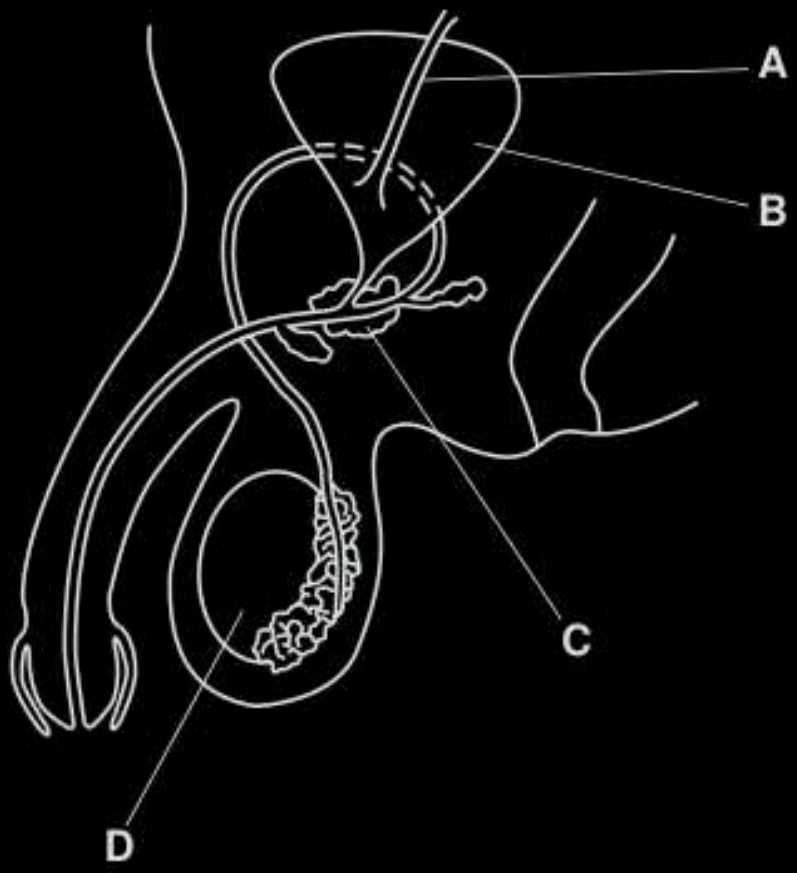
D an increase in respiration

May/June 2008 · Paper 1 · Variant 1 · Question 36

Correct answer: C

36 The diagram shows the male reproductive and urinary systems.

Which structure produces the fluid part of semen?



May/June 2008 · Paper 1 · Variant 1 · Question 37

Correct answer: A

37 What causes sickle cell anaemia?

A a change in the structure of a gene

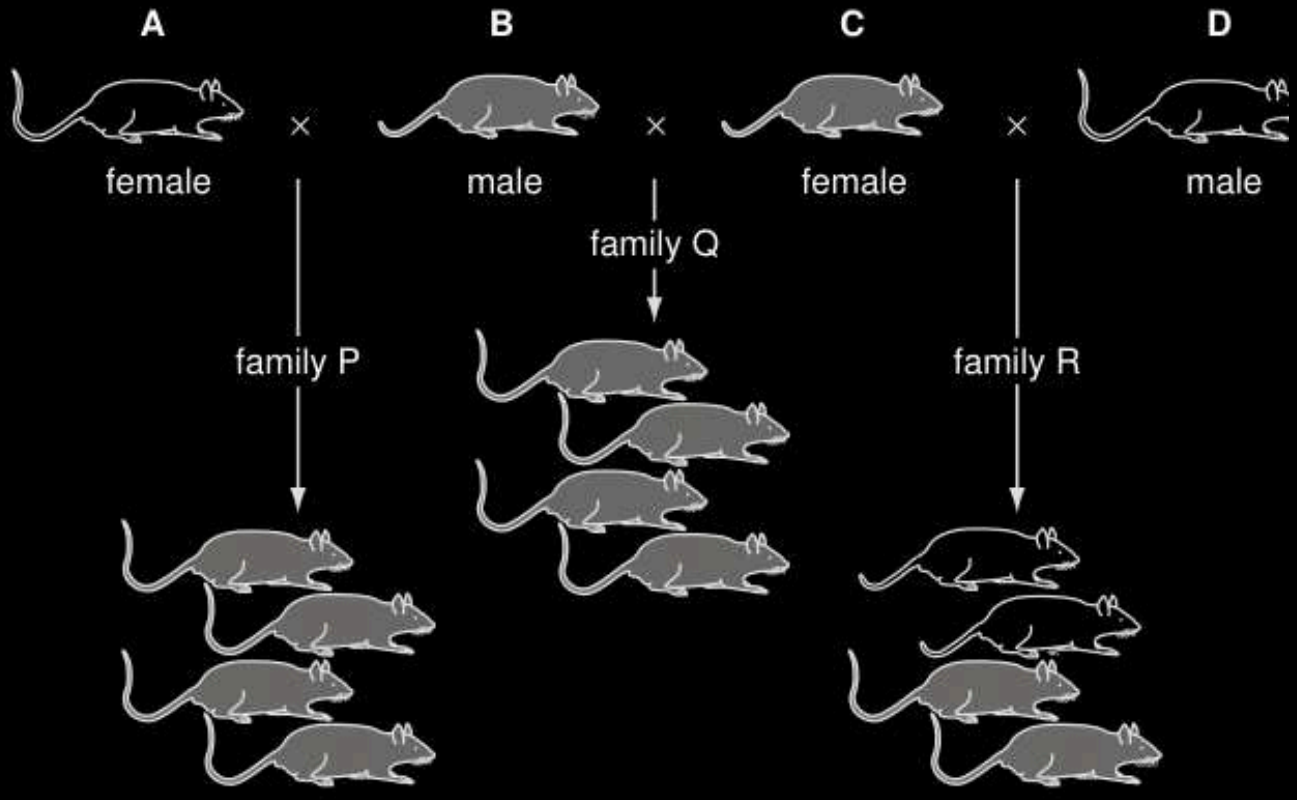
B an iron-deficient diet

C infection following a bite from a mosquito

D loss of blood following an accident

May/June 2008 · Paper 1 · Variant 1 · Question 38

Correct answer: C



May/June 2008 · Paper 1 · Variant 1 · Question 39

Correct answer: A

39 Two fruit flies with normal wings were mated with each other.

For every three offspring with normal wings, there is one with short wings.

What are the genotypes of the parents?

	female	male
A	Nn	Nn
B	Nn	NN
C	NN	Nn
D	Nn	nn

key

N = allele for normal wings

n = allele for short wings

May/June 2008 · Paper 1 · Variant 1 · Question 40

Correct answer: D

40 What is a result of natural selection?

A cattle that produce large quantities of milk

B crop plants that are resistant to disease

C grapes that develop no seeds

D mosquitoes that are resistant to insecticides

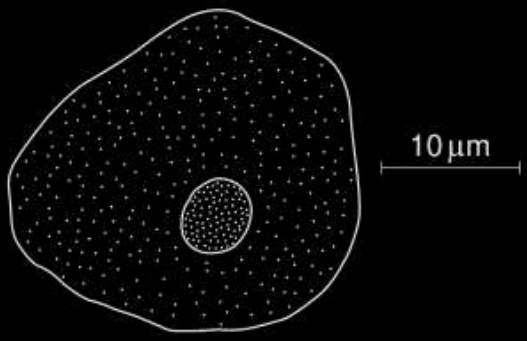


Fig. 2.1

(a) Name the chemical found in the nucleus that controls the production of protein.

..... [1]

Fig. 2.2 shows a gamete from the same person.

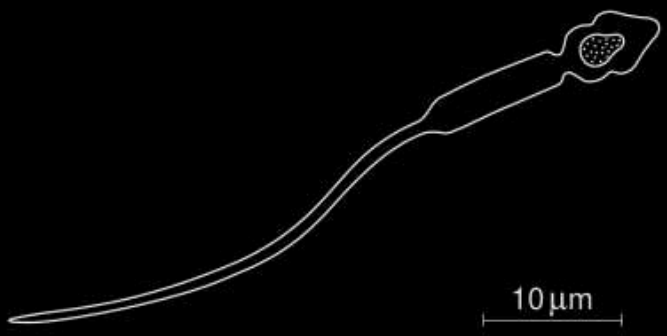


Fig. 2.2

(b) Describe how and explain why the two cells differ in appearance.

.....

.....

.....

.....

..... [4]

(c) (i) State two ways in which the nucleus of the gamete differs from the nucleus of the cheek cell.

For Examiner's Use

1. .....

2. ..... [2]

(ii) Explain why it is important that the two nuclei are different.

.....

.....

.....

..... [3]

[Total: 10]

2 (a) DNA ; [1]

(b) (no requirement for insistence on comparatives)
(how) ref. tail / no tail ;
ref. amount of cytoplasm (**R** no cytoplasm for sperm) ;
ref. to shape / length of cells / head or acrosome in sperm ; (**R** round)
ref. to position / size / shape of nucleus or nuclei ; (max 3)

(why) (**A** v.v. for cheek cell where relevant)
ref. swimming / movement / reaching ovum AW ; (**A** motility / mobility)
ref. absorption of nutrition from seminal fluid ;
ref. penetrative function of sperm ;
(nucleus at front of sperm) since only nucleus enters ovum AW ;
half the amount of (**A** less) genetic material / haploid ; (max 3) [max 4]

(c) (i) (**A** v.v. for cheek cell if clear, otherwise first mentioned assumed to be sperm)
Any two from: ½ the number, fewer or stated number of chromosomes / haploid,
½ the number of genes / amount of DNA,
either X or Y or only one sex- chromosome,
ref. shape / size difference,
ref. mitosis or mitosis (must be to how the cell is produced), ; ; [2]

(ii) (**A** v.v. for cheek cell – including negatives where relevant)
(for sperm) fusion / fertilisation ;
with another gamete / ovum / egg / nucleus ;
ref. zygote (production) ;
restoration of diploid number AW ; [max 3]

[Total: 10]

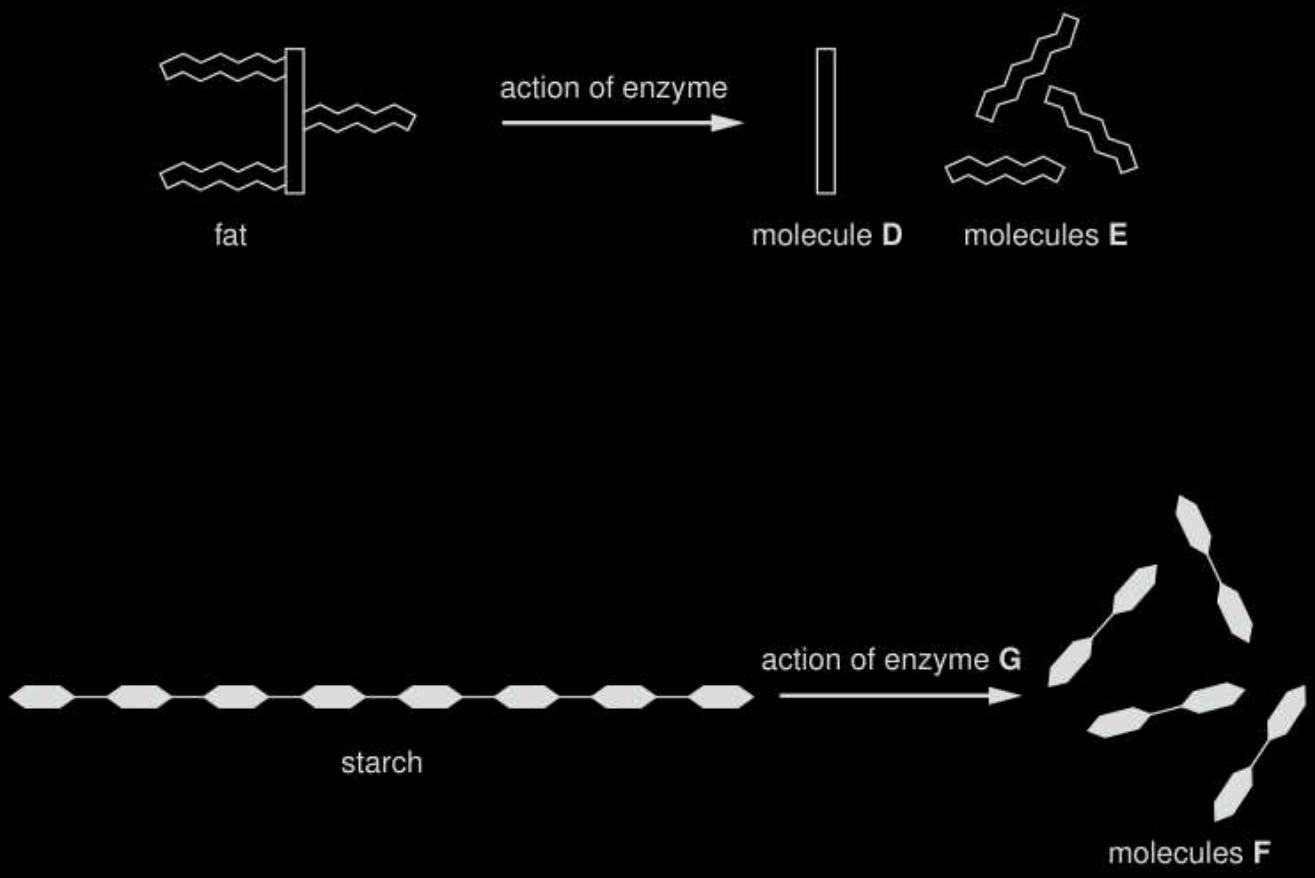


Fig. 4.1

(a) Identify the molecules shown in Fig. 4.1.

D .....

E .....

F ..... [3]

(b) Identify enzyme **G**. ..... [1]

(c) It has been found that fresh pineapple contains an enzyme that can be used to make meat more tender.

For Examiner's Use

(i) Explain why the pineapple is placed on the meat a few hours before, rather than during, cooking.

.....

.....

.....

..... [3]

(ii) Suggest the name of the enzyme and how it tenderises the meat.

.....

.....

.....

..... [3]

[Total: 10]

4 **(a)** **D** glycerol / glycerine ; ⎫ **A** one word only
E fatty acid ; ⎭ per line
F maltose / (reducing) sugar / disaccharide ; [3]
(**R** glucose / sucrose)

(b) **G** amylase / diastase / ptyalin ; [1]

(c) **(i)** strong heat / ref. high temperature AW / 60 °C or above ;
denatures / destroys / damages + enzymes ;
would not tenderise the meat AW / gives enzyme time to work AW ; [3]
(available only with ref. enzyme damage)

(ii) protease or any named ;
acts on / digests / breaks down proteins ;
changing them to peptides / peptones / proteoses / polypeptides etc. ;
*smaller or simpler molecules / bonds broken / easier to chew AW ; [max 3]
(**A** softer)
(* this mark available if enzyme chosen was lipase)

[Total: 10]

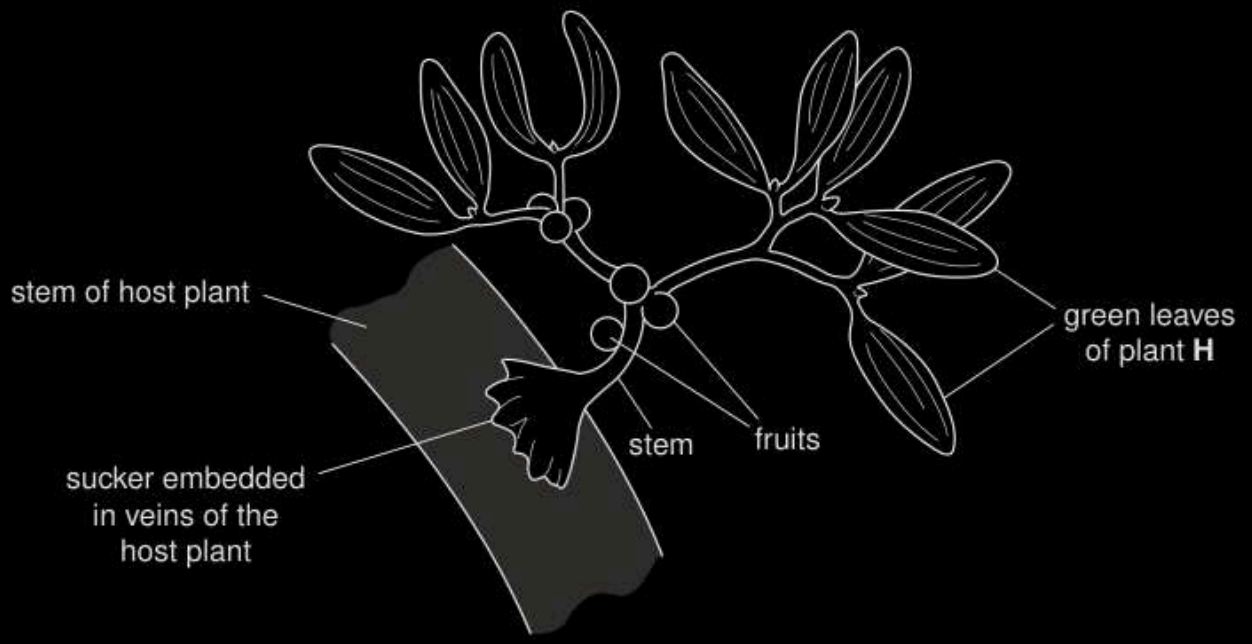


Fig. 5.1

(a) (i) Name an organ, found in free-living plants, that is absent in plant **H**.

..... [1]

(ii) Name the process used by plant **H** to manufacture its carbohydrates.

..... [1]

Plant **H** obtains one raw material for food manufacture from the air and other requirements, as well as nutrients, from its host.

(b) (i) Suggest three substances it absorbs from its host, and for each, state its use in Plant **H**.

substance 1 ....., use .....

substance 2 ....., use .....

substance 3 ....., use .....
[6]

(ii) Suggest the name of the tissue in the host plant which supplies these substances to plant **H**.

..... [1]

[Total: 9]

Section B

Answer **all** the questions including questions 6, 7 and 8 **Either** or 8 **Or**.

Write your answers on the separate answer paper provided.

5 **(a)** **(i)** roots / root system (**R** root hairs) ; [1]

(ii) photosynthesis ; [1]

(b) **(i)** Any **three** from:

water ;	for photosynthesis / turgor / transport solvent / hydrolysis / enzyme activity ; (**R** transpiration)
magnesium (ions) ;	for chlorophyll manufacture ; (**R** photosynthesis / chloroplasts)
nitrates ;	for making amino acids / proteins / enzymes / growth ;
calcium (ions) ;	for making cell walls / binding cells together AW;
other correct ion ;	and correct function ;
(e.g. P, K)	(P for growth or respiration, K for fruits and guard cell function)

*sucrose / CHO / sugar / amino acids ; and correct function for example; [max 6]

(NB 'salts for growth' = ONE mark for the line, but not available if a named salt / ion has also been given. 'Growth' can score ONCE ONLY on right hand-side.)

(ii) xylem (* if this mark scored, then xylem + phloem) ; [1]

[Total: 9]

[Section A Total: 50]

Section B

May/June 2008 · Paper 2 · Variant 1 · Question 6

6 Although skin is a waterproof structure, a few chemicals are able to pass through the tissues of the skin. When a person places a finger in a solution of one of these chemicals, it is possible for that chemical to enter the circulatory system and be carried to the tongue. The person then experiences the sensation of taste.

(a) Describe the pathway followed by this particular chemical from the finger until it reaches the tongue. [7]

(b) Describe the part played by the nervous system to enable the person to experience the sensation of taste. [3]

[Total: 10]

6 **(a)** #diffusion ;
epidermis / any layer of correctly named ;
dermis ;
#tissue fluid / plasma ;
#capillary ;
---
venule / vein / vena cava ;
A
*R.A + R.V. ;
---
correct ref. pulmonary circulation (either artery or vein) ;
B
lungs ;
---
*L.A. + L.V. ;
C
aorta / artery / arteriole ; [max 7]

(marks for *A*, *B* and *C* can be awarded if individually correct within an otherwise confused account)
(* or one for heart – also available within a confused account)
(# mark available anywhere so long as in correct context)

(b) stimulus / stimulates (**R** detects) ;
receptor / taste bud / sensor / nerve or sensory endings ;
sensory neurone (**R** nerve) ;
impulses (**R** messages) ;
brain / C.N.S. (ignore refs to spinal cord) ; [max 3]

[Total: 10]

October/November 2007 · Paper 1 · Variant 1 · Question 1

Correct answer: A

1 When a red stain is added to a culture containing both living and dead cells, only the dead cells take up the stain.

Which structure prevents the stain entering the living cells?

A cell membrane

B cell wall

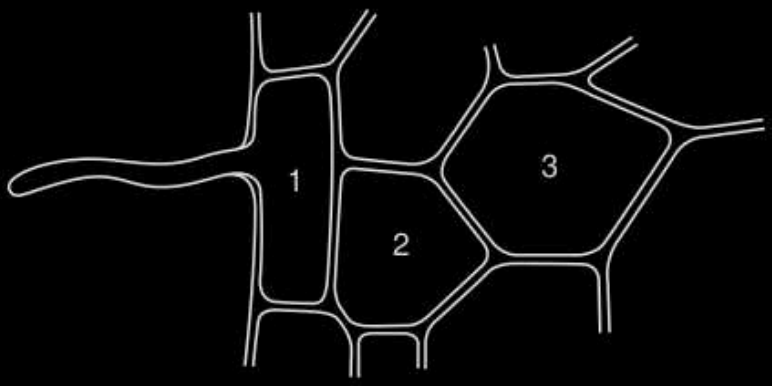
C cytoplasm

D vacuole

October/November 2007 · Paper 1 · Variant 1 · Question 2

Correct answer: A

2 The diagram shows some cells in a root of an actively growing plant.



How does the water potential of the cell marked 2 differ from the water potential of the cell marked 3 and how do mineral ions mainly enter the cell marked 1?

	water potential of the cell marked 2	mineral ions mainly enter the cell marked 1
A	higher than cell 3	by active transport
B	higher than cell 3	by diffusion
C	lower than cell 3	by active transport
D	lower than cell 3	by diffusion

October/November 2007 · Paper 1 · Variant 1 · Question 3

Correct answer: D

3 The mass of a cube of fresh potato is found. It is then placed in a test-tube containing a dilute solution of sucrose. After a day, its mass has increased.

Which process has occurred and what has happened to the concentration of the sucrose in the solution in the test-tube?

	process	sucrose concentration
A	active transport	decreased
B	active transport	increased
C	osmosis	decreased
D	osmosis	increased

October/November 2007 · Paper 1 · Variant 1 · Question 4

Correct answer: D

4 Which statement about enzymes is essential to the lock and key hypothesis?

A Enzyme molecules are catalysts.

B Enzyme molecules can be damaged by high pH values.

C Enzyme molecules can be damaged by high temperatures.

D Enzyme molecules each have a distinct shape.

October/November 2007 · Paper 1 · Variant 1 · Question 5

Correct answer: B

5 Magnesium is an essential element for

A the formation of cell walls.

B the formation of chlorophyll.

C the formation of proteins.

D the process of cell division.

October/November 2007 · Paper 1 · Variant 1 · Question 8

Correct answer: C

8 A person tries eating a diet consisting only of lettuce leaves and water.

Which condition might develop?

A constipation

B heart disease

C rickets

D scurvy

October/November 2007 · Paper 1 · Variant 1 · Question 10

Correct answer: C

10 A protein solution is tested using three different reagents.

Which set of results is obtained?

	iodine solution	Benedict's solution	biuret solution
A	black	blue	blue
B	black	red	blue
C	brown	blue	purple
D	brown	red	purple

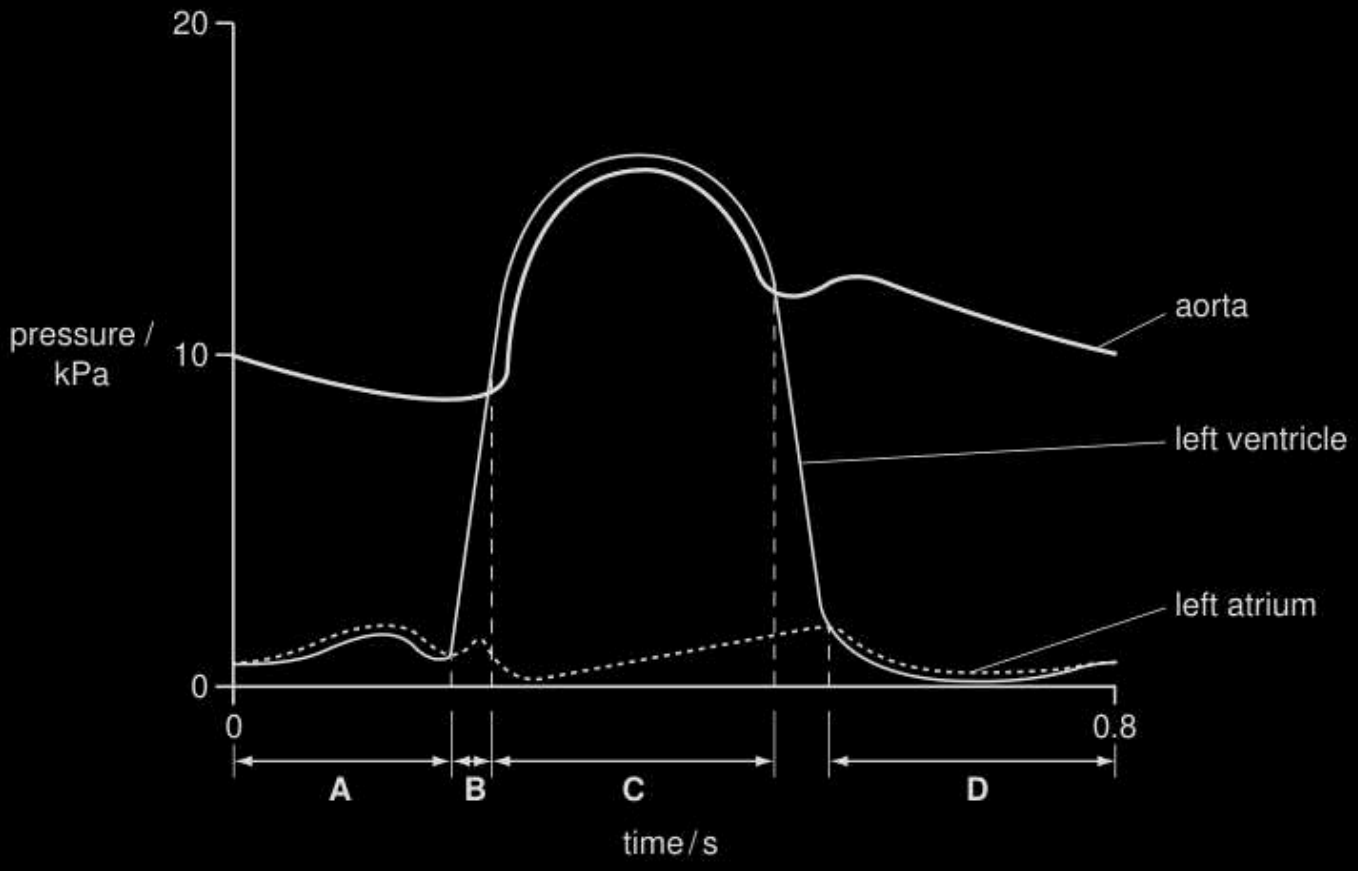
October/November 2007 · Paper 1 · Variant 1 · Question 12

Correct answer: C

12 Which adaptations of a root hair cell make it suitable for water uptake?

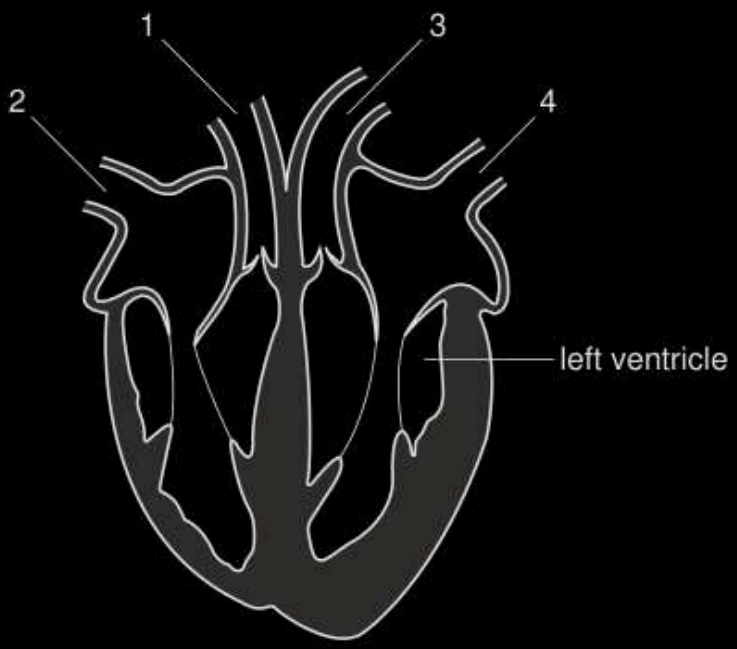
	partially permeable cell membrane	surface area to volume ratio of the cell
A	absent	high
B	absent	low
C	present	high
D	present	low

Correct answer: B



October/November 2007 · Paper 1 · Variant 1 · Question 14

Correct answer: D



What are the functions of the numbered blood vessels?

	carries blood to body	carries blood to lungs	carries blood from lungs	carries blood from body
A	1	2	3	4
B	1	3	4	2
C	2	4	3	1
D	3	1	4	2

October/November 2007 · Paper 1 · Variant 1 · Question 15

Correct answer: B

15 After muscular exercise, which blood vessel carries the **lowest** concentration of carbon dioxide?

A hepatic artery

B pulmonary vein

C renal artery

D vena cava

October/November 2007 · Paper 1 · Variant 1 · Question 17

Correct answer: C

17 Which do the cilia do in the bronchi of the lungs?

	trap bacteria	move mucus out of the lungs
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = function of the cilia

✗ = not a function of the cilia

October/November 2007 · Paper 1 · Variant 1 · Question 20

Correct answer: D

20 Urea is removed by the kidneys.

What is this process called?

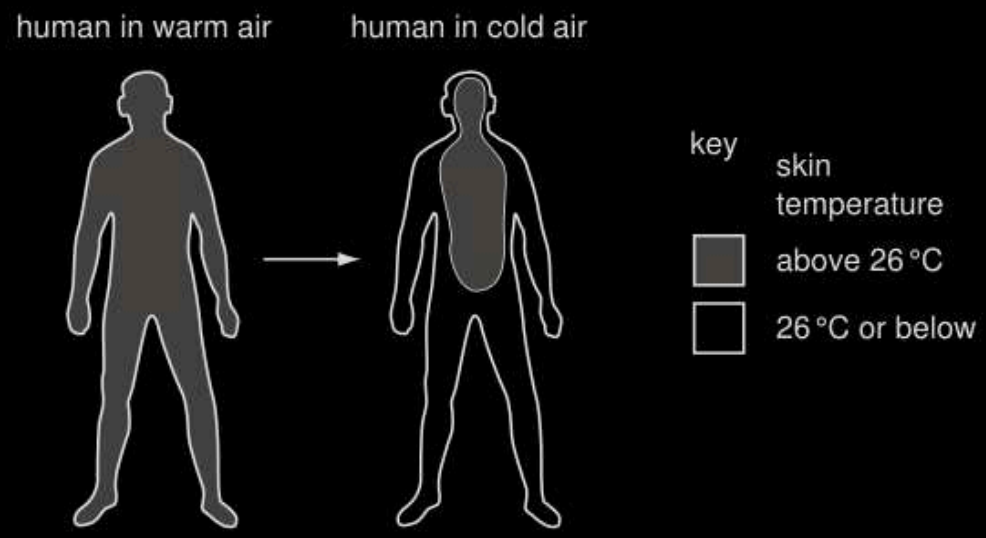
A dialysis

B diffusion

C egestion

D excretion

Correct answer: A



What causes the observed change in skin temperature on exposure to cold air?

A less blood flowing just below the skin

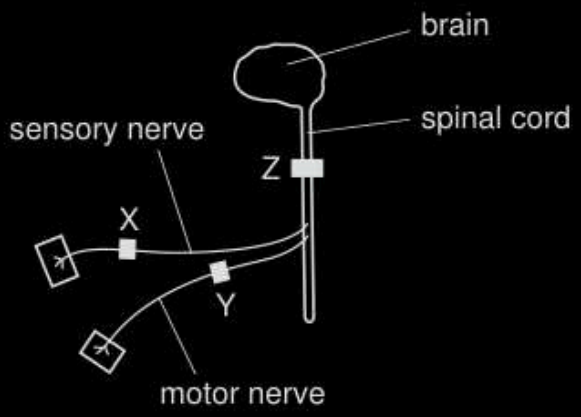
B less blood going to the heart and lungs

C more blood flowing just below the skin

D more blood going to the heart and lungs

October/November 2007 · Paper 1 · Variant 1 · Question 22

Correct answer: C



Of four men, one had no anaesthetic block and the other three had only one anaesthetic block at X, Y or Z.

One of the men can move his leg in response to a pinprick, but does not feel it.

Where is the anaesthetic block?

A block is at X

B block is at Y

C block is at Z

D no block

October/November 2007 · Paper 1 · Variant 1 · Question 23

Correct answer: A

23 Which is a function of adrenalin?

A to convert glycogen into glucose

B to decrease the breathing rate

C to increase the rate of peristalsis in the ileum

D to stimulate cells of the liver to take up glucose

October/November 2007 · Paper 1 · Variant 1 · Question 24

Correct answer: B

24 What happens to these structures when focussing on an object near the eye?

	ciliary muscles	suspensory ligaments
A	contract	tighter
B	contract	looser
C	relax	tighter
D	relax	looser

October/November 2007 · Paper 1 · Variant 1 · Question 27

Correct answer: A

27 New yoghurt can be made by adding a small amount of old yoghurt to some fresh, sterile milk.

What does the old yoghurt provide that is essential for this process?

A bacteria

B protein

C sugar

D vitamins

October/November 2007 · Paper 1 · Variant 1 · Question 31

Correct answer: A

31 Some samples are taken from a human patient.

Which sample could be examined to find out whether the patient is infected with the malarial parasite?

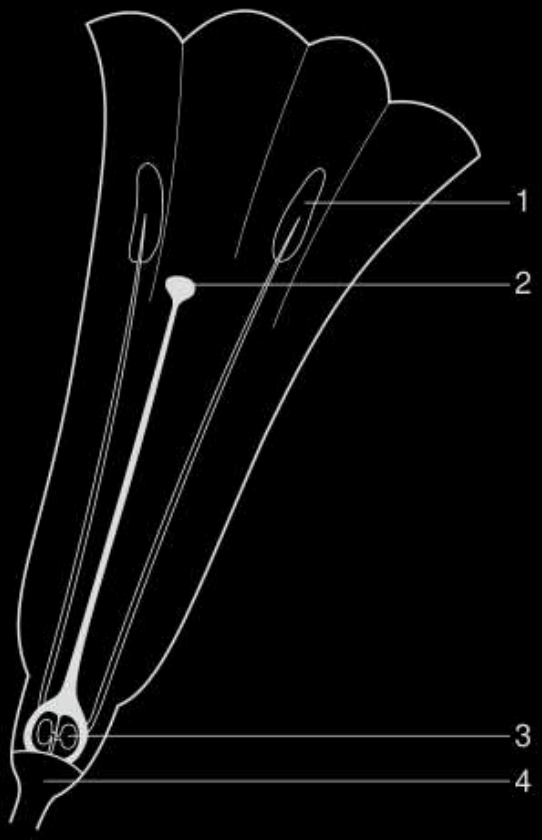
A red blood cells

B saliva

C urine

D white blood cells

Correct answer: C

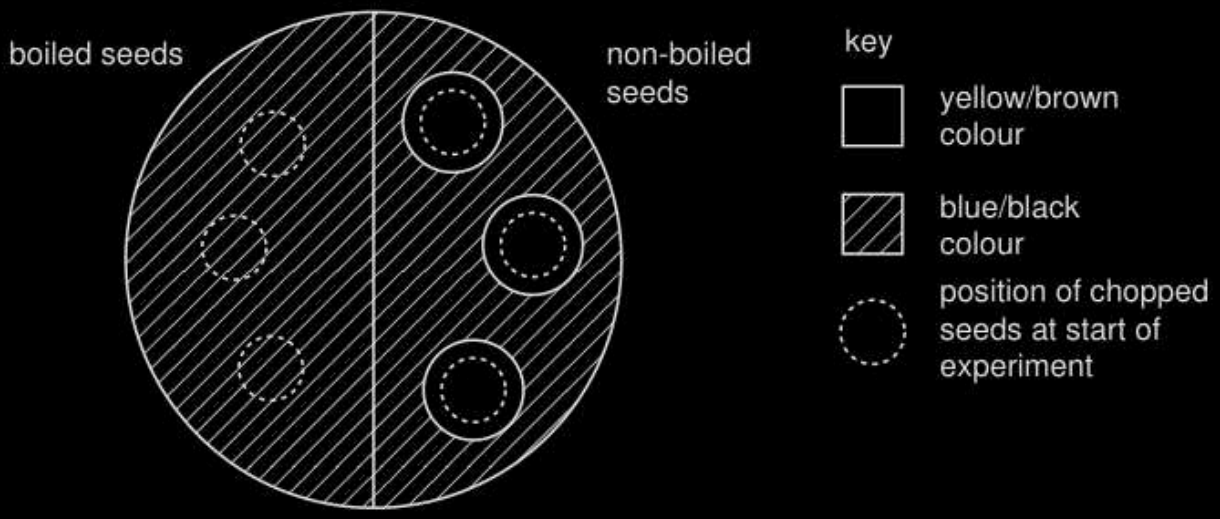


In which structures are haploid nuclei formed by reduction division?

A 1 and 4 **B** 2 and 3 **C** 1 and 3 **D** 2 and 4

October/November 2007 · Paper 1 · Variant 1 · Question 34

Correct answer: D



What is the explanation for the results with the non-boiled bean seeds?

A They absorb iodine.

B They absorb starch.

C They contain acid.

D They contain amylase.

October/November 2007 · Paper 1 · Variant 1 · Question 36

Correct answer: A

36 What would be the result of cutting the sperm ducts on the right and left sides in a man?

A He would become sterile.

B He would be unable to develop sperms.

C He would be unable to pass urine.

D Male sex hormones would no longer circulate in the blood.

October/November 2007 · Paper 1 · Variant 1 · Question 37

Correct answer: C

37 The inheritance of the ABO blood groups depends on three alleles I^A, I^B and I^o.

I^A and I^B are codominant, I^o is recessive.

What are the possible genotypes for a man of blood group A?

A I^AI^A only

B I^AI^A, and I^AI^B only

C I^AI^A, and I^AI^o only

D I^AI^A, I^AI^o and I^AI^B

October/November 2007 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 Which statement is true of a dominant allele?

A It cannot undergo mutation.

B It gives a greater chance of survival than a recessive allele.

C It gives the same phenotype in heterozygotes and homozygotes.

D It is only responsible for male characteristics.

October/November 2007 · Paper 1 · Variant 1 · Question 39

Correct answer: B

39 What determines the sex of a child?

A chromosome content of the ovum

B chromosome content of the sperm

C number of days between ovulation and fertilisation

D number of days between fertilisation and implantation

Correct answer: D

40 A homozygous short-winged fly is crossed with a homozygous long-winged fly.

Which phenotypic ratio would result?

A 1:1 **B** 1:2:1 **C** 3:1 **D** all similar

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

1 Fig. 1.1 shows a plant in a beaker containing a solution of dye. Fig. 1.2 shows the tissues of the same plant as they would appear if a cross section is taken at X.

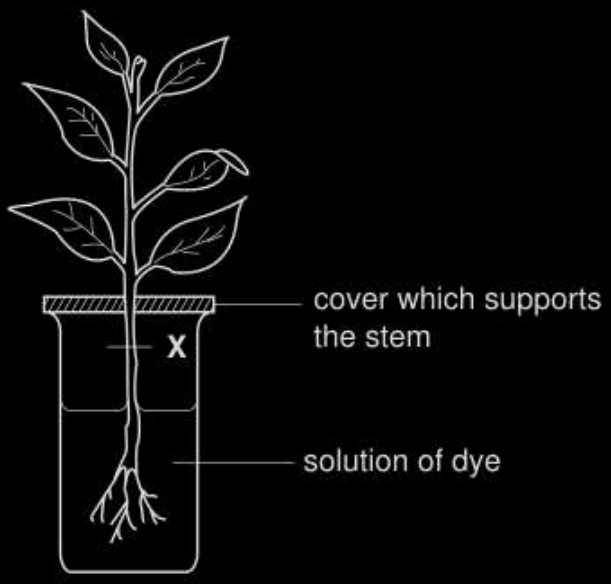
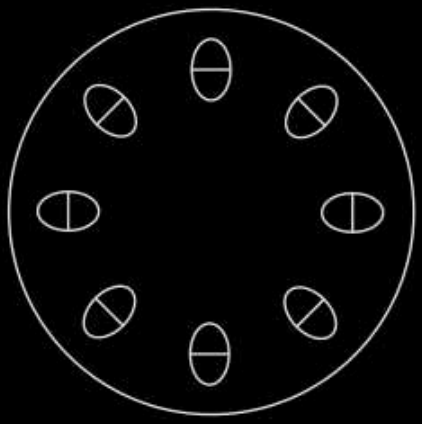


Fig. 1.1



cross section at **X**, magnified

Fig.1.2

(a) **(i)** On Fig. 1.2, shade the tissue that would be most heavily stained by the dye. [1]

(ii) Name this tissue .....[1]

(b) Explain the processes occurring in the plant that cause the dye to move upwards through the stem.

.....

.....

.....

.....

.....[3]

Use

(c) Describe the process responsible for the uptake of the dye into the roots of the plant.

.....

.....

.....

.....

.....[3]

(d) Suggest and explain what would happen to a plant if its roots are submerged in a concentrated salt solution for six hours.

.....

.....

.....

.....

.....[3]

[Total: 11]

1 **(a)** **(i)** xylem only shaded (A shaded on only one vascular bundle);

(ii) correctly named (Ignore 'vessel'); [2]

(b) transpiration/evaporation;
water/vapour + lost from leaf/stomata/plant;
pulls/draws/pushes + water/solution
(R sucks) OR water (from transpiration) must be replaced;
ref. capillarity/root pressure AW/cohesion AW/adhesion AW; [max. 3]

(c) diffusion (R if osmosis mentioned with diffusion, ignore active transport);
ref. concentration gradient;
through (cellulose) cell walls;
cell wall permeable;
(A partially/selectively)
cell membrane is a p.p.m./allows molecules of dye to pass; [max. 3]
(R if in osmosis context)

(d) osmosis/diffusion;
water leaves cells;
water loss from plant or from plant part named;
salt solution more concentrated than cell sap/ref. water potential
(R refs to quantity of water rather than concentration) gradient;
plant wilts or described e.g. refs flaccidity/loss of turgor/of support ; [max. 3]
(A drooping, R withering)

2 In living organisms, enzymes may be used in reactions to convert large molecules into smaller ones.

(a) State the term used for all

(i) molecules on which enzymes act, ...

(ii) molecules that are formed in a reaction. .....[2]

(b) For a chemical reaction that begins in the stomach, state

(i) the original large molecule, ...

(ii) the stomach enzyme involved, ...

(iii) the smaller molecules formed. .....[3]

Fig. 2.1 is part of a graph showing how the rate of an enzyme-controlled reaction changes with increasing temperature.

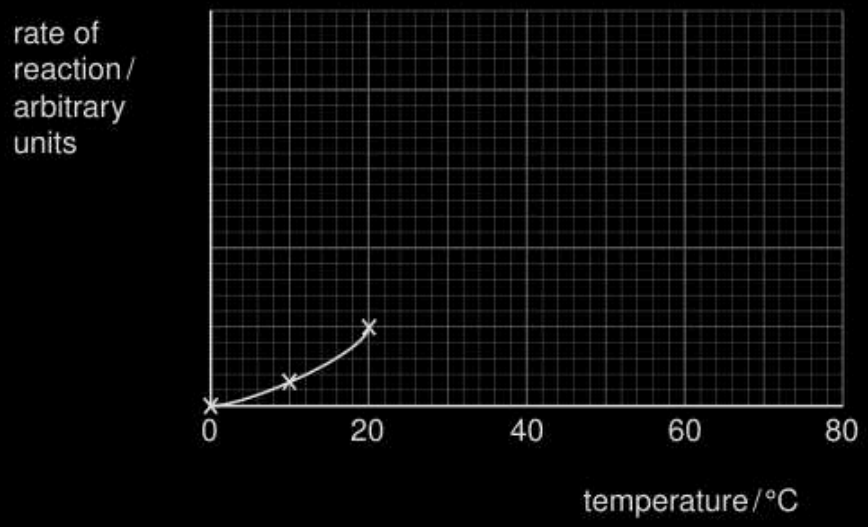


Fig. 2.1

(c) Complete the graph in Fig. 2.1 to show how the rate of this enzyme-controlled reaction changes as the temperature is increased from 20°C to 80°C. [3]

(d) Fig. 2.2 shows diagrammatically how an enzyme-controlled reaction may occur.

Use

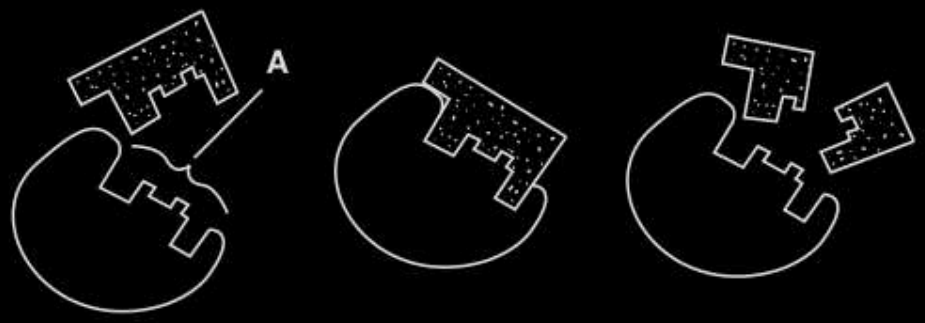


Fig. 2.2

(i) Identify region **A** on Fig. 2.2. .....[1]

(ii) State the hypothesis illustrated in Fig. 2.2.

.....

.....[1]

[Total: 10]

2 **(a)** **(i)** substrate/s;

(ii) product/s; [2]

(b) **(i)** protein (A casein);

(ii) protease/pepsin/proteolytic (A rennin if casein given above);

(iii) (poly)peptides/peptones/proteoses (A amino acids);
(this mark not available with casein/rennin option) [3]

mark independently

(c) graph rises;
reaches peak between 35 and 55 °C;
falls to zero between 50 and 80 °C;
(A vertical drop, R incurving drop) [3]

(d) **(i)** active site/place where substrate fits AW (R lock / key);
(A region/area) (A place where reaction occurs) [1]

(ii) lock and key (A words in a description); [1]

3 Fig. 3.1 shows a student involved in an experiment on breathing.

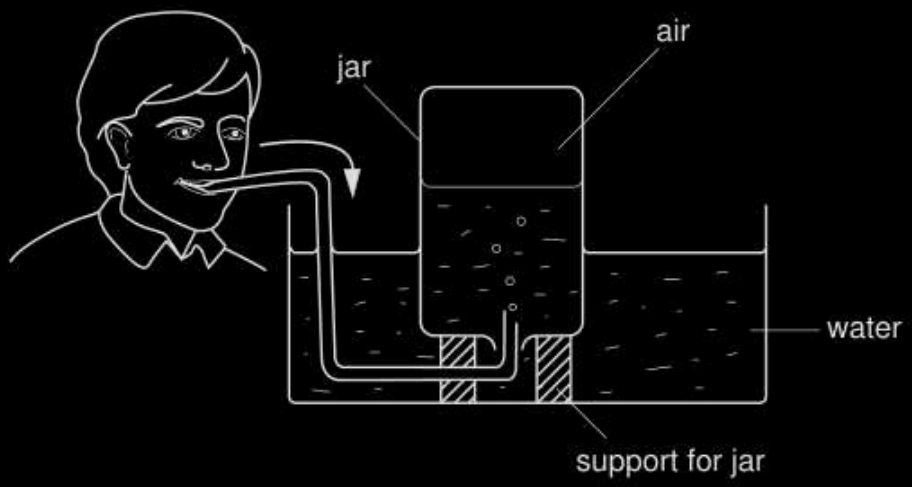


Fig. 3.1

(a) State three ways in which the air in the jar will differ from atmospheric air.

1. .....

2. .....

3. .....[3]

(b) Describe what occurs in the student's thorax to cause air to be breathed out into the jar.

.....

.....

.....

.....

.....

.....

.....[4]

(c) By referring to the cells in the student's body, suggest how the differences you mention in **(a)** are brought about.

.....

.....

.....

.....[2]

[Total: 9]

3 **(a)** Mark the first, one per line, any **three** from:

cleaner, at or near body temperature (A higher temp./warmer),
more CO_2, less O_2, more moisture AW (A saturated);;; [3]
(Ignore pathogens)

(b) (external) intercostal muscles relax;
diaphragm relaxes;
ribs / thorax moves down/in;
diaphragm domes AW;
decreased volume/increased pressure; [max. 4]

(c) *There are two routes to the two marks in this section*
Route 1: <u>respiration</u> ;

production of carbon dioxide/removal of oxygen (A oxidation)/
release of energy or heat/release of water
(R produce/generate etc.)
(A points on equation in words or symbols – need not be balanced);

Route 2: <u>named</u> specific cell (e.g. RBC);

The effect of this cell on a the content of the air in the jar; [2]
(e.g. collects/carries away AW O_2) (Ignore refs CO_2 and RBCs)

4 **(a)** Name the mineral ion required for the manufacture of the pigment in red blood cells.

.....[1]

Fig. 4.1 shows some red and white blood cells from a bird. (All cells are drawn to the same magnification.)

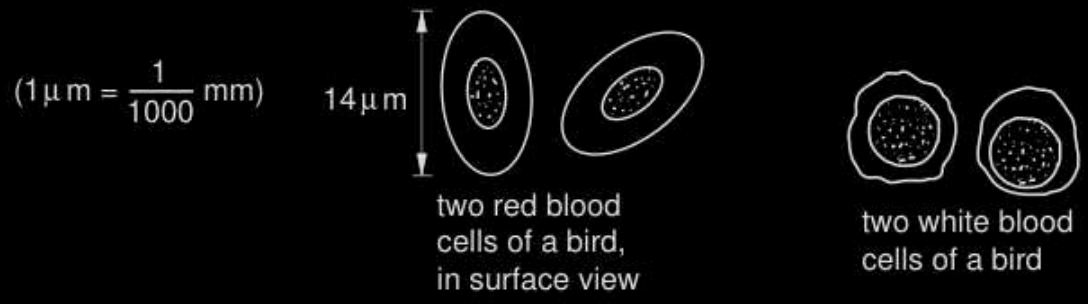


Fig. 4.1

(b) By reference to Fig. 4.1, complete Table 4.2 to show three differences between the red blood cells of birds and mammals.

Table 4.2

	red blood cells	
	birds	mammals
1		
2		
3		

[3]

Use

Fig. 4.3 shows human blood cells in a type of blood vessel **(B)** passing between body cells.

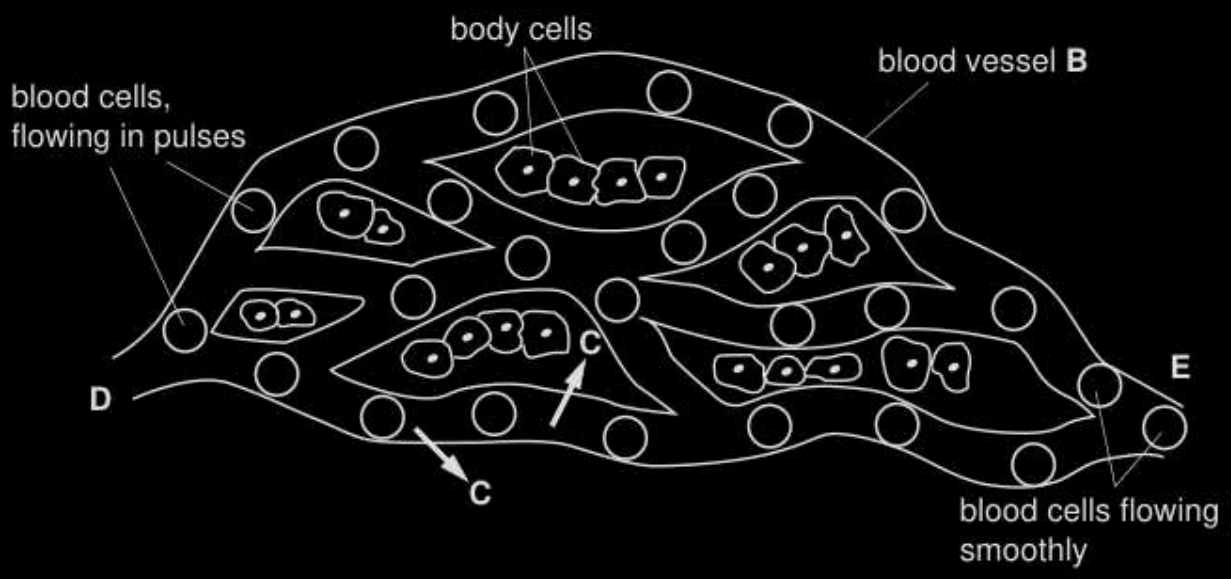


Fig. 4.3

(c) (i) Name the type of blood vessel labelled **B** and give reasons for your answer.

type of blood vessel .....

reasons .....

.....

.....[3]

(ii) Name the liquid leaving the blood vessel at **C**. .....[1]

(iii) Using information from Fig. 4.3, explain why blood is under higher pressure at point **D** than at point **E**.

.....

.....

.....[3]

[Total: 11]

4 **(a)** iron/Fe;

(b) *If the column headed 'mammals' is left blank, or if there is a clear, but inaccurate, attempt to describe mammal (as opposed to human) RBCs, then all three marks are available for correct statements re. bird RBCs .*
All comparisons must be valid pairs (R oval v. biconcave)
Mark each line separately, (R refs. to haemoglobin/surface area)

(A longer) (A shorter)
Any 3 from: nucleus + no nucleus, larger in size/smaller in size ,
larger than WBCs +smaller than WBCs, oval/egg-shaped + round isc,
biconvex/not biconcave + biconcave, (R spherical);;; [3]

(c) **(i)** capillary,; [1]

Any **two** from: blood cells in single file AW, running between body cells, walls thin/permeable/one cell thick, substances pass through (at C), vessels in the form of a network,;; [2]

(ii) tissue fluid/ECF/lymph/plasma/interstitial fluid (R blood); [1]

(iii) pulse beat + in arteries/arterioles;
greater pressure in arteries/lower pressure in veins;
ref. arteries or D nearer heart/pump/ventricle;
blood flows smoothly/no pulse + in veins/venules;
resistance offered by capillary network;
fluid lost from network; [max 3]

October/November 2007 · Paper 2 · Variant 1 · Question 6

6 **(a)** State the functions of the following structures:

(i) the kidneys,

(ii) the urethra. [5]

(b) Describe how a kidney dialysis machine performs the functions of the kidneys in a patient with kidney failure. [5]

6 **(a)** **(i)** remove urea/nitrogenous waste/uric acid (R urine);
salts/minerals/ions/toxins/hormones;
excretion;
water + in excess/ref. osmoregulation;
(filtration) from blood;

(ii) carries urine;
from bladder + to outside;
seminal fluid/sperms; [max. 5]

(b) connected to patients circulatory system/blood through machine;
(along)
blood + passed through partially (etc.) permeable/dialysing tube;
(R if blood is passing through the wall of the tube – Ignore named membrane)
diffusion/differential conc. solutes in bathing fluid/fluid renewed;
of excretory/waste products/urea [see list for **(a)(i)**];
salts/small molecules [see **(a)(i)** list)];
from blood;
large molecules (or named) stay in bloodAW;
ref bathing/washing/dialysing fluid; [max. 5]

[Total: 10]

May/June 2007 · Paper 1 · Variant 1 · Question 1

Correct answer: C

Which are animal cells?

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

May/June 2007 · Paper 1 · Variant 1 · Question 2

Correct answer: A

2 Which is an example of active transport?

A movement of glucose into the cells of the villi

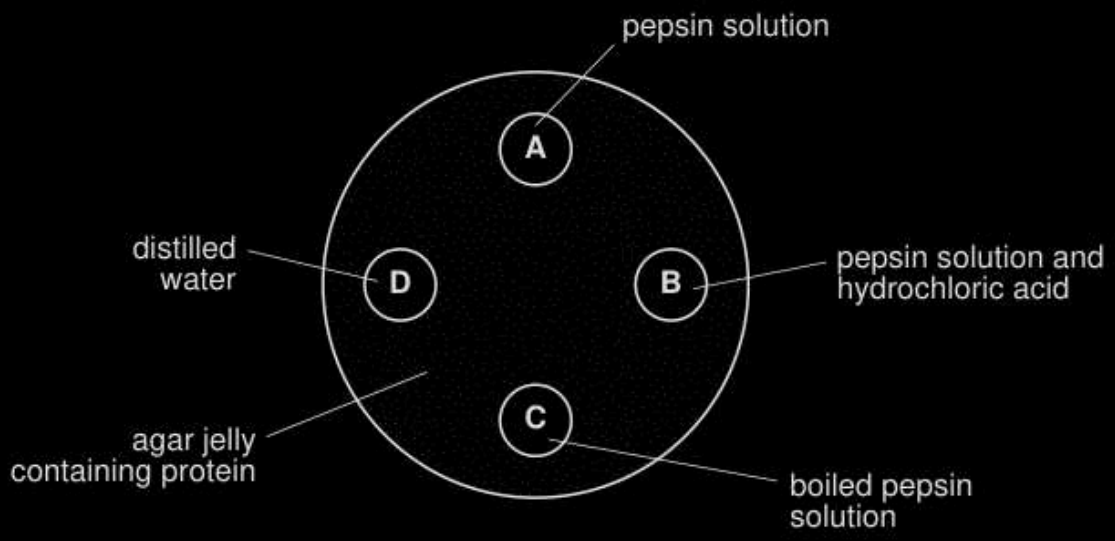
B movement of glucose molecules down a concentration gradient

C movement of ions in blood plasma

D movement of water in the transpiration stream

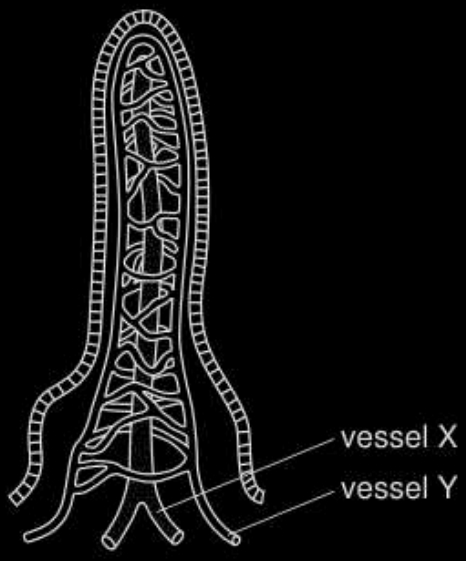
May/June 2007 · Paper 1 · Variant 1 · Question 4

Correct answer: B



May/June 2007 · Paper 1 · Variant 1 · Question 9

Correct answer: B



After a meal containing starch and oil, which substances are absorbed mainly into vessel X and which substances are absorbed mainly into vessel Y?

	vessel X	vessel Y
A	amino acids	water
B	fatty acids and glycerol	glucose
C	glucose	amino acids
D	water	fatty acids and glycerol

May/June 2007 · Paper 1 · Variant 1 · Question 10

Correct answer: A

10 Only two of the following statements accurately describe what happens in the mouth.

1 Amylase breaks down large starch molecules into smaller maltose molecules.

2 Chewing increases the surface area of food for digestion.

3 Saliva emulsifies fats into smaller droplets.

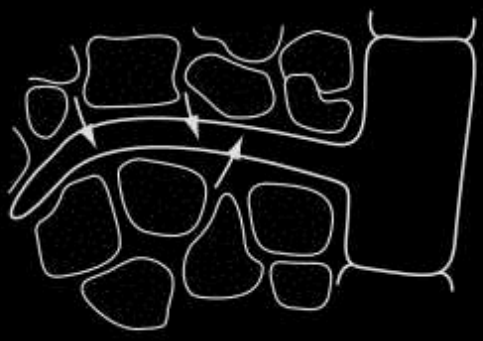
4 Teeth break up large insoluble molecules into smaller soluble molecules.

Which statements are correct?

A 1 and 2 **B** 2 and 3 **C** 3 and 4 **D** 1 and 4

May/June 2007 · Paper 1 · Variant 1 · Question 11

Correct answer: D



What will assist the uptake of ions by the root hair?

A a higher concentration of ions in the root hair than in the soil

B a low temperature in the surrounding soil

C rapid uptake of water from the soil by osmosis

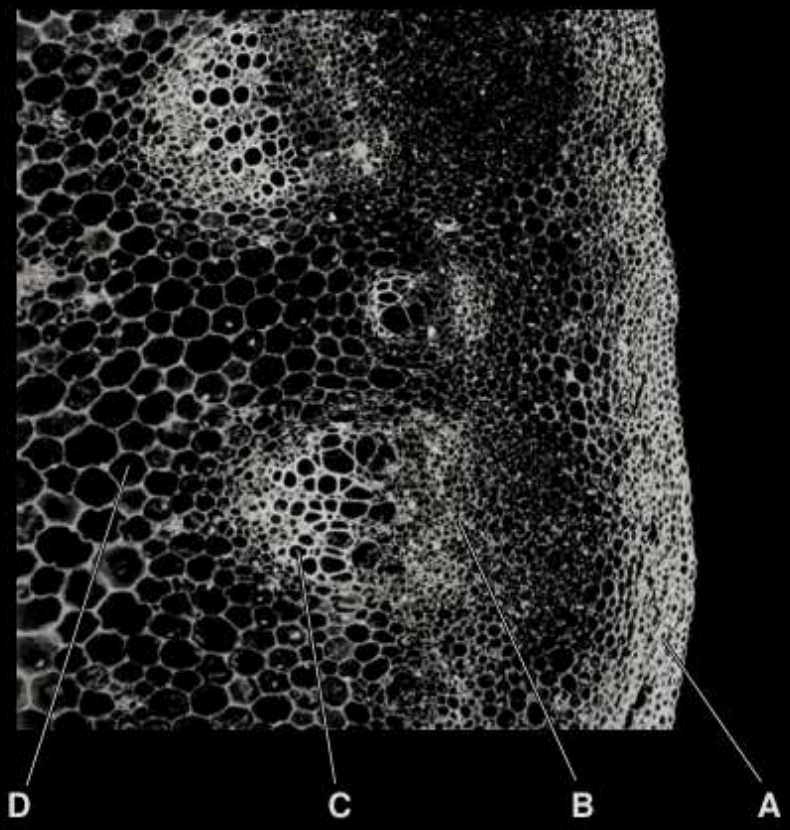
D the large surface area of the root hair

May/June 2007 · Paper 1 · Variant 1 · Question 12

Correct answer: C

12 The photograph shows a cross-section of part of a sunflower stem under the microscope.

Which tissue transports water and mineral salts?



May/June 2007 · Paper 1 · Variant 1 · Question 13

Correct answer: B

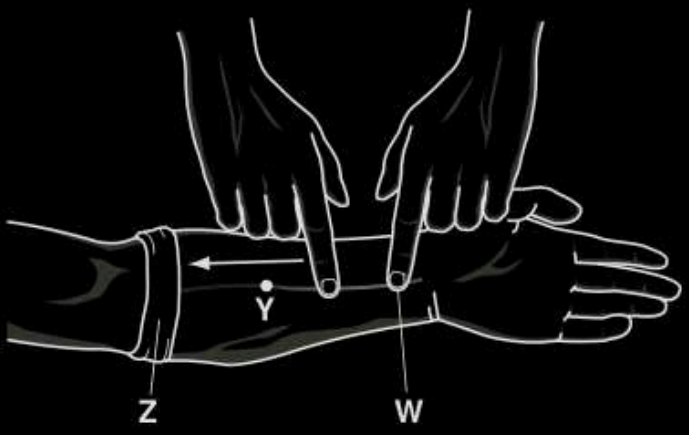
13 Which description correctly applies to blood in the pulmonary circulation?

	pulmonary artery	pulmonary vein
A	deoxygenated	high pressure
B	deoxygenated	low pressure
C	oxygenated	high pressure
D	oxygenated	low pressure

May/June 2007 · Paper 1 · Variant 1 · Question 14

Correct answer: C

14 The diagram shows the investigation of blood flow in the veins of the lower arm.



A cloth is tightly wrapped round the arm at point **Z** and the veins stand out clearly. One finger presses on the vein at **W**.

When another finger strokes the vein, as shown in the diagram, the vein lies flat between points **W** and **Y**.

Some possible explanations are listed.

1 The bandage at **Z** prevents backflow of blood.

2 The finger pressed at **W** prevents more blood entering the vein.

3 A valve at **Y** prevents backflow.

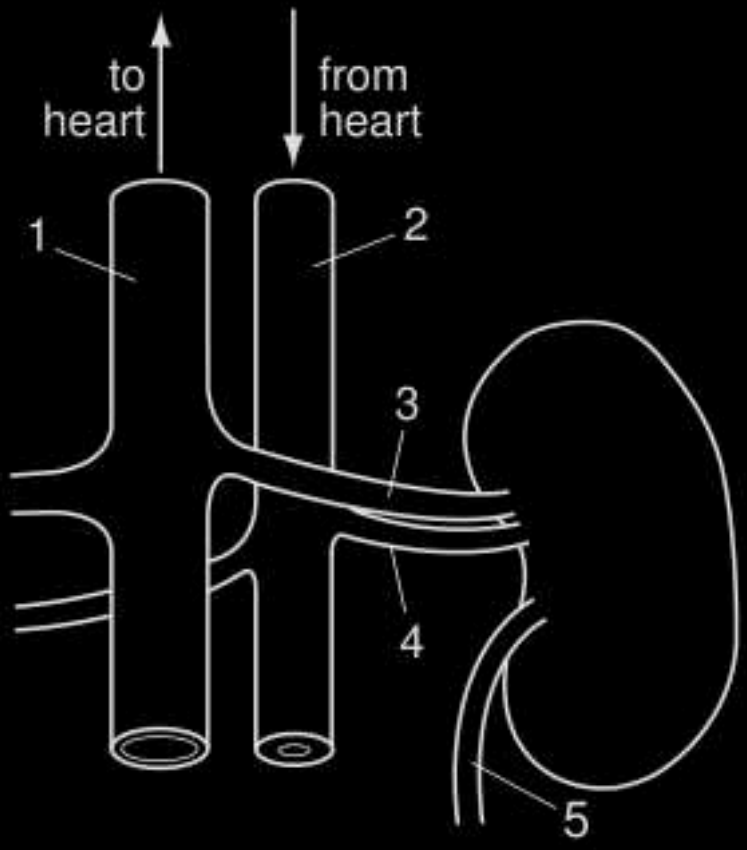
4 A valve at **Z** prevents more blood from entering the vein.

Which explanations of the vein lying flat are correct?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

May/June 2007 · Paper 1 · Variant 1 · Question 15

Correct answer: D



Which structures have the **most** and **least** urea concentrations?

	most	least
A	1	2
B	4	1
C	4	3
D	5	3

May/June 2007 · Paper 1 · Variant 1 · Question 16

Correct answer: C

16 Which process does **not** depend on respiration?

A active uptake of ions

B conduction of nervous impulses

C diffusion of glucose

D muscle contraction

May/June 2007 · Paper 1 · Variant 1 · Question 18

Correct answer: D

18 Which feature of alveoli decreases the distance over which oxygen and carbon dioxide molecules diffuse?

A Each alveolus has a large blood supply.

B Each alveolus is only 0.1-0.2 mm in diameter.

C There are approximately 150 million alveoli in each lung.

D The walls of the alveoli are only one cell thick.

May/June 2007 · Paper 1 · Variant 1 · Question 20

Correct answer: C

20 What is an example of excretion?

A release of adrenalin from the adrenal glands

B release of sweat from the sweat glands

C removal of carbon dioxide from the lungs

D removal of faeces from the alimentary canal

May/June 2007 · Paper 1 · Variant 1 · Question 21

Correct answer: D

21 Which response to a drop in body temperature does **not** involve muscle contraction?

A blood vessels narrowing

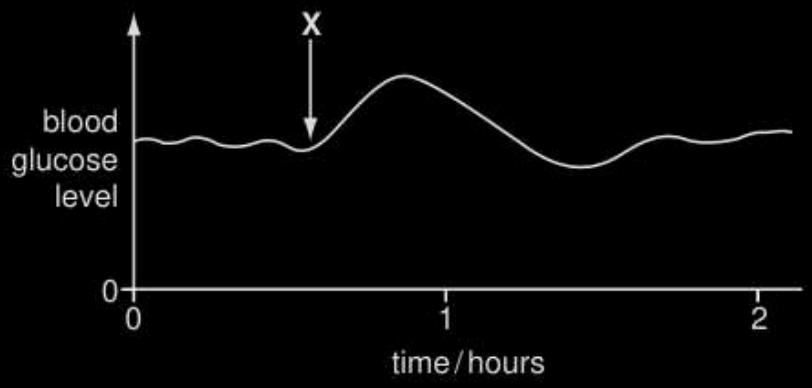
B hairs standing up

C shivering

D reduced sweating

May/June 2007 · Paper 1 · Variant 1 · Question 22

Correct answer: A



What explains the shape of the graph after **X**?

A The person has eaten a sugary food.

B The person has had an insulin injection.

C The person is suffering from diabetes mellitus.

D The person starts some physical exercise.

May/June 2007 · Paper 1 · Variant 1 · Question 23

Correct answer: B

23 In a reflex action, after the hand touches a hot object, what is the role of the brain?

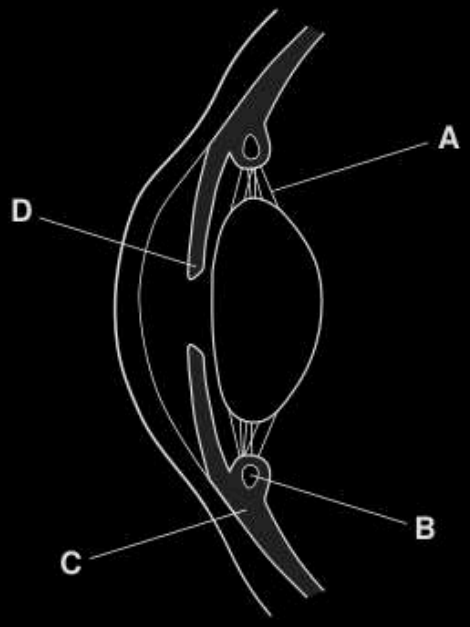
	receives information	initiates the withdrawal response
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

May/June 2007 · Paper 1 · Variant 1 · Question 24

Correct answer: B

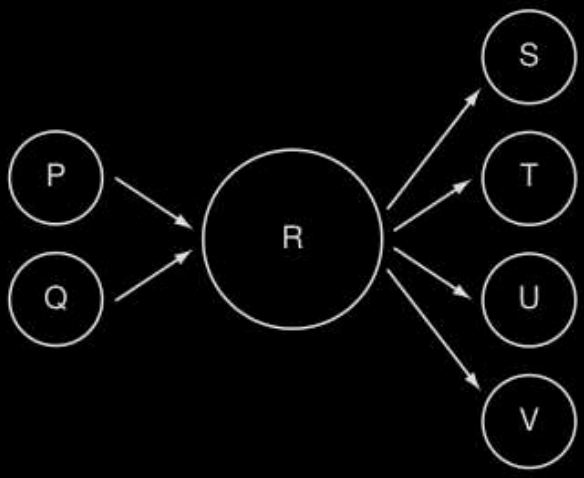
24 The diagram shows a section through a human eye.

Which structure contains the muscles which contract to produce a focused image on the retina?



May/June 2007 · Paper 1 · Variant 1 · Question 33

Correct answer: B



Which statement about the numbers of chromosomes in the cells and gametes is correct?

A The numbers of chromosomes in P and Q are different.

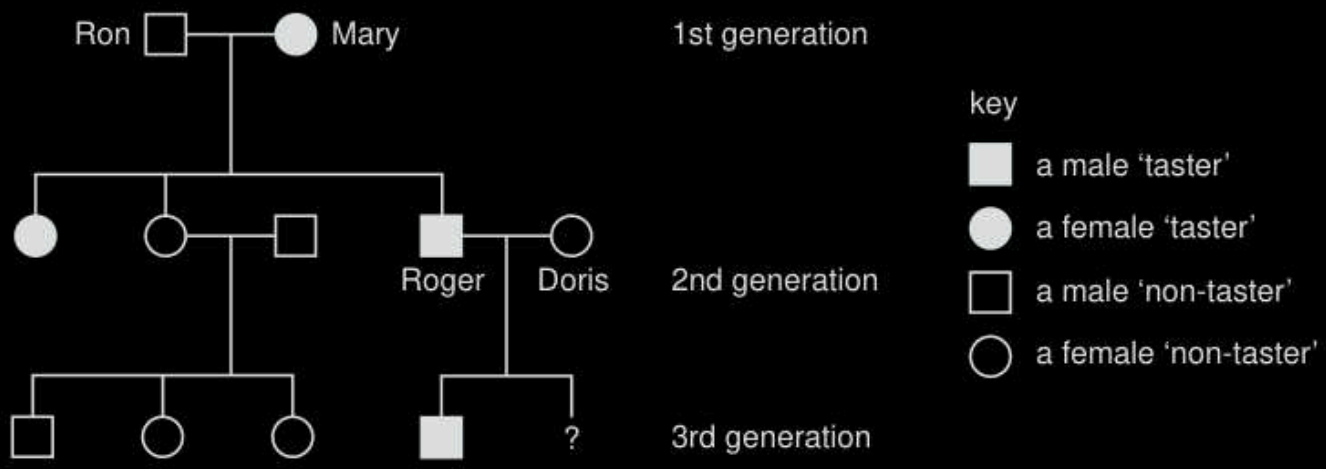
B The numbers of chromosomes in P and S are the same.

C The number of chromosomes in S is one quarter of the number of chromosomes in R.

D The number of chromosomes in T is half the number of chromosomes in Q.

May/June 2007 · Paper 1 · Variant 1 · Question 37

Correct answer: C



What is the probability of the second child of Roger and Doris being a 'non-taster'?

A 0.25 **B** 0.34 **C** 0.5 **D** 1.00

May/June 2007 · Paper 1 · Variant 1 · Question 38

Correct answer: C

38 In rabbits the allele for black hair is dominant. A heterozygous black-haired rabbit is crossed with a heterozygous black-haired rabbit.

Which phenotypic ratio would result?

A 1:1 **B** 1:2:1 **C** 3:1 **D** all similar

May/June 2007 · Paper 1 · Variant 1 · Question 40

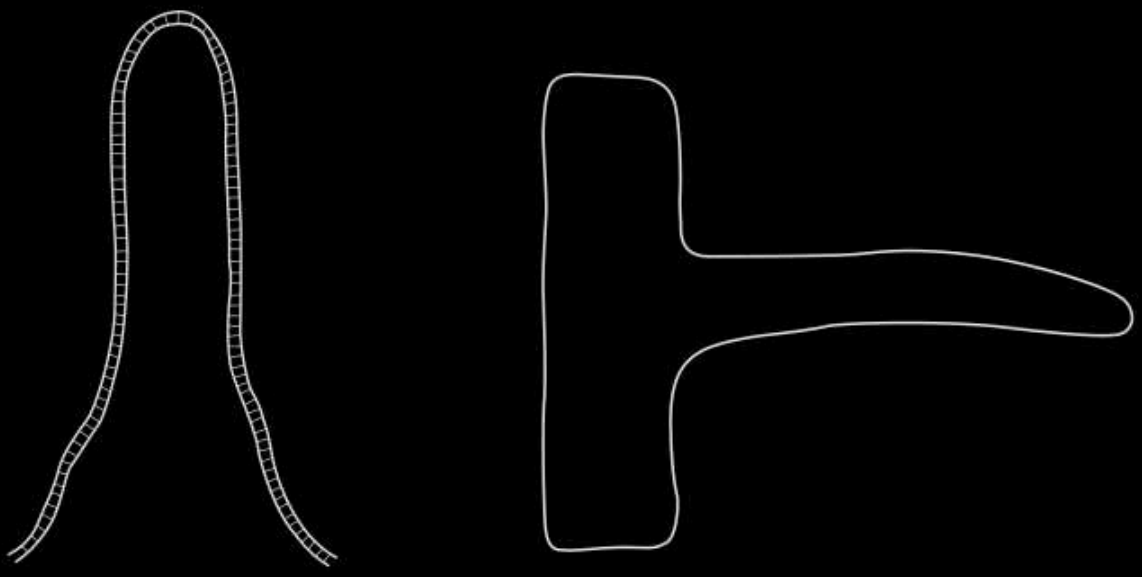
Correct answer: D

40 Two heterozygotes are crossed. Some of the offspring show the recessive characteristic.

What is the probability that the offspring that show the recessive characteristic are homozygous?

A 0.00 **B** 0.25 **C** 0.50 **D** 1.00

2 Fig. 2.1 shows two microscopic structures that are adapted for absorption, one from an animal and the other from a plant. The structures are not drawn to the same magnification.



structure **D** structure **E**

Fig. 2.1

(a) State **one** feature that these two structures have in common that helps them to carry out their function.

.....

.....[1]

(b) Identify each structure and, for each one, name two chemical substances that are absorbed by it.

structure **D**: name .....

chemical substance 1. .....

2. .....

structure **E**: name .....

chemical substance 1. .....

2. .....[6]

(c) (i) Structure **E** in Fig. 2.1 is a single cell. **Draw in and label**, on Fig. 2.1, a feature that helps the cell to carry out absorption. [1]

(ii) Explain how the feature you have drawn and labelled helps the cell to carry out absorption.

.....

.....

.....

.....

.....[2]

[Total: 10]

2 (a) large surface area (to volume ratio) / long / thin ; [1]

(b) **D** villus/i (R microvillus) ; [1]
(or fats for one mark)

mtf,opl, any **two** from: amino acids, fatty acids, glycerol, glucose,
ions AW (or named e.g.), drugs/toxins, water, vitamins (or named e.g.), ;; [2]

E root hair (cell) ; [1]

(in either order) water ;
ions / salts / minerals / named ion / oxygen (R nitrogen) ; [2]

(c) (i) feature drawn and labelled (A on extra diagram)
(R mitochondria / nucleus, but A reasons – see later) ; [1]

(ii) These must **both** be for the same feature for 2 marks, but if they do not match the label, then **0** for label.
(membrane) living ;
partially permeable AW ;
only selected molecules enter ;
allows osmosis or described (R diffusion) ;
active transport ; [max 2]

OR (ccw) permeable ;
tough / flexible ;
helps keep shape of cell ;
ref. part played in turgor AW ; [max 2]

OR (cell sap / vacuole / cytoplasm)
contains concentrated solution / ref. concentration gradient ;
for water entry ;
by osmosis / diffusion ; [max 2]

OR (for cytoplasm / mitochondria)
ref. respiration / energy release ;
for active transport ; [2]

OR (nucleus) – for **one** mark only
controls cell activites AW ; [1]
(R e.c.f. for villus)

4 Microorganisms can be grown in the laboratory on agar (a jelly-like medium) in a Petri dish. Nutrients required by the microorganisms are added to the agar.

(a) Name a nutrient that should be added to ensure the maximum rates of

(i) respiration .....

(ii) growth .....[2]

(b) Respiration and growth are enzyme-controlled processes.

A student is provided with a Petri dish containing agar with the optimum concentration of nutrients. Explain how the student could speed up the rate of growth of microorganisms growing in this Petri dish.

.....

.....

.....

.....[2]

Fig. 4.1 shows the stages in an experiment to show the effect of different antibiotics on microorganisms.

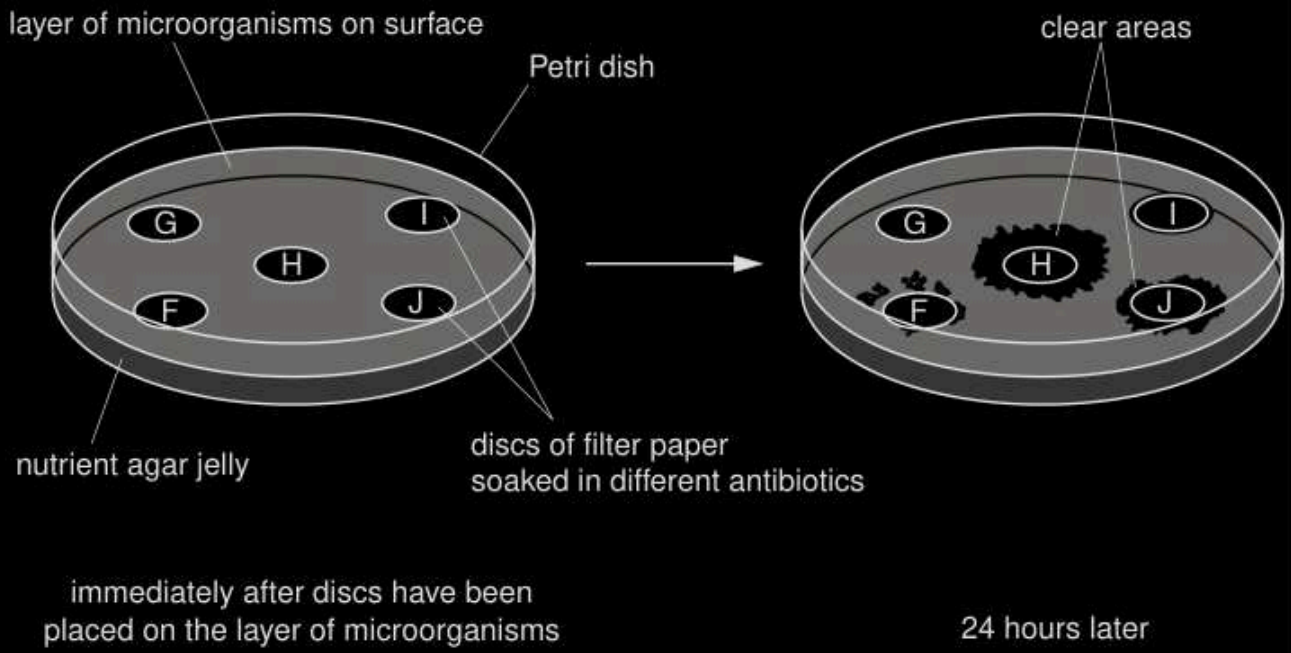


Fig. 4.1

The microorganisms originally came from the throat of a person suffering from a throat infection.

Use

(c) Using the information in Fig. 4.1,

(i) identify by letter the antibiotic which might be best for treating the infection,

..... [1]

(ii) state which type of microorganism was probably causing the throat infection and give a reason for your answer.

type of microorganism .....

reason .....[2]

(d) Antibiotic **F** used to be the most effective in treating the infection. Suggest and explain why it has become less effective.

.....

.....

.....

.....

.....

.....[3]

[Total: 10]

4 **(a)** **(i)** carbohydrate / sugar / glucose / maltose / sucrose / starch ;

(ii) protein / amino acids (vitamins / ions etc.) ; [2]

(b) incubate / oven / suitable location ;
optimum / increased / suitable + temperature or stated (25 – 50 °C) ;
suitable pH ;
correct ref. enzymes ; [max 2]

(c) **(i)** H ; [1]

(ii) bacterium (or suitable, named e.g. *Streptococcus*) ;
antibiotics effective against bacteria AW ; [2]

(d) (A e.c.f. for virus)
bacteria or microorganisms mutate ;
become resistant to antibiotic (R immune) ;
antibiotic too readily prescribed / available ;
patients not completing course ; [max 3]

5 Fig. 5.1 shows some food just before it enters the stomach and the same food as it leaves the stomach four hours later.

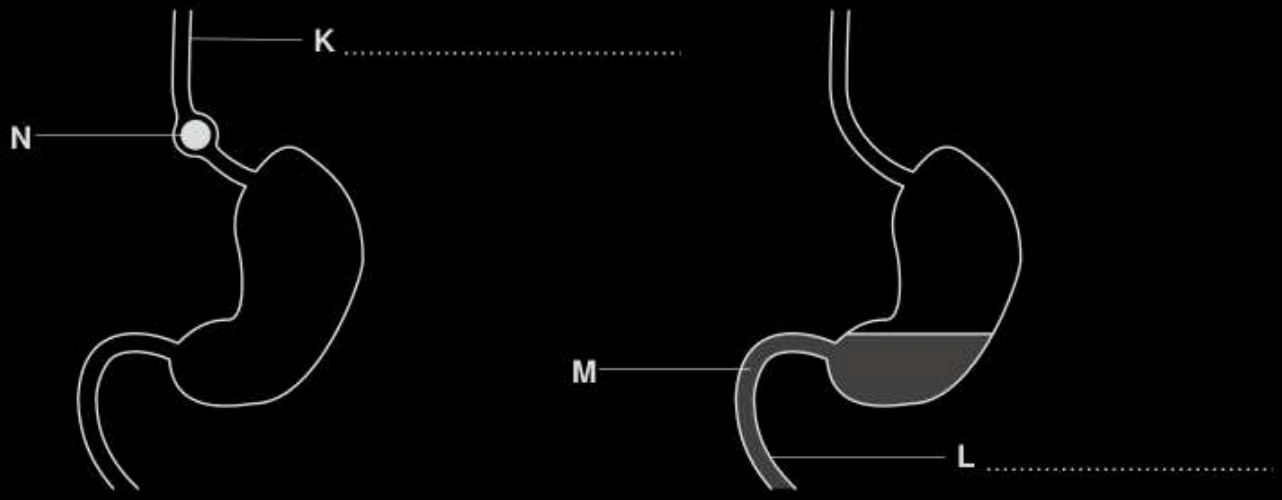


Fig. 5.1

(a) On Fig. 5.1, label structures **K** and **L**. [2]

(b) The food consisted solely of meat and potatoes. By placing ticks (✓) in the appropriate boxes in Table 5.2, show how the major components of the food compare at positions **M** and **N**.

Table 5.2

	more at M than at N	less at M than at N	almost the same at M and N
starch			
protein			
fibre			

[3]

Use

Fig. 5.3 shows the same food at **O**, and 24 hours later, at **P**.

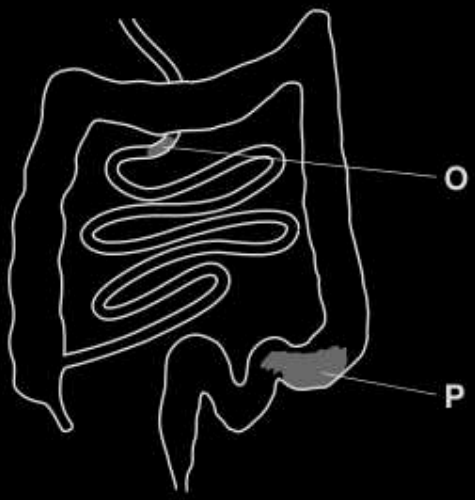


Fig. 5.3

(c) (i) Explain what has happened to the protein between **O** and **P**.

.....

.....

.....[2]

(ii) Name the region of the alimentary canal which will contain fibre in the highest proportion and give reasons for your answer.

region of alimentary canal .....

reasons .....

.....

.....

.....[3]

[Total: 10]

Section B

Answer **all** the questions including questions 6, 7 and 8 **Either** or 8 **Or**.

Write your answers on the separate answer paper provided.

5 **(a)** **K** oesophagus / gullet ;
L duodenum / small intestine ; [2]

(b) starch – almost the same ;
protein – less ;
fibre – almost the same ; [3]
(A any symbols, but R more than one per line)

(c) **(i)** digested / changed / broken down / ref. enzyme action ;
to amino acids / (poly)peptides AW ;
absorption (available only with ref. amino acids) ; [max 2]

(ii) colon / rectum / large intestine ;
no suitable enzyme / not digested ;
ref. cellulose / lignin ;
therefore no absorption / egested as faeces / used in peristalsis
(A all other things, or named substances, are absorbed) ; [max 3]

[Maximum mark for Section A: 50]

May/June 2007 · Paper 2 · Variant 1 · Question 7

7 **(a)** With reference to the circulatory system, name an example of each of the following:

(i) a cell,

(ii) a tissue and

(iii) an organ. [3]

(b) Name the main blood vessels carrying blood to and from the liver and describe any differences that there may be between the composition of the blood as it enters and leaves the liver. [7]

[Total: 10]

Question 8 is in the form of an **Either**/**Or** question. Answer only question 8 **Either** or question 8 **Or**.

7 **(a)** (R any examples which do not come from the circulatory system, mark the first in a list.)

(i) blood cell / any correct named e.g. ;

(ii) muscle or correct named e.g. (A blood / capillary) ;

(iii) heart / blood vessel or named (R capillary) ; [3]

(b) hepatic artery (A name, disregard context) ;

carries more / most oxygen / oxygenated blood ;
more or ref. to being used or destroyed in the liver + insulin / glucagon / adrenaline / hormones ;

hepatic vein (A name, disregard context) ;
carries carbon dioxide / deoxygenated (A with ref. HPV) ;
urea / ref. more amino acids in hepatic artery ;
any reasonable ref. to comparative numbers of RBCs + named blood vessel ;
(hepatic) portal vein / HPV (A name, disregard context) ;
#digested foods / any named ;
#ions / vitamins / water ;
any correct ref. to glucose levels in hepatic artery / vein ; [max 7]
(#both marks require a ref. somewhere to direction of blood flow)

[Total: 10]

October/November 2006 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 The table shows some characteristics of four types of cell.

Which cell could be a root hair cell?

	nucleus	chloroplast
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key
✓ = present
✗ = absent

October/November 2006 · Paper 1 · Variant 1 · Question 2

Correct answer: D

2 Which processes can take place in a root hair cell when oxygen is **not** available?

A active transport only

B diffusion only

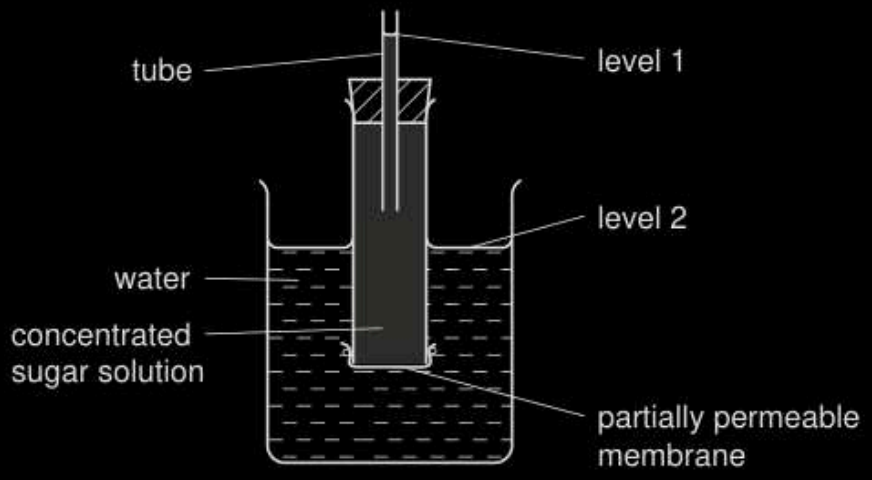
C active transport and osmosis only

D diffusion and osmosis only

October/November 2006 · Paper 1 · Variant 1 · Question 3

Correct answer: D

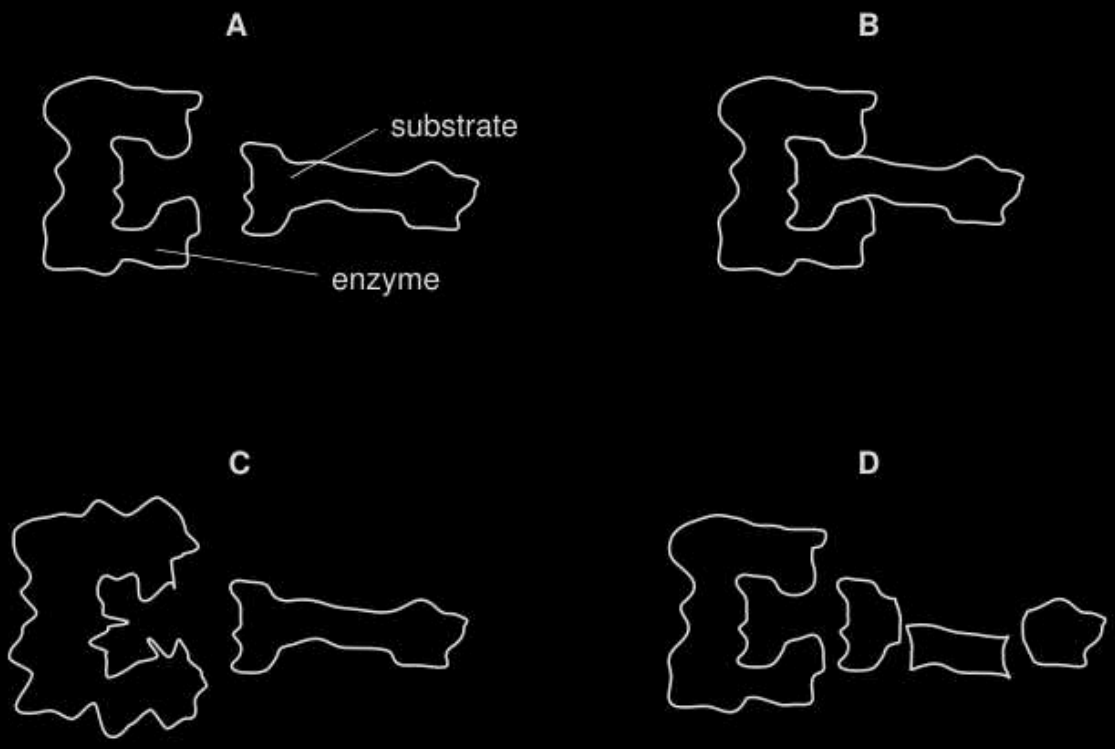
3 The diagram shows apparatus used to investigate osmosis.



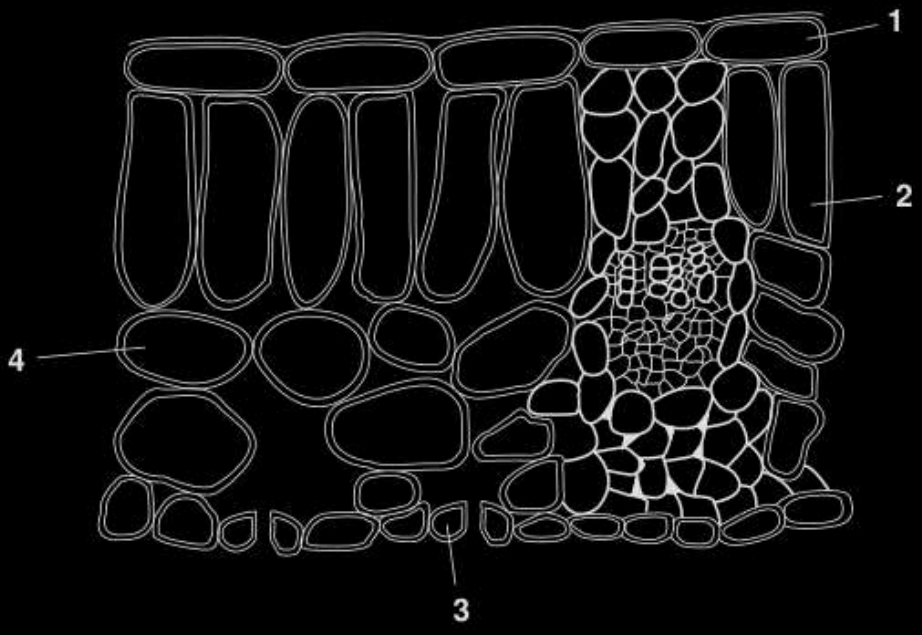
Which molecules will move across the partially permeable membrane and which changes in levels will occur?

	molecules	level 1	level 2
A	sugar	fall	rise
B	water	fall	rise
C	sugar	rise	fall
D	water	rise	fall

Correct answer: C



Correct answer: D



Which cells would contain chloroplasts?

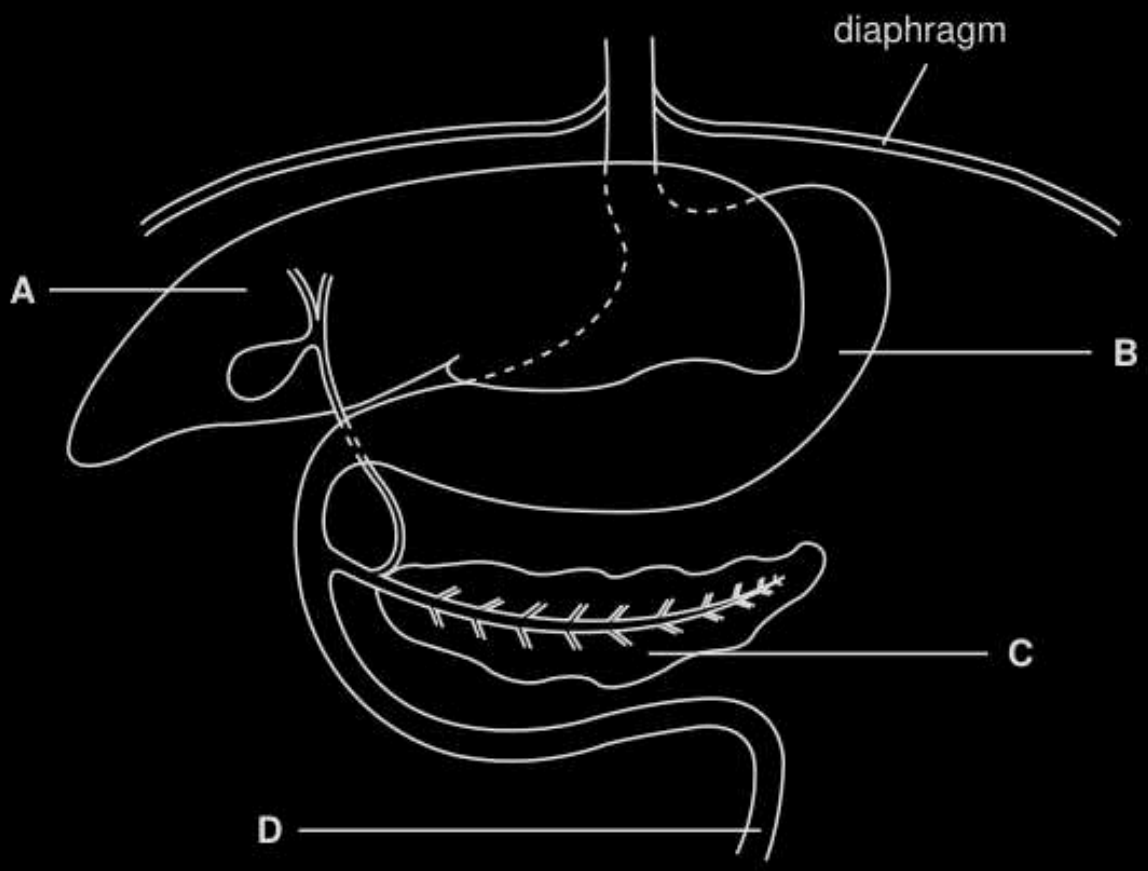
A **1** and **2** only

B **2** only

C **2** and **3** only

D **2**, **3** and **4** only

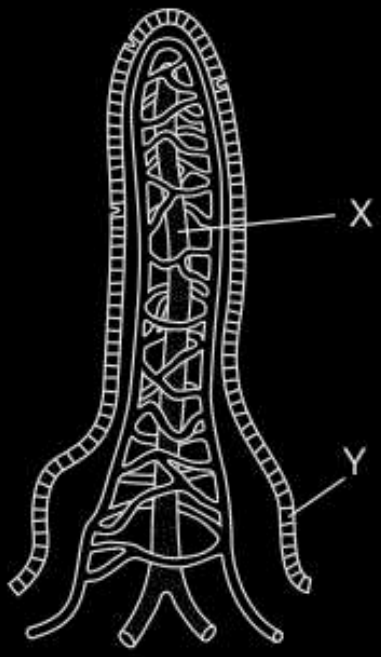
Correct answer: B



October/November 2006 · Paper 1 · Variant 1 · Question 11

Correct answer: C

11 The diagram shows a section through a villus.



What is a function of structure X and of structure Y?

	X	Y
A	to absorb amino acids	to digest starch
B	to carry blood	to secrete mucus
C	to transport fats	to secrete enzymes
D	to transport glucose	to help peristalsis

October/November 2006 · Paper 1 · Variant 1 · Question 12

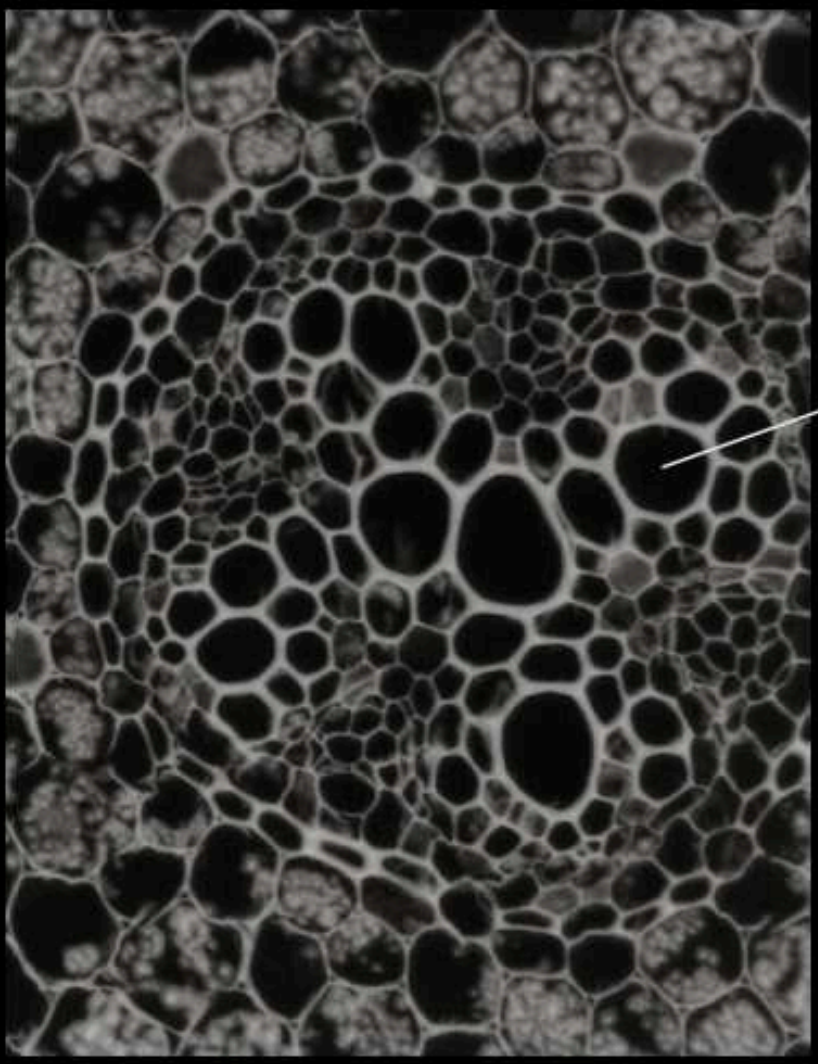
Correct answer: C

12 Which of the following environmental conditions would cause rapid transpiration?

	air	light	temperature
A	damp	bright	cold
B	damp	dim	warm
C	dry	bright	warm
D	dry	dim	cold

Correct answer: D

13 The photomicrograph shows part of a section of a plant.



Samples of the contents of cell X were tested.

What results are expected?

	Benedict's reagent	iodine
A	+	+
B	+	–
C	–	+
D	–	–

key

+ = positive result

– = negative result

October/November 2006 · Paper 1 · Variant 1 · Question 14

Correct answer: B

14 How does tissue fluid differ from blood plasma?

A Tissue fluid contains dissolved food.

B Tissue fluid contains less protein.

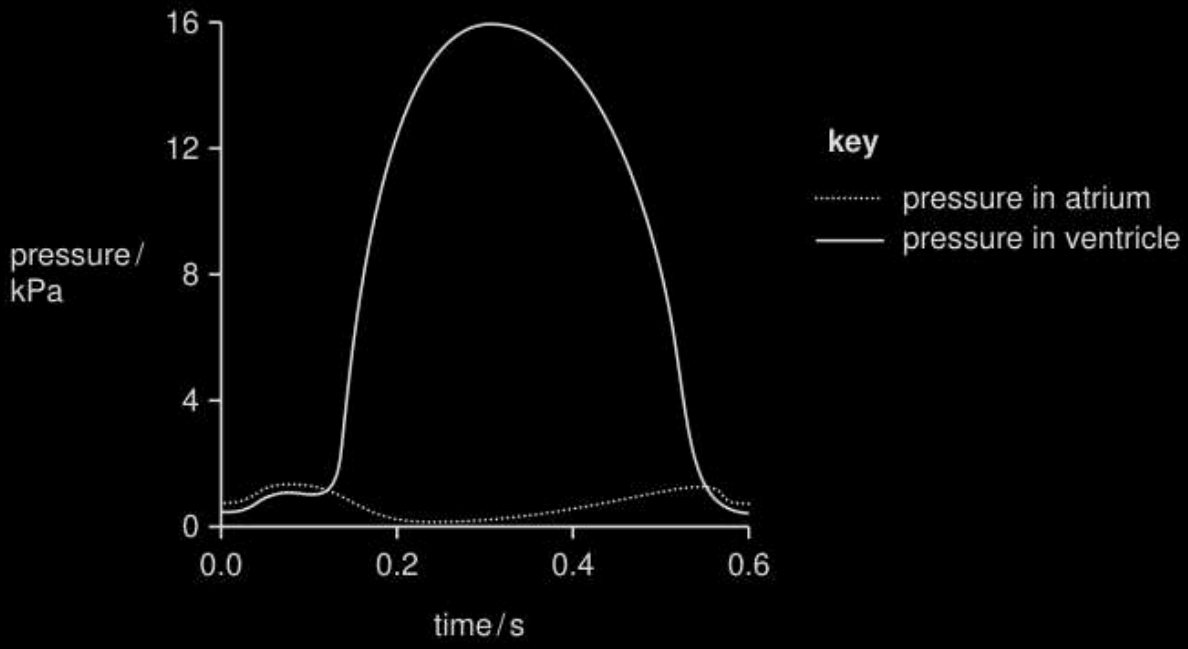
C Tissue fluid does not contain dissolved oxygen.

D Tissue fluid does not contain white blood cells.

October/November 2006 · Paper 1 · Variant 1 · Question 15

Correct answer: C

15 The graph shows pressure changes in the left side of the heart, while at rest, during a single heart beat.



What is the number of times that this person's heat beats in one minute, while at rest?

A 60 **B** 70 **C** 100 **D** 120

October/November 2006 · Paper 1 · Variant 1 · Question 16

Correct answer: A

16 Which statement explains why humans are said to have a double circulation?

A As blood circulates it passes twice through the heart.

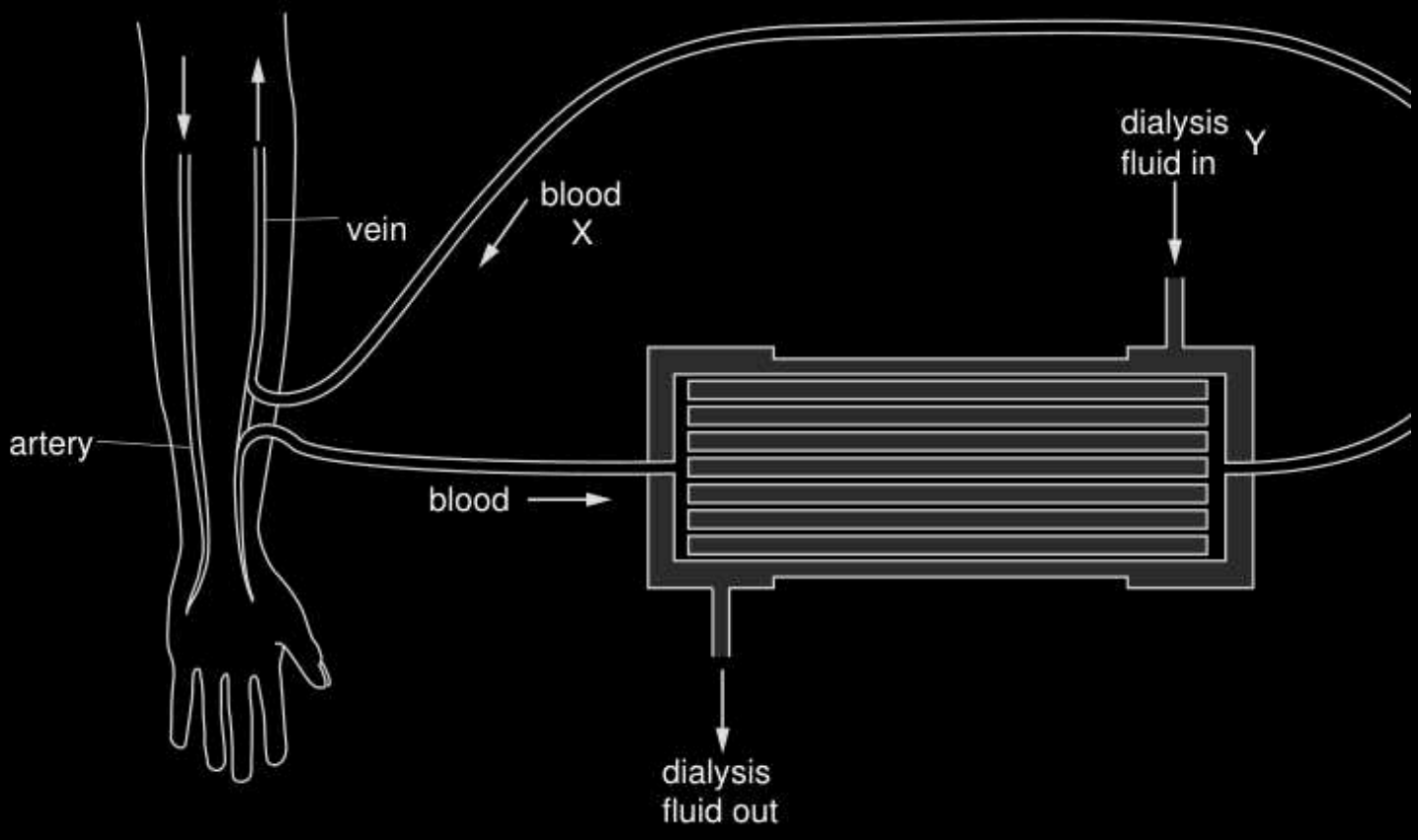
B Each side of the heart has two chambers.

C Each side of the heart has two valves.

D There are two different sets of arteries leaving the heart.

October/November 2006 · Paper 1 · Variant 1 · Question 20

Correct answer: C



Which substances have the lowest concentration at X and the highest concentration at Y?

	lowest at X	highest at Y
A	glucose	salts
B	salts	glucose
C	urea	water
D	water	urea

October/November 2006 · Paper 1 · Variant 1 · Question 21

Correct answer: C

21 The body can regulate both its temperature and the amount of water in its cells.

What are these processes?

A assimilation

B excretion

C homeostasis

D osmosis

October/November 2006 · Paper 1 · Variant 1 · Question 22

Correct answer: C

22 In temperature control of the body, which types of neurones carry information from skin receptors to the brain, and from the brain to sweat glands?

	from skin receptors to the brain	from the brain to sweat glands
A	motor	sensory
B	relay	motor
C	sensory	motor
D	sensory	relay

October/November 2006 · Paper 1 · Variant 1 · Question 23

Correct answer: C

23 What are the labels for the parts of the brain?

	1	2	3
A	cerebellum	cerebrum	pituitary gland
B	cerebrum	cerebellum	pituitary gland
C	pituitary gland	cerebrum	cerebellum
D	pituitary gland	cerebellum	cerebrum

October/November 2006 · Paper 1 · Variant 1 · Question 24

Correct answer: A

24 What structures cover the pupil of a human eye?

A conjunctiva and cornea

B conjunctiva and sclera

C cornea and retina

D retina and sclera

October/November 2006 · Paper 1 · Variant 1 · Question 26

Correct answer: C

26 Tar and carbon monoxide are present in tobacco smoke.

What are their effects on health?

	tar	carbon monoxide
A	causes high blood pressure	damages haemoglobin
B	causes high blood pressure	is addictive
C	causes lung cancer	damages haemoglobin
D	causes lung cancer	is addictive

October/November 2006 · Paper 1 · Variant 1 · Question 28

Correct answer: C

28 Using the key, which organism is a virus?

1. Has a cell wall..... go to 2
   Does not have a cell wall..... go to 3
2. Cell wall is made of chitin..... organism **A**
   Cell wall is made of cellulose..... organism **B**
3. Has a protein coat..... organism **C**
   Has a cell membrane..... organism **D**

October/November 2006 · Paper 1 · Variant 1 · Question 34

Correct answer: B

34 A plant has 20 chromosomes in its leaf cells. The plant reproduces both sexually and asexually.

What is the correct number of chromosomes in the gametes and in cells used for asexual reproduction?

	number of chromosomes	
	gametes	cells used for asexual reproduction
A	10	10
B	10	20
C	20	10
D	20	20

October/November 2006 · Paper 1 · Variant 1 · Question 38

Correct answer: B

38 Dillip and Shabnam made four statements about themselves.

	Dillip	Shabnam
1	I am a boy.	I am a girl.
2	I am 150 cm tall.	I am 153 cm tall.
3	I am not very good at games.	I am good at games.
4	My blood group is A.	My blood group is AB.

Which statements describe characteristics that show discontinuous variation?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

October/November 2006 · Paper 1 · Variant 1 · Question 39

Correct answer: C

39 Flower colour is controlled by a single pair of alleles, the allele for red colour is dominant to the allele for white colour.

A plant homozygous for red flower colour is crossed with a plant homozygous for white flower colour. All the resulting plants have red flowers (F_1 generation).

When these are crossed with each other, 18 plants were obtained. 12 plants have red flowers and 6 have white flowers (F_2 generation).

What ratio is expected in the F_2 generation and what ratio has been obtained?

	expected ratio red to white	obtained ratio red to white
A	1:1	1:1
B	1:1	2:1
C	3:1	2:1
D	3:1	3:1

Correct answer: C

40 A recessive homozygote is crossed with a heterozygote.

What will be the phenotypes of the F_1 generation?

A all dominant

B 0.75 dominant 0.25 recessive

C 0.5 dominant 0.5 recessive

D 0.25 dominant 0.5 heterozygous 0.25 recessive

1 **(a)** Describe how the products of fat digestion enter a person's blood.

.....

.....

.....[4]

Fig. 1.1 shows apparatus at the start of an experiment to investigate the digestion of an emulsion of fat droplets in water by enzyme **A**.

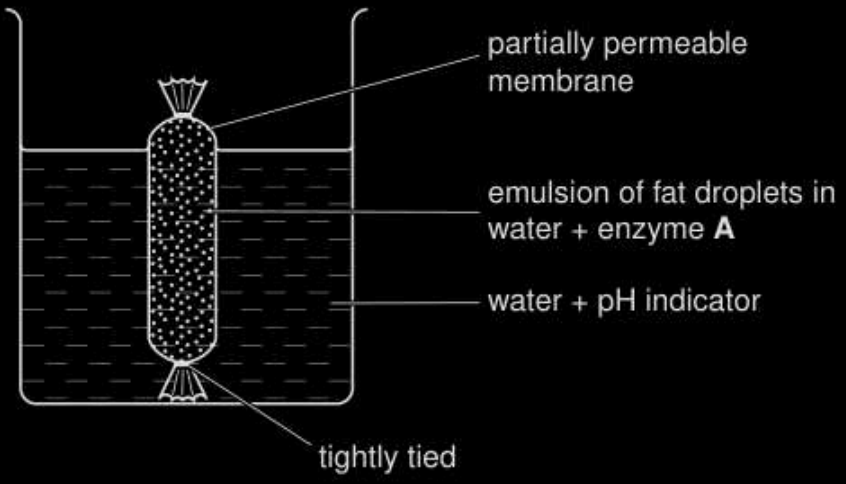


Fig. 1.1

The pH indicator is green in a pH of 7, blue when the pH is above 7 and red when it is below 7. The apparatus is kept at 40 °C for 20 minutes during which time the indicator changes from green to red.

(b) **(i)** State the identity of enzyme **A**. .....

(ii) Explain why the apparatus was kept at 40 °C.

.....

.....[2]

(c) Name the products of digestion of the emulsion by enzyme **A**.

.....[2]

(d) Describe the processes which led to the change in pH.

Use

.....

.....

.....

.....[3]

[Total: 11]

1 **(a)** (micro)villi
diffusion/or good description of
thin wall/epithelim (R ref. cell wall)
into lacteals/lymph (Ignore capillaries)
lymph returned to blood — max 4

(b) **(i)** lipase/steapsin — 1

(ii) optimum/best AW + for enzyme/lipase action (I ref. body temp) — 1

(c) fatty acids
glycerol/glycerine/propantriol — 2

(d) fatty acids/ref. smaller molecules
(can) pass through membrane/Visking tubing
concentration gradient/diffusion
ref acidity of or lowers pH of water/ref acidity of molecules — max 3

Total = 11

3 Fig. 3.1 shows how a blocked blood vessel in the heart can be by-passed using an artificial blood vessel.

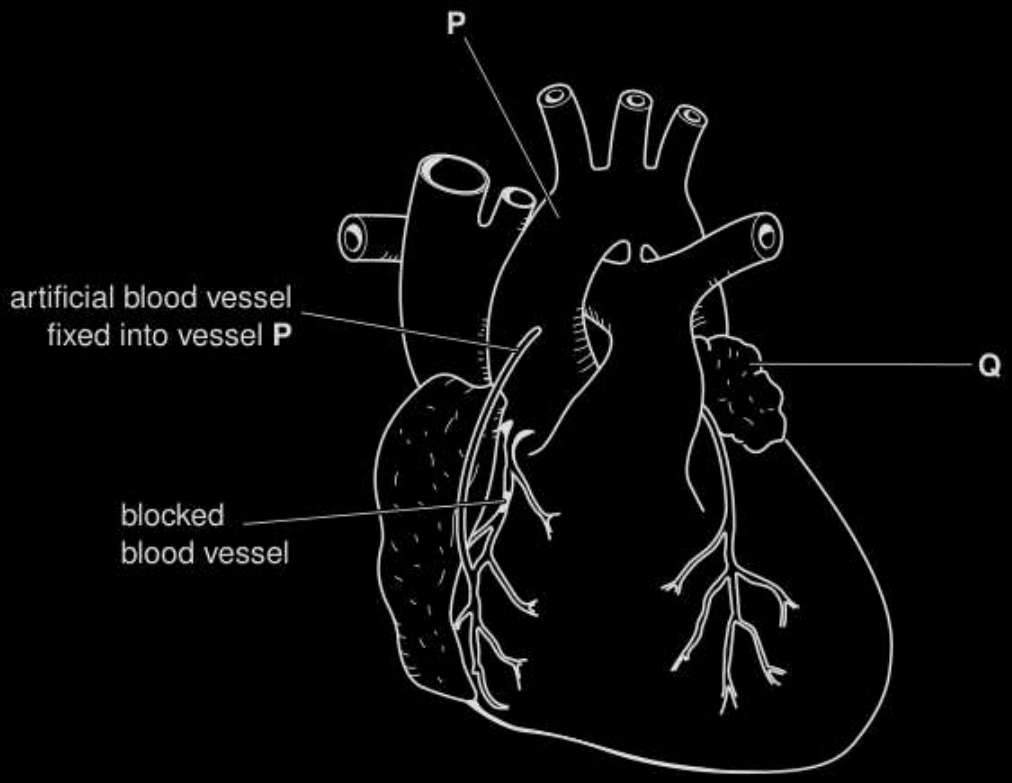


Fig. 3.1

(a) **(i)** Name the blocked blood vessel. .....[2]

(ii) Identify structures **P** and **Q** on Fig. 3.1.

P .....

Q ..... [2]

(b) Sometimes, instead of an artificial blood vessel being used for the graft, a vein is taken from elsewhere in the patient's body.

Suggest two ways in which a vein might not be as suitable for carrying blood to the heart muscle as the blood vessel in the heart before it became blocked.

1 .....

2 .....[2]

(c) Fig. 3.2 shows the same blood vessel, as in **(a)(i)** but this time the blockage is being treated with the use of a 'stent'.

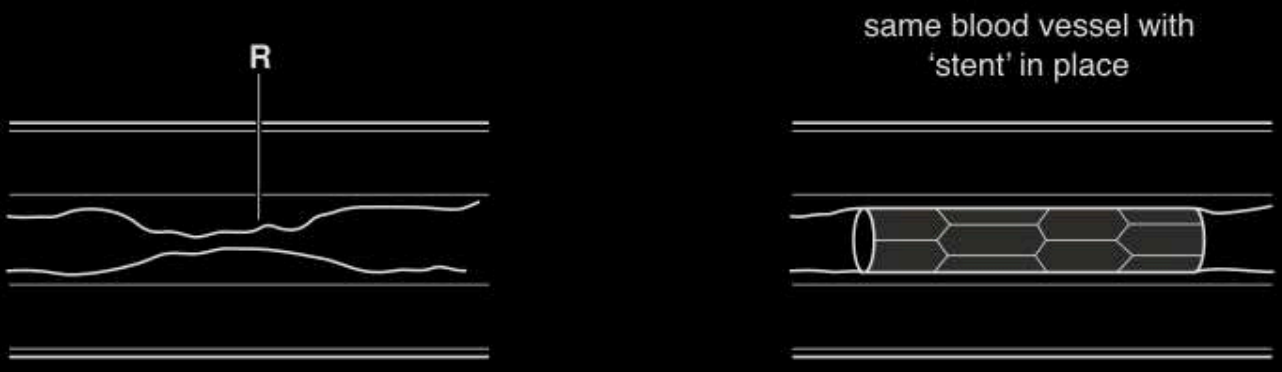


Fig. 3.2

(i) Name two components of the material that is causing the blockage at **R**.

1 .....

2 .....[2]

(ii) Suggest and explain why patients are given 'anti-platelet' drugs before inserting the stent.

.....

.....

.....[2]

[Total: 10]

3 **(a)** **(i)** coronary
artery 2

(ii) P aorta (-tic arch)
Q left + atrium/auricle 2

(b) (mark the first, one per line)
2 from: thinner or weaker + walls/valves/pressure ref.
(A less muscular + walls) 2

(c) **(i)** (mtf,opl) (A platelets)
2 from: fat/cholesterol/blood cells/clot(ted blood) 2
(A atheroma for 1 mark) (A ref. fibres/fibrin)

(ii) natural response to damage or injury is for blood to clot AW
platelets + release enzymes/cause fibrinogen to change to fibrin
therefore drug prevents clotting (or implied – platelets cause blood to clot) max 2

Total = 10

5 Fig. 5.1 shows the development of a zygote into a fetus.

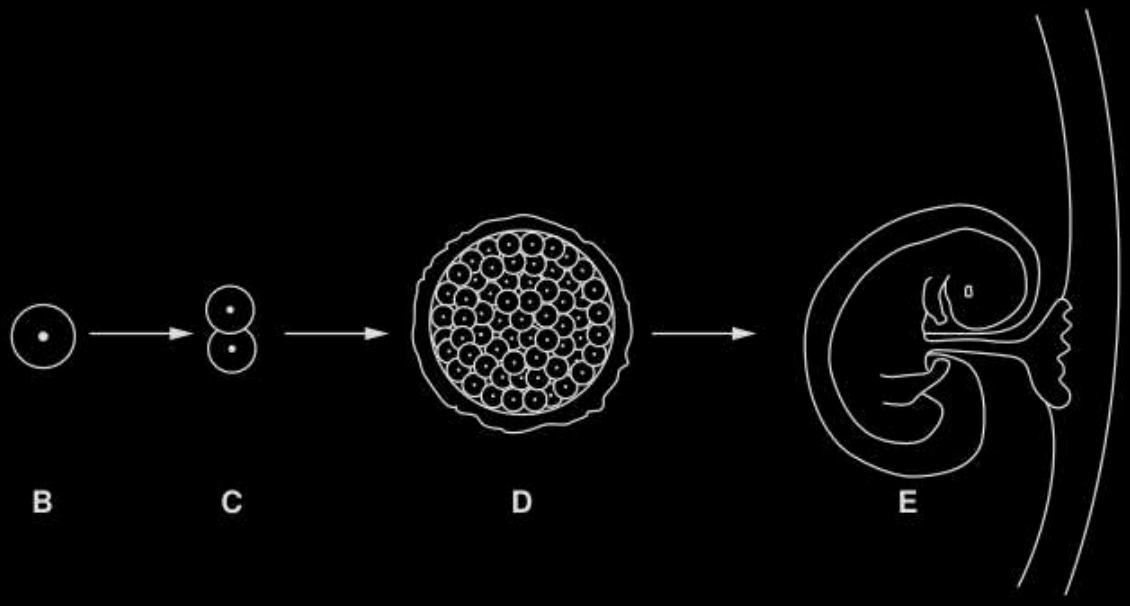


Fig. 5.1

(a) State whereabouts in the female reproductive system structure **C** would be found.

.....[1]

(b) Name the type of cell division occurring in Fig. 5.1.

.....[1]

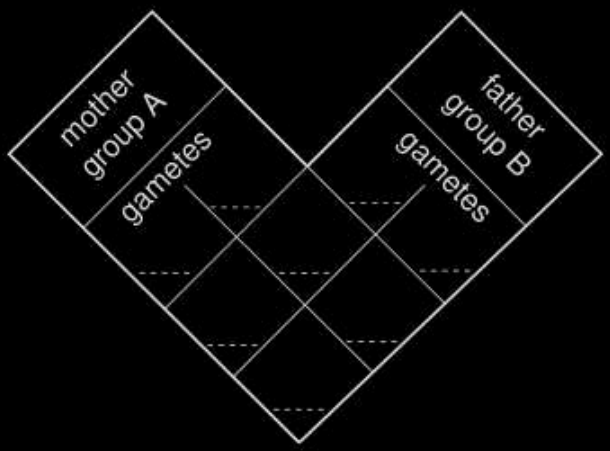
(c) Describe what happens to structure **D** to enable it to continue to develop into a fetus.

.....

.....

.....[2]

(d) If the embryo's mother had blood group A and its father had blood group B, complete the grid below to show how the embryo could have blood group O. Use I^A, I^B and I^o to represent the alleles and clearly indicate any embryo that has blood group O.



[4]

[Total: 8]

Section B

Answer **three** questions.

Question **8** is in the form of an **Either/Or** question. Only one part should be answered.

Write your answers on the separate answer paper provided.

5 **(a)** oviduct/Fallopian tube (mark the first) — 1
(A description of oviduct)

(b) mitosis (-totic) — 1

(c) implantation AW
in lining/endometrium (R wall)
of uterus/womb
differentiation AW/ref. placental devpt. (I fetal membranes) — max 2

(d) mother's gametes# shown as I^A and I^o } max 1 if wrong
father's gametes# shown as I^B and I^o } symbols used
*grid correctly filled (A e.c.f. if gametes incorrectly shown)
square I^oI^o identified as the embryo — 4
(A genetic diagram, but #ensure gametes are not shown as parental genotypes –
*this mark not available on a genetic diagram)

Total = 8

Total for Section A = **50**

| | GCE O LEVEL - OCT/NOV 2006 | 5090 | 2 |

Section B

October/November 2006 · Paper 2 · Variant 1 · Question 6

6 **(a)** State the differences between osmosis and active transport. [3]

(b) Describe an example of active transport in

(i) a plant and

(ii) a human being.

In each case, explain the importance of the transported substances. [7]

[Total: 10]

6 **(a)** (A any three facts linked to a process)
osmosis is simple diffusion
partially/selectively/semi-permeable membrane
correct refs. in each case to:
no energy/energy required
water only/ions AW or larger molecules
down/against concentration gradient
(R along) — max 3

(b) **(i)** salts ions or one named (A minerals) (R nutrients)
from soil
ref. root hairs
to make proteins/amino acids/DNA
chlorophyll (R chloroplasts)
even when scarce in surrounding soil AW
(could be ref. to concentration gradient) — max 4 for (i)

(ii) glucose
amino acids
uptake from gut
through (micro)villi
*for protein (or named) manufacture (linked to amino acids)
*for respiration/correct energy ref. (linked to glucose) — max 7 for (b) (mark 1st.2)

Or kidneys; reabsorption; 2 named salts or any 2 from glucose, amino acids, urea, salts (unspecified or one named);;
ref. osmoregulation; any one of those marked * above; — max 7 for (b)

Total = 10

May/June 2006 · Paper 1 · Variant 1 · Question 1

Correct answer: A

1 The concentration of nitrate ions in a root cell is higher than in the surrounding soil solution.

How do the nitrate ions move into the root cell?

A active transport

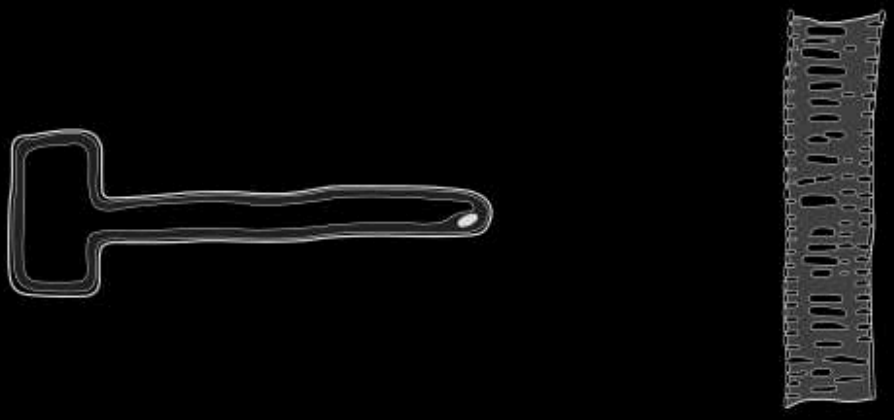
B diffusion

C osmosis

D transpiration

Correct answer: C

2 The diagram shows two plant cells.



cell X cell Y

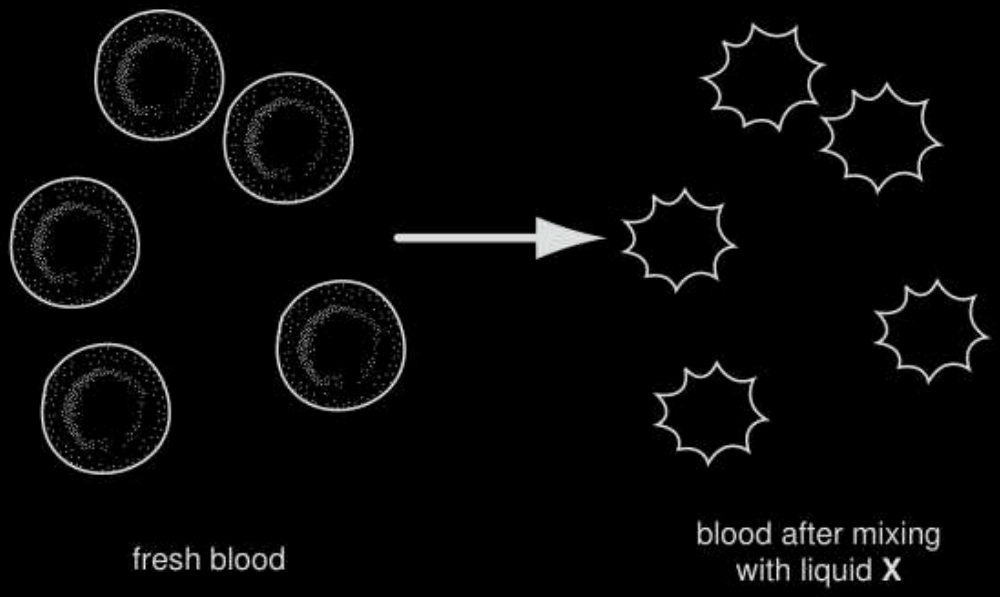
Plant cells become turgid when they take up water by osmosis.

In which of these cells does being turgid provide support?

	cell X	cell Y
A	no	no
B	no	yes
C	yes	no
D	yes	yes

May/June 2006 · Paper 1 · Variant 1 · Question 3

Correct answer: D



Which statement describes the water potential of liquid **X**?

A It is equal to that of pure water.

B It is equal to that of the cell cytoplasm.

C It is higher than that of the cell cytoplasm.

D It is lower than that of the cell cytoplasm.

May/June 2006 · Paper 1 · Variant 1 · Question 4

Correct answer: C

4 Starch is digested to maltose by the enzyme amylase.

According to the 'lock and key' hypothesis, which is the 'key' and which is the 'lock'?

	'key'	'lock'
A	amylase	maltose
B	amylase	starch
C	starch	amylase
D	starch	maltose

May/June 2006 · Paper 1 · Variant 1 · Question 7

Correct answer: A

7 What is the function of each type of plant cell?

	palisade cells	phloem cells	root hair cells
A	photosynthesis	sugar transport	ion uptake
B	photosynthesis	sugar transport	transpiration
C	transpiration	photosynthesis	ion uptake
D	transpiration	photosynthesis	sugar transport

May/June 2006 · Paper 1 · Variant 1 · Question 8

Correct answer: C

8 The table shows the nutrients in different parts of a meal.

Which food would be most useful in preventing constipation?

	food	energy kJ	protein g	fat g	carbohydrate g	fibre g
A	apple juice	163	0.1	0	9.4	0
B	ripe banana	466	1.5	0.4	27	4.9
C	salad sandwich	1054	19	7.3	27	6.1
D	toffee bar	458	2.1	3.3	19	1.1

May/June 2006 · Paper 1 · Variant 1 · Question 10

Correct answer: A

10 Tennis players often eat bananas during long matches.

Which nutrient in a banana is important during the match?

A carbohydrate

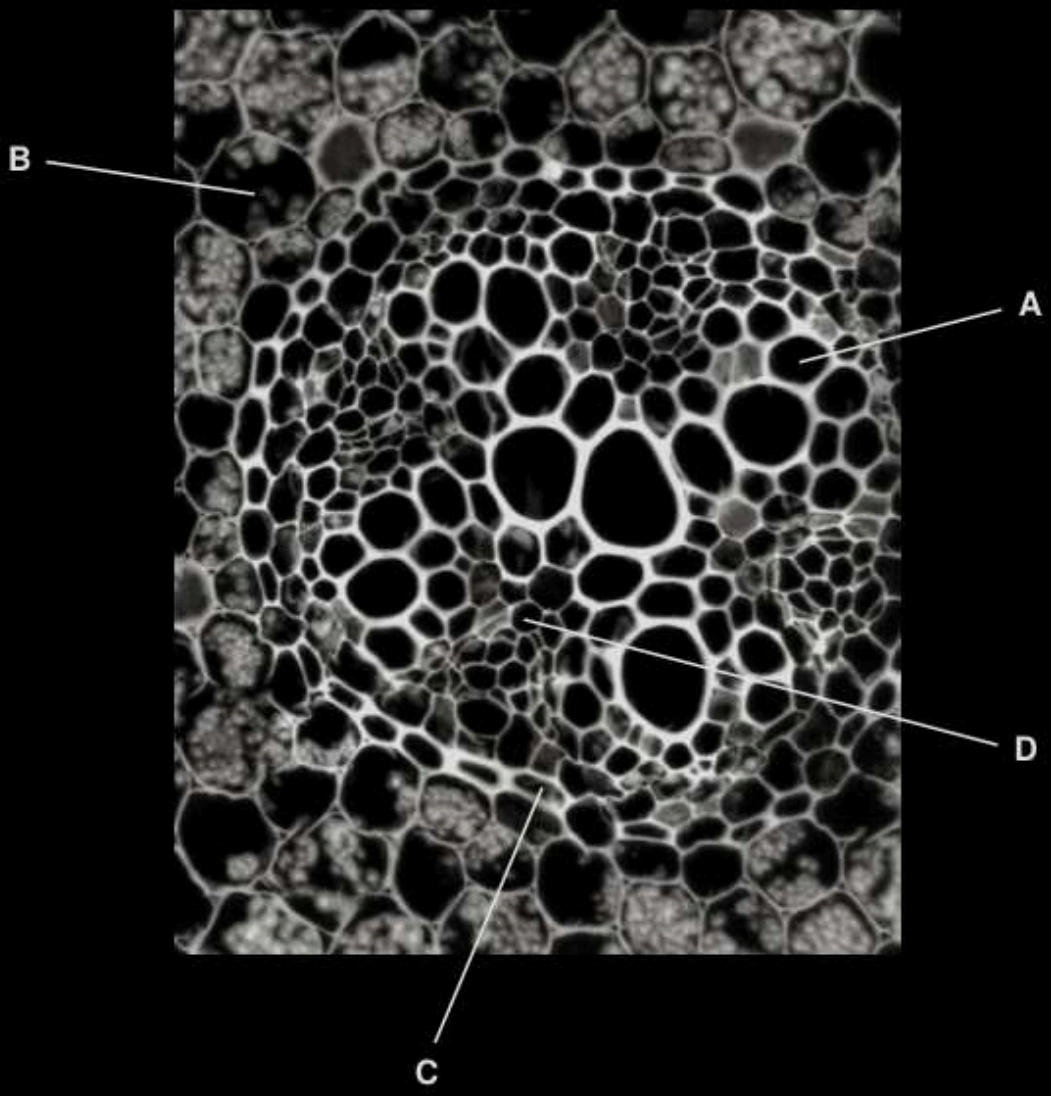
B fibre

C iron

D vitamin C

May/June 2006 · Paper 1 · Variant 1 · Question 11

Correct answer: D



May/June 2006 · Paper 1 · Variant 1 · Question 12

Correct answer: B

12 During translocation in plants, what substance is moved from which organ to which organ?

	substance	from	to
A	sucrose	anthers	stigmas
B	sucrose	leaves	roots
C	water	roots	leaves
D	water	soil	root hairs

May/June 2006 · Paper 1 · Variant 1 · Question 13

Correct answer: A

13 Which chambers of the human heart contain oxygenated blood?

A left atrium and left ventricle

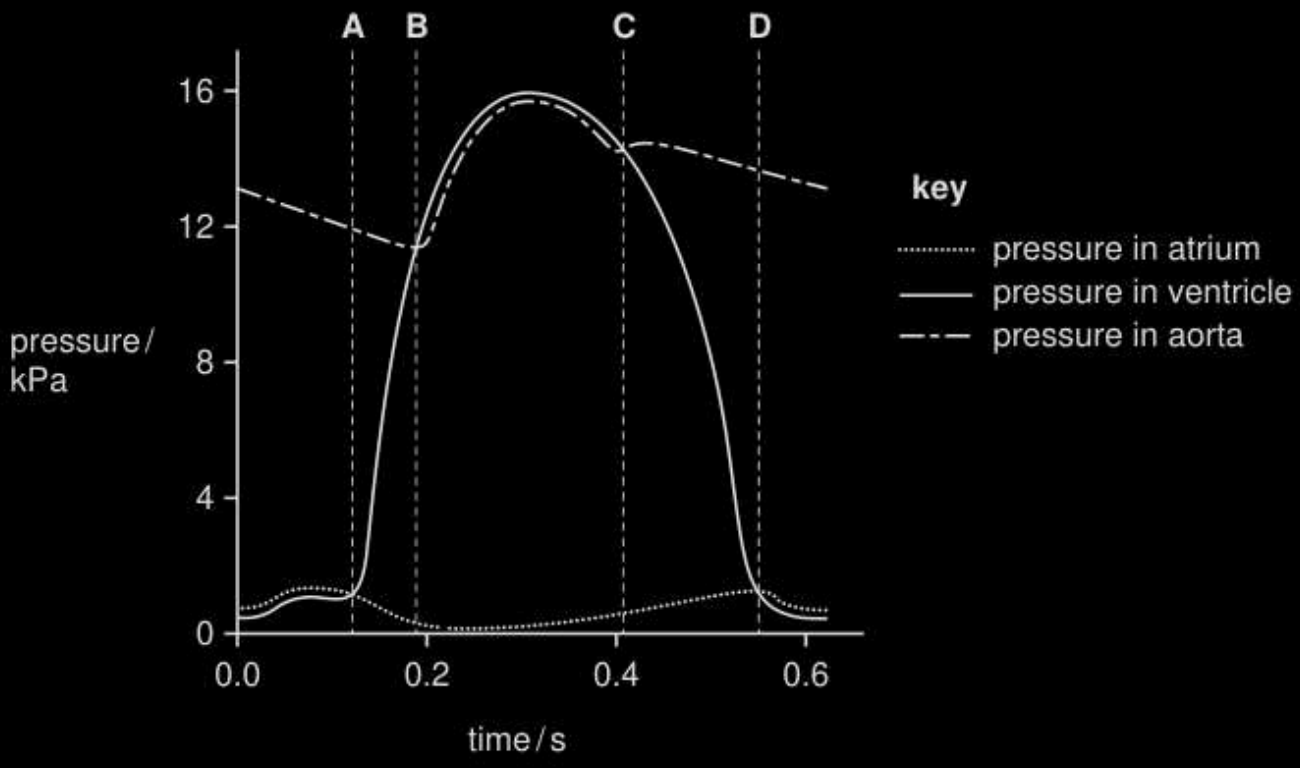
B left atrium and right ventricle

C right atrium and left ventricle

D right atrium and right ventricle

May/June 2006 · Paper 1 · Variant 1 · Question 14

Correct answer: B



May/June 2006 · Paper 1 · Variant 1 · Question 15

Correct answer: B

15 What causes the transfer of materials between capillaries and tissue fluid?

A active transport

B blood pressure

C capillarity

D osmosis

May/June 2006 · Paper 1 · Variant 1 · Question 19

Correct answer: B

19 Through which sequence does carbon dioxide pass as it leaves the lungs?

A alveolar wall → in the alveoli → blood → capillary wall

B blood → capillary wall → alveolar wall → in the alveoli

C capillary wall → blood → in the alveoli → alveolar wall

D in the alveoli → alveolar wall → capillary wall → blood

May/June 2006 · Paper 1 · Variant 1 · Question 20

Correct answer: B

20 What does the hypothalamus control?

A adrenaline secretion

B body temperature

C heart beat

D memory

May/June 2006 · Paper 1 · Variant 1 · Question 21

Correct answer: D

21 Which of these four mechanisms that lower the body temperature is the slowest to occur?

A dilation of surface capillaries

B relaxation of hair erector muscles

C secretion of sweat

D thinning subcutaneous fat layer

May/June 2006 · Paper 1 · Variant 1 · Question 22

Correct answer: A

22 When a person is frightened which responses will occur?

A	adrenaline released	heart beat increases	eye pupils dilate
B	adrenaline released	blood glucose increases	urine production increases
C	insulin released	breathing rate increases	peristalsis stops
D	insulin released	eye pupils dilate	saliva secretion stops

May/June 2006 · Paper 1 · Variant 1 · Question 24

Correct answer: D

24 Which of these substances do **not** occur naturally in the human body?

adrenaline	heroin	nicotine	penicillin
1	2	3	4

A 2 and 3 only

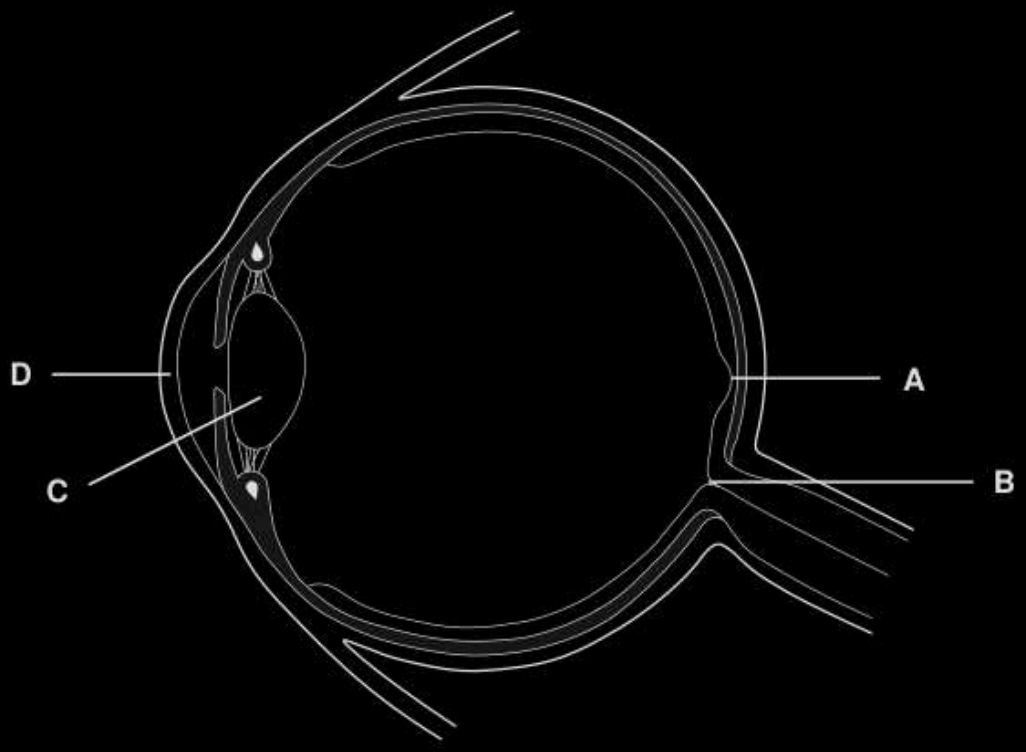
B 1, 2 and 3 only

C 1, 3 and 4 only

D 2, 3 and 4 only

May/June 2006 · Paper 1 · Variant 1 · Question 25

Correct answer: A



May/June 2006 · Paper 1 · Variant 1 · Question 26

Correct answer: B

26 'This group of organisms have long thread like bodies. Their cells contain nuclei. Some of them produce antibiotics.'

Which group of organisms is being described?

A bacteria

B fungi

C plants

D viruses

May/June 2006 · Paper 1 · Variant 1 · Question 27

Correct answer: D

27 Fermenters that are used to produce single cell protein may be closed (batch) or open (continuous).

Which set of fermenter properties is correct?

	closed	open
A	growth rate of culture remains at optimum	fermenting microorganisms must be added all the time
B	growth rate of culture remains at optimum	product can be removed all the time
C	vessel must be sterilised after each separate cycle	fermenting microorganisms must be added all the time
D	vessel must be sterilised after each separate cycle	product can be removed all the time

May/June 2006 · Paper 1 · Variant 1 · Question 28

Correct answer: C

28 Which statement is **not** correct?

A Anaerobic respiration releases less energy than aerobic respiration.

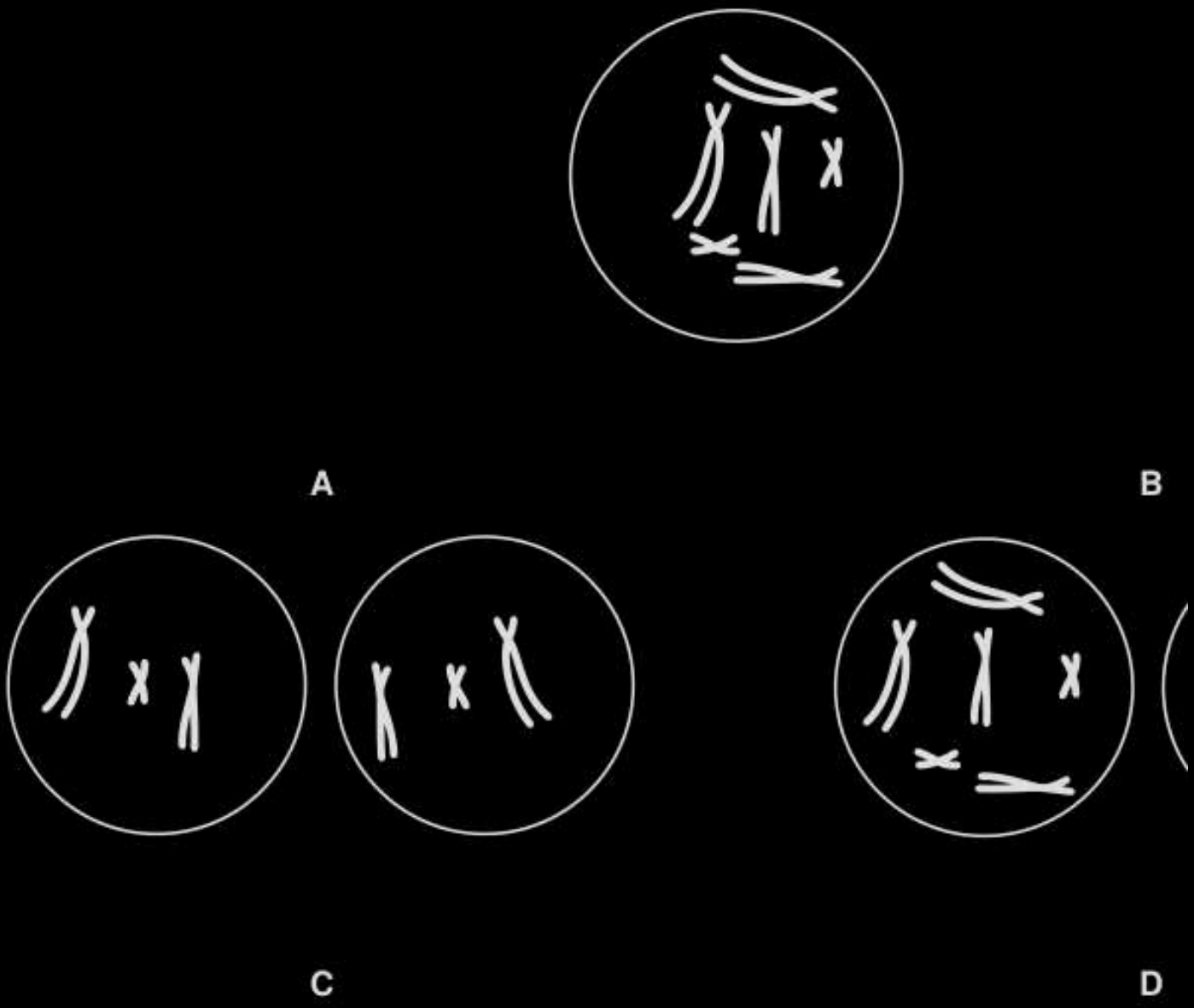
B The sun is the principal source of energy input into biological systems.

C Energy flowing through biological systems is recycled.

D Active transport is an energy-consuming process.

May/June 2006 · Paper 1 · Variant 1 · Question 33

Correct answer: B



May/June 2006 · Paper 1 · Variant 1 · Question 37

Correct answer: C

37 Which of these may be heterozygous?

A a cell which is haploid

B an allele of a gene

C an organism with a dominant phenotype

D an organism with a recessive genotype

May/June 2006 · Paper 1 · Variant 1 · Question 38

Correct answer: B

38 Cells contain long thread-like structures that carry genetic information.

What are these structures called?

A alleles

B chromosomes

C genes

D nuclei

May/June 2006 · Paper 1 · Variant 1 · Question 39

Correct answer: A

39 Farmers crossed two breeds of cattle, the Jersey from Europe and the Sahiwal from Africa. For many generations, the farmers picked out the offspring with the highest milk yields to breed the next generation.

Which phrase best describes this process?

A artificial selection

B discontinuous variation

C evolution

D natural selection

May/June 2006 · Paper 1 · Variant 1 · Question 40

Correct answer: D

40 A textbook states that because human insulin is a single …X…, its production is controlled by a single …Y… .

What are X and Y?

	X	Y
A	carbohydrate	DNA molecule
B	carbohydrate	gene
C	protein	DNA molecule
D	protein	gene

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

1 Fig. 1.1 shows the cell membrane of an animal and a plant cell.

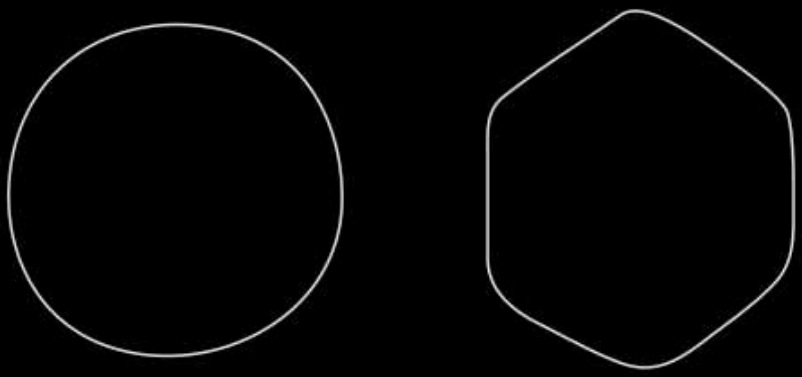


Fig. 1.1

(a) On Fig. 1.1, draw and label

(i) one other structural feature that is found in **both** cells,

(ii) **two** features found in a plant cell but **not** in an animal cell.

[3]

Use

Table 1.1 shows the average amounts of protein, fat and carbohydrate found in some plant and some animal tissues (expressed as a percentage of the total mass).

Table 1.1

	protein	fat	carbohydrate
plant tissues	5	1	30
animal tissues	20	15	1

(b) (i) Explain why there is a high percentage of carbohydrate in the plant tissues.

.....

.....[2]

(ii) Name an animal tissue that would have much more than the 1% carbohydrate shown in Table 1.1. Explain your choice of tissue.

tissue .....

explanation .....

.....[3]

(iii) Suggest reasons for the differences in the protein and in the fat content of the plant and animal tissues.

protein .....

.....

fat .....[3]

[Total: 11]

1 **(a)** *mark awarded only if structure is in a plausible position*

(i) nucleus/cytoplasm/(shown in both cells) ;

(ii) Any 2 from: chloroplast/wall/(cell) sap/membrane ;; **[3]**

(b) **(i)** photosynthesis ;

manufactures or stores CHO/sugar/glucose/cellulose ; **[2]**

(ii) liver/muscle ;

*storage/cells contain ;

*glycogen ; **[3]**
(**mark separately from liver/muscle mark*)

(iii) muscles largely protein/contain fat ;

skin largely protein ;

animal cells/tissues/skin stores fat ;

fat insulates against heat loss ; **[max 3]**

May/June 2006 · Paper 2 · Variant 1 · Question 2

2 **(a)** **(i)** State the name for chemicals that are secreted by glands into the blood and that affect organs elsewhere in the body.

.....[1]

(ii) State the term used to describe the organs that are affected by these chemicals.

.....organs [1]

(b) Listed below are a number of responses shown by the human body:

A the amount of saliva produced increases

B the concentration of blood glucose falls

C the concentration of blood glucose rises

D the pupils become smaller

E the rate of heart beat increases

Which two of these are responses to the release of adrenaline?

1. .....

2. .....[2]

Use

(c) Fig. 2.1 shows the blood glucose concentrations of three different people, **G**, **H** and **I**, over a 48-hour period.

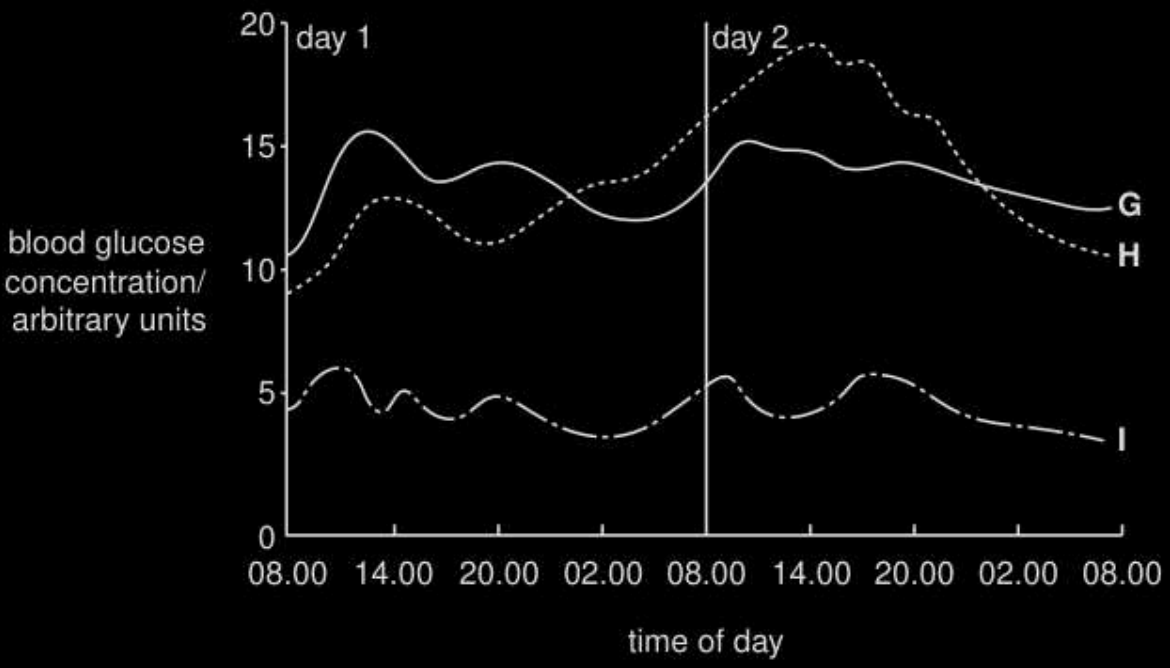


Fig. 2.1

Two of the people are diabetic. The pancreas of one of these produces small quantities of insulin. In the other, the pancreas produces no insulin. The third person is **not** diabetic.

Using the information in Fig. 2.1 and giving a reason for your answer in each case, identify

(i) the person without diabetes .....

reason: .....

(ii) the person who produces no insulin .....

reason: .....

[4]

(d) Diabetes can be treated by injection of insulin. Insulin can now also be taken by breathing it in. Suggest how insulin taken by breathing it in enters the blood.

.....

.....

.....[3]

[Total: 11]

2 **(a)** hormones ;

target ; **[2]**

(b) 1. C/blood glucose rises ;

2. E/heart beat increases ; **[2]**

(c) **(i)** I (or otherwise identified) ;

greatest control over sugar level/smallest fluctuations AW ;

at lowest (blood glucose) level ;

(ii) H ;

greatest fluctuations/little control over sugar levels ; **[max. 4]**

(d) lungs ;

alveoli/air sacs ;

diffusion ;

into capillaries ; **[max 3]**

3	(a)	pollen	;	[1]
	(b)	by insect	;	
		grain sticky/rough AW	;	[2]
	(c)	fusion AW	;	
		male and female	;	
		gametes/nuclei/sex cells	;	
		fertilisation	;	
		ref. food storage	;	
		mitosis/growth	;	
		embryo development	;	[max 4]
	(d)	(seed) dispersal (ignore refs. to wind)	;	[1]
4	(a)	any 2 from : urination/exhaling or breathing out/faeces/ bleeding or crying or vomiting	;;	[2]
	(b) (i)	higher when walking (or v.v.)/quoted figures	;	
		more energy/heat released/raises body temperature	;	[2]
	(ii)	lower when clothed (or v.v.)/quoted figures	;	
		greater humidity next to skin/(v.v.) less skin exposed/ clothes deflect or absorb heat AW	;	[2]
	(iii)	higher in sun (or v.v.)/quoted figures	;	
		higher temperatures in direct sunlight/higher rate of evaporation	;	[2]
	(c)	more energy released/respiration/work done by	;	
		muscles	;	[2]
5	(a) (i)	105	;	[1]
	(ii)	genes/alleles (A any given pair of contrasted characters)	;	[1]
	(iii)	to prevent choice/bias/so results are random	;	[1]
	(b) (i)	red + **W**	;	[1]
	(ii)	ref. both cubes and both flowers being the same/heterozygous AW/ the only way to produce both colours of offspring/gives all genetic combinations AW	;	[1]

(c) **(i)** T t + red* ;

(x) tt + yellow* ;

<u>gametes</u> ;

gametes correctly shown (need be once only for tt) ;

genotypes of offspring correctly derived (* A colour tie-up here) ; **[max. 4]**

(ii) 3 x T + 3 x t on one cube + 6 x t on the other ; **[1]**

The maximum for Section A = 50 marks

Section B

6 **(a)** active site ;

of specific shape AW ;

substrate ;

fit/are complementary ;

any ref. enzyme/substrate complex being like lock and key ;

stress on substrate molecule ;

product formed ;

also works in reverse ; **[max 5]**

(b) reaction rate increases ;

similar to key turning more often ;

more energy/faster movement of molecules ;

active site changes shape ;

proteins are denatured by heat AW ;

permanently ;

reaction stops ;

substrate no longer fits active site ;

key no longer fits lock ; **[max 5]**

[Total = 10]

7 **(a)** named e.g. of bacterial disease ;

named method of administration ;

antibiotics kill only bacteria ;

must continue with course until all bacteria are eliminated ;

named antibiotic ; **[max 3]**

(b) fermenter/vat/large container ;

culture medium ;

addition of organism (fungus or bacterium) ;

controlled temperature ;

provision of oxygen ;

conditions optimum/controlled for maximum production ;

extraction of antibiotic ;

purification ; **[max 7]**

[Total = 10]

8 E **(a)** traps/harnesses/absorbs ;

sunlight ;

energy ;

for photosynthesis ;

which makes carbohydrate AW ; **[max 4]**

(b) large surface area ;

for maximum/rapid ;

uptake of water ;

by osmosis/diffusion ;

of ions/salts/minerals ;

by active transport ;

oxygen ;

for root respiration ; **[max 6]**

[Total = 10]

(a) absorbs + quickly ;

and carries ;

oxygen ;

as oxyhaemoglobin ;

in red blood cells ; **[max 4]**

(b) large surface area ;

uptake from ileum/small intestine ;

*of amino acids ;

*of glucose ;

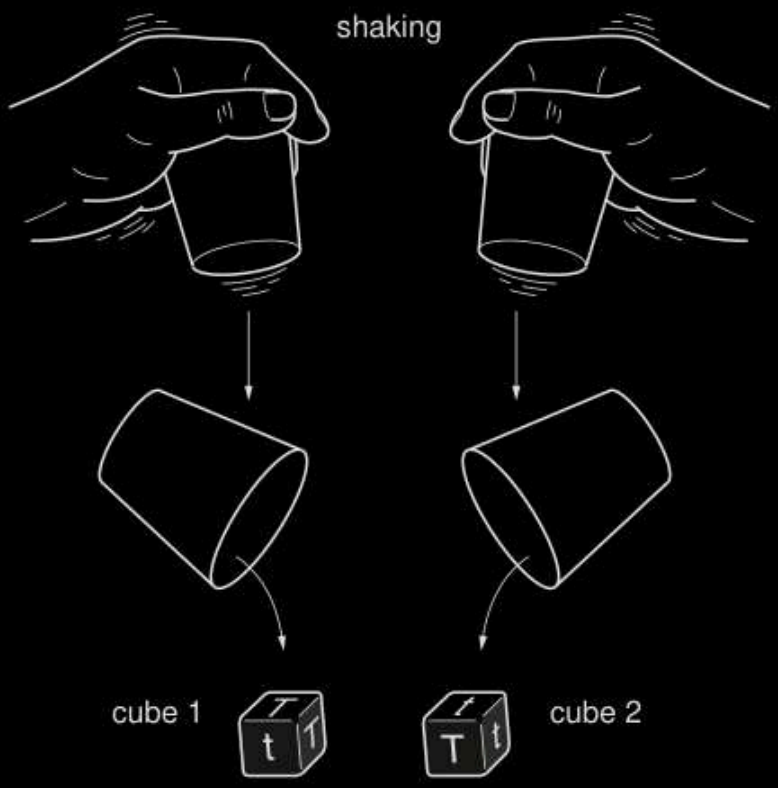
into blood capillaries ;

*fats/fatty acids/glycerol ;

into lacteals ;
(* *allow one for digested foods*) **[max 6]**

[Total = 10]

5 Fig. 5.1 shows a method used by a student to understand how characteristics are inherited when two plants of species **X** are crossed. Both cubes had three of their faces marked with the letter **T** and three with the letter **t**.



In this example, the letters appearing on the upper faces are **Tt**.

Fig. 5.1

The student shook each container and then tipped both cubes out at the same time and recorded the letters appearing on the upper faces of the cubes.

The student tipped both cubes out a total of 405 times.

(a) (i) Complete Table 5.1 to show the results obtained.

Table 5.1

letters appearing on the upper faces of the cubes	**tt**	**TT**	**Tt**
number of times each pair of letters appeared	98	.....	202

[1]

(ii) State what the letters on the faces of the cubes represent.

.....[1]

(iii) State the reason for shaking the containers.

.....[1]

Use

(b) In living plants of species **X**, items **T** and **t** control flower colour. The genetic diagrams below show a sequence of crosses (**V** and **W**) between plants of species **X**, some with red flowers and some with yellow flowers.

cross **V**

parental generation: red × yellow

↓

F_1 generation: all red flowers

cross **W**

F_1 generation from cross **V** : red × red

↓

F_2 generation : red flowers and yellow flowers

Using the information in **Table 5.1** and crosses **V** and **W**,

(i) state which colour is dominant and which of cross **V** and cross **W** could produce the results seen in Table 5.1

dominant colour, ...

cross. .....[1]

(ii) Explain why you chose this cross.

.....

.....[1]

Use

(c) (i) Using the letters **T** and **t**, draw a full genetic diagram for another cross, **Y**, between plants of species **X** shown below.

cross **Y**

parental generation: red × yellow

↓

offspring: red flowers and yellow flowers

[4]

(ii) In the method shown in Fig. 5.1, the student put **T** on three faces and **t** on the other three faces of each cube, so that cube 1 had **T**, **T**, **T**, **t**, **t**, **t** and cube 2 **T**, **T**, **T**, **t**, **t**, **t**.

State what letters the student should write on the faces of each cube to represent the cross in **(c)(i)**.

cube 1 .....

cube 2 ..... [1]

[Total: 10]

Section B

Answer **all** the questions including questions 6, 7 and 8 **Either** or 8 **Or**.

Write your answers on the separate answer paper provided.

5 **(a)** **(i)** 105 ; **[1]**

(ii) genes/alleles (A any given pair of contrasted characters) ; **[1]**

(iii) to prevent choice/bias/so results are random ; **[1]**

(b) **(i)** red + **W** ; **[1]**

(ii) ref. both cubes and both flowers being the same/heterozygous AW/ the only way to produce both colours of offspring/gives all genetic combinations AW ; **[1]**

(c) **(i)** T t + red* ;

(x) tt + yellow* ;

gametes ;

gametes correctly shown (need be once only for tt) ;

genotypes of offspring correctly derived (* A colour tie-up here) ; **[max. 4]**

(ii) 3 x T + 3 x t on one cube + 6 x t on the other ; **[1]**

The maximum for Section A = 50 marks

Section B

May/June 2006 · Paper 2 · Variant 1 · Question 6

6 **(a)** Describe how the action of enzymes can be explained by the 'lock and key' hypothesis. [5]

(b) Describe and explain the effect of a gradual increase in temperature on an enzyme-controlled reaction. You should refer to the 'lock and key' hypothesis. [5]

[Total: 10]

6 **(a)** active site ;

of specific shape AW ;

substrate ;

fit/are complementary ;

any ref. enzyme/substrate complex being like lock and key ;

stress on substrate molecule ;

product formed ;

also works in reverse ; **[max 5]**

(b) reaction rate increases ;

similar to key turning more often ;

more energy/faster movement of molecules ;

active site changes shape ;

proteins are denatured by heat AW ;

permanently ;

reaction stops ;

substrate no longer fits active site ;

key no longer fits lock ; **[max 5]**

[Total = 10]

7 **(a)** named e.g. of bacterial disease ;

named method of administration ;

antibiotics kill only bacteria ;

must continue with course until all bacteria are eliminated ;

named antibiotic ; **[max 3]**

(b) fermenter/vat/large container ;

culture medium ;

addition of organism (fungus or bacterium) ;

controlled temperature ;

provision of oxygen ;

conditions optimum/controlled for maximum production ;

extraction of antibiotic ;

purification ; **[max 7]**

[Total = 10]

8 E **(a)** traps/harnesses/absorbs ;

sunlight ;

energy ;

for photosynthesis ;

which makes carbohydrate AW ; **[max 4]**

(b) large surface area ;

for maximum/rapid ;

uptake of water ;

by osmosis/diffusion ;

of ions/salts/minerals ;

by active transport ;

oxygen ;

for root respiration ; **[max 6]**

[Total = 10]

(a) absorbs + quickly ;

and carries ;

oxygen ;

as oxyhaemoglobin ;

in red blood cells ; **[max 4]**

(b) large surface area ;

uptake from ileum/small intestine ;

*of amino acids ;

*of glucose ;

into blood capillaries ;

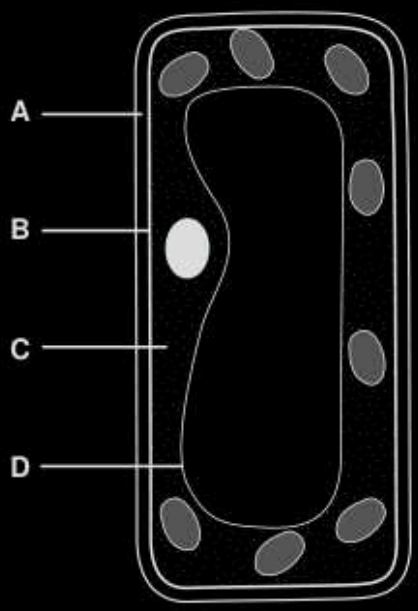
*fats/fatty acids/glycerol ;

into lacteals ;
(* *allow one for digested foods*) **[max 6]**

[Total = 10]

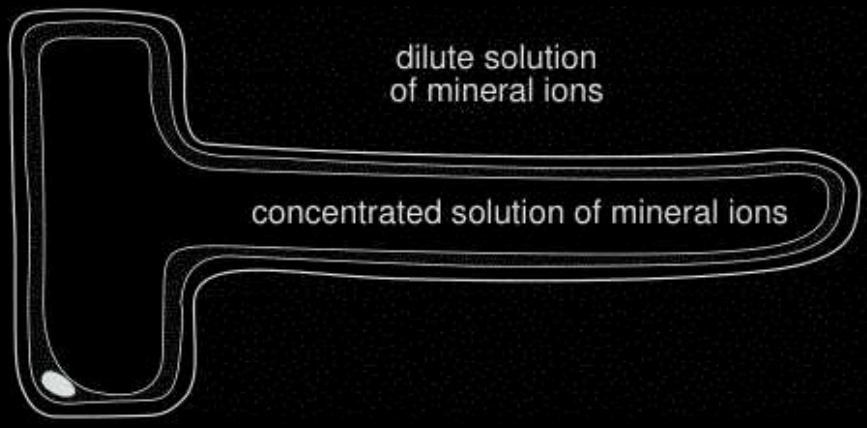
October/November 2005 · Paper 1 · Variant 1 · Question 1

Correct answer: B



Correct answer: A

2 The diagram shows a root hair, surrounded by a dilute solution of mineral ions.



Which statement is correct?

A Water molecules move into the root hair because their concentration is lower inside.

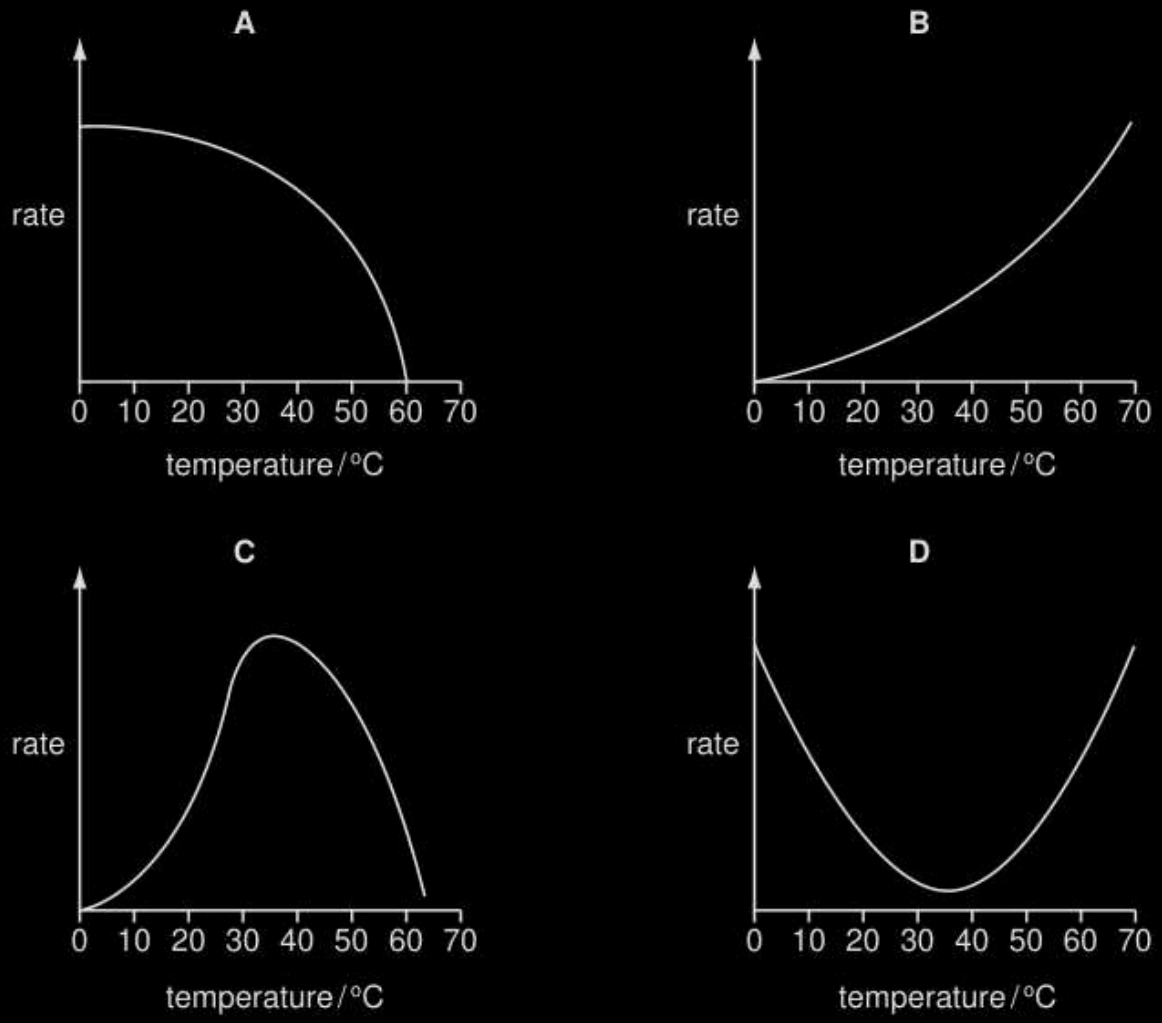
B Water molecules move into the root hair because their concentration is lower outside.

C Water molecules move out of the root hair because their concentration is lower inside.

D Water molecules move out of the root hair because their concentration is lower outside.

October/November 2005 · Paper 1 · Variant 1 · Question 3

Correct answer: C



October/November 2005 · Paper 1 · Variant 1 · Question 4

Correct answer: B

4 Which process can involve active transport?

A carbon dioxide intake through stomata

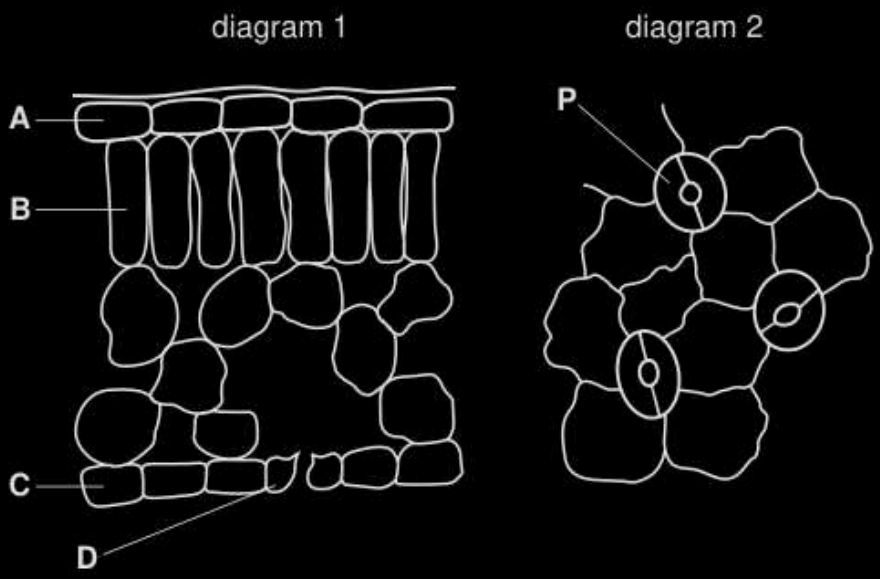
B mineral ion intake through root hairs

C mineral ion transport through xylem vessels

D water leaving mesophyll cells

October/November 2005 · Paper 1 · Variant 1 · Question 5

Correct answer: D



October/November 2005 · Paper 1 · Variant 1 · Question 7

Correct answer: B

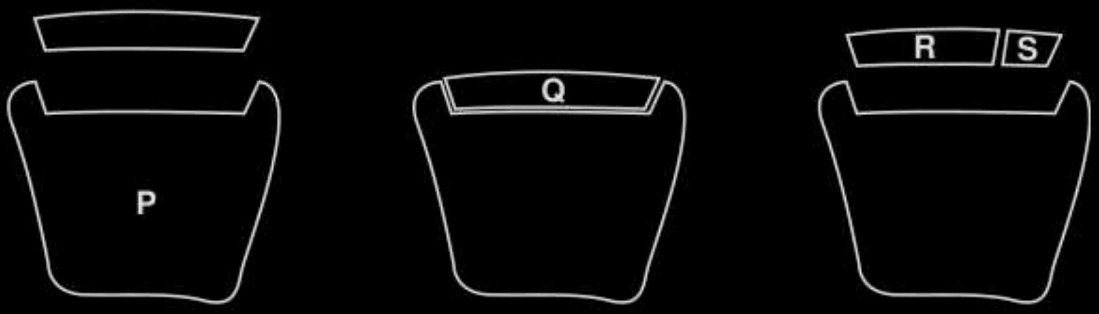
7 Four different foods were tested as shown below and the test results were recorded as positive (+) or negative (–).

Which food contained both glucose and oil?

food	Benedict's test	biuret test	ethanol emulsion test	iodine test
A	+	+	–	–
B	+	–	+	–
C	–	+	–	+
D	–	–	+	+

October/November 2005 · Paper 1 · Variant 1 · Question 10

Correct answer: A



Which are the correct labels?

	amylase	maltose	starch
A	**P**	**S**	**Q**
B	**Q**	**R**	**S**
C	**R**	**Q**	**P**
D	**S**	**P**	**R**

October/November 2005 · Paper 1 · Variant 1 · Question 11

Correct answer: B

11 Which conditions cause the fastest rate of transpiration?

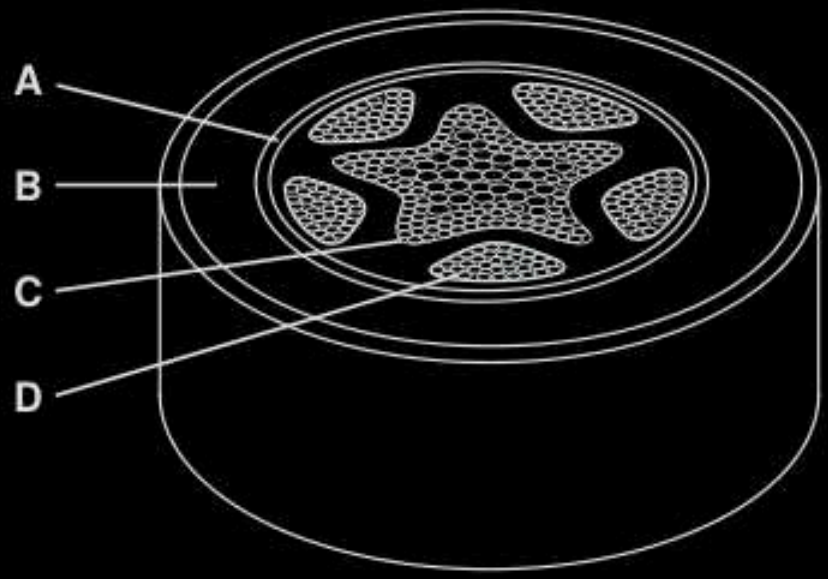
A dry and cold

B dry and warm

C wet and cold

D wet and warm

Correct answer: C

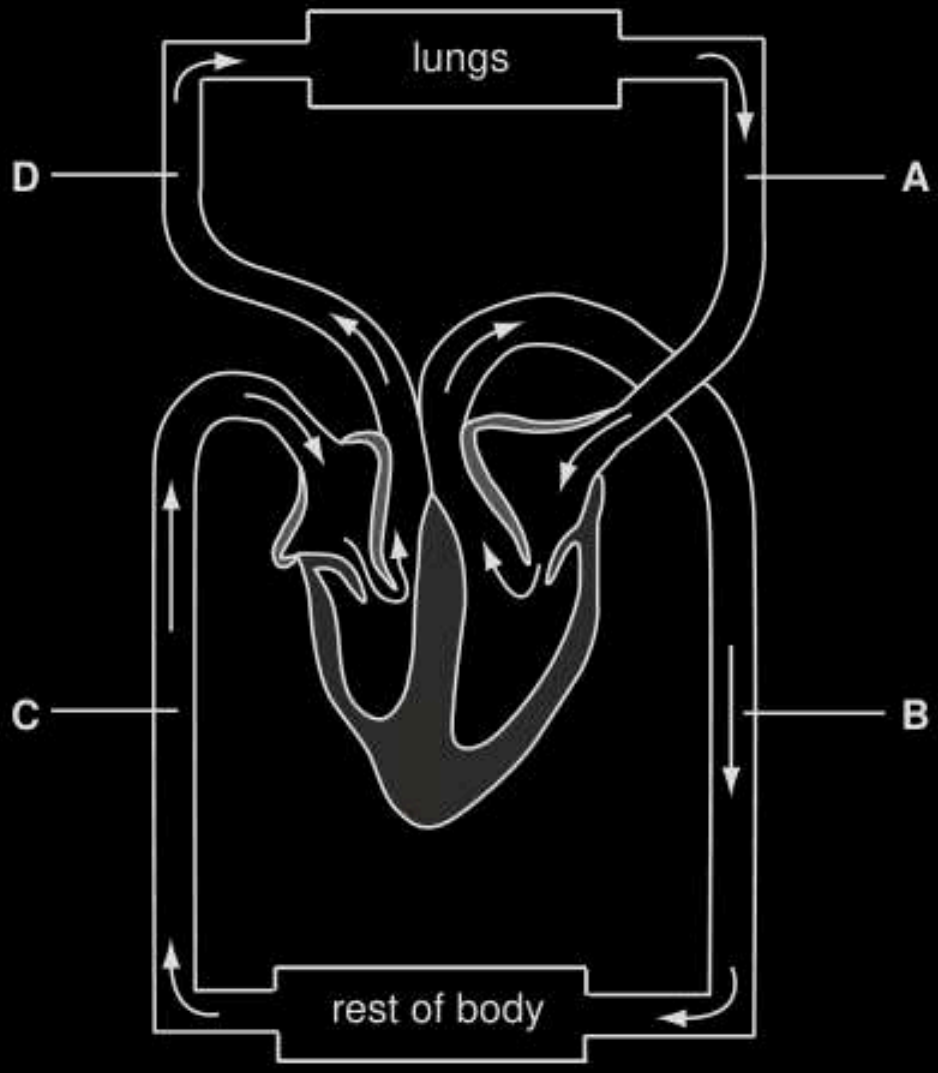


October/November 2005 · Paper 1 · Variant 1 · Question 13

Correct answer: B

13 The diagram shows part of the human circulatory system.

In which vessel is the blood pressure highest?



October/November 2005 · Paper 1 · Variant 1 · Question 14

Correct answer: C

14 Which blood vessel carries blood with the **lowest** concentration of urea?

A hepatic portal vein

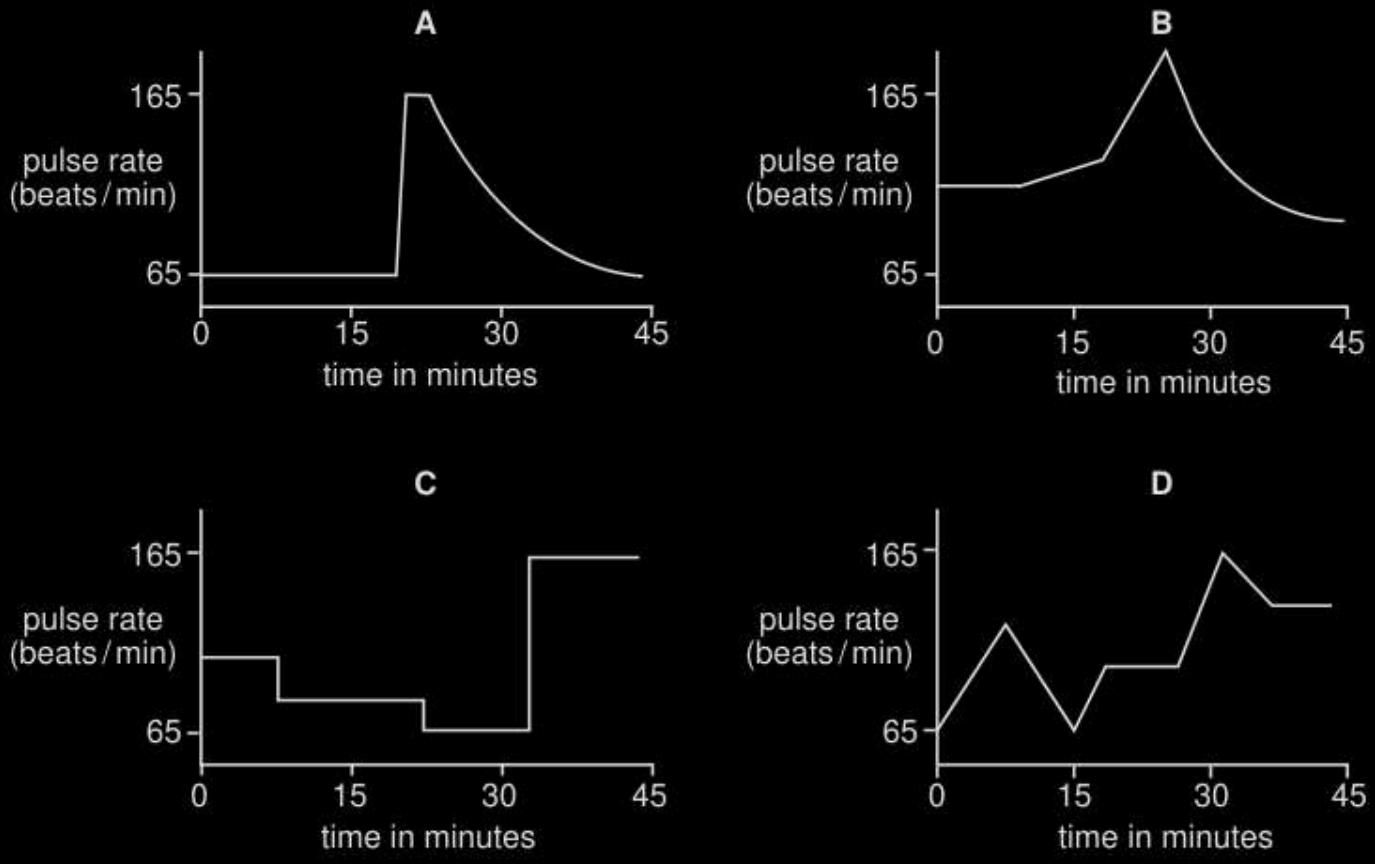
B pulmonary vein

C renal vein

D vena cava

October/November 2005 · Paper 1 · Variant 1 · Question 15

Correct answer: B



October/November 2005 · Paper 1 · Variant 1 · Question 16

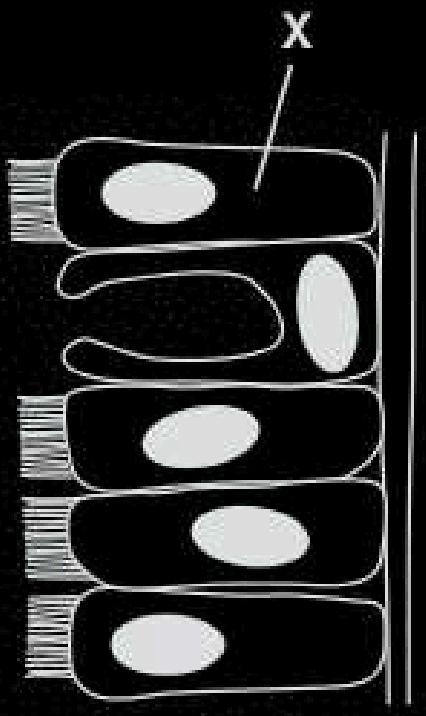
Correct answer: D

16 When the skin is cut the blood clots.

In which order would the components of the blood become involved?

	first →			last
A	fibrin	platelets	red blood cells	fibrinogen
B	fibrinogen	red blood cells	platelets	fibrin
C	platelets	fibrin	fibrinogen	red blood cells
D	platelets	fibrinogen	fibrin	red blood cells

Correct answer: B



What is the function of cell **X**?

A gaseous exchange

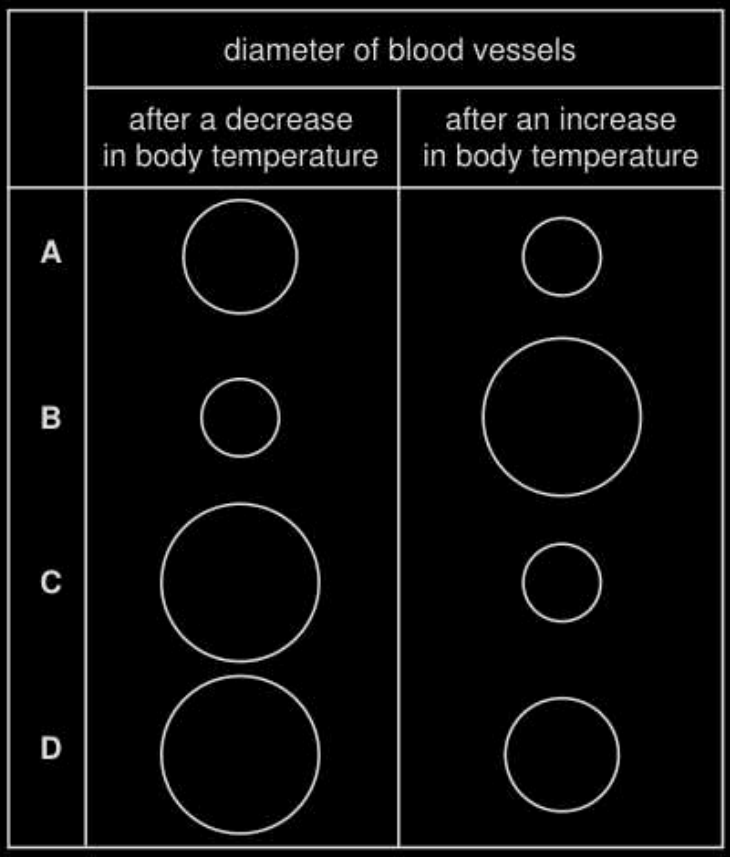
B mucus removal

C phagocytosis

D secretion of mucus

October/November 2005 · Paper 1 · Variant 1 · Question 21

Correct answer: B



October/November 2005 · Paper 1 · Variant 1 · Question 22

Correct answer: D

22 What passes through the membranes of a kidney machine?

A protein and red blood cells

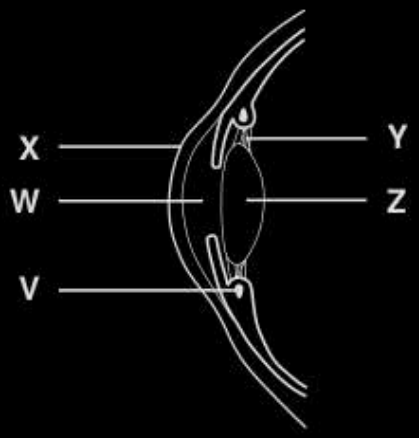
B urea and red blood cells

C water and protein

D water and urea

October/November 2005 · Paper 1 · Variant 1 · Question 23

Correct answer: D



Which parts will change to focus on a distant object?

A	W	X	Y
B	V	X	Z
C	W	Y	Z
D	V	Y	Z

October/November 2005 · Paper 1 · Variant 1 · Question 25

Correct answer: D

25 The table shows some structures found in four different cells.

Which cell is from a fungus?

cell	cell wall	cell membrane	nucleus membrane	chloroplasts
A	✓	✓	✓	✓
B	✗	✓	✓	✓
C	✓	✓	✗	✗
D	✓	✓	✓	✗

key

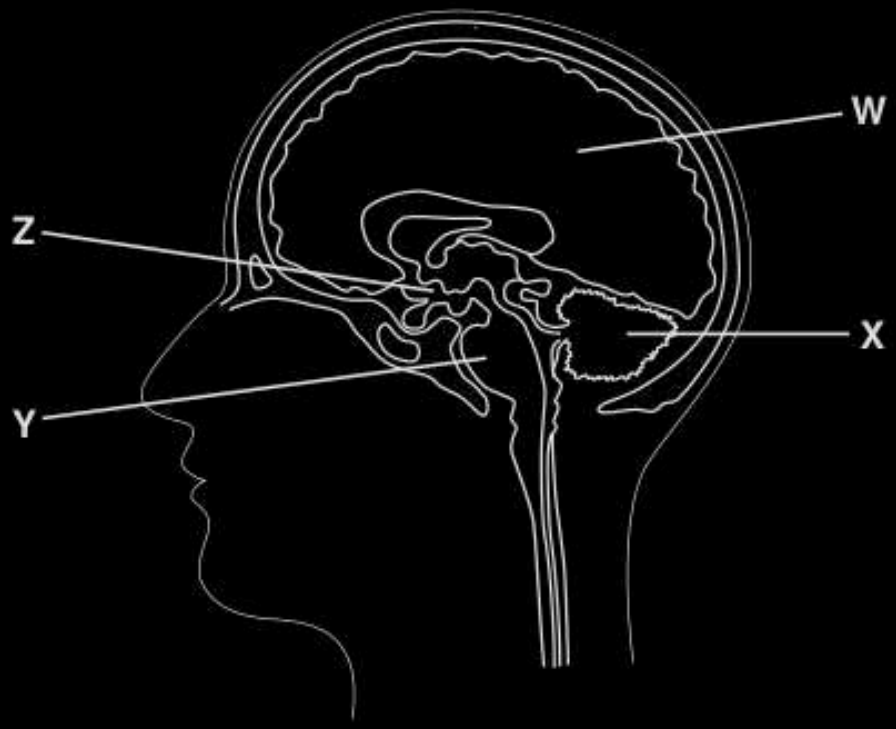
✓ = present

✗ = absent

October/November 2005 · Paper 1 · Variant 1 · Question 26

Correct answer: B

26 The diagram shows a section through the brain.



Which regions control balance, heart rate and temperature?

	balance	heart rate	temperature
A	**W**	**Z**	**X**
B	**X**	**Y**	**Z**
C	**Y**	**X**	**W**
D	**Z**	**W**	**Y**

October/November 2005 · Paper 1 · Variant 1 · Question 28

Correct answer: D

28 What is a feature of bacteria?

A They are always parasitic.

B They have a nucleus.

C They are made of hyphae.

D They reproduce by binary fission.

October/November 2005 · Paper 1 · Variant 1 · Question 34

Correct answer: C

34 The skin cells of an animal contain 8 chromosomes.

How many chromosomes will be present in each of the gametes produced by this animal?

A 16 **B** 8 **C** 4 **D** 2

October/November 2005 · Paper 1 · Variant 1 · Question 35

Correct answer: D

35 In pigs, the allele for black hair is dominant to the allele for red hair.

Two black-haired pigs mated and produced twelve offspring. Of the first eleven, eight had black hair and three had red hair.

What is the probability of the twelfth offspring having red hair?

A 0.75 **B** 0.50 **C** 0.33 **D** 0.25

October/November 2005 · Paper 1 · Variant 1 · Question 36

Correct answer: C

36 What is an advantage of feeding a baby with bottled milk rather than breast milk?

A Breast milk contains antibodies.

B It is easier to keep the milk free of pathogens.

C The baby's intake of food can be measured.

D The milk is always at the right temperature.

October/November 2005 · Paper 1 · Variant 1 · Question 39

Correct answer: B

39 Sex in humans is determined by **X** and **Y** chromosomes inherited from parents.

Which shows the chromosome inherited from the father and from the mother?

	sex of child	chromosome from father	chromosome from mother
A	male	**X**	**Y**
B	male	**Y**	**X**
C	female	**X**	**Y**
D	female	**Y**	**X**

October/November 2005 · Paper 1 · Variant 1 · Question 40

Correct answer: C

40 Which two characteristics both show continuous variation?

A eye colour and height

B gender and eye colour

C height and weight

D weight and blood group

1 Fig. 1.1 is a flow diagram which shows some of the stages in the manufacture of a type of cheese.

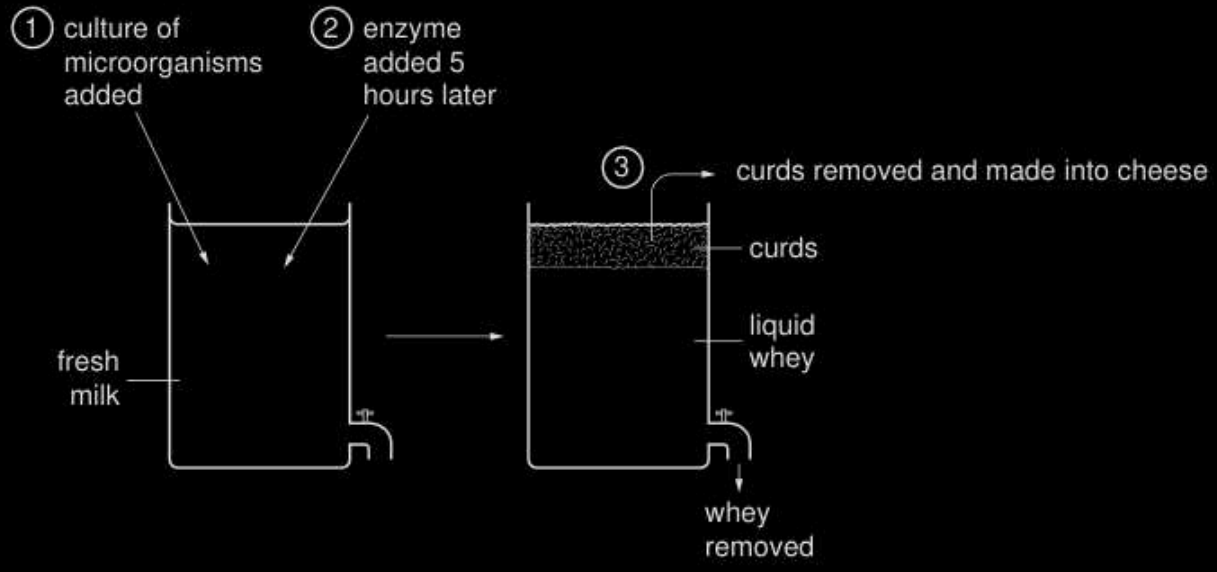


Fig. 1.1

Fig. 1.2 shows the changes in pH which occur during the first few hours of this process.

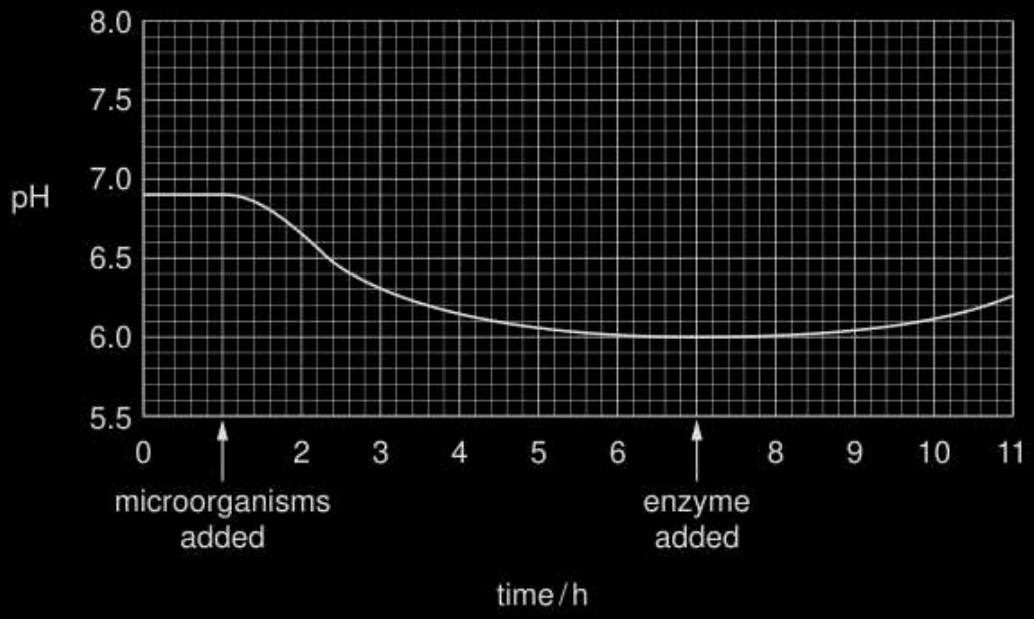


Fig. 1.2

(a) Identify the type of microorganism added to the milk. .....[1]

(b) Use the information from Fig. 1.1 and Fig. 1.2 to describe the function of these microorganisms in this process.

.....

.....

.....[3]

(c) The enzyme which curdles milk can be obtained from the alimentary canal of a young mammal. Suggest from which part of the alimentary canal the enzyme is obtained. State a reason for your answer.

part of alimentary canal .....

reason .....

.....[2]

(d) Fig. 1.3 shows how milk is treated **before** it is used to make a cheese-like product **P**. After this treatment, it undergoes a process similar to the one shown in Fig. 1.1. The end-product has a flavour and texture similar to cheese, but it is considered to be healthier to eat.

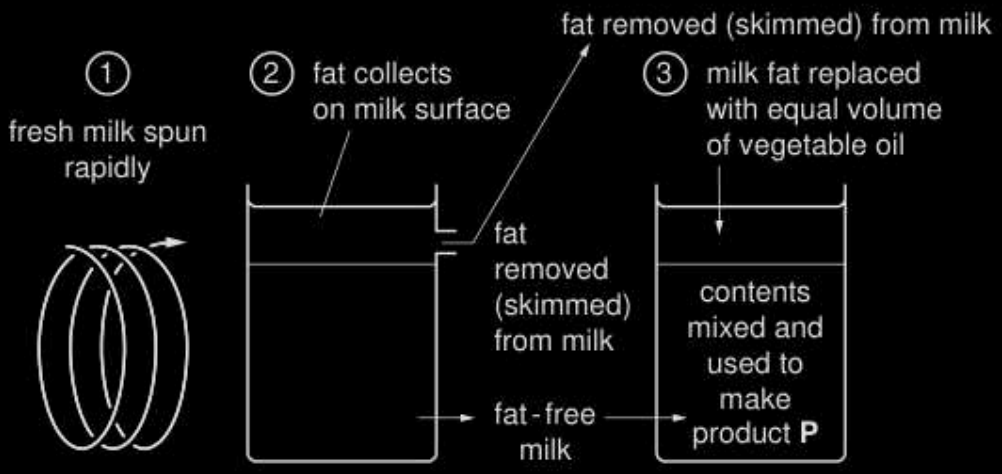


Fig. 1.3

(i) Suggest which feature of the manufacturing process in Fig. 1.3 may make product **P** healthier to eat than cheese.

.....[1]

(ii) State two possible harmful effects of eating cheese often and in large quantities.

1. .....

2. .....[2]

[Total: 9]

1 **(a)** bacterium/named bacterium ; **[1]**

(b) specific pH read from graph ;

ref. to acid/low pH ;

sours/curdles the milk ;

flavours the cheese ; **[max. 3]**

(c) stomach ;

stomach is acidic/ref. HC*l* ;

enzyme has optimum low pH AW ;

young mammals consume only milk ;

curdling increases surface area ; **[max. 2]**

(d) **(i)** the use of vegetable oil/unsaturated fat (A v.v. for animal fat) ;

(ii) Any two from: obesity or described, named circulatory disorder, dairy allergies ;; **[3]**

[9]

2 Fig. 2.1 shows the structures involved in oxygen uptake in the lungs.

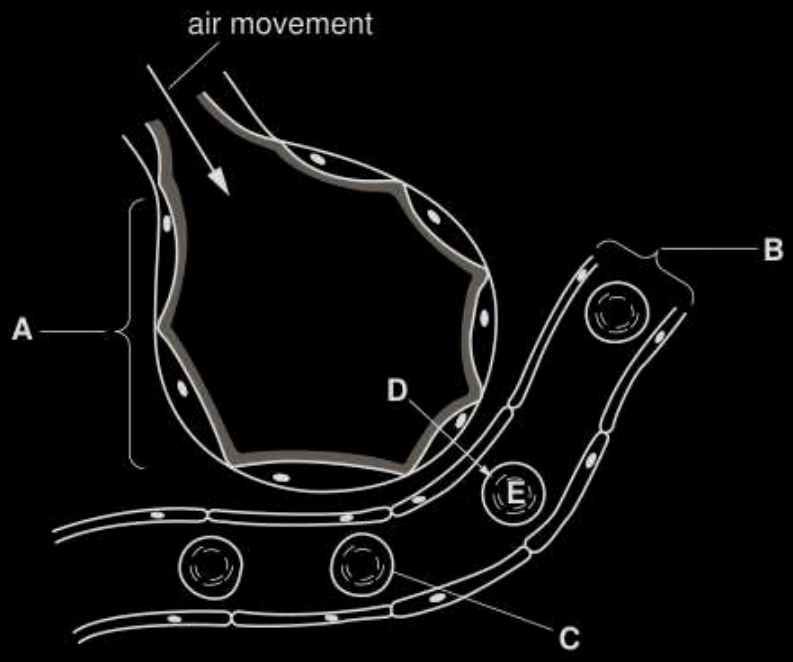


Fig. 2.1

(a) Identify structures **A**, **B** and **C** in Fig. 2.1.

A .....

B .....

C ..... [3]

(b) Each statement below describes a process that occurs during breathing.

Place a tick (✓) in the box beside each statement that describes a process necessary to cause air to move in the direction shown in Fig. 2.1.

☐	diaphragm relaxes
☐	(external) intercostal muscles contract
☐	ribs rise
☐	diaphragm rises
☐	volume of thorax decreases

[2]

(c) Describe what happens to a molecule of oxygen as it moves from **D** to **E** in Fig. 2.1.

.....

.....

.....

.....[3]

Use

(d) Table 2.1 shows the percentage of oxygen in the inspired air and expired air of a healthy person.

Table 2.1

% oxygen in inspired air	% oxygen in expired air
20.5	16.5

Suggest and explain how these figures might be different for a person whose diet had been deficient in iron over a period of several years.

.....

.....

.....

.....[3]

[Total: 11]

2 **(a)** A - alveolus/air sac ;

B - **capillary** ;

C – RBC/erythrocyte ; **[3]**

(b) one mark each for statements 2 and 3, remove a mark for each incorrect ;; **[2]**

(c) dissolves ;

diffuses ;

combines with haemoglobin/forms oxyhaemoglobin ;

ref. any structure/substance which O_2 passes through ; **[max. 3]**

(d) more oxygen in air breathed out ;

Fe needed for haemoglobin ;

less haemoglobin formed ;

less oxygen absorbed by (red) blood (cells) ; **[max. 3]**

[11]

3 Fig. 3.1 shows a tank containing animals and water plants.

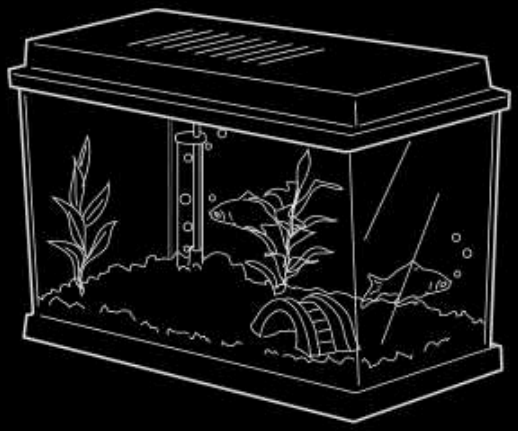


Fig. 3.1

(a) (i) Name the pigment responsible for the green colour of the plants.

..... [1]

(ii) Name the mineral ion which must be present for the manufacture of this pigment.

.....[1]

(b) (i) The animals supply the plants with a gas essential for the plants' food production.

Name the gas and the process for which it is used.

gas .....

process ..... [2]

(ii) Suggest how the animals might benefit from the presence of these plants.

.....

.....

.....

.....

.....[4]

Use

Fig. 3.2 shows a cell taken from one of the organisms in the tank.

Fig. 3.2

(c) From which type of organism was this cell taken? Explain your answer.

type of organism .....

explanation .....[1]

[Total: 9]

3 **(a)** **(i)** chlorophyll ; **[1]**

(ii) magnesium ; **[1]**

(b) **(i)** carbon dioxide/carbon(IV)oxide ;

photosynthesis ; **[2]**

(ii) supplied with oxygen ;

for respiration ;

waste product/carbon dioxide removed AW ;

camouflage ;

food/part of food chain AW ;

importance in reproduction/eggs attached to leaves AW ;

shade ; **[max. 4]**

(c) animal/fish + no CCW/vacuole/chloroplasts ; **[1]**

[9]

5 Fig. 5.1 shows a section through the heart.

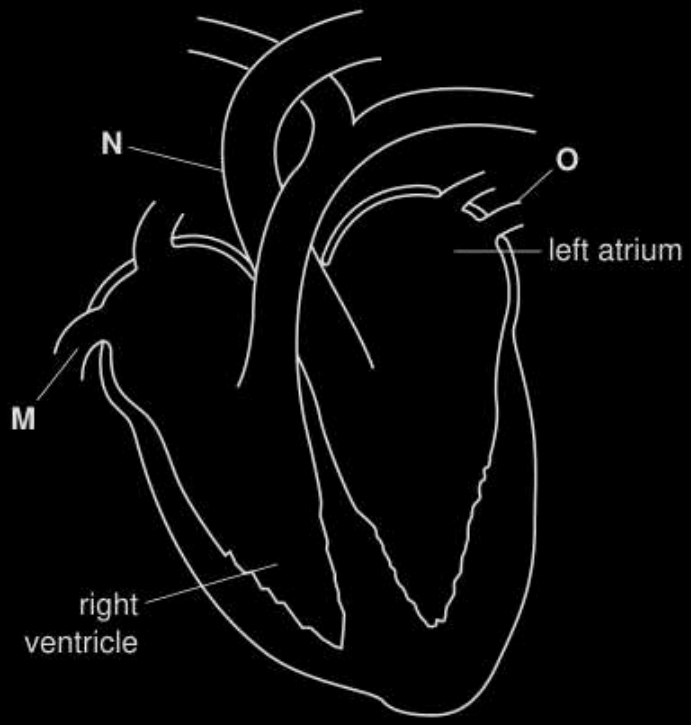


Fig. 5.1

(a) **(i)** Identify blood vessels **M**, **N** and **O** in Fig. 5.1.

M .....

N .....

O ..... [3]

(ii) Name the type of tissue which forms the major part of the heart and name the blood vessel which supplies this tissue with oxygen unless it is blocked by heart disease.

type of tissue .....

blood vessel ..... [2]

(b) On Fig. 5.1, draw, in the correct position, and label

(i) the semilunar valves,

(ii) the tricuspid valve,

(iii) the bicuspid (mitral) valve.

[4]

Use

(c) The four statements below describe some of the events that occur during the flow of blood through the heart. By placing the numbers **1** to **4** in the boxes, indicate the correct sequence of these events, starting immediately after deoxygenated blood has entered the heart and ending as the blood is sent to the lungs.

	The right atrium contracts.
	The semilunar valves open.
	The right ventricle contracts.
	The tricuspid valve closes. [3]

[Total: 12]

Section B

Answer **three** questions.

Question **8** is in the form of an **Either** / **Or** question. Only one part should be answered.

Write your answers on the separate answer paper provided.

5 **(a)** **(i)** M - (inferior) **vena cava** ;

N - (systemic) **aorta** ;

O - pulmonary **vein** ; **[3]**

(ii) muscle ;

coronary + artery ; **[2]**

(b) **(i)** semilunar valves correctly shown + label ;

(ii) & **(iii)** tricuspid correctly shown ;

bicuspid correctly shown ;

both correctly labelled ; **[4]**

(c) 1, 4, 2, 3. (3 if all correct, 2 if correct sequence, but starting in wrong place, 1 for any two in sequence) ;;; **[3]**
atrium contracts then ventricle contracts, forcing tcv closed and as pressure builds up, slv open

[12]

The **maximum** mark for Section A is **50**

Section B

6 A person is sitting in the shade reading a book when he looks at the bright sky to see an aeroplane flying past. Explain the changes in

(a) the lens of the eye, [6]

(b) the pupil of the eye. [4]

[Total: 10]

6 ref. in either **(a)** or **(b)** to reflex action ;

neurones/impulses ;

(a) **ciliary** ;

muscles + relax ;

(suspensory) ligaments ;

tighten AW ;

pull on lens AW ;

lens flatter AW ;

to focus on distant object/longer focal length ; **[max. 6*]**

(b) brighter light ;

circular iris muscles contract ;

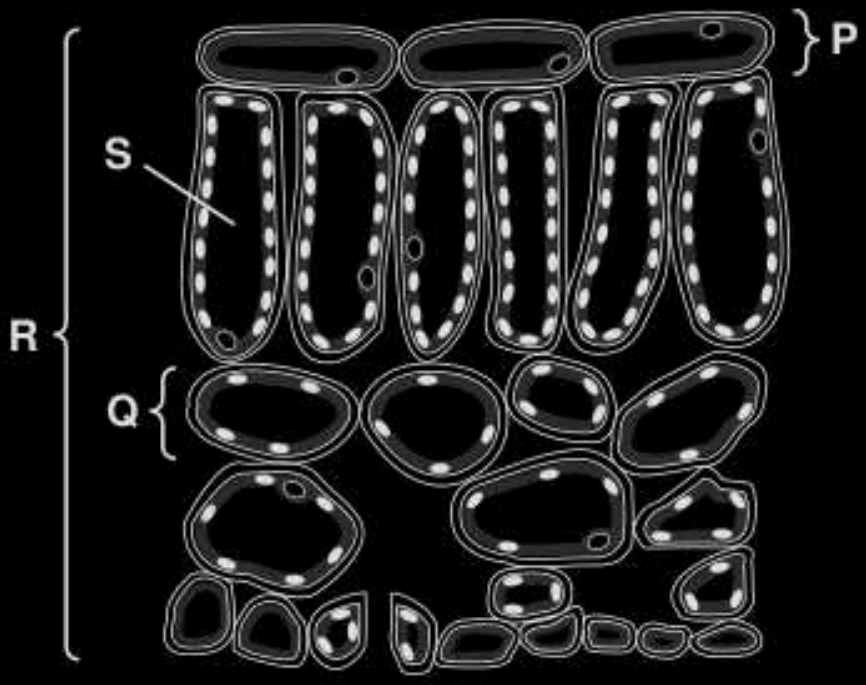
radial muscles relax ;

pupil becomes smaller ; **[max. 4*]**

(* to include either of the first two marking points) **[Total = 10]**

May/June 2005 · Paper 1 · Variant 1 · Question 1

Correct answer: C



Which is an organ and which is a tissue?

	organ	tissue
A	P	R
B	Q	S
C	R	P
D	S	Q

May/June 2005 · Paper 1 · Variant 1 · Question 2

Correct answer: D

2 What causes water to enter plant roots from the soil?

A Water potential in root hairs and the soil is equal.

B Water potential in root hairs and xylem is equal.

C Water potential in root hairs is higher than in the soil.

D Water potential in root hairs is lower than in the soil.

May/June 2005 · Paper 1 · Variant 1 · Question 4

Correct answer: C

4 Protease enzyme breaks down protein to amino acids.

In the 'lock and key' hypothesis, what is the lock and what is the key?

	lock	key
A	amino acid	protease
B	protease	amino acid
C	protease	protein
D	protein	protease

May/June 2005 · Paper 1 · Variant 1 · Question 5

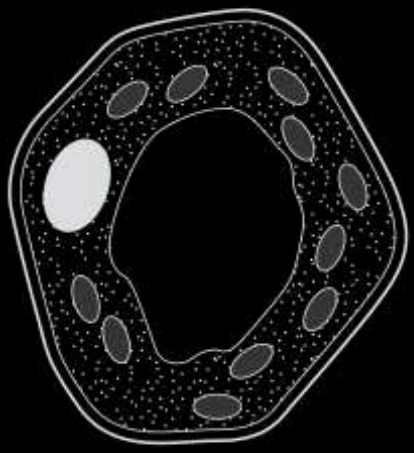
Correct answer: D

5 Where and how does carbon dioxide enter a plant?

	where	how
A	root hair cells	active uptake
B	root hair cells	diffusion
C	stomata	active uptake
D	stomata	diffusion

May/June 2005 · Paper 1 · Variant 1 · Question 7

Correct answer: B



What type of cell is it?

A epidermal cell of a leaf

B mesophyll cell of a leaf

C root hair cell

D xylem cell

May/June 2005 · Paper 1 · Variant 1 · Question 8

Correct answer: B

8 Which chemical test shows the presence of an enzyme in a biological washing powder?

A Benedict's

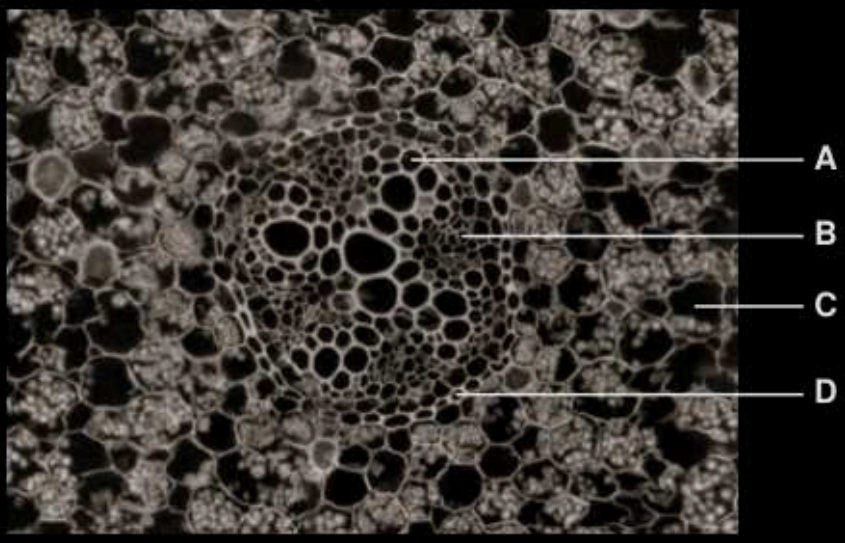
B biuret

C ethanol emulsion

D iodine solution

May/June 2005 · Paper 1 · Variant 1 · Question 11

Correct answer: B



May/June 2005 · Paper 1 · Variant 1 · Question 12

Correct answer: D

12 A young plant may wilt when dug up and re-planted in another place.

What causes this?

A The leaves lose less water.

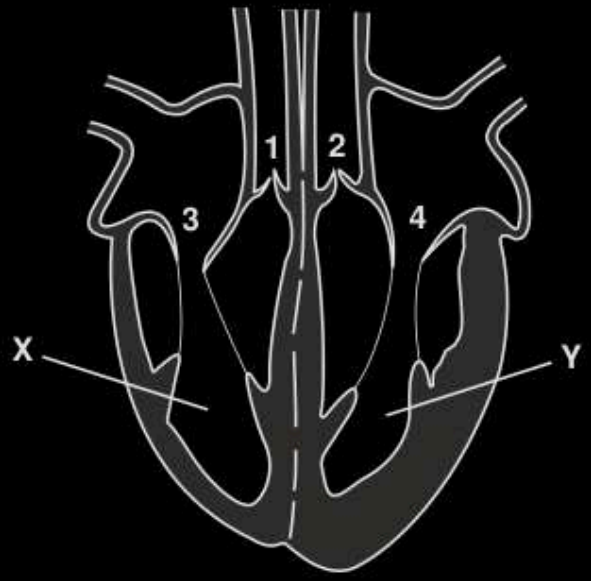
B The roots cannot take up mineral salts.

C The stem cannot transport water.

D The surface area of the root is reduced.

May/June 2005 · Paper 1 · Variant 1 · Question 13

Correct answer: C



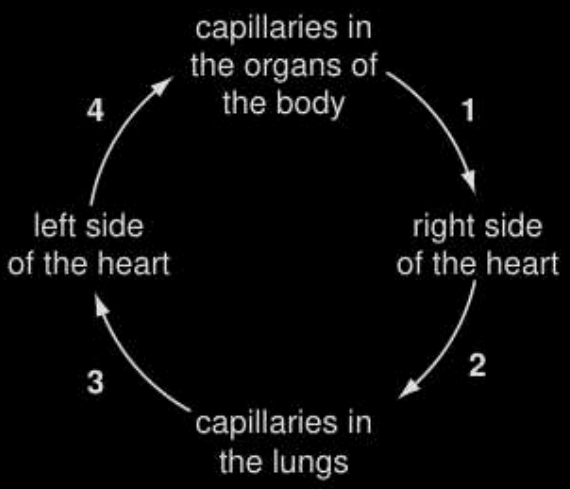
While chambers **X** and **Y** are emptying of blood, which valves are open and which are closed?

	valves **1** and **2**	valves **3** and **4**
A	closed	closed
B	closed	open
C	open	closed
D	open	open

May/June 2005 · Paper 1 · Variant 1 · Question 14

Correct answer: C

14 The diagram represents blood flow through the human body.



At which stages does the blood contain the most oxygen?

A **1** and **2** **B** **2** and **3** **C** **3** and **4** **D** **4** and **1**

Correct answer: C

15 The table shows substances passing between capillaries and tissues in a part of the body.

substance	into the capillaries from the tissues	out of the capillaries into the tissues
oxygen		✓
carbon dioxide	✓	
amino acids		✓
urea	✓	

key

✓ = does pass

In which part of the body are these capillaries?

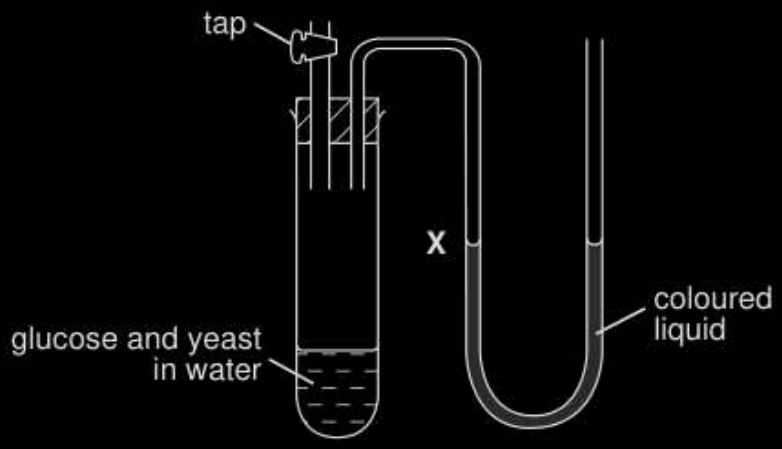
A between the alveoli

B in the kidney

C in the liver

D in the villi

Correct answer: A



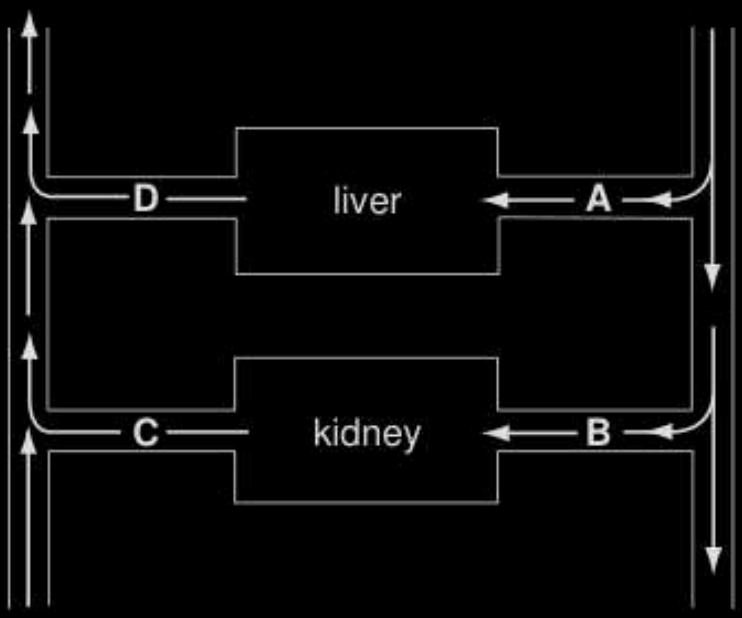
The tap is closed and the yeast respires anaerobically.

What is observed at **X** and which explanation is correct?

	observation at **X**	explanation
A	liquid level falls	carbon dioxide is produced
B	liquid level falls	oxygen is used
C	liquid level rises	carbon dioxide is produced
D	liquid level rises	oxygen is used

May/June 2005 · Paper 1 · Variant 1 · Question 19

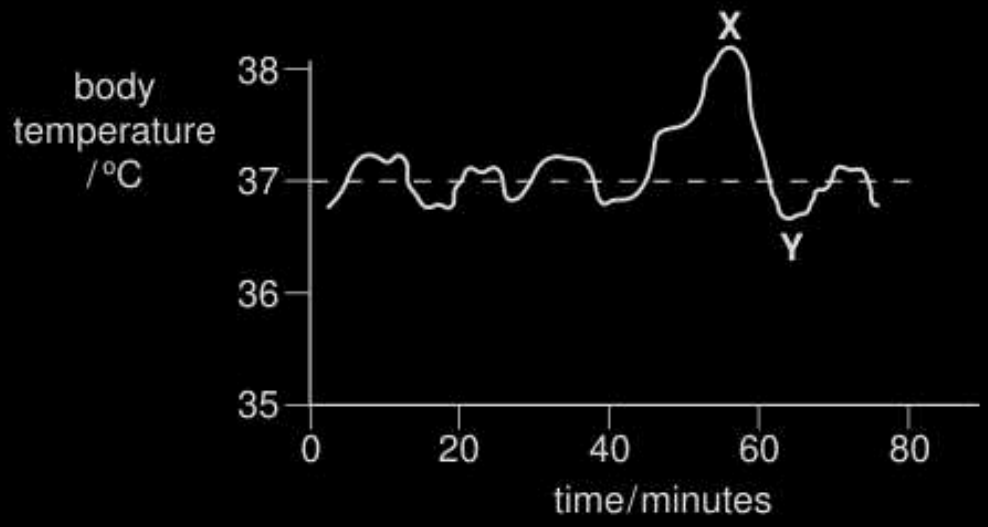
Correct answer: C



May/June 2005 · Paper 1 · Variant 1 · Question 20

Correct answer: B

20 The graph shows changes in a person's body temperature plotted against time.



What causes the change in temperature between **X** and **Y**?

A increased air temperature

B increased evaporation of sweat

C reduced blood flow through surface capillaries

D shivering

May/June 2005 · Paper 1 · Variant 1 · Question 21

Correct answer: D

21 Why does emphysema cause severe breathlessness?

A The alveoli become coated with tar.

B The cilia lining the trachea are destroyed.

C The lungs become cancerous.

D The surface area of the lungs is reduced.

May/June 2005 · Paper 1 · Variant 1 · Question 23

Correct answer: B

23 What is controlled by the medulla in the brain?

A balance

B breathing

C memory

D vision

May/June 2005 · Paper 1 · Variant 1 · Question 24

Correct answer: A

24 When a person is frightened, which substance increases the blood sugar levels?

A adrenaline

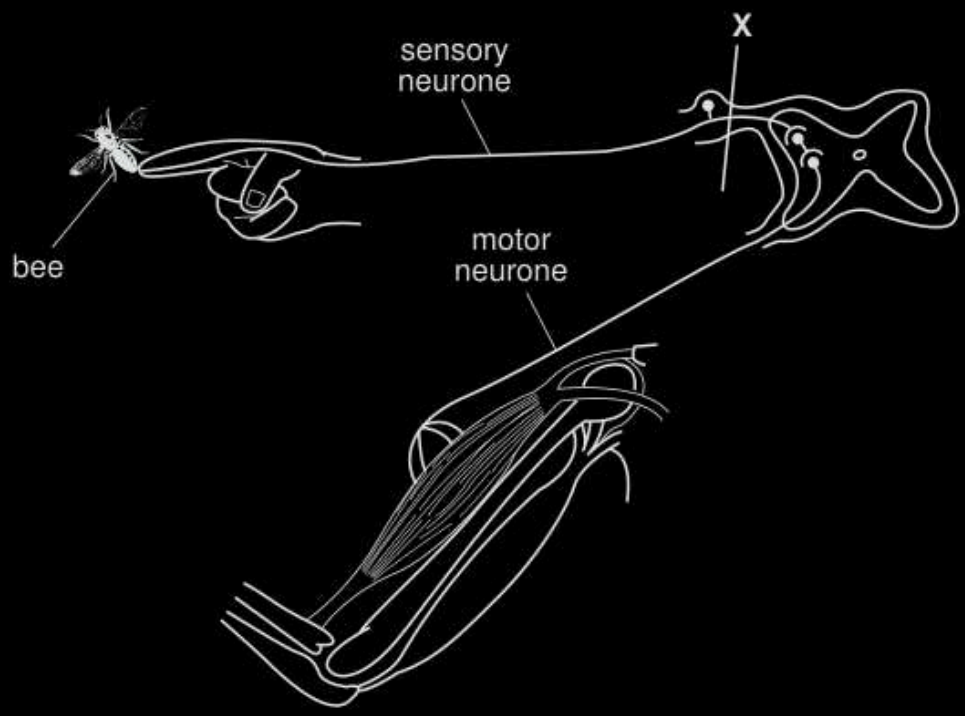
B carbon dioxide

C insulin

D lactic acid

May/June 2005 · Paper 1 · Variant 1 · Question 25

Correct answer: A



A bee stings the finger, as shown.

What are the effects of this sting on the person?

	pain felt	arm moved
A	no	no
B	no	yes
C	yes	no
D	yes	yes

May/June 2005 · Paper 1 · Variant 1 · Question 32

Correct answer: D

32 What will be the effect of increasing nitrate levels in rivers?

A Animals will absorb the nitrates and make more urea.

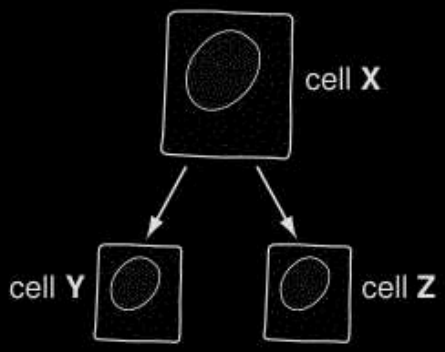
B Animals will absorb the nitrates and form more proteins.

C Plants will absorb the nitrates and make more urea.

D Plants will absorb the nitrates and form more proteins.

May/June 2005 · Paper 1 · Variant 1 · Question 33

Correct answer: C



How many chromosomes does cell **Z** contain?

A 8 **B** 12 **C** 24 **D** 48

May/June 2005 · Paper 1 · Variant 1 · Question 35

Correct answer: C

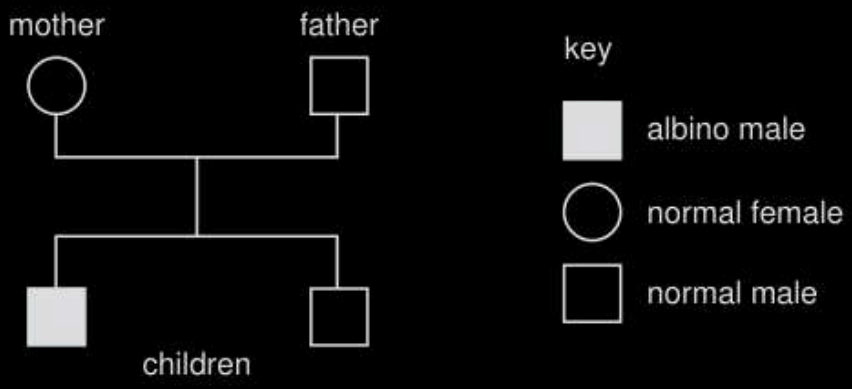
35 After sexual intercourse, sperm can survive for 3 days in the uterus and oviducts. Assume ovulation can occur any time from day 13 to day 15.

If an egg cell lives up to 2 days after ovulation, how long is the fertile phase of the menstrual cycle?

A 5 days **B** 6 days **C** 7 days **D** 8 days

May/June 2005 · Paper 1 · Variant 1 · Question 38

Correct answer: A



What are the genotypes of the parents?

	mother	father
A	heterozygous	heterozygous
B	heterozygous	homozygous dominant
C	homozygous recessive	homozygous dominant
D	homozygous recessive	homozygous recessive

May/June 2005 · Paper 1 · Variant 1 · Question 39

Correct answer: A

39 A person with Down's syndrome is born with 47 chromosomes in each of his/her cells, instead of 46.

What could cause this?

A A mutation happened during the production of the egg cell.

B More than one sperm fused with the egg at fertilisation.

C Radiation caused a change in structure of a gene in the father's sperm.

D The mother was exposed to harmful chemicals while she was pregnant.

May/June 2005 · Paper 1 · Variant 1 · Question 40

Correct answer: C

40 Huntington's disease is an inherited condition caused by a dominant allele.

A person heterozygous for the disease and a person without the disease have a child.

What is the probability that their child will inherit the dominant allele for Huntington's disease?

A 0 **B** 0.25 **C** 0.5 **D** 0.75

2 Fig. 2.1 shows some parts of an insect-pollinated and a wind-pollinated flower.

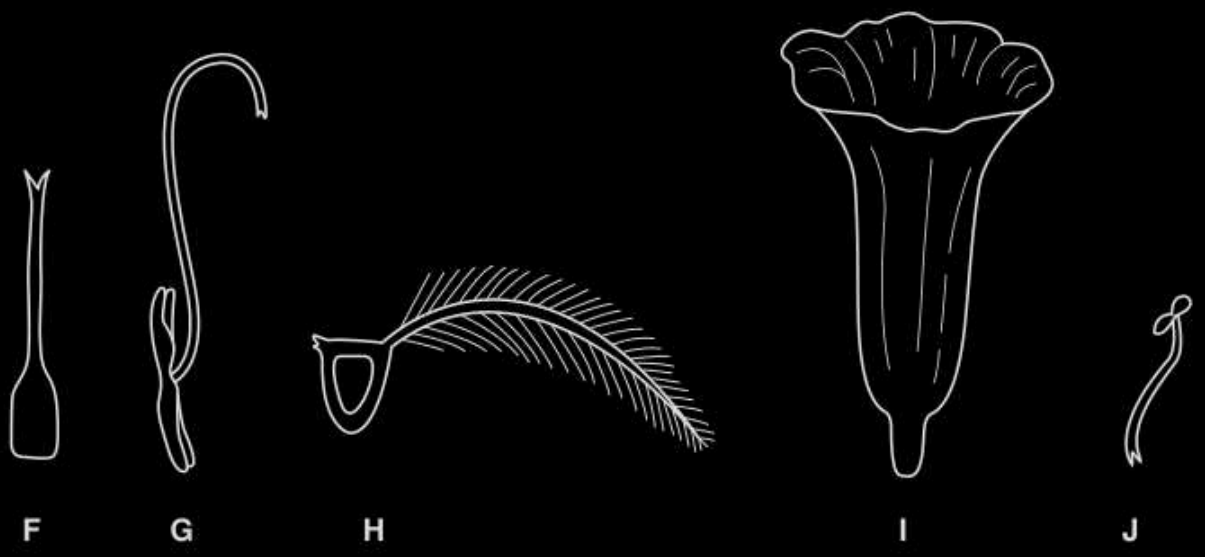


Fig. 2.1

(a) Using the letters **F**, **G**, **H**, **I** and **J**, list the parts that are from

the insect-pollinated flower: .....

the wind-pollinated flower: .....

[2]

(b) **(i)** In the table below, identify parts **H**, **I** and **J** and state the function of each.

[3]

(ii) Explain how the structure of part **H** helps it to carry out its function.

.....

.....

.....[2]

(c) Fig. 2.2 shows a pollen grain with its pollen tube.

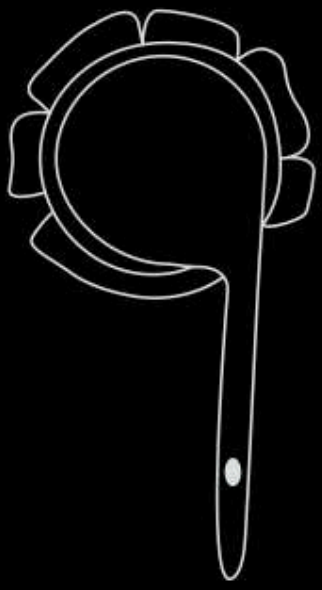


Fig. 2.2

On Fig. 2.1, use a line labelled **L** to show exactly where the pollen grain as shown in Fig. 2.2 is found. [1]

(d) State and explain the difference between a nucleus in a pollen grain and a nucleus found in a cell in structure **I**.

.....

.....

.....[2]

[Total: 10]

2 **(a)** F, I, J (mark first 3) ;
G, H (mark first 2) ; **[2]**

(b) **(i)** (H) stigma + to catch pollen/ovary + to hold ovule (A female gamete)/seed ;
(A carpel/gynoecium/pistil)
(I) petal/corolla + to attract insect/landing platform ;
(J) stamen + to produce AW pollen/male gametes ; **[3]**
(A pollen sac/anther + filament or androecium)

(ii) large surface area/sticky/hairy/feathery ;
maximum AW + collection of pollen ; **[2]**

(c) line plus label L pointing to stigma on either F or H ; **[1]**
(R more than one line unless both correct,
A line to J if identified as stigma in **(b)**)

(d) ('It' = the pollen grain)
haploid AW/diploid AW/different no. of chromosomes ;
(correct ref. to) (ignore any stated numbers)
gamete/male contribution AW/ref. reproduction/meiosis
or/not involved in reproduction/part of adult plant/mitosis ; **[2]**

[Total: 10]

3 Fig. 3.1 shows a section through a leaf with only some of the cells drawn.

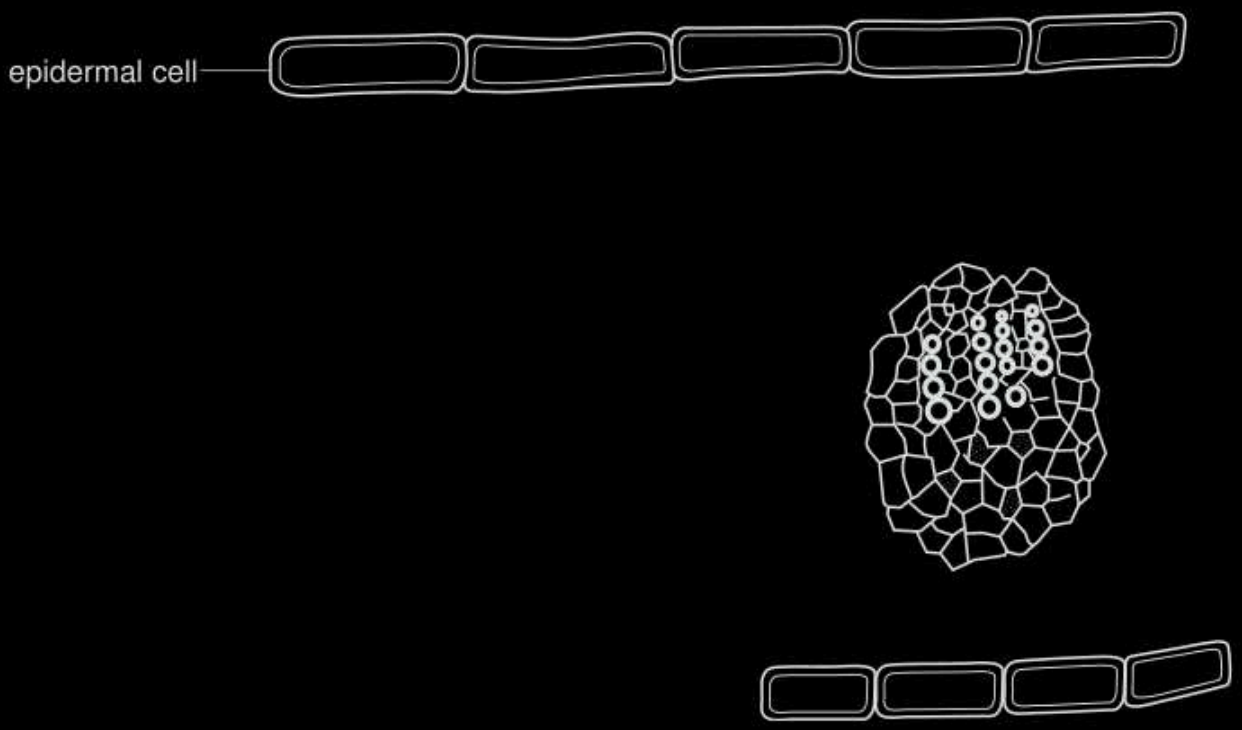


Fig. 3.1

(a) On Fig. 3.1, draw and label in their correct positions:

(i) a palisade mesophyll cell;

(ii) a spongy mesophyll cell;

(iii) a pair of guard cells;

(iv) a cuticle.

[4]

(b) On Fig. 3.1, label and name the tissue bringing water into the leaf. [2]

(c) Explain how an increase in atmospheric humidity slows down the process of transpiration.

.....

.....

.....[3]

(d) Suggest **one** reason why epidermal cells are important to a leaf.

.....

.....[1]

[Total: 10]

3 **(a)** **(i)** (palisade) 1 cell drawn, correct shape and position and labelled ;

(ii) (spongy) 1 cell drawn, correct shape and position and labelled ;

(iii) 2 guard cells drawn in (lower) epidermis and labelled ;
(R label to stoma)

(iv) (cuticle) shown as thin acellular layer and labelled ;
(A cuticle on upper or lower epidermis, R if it crosses stoma)
(max: 2 marks if no labels) **[4]**

(b) xylem correctly indicated on Fig. 3.1 ;
xylem - the name (A even if incorrectly labelled) ; **[2]**

(c) greater concentration in the atmosphere ;
water molecules ;
loss of gradient AW ;
slow(er)/less evaporation ;
slow(er)/less diffusion ; **[max:3]**

(d) (Mark the first, but ignore waterproof)
make cuticle ;
protects photosynthesising or otherwise qualified cells ;
transparent/light entry ; **[max: 1]**

[Total: 10]

4 Fig. 4.1 shows a section through a small surface wound to the skin.

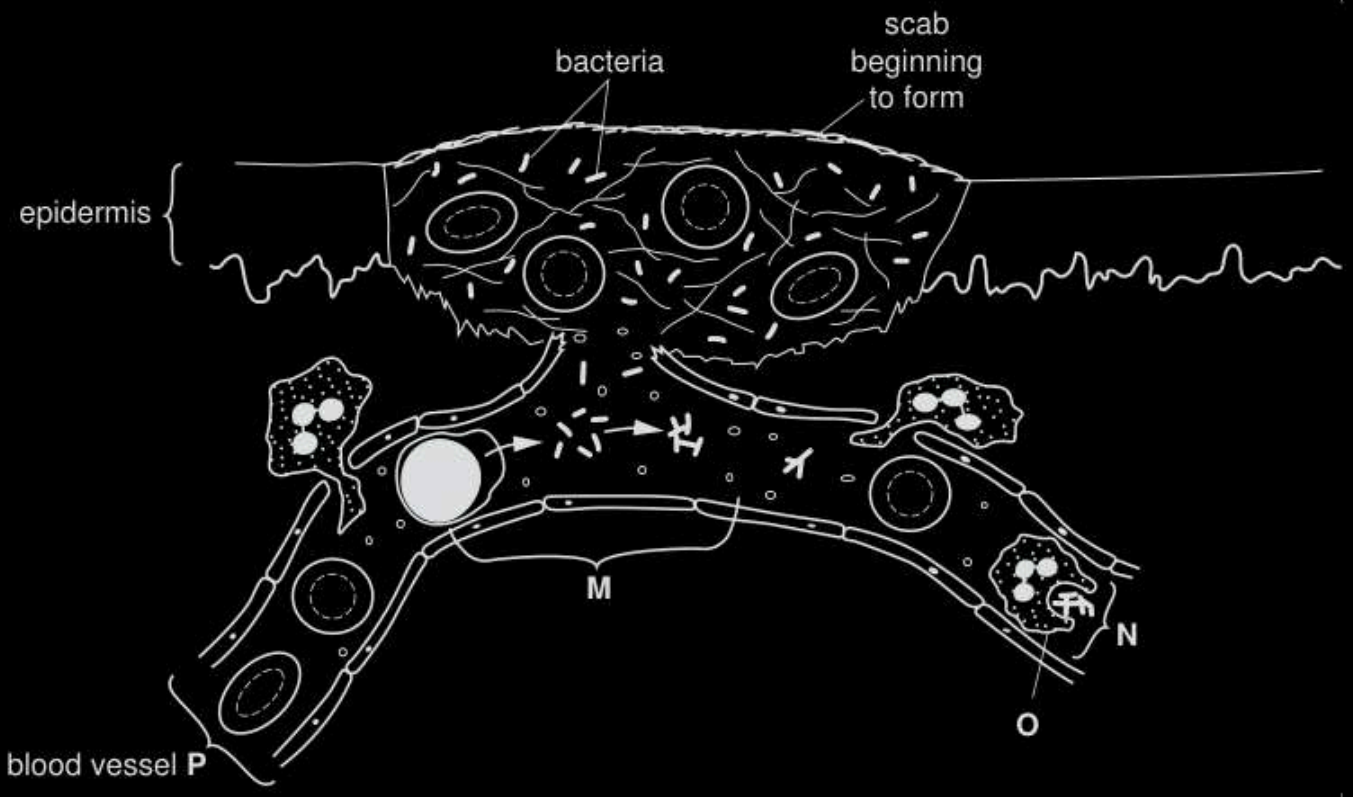


Fig. 4.1

(a) Name structure **O** and the type of blood vessel **P** shown in Fig. 4.1.

O .....

P .....[2]

(b) State two features that are characteristic of the type of blood vessel **P**.

1. .....

2. .....[2]

(c) Explain what is happening to the bacteria in Fig. 4.1.

(i) at **M**: .....

.....

.....[2]

(ii) at **N**: .....

.....[1]

Use

(d) Explain how the wound is being sealed in the region under the scab.

.....

.....

.....

.....

.....[4]

[Total: 11]

4 **(a)** (O) WBC/phagocyte AW (R lymphocyte or other BI) ;
(A polymorph/macrophage/cell membrane)
(P) capillary ; **[2]**

(b) ('It' = the blood vessel)
any two from :
walls + very thin/one cell thick ;
microscopic/narrow/small bore/allow RBCs only in single file ;
walls leaky/permeable AW/allow substances to pass through ;
(or named, A phagocytes AW)
slow blood flow ; **[max: 2]**

(c) **(i)** M - WBC/lymphocyte + antibodies ;
clumping AW (of bacteria) ; **[2]**

(ii) N - engulfing/ingesting/phagocytosis (R digesting) ; **[1]**

(d) action of platelets ;
fibrinogen to fibrin ;
clotting ;
trapping of RBCs ;
stopping bleeding ;
prevention of further bacterial entry ;
reproduction/growth of skin cells AW ; **[max. 4]**

[Total: 11]

5 **(a)** **(i)** Name the hormone, released after a meal, which controls the level of glucose in the blood.

.....[1]

Fig. 5.1 shows some organs in the body of a woman.

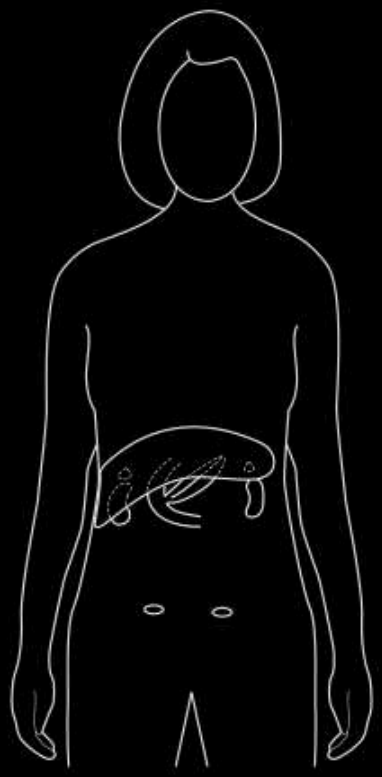


Fig. 5.1

(ii) On Fig. 5.1, label and name the gland which produces the hormone you mention in **(i)**. [2]

(iii) Name two other substances produced by this gland.

1. .....

2. .....[2]

Use

(b) The hormone in **(a)(i)** can be manufactured in the laboratory by adding a piece of human DNA to the DNA in a host cell. Fig. 5.2 shows how this is done.

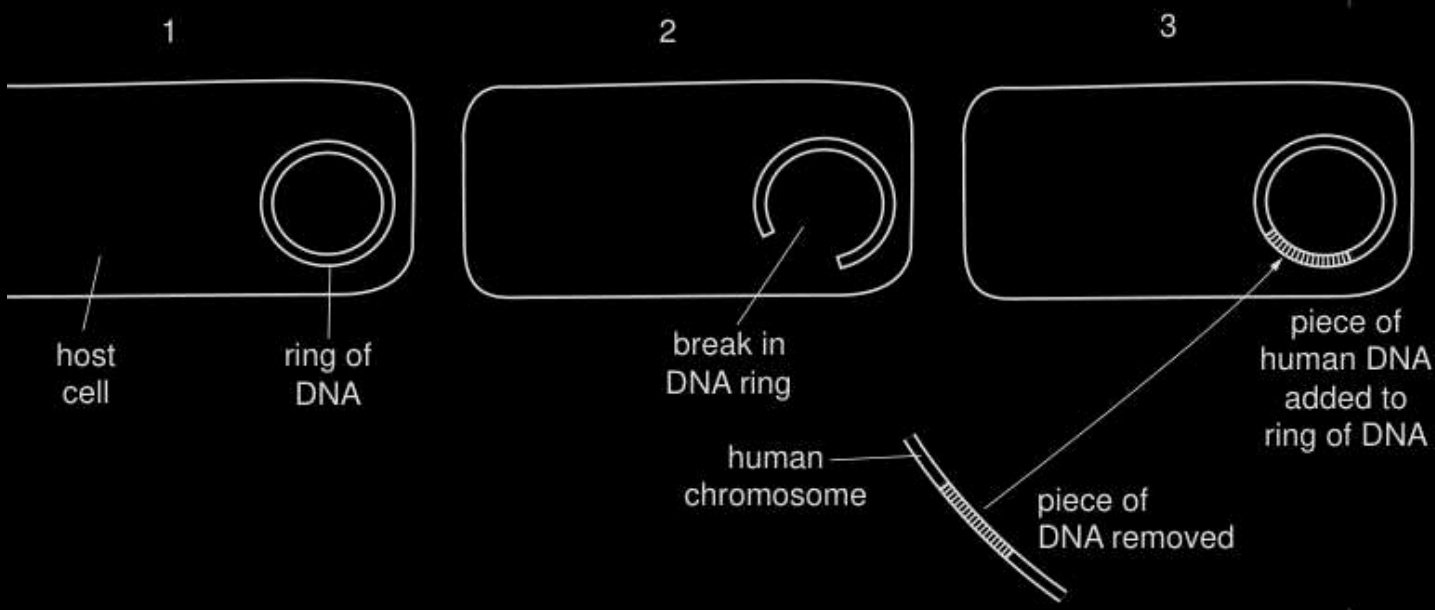


Fig. 5.2

(i) Name the type of host organism used in this process.

.....[1]

(ii) After the process in Fig. 5.2, the host cell will produce the hormone in **(a)(i)**. Describe the piece of human DNA which must be used in this process.

.....

.....

.....[2]

(c) Suggest what is used to break and rejoin the DNA molecules.

.....[1]

[Total: 9]

Section B

Answer **all** the questions including questions 6, 7 and 8 **Either** or 8 **Or**.

Write your answers on the separate answer paper provided.

5 **(a)** **(i)** insulin (if wrong hormone, allow ecf* for **(ii)** - 1 mark only) ;

(ii) pancreas labelled (line must touch or enter gland) ;
[+*] (1 mark max. if more than one organ is named and labelled so long as one is the pancreas)
pancreas named (A even if incorrectly labelled) ;
(A islets of L)

(iii) any **two** (allowing ecf from either **(i)** or **(ii)** if applicable) from :
amylase, protease, lipase, alkali/salts, glucagons/CO_2
(A enzymes, pancreatic juice, but not with named enzyme or juice constituent) ;; **[5]**

(b) **(i)** bacterium/E.coli ; **[1]**

(ii) gene/sequence AW of bases ;
for insulin/hormone production ; **[2]**

(c) enzyme or named e.g. endonuclease(s), lingase(s)/catalyst ; **[1]**

[Total: 9]

[Maximum mark for Section A: 50]

Section B

May/June 2005 · Paper 2 · Variant 1 · Question 6

6 Describe how the following processes differ from each other:

(a) diffusion and active transport, [3]

(b) excretion and egestion, [3]

(c) breathing and respiration. [4]

[Total: 10]

6 (ref. to 'It' should be taken to refer to the first in each pair)

(a)
- no energy required in diffusion/diffusion is passive ;
- energy required in active transport ;
- ref. down a concentration gradient AW in diffusion ; (R along)
- ref. against concentration gradient AW in active transport ;
- ref. respiration/ATP necessary in active transport/carriers ;
- cell/living membrane in active transport ; **[max: 3]**

(b)
- (excretion) removal via lungs/bladder/skin ; (A kidneys)
- any **two** from: toxins, nitrogenous waste, CO_2 ; (A urea/uric acid)
- metabolic AW + waste AW ;
- (egestion) removal from alimentary canal/rectum/anus/gut ;
- undigested/e.g. cellulose/lignin/fibre/roughage/faeces ; **[max: 3]**

(c)
- (breathing) muscular ;
- to move air/ventilate (R O_2, CO_2) ;
- into and out of lungs/inhale and exhale ;
- (respiration) in cells/metabolic AW ;
- *oxidation/ref. oxygen + of glucose ;
- *to release (or accepted alternatives) energy ; **[max: 4]**
- (*available on an equation - balanced if symbols used)

[Total: 10]

May/June 2005 · Paper 2 · Variant 1 · Question 7

7 **(a)** Explain how a knowledge of genetics has helped breeders to improve the quality of wool produced by sheep. [6]

(b) Explain how variation can occur in a natural population of organisms. [4]

[Total: 10]

Question 8 is in the form of an **Either**/**Or** question. Answer only question 8 **Either** or question 8 **Or**. Do **not** answer both parts of question 8.

7 (A points on annotated genetic diagrams, and points, where relevant if they use the gene transfer idea in question **5**.)

(a)
sheep with better wool ;
used for reproduction/are crossed/selective breeding ;
gene passed to offspring ;
offspring vary ;
due to mutation ;
different gene combinations ;
'artificial selection' ;
process repeated ;
over many generations ;
use of back/test cross the find homozygotes AW ; **[max: 6]**

(b)
ref. mutation/ref. change in gene of chromosome ;
spontaneous/random/sudden/abrupt ;
mutagen/named mutagen (A radiation/chemicals) ;
meiosis (R spelling if a 't' appears) ;
random mating/cross fertilisation/recombination ;
ref. environment ;
one environmental factor identified ; **[max: 4]**

[Total: 10]

8 **EITHER**

(a)
protein ;
alters/speeds up rate of (chemical) reactions ;
in (living) cells/made by cells ;
catalyst/not used up/small amounts needed ; **[max: 3]**

(b) **(i)** or **(ii)** (pH or temperature)
ref. best/optimum/fastest ;
*rate slower both sides of the optimum ;
ref. active site AW ;
change in shape (of active site) ;
substrate no longer fits enzyme (or v.v.) ;
inactive outside range/range specific AW for enzyme ;

(i) (pH only)
*symmetrical curve or described ;

(ii) (temperature only)
destruction/denaturation only at high temperatures ;
(A 60 ^{o}C + if figure given)
(denaturing) permanent AW ;
heat increases rate of molecular movement or v.v/ref. Q_{10}/
increased frequency of collision AW (or v.v.) ; **[max: 7]**

(* = marks available on graph with axes labelled) **[Total: 10]**

8 Either **(a)** Explain what is meant by an enzyme. [3]

(b) Describe and explain how

(i) pH and

(ii) temperature

affect enzyme activity. [7]

[Total: 10]

Or **(a)** Describe

(i) chlorophyll,

(ii) a chloroplast. [4]

(b) Describe and explain how

(i) temperature and

(ii) light intensity

affect the rate of photosynthesis. [6]

[Total: 10]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

8 OR

(a) (i) (chlorophyll)
a green chemical/pigment/substance/molecule ;
contains magnesium ;
traps/harnesses/collects/harvests/absorbs/converts ;
light ; **[max: 3]**

(ii) (chloroplast)
structure/organelle ;
in plant/leaf + cells or named plant cell ;
contains chlorophyll ;
contains enzymes for photosynthesis ; **[max: 3]**
for part **(a)** **[max: 4]**

(b) (in **(i)** or **(ii)**) controlled by limiting factors or described

(i) (temperature)
*rate increases with increased temperature ;
faster molecular movement ;
photosynthesis is enzyme-controlled ;
ezymes work faster with increased temperature ;
may cause water loss slowing P/S ;
guard cells lose turgidity AW/stomata close (for **(b) (i)** max 4) ;

(ii) (light intensity)
*higher the light intensity the faster the rate of P/S ;
up to a maximum ;
more energy/light absorbed AW by chlorophyll/chloroplasts ; **[max: 6]**

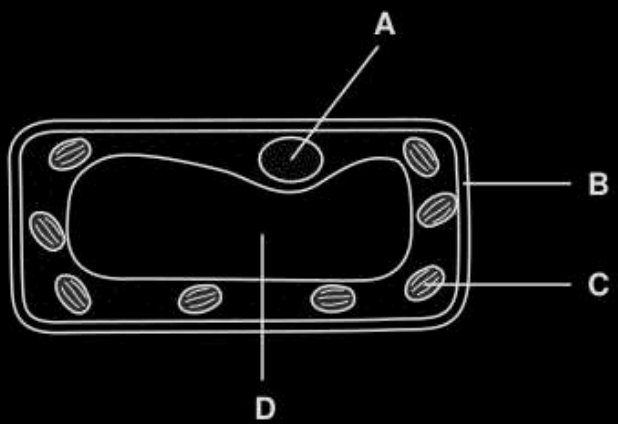
(*marks available on graphs with axes labelled) **[Total: 10]**

October/November 2004 · Paper 1 · Variant 1 · Question 1

Correct answer: C

1 A plant is grown in bright sunshine. After a few hours, a leaf is stained with iodine solution. The diagram shows what is seen when the leaf is placed under a microscope.

Which structure will be stained blue / black?



October/November 2004 · Paper 1 · Variant 1 · Question 2

Correct answer: A

2 The small intestine contains a low concentration of glucose produced from the digestion of starch.

Glucose is taken up by the cells of the villi.

How does the glucose move?

A by active transport against the concentration gradient

B by active transport with the concentration gradient

C by diffusion against the concentration gradient

D by diffusion with the concentration gradient

October/November 2004 · Paper 1 · Variant 1 · Question 3

Correct answer: C

3 A human red blood cell is placed in a strong salt solution.

In which direction does water move and what is the effect on the cell?

	movement of water	effect on cell
A	into the cell	slight increase in size
B	into the cell	cell bursts
C	out of the cell	slight decrease in size
D	out of the cell	no change in cell volume

October/November 2004 · Paper 1 · Variant 1 · Question 4

Correct answer: D

4 According to the lock and key hypothesis, which is the lock and which is the key for the enzyme lipase?

	key	lock
A	fatty acids	lipids
B	lipase	lipids
C	lipase	fatty acids
D	lipids	lipase

October/November 2004 · Paper 1 · Variant 1 · Question 5

Correct answer: A

5 What is the function of chlorophyll?

A to convert light energy to chemical energy

B to release energy from carbohydrates

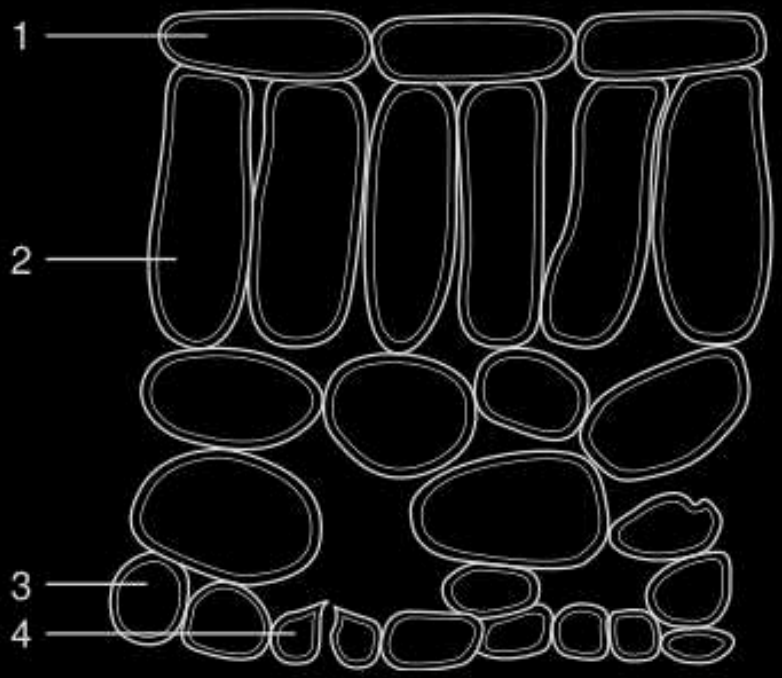
C to store magnesium ions

D to store raw materials for photosynthesis

October/November 2004 · Paper 1 · Variant 1 · Question 7

Correct answer: D

7 The diagram shows the arrangement of cells inside the leaf of a green plant. (No cell contents are shown.)



Which cells normally contain chloroplasts?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

October/November 2004 · Paper 1 · Variant 1 · Question 9

Correct answer: C

9 Which element in the molecule of urea shows that it is formed from amino acids and not from glucose?

A carbon

B hydrogen

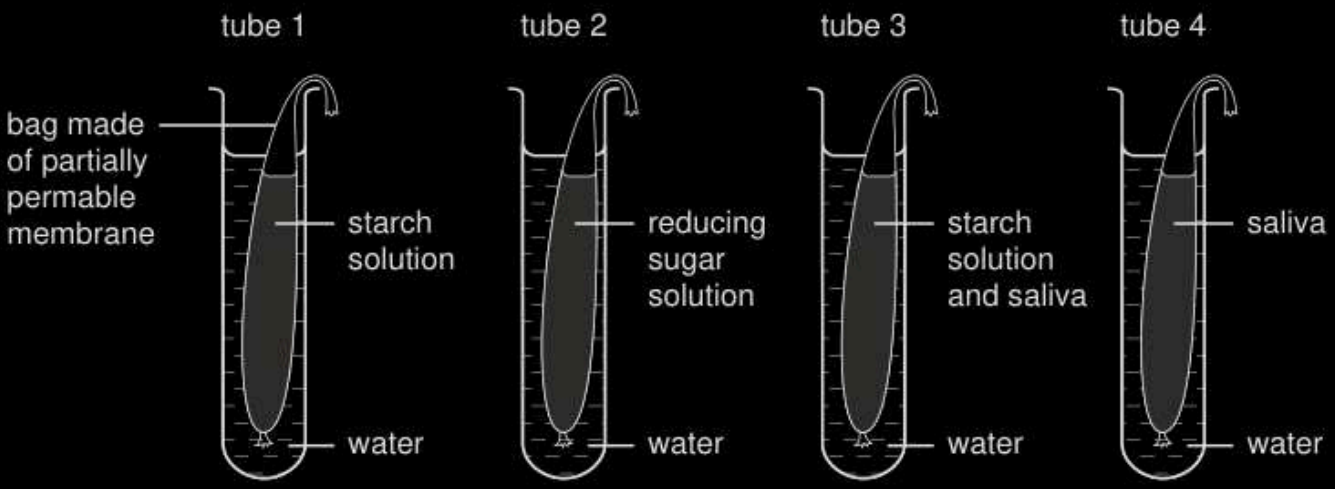
C nitrogen

D oxygen

October/November 2004 · Paper 1 · Variant 1 · Question 10

Correct answer: B

10 Four bags made of partially permeable membrane are placed in tubes as shown in the diagram.



After 20 minutes at 35 °C a sample of water from each tube, outside the bag, is boiled with Benedict's solution.

What are the results?

	tube 1	tube 2	tube 3	tube 4
A	blue	orange	blue	orange
B	blue	orange	orange	blue
C	orange	blue	orange	blue
D	orange	orange	blue	orange

October/November 2004 · Paper 1 · Variant 1 · Question 12

Correct answer: C

12 A plant is exposed to different temperatures and humidities.

Which set of conditions cause the plant to lose most water?

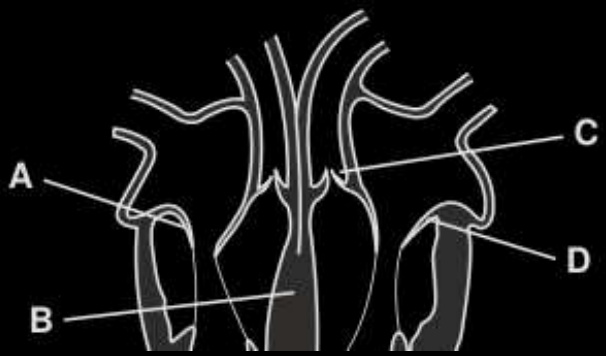
	temperature / °C	humidity / %
A	15	30
B	15	60
C	25	30
D	25	60

October/November 2004 · Paper 1 · Variant 1 · Question 13

Correct answer: B

13 The diagram shows a vertical section through the human heart.

Which structure separates oxygenated blood from deoxygenated blood?



October/November 2004 · Paper 1 · Variant 1 · Question 14

Correct answer: B

14 The coronary arteries carry blood to the

A brain tissue.

B heart muscle.

C liver cells.

D walls of alveolus.

October/November 2004 · Paper 1 · Variant 1 · Question 15

Correct answer: A

15 What is the shortest route that can be taken by the blood travelling from a leg to an arm in the human body?

A leg → heart → lungs → heart → arm

B leg → heart → lungs → liver → arm

C leg → liver → heart → lungs → arm

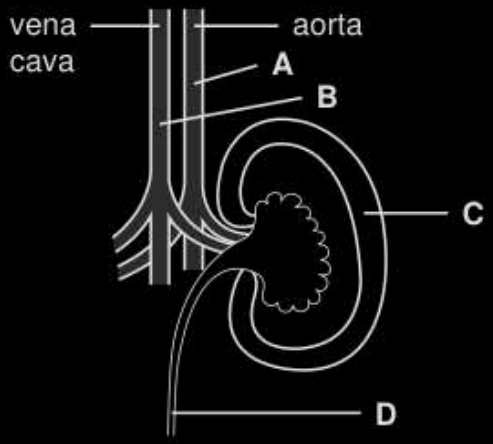
D leg → liver → stomach → heart → arm

October/November 2004 · Paper 1 · Variant 1 · Question 18

Correct answer: D

18 The diagram shows part of the human urinary system.

Where is urea **most** concentrated?



October/November 2004 · Paper 1 · Variant 1 · Question 19

Correct answer: C

19 What is an example of excretion?

A release of adrenalin from the adrenal glands

B release of sweat from the sweat glands

C removal of carbon dioxide from the lungs

D removal of faeces from the alimentary canal

October/November 2004 · Paper 1 · Variant 1 · Question 20

Correct answer: D

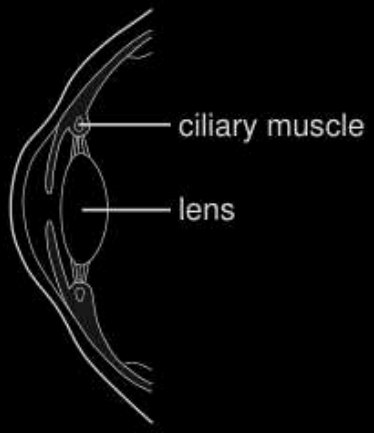
20 On a hot day how would these skin structures respond to help maintain a constant body temperature?

	sweat gland	surface blood vessels
A	decreased sweat production	contract
B	decreased sweat production	dilate (get wider)
C	increased sweat production	contract
D	increased sweat production	dilate (get wider)

October/November 2004 · Paper 1 · Variant 1 · Question 21

Correct answer: D

21 The diagram shows the front part of an eye.



Which changes take place in the eye when a person looks up from reading a book to view a distant object?

	ciliary muscle	lens
A	contracts	becomes thicker
B	contracts	becomes thinner
C	relaxes	becomes thicker
D	relaxes	becomes thinner

October/November 2004 · Paper 1 · Variant 1 · Question 22

Correct answer: A

22 A finger that touches a hot object is quickly taken away from the source of heat.

What is the role of relay neurones in this response?

A to carry nerve impulses within the spinal cord

B to generate impulses in the receptors of the finger

C to link the sense organs to the sensory neurones

D to pass nerve impulses out to the neurones

October/November 2004 · Paper 1 · Variant 1 · Question 23

Correct answer: B

23 Chemicals in tobacco smoke lead to the breakdown of the elastic tissue in the walls of the alveoli.

What is the name of this condition?

A bronchitis

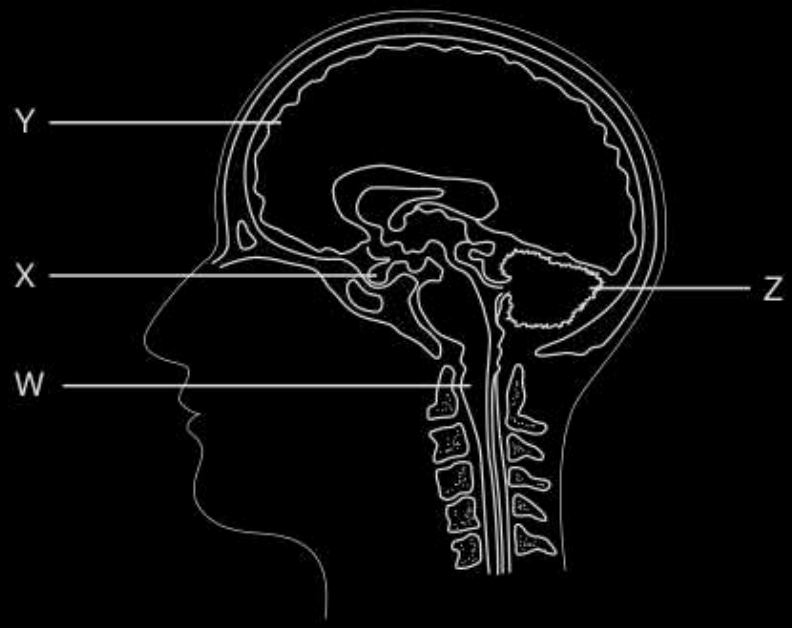
B emphysema

C heart disease

D lung cancer

Correct answer: C

26 The diagram shows a section through the head showing the brain and part of the spinal cord.



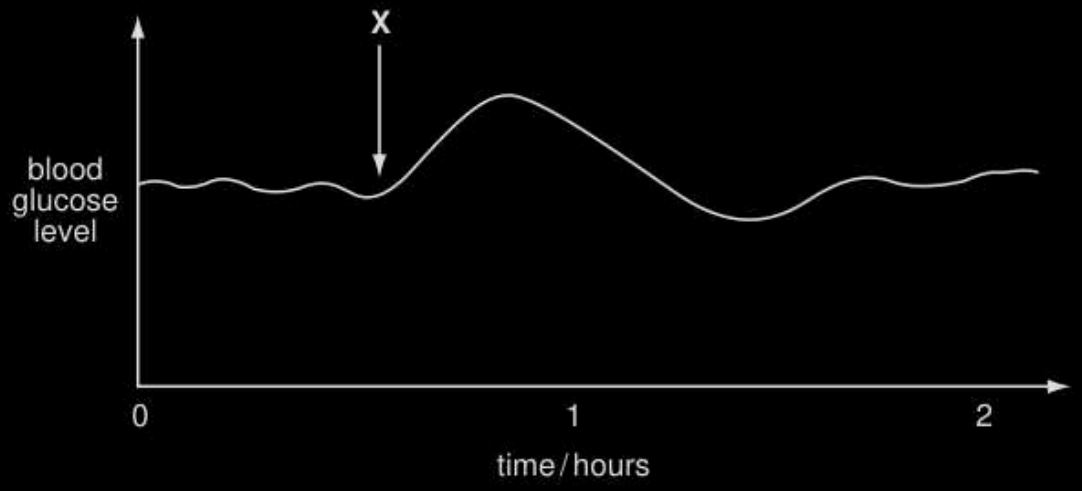
Which structures carry out the functions below?

	conscious thought and memory	control of balance and posture
A	X	W
B	W	X
C	Y	Z
D	Z	Y

October/November 2004 · Paper 1 · Variant 1 · Question 27

Correct answer: A

27 The graph shows changes in the glucose concentration in the blood of a person during two hours.



What explains the shape of the graph after **X**?

A The person has eaten a sugary sweet meal.

B The person has had an insulin injection.

C The person is suffering from diabetes mellitus.

D The person starts some hard physical exercise.

October/November 2004 · Paper 1 · Variant 1 · Question 35

Correct answer: B

35 Chimpanzees have 48 chromosomes in each normal body cell.

How many chromosomes does a chimpanzee gamete contain?

A 23 **B** 24 **C** 46 **D** 48

October/November 2004 · Paper 1 · Variant 1 · Question 38

Correct answer: A

38 What causes sickle cell anaemia?

A a change in the structure of a gene

B an iron-deficient diet

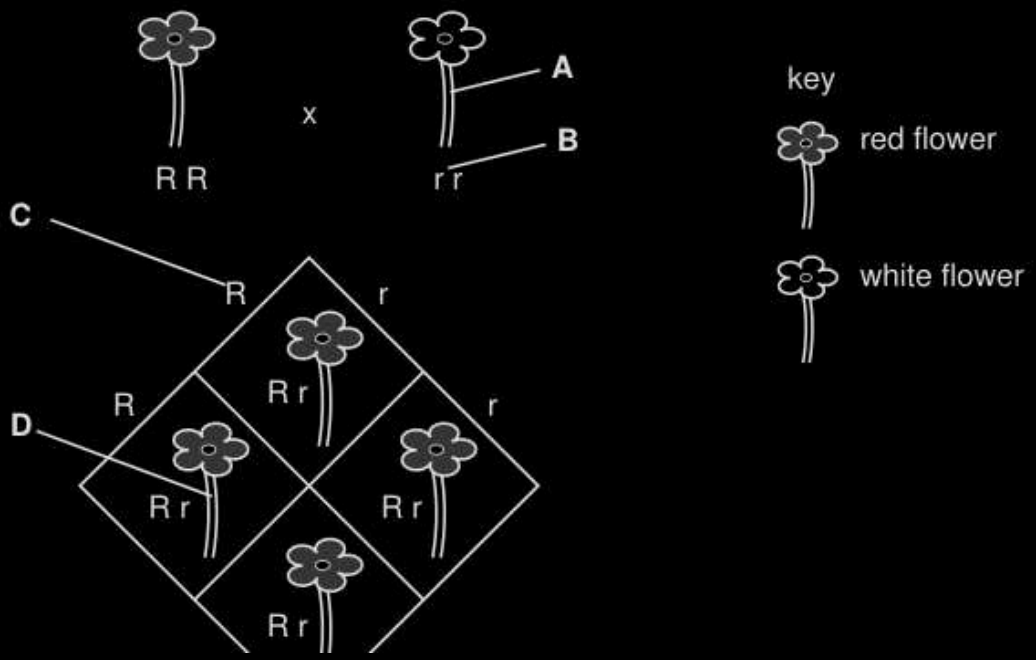
C infection following a bite from a mosquito

D loss of blood following an accident

Correct answer: D

39 The diagram shows a simple genetic cross between a red flower and a white flower.

Which represents the dominant phenotype?



October/November 2004 · Paper 1 · Variant 1 · Question 40

Correct answer: B

40 The genotype for the height of an organism is written as Tt.

What conclusion may be drawn?

A The allele for height has at least two different genes.

B There are at least two different alleles for the gene for height.

C There are two different genes for height, each having a single allele.

D There is one allele for height with two different forms.

University of Cambridge International Examinations is part of the University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

1 Fig. 1.1 shows a water plant and Fig. 1.2 shows a section through one of its leaves.

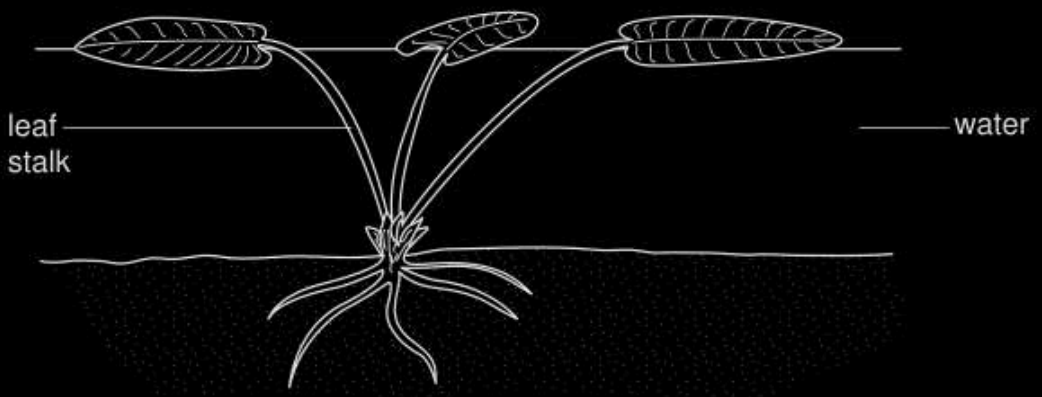


Fig. 1.1

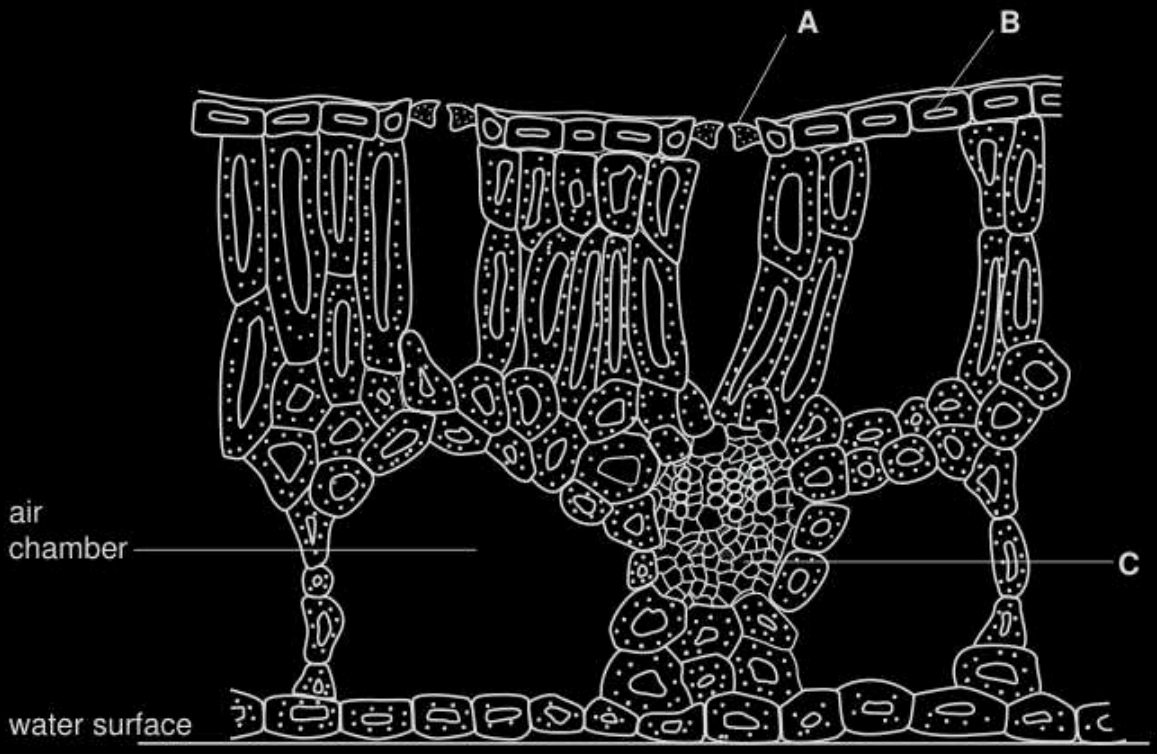


Fig. 1.2

(a) Identify the types of cell found at **A**, **B** and **C**.

A .....

B .....

C ..... [3]

(b) **(i)** Suggest a function of the air chambers in Fig. 1.2 related to maintaining the position of the leaf.

.....

(ii) State a different function of the air chambers which helps photosynthesis.

.....[2]

(c) State three ways in which the structure of this leaf differs from that of a typical plant growing on land.

1. .....

2. .....

3. .....[3]

(d) Fig. 1.3 shows the water plant from Fig. 1.1 as it looks when it is **not** in the water. Fig. 1.4 shows a plant growing on land.

Fig. 1.3

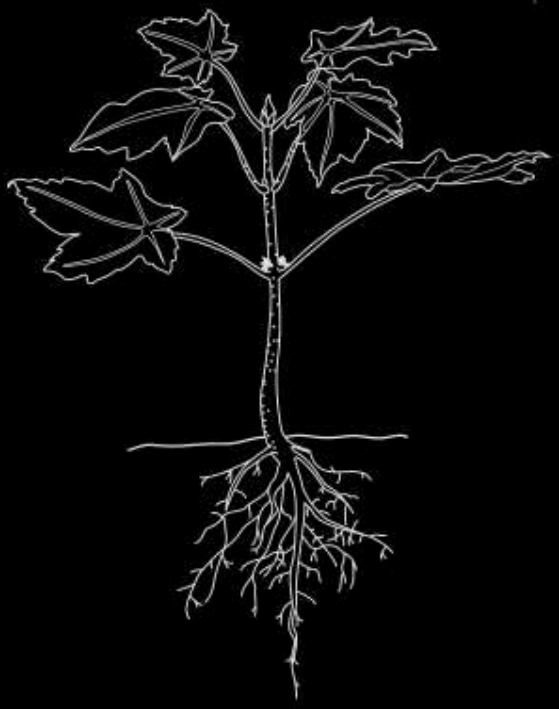


Fig. 1.4

Suggest an explanation for the difference in the strength of the leaf stalks of the water plant (Figs. 1.1 and 1.3) and the land plant (Fig. 1.4).

.....

.....

.....[2]

[Total: 10]

1 **(a)** **A** - guard cell ;

B - epidermis/al cell (**R** lower epidermis) ;

C - phloem/sieve tube (**A** companion) ; **3**

(b) **(i)** allows leaf to float AW/(maximum) exposure to light* ;

(**R** support unqualified)

(ii) diffusion/movement/collection/source/provides/gives AW + CO_2

OR (maximum) exposure to light* (*once only)

(Ignore references to oxygen, but **R** O_2 references if they refer to respiration) ; **2**

(**R** absorbs/takes in/references gas exchange)

(c) (Ignore references to leaf stalks and to spaces not interconnected)

stomata/guard cells (mainly) on upper surface AW ;

(or v.v.)

air spaces/chambers + palisade cells (or pos^{n} described) ;

chloroplasts/chlorophyll in epidermis (**R** upper epidermis) ;

reference cells in clumps v. cells loosely packed AW/
air chambers v. intercellular spaces/
large spaces v. small spaces (**R** more/fewer spaces) ;

no cuticle on lower surface ;

reference quantity of chloroplasts/chlorophyll in spongy cells ;
max. 3

(d) less/no + thickening/lignin/xylem/woody (or v.v.) ;

(R unqualified references to hard/rigid)

no need for support/support from water (or v.v.) ; **2**

(**A** floats on)

Total 10

2 In an experiment, a person looked at the same light source from various different distances. The diameter of their pupil was measured at each position. Fig. 2.1 shows how the diameter varied.

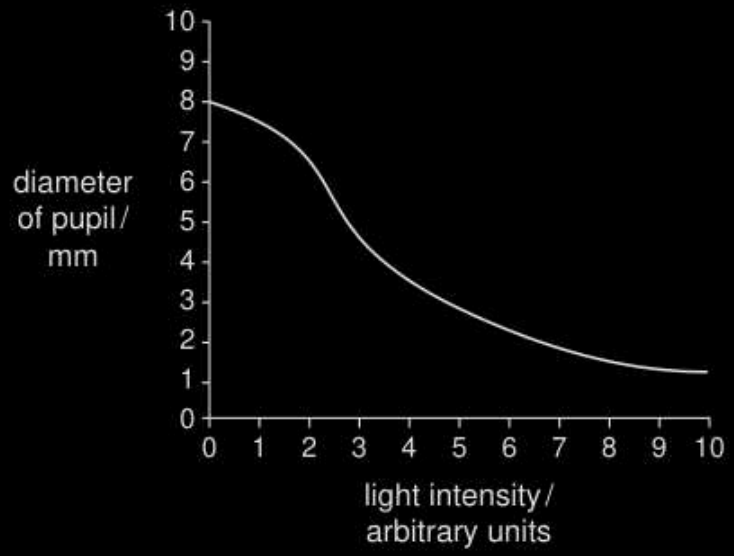


Fig. 2.1

(a) Describe the relationship between light intensity and pupil diameter shown in Fig. 2.1.

.....

.....

.....[2]

(b) What type of response accounts for this change in pupil diameter?

.....[1]

Use

(c) Explain how the response is brought about as the light intensity is increased from 2 to 4 arbitrary units.

.....

.....

.....

.....

.....

.....

.....[5]

(d) Albino people lack colouring material (pigment) in their bodies.
Suggest why albino people should avoid looking at bright lights.

.....

.....

.....

.....

.....[3]

[Total: 11]

2 **(a)** large(r) diameter at low light intensity/or v.v. ;

(**A** bigger/inversely proportional or description) (**R** proportional unqualified)

fastest rate of change around 2 - 4 a.u./
slowest rate of change/levels off at 7 - 10 a.u. ; **2**

(b) reflex/autonomic/automatic/involuntary ; **1**

(**R** spinal/conditioned)

(c) light sensitive/receptor (cells) or named/retina ;

neurones/nerve cells or fibres (**A** optic nerve) ;

impulses ;

contraction + circular muscles (**R** if reference ciliary) ;

relaxation + radial muscles (**R** if reference ciliary) ;

correct reference iris ;
max. 5

(d) no colour/pigment in iris/choroid (**R** eye) ;

permits internal reflection AW of light/too much light enters
eye/received by retina (**A** no shading/shielding/protection for retina) ;

damage to retina/receptors/light-sensitive + cells/visual impairment AW
(**R** damage to eyes) ; **3**

Total 11

3 Fig. 3.1 represents four genes on a part of the X chromosomes from a body cell of a woman.

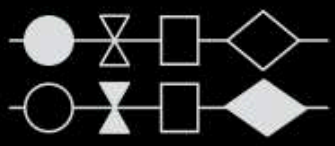


Fig. 3.1

(a) In the space below, draw these genes as they might appear in one of her ova (gametes).

[3]

(b) Fig. 3.2 shows the same section of an X chromosome in another cell of the same person in which the structure of one of the genes has changed.

Fig. 3.2

(i) State the name given to such a change in a gene.

.....[1]

(ii) Suggest a possible cause of such changes in genes.

.....[1]

(c) Fig. 3.3 shows the alleles of a person with blood group A.

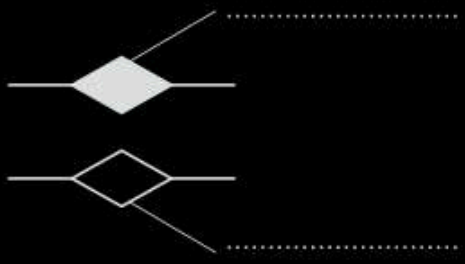


Fig. 3.3

Use

The symbols I^A, I^B and I^o can be used for the alleles controlling blood groups.

(i) Using these symbols, label the alleles in Fig. 3.3. [2]

(ii) Explain why a person with blood group AB would not be expected to have an offspring with group O blood.

.....

.....

.....

.....

.....[3]

[Total: 10]

3 **(a)** one chromosome shown - in a string (mark the first) ;

genes matching in shape and sequence (**A** reversed) ;
(the appropriate 4 may be selected from a string of more than 4)

gene 3 not shaded (all others must be uniform black or white) ; **3**
(gene 2 if the chromosome has been reversed)

(b) **(i)** mutation (ignore reference chromosome) ; **1**

(ii) mutagen (or named)/reference change in DNA structure ; **1**

(**A** any plausible e.g. - radiation or named (α -/γ -/X-rays)/chemicals /u.v./sunlight/carcinogens/smoking/viruses) ;

(**R** heat/infra-red/disease)

(c) **(i)** I^A ;

I^o (allow in either order) ; **2**

(ii) O/I° from partner/offspring must be I° I° or OO ;

A/I^A or B/I^B from the person/person cannot supply I°/O ;
(must have reference to both alleles)

I^A and I^B are dominant* (to I°) / I° recessive* (to both) ; **3**
(*AW) (**A** references to A/B/O without I)

Total 10

6 **(a)** Explain how the heart functions as a pump and keeps blood flowing in only one direction. [7]

(b) Explain how the difference in the pressure of the blood in the pulmonary artery and in the aorta is related to **(i)** the structure of the ventricles and **(ii)** where the blood is going. [3]

[Total: 10]

6 **(a)** correct reference atria(um)/auricle(s) ;

correct reference ventricle(s) ;

muscles/muscular + contract(ion) (**R** pushing/forcing pumping - in Q.) ;

reference thickness of ventricular compared with atrial walls ;

atrio-ventricular/identified valve(s) (open) + blood passes ;

close + to prevent return of blood ;

tendons/cords/(**R** heartstrings) + action/function of ;

reference aortic valves + their action (**A** close prevent backtflow) ;

cycle repeated/idea of co-ordinated action; ;

max. 7

(b) right (ventricle) wall thinner/left (ventricle) wall thicker OR reference less/ more muscle OR weaker/stronger contractions ;

(**A** smaller—Larger)

(pulmonary) shorter distance to travel (**A** only to the lungs) (or v.v.) ;

little work to do against gravity (the idea of) (or v.v.) ;

avoidance of damage to lung capillaries/low pressure required in lungs ;

(body) high pressure for kidney filtration ;

oxygen/glucose to brain ;

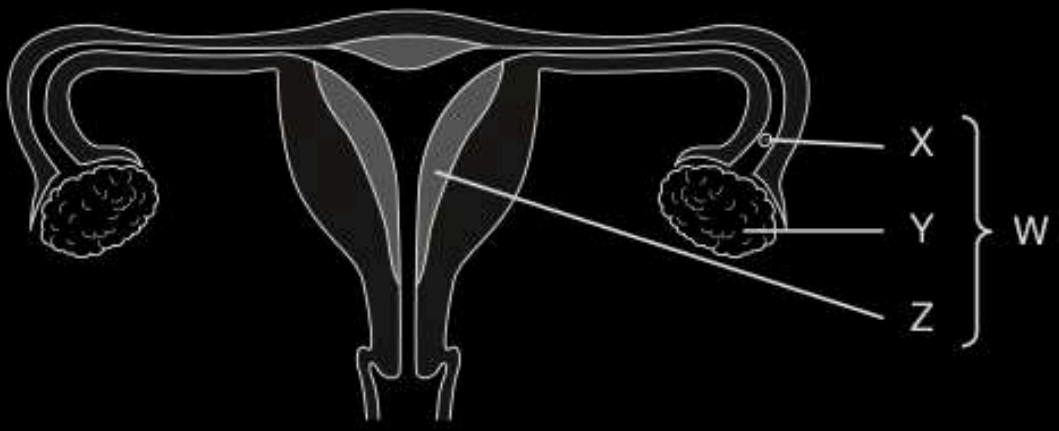
max. 3

Total 10

May/June 2004 · Paper 1 · Variant 1 · Question 1

Correct answer: B

1 The diagram shows the female reproductive system.



Which level of organisation are the structures W, X, Y and Z?

	cell	tissue	organ	organ system
A	W	Y	X	Z
B	X	Z	Y	W
C	Y	X	Z	W
D	Z	W	Y	X

May/June 2004 · Paper 1 · Variant 1 · Question 2

Correct answer: A

2 Which is an example of active transport?

A movement of glucose into the cells of the villi

B movement of glucose molecules down a concentration gradient

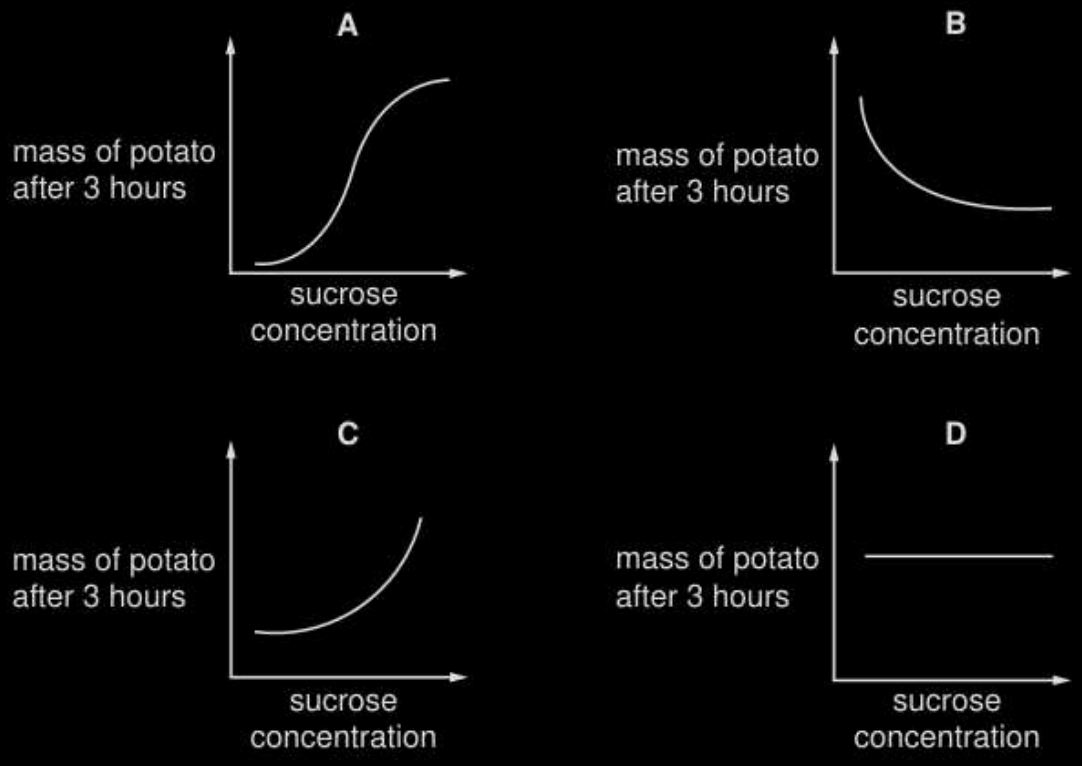
C movement of ions in blood plasma

D movement of water in the transpiration stream

Correct answer: B

3 Similar shaped pieces of potato are placed in sucrose solutions of different concentrations. After three hours, the mass of each potato piece is measured.

Which graph shows the results of this experiment?

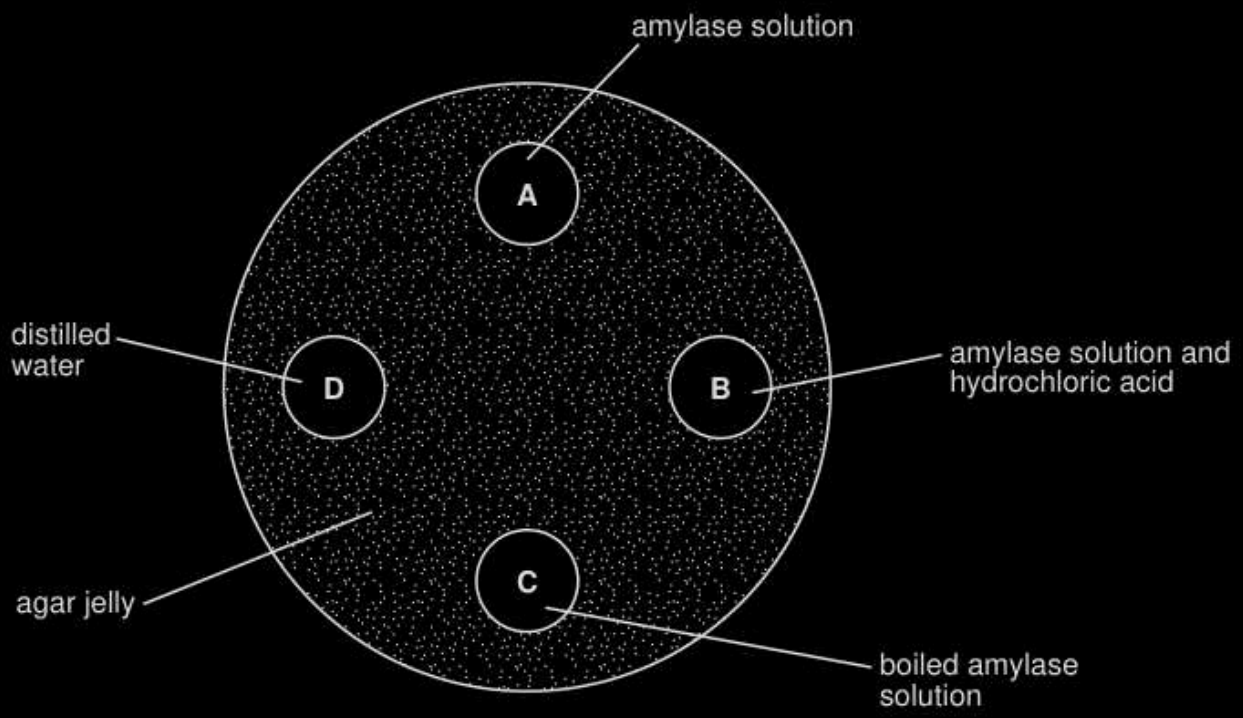


May/June 2004 · Paper 1 · Variant 1 · Question 4

Correct answer: A

4 A dish is filled with agar jelly containing starch. Four holes are cut in the jelly and each hole is filled with the different substances shown.

Which hole will be surrounded by the largest area without starch after 30 minutes?



May/June 2004 · Paper 1 · Variant 1 · Question 8

Correct answer: C

8 Two samples of food are tested. The results are shown in the table.

	test used		
	iodine solution	Benedict's test	biuret test
sample 1	brown	orange	blue
sample 2	blue / black	blue	violet

What do these results show?

A Sample 1 contains starch and sugars.

B Sample 1 contains starch only.

C Sample 2 contains starch and protein.

D Sample 2 contains protein only.

May/June 2004 · Paper 1 · Variant 1 · Question 9

Correct answer: A

9 The pH in the mouth decreases after eating.

Which statement explains the decrease in pH?

A Bacteria release acids when respiring food substances.

B Enzymes in saliva release acids during digestion.

C Food substances become alkaline when chewed.

D Salivary glands release an alkaline solution.

May/June 2004 · Paper 1 · Variant 1 · Question 11

Correct answer: D

11 When a leafy shoot is placed in a solution of dye, which part becomes most heavily stained?

A guard cells in the leaves

B palisade cells in the leaves

C phloem cells in the stem

D xylem vessels in the stem

May/June 2004 · Paper 1 · Variant 1 · Question 13

Correct answer: D

13 What is the correct route for blood flow in a human?

A left atrium → left ventricle → lungs → right ventricle → right atrium

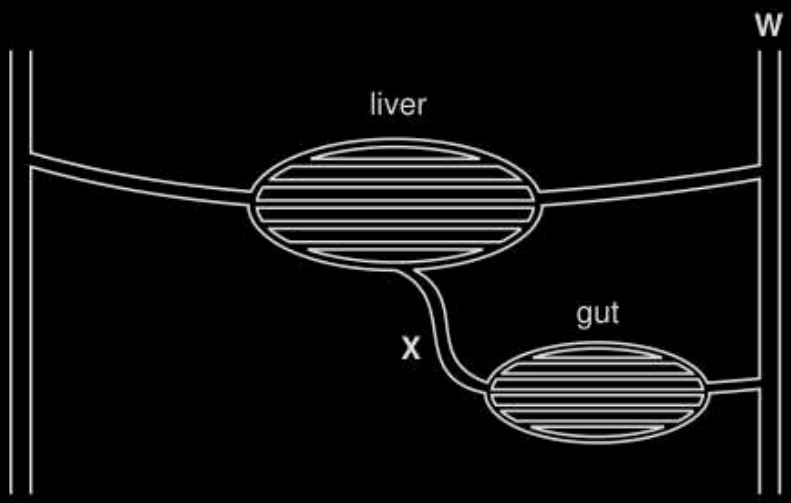
B left atrium → left ventricle → right ventricle → right atrium → lungs

C right atrium → right ventricle → left ventricle → left atrium → lungs

D right atrium → right ventricle → lungs → left atrium → left ventricle

Correct answer: B

14 The diagram shows the liver and its blood supply.



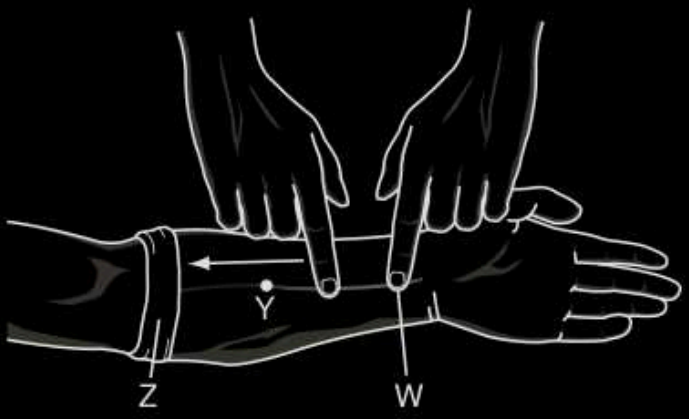
What is the name of blood vessel **X** and which substance increases in concentration between vessels **W** and **X**?

	name	substance
A	hepatic artery	carbon dioxide
B	hepatic portal vein	carbon dioxide
C	renal artery	oxygen
D	renal vein	oxygen

May/June 2004 · Paper 1 · Variant 1 · Question 15

Correct answer: C

15 The diagram shows the investigation of blood flow in the veins of the lower arm.



A cloth is tightly wrapped round the arm at point Z and the veins stand out clearly. One finger presses on the vein at W.

When another finger strokes the vein, as shown in the diagram, the vein lies flat between points W and Y.

Some possible explanations are listed.

1 The bandage at Z prevents backflow of blood.

2 The finger pressed at W prevents more blood entering the vein

3 A valve at Y prevents backflow.

4 A valve at Z prevents more blood from entering the vein.

Which explanations of the vein lying flat are correct?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

May/June 2004 · Paper 1 · Variant 1 · Question 16

Correct answer: D

16 After muscular exercise, which blood vessel carries the most carbon dioxide?

A aorta

B hepatic artery

C pulmonary vein

D vena cava

May/June 2004 · Paper 1 · Variant 1 · Question 17

Correct answer: B

17 Which process, occurring in the human body, does **not** involve energy from respiration?

A contraction of heart muscle

B diffusion of oxygen from the alveoli into the blood

C digestion of bread

D maintaining a constant body temperature

May/June 2004 · Paper 1 · Variant 1 · Question 19

Correct answer: A

19 Which substance has the same concentration in dialysis fluid as in blood?

A glucose

B protein

C salts

D urea

May/June 2004 · Paper 1 · Variant 1 · Question 20

Correct answer: C

20 What happens when the core temperature of the body increases?

	diameter of surface blood vessels	urine production
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

May/June 2004 · Paper 1 · Variant 1 · Question 21

Correct answer: A

21 Which diagram shows how light from a distant object is focused on the retina to form a clear image?

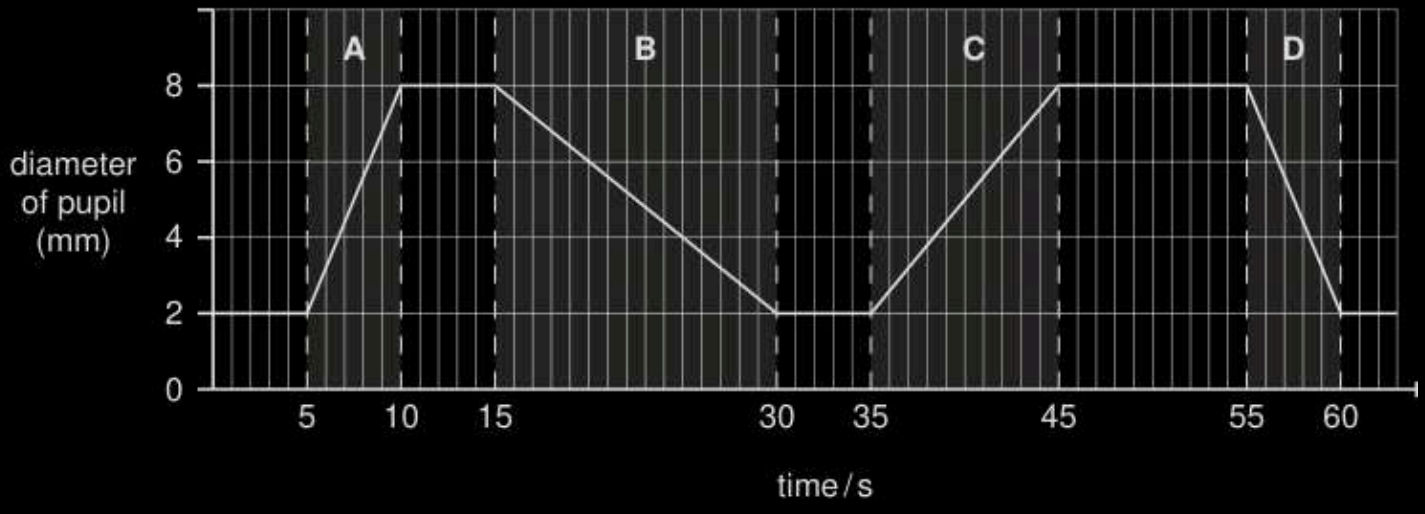
A **B** **C** **D**

May/June 2004 · Paper 1 · Variant 1 · Question 22

Correct answer: D

22 The diameter of a person's pupil is measured as the light intensity is varied.

During which time period does the light intensity increase fastest?



May/June 2004 · Paper 1 · Variant 1 · Question 25

Correct answer: C

25 Which organ is damaged as it breaks down alcohol?

A brain

B kidney

C liver

D stomach

May/June 2004 · Paper 1 · Variant 1 · Question 34

Correct answer: C

34 Which statement about sexual reproduction is correct?

A All types of organism reproduce by this process.

B Many cells of one type fuse with a single cell of another type.

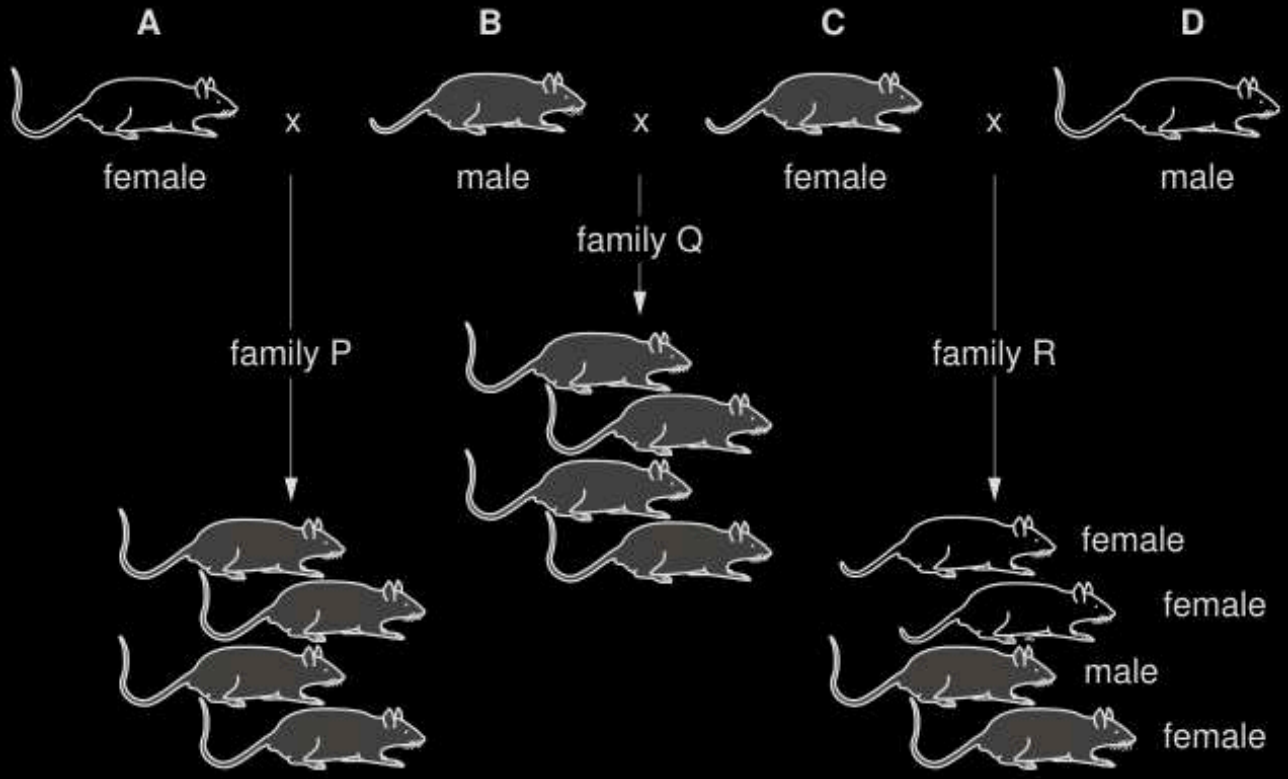
C Nuclei of two specialised cells fuse together.

D Parents produce genetically identical offspring.

Correct answer: C

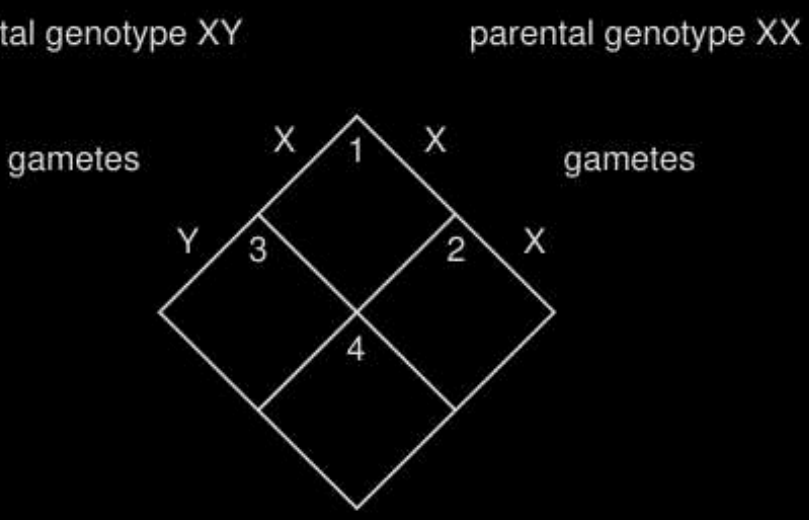
39 The diagram shows the inheritance of coat colour in mice.

Which mouse is heterozygous for coat colour?



Correct answer: D

40 The diagram shows the inheritance of sex in humans.



Which sex are the offspring in boxes 1, 2, 3 and 4?

	1	2	3	4
A	boy	girl	boy	girl
B	boy	girl	girl	boy
C	girl	boy	girl	boy
D	girl	girl	boy	boy

2 In an experiment to investigate starch production by a plant, three similar plants, each with variegated (green and white) leaves were set up as shown in Fig. 2.1.

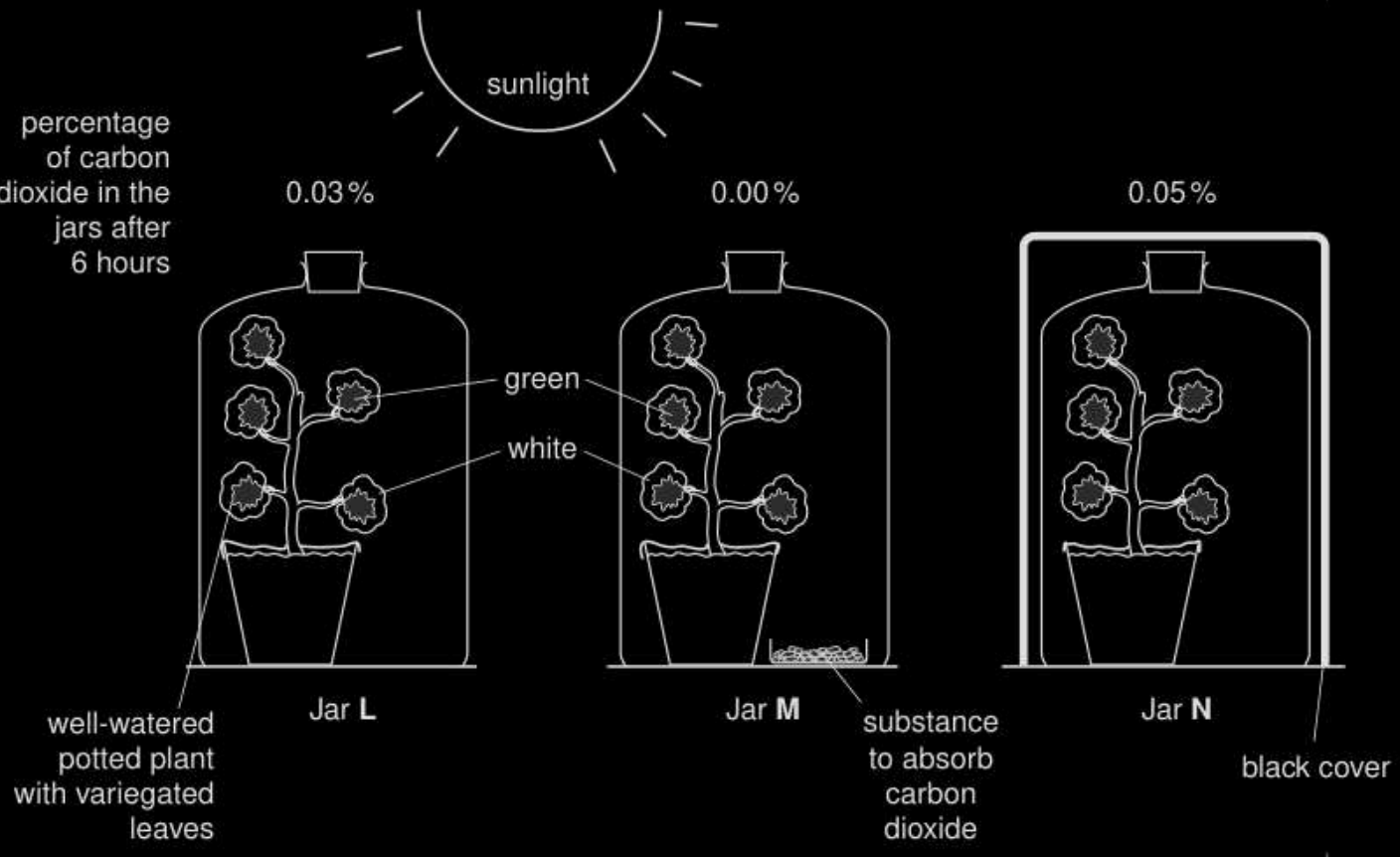


Fig. 2.1

(a) Name the process that produces starch in the leaves.

..... [1]

(b) At the start of the experiment, each jar contained atmospheric air.

Name a gas, other than oxygen and carbon dioxide, which was present in the air inside the jars.

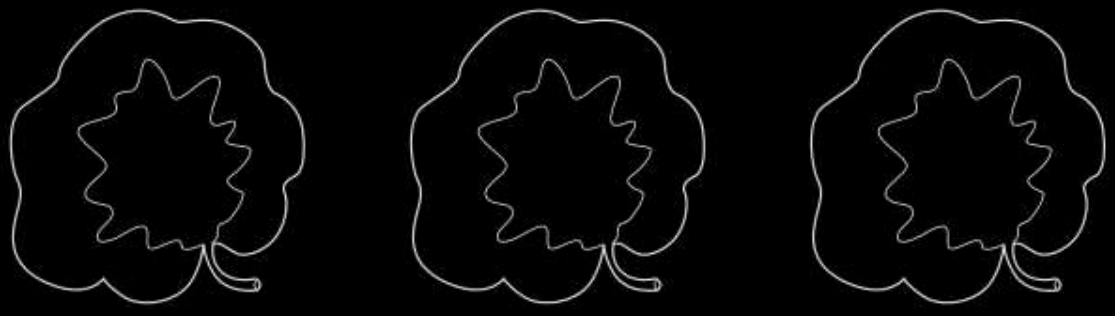
..... [1]

(c) Explain how the conditions in Jar **L** make it a control.

.....[1]

Use

(d) At the end of the experiment, a leaf was taken from each plant and tested for the presence of starch. On the outlines in Fig. 2.2, **clearly label** the colours of each leaf after the starch test. Do **not** colour in the leaves.



leaf from **L** leaf from **M** leaf from **N**

Fig. 2.2

[3]

(e) When the air was first trapped under the jars, it contained 0.04% carbon dioxide. For each of the jars, explain why this percentage has changed by the end of the experiment.

Jar **L** .....

.....

Jar **M** .....

.....

Jar **N** .....

.....[6]

[Total: 12]

2	(a)	photosynthesis (Ⓐ condensation reaction AW);	;	1
	(b)	nitrogen/water vapour/valid e.g.(® hydrogen, ignore symbols)	;	1
	(c)	contains all required/CO_2, H_2O and light + for P/S AW	;	1
		(Ⓐ yellow/brown/iodine colour, R white)		
	(d)	(L) blue/black centre + orange AW round outside	;	
		(M) orange (or colour e.c.f.) AW all over	;	
		↕ (Ⓐ blue/black centre if ref. to plant not destrached)		
		(N) orange (or colour e.c.f.) all over	;	3
		N.B. For all of (d), Ⓐ heading to leaf as label for whole leaf. Something must be written on/above leaf to score except if colours used. Colouring only = max 2.		
	(e)	(L) photosynthesis/CHO (or named) production	;	
		uses up CO_2	;	
		(M) respiration	;	
		CO_2 released AW	;	
		Absorbed (by substance)	;	
		(N) respiration/noP/S	;	
		CO_2 released AW	;	max. 6
				Total [12]

3 Fig. 3.1 shows a diagram of the human brain and Table 3.2 shows the functions of some parts of the brain.

Fig. 3.1

Table 3.1

part of brain	function
P	controls body temperature
Q	is the master hormone-producer
R	controls unconscious activities such as heart-beat
S	helps to control balance and give co-ordination
T	memory storage and conscious behaviour

(a) Label Fig. 3.1 using the letters **P** to **T** from Table 3.1. [5]

(b) One of the hormones produced by **Q** regulates growth and the development of the reproductive organs.

(i) Explain how a hormone made in the brain can have its effect in the reproductive organs.

.....

.....[1]

(ii) Suggest possible effects on a child of the region **Q** producing unusually high amounts of this hormone.

.....

.....

.....[3]

[Total: 9]

3 **(a)** 5 parts correctly labelled

same label to 2 Different parts = 0,

2 labels to same part -mark the correct one.

(line must not stop short)

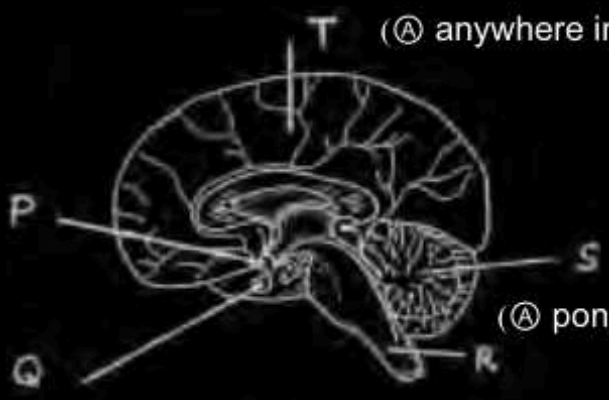


Fig. 3.1

(Ⓐ anywhere in cerebrum)

(Ⓐ pons/brain stem)

;;;;; **5**

(b) **(i)** (carried by) <u>blood</u>/<u>plasma</u> ; **1**

(ii) early maturity AW ;

tall AW/<u>rapid</u> growth ;

early appearance of any 2 secondary sexual characteristics ;; **max. 3**

Total [9]

4 Fig. 4.1 is a flow-diagram showing the pathways taken by oxygen and carbohydrate from their absorption into a mammal's blood to their use in the liver.

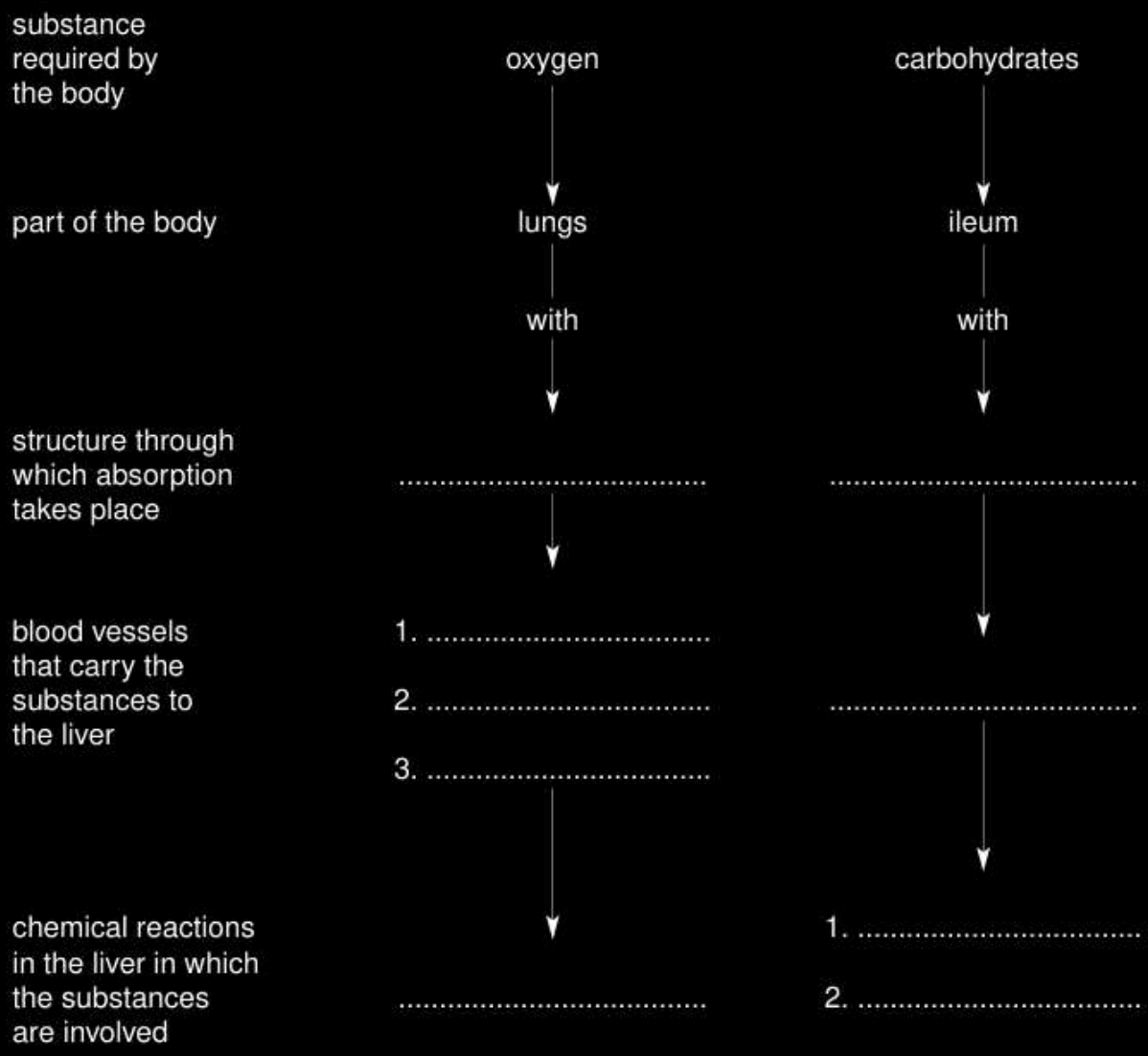


Fig. 4.1

(a) By filling in the spaces, complete Fig. 4.1 to state

- the structures involved
- the blood vessels used
- what happens in the liver cells. [8]

(b) State a waste product formed from proteins in the liver. .....[1]

(c) Suggest

(i) a chemical element present in the waste product you mention in **(b)** that is also present in a fat;

.....

(ii) a chemical element present in the waste product that is **not** normally found in a fat.

..... [2]

[Total: 11]

4 **(a)** alveoli/capillaries/air sacs ; (micro) villi ;

1 pulmonary vein ;
2 aorta } (in any order) ; (H).P.V. ;
3 hepatic artery ;

respiration/oxidation of glucose ; 1/2 respiration ;

1/2. conversion from/to glycogen ; **max. 8**

(b) urea/uric acid (Ignore nitrogenous waste) ; **1**

(c) **(i)** H(hdrogen/C(arbon/O(xygen)q ;
(ii) N(itrogen) ; **2**

Total [11]

May/June 2004 · Paper 2 · Variant 1 · Question 6

6 **(a)** Explain how xylem is suited to its functions in a plant. [7]

(b) Suggest why some insects that are parasitic on plants obtain their food from the phloem, rather than from the xylem. [3]

[Total: 10]

6	(a)	long/root to leaves or stem	;	
		narrow/thin/capillary-like	;	
		pipe-like/hollow/tubular/no end-walls/no cytoplasm/continuous	;	
		water carriage	;	
		mineral s(alts)/ions/nutrients	;	
		thickened/strengthened/lignin (® strong/hard/rigid)	;	
		significance of position in root/stem	;	
		support AW/prevents collapse of vessel	;	
		prevents tearing/spreads out + leaf	;	**max. 7**
	(b)	sugar/sucrose/CHO* (® starch/glucose) ↕ (Ⓐ first two in a list) amino acids* (* or v.v. for saying not present in xylem)	;	
		for energy	;	
		and growth	;	
		phloem unthickened or softer AW/insects can penetrate wall	;	
		nearer the outside	;	**max. 3** **Total [10]**

May/June 2004 · Paper 2 · Variant 1 · Question 8

8 Either

(a) Explain how nitrogen in the muscle protein of a herbivore may be re-cycled to form protein in another herbivore some years later. [7]

(b) Explain how the activities of some bacteria form a part of both the carbon and nitrogen cycles. [3]

[Total: 10]

Or

(a) Explain what is meant by the terms

(i) *gene*;

(ii) *allele*.

[4]

(b) Describe the part played by genes in the process of evolution. [6]

[Total: 10]

8E	**(a)**	breakdown/decay/decomposed AW	;	
		urea/<u>dead</u> animal	;	
		by bacterial/fungi/<u>named</u> decomposer (Ⓐ saprotroph etc.) (Ⓐ denitrifying bacteria)	;	
		(protein) to amino acids ↕ (Ⓐ fixation to salts by lightning/to amino acids by N_2-fixing bacteria)	;	
		(amino acids) to salts (or named) (Ⓐ ammonia)	;	
		absorbed by plants (® if NH_3, proteins, amino acids)	;	
		for protein manufacture/ref. protein in plants	;	
		eaten by animals	;	
		digestion + absorption	;	
		<u>assimilation</u> (® turned into protein)	;	**max. 7**
	(b)	bacteria/they + stated activity in N-cycle (e.g. fixation / decomposition / (de)nitrification)	;	
		respire	;	
		release carbon dioxide	;	
		(CO_2 + nitrates) – starting point for protein synthesis	;	**max. 3**
				Total [10]

8O (a) (gene)
a section of DNA/chromosome ;

controls production of a protein/or a characteristic or e.g. ;
(Ⓐ feature/phenotype)

can be copied ;

passed on /(unit of) inheritance ; **max. 3**
(for gene)

(allele)
(Ⓐ sort/type)
a form of a gene/ref. upper + lower case letters, or e.g./
pair of phenotypic examples ;

on homologous AW chromosomes/at same locus AW ;
(Ⓐ a pair of)

the idea of dominance/recessiveness/codominance/
can have different effects ; **max. 4**
[for **(a)**]

(b) are inherited/ref. reproduction ;

ref. mutation/change in gene ;

producing variation/differences/changes in appearance or in
behaviour or in phenotype ;

advantageous/useful/better adaptation ;

survival ;

change in environment ;

long period of time ;

change in phenotype ;

ref; competition ;

ref. natural selection ; **max. 6**

N.B. Accept and apply scheme as appropriate to specific examples.

Total [10]